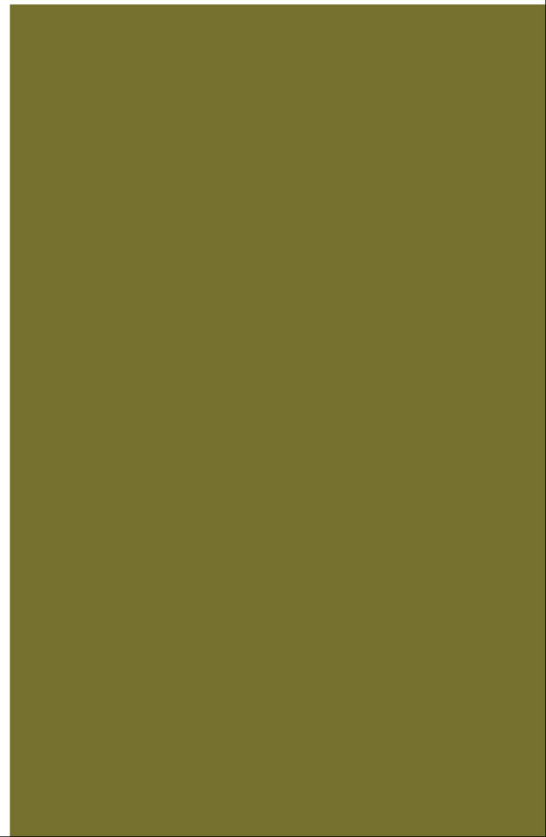
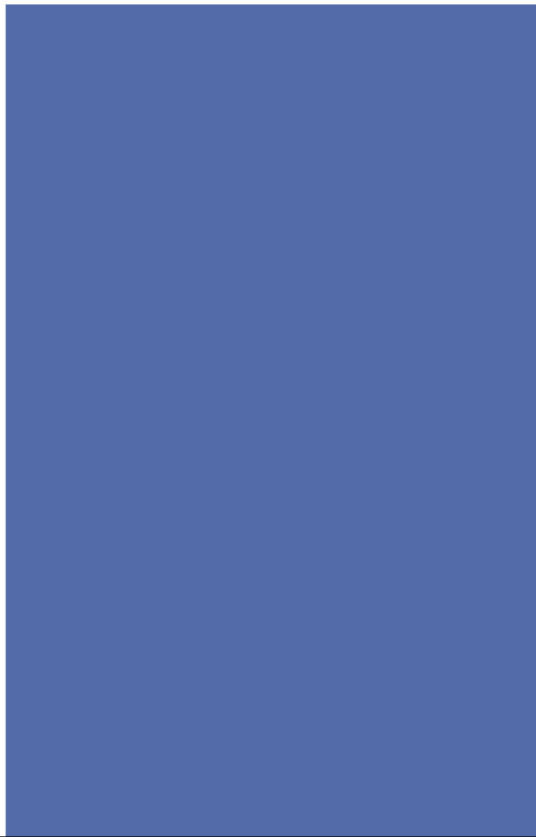


Robert C. Eklund | Gershon Tenenbaum
Editors

Encyclopedia of
Sport and
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Psychology



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Introduction

Sport psychology and exercise psychology are two closely interrelated fields of scientific study and professional practice. Human movement is central in both instances. In general, competitive physical activities characterized by athleticism and/or physical skill are typically categorized as sport, particularly when governed by rules and conventions to ensure fair competition and reasonably clear and consistent determinations of competitive outcomes. Exercise, again speaking broadly, involves planned and regular engagement in physical activity for the purpose of improving or maintaining physical fitness. The categories of “sport” and “exercise,” however, do not encompass the entirety of possibilities available in physical activity. This is a matter that is fairly obvious given that not all physical activity inherently, or necessarily, involves competing or efforts to improve or maintain fitness. Research on both sport psychology and exercise psychology has tended to be inclusive and definitely elastic on this account as long as the participants are engaged in physical activity. Nonetheless, as noted in Diane Gill and Erin Reifsteck’s entry on the history of exercise psychology, sport psychology and exercise psychology have essentially always been deeply interconnected despite the relatively recent interest in making more emphatic subdisciplinary distinctions. The ambiguities involved in making these distinctions have generally been addressed by acknowledging that the psychological aspects of particular subcategories of physical activity warrant recognition—that the uniquenesses are not entirely trivial even if the commonalities are very substantial.

Psychological factors have long been implicated as influencing, and being influenced by, involvement in sport and exercise. As discussed in entries on the histories of sport psychology and exercise psychology, and as is evident in the

timeline included in the encyclopedia’s appendix, commentary connecting engagement in human movement activity and psychological factors can be identified across millennia. The emergence of systematic scientific study of the psychology of human movement endeavors, however, is relatively recent. In the last half century, the acceleration, accumulation, and span of quality research-based knowledge on human behavior in the area has been impressive. The research knowledge conveyed in the encyclopedia stems from the many areas and traditions of psychological inquiry pursued in sport and exercise psychology. Findings from investigations involving participants relevant for the given research questions being examined in experimental, quasi-experimental, and descriptive correlational designs as well as findings from qualitative, naturalistic, and observational inquiry are featured across entries in this compendium.

Although sport and exercise psychology is fundamentally grounded in scientific study of human behavior in physical activity settings, it is also inherently an area of applied practice. Simply seeking to acquire knowledge and understanding on psychological influences and consequences of sport and exercise involvement without any applied application would be an incredibly sterile endeavor. In fact, the growth and acceptance of the practice of applied sport and exercise psychology has been no less impressive than the accelerating growth of research-based knowledge. Importantly, the developments in sport and exercise psychology research and applied practice have been synergistic in nature with both research informing practice and practice informing research in some meaningful degree. Clearly there is variation in opinions on whether or not the science of sport and exercise psychology is too distant from (and hence of limited relevance to) applied practice in the area, as well as whether or not applied practice

in sport and exercise psychology is too distant from, and insufficiently grounded in, research in the area. It is, nonetheless, reasonable to suggest that the applied and research spheres within sport and exercise psychology are not entirely disconnected. Certainly, however, the days when athletes and coaches shunned the presence of applied sport psychologists are now largely in the past. Applied sport psychologists now enjoy more generalized acceptance and valuing of their potential contributions. There is now broader recognition than in past decades that individuals with relevant sport psychology expertise can offer avenues for optimizing both on-field performance and athlete health and well-being. Although not everyone would agree, we suggest that this recognition has also opened doors for more extensive and intensive research in applied settings, and the fruits of that applied research have increasingly provided credibility and greater trust in the applied practice of sport psychology. Certainly parallel developments and gains in professional acceptance can be observed for applied exercise psychologists as well, and for the vibrancy brought by healthy synergies across research and applied endeavors.

Rationale for the *Encyclopedia of Sport and Exercise Psychology*

Given the previously mentioned rapid growth and acceptance of research and applied practice during the relatively short history of formal recognition of these subdisciplinary areas, it is not surprising that a variety of substantial scientific journals and a cavalcade of books and scholarly resources have also come into existence. Nonetheless, a single authoritative, comprehensive resource explicating and supporting exploration of topics, constructs, issues and theories in the field had not yet been compiled at the time that we started contemplating the possibilities and challenges involved in undertaking such a tantalizing project. This two-volume *Encyclopedia of Sport and Exercise Psychology* is the result of our contemplations and the input received from the membership of the stellar advisory board assembled to guide our efforts in this daunting task. Of course, the real substance of the compendium lies in the 300+ authoritative entries written by researchers with expertise particular to the topics of focal

interest in each instance—researchers and practitioners who are typically among the world leaders in their areas of expertise. The notable enthusiasm of these authors for this project was gratifying. We believe that readers will find the results of the authors' efforts to be edifying in their utility for answering questions of interest and in expanding understanding of the state of knowledge in the area.

Questions on sport and exercise participation relating to psychological matters are myriad. For examples: How do athletes overcome fears, mental blocks, or injuries and how do they deal with stress and anxiety? Does engaging in exercise really make people feel good? What influence do competitors, teammates, audiences, parents, and coaches have on athletic performance? How do I get my spouse, children, friends, patients, or colleagues to be more physically active? What specific psychological techniques have proven effective in mental training for peak performance, maintaining concentration, motivation, and competitive drive? How can an athlete enhance his or her commitment to a training regimen, or how might the average person better adhere to a regime of exercise intended to advance health and well-being? What are the psychological risks and consequences associated with serious competition, training “too much,” using performance-enhancing drugs, or training under aversive conditions? Does exercise really build self- and body-esteem? How does emotional experience in sport and exercise play into the behaviors exhibited by physical activity participants? Does sport involvement build character or does it really undermine moral development? How do coaches provide effective leadership for their athletes? Though final and complete answers to such questions may never be entirely at hand, the existing knowledge base is substantial and hence answers—some tentative, others more maturely concrete—are available in many areas. The *Encyclopedia of Sport and Exercise Psychology* was assembled to provide readers with a single authoritative, comprehensive resource to explore state-of-the-art scientific knowledge in seeking answers to the array of possible questions that might be asked, as well as to spur interest in areas beyond the specifics of any particular question that someone might have initially turned to the encyclopedia for an answer. The entries contained in this two-volume compendium explore the theory, research, and application

of psychology as they relate to sport and exercise behavior.

The Audience of the Encyclopedia

The entries in this compendium convey information in a manner that is open, accessible, and relatively jargon-free. This stands in stark contrast with other available primary and secondary resources reporting scientific findings in the field where intimate knowledge of the specific topic is required to fully understand the issues and findings being addressed. In the encyclopedia, effort has been invested by the authors, advisory board members, and editors to ensure that the information is accessible to any reasonably well-educated reader having an inquiring mind regardless of the topical focus of that education. Readers with backgrounds in psychology (particularly health, exercise, and sport psychology), physical education, kinesiology, athletic training, and related fields will likely find the information more accessible than readers having little or no topical exposure to the area, but overall the entries have been shaped to convey information to as broad an audience as possible. The intent in having authors convey information in this manner was to help a wide array of interested readers to better understand human behavior in sport and exercise settings while avoiding the pitfalls engendered by an overly ponderous reliance on scientific jargon or area-specific terminology.

Content of the Encyclopedia

The entries in the encyclopedia span the spectrum of topics in the scientific study and applied practice of sport and exercise psychology, as well as on matters of foundational importance. The content of the encyclopedia can reasonably be captured within the 18 topical categories identified below.

- Career Transition
- Certification, Credentialing, and Roles of Sport and Exercise Psychologists
- Disability
- Emotion
- Exercise Health
- Group Dynamics
- History and Foundation

- Leadership
- Morality, Aggression, and Ethics in Sport
- Motivation
- Motor Control
- Perception and Cognition in Sport
- Personality and Psychological Characteristics in Sport
- Psychobiology
- Psychological Skills and Interventions
- Psychosociocultural Considerations
- Self-Concept, Self-Perceptions, and Identity
- Youth Sport

It is important to note that these categories are not mutually exclusive. Many individual entries cut across more than one topical area and some cut across many. Emotional experiences in sport and exercise, for example, have motivational implications, but motivation and emotion are not topically one and the same thing. Moreover, both also impact upon and are shaped by self-concept, self-perceptions, identity, and so on but not necessarily in the same ways. Identification of topical categories that were entirely independent of one another would have been impossible simply because of the nature and complexity of the human psyche. It should also be intuitively obvious that some categories are much broader than others. Motivation, for example, is fundamental to all human behavior, but commentary on, for example, disability or youth sport, is inherently focused more specifically.

Entries relating to these topics range from approximately 1,000 to 3,000 words in length. Some are very broadly focused and theoretical in nature (e.g., entries describing key psychological theories employed in sport and exercise psychology) while others are much narrower and more concrete in focus (e.g., use of humor in sport or on the nature of preperformance routines). Specific entry lengths were largely dictated by topical breadth and/or extent of available scientific or professional practice evidence.

Organization of the Encyclopedia

The 300-plus signed entries appearing in the encyclopedia are organized alphabetically for easy location of specific entries. A thematic Reader's Guide is provided in the front matter so that the specific entries can be identified relative to the

primary topics they were associated with during development of the encyclopedia's content. Cross-references within the encyclopedia are provided with all entries to ensure topically related entries are brought to the reader's attention. Moreover, suggestions for further readings are provided with every entry so that avenues for deeper topical exploration are readily identifiable to satisfy the interested reader. Finally, a representative timeline on the History of Sport and Exercise Psychology is provided in the Volume 2's appendix.

How the Encyclopedia Was Created

This two-volume encyclopedia is the manifestation of a multistep process. Our initial considerations were focused upon developing a suitable keyword "skeleton" to serve as the roadmap for our efforts to create the encyclopedia. The first step in creating this skeleton involved collation of subject index entries from prominent textbooks in sport and exercise psychology, motor learning, and motor control, as well as article keywords appearing in prominent journals in sport and exercise psychology. This list of candidate keywords relating to constructs, theories, effects, and so on was subsequently reduced to a more manageable length by the removal of obvious redundancies, and through omission of potential keywords deemed too idiosyncratic or obscure for inclusion. We completed this initial phase of the process by conducting an "intuitive factor analysis," subjectively lumping potential keywords together that seemed to have topical coherence. Armed with this array of potential keywords and a potential topical organization to guide our invitations, a seventeen-member advisory board was recruited. As an aside, recruitment of the advisory board was far easier than expected given the amount of work involved in the project. Seventeen of 18 scholars on our list of candidates accepted our invitations—most were accepted immediately and the balance, a small minority, were accepted following relatively brief periods of contemplation.

The keyword lists generated by the editors for each potential topical area were first forwarded to the advisory board member recruited for his or her expertise in that area. The feedback obtained in this process was used to refine each of the area listings as well as the foci of identified topical

areas themselves. Not surprisingly, suggestions for some keyword additions were common across several topical areas. Subsequent to making adjustments to these sublists, a single overall list of keywords was forwarded to all advisory board members so that they could comment on the totality of potential entries as well as the placement of keywords within topical areas. More excellent feedback was obtained in the course of a couple of iterations of this process, after which the final working "skeleton" keyword list was settled upon. Adjustments to the overall list after that point were relatively rare and minor and occurred only as terms, concepts, constructs, and so on were tripped across, or identified by contributing authors as really "needing" to be added. From that point forward, the process of creating the encyclopedia involved "putting meat" on the 300-plus keyword "skeleton" by securing authors to write the entries. This process was greatly facilitated by the stature and "connectedness" of our advisory board members.

Acknowledgments

The *Encyclopedia of Sport and Exercise Psychology* came to fruition only as a result of the committed involvement of many people. Completion of a project of this magnitude and complexity cannot occur in a timely manner otherwise. We acknowledge this fact and extend our sincerest and deepest appreciation to everyone who contributed to this project. We particularly thank the advisory board members identified in the front matter of the encyclopedia because their input and efforts were crucial at every stage of the process. They contributed to identification of topics and entry keywords in the developmental process, facilitated the identification and securing of authors of suitable expertise and stature, and subsequently provided feedback to those authors on entries submitted within their topical areas. They all have our gratitude for their Herculean efforts. The many authors who invested their time and expertise in this project also have our gratitude. They are all identified by name in conjunction with their entries. Their willingness to join us in this project was both bracing and satisfying. We also deeply appreciate the terrific support we received from "our" team on this project at SAGE: Jim Brace-Thompson,

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Robert C. Eklund
Gershon Tenenbaum

A

ACHIEVEMENT GOAL THEORY

Achievement goals refer to the aim, purpose, or focus of a person's achievement behavior. These goals are dynamic cognitive entities representing future-based possibilities that respond to changes in the person as well as the situation. They do not refer strictly to the level of aspired performance (as in the goal-setting literature) but, rather, to how people evaluate their competence or incompetence and orient their behavior accordingly. Achievement goal theory has emphasized the role of achievement goals in regulating a wide variety of affective, behavioral, and cognitive outcomes during people's competence pursuits. More recently, research on achievement goal theory has developed the hierarchical model of achievement motivation, which integrates a variety of achievement motivation theories. This entry defines key constructs in achievement goal theory, reviews major models of achievement goals, reviews the consequences of different achievement goals, and explains how achievement goals have been integrated with other theoretical approaches in the hierarchical model of achievement motivation.

Key Constructs in Achievement Goal Theory

Achievement goal theory posits three major explanatory constructs: *states of goal involvement*, *goal orientations*, and *goal climates*. A person's state of achievement goal involvement reflects the aim, purpose, or focus of the achievement behavior in

a specific context at a particular moment in time. States of goal involvement can change over time, so it is very important to understand their contextual and temporal frame of reference. The narrower and more specific the frame of reference, the closer the construct will be to the theoretical conception of a goal as a dynamic cognitive entity. The predictive power of states of involvement varies based on the correspondence between the frame of reference for the goal and the outcome. For example, goals for today's practice session will be more effective at predicting outcomes in today's practice than in predicting outcomes over the course of the 6 months ahead. Unfortunately, research on states of achievement goal involvement is limited, in part because of the challenges associated with assessing goals in relevant performance contexts repeatedly over time.

An individual's achievement *goal orientation* reflects one's typical state of achievement goal involvement over time within a particular context. Goal orientations can be thought of as the central tendency of a distribution representing a person's states of involvement for an activity sampled repeatedly over time. Goal orientations are thought to be more stable than states of goal involvement because they are contextually—but not temporally—specific. Goal orientations emerged as a preferred level of analysis in achievement goal research because they can be assessed via self-reports on a single occasion and do not require repeated assessment to capture a process.

Despite the relative ease of assessing goal orientations, a noteworthy limitation of this construct

is that it cannot distinguish goal-based variability from variability associated with other stable individual differences (e.g., motives). For example, a person who has a strong fear of failure will tend to adopt some achievement goals more than others, and the difference between fear of failure-based variability and variation due to the actual aim, purpose, or focus of that person's achievement behavior is blurred. Likewise, goal orientation ratings may be tainted by situational factors at the time the measure is completed. Notwithstanding these limitations, the achievement goal orientation construct has provided a very accessible entry point for testing the propositions of achievement goal theory, and the vast majority of the research on achievement goals has focused on goal orientations.

Finally, achievement *goal climates* represent the situational cues that lead people to adopt different states of involvement. Joyce Epstein proposed a set of TARGET structures within achievement contexts, which influence the goals that people adopt. This model is derived from research on academic achievement motivation but also provides a rich conceptual framework for dissecting the motivational climate in physical activity settings.

The TARGET acronym stands for the *task, authority, recognition, grouping, evaluation, and timing structures* in a particular situation. *Task structures* refer to the type of task that people are working on. Tasks that are perceived as challenging will evoke different goals than tasks that are perceived as repetitive and boring. *Authority* refers to how decisions were made about the task. Activities that are perceived as more autonomously selected by participants evoke different goals than those that are selected by leaders such as coaches or teachers. *Recognition* refers to the procedures used to recognize performers' accomplishments. Praise and criticism that are provided privately evoke different goals than similar feedback that is provided publicly. *Grouping* refers to whether participants are assigned to groups of similar or diverse ability levels. Groups formed with the intention of having a diverse range of abilities evoke different goals than groups formed with the intention of creating status differences between groups. *Evaluation* describes how participants' competence is evaluated and recognized. Evaluations that emphasize improvement evoke different goals than evaluations that emphasize normative comparisons. *Timing* refers to whether

the time demands promote a focus on learning or immediate performance. It is worth noting that the perceived motivational climate (i.e., TARGET structures) may differ from an objective assessment of the motivational climate. As in other social cognitive approaches, it is the perceived climate that is held to influence states of goal involvement.

Structure of Achievement Goals

The structure of and terminology for describing achievement goals have evolved considerably over the past 3 decades. Goals were originally proposed to account for differences between mastery and helpless responses to failure. People who exhibited a mastery response to failure (i.e., attributions to low effort, increased persistence, greater enjoyment, and enhanced problem solving) were thought to exhibit this response because they focused on learning and improving. In contrast, people who exhibited a helpless response to failure (i.e., attributions to low ability, decreased persistence, less enjoyment, and reduced problem solving) were thought to exhibit this response because they focused on protecting themselves from undesirable social comparisons. These goals became the focus of the dichotomous model of achievement goals.

Dichotomous Achievement Goals

The dichotomous model of achievement goals emphasizes two primary definitions of competence. *Task goals* emphasize learning and improving, whereas *ego goals* emphasize outperforming others. Task and ego goals have also been referred to as mastery-learning and performance-competitive goals, respectively.

Based on research in sport and physical education contexts, task goal orientations are associated with greater enjoyment, interest, pleasant emotional experiences, prosocial behavior, commitment, perceived improvement, intrinsic motivation, moral functioning, life satisfaction, and satisfaction with coaching and competitive results. Ego goals are associated with consequences such as anxiety, worry, competitiveness, public self-consciousness, motivation for social status and recognition, extrinsic motivation, hypercompetitive attitudes, and antisocial behavior. Relations between dichotomous goal orientations and perceived competence are more equivocal in the literature and require further attention. Both task and

ego goal orientations positively predict perceptions of task and ego goal climates, respectively.

Goal climates have also been linked with a number of affective, behavioral, and cognitive outcomes. This literature has emphasized the distinction between mastery (i.e., task) and performance (i.e., ego) motivational climates without attending much to the *approach–avoidance* distinction that became the focus in subsequent extensions of the goals model. It has also focused on physical education settings. Compared with control groups and groups exposed to performance goal climates, interventions that created mastery goal climates in physical education classes have been associated with more positive attitudes toward the activity, greater enjoyment, increased health and fitness, improved skill, lower anxiety, and greater use of effective learning strategies compared with performance goal climates.

Trichotomous Achievement Goals

The trichotomous model of achievement goals was noted as a possibility in early writing on the dichotomous model of goals but only received empirical attention in the 1990s when it was used to explain an inconsistent finding in the literature. Specifically, relations between performance (or ego) goals and intrinsic motivation varied considerably, from negative to nil, across studies. Scholars posited that these goals had no influence on intrinsic motivation when they were approach oriented but reduced intrinsic motivation when they were avoidance oriented. This hypothesis was supported by a meta-analysis of existing studies and a series of experiments. Subsequent research has identified a number of other variables with different relations to performance goals depending on whether those goals are approach or avoidance oriented. Thus, the trichotomous model of achievement goals includes *mastery goals*, *performance–approach goals*, and *performance–avoidance goals*. These goals reflect aims of learning and improving, outperforming others, and not being outperformed by others, respectively.

In the physical domain, the added value of distinguishing performance–approach from performance–avoidance goals in physical activity was established in experimental research. These studies manipulated the instructions given to participants to induce particular goals. Participants who received mastery or performance–approach

goals exhibited greater intrinsic motivation than participants who received performance–avoidance goals, and these relations were not due to differences in perceived competence. The latter finding was particularly important because early critics of expanding the dichotomous model of achievement goals speculated that the approach–avoidance distinction was simply a reflection of differences in perceived competence—a hypothesis that was clearly refuted by these studies. These findings were instrumental for introducing the approach–avoidance dimension to achievement goal theory in physical activity.

2x2 Model of Achievement Goals

Shortly after the introduction of the trichotomous model of achievement goals, the model was expanded to a fully symmetric 2x2 model of achievement goals. This model outlined four possible achievement goals based on (a) the *definition of competence* and (b) the *valence of the competence-based incentive*. The two levels for definitions of competence were consistent with the dichotomous and trichotomous models. Mastery goals defined competence in reference to the self or absolute standards, and performance goals defined competence in reference to normative standards. The two competence-based possibilities that orient people's behavior in achievement situations are success (which people typically approach) and failure (which people typically avoid).

A 2x2 taxonomy emerges from the combination of the definition of competence with the valence of the incentive. The *performance–approach* and *performance–avoidance* goals in this model mirror those goals in the trichotomous model. Mastery goals from the trichotomous model are specified as *mastery–approach goals* in the 2x2 model. These goals involve striving to improve or execute a skill as it should be executed (i.e., self- and absolutely referenced competence, respectively, with a focus on being successful). The new addition to the 2x2 model involves *mastery–avoidance goals*, which describe strivings not to have one's skills deteriorate or not to make mistakes (i.e., self- and absolutely referenced competence, respectively, with a focus on not failing). One of the major objectives for researchers interested in the 2x2 model over the past decade has involved evaluating whether the added complexity of this model is warranted in relation to increases in predictive power. Following

is a summary of bivariate relations between each goal and a variety of affect, behavioral, and cognitive consequences in physical activity; this summary does not account for any overlap between goals due to a shared definition of competence or valence.

Mastery–approach goals have been linked with affective consequences such as heightened enjoyment and positive affect, and reduced boredom. Behaviorally, these goals are associated with increased fitness, perceived exercise intensity, physical activity participation and, in some studies, even performance (typically in timed races or similar tasks). They have also been linked with reduced self-handicapping. From a cognitive standpoint, mastery–approach goals have been linked with (a) greater meta-cognitive regulation, relative autonomy, intrinsic motivation, perceived usefulness and importance of the activity, effort, tolerance, help-seeking, situational interest, satisfaction, intentions to continue in sport, preferences for the activity, and utility value, and (b) reduced amotivation.

Mastery–avoidance goals have been linked with mixed affective consequences such as increased enjoyment and increased negative affect. Behaviorally, these goals are associated with increased fitness, shuttle run performance, and self-reported intensity of activity. From a cognitive standpoint, mastery–avoidance goals have been linked with stronger preference for the activity, tolerance, effort, physical activity participation, and intentions to participate in sport.

Performance–approach goals have been linked with affective consequences, such as increased enjoyment and positive affect, and decreased boredom. Behaviorally, these goals are associated with physical activity and performance (typically in timed races or similar tasks). From a cognitive standpoint, performance–approach goals have been linked with greater meta-cognitive regulation, effort, introjected and external regulation, intentions to participate in sport, value (intrinsic and utility), and satisfaction.

Performance–avoidance goals have been linked with affective consequences such as increased enjoyment and negative affect as well as decreased boredom. Behaviorally, these goals are associated with fitness, self-handicapping, and physical activity participation. From a cognitive standpoint, performance–avoidance goals are associated with interest; identified, introjected, or external

regulation; and intentions to continue in sport as well as decreased satisfaction with sport.

Even a cursory review of this list reveals that some outcomes (e.g., enjoyment) are associated with all four goals in the same way. This finding reflects, at least in part, the overlap between goals. It also suggests that a general achievement goal profile elevation may be contaminating some of these results, that is, people may be high or low in all goals because of some other factor such as the importance of the domain to them. Shifting the focus of future research from achievement goal orientations to states of involvement will help to resolve this issue. Notwithstanding this limitation, clear nomological networks for each goal are emerging. Taken as a whole, these findings indicate that mastery–approach goals have unequivocally desirable consequences, and the other goals have mixed profiles, with performance–approach goals appearing preferable to mastery–avoidance goals, which seem preferable to performance–avoidance goals.

Although the 2x2 model of achievement goals is fairly comprehensive, there may be reasons to make further distinctions between goals. For example, scholars have debated the merits of differentiating between mastery goals based on a self-referenced definition of competence and mastery goals based on a task-referenced definition of competence. Any such expansions of the 2x2 model of achievement goals must ensure that the new goals are grounded in competence-based incentives and must demonstrate that the explanatory or predictive power of the new goals is sufficient to offset the added complexity of the model.

Hierarchical Model of Achievement Motivation

The hierarchical model of achievement goals was proposed at the beginning of the 21st century to integrate the classic, motive-based approaches and contemporary, goal-based approaches to studying achievement motivation. This model posits goals as the proximal regulators of affect, behavior, and cognition during competence pursuits. It also proposes that many established achievement motivation constructs in other approaches predispose people to adopt certain goals. These goal antecedents include achievement motives, self-related constructs (e.g., perceived competence, self-theories of ability), neurophysiological predispositions, and the motivational climate in a given situation.

For example, mastery–approach goals are more likely to be adopted by people with high need for achievement, perfectionistic strivings, perceived competence, incremental theories of ability, and approach motivational temperaments and in situations that create a mastery motivational climate. In contrast, performance–avoidance goals are more likely to be adopted by people with high fear of failure, perfectionistic concerns, entity theories of ability, avoidance motivational temperaments, and low perceived competence and in situations that create a performance motivational climate.

One benefit of this hierarchical, antecedent–consequence model of achievement goals is that it has increased conceptual precision in research on achievement goal theory. Researchers now tend to differentiate between factors that lead to goal adoption and those that are produced by goal adoption. Although the vast majority of this literature is nonexperimental and does not provide a strong basis for causal claims, coherent and logical conceptual articulations of relations within the antecedent–consequence framework are contributing to theoretical clarity. Another trend worth noting in recent literature is a slight increase in the number of longitudinal research studies that investigate how goals fluctuate within people over time. This approach is valuable because it shifts the focus from goal orientations to states of involvement and removes threats from confounding third variables that may be antecedents of goals (e.g., fear of failure). Finally, as evidenced by the preceding review, the majority of the research on achievement goal theory as it relates to physical activity has been conducted in the context of sport and physical education. Exercise and rehabilitation contexts have received considerably less attention but also involve competence pursuits and therefore may be useful contexts for understanding the role of achievement goals in regulating affect, behavior, and cognition.

Conclusion

In sum, achievement goal theory has developed considerably since it was first proposed over 3 decades ago. Extensions of the theory have met with resistance in some quarters, but the associated debates have clarified differences in assumptions and contributed to a more precise articulation of the theory. The hierarchical model of achievement motivation represents the current

state of the science. It is a significant development because it (a) links multiple levels of analysis in motivation from the neurophysiological to the social; (b) integrates diverse theories of achievement motivation; (c) differentiates the causes of goals from their effects, at least conceptually; and (d) specifically addresses the factors that energize and orient achievement strivings.

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See also Achievement Motive Theory; Attribution Theory

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ACHIEVEMENT MOTIVE THEORY

Competence is a recurring theme in human movement whether the setting is sport, exercise, or rehabilitation. From the earliest days of life, people strive to feel effective in their unfolding interactions with the environment and, throughout the lifespan, people's well-being is compromised when this need is thwarted. Despite the apparent universality of competence motivation, there are clear differences in the ways that people pursue competence. Achievement motivation theories strive to explain the processes that initiate, direct, and maintain achievement behavior. Achievement motives are relatively stable individual differences that influence people's motivation during competence pursuits. This entry defines achievement motives, provides an overview of how they develop, reviews implicit and explicit measures of motives, and summarizes documented outcomes of achievement motives.

Achievement Motives

The earliest achievement motivation theories focused on people's aspired level of behavior and the perceived utility of different behaviors as indicated by the expectancies and values that people associated with those behaviors. A key assumption underlying these approaches was that people make conscious and rational choices about their achievement behavior. Around the same time that the expectancy-value theories were emerging,

personality psychologist Henry Murray proposed a system of psychological needs that included *achievement* (a need for efficiency and effectiveness) and *infaavoidance* (a need to avoid humiliation and to refrain from action due to fear of failure). These needs were proposed to explain individual differences in behavior. David McClelland and colleagues in several works subsequently introduced the construct of achievement motives to account for individual differences in people's achievement behavior under similar conditions. Motives represented the strength of associations between environmental cues (e.g., competence pursuits where success or failure are possible) and learned affective responses to those cues. Given the importance of competence to the self, self-evaluative emotions such as pride and shame provided logical affective bases for these motives.

Two motives were proposed based on people's orientation toward or away from competence-based incentives during competence pursuits. The *motive to approach success* (sometimes referred to as hope for success) described individual differences in people's tendency to experience anticipatory pride while engaged in a competence pursuit. The *motive to avoid failure* described individual differences in people's tendency to experience anticipatory shame while engaged in a competence pursuit. Over time, these motives have been referred to as the need for achievement and fear of failure, respectively.

The anticipatory pride and shame involved in motives are instrumental in energizing and orienting achievement behavior. For example, pride fosters persistence and heightened engagement in goal pursuit, promotes long-term achievement, stimulates interpersonal expressiveness, engages flexible social behavior, and contributes to the development of social capital over time. Shame, on the other hand, motivates withdrawal. It can promote appeasement or aggressive behaviors depending on how people regulate their shame. From an achievement motivation perspective, shame will undermine persistence and create difficulties in achieving long-term goals.

In their achievement motivation theory, McClelland et al. posited that a person's motivation is influenced by a tendency toward success and a tendency away from failure. Each of these tendencies was represented as the product of the person's perceived probability of succeeding (or failing), the value of the reward for succeeding

(or the punishment for failing), and the person's motives. Thus, motives were proposed to refine basic expectancy-value predictions and elaborate on interindividual variation in behavior. Over time, the independence of *approach* and *avoidance* motivation became apparent, and the idea that approach and avoidance tendencies produced a single resultant motivation orientation was abandoned. In contemporary research, the approach- and avoidance-based achievement motives are typically treated as independent predictors of motivational outcomes, and their direct effects on motivational outcomes receive more attention than their interactions with expectancies and values.

Development of Achievement Motives

Research on the development of self-evaluative emotions informs our understanding of how the two achievement motives develop. Self-evaluative emotions are a unique class of emotions because they are not present at birth and require some cognitive development before they appear. The necessary cognitive milestones include (a) the development of a sense of self (typically 15–24 months of age); (b) the internalization of rules, standards, or goals for desired behavior (typically 24–41 months of age); and (c) the evaluation of oneself in relation to those internalized rules, standards, or goals (as early as 30 months of age). If children appraise that they are complying with the rules, standards, or goals that they internalized through socialization, they should feel proud. On the other hand, if they appraise that they are not complying with those rules, standards, or goals, and attribute that deviance to a personal flaw (as opposed to a simple behavioral error), they should feel shame. As children develop a history of experiencing pride and shame in their competence pursuits, those competence pursuits increasingly become cues for anticipatory pride and shame. This association and anticipatory affective response is the basis for achievement motives. With roots in such foundational self-evaluative processes, these motives are likely to generalize across achievement contexts and should not be domain specific.

Measuring Implicit and Explicit Motives

One major controversy in the achievement motives literature concerned the measurement operations used to assess these motives.

Originally, McClelland and his colleagues adapted the Thematic Apperception Test and asked participants to view ambiguous images of people involved in competence pursuits, and to write a story about the image (e.g., What is happening? What happened previously? What will happen next?). Content from the ensuing narratives can be coded using different schemes for the need for achievement or fear of failure, which was called *hostile press* in an early scoring system. This procedure has been refined in the contemporary picture story exercise. This fantasy-based assessment procedure is relatively time-consuming and requires extensive training, so a number of researchers attempted to develop parallel self-report measures that could be administered quickly and easily without sacrificing (and possibly even enhancing) validity. When scores from self-reported achievement motives were compared with scores from projective tests, the correlations were unexpectedly small.

After extensive debate over which score was the more valid measure of achievement motives, researchers concluded that both scores were valid and that the difference reflected differences in the motivational systems that were assessed. This discovery contributed to the distinction between implicit and explicit motivation. *Implicit achievement motives* are rooted in affective arousal and are reflected in scores from the projective, fantasy-based measures. *Explicit achievement motives* involve cognitive elaboration and are reflected in scores from the self-report measures. Implicit motives are posited to predict spontaneous, nondeclarative outcomes that may be regulated outside of a person's conscious awareness (e.g., procedural learning), whereas explicit motives are posited to predict declarative outcomes of which people are self-consciously aware (e.g., enjoyment). Although the implicit or explicit distinction is still common in the achievement motivation literature, other models refer to these dual processes as *impulsive or automatic* and *reflective or controlled* processes, respectively.

Overall, the motivational taxonomy at the heart of motive-based approaches can be summarized as a 2x2 taxonomy. Both approach- and avoidance-based achievement motives exist at implicit, impulsive, or automatic levels and explicit, reflective, or controlled levels of analysis. The vast majority of research on achievement motives in physical activity contexts has employed explicit motivation

measures so relatively little is known about implicit measures of motivation in these contexts.

Within the context of sport, most of the research on achievement motives has focused on fear of failure. This research often equated fear of failure with performance anxiety although it is now clear that athletes may experience anxiety over threats other than failure (e.g., injury, success). This distinction is important because contemporary models of emotion hold that emotions reflect people's ongoing adaptational struggles. Thus, the possibility of failure by itself is insufficient as a stimulus for activating a person's fear of failure because the meaning of failure can vary considerably from one person to the next.

The cognitive-motivational-relational theory of emotion has been applied to understand the meaning of failure. In a series of studies, five major aversive consequences of failing have been identified. These include *shame and embarrassment*, *devaluing one's self-estimate*, *having an uncertain future*, *upsetting important others*, and *having important others lose interest*. Beliefs in each of these consequences are strongly correlated so there appears to be a general fear of failure that underlies beliefs in each of these specific consequences. Beliefs that failure leads to shame and embarrassment are most closely related to the original definition of fear of failure. Perhaps not surprisingly, this belief was also the most strongly associated with the general fear of failure and seems to be the most relevant for achievement motivation. It is clear that all five of the beliefs are strongly associated, and collectively they provide a better representation of the universe of the fear of failure domain than does any single belief by itself.

Consequences of Achievement Motives

Achievement motives influence people's lives in a variety of ways, although more attention has focused on the impact of fear of failure than on need for achievement. In academic contexts, fear of failure has been linked with decreased moral functioning and increased attention-seeking behavior. From a health perspective, fear of failure is positively associated with anorexia, anxiety, depression, and headache disorders. College students who present for counseling services frequently cite fear of failure as a problem that interferes with their lives and academic performance.

Research within the context of sport has documented that young athletes report fear of failure as a salient source of stress and a reason for dropping out of sport. Athletes have also attributed their use of ergogenic drugs to their fear of failure. Officials, such as umpires and referees, cite fear of failure as a common reason for burnout and turnover in their work.

The most well-established consequences of achievement motives involve achievement goals. People with a strong need for achievement tend to adopt *approach-valenced achievement goals* such as *mastery-approach* (focused on learning and improving) and *performance-approach goals* (focused on outperforming others). People with a strong fear of failure tend to adopt avoidance-valenced achievement goals such as *mastery-avoidance* (focused on not making mistakes or getting worse) and *performance-avoidance goals* (focused on not being outperformed others). People who fear failing may also adopt an approach-to-avoid strategy whereby they adopt performance-approach goals because demonstrating normative competence provides immediate, albeit short-lived, evidence that one is not incompetent. Each of these achievement goals has important consequences for achievement behavior. Over time, consistent patterns of achievement goal involvement contribute to the achievement differences between people with different motive profiles.

Conclusion

Achievement motives are useful theoretical constructs for explaining factors that energize and initially orient achievement behavior. These motives emphasize fundamental differences between people in their approach and avoidance tendencies during competence pursuits. They remain a relevant component of contemporary achievement motivation theories by virtue of their role in predisposing people toward characteristic achievement goals during their competence pursuits. These constructs will remain useful as motivation theories develop more nuanced explanations of how people respond in the different psychological contexts of their competence pursuits.

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See also Achievement Goal Theory; Automaticity; Implicit Attitudes; Dual Process Theory; Expectancy-Value Theory

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ADAPTATION

The term *adaptation* has been integrated within the sport psychology literature, from as early as 1986. Initially mentioned in relation to elite athlete retirement, adaptation is a broad term associated with monumental change in the athlete's life. People experience stress in their lives, and at certain times stress reaches a threshold, after which one must make decisions to alleviate that stress and reestablish psychological balance. When the performer integrates the necessary information about a significant stressor or the accumulation of several smaller stressors, that person can begin to establish or reestablish control, ideally culminating in adaptation.

Adaptation has been discussed in relation to contributive pathways, comprised of *understanding*, *control*, *self-enhancement*, *belonging*, and *trust*. Understanding facilitates adaptation when one gains an accurate appreciation of the stress episode, in advance of action. Control is sought through either direct or indirect means. Self-enhancement encompasses decisions that lead to

better performance via purposive effort and learning. Trust delineates the performer's belief that social support within the performance context holds the performer's best interests in mind, that supporters are creditable, and that they will act when assistance is needed. Belonging, like trust, is a social pathway. However, through belonging one facilitates social affiliations that in turn make trust more likely. Any of these five adaptation pathways can segue to a larger adaptation process, and so to the outcome of adaptation.

Adaptation interventions are temporal in nature, with at least a search for understanding preceding all other pathways. However, understanding is not necessarily acquired in its totality before the performer begins an adaptation process. The rule of thumb, however, is for the performer to seek as much detail about the stress episode and how one might engage in the process when the information is most needed. For example, the elite junior athlete who is drafted to a professional ice hockey, baseball, or football team will encounter several stressors that catalyze an adaptation process. The performer's understanding of the previously unfamiliar in such circumstances would include the coach's performance expectations, social norms that build relations with one's teammates, relocation to a new city and a new sport environment, media demands, fan expectations, and a significant change in financial status. Learning of these new contextual changes and beginning to engage in effective responses that lead to restored ease, the athlete can perform at the optimum. Hence, the process can generally be regarded as the move into, the move through, and the move onward from stress, with adaptation.

There are cases when performers maladapt, meaning they engage in an incorrect psychological process that manifests in rumination, apathy, or ego-protective thinking. It is human tendency to seek understanding throughout the adaptation process. However, how one chooses to view the stressor, via personal reflections and social support resources, will help determine whether the stress episode is resolved and the performer reestablishes psychological balance. There are, indeed, cases where an athlete's best performances are early in the career, with results waning at the juncture when the athlete should be performing at potential. When athletes encounter barriers during formative points in their athletic careers and there is incomplete or inaccurate understanding

as a pattern, the athletes engage in negative contemplation and impede their own career development. The objective is for athletes and those who work with them to identify the barriers to performance in advance of the challenge, take only as much time and effort as is necessary to resolve the stress episode, and then reengage with correctly chosen personal and social support strategies. With the process of adaptation accomplished, the athlete gains (or regains) efficacy in personal abilities to resolve stress episodes in advance of further—inevitable—challenges.

In a similar vein, people who engage in exercise and move toward healthier choices can also experience adaptation processes when life changes are required, be these changes foreseen or unforeseen. The process would pertain to cardiac patients and people suffering from obesity who are seeking to begin a targeted exercise program. The identification of what is expected of oneself in relation to the program, and also of one's barriers in advance of the required behavior change, would increase the likelihood that personal and social support strategies are correctly chosen and implemented, leading to *good change*.

Up to the present day, formal adaptation research in sport and exercise psychology has included investigations into the adaptation processes of National Hockey League athletes during various stages of a professional sport career; immigrant athletes performing in major league baseball; Olympians; elite amateur cyclists; and indigenous developmental, elite amateur and professional athletes. These investigations have been for the most part qualitative, with several projects positioned as atheoretical, and others reflecting various adaptation, career, or life transition frameworks. This author proposes that adaptation research, reflecting its reality in application, should include cross-cultural investigations, community sport contexts, developmental sport, youth sport, sport for the elderly, and interventions in exercise and health settings with at-risk cohorts. Indeed, every performer experiences adaptation processes, though not all adaptation attempts are successful. Through a more systematic approach to research and practice, the sought after outcome of adaptation can become the performer's effective resolution to stress episodes in sport and exercise contexts.

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See also Anticipation; Burnout; Career Transitions; Choking; Competition; Coping; Decision Making; Effort; Injury, Psychological Susceptibility to; Resilience

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ADAPTED PHYSICAL EDUCATION

Physical education (PE) is considered an important vehicle for the promotion of physical activity, psychosocial development, and teaching dance, games, and sports skills. Unfortunately, children with disabilities are often inactive and socially isolated during PE despite laws requiring children with disabilities to be included in general educational schools, PE not excepted. This entry centers on the psychosocial and educational research in four related areas: PE teachers, peer tutors, able-bodied classmates of children with disabilities in PE, and children with disabilities.

Most teachers find that including children with disabilities in PE is challenging because of the need to change activities and adapt how they are implemented and evaluated. Many PE teachers report that they lack knowledge on disability conditions and have limited experiences in teaching children with disabilities. Because many school districts do not hire adapted physical education specialists, general PE teachers are required to

teach children with disabilities. Hence, PE teachers' lack of knowledge and experience is likely grounded in the limited to nonexistent adapted PE training they receive while in teacher preparation programs. In turn, their limited experiences lead to feelings of low perceived competence. In addition to a lack of perceived competence, researchers have also documented that teachers often have negative attitudes toward including children with disabilities into their general PE classes, although other reports suggest teachers tend to have more mixed attitudes. Teachers also tend to grade children with disabilities differently by emphasizing participation and effort versus fitness and skill development. Whether such grading differences reflect reasonable accommodations of children's limitations versus lowered expectations of their abilities is unclear from the literature.

In brief, most of the research on PE teachers of children with disabilities indicates they feel their professional preparation is inadequate. Hence, they lack the ability to effectively accommodate children with disabilities into their general PE classes. It would seem reasonable that providing support for PE teachers in the form of adapted physical education (APE) specialists who have received extensive training in adapted physical activity would be helpful. Unfortunately, very little research has been done in this area, and the limited research often involves case studies or very small samples. For instance, many physical education classes might have only two or three students with disabilities in them. Nonetheless, researchers conducting studies in this area have suggested that APE teachers can enhance student outcomes.

In addition to research on support offered by trained APE specialists, PE teachers are also helped by trained peer tutors and teacher aides. Although only a handful of researchers have examined the effectiveness of peer tutors, the results have consistently established that peer tutors are effective. Children with disabilities working with peer tutors have increased their physical activity engagement and in some cases enhanced motor skill development. Added benefits also include increased social interaction and subsequent feelings of belonging. Based on limited research, teacher aides, while lacking training in PE, appear to be beneficial, especially if they are paired up with peer tutors.

Although it might be assumed that APE specialists may lack the same concerns as general PE teachers, APE specialists have noted they often lose

access to the gymnasium or have to share it, which limits the space they have to work with. Lack of adequate functional equipment is also a complaint of APE teachers. Many APE teachers travel from school to school and may teach students ranging from preschool to high school at 4 to 14 different schools. To ensure they have adequate equipment, they often buy their own and transport it back and forth between schools. The constant travel while carrying equipment is often quite tiring. Similar to general physical education (GPE) teachers, APE specialists also receive support from teacher aides. Unfortunately, they have sometimes reported aides as apathetic and as exhibiting negative behaviors. Some APE specialists have also reported that they perceive themselves as disrespected and do not feel valued or part of the *family* of teachers at the schools where they teach. It should be noted, however, that the above concerns were not universally shared by all study participants. For instance, some APE teachers felt quite respected by colleagues and have very dedicated and competent teacher aides.

When GPE teachers are provided with teaching support from peers, teacher aides, or APE specialists, it is thought that any negative impact on children without disabilities as a result of inclusion is negated or minimized. Historically, a major concern arising from the inclusion of children with disabilities in general PE is that the educational experiences of children without disabilities may suffer. Accordingly, many scientists have examined how children without disabilities are affected by inclusion. In situations where children do not have significant disabilities, such as mild developmental delays, the experiences of both able-bodied and disabled children are similar; they experience comparable levels of physical activity and on-task behavior. In another study, researchers examined 15 students with disabilities and 20 students without as they participated in a regular game of volleyball and an adapted game of volleyball. All students enjoyed success, expressed fun and interest, and were physically active. The one caveat to these findings was that some of the older students (i.e., 12 years old) were not particularly keen on the adapted version of volleyball that included a smaller court and net. Despite the portrayal of a successful adapted PE class where all students do well, there is a less optimistic line of research examining the social dynamics of adapted PE classes.

In contrast to the reported results, more mixed findings have been reported on the experiences of nine elementary-age children who participated in inclusive PE classes. The scientists reporting the study categorized the children's experiences into *good* and *bad* days. Children reported having bad days if they experienced being teased. While teasing was seen as an example of being rejected, the children were also ignored and neglected. Finally when able-bodied classmates seemed to focus on children's disabilities, those children felt objectified and seen as curiosities. Being seen as objects of curiosity or rejected and neglected were all classified as forms of social isolation. A second major theme was represented by instances where the children perceived that their abilities were questioned and devalued. It was not uncommon for their mental as well as their physical abilities to be questioned. Finally, in the third theme, there were situations where the children were inactive or had minimal levels of participation. Common reasons were environmental barriers like a grass field that inhibited wheelchair access, lack of teacher and classmate support, as well as outright neglect. In summary, when children could not be active, had their capabilities questioned, or were socially marginalized, they experienced these situations as constituting a bad day in PE.

Fortunately, the same participants also described how they had good days in PE. There were times when classmates were quite encouraging and helpful. For example, during relay races children with disabilities remarked on how they were cheered on. Moments like those described above were categorized as promoting a sense of belonging. When participants were engaged in PE, they recognized they were reaping the intended benefits of PE, such as developing skills, enhancing their health, and learning fitness and health concepts. The recognition that they were obtaining benefits just like the able-bodied children represented having a good day in PE. One final theme indicative of having a good day was the intrinsic (pride) and extrinsic (compliments) rewards they experienced from being successful and demonstrating their skills.

In a research effort similar to the above study, Donna Goodwin sought to understand if all children with disabilities perceived offers of assistance positively. She found that children with disabilities viewed some forms of help as supportive and other types of help as threatening. There were a number

of key differences between the types of support seen as threatening versus supporting. First, support that was pragmatic or functional and helped the children engage in a game was seen as supportive—being pushed in the wheelchair across a grass field to reduce travel time when the alternative was wheeling all the way around the field. When children without disabilities did not ignore children with disabilities and encouraged their sport participation, this was perceived as being done to secure control over their moves.

In contrast, threatening help was that which was provided unilaterally, where the helper refrained from asking if the help was needed. If help was offered when it was not needed, it was seen as interfering and threatening to children's sense of independence. Furthermore, some enactments of help were dangerous, such as pushing a child's wheelchair too fast and causing the child to fall. Finally, when children viewed offers of help as tantamount to an indictment of their capabilities, it was seen as a threat to their self-worth and interpreted negatively. Individuals with disabilities have to weigh the short-term task-oriented benefits, such as being helped across a grass field quickly, against the more long-term self-oriented costs, such as a missed opportunity to experience pride in independently wheeling around the entire field.

The authors of this research suggested that their findings have some bearing on the findings presented earlier on peer tutors. Given the various interpretations that children with disabilities might make about whether help is beneficial or harmful, it would be of value if PE teachers understood these complexities. Peer tutors and teacher aides could then be informed about the various ways their help might be interpreted and act accordingly.

In summary, researchers of inclusive PE have documented a mixed and complex picture of teachers, support personnel, and children's experiences. Teachers often feel unprepared and hence are reticent about engaging with and teaching children with disabilities. While support personnel, such as peer tutors, teachers' aides, and adapted PE specialists, have the potential to enrich the experiences of children with disabilities, their own lack of training and challenges can compromise their ability to deliver quality education. Finally, children with disabilities have mixed experiences in PE resulting from reliance on the quality of instruction they obtain from educators and the quality

of the peer interactions they experience with their classmates. Clearly, adapted PE can be improved.

Jeffrey J. Martin

See also Disability; Disability and Exercise; Disability and Sport

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ADDICTION, EFFECTS OF EXERCISE ON

Exercise has been proposed as a potential treatment to help people quit smoking and, more recently, to treat addictions to alcohol and other drugs of abuse. This entry discusses the rationale and empirical support for the use of exercise as a treatment for addiction.

Rationale

Exercise has been proposed as a stand-alone or supplementary treatment for addiction. The focus has been on exercise as a potential treatment to assist with successful quitting and prevention of relapse among those who are initially motivated to discontinue use of the addictive substance, rather than to produce intentions to quit among those who are not initially motivated. The rationale for exercise as a potential treatment for addictive

substances is predicated on a number of potential mechanisms, including potential effects of exercise on (a) affective states, (b) cravings or urges to use the addictive substance, and (c) reduced concerns about postcessation weight gain.

Affective States

Withdrawal from addictive substances typically results in chronic negative shifts in affective valence (i.e., negative mood states). It has been proposed that exercise might help to prevent relapse by attenuating chronic negative mood states or providing a substitute means for producing an acute positive shift in affective valence. This proposed pathway was predicated on early research findings supporting the general notion that “exercise feels good.” More recent research has shown that the effect of exercise on affective states (e.g., emotion, mood, pleasure or displeasure) is more complex. It now appears that affective response *during* exercise varies greatly based on a number of variables, including personal factors, such as exercise history, fitness, health status; exercise setting and mode; and, perhaps most importantly, exercise intensity. Nonetheless, across populations, settings, and exercise modes and intensities, people generally tend to have a positive shift in affective valence (feel more pleasure or less displeasure) immediately *after* an acute session of exercise. Additionally, there is accumulating evidence that engaging in a program of regular exercise for at least 8 weeks results in reduced depressive symptoms among clinically depressed adults. Thus, given these caveats, there is reason to believe that exercise might serve as a treatment for addictive substances through its influence on affective states.

Craving

A second potential pathway through which exercise may serve as a treatment for addictive substances involves the effects of exercise on acute cravings or urges for the addictive substance. It has been posited that acute exercise may serve as a substitute or distraction for use of the addictive substance during a craving or urge, particularly to the extent that it is difficult—if not impossible—to use the addictive substance while exercising.

Concerns About Weight Gain

Another potential pathway through which exercise might serve as a treatment for smoking

cessation is through its effects on concerns about postcessation weight gain—a common barrier to smoking cessation, especially among women. Indeed, women who are concerned about postcessation weight gain are less likely to attempt smoking cessation, less likely to successfully quit smoking, more likely to relapse postpartum, and more likely to drop out of smoking cessation programs. Exercise plays a major role in preventing weight gain, as a moderate increase in one's level of exercise can minimize weight gain in women who have quit smoking. The ability for exercise to combat weight gain in the general population and particularly during smoking cessation is likely to make it especially attractive for women smokers who are concerned about gaining weight during their smoking cessation attempt.

Evidence

Exercise as a Smoking Cessation Treatment

Numerous research studies have shown that a single exercise session has favorable effects on immediate changes in affective states and cigarette cravings. However, the optimal mode, intensity, and duration of the exercise stimulus remain unclear. Moreover, the duration of the effects of a single session of exercise on affect and cravings remains uncertain.

While outcomes of studies examining a single session of exercise on affect and cravings have been generally positive, results from the first 14 randomized controlled trials conducted have been equivocal, with only a single study showing positive effects of exercise on rates of successful smoking cessation at the end of the trial. Some problems with these studies are (a) a lack of adherence to the exercise treatments; (b) an inability to objectively verify adherence to the exercise program; (c) lack of data on who continues to exercise during the study follow-up period, following the initial exercise treatment (usually about three months); (d) high drop-out among study participants; and (e) lack of data on the potential mechanisms of treatment (see above). Thus, more research is needed to determine whether exercise is an effective treatment for smoking cessation, and if so, the optimal dose.

Exercise as a Treatment for Addiction to Alcohol and Other Drugs of Abuse

Research on exercise as a treatment for alcohol and other drugs has thus far been limited to studies

conducted on animals and a few small-scale studies among humans. While the existing research shows promise, no large-scale clinical trials have been conducted in humans. Thus, it remains unclear as to whether exercise will serve as an effective treatment for addiction to alcohol and other drugs

Conclusion

There is a strong rationale for exercise as treatment for addiction to various substances. Findings from research on animals and small-scale studies with humans are supportive of the rationale. However, outcomes of randomized control trials have been equivocal for smoking cessation and are yet to be conducted for alcohol and other drugs of abuse.

David M. Williams

See also Alcohol Abuse; Affective Responses to Exercise; Drug Use and Control; Multiple Behavior Change; Obesity; Stress Management

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ADHERENCE

Many of the benefits of exercise come through sustained participation. Unfortunately, it is difficult for sedentary individuals to start an exercise program, and of those that do approximately 50% on average will drop out in the first 3 to 6 months. Furthermore, after 12 months most people who started a new exercise program will be sedentary again. As a result, a large body of literature is available on the degree to which different factors contribute to *exercise adherence*, defined as the degree to which an individual is able to sustain an

exercise program. Among the many psychological factors that may contribute to exercise adherence, three broad categories that have received extensive research attention are outcome expectations, social influences, and perceptions of control.

Outcome Expectations

Outcome expectations include the value that an individual puts on the outcomes associated with regular exercise as well as the likelihood that the person believes the outcome will occur. Outcome expectations can be positive (reduce the risk of cardiovascular disease) or negative (sore muscles or injury). The literature is mixed on the relationship between outcome expectations and exercise adherence. Still, most literature supports the notion that having positive outcome expectations is a necessary, though potentially insufficient, characteristic of exercise adherence. Further, when evaluating what value a person places on an outcome, one must take into account age and other demographics. Values vary significantly depending on the person involved. For example, older adults may place a higher value on perceived health and longevity than do younger adults. Older women may place more importance on the social aspects of exercise, while younger women may find physical activity as an important way to control weight. Value expectations may also differ between demographic groups. For example, Caucasian women generally may value physical activity as a method to lose weight, while Latina women may positively associate overweight with a healthy ideal and, therefore, not value physical activity from a weight-loss perspective. Finally, there is some evidence that the expected timing of the outcome is important. Specifically, if someone expects to lose 10 pounds in 1 week because of a new exercise program, it will be demotivating when that outcome does not happen when one wants it to happen.

Social Influences

Like outcome expectations, social influences are included in most theoretical models developed to predict exercise adherence. Social influences can include social support from a family member or friend, group norms or cohesion in an exercise class, or social environmental factors related to economic status or culture. The way people perceive social support can have positive and negative influences on exercise adherence. Positive social

support from family members, friends, and others can include opportunities to exercise with the person, planning activities around exercise, and giving encouragement for the person to continue exercising. However, even these kinds of activities can negatively influence exercise adherence if the person receiving the support feels pressured. Social norms can also have a positive or negative impact on exercise adherence. Within an exercise class, norms and a sense of cohesion result in increased adherence. Similarly, normative beliefs related to people who are important to us or have expertise can influence adherence. Normative beliefs are formed when a person's perception that other people who are personally important think the individual should engage in a specific behavior. For example, if a son believes that his mother is convinced track is the safest form of exercise for him, this belief will influence the subjective norm of the son, who may be apt to run instead of playing baseball, football, or another team sport. Alternatively, an individual who lives in a house full of other sedentary people may have a hard time initiating and sustaining an exercise regimen if the following is taking place: (1) subtle teasing ("Look at you wanting to be Mr. Universe"), (2) complaints about the time spent exercising, and (3) instances of sabotage.

Perception of Control

When people feel they have control over a situation, they are more likely to participate in a given behavior. Perceptions of control can be thought of as expectations related to one's ability to complete a task. Self-efficacy is the most commonly studied control belief—the belief that one has the ability or the competency to complete a certain action. It is predictive of attendance of exercise classes, and like value expectancy, can be influenced by demographics or age. Older adults who may be afraid of injury could have lower self-efficacy than younger adults; however, researchers have hypothesized that methods to increase self-efficacy and exercise adherence include (1) having participants experience personal successes with their exercise regimen; (2) providing vicarious experiences, or modeling exercise behaviors; (3) integrating social and verbal persuasion; and (4) monitoring physiological states (soreness, sickness, etc.).

Although many people are likely to not stick with exercise programs and regimens, these factors

may improve exercise adherence. Outcome expectations, social influences, and perceptions of control are closely related to the intentions and goals of a person. Indeed, these broad categories are interrelated. Outcome expectations are typically lower when participants do not value the outcomes of regular exercise. Also, when one's self-efficacy is low, the likelihood of valued outcomes also decreases. Conversely, when social influences are supportive of exercise, then perceptions of control increase. All of these factors interact within the broader ecological milieu. Laws and policies related to ensuring safe biking, opportunities for children to walk to school, and traffic calming all contribute to exercise adherence, sometimes acting through outcome expectations, social influences, and perceptions of control and other times directly impacting an individual's opportunities for exercise.

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See also Control Theory; Enjoyment; Expectancy-Value Theory; Health Promotion; Satisfaction; Self-Efficacy

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AFFECT

Affect, also referred to as *core affect*, is the basic substrate of consciousness, its most elementary constituent. It is the constant readout of human feeling. Affect has a distinctive experiential quality that does not consist of nor require cognition or

reflection. It is an inherent and necessary ingredient of emotions and moods; it is what gives these states affective *color*. However, affect is not only present during emotions and moods. Rather, it is always accessible to conscious awareness, although its experiential nature and intensity constantly fluctuate in response to internal and external stimuli.

Emotions and moods are considerably more complex and multifaceted constructs than affect. For example, the emotion of anxiety includes, besides unpleasant affect, a redirection of attentional and cognitive resources, attribution to the eliciting stimulus (a remark by a supervisor alluding to possible layoffs); a cognitive appraisal of threat (to one's life, social status, or self-image); a pattern of physiological changes (activation of the autonomic nervous system and the main neuroendocrine stress axes); observable changes in behavior (nervousness; altered voice modulation, facial expressions, muscular tics; exaggerated mannerisms); characteristic cognitions (distractability, thoughts of failure and negative consequences); and coping efforts (a search for solutions or sources of support). Similarly, moods also include multiple components besides affect. Although this may not always be immediately apparent, moods like emotions are also *about something*. They follow an eliciting stimulus, although the stimulus might have occurred long before the mood. They also require a cognitive appraisal, although what is appraised might be something as unspecific as one's life overall or one's place in the universe. In contrast, affect is not about something—it is not directed at anything; it is noncognitive, and it does not require an antecedent cognitive appraisal.

Examples of affect include pleasure, displeasure, energy or vigor, tiredness or fatigue, tension or distress, and calmness or relaxation. It is important to recognize that these affective states may occur by themselves or embedded as essential ingredients within emotions or moods. For example, when one is injured, the displeasure of pain is an affective reaction. It occurs instantaneously and automatically, without any need for cognitive recognition, evaluation, and interpretation. When one feels exhausted, drained of energy after a strenuous run on a hot and humid day, the sense of exhaustion is an affective state that stems directly from the physiological condition of the body, without any need for cognitive mediation. Similarly, the feelings of energy, invigoration, and

revitalization that a physically fit individual may experience after a great workout are also affective states.

On the other hand, the fear that a patient in cardiac rehabilitation feels during the first exercise session after a heart attack is an emotion; it includes affect at its core (fear is unpleasant) but it is more than that. There are memories of the heart attack, a fixation on the somatic sensations elicited by exercise in search for anything suspicious, an appraisal of threat due to the severity and the unpredictability of the situation, an accentuation of the physiological stress response to exercise, and avoidance tendencies. Likewise, the pride experienced by a formerly sedentary individual after being able to walk continuously for 30 minutes is also an emotion; there is again affect at the core (pride is pleasant), but there are several additional components. These include, for example, a cognitive appraisal of achievement (one has succeeded in attaining an important and challenging personal milestone) and characteristic behavioral manifestations (smiles, happy vocal expressions, arms raised in celebration). Put differently, when one says “I feel great that I was able to finish my first marathon,” the “I feel great” part is a reference to affect. The “that I was able to finish . . .” part is a reference to a cognitive appraisal that qualifies this state as an emotion.

Conceptualization and Measurement of Affect

As an object of scientific study, affect has been approached from two perspectives. Some investigators have considered each affective state as a separate entity, independent of all others. Examples of this *distinct-states* or *discrete-states* approach are questionnaires that assess various assortments of states (tension, fatigue, vigor, energy, revitalization). Such questionnaires are developed on the basis of factor analyses followed by orthogonal rotations, based on the assumption that the resultant factors are statistically independent of one another.

A different perspective considers affective states as systematically interrelated; some states are similar to others, some are unrelated, and some are antithetical. So, since the early twentieth century, investigators have been searching for a core set of dimensions that could explain these differences and similarities among affective states. Although various such dimensional models have been

proposed, the first two dimensions are usually the same. The first dimension, and the one that accounts for a larger portion of the variance, is pleasure-versus-displeasure (also termed *affective valence* or *hedonic tone*). The second dimension is low-versus-high perceived activation (also termed *arousal*). These two dimensions are bipolar and orthogonal to each other. So, when used in combination, they can be thought of as a Cartesian coordinate system in which one can place the various affective states, depending on the degree of pleasure or displeasure and perceived activation they entail. For example, there are states that combine pleasure and high activation (energy, vigor), displeasure and high activation (tension, distress), pleasure and low activation (calmness, relaxation), and displeasure and low activation (tiredness, boredom). In psychology, this two-dimensional model of affect is known as the *affect circumplex*. Alternatively, researchers have also used a version of this coordinate system rotated by 45°. In that case, the dimensions extend from pleasant high activation (termed high *positive activation*) to unpleasant low activation (termed low *positive activation*) and from unpleasant high activation (termed high *negative activation*) to pleasant low activation (termed low *negative activation*).

Regardless of which rotational variant is used, these dimensional models of affect can serve as encompassing maps of affective space. By gathering information about each respondent's position on only two dimensions, investigators can get a meaningful representation of the respondent's affective state and, with repeated assessments, track the individual's movement in response to experimental manipulations, including sport- and exercise-related stimuli.

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See also Affective Disorders; Affective Responses to Exercise; Emotional Reactivity; Emotional Responses; Energy, Effects of Exercise on; Hedonic Theory

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AFFECTIVE DISORDERS

Affective disorders, also known as mood disorders, are clinical psychological disorders. The most common affective disorders are major depressive disorder, dysthymic disorder, bipolar disorder, and cyclothymic disorder. A core feature of these disorders is dysfunction in emotion processing and neurohormonal regulation leading to subjective feelings of sadness, depressed mood, and loss of pleasure in things normally pleasurable (anhedonia) for 2 weeks or more. These symptoms must also subjectively impair the fulfillment of social or occupational responsibilities. Additional possible symptoms include cycling episodes of mania in bipolar disorder; insomnia or hypersomnia; feelings of worthlessness, guilt, suicidal thoughts; and psychomotor agitation (restlessness, pacing) or psychomotor retardation (fatigue, tiredness). Affective disorders often co-occur with anxiety disorders, such as panic disorder, generalized anxiety disorder, posttraumatic stress disorder, and social phobia. Women are at greater risk than men for the development of both affective and anxiety disorders.

Diagnosis of an affective disorder, mood disorder, or anxiety disorder requires an extensive in-person interview with a licensed clinical psychologist or psychiatrist to establish whether criteria for the diagnosis are met. A score on a self-report survey of depression or anxiety symptoms, even when administered by a licensed clinician, is not sufficient for diagnosis. The primary diagnostic criteria have been set forth by the American Psychiatric Association (APA) in the *Diagnostic and Statistical Manual for Mental Disorders, 4th Edition, Text Revision* (DSM-IV-TR) and by the World Health Organization in the *International Statistical Classification of Diseases and Related Health Problems*, 10th revision (ICD-10). Adherence to these diagnostic

standards have been difficult in the fields of sport and exercise psychology, as few in the field have the necessary credentials or lack the financial or collegial resources.

Nevertheless, it is critical to understand the effects of leisure-time physical activity and acute and chronic exercise on affective and anxiety disorders. The focus here will be on the use of exercise as a treatment intervention among individuals diagnosed with affective or anxiety disorders. However, it is also important to understand how these disorders may affect physical activity behavior in general. Symptoms of depression are associated with lower levels of physical activity and inhibition of behavioral activation. For example, feelings of hopelessness and fatigue are difficult to overcome, and, as such, these patients experience difficulty in engaging in effortful tasks. In addition, among athletes, there is evidence that a core feature of the staleness syndrome (as a result of overtraining) is depressed mood, and the symptoms of staleness map directly onto the diagnostic criteria for a major depressive episode. Monitoring depressed mood in athletes may be a method to help avoid staleness during an overtraining period.

Major Depressive Disorder

Epidemiological studies consistently indicate that greater levels of physical activity or cardiorespiratory fitness are related to reduced risk for the future development of major depression in both men and women. Engaging in regular physical activity provides protection against symptoms of depression, compared with being sedentary. However, there is not strong evidence for a dose response effect, so greater levels of physical activity are not necessarily more protective. Exercise training has been shown to be an effective treatment for major depression. Both aerobic exercise (walking, jogging) and resistance exercise, compared with a wait list or non-exercise control condition, have been shown to effectively reduce symptoms of depression and result in a remission in symptoms of depression. Walking or jogging exercise interventions, 4 to 6 months in duration (but longer is better), have been shown to be as effective as antidepressant drug therapy and cognitive behavioral therapy compared with a placebo. Exercise is a good treatment option or adjuvant to treatment for major depression; however, adding exercise

training to pharmacologic or cognitive behavioral therapy does not produce synergistic effects. The behavioral deactivation and extreme feelings of hopelessness and fatigue present clear challenges to the initiation of and adherence to a physical activity or exercise training program, although one possible advantage of exercise training over pharmacotherapy is that remission of symptoms may persist for a longer time after the exercise and drug treatments have ended.

Bipolar Disorder

Although there is increasing interest in using exercise as a treatment in bipolar disorder, and high-functioning bipolar disorder patients report exercise as one of many methods they use to help maintain emotional stability, very little empirical research and no clinical trials for exercise have been conducted in patients with bipolar disorder. One study has shown that markers of cardiovascular disease risk can be improved with exercise in patients diagnosed with bipolar disorder, but it is unknown if exercise can improve the core symptoms of bipolar disorder.

Panic Disorder

Patients diagnosed with panic disorder tend to be less physically active than their healthy counterparts. This may be due, in part, to feelings of discomfort experienced during exercise. The physiological arousal due to exercise (increased heart rate and respiration) is similar to the core symptoms of a panic attack, and thus may be avoided. Another reason may be due to false beliefs that exercise will cause a panic attack. The evidence, however, clearly indicates that exercise is safe for people diagnosed with panic disorder and exercise, even at maximal capacity, does not cause panic attacks. The very few documented instances of panic attack during exercise can be viewed as chance occurrences relative to the number of documented exercise and physical activity sessions that did not involve a panic attack. Exercise training is known to be a very useful treatment for panic disorder and can be useful as a cognitive restructuring tool ("I can sweat and breathe hard and my heart can beat very fast, and it does not mean I am about to die or that I am going crazy"). Exercise is comparable to pharmacological treatments for reducing clinician rated symptoms of panic disorder. However, the combination of drug

treatment with exercise training does not produce a synergistic effect.

Generalized Anxiety Disorder

There is epidemiological evidence that greater levels of physical activity or cardiorespiratory fitness are related to reduced risk for the future development of anxiety disorders. However, very few studies have tested the effects of exercise as a treatment for generalized anxiety disorder (GAD). In two clinical trials, both aerobic exercise and resistance exercise resulted in significant symptom reductions compared with a wait-list control condition. Exercise has not currently been compared with drug treatments or other treatment methods. Additionally, the affective experience during or immediately after exercise in GAD patients has not been examined. There is very little information about how a single session of exercise affects symptoms in people clinically diagnosed with affective or anxiety disorders.

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See also Affect; Emotional Reactivity; Emotional Responses; Mental Health; Psychological Well-Being; Stress Reactivity

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AFFECTIVE RESPONSES TO EXERCISE

Exercise can influence how people feel. This observation has attracted considerable research attention in the last 50 years. There are several reasons for this. First, if exercise can improve how people feel, this could have significant implications for mental health. Disorders impacting mood (depression, dysthymia, bipolar disorder) and anxiety (generalized anxiety, phobia, posttraumatic stress) are prevalent and can have a devastating effect on quality of life for sufferers and their families. Moreover, standard therapies such as pharmacotherapy and psychotherapy are costly and not always effective. Psychoactive drugs in particular can have several undesirable side effects. Against this backdrop, exercise offers the promise of an intervention that can be effective (by some estimates, at least as effective as the standard forms of therapy), inexpensive, free of undesirable side effects, and associated with many additional benefits for the body (e.g., reduced cardiovascular risk) and mind (e.g., reduced risk for dementia).

Second, people engage in various unhealthy lifestyle behaviors to regulate how they feel. For example, they consume caffeine and sugary snacks to feel more energized and they smoke cigarettes or drink alcohol to calm the nerves and relax. In pursuit of a feel-better effect, some people even engage in illicit and dangerous activities, such as abusing psychotropic drugs. Over time, these behaviors may lead to serious problems, from obesity and diabetes to chronic cardiorespiratory conditions to life-endangering addictions. Therefore, it would be desirable to replace these behaviors with an alternative that has the same affect-enhancing properties without a negative impact on health. Exercise engages some of the same brain mechanisms targeted by widely abused chemical affect regulators, such as dopamine, endogenous opioids, and endocannabinoids; can increase perceived energy and calmness; and has positive, rather than negative, effects on overall health.

Third, the low adherence to exercise represents a major public health problem. Most people who become physically active either do not exercise regularly or quit. Although most contemporary theories assume that nonadherence and drop-out are the result of a rational decision-making process, these phenomena may also be driven by affective processes. Affect is a powerful motive in human

behavior. People may adhere to exercise if their affective responses are positive and may drop out if their affective responses are consistently negative. This possibility, which has received empirical support, offers researchers new insight into the mechanisms underlying exercise behavior.

History

The first studies on affective responses to exercise appeared in the late 1960s. The typical methodological approach consisted of administering a questionnaire of mood (such as the Profile of Mood States) or anxiety (such as the State-Trait Anxiety Inventory) shortly before and after an exercise bout. At the time, very few questionnaires were designed to assess nonclinical forms of how people feel. Therefore, the limited availability of measures dictated the dependent variables being studied. Consequently, those variables might or might not have been the most relevant, raising the possibility that changes also occurred in variables other than those being measured. The samples of respondents typically consisted of conveniently accessible groups, such as young, healthy, physically active, and fit university students. The intensity of exercise was rarely monitored via objective means (electrocardiography or expired gases) and, when it was standardized across participants, the method was often based on estimated, rather than directly measured, maximal exercise capacity (typically, age-predicted maximal heart rate, known to result in considerable errors). Despite these methodological limitations, which were consistent with a nascent line of research, early studies provided voluminous evidence of an exercise-associated anxiolytic and mood-enhancing effect.

Mechanisms

In the 1980s and 1990s, along with numerous replications of the anxiolytic and mood-enhancing effects across different settings, samples, and types of exercise, research attention turned to mechanistic hypotheses. These included proposals that exercise makes people feel better because (a) they perceive that they are doing something challenging and, at the same time, beneficial (the mastery hypothesis); (b) it provides an opportunity to temporarily escape the stresses and hassles of daily life (the distraction or time-out hypothesis); (c) it provides an opportunity for enjoyable social

interaction (the social interaction hypothesis); (d) it corrects imbalances in monoaminergic neurotransmission that are associated with negative affectivity (the monoamine hypothesis); (e) it promotes the release of peripheral and central endogenous opioids (the endorphin hypothesis); and (f) it raises core temperature, which creates a sense of relaxation or exhilaration (the thermogenic hypothesis).

The conclusions from these investigations have been mixed. What seems clear is that no single explanation can provide an exclusive account of the reasons why exercise can make people feel better. Studies on the mastery hypothesis have demonstrated that participants whose physical confidence is strengthened report feeling better than those whose physical confidence is weakened. On the other hand, the thermogenic hypothesis has been largely discredited, with studies demonstrating that elevations in core temperature during exercise are associated with feeling worse, not better. The endorphin hypothesis continues to hold promise. However, interest in this idea has declined following a string of studies that produced conflicting results and, thus, confusion and frustration among researchers. However, upon closer analysis, the inconsistencies can be attributed to methodological weaknesses, which, in turn, could be due to the lack of interdisciplinary expertise on the physiology and pharmacology of the endogenous opioid system. The distraction and social-interaction hypotheses may provide partial explanations, but there are caveats for both. Specifically, while other distracting activities may also produce a feel-better effect, exercise often produces changes that are qualitatively different. For example, while a session of meditation or a period of quiet rest may primarily induce relaxation, a typical response to a bout of moderate-intensity exercise consists of an increase in perceived energy during and immediately following the bout and, only later, an increase in relaxation compared to baseline. Furthermore, while an enthusiastic and supportive social group can enhance the positive affective response to exercise, an indifferent group may experience no effect and a group or an exercise leader perceived as critical can have a negative effect. Moreover, studies have shown that people can feel better even when they exercise in an empty room while staring at a barren wall. The monoamine hypothesis remains viable, with findings showing that monoamines (serotonin, dopamine) may be implicated in the

feel-better effect. However, at least for now, this research is limited to experimental animals, with all the interpretational challenges that this entails, that is, inability to directly extrapolate from observable animal behavior to subjective human feelings.

Mechanistic research has now moved in some notable new directions. First, studies have begun exploring associations between affective responses to exercise and neurotransmitter dynamics. Advances in positron emission tomography have made it possible to quantify exercise-associated changes in receptor occupancy in the human brain. Second, research is emerging on the role of endocannabinoids, a class of substances discovered relatively recently, that are extensively involved in reward. Both experimental studies with animals and preliminary correlational studies of peripherally circulating endocannabinoids in humans suggest that these substances may add one more piece to the mechanistic puzzle. Third, research is examining the role of exercise-upregulated neurotrophic factors in anatomical adaptations in the human brain that may be associated with how people feel. While chronic psychological stress is associated with reduced synthesis of neurotrophic factors and reduced volumes of brain structures involved in emotion and mood regulation, exercise is among the most potent known stimuli for the upregulation of these neurotrophic factors.

Beyond the Feel-Better Effect

Critics express skepticism about the ability of exercise to make people feel better based on a simple but intriguing argument: If exercise could, in fact, make people feel better, would most people be sedentary? Research based on a new methodological platform is beginning to show that the feel-better effect, while feasible, is neither automatic nor guaranteed for everyone. It should more accurately be described as conditional.

One of the methodological innovations was the introduction of measures that tap the main dimensions of affect, as opposed to a few discrete affective states. Theoretically, the advantage is no major variant of affective experience resulting from exercise (including negative variants) can go undetected. A second aspect of the revised methodology is the timing of affect assessments. It became clear that, by measuring only before and after the exercise bout, the shape of the affective response could be misrepresented. For example, depending on the

intensity of exercise, pleasure could be reduced during exercise but rebound postexercise. However, if affect is assessed only before and after the bout, one could conclude that the only change was a pre-to-post increase in pleasure. Thus, newer studies have employed repeated assessments of affect, both during and after the bout. Thirdly, newer studies use more accurate methods for standardizing exercise intensity, reducing error variance and increasing statistical power. The measurement of expired gases has become common practice. Furthermore, several laboratories base the standardization of intensity on the more laborious but more meaningful practice of identifying physiological markers, such as the ventilatory or lactate threshold and the respiratory compensation point. These markers differ among individuals, even of the same sex, age, health status, activity habits, and aerobic capacity. Research suggests that exercising at intensities slightly above and below these markers may be associated with considerable differences across several physiological systems as well as differences in affective responses. Finally, once it became clear that affective responses varied between individuals, even in response to the same, well-standardized, exercise stimulus, it also became apparent that analyses of change restricted to the level of entire groups could be misleading. This is because subgroups within the same sample may respond in different directions (e.g., increased versus decreased pleasure). Thus, it is possible for two subgroups to exhibit changes of equal magnitude but in opposite directions, resulting in a group mean that appears unchanged over time. In such cases, the sample mean fails to reflect the actual response of individuals, becoming merely a statistical abstraction. To address this problem, in newer studies, change is examined both at the level of the entire sample and at the level of individuals and subgroups.

The conclusion from studies based on this revised methodology is that the feel-better effect represents only one aspect of the multifaceted exercise–affect relationship. Interindividual differences are prevalent and reductions in pleasure are common. For example, obese and inactive middle-age women report declines in pleasure across the entire range of exercise intensity.

Affective Responses and Exercise Prescription

The optimization of affective responses to exercise is gradually being adopted as one of the pillars

of exercise prescription guidelines, alongside the maximization of biological adaptations like gains in fitness and health and the minimization of risk. Exercise practitioners are advised to systematically monitor the affective responses of participants and to regulate exercise intensity to ensure that affective responses remain positive or at least nonnegative. This can be achieved by (a) allowing participants to self-select their intensity, in order to engender a sense of perceived autonomy and self-efficacy; and (b) ensuring that intensity does not greatly exceed the ventilatory threshold (which can be estimated without instruments as the level of intensity that brings about a perceptible increase in the frequency and depth of ventilation and a subjective characterization of perceived exertion as “somewhat hard” or “hard”). Maintaining proper hydration and comfortable ambient temperature and humidity levels is also important.

Furthermore, it is crucial to recognize that the relationship between exercise intensity and affective responses is influenced by individual differences. Because of a combination of genetic and epigenetic factors, people develop varied preferences for levels of exercise intensity and different degrees of tolerance to intense exercise. These differences influence the affective responses that individuals experience at different intensities. Although a standard method of tailoring exercise intensity to individual levels of preference and tolerance has yet to be developed, practitioners should keep in mind that what was pleasant for one participant may not be pleasant for another.

Finally, it is advisable to maintain a social environment in which participants can feel confident and secure. The presence of other exercisers who appear to be of superior fitness or an exercise leader who emphasizes skill, appearance, or interpersonal comparisons could induce social-evaluative and self-presentational concerns.

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See also Addiction, Effects of Exercise on; Affect; Emotional Responses; Energy, Effects of Exercise on; Hedonic Theory; Runner's High

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AFFIRMATIONS

Affirmation is the act of reflecting on core aspects of the self, such as important values, relationships, and personal characteristics like religion, music, or sports. Previous research shows that self-affirmation interventions can reduce psychological and physiological stress and defensiveness, while boosting personal responsibility and performance. Self-affirmation interventions and theory have promising applications in sports and exercise, including facilitating achievement and helping individuals respond adaptively to setbacks.

Self-Affirmation Theory

The social psychologist Claude Steele proposed self-affirmation theory in 1988. It holds that individuals are motivated to maintain self-integrity: a sense that one is a person of worth, morally adequate and effective at making changes in one's life. There are many routes to self-integrity, and affirmations of the self in one part of life (e.g., reflecting on being a good father) can buffer threats in other parts of life (e.g., poor performance). Affirmations in the context of threat can protect the self and allow people to respond with reduced stress and defensiveness because they are reassured that they possess integrity and worth.

When an event such as a sports loss or failure to complete a workout regimen threatens a valued self-image (e.g., being a good athlete or motivated exerciser), people are at risk of responding

defensively by rejecting responsibility or giving up. If, however, the person affirms an important personal value before the threat, their sense of moral adequacy and efficacy can be reinforced and protected. Within social psychology, interventions involving values affirmations often take the form of having individuals reflect and write briefly about an important personal value such as relationships with friends and family. Writing about important personal values can fulfill the global need for self-integrity and enable people to constructively respond to threatening events.

Reduction of Defensive Strategies

Sport and exercise present psychological threats like the fear of low performance that can impact one's personal and public image. There is empirical evidence that people can respond to these threats by construing situations as less threatening to personal worth and well-being. For example, athletes may use defensive strategies such as attributing more internal causes for success than for failure (e.g. "I won because of my ability," but "I lost because of the weather": self-serving biases); denying their team's responsibility for a negative outcome or exaggerating their role in victory (group-serving biases); or claiming handicaps (e.g., claiming back pain before a competition to have an excuse for failure or to enhance credit for success: claimed self-handicapping). These defensive strategies help maintain self-integrity by reducing threats but can limit achievement when personal responsibility is denied and failure is attributed to external causes. Self-affirmation can reduce engagement in these maladaptive strategies.

For instance, a field study demonstrated how self-affirmation can lower athletes' engagement in self-handicapping strategies. Claimed self-handicapping was assessed before and after an affirmation intervention. First, coaches asked their athletes to report to what extent handicaps such as physical pain or stress could disrupt their training. Using a classic self-affirmation study design, athletes assigned to an affirmation condition ranked a list of values (e.g., relationships with friends) from the most important to the least important, and then wrote an essay about their most important value. Athletes in a no-affirmation control condition ranked the same values but wrote an essay on why their least important value might be important to someone else. Athletes in the affirmation

condition claimed fewer handicaps after the intervention (no difference in the control condition).

Field studies with athletes immediately after competition examined their attributional patterns for victories and defeats. The studies demonstrated that an affirmation manipulation reduced self-serving and group-serving attributional biases. Without affirmation, winning team members claimed that their efforts and their team's efforts were more responsible for the outcome of the game than losing team members'. These findings were observed for players as well as nonplayer fans, such that collegiate fans were less defensive in their attributions about their team's outcomes when they affirmed a value central to their university. In health psychology, affirmed individuals are less defensive and more open to learning about their health risks, and more likely to take behavioral steps to address drinking, diabetes, or excessive weight. One study found that overweight women who completed a self-affirmation lost more weight than women in a control condition, suggesting that the threat and stress stemming from their appearance may have hindered their attempts to diet and exercise.

Reduced Stress

Self-affirmation can reduce physiological and psychological stress responses. Compared to a control group, participants who affirmed personal values by reporting their thoughts and feelings about an important value had lower salivary cortisol responses, a marker of stress, in a stressful laboratory task. In a longitudinal study, compared to control students who had a marked increase, students who affirmed personal values 2 weeks prior to an academic evaluation did not have increased cumulative epinephrine levels from baseline (an indicator of stress measured in urine).

Increased Performance

Whereas threat depresses performance, affirming core values could alleviate threat and improve performance. In both laboratory and field studies, self-affirmations have improved academic performance among people confronting a negative stereotype about their ability; for example, it improved the academic performance of African American and Latino American, but not White, students in mixed middle schools in the United States. These effects persist for years by changing the narrative students tell themselves about their

ongoing experience, thereby instigating recursive processes and positive feedback loops.

Conclusion

In sum, sports research demonstrates that self-affirmation reduces athletes' defensiveness, whereas other research shows that it helps address health problems, reduces stress responses, and boosts academic performance. Future research should address the specific effect of self-affirmation on the stress, performance, and commitment to a training regimen among both athletes and exercisers.

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See also Attribution Theory; Identity; Self-Appraisal/Assessment/Perception; Self-Handicapping; Self-Presentation; Stereotype Threat; Team Attributions

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AGGRESSION

Aggression has a long history in both sport and nonsport contexts. There is some variation in the definitions of aggression employed by different people. However, it is commonly agreed that *aggression* is a verbal or physical behavior that is directed intentionally toward another individual and has the potential to cause psychological or physical harm. In addition, the target of the behavior should be motivated to avoid such treatment. Typically, definitions of aggression incorporate the

notion of *intent to cause harm*; that is, for behavior to be classified as aggressive, the perpetrator must have the intent to harm the victim. However, strict behavioral definitions of aggression exclude the term *intent* because it refers to an internal state, which cannot be observed.

Aggression has been distinguished between *instrumental* and *hostile*. Instrumental aggression is a behavior directed at the target as a means to an end, for example, injuring a player to gain a competitive advantage, or late tackling to stop an opponent from scoring. Thus, instrumental aggression is motivated by some other goal. In contrast, hostile aggression is a behavior aimed toward another person who has angered or provoked the individual and is an end in itself. Its purpose is to harm for its own sake, for example, hitting an opponent who has just been aggressive against the player. Hostile aggression is typically preceded by anger. Instrumental aggression, in pursuit of a goal, is not normally associated with anger and, in sport, is far more frequent than hostile aggression. In both types of aggression, a target person is harmed, and the harm can be physical or psychological.

In this entry, the construct of aggression is presented. First, the distinction is made between aggression and assertion, and difficulties with the notion of intent in the definition of aggression are discussed. Then measures of aggression are outlined followed by factors associated with aggression in sport.

Aggression, Assertion, and Intent

In sport, the word *aggressive* is often used when *assertive* is more appropriate. For example, coaches describe strong physical play as aggressive, when this type of play is actually assertive; it is within the rules of the game and there is no intention to cause harm. The difference between aggression and assertion lies in the intention to harm. If there is no intent to harm the opponent, and the athlete is using legitimate means to achieve goals, the behavior is assertive, not aggressive. When one is being assertive, the intention is to establish dominance rather than to harm the opponent. Behaviors such as tackling in rugby, checking in ice hockey, and breaking up a double play in baseball may be seen as assertive as long as these are performed as legal components of the contest and without malice. However, these same actions would represent aggression if the athlete's intention was to cause injury.

It is often difficult to distinguish aggression from assertion in sport. Although assertive behaviors are forceful behaviors that are not intended to injure the victim, by their nature, they may result in unintended harm to the athlete's opponent. In addition, some sports involve forceful physical contact, which *has the capacity* to harm another person, but this contact is within the rules of sport. Assertive behaviors have also been labeled *sanctioned aggression*. Thus, sanctioned aggression is any behavior that falls within a particular sport's rules or is widely accepted as such: for example, using the shoulder to force a player off the ball in soccer and tackling below the shoulders in rugby. Examples are combat sports, such as judo, karate, and wrestling, and team contact sports, such as rugby, ice hockey, American football, and lacrosse. Perhaps the confusion between assertion and aggression arises because both have the capacity to harm the target, although, as noted earlier, only aggression involves intention to harm.

Incorporating the notion of intent in definitions of aggression has the difficulty of establishing which behavior is aggressive. This is because the only person knowing whether there is intent to cause harm is the person who carries out the action. Two features of definitions of aggression that have not been questioned are the capacity of behavior to cause harm and the intentional (nonaccidental) nature of the behavior.

The Measurement of Aggression

The notion of *intent*, which is part of most definitions of aggression, has created difficulties in the measurement of aggression. Therefore, many studies have operationally defined and measured aggression without considering intent, or the reasons for the behavior. A very common aggression measure in the laboratory context is administering electric shocks, which is known to hurt the participant. Thus, aggression is reflected in the intensity of the shock administered. Other studies used delivering an aversive stimulus, for example a loud noise, as their measure of aggression.

In the sport context, aggression has been measured in a variety of ways, such as number of fouls, coach ratings, penalty records, as well as using self-reports and behavioral observation. In studies of behavioral observation, instrumental and hostile aggression have been measured. *Instrumental aggression* has been operationally defined as aggression occurring during game play

and involves opponent-directed physical interactions that contribute to accomplishing a task. In contrast, *hostile aggression* has been operationally defined as physical or verbal interactions aimed at various targets but not directly connected to task accomplishment; these behaviors are directed at opponents, teammates, or referees. For example, in handball, repelling, hitting, and cheating have been coded as instrumental aggression, and insulting, threatening, making obscene gestures, and shoving against opponents, referees, teammates, and others have been coded as hostile aggression. Aggressive behaviors (e.g., late tackle, hitting, elbowing) have also been measured as part of the construct of *antisocial behavior*, which has been defined as behavior intended to harm or disadvantage another individual and has considerable overlap with aggression.

Other studies have used athlete self-reports to measure aggression, either by presenting them with a scenario that describes an aggressive behavior and asking about their intentions or likelihood to aggress, or by asking them to respond to a number of items measuring aggressive or antisocial behavior. Self-described likelihood to aggress has been used as a proxy for aggression. In these studies, participants are presented with a scenario in which the protagonist is faced with a decision to harm the opponent to prevent scoring and they are asked to indicate the likelihood they would engage in this behavior if they were in this situation. Finally, aggression (e.g., trying to injure another player) has been measured as part of antisocial behavior in sport.

Why Aggression Occurs

Aggression has a long history in both mainstream psychology and sport psychology. One view is that aggression results from *frustration*. In sport, frustration can occur for a variety of reasons: because of losing, not playing well, being hurt, and perceiving unfairness in the competition. Frustration heightens one's predisposition toward aggression. Contextual factors come into play so that the manner in which an individual interprets the situational cues at hand best predicts whether this athlete, or spectator, will exhibit aggression.

Some theorists view aggression as a learned behavior, which is the result of an individual's interactions with personal social environment over time. Aggression occurs in sport where an

athlete's expectancies for reinforcement for aggressive behavior are high (receiving praise from parents, coaches, peers), and where the reward value outweighs punishment value (gaining a tactical or psychological advantage with a personal foul). Situation-related expectancies, such as the time of game, score opposition, or the encouragement of the crowd, also influence the athlete in terms of whether this is deemed an appropriate time to exhibit aggression.

A number of individual difference factors have been associated with aggression. Three of them are *legitimacy judgments*, *moral disengagement*, and *ego orientation*. When athletes judge aggressive and rule-violating behaviors as legitimate or acceptable, they are more likely to be aggressive. Moral disengagement refers to a set of psychosocial mechanisms that people use to justify aggression. Through these justifications, athletes manage to engage in aggression without experiencing negative feelings like guilt that normally control this behavior. For example, players may displace responsibility for their actions to their coach, blame their victim for their own behavior, claim that they cheated to help their team, or downplay the consequences of their actions for others. Finally, individuals who are high in ego orientation feel successful when they do better than others; they are preoccupied with winning and showing that they are the best. These players are more likely to be aggressive in sport.

Social environmental variables are also associated with aggression. One of them is the performance motivational climate, which refers to the criteria of success that are dominant in the athletes' environment. Through the feedback they provide, the rewards they give, and, in general, the way they interact with the players, coaches make clear the criteria of success in that achievement context. As an example, when coaches provide feedback about how good a player is relative to others and reward only the best players, they create a performance motivational climate, sending a clear message to athletes that only high ability matters. Players who perceive a performance climate in their team are more likely to become aggressive.

Conclusion

Aggression is a construct with a long history and considerable debate around its definition, primarily due to the difficulties of determining whether

the perpetrator has the intention to harm the victim when acting in a certain way. Aggression can be instrumental or hostile. Many sports involve forceful play, which could result in an injury. However, if players do not intend to harm the opponent, this play is considered as an assertive act, not an aggressive one. Finally, several individual difference and social environmental factors have been associated with aggression in sport.

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See also Moral Behavior; Moral Disengagement

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ALCOHOL ABUSE

Hazardous alcohol use is a significant health problem that affects many people. In the United States, almost 10% of the population will meet past-year diagnostic criteria for either alcohol abuse or alcohol dependence, with the highest rates occurring among college students and other young adults. Alcohol use disorders co-occur with mental health problems like depression, anxiety, and other substance use disorders, and can cause a variety of physical ailments. According to Bouchery, Harwood, Sacks, Simon, and Brewer (2011), the economic cost of alcohol use disorders in the United States is approximately \$223.5 billion each year.

Despite the fact that alcohol use is known to be harmful toward athletic performance, rates of alcohol use are relatively high among some groups of athletes. This entry compares rates of alcohol

use between athletes and nonathletes, discusses sport-related factors that might impact alcohol use among athletes, and highlights effective intervention and prevention strategies.

Rates of Alcohol Use Among Athletes

A number of studies across several countries have shown adolescent athletes consume alcohol at rates similar to or higher than peers. Findings from several recent studies, though, suggest the relationship between sport participation and alcohol use among adolescents is impacted by other factors. One national study of U.S. adolescents found self-reported rates of heavy drinking and drinking and driving in the past 30 days were higher for male athletes versus male nonathletes. In contrast, female athletes reported lower rates of ever using alcohol or use within the past 30 days versus female nonathletes. Another national, longitudinal study found that adolescents within the United States whose extracurricular activities included only sports display accelerated rates of alcohol use and alcohol-related problems. In contrast, involvement in sports and extracurricular academic activities was associated with a deceleration in alcohol use and related problems. Additionally, a national study of Norwegian high school students found participation in collaborative team sports like soccer was associated with an increase in alcohol intoxication over time, but participation in endurance sports like running was associated with a decrease in alcohol intoxication over time. Thus, the answer to the degree to which sports participation among adolescents is a risk or protective factor for alcohol use is not a simple one, but is instead often contingent upon a variety of factors.

Research examining the relationship between sport participation and alcohol use among college athletes in the United States has provided clear evidence that athletes tend to consume more alcohol than nonathletes. For example, in three national studies with sample sizes ranging from 12,777 to 51,483, the researchers reported past 2-week binge drinking rates of 57% to 62% and 48% to 50% among male and female college athletes, respectively. These percentages were approximately 15 points higher than corresponding rates for nonathletes. Similar patterns emerged for other measures of alcohol use, such as frequent binge drinking and average number of drinks per week. As one might

expect given these differences in heavy drinking rates, college athletes were also more likely than other students to experience problems from alcohol like impaired academics, trouble with the authorities, and participation in behaviors later regretted. There is also evidence to suggest that college students who engage in recreational sports like club teams and intramurals are more at risk for excessive alcohol use than other students.

Relatively few studies have examined rates of alcohol consumption among professional or other elite athletes, particularly in terms of comparing them with relevant nonathlete groups. Those that have been conducted suggest rates of alcohol use among adult elite athletes are higher than general population rates. Research is also lacking on rates of alcohol use disorders among athletes. However, it is likely that rates of alcohol abuse and dependence are particularly high among some groups of athletes, especially those where evidence suggests they experience more alcohol-related problems than others (e.g., college athletes).

Sport-Related Factors and Alcohol Use

Researchers have identified numerous factors that increase the likelihood of hazardous alcohol use in the general population, including demographic characteristics, genetic factors, personality variables, environmental factors, and a host of other individual, interpersonal, and contextual variables. The impact of such factors is presumably consistent between athletes and nonathletes, but researchers have also identified several sport-related factors that may increase the likelihood of heavy drinking among athletes.

There is a clear cultural link between athletics and alcohol use in many countries. Alcohol beverage companies advertise heavily during televised sporting events and provide key sponsorship for many sporting leagues. In some countries, alcohol companies even provide direct sponsorship for individual teams and players. Research has shown that athletes receiving alcohol industry sponsorship report higher rates of hazardous drinking than those who do not receive such sponsorship. Other research has documented an association between exposure to alcohol advertising and subsequent alcohol consumption. It is therefore possible that athletes are more likely than others to be influenced by the advertising or sponsorship efforts of alcohol beverage companies.

A second set of factors that may be associated with heavy alcohol use among athletes is a particular susceptibility to the positive and negative reinforcing aspects of alcohol. For example, the personality trait of sensation seeking has been shown to be positively associated with alcohol consumption, and several studies have shown that athletes are more likely than others to report high levels of this trait. Similarly, a number of writers have suggested that some groups of athletes experience especially high levels of stress and other pressures, such as college athletes attempting to balance the demands of athletics and academics. Such individuals are thought to be particularly prone to using alcohol as a negative reinforcing coping strategy (e.g., reducing stress, distracting from life's problems), although research studies have not provided convincing support that this is in fact the case. There may be other factors that are associated with both the likelihood of participating in athletics and the likelihood of engaging in at-risk alcohol use.

Increased access to alcohol may also account for heavier drinking rates among athletes in comparison with the general population. Athletes at many competitive levels often have more social opportunities involving alcohol than others. For example, college athletes are usually among the most popular students on campus, and therefore have ample opportunities to attend parties or other gatherings where they will be provided alcohol. Similarly, many athletes socializing in public establishments, particularly those who are recognizable in their communities, will experience the phenomenon of others wishing to buy them drinks or otherwise supply them with alcohol. Thus, heavy drinking among some athletes may be partially explained by relatively easy access to a supply of free or low-cost alcoholic beverages, which would be consistent with basic behavioral economics theories.

A final factor that may impact at-risk drinking among athletes involves their seasonal calendar cycle. Research suggests athletes tend to limit alcohol use during their competitive seasons, but drinking rates increase in the off-season. Some athletes may engage in particularly heavy drinking during the off-season believing (a) they are not harming their athletic performance since they are not in-season; and (b) they have to take advantage of a limited timeframe that does not involve regular practices, games, and accountability to coaches. Such a spike in heavy drinking can lead

to increased likelihood of a host of severe alcohol-related consequences.

Interventions for Hazardous Drinking Among Athletes

Unlike both recreational and performance enhancing drugs, regular testing for the presence of a substance is not a logistically feasible deterrent for alcohol use among athletes. Thus, it is particularly important to explore alternative strategies for preventing harmful alcohol use among athletes. Several effective treatments have been identified for individuals experiencing alcohol use disorders, including cognitive behavioral therapy, twelve-step facilitation therapy, and behavioral family therapy. Athletes experiencing significant problems with alcohol should be referred to settings where intensive treatment could be provided.

It is also important to provide interventions to those who may be at risk for experiencing alcohol-related problems but whose current alcohol use habits do not necessarily warrant extensive treatments. Over the past 10 to 15 years, clinical researchers have examined the efficacy of brief interventions in reducing harmful alcohol use. One of the most popular and efficacious approaches involves a single-session model where the clinician uses a motivational interviewing-based style and provides personalized feedback about one's drinking habits. Motivational interviewing is designed to increase an individual's motivation to change behavior by exploring and resolving ambivalence regarding change, and this process can be facilitated by receiving personalized information on one's drinking habits. Commonly included pieces of personalized feedback include social norms information (how one's own alcohol use and perceived typical alcohol use among others compares to actual population norms), a summary of alcohol-related risks or problems experienced, and possible genetic risk for an alcohol use disorder. More recently, researchers have explored the efficacy of personalized feedback-only interventions where the feedback is provided without one-on-one clinician contact. Three studies have examined the effects of these interventions specifically among athletes, all of which showed positive effects in terms of reducing alcohol consumption relative to control conditions. One of the studies included feedback that was targeted specifically for athletes (e.g., the impact of alcohol use on one's athletic

performance), which was shown to be more effective than personalized feedback that did not include the athlete-targeted information. Other studies have provided promising support for interventions focusing exclusively on correcting misperceptions of drinking norms, although they have been limited by the lack of proper control conditions.

An important gap in the literature on the efficacy of alcohol-related interventions among athletes is that the effectiveness of interventions delivered via sporting organizations themselves has not been examined. There are, though, potentially promising avenues that could be explored in this area. For example, research has supported the efficacy of brief advice interventions delivered by physicians and parent-based interventions, both of which could be modified to be delivered by athletic organization personnel like coaches and team doctors. It may also be possible to integrate alcohol interventions into the context of a team's sports medicine staff, which would be advantageous given the degree to which athletic trainers and other sports medicine staff are often the ones working most closely with athletes on a variety of health-related issues.

Conclusion

Some groups of athletes are particularly at risk for excessive alcohol use that can lead to a variety of negative or harmful outcomes, including impaired athletic performance. Researchers and theorists have explored several sport-related factors that might serve to heighten the risk of heavy drinking among athletes, although the specific ways in which many of these factors impact alcohol use are not well understood. Fortunately, a number of interventions exist that can either reduce or prevent problematic alcohol use, including brief models that have been shown to be efficacious specifically among athletes.

Matthew P. Martens

See also Moral Behavior; Performance-Enhancing Drugs; Recreational Drugs

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ANTICIPATION

In sport and exercise psychology, *anticipation* usually refers to the ability to quickly and accurately predict the outcome of an opponent's action before that action is completed. Skilled athletes can use bodily cues to anticipate outcomes at earlier moments in an action sequence than can unskilled athletes, allowing them more time to perform an appropriate response in time-stressed tasks. A basic understanding of anticipation requires a comprehension of how skilled athletes anticipate actions, how anticipation is best tested, and what the practical implications are for training anticipation.

How Skilled Athletes Anticipate Actions

Anticipation is most commonly tested by occluding vision at a critical point in an action sequence,

after which the observer must predict the action outcome. For instance, a tennis player may observe an opposing player performing a serve, but at the moment of racquetball contact, vision is occluded, and the receiver must predict the direction of the serve. Skilled athletes in a wide range of sports do better than lesser skilled performers in these tests, including in tennis, soccer goalkeeping, squash, and batting in baseball and cricket. Occlusion is achieved in the laboratory using edited video footage or in the field by using liquid crystal glasses that quickly and selectively occlude vision.

Skilled athletes anticipate action outcomes based on events presented earlier in a movement sequence, providing a distinct advantage for sports skill that must be performed under severe time constraints. The selective occlusion of different body segments (e.g., the arms or legs) in video displays has shown that experts—when compared with novices—rely on the movement of body segments that are more remote from the end effector. For example, novice badminton players typically anticipate based on the movement of the opponent's racquet, whereas skilled players use the movement of the opponent's racquet *and* arm. Attention toward the arm provides a temporal advantage, as movement of the arm precedes the movement of the racquet. The expert advantage in anticipation is based on sensitivity to kinematic movement patterns, rather than to figural or contextual cues. Point-light displays replace video footage of an opponent with a series of isolated points of light located at critical joint centers; expert–novice differences in anticipation are replicated when athletes view these displays. Evidently, skilled athletes have developed the ability to understand the consequences of the underlying kinematic movement pattern of their opponents. It is likely that this skill has developed not only as a consequence of *observing* these movements but also by skilled athletes *performing* the same actions. Perception may share a mutual form of neural programming with the production of action; recent work has shown that anticipation of an action relies on the same brain region that is used when generating the same action.

Testing Anticipation

Anticipation can be tested using a range of different display stimuli and responses. While it is most favorable to use conditions that accurately reflect

those found in the natural environment, the need for consistency and control in testing conditions means that this is not always possible. Skilled athletes outperform lesser skilled players in simulated conditions; however, the degree of superiority will be an underrepresentation of the true ability that would be found in the natural environment.

Display Stimuli

Video simulations allow anticipation to be tested in a very reliable and repeatable manner, though they often lack the size, contrast, or depth information available in real life. Liquid crystal occlusion goggles allow anticipation to be tested in the performance setting; this improvement in display fidelity usually leads to a commensurate increase in the size of the expert–novice difference.

Perception-Action Coupling

Athletes make perceptual predictions in most tests of anticipation (e.g., verbal or pen-and-paper); however, the separation of perception from action may miss an important element of sporting expertise. It is likely that perceptual responses test only the vision-for-perception neurological pathway; in contrast, skilled athletes rely on a specific vision-for-action pathway to produce real-time movements in the natural setting. Accordingly, it has been found that movement-based responses provide a better assessment of skilled anticipation than purely perceptual responses do.

Practical Implications

Facilitation of Performance

Skilled athletes use prerelease information to facilitate early and appropriate body positioning, rather than to stipulate the exact location the ball or target will arrive. This allows for better use of postrelease information to engender successful interception. The kinematic movement pattern of the opponent is also used to coordinate the timing and movement of an athlete's response. The importance of anticipatory skill suggests the need for advance information to be present in the training environment to optimize learning; for example, this principle has been used to argue against the use of ball projection machines, as they remove the kinematic movement information essential for anticipation.

Training Anticipatory Skill

Perceptual training programs are used to improve the anticipatory skill of developing athletes. These programs expose learners to a high volume of action sequences (usually occluded), observed either using video displays or in the field setting and often accompanied by some form of guiding information to accelerate skill acquisition. Perceptual training generally leads to an improvement in anticipatory skill, though there is conjecture about the most effective forms of training. Intuitively, practitioners have sought to provide learners with explicit information about how they should search for and interpret kinematic cues. More recent work suggests that implicit means of training, which guide attention without the provision of explicit rules, may enhance the likelihood that a skill is retained and may render the skill more robust under stress.

David L. Mann

See also Cue Utilization; Decision Making; Expertise; Perception; Skill Acquisition

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ANTISOCIAL/PROSOCIAL BEHAVIOR

See Moral Behavior

ASSIMILATION

Assimilation refers to the integration of one culture into another. This integration may include changes in cultural characteristics such as language, appearance, food, music, and religion

among other customs. Cultural values and beliefs also influence this integration of cultures. Assimilation is relevant to sport performance in that sports occur in the context of culture, society, and politics. In addition, each sport has its own culture and characteristics that may reflect or be in contrast with the values and beliefs of society and the individual athletes that play a particular sport. As such, an athlete's cultural milieu may interact with that of the sport and society in which the sport occurs to influence performance. For instance, a sport that focuses on individual performance and recognition may not fit well for an athlete from a culture that emphasizes teamwork and humility.

Assimilation may influence the type of sport an athlete selects and the roles assumed within sports and may also manifest in interactions and relationships with teammates and coaches. Typically, cultural assimilation involves an underrepresented (minority) group integrating into a dominant group's culture. An athlete from rural American Samoa playing American football at an urban college in the United States, for example, might integrate into the dominant cultural group of the city in which he plays by adopting the cultural practices of that region. However, the preceding example and description represent a unidirectional and oversimplified version of the concept of assimilation, as they suggest that assimilation necessarily occurs in the direction of the dominant group and that it is a linear process of change. Assimilation can also occur from the direction of the dominant to the underrepresented group, though this is less common. For example, a White athlete playing American football at a predominantly Black college may assimilate toward the culture of his fellow students, campus, and teammates, which may not reflect the dominant culture per se. Moreover, assimilation is not a process that is linear with a finite endpoint of *being assimilated*; rather it is a process that is constantly evolving. Individual athletes may assimilate to varying degrees based on situational, historical, and other factors. Thus, a Cuban immigrant playing professional baseball may be less assimilated around other Cuban or Latin American teammates or coaches, whereas he may be more assimilated around teammates and coaches from other cultures. In other words, assimilation is an evolving process that may be state dependent and also reflect the evolution of an athlete's assimilation.

The way in which assimilation is initiated may also affect how it is perceived, thereby influencing the athlete in negative or positive aspects or both. Assimilation may occur voluntarily or involuntarily: A soccer player from North Africa may be traded to a team in Russia, with little control over the decision. In contrast, another athlete may decide to immerse himself or herself in a culturally different environment to expand life experience. The perceived level of control over assimilation may affect how an athlete responds to it.

Enculturation and Acculturation

Regardless of the mechanism for assimilation, there are two primary components to assimilation: (1) *enculturation*, or the level to which someone adheres to primary cultural beliefs, values, and customs, and (2) *acculturation*, or the level to which someone adopts dominant or other cultural beliefs, values, and customs. One might think of complete enculturation and acculturation as extreme endpoints on a balancing continuum (see Figure 1). Most athletes' assimilation will balance between some percentage of acculturation and enculturation that together equals 100%; for example, 75% enculturated, 25% acculturated. This percentage is based on which side the cultural characteristics discussed earlier reside. An athlete's assimilation balance may shift based on the situation and may change over time to reflect life experiences.

Effects of Assimilation

Assimilation can sometimes result in acculturative stress (stress related to adapting to another culture) and may adversely affect performance in sport. In

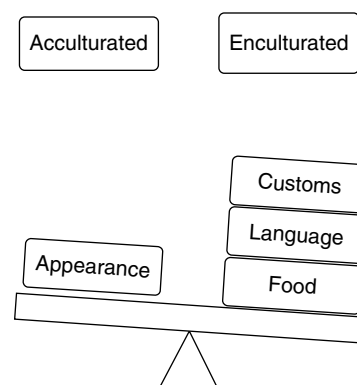


Figure 1 Balancing the Assimilation Continuum

Source: Anthony P. Kontos.

extreme cases, if not addressed, acculturative stress may develop into depression, anxiety, or hostility. These negative effects of assimilation may be more salient for athletes who are highly acculturated and then return to their own culture. When athletes return to their own culture, they may be perceived as having sold out their own culture for the dominant culture. In contrast, a recently immigrated athlete or one who is highly enculturated may struggle to adapt to a new culture and the athletes from that culture. As a result, such an athlete may be isolated in the new cultural environment. However, for some athletes assimilation may not play any role at all in creating stress or adversely affecting sport performance. For these athletes, assimilation may even help alleviate stress and anxiety by allowing them to fit in better and feel more comfortable in an unfamiliar culture.

Conclusion

Assimilation is nonlinear and constantly evolving through both direct and indirect experiences with one's own and other cultures. An athlete's level of enculturation and acculturation occurs on a balanced continuum that may shift back and forth based on the situation and cultural context, as well as the evolution of the athlete's assimilation process. It is important to point out that assimilation does not typically involve a purposeful integration of another culture. In fact, assimilation may simply occur as a product of direct or indirect exposure to and familiarity with another culture over time. As such, many athletes are unaware of how assimilated they are or of assimilation's potential role on performance and sport. Therefore, assimilation should be viewed as neither positive nor negative but rather as an evolving process that is highly individualized for each athlete.

Anthony P. Kontos

See also Cultural Competence; Cultural Safety; Diversity; Ethnicity; Multiculturalism; Racism/Whiteness; Stereotyping, Cultural/Ethnic/Racial

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ATHLETE LEADERSHIP

An *athlete leader* in sport is defined as an individual who holds a formal or informal leadership role within a team and influences other group members in the pursuit of common objectives. Researchers suggest that approximately one quarter of athletes occupy some form of leadership role within a team, and highlight the potential importance of athlete leadership toward positive group functioning as well as the need for a more thorough understanding of the topic. The following entry briefly highlights the characteristics, types, and functions of athlete leaders, as well as other important variables associated with the presence of these individuals on sport teams.

Characteristics of an Athlete Leader

Research generally demonstrates that individuals possessing athlete leadership roles within a group have similar characteristics. Maureen Weiss and colleagues Molly Moran and Melissa Price revealed that athlete leaders self-report greater friendship quality and peer acceptance, while Joseph Bucci, Gordon Bloom, Todd Loughhead, and Jeffrey Caron found that these individuals have a stronger work ethic, desire for high performance, and respect from teammates. The latter researchers also noted that possessing these characteristics leads to positive relationships with both coaches and teammates, helping foster effective levels of communication within the overall group structure. However, despite the importance and prevalence of the social psychological characteristics noted above, the most consistent characteristic of athlete leaders found in previous research pertains to sport-related competence; in other words, athletic ability is positively associated with ratings of athlete leadership.

Types of Athlete Leadership

To garner an understanding of how athlete leadership manifests itself in the sport context, it is

important to understand the emergence of the leadership role, as well as the extent to which leadership behaviors influence group members.

Formal and Informal Athlete Leadership

The formal athlete leader represents a role that is prescribed by another individual within the group or sport organization, usually a member of the coaching staff. This type of leader is highly visible within the group and is assigned specific responsibilities. A common example of a formal athlete leader in sport is the team captain. In many instances, the coach selects the captain of the team and, within certain sports, the occupant of this role may even be formally designated by such things as a C on the uniform or an armband to wear during competitive matches.

Conversely, an informal athlete leader emerges as a function of (a) group interaction, (b) distinct group needs, and (c) the personality traits of individual athletes. This type of athlete leader acts in a way that often complements the style of an established leader within the group (the formal athlete leader). However, the informal leader emerges naturally, without designation by another group member or the organization. For example, an athlete may assume the role of an emotional leader to help rally a team around its goals. This potential informal athlete leader may exist on teams in which the formal leader is more task driven (less expressive in nature); however, in other teams this informal role may be redundant if the formal leader possesses high interpersonal attraction and engages in socially supportive behaviors.

Team- and Peer-Level Athlete Leaders

In addition to the level of formality, athlete-to-athlete leadership is categorized in terms of the scope of influence held by each individual. Todd Loughhead, James Hardy, and Mark Eys discussed the presence of both team- and peer-level leaders. First, a team-level leader is more influential and is identified as an athlete leader by the majority of the team (over 50% of the membership). Examples include veteran players who provide valuable advice to all members of a group at different times or who are highly vocal (in a productive manner) during team meetings. It is also worthwhile to note that the team-level athlete leader likely emerges through the formal leadership process discussed previously.

In contrast, peer-level leaders represent those who are identified by a lower percentage of the team (less than 50%). Equally important in a team setting, peer-level leaders exert individualized influence on a small number of athletes. An example of this type of leader is an individual within the team who acts as a mentor to two or three less-experienced teammates. Although other team members may never be influenced by this specific individual, the inexperienced athletes may view the mentor as a very important leader.

Functions of Athlete Leadership

Athlete leadership roles are often differentiated by their specific functions within the group. These functions revolve around both internal and external activities. Athlete leaders who attend to the internal functions focus on the (a) task or (b) social related activities of the group.

- *Internal task functions.* Task-related functions represent the behaviors executed by a leader surrounding a group's instrumental objectives, such as the performance of the sport team. The behaviors oriented toward this function, for example, influence group members to perform to the best of their abilities and to coordinate effectively with their teammates.

- *Internal social functions.* Social-related functions represent the behaviors executed by a leader surrounding interpersonal relations and optimal team unity. An example of an athlete leader concerned with the social activities of the group would be someone who works to resolve conflict or plans events aimed at bonding the members of a team together.

- *External functions.* Athletes can also serve the function of leading their team in external activities. Todd Loughhead et al. described the external athlete leader as one who leads and represents the group outside of the competitive setting. Specifically, an external athlete leader helps a team cope and adapt to the surrounding environment. As examples, individuals who represent the team at different community-driven events or speak to the media on behalf of the group undertake external functions.

It is important to note that the various functions of an athlete leadership role, although distinct from one another, are not necessarily performed by

different people. One individual may have the ability to occupy a leadership role that is solely task-related, whereas another individual may occupy a leadership role that executes all three functions.

Correlates of Athlete Leadership

Interesting insights have been yielded with respect to the presence of athlete leadership in sport. The following sections briefly highlight both individual (satisfaction) and team level (group cohesion and collective efficacy) correlates of athlete leader behavior.

Athlete Satisfaction

Following research that linked leadership behaviors of the coach with athlete satisfaction, Mark Eys, Todd Loughhead, and James Hardy demonstrated that athletes who perceived a balanced dispersion of athlete leaders across the aforementioned functions (approximately equal number of leaders focusing on task, social, and external objectives) had higher satisfaction with their sport experiences. Specifically, these athletes were more satisfied with team performance and the degree of integration of team members than those who perceived a relative imbalance with respect to the focus of athlete leaders on their team, as with a high number of task leaders but low numbers of social and external leaders.

Group Cohesion

Athlete leadership is also related to the perceptions of cohesion experienced by group members. Two studies provide different perspectives on the relationship between these variables. First, Price and Weiss found that adolescent female soccer players who self-reported higher leadership abilities with respect to instrumental and prosocial behaviors perceived greater task and social cohesion on their teams. Furthermore, those who were rated higher by their teammates in the same leadership abilities perceived greater social cohesion.

In a second study, Hardy, Eys, and Loughhead were interested in the links between the percentage of athlete leaders on sport teams (i.e., dispersion) and group members' perceptions of cohesion. Their findings indicated that the dispersion of leaders focused on task functions was negatively related to perceptions of task cohesion. In other words, a more constrained number of task-focused

leaders was associated with increased group unity. Furthermore, these researchers found that intrateam communication mediated this relationship. The interpretation of these results was that a lower number of task leaders leads to better quality communication in terms of consistency, clarity, and overall effectiveness that, in turn, is associated with more positive perceptions of task cohesion.

Collective Efficacy

Finally, athlete leadership is linked to members' beliefs regarding the group's ability to carry out required tasks, as well as its general ability to perform at a high level. Price and Weiss revealed that the self-reported instrumental and prosocial leadership behaviors of athletes were positively related to their indications of efficacy related to the team's ability, unity, effort, preparation, and persistence.

Conclusion

Overall, athlete leadership represents an important aspect of group functioning. This entry has briefly highlighted the characteristics, types, and functions of athlete leadership. Future research, however, must continue to build upon the current breadth of knowledge regarding the influence of athlete leaders within sport. In doing so, sport psychology researchers can continue moving toward unearthing different individual and group level variables that are related to both effective and ineffective cases of athlete leadership. Furthermore, another future step can involve translating this knowledge into the group exercise setting to test similar relationships to those found in sport like individual satisfaction and group cohesion and determine if exerciser-to-exerciser leadership influences important outcomes such as physical activity adherence.

Mark Eys, Colin McLaren, and Robyn Bertram

See also Cohesion; Friendships/Peer Relations; Leadership in Sport: Multidimensional Model; Leadership in Sport: Transactional and Transformational; Mentoring; Roles

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ATTACHMENT THEORY AND COACHING

The aim of attachment theory has largely been to explain how relationships with parents in childhood have such a persistent effect on personality development. The focus of attachment theory has subsequently been extended from child to adolescent and adult development and social relationships within the context of both contemporary personality and social psychology. Attachment has been viewed as a natural phenomenon sought by all human beings. Subsequently, the theory postulates that, whereas successful bids for proximity and connectedness with warm, kind, dependable, and encouraging attachment figures are important for optimal functioning, the loss of such proximity and connection can be a natural source of distress and psychosocial dysfunction.

In sport and exercise psychology, attachment theory has recently been used to understand (a) how athletes and coaches perceive and cope with fears and anxieties, injuries, and performance slumps; and (b) how personal relationships (parent–athlete or parent–child) and social relationships (coach–athlete, athlete–athlete) help their members to either flourish or diminish. The appeal of attachment theory to explain a whole host of research questions has sparked research within sport and exercise psychology that has the potential to advance theory, measurement, and practice,

as seen in work by Sam Carr and Neil Fitzpatrick and by Louise Davis and Sophia Jowett.

Attachment Figures, Interactions, and Relationships

Attachment figures are not just any close relationship member. Attachment figures are special individuals to whom, for example, an athlete turns when assistance, encouragement, and cooperation are needed. In the context of sport, coaches serve as attachment figures when they allow their athletes to act independently. Athletes, in these cases, are encouraged to explore and discover new techniques, skills, or competition in the knowledge that their coaches are near and they can reliably provide protection, comfort, encouragement, or relief should the athletes need it (failing to execute the technique or the skill, failing to qualify, losing a major championship, getting injured during a task). While there are numerous interpersonal behaviors that cannot be characterized as *attachment interactions*, such as organizing a training session or providing feedback and criticism, there are other interpersonal behaviors that can be characterized as attachment interactions. In such interactions, the expectation is that the athlete would feel a degree of threat or distress, thus compelling that athlete to seek comfort and support from the coach. In competitive sport especially where the stakes are high, athletes undergo numerous stressful situations, including dealing with complicated technical routines, serious injuries, team selections, career transitions, or personal circumstances such as the loss of a loved one, financial problems, and social pressures—all of which could require them to seek comfort and support from people they trust. The *attachment bond* becomes apparent when coaches are shown to be available, sensitive to the needs of their athletes, and responsive to the athletes' feelings of threat, worries, distress, hurt, or bids for proximity when in need.

Attachment Working Models

According to attachment theory, variations in attachment figure responses to an attached individual's bids for proximity, connectedness, and protection are capable of progressively producing lasting changes in how attached individuals function personally and interpersonally. For example, over time, athletes (attached individuals) have a host of interactions with their coaches (attachment

figures) and the quality and type of responses of these interactions are stored in athletes' long-term memories. This stored knowledge takes the form of *internal working models* (IWMs) and allows, for example, athletes to predict future interactions with their coaches and guide their behavior, cognitions, and feelings. Specifically, IWMs are capable of orienting individuals in specific ways toward their own self and toward their close others. On one hand, *IWM of the self* reflects how worthy one feels in obtaining assistance when in need (I am a valued member), and on the other hand, *IWM of others* refer to whether an individual can expect assistance, including responsive and caring behavior, from an attachment figure in times of stress (I expect or don't doubt coaches' understanding and help). According to attachment theory, IWMs are thought to account for the long-term effects on individuals' personality functioning as this relates to attachment interactions during infancy, childhood, adolescence, and adulthood. An individual's *attachment style* reflects the most constantly accessible IWM, and thus how typically an individual functions within one's attachment system at a local level and within a specific relationship like the coach-athlete relationship or at a global level and across a number of relationships in general attachment style.

Attachment Styles

There are three most commonly referred to attachment styles: *secure*, *anxious*, and *avoidant*. Individuals with a secure attachment style display confidence in the availability of close others like a coach for comfort and support in times of need. Those who display an anxious attachment style have a desire for proximity and intimacy to an attachment figure even in nondistressing conditions. Under stressful situations, they can display excessive distress and may withdraw in anger even if support is offered from the close other on the basis that the support offered is not good enough. Finally, individuals with an avoidant attachment style display little distress during stressful events and few attempts at maintaining contact with an attachment figure. As mentioned earlier, attachment styles are thought to be determined by an individual's IWM of both the self and others. Individuals with a secure attachment style typically have an expectation that assistance should be available in times of need and that they

are worthy of such assistance; this helps them develop and maintain a positive IWM of themselves and others. Conversely, individuals with insecure attachment styles typically expect either inconsistent assistance (anxious), or no assistance at all (avoidant), in times of need; this results in the development and maintenance of feelings of unworthiness regarding others' care and affection while also becoming suspicious of any affection they may receive. This leads individuals with anxious attachment to display a negative IWM of themselves, and for those with an avoidant attachment style to display negative IWMs of others.

Lifespan Perspective of Attachment

It is important to note that although an individual's attachment style could remain stable over time, this can be dependent on the caregiving environment or attachment figure. If, for example, an individual's attachment bond with the coach supports and reinforces one's dominant or currently most active IWM, the stability of an attachment style would be expected. However, if an individual develops an attachment bond (e.g., with the coach) that is substantially different from attachment bonds formed in the past (predominantly by the primary caregiver, such as a parent or indeed a previous coach), then there is a chance of altering IWMs (tendencies to anticipate, attend to, interpret, and recall behavior) and, in turn, attachment style. For example, an athlete who has constantly received inconsistent support and responsiveness from previous attachment figures, such as parents, teachers, or coaches, is likely to develop a negative IWM of self and thereby bring about an anxious attachment style. However, the intense and enduring support, responsiveness, and encouragement demonstrated by a new coach over a period of time can foster a much more positive intrapersonal outlook, which can potentially lead to an alteration of the athlete's IWM and consequently the adoption of a secure attachment style within the coach-athlete relationship.

Attachment Theory in Sport

Research within the broader discipline of social psychology on adult attachment has demonstrated the importance of secure attachments for relationship quality and functioning as well as psychological well-being (for a comprehensive review, see Mikulincer & Shaver, 2007). In sport and

exercise psychology, research is gathering momentum in this area. At present, the findings suggest that athletes displaying insecure attachment (anxious or avoidant attachment) are likely to report poor relationship quality with the coach, less sport satisfaction, greater risk of developing an eating disorder, low perceptions of basic psychological need satisfaction, and ill-being (e.g., lack of vitality, negative affect, depression) when compared to more securely attached athletes. Moreover, findings highlight that the satisfaction of basic psychological needs, such as connectedness, competence, and autonomy, can potentially explain the association between insecure attachment and well- or ill-being. It would appear that while security attachment can satisfy basic needs and thus lead to athletes' well-being, insecure attachment can thwart basic needs and lead to athletes' ill-being. These studies started to underline the significance of attachment theory in understanding important research questions of practical significance for effective coaching and successful athletic performance.

Coaching Practice

Given that coaching is an interpersonal affair revolving primarily around the coach and the athlete, it is important that both coach and athlete have an appreciation of the potential impact that their attachment styles can have on their own and the other's behaviors, thoughts, and feelings. As described earlier, an avoidant attachment style is reflected in an individual's being heavily self-reliant and uninterested and unwilling to connect with others. If an athlete appears to be distant to the coach when interacting in training or competitions, it does not mean that that athlete is being indifferent toward sport or dislikes the coach; it may simply be a reflection of the athlete's personality. Athletes with insecure attachment styles may be misunderstood by their coaches, and thus coaches may decide to write them off or not give them the time and attention they require to progress. Coaches who recognize their athletes as insecure can, instead of adopting a similar pattern of behavior (e.g., less connection, responsiveness) that only serves to strengthen the athlete's IWM and further reinforces the athlete's negative IWM, try creating a positive, caring, and nurturing environment that is consistently unambiguous. Ultimately, coaches' persistent positive interpersonal behaviors are more likely to help athletes satisfy basic

psychological needs and promote their well-being. Happy and satisfied athletes are more likely to persist and achieve in their chosen sports. Thus, coaches who (a) behave in an autonomy-supportive manner, while avoiding controlling behaviors, allow the athlete to take on a more integral role in their training and competition; and (b) develop positive relationships characterized by trust, respect, appreciation, commitment, readiness, reassurance, and security are more likely to help their athletes recognize that they are being valued and cared for. Such enduring, supportive, responsive, and encouraging interactions may begin to lay the foundations for a substantial change in the insecure athletes' IWMs, leading to a more secure attachment style for that athlete within the coaching context. Theory and research suggest that secure attachment styles are more advantageous for optimal functioning at both intrapersonal and interpersonal levels. Given that sporting success is the product of the combined interrelation between the coach and the athlete, as typically neither of them can do it alone, understanding their personalities can make the journey to achieving performance success much happier, easier, and rewarding.

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See also Caring Climate; Coach–Athlete Relations; Coaching Efficacy; Friendships/Peer Relations; Interdependence Theory and the Coach–Athlete Relationship

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ATTENTION THEORY

Every waking moment we face an important selection problem. How do we pick some information for further processing while ignoring almost everything else? This problem is not easy to solve given the brevity and fragility of our *working memory* (the mental system that regulates our conscious awareness) and the all but unlimited array of information available to us—not only from the external world but also from the internal domain of our own memory and imagination. So, the mind has developed a system that helps us select some information for further processing while blocking out other information. This system is called *attention*—a term that denotes the process of exerting mental effort on specific features of the world around us or on our own thoughts and feelings. For example, in sport, making a conscious effort to listen carefully to a coach's instructions before a match involves attention. Similarly, a soccer goalkeeper who is preparing to defend against a corner kick from the opposing team must pay attention to the flight of the incoming ball while disregarding the movements of the players in the penalty area. These two examples show that the ability to focus on what is most important in any situation while ignoring distractions is vital for success in sport. But when we delve deeper into the psychology of attention, a number of questions arise. Are there different types of attention? If so, how can these different attentional processes be measured? Finally, what theories best explain how the attention system works? The purpose of this entry is to answer these and other relevant questions.

Describing Attention: Nature and Types

For well over a century, cognitive psychologists (researchers who study how the mind works in seeking, storing, and using knowledge) have investigated attentional processes. From this research, three different types of attention have been identified.

First, *concentration* refers to a person's decision to invest mental effort in what is most important in any situation (as in the example above of listening to a coach's instructions). Next, *selective attention* is the perceptual ability to zoom in on task-relevant information (the flight of the ball) while ignoring distractions (the movement of players). Finally, *divided attention* refers to a form of mental time-sharing ability whereby people can learn, as a result of extensive practice, to execute two or more concurrent skills equally well. To illustrate, a skilled basketball player can dribble the ball while simultaneously looking around for a teammate who is in a good position to receive a pass. In summary, *attention* is a multidimensional term that refers to at least three different cognitive processes—concentration or effortful awareness, selectivity of perception, or the ability to coordinate two or more skills at the same time. Having explained what attention involves, let us now consider how attentional processes can be measured.

Measuring Attention

In general, cognitive sport psychology researchers have developed three main strategies for the measurement of attentional processes. These strategies include the *psychometric approach* (the use of psychological tests to measure individual differences in people's attentional skills), the *neuroscientific approach* (measurement of the brain processes underlying attention), and the *experimental approach* (which involves testing people's ability to perform two or more tasks at the same time in laboratory settings).

The Psychometric Approach

The psychometric approach involves the use of standardized paper-and-pencil tests in an effort to measure individual differences in attentional processes in athletes. Such tests are both administratively convenient, as they are usually quick and easy to use, and amenable to empirical validation. To illustrate this approach, Robert Nideffer developed the Test of Attentional and Interpersonal Style (TAIS), which is based on a model that postulates that people's attentional focus varies simultaneously along two independent dimensions—width and direction. With regard to width, Nideffer proposed that attention can range along a continuum from a broad focus (where one is aware of many stimulus features at the same

time) to a narrow one (where irrelevant information is excluded effectively). Attentional direction refers to the target of one's focus: whether it is external or internal (see also the "Attentional Focus" entry). These dimensions of width and direction may be combined to yield four hypothetical attentional foci: (1) broad external, (2) broad internal, (3) narrow external, and (4) narrow internal. These various types of attentional focus can be illustrated as follows. First, a broad external focus is required for sport skills that involve the ability to read a game and quickly assess a situation for relevant information. For example, a good mid-field player in soccer must be able to quickly size up the best passing options available having gained possession of the ball. Second, a narrow external focus is necessary when an athlete locks on to a specific target in the environment. To illustrate, a golfer needs this type of focus when looking at the hole before putting. Third, a broad internal focus is necessary when a tennis player develops a general tactical game plan for a forthcoming match. Last, by contrast, a narrow internal focus is demanded when a gymnast mentally rehearses a skill such as a backflip while waiting to compete.

Nideffer speculated that optimal athletic performance is possible only if performers manage to ensure that their attentional focus matches the specific requirements of the sport skill that they are required to execute. Some evidence to support this idea comes from a study of the attentional strategies used by former Olympic cyclists to cope with the pain of physical exertion in their sport. Briefly, these cyclists reported that they had regularly used all four types of focus when competing. For example, they used a broad external focus when concentrating on getting to the finish line and a broad internal focus when concentrating on their body movements. Similarly, they used a narrow external focus when trying to keep up with the riders ahead of them and a narrow internal focus when concentrating on maintaining a smooth pedaling stroke. Unfortunately, despite their intuitive plausibility, Nideffer's theory and his Test of Attentional and Interpersonal Style have been criticized on a number of grounds. His theory does not distinguish between task-relevant and task-irrelevant information in sport settings. Similarly, the Test of Attentional and Interpersonal Style is questionable because it assesses people's *perceived* rather than their actual attentional skills. Despite its limitations, however, the psychometric approach

has made progress recently with the development of self-report tests that purport to measure athletes' susceptibility to *internal* distractions or thoughts and feelings that divert focus away from its intended target (see also the "Concentration Skills" entry). In sport psychology, for example, researchers have devised psychometric measures of *cognitive interference* (task-irrelevant, self-preoccupied thinking) in athletes. Although these measures seem promising, however, they require comprehensive evidence of validity before they can be recommended for use in applied settings.

The Neuroscientific Approach

The second approach to measuring attentional processes in athletes comes from cognitive neuroscience—a field that is concerned broadly with the identification of the neural substrates of mental processes. Typically, this approach involves a range of neuroscientific techniques in an effort to reveal which parts of the brain are activated when an athlete is paying attention to a sport-related task. Among the most popular of these techniques are electroencephalography (EEG), a method that records electrical activity in the brain using special electrodes placed on the scalp; functional magnetic resonance imaging (fMRI), a method that detects changes in the activity of the brain by measuring the amount of oxygen brought to a particular location within it; and transcranial magnetic stimulation (TMS), a method in which the functioning of a specific area of the brain is temporarily disrupted through the application of pulsating magnetic fields to the skull using a stimulating coil. Using EEG measures, sport psychology researchers have discovered cerebral asymmetry effects (differences between the activities of the two hemispheres of the brain) in athletes involved in target shooting. Specifically, it seems that just before expert archers and pistol performers execute their shots, their EEG profiles tend to reveal a distinctive shift from left-hemisphere to right-hemisphere activation. This shift may indicate a deliberate suppression of self-talk (what one says to oneself silently) on the part of the shooters in an effort to achieve a truly focused state of mind. Another finding—this time from fMRI research—is that compared to expert players, novice golfers have difficulty in filtering out irrelevant information when imagining the appropriate shot to play in a golf simulation task. More recently, TMS studies have shown that

expert basketball players have developed finely tuned *resonance* mechanisms that enable them to simulate and predict other players' actions.

Taken together, neuroscientific techniques have a major advantage over their psychometric counterparts in yielding objective data on biological processes that can be recorded *while* the athlete is performing skills. Unfortunately, key drawbacks associated with such neuroscientific measures are excessive cost and impracticality of use in everyday sport settings.

The Experimental Approach

The third approach to measuring attentional processes comes from experimental psychology and is called the *dual-task paradigm*. This method is based on the capacity theory of attention (explained below) and investigates how well people can divide their attention between two concurrent tasks like reading and listening to music. More precisely, it compares people's performance when executing the tasks separately with that when executing the tasks simultaneously. The logic of this method is as follows. If people's performance on one or both tasks is *worse* when the tasks are executed *simultaneously* as compared with when they are performed separately, then these two tasks must be interfering with each other—suggesting that they are competing for the same limited pool of attentional resources in the brain. However, if the two tasks can be performed as well simultaneously as individually, then at least one of these tasks must be automatic, hence making minimal demands on available attentional resources.

When the dual-task paradigm is used in sport psychology, the primary task usually consists of a self-paced skill (one that can be performed without interference from others such as target shooting in archery), whereas the secondary task tends to involve monitoring the environment for a signal such as an auditory tone. To illustrate this approach, a study was designed to investigate the effects of anxiety on skilled performance. The researchers used a dual-task design to examine the effects of manipulating people's focus of attention on performance on a driving simulator under low- or high-anxiety experimental conditions. Attention was directed explicitly to either the participants' own driving performance or to a distracting secondary task. In this study, participants were required to perform a primary task

(simulated rally driving) as fast as possible while responding as accurately as possible to one of two theoretically-derived secondary tasks. The skill-focused secondary task required participants to respond to an auditory tone by indicating at that moment whether their left hand on the steering wheel was higher, lower, or at the same height as their right hand. The distraction secondary task required participants to remember the pitch of an auditory tone presented while they were driving. Results showed that racing performance effectiveness (as measured by lap times) was maintained under anxiety-provoking conditions—although occurred at the expense of reducing processing efficiency. Unfortunately, despite its ingenuity, the dual-task paradigm has not been used widely to measure attentional processes in athletes—mainly because of the difficulty in locating pairs of attentional tasks that engage athletes' interest in laboratory settings.

Explaining Attention: Theories and Issues

Cognitive psychologists have developed three main theories of attention: *filter theory*, *capacity theory*, and *spotlight theory*.

Filter Theory

The first modern theory of attention, proposed by Donald Broadbent in the 1950s, was based on a series of laboratory experiments on selective listening tasks. Two key assumptions of this theory were that people are limited in their ability to process information, and there must be a mechanism that facilitates the selection of some information while inhibiting the selection of competing information. To explain this mechanism, Broadbent drew an analogy between attention and a filtering device or bottleneck that restricts the flow of information into the mind in accordance with a set of criteria. He proposed that just as the neck of a bottle restricts the flow of liquid, a hypothetical filter in the mind limits the quantity of information to which we can pay attention at any given time. In addition, Broadbent suggested that although multiple channels of information reach the filter, only one channel is permitted to pass through for further information processing. Unfortunately, Broadbent's filter model of attention soon ran into difficulty. Attentional researchers failed to agree on either the location or timing of the filter. Also, this model could not explain the *cocktail party*

phenomenon. Imagine that you are at a noisy party and trying to pay attention to a conversation. Suddenly, you hear your name being mentioned in another conversation somewhere else in the room. How can Broadbent's theory explain the fact that you recognized your name? Clearly, if you heard your name being mentioned, then it could not have been blocked by the hypothetical filter in the first place. Problems like these hastened the demise of filter theory.

Capacity Theory

Capacity or resource theory was developed by Daniel Kahneman in an effort to explain the mechanisms underlying divided attention (explained earlier in this entry). This theory postulates that attention is analogous to a pool of mental energy that can be allocated to tackle various processing demands according to certain strategic principles—such as the influence of the performer's arousal level. For example, people usually have a greater amount of attentional capacity available to them when they are fully alert than when they are sleepy. Also, the notion of automaticity suggests that the more practiced a task is, the more automatic (or unconscious) it becomes and the fewer attentional resources it requires. One implication of this latter notion is the prediction that expert athletes are especially vulnerable to distractions because they have considerable spare attentional capacity due to the largely automated nature of their highly practiced skills.

A recurrent weakness of resource models of attention, however, stems from the possible circularity of some of their terminology. For example, there is a logical flaw in the explanation offered by resource theorists for people's inability to divide their attention successfully between two concurrent tasks. Specifically, the assumption in such cases that the resources of some central attentional capacity system have been exceeded by the joint demands of these concurrent tasks is circular in the absence of an independent measure of attentional resources.

Spotlight Theory

According to spotlight theory, as developed by researchers such as Michael Posner, selective attention or our ability to zoom in on some aspects of a stimulus while ignoring other features of it, resembles an adjustable mental beam that we shine

at things that are important to us at any given moment. These targets of our concentration beam can be external (in the world around us) or internal (in the private domain of our own thoughts and feelings). Aiming one's golf drive at a specific target on the fairway, for example, involves an external focus of attention, whereas concentrating on the rhythm or feeling of one's golf swing involves an internal focus of attention. This idea of attention as a mental spotlight appeals to athletes and coaches for two main reasons. First, it shows us that our concentration can never be really lost because our mental spotlight is always shining *somewhere*—either at external or internal targets. But lapses of attention can occur whenever we focus on the *wrong* target (something that is irrelevant to the task at hand or that lies outside our control). In such cases, our performance is likely to deteriorate (see also the "Concentration Skills" entry). Second, the spotlight theory of attention suggests that athletes can exert voluntary control over where they shine their mental beam. In other words, athletes have choice over their attentional focus (see earlier discussion of Robert Nideffer's theory).

Unfortunately, although the spotlight metaphor of attention is intuitively appealing, it has three main weaknesses. First, spotlight theorists have largely neglected the issue of what lies *outside* the beam of one's conscious attention; they have tended to ignore the possibility that *unconscious* factors, such as ironic or counterintentional effects, can affect people's attentional processes in certain situations, as when they are cognitively overloaded (see the "Concentration Skills" entry). Second, spotlight theory has historically been concerned with external targets (stimuli in the external world) rather than the internal distractions that often afflict athletes. Finally, spotlight theory has little to say about the influence of emotional factors like anxiety on attentional processes (but see the "Choking" entry).

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See also Attention Training; Attentional Association and Dissociation; Attentional Focus; Automaticity; Choking; Concentration; Concentration Skills

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ATTENTION TRAINING

The term *attention* denotes the process by which we exert mental effort in focusing either on specific features of the world around us or on our own thoughts and feelings. An example would be a soccer goalkeeper who is preparing to defend against a corner kick from the opposing team must pay attention to the flight of the incoming ball while disregarding a variety of distractions arising from the movements of players in the penalty area and from any irrelevant thoughts he or she might have at that time. Clearly, the ability to focus on what is most important in any situation while ignoring a multitude of distractions is vital for successful performance in sport. Accordingly, researchers in sport and exercise psychology have developed a variety of practical strategies that purport to improve attentional skills in athletes. Although these strategies differ in the extent to which they have been validated by empirical evidence, they share a common purpose—namely, to help sports performers to achieve a focused state of mind in which there is no difference between what they are thinking about and what they are doing.

Specifying Performance Goals

Psychologists commonly distinguish between result goals, such as the outcome of a sporting encounter, and performance goals, or specific actions that lie within the athlete's control, such as attempting to achieve 100% accuracy in one's first serve in tennis. Some evidence has accumulated to suggest that athletes who specify performance goals while competing can improve

their concentration skills. To illustrate a practical implication of this finding, a golfer could strive to improve concentration during a round by focusing on maintaining a slow, rhythmic swing on every shot rather than worrying about the score. Theoretically, performance goals should enhance athletes' concentration because they encourage them to focus only on task-relevant information and controllable actions. Support for this conjecture comes from studies that have contrasted psychological variables associated with sports performers' best and worst competitive performances. One finding from these studies is that collegiate athletes reported performing worst when they were preoccupied by result goals but best when they focused on performance goals.

Adhering to Preperformance Routines

Sport is a highly ritualized activity. To illustrate, most top-class athletes display characteristic sequences of preparatory actions before they perform key skills: Golfers tend to waggle their clubs a consistent number of times before striking the ball and tennis players tend to bounce the ball a standard number of times before serving. These preferred action sequences or repetitive forms of behavior are called *preperformance* routines and are typically followed prior to the execution of self-paced skills—actions that are carried out largely at the performer's own speed and without interference from other people.

As in the case of performance goals, preperformance routines should facilitate concentration because they encourage athletes to stay in the present moment, focusing only on specific, controllable actions. Implementing this theory, many soccer goalkeepers go through identifiable prekick routines in an effort to block out external distractions such as hostile noise that is directed at them by supporters of opposing teams. Unfortunately, despite their potential value as attentional strategies, preperformance routines often overlap with superstitious rituals in the minds of athletes. So, what exactly is a superstition?

In psychology, a *superstition* can be defined as the belief that, despite evidence to the contrary, certain behavior is causally related to certain outcomes. Research shows that athletes are notoriously superstitious—perhaps because of the capricious nature of success in their field. For example, in competitive matches, the tennis star Rafael Nadal

appears to believe that he must have two water bottles beside the court, perfectly aligned and with the labels facing the baseline. Psychologically, routines and superstitious behavior differ on two key criteria—control and purpose. The essence of superstitious behavior is the belief that one's fate is governed by factors that lie *outside* one's control. But the virtue of a routine is that it allows the player to exert complete control over preparation. Indeed, players often shorten their preperformance routines in adverse circumstances—as happens, for example, when a competitive event is delayed unexpectedly. Unfortunately, the converse is true for superstitions. They tend to grow *longer* over time as performers chain together more and more illogical links between behavior and outcome. The second criterion that may be used to distinguish between routines and rituals concerns the technical role of each behavioral step followed. To explain, whereas each part of a routine should have a rational basis, the components of a superstitious ritual may not be justifiable objectively.

Returning to routines, at least three psychological factors have been postulated to explain their apparent efficacy in enhancing athletic performance. First, some theorists believe that routines are effective because they encourage athletes to prioritise task-relevant information over other available stimuli, such as penalty takers in soccer who may follow prekick routines to remind themselves to focus only on the target that they are aiming at rather than on distractions such as the movements of the goalkeeper. Second, routines may be useful because their sequential nature helps athletes concentrate on the present moment rather than on past events or on possible future outcomes. Finally, paying attention to each step of a routine involves conscious mental effort and consumes short-term or working memory resources. Therefore, adhering to a preperformance routine may prevent athletes from devoting too much attention to the mechanics of well-learned skills—a habit that can unravel automaticity and precipitate skill failure in certain circumstances. In short, routines may work by helping athletes to suppress the type of inappropriate conscious control that often occurs in pressure situations.

Some empirical evidence to support the attentional benefits of preperformance routines has emerged in recent years. Research has shown that amateur international golfers reported using routines explicitly for focusing purposes such as

attempting to *switch on and off* when required and trying to stay in the present while playing.

Using Trigger Words as Cues to Concentrate

Many athletes talk to themselves covertly as they train or compete—usually in an effort to improve their concentration and performance. Such self-talk, or what athletes say to themselves silently, may involve praise, criticism or instruction. An example of the instruction type of self-talk is the use of *trigger words*, which are short, vivid, and positively phrased verbal reminders to cue athletes to focus on a specific target or to perform a given action. An example would be gymnasts may say the word *forward* silently to themselves as a reminder to push their bodies upward when executing a floor routine. A graphic example of athletes' use of trigger words occurred during the 2002 Wimbledon ladies' singles tennis final between the Williams sisters, Serena and Venus. During this match, Serena (who defeated Venus 7–6, 6–3) was observed by millions of television viewers reading hand-written notes as she sat down beside the court during the change-overs between games. Afterwards, she explained that she had written these notes to herself as instructional reminders to *hit in front* or *stay low* during the match. Another example of trigger words in action occurred recently when Paula Radcliffe, who holds the current women's world record time for the marathon, advocated a trigger word strategy involving silently counting her steps in an effort to maintain her concentration in a race.

Does empirical research support the claim that trigger words can improve attentional skills? Unfortunately, although studies have shown that instructional self-talk is one of the strongest predictors of successful performance among U.S. Olympic athletes, there is a dearth of empirical evidence on the efficacy of trigger words as concentration cues. However, it seems plausible that instructional self-talk could enhance attentional skills by reminding athletes about what to focus on in a given situation. For example, novice golfers may miss the ball completely on the fairway in the early stages of learning to swing the club properly. So, in an effort to overcome this problem, golf instructors may advise learners to concentrate on *sweeping the grass* rather than hitting the ball. This evocative trigger phrase ensures that learners focus down on the ball instead of looking

up to see where it went after they had struck it. Research suggests that for optimal effectiveness, trigger words should be short, vivid, and positively phrased. They should also emphasize positive targets (what to aim for) rather than negative ones (what to avoid).

Mental Practice

In psychology, the term *mental imagery* refers to the cognitive simulation process by which we can represent perceptual information in our minds in the absence of appropriate sensory input. If you close your eyes, you should be able to *see* yourself in your mind's eye throwing a bright yellow tennis ball up in the air (a visual mental image), *feel* yourself catching it (a kinesthetic mental image) and then bouncing it (a motor mental image). In sport, many athletes use motor imagery or *mental practice* to cognitively rehearse their skills. This technique involves seeing and feeling a skill in one's imagination before actually executing it. Jenson Button, the Formula One star who won the 2009 World Drivers' Championship, regularly rehearses his gear shifts by sitting on a Swiss ball with a steering wheel in his hands—imagining himself navigating the course. Although there is considerable empirical evidence that such mental practice can facilitate skill learning and competitive performance, their value specifically as a concentration tool remains unproven. Anecdotally, however, mental imagery is used widely by performers to improve their focusing skills. Swimmer Michael Phelps, who has won more Olympic gold medals than any other athlete in history, uses imagery to see and feel his strokes and turns before a race. Such use of imagery helps sports performers to prepare for various hypothetical scenarios, thereby ensuring that they will not be distracted by unexpected events. However, this hypothesis has not been tested empirically to date. Therefore, despite the fact that mental imagery is known to improve athletic performance, its status as a concentration technique is uncertain.

Simulation Training

The term *simulation training* refers to the idea of practicing under conditions that replicate key aspects of an impending competition. Certain football teams have tried to simulate the noisy conditions that they expect to encounter in opposing teams' stadia by training on their home grounds using giant screens playing loud music

and recordings of rival fans cheering. Intuitively, it seems plausible that simulation training could help skilled performers concentrate because research suggests that people's recall of information is facilitated by conditions that resemble those in which the original encoding occurred. Based on this principle, the simulation of competitive situations in practice should lead to positive transfer effects to the competition itself. Another advantage of adversity training is that it may counteract the tendency for novel or unexpected stimuli to distract athletes in competition. Interestingly, simulation training was used by the renowned swimming coach Bob Bowman, who admitted deliberately breaking the goggles of Michael Phelps during practice so that he could learn to swim calmly without them, if necessary, in a competition. Remarkably, this situation actually arose in the 2008 Olympics when Phelps won the 200-meter butterfly event even though his goggles had been broken for the last 100m of the race.

To summarize, the ability to pay attention to what is most important in any situation while ignoring distractions is vital for successful performance in sport. We have reviewed five attentional strategies that are intended to help athletes achieve a focused state of mind in which there is no difference between what they are thinking about and what they are doing. Empirical research evidence is available to support the efficacy of four of these strategies—setting performance goals (or actions that lie under the control of the athlete), adhering to preperformance routines, using trigger words (or covert verbal cues), and engaging in mental practice (or seeing and feeling a skill in one's imagination before executing it). The fifth attentional strategy—simulation training—has a plausible theoretical rationale but currently lacks relevant empirical validation.

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See also Attention Theory; Automaticity; Choking; Concentration; Concentration Skills

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ATTENTIONAL ASSOCIATION AND DISSOCIATION

Coping with acute exertional sensations during physical effort expenditure requires optimal use of attentional resources. *Association* and *dissociation* are two broad attentional strategies for coping with exertional stimuli during effort expenditure. Association represents the shift of the attentional focus inward (to somatic sensations), and dissociation represents the shift of the attentional focus outward (away from somatic sensations). Association and dissociation, therefore, correspond to internal and external foci, respectively.

Comparative research findings indicate that both association and dissociation can be effective and ineffective. Specifically, depending on several characteristics inherent in task and effort conditions either strategy was shown to be more or less potent. There is a general consensus that associative strategies may help performance in competitive events such as long-distance running, while dissociative strategies may help adherence in noncompetitive physical activity settings such as a routine treadmill walk.

Associative Versus Dissociative Strategies

Associative strategies increase awareness of physiological cues, such as breathing, heart pounding, and muscle pain, leading to increased negative affective responses during physical activity. Nevertheless, association also enables better effort monitoring and self-regulation, thereby allowing increased time on task as well as greater performance efficiency, performance outcome, and injury prevention. Dissociative strategies, on the other hand, are closely linked to a number of task-related variables, including task-related pleasure, confidence, feelings of ease, and motivation. Moreover, dissociating helps lower perception of

fatigue and exertion, thereby allowing reduced sense of task difficulty.

In sum, because associative strategies correspond to an inward focus of attention, their use increases awareness of somatic cues during effort expenditure. In contrast, the use of dissociative strategies decreases awareness of somatic cues, helps curb the physiological stress posed by the effort, and overall results in a greater potential for enhancing the physical activity experience.

It is evident that, while an associative attentional focus is more beneficial for performance enhancement during competitive events, a dissociative attentional focus is of greater value for relatively untrained individuals or during noncompetitive events. Distraction from aversive sensations via the use of dissociative strategies is, however, limited and closely depends on the effort intensity (workload).

Attention Inflexibility and Attention Threshold

As a limited capacity, attention is compromised in its flexibility to cope with physiological sensations as the effort expenditure increases. Specifically, at the onset of effort expenditure or at submaximal workloads, attentional focus is flexible: It can be switched back and forth effortlessly between associative and dissociative foci. Increase in physiological stress, however, leads the system to be challenged and compromised in its attentional flexibility. Specifically, as the workload gets harder or time on task increases, attention shifts inwards, preventing the system from distracting. The point at which attention loses its flexibility to shift between dissociation and association is termed the *dissociative/associative (D/A) attention threshold*. The concept entails that, once a subjective perceptual threshold relative to perceived physiological cues is exceeded, attention shifts from a dissociative focus to an associative one. Thus, as this perceptual threshold is attained, dissociative strategies are compromised in their capacity to distract from aversive cues of fatigue. The latter results in a final tuning into an associative focus, hence a growing concentration on the acute stress of exertion. The attention threshold occurs in parallel to the aerobic–anaerobic transition. Thus, when the physical effort expenditure is maintained below the attention threshold, effort can be sustained significantly longer than when the effort expenditure is at or above the aerobic–anaerobic transition

levels. Specifically, a shift from dissociative to associative focus of attention takes place as the workload intensity begins exceeding approximately 50% of the individual's maximal capacity. In fact, when individuals report equal employment of associative and dissociative strategies, associative strategies gradually take over rendering the cessation of the effort imminent. The issue of attention inflexibility and interventions to address it are the subject of a growing number of studies.

Methods for Extending the D/A Threshold

Distractive capabilities of dissociative strategies are compromised once the D/A attention threshold is exceeded, that is, as marked by the final tune into the associative attention focus. Successful attempts to delay the occurrence of the D/A threshold may aid in extending effort expenditure. To that aim, exposure to polysensory feedback including the use of auditory stimuli such as music, use of olfactory stimuli such as differential odorants, and use of mental imagery and virtual reality, have been shown to help postpone the final tune into the associative focus—but at low to moderate workload intensities only. Because the optimal use of attentional resources is central to sport and exercise performance and adherence, testing the effectiveness of further modalities designed to address the issue of attentional inflexibility remains important.

Gershon Tenenbaum and Selen Razon

See also Adherence; Attention Theory; Attention Training; Attentional Focus; Attention–Performance Relationships; Choking; Concentration

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ATTENTIONAL FOCUS

The effect of an individual's focus of attention on motor performance and learning has long intrigued both researchers and practitioners. Numerous studies have shown that what a performer focuses or concentrates on while executing a motor skill has an impact how well that person performs the skill. An important distinction is that between an *external focus* of attention on the intended movement effect, as on an implement versus an *internal focus* on body movements. These foci have differential effects on motor performance—sometimes seen almost immediately—as well as more long-term effects on motor learning. There is considerable evidence for the superiority of an external focus of attention with respect to both the effectiveness and efficiency of movements.

Movement Effectiveness

Numerous studies have assessed movement effectiveness as a function of attentional focus using outcome measures such as deviations from a balanced position, the accuracy in hitting a target or producing forces, or movement speed. The first study, by Gabriele Wulf and Rebecca Lewthwaite, demonstrated learning advantages of instructions inducing an external relative to an internal focus of attention using dynamic balance tasks. The results showed that the learning of those tasks was enhanced when participants' attention was directed to the movements of the platform on which they were standing as compared to the movements of their feet. Since then, numerous researchers have replicated the benefits of an external focus for other tasks.

Learning advantages of an external focus have also been shown for many sport skills, such as hitting golf balls, shooting basketball free throws, throwing darts, serving in volleyball or kicking in soccer, kayaking, swimming, and running. For

example, the accuracy in hitting golf balls is greater when performers focus on the swing of the club or on the intended ball trajectory (external focus), rather than on the swing of their arms or on their wrists (internal focus). Similarly, in dart throwing, accuracy is increased with an external focus on the flight of the dart or bull's eye. In swimming or running, performance (speed) is enhanced when performers focus on the force they are exerting against the water or ground as opposed to the movement of their arms or legs exerting the force, respectively. When the goal is to produce a certain amount of force, concentrating on the device, such as a weight bar or force platform, against which the force is exerted generally results in greater accuracy than focusing on the effector, such as the arms or legs.

Interestingly, when the effectiveness of external and internal focus conditions is compared to that of control conditions without focus instructions, performance in the control condition is typically similar to that with internal focus instructions, and external focus instructions result in more effective outcomes than both. One possible reason for this result is that people spontaneously focus on their body movements if they are not specifically asked to adopt an external focus. The only exception to this pattern of results is sometimes seen with highly skilled athletes who show similar performances in external focus and control conditions but degraded performance with an internal focus. This pattern of results suggests that—aside from relatively rare cases in which performance is already highly automatized (see below)—an external focus *enhances* performance or learning.

An intriguing finding is the so-called *distance effect*. Some studies have compared the effectiveness of external foci that differed with respect to the distance of the intended movement effect from the body. For example, a task may involve a balance platform that has markers attached to it, and learners are asked to concentrate on the markers while keeping the platform horizontal. Performers whose markers are placed at a greater distance from their feet show more effective balance learning than performers who are asked to focus on markers that are closer to their feet. Greater benefits of more distal relative to more proximal external foci have also been found for kayaking, golf, dart throwing, and long jump. Thus, concentrating on a movement effect that is more remote from one's body movements seems to be even

more advantageous than a focus on an effect that is closer to the body.

Overall, the benefits of an external compared to an internal focus have been shown not only for a variety of skills but also levels of expertise and age groups, as well as healthy individuals and those with motor impairments (for a review, see Wulf, 2012).

Movement Efficiency

If the same movement outcome is achieved with less energy, the movement is considered more efficient. While some studies have used direct measures of efficiency, such as muscular activity or oxygen consumption, others have used more indirect measures such as maximum force production, movement speed, or endurance to examine the effects of attentional focus. If more effective outcomes (e.g., greater forces) are achieved with the same (physical) resources, they reflect greater movement efficiency as well. There is converging evidence that an external relative to an internal focus optimizes movement efficiency.

Muscular Activity

For weight lifting tasks, electromyographic (EMG) activity has been found to be reduced when performers concentrate on the weight they are lifting (external focus) as compared to their arms or legs that are lifting the weight (internal focus), or compared to lifting without an instructed focus (control conditions). An external focus has been shown to be associated with greater efficiency *within* muscles, as only the necessary motor units are recruited, whereas superfluous recruitment of larger motor units is seen when performers adopt an internal focus. In addition, fewer cocontractions between the agonist and antagonist muscles occur with an external focus, indicating a more efficient coordination *between* muscle groups. As a result, performers are able to execute more repetitions with the same weight or produce greater maximum force (see below) with an external focus.

Reduced EMG activity has also been found for tasks requiring accuracy such as free-throw shooting in basketball or dart throwing, when performers concentrate on the target as opposed to their arm. Interestingly, with an internal focus, increased EMG activity is seen not only in those muscles on which the performer focuses but also in other muscle groups—indicating that a focus

on one part of the body *spreads* to other muscle groups. Thus, movement inefficiency is increased at a more general level, presumably contributing to the greater observed inaccuracies in the movement outcome with an internal focus.

Oxygen Consumption and Heart Rate

If the same (or more effective) movement outcome is achieved with less muscular activity, cardiovascular responses should be lowered as well. Indeed, studies confirm this assumption. In experienced runners who ran on a treadmill at a certain speed, oxygen uptake was reduced with an external focus (on the surrounding environment) relative to internal foci (on running technique or breathing). Furthermore, lower heart rates have been found when exercisers adopted an external relative to an internal focus while performing sit-ups.

Maximum Force Production

Producing maximum forces requires optimal muscle fiber recruitment within muscles and optimal activation patterns among muscles. Studies examining maximum force production demonstrate differences in muscular coordination as a function of attentional focus. Greater maximum forces with an external focus have been found for isokinetic contractions (e.g., biceps contractions on a dynamometer) and for dynamic tasks such as vertical jumps and standing long jumps. Also, complex tasks such as discus throwing benefit from an external focus (discus) as opposed to an internal focus (throwing arm or hand).

Speed and Endurance

Movement speed has been found to be increased for an external focus compared with an internal one, presumably due to the greater movement efficiency associated with an external focus. The tasks used to demonstrate this advantage have ranged from functional reach tasks in persons following cerebrovascular accident (stroke) where a focus on the manipulated object resulted in more fluid and faster motions than a focus on the grasping hand, to riding a paddle boat powered by pedals (Pedalo) in young healthy adults, where focusing on the boards under their feet led to higher speeds than focusing on the feet themselves.

In longer duration tasks in which fatigue is a limiting factor, the adoption of an external focus can enhance performance as well. In tasks that

require the production of submaximal or maximal forces over a longer period of time, an external focus enables performers to maintain those forces longer or to produce greater force in a given period of time (e.g., 10 secs). Performance on sprinting and agility tasks involving running and turning components has been shown to be enhanced by external focus instructions, relative to internal focus or no instructions. Similarly, (intermediate) swimmers swim faster when they focus on pushing the water back as opposed to pulling their hands back.

Also, the attentional focus of exercisers has an influence on muscular endurance. Trained individuals performing various exercises like bench press or squat lift were able to complete a greater number of repetitions to failure with a given weight when instructed to focus on the weight as opposed to the movements of the limbs involved or without focus instructions. In a study using an isometric force production task (wall-sit), participants' ability to hold the posture increased with an external focus (imaginary horizontal lines between their hips and knees), relative to an internal focus on the horizontal position of their thighs.

Mechanisms Underlying the Attentional Focus Effect

An internal focus induces a conscious type of control, causing individuals to constrain their motor system (*constrained action hypothesis*) and interfere with automatic control processes that have the capacity to control movements effectively and efficiently. In contrast, an external focus promotes an automatic mode of control. Adopting an external focus allows unconscious, fast, and reflexive processes to control the movement, with the result that the desired outcome is achieved effectively and relatively effortlessly. Several converging lines of evidence support the notion that an external focus facilitates automaticity relative to a focus on body movements. For example, attentional-capacity demands have been shown to be reduced with an external focus. High-frequency movement adjustments while balancing with an external focus indicate the use of reflex-based feedback loops, while slower and more conscious feedback loops seem to be utilized with an internal focus. Other findings, such as the reduced EMG activity seen with an external focus, also indicate that an external focus allows individuals to perform at the higher, more

advanced level—essentially reflecting a speeded learning process.

Conclusion

The benefits of an external compared with an internal focus have been shown for a variety of skills, ranging from pressing keys to driving golf balls. They have also been found for different levels of expertise, ranging from novice to skilled performance; for people of various age groups, ranging from children to older adults; and for healthy people as well as those with injuries or with motor impairments (stroke, Parkinson's disease, intellectual disabilities). Performance-enhancing effects of an external focus are often seen immediately, but practice with an external focus also has more permanent effects on motor skill learning. These effects are seen in movement effectiveness like accuracy, consistency, and balance as well as movement efficiency like EMG, heart rate, force production, speed, and endurance. Adopting an external focus during practice promotes movement automaticity and accelerates the learning process. In practical settings, coaches, trainers, and physical therapists often use less-than-optimal instructions and feedback by referring to body movements, thus inducing an internal focus. Rewording instructions to direct performers' attention to the intended movement outcome, and away from body movements, has the potential to significantly enhance performance and learning.

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See also Attention Theory; Attention Training; Attentional Focus; Attention–Performance Relationships; Feedback

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ATTENTION–PERFORMANCE RELATIONSHIPS

The desire to perform as well as possible in situations with a high degree of (personally felt) importance is thought to create *performance pressure*. Paradoxically, despite the fact that performance pressure often results from aspirations to function at one's best, pressure-packed situations are where major performance failures may be most visible. The term *choking under pressure* has been used to describe this phenomenon. *Choking* is defined as performing more poorly than expected given one's skill level and is thought to occur across many diverse task domains where incentives for optimal performance are at a maximum. Two of the most common examples of situations in which choking under pressure can occur are sport competitions and exams or tests.

Many different theories have been put forth to explain how and why choking under pressure occurs with the ultimate goal of developing interventions and training methods to prevent it. These theories can be roughly divided into three categories: drive theories, biomechanical theories, and attentional theories. *Drive theories*, which are primarily descriptive in nature, propose that there is an optimal level of physiological arousal for each skill that we perform. *Biomechanical theories* provide hypotheses about how the kinematics and motor control strategies involved in skill execution (e.g., control of the velocity of the racquet head in a tennis serve) change in response to pressure. Finally, *attentional theories* seek to describe the cognitive processes governing pressure-induced failure—how pressure changes the attentional mechanisms and memory structures supporting performance.

The remainder of this entry focuses on the major attentional theories.

Distraction Theory

Distraction theory is one of the primary attentional theories of choking under pressure. This

theory assumes that every performer has a limited amount of information processing capacity that can be devoted to the execution of one's skill. Information processing involves multiple resources, including perception, working memory, and attention. Successful skill execution requires accurate perception of objects in environment such as judging the speed at which a ball is approaching or discriminating between opponents and teammates on a basketball court. Working memory is a short-term memory system that maintains, in an active state, a limited amount of information with immediate relevance to the task at hand (e.g., a phone number you are about to dial) while preventing distractions from the environment and irrelevant thoughts. Attention is a resource that allows us to enhance the processing of relevant objects or locations in our environment (analogous to the way in which a spotlight allows one to see more clearly in small areas). Resource allocation models of human performance typically assume that each of these resources is capacity limited.

As an example of these information processing capacity limits in sport, consider an attacker in soccer processing the locations of opponents and teammates on a pitch. In this situation, there are only a limited number of players that the attacker can focus attention on in a given instant and there are only limited number of player locations that can be held in working memory. Therefore, anything else in the performer's environment that uses processing resources, such as a fan drawing the player's attention by yelling an insult, may require that resources are taken away from skill execution, which can lead to degraded performance (e.g., the attacker passes the ball to an opponent). In this example, player locations are an example of task-relevant information, an information source needed for successful skill execution, while the words spoken by the fan are an example of task-irrelevant information, a source that does not aid skill execution.

According to distraction theory, the introduction of performance pressure creates additional sources of task-irrelevant information that could potentially draw processing resources away from skill execution. These information sources primarily come in the form of negative thoughts or worries about the outcome of the action being performed and its potential consequences. Examples of such worries could include losing a large monetary prize, being booed by fans, letting down

one's teammates, or failing a class in school. These negative thoughts are hypothesized to sap both working memory (as the performer plays out the disastrous outcomes in their mind) and attentional resources (as the performer's attention is shifted from the external environments to these internal thoughts). Another way to conceptualize this effect is that pressure serves to change a single task situation where the performer is only required to execute one particular skill into a multitasking situation in which the performer must do two things at once. In sum, in distraction theory, performance pressure serves to distract the performer and draw processing resources away from the task at hand. As described below, this account is the polar opposite to explicit monitoring theory, which proposes that pressure serves to increase the amount of attention and working memory resources devoted to skill execution.

Processing Efficiency Theory

Does the distraction created by the negative thoughts and worries always lead to performance failure? A related theory, called *processing efficiency theory*, suggests this might not always be the case. Similar to distraction theory, processing efficiency theory proposes that pressure serves to draw processing resources away from task-relevant information; however, it further proposes that decreases in resources available to support skill execution can be partially or fully compensated for by an increase in the effort devoted to the task. Therefore, the main outcome predicted by processing efficiency theory is that *skill efficiency* (defined as the ratio of performance to the amount of effort exerted) will decrease under pressure. The extent to which associated declines in performance also occur will depend on how much additional effort is exerted by the performer. In any case, both theories propose the same mechanism for the effects of pressure with the difference between them being the degree to which performance is affected in the end.

Because it is based on the assumption that information processing capacity is limited, a strong prediction made by distraction theory is that pressure-induced performance failures will be greatest for skills that normally in nonpressure situations require a large amount of working memory and attentional resources. When a task has very low processing demands, it follows that a performer will have more available capacity, that is, will have

more working memory and attention resources available for handling task-irrelevant information, as compared to a skill with high processing demands. Therefore, when a low demand skill is performed under pressure, it is less likely that the additional resources required to process distracting thoughts and worries will cause an overloading of processing capacity. Instead, it is possible that the performer will be able to successfully process both task-relevant and task-irrelevant information. This prediction has received strong support from research involving academic test anxiety. For example, it has been demonstrated that individuals who become highly anxious during test situations, and consequently perform at a suboptimal level, often divide their attention between task-relevant and task-irrelevant thoughts more so than those who do not become overly anxious in high pressure situations. However, suboptimal performance only appears to occur for test problems with high working memory demands like a difficult math problem, while performance for low-demand problems is relatively unaffected.

Thus, there is evidence that pressure can compromise working memory resources, causing failure in tasks that rely heavily on this system. But, not all tasks *do* rely heavily on working memory. Specifically, the types of high-level motor skills that have been the subject of the majority of choking research in sport (well-learned golf putting, baseball batting, soccer dribbling) are thought to become proceduralized (unconscious, automatic) with practice. Proceduralized skills do not require constant online attentional control and are in fact thought to run largely outside of working memory. Such skills, then, should be relatively robust to conditions that consume working memory resources as distraction theory proposes. However, these types of skills may be sensitive to other attention-induced disruptions under pressure. A second class of theories, generally known as explicit monitoring theories, has been used to explain such failures.

Explicit Monitoring Theories

Explicit monitoring theories suggest that pressure situations raise self-consciousness and anxiety about performing correctly. This focus on the self is thought to prompt individuals to turn their attention inward on the specific processes of performance in an attempt to exert more explicit monitoring and control than would be applied in

a nonpressure situation. Explicit attention to step-by-step skill processes and procedures is thought to disrupt well-learned or proceduralized performance processes that normally run largely outside of conscious awareness. This proposal is based on the assumption that the processing demands for complex motor skills change systematically during skill acquisition.

For example, consider a golfer attempting to make a putt. A performer relatively new to the sport is likely to have been given a lot of explicit instructions about how to putt effectively (e.g., keep head down, use a smooth backstroke, keep your eye on the ball, etc.). Executing a skill based on these types of instructions is assumed to require a large amount of working memory (the golfer must actively hold in working memory all these instructions and the order they should be executed) and attentional resources (the golfer must focus attention on the position of the head, hands, etc. to determine consistency with the instructions). Through extensive practice, it is assumed these instructions become internalized (or *proceduralized*) so that expert performance is guided by a set of motor programs, and procedures, once initiated, can run without the need for working memory or attention. How to perform the skill is no longer held actively in working memory but rather it is stored in *muscle memory*. And the performer no longer needs to consciously monitor the position of the body by focusing attention because the execution of movement is now controlled by unconscious, automatic processes. Instead, more attentional and working memory capacity can be devoted to processing external or strategic information.

In explicit monitoring theory, it is assumed that not only is attention to the execution of a complex motor skill not required, but also that it can be harmful for performance if it does occur. It is argued that directing one's attention to a well-learned skill effectively serves to disrupt the highly efficient and automatic motor procedures developed through practice. Instead, if skill execution is controlled consciously in a step-by-step manner using a large amount of attentional and working memory resources, performance becomes slow, nonfluent and error prone, outcomes typical of novice performance. Evidence for explicit monitoring theory is provided by dual-task experiments, which require a performer to execute a skill—such as putting or batting—while simultaneously

performing a secondary task designed to reorient attention toward skill execution (judging the angle of the putter head or direction of bat movement) or away from skill execution (judging the pitch of an irrelevant sound). Consistent with explicit monitoring theory, it is typically found that expert performance suffers for tasks that direct attention to skill execution and is relatively unaffected by tasks that direct attention away from it. Also consistent with the theory, novice performers typically show the reverse pattern where performance is harmed by irrelevant dual tasks and relatively unaffected (or in some cases actually improved) by skill-focused dual tasks.

Reinvestment Theory

One type of explicit monitoring theory, called *reinvestment theory*, suggests that the specific mechanism governing explicit monitoring is *dechunking*. Pressure-induced attention to execution causes an integrated or proceduralized control structure that normally runs off without interruptions to be broken back down into a sequence of smaller, independent units—similar to how the performance was organized early in learning. Once dechunked, each unit must be activated and run separately. Not only does this process slow performance, it creates an opportunity for error at each transition between units that was not present in the integrated control structure.

Explicit Monitoring and Distraction Theories Compared

Explicit monitoring and distraction theories essentially make opposite predictions regarding how pressure exerts its impact. While distraction theories suggest that pressure shifts needed attention away from execution, explicit monitoring theories suggest that pressure shifts too much attention to skill execution processes. Can both theories be correct? One possibility is that performance pressure creates two effects that alter how attention is allocated to execution: (1) Pressure induces worries about the situation and its consequences, thereby reducing working memory capacity available for performance—as distraction theories would propose; and (2) at the same time, pressure prompts individuals to attempt to control execution in order to ensure optimal performance—in line with explicit monitoring theories. This suggests that how a skill fails is dependent on performance

representation and implementation. That is, skills that demand working memory will fail when pressure consumes the resources necessary for performance, while proceduralized skills that run largely outside of working memory will fail when pressure-induced attention brings such processes back into conscious awareness. Therefore, it is perhaps better to think of these two theories of pressure-induced failures of performance as complementary rather than competing.

It is important to note that it does not seem to be merely a cognitive versus motor distinction that predicts how a skill will fail under pressure. That is, just because one is performing an academically based, cognitive task does not mean this task will show signs of failure via pressure-induced distraction. And, likewise, sports skills do not necessarily fail via pressure-induced explicit monitoring. Rather, it appears to be the manner in which skills utilize on-line attentional resources that dictates how they will fail (though often, this is related to skill domain). Thus, sports skills that make heavy demands on working memory, such as strategizing, problem solving, and decision making (skills that involve considering multiple options simultaneously and updating information in real time), will likely fail as a result of pressure-induced working memory consumption—similar to a working-memory-dependent academic task. In contrast, motor skills that run largely outside of working memory—for instance, a highly practiced golf putt or baseball swing—will fail when pressure-induced attention disrupts automated control processes.

Explicit monitoring and distraction theories have very different implications for how to prevent choking under pressure. According to explicit monitoring theory, choking is best prevented by not allowing a performer to turn attention inward and explicitly control movements. Two promising interventions for achieving this end have been identified. First, it has been shown that allowing a performer to become accustomed to the desire to turn attention inward during practice (through the use of videotaping or some other means of evaluation that tends to induce self-consciousness) can help prevent choking from occurring during a subsequent competition. A more radical solution, based on the reinvestment theory described above, is to change the way a performer acquires a skill: If the performer acquires less knowledge about how to explicitly control a skill (e.g., a golfer is not given instructions about where to place the feet, hands,

etc.), it is less likely he or she will switch to this control mode under pressure. Indeed, research evidence suggests that skills acquired implicitly, as in learning by doing as opposed to following instructions, are less prone to choking under pressure.

The primary means for remedying pressure-induced failure, according to distraction theory, is to increase the amount of processing resources devoted to task-relevant information when a performer is placed in a high pressure situation. This could be achieved in different ways. First, as discussed above with reference to processing efficiency theory, it is theoretically possible to reduce (or eliminate) the effects of pressure on performance by increasing effort. Studies that incorporated effort measurement under pressure have shown that this does indeed occur for many performers, suggesting that learning to put more effort into one's skill can be an effective means of handling pressure. An alternative remedy is to reduce the level of processing demands required for the task-irrelevant worries and negative thoughts that can occur under pressure. Research has shown that this can be achieved by having the performer explicitly verbalize (or write down) the negative thoughts immediately prior to performing the high-pressure skill.

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See also Attention Theory; Attentional Focus; Automaticity; Choking; Chunking/Dechunking; Concentration; Information Processing; Memory; Motor Control

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ATTRIBUTION THEORY

Attributions are explanations about why particular performances or behaviors have occurred. When faced with important, negative, novel, or unexpected events, individuals search for meaningful explanations for the causes of those events. In this regard, it is widely acknowledged that attributions are an area of importance in the field of applied psychology because of their implications for motivation and emotion. This entry discusses the historical theories that have provided the framework for research into, and application of, attribution theory in sport and exercise settings, then moves on to highlight recent theoretical advances, assessments of attributions, empirical evidence for attribution theory, and the application of attribution theory.

Historical Theories

If a particular individual were to be singled out as the founder of the scientific study of attributions, it would be Fritz Heider. He proposed that people explore explanations for events or behaviors to increase their control over the environment and to satisfy a desire to understand and gain knowledge about the world. His insights provided the impetus for numerous theories and investigations. Three of the most well known are Edward Jones and Keith Davis's correspondent inference theory, Harold Kelley's covariation model, and Bernard Weiner's attributional theory of achievement motivation and emotion.

Correspondent inference theory proposes that people infer dispositions and intentions of others, as a result of observing their behavior. That is, behavior is seen as corresponding to or reflecting an underlying disposition of the actor. In general, correspondence is high when behavior is atypical and has clear implications. In other words, behavior is informative to the extent that it is seen to involve choice among alternatives. Whereas correspondent inference theory focused mainly on person perception (or attributing to others), the covariation model made contributions to our understanding of self-perception. The covariation

model suggests that people arrive at a cause for an event by processing information about whether accompanying conditions and circumstances vary or not as the event changes. According to the model, people use three types of information—consistency, distinctiveness, and consensus—to verify whether they have correctly linked causes and effects.

The approach that has had most influence on attribution research in sport and exercise psychology is the attributional theory of achievement motivation and emotion. A central premise of this theory is that there is a dimensional structure underpinning the explanations people give for events. Building on previous works in this field, Bernard Weiner, a key figure in conceptualizing the attributional categories into dimensions, initially identified four main attributions (or attribution elements), namely, ability, effort, task difficulty, and luck; these were then classified into two dimensions, locus of causality and stability. *Locus of causality* refers to whether the cause is inside (internal) or outside (external) the person; *stability* refers to whether the cause will (unstable) or will not (stable) change over time. Later deductive theorizing led to the identification of a third dimension, *controllability*, referring to whether the cause could be viewed as controllable or uncontrollable.

Recent Theoretical Advances

Despite warnings by a number of researchers that the three dimensions of locus of causality, stability, and controllability may not be appropriate for all types of situations (e.g., sport and exercise), until recently, there has been little effort to examine alternative approaches. In 2005, Tim Rees, David Ingledew, and Lew Hardy published an article that examined the congruence between theory, research, and practice of attributions in sport. On the basis of their observations, they proposed a broader conceptual approach to our understanding of attributions. They encouraged (a) a focus on the controllability dimension, (b) an expanded conceptualization of generalizability attributions (in addition to stability, examining globality and universality attributions), and (c) exploring interactive effects of controllability and generalizability attributions upon performances or behaviors.

The focus on controllability is based on empirical and theoretical grounds. Empirically, studies

in a wide array of domains have shown that perceived personal control is an important psychological predictor. For example, within general social psychology, perceived personal control has been shown to affect a person's level of depression, loneliness, and shyness, and in sport an athlete's level of subsequent self-efficacy. Theoretically, the need to exert control over future events was foundational to early attribution theorizing, with Harold Kelley commenting that the purpose of causal analysis is effective control. Further, drawing upon attribution theory, the expectancy of future uncontrollability is at the heart of Lyn Abramson, Martin Seligman, and John Teasdale's reformulation of the learned helplessness model: The *expectancy of future uncontrollability* is the most direct determinant of helplessness.

The second proposal is that, in addition to the stability dimension, attribution research in sport and exercise should examine the dimensions of globality and universality. The addition of globality refers to whether the cause affects a wide range of situations with which the person is faced (a global attribution) or a narrow range of situations (a specific attribution); universality refers to whether the cause is common to all people (a universal attribution) or unique to the individual (a personal attribution). This leads to an expanded conceptualization of generalizability: In addition to whether causes generalize across time (stability), attribution research should examine whether causes generalize across situations (globality) or all people (universality).

The final proposal is that researchers need to move beyond testing solely main effects of attributions to exploring interactive effects of controllability and generalizability attributions upon performances or behaviors. Until recently, the focus of much attribution research had been upon main or independent effects of attribution dimensions. To model *generalizability*, however, implies the need to consider *interactive* effects. Interactions of attribution dimensions may well be important because attributing, for example, less successful performances to uncontrollable causes may only lead to negative effects when causes are also considered to be stable (unlikely to change over time), global (likely to affect a wide range of situations), or personal (unique to the individual). For example, an athlete attributing poor performance to poor concentration might say, "There was nothing I could do about it" (an uncontrollable

attribution), together with “and this will never change” (a stable attribution), or “and this affects a lot of situations I find myself in” (a global attribution), or “and this only happens to me” (a personal attribution). In this instance, the athlete might well be expected to experience poorer subsequent performance. Conversely, higher levels of performance would be expected if the athlete were to combine this uncontrollable attribution with “but this will change” (an unstable attribution), or “however, this only affects a few situations I find myself in” (a specific attribution), or “but this affects everyone, not just me” (a universal attribution).

Assessing Attributions

Various methods have been employed in the measurement and categorization of attributions. *Open-ended methods* involve the researcher categorizing the oral replies of participants to open-ended questions. *Derived score methods* require the participant to rate reasons for a success or failure on five-point scales for different elements (e.g., ability, effort, task difficulty) related to the attribution dimensions. Problems can arise, however, when researchers try to summarize attributions along dimensions or otherwise assume the dimensional categories of attributions. Recent attribution measures, such as the Causal Dimension Scale (CDS), the Causal Dimension Scale II (CDSII), and the Measure of Controllability, Stability, Globality, and Universality Attributions (CSGU), have used *direct rating methods*. These methods require the participants to state their reasons for the event and then map those reasons onto items referring to attribution dimensions.

The CDS was developed to reflect Weiner's three-dimensional model of attributions. A number of methodological criticisms have been leveled at the CDS, in particular concerns over the nature of the controllability subscale, which contains items referring to *controllability*, *responsibility*, and *intentionality*. The revision of the CDS (the CDSII) focused on changes to the controllability items, with all six of the items from the locus of causality and stability subscales left unaltered. In the CDSII, the controllability dimension was subdivided into personal control (control by the actor) and external control (control by others). Based upon recent proposals in the sport

and exercise psychology literature, measures of attributions should include four scales for controllability and the generalizability dimensions of stability, globality, and universality. Following this, researchers developed the CSGU, a 16-item, 4-factor measure of attributions that assesses the four dimensions of controllability, stability, globality, and universality.

Empirical Evidence for Attribution Theory

A number of research articles have been published, exploring the following key propositions of the attributional theory of achievement motivation and emotion: (a) the locus-of-causality dimension affects intrapersonal emotions including pride and shame; (b) the stability dimension affects expectations of future success and feelings of hopefulness; and (c) the controllability dimension affects interpersonal emotions including blame, anger, and pity. Despite empirical support, these propositions are limited to the extent they explore the effects of attribution dimensions in isolation. In other words, the proposals focus solely upon main effects of attribution dimensions upon outcomes.

Recent research has reported the interactive effects of attribution dimensions upon outcomes. Collectively, the research demonstrates that the independent effects of attribution dimensions upon outcomes—such as emotions, self-efficacy, and performance—may be conditioned by interactive effects. For example, research has demonstrated that, following failure, higher levels of perceptions of controllability may only lead to beneficial effects if causes are also considered to generalize, such as across time (interaction of controllability and stability).

Application of Attribution Theory

Attributional retraining involves attempts to change maladaptive explanations for outcomes (attributing failure to uncontrollable or stable causes) toward more adaptive explanations (attributing failure to controllable or unstable causes). Although there is discrepancy in the literature regarding whether negative behaviors or outcomes should be perceived as inside the individual (internal locus of causality) or outside the individual (external locus of causality), there is agreement in regard to the need to perceive such causes as controllable. For example, following

failure like a loss in sport or a relapse from a positive behavior like smoking a cigarette following a period of abstinence, an individual is encouraged to attribute the negative outcome or behavior to causes that are within one's control. In situations where such causes are perceived as outside one's control (i.e., a maladaptive attribution), the procedure of attributional retraining involves practitioners manipulating causes toward aspects that are within one's control.

Conclusion

Attribution theory has a rich history in general (social) psychology, and has significantly influenced applied practice in sport and exercise settings. Recent theoretical advances, supported by empirical evidence, suggest a promising future for attribution theory to continue as a principal theory in sport and exercise psychology.

Pete Coffee

See also Cognitive Restructuring; Feedback; Positive Thinking; Psychological Skills Training; Self-Fulfilling Prophecy; Team Attributions

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AUDIENCE EFFECT

See Social Processing Effects

AUTOMATICITY

Automaticity is the ability to execute a skill using no (or very few) information processing resources: attention and working memory. When a skill can be executed in this fashion, the performer has resources available to process other sources of information not directly required for the task. Automaticity is thought to be a hallmark of expert performance that is acquired through learning and extensive practice. A breakdown of automaticity resulting from a performer turning attention inward toward skill execution is thought to be one of the primary causes of choking under pressure (see also the “Attentional Focus” entry).

When one performs a skill, there are different modes of control that can be used. At one extreme, commonly called *controlled processing mode*, a performer executes an action by following a series of explicit steps that are held in working memory and by focusing attention on each part of the action. Each stage during skill execution is consciously controlled and monitored. For example, when executing a complex movement in sports, the performer must remember the instructions to position different body parts and focus attention on these parts to determine if positioning is correct. In this performance mode, the information-processing demands of the skill are very high and there are few, if any, leftover resources available for processing task-irrelevant information. At the other extreme, commonly called *automatic processing mode*, skill execution relies on motor programs or procedures that, once initiated, run without the use of attentional or working-memory resources. The skill is executed unconsciously as it is thought to involve *muscle memory* (well-developed internal commands for how the different body parts should be moved) rather than high-level cognitive control. In this performance mode, the performer has information processing

resources available for handling task-irrelevant information. Planning a play to be executed on the next possession in basketball is an example of a highly controlled task, whereas running down the court is thought to be a highly automatic one.

As first proposed by Paul Fitts, the controlled and automatic processing modes are thought to be characteristics of the early and late stages of skill acquisition, respectively. For novice performers, it has been proposed that skill execution requires that attention be paid to each component stage of the motor act (in golf, back-swing, down-swing, club-head angle in putting). At this level of performance, referred to as the *cognitive* or *declarative* stage, it is assumed that skill execution depends on a set of unintegrated control structures that must be held in working memory and attended in a step-by-step fashion. The attentional and memory requirements result in the slow, nonfluent, and error-prone movement execution that is characteristic of a novice performer. As expertise develops through practice and the performer reaches the highest stage of skill execution (the *autonomous* or procedural stage), it has been proposed that the role of attention and working memory in performance changes dramatically. As procedural knowledge develops, the conscious step-by-step control of execution is no longer required. Instead, skill execution is assumed to operate by fast, efficient control procedures that function largely without the assistance of working memory or attention. At this final stage, it is proposed that skill automaticity has been achieved. It is frequently also proposed that automatic skills are encapsulated such that, once they are initiated, it is difficult for the performer to inhibit or disrupt them.

A key advantage that automaticity is thought to give an athlete is the availability of attentional and working-memory resources to process information in the environment not directly related to movement control. First, the performer is more likely to detect situational or contextual information that is not required for movement control but may serve to improve performance success. If an expert footballer does not need to actively monitor the position of the body when taking a penalty kick, that player will have processing capacity available to attend to the movements of the goalkeeper. Second, the performer is less likely to be affected by the introduction of stimuli completely

irrelevant to the task like a fan yelling an insult or waving hands in the background.

Theories of skill acquisition and automaticity make several predictions about the nature of performance (and the supporting attentional mechanisms and memory structures) as a function of skill level. For example, if it involves the automatic processing mode, expert performance should be (a) unaffected by the introduction of an irrelevant secondary task that requires attentional or working memory resources, and (b) associated with a poorer ability to verbalize the stages involved in skill execution (*expertise-induced amnesia*) since these stages are not attended or held actively in memory. For the most part, research evidence is consistent with these predictions; however, for most tasks, it has also been shown there is some decrement in performance when a highly demanding secondary task is introduced. Therefore, it may not be the case that automatic processes are completely resource free; rather, automaticity may be best characterized by a reduction in the processing resources required.

Another important prediction of skill acquisition and automaticity theories concerns pressure-induced failures of performance (choking under pressure). One of the dominant theories of choking, called *explicit monitoring theory*, proposes that pressure can cause the deautomization of well-learned skills. This occurs because an increase in self-consciousness and anxiety about performing well causes an athlete to turn attention inward on the specific processes of performance in an attempt to exert more explicit monitoring and control. This increase in skill-focused attention is thought to disrupt the automatic motor procedures.

Rob Gray

See also Attention Theory; Attentional Focus; Attention-Performance Relationships; Choking; Information Processing; Memory; Motor Control

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AUTOMATICITY: EVALUATIVE PRIMING

Priming effects occur when the processing of a target stimulus is influenced by a preceding stimulus on the basis of a relationship between prime and target. Processing of the target word *surgeon*, for example, may be facilitated by the prime word *injury*. The priming effect in this example is based on associative and nonevaluative semantic relationships between prime and target, and is called *semantic priming*. Evaluative priming, on the other hand, focuses on evaluative (positive or negative) relationships between primes and targets. In an evaluative priming paradigm, a positive or negative target (delight) is preceded by a prime of the same valence (healthy) or of the opposite valence (failure).

Usually, participants in these reaction-time tasks are asked to decide whether the target denotes something positive or negative. Responses are typically faster and more accurate when the prime and target share the same valence (healthy delight). Thus, presentation of *healthy* as the prime is likely to automatically activate a positive evaluation. If the target word that is subsequently presented is also positive (delight), then the individual is likely to respond quickly and accurately to the adjective. If the target word is negative, however, response accuracy and speed are likely to be compromised. Consequently, evaluative priming effects are based on a significant interaction between the valence of the prime and the valence of the target.

Automatic Evaluation: A Consistent Finding

By using a short timeframe (called *stimulus onset asynchrony*) between the presentation of primes and targets, research on evaluative priming has found that evaluative information about stimuli is activated very quickly (automatically) upon presentation of the prime. Also, the evaluative priming effect has been replicated numerous times using a variety of experimental stimuli. The effect has emerged, for instance, when the primes consist of words, pictures, or odors. Moreover, the effects have been evidenced when the response task given to participants is to evaluate the target ("Is the target word positive or negative?"), when they are asked to pronounce the targets, or when

they are required to generate a motor response task to the primes.

Compatibility Between Automatic Evaluation and Self-Report Evaluation

Traditionally, to gauge people's evaluation of an object or concept, researchers have utilized self-report measures that invite conscious introspection among participants. One of the most significant benefits of evaluative priming studies, however, is that they allow researchers to tap fast evaluative processes that often occur without, and sometimes against, the respondents' intentions. For research on socially sensitive topics, such as exercise and body image, significant differences might emerge between these automatic evaluations and self-report evaluations. Therefore, social stigmatization acts as a possible moderator to the relationship between self-report and automatic evaluations. Another possible moderator to this relationship is evaluation strength. In other words, people who possess strong automatic evaluations of concepts are also likely to report strong, compatible evaluations in self-report questions. If one has a strong passion for basketball, for example, the individual is likely to report congruent positive evaluations of the sport on both self-report measures and implicit measures.

Research on Consequences of Automatic Evaluation

There are two streams of literature on the consequences of automatic evaluation. One of these streams relates to the question of whether primed evaluations are predictive of approach or avoidance behaviors toward that stimulus. In other words, do automatic evaluations increase one's tendency to behave in particular ways toward the priming stimulus? Research is now emerging to suggest that these evaluations can significantly influence one's reactions toward the stimulus itself. Another stream of research focuses on the implications of a primed evaluation for the subsequent processing of unrelated stimuli. There is a possibility that interpretation of ambiguous stimuli can be guided by previous exposure to positively or negatively evaluated primes. In fact, past work has established that interpretation of words such as *beat* can be guided by strongly valenced primes, such that people are more likely to interpret the word as meaning *rhythm* after a positive prime

and *hit* after a negative prime. Collectively, these findings suggest that one's automatic evaluations of sport or exercise are likely to predict the individual's approach or avoidance of these activities, as well as influence the interpretation of information about them. Someone who possesses a negative automatic evaluation of exercise, for instance, is likely to reject an offer to join an exercise class without giving much thought to the offer. Also, that person will be likely to bias any ambiguous information in an exercise advertisement (or other communication) in a negative manner.

Mechanisms

Evaluative priming effects are most often discussed in terms of a spreading activation within a participant's semantic network. According to the spreading activation hypothesis, priming *success*, for example, would make the concept of all positively valenced representations more accessible. These salient representations are then more readily used to encode subsequent information. Although the spreading activation account to evaluative priming has conceptual appeal, it has also been criticized by some scholars. One of the key criticisms has been that the spreading activation account is based on the *fan effect*. If a prime stimulus activates many concepts in memory (as it would in the context of evaluative priming), the amount of activation should disperse over many pathways, making the activation of any one representation quite minimal. Other mechanisms for evaluative priming have been forwarded, but more research is needed to establish which of them offer the best explanation for these effects.

James Dimmock

See also Affect; Automaticity; Automaticity: Implicit Attitudes; Priming

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AUTOMATICITY: IMPLICIT ATTITUDES

Implicit attitudes, also referred to as automatic evaluations, are immediate and spontaneous evaluations of objects, groups, or behaviors as being pleasant or unpleasant. They occur unintentionally within a fraction of a second after exposure to the stimulus, preceding any reflective deliberation. Typically, these attitudes are inferred from systematic variations of performance on indirect priming or categorization tasks. Implicit attitudes can collaborate or conflict with more controlled and deliberated explicit attitudes to influence decisions and behaviors. It has been shown that implicit attitudes predict physical activity behavior, but little research has been conducted on implicit attitudes in the context of sport. This entry briefly overviews (a) measurement, (b) attitudinal concordance with explicit attitudes, (c) proposed origins, and (d) proposed consequences of implicit attitudes. Special emphasis is placed on implicit attitudes in the context of physical activity and sport.

Measurement

Implicit attitudes are typically measured indirectly via performance tasks because self-assessment (self-report) is susceptible to inaccuracy due to people's lack of awareness or willingness to report their implicit attitudes. The most common methods for implicit attitude assessments are priming procedures or categorization tasks, such as the Implicit Association Tests (IAT). Given the nature of response latency tests, error variability can be easily introduced (e.g., distraction, sneeze, fatigue). This, among other concerns about validity and reliability of the measures, remains an important issue in the field.

Attitudinal Concordance With Explicit Attitudes

Research suggests implicit attitudes are related, but distinct, from explicit attitudes. *Implicit attitudes*

are automatic, efficient, rapid, and goal-independent evaluations that involve the neural circuits of the amygdala and right insula; *explicit attitudes* are effortful, deliberative, and controlled evaluations that involve areas of the frontal cortex. Explicit attitudes may serve to override or validate implicit attitudes through reflective cost-benefit analyses, especially when ample motivation and opportunity to deliberate is present. Implicit and explicit attitudes typically show weak positive correlations within studies, although the magnitude of the relation differs as a function of the attitude being assessed. For example, implicit and explicit attitudes are largely divergent for socially sensitive topics where impression management may occur, as with racial prejudices, but less divergent for topics toward which people have little motivation to disguise their attitudes, as with preference for flowers. The magnitude of the relation between implicit and explicit attitudes also is a function of study design. Correlations are strongest when there is strong convergence of the conceptual and format structure of the implicit and explicit measures, for example. Implicit and explicit attitudes toward physical activity have been shown to be independent of one another, but little is known about concordance of implicit and explicit attitudes in the context of sport.

Proposed Origins

The origins of implicit attitudes are not fully understood; however, there is research connecting implicit attitudes to early emotional responses to the stimuli, such as averseness of the first cigarette smoked and to more recently learned affective associations, such as priming associations between valence and object. Implicit attitudes tend to show low or moderate stability across time, supporting the notion that they comprise both trait- and state-like components. Although some aspect of implicit attitudes may be resistant to change, it seems that appropriate environmental pairings have potential to counter these learned associations. Most theories propose that implicit attitudes are some manifestation of both early and recent experiences, but it has also been proposed that implicit attitudes represent knowledge of cultural biases, rather than personal attitudes. Very little is understood about the origins of implicit attitudes in the context of physical activity and sport.

Consequences

Implicit attitudes have been shown to have physiological consequences, such as an elevated heart rate response to stimuli; cognitive consequences, such as phobias and judgments; and behavioral consequences, such as prejudice and consumer and voting behavior. Within an evolutionary context, implicit attitudes have been proposed as a survival strategy to quickly differentiate hostile from friendly stimuli. Favorable implicit attitudes elicit immediate approach behavior, whereas unfavorable implicit attitudes elicit immediate avoidance behavior. Implicit attitudes seem to guide behavior without people's conscious awareness. Implicit and explicit attitudes have been shown to predict unique portions of variance of the same behavior and to interact to predict behavior. Additionally, a double-dissociative connection between attitudes and behavior has been supported in which implicit and explicit attitudes predict behaviors under different circumstances. Whereas explicit attitudes tend to direct volitional behavior or behavior when people have ample time and resources to deliberate, implicit attitudes tend to direct spontaneous behaviors or behaviors when people are under time pressure, distracted, or unmotivated for reflective deliberation. Physical activity and fitness level have also been shown to be related to implicit attitudes above and beyond explicit motivational constructs. Again, little is known about how implicit attitudes may impact outcomes in the competitive or training contexts of sport.

Conclusion

Implicit attitudes have been shown to influence peoples' physiology, judgments, and behavior independently of explicit attitudes. It has been established that implicit and explicit attitudes toward physical activity are distinct, and that implicit attitudes predict physical activity behavior; however, little is known about the origins or malleability of these attitudes. Implicit attitudes within the context of sport remain largely uninvestigated. The potential for implicit attitudes to serve as an avenue for physical activity and sport interventions needs to be clarified with future research.

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See also Automaticity: Evaluative Priming; Dual-Process Theory; Habit

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AUTONOMIC NERVOUS SYSTEM

The central nervous system (CNS) is composed of the brain and spinal cord. The CNS receives sensory information from the peripheral nervous system and controls the body's responses. The peripheral nervous system involves all of the nerves outside of the brain and spinal cord that carry messages to and from the CNS. The peripheral nervous system has two branches: the somatic nervous system and the autonomic nervous system. The somatic nervous system controls skeletal muscle as well as external sensory organs such as the skin. This system is described as voluntary because the responses can be controlled consciously, although reflex reactions of skeletal muscles are an exception.

The autonomic nervous system is the branch of the peripheral nervous system that controls involuntary actions, such as pulmonary respiration (breathing), heart rate and the force of contraction of the heart, and vasoconstriction and

dilation (widening and narrowing of blood vessels) that impact blood pressure. This system is further divided into two systems: the sympathetic system and the parasympathetic system. The sympathetic division of the autonomic nervous system regulates flight-or-fight responses. This division is responsible for tasks such as relaxing the bladder, increasing heart rate, and dilating eye pupils. The parasympathetic division of the autonomic nervous system supports homeostasis and conserves physical resources. This division performs tasks such as controlling the bladder, slowing down heart rate, and constricting eye pupils.

Like other nerves, those of the autonomic nervous system convey their messages to the appropriate end organs (blood vessels, viscera, etc.) by releasing transmitter substances to which the receptors of the target cells are responsive. The most important of these transmitters in the autonomic nervous system are acetylcholine and norepinephrine. In the parasympathetic system, acetylcholine is responsible for most of these transmissions between the afferent (leading toward the CNS) and efferent nerves (leading away from the CNS) and between the efferent nerve endings that innervate cells or organs. Acetylcholine also serves to transmit nerve-to-nerve messages in the afferent nerves and the brain centers of the sympathetic nervous system. Norepinephrine is released from sympathetic nerve endings to end organs or cells, except at the sweat glands where acetylcholine is released. Epinephrine is also released from the adrenal medulla in response to sympathetic activation.

The rate and strength of cardiac contractions is under the predominant control of the sympathetic nervous system. Sympathetic stimuli from cardiac nerves cause acceleration of the heart rate. Increases in heart rate are also complemented by simultaneous reduction in the parasympathetic stimuli via the vagus nerve, which is responsible for slowing the heart rate. Thus, heart rate responses are often used as indirect measures of sympathetic and parasympathetic activity (autonomic nervous system activation).

Situations such as emotional excitement, fear, apprehension, psychological distress, panic reactions, and fight-or-flight stimuli activate the sympathetic nervous system. In addition, the activation of the autonomic nervous system, often demonstrated in preparation for and during activity, has been an area of investigation in sport

and exercise physiology. Noninvasive measures of autonomic function (heart rate, blood pressure, galvanic skin response, etc.) have been used to investigate the role of the peripheral nervous system in human performance, infer affect and infer changes in affect during and following physical activity, understand the effects of overtraining, and explain potential positive adaptations that occur in parallel with improvements in fitness. In summary, biofeedback studies have demonstrated a relationship between effective control of autonomic function and human performance, a number of affective states have been shown to correlate with autonomic responses, overtraining seems to be associated with parasympathetic dysfunction, and improvements in fitness are associated with enhanced parasympathetic tone and diminished sympathetic activity. More specifically, the changes in response to aerobic training seem to occur primarily due to an alteration in the balance of sympathetic and parasympathetic activity. With training, several adaptations, including enhanced parasympathetic tone and increased stroke volume, explain the reduction in resting heart rate. At rest the enhanced parasympathetic activity seen in trained individuals leads to a reduction in heart rate; however, when presented with a mental challenge, trained individuals demonstrate greater vagal withdrawal and possibly an enhanced sympathetic response compared with untrained individuals. This adaptation seems to be beneficial for cardiovascular health.

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See also Biofeedback; Neurofeedback; Cardiac Function; Electromyography (EMG); Eye Movements/Gaze; Neuroscience, Exercise, and Cognitive Function; Overtraining Syndrome; Psychophysiology; Stress Reactivity

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AUTONOMY-SUPPORTIVE COACHING

Motivation is one of the foundations of successful sport performance, and coaches play a critical role in developing or undermining this attribute in their athletes. The techniques coaches use to instruct and motivate their athletes can influence whether athletes learn and achieve at a high level, develop a strong sense of confidence, enjoy their experience, and persist in their sport over a long period of time.

Self-Determination Theory

In 1985, Edward Deci and Richard Ryan wrote their influential book, *Intrinsic Motivation and Self-Determination in Human Behavior*. In this text, and numerous subsequent publications, Deci and Ryan outlined their theory of human motivation, *self-determination theory*. This theory, often known as SDT, has guided much research examining the influence that coaches can have on their athletes' motivation and subsequent outcomes in sport.

Deci and Ryan suggested that motivation is multidimensional, meaning that a person can be simultaneously motivated by multiple factors. Central to SDT is the distinction between self-determined and controlled forms of motivation. To be *self-determined* means to act with a sense of self-direction and choice. An athlete who is self-determined participates because that athlete finds sport enjoyable or interesting (intrinsic motivation) or values the benefits of sport participation. In contrast, motivation that is fueled by pressure from others or pressure from within (guilt) is referred to as *controlled motivation*. A growing body of research has provided evidence regarding the benefits of being involved in sport for more self-determined reasons. For example, athletes whose participation is driven largely by self-determined factors as opposed to controlled forces perform at a higher level; experience more positive emotions; use positive coping strategies in stressful situations; and invest higher levels of concentration, persistence, and effort than athletes with lower self-determined motivation and stronger controlled motivation. In comparison, athletes with high levels of controlled motivation tend to experience a variety of negative outcomes such as drop-out from sport, burnout, antisocial behavior, anxiety, and negative affect.

Given our understanding of the effects of controlled and self-determined types of motivation on the athlete sport experience, a natural question that arises is: How do athletes' self-determined and controlled motivations develop? According to SDT, self-determined motivation is most likely to develop when three basic psychological needs, known as autonomy, competence, and relatedness, are fulfilled. Autonomy represents the need to feel personal control over one's actions. Competence reflects the need to feel effective. Relatedness represents the need to feel connected with others and a secure sense of belonging. There is evidence that athletes require all three psychological needs to be satisfied in order for optimal motivation and well-being to develop. However, there is also research that suggests that some needs may be more important in certain circumstances. For example, competence may have a greater impact than relatedness on elite athletes' motivation.

Coaching Behaviors and Athlete Motivation

When coaches' actions support their athletes' needs, self-determined motivation will develop. When coaches' behavior undermines these needs, athletes are likely to experience increased controlled motivation. Therefore, it is important to understand the specific aspects of coaching behaviors that are positively or negatively related to athlete motivation.

Coaches' behavior can be classified in a variety of ways. When exploring motivation from a SDT perspective, coaches' provision of psychological needs support is often examined. Specifically, support that pertains to each of the three psychological needs has been investigated: autonomy support, competence support, and relatedness support. When a coach acts in an autonomy supportive way, the coach considers the athletes' perspective, provides appropriate and meaningful feedback, and offers opportunities for choice, while at the same time minimizing the use of pressures and demands to control the athletes. The process of creating an autonomy-supportive climate requires considerable skill, particularly considering the authoritarian role that coaches have often been expected to play in the past. Even within the somewhat rigid structure of many organized sports, coaches can take steps to create an autonomy-supportive climate for their athletes. Specifically, researchers suggest a number of key practices that would help a coach become autonomy supportive:

1. Provide choice—athletes making decisions about some aspects of a training session.
2. Provide a rationale for tasks, limits, and rules—explaining the reasons behind key coaching decisions.
3. Inquire about and acknowledge athletes' feelings—getting to know athletes as people first and athletes second; acknowledging that some training drills may be repetitive or tedious.
4. Promote athlete responsibility—allowing athletes to create and deliver a training drill.
5. Provide noncontrolling competence feedback—having constructive feedback that is *solution* focused rather than *problem* focused.
6. Avoid guilt inducing criticisms and controlling statements—providing critiques that focus on the behavior, not the athletes' character.
7. Limit ego involvement—encouraging athletes to improve their own performance, avoiding intrateam rivalries and social comparisons.

Studies exploring the relationship between coach autonomy support and athlete outcomes suggest a positive relationship exists. For example, various studies report that athletes who perceived their coach to be autonomy supportive displayed greater levels of psychological and physical well-being, self-determined motivation, sport persistence and adherence, enjoyment, and positive appraisal for their sports participation.

Coaching research based on SDT has largely focused on autonomy support. However, competence and relatedness support have also received some attention. A coach who supports the athletes' sense of competence by helping his athletes set clear and realistic performance targets, for example, ensures those athletes have the necessary information and experience to develop and progress. Additionally, a coach who displays high levels of involvement with relatedness support ensures the athletes feel a sense of belonging and connection. Research suggests all three types of coach support (autonomy, competence, and relatedness support) create a favorable environment for the satisfaction of athletes' needs.

Unfortunately, the very nature of organized sport can lend itself to the development of a controlling climate that undermines athletes' needs for autonomy, competence, and relatedness. Offering rewards with a condition attached, such as an activity if athletes satisfy the coach's expectations,

is a common practice. Also, athletes' schedules are often planned well in advance, thus creating rigid deadlines for training and performance. Additionally, athletes' choice is often limited, particularly in youth sport, as coaches take ownership for developing strategies that they believe will provide their athletes with a competitive advantage. Hence, it is unsurprising that many coaches tend to favor a controlling style, whereby their athletes feel pressured to think, feel, or behave in a set way. As a result, athletes will often comply with, but may not fully support, requests from the coach, which contributes to the development of controlled motivation (non-self-determined). Coaching behaviors that are viewed as controlling include the following:

- offering rewards such as guaranteed selection to motivate athletes if they put all their effort into training;
- using feedback that pressures athletes to continue with their behavior;
- making demands concerning aspects of an athlete's life not associated with sport participation;
- intimidating athletes by using verbal abuse or threats;
- promoting rivalry among athletes; and
- withholding affection and attention if athletes don't perform to an expected standard.

Unfortunately, little is known about the impact of these controlling coach strategies. A number of studies suggest that athletes who viewed their coach to predominantly display autocratic and controlling behaviors reported less self-determined motivation and greater levels of controlled motivation. Recent research indicates that athletes who view their coach as using controlling strategies report low levels of psychological needs satisfaction. In turn, these low levels of psychological needs satisfaction have been associated with negative athlete outcomes, such as disordered eating behaviors, depression, burnout, and negative affect. These findings suggest that although controlling behaviors may sometimes appear to be adaptive in that they prompt desired behaviors and performance outcomes in the short term, these behaviors may contribute to negative outcomes in the long term.

It is important to note that a coach can display elements of controlling and needs-supportive

behavior. A coach may frequently issue additional physical exercises to the athletes as a discipline strategy (controlling behavior) but may often provide opportunities for player input about other aspects of training (autonomy-supportive behavior). It is also important to note that the lack of needs-supportive behaviors does not automatically necessitate the presence of controlling coach behaviors. The lack of needs support might, for instance, simply be a sign of a more neutral rather than a controlling style. Nonetheless, in terms of the impact of these different coaching styles, needs support versus controlling behavior, the majority of studies indicate that needs-supportive behaviors are associated with self-determined motivation and positive athlete outcomes.

Autonomy-Supportive Approaches

These research findings provoke an important question as to whether it is possible for a sports coach to learn *how* to become more needs-supportive and less controlling. While a needs-supportive coach training program has yet to be experimentally tested, preliminary evidence demonstrates that needs-supportive practices can be employed by coaches working with elite athletes. Clifford Mallet described an autonomy-supportive approach when coaching two Australian relay teams competing at the 2004 Athens Olympic Games. Some of his autonomy supportive strategies included (1) providing choice to the athletes in a number of management and performance areas; (2) providing a rationale for executive decisions; (3) actively seeking suggestions, opinions, and feedback from athletes and their personal coaches; and (4) encouraging athletes to take personal responsibility for their learning. Mallet concluded that by adopting this autonomy supportive approach he promoted their levels of self-determined motivation. This preliminary evidence suggests that needs-supportive coaching is feasible, even in elite sport settings.

Additional evidence supporting the potential efficacy of needs-supportive coach training programs comes from intervention studies designed to improve coaches' interpersonal skills. Although not focused specifically on needs support, these studies provide evidence that coaches can learn to alter their interpersonal behaviors when working with athletes. For example, coaches can learn to (a) provide instruction and encouragement, (b) avoid sarcasm or degrading comments, (c) establish clear

expectations, and (d) avoid nagging or threatening athletes. Promisingly, coaches who participated in these training programs were evaluated more positively by their athletes compared with athletes whose coaches did not take part. Additionally, a variety of positive athlete outcomes were reported. For example, compared with controls, athletes of the trained coaches displayed lower levels of anxiety and drop-out, higher levels of healthy motivation and fun, and a greater liking for their teammates.

Finally, research outside of the sport setting indicates that authority figures like teachers, managers, and health care professionals can successfully alter their interactions to become more needs supportive after attending a training program and this increased support had positive effects on subordinates' motivational outcomes. Coach education research has taken this evidence on board and preliminary guidelines are now available and can help coaches learn how to be needs supportive in their interactions with their athletes.

Conclusion

There is a great deal of evidence that needs-supportive coaching is associated with more positive forms of motivation and better experiences for athletes. Preliminary data suggests that sport coaches can learn to effectively apply needs-supportive coaching principles. Further research is required to

test the effects of coach training programs specifically designed to promote needs support and lessen controlling coaching.

Chris Lonsdale and Edel Langan

See also Coach–Athlete Relations; Coaching Efficacy; Leadership in Sport: Multidimensional Model; Relational Efficacy Beliefs in Coach–Athlete Relations; Self-Determination Theory

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B

BASIC EMOTIONS IN SPORT

Emotion is a central feature of many sporting events. Athletes, as well as supporters, can experience many emotions, including joy, sadness, anger, fear, anxiety, shame or guilt, and pride. Most emotion theorists argue that emotions have the power to motivate and regulate cognitions and behaviors in sport. To understand the antecedents, experience, and consequences of emotions, theorists have attempted to classify emotions into various categories. One central question is whether some emotions should be considered basic. That is, are some emotions fundamental to human experience? To answer this question, it is important to consider the criteria to identify such basic emotions, the functions of basic emotions, and whether basic emotions be modified by social learning.

The discussion about basic emotions is challenging because there is much disagreement among emotion theorists on many critical issues. Some theorists believe basic emotions are linked to evolutionary development (phylogeny), whereas others hold that emotions are social constructions and reflect only social learning through shared experiences across cultures and the past history of the person (ontogeny). These views, of course, influence the criteria used to identify basic emotions. There are others, however, who reject the notion of basic emotions all together. Rather than address all these competing perspectives, this entry will attempt to capture the key features of basic emotions that are common to several prominent theorists, such as

Paul Ekman, Robert Levenson, Robert Plutchik, and Carroll Izard. For alternative views, the reader is directed to publications by Andrew Ortony and Terence Turner, and by Jesse Prinz.

Criteria for Basic Emotions

Basic emotion theorists tend to agree on a number of central criteria for basic emotions: They have been shaped by evolutionary history; they are associated with subcortical brain structures; they are triggered by prototypical evolutionary meaningful stimuli; they have a rapid onset with limited higher order cognitive involvement; they have potentially adaptive functions based on evolutionary needs; and they have specific neural, expressive, and physiological mechanisms. For example, Robert Levenson has argued that for emotions to be considered basic, they must meet three general criteria. First, basic emotions have distinct behavioral, expressive, and physiological responses. Second, basic emotions are hardwired in the brain, although the emotion systems are plastic, such that learning over the lifespan may modify and enhance the emotional response. Third, basic emotions have specific functionality, in that each emotion is triggered by an affect program that rapidly recognizes a survival-critical situation and produces an action to enhance survival. Paul Ekman, a leading figure on basic emotion theory and research, has identified numerous criteria that are generally consistent with Levenson's three general criteria. The criteria for basic emotions according to Ekman include distinctive universal signals, distinctive physiology,

automatic appraisals, distinctive universals in antecedent events, presence in other primates, capable of quick onset, brief duration, unbidden occurrence, distinctive thought memories and images, and distinctive subjective experiences.

Identifying which specific emotions are basic is challenging because not all the criteria are readily testable in humans; nor do theorists agree on all criteria. Thus, it is not surprising the number of emotions and specific emotions identified by various theorists ranges greatly, and theorists may even add or delete emotions from their lists over time. Carroll Izard identified seven basic or primary emotions, for example, whereas Paul Ekman identified six basic emotions in 1972 (but expanded his list to 15 emotions in 1999). Overall, there seem to be five discrete emotions—fear, happiness and joy, sadness, anger, and disgust—that most theorists agree upon and include on their lists of basic emotions. Other commonly identified basic emotions are contempt, surprise, and interest. Ekman makes the argument that each basic emotion term should be viewed as being part of a basic emotion family; thus there are many emotions that could be considered basic in his view. For example, sadness would include terms such as *distress* and *anguish* within its emotion family.

Do Basic Emotions Have Adaptive Motivational Features in Sport?

To determine whether basic emotions have adaptive motivational features in sport, we need to consider if the behavioral and physiological responses associated with each emotion can be regulated by higher order cognitions and whether social learning can modify the antecedent triggers and behavioral responses. Most emotion theorists believe that the emotion process can be modified by social learning and can be regulated by higher order cognitive functioning. Carroll Izard suggests that basic or primary emotions are only present in infants and that most emotional experience in youth and adults is governed by emotional schema involving an interface between emotion and cognitive brain structures. Others such as Ekman believe that basic emotions exist across the lifespan but are modified by the life experiences of the person. He believes that learning can change aspects of the emotion process like antecedent triggers and behavioral expression, and that cognition helps regulate the emotion response. Ekman, however, believes that

such learning would be biologically primed by the basic emotions. Fear, for example, should be triggered by prototypical stimuli that result in rapid automatic appraisal and trigger an evolutionary response—specific facial expression, high physiological arousal, blood shunting away from hand and toward legs, urge to flee. Through experience and observations, athletes can learn that certain situations, such as falling off a gymnastic beam, facing large and powerful opponents, skiing icy steep slopes with surrounding cliffs, or trying to hit a 95 mph fastball are dangerous. These stimuli can become incorporated into the evolutionary affect program as prototypical events for fear, although the way in which this occurs is not clear. Thus, these stimuli can rapidly trigger a fear response. The interface of the basic emotion system with higher level cognitive systems, however, can either enhance or inhibit the emotion response. The fear emotion can activate fear-related memories and learned behaviors, thus increasing the strength of the fear response. Conversely, higher order cognitions can also activate a number of potential coping strategies that help regulate the fear response as well as formulate solutions to manage the situation. It should be noted that the fear response may not be maladaptive, especially if it causes athletes to flee or avoid a situation that exceeds their resources and endangers their lives.

Basic emotions may have constructive and destructive functions in sport. Emotions can influence short-term individual performance, as well as persistence and engagement over time. Basic emotions can be rapidly triggered with minimal awareness, so specific emotional responses might interfere with effective performance through disruptive effects on physiological, motoric, and cognitive functioning. Moreover, the basic emotion response may overwhelm higher order cognitive functioning when a situation is extreme. The inability to override or reduce the responses to specific basic emotions may even cause the athlete to attack opponents, coaches, officials, or even fans. The news is replete with many examples of athletes losing emotional control. On the other hand, basic emotion responses may facilitate performance under some conditions when the behavioral and physiological responses augment the necessary resources needed for success. Motivational research also suggests that participation in sport is facilitated by positive emotions, such as happiness, joy, and enjoyment. Many motivational models used in sport consider

the impact of basic emotions, such as happiness and sadness, along with related emotional states, such as enjoyment and interest.

Basic emotions are also important in social functioning in sport because they can signal significant information to others. Repeated expression of basic emotions, such as happiness, joy, and anger, in specific sport situations, allows others to understand what sport situations are important. Social functioning in sport also depends on the ability to recognize basic emotions in others such as teammates, opponents, coaches, and fans. For example, when an athlete is angry, with corresponding facial and bodily (and maybe voice) expression signals, others are alerted to not only the potential reasons for the anger but also to potential consequences. Others begin to regulate their behavior in response to the anger. Athletes can, however, learn to inhibit signals, as the demonstration of specific emotions may not be socially appropriate or may interfere with sport performance. Athletes can also learn to send false signals. German sport researcher Dieter Hackfort has written about how athletes show, hide, or fake emotions for multiple reasons, such as to influence officials and fans, deceive or irritate opponents, and motivate teammates.

Research, as well as anecdotal evidence in sport, generally supports the basic tenets associated with basic emotions. Happiness and joy are associated with the attainment of important goals and the demonstration of competence. When happy, athletes are more likely to smile and embrace significant others. Loss is more likely to lead to sadness, with tell-tale facial expressions, occasional crying, initial behavioral isolation, and the urge to seek comfort from others. Anger is generated by the frustration of important goals. Expressions include flushed face, furled brow, lips curled back to expose teeth, increased heart rate and blood pressure, and the urge to attack. Fear is triggered by the appraisal of impending danger related to physical or psychological harm. Athletes are more likely to behaviorally freeze and tremble or engage in disengagement (flight) behaviors. Obvious physical reactions also can include increased respiration and perspiration, combined with facial expressions of eyes open, eyebrows raised, and mouth slightly open. In extreme cases of fear, bowel and bladder emptying may occur. For the last of the commonly agreed upon basic emotions (i.e., disgust), there is neither much empirical research nor much anecdotal evidence in sport.

Although basic emotions are likely to be hard-wired, there is no doubt that social learning and culture (and sporting culture) has a large impact on the appraisal and expression of basic emotions. Anger is a classic example. Anger is not always triggered by goal frustration that threatens the self either psychologically or physically but can also involve rapid evaluation of whether the offending other is violating normative rules and has control over the offending behavior. For example, an athlete may feel anger toward an opponent who appears to be cheating and violating the normative rules of the game. Even if the initial neural or physiological components of anger are initiated, the expression can also be modulated by higher cognitive processes. These cognitions can include evaluation of who was to blame (self: internalized anger; other: externalized anger), coping potential (what actions are possible?), and acceptable actions (determined by general and sport culture). The influence of higher order cognitions in the basic emotion process helps to explain individual differences as well as differences across sports. These higher order cognitions explain why an athlete may try to ignore the cheating opponent's behavior, or may instead tell the coach or the referee about the opponent's apparent cheating, resisting the urge to directly attack the opponent.

Conclusion

The study of basic emotions is full of controversy. For those who support the existence of basic emotions, there is a general consensus that they are rooted in evolutionary history, are associated with subcortical brain structures, and have important motivational functions. Basic emotional processes are influenced over the lifespan by social learning, with higher order cognitions influencing emotional response and emotional regulation. Basic emotions can facilitate or hinder performance and social functioning in sport. Athletes can learn to regulate the expression of basic emotions, but such regulation can be overwhelmed in extreme situations. Understanding how social learning is integrated into the brain mechanisms regulating basic emotion has critical implication for emotional regulation interventions.

Peter R. E. Crocker

See also Affect; Choking; Coping; Emotional Schemas; Self-Conscious Emotions; Stress Management

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BIOFEEDBACK, NEUROFEEDBACK

In recent years, there has been an increase in research and practical applications concerning the use of biofeedback training for athletic performance enhancement. It is important to be aware that the terms *biofeedback* and *biofeedback training* are not identical, and therefore need to be defined separately. Biofeedback (BFB) is the output from an electronic device with sensors and electrodes, which allows immediate and objective measurements of personal biological functions. This information is monitored on a computer screen and is transformed into auditory and visual signals, helping in learning to regulate body–mind activity in line with a basic psychophysiological principle. The main idea of this principle is that for every physiological change there is a parallel change in an emotional or mental component of human behavior, and vice versa. Biofeedback training (BFBT) is a process of learning psychophysiological

self-regulation skills using information from a feedback device. Possible models of BFBT are as follows:

- *One-to-one model*: Sport psychologist or consultant and athlete work together throughout the sessions.
- *Group model*: Sport psychologist or consultant and several athletes with BFB equipment for each athlete work in one laboratory.
- *Home practice*: Athlete trains with portable BFB equipment at home, following sport psychologist or consultant's written instructions.

Biofeedback Modalities

Biofeedback modalities refer to the types of physiological indicators used for feedback during BFBT:

1. *Cardiovascular* or *heart rate* (HR) feedback measures heart activity using beats per minute (bpm) by electrocardiography. In addition, practitioners use measurements of interbeat interval (IBI), the time between heart beat and msec-heart rate variability (HRV). HR generally increases with an increase in stress, and vice versa.
2. *Muscle* or surface *electromyographic* feedback measures muscle activity in microvolts (μV). In cases of an increase in stress, electromyography (EMG) usually increases, as does muscle tension, and vice versa.
3. *Temperature* or *thermal* (T) feedback is a measurement of skin temperature as an indication of peripheral blood flow. Temperature feedback is measured in Fahrenheit (F) or Celsius (C) degrees. During an increase in stress, skin temperature decreases, and vice versa.
4. *Blood pressure* (BP) feedback is the measurement of the force exerted by circulating blood on the walls of blood vessels. BP is measured in terms of the systolic pressure over diastolic pressure (mmHg), and it usually increases with increases in stress.
5. *Electrodermal activity* (EDA) feedback is a measurement of changes in the electrical activity of the skin surface. It can be measured by skin resistance (historically known as *galvanic skin response* [GSR], units of kohms, $k\Omega$), by skin conductance (SC, units of microsiemens, μS), and by skin potential (SP, units of millivolts,

mV). SC is the preferred method in psychophysiology research. SC increases with higher arousal, and vice versa.

6. *Respiration* (R) feedback is a measurement of the respiration rate, which is the number of breaths taken within a set amount of time (breaths per minute). Respiration rate can be recorded using a tensometric sensor inserted into a cuff that is fixed around the individual's chest. R increases with increased stress, and vice versa.
7. *Electroencephalograph* (EEG) feedback, more recently known as *neurofeedback* (NFB), is a measurement of brain wave activity. EEG frequency groups are: Delta (1–4 Hz), Theta (4–7 Hz), Alpha (8–12 Hz), Beta (13–36 Hz), and Gamma (36–44 Hz). The response range under stress increases from sleep states (Theta waves) to excitement (Beta waves).

Research and Applications

Biofeedback research and applications can be dated back to the 1960s, when the focus was on the medical and clinical field. The primary modalities used in research in health care and cases of trauma and injury were EMG, EEG, and HR, especially in the laboratory setting.

The main goals of the BFB studies in this period were to demonstrate self-regulation skills and symptom reduction. Moreover, researchers attempted to describe an optimal model of BFBT and its applications in clinical practice. Later, in 1975, Leonard Zaichkowsky, the pioneer of biofeedback application in sport, described the positive effect biofeedback had on athletic performance. Research on BFB (usually EMG) efficiency in sport focused on reducing anxiety, increasing muscle strength, and decreasing muscle fatigue and pain in different kinds of sports. The EEG BFB (neurofeedback) studies focused on psychomotor efficiency and skilled motor performance; for example, enhanced performance in golf, shooting, and archery was shown to be associated with an increase in alpha (decreased cortical activation) activity in the left hemisphere. An additional trend in research and application of BFBT in sport was to include BFBT as part of a larger package of psychological interventions. A major limitation of this period was the inability to transfer the results from sterile laboratory conditions to a field setting and integrating BFBT into the training process.

The Wingate Five-Step Approach (W5SA) was developed to overcome this limitation.

The Wingate Five-Step Approach (W5SA)

The W5SA consists of five steps and a self-regulation test (SRT). Three of the steps are provided in a laboratory setting and the last two under field and training conditions. The steps are as follows:

1. *Introduction:* The main goal is to teach the athlete basic self-regulation techniques, such as relaxation, imagery, self-talk, and concentration. At the end of this step, EMG, HR, and GSR BFB are used as part of the athlete's mental training, especially the use of auditory or visual signals. The duration of this step can last up to 10 to 15 sessions, while each session can last approximately 45 to 50 minutes.

2. *Identification:* In this step, most of the work that is done focuses on identifying and strengthening the most suitable biofeedback modality. Different sports require different biofeedback modalities. For example, in combat sports, EMG and GSR BFB are the most efficient modalities for measuring and practice. In contrast, EEG and HR are more suitable for sports such as golf, archery, and shooting. Additionally in this step, personal characteristics and the specific demands of the sport discipline should be taken into account. The duration of this step is approximately 15 sessions, each lasting 45 to 50 minutes.

3. *Simulation:* The main goal of this step is to practice self-regulation and concentration skills in the laboratory setting using BFB in the natural environment, such as watching video scenes under competitive situations. In addition, the basic self-regulation techniques are modified according to the sport's demands. In this step relaxation is shorter, imagery is set according to competition length, and self-talk is specific to the competitive situation. This step lasts about 15 sessions, each of them approximately 50 to 60 minutes.

4. *Transformation:* The athlete mentally prepares for the upcoming competition. All the skills that the athlete learned and rehearsed in the laboratory are transferred and integrated into the actual training setting by using a portable BFB device. This process allows the athlete to be prepared for future competitions by acquiring self-regulation abilities and mental readiness. This step lasts about

15 sessions and is provided during different parts of the training.

5. *Realization*: In this step, BFB is used in different settings, such as the field, pool, gym, bus, and boat at sea. In addition, the BFB is accompanied with relaxation, concentration, and imagery, which are used in different parts of the competition, such as before the start, between combats, after warm-up, and at the end of competition. The process of this step begins by applying the skills during less significant competitions, and gradually requesting the athlete to apply the skills in more significant events.

Self-regulation test (SRT) is given to the athlete before the beginning of the program and before each step. The goal is to examine the self-regulation level of the athlete. Recently, practitioners have been integrating the W5SA with the athlete's physical training, incorporating it within the periodization concept.

Neurofeedback

Neurofeedback (NFB) or EEG BFB relates to changes in electrocortical activity of the brain as the physiological measure of emotional and cognitive processes. The historical development of NFB began in clinical settings using alpha-type training. Later, research focusing on performance enhancement using neurofeedback of alpha activity in the left hemisphere, which is usually associated with expertise. Moreover, some research suggested that alpha range can be used as an indicator of attention processing. In sport, NFB is applied for the purpose of optimizing athletic performance by the athlete being able to master self-control and self-regulation strategies. For example, the first NFB study in sport examined the left–right hemispheric activity in rifle marksmanship. Neurofeedback training in sport is used in models for decreasing arousal or increased relaxation skills (alpha–theta type training) and for increasing attention–performance relationships (theta–beta type training).

BFB in Exercise Psychology

The major goal of using BFB in exercise psychology is to understand exercise behavior and the mechanisms that have an effect on participation rate and exercise adherence. In line with this goal, research has focused on the effect of self-monitoring and self-regulation of exercise intensity

for fitness enjoyment, as well as adherence rate. Results have indicated that the positive effects of BFBT on achieving optimal exercise intensity for health and well-being can be accomplished during a relatively short time period.

Professional BFB Organizations, Journals, and Equipment

Professional Organizations

There are four major international BFB organizations that provide annual meetings, workshops, and publications: (1) The Association for Applied Psychophysiology and Biofeedback (AAPB) in North America (www.aapb.org); (2) The Biofeedback Foundation of Europe (BFE; www.bfe.org); (3) The International Society for Neurofeedback and Research (ISNR; www.isnr.org); and (4) The Biofeedback Certification International Alliance (BCIA; www.bcia.org), which provides biofeedback certification according to the Professional Standards and Ethical Principles.

Professional Journals

There are two major scholarly journals: (1) *Applied Psychophysiology and Biofeedback* (APB), formerly known as *Biofeedback and Self-Regulation*, and (2) *Biofeedback*. Both are supported by the AAPB.

BFB Equipment

There is a variety of biofeedback equipment that can be used in laboratory and field settings, depending on the intervention goal. Advanced technology and awareness of the possibilities of BFBT have led to the innovative and improvement of BFB electrodes, noncontact monitor tools, telemetric systems, and computer programs. For more detail, see the chapter “Psychophysiology: Equipment in Research and Practice,” by Derek T. Y. Mann and Chris M. Janelle in A. Edmonds and G. Tenenbaum (Eds.), *Case Studies in Applied Psychophysiology: Neurofeedback and Biofeedback Treatments for Advances in Human Performance* (2012).

BFBT has many potential applications, such as in education, music, and the military.

Boris Blumenstein and Iris Orbach

See also Concentration Skills; Electroencephalograph (EEG); Electromyography (EMG); Imagery;

Psychological Skills Training; Psychophysiology;
Relaxation; Self-Regulation; Stress Management

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BIOPSYCHOSOCIAL MODEL OF INJURY

There are several prevailing models that connect psychological factors and sport injury, each with slightly different perspective on relevant biological, psychological, or social factors. Moreover, prevailing models have typically focused either on incorporating psychosocial factors in predicting and preventing sport injury incidence or upon identifying psychosocial factors associated with injury recovery and rehabilitation outcomes. Extensive reviews of these models have been written by Jean Williams and Mark Andersen, Britton Brewer, and Diane Wiese-Bjornstal. This entry provides an overview of the mediating biological pathways and physiological mechanisms linking psychosocial stress to athletic injury and, where appropriate, other adverse health consequences.

Both historical and recent conceptualizations of psychosocial stress posit wide-ranging biological effects on health. In this regard, Frank Perna and colleagues have argued that emotional, behavioral, and physiological aspects of stress response must be considered together with attention disruption to fully capture potential pathways mediating the relationship between psychosocial stress and adverse health including athletic injury. Additionally, since the physical demands of training volume and intense exercise required in competitive athletics are known to have adverse temporal effects on immune, neuroendocrine, and skeletal muscle repair response, a primary tenet of a biopsychosocial model of injury is that psychosocial distress may act synergistically with high-intensity, high-volume sports training to widen a window of susceptibility to illness or injury.

The biopsychosocial model of stress athletic injury and health (BMSAIH) is offered below to illustrate pathways between stress demands and athlete health (see Figure 1). The BMSAIH expands the Andersen and Williams model of psychosocial stress and athletic injury in three essential ways: (1) It clarifies mediating physiological pathways between athletes' stress response and adverse health outcomes (e.g., sport injury); (2) it considers other health outcomes and behavioral factors that impact sport participation as well as injury; and (3) it integrates the impact of exercise training upon athletes' health.

The BMSAIH should be considered an independent extension of Andersen and Williams's classic model of stress and athletic injury, which is reviewed elsewhere (see also the entry "Injury, Psychological Susceptibility to"). Similar to other generic models of stress and adverse health consequences, the Andersen and Williams model posited that a stress response mediated the effect of stressor(s) on the health outcome, athletic injury, with the stress response being composed of physiological and cognitive features (attentional perturbations such as peripheral narrowing) thought to predispose an athlete to injury. While the stress response was conceptualized as being both physiological and cognitive in nature, the original model and the preponderance of studies have principally only researched cognitive features (disturbances in attention and recognition of sport-related cues) thought to predispose an athlete to injury. Yet, the relationship between psychosocial stress and athletic injury appears stronger for overuse injuries

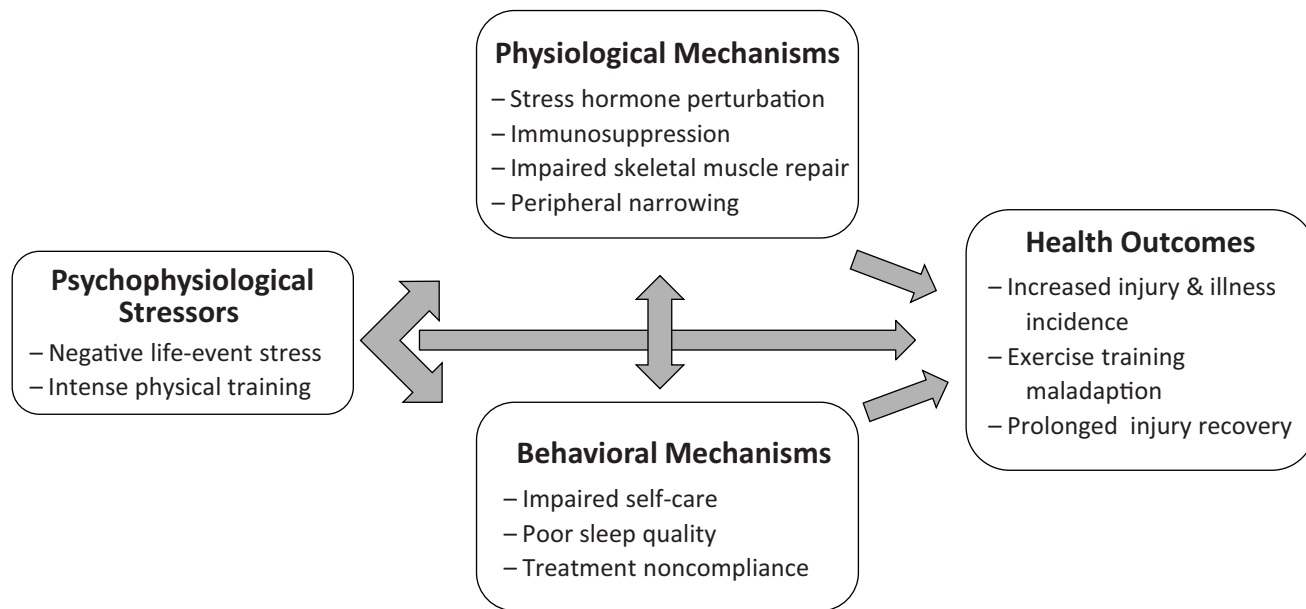


Figure 1 A Biopsychosocial Model of Stress and Athletic Injury and Health (BMSAIH)

Source: R. N. Appaneal & F. M. Perna.

that are likely less dependent on cognitive processing and more likely related to physiological processes affecting exercise training adaptation and recovery. The stress-injury model also proposed a number of factors, such as personality, history of stress, and coping resources, that may moderate a stress response, but there was less clarity and exposition of mediating pathways, particularly those related to stress physiology, by which stress response may increase risk of injury.

Stress Response Mechanisms

Negative life stress and attendant emotional distress, through autonomic nervous system (ANS) response pathways, is thought to exacerbate the adverse effects of prolonged high-intensity and high-volume exercise. Specifically, psychological stress causes activation of the ANS yielding the release of catecholamines like epinephrine, nor-epinephrine, neuropeptides, and glucocorticoids (cortisol). Because target organs, such as heart, vasculature, muscle, and immune tissue, contain efferent nerves and have receptors for catecholamine and glucocorticoids, ANS activation affects target organs either by direct innervation of the parasympathetic and sympathetic nervous systems (PNS and SNS, respectively) or by hormonal action via activation of the hypothalamic-pituitary-adrenal-cortex

(HPAC) and cortisol release. Cognitive features related to a person's appraisal of the severity of a stressor and the capacity to cope with a stressor is known to exacerbate or prolong emotional reactivity and concomitant physiological response. Direct innervation, particularly by the SNS and catecholamine release, and hormonal action (e.g., cortisol release) thus provide the mechanistic links used to explain how the brain and associated cognitive-affective processes attendant to psychological stress may influence stress physiology and other physical systems, particularly those of relevance to athlete health like immune and skeletal muscle.

Because the ANS may operate either through direct innervation or hormonal action, there are a variety of possible neuroendocrine and hormonal patterns (differential responses to stress) that have evolved to characterize the stress response since Hans Selye first proposed the general adaptation syndrome, an undifferentiated response to stress. Moreover, specific ANS patterns are known to be influenced by features of cognitive appraisal particularly being dependent if an individual views a life event as a negative stressor or as a challenge. Thus, it is recognized that ANS activation in response to stress is not always deleterious, and indeed, is necessary. For example, it has long been known that ANS activation, particularly of the SNS, is associated with performance of physical

and cognitive tasks. Similarly, HPAC activity and cortisol release are known to potentiate the actions of epinephrine in the completion of physical tasks and, when not prolonged, cortisol initiates a cascade leading to anabolic activity necessary for muscle repair following intense exercise.

A thorough discussion of possible hormonal response patterns in response to stress is beyond the scope of this entry but can be found in work by Trent Petrie and Frank Perna. In short, three hormonal patterns are thought to result in adverse health effects. These are thought to occur when individuals (a) experience frequent negative life stress and concomitant frequent activation of the SNS and HPAC pathways involving principally the release of epinephrine and cortisol to a lesser extent, (b) are hypersensitive to stressors and have an atypically high SNS and epinephrine response out of proportion to the magnitude of a stressor, and (c) experience chronic stressors and emotional distress with a prolonged activation of the HPAC and heightened cortisol release. The latter scenario involving measurement of enduring effects of negative life events, either in isolation or with their association with cortisol response, has received the most attention.

In addition to stress mechanisms described above, HPAC activation and possible concomitant behavioral disruptions (e.g., sleep disturbance) resulting from heightened negative affect may act in synergy with the demands of heavy exercise to increase risk of illness and injury. For example, poor sleep has been associated with prolonged elevation in evening cortisol, immune decrements, and lessened growth hormone release, all of which may inhibit muscle repair following acute exercise. Negative emotion-linked increases in stress hormones (cortisol) and behavioral perturbation may thus widen or prolong a window of susceptibility for illness and injury that is created by high-intensity and high-volume training. That is, psychological distress may impair muscle growth and repair processes by prolonging the presence of post-exercise catabolic hormones like cortisol that also impair immunity and inhibit the secretion and action of anabolic factors, such as growth hormone and insulin-like growth factors. Due to cortisol's immunosuppressive and muscle catabolic effects, prolonged training-induced cortisol elevation may create favorable conditions for viral infection, athletic injury, and exercise training maladaptation.

Similarly, elevated negative mood state, particularly fatigue and depression, has been used to identify overtrained athletes. Depressed mood has also been related to impaired immune function, and this effect may be mediated by cortisol elevation. Although the physiological strain of exercise training is largely responsible for cortisol and mood fluctuation, studies suggest that psychological life-event stress (LES) may also modulate cortisol and health parameters in athletes. For example, elite athletes with high LES, in comparison to low LES athletes, have been reported to experience prolonged post-exercise cortisol elevation, which was prospectively correlated with a greater frequency of physical symptom, such as muscle pain, back tightness, or flu-like symptoms. Elevated LES has also been prospectively related to increased risk of athletic injury and viral infection (e.g., common cold). In essence, the BMSAIH suggests that health effects of psychosocial stress would be most pronounced when an athlete was either in or just removed from a high-volume or high-intensity period of training, and that sport training cycle should be considered to optimally measure the association between psychosocial stress and athletic injury and adverse health.

Taken together, the literature indicates that independent from intensity and volume of sports training, psychological stress likely contributes to athletes' neuroendocrine and immune activity, which may be a mediating pathway linking stress to increased risk for adverse health effects including athletic injury. However, the effects of psychosocial stress on health may be more pronounced during periods of high-volume and intense exercise training. Because athletes must train at high volumes and intensities to make the physiological adaptations necessary for competitive success, commercially available exercise recovery monitoring systems have appeared on the competitive sports milieu and have been extensively reviewed by Michael Kellman. These systems include the assessment of physical, emotional, and social-contextual stress responses, as athletes must maintain a delicate balance between demands and recovery to stay healthy, avoid injury, and ultimately perform optimally. However, athletes' efforts to balance training and recovery occur within a sport culture that often undervalues the importance of psychosocial factors, and perhaps also recovery. As a result, the culture of

competitive sport reinforces an imbalance between psychosocial and sport training factors, which ultimately may affect training adaptation, injury, and overall health. Sport psychology personnel trained to assist athletes with life-event, precompetitive, and post-injury stress are also widely available. However, the efficacy of psychosocial intervention to reduce injury risk, facilitate exercise-training adaptation, and enhance athlete health has been less extensively investigated.

Psychological Interventions

In the athletic domain, cognitive-behavioral stress management (CBSM) intervention in the form of psychological skills training is well known to reduce competitive anxiety and enhance athletic performance. CBSM has also been shown to reduce pain and speed recovery and return to play following arthroscopic surgery among injured recreational athletes. In a handful of randomized controlled trials, CBSM improved exercise training adaptation in the form of lessened fatigue, depressed mood, and cortisol curve during a high-volume training period among competitive rowers. Further, CBSM intervention has also been shown to reduce actual incidence of athletic injury and illness. One of these trials conducted by Perna and colleagues specifically tested if intervention effects on cortisol and affect mediated health outcomes. Findings demonstrated that rowers in a CBSM condition had significantly fewer days injured or ill and half the number of health and training center visits compared with controls. Mediation analyses indicated that modulation of affect and cortisol accounted for approximately one half and one fifth, respectively, of the intervention effect on days injured or ill. Though limited, the extant literature provides compelling support for the potential value of intervention to lessen risk of athletic injury and illness and to facilitate sports-training adaptation. Future research should seek to further elucidate the physiological pathways underpinning the psychosocial stress and health relationship, and explore biopsychosocial mechanisms that may be responsible for intervention effects on athlete health.

Renee N. Appaneal and Frank M. Perna

See also Autonomic Nervous System; Injury, Psychological Susceptibility to; Stress Management; Stress Reactivity

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BODY AWARENESS

Body awareness is described as awareness of, and attentiveness to, one's internal bodily processes and sensations. It is a sensitivity to normal bodily states that is separate from emotion yet originates

from sensory proprioception and introspection and entails one's focus of attention toward the self.

Arguably, the most common perspectives used to understand body awareness are self-objectification, self-consciousness, and arousal. There is debate whether body awareness involves a somatic component of arousal or is distinct from somatic complaints. Some researchers have defined body awareness as separate from both emotion and somatic symptoms, whereas many sport and exercise researchers tend to define body awareness as somatic arousal.

From the perspective of self-objectification, it is argued that individuals, and in particular women, adopt an observer's perspective toward their bodies, and this objectification leads to an insensitivity to internal body cues. Also, individuals may be so vigilantly aware of the social cues and their outer body appearance that they deplete perceptual resources necessarily to attend to internal body sensations.

Similarly, researchers suggest that individuals who experience exaggerated self-consciousness or preoccupations with self will also deplete resources to attend to internal cues. *Private self-consciousness*, which is defined as the ability to introspect and pay attention to one's inner thoughts and feelings, has been used as a measure of body awareness in sport and exercise psychology research given the limited measurement tools available.

There is consistent evidence that women are less likely than men to attend to bodily cues and internal physiological cues, such as heartbeat, stomach contractions, and blood-glucose levels. Women are less likely to use these cues in determining how they feel, and these cues are less likely to be determinants of their subjective experiences compared to men.

Many mind-to-body approaches are used to help enhance body awareness, including yoga and tai chi, mindfulness-based meditation, and mental training for sport. In yoga, the nonjudgmental awareness of the body helps to emphasize responsiveness to body sensations, while also fostering physical challenge. Resistance-training exercise has also improved body awareness when examined in a pre- and post-study of college students. The proprioceptive and interoceptive training within sport psychology mental training programs are also important for enhanced body awareness. Breathing and progressive relaxation exercises within multimodal competitive stress and anxiety

mental training programs are also used to enhance body awareness.

From a theoretical perspective, reducing body shame, anxiety, and self-objectification through intervention strategies, such as cognitive behavioral therapy and cognitive dissonance, may also help increase body awareness.

There is some debate on the adaptive or maladaptive features of body awareness. To some researchers, body awareness leads to maladaptive cognitions, such as somatosensory amplification, distress and anxiety, and somatization. These maladaptive perspectives suggest that heightened body awareness can be an impetus to eating disorders and maladaptive dieting or exercise behaviors. Alternatively, other researchers and practitioners argue that an ability to recognize subtle bodily cues leads to adaptive behavioral strategies to manage such body cues. Sport psychology researchers and practitioners often fall into this latter frame of thought and report the benefits of body awareness for competition and training.

Specifically, researchers have found that body awareness increases prior to sport competition and that this response is adaptive to successful performance. Strong associations between body awareness and state anxiety have been reported among athletes. Among individuals practicing yoga, body awareness was associated with lower perceptions of self-objectification and higher body satisfaction. High body awareness has also been consistently linked with lower incidence of disordered eating attitudes and behaviors in athletes.

Drawing from the injury and pain literature, it is also possible that increases in body awareness can help in the management of pain and facilitate sport injury rehabilitation. The most plausible argument explaining the mechanism stems from a distraction or attentional redistribution hypothesis such that focusing on body sensations and cues will distract from exercise-induced symptoms and pain.

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See also Body Image; Self-Conscious Emotions; Self-Objectification

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BODY DISSATISFACTION

Body dissatisfaction is the negative subjective evaluation of one's body as it relates to body size, shape, muscularity or muscle tone, weight, and fitness. Body dissatisfaction is considered to be an important negative affective factor related to body image. Typically, dissatisfaction involves a perceived discrepancy between one's current body and one's ideal body that fosters negative emotions and discontent. Body dissatisfaction has been viewed as normative and has received growing research attention during recent decades. The surge in popularity is due in part to the increasing prevalence worldwide, as well as implications for the development of a range of maladaptive behaviors and emotions, such as decreases in self-esteem, self-regulations, physical activity, happiness, optimism, pride, and increases in disordered eating, depressive symptoms, and body-related shame and guilt.

Sociocultural Effects

Pressures for women to be thin and fit, and for men to be lean and muscular, can originate from numerous sources, including the media, parents, siblings, partners, and peers. These sources may provide direct or indirect pressures to attain the desirable physique.

Sociocultural pressures to attain a socially desirable physique are considered important risk factors for body dissatisfaction. In particular, media awareness, knowledge of ideals as presented by the media, or actions taken or comments made by family, partners, and friends to encourage a socially desirable physique are important facets of the sociocultural pressure to conform to the ideal

body. Researchers have concluded that even short-term exposure to idealized media images of men and women's bodies can lead to increased body dissatisfaction in both sexes. Theory and research also support modeling and negative communication as vehicles through which family and peers may influence body dissatisfaction. Nonetheless, the relationship between mass media and body dissatisfaction is complex, multiply determined, and bidirectional. It is important to recognize that a number of individual differences (internalization of the ideal physique, social comparison tendency, identification with models, appearance information, and critical body image processing) may moderate sociocultural pressures.

Gender Differences

One of the most consistent findings in the literature is that women are significantly more dissatisfied with their bodies than men. Higher prevalence rates are reported for women compared to men across the lifespan and geographic region. Upward of 90% of girls and women are dissatisfied with at least one aspect of their physiques, with elevated body weight and size typically ranking the highest. Nonetheless, body dissatisfaction among men is on the rise. Some researchers have reported that over 90% of men also experience some degree of body dissatisfaction. Contrary to the findings for women, men's body dissatisfaction focuses on muscularity and involves both ends of the weight continuum. That is, some men want to lose weight, while others want to gain weight.

Despite apparent gender differences, it is important to note that the majority of body dissatisfaction measures focus on evaluations of weight and shape at the expense of muscularity. Therefore, most measures of body dissatisfaction do not adequately assess typical concerns of men. It is difficult to draw concrete conclusions regarding gender differences in body dissatisfaction until the full spectrum of appearance and fitness evaluations represented by both men and women is considered.

Age

Body dissatisfaction can manifest at a very early age. Survey findings suggest a significant proportion of young children express dissatisfaction with their bodies, but levels in childhood are relatively low compared with adolescence and adulthood. Generally, girls experience heightened body

dissatisfaction at puberty, which intensifies during adolescence. The normal physical changes of increased weight and body fat push girls further away from the cultural ideal of a thin and fit physique. Typically, dissatisfaction in women remains relatively stable throughout adulthood. In older adulthood, some women report heightened dissatisfaction with physical functioning aspects of their bodies in addition to age-related deficits in appearance. The impact of aging on men is less consistent. Boys go through a short phase of relative dissatisfaction with appearance in early adolescence, but the physical changes associated with puberty shortly bring them closer to the masculine ideal (increased height and muscularity, broader shoulders). Similar to women, dissatisfaction plateaus during adulthood; however, some men may experience a period of discontent around middle age (“male menopause”). Similar to their female counterparts, males place greater investments in fitness and health as they age.

One of the issues in comparing different age groups with each other using a cross-sectional design is that historical or cultural ideals and experiences vary across time. Research in this area needs to use longitudinal studies that track cohorts over time as they age.

Sexual Orientation

Given the emphasis on appearance within the gay subculture, considerable research has shown that gay boys and men constitute a group particularly vulnerable to body dissatisfaction. Homosexual men tend to report higher levels of body dissatisfaction compared with heterosexual men, whereas homosexual women have reported less body dissatisfaction than heterosexual women.

Ethnicity

There are reported ethnic differences in body dissatisfaction among individuals from Western countries. This paradigm of ethnic differences suggests one’s meaning of the body is based on cultural and social group contexts. Most studies investigating ethnic differences concentrate on women. Body dissatisfaction is most frequent in Caucasian women and less frequent in Black women; however, this difference is small. Although research published in the 1990s reported that Hispanic and Asian women are typically more satisfied with their bodies than Caucasian women, new evidence

suggests that there are minimal or no differences between Caucasian women and women of other ethnic origins. Researchers targeting variations in body dissatisfaction across ethnic groups indicate that Hispanic women are slightly more dissatisfied with their bodies compared to Black women. There is less work on ethnic differences in men’s dissatisfaction, although there is general agreement that Black men report higher levels of satisfaction than Caucasian men and have heavier body ideals for women and men than do Caucasian men. Overall, there may be less ethnic difference in dissatisfaction than was once thought.

Relationship Status

Generally, people in stable, long-term relationships are more satisfied with their bodies than those who are single. This applies to all ages. With regard to less formal relationships, body dissatisfaction is more prevalent among adolescents who report lower quality of friendship and perceive less social support and less acceptance by peers. Also, women across the lifespan have reported quality peer relationships and social support protect them from body dissatisfaction.

Pregnancy

Several studies have indicated that pregnant women have more positive views about their body than nonpregnant women. Even though pregnant women may still value the thin and fit cultural body ideal, their concerns about failing to match this ideal are typically reduced during this life event. Nonetheless, postpregnancy introduces many body image concerns focused on excess weight and lack of muscle tone. The prevailing emphasis on achieving prepregnancy weight and body shape may exacerbate body dissatisfaction.

Body Dissatisfaction, Sport, and Physical Activity

Generally, body dissatisfaction can act as both a motivator and deterrent for sport and physical activity participation. Several researchers have noted that participating in sport and exercise may serve as a protective function against feelings of body dissatisfaction. Physical activity can also improve older adults’ perceptions of the body by increasing perceptions of mastery of the body and refocusing attention onto health, fitness, and

body function and away from concerns about physical appearance. Nonetheless, the relationship between body dissatisfaction and physical activity varies with the sport and cultural context. Some aesthetic sports, such as gymnastics, cheerleading, and ballet, place high importance on the culturally derived ideal body. In these cases, it is not uncommon for individuals to experience higher levels of discontent related to their body. Furthermore, some athletes report lower body dissatisfaction in the context of their sport, whereby their physiques have functional sport-specific value but greater body dissatisfaction outside of sport, where their athletic physiques are inconsistent with ideal societal standards of appearance and body shape. Less is known about the dose-response relationship between physical activity and body dissatisfaction.

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See also Body Image; Self-Discrepancy; Social Comparison

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BODY DYSMORPHIC DISORDER, MUSCLE DYSMORPHIA

Most people would like to change something about their physical appearance, and this *normative discontent* is not usually indicative of a serious body image issue. However, some individuals may feel extreme preoccupation with an aspect of their appearance: they perceive to be flawed. Typically,

this perception is inaccurate or exaggerated and indicative of body dysmorphia.

Body Dysmorphic Disorder

Characterized as a somatoform disorder in the *Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision (DSM-IV-TR)*, *body dysmorphic disorder* is described as a preoccupation with an imagined defect in appearance, which causes severe distress and impairment in daily functioning. Body dysmorphic disorder tends to co-occur with other psychiatric conditions, such as obsessive-compulsive disorder, depression, substance abuse, and eating disorders. The disorder is prevalent in settings where a high importance is placed on physical appearance, such as sport and exercise contexts and in particular aesthetic sports.

Individuals with this disorder are overcome with constant preoccupations that aspects of their appearance are deformed, when in reality, the perceived flaw is minimal or non-existent. Individuals tend to focus on a few body areas and spend much of the day thinking about the perceived flaws. These individuals typically have low self-esteem and are prone to rejection, low self-worth, and shame. Individuals tend to exhibit delusions of reference, which involves thinking that other people focus on and mock one's perceived flaws and defects. These individuals are highly motivated to examine, improve, seek assurance, and hide the perceived flaw and respond by engaging in obsessive-compulsive behaviors. In competitive sport settings, symptoms may manifest as withdrawal from teammates and constant need for reassurance from teammates and coaches. These coping behaviors may extend to excessive dieting, compulsive exercising, and seeking plastic surgery.

The etiology of body dysmorphic disorder is complex and multifactorial and includes genetic, neurobiological, sociocultural, and psychological influences. Particularly in competitive sport and exercise settings, sociocultural influences play a large role, including strong pressures from coaches, trainers, parents, and even media influences. For example, a genetically predisposed adolescent elite gymnast who presents with high tendencies for perfectionism may be heavily influenced by social pressures, and be at high risk for developing body dysmorphic disorder. Despite the probable influence of social and cultural factors,

clinical features of body dysmorphic disorder are similar across different cultures, even though typically body image concerns are more prevalent in Western societies.

Symptoms of body dysmorphic disorder initially present themselves during adolescence; however, most individuals are not diagnosed for an extended period of time after initial onset because of shame and embarrassment associated with discussing the preoccupations. Aside from difficulties in diagnosis, treatment for body dysmorphic disorder is also challenging. Treatment options include pharmacotherapy, particularly the use of serotonin reuptake inhibitors, and cognitive behavioral therapy, focusing on exposure and systematic desensitization.

Muscle Dysmorphia

Body dysmorphic disorder is equally prevalent in males and females; however, a subset of the disorder, *muscle dysmorphia*, is reported more frequently among males. Muscle dysmorphia is a chronic preoccupation with insufficient muscularity and inadequate muscle mass. Individuals presenting with muscle dysmorphia perceive themselves as much thinner than they actually are, and experience pressure to increase muscle mass and strength, despite possessing a much higher muscle mass than the average male. This condition involves excessive attention to muscularity, distress over presenting the body to others, extreme weight training, and focus on diet. Impaired function in daily life is also an outcome of these compulsive behaviors, along with a high risk of abusing physique-enhancing supplements and drugs, particularly anabolic steroids.

Individuals with muscle dysmorphia experience heightened shame with their preoccupations and engage in physique protection by hiding perceived defects and avoiding situations of physique exposure. For example, individuals may avoid busy times of training at the fitness center to avoid being seen by muscular weight trainers or wear loose clothing to hide the shape and size of their physiques. Researchers have indicated that athletes who are body builders and weight lifters are particularly susceptible to muscle dysmorphia and are at significant risk of anabolic steroid abuse. In competitions where physique-altering drugs are prohibited, individuals are at an increased risk for developing eating disorders and manipulating

resistance training programs to achieve higher muscle mass while maintaining leanness.

Various theoretical frameworks have been employed to understand the complexity of muscle dysmorphia. Psychological theories posit that individuals strive for high muscularity to compensate for feelings of inadequacy, low self-esteem, and issues with masculinity identity. Sociocultural theories suggest that individuals with muscle dysmorphia strive for muscular physiques to attain societal and media-driven ideals that equate masculinity with muscularity. Sociocultural theories may be useful to explain muscle dysmorphia in elite athletes and the prevalence of similar body-related disorders in sport culture. Athletes are more susceptible to muscle dysmorphia if they are involved in sports that predominantly require strength and power, such as weight lifting, or aesthetics involving muscularity (e.g., body building).

Significant stigma surrounds psychiatric disorders like body and muscle dysmorphia, especially among athletes. In sport and exercise settings especially, psychoeducation is important to increase awareness and diminish shame surrounding having these disorders. Informed coaches and trainers can play an important role in preventing, identifying, and aiding in treatment of body and muscle dysmorphia. Treatment options in sport settings are best dealt with using a biopsychosocial model, which uses pharmacological and psychological treatment, while respecting the importance of the social and cultural sport environment in which these disorders thrive.

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See also Self-Discrepancy; Social Physique Anxiety

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BODY IMAGE

In modern Western society, there is increasing pressure for females to be thin, fit, and well-toned, and for males to be muscular and lean, regardless of age. These ideal shapes tend to be difficult to achieve for most people because they have become progressively unrealistic, and with the ever-increasing obesity rates, predominantly unattainable. Consequently, many individuals have become aware that their bodies fall short of the idealized physique and experience what researchers have coined as *normative discontent*. Researchers have shown that this normative discontent is associated with impaired psychological and physical well-being, and that it can be a precursor for maladaptive behavioral patterns, such as disordered eating and dieting; self-induced vomiting; use of diet pills, laxatives, and diuretics; physical inactivity or compulsive exercising; and smoking.

Body Image Defined and Measured

Although there is no universally accepted definition of body image, it is generally explored as one's feelings and views about one's physical features and attractiveness. Expanding on this definition, current conceptualizations of the multidimensional nature of body image suggest there are perceptual, cognitive, affective, and behavioral dimensions.

The perceptual dimension reflects the mental representation or reflections that an individual has of one's body appearance and functions. It relates to the level of (in)accuracy between a person's perceived characteristics and the actual characteristics, either in relation to specific body parts or to the body as a whole. Perceptual body image is measured using body image accuracy techniques, which can include having a person manipulate the size of a body on a computer screen to fit one's body shape and size, using a compass-like device to indicate how wide the individual perceives the body to be, or illustrating the perceived body shape and size on a figure drawing. Using these techniques, it is clear that one's perception of oneself can be quite different from one's actual self. The cognitive dimension reflects an individual's thoughts, beliefs, and evaluations of personal body appearance and function. This dimension has often been studied using questionnaires or interviews targeting the assessment of an individual's

description and satisfaction or dissatisfaction with one's body shape, size, and function. The affective dimension reflects the body-related feelings and emotions that an individual experiences. Commonly assessed emotions that stem from one's body size, shape, and function include anxiety (e.g., social physique anxiety), shame, guilt, pride, embarrassment, and worry. These emotions are measured using self-report questionnaires and in individual interviews. Last, the behavioral dimension relates to the behavioral manifestations that occur as a result of one's positive or negative perceptions, affect, and cognitions focused on one's body size, shape, and function. Specific behaviors may include avoidance; actions used to divert attention away from the body; lifestyle actions, such as dieting and physical activity; steroid use; cosmetic surgery; and body checking strategies, such as weight monitoring, measuring areas of one's body (e.g., dimension of wrist, ankle, thigh, waist), or pinching oneself as a measure of fatness.

In general, each of these body image dimensions can have positive or negative connotations. Positively valenced body image in any dimension may infer accurate perceptions, positive or satisfied thoughts and feelings about the body, and predominantly adaptive actions or behaviors. Negative body image refers to distorted perceptions, predominantly negative thoughts and feelings, and maladaptive actions or behaviors that are driven by physical self-evaluations. Furthermore, when people have distorted, negative views of themselves, negative thoughts, beliefs, evaluations, and emotions may arise, which in turn can lead to what are considered unhealthy behaviors (e.g., crash dieting, excessive exercising). Thus, each body image dimension is related and important to consider in research and practice.

The Development of Body Image

Body image begins as early as preschool through socialization processes; awareness of attractiveness; social comparison; and exposure to body-related prompts, such as dolls and figurines, images, and television advertising and programming. Drawing from sociocultural perspectives, the most common determinants of body image include individual characteristics like sex, age, and weight status; interpersonal experiences with peer groups, social environments, interactions, and social comparisons; social norms and pressures,

such as societal beliefs, body-related stereotypes, media messages, and the fashion industry; and life events like puberty, early or late maturation, or illness. Some of these factors can significantly increase the risk of negative body image, while others can protect against negative body image. For example, younger people and obese people are especially likely to report negative body image. Going through puberty can influence body image because of the physical changes that occur. Based on the tripartite model of body image development, parents, siblings, peers, and other persons in one's support network can directly or indirectly play a key role in the development of negative body image through social disapproval or rejection, unsupportive behaviors, and negative social interactions. Receiving criticisms from peers, for example, can trigger negative body image. Conversely, the degree to which people sense that significant others support them can promote positive body image. Drawing on Leon Festinger's social comparison theory, comparisons can have a positive or negative effect on body image such that individuals who make upward body-related comparisons to others (the bodies they focus on are closer to their idealized bodies than their own or considered better in some way) are more likely to have negative self-evaluations, and in turn poorer body image. In contrast, individuals who make downward comparisons tend to feel better about the appearance and shape of their own bodies.

Body Image and Exercise

Two general approaches have been taken to study the relationship between body image and exercise. In the first approach, researchers have been concerned with determining if body image leads to engagement or avoidance of exercise. In the second approach, researchers have been concerned with determining whether exercise has a beneficial effect on body image. Researchers have shown support for both these approaches.

Body Image as an Antecedent of Exercise

Exercise is often viewed as a strategy to control weight, which may improve body image. Indeed, many people turn to exercise to try to achieve their body ideals and improve their body images. It is well documented that people's motives to engage in exercise often reflect appearance-related motives. Specifically, common reasons given for

exercising include weight control, physical attractiveness, muscle tone, and fitness. However, in spite of finding that body image may be a driving force that motivates people to initiate exercise, it is important to note that theories of motivation outline the importance of engaging in a behavior for intrinsic reasons (e.g., positive health benefits, enjoyment, and fun) since it has the greatest potential for resulting in long-term adherence. From this perspective, body image motives might be a key strategy to get people to initiate exercise but may be insufficient to motivate continued involvement, especially if people do not observe the desired changes in their physical appearance.

In addition to exercise motivation, the associations between body image and other exercise-related outcomes, such as setting preferences, effort, and behavior have been examined. Some researchers have documented that people who report high levels of negative body image engaged in exercise more often than those who report low levels of negative body image. In contrast, other researchers have shown that people who report high levels of negative body image are less likely to exercise. This has led to the notion that body image can be either a motivator or deterrent to exercise. However, it is important to note that the majority of the research using this approach has been cross-sectional, and thus, claims of causation are inappropriate. Furthermore, these equivocal findings demonstrate the complexity of understanding the association between body image and exercise behavior. In an effort to explain these inconsistent findings, researchers are increasingly highlighting the need to consider variables that can affect the direction or magnitude of the association between body image and exercise (i.e., moderators) in future research. Some emerging candidate moderator variables include sex and gender stereotypes, self-objectification, social support, motivation, and emotion.

Body Image as an Outcome of Exercise

The hope of improving body image to reduce the negative consequences lies in the development of effective and efficacious interventions. Psychologists have successfully adopted and used cognitive behavioral techniques to improve body image. In parallel, many exercise researchers have examined the effect of exercise on body image due to the numerous health benefits of exercise.

Typically, these researchers have instructed study participants to engage in aerobic training, strength training, or a combination of both, and examined whether body image improves following either an exercise bout or entire program spanning several weeks. The results have been positive, suggesting a beneficial role of exercise in improving body image in both males and females. That is, exercise improves body image immediately after a workout session, and there is also some improvement in body image as a result of long-term participation in an exercise program. Also, both intensity and type of exercise seem to be important features to consider when studying body image as an outcome. Specifically, aerobic training (e.g., jogging, running) and strength training (e.g., weight lifting) improve body image, and the more intense the exercise, the greater the effect on body image. One caveat to this finding is that some researchers have found strength training may exacerbate body image concerns among women.

In light of the evidence, researchers have provided some insight into how exercise might improve body image. They have suggested that at least some of the positive effects of exercise on body image may stem from improved physical fitness, increased awareness of physical capabilities, and increased self-efficacy. For example, body image may improve after exercise because people feel more efficacious by mastering a new skill or more autonomous by gaining some control over their bodies. Thus, the general inference is that exercise interventions are promising. Furthermore, both exercise and cognitive behavioral therapy have been associated with improvements in body image; however, it is unclear if the combination of exercise with cognitive behavioral therapy is more effective than the use of either intervention alone since relatively little research addressing this possibility has been conducted.

Conclusion

Although research on body image in the field of sport and exercise psychology has come a long way, the definition and measurement of body image is an ongoing source of discussion in the literature. An important challenge faced by researchers examining body image in mainstream and sport and exercise psychology is to establish a comprehensive, precise, and universally accepted definition of body image that can be adopted by all researchers. This

would have important implications for the way in which researchers conduct their studies and the refinement of current body image measures, and would enable direct comparisons of findings across studies—an important endeavor if researchers are to increase their understanding of body image and its association with biological, psychological, social, and behavioral factors.

Jennifer Brunet

See also Body Dissatisfaction; Body Dysmorphic Disorder, Muscle Dysmorphia; **Body Self-Esteem**; **Social Physique Anxiety**

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BODY SELF-ESTEEM

Self-esteem, considered synonymous with self-worth, is a global and relatively stable construct that reflects a person's evaluation about self-concept, that is, the set of beliefs and cognitions about one's qualities, character, roles, and attributes. Self-esteem is a complex and multifaceted concept of the self, and arguably one of the most important facets of self-esteem in Westernized countries is a sense of self that is focused on the

body or physique. Specifically, body self-esteem is a strong predictor of global self-esteem when individuals value how they look and feel physically—meaning how one feels about one's physique, form, and function is a predominant guide to how one feels about oneself more globally. *Body self-esteem*, also identified as physical self-worth or physical self-esteem, is defined as an evaluation about the appearance and functioning of one's body. Theorists consider that body self-esteem develops as a result of the evaluative perceptions that arise from the different domains, such as perceived sport competence, physical condition, attractiveness, and weight concern. In turn, these arise from the different subdomains. For instance, positive self-evaluations of upper body strength contribute to the corresponding subdomain of physical strength, which in turn would enhance body self-esteem. On the other hand, negative evaluations of soccer competence will decrease the related subdomain of sport competence, and then in physical self-esteem. In addition, it has been argued that although body self-esteem may be relatively stable over time, specific self-evaluations may fluctuate depending on the situational cues—an individual may feel better about personal physical stamina after completing a 10-mile run or worse if unable to complete this run.

Researchers suggest that it is during adolescence that body self-esteem becomes particularly salient as a result of the physical and social changes associated with this developmental stage. Furthermore, researchers have also shown that there are gender differences in body self-esteem during this period and across adulthood, whereby men have reported higher levels of body self-esteem than women. While there are various potential reasons for this, many researchers have proposed that the greater sociocultural pressures on women to be thin, their inability to attain this standard, and the negative stereotyping of those whose current body is discrepant from the idealized standard, may partly explain this difference. Nonetheless, high body self-esteem is desirable in both sexes because it has a powerful influence on men and women's psychological, physical, and social well-being. Essentially, individuals who have higher levels of body self-esteem are more likely to report higher well-being in these domains. As a result, researchers have highlighted the importance of exploring factors that may increase or maintain body self-esteem

because it has public health importance. One such factor is exercise.

Body Self-Esteem and Exercise Research

Given that exercise has been consistently associated with psychological benefits, there has been an enduring interest in establishing the potential benefits of exercise as a strategy for the enhancement of self-esteem, and more specifically body self-esteem. With the use of different measures like specific cognitive and affective questions (Body Esteem Scale) and figure ratings (silhouettes that gradually increase in size and shape), researchers have found that exercise and body self-esteem are reciprocally related across the lifespan and in many diverse populations. This means that individuals suffering from low body self-esteem will be less likely to be physically active, and that individuals being more physically active show higher body self-esteem levels.

Often in research studies focused on body self-esteem improvements, the exercise parameters assessed have differed markedly. Specifically, the duration and intensity of training sessions, the number of training sessions per week, or the length of the training program tested varied from one study to another. It is possible that body self-esteem improvements found following exercise might be restricted to one particular type (aerobic training, strength training) or intensity (light, moderate, vigorous). However, at this point, it is not possible to provide an indication for the specific doses or types of exercise necessary to enhance body self-esteem. Thus, these parameters of exercise seem to be an important consideration for future research.

Another important problem is that some researchers are still using measures of global self-esteem such as the Rosenberg Self-Esteem Scale. This approach implicitly assumes that the overall concept of self-esteem would be related to exercise, which may not be the case, and lead to the erroneous conclusions that exercise is ineffective in improving self-esteem. Although it is unclear the extent to which the inclusion of global measures in previous studies reduced the associations found between exercise and body self-esteem, the current availability of body specific self-esteem measures provides an opportunity to examine the associations between body self-esteem and exercise to avoid the risk of making false conclusions regarding the magnitude of the association. Therefore, it

is important to move beyond global measures and adopt a more systematic assessment of body self-esteem since exercise might have a more meaningful influence on this aspect of self-esteem without affecting others related to global self-esteem.

Nevertheless, one conclusion that has come out consistently from this research is that people who report lower levels of self-esteem and body image before beginning an exercise program report the greatest improvements in body self-esteem. This is likely because people with higher initial levels of self-esteem have less to gain than those with lower initial levels. Furthermore, although males and females across the lifespan have reported increases in body self-esteem following exercise, there is evidence that youth and middle-age adults benefit the most. Last, the magnitude of the effect produced by exercise on body self-esteem has been shown to be greater for people who are considered overweight or obese (body mass index ≥ 25.0 kilograms/meter squared); suffering from an illness or disease; or have low self-confidence, poor body image, or low self-concept. Accordingly, these personal characteristics, as well as other potential effect modifiers, should be considered when exercise is to be used as a strategy to enhance body self-esteem.

Possible Reasons for the Benefits of Exercise on Body Self-Esteem

To date, few efforts have been made to understand how exercise increases or maintains body self-esteem. Delineating the pathways, also known as mediating variables, through which exercise might exert its effect on body self-esteem has potential for developing interventions, especially since many researchers assume that exercise does not directly increase self-esteem. For instance, if improved physical fitness is identified as a factor that is affected by exercise, and in turn increases body self-esteem, then exercise interventions directed to increasing physical fitness might increase the likelihood of enhancing body self-esteem. As a result, some researchers have placed a particular emphasis on elucidating the psychophysiological pathways through which exercise might lead to enhanced body self-esteem.

Some of the explanations offered in the literature for how exercise might enhance body esteem have a cognitive or psychological foundation, such as those presented in the Exercise and Self-Esteem Model (EXSEM). The EXSEM,

a hierarchically organized model that links exercise to physical self-perceptions and global self-esteem, represents a practical framework that has helped researchers understand how exercise influences body self-esteem. Based on the most recent version of the EXSEM, exercise can influence perceptions of self-efficacy (i.e., people's judgments regarding their ability to become or remain active), which in turn influence physical self-perceptions, such as physical condition, physical competence, body attractiveness, and physical strength. Finally, physical self-perceptions, as well as physical acceptance or the extent to which individuals accept their bodies, are believed to then influence body self-esteem, which is related to global self-esteem. Cross-sectional and longitudinal research based on the EXSEM model has shown that global self-esteem can be influenced both directly and indirectly by exercise through its influence on physical competence and acceptance. Moreover, there is also some support that exercise can increase perceptions of personal control (people's belief regarding the extent to which they are able to control or influence their behaviors and outcomes) and self-schemata (set of beliefs and ideas people hold about themselves), and this may presumably increase self-esteem. For example, as people exercise more often and regularly, they may gain confidence if they are successful at trying new activities or if they master the skills required to perform their exercises. These feelings of physical confidence may increase their body self-esteem, resulting in improvements in global self-esteem.

Other explanations put forward in the literature have a physiological foundation. For instance, improvements in physical parameters (e.g., body composition, body weight, shape, fitness) may account for the positive changes in body self-esteem resulting from exercise. However, the literature examining these various pathways is scarce, preventing researchers to draw firm conclusions. Thus, insofar as the pathways through which exercise might affect body self-esteem are concerned, more studies are needed to identify mediating variables that explain the effect exercise has on body self-esteem. Based on current theorizing, it is clear that there are likely multiple mediators that should be considered in seeking to develop interventions with stronger and more sustainable effects on body self-esteem. For instance, if self-efficacy and body composition are identified as mediators, it would

be logical to develop interventions that target these particular variables.

Conclusion

Body self-esteem is a multifaceted construct that is central to people's sense of self concept and worth, well-being, and development. Researchers have provided empirical evidence to support the notion that exercise can be a key strategy used to enhance body self-esteem, as well as global self-esteem. However, the research conducted to date has highlighted that the relationship between these two variables is complex and requires further investigation. Indeed, two important tasks facing researchers at the present time are to determine which exercise parameters (i.e., type, dosage) are the best to enhance body self-esteem, and identify the psychophysiological pathways by which exercise exerts its effect on body self-esteem.

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See also Body Dissatisfaction; Body Image; Self-Acceptance; Self-Appraisal/Assessment/Perception; Social Physique Anxiety

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behaviors, and habits in daily life and on the athletic field. Brain cells called *neurons* form a communication network that serves as the foundation of information processing in the brain. The neural network of the brain holds the capacity to rearrange and strengthen communication efficiency. It is through this process of rearrangement (neural plasticity) that we can experience changes in the way our minds think, feel, and act. This includes everything from changing your backswing in golf or tennis, to developing a new mental routine for shot preparation, or restoring function following biomechanical or nervous system injury. Thus, optimal performance, skill learning, and recovery are achieved when the capacity for neural plasticity is maximized. Research has shown that physical exercise increases the brain's capacity for plasticity, reflected in part by changes in brain structure and function following exercise training in animal models and humans. Since aging results in gradual neurodegeneration, or loss and dysfunction of brain cells, and decreased neuronal plasticity, aged samples (e.g., 60–80 years) form a platform for studying methods to increase brain plasticity. This entry reviews research on exercise effects on mental performance and the brain and highlights results with aged samples.

Exercise's Influence on Cognitive Performance

Since the 1960s, studies have shown that physically more active people perform better on tests of cognitive performance compared with physically less active peers. The first studies examined performers' ability to successfully complete dual tasks by primarily measuring simple response time, the speed to respond quickly and accurately to a flash of light, and discrimination time, the speed to press one key if stimulus A appears and another key if stimulus B appears. Over time, more complex cognitive processes have been examined, such as the ability to switch between tasks, the ability to selectively pay attention and block out distractions, or the ability to inhibit automated responses or habits. Importantly, in these early studies the physically active groups were comprised primarily of competitive athletes. This may present interpretive problems due to the possibility of self-selection, such that athletes may seek and continue sports participation, in part, because of their natural superiority in cognitive processes that benefit sport performance, such as fast response time or the

BRAIN

Neural plasticity is the mechanism by which the brain encodes experience and learns new skills,

ability to focus amidst distraction. Thus, the best understanding of the effects of physical activity and exercise on mental performance and the brain comes from studying samples that have been well-matched on all characteristics other than physical activity level.

Several reviews have now quantified the effects of physical exercise on mental performance across studies in meta-analyses. Meta-analyses attempt to aggregate results from many studies and group results from similar variables together for the purpose of identifying and comparing replicable effects across studies at either the level of measurement (Does the effect replicate for a specific task?) or construct (Does the effect replicate across tasks that are all theoretically deemed to measure the same construct?). For example, the effect size of exercise training on simple response time could be calculated in different studies that included training and pre- and post-tests of simple response time, and then an average could be computed across studies to determine if exercise results in a consistent improvement independent of any one study or laboratory.

Meta-analyses that have examined the question of how exercise training affects different domains of mental performance have demonstrated that exercise has a small to moderate effect on a range of cognitive abilities across the lifespan. In older adults, consistent benefits have been shown in *speed of processing*, as in simple response time; *visuospatial and selective attention*, such as the ability to compare line drawings or to selectively attend to stimuli or objects in the environment without distraction; *executive function*, a set of abilities related to inhibiting unwanted actions, multitasking, or juggling information in one's mind such as mentally carrying out long-division; and *declarative memory*, which refers to the ability to remember previous events like the face and name of someone you met at a party last week. Across studies with older adults ages 55 to 80 years, several moderating variables have been identified that may result in greater effects of exercise on mental performance. In regard to exercise type, a combination of strength and aerobic training seems to result in greater effects than either alone. In regard to participant characteristics, women seem to benefit more than men and participants between ages 66 and 70 years may benefit more than younger or older adults. In regard to duration, 30- to 45-minute exercise sessions over 6 months

have produced a larger benefit than shorter training periods. Since many of these studies included previously sedentary participants, it does not take long for benefits to occur. Yet the question of how long benefits last and what type of exercise is optimal for maintenance of cognitive benefits is open for future research. It is also important to note that moderating variables such as these remain an active area of research in exercise neuroscience.

Exercise's Influence on Human Brain Structure

Exercise impacts performance in part through enhancement of structural properties of the brain. For example, aging typically results in shrinkage of brain volume in the frontal and temporal association cortices. This can be measured using *in vivo* brain imaging technology called magnetic resonance imaging (MRI). However, studies have shown that greater physical activity (e.g., distance walked) or moderate aerobic exercise training over 6 months (walking at 60%–70% HR max) among older adults is associated with greater gray matter volume in the frontal and temporal cortices and greater white matter volume in the frontal cortex. Gray matter refers to where neurons expend their energy for information processing and form their connections with communication points called *synapses*. White matter represents the part of neurons that transmit neuronal activity between different areas of the brain and is composed primarily of myelin, which insulates the transmission “wires” (known as *axons*) of neurons and increases the speed of neural communication.

Increases in gray matter from exercise could therefore be from increases in the number of connective branches a neuron forms to communicate with other neurons. In some brain regions like the hippocampus, exercise may actually accelerate normal generation of new neurons (*neurogenesis*). In contrast, changes in white matter could result from increased myelination production or repair or from increases in the number of axons that branch out from the neuron. Increases in the number and thickness of blood vessels could also contribute to increases in brain volume as measured in humans; blood vessels traverse through gray and white matter and are not well identified on typical brain scans that have been used in most studies to date. However, there is evidence that exercise training increases cerebral blood flow in the hippocampus in humans, which is consistent

with animal studies. One reason enhanced blood flow is important is because energy for neuronal processing, and therefore information processing, is transmitted to brain cells through increases in blood flow. Therefore, greater resting cerebral blood flow is thought to predict greater responsiveness to the energy demands of information processing.

In sum, while aging results in gray and white matter volume decline in the frontal and temporal association areas, aerobic exercise has been shown to attenuate this atrophy through mechanisms of neuroplasticity that increase the connective branching of neurons, volume of insulating myelination, density of synaptic connections, and through increased birth and survival of brain cells in the hippocampus. Future research will continue to examine the cellular and molecular mechanisms of changes in human brain volume after exercise training.

Exercise's Influence on Human Brain Function

Contrasted to brain structure, brain function refers to how well neurons and their support system can coordinate activity to support ongoing thoughts, emotions, perceptions, and behaviors. Using MRI, the effects of exercise on brain function have been studied by either examining how well different parts of the brain respond to demand for information processing, which we call task-evoked functional MRI (fMRI), or by examining how well different regions in the brain activate in teams (functional networks) that we know support coordinated mental performance. Some studies have also used more direct neuronal stimulation methods like transcranial magnetic stimulation (TMS) to study the link between regular exercise and synaptic plasticity.

When examined with task-evoked fMRI, aging studies have examined activation during executive function tasks. Executive function tasks are of interest because they are known to engage the prefrontal cortices, which are areas of the brain that become dysfunctional with increasing age. In turn, studies have found that more aerobically fit older adults have more prefrontal brain activation during executive function performance. For example, one study found that greater aerobic fitness was associated with greater prefrontal activation during the Stroop task, which requires responding to the ink color of a word regardless of what the

word says. Because of the automaticity of reading, the Stroop task is cognitively demanding and it requires coordinated brain activity in prefrontal and visual cortex. Importantly, greater fitness was only associated with greater prefrontal activity and not visual cortex activity.

Similarly, a training study found that 6 months of walking training in sedentary older adults resulted in increased prefrontal cortex activity during a task requiring attentional focus and inhibitory control, and that greater prefrontal activity was coupled with greater task performance. These studies support that aerobic exercise benefits mental performance in part through enhancement of prefrontal cortex function. Recent research also supports a beneficial effect of resistance training on brain activation associated with inhibition and memory processes that rely on areas outside the prefrontal cortex, suggesting resistance training may play a complementary role to aerobic training in supporting brain function across the lifespan.

Evidence also exists demonstrating that aerobic exercise is associated with greater coordination of brain activity—in regard to both broad brain networks and to synaptic plasticity in specific neuronal circuits. Brain networks are teams of physically distant regions that work in coordination and provide a system for the brain to carry out highly specific, local processes that feed up to coordinated, complex processes; neural plasticity is the foundation for these functional networks to maintain coordinated teamwork. In one study, older adults with greater aerobic fitness had greater functional coactivation in a brain network known as the *default network*, whose deterioration has significant implications for cognitive aging, risk for dementia, and a host of developmental psychiatric disorders. Exercise effects were strongest in the lateral and ventromedial prefrontal regions and the temporal cortex, including the hippocampus. Importantly, this research also suggests that greater functional coordination in the default network is associated with some of the cognitive benefits that are linked to aerobic fitness, suggesting this network may be an important component of how exercise improves cognition and decreases risk for dementia in late life. It may also suggest that exercise would be beneficial for developmental disorders related to impaired default network function.

Finally, there is evidence that greater aerobic fitness is associated with greater TMS-induced

synaptic plasticity. The basis of learning is the brain's ability to form new neural connections or to strengthen existing pathways based on experience. One way to study this is to pair stimulation of a hand muscle with electromagnetic stimulation of a corresponding region of motor cortex. The capacity for synaptic plasticity in this circuit can be measured by the increase in reactivity of the hand muscle to activation of the motor cortex following paired training. One study showed that more active adults had greater synaptic plasticity in the specific motor circuit studied. Although this was a cross-sectional study, it presents complementary evidence for the link between aerobic fitness and enhanced synaptic plasticity that may be a generalizable mechanism for the effect of exercise on coordinated brain function and improved learning and performance.

Overall, there is exciting evidence for exercise's potential to attenuate age-related brain dysfunction, and these results have implications for improving the brain's capacity to learn and respond adaptively to injury at any age. However, the mechanisms for how this happens are not fully understood and future research should be guided by the need to understand the cellular and molecular basis of these benefits.

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See also Brain Imaging; Cognitive Function; Information Processing; Interventions for Exercise and Physical Activity; Learning; Neuroscience, Exercise, and Cognitive Function; Sedentary Behavior; Skill Acquisition

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BRAIN IMAGING

Neuroimaging includes various techniques that either directly or indirectly image the structure and the function of the human brain. Thus, neuroimaging can be divided into two categories: structural imaging and functional imaging.

Structural imaging examines the structure of the brain (like gray and white matter) and the possible changes that occur in these structures with factors such as learning and aging. Popular methods to investigate changes in brain structure include voxel-based morphometry (VBM), which enables investigation of changes in the brain's anatomy, and diffusion tensor imaging (DTI), which enables examination of image neural tracts by measuring the restricted diffusion of water in the brain.

In contrast, functional imaging is used to observe the working brain. Functional brain imaging offers new insights into topics that lie at the heart of sport psychology. For example, research on motor imagery has reached a new level demonstrating that motor imagery is based on neural activation of core motor areas in the brain. This widely accepted finding has dramatically influenced approaches to motor rehabilitation.

Imaging of the living brain has to deal with the fundamental problem of the scale of observation. Research on the mirror neuron system (MNS), for example, is based on single-cell recordings in the monkey brain. However, cognitive neuroscience is typically interested in examining the relevance and interconnectivity of defined whole brain areas during specific tasks. In the case of the MNS, the role of the parietofrontal circuit for action recognition has been uncovered.

Functional imaging enables researchers to identify brain regions whose activation is associated with specific action-linked processes, such as action observation or action imitation processes. Possible methods for this are positron emission tomography (PET) and functional magnetic resonance imaging (fMRI). In PET studies, radioactive-marked molecules (e.g., radioactive-marked glucose) are administered into the participant's blood right before the study. The tomograph

detects the radiation and therefore shows exactly where the molecules are being used in the brain. On the other hand, fMRI does not need an injection and is based on the different magnetic properties of the human blood. Both methods examine metabolic brain activity.

In sports and motor neuroscience, most published work uses fMRI because it has good temporal and spatial resolution properties. PET is used less often for research because radioactive-marked molecules have to be administered. Therefore, the fMRI will be described in more detail.

In humans, fMRI has proven to be an efficient method to study task-relevant brain activation. The resting brain is not silent and shows neural activity even during sleep. For this reason, fMRI studies attempt to understand brain activation by examining differences of brain activities between two or more tasks, such as action observation and motor imagery. Research on the functions of brain areas for specific tasks relies heavily on cytoarchitectonic results (structural information about anatomical regions of interest in the brain, based on the cellular composition) and on research with patients with defined cortical lesions.

To map neural activity, fMRI uses the change of blood oxygen flow within the brain. More precisely, the measurements rely on the different magnetic properties of oxygen-rich and oxygen-poor blood. Oxygen-rich blood is diamagnetic and therefore has less impact on the magnetic field, whereas oxygen-poor blood is paramagnetic, which leads to stronger interferences in the magnetic field. Thus, the strength of the measured signal depends on the degree of the oxygenation of the blood. The dependency between the image quality and the oxygen saturation of the blood is called *blood oxygenation level dependency* (BOLD). Changing blood flow and the related BOLD response is directly associated with neural activation in a certain brain region.

During fMRI scanning, it is necessary for participants to lie in a strong, permanent magnetic field with high homogeneity. Certain nuclei in the human body, the hydrogen nuclei, provide magnetic properties. Being in a strong magnetic field, hydrogen nuclei behave like a compass needle; they all align with the magnetic field. During fMRI scanning, radiofrequency impulses are applied to the aligned magnetic system. This results in a change of the orientation of the hydrogen nuclei. After the radio pulse ceases, the hydrogen nuclei

return to their original orientation by emitting energy, which is detected by an antenna of the system. The source of this signal is specified by magnetic field gradients that vary the strength of the magnetic field and hence allow determination of the specific signal source and position. The position of the brain in the magnetic field is defined at the very beginning of the experiment. Therefore, it is crucial for the later analysis of the data that the participants do not move their head during the experiment. Otherwise a mislocalization of a detected increased activation may be possible.

Experimental Designs

Generally, science starts with a research question that in turn generates (neuroanatomical) hypotheses, which can then be tested by performing an experiment. For fMRI, the experimental strategy is to observe the brain's response (the BOLD response) to certain kinds of stimulation: for example, an observation task with different body movements. Over the last decade, three design types have dominated fMRI studies: the *blocked design*, the *event-related design*, and the *mixed design*. These designs vary in terms of stimulus presentation and timing. The blocked design is characterized by presenting a time interval with stimuli of only one condition, alternating this with intervals representing stimuli of other conditions. The main advantage of this type of paradigm is increased statistical power and robustness. In contrast, the event-related design presents random short-duration events drawn from the different conditions within the experiment, providing superior temporal resolution characteristics. This approach permits the temporal characterization of BOLD signal changes. A mixed design contains features of both these design types.

After completed data collection, the critical question is whether there are differences or commonalities between the different experimental conditions. To test for this, several types of comparison are possible. One central comparison strategy is the subtraction method, in which the BOLD response for the experimental condition has subtracted from it the BOLD response acquired from the control condition. The factorial strategy is an alternative to the subtraction strategy in which all experimental conditions are processed as experimental factors. This strategy also allows testing for interactions between the conditions. Some

experimental tasks show different levels of difficulty. Given this, a parametric design can be used to test whether there is an increase of the BOLD effect that systematically varies with an increase of task difficulty. Each of the comparison strategies aims to detect differences between experimental conditions. In contrast, a conjunction analysis offers the possibility to detect the commonalities between the BOLD patterns of two conditions by calculating the intersection between the two conditions.

Implications

Functional magnetic resonance imaging (fMRI) has already had a strong impact on research in fields, such as action observation, motor imagery, and attention, and has great potential to impact other key topics in sport psychology and motor control as interactive actions, emotion, and empathy. Recently, imaging genetics has started to reveal new directions for brain imaging. Genes have an effect on neural activity on the molecular level. Different concentrations of neurotransmitters moderate neural activity in different cognitive tasks. Brain imaging may help to elucidate this complex interaction between genes and neural activity.

The striking development of functional brain imaging has been driven by the technical advances of the last 20 years; fMRI has become a standard tool in cognitive neuroscience. It is complemented by magnetoencephalography (MEG), which records magnetic fields produced by electrical currents in the working human brain; near infrared spectroscopy (NIRS), which measures changes in cerebral blood flow, similar to fMRI but vulnerable to movement, only useful on the cortex, and does not reach deeper regions; and electroencephalography (EEG), which measures electrical activity along the scalp. EEG also offers tools for functional brain imaging with low-resolution brain electromagnetic tomography (LORETA). These methods differ with respect to the fundamental limitations concerning the range of active movements feasible during data recording, with EEG and NIRS offering an advantage in this regard.

There has been no doubt that the advent of new methods of brain imaging, data recording, and data analysis has facilitated progress in understanding cognitive processes. Neuroimaging must build on, rather than replace, the importance of

well-designed research with strong theory-driven hypotheses.

Jörn Munzert and Britta Lorey

See also Biofeedback; Neurofeedback; Brain; Cognitive Function; Imagery; Perception; Psychophysiology; Social Neuroscience

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BREATHING EXERCISES

Breathing strategies are often used as the basis for several advanced relaxation techniques, including progressive muscular relaxation, meditation, and calming imagery. However, breathing strategies act as an important technique in their own right to help physically and mentally relax the performer. Provided here is a synopsis of breathing exercises, based on the guidelines by Jean M. Williams, that includes diaphragmatic breathing, rhythmical breathing, and sighing when exhaling as strategies commonly advocated by applied sport psychologists to help relax athletes.

Diaphragmatic Breathing

Sometimes called complete breathing, diaphragmatic breathing is a physical relaxation strategy oriented around filling the lungs to capacity from bottom to top and emptying the lungs in a slow,

controlled, and complete manner. Athletes should use a deep, long, and slow inhalation through the nose to completely fill the lungs. To help athletes achieve this, applied sport psychologists should ask the athlete to view their lungs as a three-section cavity. First, the athlete should inhale and simultaneously relax the stomach, pulling their diaphragm muscles down to fill the lower section of their lungs. Second, the rib cage should be lifted, the chest cavity expanded to fill the mid-section of the lungs, and then the shoulder blades widened to fill the lungs to the top. A slow, continuous, smooth, and controlled inhalation should be used throughout by the performer. The exhalation phase is just as important to invoking a relaxed state; here, the athlete should reverse the process described above by exhaling the top of the lungs first, then the chest cavity, and finally the lower section. Emphasis should be given to expelling air from the bottom of the lungs to complete a full exhalation, with a reduction in muscular tension and anxieties being associated with the air leaving the lungs. Throughout the exhalation phase, the muscles of the diaphragm and those surrounding the stomach should be used to help push the air from the lungs to facilitate a complete emptying of lung capacity. Athletes should be asked to associate a feeling of calmness and quietness with the final exhalation stage of the breath cycle to help invoke a completely relaxed state.

Rhythmical Breathing

Rhythmical breathing acts as an extension to diaphragmatic breathing where ratios of holding one's breath during inhalation and exhalation are suggested to control breath rate and help keep physical tension and mental anxieties under control. To create a symmetrical rhythm within their breathing cycle, athletes should inhale to a specific count of time (e.g., 3 seconds), hold their breath for the same time count, exhale to the same time count, and pause for the same count before repeating the cycle. Once proficient in this symmetrical rhythmical breathing technique, a different ratio of time count between inhalation to exhalation should be explored to help invoke a more relaxed state. For example, using a ratio of 1:2, athletes should take a symmetrical rhythmical first breath (i.e., a 1:1 ratio, inhale for 3 seconds and exhale for 3 seconds); then, on the next inhalation, they should inhale to the same count of 3 seconds and extend

the exhalation phase over a 6-second period (i.e., a 1:2 ratio). To achieve this, the performer should be encouraged to exhale more slowly and with greater control and awareness of this phase of the breath cycle. Alternative ratios should be explored by the performer to help create greater control and awareness of the inhalation–exhalation phases of the breath cycle to help create a powerful sense of control over the rhythm of breathing and thus the relaxed state of the athlete. The benefits of adopting a rhythmical breathing rate include lowered heart rate, better oxygen utilization, and lowered muscular tension and mental anxiety.

Sighing When Exhaling

This breathing strategy is a very simple, quick approach that can be used to help relax athletes. Specifically, if tension is perceived by athletes, they should exhale fully and powerfully through the mouth with an audible sigh, and then inhale slowly and quietly through the nose. The performer should repeat the process using the muscles around the rib cage to fully expel the air in the lungs so that tension within the body is reduced.

Conclusions

The strategies outlined in this entry illustrate how breathing control can help athletes achieve a relaxed state. In general, studies testing intervention effects have noted the positive effects of breathing strategies on enhancing performance. Further, these findings are relatively consistent across different sports and different competitive level athletes, from elite to non-elite. Additionally, studies that have assessed the psychological skill use of athletes have noted that elite performers tend to use relaxation strategies (including breathing exercises) more readily when preparing for performance than their non-elite counterparts. However, research has yet to establish whether one particular breathing strategy or a specific combination of various breathing strategies is more effective than another at helping to relax the athlete. Therefore, the broad recommendations that emerge from the applied research are that applied sport psychologists should teach the performers a range of breathing strategies and allow them to explore which is most useful for them given their needs. Performers should practice these breathing strategies in nonsporting stress-inducing situations (e.g., waiting in line in a shop, dealing with

a difficult situation at work) before progressing to test the efficacy of the given strategy in a sporting context. Ideally, the strategies should be practiced in training environments before being utilized in the competitive arena.

Owen Thomas and Richard Neil

See also Centering; Mindfulness; Relaxation; Stress Management

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BURNOUT

Modern sport culture is replete with qualities thought to make burnout prevalent, including high training volumes and competitive demands, near year-round training, and in some sports, specialization at young ages. Given this sport landscape and the concerns raised by sport scientists and others involved in the sport community (coaches and administrators), the importance of athlete burnout is now widely recognized. This entry defines athlete burnout, describes its epidemiological significance, discusses potential causes, and concludes by addressing preventive strategies.

Defining Athlete Burnout

The term *burnout* is used in a variety of contexts in everyday discourse. Although some of these meanings converge with scientific uses of the term, others do not. Burnout is sometimes used synonymously with the term *sport drop-out*. However, not all athletes who drop out of sport do so because of burnout; athletes may leave sport for any of a myriad of reasons. Although some athletes who do

experience burnout will discontinue sport, others may maintain their involvement. Others may compete at a lower, less demanding, level or they may choose to participate in a different sport. Thus, the term *burnout* should not be used interchangeably with the term *sport drop-out*.

Within the scientific community, burnout has a more precise meaning than that often used in everyday language. The most widely accepted definition casts athlete burnout as a psychological syndrome of emotional and physical exhaustion, a reduced sense of accomplishment, and sport devaluation. Athletes experiencing burnout may be emotionally exhausted from dealing with continual stresses of competition, training, and other demands for their time (work, school). They may also feel physically exhausted from high training loads. Another key factor involved in burnout is a reduced sense of accomplishment in which athletes question and doubt their sport skills and ability to be successful. They may feel they are training hard yet making minimal progress toward their goals. Finally, sport devaluation is represented by a psychological detachment from sport in which athletes may stop caring about sport and their performance to the point of developing a resentful attitude toward sport.

Epidemiological Significance

Although the modern culture of sport has characteristics that suggest burnout may be on the rise, the actual prevalence of burnout is unknown. Research surveying current athletes suggests that a low percentage of athletes (e.g., 2%–10%) have relatively high scores on a self-report burnout measure as assessed via the Athlete Burnout Questionnaire, or psychological characteristics suggesting they may be experiencing burnout. Research surveying current athletes suggests that a low percentage (e.g., 2%–10%) have characteristics suggesting they are experiencing burnout. However, surveying current athletes may underestimate its occurrence as some athletes who experience burnout may have left sport or have been absent from practice when the questionnaire was administered.

Even if burnout is not very prevalent, it has epidemiological significance because of the millions of sport participants across the globe and the negative toll it has on individuals suffering from it. Although minimal research has examined

its consequences, burnout can potentially have negative impact on all spheres of an athlete's life, both within and outside of sport. Within sport, burnout is thought to lead to performance decrements, decreased motivation, and possibly sport discontinuation. Beyond sport, it can hurt physical and psychological well-being as well as negatively impact personal relationships.

If athletes do experience burnout, it is not something that will dissipate after a short break from sport—rather burnout is often chronic in nature. Consequently, it is important to structure sport in a way that prevents burnout rather than attempting to treat it once it occurs given its relatively enduring state. Developing effective prevention strategies is predicated on first understanding what causes burnout.

Potential Causes of Burnout

It is widely accepted that burnout is a reaction to chronic stress and occurs when demands associated with sport participation exceed or tax an individual's resources over an extended time period. Thus, burnout is linked to an imbalance between demands and resources.

Demands involve all the stressors involved in sport. These stressors, at least in many sports, stem from the physical demands associated with training. In some cases, burnout may be the result of overtraining characterized by overly high training volumes (duration x intensity) coupled with inadequate recovery. In addition, the time demands associated with sport can also be a contributing factor to burnout, in which athletes feel that sport takes too much time and results in their missing out on other life opportunities. Finally, the pressure associated with competition may be another source of stress associated with burnout.

A variety of external influences, such as pressure from coaches and parents, may also be sources of chronic stress associated with burnout. For example, overinvolved parents may create excessive pressure that predisposes athletes to burnout. On a more subtle level, parents who are supportive of their child's sport experience, but whose family life centers around sport, may also predispose athletes to burnout. Additional parental characteristics associated with burnout include setting high standards for their children coupled with being critical of their children and their performances.

Finally, coaches, through their leadership style and interactions with athletes, may also create a sport culture or team atmosphere that may make burnout more likely. Athletes who play for coaches who are perceived as being socially supportive, empathetic, and who provide praise, instructions, and training and a democratic coaching style have lower burnout compared with other athletes. In contrast, autocratic and aversive-style coaches who create a fear of failure in athletes may make burnout more likely.

Finally, the demands associated with sport can come from internal sources. Some athletes have personality qualities that make them vulnerable to burnout. For example, athletes who are perfectionistic, characterized by excessively high performance standards and self-doubt, are at risk of burnout. Related to perfectionism, athletes who base their self-esteem on performance accomplishments are more likely to experience chronic stress and burnout. Also, athletes who are pessimistic are more likely to experience burnout compared to those who are optimistic. Finally, athletes who are *trait anxious*, defined as predisposition to experience high levels of anxiety, are more likely to experience burnout than those with low trait anxiety.

Although burnout is a response to chronic stress, not all athletes who are in demanding sport environments experience burnout. Coping resources also play a role in the burnout process. Resources are the internal and external factors athletes have available that help them effectively manage stress. They include external factors such as social support and participation in activities that facilitate recovery. They also include internal resources such as self-awareness, strong self-regulatory skills, and effective lifestyle management skills that include healthy eating habits and good sleep habits. For example, having a good life balance wherein athletes are involved in more than just sport and potentially the strong use of mental skills training techniques may also serve as coping resources. In addition, athletes who experience low levels of life stress outside of sport are theoretically less vulnerable to burnout compared with those who experience a great deal of stress outside of sport.

At this point, it is widely recognized that burnout, especially exhaustion, is a reaction to chronic stress. Consequently, identifying stress related factors associated with burnout is important to understanding this phenomenon. However, there is more to the burnout process than a simple reaction to

chronic stress. Burnout is only experienced when highly committed athletes become disillusioned and frustrated with their sport involvement. Given that burnout is intricately connected to commitment and motivational processes as well as stress, researchers have developed a commitment perspective on burnout, drawing from the organizational psychology and relationship literatures.

On a positive note, athletes can be committed to sport because they are passionate about it. These athletes want to be involved, find it enjoyable, and concomitantly experience high benefits and low costs. Because of their favorable outlook, they are likely to invest a great deal of time and energy into sport and perceive that it is more attractive than alternative options. These athletes are not theoretically likely to burn out as they experience enjoyment-based commitment.

There is another side to commitment. Although commitment can be influenced by positive pulls (e.g., passion, enjoyment, satisfaction), it can also be affected by nonpositive pushes (e.g., too much invested to quit, lack of attractive alternatives, social pressure to continue involvement). In other words, athletes can be committed for a combination of reasons related to *wanting to be* involved and feeling they *have to be* involved. Entrapment-based commitment occurs when athletes begin to have a more negative view toward sport (e.g., decreasing positive pulls) but maintain their involvement because they feel they have to continue (i.e., increasing nonpositive pushes). These athletes feel they are trapped and stifled by sport while missing out on other life opportunities. This is evident by decreasing enjoyment coupled with decreasing benefits and increasing costs. Despite this, they maintain involvement because of feeling locked into the role of being an athlete. They may feel there is too much invested to quit; perceive few attractive alternatives to being an athlete; or perceive that other people, such as coaches, teammates, or parents, expect them to maintain their involvement.

In addition to these sources of entrapment, two additional factors that may result in athletes, especially adolescent sport participants, feeling trapped by sport include a unidimensional identity and low perceived control. In normal development, adolescents sample a variety of activities and roles in the process of forming their personal identities. The teenage years are also characterized by the development of personal autonomy. However, sport

participation, especially high-level involvement, can result in athletes prematurely developing a unidimensional identity, which increases the risk of burnout. In addition, although they may have chosen initially to participate, in some situations, adults control their sport experience, resulting in feelings of low control over the sport involvement. Having a unidimensional identity whereby their sense of self is based exclusively on being an athlete, as well as low perceived autonomy, may result in athletes experiencing sport entrapment and their feeling trapped into the role of being an athlete. This theoretically increases the risk of burnout.

Converging with a commitment perspective, athletes who are passionate about sport view it as important and invest a great deal of time and energy into sport. Much like commitment, passion is not something athletes simply have or do not have, rather there are different types. One type is an obsessive passion for sport that is associated with higher burnout scores, at least in some research. The other type, harmonious passion, is characterized by a more intrinsically motivated type of passion associated with lower burnout scores.

Given that burnout is linked to an erosion of motivation, researchers are using common motivation theories, such as self-determination, and achievement motivation theories to better understand the burnout process and what potentially might predispose athletes to it. According to self-determination, the fulfillment of basic psychological needs, including perceived competence (positive perception of skills and abilities), autonomy (sense of say and control over their sport involvement), and relatedness (sense of belonging and acceptance), is associated with higher levels of well-being. The fulfillment of these basic needs is also connected with quality motivation such as high levels of intrinsic motivation, whereby athletes participate for the inherent pleasure and satisfaction derived from sport participation. In contrast, need thwarting is associated with indices of ill-being, including burnout. Burnout and the lack of need fulfillment are also associated with low-quality motivation. On the extreme level, this can include being amotivated (without motivation). On a less extreme level, lower quality motivation characterizes athletes who participate not because they want to but, rather, because they feel they have to be involved in sport. This can be due

to either external pressure by a coach or parent or internal pressures of feelings of obligation to remain involved.

Another common motivation theory, achievement motivation, which has been used to understand burnout, focuses on whether athletes and the team atmosphere are mastery oriented or outcome oriented. With a mastery team climate, success is defined in terms of effort, learning, and improvement. In contrast, outcome-oriented team climates focus on social comparison and doing better than others. In a mastery-oriented climate, mistakes are viewed as part of the learning process, whereas in an outcome-oriented climate, they are viewed negatively and punished. Although studying burnout from an achievement goal perspective has not received extensive investigation, mastery-oriented team climates are generally thought to be associated with lower burnout scores compared to athletes who view the team climate as more outcome oriented.

Preventing Burnout

Nearly all of the scientific literature on burnout has been either correlational or qualitative in nature. The focus of the correlational research has been to examine the association of scores on a burnout measure with other variables that are theoretically related to or potential causes of burnout. The qualitative studies have focused on in-depth interviews of athletes who experienced burnout to better understand the burnout process. At this point, very few, if any, studies have evaluated the effectiveness of interventions designed to prevent or treat burnout. Thus, the knowledge based on intervention strategies is not well developed and comments on preventive strategies are provided tentatively.

As a starting point for understanding potential interventions designed to minimize burnout, public health frameworks provide a launching pad. Primary prevention strategies involve changing the sport culture or environment to eliminate or modify factors that potentially cause burnout. Interventions designed to help individuals manage or cope with the stress associated with sport are titled secondary prevention. Finally, interventions helping athletes already suffering from burnout are regarded as a tertiary prevention strategy with a focus on treatment or rehabilitation. As stated in the adage “an ounce of prevention is worth a

pound of cure,” interventions designed to prevent burnout are more effective than treating burnout once it occurs given its chronic nature.

At this point of knowledge development, one viable strategy to prevent burnout is to target theory-based variables associated with burnout in the intervention design. These can range from individual characteristics associated with stress-related processes (e.g., perfectionism) to the social-organizational structure of sport (e.g., coach and parent behaviors, how sport is structured, and training demands and recovery).

Given that burnout is a reaction to chronic stress, a common belief is that teaching athletes stress management skills will help prevent burnout. For example, helping athletes learn effective time management as well as lifestyle management skills will help them deal more effectively with the demands of being an athlete. In addition, an increased focus on recovery activities, as well as helping athletes to achieve a balanced lifestyle, will also help prevent burnout. Mental skills training techniques, such as effective goal setting, self-talk, and relaxation skills may also be effective. If athletes can learn to effectively cope with stress, then burnout will be less likely.

Although stress management strategies have a role in preventing burnout, it is premature to conclude that teaching athletes stress management skills geared at the individual will be the most effective intervention approach. In fact, researchers in organizational psychology argue that teaching individuals stress management strategies have not been very effective in reducing burnout. This is because social-environmental factors have a larger role in work burnout than individual factors. The same is likely true for athletes. Thus, interventions that target the sport environment will be more efficacious than those that target the individual and focus on helping athletes effectively manage stress. Taken one step further, some scholars suggest that teaching athletes how to cope with stress is analogous to treating burnout with a bandage. Rather than addressing the underlying cause of burnout, which is how sport is structured, teaching stress management only addresses the surface of the problem.

In addition to teaching stress management as part of life skill development, interventions need to address social-environmental modifications designed to create a more positive sport experience for athletes. Consequently, commitment and

motivation theories should play a role in designing effective interventions targeting burnout. For example, strategies could be developed to help ensure that sport fulfills the basic psychological needs of perceived competence, autonomy, and relatedness. In addition, coaches who create a mastery-oriented team climate will help prevent burnout. Sport could be structured in a way that empowers athletes by developing multifaceted identities and that gives them control over their sport experiences. Finally, interventions designed to enhance enjoyment-based commitment and minimized feelings of entrapment should be central components of interventions. Given that burnout is a complex process, interventions that are multimodal in nature will be the most effective.

Conclusion

Although most athletes do not experience burnout, it is nonetheless a significant issue within contemporary sport culture. Although the term *burnout* conjures a variety of images, it is best defined as a psychological syndrome involving exhaustion, sport devaluation, and a reduced sense of accomplishment. Burnout is a complex issue that involves both stress- and motivation-related processes. Given its complexity, research addressing antecedents, underlying processes, and

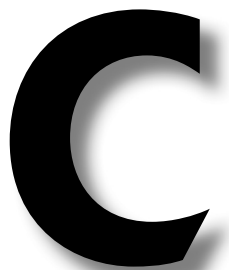
consequences associated with burnout will serve as the foundation for designing effective interventions. Interventions designed to prevent burnout should be multimodal and target both stress and motivation processes.

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See also Affective Disorders; Emotional Reactivity; Injury, Psychological Susceptibility to; Overtraining Syndrome; Stress Reactivity; Underrecovery Syndrome

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CALO-RE TAXONOMY OF BEHAVIOR CHANGE TECHNIQUES

The Coventry, Aberdeen, and London—Refined (CALO-RE) taxonomy of behavior change techniques builds on initial work on classifying psychological techniques used in intervention to change behavior, with a particular emphasis on physical activity and healthy eating. The taxonomy aims to provide a common language for the organization, identification, and adoption of behavior change techniques in interventions. The taxonomy is a tool for researchers designing effective interventions that work and practitioners wishing to identify the techniques that will be most effective in changing physical activity behavior. The taxonomy is based on Charles Abraham and Susan Michie's initial taxonomy of behavior change techniques refined to eliminate inconsistencies. The taxonomy provides a reference guide for each technique and the psychological constructs it purports to change. Forty-three techniques are specified in CALO-RE. Each has a specific definition derived from systematic reviews of intervention research independently coded and verified by leading experts. The definitions also specify exclusions and exceptions ensuring that the techniques do not overlap. The CALO-RE taxonomy does not specify the theories from which the techniques are derived; this information was supplied by Abraham and Michie. The taxonomy provides a standardized set of common terms to facilitate understanding of interventions, enable better evaluation of the effectiveness

of specific intervention techniques, and provide insight into the psychological mediators that explain the process by which the technique works in changing behavior.

Behavior Change Theory and the Taxonomy

A vast array of psychological factors, such as self-efficacy, attitudes, motivation, intentions, and risk perceptions, have been found to be linked to physical activity behavior. Such antecedents are important as it is assumed that such variables can be manipulated or changed through techniques communicated to individuals by various means like one-to-one consultations or via the media. While interventions targeting psychological factors have led to increased physical activity participation, their effectiveness has been shown to be relatively modest. A key reason is that interventions do not adequately identify the intervention techniques that will be successful in changing the psychological constructs known to be correlated with physical activity. This means that the techniques used may not be completely effective in changing behavior as they do not lead to a change in the psychological constructs, known as psychological mediators, associated with physical activity. Another problem in intervention research is that the reporting of the intervention components adopted to change physical activity behavior is inadequate. This limits the extent to which other researchers will be able to replicate the findings and limits the inferences that those aiming to synthesize research can make with regard to the effectiveness of specific techniques in changing the psychological constructs and

changing behavior. A final problem is that many interventions adopt multiple behavior change techniques that seek to target multiple psychological mediators. While adoption of multiple techniques can be effective in bringing about behavior change, it limits researchers' ability to identify which of the techniques was responsible for bringing about the change. Researchers will, therefore, be unable to arrive at a definitive conclusion as to which of the intervention techniques are *doing the work* in changing behavior.

These issues have led to calls for (1) identifying and classifying the intervention techniques that target specific antecedents of physical activity through a mapping process, (2) improved reporting of behavior change techniques within reports of intervention trials, and (3) improvements in the design of interventions to provide tests of individual techniques identified in the mapping process on physical activity behavior change. Recent developments in the taxonomy of behavior change techniques have led to a direct mapping of specific techniques on to the psychological antecedents of behavior change. This has provided researchers and practitioners with a menu of strategies from which to choose when designing interventions as well as identifying any overlaps and gaps.

Methods and Development

The lists that Abraham and Michie used to develop their taxonomy mixed general theoretical approaches, modes of delivery, and intervention settings. Therefore, a refined taxonomy of 26 clearly defined behavior change techniques was developed so as to overcome these issues and allow reliable coding of interventions. Three systematic reviews were then completed to assess whether the behavior change techniques could be used to identify core components of behavior change interventions. The 26 behavior change technique definitions resulted in 93% agreement between raters, and acceptable interrater reliability levels for most of the definitions.

Since the publication of the original behavior change technique taxonomy, which has been widely adopted and used in reviews, researchers have identified further opportunities to improve the classification. Researchers at Coventry, Aberdeen, and London collaborated to identify limitations, such as lack of clarity or disagreement between raters, with Abraham and Michie's

original taxonomy, and to introduce further classifications. Research teams adopted an iterative process of coding one or two research articles, calculating interrater reliability coefficients, and revising the taxonomy accordingly through group discussion. This process resulted in the CALO-RE taxonomy, which has fewer conceptual problems and less overlap between items and is more comprehensive.

The Techniques

The 40 behavior change techniques identified in the CALO-RE taxonomy define and differentiate techniques, offering researchers and practitioners a clear guide as to which techniques may be adopted to change physical activity behavior in interventions. There is no guidance on which techniques to adopt or which will be more effective; the purpose is for identification and classification only. An outline of each technique is provided in the next section along with exercise-specific examples of what each technique might look like in an intervention to promote exercise and physical activity.

1. Information provision (general). In this technique, general (not specific) information is provided about physical activity and its possible outcomes. (A poster might provide a statement that "physical activity is enjoyable and fun for all.")

2. Information provision (to the individual). In this technique, specific information relevant to the individual about physical activity and its possible outcomes, consequences, benefits, and costs is given. (A physician might inform a patient that "if you participate in regular physical activity, it will help you to get your weight down and help you to feel less stressed at work.")

3. Information provision (others' approval). In this technique, information is provided about what others might think of their physical activity. (An exercise specialist might tell a referred patient: "You will find that your family and friends will be very supportive of your new exercise program; they will see a fitter, healthier you!")

4. Information provision (others' behavior). In this technique, information concerning what others typically do with respect to exercise is provided. (A leaflet might inform a potential middle-age

person thinking of taking up physical activity that “people over 50 years of age often take up exercise having not ever exercised in the past.”)

5. Goal setting (behavior). In this technique, encouragement to begin or maintain behavior change is provided. It does not involve precise planning to do the behavior. (A goal may be to “exercise more next week.”)

6. Goal setting (outcome). In this technique, the individual is encouraged to set general goals achievable through performance of the behavior, but distinct from the behavior itself. (A consultant might highlight blood pressure reduction and losing weight as achievable goals derived from regular exercise.)

7. Action planning. In this technique, detailed plans are made including when (e.g., frequency) and where (e.g., in what situation) to act. It is vital that there is a clear link between plans and behavioral responses to specific situational cues. Such plans are often expressed in *if-then* formats. (A business executive’s plan might be, “If it is 5 p.m. and everyone is starting to leave the office, then I will collect my gym bag and head for the fitness center.”)

8. Identifying barriers and problem resolution. In this technique, after the formation of a clear plan, individuals are tasked with identifying possible barriers to performance and solutions to the possible problems. Barriers may be cognitive, emotional, social, or physical (“I feel too tired to exercise on Fridays—therefore, I will go to sleep earlier on Thursday nights.”)

9. Setting graded tasks. In this technique, the target behavior is broken into smaller, more manageable tasks, allowing successful progression in small increments—for instance, writing down a sequence of small steps to accomplish the overall behavior over time.

10. Review of behavioral goals. In this technique, individuals are prompted to review the successful accomplishment of previously set goals and contingencies and further plans made for instances in which goals were missed. (An individual may report not being able to exercise 5 times a week due to other commitments, so reschedules allotted

times to exercise at more convenient times or fits it into a work schedule by walking to work.)

11. Review of outcome goals. In this technique, an individual will review the attainment of previously set outcome goals and be given an opportunity to revise the goals. (At a routine check-up, a physician might encourage an exerciser to revise the blood pressure or weight goal and modify the physical activity regime accordingly to enhance chances for success.)

12. Effort or progress contingent rewards. In this technique, the person uses rewards or praise for *attempts* at achieving the goal. This is not reward for engaging in the behavior itself and is not contingent on actual success. (A trainer might say to an exerciser, “Well done, you have really tried hard to attain your goal of three visits to the gym this week.”)

13. Successful behavior-contingent rewards. In this technique, rewards for successful performance of the target behavior are provided. The reward may be material or verbal but must be based explicitly engaging in the behavior itself—for instance, providing a reward or incentive for doing some physical activity, regardless of outcome.

14. Shaping. In this technique, graded contingent rewards are provided for movement toward completion of the target behavior. The individual may reward oneself for any increase in physical activity behavior in the first instance (e.g., jogging for 10 minutes as opposed to no activity). The reward schedule becomes progressively more restricted in later weeks (e.g., rewarding only for 20 minutes of jogging).

15. Generalization of target behavior. In this technique, the person is encouraged to find opportunities to try it in other situations, to ensure the behavior does not become situation specific. (After following a walking program for 2 weeks, a personal trainer will ask the client to try swimming for the same amount of time at the local swimming pool.)

16. Self-monitoring of behavior. In this technique, the person is asked to keep a detailed record of activity and use it as a means to change or modify behavior. This could take the form of a

questionnaire or diary focusing on duration, time, and situation in which the physical activity was attempted or completed.

17. *Self-monitoring of behavioral outcome.* This technique is similar to point 16, but focus is on measurable outcomes of the behavior (e.g., blood pressure or weight reduction).

18. *Focus on past success.* In this technique, individuals reflect on successful past experience with physical activity as a means to increase motivation to be active in the future. (A person is encouraged to list or write down past successful experiences with physical activity.)

19. *Provide feedback on performance.* In this technique, individuals receive feedback regarding a recent physical activity success with the aim of increasing motivation to be more active in future. This may take the form of an exercise trainer commenting on, supporting, or critiquing a client's physical activity goals.

20. *Informing when and where to perform the behavior.* In this technique, individuals are offered advice and ideas on when and where physical activity could be performed (e.g., providing suggestions on local exercise classes or gyms, local recreation parks, or even to and from the workplace).

21. *Instruction on how to perform the behavior.* In this technique, a person is instructed on exactly how to effectively perform a behavior (e.g., advice on technique in the gym or instruction on correct frequency or duration of cycling to work).

22. *Demonstrate behavior.* In this technique, the person is shown how to perform an activity. (A gym instructor might give a customer a demonstration of a particular exercise or piece of equipment.)

23. *Training to use prompts.* In this technique, individuals are given instruction on use of cues as a reminder to perform a behavior. (Encouraging exercisers to use frequently occurring everyday events like a particular time of day or mobile phone alerts, reminds them of the need to initiate their physical activity routine.)

24. *Environmental restructuring.* In this technique, the individuals are prompted to make changes to their environment in order to facilitate

changes in behavior (e.g., informing friends so that they may help prompt physical activity or removing tempting snacks or treats to help maintain weight loss).

25. *Agreement of behavioral contract.* In this technique, a written agreement between the individual and the practitioner with respect to behavior change is established. (A trainer and client may sign a contract explicitly stating the agreed activities or exercises so there is an explicit record and a sense of commitment on the part of the practitioner and client.)

26. *Prompt practice.* In this technique, the person is reminded to rehearse and repeat the behavior, or situations that lead to the behavior, helping make the behavior more automated or habitual (e.g., providing people with means to rehearse when they are going to do their exercise routine).

27. *Use of follow-up prompts.* In this technique, a set of reminders is delivered to a person that has started a behavior change routine in order to help remind them to continue. Over time, as the person becomes better at performing the behavior, reminders and prompts are reduced. These could include providing people with a personal alarm, e-mail, text message, or other reminder to help them recall their physical activity regimen or goal.

28. *Facilitate social comparison.* In this technique, individuals are encouraged to draw comparisons with others' behavior to increase motivation through modeling. (A person is encouraged to observe other regular exercisers to provide a positive example of technique or commitment.)

29. *Plan social support.* In this technique, the person is prompted to elicit social support from other people and close relations in order to facilitate successful completion of the behavior. This may take the form of joining clubs or groups involved in physical activity.

30. *Prompt identification as role model.* In this technique, the person is encouraged to view oneself as an example or role model to others for the behavior. This includes opportunities for the person to persuade others to adopt the behavior because of the benefits inherent in it (e.g., urging friends and family to engage in more physical activity).

31. Prompt anticipated regret. In this technique, expectations of shame, regret, or guilt for failure to accomplish the goal are induced. (The exercise specialist might encourage an exerciser to think about how guilty one would feel if one missed the next gym session.)

32. Fear arousal. In this technique, fear-inducing information aimed at motivating change is provided. (If weight loss is a goal, practitioners might highlight the health risks of being overweight or obese.)

33. Prompt self-talk. In this technique, the person is encouraged to use self-talk before and during activity to provide verbal encouragement and support. (An exercise practitioner might encourage individuals to use mantras or motivational words when they are finding their exercise routine difficult.)

34. Prompt use of imagery. In this technique, the person is provided with instruction on how to use visualization techniques and imagery to facilitate successful completion of the behavior (e.g., imagining completing a given walking distance or lifting heavier weights).

35. Relapse prevention. In this technique, the person is prompted to focus on situations or occasions in which one may relapse, and then develop methods to increase chances of success. (A person following a jogging routine may highlight bad weather as a possible barrier to maintaining the exercise; encouragement to use a treadmill in the gym on cold or rainy days would be a solution.)

36. Stress management. In this technique, the person is encouraged to focus on reducing related stress and improving emotional control in order to reduce these as a barrier and promote health. (A counselor or helper may provide therapy to an individual attempting to increase activity levels in order to reduce the effect of negative emotions on their behavior.)

37. Motivational interviewing. In this technique, specific interviewing techniques to prompt changes by minimizing resistance and resolving ambivalence to change are used. (A therapist may need to express empathy and provide opportunities for the client to express personal reasons to take up exercise.)

38. Time management. In this technique, the person is assisted in managing time efficiently to be able to engage in the desired activity (e.g., using a diary or organizer to plan time).

39. Communication skills training. In this technique, individuals are directed toward improving communication skills and improving interactions with others concerning the behavior. This often involves group work and focuses on listening skills and assertive oration. (An exercise specialist may encourage clients to engage in brief role play, encouraging an exercise partner to come to the gym or go to the local swimming pool with them.)

40. Stimulate anticipation of future rewards. In this technique, individuals are encouraged to consider future rewards associated with the outcome(s), without necessarily reinforcing behavior change (e.g., getting people to consider the possible gains of exercise, including extrinsic rewards like money and intrinsic rewards like satisfaction).

Implications of CALO-RE

The CALO-RE taxonomy provides a comprehensive and standardized protocol for the identification, reporting, and appraisal of behavior-change interventions for health behaviors, including physical activity. The taxonomy provides a common language for the scientific communication of theoretical-based interventions, which not only helps researchers effectively describe, interpret, and code behavioral-change strategies in interventions that directly link to theoretical constructs, but also establishes a basis for practitioners to accurately evaluate and apply behavior change techniques derived from previous research. The taxonomy assists in the systematic accumulation of knowledge of behavior change techniques from previous research trials and improves the precision for the implementation of multiple theoretical frameworks applied for the promotion of physical activity.

The development of the taxonomy offers a number of avenues for future research and applied practice in behavior-change interventions for physical activity. First, CALO-RE provides a rigorous and systematic procedure that helps correctly map theoretical constructs into behavioral change techniques, so it becomes more realistic to test the effectiveness of individual theoretical components within an intervention. Second, interventions can be optimized by identifying and eliminating

overlapping or redundant elements of the behavioral change techniques driven by multiple theoretical components or frameworks. Third, the enhanced connection between theoretical knowledge and behavioral change techniques may enable researchers to reveal the underlying mechanisms of the intention and behaviors of physical activity.

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See also Enjoyment, as a Mediator of Exercise Behavior Change; Interventions for Exercise and Physical Activity; Multiple Behavior Change; Self-Efficacy; Self-Regulation; Social Cognitive Theory

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CARDIAC FUNCTION

Cardiac function refers to the contraction of cardiac muscle, which works as a pump to send blood throughout the circulatory system to provide adequate blood flow to organs and tissues. The heart achieves this by contracting its muscular

walls around a closed chamber to generate sufficient pressure to propel blood from the left ventricle, through the aortic valve, into the aorta, and throughout the body. To accomplish this, the heart must maintain systolic and diastolic blood pressure and cardiac output (amount of blood ejected from the heart in one minute). Each time the heart beats, a volume of blood is ejected. This stroke volume (SV), times the number of beats per minute (heart rate [HR]), equals the cardiac output (CO). Normal, resting CO differs among individuals of different size. The resting CO of someone who weighs 240 pounds is greater than the cardiac output found in a person who weighs 120 pounds. Thus, measured values for CO are expressed as a flow (liters per minute [L/min]) per body surface area (m²). When cardiac output is expressed in this way, it is termed *cardiac index* and has the units of L/min/m². The surface area is estimated from calculations based on body weight and height (body surface area = 0.007184 × W^{0.425} × H^{0.725}). Cardiac index normally ranges from 2.6 to 4.2 L/min/m².

There are numerous measures that can assess cardiac function. More extensive analyses include simple and noninvasive to more complicated and invasive tests of cardiac function. Noninvasive tests include chest x-ray analysis, electrocardiography (ECG), and echocardiography. Invasive tests include cardiac catheterization, Thallium scanning, and for patients with coronary artery disease, pharmacologic or exercise stress ECG, pharmacologic or exercise stress myocardial perfusion imaging (MPI), electron beam computed tomography (EBCT), and positron emission tomography (PET). These techniques allow for the determination of global cardiac muscle function analyses, volumetric analyses (calculation of left ventricular functional parameters like end-diastolic volumes and ejection fraction), and regional wall motion abnormalities. In addition, the function of the heart as a pump can be determined by the force of ventricular muscles during systole and is directly affected by the preload and contractility of the heart. Preload is the *load* on the cardiac muscle during diastole (relaxation). This preload (end-diastolic volume) alters the ability of the heart to change its force of contraction and therefore increases stroke volume. This response to changes in venous return is called the Frank-Starling mechanism. Contractility can also be increased with an increase in calcium release into the cell. This can occur in response to catecholamines (epinephrine

and norepinephrine), which are released by the sympathetic nervous system. Cardiac activation is often a result of sympathetic activation from the central nervous system.

Blood pressure indirectly provides a measure of cardiac function. As blood is pumped out of the left ventricle into the arteries, pressure is generated. The mean arterial pressure (MAP; the average pressure within an artery over a complete cycle of one heartbeat) is determined by the CO, systemic vascular resistance (SVR), and central venous pressure (CVP): $MAP = (CO \times SVR) + CVP$. At normal resting heart rates, MAP can be approximated by the following equation: $MAP = DBP + 1/3(SBP-DBP)$. MAP can indicate a level of cardiac function at rest.

In sport and exercise psychology, measures of cardiac function provide specific information regarding the effects of stress, both physical and psychological, on cardiac function. Elevations in cardiac function in physically healthy individuals indicate sympathetic nervous system activation. This can occur in response to perceived threat in a competitive challenge or in response to meeting the physical demands of movement. In addition, measures of cardiac function are often used to identify the physiological demand or intensity of an activity. This can be done to ensure that the activity is appropriate and safe and to examine changes in cardiac function following physical training, treatment, or an intervention.

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See also Autonomic Nervous System; Biofeedback/Neurofeedback; Neuroscience, Exercise, and Cognitive Function; Psychophysiology; Stress Reactivity

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CAREER ASSISTANCE PROGRAMS

In order to prepare athletes for and support them during transitions, career assistance programs (CAPs) have been developed by elite sport organizations like a national sport governing body, universities, and private organizations. These CAPs generally consist of an integrated and comprehensive combination of workshops, seminars, educational modules, individual counseling, and referral networks providing individualized or group-oriented multidisciplinary support services to athletes with regard to their athletic participation, developmental and lifestyle issues, and educational and vocational development. Target groups for CAPs include prospective junior athletes, student-athletes, elite senior athletes, and retiring or retired athletes.

Some of the early established CAPs include the International Olympic Job Opportunities Program (OJOP), the Canadian Olympic Athlete Career Centre (OACC), and the Athlete Career and Educational Program (ACE). The OJOP was initiated in Australia, South Africa, and the United States to develop and create job opportunities for (potential) Olympians, by identifying job positions and providing a professional network, career analysis services, personal aptitude tests, and interview skills training. The OACC assisted athletes through the retirement transition by providing career and education planning, such as preretirement planning, clarification of career planning needs, life skills training, transition workshops, and a shadow program. The ACE provides career and education services for Australia's elite athletes, including career counseling and planning, personal development training courses, educational guidance, employment preparation, career referral networks, transitional support, online services, referrals, and lifestyle management. High Performance Sport New Zealand's Athlete Life Programme allows its elite athletes to work with an athlete life advisor to manage their sport lifestyle, career and education, personal leadership skills, and finances to minimize constraints and maximize opportunities that have the potential to impact sport performance.

During the past decade, several countries around Europe also established CAPs. In the United Kingdom, UKSport developed the Performance Lifestyle support service providing

athletes the necessary skills to cope with the special demands of being an elite performer and to better prepare them for their life after sport. The British Athletes' Lifestyle Assessment Needs in Career and Education (BALANCE) was designed to identify adjustment difficulties on 12 specific factors related to career retirement, including identity as an athlete, degree of occupational planning, transferable skills, the availability of career transition support services, experience with transitions, and social support and mentoring. In France, the National Institute of Sport, Expertise, and Performance (INSEP) developed a project (*Projet de vie du sportif de haut niveau*), which includes career management support, support aimed at supporting elite athletes' social and cultural life, vocational support, and lifestyle management. In Belgium, the Vrije Universiteit Brussel developed the career support program (*Carrièrebegeleiding*) for Bloso, the Flemish governmental administration in charge of elite sport. This CAP is based on a developmental lifespan perspective on the athletic career delineating normative (predicted and anticipated) transitional challenges that athletes will face throughout, as well as after, their athletic career and that will influence their development. It provides career assistance services focusing on the major transitions and career stages of the athletic (high-ability, elite athletes), the academic (primary, secondary, higher), the vocational (elite student-athletes, professional athletes), and the postathletic career (retiring and retired athletes). In 1988, the Vrije Universiteit Brussel was also the first Belgian university to establish a top-level sport and study department focused exclusively on supporting elite student-athletes to achieve academic and athletic excellence by providing services related to education and career management and life skills training. In Sweden, a particular emphasis is put on the combination of elite sports and education through a close collaboration between elite sport and educational institutions (e.g., national and certified elite sport schools). In the Netherlands, the Dutch Olympic Committee and Dutch Sports Federation (NOC*NSF) focus on providing services related to the combination of school and elite sport with specialized secondary and vocational schools and centers for elite sport and education, as well as for coping with career transitions and the postathletic career. This includes the project *Goud op de werkvloer* (Gold on the work floor), which, in collaboration with a private employment firm, aims at

optimizing the vocational development of (retired) elite athletes and includes an active mediation between athletes and employers via, among others, specific job interview sessions for athletes, and the provision of flexible work conditions. In Spain, centers of high performance sport (Centro de Alto Rendimiento; CAR) provide specific services to talented and elite athletes. For example, the CAR Sant Cugat assists athletes in combining training and education (mentoring, tutoring), with access to employment (working experience) and personal development (retirement, finances). In Finland, the Finnish Olympic Committee collaborates with the Ministry of Education and Culture and a private employment firm in the Athlete Career Program (ACP). This CAP supports athletes in their combination of education and an athletic career and in (eventually) entering the labor market, as well as in finding a balance between sport and education and other areas of life and in coping with transitions during their athletic career.

Finally, it is noteworthy that the significance of CAPs has also been recognized at the European level. As part of a hearing on future European Union policy on sport, members of the European Parliament discussed the need to establish career assistance services for talented, elite, and retired athletes. Using the developmental lifespan perspective on the athletic career, recommendations and good practices were presented on the combination of a sporting career with education or work, on athletic retirement, and on the postathletic career. In parallel, the Sport Unit of the Directorate-General for Education and Culture of the European Commission developed, in consultation with European experts on elite sport, career development and career assistance services, specific guidelines to support the dual career of talented and elite athletes in the European Union member states. These include guidelines on, among others, sport academies and high-performance training centers, education (secondary, vocational, higher, and distance learning), health (medical support), employment (combination of work and sport; transition to a new job), finances (scholarships), and supporting services (career assistance; educational guidance).

Career Assistance Services

In light of the growing need for the provision of services to athletes, CAPs have been established

worldwide. A survey among 27 CAP providers from Europe, North America, Oceania, Asia, and Africa showed that career assistance services generally served two aims: (1) providing assistance with transitions occurring during the athletic career and (2) providing assistance with the end of the athletic career and transition into the postathletic career. The services provided by these CAPs generally included education management, life skills training and lifestyle management, career management, financial management, and health management.

Sport psychologists were found to be part of the service provision at these CAPs, and were in general related to performance issues, life skills, and lifestyle management. In 82% of these CAPs, services were directed toward senior athletes, 70% toward retired athletes, and 59% toward junior athletes. The services provided by sport psychologists to senior athletes were related to goal setting (in 78% of CAPs), time management (in 74% of CAPs), media skills (in 70% of CAPs), coping and organizational skills (both in 67% of CAPs), and communication (in 63% of CAPs). With retired athletes, sport psychologists worked on the transition out of elite sport (in 74% of CAPs); goal setting (in 44% of CAPs); and communication, time management, problem solving, and organizational skills (all in 41% of CAPs). Sport psychologists provided junior athletes with services related to goal setting and time management (both in 63% of CAPs), communication (in 58% of CAPs), and media (in 56% of CAPs). Finally, sport psychologists were also found to provide clinical services to senior (in 41% of CAPs), retired (in 37% of CAPs), or junior athletes (in 33% of CAPs).

These findings reveal that sport psychology support provision aimed at enhancing junior, elite, and retired athletes' coping with transitional challenges has become an essential part of career assistance services.

Conclusion

The developmental lifespan perspective on the athletic career delineates the normative transitional challenges athletes will face throughout as well as after their athletic career. In order to enhance athletes' ability to cope successfully with these transitions, career assistance services should be provided in a structured way, based upon a developmental (from young to former elite athlete) and holistic (development in different domains) approach.

Elite sport governing bodies, such as national Olympic committees and international sports federations, are encouraged to take leadership in not only acknowledging the influence of career transitions on elite athletes and Olympians, but also in the provision of multilevel career assistance via well-structured multidisciplinary CAPs.

Sport psychologists have been shown to play a significant role in the provision of career assistance services to talented and elite, as well as retired, athletes. Building on the provision of psychoeducational services aimed at enhancing athletes' coping with normative transitions (e.g., junior-to-senior transition, first national team selection, transiting out of secondary education, athletic retirement), sport psychologists can already provide for the long-term career development needs of talented, elite, and retired athletes. Further efforts should be made to develop sport psychology services assisting athletes to successfully prepare for and cope with nonnormative transitions, such as a season-ending injury, unanticipated deselection from the team, or loss of a personal coach.

Finally, while CAPs have generally been developed for athletes, elite sport governing bodies as well as sport psychologists should envisage providing career assistance services to elite coaches.

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See also Career Paths; Career Transitions; Developmental Considerations; Identity; Parenting; Roles; Talent Development; Transition

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CAREER PATHS

For individuals seeking a career in sport psychology (SP), there are numerous options for those who have earned master's and doctoral degrees in SP. Most individuals with master's degrees work in athletic academic advising, as teachers or coaches, as licensed professional counselors (LPCs) within certain specialized areas, or go on to pursue a doctoral degree. The two primary career paths for those with doctoral degrees are in the clinical or counseling psychology profession or in research and teaching in academia.

A master's degree in SP may lead to one of five main options (excluding pursuing a doctoral degree).

1. *Athletic academic advising.* Most U.S. universities are members of the National Collegiate Athletic Association (NCAA). All universities in the NCAA's Division I are required to operate athletic academic advising entities, which provide academic counseling for their student-athletes. Athletic academic advisers work with student-athletes on course scheduling, study skills, and time management, but may also address a variety of life skills with their student-athletes. The National Association of Academic Advisers for Athletics (www.nfoura.org) is the main professional association for athletic academic advising. An SP's background is an excellent preparation for this career.

2. *Teaching and coaching.* Depending on one's academic interests (psychology or other areas), an SP degree allows one to work potentially more effectively as a school teacher, coach, or both and may allow for private practice work in performance enhancement consulting. The master's degree may also potentially increase one's salary by having obtained a postgraduate degree.
3. *Fitness and performance training for the military.* A relatively new and very exciting career route is with the military through the Comprehensive Soldier Fitness–Performance and Resilience Enhancement Program. These individuals work at U.S. Army bases with soldiers to better prepare them for their combat experience, as well as with the soldiers and families when the soldiers return from combat (Wounded Warriors; transitioning back into the community).
4. *Licensed professional counseling.* Working as an LPC (termed differently in different states) requires a clinical or counseling psychology master's level degree. The trained individual is licensed, certified, or accredited, depending on the state, to work within psychology, and get paid, including third-party reimbursement.
5. *Consulting.* Positions developed by individuals to suit their skills—niche creation—may include performance enhancement consulting in educational sport psychology work with athletes and recreational sport participants. This could also involve work in a health club or fitness center, in a research position, or as a wellness or life coach. The possibilities are only limited by an individual's imagination.

Career paths for holders of a doctoral degree center on work as a licensed psychologist or work in academia. Obtaining a doctorate, either a doctor of philosophy (PhD) or a doctor of psychology (PsyD) degree in clinical or counseling psychology provides the clearest route toward becoming a licensed psychologist and being able to work competently, ethically, and effectively as a sport psychologist. Such individuals may be found in private practice (individual or group practice) or in a more formal position. Individuals in private practice generally see clients across a number of areas within clinical or counseling psychology and develop the sport component of their practice as

an added area of expertise. Although very few individuals have been able to become sport psychologists with 100% of their clientele being sport and exercise participants, it remains a possibility. Such professionals see clients, individually or in groups, for areas that can encompass performance-related skills such as goal setting, arousal control, imagery, attentional control, and self-talk but also clinical issues such as eating disorders, substance abuse, anger management, relationship issues, and so on. They may conduct workshops on a variety of sport psychology-related topics for teams. It is more common for psychologists to work with nonsport populations for part of their practice and sport participants for the other part. This may be by choice or because of client demand in the geographical area in which the psychologist practices.

Making a living as a psychologist in private practice is challenging, and many individuals opt for more formal positions such as in a university setting, perhaps within an athletics department (there are a limited number of these opportunities), or more often, as part of a university or college counseling center working part of the time with athletes. There are many different models for how this can work, such as half-time in the counseling center and half-time in athletics. The more elite positions are those as sport psychologists with the United States Olympic Committee (there are six such positions in the United States, although there are, of course, positions available with Olympic teams in other countries) or with professional teams. However, even though there are hundreds of professional teams across the various sports and levels (including minor leagues), there are very few sport psychologists who have full-time positions. More commonly, sport psychologists in private practice (or in academia) consult with such teams as needed.

What qualifies licensed clinical or counseling psychologists to call themselves sport psychologists? Although ethical standards require one to only work within one's area(s) of competence, some clinical or counseling psychologists believe that, because they are avid exercise and sport participants, this automatically provides them with the expertise to work as a sport psychologist. However, although they may be excellent clinical or counseling psychologists, there is a body of knowledge and research that the profession of SP believes is necessary to work effectively

as a sport psychologist. Division 47 (Exercise and Sport Psychology) of the American Psychological Association is working on guidelines to facilitate developing one's competency in SP. For the moment, the process of becoming a Certified Consultant within the Association for Applied Sport Psychology (CC-AASP) provides the clearest path for developing a level of competency. The CC-AASP process does not have a clinical or counseling focus but rather a performance-enhancement focus. The CC-AASP certification process, as well as AASP ethical guidelines, specify the performance enhancement focus of this certification. Thus, clinical or counseling psychologists and academic sport psychologists are aware of the limitations and extent of this certification.

There are a few clinical or counseling psychologists who work in academia, generally within psychology departments. The majority of individuals within academia in SP are found through the kinesiology route, which is another primary career path for those more interested in an academic career. Primarily within kinesiology (or exercise and sport sciences or physical education or other such names), individuals with doctoral degrees in sport psychology from such departments remain within the discipline and teach and conduct research. Many of these individuals have consulting practices outside of their primary job within academia. A few of these positions are found in psychology departments, but these individuals have usually earned their degrees in psychology departments. These academic sport psychologists teach a variety of SP related courses at the undergraduate and graduate levels, and conduct research across a wide variety of topics within the field. They also engage in service to their university as well as to the profession (becoming an officer in a sport psychology organization such as AASP).

Some individuals work as performance psychologists, some with clinical or counseling psychology degrees and many as licensed psychologists, in occupations where performance is critical to success. These individuals may work with companies such as Cirque du Soleil, the New York City Ballet, Wall Street traders, or with medical school students, police, and firefighters. Although this encyclopedia focuses on sport psychology, it should be noted that many of the skills learned within this field are generalizable across other performance settings and, depending on one's background, one can find success in these other

performance domains. A number of books are available by those who have worked in these specialized settings, particularly dance and the performing arts. This area has become known as performance psychology, to acknowledge the emphasis on performance in general (in a variety of performance domains, especially the performing arts) rather than sport per se.

While the focus of many individuals (and most books in the field) is on sport psychology, particularly with high-level and elite athletes, there will be many opportunities in the future for work within exercise psychology. Increased attention is being paid to the high prevalence of obesity and sedentariness in the general public, and exercise psychologists can work effectively within health care settings to facilitate motivation and adherence to exercise and physical activity and counteract this trend, now widely recognized as a public health issue. These opportunities may come through academia or in private enterprise within health clubs or fitness centers, corporations, insurance companies, and other entities.

Although there are a few undergraduate programs in sport psychology, the main route to a career in the field, as noted earlier, is through graduate study. To become a clinical or counseling psychologist, the recommended route is to obtain a PhD or PsyD in clinical or counseling psychology, either as the next step after one's undergraduate program or after obtaining a master's in sport psychology from a kinesiology department. The academic route generally finds individuals with master's degrees in kinesiology or clinical or counseling psychology followed by a PhD in kinesiology, specializing in sport and exercise psychology. These individuals often work as performance enhancement consultants.

Deciding on the best career path can be a challenge, although the availability of numerous paths to one's goal can be advantageous for individuals with a variety of backgrounds. Full descriptions of a variety of educational programs are available, as well as other resources to assist the prospective sport psychologist, including extensive coverage of career opportunities in exercise and sport psychology and personal accounts by professionals in the field (see Further Readings).

Michael L. Sachs

See also Credentials

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CAREER TRANSITIONS

Career transitions are turning phases in athletes' career development. Therefore, they should be first defined in terms of their place in the career context. The broadest career term in psychology is a *life career* that encompasses an individual's life-long development and achievements in various activities and spheres of life that unfold in a particular historical and sociocultural context. In sport psychology, an *athletic career* is defined as (1) part of and a contribution to an athlete's life career, (2) a multiyear competitive sport involvement voluntarily chosen by an athlete and aimed at achieving a personal individual peak in athletic performance in one or several sport events, and (3) a sequence of career stages and transitions.

These definitions highlight the various views and facets of athletic careers. The first definition is based on a *whole person approach* that recognizes the athlete as a person doing sports and having broader life issues, such as education, work, family,

friends, and other. The second definition emphasizes a free career choice, a long-term commitment, striving for upward career movement, and a possibility of having a specialized or a generalized career. In specialized careers, athletes compete in one sport during the entire career, whereas in generalized careers, they practice and compete in several sports, either simultaneously or successively. Athletic careers can also be of different levels, such as local, national, international, and professional, depending on the highest level in competitions achieved by the athlete. International amateur and professional careers are also called *elite careers*. In the third definition, an athletic career is viewed as a developmental process that includes proceeding through career stages and coping with *career transitions as turning phases* in development.

As noted by Dorothee Alfermann and Natalia Stambulova, models that describe athletes' career development identify the following *athletic career stages*: initiation, development, mastery or perfection, maintenance, and discontinuation of competitive sport involvement. In these models, the stage-by-stage transitions are marked by changes in (a) athletes' perception of and attitude toward sport and the degree to which they identify themselves with the athletic role; (b) time and energy investment into sports; (c) athletes' social environment and the role of coaches, parents and family, and peers in career development; and (d) the degree to which they have to sacrifice in other spheres of life to accomplish their sport goals.

The aforementioned whole person approach contributes to a *holistic lifespan perspective* in understanding athletes' career development as a multidimensional process combining stages and transitions in their athletic, psychological, psychosocial, and academic-vocational development.

Transition Process and Outcomes

The transition process, the factors involved, and the transition outcomes are considered in the various career transition explanatory models. For example, in Stambulova's athletic career transition model, a transition is viewed as a process of coping with a set of *transition demands*, in which athletes use various *coping strategies* like planning, practicing more than their opponents, or seeking professional support. Effectiveness of coping is thought to be dependent on a dynamic balance between the coping resources and barriers. *Resources*

consist of the various internal and external factors that facilitate the transition, such as previous athletic and personal experiences or social and professional support available), while *barriers* include the various internal and external factors that interfere with the coping process, such as low self-efficacy, poor coaching, low quality equipment, or insufficient financial support. The model entails two primary transition outcomes: successful transition and crisis transition. *Successful transition* is a result of effective coping when the transition demands and the athletes' coping resources and strategies are evenly matched. *Crisis transition* is a result of ineffective coping when low resources, high barriers, and inappropriate coping strategies make it difficult to meet the transition demands. Crisis is also conceptualized as a transition that the athlete is unable to cope with independently and perceives a need for transition intervention. Furthermore, according to the model, the crisis transition can have two possible secondary outcomes: *delayed successful transition* (in case of effective intervention) and *unsuccessful transition* (in case of ineffective or lacking intervention) associated with *negative consequences* of avoiding coping with the crisis (e.g., premature dropout, neuroses, overtraining, eating disorders, and substance abuse). Career transition interventions outlined by the model include crisis-prevention, crisis-coping, and negative-consequences-coping interventions.

Major Types of Athletes' Career Transitions

Athletes' transitions are classified into athletic and nonathletic, as well as into normative and nonnormative.

Normative athletic transitions are relatively predictable and based on the logic of athletic development. Thus, the transition into organized sport, the transition to the development stage or more intensive training in chosen sport, the transition to the mastery stage or from junior to senior sports, the transition to professional sport, the transition to maintenance career stage, and the transition to the post-sport career are all examples of the normative athletic transitions.

Normative nonathletic transitions refer to athletes' transitions in psychological, psychosocial, and academic-vocational development, such as the transition from childhood to adolescence, the transition from living at home to living independently,

the transition to college or university, and the transition to the workplace.

Nonnormative transitions are less predictable, such as transitions caused by injury, divorce, moving abroad, or changing team or coach.

The predictability of normative transitions creates an opportunity for athletes to prepare for them in advance, while the low predictability of nonnormative transitions explains why athletes find these more difficult to deal with.

Normative Athletic Transitions

Among the normative athletic transitions, the most research has been done on the *transition from the development to the mastery stage* (also known as the transition from junior to senior and elite sports) and the *transition to the post-sports career* (athletic retirement).

The growing interest in the junior-to-senior transition can be explained by both the difficulty and the importance of this transition for athletes wanting to achieve an elite or professional level in sports. Vanden Auweele et al. showed that only 17% of elite junior athletes made a successful transition to senior or elite sports in 5 years of observation. International research showed that athletes perceive this transition as a big step associated with much higher standards in practice and performance than they experienced before. Issues outside sports are also very important, with studies and social aspects proving the most demanding. Athletes' ambitions to succeed in this transition and meet the expectations of significant others, together with uncertainty about success in coping, lead to high stress and increased sensitivity to social influences. Therefore, social support, especially from coaches, plays a pivotal role in the transition process. Coaches believe that coping strategies such as thoughtful problem solving, acceptance of responsibility, self-control, and positive reappraisal are beneficial to the transition success. Research also confirmed that successful coping with the junior-to-senior transition is associated with athletes' personal maturation and identity development.

The transition to the post-sport career or athletic retirement is the one inevitable transition that mixes athletic context reasons for termination in sports, athletic career satisfaction, and support from sport organizations with nonathletic context relevant to starting a new life after sports. Retired

athletes must accept retirement and adjust to the status of a former athlete, start or continue studies or work, reconsider their personal identity, and renew their lifestyles and social networks. The status of former elite athletes is typically lower than that of active sports heroes. Therefore, former athletes need to adjust to a substantial decrease in social recognition and support. Many of them channel their energy into education or work. Starting a new professional career is not only important for making a living, but also for the development of a new professional identity. Many athletes confirm that their social lives change greatly during this transition. They often keep their sport friends but need to involve more nonsport people into their social networks. Family becomes a very important part of the renewed lifestyle, especially if the athletic retirement transition is associated with the transition to parenthood. Athletes who plan retirement in advance, receive their education while still in sport, have multiple personal interests and identities, keep good family relationships, and feel in control of their lives usually adjust successfully to their post-sport careers within months. About 15% to 20% of athletes experience this transition as a crisis accompanied by unemployment, separation or divorce, health problems, alcohol abuse, and feelings of being empty, alone, and forgotten.

Normative Nonathletic Transitions

In terms of normative nonathletic transitions, researchers focus mainly on the transition from adolescence to young adulthood (around 18–19 years old), emphasizing the athletes' identity formation, and on their transitions within the educational system.

Adolescence is known as a crucial period for young people to answer the question "Who am I?" integrating their social roles and related identities. For athletes, their athlete role and athletic identity or the degree of personal identification with the athlete role are involved in this process. Researchers showed that high athletic identity works as a resource during most of the athletic career transitions (e.g., the junior-to-senior transition), but it turns into a barrier in the transition to the post-sport career. Athletes who develop multiple personal identities—athlete, student, employee, family member, friend—are less at risk for the athletic identity foreclosure (single-minded focus on sport) and, consequently, for the identity

crisis (self-misinterpretation) in the transition to the post-sport career.

The identity research in athletes has stimulated an interest in the so-called *dual career* studies that focus on athletes combining sport with education and their transitions within the educational system. Relevant research deals with student-athletes at sport classes, sport boarding schools, and colleges and universities. These studies emphasize the benefits of athletes' dual careers, such as balanced life-style, reduced stress, good conditions for developing life skills, and higher employability after sport, but also highlight the related demands and challenges, especially when athletes experience simultaneous transitions to a new level in both education and sport. For example, North American research on the student-athletes' transition to college showed that the incoming students have to meet new academic requirements and the challenges of living far from home, create relationships, manage their time and energy, as well as adjust to demanding sport participation at the intercollegiate level, new teammates, and coaches. They also should be ready to cope with athletic career termination upon graduation from the college. European dual career studies are marked by the holistic life span perspective in investigating student-athletes' sport, education, and private life issues.

Nonnormative Career Transitions

As mentioned above, the nonnormative transitions are less predictable and more idiosyncratic than the normative ones. This type of athletes' transitions has been acknowledged but not extensively studied. Examples of athletic nonnormative transitions include injury; overtraining; deselection; moving abroad to play sport; and changing a club, a coach, or a sport partner. Selection for and participation in the Olympic Games are also seen as nonnormative athletic transitions. Athletes' quasi-normative (expected in principle, but not in terms of their timing in life) or nonnormative transitions outside sport are marriage, divorce, parenthood, moving to a big city, immigration, and others. The lower the predictability of a non-normative transition, the more often athletes need professional assistance.

Helping Athletes in Career Transitions

The holistic lifespan perspective is currently an influential guide in *career assistance*, which is a

rapidly developing discourse in applied sport psychology aimed at helping athletes with various issues related to their careers inside and outside of sports.

Helping athletes to cope with athletic and non-athletic, normative and nonnormative transitions is a central aspect in career assistance provided by career assistance programs or by private sport psychology consultants. Career transition interventions are planned based on the career development and transition frameworks, as well as on the thorough investigation of the athlete-client's background, current situation and needs, and future plans (for further information see Stambulova, 2012).

Two major perspectives in career transition interventions, the preventive-supportive and the crisis-negative consequences coping perspectives, have been identified.

Preventive-Supportive Perspective

The *preventive-supportive perspective* covers interventions aimed at enhancing athletes' awareness about forthcoming or current transition demands and aiding in the timely development of all the necessary resources for effective coping. These interventions may improve the athlete's readiness for a normative career transition or support the athlete during the transition process. The following are the brief definitions of relevant intervention types.

- **Career planning interventions** are counseling interventions aimed at helping athletes increase their self-awareness; set realistic career goals bridging their past, present, and future; and prepare in advance for the forthcoming transitions.

- **Life development interventions and life skills training** consist of needs assessment, education, and training with regard to sets of transferable life skills, such as effective communication, dealing with success and failure, time, energy, and stress management, that are applicable in both sports and other spheres of life.

- **Lifestyle management interventions** consist of counseling, education, and training aimed at helping athletes combine sport and other activities in life, prioritize between them, and manage time and energy in a way that helps athletes maintain good health and well-being.

- **Identity development interventions** include counseling, education, and training aimed at helping athletes in self-exploration and development of multiple personal identities to minimize the risk for the athletic identity foreclosure and one-sided development.

- **Cultural adaptation interventions** consist of needs assessment, counseling, education, and training aimed at helping athletes adjust to the new sociocultural environment when moving to another country to play sport or to do both sport and studies. Practitioners help athletes increase their awareness of the new culture and find consensus between their previous values, perceptions, and habits and ones required and expected by the new culture.

Crisis–Negative Consequences Coping Perspective

The *crisis–negative consequences coping perspective* covers interventions assisting athletes in analysis of their crisis or traumatic situations and finding the best available ways to cope. This perspective is represented by the following interventions:

- **Crisis-coping educational interventions** are aimed at helping athletes analyze the crisis situation, generate alternatives in coping, make an action plan, and increase their self-efficacy to cope with the crisis. These interventions are useful at the beginning phase of the crisis when athletes feel distressed and disoriented but still do not experience any clinical symptoms.

- **Clinical interventions** are applied when athletes experience clinical symptoms related to overtraining, neuroses, psychosomatic illnesses, substance abuse, negative identities, eating disorders, anger and aggression, grief, clinical depression, and suicidal thoughts, which often are negative consequences of not coping with previous or current transitions. These are mostly counseling interventions based on various psychotherapeutic approaches, such as psychoanalysis, existential therapy, or cognitive-behavioral therapy.

Tracing various types of career transition interventions, it is easy to see some obvious overlapping among them. For example, exploring new social roles as a part of identity development interventions also contributes to the athlete's lifestyle management and life skills training. Such overlaps are not surprising because all the interventions deal

with various aspects of the transition process and adopt the whole person approach. These overlaps also facilitate combining various types of interventions to best meet the particular needs of each athlete.

Conclusion

The career transition area of research and practice in sport psychology has been built and structured mainly over the last two decades. Major developments in this area include sport-specific definitions of key concepts; classifications of athletes' transitions, related frameworks, and interventions; the holistic lifespan perspective and the body of knowledge about athletes' transitions and factors involved; career assistance; and practical experiences with related principles, values, intervention strategies, and tools. Future challenges for the topic can be seen in more ecological and culturally sensitive research on athletic and non-athletic, normative and nonnormative transitions using various methodologies and designs, such as narratives, longitude, cross-cultural, intervention, and case studies, and also in further promotion of career assistance services among athletes and coaches.

Natalia Stambulova

See also Career Paths; Coping; Talent Development; Transition

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CARING CLIMATE

Understanding and ultimately optimizing the experience of individuals involved in physical activity is a focal point in sport and exercise psychology. From a social cognitive perspective, perceptions of the psychological climate represent an individual's perceptions of what is valued, emphasized, and promoted in a particular setting. Recently, Maria Newton and colleagues have identified a facet of the psychological climate termed *the caring climate*. This entry will define and discuss the conceptual underpinnings of the caring climate, review the existing empirical literature, and provide suggestions for future research.

Newton and colleagues define the caring climate as the extent to which individuals perceive a particular setting to be interpersonally inviting, safe, supportive, and able to provide the experience of being valued and respected. Conceptualization of the caring climate is firmly ensconced in the work of educational philosopher Nel Noddings, who describes caring as a state of awareness by the one-caring characterized by engrossment and motivational displacement for the cared-for. Engrossment refers to open and nonselective receptivity demonstrated by fully accepting and listening to the cared-for. Motivational displacement refers to emphasizing the goals and aspirations of the cared-for in lieu of one's own needs. Thus, caring is relational in nature and characterized by

the one-caring taking time and energy to express authentic concern and connection with the cared-for.

In sport and exercise settings, coaches, group exercise leaders, personal trainers, and fitness specialists predominantly play the role of the one-caring while participants (teammates, clients) are the cared-for. Importantly, however, perceptions of a caring climate occur reciprocally and among all participants within a setting; members of a team can care for other teammates, opponents, and their coach (acting as the ones-caring).

Conceptually, perceptions of a caring climate are proposed to be positively associated with motivational striving, moral engagement, quality of relationships, and mental health. Perceptions of a caring climate are somewhat similar to participants' sense of belonging and social and relatedness goals emphasized in a particular setting but are philosophically different. The caring climate taps into a social-emotional sense of safety, closeness, and the feeling of being compelled to be of assistance to others, all of which make caring a distinct construct.

Emerging literature suggests that caring is an important element of the psychological climate of a setting and influential in understanding and maximizing the experience of individuals involved in physical activity. Newton et al. (2007) created the Caring Climate Scale (CCS), which measures the extent to which individuals perceive a specific context to be caring. Example items include "the leaders care about kids" and "everyone likes kids for who they are."

Perceptions of a caring climate are related to indicators of positive engagement and developmental outcomes in sport. Mary Fry and Lori Gano-Overway reported that athletes who perceived their team to be caring were more likely to have positive attitudes toward their teammates and coach, indicate greater enjoyment of and commitment to their sport, and were more likely to report acting in a caring manner toward others. In a sample of primarily African American underserved adolescent athletes, Daniel Gould, Ryan Flett, and Larry Lauer (2012) reported that perceptions of a caring climate were linked to positive developmental outcomes, most notably teamwork and social skills.

Studies have established links between perceptions of caring climate and mental well-being and social behaviors, while also providing evidence

for possible mechanisms that explain how caring operates to influence outcomes. Fry and colleagues found that perceptions of a caring climate have been associated with greater hope and happiness and less depression and sadness as well as increased prosocial behaviors and decreased antisocial behaviors. Gano-Overway and colleagues reported that a perceived caring climate influenced mental well-being and social behaviors through aspects of personal efficacy. When individuals are cared for, they are better able to monitor and regulate their emotions and feel more capable of empathizing with others. These beliefs result in greater personal well-being as well as more adaptive social behaviors.

While a majority of the research has focused on participant outcomes associated with perceptions of a caring climate, an interesting study by T. Michelle Magyar et al. explored the predictors of leaders' self-reported ability to create a caring climate. Leaders' emotional intelligence, specifically their ability to control their emotions and efficacy for implementing instruction, were significant and positive determinants of their own personal caring.

The importance of a caring climate in exercise settings is yet unknown and a warranted focus of future research. Extending the leadership findings is another intriguing area of future research. Lastly, well controlled intervention studies would be most beneficial in identifying the impact of creating a caring climate.

Maria Newton

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CENTERING

Athletes are often faced with a variety of factors that can throw off their focus on the athletic task, such as worry, unexpected events, and physical exhaustion. Therefore, it is important for athletes to be able to focus or refocus their attention on the athletic task, despite frequently changing circumstances. In this regard, *centering* is a process by which individuals direct their attention to a central portion of the body while engaging in deep rhythmic breathing, with the purpose of merging one's attention and one's center of gravity, and thus gaining or regaining focus.

As a technique, centering was historically incorporated into the martial arts as long ago as the first century BCE. At that time, centering emphasized focused thought, muscle relaxation, and rhythmic breathing. Centering has been used more recently in modern-day sports. By helping athletes focus attention on a particular central part of the body (such as the abdomen) as they engage in deep rhythmic breathing patterns, athletes can achieve a feeling of physical and mental groundedness and togetherness, which is thought to subsequently promote enhanced focus to the athletic challenges they face. When centered, a sense of *quiet mind* is created, as diverse thoughts and emotions are replaced by a narrow focus on one's own breathing and body. The narrowing of focus helps block out distractions both within the athlete (e.g., worry) and in the environment (e.g., crowd noise) and allows the athlete to concentrate more fully on the task.

As with any other skill, effectively developing the capacity to center oneself requires training and practice. Simply attempting to occasionally get

focused is not sufficient for the development of centering skills. For most individuals, learning to center one's body and mind takes concerted effort. As such, athletes are encouraged to frequently practice centering (in a variety of contexts, and even when tasks do not require a centered state) until centering competencies are developed. Centering training usually begins with 20- to 30-minute exercises, but, as greater competencies evolve, the time required to attain a centered state can be systematically reduced to several seconds. The technique can be used with eyes open or closed, and in a lying, sitting, or standing position. Therefore, once mastered, centering can be employed by athletes at any moment and in any context just prior to movement. For example, a golfer can quickly center prior to putting and a hockey player can reach a centered state just prior to a face-off.

During sport performance, there are two primary values of centering. The first value is to help the athlete minimize internal and external distractions by transitioning attention away from past-, present-, or future-oriented thoughts, emotional states, performance concerns, or plans, and toward the present moment. The second value is to help the athlete remain focused on the athletic task at hand. It is important to state that centering is often misconstrued as a method of relaxation, as the goal is often incorrectly assumed to be a reduction or elimination of emotions such as anxiety or anger or tense bodily states. Yet while the individual may feel more relaxed after centering, any increase in relaxation would be merely a secondary consequence. In actuality, centering is more similar to mindfulness meditative exercises, and in fact, centering historically originated from the same Asian philosophical traditions as mindfulness.

Mindfulness is a state of nonjudgmental awareness and acceptance of internal experiences, and mindfulness exercises help individuals consistently engage in behaviors that matter to them rather than abandoning their goals and athletic or general life values when negative thoughts, feelings, or physical sensations get in their way. In the proper use of centering, the individual is actually not seeking a state of relaxation or a reduction of any particular thoughts, emotions, or physiological states, but rather is seeking a focused preparatory state for competitive performance. From this perspective, centering is a means of self-regulating attention by bringing the mind to a quiet place in which attention is directed away from distractions and onto

the body. Once attention is placed on the body and distractions are minimized, the athlete is then better able to focus on the competitive moment and required athletic tasks. While centering can be used as a stand-alone technique, it is often incorporated into a more comprehensive mindfulness training practice. Yet regardless of whether it is used alone or as a component of mindfulness training, centering can have a marked positive impact on the attentional processes of the athlete and subsequent athletic performance. In fact, research has found that a centered focus of attention is highly correlated with flow states and athletic performance success.

Zella E. Moore

See also Attention Training; Breathing Exercises; Flow; Mindfulness; Relaxation

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CERTIFICATIONS

Professional certification serves to identify individuals who have obtained or maintained qualifications to perform a specific work responsibility or task. Furthermore, certification indirectly serves to safeguard the public interest by assuring that the public can identify qualified professionals. In the United States, the Association for Applied Sport Psychology (AASP) certifies master's and doctoral trained professionals as Certified Consultants (CC-AASP). CC-AASP professionals are qualified to assist with performance issues that affect people in all areas of sport and exercise. Although other certifications in sport psychology exist, this entry is focused on the CC-AASP certification because of its connection to the largest nonprofit international organization (AASP) committed to

promoting the *science* and *practice* of sport and exercise psychology, and because the CC-AASP certification requirements are consistent with the American Psychological Association (APA) Division 47 Proficiency in Sport Psychology.

Certification and Licensure

Certification is the process by which a nongovernmental organization, in this case AASP, grants recognition to an individual who has met predetermined qualifications specified by that organization. The process of certification is voluntary. Certified individuals have demonstrated the level of knowledge and skill required in the profession, and the certification serves to identify the occupation, role, and skills to the public and stakeholders. Professional practice can be governed by legal requirements. When this is the case, professionals in practice must be licensed. Licensure is given to one by the state. *Licensure* is the state's grant of legal authority, pursuant to the state's police powers, to practice a profession within a designated scope of practice. States define, by statute, the tasks and scope of practice of a profession and provide that these tasks may be legally performed only by those who are licensed. Licensure prohibits anyone from practicing the profession who is not licensed, regardless of whether or not the individual has been certified by a private organization. To date, the scope of practice defined by AASP certification is not governed by legal requirements.

The most common services provided by CC-AASP professionals include educating individuals, groups, and organizations about the role of psychological factors in sport, exercise, and physical activity, and teaching participants specific mental, behavioral, psychosocial, and emotional control skills in these physical contexts. To date, these activities are not governed by legal requirements, thus there is no requirement for practicing professionals to be licensed to provide these services. Moreover, APA Division 47 Exercise and Sport Psychology has also identified the specific knowledge and training necessary for ethical professional practice in the exercise and sport context. The requirements for CC-AASP and the Division 47 Proficiency in Sport Psychology are consistent and provide clear descriptions of a model for appropriate training for service providers in applied sport and exercise psychology.

Certification assists in defining the professional and ethical responsibility of individuals

participating in consulting in applied sport and exercise environments. In addition, certification offers the public a definition and example of the training necessary for quality service. This in turn provides greater public understanding of the importance and possible impact of the application of psychology to sport and exercise. An example of this important role for CC-AASP certification is that the United States Olympic Committee Sport Psychology Registry requires CC-AASP certification for inclusion on the registry to work with U.S. Olympic athletes.

Requirements for Competent and Ethical Practice

CC-AASP and Division 47 Proficiency in Sport Psychology requirements specify that extensive disciplinary knowledge in sport and exercise psychology is required for professionals to competently practice applied sport psychology. Although consistent, the two sets of requirements differ in that CC-AASP requirements rely upon clearly defined necessities in coursework, whereas the Division 47 Proficiency requirements specify content area necessities for the ethical practice in sport psychology. For both the CC-AASP and the Division 47 Proficiency in Sport Psychology requirements, professionals must have knowledge in the following areas: professional ethics and standards; biomechanical or physiological bases of physical activity; historical, philosophical, social, or motor behavior bases of physical activity; psychopathology and its assessment; counseling skills; research design, statistics, or psychological assessment; biological bases of behavior; cognitive-affective bases of behavior; and individual behavior. For CC-AASP, the majority of the courses must be taken at the graduate level and the individual must have completed a master's or doctoral program. Furthermore, adequate training must include demonstrated competence in the field with mentored structured experiences. Finally, applied sport psychology professionals must show evidence of continued learning to maintain professional expertise. A process of certification renewal exists in the case of CC-AASP to support this end. One important side note is that the Division 47 Proficiency in Sport Psychology has included the knowledge of organizational and system aspects of sport consulting as a content area that is not inherently embedded within the specified coursework requirements for CC-AASP. Nonetheless, the overall uniformity across CC-AASP and Division

47 Proficiency in Sport Psychology requirements supports a clear and consistent depiction of the professional expertise required to practice applied exercise and sport psychology.

Commentary on Necessity for Clearly Defined Expertise and Practice

The AASP certification process initiated in 1989 and has certified over 250 professionals. One could argue that, for a licensed mental health care practitioner, there is little motivation to become CC-AASP certified. Yet there is growing interest in Applied Sport Psychology, and ethical practitioners committed to competent professional practice continue to seek opportunities to learn more about sport and exercise psychology and document professional credentials. A goal would be to increase the number of Certified Consultants so that the number reaches a critical mass that provides an awareness of this credential to the general public. One challenge that exists for professionals in the field is the difficulty in meeting specific coursework requirements. As the academic discipline continues to grow, there is the possibility for accessing online courses. In addition, AASP may consider approving continuing education credits to fulfill knowledge requirements. These as well as other options must be sensitive to maintaining the quality and rigor of the certification coursework requirements. Aoyagi, Portenga, Poczwardowski, Cohen, and Statler (2012) have argued that another possibility for increasing access and feasibility of CC-AASP certification for professionals in the field is to consider a certification exam. Again, the intent would not be to reduce the rigor of the certification process; in fact, those seeking the exam option could still be required to have a graduate degree and specific coursework. However, the exam option would allow for standardized assessment of the core content knowledge while offering a more accessible option for practitioners. An exam might offer a feasible alternative to the current process without diminishing the quality or meaningfulness of the certification.

International Developments in Professional Practice

There is growing international interest in defining the appropriate knowledge and experience for the professional practice of applied exercise and sport psychology. The terms used to identify and define

professional practice and credentials (*accredited, chartered, registered, certified, licensed*) vary from country to country. However, the criteria identify the unique professional expertise required in sport and exercise. For example, the British Association of Sport and Exercise Sciences has an accreditation for sport psychology with the title *Accredited Sport and Exercise Scientist*. In addition, the British Psychological Society Division of Sport and Exercise Psychology has criteria for individual professionals to become Chartered Sport and Exercise Psychologists. Furthermore, the Australia Psychology Society College of Sport Psychologists has qualifications beyond those required for basic registration that requires a minimum of 6 years of university training, plus 2 years of supervised practical experience in sport psychology. It is evident that credentials for the expertise are in demand and that this trend is likely to continue and extend across the globe.

Certification provides a clear definition of the qualifications necessary to practice sport and exercise psychology. CC-AASP certification is open to licensed clinical professional counselors, licensed clinical social workers, individuals with master's level (MS, MA, MEd, or EdM) counselor licensure, individuals with doctoral degrees (EdD, PsyD, or PhD) generally from Council for Accreditation of Counseling and Related Educational Programs (CACREP) or APA accredited programs, or individuals from MS or PhD programs in Exercise and Sport Psychology. CC-AASP is a credential (not a licensure) from a professional organization that oversees standards in practice and training. CC-AASP without licensure can limit professional opportunities to those that focus on performance enhancement or mental skills coaching. However, CC-AASP certified professionals who also hold a license are well equipped to help a range of individuals with a range of mental health problems often linked to human performance decrements. The CC-AASP credential is becoming increasingly meaningful within the world of sport and exercise psychology, and is often preferred by employers seeking sport and exercise psychology expertise.

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See also Consultant Effectiveness, Assessment of; Consulting; Credentials; Ethics; Services; Supervision; Training, Professional

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CHARACTER DEVELOPMENT

The term *character* has a long, complex, and controversial history. An old adage holds that “sport builds character” and historically the inclusion of sport programs in educational settings often has been justified through appeal to their supposed character-building efficacy. The following entry elaborates on the meaning of character and discusses how sport researchers have dealt with this complex concept.

What Is Character?

Etymologically, the term *character* comes from the Greek *charaktēr*, meaning “indelible mark.” Over the centuries, the term evolved and took on various and shifting connotations, yet it continued to carry the meaning of enduring qualities that mark a person. Today, there is no shared agreement on the definition of character. Some, for example, use it simply as a synonym for personality, while others promote a more narrow interpretation of character, limiting it to the moral dimensions of personhood.

Following the lead of the Character Education Partnership, the nation’s leading umbrella organization of character educators, we elaborate a middle-road view of character that defines it more broadly than moral character yet not as broadly as including all dimensions of personality. For present purposes, let us define character as the set of psychological characteristics that motivate and enable the individual to seek truth and goodness. Character is rooted in desire for what is true, right, and good, and a honed ability to translate that desire into effective action. The term *virtue* is often used to describe a quality of character when it is well established. In turn, let us describe character development as the learning and development that individuals undergo as they acquire and solidify the requisite dispositions, competencies, and skills of character.

Character is evident in action, but a person’s character does not fully determine behavior. Action arises from complex interactions among personal and situational influences, and sometimes temptations or pressures may overpower the tendencies that arise from one’s dispositions. Nonetheless, research has demonstrated that character can be an important influence on behavior across contexts and situations.

Character can be elucidated in terms of four forms or dimensions of character: intellectual character, moral character, civic character, and performance character. All four are rooted in stable dispositional tendencies. They are elaborated below.

Intellectual Character

In his influential book, *Intellectual Character*, Ron Ritchhart describes intellectual character as a set of dispositions that both shape and motivate intellectual behavior. The dispositions that make up intellectual character include curiosity, open-mindedness, and truth seeking. Those with well-developed intellectual character are curious about the world around them, think critically, and are constantly seeking a deeper understanding. Athletes with exemplary intellectual character, for example, may ponder questions such as how sports influence broader society; what responsibilities elite athletes bear; how race, gender, and class limit opportunities; and so on. While it may be hypothesized that intellectual character contributes to a range of positive actions in sport settings,

sport researchers have left this dimension of character largely unexamined.

Moral Character

When people talk of character, most often they mean moral character. Moral character refers to those psychological characteristics that motivate and enable the individual to function as a competent moral agent. These include moral reasoning competency and perspective-taking ability, empathy and moral identity, value hierarchies, and moral belief systems. At root, moral character is grounded in a basic and fundamental desire to do what is good and right. A person of high moral character is one who has a disposition to focus on moral dimensions of situations and act consistently with considered moral convictions.

Often, moral character is most evident in situations where values collide and choices need to be made. For example, an athlete may believe that cheating is wrong and simultaneously want to gain the benefits of victory. If moral character is strong, the moral concerns will take precedence over the more self-aggrandizing thoughts. It is important to recognize, however, that even a person with well-developed moral character may not be completely consistent in thought and action. All people, it is reasonable to assume, fail to resist the temptation to depart from their better selves on occasion (see “Performance Character” below).

Civic Character

Civic character refers to those psychological characteristics that motivate and enable the people’s active, prosocial involvement in the groups to which they belong. A person with strong civic character tends to be engaged in communities and society, seeking to improve them. People with strong civic character tend to think in terms of shared interests and the common good. An athlete who seeks voluntary leadership roles on the team, not for selfish reasons but to serve, is likely to have well-developed civic character.

Performance Character

A dimension of character that has received growing attention in recent years is performance character. This dimension of character refers to all of those psychological characteristics that motivate and enable the individual to function effectively, efficiently, and competently as a value-pursuing

agent. People with strong performance character have an increased likelihood of accomplishing their goals. Performance character includes such qualities as determination, perseverance, resilience, optimism, courage, loyalty, and attention to detail. Performance character reflects skills of self-management. It is important to recognize, however, that the qualities that make up performance character, while sometimes called *virtues*, are in themselves neither good nor bad. Is it good, for example, to be loyal to an ignoble leader or persistent in trying to rob a bank? The value of performance character depends on the goals served. When combined with a well-developed moral character, performance character enables a person to resist distractions and temptations and act effectively in the moral domain.

Character Development

How the multiple dimensions of character develop is a topic of considerable dispute among developmental and educational psychologists. The two most influential schools of thought have their origins in behaviorist and constructivist approaches to psychology.

The Behaviorist Approach to Character Development

According to behaviorists, and those influenced by them (such as social learning theorists), character development is primarily under the molding influence of the environment. From this perspective, the dispositions that constitute character are products of reinforcement contingencies and modeling opportunities. For example, if an athlete frequently gets in fights with opponents, it is because the athlete has learned that the negative consequences to fighting are outweighed by the rewards like an intimidated opponent. Similarly, if an athlete upholds the ideals of sportsmanship, it is because that athlete, through years of experience, has learned that there are significant benefits, such as an enhanced reputation and praise from others, from acting in a prosocial manner.

Character educators who rely predominantly on the perspectives of behaviorists and social learning theorists place a great deal of emphasis on the role of authority figures, such as parents and coaches. It is these people who have the greatest power to control how rewards are distributed. By modeling desired behaviors and managing the costs

and benefits that follow actions, behavior can be shaped and eventually dispositions established and reinforced.

In addition, direct teaching of character virtues is often advocated by behaviorists and neobehaviorists. For example, coaches are encouraged to explain what *respect* means, model it themselves, and then shape the behavior of their athletes by rewarding respectful behavior and punishing deviations. Critics maintain that the strategies recommended by behaviorists may work to modify behavior, but only in the short term. What happens when the modeling and reinforcements change? What happens, for example, when a new coach arrives who rewards winning rather than respectful behavior? Since the behaviorist's strategies don't address the core beliefs and values of the learner, they may create situational conformity to expectations without genuinely influencing character.

The influential work of Albert Bandura provides something of a bridge between the behavioral and social learning traditions of psychology and the constructivist approaches to be discussed next. His theoretical roots extend deeply into social learning theory, but over the past couple of decades he has increasingly incorporated cognitive mediators of behavior. Like other behavioral and social learning theorists, character, for Bandura, is rooted in environmental experiences, and society is the ultimate arbiter of what counts as moral. Bandura extends the paradigm, however, by suggesting that the person can develop and employ self-management skills that can provide stability to character despite changing contextual influences. Cognitive processes, however, can also derail character. Bandura discusses at length the cognitive processes of *moral disengagement* that function to provide self-justifying excuses for antisocial behavior.

The Constructivist Approach to Character Development

The constructivist approach to character development maintains that the child is not a blank slate ready to be written on through modeling and reinforcement. Rather, the child is a *meaning maker* who actively interprets experience and selects what aspects of the environment to attend to. When, for example, an adult rewards a child, the effect is not automatic. Rather, the child interprets the informational value of the adult's action and may

choose to comply with or rebel against the effort to shape personal behavior. Children's character is influenced as much by the child's own choices and initiative as through such external sources as modeling and reinforcement.

Character educators who draw from constructivist theories engage with the child on the level of meaning and relationship, not just the level of reward and punishment. They emphasize such processes as dialogue, group dynamics, and building communities where trust and care are nurtured. The foundation of character is in positive relationships where personal dispositions are forged through enduring patterns of interaction. When a person feels known, valued, included, and empowered, the person is more open to positive social influence.

Character Development and Sport

It is clear that the physical activities of sport—throwing, kicking, or hitting balls, running, jumping, and so forth—do not have any significant impact on character development. To the extent that sport experiences contribute to the development of character, it is because of the opportunities they provide to set and pursue goals, build meaningful and purpose-driven relationships, cooperate to achieve shared aims, experience and manage a range of intense emotions, and test ethical commitments through the challenges of competition. Perhaps more than any other widely shared activity, sports can provide many young people with a chance to build nurturing relationships with both peers and adults.

As historians and sociologists have pointed out, the specific virtues that sport is claimed to nurture have varied considerably from one historical period to the next. In 19th-century England, for example, team sports were a regular part of life in many boarding schools because it was believed that they would build those qualities useful to the elites of the far-flung British Empire. It was claimed that sports would develop leadership, strategic thinking, loyalty, courage, and other aspects of *manliness*. By the early 20th century in the United States, the character-building ideology continued, but with a different cast of virtues. With a large influx of immigrants and a growing urban economy, sports were seen as vehicles to prepare a compliant workforce with such dispositions as obedience, discipline, patriotism, and self-sacrifice.

The changing definitions of character highlight the need for researchers to carefully and operationally define the term. To date, most research on character in sport has focused on the moral dimension.

Research on specific aspects of moral character, such as moral reasoning capacity, has shed some light. There has been no research to support a positive role for sports in advancing moral reasoning, apart from a small number of intervention studies. A few studies have suggested that participation in some sports, particularly at the college level, may have negative effects on moral reasoning, though inconsistent results and methodological concerns prevent this from being considered a verified conclusion. The research on what has been called the *professionalization of values*, though, has demonstrated that the longer one participates in sport and the higher the competitive level, the lower *playing fair* is as a value relative to winning.

Research related to dimensions of performance character is more encouraging, but the evidence is indirect. For example, based on longitudinal investigations, it appears that participation in high school sports is associated with later educational attainment and lower drop-out rates. One likely explanation for these positive outcomes is that sports develop qualities like goal setting, time management, delay of gratification, sequential thinking, initiative, concentration, and attention focusing. These skills and dispositions, which are core to performance character, can lead to greater success in both sports and school. Additional research is needed, however, to investigate directly the hypothesized links and to examine potential mediating effects of such variables as sport type, coaching styles, and team culture issues.

One encouraging finding is that when character development is an intentional and explicit aim of physical activity programs and when leaders are trained to implement it, positive results can be obtained. Don Hellison, for example, has developed an influential approach to character development through physical activity and sport. In his model for developing personal and social responsibility, he combines a focus on both moral character and the self-control dimensions of performance character.

No doubt, many practitioners will continue to list character development as one goal for sports participation, and researchers will continue to investigate the promises and pitfalls of such a goal.

To move forward, both theory-based practitioners and practice-based researchers will need to work diligently to clarify their terms and elucidate their theoretical framework. Character will continue to be a disputed concept, but an important one nonetheless.

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See also Fair Play; Moral Development; Moral Judgment; Moral Reasoning; Self-Regulation

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CHEATING

Cheating in sport is a violation of an explicit or implicit promise to follow the rules of a sporting activity. This promise is an honorable action that an athlete takes to assure the opponent, the fans, and all interested parties that the rules will be followed. An example of an explicit promise is the International Olympic Committee oath for athletes. The oath is taken by one athlete representing all athletes from their home nation. The oath states,

In the name of all the competitors, I promise that we shall take part in these Olympic Games, respecting and abiding by the rules which govern them, committing ourselves to a sport without doping and without drugs, in the true spirit of sportsmanship, for the glory of the sport and the honour of our teams. (<http://registration.olympic.org/en/faq/detail/id/28>)

An implicit promise to follow the rules is the unspoken, accepted conditions that a player agrees to do when playing. In many amateur sport competitions, the competitors, captains, or coaches meet before the game begins and shake hands. The handshake is an implied agreement that each will play well, will play by the rules, and wishes the other good luck.

Categories of Rules in Sport

The rules that athletes and coaches promise to follow come in three different categories, and athletes and coaches have numerous creative ways to violate intentionally both the spirit and the letter of these rules. The rules are *constitutive*, *regulatory*, and *sportsmanship*.

The constitutive rules give order and direction as to how the game is to be formed and how it is played. The constitutive rules are norm referenced and give boundaries to the game. For example, consider the rule as to how long a game is played. In softball, three outs per team make an inning, seven innings for a game; in American football, four quarters make a game, and each quarter is a specific number of minutes of play; in amateur basketball, 60 minutes is the allotted period of play divided into two halves.

The regulatory rules maintain and temper the integrity of play within the constitutive rules. Regulatory rules are very useful but less fundamental. For example, in softball a player who is at bat cannot run to a base until a pitched ball is either hit or the ball reaches the catcher's glove and four balls are called by the umpire. In American football, a player cannot leave the line of play until the offensive center hikes the ball to the quarterback. In amateur basketball, a player cannot advance the ball down the court without dribbling or passing the ball.

Players accept the constitutive and regulatory rules as the guideline and give an implicit or explicit (verbal agreement) promise to follow these rules. Inadvertent violation of the playing rules, if

caught by officials, results in a foul or rule break in the game. Both the constitutive and regulatory rules are usually governed by umpires, referees, or game officials. Violation of a rule is not cheating. However, intentional violation to break the rules without being caught is a violation of the promise to accept the consequences of breaking a rule. An example of cheating of constitutive rules in American football is the faked injury to stop the play clock. If a referee deems an athlete is injured, the referee must stop the play clock until the injured athlete is attended to and removed from play. Even though a rule exists against faked injuries, a team may conspire to have an athlete pretend to be injured, especially if the play clock is running out. The time clock stops, the player is treated and removed from play, and in the meantime the team uses the extra time to set up the next play and perhaps win. An example of cheating of the regulatory rules is the baseball pitcher altering the condition of the ball. The regulatory rules are specific as to the size, shape, and condition of the ball. Pitchers often change the condition of the ball by scuffing the surface with an abrasive located in their glove, hand, or pocket, or place goo on the ball.

Sportsmanship and Gamesmanship

A player, coach, and team also implicitly promise to follow the spirit of sportsmanship, which includes accepting the philosophy of fair play: the reason for rules' existence. Fundamentally, the philosophy of fair play is to offer each team and competitor an equal opportunity to be successful. However, coaches often break their implied promises through clever and artful techniques. For example, in 1928 a football coach tried to find a method for his team to be more competitive on the field of play. He looked through the rule book and found no rules that limited placing artifacts on the players' jerseys, except a number on the back of the jersey. He cut footballs into halves, and sewed one half of a football on the front of all the backfield players' jerseys. When his quarterback handed the ball off to a backfield member, everyone in the backfield had a football on his uniform. Because no rule existed to prevent the artful strategy, the coach violated the spirit of fair play in the ideal sense and he cheated.

The artful strategy of trying to get around the rules or push the rules beyond the letter is called gamesmanship. Gamesmanship often borders on

cheating. Most players accept some gamesmanship that occurs; however, there is an arbitrary line when gamesmanship becomes cheating. In officiated games, players learn to push that arbitrary line and not only play a game against their opponent but also play another game against the official. The players use the rules to their advantage and challenge the official to catch them violating the letter and spirit of the rules. Players learn how much they can get away with when pushing the rules by learning how tight the game will be called by the official(s). Officials therefore must know the rules as well as know the ethos of how the game is played. The *ethos*, the essential nature of the game and how rules are interpreted, gives direction as to how much latitude the official uses in determining how closely the rules are followed. For example, basketball is a no contact game; however, there is a great deal of contact that is accepted as part of playing the game. The ethos directs the officials to decide how much contact is accepted and what is not accepted. Within this game climate, knowing the difference between violating a rule and cheating becomes fuzzy. The players and officials thus rely greatly on the written rules, the spirit of the rules, and their game-to-game experience in deciding the difference.

Intentionally violating an organizational or institutional rule that governs the preparation for the game, including player training, is governed by a different set of rules. These sportsmanship rules are established by the governing body that oversees how games are played. The goal is to establish a culture of fair play in which all players come to the game with a fair chance of being successful in playing and winning the contest, all of which is also very arbitrary, depending on each team's resources and genetics of the players themselves. A bobsled team from the Caribbean island of Trinidad would have little chance at success considering that no bobsled track exists in Trinidad. However, Trinidad has fielded a bobsled team to the Olympic Games. The team's chance of success is not necessarily fair. Players thus are members of teams and both the player and the team agree explicitly to follow the rules established by the organization in preparation and training as well as the actual playing of game. When players and their teams intentionally violate these rules, cheating occurs.

The same issue occurs in the constitutive rules and regulatory game rules—players and teams work the rules to gain an advantage; rules then

become more explicit and officials become more aggressive in monitoring whether players and teams are following the rules. A case in point is the rule against players ingesting or injecting banned performance-enhancing drugs or supplements to gain a physiological advantage. A player is always limited by genetic predisposition, and no matter how much practice takes place, that limit cannot be overcome without some external chemical aid. Thus, some players try to gain that additional advantage beyond their own physiological makeup and choose to take chemical supplementation to overcome their own limitations. The practice is almost as old as competitive sport. Since the purpose of play is about fair play, meaning that everyone should have the same degree of chance to perform well, most if not all governing bodies have specific lists of acceptable and unacceptable supplement usage. In order to monitor players, the organizations demand testing—including urine samples and in some cases blood samples—to check if players are following the rules. Whole industries have been developed to test, track, record, and monitor players; the World Anti-Doping Agency is one of them. However, again the problem is gamesmanship. The players are constantly looking for an advantage, and since each human body has limitations, the players are looking for nutritional or chemical supplementation to overcome physiological and genetic limitations. The line again becomes fuzzy—which supplements are acceptable and which are not.

All players supposedly are acquainted with cheating, but in some cases they choose to manipulate and stretch the rule. The way sports are played and played out often makes it difficult to admit that cheating is the intentional violation of a promise to play by both the spirit and the rules of the game. However, cheating is always the intentional action of breaking a promise and violating the rules.

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See also Character Development; Ethics; Moral Development; Moral Reasoning; Moral Values and Attitudes; Sportspersonship

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CHOKING

Researchers, spectators and performers alike are often intrigued, and sometimes shocked, if motivated people perform dramatically worse than usual in important, high-pressure situations. This can happen not only in sports but in almost any domain in which people are specifically motivated to do well, be it music, surgery, academic examinations, or even a driving test.

Choking, or *choking under pressure*, is a metaphorical expression frequently used in sport settings to describe occasions when severe unexpected decrements occur in performance under anxiety-provoking situations. Such occasions potentially can be invoked by the presence of an evaluative audience, the possibility for high reward, competition, or factors unique to each individual choker. Examples of choking in sport tend to be memorable and sometimes sports evolve their own domestic expression for the phenomenon.

There is no agreed definition of what it means to choke. Early definitions suggested that *choking* is inferior or suboptimal performance in response to high motivation to perform well, but recent researchers have argued that such a definition does not capture the acute, dramatic declines in performance that go beyond merely suboptimal performance. Most examination of choking has derived from empirical studies, which are unable

to re-create the factors that cause choking in each individual, so some researchers have highlighted the need for a more idiographic, qualitative approach that is ecologically valid.

Why Does Choking Occur?

It seems likely that choking arises as a stress response associated with very high motivation to perform well. It also has been suggested that a choking event can have a long-term negative psychological impact on performers, leaving them vulnerable to further choking events in the future.

Historically, arousal theories have been used to explain performance decrements in anxiety-provoking situations, but these are merely descriptive models, which link arousal or drive with performance. They offer little explanation of the underlying mechanisms that result in performance decrements (or indeed performance peaks). The best known of the theories is probably the Yerkes–Dodson law (better known as inverted-U theory), which suggests that improved performance accompanies increased arousal until intermediate levels of arousal are reached. Further increases in arousal result in performance decrements. Such a model has a number of shortcomings with respect to the phenomenon of choking. For example, the model implies that if arousal becomes too high and performance declines, a performer can regain the previous level of performance merely by reducing arousal. The model also implies that changes in performance level are smooth and gradual as arousal increases (or decreases), which does not explain the acute, dramatic drops in performance that typify choking. To resolve these issues, some researchers have turned to René Thom's theory of catastrophes, which tries to explain abrupt shifts or *bifurcations* in behavior (e.g., landslides, bridge collapses, and even capsized ships) that are associated with relatively small, continuous changes in contributing factors. In particular, the researchers have examined cusp catastrophes to elucidate how physiological arousal and cognitive anxiety (worry) interact to cause high or low levels of performance in sport. The model suggests that high cognitive anxiety couples with increasing physiological arousal to cause abrupt catastrophic drops in performance or *chokes*.

A shift in emphasis has seen attention theories exploited to better understand the cognitive mechanisms that underlie the effects of performance

pressure on performance. These theories propose that under anxiety-provoking conditions there may be a disruptive reallocation of attention, which subsequently triggers performance decrements. To a certain extent this approach is underpinned by *cue utilization theory*, which suggests that changes in attentional focus caused by performance pressure influence the degree to which an individual is able to process task-relevant information but not task-irrelevant information. Two current models that exploit attention theories propose that performance can be disrupted because attention either becomes distracted away from important aspects of performance or because attention becomes overly attracted to skill-focused aspects of performance.

Distraction Models

Distraction models propose that performance pressure causes a shift in attention, which results in unnecessary processing of information irrelevant to performance of the task, such as worry and self-doubt. Working memory is thought to support the storage and active maintenance of information over short durations and, as a consequence, to be involved in processing information related to task performance. Processing additional irrelevant information is therefore likely to decrease the attention resources available to working memory and subsequently compromise its ability to process task-relevant information that is crucial for effective performance.

One particular theory, which offers insight to how distraction mediates the relationship between anxiety and cognition in sport (although it was not developed specifically for sport) is *attentional control theory* (ACT). ACT is an extension of *processing efficiency theory* (PET), proposed by Michael Eysenck and his colleagues. ACT proposes that anxiety reduces goal-directed focus but increases stimulus-driven focus, allowing attention to more easily be distracted to threat-related stimuli and the processing of threat-related stimuli to be less easily inhibited. Unnecessary processing of threat-related internal stimuli like worry about the consequences of performance failure or external stimuli like irrelevant environmental factors can harm performance, but the theory proposes that performance effectiveness can be preserved by increasing task-related effort in order to augment the processing resources dedicated to the task. Consequently,

anxiety may not always reduce the quality of performance.

To a certain extent, this theory seems feasible. For example, there is no doubt that most soccer players become highly anxious before a penalty shootout at the World Cup. Their attention is distracted to threatening internal stimuli, such as “What will the press say if I miss?” or to threatening external stimuli, such as sudden realization that the goal is smaller than they recall (or the goalkeeper taller). Fewer resources are thus available to allocate to the task at hand and the probability of failed performance increases dramatically. But not all players do fail in penalty shootouts. Some players seem to counteract the effects of pressure on their penalty-taking performance with intense effort. Others appear unable to compensate; they choke, as the newspapers will later claim.

A problem with distraction theories, however, is that evidence from research on nonmotor, cognitive tasks such as mathematics problems, suggests that those tasks that rely heavily on working memory are most affected by worry associated with pressure. Most examples of choking in sport involve performance of well-practiced skills that do not rely heavily on working memory because they are procedural and run automatically. Consequently, the performance of skills in sport should be relatively immune to depleted working memory resources. A second explanatory model has therefore emerged, which suggests that pressure heightens self-consciousness about performing successfully, causing attention to become skill focused. Theories that take this approach have a wide variety of nomenclature that includes self-focus theory, explicit monitoring theory, conscious processing hypothesis, the constrained action hypothesis, and the theory of reinvestment.

Skill-Focused Models

Skill-focused models intimate that well-practiced skills in sport (and in other movement-related domains) can be disrupted not by depletion of working memory resources but by the misappropriation of spare working memory resources to consciously control performance that is best left to run automatically. In order to ensure that performance is effective, the performer explicitly tries to control the process of skill execution, which disrupts or inhibits automatic control processes and usually results in poor performance. Roy

Baumeister probably described this best when he simply argued that consciousness does not have access to enough of the task-relevant knowledge needed to run automated skills effectively.

Some of the best evidence for the model comes from studies that have shown that expert performers tend to be able to preserve the high quality of their performance when carrying out a concurrent task (such as tone recognition) that takes attention away from execution of the skill. Well-practiced skills require minimal working memory resources for effective performance, leaving resources spare for the concurrent task. Skilled performers, however, tend to display disrupted performance when carrying out a concurrent task that requires attention to execution of the skills (e.g., a task that requires indication of the position of the bat when a tone sounds). Evidence from studies, using the same or similar paradigms, shows that under pressure skilled performers also become more accurate at the secondary task (e.g., indicating the position of the bat when the tone sounded), indicating that pressure causes increased skill focus.

It has been argued that distraction models and skill-focus models lead to opposing predictions, but it is unlikely that the models are independent. A shift in attention under pressure, which disrupts automaticity by unnecessary processing of declarative knowledge that is redundant for performance, appears to encompass both distraction and skill-focus models.

Personality and Performance Under Pressure!

Some performers may be more susceptible to choking than others. Specific dispositional traits (relatively stable personality characteristics) may predispose performers to choking. Attentional control theory, for example, suggests that high trait anxious individuals are more likely than low trait anxious individuals to display disrupted performance under pressure, although these findings are confined to test-anxiety situations, rather than performance in sport. It has been suggested that high levels of chronic anxiety might not even be relevant in sport as these relatively public settings may not attract individuals with such traits.

One personality trait associated with skill-focus theories is self-consciousness. It has been claimed that people who are generally high in self-consciousness are likely to perform better under pressure than people who are low in self-consciousness, because they are accustomed to

performing with the high levels of self-focus that pressure causes. It is unclear whether this proposal is correct, given that some evidence shows that people generally high in self-consciousness are more likely to choke than people low in self-consciousness. The *theory of reinvestment* argues that not only self-consciousness but also the inclination to consciously control performance is subject to individual differences. Psychometric measures of a general disposition for reinvestment, or for movement specific reinvestment, have been developed. Both make reference to personality traits that include self-consciousness, rumination, and even slips of action, and both show that the performance of people who score high on such metrics is generally more negatively influenced by pressure than the performance of people who score low on such metrics.

Protecting Performance Under Pressure

Techniques that divert attention away from conscious control, perhaps to an external rather than an internal focus, or that improve the ability of the performer to maintain the correct focus of attention, may help to prevent choking. Methods that train control of attention, such as emotion control training or hypoeegoic self-regulation (in which an individual learns deliberately to relinquish conscious control of behavior), potentially can help performers maintain the fluid automaticity of their performance under pressure. Alternatively, training that acclimatizes performers to heightened self-consciousness may inoculate them against pressure. Ritualized behaviors or routines may swamp working memory and prevent skill focus, as may concurrent secondary tasks. There is even evidence that performing faster than usual when under pressure may help performers to avoid skill focus, although as anyone who has ever tried this approach will attest, it is not so simple in the heat of battle. A more enduring approach may be to use implicit motor learning techniques to curtail construction of a task-relevant declarative knowledge base. The integrity of automatic control processes should be more secure if task-relevant declarative knowledge cannot be extracted from long-term memory for conscious control purposes.

Research directly or indirectly related to choking has expanded considerably over the past three decades. Investigation has shifted from a passive to a more active approach, progressing from qualitative discussion about entertaining anecdotal

reports of performance failure under pressure to concrete empirical research. As long as observers continue to be intrigued and shocked when motivated people perform their skills dramatically worse than usual when the stakes are high, the research interest is likely to continue.

Rich Masters and Neha Malhotra

See also Attentional Focus; Attention–Performance Relationships; Automaticity; Competition; Effort; Expertise; Motor Learning; Skill Acquisition

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termed *chunking*. *Dechunking* refers to when well-developed chunks are broken down into a number of smaller chunks or bits of information. Chunking is a fundamental building block of human memory and underlies the learning of perceptual and motor skills. Dechunking is a mechanism that potentially underlies suboptimal performance. In this entry, the process by which information is chunked during learning is explained in general terms and in relation to both perceptual and motor skills. The process and implication of dechunking are then briefly discussed.

To operate effectively, humans must utilize information in the performance environment, but the human information-processing system has limitations. Specifically, the capacity to store and retrieve verbally coded, declarative knowledge is finite and small, and serial processing of such information is time consuming. Despite these limitations, people can successfully execute perceptual–motor skills in environments that are often complex, dynamic, and time constrained. Indeed, experts are defined by their ability to overcome the limitations of the information processing system. Their decisions depend upon rapid processing of large quantities of perceptual information. Their motor skills are executed in a smooth and automated fashion. Chunking underlies these capabilities.

Stage models of skill acquisition propose that early in learning bits of information pertaining to a task are stored, retrieved, and processed as declarative knowledge by a short-term information processor (working memory), which interfaces with long-term memory. With repeated on-task experience, meaningful associations emerge between recurring bits of information that are grouped together and coded as chunks. Over the course of practice, between-chunk associations develop and higher and higher level chunks are compiled, until large amounts of the information pertaining to a task are hierarchically organized into a single representative chunk. The demand on information processing resources lessens as each level of the hierarchical structure is formed, until higher level chunks take a procedural form that no longer requires the retrieval and processing of declarative knowledge. In the domain of sport and exercise, chunking takes two distinct forms: the chunking of perceptual information for response selection, and the chunking of the motor information needed to organize and execute a successful response.

CHUNKING/DECHUNKING

Cognitive psychologists argue that over time, task-related knowledge is organized into information-rich representations or chunks, a process

The strategies and rules inherent within dynamic activities like team sports generally result in the availability of structured perceptual information, such as the relative position of teammates and opponents or the mechanics of an opponent's movements. Visual elements that meaningfully correlate gradually become recognized as distinct patterns. With increased exposure, relations between these patterns emerge and so they are compiled together, and from then on recognized as a single pattern. Empirically, the chunking process has been demonstrated by findings that show experts have superior capacity to recall patterns of game play after only a brief presentation of a structured, rather than transitory, unstructured game scenario. Chunking allows experts to quickly and accurately recognize patterns from within the unfolding events and this, in turn, permits both the rapid and accurate selection of an appropriate motor response.

A motor response is the outcome of the ordered execution of a sequence of movement components. In the early stage of motor learning, the sequencing and control of independent movement components is typically controlled at a conscious level, placing high demands on the available information processing resources. Consistent repetition of a pattern of movement through practice causes adjacent movement components to coalesce into chunks. Chunks in motor learning are defined by pauses between successive movement components. For example, sequential key pressing experiments show that latencies between successive key presses decrease with practice in a clear and consistent fashion such that longer pauses demarcate chunk boundaries. Pauses between adjacent motor chunks result in a segmented or jerky motor action. Continued practice builds a hierarchical structure, in which fewer chunks comprising longer sequences of movement components exist. As a result, the movement is both less jerky and places fewer demands on information processing resources. Eventually, all movement components are incorporated into a single representative motor chunk, which, when selected, results in the smooth, automatic production of complex sequences of movement.

Compared to the concept of chunking, dechunking is a less well established phenomenon. As chunking is associated with the learning or progression of skills, dechunking may be used to explain occurrences of skill regression or skill

failure. Declarative knowledge accrued during learning likely remains retrievable from long-term memory after it has been chunked at a higher level. The retrieval and processing of previously chunked declarative knowledge represents the dechunking of optimal and automated representations and regression to a less efficient and less fluid mode of control. In the competitive environment, contingencies such as performance anxiety may trigger dechunking and result in skill failure. Alternatively, in the training environment athletes may purposefully attempt to revise the composition of high-level representations by first dechunking and then rechunking information.

In summary, the chunking of information is a consequence of the practice that underlies skill learning. It allows well-practiced performers to operate proficiently despite the temporal and capacity constraints of the information processing system. Dechunking describes a return to the retrieval and processing of hitherto chunked declarative knowledge.

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See also Choking; Information Processing; Motor Learning; Pattern Recognition and Recall; Skill Acquisition

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COACH–ATHLETE RELATIONS

The coach–athlete relationship is a unique interpersonal relationship characterized by mutually and interconnected thoughts, feelings, and emotions between an athlete and a coach. There are different types of coach–athlete relationships, including traditional coach–athlete dyads (the

coach and athlete are not related in any way other than their coaching relationship), married coach–athlete dyads, and family coach–athlete dyads.

The coach–athlete relationship is of importance to both the athlete and the coach as an effective coach–athlete relationship is crucial to the achievement of successful performance and interpersonal satisfaction. For example, research has indicated the quality of the coach–athlete relationships can influence coaches' passion for coaching, coaches' motivation and the motivational climate they create, athletes' perceptions of self-concept and self-esteem, levels of intrinsic motivation, and perceptions of performance. Further, the quality of the coach–athlete relationship can influence athletes' perceptions of fear of failure, and the likelihood of burnout or drop-out.

Understanding Coach–Athlete Relationships

Initial research examining coach–athlete relationships almost exclusively focused upon utilizing models of leadership to illustrate how coaches' behaviors and actions influence athletes' actions and behaviors. Packinthan Chelladurai's Multidimensional Model of Leadership is one theory that has been widely used to examine coach leadership and the subsequent influence it has upon athletes. This model consists of three aspects of coaching behavior: actual leader behavior, preferred leader behavior, and required leader behavior. Actual leader behavior incorporates the behaviors a coach displays based upon his experience, personality, and ability. Preferred behaviors are the behaviors that athletes would prefer the leader to display. Preferred behaviors can be influenced by various factors such as athlete's age and gender. For example, younger or less experienced athletes may prefer higher levels of relationship-oriented behaviors from coaches than older or more experienced athletes. Similarly, although research is not conclusive, male athletes may prefer more autocratic coach behaviors than female athletes. Required behaviors are behaviors that coaches employ due to environmental or situational constraints and demands. This model suggests that athletes' performance and satisfaction are attributed to the degree of congruence among the three aspects of leader behavior.

Taking a relational perspective to examining coach–athlete relationships, Sophia Jowett developed the 3 + 1C model. This model proposes that

effective coach–athlete relationships are characterized by four interrelated dimensions: *closeness*, *commitment*, *complementarity*, and *coordination*. Closeness accounts for the emotional aspect of the relationship. It refers to the interpersonal feelings between the coach and the athlete, and includes aspects such as trust, respect, and caring for each other. Commitment describes the cognitive component of the relationship and reflects the coach's and athlete's thoughts regarding their intentions to maintain a close athletic relationship over time. Complementarity refers to the behaviors part of the relationship. Complementarity reflects the coach's and athlete's cooperative interactions during training and competition. Finally, coordination accounts for the coach's and athlete's shared perspectives, which are developed as a result of open channels of communication. The quality of the relationship is influenced not only by how the coach and athlete directly perceive their relationship but also by how they believe each other perceive the connection.

Conflict in the Coach–Athlete Relationship

Conflict has been defined as experiencing discord between oneself and significant others. Two of the main reasons conflict arises between coaches and athletes are an incompatibility in the relationship and either party breaking the rules of the relationship. Incompatibility can arise within the coach–athlete relationship when differences emerge regarding goals, beliefs, attitudes, values, or behaviors. For example, if a coach and an athlete are striving to achieve different goals there is potential for conflict to arise as each party is working toward a different outcome. Similarly, changes in circumstances may contribute to a coach or an athlete breaking the agreed upon rules of their relationship. Such rules are the expectations for behavior each party holds for the other party, themselves, and the overall relationship. Such conflict can arise or be enhanced if the relationship is lacking in terms of open, honest communication.

Coach–Athlete Relationships in Youth Sport

The coach–athlete relationship within youth sport is often complicated by the involvement of parents. When children begin to take part in sport, parents are often heavily involved and coaches play a more marginal role. However, as children progress in sport, they increasingly rely on their coach as more

hours are spent training. At this time, the role of parents often changes as parents assume a more background role. The changing roles between parents and coaches can be associated with conflict or power struggles. If such conflict arises, it can cause tension within the coach–athlete and parent–athlete relationship. Children can also become unhappy or distressed if they become caught in the conflict.

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See also Autonomy-Supportive Coaching; Caring Climate; Interdependence Theory and the Coach–Athlete Relationship; Parenting; Talent Development

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COACHING EFFICACY

Self-efficacy judgments are domain-specific beliefs held by individuals about their ability to successfully execute differing levels of performance given certain situational demands (a situation-specific self-confidence). *Coaching efficacy* is the extent to which a coach believes in the personal capacity to affect the learning and performance of the athletes. Efficacy beliefs of a head coach play a central role in several broader theoretical models of effective sport coaching. This entry provides a broad overview of related research and first introduces a conceptual model of coaching efficacy. Measurement models for coaching efficacy are then described. Potential sources and proposed outcomes of coaching

efficacy are then reviewed, and the entry concludes with a discussion of future research directions.

A Conceptual Model of Coaching Efficacy

For as long as there has been organized sport, coaches have played a key role in the motivation and performance of athletes. Perhaps surprisingly, it was not until 1999 that a formal conceptual model of coaching efficacy was put forth by Deborah Feltz, Melissa Chase, Sandra Moritz, and Phillip Sullivan. The proposed conceptual model of coaching efficacy was intended to address a gap in the sport-oriented literature and was based, in part, on Albert Bandura's self-efficacy theory as well as the teacher efficacy literature. An important caveat of the original conceptual model was that it focused on high school and lower division collegiate coaches.

The conceptual model of coaching efficacy consisted of three key elements: (1) proposed sources of coaching efficacy information, (2) dimensions of coaching efficacy, and (3) proposed outcomes of a coach's efficacy beliefs. Since 1999, confirmatory evidence has been provided for many of the core facets of the conceptual model of coaching efficacy. Evidence has also been provided that has prompted modifications to the initial framework. Other populations such as youth sport coaches have also since been studied, which has further contributed to a continually evolving framework.

Measurement of Coaching Efficacy

The coach must competently and confidently perform multiple roles to be an effective leader. These roles often include but are not limited to that of teacher, motivator, strategist, organizer, physical conditioning trainer, and character builder. This complex and multifaceted task suggests that a coach's own belief in the personal ability to successfully execute core tasks may be best reflected within a multidimensional measurement model. A standard first step in multidimensional measurement is to operationally define the most general construct of which multiple dimensions are thought to exist. Coaching efficacy, broadly defined, is the extent to which a coach believes in the personal capacity to affect the learning and performance of the athletes. Because coaching efficacy is a belief, for which only imperfect observed indicators can be collected, measurement of the construct generally has taken a latent variable approach.

Deborah Feltz et al. put forth the first formal measurement model (theory) of coaching efficacy in 1999: the Coaching Efficacy Scale (CES). The multidimensional model for the CES posited that four specific efficacies were related to one another and defined coaching efficacy. *Motivation efficacy* was defined as the confidence coaches have in their ability to affect the psychological mood and psychological skills of their athletes. *Game strategy efficacy* was defined as the confidence coaches have in their abilities to lead during competition. *Technique efficacy* was defined as the confidence coaches have in their instructional and diagnostic skills. *Character building efficacy* was defined as the confidence coaches have in their abilities to influence the personal development and positive attitude toward sport in their athletes.

An investigation published in 2005 regarding the psychometric properties of measures derived from the CES concluded that several modifications could be made to increase the precision of coaching efficacy measures produced by the CES. Two of the major suggested modifications were to delimit subsequent versions of the CES by level coached, and to make a few structural changes to the CES itself. In 2008, the Coaching Efficacy Scale II–High School Teams (CES II–HST) was published, and in 2011, the Coaching Efficacy Scale II–Youth Sport Teams (CES II–YST) was published. Within both the CES II–HST and the CES II–YST and as compared with the CES, a new dimension of coaching efficacy, physical conditioning efficacy, was added, revised operational definitions for two of the previous dimensions were put forth, and a majority of new or revised items were developed. *Physical conditioning* was defined as the confidence a coach has in the ability to prepare athletes physically for participation in the sport. *Technique efficacy* was redefined as the confidence a coach has in the ability to use instructional and diagnostic skills during practices. *Character building* was redefined as the confidence a coach has in the ability to positively influence athletes' character development through sport. Evidence has been provided, for both the CES II–HST and the CES II–YST, that the proposed measurement model closely and reliably resembles real data for both male and female coaches.

Sources of Coaching Efficacy

Albert Bandura proposed that self-efficacy beliefs are based on the complex cognitive processing of

approximately four general categories of potential sources of information: (1) past performance accomplishments, (2) vicarious experiences, (3) verbal persuasion, and (4) physiological or emotional arousal. Past performance accomplishments have typically been identified as the strongest source of efficacy information. Deborah Feltz et al. worked from Bandura's general framework and put forth potential sources of coaching efficacy information, which included coaching preparation, coaching experience, prior success in coaching, perceived ability of team, and social support from various stakeholders (e.g., athletic director, community, students, faculty, and parents). Subsequent research has since built upon this aspect of the conceptual model of coaching efficacy and has added a few potential sources of coaching efficacy information. These include assistant coaching experience, relevant athletic experience of the coach, perceived improvement of athletes, social support from athletes, and imagery use by the coach.

Each of the proposed sources of coaching efficacy information outlined in the previous paragraph has been found to be an important predictor of at least one of the proposed dimensions of coaching efficacy at one or more levels of competition. There is some evidence, however, that the degree of importance for some of these sources of coaching efficacy information may vary somewhat by level coached and or gender of the coach. For example, coaching preparation and coaching experience have been found to be reliable sources of efficacy information at both the high school and youth sport levels but less so for collegiate coaches at Division II and III. Prior success (e.g., won–loss record in the previous year or career winning percentage) has been shown to be an important source of efficacy information for female coaches but less so for male coaches at the high school level. Finally, the current proposed collection of proposed sources of efficacy information appear to be more salient, on average, for some dimensions of coaching efficacy, such as game strategy efficacy, technique efficacy, and motivation efficacy, than for other dimensions of coaching efficacy like character building efficacy. That said, it is important to note that even for the most understood dimensions of coaching efficacy, most of the variability in even these dimensions has yet to be explained. This suggests that there may be more work to be done to discover new sources

of coaching efficacy information in order to get a fuller understanding of the complex cognitive processes that underlie these beliefs.

Outcomes of Coaching Efficacy

Albert Bandura proposed that self-efficacy beliefs often are the primary determinant of people's behavior and thought patterns. Self-efficacy beliefs affect the choices a person makes (Do I attempt to take on a particular task?), the effort a person dedicates to a particular task (Will I give my best effort to this task?), and the persistence a person displays while working toward completion of a particular task (Given a setback, will I redouble my efforts toward accomplishing this task?). Self-efficacy beliefs also affect the goals that a person sets (Will I select a goal that pushes me to perform my very best?), the attributions a person makes (Given a setback, will I identify internal or external explanations for the outcome?), and the emotional reactions a person has (Given that I almost, but not quite, accomplished my goal and that I no longer have the opportunity to pursue this goal, will I feel proud of what I did accomplish?).

Deborah Feltz et al. worked from Bandura's general framework and highlighted potential outcomes of coaching efficacy, which included coaching behavior, player and team satisfaction, player and team performance, and player and team efficacy. More efficacious coaches as compared to less efficacious coaches, on average, are theorized to provide more effective feedback (e.g., more of a focus on encouragement and inclusive leadership and less of a focus on organizational issues and autocratic leadership) to athletes or teams, be more appropriately committed to coaching and suffer less burnout from coaching, have more satisfied athletes or teams (e.g., want to play for the same coach again next year), have athletes or teams more frequently perform at or above their current ability level, and serve as positive source of efficacy information for their athletes or teams (e.g., the coach is perceived as confident in the athlete or team). Interestingly, to date, motivation efficacy and character building efficacy have generally been the most consistent predictors of positive outcomes for athletes or teams. Thus, the more relationship-based dimensions of coaching efficacy as compared with the more technical dimensions of coaching efficacy, such as technique efficacy and game strategy efficacy, may be especially important to athletes

or teams. The predictive ability of physical conditioning efficacy has yet to be systematically studied.

Future Directions

Self-efficacy theory provides a strong theoretical base for the construction of the coaching efficacy construct. The conceptual model of coaching efficacy serves as a very useful and flexible sport-specific theoretical framework. The measurement of coaching efficacy has improved yielding better model-data fit in recent years, in part, because of the recent development of different instruments based on level coached. Clearly there exists a solid basis from which future scholarship can proceed in some way. Future directions that may prove especially fruitful may include a sharper focus on relevant subpopulations, such as gender of both the coach and the team and level coached, with regard to potential sources and outcomes of coaching efficacy, and longitudinal research designs that describe the dynamic nature of coaching efficacy.

Nicholas D. Myers

See also Collective Efficacy; Implicit/Self-Theories of Ability; Interdependence Theory and the Coach–Athlete Relationship; Modeling; Relational Efficacy Beliefs in Coach–Athlete Relations; Self-Efficacy; Social Cognitive Theory

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COGNITIVE CAPABILITIES

In the domain of sport, the term *cognitive capabilities* refers to the athlete's aptitude to process, evaluate, select, and compare information. Cognitive capabilities are encompassed in the cognitive system and serve as a linkage between the perceptual and motor systems. Thus, these assume the role of interpreters, translating environmental stimuli into meaningful patterns for further processing. The main function of these analytical mental skills is problems solving and decision making. The list of cognitive capacities is broad and includes, but not limited to memory, perception, attention, anticipation, and situational assessment.

An example from tennis is provided here to demonstrate and clarify the importance of cognitive capabilities in the sport domain: A player returning a tennis serve is required to process environmental information, which stems from the player serving the ball (the height and positioning of the ball, hip angle, arm and racquet motion, etc.) and other potentially salient sources of information (prevailing wind direction, the court characteristics, the match score, etc.). The capacity to efficiently analyze this information and anticipate the final location, speed, and spin of the ball, and generate plausible return options) for instance, is indicative of the returner's cognitive capabilities.

Measuring and Capturing Cognitive Capabilities

The exploration of cognitive components evolved from cognitive psychology and the information-processing approach, in which the computer processor was used as a metaphor for the black box (the human processor). Thus, similarly to computers, cognitive capabilities were traditionally measured by speed (e.g., reaction time), quantity (e.g., size of knowledge base), and quality (e.g., decision optimization) information. These three cognitive characteristics systematically distinguish among athletes of different skill levels, even more so than

physical and technical skills. Although sport activity is primarily defined as a motor task, the cognitive components appear to be the decisive factor in the attainment of expertise.

As a result of the limitations in observing cognitive capabilities, a combination of creative research designs and state of the art technology are required to study these processes. Initially, researchers focused on measuring basic memory functions by utilizing pattern recognition and recall tasks (see also the entry "Pattern Recognition and Recall"). These set of studies attempted to capture athletes' ability to recognize and recall static and dynamic game patterns and structures. Although experienced, skilled athletes exhibited superior performance over novices on these tasks, ensuing studies provided evidence that these capabilities are relatively easy to train and do not afford information to the underlying mechanisms leading to the acquisition of advanced cognitive capabilities. Thus, research paradigms and designs shifted to the examination of more domain-relevant and representative tasks, which have been used in the past 30 years.

An innovative assessment tool, the temporal and spatial occlusion paradigm, explores athletes' anticipatory capability in representative situations (see also the entry "Anticipation"). The paradigm consists of a domain-specific, sport action video sequence, presented to athletes of different skill levels. In the spatial occlusion paradigm, different areas of the display are blocked for part or the duration of the entire action sequence. While using temporal occlusion, the action sequence is occluded at a certain point in time and usually remains occluded until the end of the sequence. Participants are then required to predict the following events of the sequence, based on the limited information they received. Thus, by observing differences in performances among spatial areas and temporal points, insight is given to the key environmental stimuli needed for successful anticipation. Additionally, under this paradigm, skilled athletes are able to predict opponents' intentions early and accurately by utilizing cues more efficiently (see also the entry "Cue Utilization").

To assess visual search behaviors that lead to anticipation and enhance decision making, eye-tracking technologies are used. The introduction of this equipment enables researchers to measure variables, such as fixation areas, fixation duration, frequency, and search order or sequence.

Comparison between successful and unsuccessful performances allows researchers to identify effective search strategies and gaze behaviors. Although this technology provides detailed and imperative information, there are several limitations that should be considered: the calibration of the instruments, inference from gaze behaviors, size and mobility of the equipment, and connection issues.

Finally, with the use of retrospective and concurrent verbal reports, researchers are able to explore the intact thought process from stimuli detection to action generation. In this process-tracing research method, athletes provide detailed reports of their thoughts during or after performance. This procedure enables gaining insight and access to cognitive processes.

All of the above-mentioned capturing and measuring paradigms make use of the expert–novice paradigm in which performance comparisons were used to infer on the cognitive capabilities necessary for achieving successful performance. Extant research using these paradigms resulted in abundant and valuable knowledge related to skill level differences in cognitive capabilities.

Expert–Novice Differences

Expert athletes are able quickly and accurately to anticipate future events. They are capable of assessing a situation efficiently by integrating in-vivo information from the environment with stored information like knowledge base and long-term memory, utilizing an advanced memory structure termed *long-term working memory*. Specifically, long-term working memory facilitates the acquisition of mental representations and planning. Thus, by relying on advanced cognitive capabilities, elite athletes display superior decision-making skills. These decisions are characterized by flexibility, creativity, and adjustment to dynamic situations.

Cognitive capabilities are often divided into different sequential processes. Visual search behavior is the initial process and refers to where athletes gaze and what they attend to. The two prevailing strategies are context (attending to a large visual field) and target (attending to a narrow visual field) control. Novice athletes cannot utilize context control strategies because of limitations in the amount of information they can process. On the other hand, experts have the flexibility to shift from context to target-control strategies depending on environmental demands.

Furthermore, experts exhibit advanced cue utilization skills. This allows them to incorporate complex and expansive cues in the anticipation process, and predict what will occur next with higher probability and confidence. Skill level differences are also evident in the option generation and prioritization process. Primarily, experts generate more relevant options and efficiently prioritize them. This cognitive advantage is subsequently reflected in the action selection process, where the expert athlete is able to determine the action that will result in the best possible outcome. More importantly, experts have the capability to alter a previously selected action pattern on the fly. Finally, elite performers evaluate their performance with better precision, compared to low-level performers, and hence can detect errors in their performance (process and outcome) and adjust their actions accordingly in the upcoming situations.

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and Oscar Gutierrez*

See also Anticipation; Cue Utilization; Expertise; Information Processing; Pattern Recognition and Recall

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COGNITIVE FUNCTION

Cognitive function refers to the mental process of knowing or thinking. It involves all aspects of mental processes that enable individuals to perceive, recognize, process, and understand thoughts. Specifically, cognitive function involves processes such as action, attention, memory, learning, reasoning, planning, problem solving, decision making, and communication. These cognitive functions are not only important to daily life but

are also recognized as main components in health-related quality of life.

Cognition: A Lifespan Spectrum

Although the major brain areas that are the core resource of cognition are set at birth, along with emerging and forming networks of brain cells, cognition and intelligence continuously develop throughout childhood, adolescence, and into young adulthood, thereafter declining gradually after adulthood. For example, with the exception of vocabulary, multiple cognitive functions, including the speed of information processing, reasoning, and memory, peak at 20 years of age and then decline linearly until the age of 70 years. Generally, compared to the entire population, an individual's cognitive function at 20 years is above the 75th percentile, at 50 years it is near the 50th percentile, and at 70 years, it is in approximately the 20th percentile. This decline of cognitive function is linked to cognitive impairment in later life, such as dementia and Alzheimer's disease.

As one can easily notice, the rate of cognitive development during early life and cognitive decline later in life exhibit marked interindividual variability. Environmental factors and an individual's experiences, including physical activity, intellectual engagement, social interaction, nutrition, and self-confidence, can positively influence these cognitive changes. Participation in physical activity is particularly emphasized in both lifespan phases by numerous social and health institutions and organizations, because increased physical activity has been related to multiple benefits affecting physical and mental health in children and older adults.

Physical Activity and Cognition: Different Perspectives

Early and Large-Scale Observational Studies

Physical activity and cognition were first investigated by Waneen Spirduso, Karen Francis, and Priscilla MacRae in the 1970s. While the researchers found that older adults demonstrated a longer reaction time of information processing in response to several cognitive tasks compared with younger adults, they also found that both older and younger adults showed differences based on participation in sports. The older adults who participated in sports regularly not only had a greatly increased cognitive speed compared to older

sedentary adults, but also had similar cognitive performances as young sedentary adults, suggesting that participation in sports positively moderated the age-related cognitive decline.

The positive associations between physical activity and cognitive performance have also been shown in large-scale epidemiological studies. Generally, studies of older adults based on retrospective reports of physical activity have suggested protective effects against cognitive decline and cognitive impairment. However, these subjective measurements of physical activity are somewhat imprecise, and information regarding the modality, intensity, frequency, duration, and length of participation in physical activity is often deficient.

A study focusing on physical fitness, rather than physical activity, which involved approximately 900,000 California schoolchildren, demonstrated that increased physical fitness was significantly positively associated with enhanced reading and mathematics scores, regardless of gender and grade. Although a cause-effect relationship cannot be established from observational studies, the positive correlation between physical activity and cognition is promising.

Interventional Studies

Using experimental designs, particularly with randomized assignment into an exercise group or non-exercise control group, some studies have provided stronger evidence for a cause-effect relationship. Interestingly, while an older adult group with long-term physical activity improved in general cognitive performance, such as speed-, spatial-, controlled-, and executive function-related cognition compared with a sedentary control group, physical activity training also provided the greatest benefit to executive function. Executive function is a high-order type of cognition that is used to regulate, control, and manage multilevel basic information processing for purposeful and goal-directed behaviors. Therefore, older adults who participated in long-term physical activity demonstrated improved cognitive function in general, and selectively, particularly in executive function.

Long-term exercise may also benefit cognitive performance in overweight children. Research has indicated that participating in an exercise program 5 days per week over 15 weeks for 40 minutes per day at moderate intensity improves the planning aspect of cognition. It should be noted that

performing this exercise protocol for 20 minutes per day had only a limited effect, suggesting that the duration of 40 minutes is the minimum threshold for cognitive function to be enhanced among children. Physical activity in school-based physical education, play during recess, classroom physical activity, and extracurricular physical activity, has either been found to be positively related to academic attitude, behavior, and achievement or to have no relationship, suggesting that participation in physical activity at school does not negatively affect student achievement.

Chronic Exercise Versus Acute Exercise

Besides studies that target long-term physical activity, known as *chronic exercise*, the effect of single bouts of physical activity, known as *acute exercise*, on cognitive function has also been investigated. Generally, when performing acute exercise, individuals with higher fitness and those who do 20 minutes or more of exercise per session show greater improvements in cognitive performance than those with low fitness and those who perform exercise of shorter durations. Similar to chronic exercise, which has been shown to benefit many types of cognition including information processing and attention, acute exercise also has a positive effect on executive function. Although the effect of acute exercise on cognition is of a somewhat short duration, higher but not extreme exercise intensity appears to prolong the positive effect to 60 minutes. The beneficial effects of acute physical activity on cognition have recently been extended to overweight children and children with attention deficit–hyperactivity disorder.

Although both chronic and acute exercise benefits cognitive function, the mechanisms are likely to be different. According to the cardiovascular fitness hypothesis, the improvement in cognitive performance following chronic exercise is mediated by gains in cardiovascular fitness. In support of this hypothesis, studies using sophisticated neuroscientific and psychophysiological methods (e.g., magnetic resonance imaging or event-related potentials) have indicated that both older adults and children with high fitness levels show larger brain volumes in areas implicated in cognitive function (including gray matter, white matter, specific cortical structures like the prefrontal cortex, and subcortical structures like the hippocampus) than their counterparts with low fitness levels. In contrast, the mechanism linking acute exercise and

cognition is believed to be exercise-induced arousal or enhanced allocation of attentional resources and improved efficiency of stimulus evaluation in response to a given cognitive task.

Exercise Mode

Exercise-cognition research primarily emphasizes aerobic exercise, possibly because this exercise mode is linked to cardiovascular fitness. However, a recent line of investigation has evaluated the effect of exercise types on cognitive function. Particularly with respect to older adults, studies have demonstrated that resistance exercise training may also improve cognition. It is speculated that the ability to improve cognition through resistance exercise is based upon the increase of certain neurotrophic factors, such as insulin-like growth factor I (IGF-I), which are upregulated in response to muscle contraction. IGF-I is not only negatively related to aging but has also been linked to the promotion of neurogenesis, neuronal survival, and synaptic plasticity, which collectively can lead to improved cognitive performance. Furthermore, given that acute resistance exercise elevates physiological arousal similar to acute aerobic exercise, it is believed that acute resistance exercise can also benefit cognition. Indeed, positive effects of resistance exercise have been observed in studies evaluating executive function among young and middle-age adults.

Coordinative exercises, such as gymnastics, martial arts, soccer, or dance, have also been linked to executive function in kindergarten children. It is believed that the adaptation of coordinative exercise, which involves complex movement and cognitive demand for goal-directed behavior, may improve neuromotor abilities in both the peripheral and central levels, which in turn may enhance cognitive performance. In addition to cardiovascular fitness, coordinative exercise may also advance cognitive functions by increasing motor fitness, including movement speed, balance, motor coordination, and flexibility. In fact, individuals with better motor fitness have demonstrated to activate more cognition-related brain regions during the cognitive tasks.

Conclusion

Although cognitive functions develop and deteriorate during an individual's lifespan, participation in physical activity may alter the rate of these

changes. Large-scale observational and interventional studies have revealed a positive relationship between physical activity and cognitive function in older adults, as well as a positive correlation between school-based physical activity and academic achievement in children. In addition, chronic exercise has been shown to impact cognitive functions in general and also, more selectively, executive functions. Similarly, beneficial effects of acute exercise have also been identified, although the magnitude of these effects can be moderated by fitness status, exercise duration, exercise intensity, and the type of cognitive task. Notably, the biological mechanisms associating chronic and acute exercise with cognition are likely to be different. Along with the beneficial effects of aerobic exercise on cognition, studies have also demonstrated the potential of resistance exercise and coordinative exercise for enhancing cognitive activity. Taken together, this body of evidence indicates that physical activity has beneficial effects on cognitive function throughout an individual's lifespan.

Yu-Kai Chang

See also Brain; Brain Imaging; Decision Making; Information Processing; Memory; Mental Health; Neuroscience, Exercise, and Cognitive Function; Psychophysiology

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COGNITIVE RESTRUCTURING

Cognitive restructuring is a technique that is commonly taught to athletes by sport psychologists in which self-defeating thoughts and negative self-statements are identified and substituted with positive, adaptive self-statements, and coping thoughts. Cognitive restructuring was originally developed in clinical settings and has since been used by practitioners in various contexts (including sport) to address a range of issues related to performance, social anxiety, maladaptive perfectionism, aggressive behavior, depression, and low self-esteem.

Generally speaking, cognitive restructuring has been examined as part of a package of mental interventions. These typically include the setting of effective goals, visualization, relaxation strategies, and self-talk. Invariably, psychological skills packages are associated with improved athletic and sporting performance. For instance, the positive effects associated with psychological skills packages have been reported in sports as diverse as baseball, basketball, boxing, figure skating, golf, gymnastics, karate, skiing, tennis, and volleyball. Pertinently, systematic reviews of the effects of psychological skills on performance indicate that cognitive restructuring interventions often have large positive effects on performance, suggesting these interventions are particularly effective. As yet, few studies have involved direct tests of the mechanisms that might account for the positive performance effects associated with cognitive restructuring. However, there is a large body of evidence indicating that cognitive restructuring is associated with improvements in a variety of psychological variables such as increased self-efficacy, reduced cognitive and somatic anxiety, an increased ability to cope, as well as increased effort and motivation to succeed. As a word of caution, research suggests that the skills required to deliver cognitive restructuring interventions require many

hours of training to master. Practitioners considering using cognitive restructuring should bear this in mind prior to delivering such interventions.

Cognitive restructuring is often described as a technique in its own right. In this way, cognitive restructuring involves four sequential steps: (1) identifying the individual's negative thoughts or self-statements during problematic situations, (2) identifying coping self-statements and rehearsing them, (3) replacing negative self-defeating statements with coping self-statements, and (4) identifying and rehearsing positively reinforcing self-statements. However, it is worth noting that cognitive restructuring is more commonly used within a rational emotive behavior therapy (REBT) or a cognitive-behavior therapy (CBT) framework.

Albert Ellis developed REBT in the 1950s. At its core, REBT attempts to apply rationality and logic to a person's beliefs. Rational beliefs, which are thought to be at the core of psychological health, are flexible, consistent with reality, logical, and self-enhancing. Irrational beliefs, which are thought to be at the core of psychological disturbance, are rigid, inconsistent with reality, illogical, and self-defeating. In the context of sport, irrational beliefs often underlie much of the stress and resulting self-defeating thoughts and feelings experienced by athletes prior to or during athletic performance. REBT teaches that it is the irrational beliefs and thinking, and not the event or circumstance that contributed to the irrational beliefs, that lead to negative emotions. Sport psychology practitioners can help athletes reduce their self-caused negative emotions by enabling them to identify and dispute their irrational beliefs via a process known as ABCD cognitive restructuring:

- A: Athletes are asked to keep a diary of daily events that generate negative emotions. The athlete is asked to describe the facts of the events as they occurred.
- B: The athlete is to record in the diary the exact content of the dysfunctional self-talk (said out loud or silently in private) that followed the activating events.
- C: The athlete then records the resulting emotional or behavioral responses.
- D: After completing the ABC steps across a designated number of days, the final step requires the athlete to identify which aspects of the self-talk are irrational or distorted and substitute more rational and productive thoughts in their place.

This process is practiced repeatedly until the substituted rational statement is internalized.

Cognitive-behavioral therapies are often an amalgamation of cognitive and behavioral procedures. While some treatments may emphasize cognitive more than behavioral techniques, and vice versa, these interventions typically share an appreciation for basic learning principles *and* the role that cognitions play in human behavior and affective experience. When cognitive restructuring is used within a CBT framework it is often considered a discrete part of a wider intervention. This is because cognitive-behavioral therapies are more directive, educational, and future focused than traditional psychotherapies. Cognitive restructuring is one of the skills taught to the client as part of the therapy but it is often accompanied by other cognitive-behavioral techniques like relaxation training.

James Hardy and James Bell

See also Affirmations; Positive Thinking; Self-Talk; Thought Stopping

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COGNITIVE STYLES

Broadly defined, *cognition* refers to mental operations involving information processing and thus includes processes such as perception, problem solving, memory recall, and decision making. The term *cognitive styles* refers to the different

approaches people characteristically use in undertaking cognitive tasks. Considered to be a personality trait and representing both nature and nurture effects, cognitive styles are thought to influence individuals' values, attitudes, and social interactions. Examples of cognitive styles include (a) reflectiveness versus impulsiveness, (b) cognitive complexity versus simplicity, and (c) tolerance versus intolerance for unrealistic experiences. Individuals are thought to differ in their quickness to respond to stimuli, their ability to process complex cognitive information, and their acceptance of unexpected experiences. To explain variability in cognitive styles, scholars from various domains have generated a number of theoretical and applied explanations addressing differences in (a) hemispheric lateralization, (b) perception and information processing, (c) problem-solving approaches, and (d) field dependence or independence. Aligned with the idea of cognitive styles, there is the notion of learning styles, specifically, (a) different types of learning and (b) idiosyncratic modes of intelligence. This entry presents research findings on both cognitive and learning styles and their application to the sport psychology domain.

Neuroscience research has revealed that the right hemisphere of the brain controls holistic and pictorial processing while the left hemisphere is more centrally involved in analytical and logical operations. Accordingly, peoples' cognitive styles are expected to vary on a continuum ranging from primarily logical-analytical (left-hemisphere dominant) to predominantly holistic-pictorial functioning (right-hemisphere dominant). Furthermore, individuals' cognitive styles may be further categorized into two orthogonal dimensions: (1) holistic-analytic and (2) verbal-imagery. The holistic-analytic dimension refers to how individuals gather and store structured information. Holists retain a gestalt, global conceptualization, of information, whereas analytics deconstruct information into its subcomponents. The verbal-imagery dimension refers to how individuals decode structured information. Verbalizers decode information by using words and verbal associations, whereas visualizers create mental pictures to represent structured information. These styles or individual differences typically exist on a continuum, and most people do not fall clearly into one or other discrete style classification.

Moreover, people perceive and process information using different methods. Activists learn

through active experimentation and by proactively engaging in new experiences. Theorists generate hypotheses and then engage in analytical thinking and deductive reasoning. Pragmatists prefer methods based on measurable goals and practical outcomes. Reflectors, in contrast, engage in conscious and deliberate processing before comprehending and storing new information. Theoretically, therefore, activists are likely to learn best when exposed to new experiences; theorists optimize their learning experiences when presented with testable concepts; pragmatists prefer simulations of real scenarios; and reflectors excel when working on observational reports and analyses.

People also differ in their tendencies toward solving problems by engaging in (a) adaptive or innovative solutions, and (b) convergent or divergent thinking approaches. Adaptors utilize existing paradigms to generate solutions for a given problem, whereas innovators create new paradigms to problem solving. Similarly, convergent thinkers generate a singular and accurate response to a given problem, whereas divergent thinkers generate multiple responses. Additional variability in information processing among human beings may also be explained through a tridimensional approach based on (1) emotional or relational, (2) mental, and (3) physical dimensions. The relational or emotional dimension refers to knowledge primarily gained through social interactions and verbal and nonverbal communication. The mental dimension pertains to information acquired via deliberate thinking, abstract conceptualization, objectification, and reflection. The physical dimension represents knowledge apprehended via action-oriented skills, such as playing a sport or musical instrument.

Field dependence or independence is another extensively studied cognitive style. According to cognitive control theory, field dependent individuals rely on external cues and tend to generate a macroglobal view of a given context. Conversely, field independent individuals rely on internal cues and focus on identifying detailed information of a given context. Furthermore, field-dependent individuals have a greater social orientation than field-independent individuals, who are more likely to be introverted. Although not conclusive, research has shown that demographic, cultural, and situational factors may also influence people's reliance on field dependence or independence styles. For example, children and older adults are more likely to be field-dependent than younger adults. Also,

Western societies are thought to be primarily field-independent, whereas non-Western cultures have been described as predominantly field dependent. Finally, research suggests that people may also vary their preferred cognitive style from task to task.

Research in the sport and exercise psychology domain has revealed personal differences in athletes' cognitive styles. For example, Robert M. Nideffer's theory of attentional and personal styles suggests that people vary in their preferred attentional focus. More specifically, athletes have been shown to vary in both the width (broad to narrow) and direction (internal or external) of their preferred attentional style. Furthermore, research findings in the athletic domain confirmed the central tenets of attribution theory, as different sport actors (athletes, coaches, or referees) have been shown to vary in their causal attributions accounting for their success or failure. Finally, studies assessing learning and motor performance, as a function of field dependence or independence, have not been conclusive. This is congruent with current understanding of expert performance in sports, which suggests that reliable superior performance is dependent upon the task- and domain-specific cognitive skills.

Cognitive styles have also been linked to learning preferences. More specifically, people tend to prefer some learning modalities, such as visual, auditory, and tactile or kinesthetic, over others. Visual learners rely primarily on diagrammatic and pictorial information to gain knowledge of a particular subject. Auditory learners gain knowledge by primarily attending to spoken and written information. Kinesthetic or tactile learners acquire new knowledge by experiencing new activities and performances. Accordingly, kinesthetic or tactile learners should be given the opportunity to try out their new skills. Auditory learners may benefit from class discussions, as well as thinking and reading aloud exercises. Learning styles, however, may not be as discretely different as originally categorized.

Recommended instructional activities for visual learners include use of maps, videos, and presentations. Finally, a combination of multimethods like visual auditory and kinesthetic methods may facilitate learning across the board. However, it is important to note that learning styles may not be as discretely different as originally categorized.

Congruent with the idea of preferred learning styles, psychologist Howard Gardner has proposed

seven distinct intelligence modes underlying cognitive variability among human beings. Specifically, Gardner's model is grounded on the notion that people possess idiosyncratic mental capabilities represented by the following seven domains: (1) visual and spatial, (2) verbal and linguistic, (3) logical and mathematical, (4) bodily and kinesthetic, (5) musical and rhythmical, (6) interpersonal, and (7) intrapersonal intelligence. According to this view, creative architects primarily operate under visual and spatial intelligence mode; accomplished writers rely on verbal and linguistic intelligence mode, and so forth.

In summary, people vary in their preferred methods of processing information. For example, hemispherical dominance may explain why some individuals are more analytical than others. Personality tendencies may explain why some people are labeled activists and others theorists. Moreover, people vary in types of intelligence: field dependence, independence, and preferred attention and attribution styles. Research findings revealed that some individuals learn through visual channels while others primarily rely on their listening skills. There are also innovators, adapters, convergent, and divergent thinkers. This extensive variety of cognitive styles and abilities illustrates the complexity of designing optimal learning environments. Therefore, educators and applied professionals, such as coaches, should consider people's cognitive idiosyncrasies when preparing their instructional activities and selecting domains of expertise. Consideration of multimethods may facilitate learning across the board and retention of new and difficult information.

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See also Decision Making; Information Processing; Perception

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COGNITIVE TASK ANALYSIS

Cognitive task analysis (CTA) refers to a suite of scientific methods designed to identify the cognitive skills, strategies, and knowledge required to perform tasks proficiently. The goal of CTA is to use this information to improve instruction, training, and technological design (e.g., decision aids) for the purposes of making work more efficient, productive, satisfying, and of a higher quality, or to accelerate proficiency.

Background and Historical Development

CTA has a long history, spanning multiple communities of practice, ranging from those studying perception and thinking, to those studying the behavioral aspects of work and, subsequently, skilled performance in applied settings. Prior to the 1980s, these communities included (in approximate chronological order) those practicing introspection, applied and industrial psychology, task analysis, ergonomics, human factors, and instructional design.

In the early 1900s, physical labor dominated many aspects of work, and physical issues, such as fatigue and injury risk, were of concern. Accordingly, task analytic methods were often behaviorally oriented—designed to decrease inefficiencies and to increase productivity. Classic examples of task analysis include Frederick W. Taylor's time-motion studies of factory workers, and Frank and Lillian Gilbreth's study of bricklayers. Although worker behavior was emphasized during early task analyses, the decomposition of tasks using such methods rarely excluded cognitive aspects of work. However, over time, as work became more reliant on higher order cognition, the focus of task analysis shifted.

Despite historical antecedents, the term CTA did not emerge until the 1980s, when it was used to understand the cognitive activities involved in man-machine systems. A key turning point was a Nuclear Regulatory Commission workshop in 1982 on Cognitive Modeling of Nuclear Plant Operators, attended by cognitive and human factors psychologists David Woods, Donald Norman, Thomas Sheridan, William Rouse, and Thomas Moran. The changing nature of technology in the workplace and the increasing complexity of work systems resulted in a greater emphasis on cognitive

work and, therefore, a greater need for cognitively oriented task analytic methods.

From the 1980s, a system-oriented perspective to CTA prevailed, focused on understanding adaptive cognitive work in complex contexts. This *joint-cognitive systems* or *sociotechnical systems* perspective viewed cognitive work as an embedded phenomenon—inextricably tied to the context in which it occurs. In naturalistic contexts, (a) systems may comprise multiple (human and technological) components; (b) task goals are frequently ill-defined; (c) planning may only be possible at general levels of abstraction; (d) information may be limited; and (e) complexity, uncertainty, time-constraint, and stress are the norm. Cognition in these contexts is often emergent or distributed across individuals—and technology. Moreover, cognition is also constrained by broader professional, organizational, and institutional contexts, which influence the strategies, plans, goals, processes, and policies employed. Consequently, CTA has evolved as a means to study adaptive, resilient, and collaborative cognition in simulated environments and, especially, in the field.

CTA has been championed by the cognitive systems engineering and naturalistic decision-making communities, and cognitive scientists studying expertise, including cognitive anthropologists and ethnographers. The range of CTA methods is vast. Rather than present an exhaustive list of methods, the reader is referred to Beth Crandall, Gary Klein, and Robert R. Hoffman's 2006 book *Working Minds*. A more detailed description of one particular method, the Critical Decision Method, is provided later in this entry.

In general, CTA is most synonymous with knowledge elicitation and cognitive process-tracing data collection methods, of which there are several classes: self-reports, automated capture-based techniques, observation, and interviews. As with all empirical methods, there are strengths and limitations to each method. Self-reports, such as questionnaires, rating scales, and diaries, permit efficient data collection and interpretation, which can be automated by computer. However, valid psychometric scale development takes time and effort. Moreover, self-reports do not afford the opportunity for additional exploration of the data, which may be problematic given the possibility of participant self-presentation or amotivation.

Automated capture includes computer-based tasks or those implemented in simulated task

environments (e.g., situation awareness global assessment technique; temporal or spatial occlusion). These methods allow direct capture of important cognitive behavior in situ, such as making a specific prediction or decision. However, they may require supplementary methods (e.g., *a priori* goal-directed task analysis, verbal reports, eye movements) to generate design recommendations. Additionally, they are costly to establish, require extensive system and scenario development, and have limited flexibility.

Observation like ethnographic immersion and shadowing provide first-hand information about how events unfold (and permits post hoc verification of information elicited via other methods). However, as with many CTA techniques, it requires domain knowledge to identify useful targets for data coding and interpretation. Observation is not always feasible (e.g., low-frequency, life-threatening events), can be intrusive and, per the Hawthorne effect, can change the behavior of those being observed.

Structured and semistructured interview techniques (e.g., *critical decision method* and *crystal ball technique*) that employ directed probes can reveal information that other methods would not. In addition to conducting naturalistic studies, interviews and observations can be combined with experiment-like tasks (e.g., 20 questions, card sorting) to generate useful insights into the cognitive processes underlying superior performance. However, trained interviewers are required, interviewees' memory is fallible, and the validity of retrospective reports has been questioned.

The Critical Decision Method

The critical decision method (CDM) was developed by Gary Klein, Robert Hoffman, and colleagues and is adapted from the critical incident technique (CIT) developed by John Flanagan and colleagues (including Paul Fitts)—which was designed to generate a functional description of, and identify the critical requirements for, on-the-job performance in military settings. Like the CIT, rather than probe for general knowledge, the CDM is a case-based, retrospective, semistructured interview method. It uses multiple *sweeps* to elicit a participant's thinking in a specific, nonroutine incident, in which they were an active decision maker and played a central role. A particular goal of the CDM is to focus the interviewee on the parts of the incident that most

affected their decision making and to elicit information about the macrocognitive functions and processes—such as situation assessment, sense-making, (re)planning, and decision making—that supported proficient performance. Although elicitation is not limited to events that could be directly observed, the use of skilled individuals and specific, challenging, and recent events in which they were emotionally invested was hypothesized to scaffold memory recall based on their elaborate encoding of such incidents.

In the first sweep of the CDM, a brief (e.g., 30- to 60-second) outline of a specific incident is elicited (see below for pointers on framing the interview). In a second sweep, a detailed timeline is constructed to elaborate on the incident. This should highlight critical points where the interviewee made good or bad decisions; the goals or understanding changed; the situation itself changed; the interviewee acted, failed to act, or could have acted differently; or relied on personal expertise. Once delineated, the timeline is restated back to the interviewee to develop a shared understanding of the facts and to resolve inconsistencies.

In the third, progressively deepening, sweep, the interviewer tries to build a contextualized and comprehensive description of the incident from the interviewee's point of view. The goal is to identify the interviewee's knowledge, perceptions, expectations, options, goals, judgments, confusions, concerns, and uncertainties at each point. Targeted probe questions are used to investigate each point in turn. The CDM provides a list of useful probes for many different types of situations. For instance, probes for investigating decision points and shifts in situation assessment may include: "What was it about the situation that let you know what was going to happen?" and "What were your overriding concerns at that point?" Probes for investigating cues, expert strategies, and goals may include: "What were you noticing at that point?" "What information did you use in making this decision?" "What were you hoping to accomplish at this point?"

A fourth sweep can be used to gain additional insight into the interviewee's experience, skills, and knowledge using *what if* queries, for instance, to determine how a novice may have handled the situation differently. Although some probes require reflection on strategizing and decision making, which increases subjectivity and may affect reliability, they provide a rich source for hypotheses

formation and theory development. With subsequent analysis, these data can be leveraged into design-based hypotheses, for training or technology development to aid future performance and learning.

From Elicitation to Design

To meet the goal of communicating results (e.g., from a set of CDM interviews) for improving system performance, CTA embraces a range of data analysis and knowledge representation methods. Quantitative and qualitative methods are leveraged to understand the data, including hit rates, reaction times, thematic or categorical analyses, chronological or protocol analyses, and conceptual and statistical or computational models.

As with all qualitative analyses, the goal of the CTA practitioner is to unravel the story contained in the data. After conducting data and quality control checks, analysis is used to identify and organize the cognitive elements so that patterns in the data can emerge. Importantly, elements should be informed by asking cognitive questions of the data, such as: What didn't they see? What information did they use? Where did they search? What were their concerns? What were they thinking about? At this stage, knowing the data is key and organizational procedures—such as categorizing, sorting, making lists, counting, and descriptive statistics—can be used to assist in this effort.

Once individual elements have been identified, the next goal is to identify the higher order data structure that describes the relationships between elements. This is done by generalizing across participants to describe regularities in the data (e.g., by looking for co- and re-occurrences, or their absence, that might signify a pattern); organizing elements into inclusive, overarching formats (e.g., create tables of difficult decisions, cues used, and strategies employed); looking for similarities or differences across groups (e.g., cues used by experts but not novices); and using statistical analyses to examine differences and relationships.

Following the data analysis, knowledge must be represented in a useful form. Fortunately, many forms of representation exist as a natural part of the analysis process. However, representations developed early in the process will be data driven, whereas those developed later should be meaning driven. Narrative-based representations

(story summaries) can extend participants' verbalizations to highlight what is implicit in the data. Graphical chronologies like timelines that retain the order of events can be used to represent the multiple viewpoints of team members and permit subjective and objective accounts to be linked. Data organizers, such as decision requirements tables, permit multiple sources of information to be synthesized to form an integrated representation. Conceptual mapping tools (e.g., <http://cmap.ihmc.us>) allow knowledge structures to be graphically and hierarchically represented. Process diagrams like a decision ladder represent cognition in action and provide insights into aspects of cognitive complexity that might otherwise appear to be simple.

The selection of CTA methods should be driven by framing questions, including: What is the primary issue or problem to be addressed by the CTA? What is the deliverable? Inexperienced CTA practitioners frequently underspecify the project and adopt a method-driven, rather than problem-focused, approach. To overcome these biases and to direct project resources efficiently, practitioners should become familiar with the target domain, the study of micro- and macrocognition, and the range of CTA methods available, and they should conduct preliminary investigations to identify the most cognitively challenging task components.

Translating CTA into actual design is often the least well-executed and most ambiguous step. However, it need not be. Making explicit what will be delivered (training plan, intervention, or decision aids) and agreeing on this with the target users permits the original goal—positively changing (system) behavior—to be attained. To do this effectively, however, the CTA practitioner needs to identify the stakeholders involved, understand their goals and needs, and determine the intended use of the deliverable. Frequently, data exist that can help frame a CTA project—for instance, behavioral task analyses or documented evidence about training or system inadequacies. Generating outcomes without reference to these issues will likely result in design recommendations or tools that do not generate an effective—or even adequate—solution to the problem. Ultimately, the goal of any CTA is to use mixed methods to generate products that leverage expert data in a way that can improve performance.

Paul Ward

See also Cue Utilization; Decision Making; Expertise; Human Factors; Knowledge Structure; Simulation Training; Situational Awareness; Verbal Protocols

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COHESION

Cohesion represents the degree to which task and social bonds exist among group members, as well as the strength of individuals' attractions to the task and social activities of the group. This entry briefly highlights the history, characteristics, conceptualization, measurement, and correlates of cohesion within sport and exercise environments.

History

Interest in and discussion surrounding the concept of cohesion has a long history across a wide variety of academic domains, including sport, exercise, business, education, and the military. This has resulted in the development of a number of definitions, conceptualizations, and assessment tools. As examples, previous research has proposed that cohesion is represented by the factors keeping

members in the group, the degree to which the group can resist disruption, the commitment displayed by group members, and individuals' attractions to other members and the group's tasks.

Characteristics

Despite the varied historical approaches to understanding cohesion, the 1985 work by Albert Carron, Neil Widmeyer, and Larry Brawley served to provide a strong foundation for research within the fields of sport and exercise psychology, and highlighted four characteristics of group cohesion:

Multidimensional. There are a number of facets to consider with respect to cohesion, including the orientation of the social cognitions individuals hold about their group (*perceptions of group integration* and *personal attractions to the group*), as well as the focus of those perceptions (*task and social objectives of the group*). Importantly, groups within one context like sport or across contexts like sport versus business may differ in terms of which factors are most important to group unity.

Dynamic. Another assumption about the nature of cohesion is that it is dynamic. Essentially, cohesion can change over the course of group development. From a practical perspective, this is encouraging as it suggests that effective interventions could facilitate more positive perceptions of group unity.

Instrumental. Whether for task or social objectives, groups form with some purpose in mind. Furthermore, individuals hold perceptions about the unity with which the group pursues these objectives.

Affective. The interactions and communication that occur in groups result in social relationships among members and a larger sense of social cohesion.

Conceptualization

A well-accepted conceptual model of cohesion was advanced by Carron et al. in 1985 in conjunction with the development of their Group Environment Questionnaire. This model draws distinctions with respect to the two aspects of cohesion outlined previously (refer to the *multidimensional* characteristic of cohesion). The first considers

the general objectives of the group, and broadly assumes that these objectives can be classified as being either task or social in nature. Task objectives refer to those that focus on the productivity and performance of the group. In contrast, social objectives refer to those that focus on maintaining group harmony and interrelationships among the members.

A second distinction that is made in the conceptual model considers cohesion from the perspective of the group as a whole (group integration), as well as individual members' attractions to the group. Group integration refers to the degree of bonding, closeness, and unity displayed by the members, while individuals' attractions refer to the personal motivations and feelings that draw and keep members within the group.

Consideration of the general objectives of the group (task vs. social), as well as the group versus individual perspective, results in a four dimension model of cohesion: (1) group integration–task, the degree to which the group is united around task objectives (GI-T); (2) group integration–social, the degree to which the group is united around social objectives (GI-S); (3) individual attractions to the group–task, group members' personal perceptions about their involvement with task aspects of the group (ATG-T); and (4) individual attractions to the group–social, group members' personal perceptions about their involvement with social aspects of the group (ATG-S).

It should be noted that recent research has highlighted that the above conceptual model is most relevant to an adult population. Studies conducted with younger sport participants provided evidence that these individuals do not distinguish between *group integration* and *individual attractions to the group* perspectives. As a result, it has been suggested that researchers differentiate cohesion perceptions of children and youth based only on the general objectives of the group, that is, task versus social cohesion. This is reflected in the measures that have been developed in recent years, which are summarized in the subsequent section.

Measurement

A number of assessment tools exist that capture perceptions of cohesion held by a variety of different groups. The most recent versions are presented:

- *Group Environment Questionnaire* (GEQ). Developed in 1985 by Carron and colleagues, the GEQ directly corresponds with the four dimension conceptual model outlined earlier.
- *The Youth Sport Environment Questionnaire* (YSEQ). Early cohesion studies periodically assessed cohesion perceptions held by youth sport participants via the GEQ. However, researchers noted that some of the psychometric properties of the GEQ were questionable when used with this population. Others postulated that the advanced level of readability, use of negative phrasing for some items, and decreased applicability of the conceptual model may partially explain the experienced difficulties. As a result, Mark Eys, Todd Loughhead, Bert Carron, and Steven Bray developed the YSEQ in 2009 specifically to assess youth (approximately 13–17 years of age) perceptions of cohesion.
- *The Child Sport Cohesion Questionnaire* (CSCQ). Further extending the ability to assess sport participants' perceptions of cohesion, Luc Martin and colleagues developed an assessment tool relevant to children 9–12 years of age.
- *Questionnaire sur l'Ambiance du Groupe* (QAG). The Group Environment Questionnaire has been translated into several languages. The work conducted by Jean-Philippe Heuzé and Paul Fontayne in 2002 highlighted the difficulties in translating the meaning of specific questions across cultures and languages, but also reinforced the salience of the four dimension conceptual model of cohesion for adult populations.
- *The Physical Activity Group Environment Questionnaire* (PAGEQ). With the intention of gauging *exercisers'* perceptions of group cohesion within physical activity contexts, Paul Estabrooks and Bert Carron developed the PAGEQ. Its structure is similar to the GEQ in terms of dimensionality and response options.

Correlates

The availability of assessment tools and interest in group dynamics has yielded a number of insights pertaining to the antecedents and consequences of cohesion. The following sections briefly highlight two major correlates of interest, performance and adherence, and list a number of individual and group variables that have been linked with cohesion.

Performance. The cohesion–performance relationship has been a major research interest for the field. Substantial evidence exists that this relationship is positive and bidirectional, that is, greater cohesion leads to better performance and better performance leads to greater cohesion. Furthermore, this relationship exists at all competitive levels, from recreational to competitive levels, and across sport types (e.g., interdependent versus independent sports). Finally, of all the moderators examined, only gender appears to moderate this relationship. Specifically, the positive cohesion–performance link is stronger for female athletes.

Adherence. Another major variable of interest is the adherence of sport and exercise group members. The sport studies that have been conducted suggest that members of highly cohesive teams drop out less during the competitive season, show up to games and practices on time, expend more effort, and indicate stronger intentions to return to their team the following season. Furthermore, athletes' actual return to their sport team the following year has been predicted by their perceptions of group cohesion in the previous season. Similar findings have been demonstrated in exercise contexts, in that those individuals who participate in a cohesive group environment are more likely to adhere to their exercise regimens and accrue positive outcomes.

Individual variables. Perceptions of greater group cohesion tend to have positive associations with important variables at the individual level. As examples, cohesion is positively associated with athletes' sport satisfaction and negatively associated with perceptions of state anxiety and depression.

Group variables. In addition to team performance, stronger perceptions of cohesion relate to other group-oriented concepts, such as improved role involvement like greater role clarity, normative expectations, and collective efficacy, as well as lower levels of group conflict.

Potential Downsides of Cohesion

While the presentation of group cohesion highlights the many positive associations that cohesion has with important sport and exercise variables, it is also critical to consider whether high cohesion

creates challenges for groups. It is possible for highly cohesive groups to be more likely to have issues with groupthink (tendency for a group to apply normative pressures on its members within decision-making processes), deindividuation (loss of personal identity and self-awareness), and social loafing. In 2005, James Hardy and colleagues explored whether athletes perceived downsides of highly cohesive sport teams; their findings indicated that over half of the athletes identified specific issues on this topic. For example, a number of participants noted that high social cohesion among team members may challenge their ability to focus and commit to task goals.

Mark Eys

See also Conflict; Friendships/Peer Relations; Group Characteristics; Group Formation; Interdependence Theory and the Coach–Athlete Relationship; Roles; Shared Mental Models; Team Building

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COLLECTIVE EFFICACY

Albert Bandura defined *collective efficacy* (CE) as a group's shared belief in its conjoint capabilities to organize and execute the courses of action required to produce given levels of attainments, that is, situation-specific confidence in a group's ability. A commonly used definition of CE in sport and exercise is a group's (e.g., a sport team or an exercise group) belief in its ability to produce given levels of attainment. Whether in sport, exercise, or human behavior more broadly, CE is believed to be a primary determinant of group behavior and thought patterns that occur within groups. The content of this entry focuses mainly on CE in sport due to spatial limitations (but can easily be generalized to other settings such as exercise), provides a broad overview of related research, and is organized as follows. First, a distinction between self-efficacy (SE) and CE is provided. Measurement of CE then is described. Potential sources and proposed outcomes of CE then are reviewed. Future directions conclude this entry.

Self-Efficacy Versus Collective Efficacy

Like SE, CE beliefs focus on *can do* and not *will do* perceptions that reside within individuals. A key distinction between these two types of efficacy beliefs is the target of the efficacy appraisal or unit of agency where SE is focused at the individual level (my confidence in *my ability* to . . .) and CE is focused at the group level (my confidence in *my group's ability* to . . .). The relevance of CE depends, in part, on the presence of a sufficient level of interdependence within the group with regard to the performance task.

As an example, which will be used periodically throughout the remainder of this entry, suppose that female athletes are nested within female teams and that both SE and CE are of interest in relation to athlete-level and team-level performance, respectively. A team member can have beliefs about her own ability in relation to a particular individual performance task (SE for individual performance) that may differ from her beliefs in the team's ability in relation to a particular team performance task (CE for team performance). Similarly, individual athletes within the same team may differ markedly in their assessment of the team's ability in relation to a particular team

performance task (within-group variability of CE for team performance). For the purpose of this entry, let us assume that CE and team performance are of primary interest. It should be noted, however, that both types of efficacy beliefs and both types of performance tasks could be studied simultaneously and that doing so would probably better reflect actual behavior and thought patterns that occur within interdependent groups in practice.

Measurement of Collective Efficacy

Kevin Spink and Deborah Feltz encouraged the study of CE in sport in the early 1990s, several years after the study of SE in sport was well established. These two seminal studies did not define, and therefore measure, CE in the same way. In one of the studies, the definition of CE focused on the fact that groups often have collective expectations for success (e.g., What placing do you expect to attain in this performance task?). In the other study, the definition of CE focused on a group's belief in its conjoint capabilities to produce given levels of attainment (e.g., Rate your confidence right now that your team can outskate your upcoming opponent team). Both of these studies used a task-specific measurement approach, where unique items were developed with respect to the specific performance task under study. From these seminal studies, CE in sport has since been assessed in a variety of ways.

There are at least four task-specific methods that have been used to assess CE in sport. The first method involves measuring each athlete's SE beliefs in relation to a particular performance (e.g., how confident are *you in your ability* to . . .) and then combining team members' SE measures in some way to form a single CE ($\rightarrow$ CE) measure ($SE \rightarrow CE$) for each team. The second method involves measuring each athlete's CE beliefs (assumed to reside in the individual athlete; CEI) in relation to a particular performance (e.g., how confident are *you in your team's ability* to . . .) and then combining team members' CE measures in some way to form a single CE measure ($CEI \rightarrow CE$) for each team. The third method involves measuring each athlete's perception of the team's CE beliefs (assumed to reside in the team; CET) in relation to a particular performance (e.g., how confident is the *team in its ability* to . . .) and then combining team members' CE measures in some way to form a single CE measure ($CET \rightarrow CE$) for each team. The fourth

method involves using a group discussion format to obtain a single, group-level (CEG) measure of CE. The CEI→CE method and the CET→CE method have become the most widely used and are largely accepted as valid. A task-invariant instrument, the CE Questionnaire for Sports, also exists but this instrument appears to be used relatively infrequently to date.

Sources of Collective Efficacy

Albert Bandura proposed that CE beliefs, just like SE beliefs, are based on the complex cognitive processing of approximately four general categories of potential sources of information: past performance accomplishments, vicarious experiences, verbal persuasion, and physiological or emotional arousal. Neither vicarious experience (e.g., team modeling) nor physiological or emotional arousal (e.g., team burnout) have been formally studied as potential sources of CE information in sport. Researchers in sport psychology have identified several sport-specific sources of CE information and a few of these are briefly summarized below.

Past performance accomplishments have typically been identified as the strongest source of CE information and have been the most studied to date. There is evidence that past performance, in addition to directly affecting CE, may also affect SE, which in turn can affect CE. Another efficacy belief that may serve as a source of CE information is role efficacy. Role efficacy has been defined as an athlete's confidence in his or her capabilities to successfully carry out formal interdependent role responsibilities within a group.

Verbal persuasion has been shown to be an effective way to increase CE. For example, a motivational pregame speech delivered by a coach has been shown to increase CE. More generally, teams that exist within a more task-oriented motivational climate as compared to teams that exist within a more ego-oriented climate are believed to have greater CE due, in part, to the positive reinforcements that exist in such environments for more malleable outcomes than won-loss (e.g., improvement, hard work, teamwork). Finally, a coach's behavior (e.g., communication style) in general is a key source of CE information.

There are a few proposed sources of CE information in sport that do not fit neatly into any of the four general categories of potential sources of information but may nonetheless be important.

Team size has been shown to be an important factor but the direction and magnitude of the relationship seems to depend strongly on the specific type(s) of performance task(s) being studied. Length of time that a team has been together is another potentially important factor given that CE beliefs take time to form (recall, for example, the role of past performances). Length of time a team has been together may also be related to more general team development issues, which in turn may be related CE beliefs.

Outcomes of Collective Efficacy

Albert Bandura proposed that CE beliefs often are a primary determinant of group behavior and thought patterns that occur within groups. CE beliefs affect the choices a group makes (Do we attempt to take on a particular task?), the effort a group dedicates to a particular task (Will we give our best effort to this task?), and the persistence a group displays while working toward completion of a particular task (Given a setback, will we redouble our efforts toward accomplishing this task?). CE beliefs also affect the goals that a group sets (Will we select a goal that pushes our team to perform our very best?), the attributions a group makes (Given a setback, will we identify internal or external explanations for the outcome?), and the emotional reactions a group has (Given that we almost, but not quite, accomplished our goal and that we no longer have the opportunity to pursue this goal, will we feel proud of what we did accomplish?).

The effect of CE on team performance or team functioning has been the most studied outcome in sport. Much of the earlier work was done in controlled laboratory settings, while much of the more recent work has been done in less controlled field settings. A key advantage of the laboratory setting stems from the ability to randomly assign participants to groups (e.g., high efficacy vs. low). A key advantage of the field setting stems from the ability to collect data from real teams that are engaging in a task that is likely to be important to the team members. Regardless of the setting, a positive effect of CE on team performance has typically been observed.

Potential team-level outcomes of CE other than team performance have also been studied and a few of these are briefly summarized below. Team cohesion and CE have frequently been studied

simultaneously. Team cohesion has been defined as a dynamic process that is reflected in the tendency of a group to stick together and remain united in the pursuit of its instrumental objectives or for the satisfaction of member affective needs. In general, stronger CE beliefs have been found to be a positive predictor of team cohesion. CE as a predictor of team attributions has also been studied. In general, athletes who have strong CE beliefs also believe that their team's performance is something that could be controlled by the team, that is, the attribution for the outcome was internal to the team.

Future Directions

SE theory provides a strong theoretical base for the construction of the CE construct. Albert Bandura's model of CE serves as a very useful starting point for the nature, sources, and consequences of CE in sport. The definition and therefore the measurement of CE in sport have both become more consistent across time. Clearly there is an emerging basis from which future scholarship likely can proceed in some way. A future direction that may prove fruitful is to develop sport-specific and conceptual model(s) of CE that expand potential sources of CE information as well as potential outcomes of CE.

Nicholas D. Myers

See also Coaching Efficacy; Cohesion; Group Characteristics; Interdependence Theory and the Coach–Athlete Relationship; Relational Efficacy Beliefs in Coach–Athlete Relations; Self-Efficacy; Social Cognitive Theory; Team Building

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COLLECTIVISM/INDIVIDUALISM

Clients in sport and exercise psychology contexts are from a diversity of cultural backgrounds. The terms *individualism* and *collectivism* have recently been integrated within sport psychology as elements of one criterion by which to understand the client's cultural standpoint. Individualism has often been associated with mainstream clients who are white and from the Western hemisphere. In contrast, people generally characterized as collective in their approach to self and world have often been depicted as South American or Asian in origin, or as having been born and raised in collective-minded communities such as an Israeli kibbutz or an aboriginal reserve. To assume, however, that every athlete born and raised in a collective community must be group minded, or that every client socialized in an individualist context must be driven by personal objectives would of course be too simplistic.

Socialization

A person's inclination to view the self or collective at the center of interpersonal exchanges is a matter of many personal factors and not necessarily the effect of such broad considerations as one's geographic origin, skin color, or religion. For example, there are white, mainstream, North American athletes who are collectively inclined, just as there are fitness enthusiasts from Latin American or Asian cultures with an individualistic mindset as they approach their physical exercise. Clients must be understood in their own right and not on the basis of stereotypes. However, it should also be observed that some clients are inclined toward the values of

individualism, where others are somewhat or fully collective in their approach to performance.

Individualism

What differentiates someone with an orientation that begins with self from another with a collective inclination? A person who is situated more toward individualism in one or more areas of life would often use language that places one's self at the center of discussions, such as when the elite athlete is interviewed by the media and attributes personal accomplishments exclusively to personal efforts and personal abilities. Individualism is also typified through the use of theoretically informed objectives, including self-determination, self-efficacy, self-concept, and self-esteem, as well as the behaviors that manifest the intentions underpinning each term. Decisions are often informed more by personal objectives than those from within one's communities of affiliation, whether geographic, peer group, religious, or professional. Personal goals are derived from personal interests, ending in personalized accomplishments.

Collectivism

The collectively inclined person, in contrast, seeks position in relation to one or several groups, with the views and practices of the group(s) informing, and in some instances, determining personal decisions and behaviors. The decision to become an elite athlete might be informed by how that career path contributes to those living in one's community. Thereafter, the collectively inclined client might opt to return to that community and share experiences and skills with others for the collective's betterment, such as when a national team athlete or professional athlete returns to the cultural community to become a coach, or when one studies physical education and then returns to one's locale to work a physical educator. Hence, the person's decision to act in a certain manner or develop targeted skills will align with the development of the group as a whole and also informed by the community's stakeholders. The importance of social support, both in its reception and provision, becomes particularly salient for people socialized with a collective mindset.

Research Applications

There are research strategies that illustrate inclinations toward individual and collective values.

Consider qualitative strategies that reflect an emphasis on the individual, including personal interviews and single-subject case studies. Practices that are individualist are best used with participants holding like-minded cultural norms. When working with people who are more collectively inclined, one might consider group interviews to encourage group exchange and the convergence of ideas. Within research, the objective would be either to conceive of strategies that match where participants reside on the aforementioned continuum or at least to approach research studies with an awareness that orthodox research practices for individualist participants might cause discomfort among collectively inclined participants, and vice versa.

Conclusion

There are strategies that could be employed to better understand where the client resides in terms of self in relation to the collective. Perhaps the most basic suggestion is to listen with care to the client in terms of how the client describes, and therefore positions, him- or herself in relation to others. When motives seem to be informed by a community perspective, the sport or exercise psychologist can seek further clarification about where the client is from, what the values are within the community of origin, and whether these values are held dear by the client. Conversely, also through listening, when the client's description of self centralizes personal accomplishments, again, the practitioner might follow up with questions that clarify the client's origins and socialization.

Robert J. Schinke

See also Assimilation; Attribution Theory; Cultural Competence; Cultural Safety; Decision Making; Diversity; Race; Self-Presentation; Self-Talk; Team Communication

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COMMITMENT

Sport commitment is a central motivational construct because it goes right to the heart of athletes' persistent pursuit of their sport. Simply put, it is a psychological state explaining why athletes do what they do. There are two types of sport commitment: enthusiastic and constrained. *Enthusiastic commitment* (EC) is the psychological construct representing the desire and resolve to persist in a sport over time. Conceptualized recently to provide a more complete picture of commitment, *constrained commitment* (CC) is the psychological construct representing perceptions of obligation to persist in a sport over time. Knowledge of the sources of both types of commitment is critical to understanding the commitment process. These sources are presented as part of the sport commitment model (SCM) detailed next.

The SCM explains the complex process of commitment by encompassing the sources of both types of commitment, and their predicted behavioral consequence: actual persistence in one's sport. The SCM has been tested, modified, clarified, and expanded from the time it was first introduced in the literature in 1993. Many researchers have contributed to this progress, and using the combination of quantitative survey and qualitative

interview methods has increased significantly the depth with which we understand what it is that creates commitment.

The sources of commitment are presented in this entry in the following manner. First, every source in the SCM is defined. Next, the hypothesized or explored relationships between the sources and EC and then CC are summarized.

Sources of Commitment: Construct Definitions

Sport Enjoyment—the positive emotional response to a sport experience that reflects generalized feelings of joy.

Valuable Opportunities—important opportunities that are only present through continued involvement in a sport, such as traveling to competitions or playing in a future World Cup.

Personal Investments-Loss—personal resources put into a sport that cannot be recovered if participation is discontinued. Example investments include time and effort expended.

Personal Investments-Quantity—the amount of personal resources put into a sport.

Emotional Social Support—the encouragement, caring, and empathy received from significant people to the athlete such as parents, coaches, and teammates.

Informational Social Support—the provision of useful information, guidance, or advice received from significant people.

Desire to Excel-Mastery Achievement—striving to improve and achieve mastery in a sport.

Desire to Excel-Social Achievement—striving to win and establish superiority over opponents in a sport.

Other Priorities—attractive or pressing alternatives that conflict with continued sport participation such as work, educational pursuits, and family.

Social Constraints—social expectations or norms that create perceptions of obligation for the athlete to remain in a sport; for example, anticipating the disappointment of parents or coaches if the athlete dropped out of the sport.

Sources of Commitment: SCM Predictions

Sources of Enthusiastic Commitment

The SCM hypothesizes that many sources are positive predictors of EC and, thereby, strengthen

it, while one source is negatively related to EC and, hence, lessens it. The following sources are contended to strengthen EC: sport enjoyment, valuable opportunities; both types of personal investments (loss and quantity), social support (emotional and informational), and desire to excel (mastery and social achievement); and social constraints.

SCM conceptualizes other priorities to be negatively related to EC and, therefore, the source lessens or chips away at this type of commitment. In fact, to keep this negative effect from happening, rich interview data show that many elite athletes, and their families, actively employ strategies to eliminate any conflict with other priorities in their lives. This allows playing to remain the top priority.

Sources of Constrained Commitment

The research is limited when CC is studied within the general framework of the SCM. Hence, with three exceptions, the current sources of EC will need to be explored to see how they relate to CC. In a few studies, personal investments, other priorities, and social constraints were found to be positive predictors of CC. Future research will help us understand why some athletes feel trapped in their sport.

Progress with the SCM continues. Based on what has been learned from research to date, a significantly updated English version of the Sport Commitment Questionnaire has been developed, and adapted into Spanish and Portuguese to stimulate cross-cultural research. There also is considerable interest in adapting the questionnaire into other languages.

Tara K. Scanlan and Graig M. Chow

See also Adherence; Effort; Enjoyment; Talent Development

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COMPETENCE MOTIVATION THEORY

Competence motivation theory is a conceptual framework designed to explain individuals' motivation to participate, persist, and work hard in any particular achievement context. The central thesis of the theory is that individuals are attracted to participation in activities at which they feel competent or capable. The theory can be used by researchers and practitioners in sport and exercise psychology fields to identify why and how children, adolescents, and adults can be encouraged to participate and to exert effort in these achievement contexts.

In the following entry, the research and theory on competence motivation within the physical domain are reviewed. This begins with a brief historical overview of the theory and its constructs. Following that, the results of the research on the following segments are summarized: (a) correlates of competence motivation, (b) developmental trends in perceived competence, and (c) the impact of significant others on competence motivation.

Historical Overview

Most scholars identify Robert White's classic paper on motivation reconsidered as the forerunner of competence motivation theory. In this 1959 publication, White coined the term *effectance* and

defined it as a tendency to explore and influence one's environment. White argued that organisms are intrinsically motivated to engage in interactions with their physical and social environments. If such attempts result in success (production of an observable effect on the environment), then that individual receives intrinsic rewards like feelings of efficacy and pleasure and is motivated to continue effectance efforts. White's theory of competence motivation was considered a novel approach in that it differed significantly from the traditional drive theories of human behavior and from the psychoanalytic instinct theory proposed and popularized by Sigmund Freud.

In the late 1970s, Susan Harter extended White's theory to develop a more complete framework that she initially identified as effectance motivation theory but was later more commonly referred to as competence motivation theory. Consistent with White, Harter also centered enjoyment as the reason why individuals are motivated to interact with their environment, but she added a number of other components. First, she introduced the idea that individuals' effectance or competence motivation can vary across achievement domains (e.g., cognitive, physical, social). Within each domain, individuals are motivated to engage in mastery attempts for the purpose of developing or demonstrating competence. If their mastery attempts result in success at an optimally challenging task and if they receive socioemotional support from significant individuals for such task success, then they will experience perceptions of competence (belief in their abilities in that domain) along with perceptions of performance control (belief in their ability to control their performance). High perceptions of competence and control, in turn, result in feelings of pleasure that lead to maintenance of or increases in effectance (competence) motivation. In contrast to White, Harter also proposed a negatively oriented path that specified that individuals who engage in mastery attempts but meet with failure at optimally challenging tasks or lack of reinforcement or disapproval from significant social agents will experience decreased perceptions of competence and control in that achievement domain, along with anxiety and shame. This combination of events will lead to decreased effectance motivation in that particular domain.

Harter also added a developmental dimension to her theoretical framework by suggesting that children who are successful in their initial mastery

attempts and who get positive and effective reinforcement from significant adults will (with sufficient cognitive maturation) internalize both a self-reward system and a set of mastery goals. Due to this internalization of standards for optimal challenge in that domain, such adolescents will no longer be dependent on social agents to evaluate their performance or to motivate them to continue their mastery attempts. Correspondingly, children who are either continually unsuccessful in their early mastery attempts or who get disapproving or no feedback from significant adults will not only develop low perceptions of competence and control in that achievement domain but will continue, as adolescents and adults, to be dependent on external sources both to evaluate their performance and to motivate them to continue their participation in that domain.

The publication of Harter's early work stimulated much research during the next 2 decades in the academic, social, and physical domains. More recently, interest in competence motivation theory has waned somewhat, likely because the construct of competence has been subsumed within other theories of motivation, such as achievement goal and self-determination. However, in 2005, Andrew J. Elliot and Carol Dweck proposed a more central role for competence and recommended that the term *achievement motivation* be changed to *competence motivation*. Their arguments were based on the idea that competence motivation (a) is broadly present in daily life activities, (b) has a large and influential effect on individuals' emotional and psychosocial well-being, (c) is operative across the lifespan (from infancy through older age), and (d) has relevance across cultures.

Research on Competence Motivation Theory in the Physical Achievement Domain

Correlates of Competence Motivation

The initial studies by researchers in sport psychology provided support for the model itself but also for the importance of competence as a motivational construct. Specifically, individuals who perceive themselves as having high competence in any particular sport or physical activity context exhibit higher intrinsic motivation to participate in that activity and experience more positive affective reactions (e.g., pleasure, satisfaction, enjoyment) when participating than do their peers who hold lower perceptions of competence.

Correspondingly, perceived competence levels can also affect or predict individuals' behavior. Children and adolescents in a variety of physical contexts (e.g., physical education classes, sport teams) who perceive themselves as having high competence are more apt to continue their participation and to exhibit higher levels of effort, persistence, and preference for more challenging tasks, whereas their peers who perceive low levels of competence exhibit lower levels of such task-oriented behaviors and are, subsequently, at greater risk of discontinuation. More recent studies have indicated that perceptions of competence are also tied to physical activity behaviors; children with high perceived physical competence exhibit higher frequency and intensity of daily physical activity levels. Such connections have also been demonstrated in intervention studies; children who successfully engage in physical activity programs show increases in perceived physical competence, which, in turn, increases both their motivation to be physically active and their actual physical activity levels. Interestingly, perceptions of physical competence have also been shown to be relevant for older adults, as those who have high perceptions of physical competence are more physically active and also exhibit higher perceived quality-of-life attitudes than do their peers who have lower perceptions of physical competence. Finally, high perceptions of competence in individuals across the lifespan have been consistently linked to higher levels of global self-worth or self-esteem. These results, again, point to the importance of individuals possessing high perceptions of ability or competence in at least one valued achievement domain in order to have an overall high regard for the self.

Developmental Trends in Relation to Perceived Competence

Two lifespan changes are particularly relevant to the physical domain. First, the number of domains in which individuals evaluate their competence appears to increase with age. That is, young children (4–7 years) tend to perceive competence in only two domains (general competence and social competence). However, older children (8–13 years) exhibit a somewhat more diversified perception of competence that spans five different achievement domains (academic, athletic, social, physical appearance, and behavioral). During the adolescent years, the number increases to include

three additional ones (close friendship, romantic relationships, and job competence). Other, smaller, developmental changes continue to occur across the adult years.

Second, research in both the cognitive and physical domains has indicated that the sources of information that individuals use to evaluate their competence change with increased maturation. For younger children (4–7 years), the primary sources of information are few in number and very concrete, for example, feedback from significant adults and simple task accomplishment. Thus, such a child might say, "I am really good at throwing because I can throw this ball all the way to that wall," or "I know that I am a really good thrower because my teacher says so." During the middle to late childhood years (7–12 years), the number of sources children use to evaluate their competence increases somewhat, but the sources remain quite concrete. It is during this period that peer comparison and performance outcomes (winning or losing) become more relevant. During the adolescent years (13–18 years), the number of sources again increases, and most adolescents have the cognitive capacity to use a broader range of sources that include both internal (e.g., achievement of self-set goals) and external (e.g., feedback from others, performance outcomes, peer comparison) sources. Going back again to the early Harter model, it may be important for adolescents to reach a stage where they do use multiple sources and where they are able to use more internalized sources like personal satisfaction rather than remaining dependent on single sources like peer comparison or on externally available sources such as coaches' evaluation of their performance.

The Impact of Significant Others on Competence Motivation

As indicated in the original Harter model, significant others, such as parents, teachers, coaches, and peers play a major role in the development of individuals' perceptions of physical competence and effectance motivation. In particular, parents may serve as the first important social agents. They can, for example, affect their children's perceptions of competence through modeling (children observing their parents' own engagement in, and enjoyment of, mastery attempts). However, an equally important role that parents may play is in the feedback they provide in response to their child's early

engagement in mastery attempts. Obviously, positive feedback like “Good work, Maria!” is better than negative or no feedback. But, it is also the content of parents’ feedback that appears important. Specifically, a child who participates for the first time in a tee-ball league will learn from the parents’ feedback not only whether mastery attempts in such a domain are valued but also *how* the performance should be evaluated. That is, does the parent provide feedback that evaluates the child’s performance based on (a) the performance outcome (“Good job, Susan. You got to first base again.”), (b) peer comparison (“You have to work harder in practice. You let Chase beat you again today.”), or (c) personal mastery of a skill (“Good work in practice today, Joshua. I see that you learned how to catch the ball.”)?

A second group of adults who can have a significant impact on individuals’ perceptions of competence and effectance motivation in the physical domain are coaches, physical education teachers, and physical activity directors. Although positive feedback from such individuals is certainly more facilitative of high perceived competence in students and athletes, teachers and coaches also need to ensure that such positive feedback is appropriate. That is, for very young children (4–7 years), positive and general feedback (“Good job, Enrico!”) may be good. But for older children and adolescents (12 years and up), coaches and teachers should be sure that their praise is appropriate. That is, if a coach provides a 12-year-old player who just got to first base on a pitcher’s error with exaggerated praise (“Wow, Robert, that was truly just an excellent performance.”), the athlete’s competence may not be enhanced as he may realize that the praise is not consistent with the performance. In such a situation, the receiver of that inappropriate praise may actually perceive low competence as he assumes that the coach thinks he is really bad at batting if that is the level of performance that is rewarded. Thus, the coach could better respond by providing more contingent praise (“Robert, you really did a good job in not swinging at those bad pitches.”)

In addition, coaches and teachers who provide skill-relevant, informational, and corrective feedback to athletes in response to their performance attempts (“That was a good hit, Samantha. But next time, you should extend your elbow a little bit more. That will give you even more power.”) will do more to enhance young athletes’ perceptions of

competence and control than will feedback statements that are evaluative only and which provide no corrective information (“That was a good play, Samantha,” or “You struck out again, Domingo!”)

Another social group that may be particularly important in regard to children’s and adolescents’ perceptions of competence in particular achievement domains is peers (friends, classmates and teammates). Research studies have demonstrated that peer acceptance (the degree to which children and adolescents feel accepted and valued by their peers) and the number and quality of friendships can enhance perceptions of competence and can result in continued motivation to participate in that achievement activity. Correspondingly, the type of feedback that peers provide has also been shown to affect individuals’ levels of competitive anxiety, sport motivation, and enjoyment.

Conclusion

The central thesis of competence motivation theory is that individuals are attracted to participate in activities at which they feel competent or capable. In physical activity domains, then, if the goal is for people to be motivated to be physically active or to strive for performance excellence, it will be necessary to design environments that will enhance their perceptions of competence. Based on the research and theory to date, enhanced perceptions of competence can be achieved when individuals experience success at optimally challenging tasks and when they receive positive, encouraging, consistent, and information-based feedback from significant others within that environment.

Thelma S. Horn

See also Achievement Goal Theory; Achievement Motive Theory; Hierarchical Self; Self-Determination Theory

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COMPETITION

Competition is often described as a contest, or a process of contesting, between two or more parties (organisms, individuals, or groups) for a scarce resource or good. The scarcity can result from nature or history, such as competition for limited food, or it can be created artificially, such as the good of winning a game. As the term is most often used, competition is the opposite of cooperation. Since competition is a central dynamic in most if not all sports, competition is of keen interest to sport scientists. In this entry, various forms of competition are described, the social science research on competition is reviewed, and theory and research on competitiveness is summarized.

What Is Competition?

Competition comes in many forms and is relevant to many fields of study. Economic, political, biological, evolutionary, and sport competition, for example, each have unique characteristics, along with some commonalities. Competition occurs between and among nonhuman organisms, individuals, teams, organizations, cultures, and nations, to name only some of the most obvious. Competition can be direct or indirect, subtle or intense, formal or informal. It can be zero sum, resulting in winners and losers, or allow for many variations of outcome distributions.

To define competition, most social psychologists have focused on the goal structure of situations. Morton Deutsch, for example, defined *competition* in terms of “negative social interdependence.” A situation is competitive if the progress or success of one party interferes with the progress or likelihood of success of other parties.

Rainer Martens proposed a definition of competition from a sport psychology perspective. For Martens, competition is a form of social evaluation; it entails comparison of individual or team performances against one another or against an objective standard of excellence. In Martens’s view, people compete because humans have an innate motive to evaluate their ability. A competitive situation is one that allows for the evaluation process through providing a rule structure that (a) defines the goal toward which action is directed and (b) defines the allowable means to try to achieve the goal.

Competition: The Social Science Research

Psychologists have studied the effects of competition for more than a hundred years. In the late 1890s, in what is often called the first sport psychology study, Norman Triplett found that cyclists performed faster when they were racing against other cyclists than when they were racing against the clock. Despite this promising initial start, subsequent research has demonstrated that for most types of tasks in most situations, competition leads to poorer performance. The exceptions are simple tasks or rudimentary skills, the performance of which can be facilitated through competition.

In his extensive if somewhat polemical review of the literature in *No Contest: The Case Against Competition*, Alfie Kohn concluded that competition is associated with a range of negative outcomes. As noted above, in most situations it hinders optimal performance. Competition, both interpersonal and intergroup, has also been associated with increased hostility, aggression, and prejudice. It tends to interfere with developing positive and stable self-esteem; it creates stress and anxiety, along with their negative health consequences; and it promotes conformity and reliance on external evaluation.

Many researchers, such as David and Roger Johnson, acknowledge the extensive literature demonstrating these negative consequences, but still maintain that competition can be useful. Writers in this tradition seek to distinguish healthy from unhealthy, beneficial from destructive, and productive from harmful forms of competition. Unfortunately, the distinctions are often unclear and are based more on differences of degree than of kind.

A different approach has been advanced by David Shields and Brenda Bredemeier, who offer a qualitative distinction between competition

and the forms of contesting that lead to negative outcomes. They define *competition* as striving for excellence in a contest situation. According to this view, competition entails a combination of a contest structure with a personal orientation toward seeking excellence through the mutual challenge that opponents provide to each other. They introduce a new term, *decompetition*, to describe a situation in which a contest is metaphorically understood as a battle for superiority or extrinsic reward. They suggest that it is the different motivations and goals associated with decompetition that account for the negative findings prevalent in the literature.

What Is Competitiveness?

When people compete, they strive to obtain a limited goal and to compare favorably to others similarly striving. In sports, the contest-specific goal, of course, is to win. The amount of preparation, focus, determination, and energy that a person puts into striving to win is often referred to as their competitiveness. Competitive people have a strong desire to achieve and succeed, thereby demonstrating their competency to themselves and others. There are two main instruments that have been developed to measure competitiveness in sport.

Robin Vealey used achievement-goal theory to help her define and operationalize competitiveness. Correspondingly, competitiveness can take two different forms, depending on what goal the athlete is trying to achieve. Athletes can be outcome oriented (winning) or performance oriented (performing masterfully) or both. When Vealey developed the Competitive Orientation Inventory (COI), she placed the two orientations in opposition. The format of the instrument consists of a 16-cell matrix in which four levels of performance (very good, above average, below average, and very poor) are intersected with four outcomes (an easy win, a close win, a close loss, and a big loss). For each of the resultant 16 cells, the athlete indicates their level of satisfaction. Those athletes who indicate high satisfaction when they win, irrespective of how well they played, score high on outcome-oriented competitiveness; in contrast, those who are most satisfied with playing well score high on performance-oriented competitiveness.

Another widely used assessment of competitiveness, called the Sport Orientation Questionnaire (SOQ), was developed by Diane Gill and Thomas Deeter. The measure assesses three orientations:

competitiveness, goal orientation (a focus on personal standards), and a win orientation. The concept of *competitiveness* is defined as a desire to enter and strive for success in sport competition. For example, respondents are asked to indicate their extent of agreement with “I thrive on competition,” and “The best test of my ability is competing against others.”

Researchers have investigated competitiveness both as a dependent and as an independent variable. For example, researchers have generally found male athletes to be more competitive than female athletes, elite athletes to be more competitive than recreational athletes, and U.S. athletes to be more competitive than those from more collectivist cultures. In turn, competitiveness seems to predict sport involvement, poor sportsmanship, higher alcohol consumption, and sexually aggressive attitudes.

*Brenda Light Bredemeier and
David Light Shields*

See also Achievement Goal Theory; Aggression; Conflict; Interdependence Theory and the Coach–Athlete Relationship

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CONCENTRATION

In sport psychology, concentration refers to focusing on sensory or mental events coupled with mental effort. It therefore relates primarily to the selective attention dimension in which individuals are able to selectively process some sources of

information while ignoring others. It has its roots in the oft-cited quote from William James's 1890 book *The Principles of Psychology* (Vol. 1):

Everyone knows what attention is. It is the taking possession by the mind, in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought. Focalization, concentration, of consciousness are of its essence. It implies withdrawal from some things in order to deal effectively with others. (pp. 403–404)

Within sport psychology, research interest in concentration is predominantly focused in two areas. First is the study of skill failure under pressure, or choking; there is strong evidence that skill failure can result from *reinvestment* of conscious control processes. Second, researchers have investigated the efficacy of external and internal foci of attention during learning and in skilled performance.

Measurement

Researchers have used a range of psychophysiological tools, as well as self-report measures, to infer attentional states. These include heart rate measurement, electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and ratings of the extent of focus on a particular information source in experimental paradigms. In addition, dual-task paradigms have been used to measure performers' awareness of task-relevant and extraneous information, most notably in the study of choking. Self-report measures include the Test of Attentional and Interpersonal Style (TAIS), which was devised by Robert Nideffer to assess a range of personality measures, a subset of which were linked to his proposed two-dimensional model of attention. This model distinguishes between direction of attention (external, internal) and breadth of attention (broad, narrow) creating four combinations. The scale contains groups of items to measure positive attributes of attention (effective integration and effective narrowing) and items to measure negative attributes (overloading and underinclusion). Other scales that are more specifically focused include the Self-Consciousness Scale, which measures individual propensity for attending to the self, and the Reinvestment Scale (and its movement-specific and decision-specific versions), which measures the tendency to exert conscious control over movements or the decision making process.

Concentration and Choking

There is good evidence that skill failure in self-paced skills such as golf putting, basketball free throws, rugby goal kicks, and static target shooting is caused by the performers concentrating on deliberately and consciously controlling their movements. Much of the research work examining this issue has assessed the performer's attentional state in the few seconds prior to skill execution using a variety of experimental paradigms. These experimental paradigms have included cross-sectional designs comparing individuals of different levels of ability or experience, and within-participant designs in which the performer's concentration is directly manipulated via explicit instruction or task-related instructions. Evidence from performance measures and indirect measures of attention indicates clear differences between high-skilled and less-skilled performers, and that skilled performance breaks down when individuals reinvest conscious control over largely automatic actions. For example, explicit monitoring of motor skills has been shown to have a detrimental effect on skilled performers but does not impair, and sometimes facilitates, novice performance. By contrast, experts are able to perform a concurrent secondary task with minimal impairment to primary task performance, whereas novice performance tends to get worse. Relatedly, there is within-group evidence that skilled performers' ability to report task-relevant and extraneous information relates to how well they have been performed on preceding trials. For example, baseball batters are better able to report bat position during a performance slump than when on a streak of good performance, implying their concentration is more directed to the swing motion during slumps.

A second line of research into concentration in sport concerns the efficacy of adopting either an external focus (on an implement or movement effects) or internal focus (on body movements) during learning and skilled performance. There is now a considerable body of evidence across a variety of tasks and using different age groups that an external focus is preferable for learning and evidence that this also extends to skilled performance. Further, the effect is moderated by distance so that a more distal external focus is more beneficial than a proximal focus. By studying task outcome alongside the movement kinematics and EMG activity, improvements in movement efficiency as

well as accuracy have been reported resulting in more rapid skill acquisition and better retention. The *constrained action hypothesis* has been proposed to explain these findings, referring to the self-organizing nature of the motor system and the potential for an external focus of attention to facilitate high frequency unconscious adjustments. Conversely, an internal focus of attention is hypothesized to constrain the self-organizing nature of the motor system as the performer attempts to actively monitor and control the system.

Conclusion

The term *concentration* embodies one element of the multidimensional construct of attention. In the sport psychology literature, the term has been particularly important in the study of skill failure under pressure and in the realm of skill acquisition. Concentration strategies are a key component of the mental routines performers employ before executing self-paced skills. Proposed interventions for aiding performance, particularly under pressure, support the view that the information on which a performer concentrates in the few seconds before skill execution is a key determinant of performance.

Robin C. Jackson

See also Attention Theory; Attentional Focus; Choking

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CONCENTRATION SKILLS

Several sources of evidence reveal that concentration, or the ability to focus on what is most important in any situation while ignoring distractions, is vital for success in sport. First, anecdotally, lapses in concentration can mean the difference between winning and losing at the Olympics. For example, at the 2008 Games in Beijing, rifle shooter Matthew Emmons missed an opportunity to win a gold medal in the 50m three-position target event due to a lapse in concentration. Leading his nearest rival Qiu Jian as he took his last shot, Emmons lost his focus and inexplicably misfired, finishing without a medal (in 4th place). A second source of evidence on the importance of concentration comes from research showing that the capacity to become absorbed in the present moment is a key component of peak performance experiences in sport (see also the entry “Flow”). Finally, experimental evidence suggests that athletes who have been trained to focus on task-relevant cues tend to perform better than those who have not received such training. Despite the preceding evidence, however, psychology researchers have struggled to answer two significant questions in this field. First, if concentration is so important, why do athletes appear to lose it so easily in competitive situations? Second, what are the building blocks of effective concentration skills in sport?

Why Do Athletes Lose Their Concentration?

According to a conceptual approach called *spotlight theory* (see also the entry “Attention Theory”), paying attention is like shining a mental beam at targets located either in the external (real) world or in the internal world of our thoughts and feelings. It is like the head-mounted torch that divers and miners wear in dark environments. Wherever they look, their target is illuminated. A practical implication of this theory is that concentration cannot be lost, although one’s mental spotlight can be diverted easily to a target that is not relevant to the task at hand.

In competitive sport, an abundance of distractions can threaten to divert an athlete’s mental

spotlight from its intended target. Normally, these distractions fall into two main categories: external and internal. Whereas external distractions are objective stimuli that exist in the world around us, internal distractions include a vast array of thoughts, feelings, or bodily sensations like fatigue that can disrupt our focus. Typical external distractions include factors such as spectator movements, sudden changes in ambient noise levels (e.g., the click of a camera), gamesmanship (e.g., trying to block a goalkeeper in soccer from seeing the ball at a corner kick) and unpredicted weather conditions (e.g., tennis players may get distracted when windy conditions affect their ball toss before serving). Typically, these distractions impair athletic performance. For example, the marathon runner Vanderlei De Lima was leading the race in the 2004 Olympics in Athens when an unstable spectator suddenly jumped out from the crowd and wrestled him to the ground. Stunned and distracted, De Lima eventually finished third in the event. Internal distractions stem mainly from thoughts such as wondering what might happen in the future, regretting what has happened in the past, or worrying about what other people might think, say, or do. A classic example of a costly internal distraction occurred in the case of the golfer Doug Sanders, who missed a putt of less than 3 feet, which prevented him from winning the 1970 British Open championship in St. Andrews, his first major tournament. Remarkably, Sanders's attentional lapse was precipitated by thinking too far ahead: He had made a victory speech in his mind before his final putt had been sunk. Unfortunately, little research has been conducted on the mechanisms by which internal distractions disrupt performance. Nevertheless, Daniel Wegner developed a model that attempted to explain why attentional lapses occur *ironically*, or precisely at the most inopportune moment for the person involved. Briefly, this model postulates that when our working memory system (which regulates our conscious awareness) becomes overloaded because we are anxious or tired, a particular form of ironic distractibility may occur. Specifically, trying *not* to think about something may paradoxically *increase* the prominence of this very thought in our consciousness. According to Wegner, this increased awareness of the very thing we try to suppress is called a *rebound effect* and applies to *actions* as well as thoughts. For example, in sport, ironies of action are evident when anxious penalty takers in

football try *not* to kick the ball at the goalkeeper but end up doing so or alternatively, when tired golfers try *not* to overshoot the ball when putting, but again, end up doing so.

Building Blocks of Effective Concentration Skills

Research suggests that there are at least four building blocks of effective concentration skills in sport.

Deciding to Concentrate

To begin with, athletes must make a conscious decision to invest mental effort in their performance. Just as researchers have distinguished between deliberate practice (a purposeful activity in which the learner tried to improve a specific skill under the guidance of an expert coach) and mindless practice (whereby the learner flits from one skill to another without any plan), we can distinguish between *deciding* to concentrate and merely hoping that it will happen. To implement this principle, many athletes establish imaginary *switch on* and *switch off* zones in their competitive environment. For example, entering the locker room before a game may serve as a trigger for footballers to turn on their concentration before a match. Likewise, in an intense game, tennis players may use a between-points routine of wiping their face with a towel at the back of the tennis court to safely switch off from thinking about the previous point.

Having Only One Thought at a Time

Next, the *one thought* principle is the idea that one can focus consciously on only one thing at a time. This principle comes from research on the bandwidth of attention or the number of items in working memory on which one can focus effectively. Studies on this topic suggest that people's focus of attention may be limited to just one item. Extrapolating from this finding, it seems plausible that the ideal thought for a sports performer should be a single word that triggers the appropriate feeling or tempo of the action to be executed (e.g., *smooth* for a golf swing) rather than a complicated set of instructions (e.g., bend your knees and go from low to high) (see also the entry "Attention Training").

Focusing Only on Factors Under One's Control

Third, research shows that people's concentration wanders when they think too far ahead or

otherwise focus on factors that are irrelevant to the job at hand. To counteract this tendency, athletes should focus only on actions that are under their control; for example, a penalty taker in soccer should focus only on where the player wants the ball to go, not on the position of the goalkeeper.

Focusing Outward When Nervous

The final building block of effective concentration is the idea that anxious athletes should focus *outward* on actions, not inward on doubts. This outward focus is necessary because anxiety tends to make people self-critical and hypervigilant, primed to detect any sign of what they may fear.

Aidan Moran

See also Attention Theory; Attention Training; Concentration; Flow

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other's existence, and were transported separately to opposite sides of the camp. This segregation enabled the initiation of the first of three experimental stages: *in-group formation*, *friction*, and *integration*. In the first stage (in-group formation), through camping situations, the researchers witnessed spontaneous friendship choices and the development of status hierarchies (e.g., leadership roles, group norms) within the two groups. After a week, the groups were made aware of each other's presence in the camp and quickly developed an us-versus-them mentality (in-group favoritism and out-group rejection). This marked the beginning of the second stage (friction) in which direct competition activities involving camp chores and sports were implemented. Initially, these activities led to minimal tension between groups; however, in a short time, this tension escalated to the point where the researchers were required to intervene. Interestingly, this intergroup hostility also resulted in greater intragroup solidarity. In order to decrease intergroup conflict, the third stage (integration) involved a variety of noncompetitive scenarios in which the groups played games and ate meals together. When these scenarios failed to decrease hostilities, the members of both teams were required to engage in a series of collaborative tasks that required the pursuit of a series of *super-ordinate goals*, and only when members of both teams worked together in this way did the tension and intergroup conflict dissipate.

The Robbers Cave example is a useful representation of what is known as intergroup conflict. Conflict can be generally described as a situation in which one party believes its goals or objectives to be adversely influenced by another. Conflict involves a minimum of two individuals and refers to disagreements or friction involving verbal or nonverbal actions and emotions. This can occur between members of opposing groups (intergroup) or between members within a group (intragroup). In light of the pervasiveness of groups across diverse life contexts, it is perhaps unsurprising that individuals are constantly placed in situations where collaboration and competition, either within or between groups, is necessary; and where interactions are inevitable, so too, is conflict. Sport represents one prominent social context in which conflict is particularly evident.

In sport, a common criterion for success is to defeat one's opponent. Naturally, this leads to competition and occasionally highly overt

CONFLICT

In 1954, Muzafer Sherif, O. J. Harvey, B. Jack White, William R. Hood, and Carolyn W. Sherif undertook a project that allowed them to examine inter- and intragroup relationships in a naturalistic setting. This classic field-based experimental study, known as the Robbers Cave Experiment because of its location (Robbers Cave State Park in Oklahoma), involved 22 boys (mean age 11 years, 1 month) with similar appearances and backgrounds. The children were assigned to one of two groups (they named themselves the "Rattlers" and "Eagles"), with each group unaware of the

intergroup conflict between the competing parties. Intragroup conflict on the other hand, is often much less obvious. Although team members share common goals and must interact in order to achieve these goals, individuals frequently compete with one another for personal accolades or recognition within the team. Such intragroup conflict can develop from interactions between teammates as well as those that occur between a coach and athletes. The former (intragroup conflict between teammates) represents the primary focus of this entry. The relationships (and potential incongruence) between coaches and their athletes are covered in greater detail within other entries in the encyclopedia. (See, for example, "Attachment Theory and Coaching"; "Autonomy-Supportive Coaching"; "Coach-Athlete Relations"; and "Relational Efficacy Beliefs in Coach-Athlete Relations.")

Conceptual Framework for Intragroup Conflict

A prominent conceptual framework that differentiates between types of intragroup conflict that may exist was advanced by the organizational psychologist Karen Jehn. Based on her model, intragroup conflict is considered to be multidimensional in nature, consisting of relationship, task, and process dimensions. *Relationship conflict* refers to incompatibilities or disagreements relating to social attachments. *Task conflict*, on the other hand, results from differences in beliefs or opinions in relation to the instrumental (performance-related) objectives of the group. Finally, *process conflict* involves controversies or arguments relating to *how* established objectives will be carried out. This occurs when parties have differing beliefs and expectations with regard to the roles and responsibilities for different members of the group and the processes by which they carry out their specific assignments. In the organizational psychology literature, the process conflict dimension has been the focus of debate, with some suggesting that it is not a distinct dimension in itself but, rather, represents a specific form of task conflict. The other two dimensions, task and relationship conflicts, however, are widely accepted.

In the sport literature, even though research on intragroup conflict is in its infancy, the presence of these two dimensions has also been supported. Jehn contended that relationship conflict

is detrimental to group processes and outcomes, whereas task conflict is not necessarily debilitating, and if resolved may in fact result in improvements in team performance. This is also the case in sport settings. To elaborate, task conflict can provide an outlet for athletes to voice their opinions and provide insight into different viewpoints and beliefs. Take, for example, a pitcher and a back catcher in baseball who disagree on the type of pitches that should be thrown. If task conflict arises, and both parties are able to voice their opinions, a superior strategy may be derived and consensus can subsequently improve the commitment from both athletes with regard to the pitches chosen.

Research with youth sport participants has found that friendship conflict (similar to relationship conflict) is positively correlated with athlete anxiety, and negatively correlated with perceptions of friendship quality. Furthermore, research with youth girls has found the presence of friendship conflict to inhibit the development of friendships and social support. At the university level, athletes have been found, in general, to be adversely affected by intragroup conflict. They indicate that task and relationship conflicts are inevitable in the sport domain, and the tendency for these conflicts to arise increases when athletes exhibit self-centered behavior in relation to both the task (e.g., not involving certain teammates in practices or game systems) and relationships (e.g., not reciprocating support offered by teammates). Although both types of conflict have been found to hinder team performance, in certain instances, if task conflict is moderate in nature and managed at an early stage, it can have positive consequences.

In recognition of this potential facilitative role for task-related conflict, a preliminary study by Phillip Sullivan and Deborah Feltz distinguished between *constructive* and *destructive* conflict styles in sport teams. An example of a constructive conflict style might involve an individual tackling disagreements by actively involving all affected parties and working to derive a mutually beneficial result, whereas a destructive conflict style might involve teasing a teammate or making the teammate feel guilty. The results of this work indicated that although destructive conflict was negatively related to task and social cohesion, constructive conflict (in particular by using integrative tactics) was positively related to social cohesion. This is significant because cohesion is positively related

to many important psychological variables, such as participant satisfaction, motivation, and adherence.

Strategies for Preventing and Resolving Intragroup Conflict

In acknowledgement of the potential deleterious effects of destructive intragroup conflict (especially relationship conflict), the *prevention* (proactive approach) and *resolution* (reactive approach) of these debilitating intragroup dynamics becomes increasingly important for optimal group functioning. However, because conflict can arise for many different reasons and in many different situations, there is no one best established method to alleviate such conflict. Nevertheless, there is evidence from the group dynamics literature of the existence of effective conflict prevention and resolution strategies. The most ideal situation is to *prevent* elevated levels of intragroup conflict from developing in the first place. Because conflict arises when differing parties perceive their objectives as being negatively influenced or impeded by another party, one method to successfully prevent intragroup conflict involves targeting athletes' understanding of their own and their teammates' preferences for interdependent interaction. By having open lines of communication and conducting *role playing* activities, team members are able to see (a) how their personal preferences directly affect their teammates, and (b) why their teammates might act in different ways to themselves. By helping athletes to better understand themselves and their teammates (empathy is developed), they are better able to adapt and connect with one another.

Another successful way to prevent intragroup conflict is through the implementation of peer-mentoring systems. During a sport season, athletes are often in close proximity to one another and although this can foster positive social interactions and team cohesion, there is also the potential for increased task and relationship conflict. By assigning partners, athletes are encouraged to provide feedback to one another, discuss situations that may cause conflict, and consider strategies to prevent such interpersonal conflict. Through this process, athletes are better equipped to handle and address the idiosyncrasies of their teammates.

Nevertheless, when conflict does arise and is allowed to go unresolved, it has the potential

to intensify and cause problems for the team at important times during the season. Identification and intervention at an early stage are important for two reasons. First, emotions are not as intense at an early stage and resolution is easier than if the conflict is allowed to fester. Second, and based on the literature that suggests there are positive benefits relating to conflict resolution, if conflict is resolved early, there is potential for the added advantage of different viewpoints and strategies to be voiced that could eventually result in improved group functioning.

From an interventionist perspective, conflict resolution can be approached in a negative or positive manner, and by taking a passive or active approach, resulting in four types of conflict management: (1) *avoiding* (negative and passive), (2) *fighting* (negative and active), (3) *yielding* (positive and passive), and (4) *cooperating* (positive and active). The latter (cooperating) of these has generally been found to be the most effective in guiding conflict resolution. One strategy for resolving conflict that utilizes a positive and active approach is through the use of a third party mediator. Ideally, this mediator should be a formal (captain or assistant captain) or informal (senior athlete) leader who is not directly involved with either of the conflicting parties. It is also possible, however, that using teammates as mediators can lead to further conflict because the mediator can be forced to take sides. In certain instances, a coach can act as a mediator, although on occasions athletes may be reluctant to involve the coach in this process. An alternative is the use of a sport psychology consultant. These options should be implemented based on the needs of the athletes or team as well as the context of the conflict.

Another somewhat related conflict resolution strategy involves clear-the-air team meetings. The difference between team meetings and the third party mediator approach is that the whole team participates in the discussion rather than only the parties involved. These meetings can be executed with (a) only the athletes (without an authority figure), (b) a coach as the mediator, or (c) a sport psychology consultant as the mediator. In these sessions, athletes are encouraged to voice their opinions and discuss issues pertinent to the group. Through this process, athletes take ownership of the conflict resolution and develop perceptions of autonomy and empowerment.

Conclusion

To summarize, intragroup conflict is inevitable in sport. Based on the extant literature, conflict can be resolved and the negative outcomes can be minimized if handled properly. In addition, the resolution of task conflict in certain instances can lead to positive outcomes such as increased teammate understanding and cooperation and, subsequently, group functioning. In order for this to occur, conflict should be handled at an early stage and ideally in a collaborative manner.

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See also Cohesion; Collectivism/Individualism; Competition; Friendships/Peer Relations; Self-Categorization Theory; Social Identity Theory; Team Building; Team Communication

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those around us. Research into the nature of this phenomenon was popularized by prominent social psychologists in the early to mid-20th century such as Muzafer Sherif and Solomon Asch. Today, there exists a well-established literature base regarding the different types of conformity that individuals display, the conditions under which conformity is (and is not) likely to occur, and the various desirable and undesirable consequences of conforming behavior.

Types of Conformity

Two interrelated types of conformity are described in the literature, which differ according to their underlying cause. On the one hand, *informational* conformity occurs when we are uncertain about the most appropriate behavior in a given situation, and we conform to those around us in order to bring about correct courses of action. Imagine, for instance, the uninitiated exerciser who observes more experienced individuals using the weights machine in the gym, before trying to mirror their technique when the time comes to use the equipment oneself. In the case of informational conformity, individuals most often look to put their trust in those who are viewed as highly credible, and tend to genuinely believe that they are adopting the most effective strategy. In sport, informational conformity has been documented in relation to performance appraisals, whereby gymnastics judges have been shown to display more consistent scoring patterns when they have been made aware of one another's previous ratings (even despite knowing their subsequent scores will not be revealed). On the other hand, *normative* (or instrumental) conformity is underpinned by a desire to integrate with others and occurs when we acquiesce in order to be liked or accepted. In this case, individuals may not have genuine faith that their chosen behavior or belief is correct, but they conform nonetheless as a display of public compliance. For example, new players on a sports team may feel it is in their best interest to join in when their new teammates are complaining about their coach, despite having no personal grievances against the coach.

Although informational and normative conformity typically occur when one conforms to the will of the many, this is not always the case. In some instances, it is possible that the direction of influence may be reversed, and that the majority (the

CONFORMITY

Conformity refers to the process of matching one's actions or beliefs with the behavior and norms of

group) may conform to the opinions and behaviors of a single individual. This notion of *minority influence* dictates that when a high-status person makes a particularly compelling case for a course of action, that person may successfully shape the behavior of teammates. In sport for instance, it is easy to imagine how a group of rookies might willingly conform to the behavior of someone who occupies a position of authority on the team, such as a captain or senior player.

Individual and Situational Influences

From an individual standpoint, there is less pressure on high-status individuals to conform within the group. For example, members of a sports team might implicitly accept that although their flamboyant star player consistently performs well in competition, that player is not expected to practice as hard as the rest of the group through the week. There is also evidence that conformity to group norms increases when individuals hold strong perceptions of team cohesion or unity, and that gender may interact with task characteristics to shape conformity behavior. Specifically, females have been shown to conform more than males when performing “masculine” tasks, whereas males tend to display greater conformity on “feminine” tasks. In terms of situational and societal influences, people from collectivist (relative to individualist) cultures appear to be more likely to conform to others’ behavior, and it has been recognized that conformity is more likely as task difficulty increases. It has also been shown that individuals display a greater tendency to conform when the size of the unanimous majority (to which they are conforming) is large. Finally, priming research has demonstrated that the subconscious activation of affiliation goals and conformity-related concepts may also be responsible for promoting conformity. For example, simply viewing conforming (e.g., a picture of an accountant) versus nonconforming (e.g., a picture of a skinhead punk) images has been shown to impact conformity-related behavior.

Potential Benefits and Pitfalls

Conformity behavior is driven by a need for certainty and inclusion in one’s social interactions. By adopting both the *correct* (informational conformity) and *desired* (normative conformity) behavior, individuals not only bring their actions in line

with the other members of their group, they also satisfy their need for belongingness and collective identity, as well as experience improved self-esteem. In addition, they may also foster more positive social relationships by eliciting favorable responses toward themselves from the other group members.

That said, researchers have also documented what may be viewed as adaptive outcomes associated with nonconformity (a heightened perception of uniqueness and personal identity), and conformity behavior is not without its potential drawbacks. The concept of *groupthink*, which has been studied extensively in organizational and political contexts (though less so in sport), provides an interesting illustration of the dangers associated with conformity behavior. Groupthink occurs when a team’s desire for conformity interferes with the necessary processing of alternative ideas, viewpoints, and decision-making options, and several famous historical and contemporary events have been described as prototypical examples of this phenomenon: decisions associated with the 1941 attack on Pearl Harbor, the 2003 invasion of Iraq, and the escalation of the current global financial crisis. Groupthink is most prevalent within highly cohesive networks that do not possess norms encouraging free speech, dissent, expressiveness, and sharing of opinions. This concept is typically characterized by, among other things, overestimations of the group’s capabilities (“We’re a much better team than these guys.”), closed-mindedness (“They’ll never play as well against us as they did last week.”), illusions of unanimity and invulnerability (“We’re all agreed, there’s nothing to fear from this opponent.”), and indeed a pressure to conform (“Everyone else is fine, what are you worrying about?”)

Ben Jackson

See also Cohesion; Group Characteristics; Norms; Self-Categorization Theory; Status

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CONSULTANT EFFECTIVENESS, ASSESSMENT OF

Sport psychology professionals maintain an ethical obligation to ensure services are helping clients (and conversely, not harming them), and thus allowing clients the opportunity to provide feedback is a key element of effective service provision.

Components of Evaluation

There are a variety of subjective (or self-report) and objective measures available that consultants can select to evaluate the effectiveness of any consultation. Self-report forms typically include items focused on characteristics of the consultant, positive and negative experiences with the consultation, impact on team and individual performance, ratings of individual intervention components, time spent on aspects of the consultation, and specific information learned. Items can ask clients to rate their agreement or disagreement with particular statements, or they may ask clients to rate satisfaction (from “not at all satisfied” to “very satisfied”) with specific intervention components. Often, these forms allow open space for athletes to express confidential feedback regarding any aspect of the consultation. It may also be useful to solicit similar feedback from the coaching and other support staff, assuming they were involved as clients in a team consultation. These self-report data solicited from athletes and coaches can be critical to helping consultants understand if the clients’ needs are being met, and if there is a desire to continue services in the future. It is best practice to have someone else besides the consultant administer these subjective forms so clients can feel free to express their true opinions.

These subjective measures of effectiveness can be complemented with objective measures to provide a more comprehensive evaluation model. Depending on the nature of the consultation, the professional may choose a specific behavior (e.g.,

improved first-serve percentage or reduced penalty minutes) or performance indicator (e.g., defensive points against, running or swimming time in a specific event) that is a relevant dependent variable in the consultation. If the consultant intends to share this behavioral or performance data at a scientific meeting or in the form of a scholarly publication, an informed consent for research prior to data collection must be obtained. If the data remain internal to the consultation, then a brief discussion of the measurement approach and intervals for data collection is sufficient. The key dependent variables in any consultation emerge from a joint discussion between client and consultant based around the question, “How would we know if the time spent on sport psychology services was worth the effort?”

Timing of Evaluation

Most professionals suggest that postseason (or termination) evaluations are necessary to meet the minimum requirement for evaluation. However, best practice guidance from the literature suggests that an ongoing evaluation with regular tracking of performance and behavioral data is a more useful and effective approach. Thus, subsequent to a needs assessment and the identification of individual client or team needs, the parties can put into place a transparent system that allows everyone to track progress toward objective goals. Additionally, consultants may want to gather self-reported feedback from all involved parties at regular intervals throughout the consultation, at least once in the middle and once at the end of the experience.

In sum, a comprehensive approach to evaluation has the potential to generate empirical support for the efficacy of sport psychology interventions and strengthen the field as a whole. Without the checks and balances provided through a comprehensive evaluation system, sport psychology practice can seem like a trial-and-error process, and will likely prove less efficient and potentially less effective for the consumer.

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See also Consulting; Ethics; Interventions; Psychological Skills Training

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CONSULTING

Consulting may be described as a temporary relationship that is developed when an individual or entity seeks information or advice. Consultation may occur at the individual, group, or organizational level. It is intended to help the designated client function more effectively and efficiently within a specific setting.

Although consultation is a helping relationship, it differs from psychotherapy, counseling, or coaching along a number of dimensions. Psychotherapy (or counseling—used interchangeably for the purpose of this article) is often described as directed toward remediation, restoration, or recovery of former functioning. In contrast, consultation is designed to improve or enhance functioning and performance. It is intended to resolve a problem or assist in the development of performance excellence, as compared to changing an individual's overall functioning. Although the relationship between consultant and client may be an important means to the end (e.g., positive results), the relationship per se is not the focus of consultation. Psychotherapy is bounded by strict rules regarding confidentiality, and services almost always occur within a therapist's office. Issues of confidentiality within consulting are more complex, in part because multiple players may be involved; services often take place within the client's work setting.

In a strictly *medical model* of psychotherapy, the practitioner performs an assessment from which a diagnosis is derived. This in turn determines

the practitioner's plan for treatment. Treatment is hierarchical in nature: The therapist, by virtue of knowledge and regulation, is in a position of power in relation to the client or patient.

Consultation may share many of these elements, such the sequential aspects of data gathering, evaluation, and decision making in regard to the most useful intervention. As with psychotherapy, this process may take place via interview alone, or may be augmented by the use of relevant assessment instruments. At the same time, the method of interaction and presentation may be much more collaborative. Assumptions are made that the client has the requisite skills to perform the job; it is the consultant's role to assist in the specific enhancement of that job.

At times and in some settings, the term *coaching* is used interchangeably with consulting. Often, however, a coach is in the position of a teacher, imparting knowledge about specific skill development. Consulting, in contrast, usually implies that the consultant will assist in providing information or assist with application rather than specific work skill development.

Psychotherapy and coaching are often dyadic in nature, involving an interaction of two people. Consultation can be considered triadic, involving consultant, consultee, and client. The consultee may be the contracting organization, interested in having the consultant work directly with the client. Alternatively, consultants at times work directly only with the consultee concerning the client; knowledge of the client in this situation may be indirect.

As a consultant, one is typically hired or retained as an outsider rather than a staff member who is part of the organizational structure. The advantage to this role relationship is that the consultant is less bound by some of the constraints that an employee may experience; the consultant may feel freer to offer advice, recommendations, or suggestions even if these might prove unpopular. On the other hand, the system, or organization, has no obligation to accept the advice proffered.

Use of Title

In contrast to counseling, psychotherapy, or psychology, which may be regulated by law for the protection of clients, the term *consultant* is not regulated. The term is widely used; it does not imply any specific training or ethical duty. For that reason, it can be viewed as positive, neutral, or

because it is so widely used in some settings, might be considered pejorative.

Applications to Sport Psychology

A *sport psychology consultant* is a professional with expertise in understanding systemic structures, interpersonal interactions, or mental skills for optimal performance, brought in for specific assistance over a limited period of time. For example, such a consultant might work with a university athletic director to develop services that would assist athletes in stress management, might assist a coach in developing learning strategies regarding optimal methods for team teaching, might be brought in to work with a team on leadership skills or team cohesion, or might work with an athlete on mental skills development.

Within sport settings, the term *therapist* may imply psychopathology; the term *consultant* may not hold the same negative connotation. Although consulting and coaching may be similar, within sport settings the term *coach*, without modification, may be confusing. On the other hand, when modified in a specific way, such as *mental skills coach*, it may have greater acceptability.

Sport psychology consultants receive graduate training in both psychology and the sport sciences in order to offer requisite consultation with regard to cognitive and behavioral skills for performance enhancement or optimal performance. As with more generic consultation, the actual work may occur at the individual, team, community, or organizational level. Because there are various routes to competent practice, potential practitioners and potential clients in North America should be guided by the criteria for becoming a certified consultant through the Association for Applied Sport Psychology and the proficiency in sport psychology of the American Psychological Association.

Kate F. Hays

See also Decision-Making Styles in Coaching; Ethics; Expertise; Leadership in Sport: Mutidimensional Model; Leadership in Sport: Transactional and Transformational; Psychological Skills Training

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CONTROL THEORY

Concepts from *control theory* date back to the early 1900s; however, the origination of control theory is usually ascribed to the publication of Norbert Wiener's 1948 acclaimed work, *Cybernetics: Control and Communication in the Animal and the Machine*. To describe control theory simply, one could generalize that it is premised in understanding and describing self-regulating systems. In the decades following its conception, control theory and various elements of control theory were used to guide work in an array of diverse disciplines, including engineering, economics, mathematics, and medicine. It wasn't until the early 1980s that Charles Carver and Michael Scheier presented control theory within the context of human behavior and psychology, and more specifically the regulation of health behavior through health psychology. Although introduced to human behavior in the 1980s, control theory has not been readily adopted in fields of health

and exercise behavior. As described by Carver and Scheier, this could be in part because of the acceptance and prolific use of learning theories to understand human behavior, or potentially a hesitancy to consider a theory developed outside the realm of human or animal behavior. Regardless, the work of Carver and Scheier to apply control theory to human behavior and health psychology is the impetus that opens the door for application of control theory concepts in understanding exercise and physical activity behaviors. The purpose of this entry is twofold: first, to describe concepts of control theory as applied to health and specifically exercise and physical activity behaviors; and second, to discuss how these concepts have been and could be used in understanding and changing exercise and physical activity behaviors.

Although most terms used in this entry will be defined at the time of introduction, there are a few that should be more specifically discussed at this point. First, given the original work in control theory and subsequent application, it should be noted that control theory is also referred to as *cybernetic theory* or *cybernetics* in the literature. Clarification of the terms *exercise* and *physical activity* is also needed. While these terms are often used interchangeably, each is distinct. *Physical activity* refers to any movement of the body produced by skeletal muscle contraction that is associated with energy expenditure beyond a person's basal level. *Exercise*, on the other hand, is considered a subcategory of physical activity and refers to any planned, structured, and repetitive physical activity carried out for the purpose of improving or maintaining at least one physical fitness component (cardiorespiratory endurance; muscular strength, endurance, and power; flexibility; balance; body composition; movement speed; or reaction time). For this entry the term *exercise* will be used, although principles of control theory are also pertinent to explaining and changing broader physical activity behavior as well.

The Negative Feedback Loop

The foundational concept of control theory lies in what is referred to as the *negative feedback loop*. The feedback loop contains several important features, beginning with an *input function* (a point where present conditions are evaluated by an individual), also called *perception*. Information gathered during perception is then compared against a

reference value (also called the *standard of comparison*) by a *comparator*. This reference value represents a predetermined value of a condition, such as a predetermined behavioral or health goal. In this loop, the role of the comparator is similar to the concept of self-monitoring, where there is an assessment of current behaviors, any potential external influences, and progress toward an established standard (a goal). If a discrepancy is detected during this comparison between the assessment of present conditions and the reference value, a *behavior* is then performed (also called *output function*) to reduce the discrepancy by creating change in the present condition so as to have an *impact on the environment* (anything external to the person or system). This change in the environment then leads to a new perception by the individual and subsequent comparison against the established reference value. The feedback loop also includes what is referred to as *disturbance*, which is any change external to the individual or system (environment) that influences the feedback loop independently of the person's own action, specifically entering the loop through environmental impact. This feedback loop is a closed circuit whereby the environment, person (system), and behavior all influence and are influenced by one another, similar to Albert Bandura's construct of reciprocal determinism within the social cognitive theory. One important note to make here is that the purpose of the feedback loop is not to create a behavior, but rather to ensure that a predetermined, desired condition (as set by the reference value) is maintained through the comparison of the perception and reference value. Therefore, if a discrepancy is not detected, no change in behavior is needed.

An example of this feedback loop can be demonstrated by considering a person who is jogging in one's neighborhood. Once on a jog throughout the neighborhood, most people primarily aim to remain in a safe portion of the road, clear of cars and other hazards. This maintenance is usually determined and addressed through visual monitoring of the road for traffic and by establishing ample room to jog safely between regular traffic and the curb or edge of the pavement. What would happen if a large moving truck approaches the jogger or if a shoulder that was once part of the road disappears? In using control theory with either of these scenarios, the jogger would detect a discrepancy between the perception of currently jogging in the road given these disturbances and the established

safe jogging space reference value. For an avid jogger, this discrepancy would be quickly addressed by the jogger swiftly jumping off the street to continue jogging in a neighbor's front lawn until the moving truck passed or the shoulder reappeared. As is apparent in this example, during any given jog numerous disturbances that require behavioral actions to eliminate associated discrepancies could occur.

Reference Point/Standard of Comparison

Given that the feedback loop heavily relies on a standard of comparison, it is important to understand the derivation and nature of the *reference point*. As conceptualized by Carver and Scheier, the reference point is dynamic and determined within the context of William T. Powers's hierarchical system of goals. In this hierarchy, there are both higher order and lower order goals. Achievement of the lower order goals is intimately related with and essential for the achievement of the higher order goals. In this hierarchy, a higher order goal provides the reference values for feedback loops at the next lower order level. The highest order level within this hierarchical system is referred to as a *system concept*, as it represents both personal characteristics that one wants to embody as an idealized self-image and an attempt for one to minimize sensed discrepancies between how they currently are versus the ideal they desire to be. The second highest order level, *principle control*, specifies principles to use as reference in relation to the system concept; however, these are still qualities and are not specific to any one behavior. In the next lower order level in the hierarchy, referred to as *program control*, actions are identified although specific steps to accomplish these actions are not completely defined since specificity of some steps depends on situational circumstances. Subsequent lower order levels continue to monitor discrepancies between actuality and goals based on the next higher order feedback loop. Therefore, each level within the hierarchy has independent feedback loops that establish and adjust reference points for the feedback loop of the next lower order level, thereby demonstrating the dynamic nature of the goal system.

Again, the jogging example helps illustrate this hierarchical goal system. In this example, the individual is a 26-year-old female, Elizabeth,

who views herself as a good friend and daughter, which represent the *system concept*. Given this view, Elizabeth usually attempts to behave in such a manner that her perception of her present-self matches this reference point, demonstrating self-regulation within this level. One example of how Elizabeth could accomplish this is to make certain that her mother feels special. This becomes the next lower order level reference point or goal, referred to as *principle control* since it is applicable to many behaviors, and thus could happen in many different ways. To self-regulate with such a broad reference point, *program control* feedback loops are then established at the next lower order level where the principle level goal is operationalized to specific behaviors.

In this example, Elizabeth can make her mother feel special by wearing her original wedding dress for her own wedding. However, she needs to be at a lower weight so that she can wear her mother's dress, which can be accomplished by exercising daily (30–60 minutes/day). Although her behavior is generally defined at this point (exercising), it also includes a series of implicit *if-then* decisions based on the circumstances she encounters. Elizabeth's normal 45-minute jog would keep her within the perimeter of her neighborhood; however, if a close friend called and needed to talk she might leave the neighborhood and jog over to her friend's house. On a different day, it could be raining, and she might choose to go up to the school gym and ride a stationary bike instead of jogging. Each of these decisions is made to ensure that the goal of exercising 30 to 60 minutes daily is met, and each variation from the normal occurs to promote conformity to any of her other goals (e.g., not getting sick or ensuring she is a good friend). In each level of the goal hierarchy, the higher order feedback loop specifies the reference point for the next lower order level, whereby behavioral output is required for successful self-regulation in maintaining each reference point, and reference points are dynamic in that they reflect the feedback loop of the higher order goal.

Control Theory and Exercise Behavior

As previously mentioned, the application of control theory to exercise behavior is limited, although concepts found within control theory, especially those consistent with other behavioral and learning theories, such as social cognitive

theory, self-regulation and the transtheoretical model, and processes of change, are consistently applied to and related with exercise. Some of the most prominent concepts congruent with control theory that have been applied to exercise are self-monitoring, specific goal setting, goal review, the idea of tiered goals (goal staircase), and performance feedback provision. In fact, research supports self-regulatory strategies as both direct and indirect predictors of exercise behavior across the lifespan. Additionally, interventions that use self-monitoring alone or with at least one other control theory concept (defined by some researchers as immediate goal setting, performance feedback, and review of behavioral goals) have demonstrated a significantly greater effect on physical activity behavior change than those that didn't include any of these self-regulatory strategies.

Conclusion

In reading about how control theory concepts have been applied to exercise behavior, it is evident that these concepts fall into the broader theoretical context of self-regulation. Where self-regulation is defined as goal-directed behavior, exercising control over self in terms of maintaining or achieving a preferred standard, or feedback loops, it is only successful when it succeeds at both monitoring current conditions and behavior in relation to a goal (reference point) and making changes and adjustments as desired (based on a hierarchy of goals). In this context, control theory concepts are consistently supported as invaluable to understanding and initiating exercise behavior. However, given that research has heavily focused on predictors of initiating behavior, the roles of self-regulation and control theory concepts might differ when trying to understand and improve the maintenance of exercise behavior and habit formation.

In comparing control theory with many other prominent behavioral theories, two additional points need to be made. First, where many value-expectancy and learning theories postulate that behavior change is influenced by self-reinforcement via reward for goal achievement and punishment when goals are not achieved, the control theory does not. Rather, as described earlier, control theory operates on the notion that humans inherently self-regulate through the goal-striving (reference point striving), discrepancy-reduction

cycle. In this, reinforcement is only useful if it provides information (knowledge of results) that can help in the perception-goal comparison. Second, the concept of outcome expectations is also not directly addressed here. However, this idea is widely included throughout prominent theories and works in tandem with control theory.

While control theory has much overlap with other current and prominent health behavioral theories, control theories have been met with some criticism and misunderstanding. Some of this can be explained through the historical perspectives described earlier through the prominent application of learning theories and the hesitancy to consider a theory based on a machine model. However, a more macrolevel distinction between control theory and other current health behavior theories lies in the concept of reinforcement. Where many value-expectancy and learning theories postulate that behavior change is influenced by self-reinforcement via reward for goal achievement and punishment when goals are not achieved, the control theory does not. Rather, as described earlier, control theory operates on the notion that humans inherently self-regulate through the goal-striving (reference point striving), discrepancy-reduction feedback loops. In this, reinforcement is only useful if it provides information (knowledge of results) that can be integrated and used within a feedback loop.

Although prominent control theory concepts pertaining to exercise have been addressed within this entry, it should be noted that control theories include additional concepts and interested readers should see the Further Readings list below.

M. Renée Umstadtd Meyer

See also Goal Setting; Self-Monitoring; Self-Regulation; Social Cognitive Theory

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COORDINATION

Coordination is something we all take for granted—at least until it breaks down under extreme stress (as it does sometimes in competition) or following an insult or disease to the brain, or even when body parts are injured or replaced. So to understand coordination scientifically, we must somehow *make the familiar strange*—akin perhaps to the proverbial falling apple that led to Newton’s great insights about gravity. Coordination is not just physics, even though we may apply physical methods to try to understand it. Nor is it just psychology, even though methods of cognitive science such as reaction time are used to assess how coordination is planned before movements are even initiated. To understand coordination is a very deep problem, maybe as deep as understanding life itself. The reason is that coordination is not just any kind of order in space and time. It concerns how very many component parts and processes on many different levels of organization relate in an orderly fashion to produce a recognizable function or accomplish some particular task. *Coordination* may thus be defined as a *functional* ordering among interacting components in space and time. Coming in many guises, coordination represents one of the most striking features of living organisms. It’s everywhere we look—whether in the regulatory interactions among genes that affect how cells become organs; the tumbling of a bacterium; the coordinated responses of organisms to constantly varying environmental stimuli; the coordination among nerve cells and muscles that produce basic forms of locomotion; the coordination among cell assemblies of the brain that underlies our ability to think, decide, remember, and act; the miraculous coordination between the lungs, larynx, lips, and tongue that belies a child’s first word; the learned coordination among fingers and brain that allows the skilled pianist to play a concerto; the coordination of motion and emotion when making a key play in sport settings when the game is on the line; or the coordination between people like rowers in a

racing eight and players in a rugby team, working together to achieve a common goal. From the micro to the macro, from genes and cells to brains, people, and society, everything involves coordination.

Basic Principles of Coordination

Although the details of coordination are bound to be different at different levels of biological organization, for different organisms and for different functions, might there also be some basic principles of coordination that transcend these differences? The behavioral physiologist Erich von Holst certainly thought this was so. In his essay “On the Nature of Order in the Central Nervous System” (1937), von Holst surveyed the wide occurrence of three basic kinds of coordination in the neural and rhythmic activities of animals, from respiration to voluntary movements, from worms to human beings. One he called *absolute coordination*, a long-recognized form in which component parts operate with the same frequency and with specific reciprocal phase relationships, just like a marching band or people clapping after a performance. Another, extremely rare, form that von Holst didn’t give a name to concerned the complete *lack of interaction* between component parts as in the locomotion of centipedes and millipedes in whom a certain number of (middle) legs had been amputated (no reference to the game of cricket intended!). Persistent practice, von Holst thought, as in playing the piano or the violin, could also lead to complete independence among the fingers of the two hands. The third, possibly most important basic form, von Holst termed *relative coordination*. Here the activities of the individual component parts are neither completely independent of each other nor linked in a fixed mutual relationship. For example, the fins of a fish may not always oscillate at the same frequency and can flexibly slip in and out of preferred phase relationships as internal and surrounding conditions change. Relative coordination provides a glimpse of the tensions between two opposing tendencies that are present in all forms of complex coordination, the tendency of the components to keep separate (segregation) and the tendency to cooperate together (integration). Back in the days when chain reflexes were thought to govern coordinated behavior, the phenomenon of relative coordination hinted at the importance of intrinsic pattern-generating

processes in the central nervous system. Since these early days, great strides have been made in identifying the cellular mechanisms involved in neuronal circuits underlying the generation of rhythmic patterns of coordination. Moreover, in the last 30 years or so, a theoretical framework called *coordination dynamics* has emerged to explain all of von Holst's basic coordination types, mixtures among them, and more generally how coordination emerges, adapts, persists and changes in complex biological systems. Principles of coordination dynamics have been shown to govern patterns of coordination (a) within a moving limb and between moving limbs; (b) between the articulators during speech production; (c) between limb movements and tactile, visual, and auditory stimuli; (d) between people interacting with each other spontaneously or intentionally; (e) between humans and avatars; (f) between humans and other species, as in riding a horse; and (g) within and between the neural substrates that underlie the coordinated behavior of human beings as observed using modern brain imaging methods.

Key Concepts

Some of the key concepts that are allowing a deeper understanding of coordination are self-organization, collective variables, degeneracy, synergy, informational coupling, and intrinsic dynamics. *Self-organization* refers to the fact that patterns of coordinated behavior can arise solely as a result of the dynamics of the system, with no homunculus-like agent inside telling the parts what to do and when to do it. The *self* in the word comes from the fact that the system organizes itself. What is important is setting up the conditions for such self-organized pattern formation to occur. The latter is defined, not in terms of the many individual parts or degrees of freedom, but rather in terms of *collective variables* that arise as a result of the many interactions that are going on. Collective variables are low dimensional, hence simpler descriptions of a complex system. They are meaningful quantities for the system's proper functioning. Collective variables are important to identify because they span different domains, such as sensory and motor, brain and body, perception and action, which are usually defined as separate. Collective variables thus refer to the coupling between different things and processes. *Degeneracy* is an important concept in biology

and means that at every conceivable level of description, the same outcome or function can be achieved in many ways using different components and different combinations among them. Thus, for example, in coordinated movements such as reaching for a cup, many different neural pathways and muscular configurations can combine to achieve the same goal. The mechanism that nature seems to use to handle degeneracy is that it synergizes. *Synergies* are context-sensitive *functional* groupings of elements that are temporarily assembled to act as a single coherent unit. Depending on context, synergies may accomplish different coordinative functions using some of the same components (e.g., the jaw, tongue and teeth to speak and chew) and the same function using different components (e.g. "hand" writing with a pen attached to the big toe). The hallmark of a synergy is that during the course of ordinary function a perturbation to any part of the synergy is immediately compensated for by remotely linked parts in such a way as to accomplish a task or preserve functional integrity. Synergies are important because they are the functional units of coordination at all levels of biological organization. A nice example is the so-called coxless pair in rowing where each oarsman has a single opposing oar. The boat can only go straight across the river if each rower pulls his own weight. If one slacks off, the boat will go in circles and the joint goal of the pair will not be accomplished. This kind of cooperative, mutually beneficial interdependency among the interacting parts of a coupled system to achieve a common objective is ubiquitous in nature. It is a signature of *functional synergy*. Relatedly, the coordination between different things (e.g., parts of the body, regions of the brain) and between different kinds of things (e.g., the organism and the environment, two people, and so forth) depends on information exchange, usually bidirectional in nature ("I talk to you; you talk to me."). Interacting components and features can thus be coupled by material forces, by light, by sound, by touch, by smell, and by intention to accomplish an objective. Such meaningful information transcends the medium through which the parts communicate; it is context specific to the particular form that coordination takes in different task settings. In the coordinated systems of life and movement, the component parts and processes are seldom coupled purely mechanically; they are *informationally coupled*. Information is not lying out there as mere data, coded in some symbolic

form; information is meaningful to the extent that it modifies, and is modified by the intrinsic, self-organizing dynamics.

Intrinsic Dynamics and Learning

This brings us to the important question of how new patterns of coordination are learned. Much evidence now indicates that this depends on the predispositions and capabilities of the individual learner *before* learning begins. This is sometimes quite difficult for scientists to quantify. Nevertheless, such predispositions constitute the learner's behavioral repertoire at a given point in time: the learner's *intrinsic dynamics*. As any inspiring coach or teacher knows, the great benefit of identifying the learner's intrinsic dynamics is that one knows *what* to modify. Any new information (say a task to be learned, an intention to change behavior) has to be expressed in terms of the learner's intrinsic dynamics, otherwise change is not possible. Indeed, the mechanisms through which coordination changes and the nature of the change with learning itself depend crucially on the initial individual repertoire before new learning begins. In this view of coordination, information is not really information unless it modifies the dynamics.

J. A. Scott Kelso

See also Anticipation; Cognitive Function; Competition; Dynamical Systems; Generalized Motor Program; Laws of Movement Learning and Control; Motor Development; Movement; Neuroscience, Exercise, and Cognitive Function

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COPING

Coping refers to conscious and effortful cognitions and behaviors used by the athlete to manage the perceived demands of a situation. Coping is of interest to sport and exercise psychologists because athletes are constantly under pressure to perform. Athletes' and coaches' expectations, injury, performance plateaus, poor performances, equipment failure, superior opponents, skill difficulty, and audiences can all trigger a process of stress and emotion. Athletes need to manage or cope with these demands and their own physiological, emotional, and psychological reactions. The most widely used approach to studying coping in sport is Richard Lazarus's process-oriented perspective called the cognitive–motivational–relational theory of emotion (CMRT). Lazarus's CMRT is a process-oriented perspective about the relationship between a person and one's environment that combines stressors, emotions, and coping as interrelated parts. In the following section of this entry, how athletes appraise stressors in their sporting environments is discussed. Next, how athletes cope with stressors is identified and trait versus state coping, coping effectiveness, and gender differences in coping are discussed. Finally, the development of coping skills and the way in which athletes learn to use coping skills in sport is examined.

Stressor Appraisals

Stressors are subjective appraisals of internal demands, such as personal expectations or internalized performance standards, or external demands like opponent's skill or external time standards that are taxing or exceeding the athlete's personal and social resources, such as their coping skills or social support available to deal with the stressor. These appraisals can be very rapid and

automatic or more conscious and reflective and involve higher cognitive functioning. Thus, social learning, values, and beliefs, along with hard-wired emotional responses linked to basic survival can all impact the appraisal process.

There are many types of appraisals that influence the stress and emotion process. Lazarus argued that individuals make primary and secondary appraisals about events or demands. Primary appraisals involve what is at stake to an athlete, especially if the demand is relevant to the athlete's personally meaningful goals, values, beliefs, personal resources, social environments, or intentions. An assessment of these factors can produce appraisals of harm or loss (social or personal damage has already occurred), threat (there is a possibility of damage in the future), challenge (difficulties to be overcome), or benefit (there is a possibility for gain). Take an example of a female basketball player who is injured in a game. The athlete appraises that the injury will prevent her from continuing to play, perform, and receive social recognition, resulting in an assessment of harm or loss. Threat appraisal also becomes activated when she realizes that the injury might prevent her from playing in future games, endangering her sport scholarship, other sport-related goals, and her athlete identity.

Secondary appraisals involve evaluating coping options or coping potential (Do I think I can cope with this event?), the controllability of the stressor, and perceived outcomes of the event (What will happen?). In the example above of the basketball player, the athlete's secondary appraisal may involve her evaluation that she can manage some of the initial distress and pain, she has access to excellent medical and rehabilitation services, she can work hard to recover during rehabilitation, and she has some control over rehabilitation process. She is, however, uncertain about the long-term outcome. Primary and secondary appraisals made by an athlete will have a major impact on how the athlete will cope.

Appraisals result in decisions that produce a relational meaning, which is the athlete's subjective evaluation of the relationship between oneself and the environment. Emotions arise or change based on these relational meanings (subjective evaluations) that an athlete constructs about one's person-environment relationship. For example, during tryouts, an athlete may appraise the relationship between self and the competitive

environment as threatening since one may be cut from the team, resulting in the emotion of fear. If the athlete makes the team, the relationship between self and the environment has changed and the threatening situation has been resolved, resulting in the emotion of relief. Thus, athletes may need to manage a variety of negative and positive emotions, involving anger, anxiety, sadness, shame, guilt, envy, jealousy, hope, relief, fear, happiness, joy, and pride.

Stressor appraisals can vary from person to person because appraisals are influenced by the athlete's personal goals, values, and past experiences. Therefore, athletes may interpret stressors differently, and not all athletes will appraise the same situation or event as a stressor. However, some situational variables seem to consistently influence stressor appraisals such as the magnitude of the stressor or whether the stressor was anticipated or unanticipated. The magnitude of the stressor (e.g., major or minor) is related to the extent to which the event interferes with or is related to the athlete's goals and values. Minor stressors such as spilling water on one's uniform before a competition are likely to be less stressful than major stressors such as trying out for a team and being cut. Stressors that are anticipated or expected are considered less threatening than stressors that are unanticipated or unexpected. For example, consider a situation in which an athlete is facing an opponent who is stronger in ability. The athlete may expect to lose the match and, therefore, the stressor appraisal may be less threatening to goals and values. However, if the opponent is weaker in ability and our athlete does not expect to lose the competition, then a loss would be appraised as more harmful to goals and values since the loss was unexpected.

Coping Strategies

Coping refers to conscious and effortful cognitive and behavioral efforts to deal with the perceived demands of a situation. Some responses to stressful situations like defense mechanisms, crying, or yelling are considered unconscious, noneffortful, or involuntary responses, which fall outside the commonly accepted definition of coping. Researchers in sport have identified over 100 coping strategies, although many of these can be classified into more general categories. Common coping strategies in sport include arousal control, relaxation,

concentrating on goals, time management, isolation, deflection, seeking social support, increasing effort, wishful thinking, venting, refocusing, information seeking, learning about opponents, practicing, increasing training, visualizing and imagery use, humor, prayer, substance use, denial, self-talk, maintaining positive focus, positive self-talk, and positive reappraisal. Coping strategies can affect various aspects of the stress and emotion process in athletes. These strategies can help athletes change the perceptions of demands (appraisal), recognize and communicate emotions, regulate arousal and impulses associated with emotional feelings, regulate competing demands, develop effective plans and actions, seek available resources, and develop better technical and physical skills.

There are different definitions and ways of conceptualizing coping. A popular macro-level approach, based on the work of psychologists Richard Lazarus and Susan Folkman, is to categorize various coping strategies under two general coping functions: problem-focused coping and emotion-focused coping. *Problem-focused coping* refers to cognitive or behavioral actions to deal with the demands of the situation like planning, increasing effort, or seeking informational support, whereas *emotion-focused coping* refers to cognitive or behavioral actions to deal with the emotions arising from the situation, such as acceptance, positive reappraisal or reinterpretation of the event, and seeking emotional support. Recent conceptualizations of coping in sport by psychologist Patrick Gaudreau and Jean-Pierre Blondin (2002) propose a distinction between engagement coping (e.g., increased effort, relaxation, thought control), disengagement coping (e.g., venting, physical or mental disengagement) and distraction coping (e.g., mental distraction, focusing attention on other tasks). Other approaches involve a micro-analytic approach where researchers examine specific coping strategies used to deal with particular stressors.

Trait Coping Versus State Coping

There is some debate about whether athletes may have a preferred coping *style* or if coping fluctuates and changes across situations and contexts. It appears some athletes do cope in a consistent manner with similar stressors. For example, an athlete may cope with training stressors in a consistent manner, particularly if the features of the

training session are similar: same coach, same teammates, or same drills. However, there is strong evidence that athletes do not cope consistently across all situations and contexts; as an example, athletes do not cope with competition stressors in the same way they cope with training stressors. There are several reasons for a lack of stability in coping. First, the stressors might be quite different across situations. Second, even for the same type of stressor, the individual might appraise the stressor differently at different times. Stressors are appraised in terms of their relevance to the athlete's goals, values, and the relational meaning the individual constructs about the relationship between oneself and the environment. If the features of a competition are deemed to be different from training sessions (e.g., a competition is considered "more important" than a training session), then the appraisal of competition demands will result in a different relational meaning for the athlete. This appraisal is likely to trigger a different emotion requiring a different combination of coping strategies. Therefore, athletes' inconsistency in coping across situations and contexts may be due to differences in the actual stressor faced as well as appraisal of the stressor.

Researchers of competitive anxiety have also suggested that the degree to which athletes perceive they have control over a stressor is a key factor influencing stressor appraisals, emotions, and the use of coping strategies. That is, athletes who perceive a stressor is controllable will likely interpret their emotions as facilitative for performance and are more likely to be able to cope with the stressor. Conversely, athletes who do not perceive they can control the stressor will interpret their emotions as debilitating for performance and are less likely to be able to cope with the stressor. Perceptions of control are also important for predicting athletes' use of coping strategies; athletes who perceive they are in control of the appraised stressor are more likely to use problem-focused or engagement-type coping strategies. Athletes who perceive they are not in control of the appraised stressors are more likely to use emotion-focused, disengagement-, or distraction-type coping strategies. Athletes' perceptions of control may change over time as they become more accustomed to coping with particular stressors. As athletes face similar stressors, they may develop enhanced perceptions of control and enhanced beliefs about their ability to cope with the stressor effectively.

Stressor appraisals and coping do not occur in isolation and are related to athletes' social environments. Athletes may appraise stressors related to a coach, teammate, opponent, referee, or parent—pressure to perform; criticism from coaches, teammates, or parents; receiving a bad call from a referee or official; or interpersonal conflicts. Athletes also frequently report social dimensions related to coping with stressors. In terms of coping, common strategies include seeking emotional support (e.g., turning to others for comfort) and informational support (e.g., seeking advice or guidance about a problem). Social support is helpful for athletes in that it can sometimes protect or buffer athletes from appraising stressors in sport, it can decrease the severity of athletes' appraisals, or it can enhance athletes' perceptions of their ability to deal with potential stressors.

Coping Effectiveness

A number of coping strategies can be used to deal with stressors; however, not all coping actions are effective in all situations. Some measures of effective coping are achieving performance outcomes, reducing anxiety or negative emotions, alleviating the stressor (e.g., resolving the problem), and improving overall well-being. There is, however, no clear answer as to what constitutes effective coping. A specific strategy may have a short-term effectiveness but produce problems in the long term. For example, ignoring the coach may help reduce the athlete's initial distress but cause long-term problems in the athlete-coach relationship. Some researchers have suggested that problem-focused or engagement coping is associated with achievement of performance goals and is an effective approach to coping with stressors, particularly controllable stressors. Some strategies like seeking support can serve as both a problem-focused and an emotion-focused coping strategy. For instance, an athlete may seek support from a teammate in order to learn a new drill or skill (seeking informational support as a problem-focused coping strategy), while at the same time using the opportunity to talk to the teammate about frustrations (venting and seeking emotional support as an emotion-focused coping strategy).

Researchers such as Susan Folkman, Judith Moskowitz, and Hugh Richards have advocated that coping effectiveness must account for the individual's personal goals, the situational context,

and the individual's appraisal of the situation. For example, avoidance may be considered an ineffective coping strategy in some cases, yet in other circumstances it may be necessary and desirable to remove oneself from a stressful situation. Avoidance may be an effective coping strategy in the short term while the athlete determines an appropriate way to deal with the situation. By taking into account the athlete's goals, the situational context, and the athlete's appraisal of the situation, different coping strategies will be effective at different times and across different contexts. Effective coping should entail flexible coping, wherein athletes are able to use different coping strategies as situations change and unfold.

Gender Differences in Coping

Gender differences in sport coping are unclear. Some studies report no gender differences in athletes' coping, whereas others do report differences. Studies that find gender differences report different findings such as suggesting that female athletes use more support-seeking or help-seeking to deal with stressors and focus their efforts on managing their emotions, whereas male athletes are more likely to use problem-focused coping strategies, confrontational strategies, avoidance, and venting.

The issue of gender differences in coping is complex and there are several possible reasons for the equivocal findings to date. One suggestion is that males and females are socialized to cope with stressors in different ways that are considered acceptable: Females are encouraged or socialized to express emotions, whereas males are socialized to conceal emotions or engage in assertive coping behaviors. However, sport also may be considered a context that supports traditional masculine values such as competitiveness, assertiveness, and achievement. Consequently, female athletes may be less likely to cope in a traditionally feminine manner and may adopt more masculine-gendered coping behaviors.

Recent advancements in the examination of gender roles suggest two hypotheses to explain differences in male and female athletes' coping: the *dispositional hypothesis* and the *situational hypothesis*. According to the dispositional hypothesis, if male and female athletes appraise the same stressor in the same manner (e.g., equally relevant to their goals, values), gender differences in coping will emerge. For example, if a male and female

athlete both appraised coach criticism as a stressor in a similar manner (e.g., equally threatening to each athlete's personal goals and values), the male athlete might cope by venting emotions, while the female athlete might cope by seeking emotional support from teammates. The situational hypothesis, on the other hand, suggests that the reason for gender differences in coping is because of differences in the way male and female athletes appraise stressors. According to the situational hypothesis, coach criticism might be appraised in a distinctly different manner by the male and female athletes. The male athlete might appraise the coach's criticism as unwarranted and he might feel that the coach was unfair, resulting in the athlete venting his emotions. Conversely, the female athlete might appraise the coach's criticism as accurate and feel embarrassed about her performance, and she may seek support from a teammate to avoid drawing any further attention to herself. Additionally, the athlete's appraisal of the intensity of the stressor is important in determining gender differences in coping: The male athlete may appraise the coach's criticism as less severe or intense than the female athlete does, which could also explain gender differences in coping. Therefore, it is important to understand the way in which male and female athletes appraise stressors before being able to determine gender differences in coping.

Development of Coping

Coping changes as athletes age and mature, and also as they gain more competitive experience in sport. Coping is also influenced by developmental changes in cognition and through socialization practices (e.g., the way athletes are encouraged to cope with stressors in sport by parents, coaches, or peers and teammates). Early adolescent athletes appear to appraise fewer stressors and appraise qualitatively different stressors than middle adolescent athletes. For example, early adolescent athletes report stressors related to making errors, opponents, team performance, and family concerns, while middle adolescents reported stressors related to making errors, team performance, coaches, social evaluation, contractual stressors, and playing at higher competitive levels. These differences in stressor appraisals likely reflect contextual differences in the athletes' competitive environments, where older athletes may be more concerned about their future careers as competitive

athletes. Therefore, athletes' stressor appraisals change in response to changes in their competitive context.

With regard to coping, older adolescent and young adult athletes appear to use a more diverse range of coping strategies than do younger adolescent athletes. Younger adolescent athletes appear to use more behavioral coping strategies like increasing effort, seeking support, or behavioral disengagement, while older adolescent athletes appear to use more cognitive forms of coping like reflection, cognitive reappraisal, positive self-talk, or mental disengagement. There are several explanations for the emergence of a greater variety of coping strategies among older adolescent athletes. As athletes age, they develop a greater capacity for abstract thinking and the ability to use cognitive coping strategies, in addition to primarily behavioral coping strategies. Second, younger athletes may have difficulty reflecting on their coping or discussing and reporting their coping attempts on measures such as surveys, questionnaires, or in interview settings. Increased self-awareness as athletes age and develop may contribute to older adolescents' reporting of greater coping repertoires. As athletes mature and develop, their coping repertoire becomes more diverse and the use of coping strategies is more differentiated; they use coping strategies specific to the perceived demands of the situation. Coping becomes more organized and flexible compared to younger athletes, and older athletes appear to be able to reflect upon their coping and evaluate the effectiveness of their coping, potentially learning from past coping experiences to deal with future stressors.

Learning Coping Skills

Athletes can learn the necessary cognitive and behavioral skills to effectively manage stress and emotion. Learning can occur through experience and training. Research has shown that adolescent athlete coping changes across different stages of development and at different levels of competition, and older athletes seem to have a wider repertoire of coping strategies than younger athletes. Increased exposure to stressors (e.g., playing more difficult opponents, higher training loads, learning more complex strategies) may enable some athletes to learn coping skills, although the support of coaches and parents seems necessary to allow athletes to develop their coping skills in a safe and

supportive environment. Learning to cope may be associated with athletes' ability to anticipate stressors and plan ahead, for example by thinking about upcoming competitions and anticipating increased training demands. Learning to cope also may be associated with self-awareness and the ability to reflect upon past coping episodes in order to learn from prior experience. As athletes gain experience coping with stressors in sport, they may learn to apply particular coping strategies in response to the particular demands of the situation. Although some coping strategies may be more adaptive than others (e.g., substance abuse would be considered a maladaptive coping strategy), it is important to note that no single coping strategy will be universally effective in coping with all stressors. Instead, athletes should learn to apply strategies selectively based on the demands of the situation.

There are also a number of psychological skills or coping skills training programs that can facilitate the learning of coping skills. Coping training programs developed or modified for sport include stress inoculation training (athletes are given opportunities to practice a range of coping skills), COPE training (control, organize, plan, and execute coping skills), cognitive-affective stress management training (athletes develop a coping response involving relaxation and self-statements to manage stressful sport situations), self-compassion training (athletes learn to more accurately perceive and rectify maladaptive patterns of thought) and coping effectiveness training (athletes learn appraisal skills and a range of coping skills [see also the entry "Stress Management"]). Most of these programs involve an educational phase, a coping skill development phase, and a coping skill application phase. Enhancement of coping skills is also possible through many other psychological skills training and emotional regulation programs.

Conclusion

Coping represents a complex topic that is related to athletes' stressor appraisals, goals and values, emotions, beliefs, identity, and their resources and abilities to deal with the demands associated with sport. Because of the intricacy of the coping process, it is important to consider multiple antecedents and outcomes. There are situational, as well as individual (e.g., gender, developmental) differences in coping that require attention by researchers in

order to evaluate what constitutes effective and adaptive coping in sport. Athletes' coping changes with development and can be improved through interventions and training, which represents an important area for improving athletes' performance and overall experience in sport.

Katherine Tamminen and Peter R. E. Crocker

See also Basic Emotions in Sport; Stress Management

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CREDENTIALS

To call oneself a psychologist (at least in the United States and Canada), or use the terms *psychologist*, *psychological*, or *psychology*, one must be licensed in the state or province in which one practices. However, if one works in academia,

calling oneself a sport psychologist is an appropriate term to use as it is descriptive of one's role at one's academic institution. Credentials are aimed at demonstrating one's competency within the field. This competency is critical for the sport and exercise psychology profession. The term *competency* indicates that one has the educational background that provides the knowledge base to be an effective practitioner and has the experience, gained through one's educational program, to practice effectively. Consumers have the right to require competent practitioners; similarly, practitioners must demand that we and our colleagues have demonstrated levels of competence to earn the credentials discussed in this entry.

Competency underlies obtaining a credential as a sport and exercise psychologist. This competency is generally obtained through experiences in three areas. First, one obtains an academic degree, usually a doctoral (PhD, PsyD) degree, but in some cases a master's degree (MS, MA, MEd) is sufficient. Second, one would successfully complete specific coursework at an academic institution, usually as part of one's master's or doctoral degree, including psychology, kinesiology, and sport and exercise psychology-specific courses. Third, one would obtain experience within clinical or counseling settings. This would generally be mentored through applied experiences with individuals as well as teams in a variety of sports and settings. This experience provides an indication that one has obtained a base of knowledge in clinical or counseling psychology as well as the skills set and temperament to work effectively in a role as a sport and exercise psychologist. This knowledge base and skill set may encompass counseling and psychotherapy work with a variety of mental health issues, program development and evaluation, and psychological assessment.

Individuals who work as licensed professional counselors (LPC; the term used depends on the state, province, or country) have earned master's degrees in counseling or clinical psychology or related fields and are eligible for licensure, certification, or accreditation (the term used depends on the state, province, or country). They are qualified, if they meet the relevant criteria, to work as counselors and receive payment for such work, including third-party reimbursement.

Individuals with doctoral degrees may become licensed as psychologists in the United States and Canada (other countries may allow master's level practitioners to be called psychologists). This process generally requires a doctoral degree (e.g., PhD, PsyD) in psychology (or a related field), specific coursework, internship hours, and successfully passing state and national exams. States, provinces, and countries have generally similar standards, but they may differ. One must verify the specific standards for the jurisdiction in which one wishes to practice.

The primary credential in the sport and exercise psychology field is being a Certified Consultant within the Association for Applied Sport Psychology (CC-AASP). CC-AASP provides the clearest path for developing the level of competency required to work within a performance enhancement focus in sport and exercise psychology. The CC-AASP process does not have a clinical or counseling focus. CC-AASP requires attainment of numerous criteria, including a doctoral degree, 400 hours of mentored applied work (of which 100 must be directly with individuals, teams, or both), and a variety of coursework across areas within kinesiology and psychology. This coursework includes not only sport and exercise psychology specific courses but also coursework in other areas within kinesiology like motor learning and exercise physiology, as well as numerous areas within psychology, such as social, developmental, and learning. There is also the option of provisional certification with a master's degree, and full certification if one completes an additional 300 hours (above the 400) of mentored experience. There is a standard application process and review by the Certified Consultants Committee to determine whether one has met the criteria needed for CC-AASP status. There were approximately 300 Certified Consultants as of fall 2012. AASP is working toward publicizing the CC-AASP credential more widely so that potential clients (e.g., athletes, parents, coaches) will be aware of what to expect from a practitioner with CC-AASP credentials in terms of competency and experience.

The British Association of Sport and Exercise Sciences (BASES) has several accreditation programs for identifying competency within sport and exercise science, including a High Performance Sport accreditation. The basic one for sport and exercise psychologists is termed *Accreditation* and

focuses on the ethical and professional standards of BASES members practicing in the exercise and sport sciences. An individual fulfilling the accreditation criteria is called a *BASES Accredited Sport and Exercise Scientist*. This involves a similar process to that of the CC-AASP in terms of mentored-applied work and coursework. Accreditation can be obtained with undergraduate and master's degrees—a doctoral degree is not required.

Division 47 (Exercise and Sport Psychology) of the American Psychological Association has established sport psychology as a proficiency within APA. Proficiency is seen as a specialization after one attains one's doctoral degree. The proficiency focuses on competence in working with athletes on their psychological skills, well-being, developmental and social components of participation in sport, and issues related to working within sports organizations. Division 47 within APA is taking steps to develop guidelines for meeting the proficiency.

The Australian Psychological Society (APS) offers a specialty area within sport and exercise psychology (one of nine specialty areas within the APS), and established the Australian College of Sport and Exercise Psychology (a "college" within APS) among its other branches. Sport and exercise psychologists are regarded as specializing in numerous areas, including performance enhancement, stress management, overtraining, team building and leadership, and psychological assessment. Attaining full membership as a sport psychologist requires at least 6 years of university education, plus 2 years of supervised practical experience within sport and exercise psychology. It should be noted that this is different from the United States and Canada, where the doctoral degree is generally seen as required to become a psychologist. Other countries may have specialty certifications as well.

Credentials for someone working in academia would generally encompass an advanced degree like a PhD or EdD in kinesiology or psychology, with an emphasis on sport and exercise psychology. This includes the standard teaching, research, and service roles of an academician. Someone providing performance enhancement consulting services needs to be mindful of the credentials previously identified with respect to certification. It is not uncommon for academicians to work with athletes or teams at their host college or university. While academicians generally do not acquire additional credentials per se (although one can become a fellow in one's professional organizations, such

as AASP and APA), one is expected to remain current in the field. Many professions (e.g., psychology, medicine) require continuing education hours each year. While academia generally does not require these as such, one is expected to be acquainted with the literature, attend conferences, and perhaps conduct research and present and publish the work. The goal is to be knowledgeable of the current literature within sport and exercise psychology.

Offering consulting services outside the college or university setting encompasses a different level of practice. In the United States, anyone can hang up a shingle and provide services as a consultant, a therapist, or another "unprotected" term (protected terms include, e.g., the term *psychologist*). The standard caveat, *Buyer beware*, is critical. While some of these practitioners may have acceptable levels of experience, it is preferable to identify someone with the credentials identified earlier (e.g., certification, licensure) if one is looking for a professional to provide services for oneself, one's athletes, one's team, or one's organization.

Michael L. Sachs

See also Career Paths; Certifications; Consultant Effectiveness, Assessment of; Consulting; Training, Professional

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CUE UTILIZATION

Whether out of envy or admiration, people have long been fascinated by the extraordinary skills of champion athletes such as Roger Federer (tennis), Michael Phelps (swimming), and Lionel Messi (soccer). Building on this interest, recent years have witnessed increasing collaboration among researchers from cognitive psychology, sport psychology, and cognitive neuroscience in studying the mental and neural processes that underlie *expertise*—or the growth of specialist knowledge and skills as a result of effortful experience—in sport. Such research has helped scientists understand how expert athletes manage to achieve certain remarkable feats in the face of severe time constraints and rapidly changing environmental conditions. For example, how can top tennis players hit winning returns of balls that travel toward them so fast (up to 250 kilometers, or more than 150 miles per hour) and cannot be seen clearly? This paradox is intriguing because in dynamic sports like tennis, the time taken for a ball to travel from one player to another is often shorter than the combined duration of an athlete's reaction time and movement time. Thus, how do top athletes respond effectively to fast-moving balls *before* they consciously perceive them?

The answer lies in *anticipation*. Specifically, recent studies of rapid reactive sports like tennis, baseball, and cricket show that expert performers can circumvent fundamental information-processing constraints by using clues (advance cues) from their opponents' movement patterns to anticipate the direction, trajectory, and likely landing point of balls speeding toward them. Such *cue utilization* (or the ability to extract and extrapolate from task-relevant information provided by opponents' behavior) gives athletes additional time to devise and execute appropriate responses to fast-moving balls. Clearly, expert athletes in fast-ball sports appear to have a *cognitive* (anticipation-based) rather than a

physical advantage over less skilled athletes. This conclusion raises two key questions. First, what methods do psychology researchers use to investigate cue-utilization processes in athletes? Second, what are the main research findings in this field?

Investigating Visual Cue Utilization Processes in Sport

Various research methods have been used to investigate expert–novice differences in the perceptual–cognitive skills of athletes. These methods include qualitative procedures (e.g., protocol analysis, involving recording what people say as they think aloud while solving problems in their specialist sport), as well as quantitative strategies such as *occlusion* tasks, where viewers have to make sport-specific predictions based on limited information, and *eye-tracking technology*, which measures the location and duration of perceivers' visual fixations elicited while viewing slides or movie clips of sporting scenes. The latter methods, which have proved fruitful in studying visual cue-utilization processes in athletes, are explained briefly as follows.

Occlusion Tasks

Occlusion tasks require people to make predictions, such as judging the likely direction of a shot in tennis, from sport scenes in which vital information is either obscured or incomplete. By analyzing how expert athletes differ from novices in their performance on these predictive tasks, researchers can establish the relative importance of different perceptual cues to likely outcomes. Normally, occlusion tasks can be either temporal or spatial. In temporal occlusion studies, participants have to guess what happens next when viewing film sequences in which certain important temporal information (e.g., disguising the flight-path of a ball) has been deliberately occluded. In spatial occlusion studies, specific portions of the visual scene are progressively removed or occluded from view, and their effects on viewers' accuracy scores are analyzed. The assumption of this latter approach is that if there is a performance decrement when a particular spatial area of a stimulus display like the hips or shoulders of a tennis player model during a simulated serve is occluded from participants, then that *area of interest* seems likely to be especially informative to viewers.

Eye-Tracking Methods

Eye-trackers (whether fixed or mobile) are designed to record the location, duration, number, and sequence of a perceivers' visual fixations as they look at slides or video film simulations of sport-relevant information in laboratory settings. Using such variables, eye-tracking researchers draw inferences about athletes' visual search behavior. For example, the location of a visual fixation is usually regarded as an index of the relative importance of a given cue within the display being viewed. Similarly, the number and duration of athletes' fixations in a particular region are believed to reflect the information-processing demands of the information displayed in that area.

Research Findings on Cue Utilization

A considerable volume of research on cue utilization processes in athletes has been conducted by sport psychology investigators using occlusion and eye-tracking methods. Three important findings of these studies may be summarized as follows. First, expert athletes are generally superior to novices in their ability to anticipate what their opponents will do next on the basis of advance visual cues extracted from their adversaries' postural movements and from the relative motion between opponents' bodily features. One advantage of this superiority in anticipation skills is that it gives expert athletes additional time to counteract the perceived intentions of their adversaries. Second, expert athletes appear to employ more efficient visual search strategies than those of novices when inspecting visual displays in their specialist sports. For example, elite performers tend to display fewer visual fixations than novices while viewing sport scenes, but these fixations are often of longer duration than those of their less skilled counterparts. In other words, experts tend to be drawn more than novices to information rich areas of interest in sport-related displays. Finally, expert athletes tend to be superior to novices in sporting pattern-recognition—using their greater knowledge of game-specific event probabilities to anticipate how a given scenario is likely to develop over time (see also the entry “Situational Awareness”).

Despite these advances in our understanding of cue utilization processes in athletes, at least two unresolved issues remain. First, the theoretical mechanisms underlying expert–novice differences in anticipation skills remain largely unknown.

Therefore, the attempt to explain how expert athletes acquire, develop, and update their anticipatory advantages over less proficient counterparts is an urgent priority for future researchers in this field. Second, there is little consensus at present among researchers as to whether or not it is possible to effectively train perceptual–cognitive skills in athletes. As before, additional research is required to arbitrate empirically on this issue.

Aidan Moran

See also Anticipation, Expertise

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CULTURAL COMPETENCE

Sport and exercise psychology has traditionally been understood to consist of a set of skills and theoretical underpinnings distanced from culture. These skills have been taught in postsecondary educational settings and presented in authoritative textbooks. Although readers might not at first recognize what rests beneath the surface of these writings, a closer look suggests that they are in fact culturally informed and in relation to one culture, suggestive of a very specific cultural competence, loosely referred to as European in origin. Thus, the practitioner or researcher who works within a group that conforms to mainstream cultural

practices is engaged in culturally competent work, intended only for that population.

Culturally Competent Practice

Sport and exercise psychologists are becoming increasingly open to cultural sport psychologies whose approaches account for cultural uniqueness, in place of a single monolithic cross-cultural approach. There is now a greater understanding among a growing number of scholars in sport science that what motivates athletes and exercisers, in part, pertains to their cultural socialization. Underpinning much of the discussion of cultural sport psychology is the search for culturally relevant strategies for practitioners and researchers, targeting discrete groups and better understanding each group's uniqueness.

Cultural competence involves a narrower approach than multicultural competence, with one's focus placed upon what defines a given culture in terms of conventional practices. One might seek to become culturally competent in terms of the general practices of American Indians, Latin Americans, Indo-Canadians, Muslims in Kuwait, or another general grouping of people, residing in a geographic region at a specific moment in time (current practices). One might also seek to learn the current religious practices of Islam, Hinduism, Judaism, or Buddhism, generally, in advance of working with athletes who actively practice any of the above sets of beliefs.

Within sport and exercise psychology, the onus is on the practitioner and researcher to seek an understanding of the cultural other, partly in advance of consultation, and partly through an open exchange with the client and astute observation for confirmation of what ought to be appropriate cultural practices. There are several general dimensions that are found within cultural competency training. Included among these criteria are (a) what constitutes appropriate use of physical space between people during formal and informal conversations; (b) whether time is regarded in terms of a clock (clock-based time) or an event (event-based time); (c) whether university education, life education, or both types of learning are regarded as creditable; (d) eating practices; (e) appropriate dress; (f) views of gender definition, and on a continuum; (g) the practice of sustained or averted eye contact; (h) where a group resides in terms of individualism to collectivism;

and (i) how often one should speak or listen within a conversation—contingent upon the nature of the discussion and one's place within.

Courses are often provided for professionals to inform them what constitutes culturally competent behavior. A common belief is that after one has completed a formal cultural competence education module, one has gained the necessary cultural skills to conduct appropriate cultural practices with competence. However, taking a course to learn the tenets of cultural competence is only the start of a larger educational process in which one seeks to hone the necessary skills and then correctly support the cultural identity of one's client base. Hence, application of the educational module and the successful integration of the teachings determine whether one is effectively skilled and thus culturally competent.

Research Applications

Within sport and exercise psychology research, there are also important cultural practices that must inform a culturally competent exchange, building on the aforementioned dimensions. Specific methodologies might lend themselves more closely to the general tenets of a given culture. For example, within a collectively minded culture, the use of participatory action research might feature appropriate consultation processes and engaged group discussion. Research protocols might also require a feast in advance of data collection. One's methods might also vary depending on the cultural group in question. For example, the researcher might use talking circles in place of conventional focus groups when engaging with select indigenous populations. Writing practices might also vary from a conventional presentation of data to storied writings and teachings, reflecting the cultural practices of the participants and their cultural community. Cultural competence within research contexts, then, must permit opportunities to feature participant-relevant practices, leading to the featuring of diverse perspectives and voices as opposed to cultural silencing.

Conclusion

Cultural competence denotes a general approach intended to provide researchers and practitioners with a set of useful rules in advance of work with a population unfamiliar to them. The understanding and honing of these skills is meant to

undergird approaches to communication, the selection of appropriate motivational skills, and also investigative techniques. These general strategies are intended as a platform upon which specific culturally safe practices are negotiated with the intended audience. The dialectic between cultural competence and cultural safety fosters what becomes the meaningful approach, developed to centralize the identity of the client or the participant.

Robert J. Schinke

See also Certifications; Consultant Effectiveness, Assessment of; Ethics; Ethnicity; Feminism; Gender; Heterosexism, Homonegativism, and Transprejudice; Multiculturalism

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CULTURAL SAFETY

Over the course of the past 10 years, there has been considerable discussion devoted to multicultural competence within the counseling professions. These discussions are only now beginning to surface within sport and exercise psychology certification. The intent through multicultural training is to provide the intended clients with health services that more closely align with their specific cultural background. However, there is a distinction between garnering multicultural competencies and skills that pertain directly to a client in relation to the cultural experience. The divide between cultural competencies and the uniqueness of each client determines whether the services are culturally safe.

Culturally Unsafe Practices

There are a variety of culturally unsafe practices sometimes committed by the practitioner. Within the writing, one finds reference to the term *race blindness*: an act committed by people in authority who attempt to view and treat the diversity of their clients the same—using a one-size-fits-all cultural approach. There is an obvious distinction between *sameness* and *equality*, with the latter term conforming more closely to cultural safety. A second unsafe practice is for the practitioner to engage in acts of stereotyping, where clients are classified as part of larger groups, such as a nationality, race, gender, or religion. From overarching categories of classifications, the practitioner might move forward, employing general cultural guidelines, proposed as pertinent across the group. A third example of unsafe practice is for the sport and exercise psychology consultant to approach all consultation opportunities employing a mainstream sport psychology approach, mostly built from Western academics. From this third misstep, the consultant might entirely overlook the practices of a client from a marginalized background, perpetuating the marginalization.

Possible Considerations

Cultural safety requires a close match of the specific counseling skills and proposed motivational tools of the sport and exercise psychologist with the customs and practices of the client. For example, when working with a Canadian Aboriginal athlete in northern Canada, socialized with traditional indigenous customs, the sport psychology consultant might work in consultation with Aboriginal elders or enter into meetings that begin with the use of local Aboriginal sacred medicines, such as the burning of sweet grass. For each client that holds cultural values that differ from that of the sport psychology consultant, there are specific practices that need to be accounted for, outlined within another section pertaining to cultural competence. Beyond more general cultural guidelines, there must also be a closer consideration of the client's socialization and identity. The possibility of offending or disempowering a client exists within the diversity of exchanges encountered by sport psychology professionals. To offset culturally unsuitable applications, the practitioner must engage in a process of discovery that allows for

specific practices and general guidelines pertinent to the client.

Garnering a Culturally Safe Approach

There are various ways to develop a closer understanding of clients in terms of their cultural identity. The most likely strategy is to meet with the client, and during an intake interview, learn more about the socialization through direct questions and observation. Questions might bring to light the client's religion and level of adherence, view of self in relation to family and community, value in terms of formal and informal education, dietary practices, and use of wording, among a wider range of considerations brought forth with open-ended questions. Via observation, the astute practitioner can learn from the client's eye contact the extent that sustaining or averting is suitable. In addition, the client will seat or re-seat at a specific distance from the practitioner, suggesting an appropriate physical distance that in turn suggests formality or informality. Clothing would also provide some visible indication of whether the client is traditional in relation to general cultural guidelines or assimilative of specific subcultural practices. One might learn about the client's cultural standpoint, also by engaging in collateral meetings with coaches or people the practitioner is referred to as offering collateral information, by the client. From collateral interviews, the practitioner learns more about the client's affiliations and motivational views.

Research Applications

Cultural safety is also relevant within research contexts. There are instances when the applied sport researcher interviews or observes participants from minority or oppressed cultures and subcultures. When entering into these exchanges, the researcher can seek to learn the specific research practices that align with the participant's standpoint. One example might be to interview participants with a collective inclination in groups, or have people threatened by personal interviews or cross-gender exchanges via research accompanied by a family member or friend. There is uniqueness within

small local groups of participants that refine more general cultural considerations, such as among a community's early adolescents as contrasted with its elders. These details become parts of a culturally safe methodological approach, building trust and rapport with the intended participants.

Conclusion

Cultural competence is often gained through courses where people, including sport and exercise psychologists, might learn about, for example, Hispanic, Indo-American, or American Indian (indigenous) practices. Cultural competencies include general cultural practices within a larger cultural group. Cultural safety narrows cultural understanding to the local level, where the client is considered more closely in relation to his immediate context, peer affiliations, and familial socialization.

Robert J. Schinke

See also Identity; Participation Motives; Psychological Well-Being; Race; Racism/Whiteness; Self-Presentation; Sex Differences; Stereotyping, Cultural/Ethnic/Racial

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D

DECISION MAKING

Decision making (DM) is the cognitive operation of selecting a response from a range of available responses in circumstances where an action is needed. DM usually takes place while interacting with either the external environment or internal desires and requirements. Decisions may be made by an individual or a group, which mediates between the environment and the behavior or performance. DM can operate slowly under conditions that lack environmental constraints and demands but must be fast under circumstances of pressure, stress, or temporal constraints. When circumstances permit, the brain processes the information needed for DM intentionally through a perceptual–cognitive linkage.

The environmental information needed for a decision to be made is perceived by the senses, mainly the visual system in sport-related environment. The visual system enables capture of environmental stimuli and perceives patterns in the visual field, and feedforward of this information to long-term memory (LTM) allows anticipatory decisions to be made or a response decision to be primed. Such a process is slow, deliberate, and intentional. It operates under conscious control and can be modified and altered if time allows. In contrast, an emergent-type of DM is spontaneous, automatic, self-organized, and relies on established perception–action couplings. Such a DM triggers responses that are either retrievable automatically from LTM (in experts who

have accumulated many hours of practice and experience) or used randomly with the possibility of a high rate of error (in novices who have acquired limited hours of practice and situational exposure). Repeated exposure to similar situations, stimuli, tasks, and environments may turn DM operations from a deliberate–slow process into an automatic–fast process. Experience and expertise allow DM to shift from an intentional and deliberate mode into an automatic mode of operation when the environmental conditions and constraints necessitate this shift. The ability to shift between different attention and DM modes enables the cognitive system to operate efficiently by increasing the probability of making the right decision at the right time and avoiding errors, detrimental to performance.

Approaches to Decision Making

DM has been given much attention in the military, business, economic, gambling, and statistics domains. While the approaches related to the types of DM in each of these domains differ, each of these approaches has some relevance to sport. We describe each of these approaches briefly before examining the DM concept in the sport domain. The *prescriptive approach* to DM (the prescriptive theoretical normative model) views the person as a goal- or outcome-oriented creature attempting to maximize effort toward goal attainment. In gambling, where uncertainty is an inherent condition, probabilistic estimates are used to arrive at an optimal solution (e.g., the winning DM).

The *cognitive-oriented approach* to DM relies on the supposition that the person's cognitive capabilities are limited. However, repeated exposure and domain-specific experience circumvents these limitations by allowing the decision maker to efficiently capture visual and logical patterns, deliver them to LTM via the operation of long-term working memory (LTWM), and accordingly retrieve responses that are stored in a rich network of mental representations.

The *naturalistic-descriptive approach* to DM (NDA) consists of both rational and irrational processes, and incorporates personal values, morals, motivation, personal state, and emotions; all affect the personal DM process. NDM models, such as the *image theory*, *explanation-based theory*, *recognition-primed DM* (RPD), and *cue-retrieval of action* attempted to account for the DM behaviors. Heuristics (e.g., a method of solving a problem for which no formula exists, based on informal methods or experience, and employing a form of trial and error iteration) have been offered within the NDA to account for the underlying mechanisms of DM in the real world. The RPD postulates that DM consists of cue identification, situational goals, alternative action generations, and expectations for possible alterations; and all are affected by experience. The more complex the situation is, the more practice is needed for adjustment to take place. This concept is based on mental representations (knowledge structure) for guiding, monitoring, and executing the decision process. The NDA is therefore a knowledge-driven discourse, which consists of accumulating both the *declarative* and *procedural* neural circuits necessary for DM in situations that vary in complexity and certainty. The RPD within the NDA is an approach that influenced the current concepts of DM in the sport and exercise domain.

Decision Making in Sport

The approach to DM employed in almost all sport research has been heavily influenced by the RPD naturalistic concept, but modified for the unique environment of each sport. For example, DM in rifle shooting pertains to attending to internal bodily cues and pulling the trigger at the right time. DM in dynamic and fast sports, such as soccer, football, hockey, basketball, handball, volleyball, water polo, and racquetball, is dependent on

visual-spatial attention strategy (mainly visual), cue-priming, attention flexibility, selectivity, pattern recognition, anticipatory mechanisms, and the ability to assign probabilities to sequential events (to prime responses), all of which are governed by mental representations, the neural schemas containing declarative and procedural knowledge. Extensive exposure to the sport environment may result in the intentional conscious control of information processing and DM being replaced, at least in part, by more automatic control processes that allow the attention system to be more flexible and seek information from more than one source in parallel. Thus, the efficiency of the information processing system in making decisions is dependent on the richness and structure of the knowledge system (the mental representations network). The ability to encode information via the perceptual system, deliver it to the higher level processing system via LTWM, and process and retrieve responses are all a function of the extent to which the knowledge system is well developed and structured.

When an athlete chokes under pressure, a breakdown in the mental representation network occurs, and the perceptual-cognitive-motor linkage becomes dysfunctional. Specifically, under conditions of emotional or temporal pressure, the perceptual-cognitive system ceases to function appropriately and the probability of an erroneous decision being made increases substantially. Thus, coping with stress must be taken into account within the DM conceptualization.

Competitive sport events are laden with social and emotional stressors. Information processing under pressure may be affected in that attention is narrowed, which inhibits recognition and selection of essential environment cues. In turn, the cognitive system has limited resources to establish visual and meaningful patterns and prime a response. Instead, the cognitive system becomes overloaded with interfering thoughts, and attempts to control emotions are accompanied by declined self-efficacy. Under such stressful conditions, DM is expected to suffer because of the inability to prime and trigger the appropriate response, and this, in turn, is likely to result in performance decline. Existing evidence supports the notion that the quality of DM depends on how pressure is appraised and interpreted and what coping strategies and self-regulation are applied

by the athlete in an effort to maintain the operating efficiency of the perceptual–cognitive system.

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See also Anticipation; Attention Theory; Attention Training; Attentional Focus; Attention–Performance Relationships; Expertise; Information Processing

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DECISION-MAKING STYLES IN COACHING

A critical component of coaching is decision making, which is the process of selecting an alternative from among many choices to achieve a desired end. Decisions may involve the training programs; selection of team members; deployment of various strategies, practice, and tournament schedules; choice of uniforms; and such other serious and simple matters. In fact, every act of the coach dealing with the team or athletes is an instance of decision making with a profound impact on the team and the members. From this perspective, it can be confidently stated that successful coaching is in essence the art and science of decision making.

While this perspective is generally understood and agreed upon, there is considerable debate over the extent to which the coach should allow members to participate in decision making. This entry focuses on the specific issue of participation in decision making.

Decision-Making Processes

There are two processes in decision making. The cognitive process is focused on the rationality of the decision, which means selecting the best means to achieve a given end. Rationality is best assured when the decision maker defines the goal (or the problem) clearly, generates all possible alternatives to achieve the stated end, evaluates the alternatives, and chooses the best alternative. The social process refers to the extent to which members are allowed to participate in decision making. It must be stressed that the focus in the social process is not on the substance of the final decision but on who should be involved in making that decision.

Advantages and Disadvantages of Participative Decision Making

Coaches need to be aware of benefits inherent in member participation in decision making. First, there is more information and insight in a group than in an individual, which would result in the generation of more and meaningful alternatives and subsequent evaluation of them. Second, when the problem is explained and the solutions are discussed in a group setting, the members understand the problem and the rationale behind the choice of a solution. Third, members feel that it is *their* decision, and that sense of ownership would spur them to execute the decision more effectively. Finally, such participation contributes to the personal growth of the members, their feelings of self-worth and self-confidence and the development of their problem-solving skills.

Coaches must also be aware of the disadvantages of participative decision making before permitting the athletes to participate in decision making. Participative decision making takes a long time not only because of the complexity of the problem and the factors associated with it but also because members tend to engage in tangential discussions and to argue over trivial issues. A second issue is that the group may not be as effective as the best individual in the group in solving complex problems where a number of factors have to be kept in perspective and series of steps have to be taken to link all the factors involved and gain a gestalt, or overall, view of the issue. As Harold Kelley and John Thibaut have noted, a group may be more proficient than an individual in solving a crossword puzzle because the

members can suggest several words to fill up the blanks and each word can be checked against the criteria. Also, it is only necessary to be concerned with one word at a time. But construction of the puzzle involves looking at the whole set of words and associated criteria and ensuring they are all linked in a coherent and logical manner. In this case, the best individual in the group can do a better job than the group as a whole. An example of a complex problem from sport would be that of drawing up a set of plays in football. This would require that the relative abilities of team members and opponents, the sequence of events, and the various options and their consequences all be held in perspective. The coach with all relevant information is more likely to be better at this task than a group of players.

Further, the group needs to be integrated before they can effectively participate in making decisions. It must be recognized that every athlete is self-oriented and therefore would be competing with teammates for opportunities and rewards within the team context (e.g., playing time). In this scenario, it is conceivable that there could be internal rivalries and conflicts, which may not bode well for participative decision making.

It is not uncommon to hear the complaint that coaches in general are autocratic in decision making. In fact some coaches acknowledge that they are autocratic and that they need to be. In the movie *Remember the Titans*, Coach Herman Boone, referring to his football team, says, "This is not a democracy. It is a dictatorship. I am the law." In his view, coaching football is a situation that does not allow for much participative decision making. This perspective can be extended to specific problems faced by a coach in any sport. That is, some of the problems would call for participative decision making while some other problems may call for autocratic decisions. For example, there is no time for participative decision making during a timeout in basketball. Hence, the coach has to make autocratic decisions. In essence then, it is the nature of the problem to be solved that defines the extent to which the coach will allow members to influence the decision. Following the model proposed by Victor Vroom and associates, we carried out some research in the context of sport. Our research framework begins with describing the problem in terms of five attributes as follows:

1. *Quality Requirement*

The solutions to some problems have to be optimal while some other problems may not call for high-quality decisions. For example, selecting team members is more critical than the decision on the style of the uniforms.

2. *Coach Information*

Rationality of a decision is based on the quality and reliability of the information available. Thus, the extent of athlete participation in decision making is contingent on the coach not having the information and the athletes having it. If the players do not possess the information, engaging them in decision making would tantamount to "pooling of ignorance."

3. *Problem Complexity*

As noted earlier, when a problem is complex, the coach with the necessary information is more likely to make the optimal decision than the team as a whole. For example, the selection of plays for a football team should be based on the knowledge of the relative abilities of team members and opponents, the sequence of plays, and the various options and their consequences. Thus, the coach is likely to make more optimal decisions than the team.

4. *Group Acceptance*

Most of the decisions have to be implemented by the members of the team. Thus, it is necessary that the athletes accept a decision as optimal and executable. For instance, the strategy of a full court press in basketball should be perceived by the players as useful and well within their capacity. Such acceptance is easily generated when members participate in the decision to employ full court press as and when necessary.

5. *Team Integration*

This attribute refers to the quality of interpersonal relations and cohesion among the team members. Participative decision making would be fruitful only when the team is integrated. If it is not integrated, decision participation may weaken the already fragile team consensus and team spirit.

The configuration of these five attributes in a given problem situation would indicate the extent

to which the coach would allow the athletes to participate in decision making regarding that problem. The various levels of member participation (or decision styles, as we had labeled them) are described as follows.

Decision Styles

Autocratic I. The coach makes the decision using the information available at the time.

Autocratic II. The coach obtains the necessary information from relevant players and then decides. The coach may or may not tell the players what the problem is. The role played by the players is simply providing the information they have.

Consultative I. The coach consults with relevant players individually and then decides. Coach's decision may or may not reflect players' influence.

Consultative II. Coach consults with all players as a group and makes the decision, which may or may not reflect players' influence.

Group. Coach shares the problem with the players and the coach and the players jointly make the decision without the coach exercising his or her authority.

The results of two studies conducted by Pakianathan Chelladurai, Terry Haggerty, and Peter Baxter in 1989 and Chelladurai and Cheng Quek in 1995 are reported here. Our approach was to present to the respondents a set of cases representing all possible configurations of the presence or absence (or high or low) of the five problem attributes, which resulted in 32 cases ($2 \times 2 \times 2 \times 2 \times 2$). Following each case, the set of five decision styles described above was presented for the respondents to indicate the one style they would choose or prefer in that particular case.

The participants in the 1989 study were 22 coaches (males = 15; females = 7) and 99 players (males = 53; females = 46) from university basketball teams in Ontario, Canada. The coaches indicated their choice of a decision style in specific situations, and the athletes expressed their preferences of a particular decision style in the same situations. Although there were differences among the three groups (coaches, male, and female players), the AI style (coach making decision alone) was chosen more often than any other style by each group. The CII style (consultation with all players

on a group basis) was the second most popular choice in all three groups. While the participative style (G) was chosen less than 20% of the time in all three groups, the combination of AI, AII, and CI (because they involve minimal influence from the members) resulted in 64.8% for coaches, 59.4% for male players, and 57.3% for female players. The study in 1995 was concerned with the decision-style choices of 51 coaches of high school boys' basketball teams in and around Toronto, Canada. With these coaches, AI was the most preferred choice (32.5%) and CI style was the least preferred choice (9.7%).

The results of these two studies and earlier ones show that the respondents chose or preferred different decision styles according to situational characteristics. It is also noteworthy that players of both genders and coaches were influenced more or less to the same degree by the variations in the problem situation. These results support Vroom's assertion that we should view the situation as either democratic or autocratic rather than dubbing the leaders or coaches as democratic or autocratic.

Commentary

Based on the preferences of players for more autocratic decision making than participative decision making, the coaches should not be faulted for being autocratic. That is, the coaching context is relatively more autocratic than participative. However, coaches should be able to analyze a problem situation correctly and decide on the extent to allow participative decision making. Further, they should also realize that participative decisions require different skills than coaching per se. The coach should make the problem clear to the group, control the discussion to be focused on the problem than on tangential issues, and help the group reach a good decision, and in the whole process, should avoid influencing the decision in any way. Thus, the coach needs to consciously develop the skills to conduct participative decision making.

Pakianathan Chelladurai

See also Athlete Leadership; Autonomy-Supportive Coaching; Coach-Athlete Relations; Decision Making; Group Characteristics; Leadership in Sport: Trait Perspectives; Team Building; Team Communication

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DEVELOPMENTAL CONSIDERATIONS

Development refers to physical, cognitive, emotional, and psychological changes across the life span. Considering development within sport and exercise contexts allows for more realistic expectations regarding participants' attitudes, perceptions, affect, and behavior and helps us account for important differences in physical activity settings. Theories commonly emphasize that development is a result of the interaction between personal and contextual factors. While the nature–nurture debate in the past was considered winnable, presently there is strong consensus that both individual factors (e.g., biological, cognitive, psychological) and environmental factors (e.g., interpersonal interactions, social relationships, roles, cultural norms) together influence the growth and development of individuals. In this entry, discussion of developmental changes acknowledges that changes are organized in character, are based on the earlier self, and engender a sense of continuity alongside transformation.

Each theory of development provides a different lens or perspective for explaining the important aspects of development, providing additional insights into why we grow and change the way that we do. Theories differ regarding *when* development occurs. Some earlier theorists like Sigmund Freud and Jean Piaget regarded childhood and

adolescence as the primary time periods for growth, while others like Erik Erikson focus on change and development over the entire lifespan. Theories also differ regarding explanations of *how* we change and develop, with different perspectives emphasizing the driving force behind developmental change as either coming from outside or from within the individual. For example, learning theory (e.g., Albert Bandura) and behaviorism (e.g., B. F. Skinner, John B. Watson) assert that much of development is a result of the environments we are placed in, whether focusing on the behaviors that other people show or the reinforcement patterns they use. Other theorists suggested that characteristics contained within individuals are most important in influencing development, based on such biological factors as brain systems, neurotransmitters, or genes or personality traits. While research examining developmental differences in adulthood lags behind research examining development of youth, some of the most successful and well-accepted theories today embrace individual factors and contextual factors, as well as the interaction between the two, as important considerations to be made across the lifespan.

Development occurs within a context, but individuals are active in their own development and simultaneously act upon the context. For example, Uri Bronfenbrenner's bioecological model has provided a framework that has fostered a more ecologically valid (real world) consideration of development as the interrelated forces or systems (including the immediate social context and the cultural context) that influence and are influenced by the individual. A systems perspective integrates aspects of the individual as well as the context in explaining development, thus serving as a bridge in the nature versus nurture debate by acknowledging the mutually influential role of both. Other common theories in sport psychology (e.g., self-determination theory, achievement goal theory) similarly emphasize the importance of both individual factors like basic need satisfaction or goal orientations and contextual factors like autonomy-supportive behaviors or motivational climate as mutual predictors of behavior.

Development is both systematic by following general patterns within populations and idiographic by highlighting specific individual differences in pathways. Thus, variations in developmental trends will occur across individuals and across contexts. Below are several common

trends in development identified in the sport and exercise literature, including physical activity levels, reasons for participation, social development, motivational orientations, and self-perceptions. Developmental considerations regarding contextual factors are also addressed in the following sections.

Developmental Trends

Physical Activity Level

Participation in sports and any physical activity peaks in early adolescence and then steadily declines. This is problematic as physical activity represents an opportunity to curb the increasing disease rate and biological declines that accompany aging. Physical activity habits developed in youth tend to be predictive of future behaviors. Adults over 65 represent a growing population and are the most sedentary segment of the population, yet they have received relatively little attention in the sport and exercise literature. Physical activity behaviors associated with age are highly reflective of the life context and the associated norms, values, and beliefs for different age groups (e.g., socialization, education, culture, employment).

Reasons for Participation in Sport or Physical Activity

Children and adolescents most often cite enjoyment, affiliation, and competence development as primary reasons for engaging in sport. Although children typically move in and out of particular sports to sample different activities, they also cite reasons for discontinuing sport that include overemphasis on winning, issues with the coach, lack of playing time, and lack of enjoyment. As children enter adolescence, they may also discontinue sport because of increasing competition and team selection or cuts. In contrast to engaging in youth sports, youth who exercise provide reasons more closely aligned with adults. Thus, adolescents initiate exercise largely for external factors, including to lose weight, for health benefits, to satisfy other's expectations (e.g., doctor's orders), or to decrease stress.

As there are fewer opportunities for adults to reap the many benefits of physical activity through sport involvement, there is a greater emphasis on exercise participation in adulthood. Adults cite health and fitness as important reasons for

participation in physical activity. The majority of those who initiate exercise will discontinue in a relatively short amount of time. Those that do continue to exercise do so for more intrinsic reasons, such as enjoyment and developing value and identity for exercise. Competition and comparisons to others generally becomes less important with increasing age across adulthood, while self-referenced criteria are generally more consistent. Group involvement constitutes a motivating factor to both initiate and continue involvement. Adults who exercise in a group setting report more positive outcomes compared to those exercising alone. However, the majority of adults do not participate in regular exercise, demonstrating the difficulty of sustaining exercise involvement. Adults identify reasons for not exercising that reflect the demands and responsibility inherent in adulthood, including a lack of time, energy, knowledge, and value.

Although the level and intensity of physical activities generally decrease across adulthood, older adults can enjoy many physiological, psychological, and social benefits of being physically active. As physically active persons are less likely to experience cognitive decline and dementia later in life compared with sedentary persons, physical activity serves an added benefit for maintaining or reducing the risk of cognitive decline. Relevant concerns for this population tend to focus on functional movement and quality of life. Older adults' perceived ability to be active is as important as actual ability when predicting physical activity levels. Key physical activity motivators for older adults include enjoyment, social opportunities, maintaining or enhancing control over activities of daily living, and increasing self-confidence and perceptions of competence. Unfortunately, there are many barriers to participation that largely reflect lack of information and negative perceptions or attitudes toward physical activity for older adults.

Social and Emotional Development

A pervasive and persistent motivator across the lifespan is the social opportunities afforded by participation in sport, exercise, and physical activities. Initiating and maintaining social relationships and social connections constitute a core aspect of motivation and behavior in physical activity pursuits, yet affiliation alters in structure and function across the lifespan.

Childhood

The role of adults in the sport experience of children has garnered substantial attention. Parents largely direct their children's activities, exerting the primary social influence in early participation. Factors such as parent involvement, beliefs, and activity patterns are all influential on children's perceptions and behaviors regarding physical activity. However, other adult leaders like coaches and teachers also constitute important social influences. Important adults help shape the meaning of and understanding of emotions, such as pride, shame, and guilt.

Awareness of and reliance on peers increases through middle childhood and into adolescence. Young children view peers as others to share activities with, while peer group acceptance and belonging become important in middle childhood. Although the role of parents does not necessarily decline, there is a clear emergence of peers as an important source of social influence in middle childhood to early adolescence. Developing in early adolescence, interpersonal intimacy and close friendships are also important and serve an emotional function. This is increasingly important as the ability of emotions to be verbally acknowledged emerges.

Adolescence

Peer relationships, peer pressures, and general attention to one's social surroundings make adolescence a time of particular sensitivity toward peers. Peers therefore constitute a salient source of information about one's self (e.g., self-esteem, competence) and in developing attitudes, beliefs, and affective responses to sport and physical activities. Other markers of adolescent development are the formation of identity and of independence. In mid- to later adolescence, there is an increased awareness of and development of one's personal standards and expectations. Though social influences like parents and peers remain important, they are integrated with and accompanied by greater self-reflective ability and more internal judgments that reflect more abstract thoughts.

Adulthood

In the later adolescent years and into adulthood, romantic relationships are developed and constitute a primary social relationship.

Unfortunately, there is sparse information about romantic relationships in the sport and exercise literature. As salient social relationships during adulthood, romantic relationships and spouses are an important potential source of influence on sport and exercise experiences, for example serving as sources of social support for exercise. Additionally, their own children become another source of influence for adults. By middle adulthood, social networks become stronger and more stable, aiding in the increased ability to manage emotions and enhanced coping skills to deal with a variety of life changes.

Older Adulthood

Social relationships change in structure and function in older adulthood. Career, family responsibilities, and social circles decrease, while close relationships and social ties take on increased relevance. Having friends and social opportunities predicts mental health in older adulthood. Sport and exercise provide a venue for social interactions and an opportunity to make and maintain social relationships. As such, the sport and exercise context can provide an important source of social support for this population.

Motivational Orientations

Young children do not differentiate the concepts of ability and effort. Thus, children tend to view self-referenced ways of defining success such as trying hard and task completion as markers for high perceived ability. By adopting this mastery-oriented approach to their activities, children embrace effort and personal improvement as signs of success. However, as children age, they become more able to differentiate the concepts of ability and effort. Whereas previously, trying hard signaled being good at an activity, comparisons against others become more meaningful and trying hard can actually signal less perceived ability than others. By early adolescence, individuals are more fully able to adopt a performance-oriented approach that emphasizes comparisons against others as ways of defining success.

In addition, across youth, participating in sport for intrinsic reasons declines. Children's enjoyment of sport and physical activity declines, as does the focus on self-regulation. Participation for external reasons (e.g., feeling controlled by others, feeling guilty) increases across childhood and

early adolescence. These changes in motivation are linked to both contextual features as well as cognitive development.

Self-Perceptions

Research findings are inconsistent about changes in level of perceived physical competence across childhood and adolescence; however, accuracy of perceived physical competence increases across childhood and early adolescence. Young children tend to rely on important adults for competence information. In early adolescence, social comparisons are also used to form competence perceptions. Adolescents become increasingly able to use internal criteria to judge competence. The incorporation of more sources of information enables a more accurate determination of competence; as a result, perceived competence drops for some with age.

Perceptions of physical appearance are relevant from early adolescence across the lifespan. Early adolescents are keenly attuned to how others perceive them, and self-presentation concerns arise, such as social self-consciousness, sensitivity to early or late physical maturation rates, needs for social acceptance, and in particular body-image concerns. For example, girls cite reasons for not engaging in physical activities that often reflect being self-conscious of how the activity may negatively impact their appearance or social relationships. Self-presentation concerns, including body image, are associated with negative consequences, such as negative attitudes, low self-esteem, and avoidance of physical activity. Body-centered issues tend to persist across the lifespan, taking on different emphases at different life stages. There are changes in physical appearance across the lifespan, and although the importance of appearance tends to decline through adulthood, perceived physical appearance is a consistent predictor of self-esteem across the lifespan.

Contextual Factors

Youth Sports

Youth sport often aims to use a developmental model, treating the child as a whole person to be developed. However, the context of sport changes with age and level and generally shifts toward a more professional model emphasizing competition and outcome. When the motives of children do not match the experiences provided to them, such

as when a professional model is applied too early, there can be deleterious effects. As sport becomes more competitive and selective, there are not only changes to the social context but fewer opportunities for participation in organized sport. Because organized sport is a large source of physical activity for kids, as participation in organized sports declines, so do physical activity levels.

Motivational Climate or Coaching Style

Coaches are important influences in the youth sport context, as their philosophies and ways of responding to athletes influence youth athletes' psychological growth and development in sport. For example, coaches who create a mastery motivational climate provide goals and reinforcement to participants who improve over past performances, try hard, and work together. Coaches using a mastery motivational climate generally have athletes who show greater self-perceptions, more self-determined motivation, and general well-being. In contrast, coaches who use a performance motivational climate emphasize comparisons against others; youth in these contexts generally show less self-determined motivation, more anxiety, and lower physical activity.

Coaches who adopt a positive approach to providing feedback emphasize instruction, reinforcement, and encouragement toward their athletes. Utilizing the positive approach in a manner contingent with athletes' behavior predicts increased enjoyment, self-esteem, liking of coach and teammates, more prosocial attitudes and behaviors, increased likelihood of continued participation, and lower anxiety. The coach clearly provides an important influence on the development of attitudes, perceptions, and behavior during the childhood and adolescent years.

Physical Education

For youth, physical education (PE) is a physical activity context that is distinct from organized sport. Often compulsory in nature, PE includes a wider range of skill levels across youth participants. PE curricula traditionally focus on competitive, sport-based skills rather than on experiences that foster lifetime activities like fitness or recreation. This is problematic for youth who may have lower fitness, physical ability, or slower physical maturity. However, PE can serve as a context from which to develop adaptive beliefs and values that

can foster lifelong physical activity habits with the right focus. PE teachers create a leadership structure and set the tone of the environment. Teachers who provide a more supportive, cooperative environment foster more positive experiences within PE that also translate into positive outcomes beyond the PE environment.

Positive Youth Development

Organized sport is a popular activity for children and youth, in part because sport participation is expected to impart character. Sportsmanship, teamwork, leadership, and fair play represent common idealized outcomes of competitive sport participation. However, research has not confirmed these common beliefs. In fact, it appears that increased involvement in competitive sport may come at a price for some. Older athletes are more accepting of cheating than younger athletes, rule violations are more accepted as competitive level increases, and collegiate athletes use more aggression than youth. In addition, beginning around age 12, athletes use lower levels of moral reasoning in sport than in daily life. However, these outcomes only represent average experiences. There are many examples of positive outcomes in sport, but most are predicated on intentional efforts to foster youth development.

Positive youth development (PYD) programs promote personal and social assets of youth to enhance physical and mental health, foster social and psychological well-being, increase academic achievement, and develop character. PYD programs generally emphasize the need to create an engaging, positive context encouraging long-term commitment, positive relationships with caring adult mentors, and opportunities to develop life skills such as social skills and problem-solving skills and to transfer skills to new contexts. Sport and physical activity contexts provide an optimal vehicle for PYD, as they can address physical and mental health and foster social and psychological well-being. Sport-based PYD programs therefore tend to promote more growth and development than do many traditional competitive sport programs that focus on sport-specific skills and competition.

Adult Exercise Settings

Adult exercise settings vary widely. Factors that enhance the exercise experience tend to stem

from social and contextual factors. Social support and group cohesion foster positive norms, values, and overall motivation and behavioral adherence. Some exercise facilities cater to the needs of adults in order to minimize barriers to exercise. For example, it is common to provide child care; have extended hours of operation; provide multiple services in one location, such as a spa or restaurant; and offer a wide range of classes. Exercise facilities sometimes target certain groups, such as women-only gyms, in an effort to tailor the facility to the unique needs of certain segments of the adult population. Despite these efforts, exercise adherence remains a problem for most adults.

Person–Context Interactions

The trends noted earlier suggest that a sport or exercise context that emphasizes mastery, support, and developing assets will produce the most positive outcomes. However, there are certain individuals who will thrive in an environment that does not reflect these ideal conditions. For example, an adolescent who is competent, self-determined, and focused will likely thrive in a highly competitive environment (e.g., elite level competitive sports). Likewise, an adult may prefer to exercise alone despite the expectation that exercising in groups is preferable on average. Though the contextual features described earlier are generally advantageous, there is also a need to adapt contexts to specific individual or developmental needs of participants. While in general there are many changes in psychology-related variables across the lifespan, it is also clear that development needs to be viewed as an interaction between individuals in specific contexts.

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See also Character Development; Coach–Athlete Relations; Cognitive Capabilities; Ecological Theory; Friendships/Peer Relations; Motor Development; Quality of Life; Youth Sport, Participation Trends in

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DEVELOPMENTAL HISTORIES

In the context of sport, developmental histories provide information on an athlete's career with respect to practice activities related to their primary sport. In studies of elite athlete development, researchers have used carefully designed questionnaires or interview methods to ascertain information about the routes to success in sport. Methods for obtaining developmental histories vary with respect to the research question guiding their use and the breadth and depth of information required. Typically developmental histories provide insights into current and previous practice habits at various levels of detail.

Over the past 2 decades, developmental histories have become a commonly used method of ascertaining data on athletes' career histories. The interest in collecting developmental histories has stemmed largely from the seminal paper by K. Anders Ericsson, Ralf Krampe, and Clemens Tesch-Römer on the concept of *deliberate practice* and its centrality to expert performance. According to Ericsson et al., expert performance can be predicted based on the number of quality hours spent engaged in practice designed specifically to improve performance. There has therefore been a need to find methods for collecting reliable and valid data on the amount and quality of practice accrued over an athlete's career to test the validity of this proposition. Across a diverse range of sports, considerable support has been forthcoming. As a result of data collected from developmental histories, it has been shown that practice hours consistently differentiate across skill class (e.g., international versus national versus club level performers) and performance metrics (e.g., swimming or running times).

Recently there have been attempts to use developmental histories to understand how age of involvement in a primary sport impacts later success in the sport (either positively or negatively).

This has been studied with respect to ideas concerning early specialization versus early diverse involvement in many sports with an early focus on fun and enjoyment. The Developmental Model of Sport Participation, developed by Jean Côté, Jennifer Murphy-Mills, and Bruce Abernethy, has been used as a theoretical framework to study developmental pathways to success in sport. There has been mixed evidence in support of the suggestion that an early focus on play type of activities, potentially coupled with early sampling of various sports, positively impacts attainment of later success.

Data collected via developmental histories usually pertain to three general areas: *personal demographics* (including such variables as gender, age, education, family involvement in sport and living environment), *current practice profiles and levels of performance*, and *previous practice profiles and past successes*. Data are collected to inform as to the amount of hours per week spent practicing in the person's primary sport and sometimes in other sports or activities. This is usually evaluated with respect to the number of practice sessions per week, the number of weeks or months away from the sport (due to off-season or injury) as well as with respect to the current reasons for engagement in the activity (e.g., primarily for fun or for improvement). In addition, researchers have also tried to ascertain how much practice was supervised or coach-led like team practice versus self-directed or individual practice, as well as the extent of competition involvement. Once the researcher has a measure of the number of hours per week spent in practice, it is possible to estimate the number of hours accumulated in various types of practice activities across a 12-month period, that is, the number of hours per week multiplied by the number of weeks of participation (not including injury and off-season). Current levels of performance and areas of strength and weakness are then related to current amounts and types of practice.

Past levels of practice over an athlete's career are usually obtained through similar methods as for current practice but typically not in as much detail. Researchers usually solicit information about age of first involvement in the sport and age of serious commitment to the sport, and this information is either gathered for every year of involvement or in 2- to 3-year blocks (to make the recall process less arduous for the participant). In cases where data are not collected for every year of involvement,

estimates of practice in the intervening years are calculated based on the previous and subsequent year entries and on the assumption that practice gradually increased in a linear fashion.

Retrospective, recall-based methods of data collection, as is the case with developmental histories, require some demonstration of accuracy on the part of the researcher (validity and reliability checks). There have been a number of different methods used to ensure that the data are accurate. With respect to reliability, the same questions have been asked a number of times within a questionnaire or interview session, usually under slightly different guises: one question asking how many hours a week are spent in team practice versus a table requiring the person to specify the number of sessions per week they attend team practice and the average length of each session. Reliability is also gained through test–retest methods. Typically, 1 to 3 months after initial data collection, a select number of participants are asked to fill in all or parts of the questionnaire again. With respect to validity (or truthfulness) of the reports, attempts have been made to measure practice composition during actual practice sessions (e.g., via video), semistructured interviews have been coupled with questionnaire-based methods, and parents or coaches have been asked to report on current and past practice habits of the athletes.

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See also Expertise; Practice; Skill Acquisition; Talent Development; Youth Sport, Participation Trends in

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DIET DRUGS

The World Health Organization defines *obesity* and *overweight* as the excessive accumulation of body fat and warns that both conditions pose serious threats to health by increasing the risk for chronic diseases like diabetes, cancer, and cardiovascular diseases. Effective methods to counter obesity include changes in lifestyle, such as engagement in physical activity and exercise, as well as controlling one's weight through balanced dieting. Nevertheless, lifestyle modifications are not always easy to achieve. Therefore, scientists have called for effective antiobesity pharmacological treatments, or diet drugs. These drugs include both pharmacological and nonpharmacological agents that aid weight regulation by interfering with the processes of metabolism, reducing appetite, or influencing the absorption of calories or fat in the body. Pharmacological agents typically comprise substances that are purchased with medical prescription, whereas nonpharmacological agents refer to nonprescribed substances, such as dietary supplements and herbal products. While primarily intended to counter obesity, diet drugs are widely used by athletes and leisure-time exercisers to increase endurance, physical stamina, and leaner muscle mass. This entry presents an overview of commonly used diet drugs and addresses the potential role diet drugs play in weight control and athletic performance by focusing on their alleged health benefits and reported side effects.

Pharmacological Agents for Weight Loss

A wide range of diet drugs can be purchased in nutritional supplement stores, pharmacies, and through online retailers. However, regulatory authorities, such as the U.S. Food and Drug Agency (FDA), have only approved a handful of these substances. In professional sports, the World Anti-Doping Agency (WADA) and affiliated official sporting associations consider the use of diet drugs a doping practice.

Orlistat, *sibutramine*, and *rimonabant* are pharmacological agents used in dieting and weight-loss interventions. Orlistat is a gastric and pancreatic lipase inhibitor that aids weight

control by reducing the absorption of dietary fat. Sibutramine is a monoamine reuptake inhibitor initially developed to treat depression. Unlike orlistat, sibutramine stimulates thermogenesis (the production of *heat* energy in the body) and acts in the human brain to increase satiety, which is the feeling that one is full. Rimonabant is a selective cannabinoid receptor antagonist that was approved in 2006 in Europe as an aid to exercise and weight-loss interventions. Clinical trials have shown that both orlistat and sibutramine are effective in weight loss, and their effects are increased if followed by lifestyle modification, such as exercising and dietary changes. There is limited evidence about the effectiveness of rimonabant in weight-loss interventions. However, following several reports about their side effects (e.g., increased cardiovascular risk, suicide ideation), sibutramine and rimonabant were suspended and withdrawn from the markets. Orlistat is the only currently licensed drug for dieting and weight-loss interventions.

Nonpharmacological Diet Drugs

Nonpharmacological diet drugs include herbal supplements with weight-loss properties and typically serve two purposes. They provide the body with nutrients that are scarce in low-caloric diets and stimulate weight loss. These supplements can be purchased over the counter and usually come in formulas or so-called proprietary blends containing vitamins, botanical derivatives (e.g., green tea, açai berry, guarana), caffeine, or even aspirin. *Ephedrine* is a well-known herbal diet drug and stimulant derived from the Chinese plant *Ephedra sinica* and was initially used for the development of amphetamine drugs. Several nutritional supplements contain ephedrine alkaloids. These supplements are assumed to counter obesity and increase athletic performance by stimulating thermogenesis. A meta-analytic study found that ephedrine use is associated with modest short-term weight loss, but there is no sufficient data to support the long-term effects of ephedrine on long-term weight loss and athletic performance. Most importantly, the concurrent use of ephedrine with others stimulants, such as caffeine, is associated with increased heart palpitations and psychiatric, autonomic, and gastrointestinal symptoms. Ephedrine was banned by the U.S. FDA in 2004 in response to consumer reports of adverse health effects, and is included

in the 2012 list of prohibited substances issued by WADA.

Ephedra-Free Diet Drugs

Following the ban on ephedra-based products, the nutritional supplements industry introduced alternative ephedra-free diet drugs. These supplements come in formulas or so-called proprietary blends. Bitter orange or *Citrus aurantium* is assumed to be an effective and safe alternative to ephedrine, and is a rich source of adrenergic amines *synephrine* and *octopamine*. Synephrine and octopamine are similar to norepinephrine, and their thermogenic effects are heavily advertised. Although related research is still growing, the safety of octopamine and synephrine was challenged by evidence showing that their effects on cardiac function are similar to that of ephedra-based supplements. While synephrine is not considered a doping substance, octopamine is included in WADA's 2012 list of prohibited substances.

Conclusion

Dieting and weight management can effectively counter obesity. However, only a few pharmacological agents have been licensed for this purpose. Over-the-counter pharmacological and herbal-based dietary supplements with alleged health benefits and weight-control properties are easily accessible for individuals suffering from obesity, athletes, or leisure-time exercisers. Nevertheless, the safety of such products has either been challenged or still remains to be determined by empirical evidence. The development of regulation policies for diet drugs in professional sports is an ongoing and dynamic process.

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See also Body Dissatisfaction; Obesity; Performance-Enhancing Drugs

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DISABILITY

What is disability? There are many ways to answer this question. But how sport and exercise psychologists define disability has profound implications. Either intentionally or unintentionally, their perspective of disability will shape how research is carried out, what is deemed valuable in applied practice, who is considered the expert, and what is to be done to enhance the lives of people with disabilities. Thus, sport and exercise psychologists cannot ignore this question. Within sport and exercise psychology there are several ways to understand what constitutes disability; this entry discusses four of them.

The Medical Model and the Social Model

Two popular models for understanding disability are the medical model and social model. The *medical model*, or what is sometimes referred to as the individual model of disability, is based on decades of Western thinking that defines disability

as largely an individual deficit. Documented in 1980 by the World Health Organization, *disability* was defined as impairment, the loss or abnormality of psychological, anatomical, or physiological function. In this sense, disability was defined as any restriction or lack of ability (resulting from an impairment) to perform an activity in the way or within the range considered normal for a person. Therefore, disability is *caused* by parts of the body that do not work *properly*. The medical model shaped much early social and welfare policy and furthermore has, deliberately and by default, informed a great deal of sport and exercise psychology work on disability. Consider the example of spinal cord injury (SCI) and athletic identity. In such a study, participants with SCI would be labeled as having a physical disability and the dominant focus of identity and sport participation questions would be on the individual; such a project would be particularly amenable to survey research. Disability, as defined by the SCI, is the same for all participants. It is a medical problem or matter.

In recent years, the medical model has been widely criticized. Critics argue that it relies exclusively on individualistic medical definitions and biophysical assumptions of *normality*. However, normality is a highly contentious concept; normality is influenced by various cultural and historical forces that, in turn, mean what is normal in one cultural or context might be defined as not normal in another. Moreover, the medical model paints an overly negative and tragic image of people with disabilities. Individuals are depicted as defective and therefore disability is seen as a personal tragedy that should be overcome. As a result, individuals with disabilities may be pitied, stigmatized, and subject to a number of negative consequences. In relation to the previous two criticisms, the medical model has been further criticized for locating *solutions* to the *problem* of disability within the individual. By consequence, the individual's impairment needs to be cured or dealt with by health and medical professionals. Disabled bodies, rather than society, are seen as the site where interventions should take place. The focus on the normalization of the body and compliance with medical standards creates a hierarchy of power in which individuals with disabilities lose autonomy over their bodies to medical experts; failure to follow medical advice identifies an individual as deviant, potentially leading to stigma and other

negative outcomes. As a result, power and control is placed in the “expert hands” of the medical profession as opposed to people with disabilities themselves.

In the 1960s, the disability movement began to challenge the assumptions of disability put forth by the medical model; this challenge was fuelled by a new understanding of disability. The *social model* posits that disability is the result of sociostructural barriers that serve to exclude and restrict people with impairments. For example, in this model, inaccessible sporting facilities or negative attitudes from coaches produce disability as opposed to individual impairments. The social model contains several key elements. It claims that social structures and attitudes exclude people with disabilities from participation in certain activities, such as sport. Furthermore, the social model asserts that people with disabilities are an oppressed social group. It also distinguishes between the impairments that people have and the oppression that they experience within society. Therefore, the social model severs any causal link between impairment and disability. Disability is reconceptualized as having nothing to do with impairment and the body; rather, it is a social construction and a restriction of activity. Consider again the example of SCI and athletic identity. When framed from the social perspective, researchers might examine how individuals with SCI are either denied access to sporting opportunities based on physical impairments and thus cannot develop an athletic identity based on these oppressive, sociostructural barriers to sport.

The social model has had an important impact on the lives of people with disabilities, including elite athletes and people interested in engaging in leisure time physical activity. First, it enabled the identification of a political strategy to remove barriers and oppression. If people with impairments are *disabled* by society, as the social model proposed, then the priority is not to pursue a strategy of medical cure or psychological rehabilitation. Rather, it requires the removal of disabling barriers in order to promote the inclusion of people with impairments. This model has been instrumental in shaping antidiscrimination law. Indeed, the social model was a catalyst for the numerous legislative measures and policy initiatives to address the oppressive environments encountered by many persons with disabilities. For example, by law, it has meant that people with disabilities

should be able to access gyms, sport clubs, sporting stadiums, and so on.

The second impact of the social model was on people with disabilities themselves. Offering an alternative to the medical model of disability, the social model was and remains very liberating for many individuals with disabilities—it offers them a very different narrative of disability that enabled them to understand that they themselves were not at fault—society was. They did not need to change; society needed to change. They did not have to identify their impairments as tragic; rather, they could accept their body and lives as they chose. In such ways, the social model is a source of empowerment and a way to restore the autonomy of body and choice for people with impairments. The social model as a counternarrative has enabled people with hearing impairments to positively think of themselves as communicating in a different language compared to individuals without hearing impairments. As a result, they view themselves in affirmative ways and do not consider themselves as having a disability.

Critique of the Social Model

Despite the important impact the social model has had, recently this way of understanding disability has also been critiqued. First, it has been criticized as ignoring the cultural and experiential dimensions of disability. For example, with the emphasis on the removal of social barriers, matters like athletic identity, emotional regulation, stories of mental coping strategies, or psychological well-being are passed over.

Second, it is argued that the idea of a barrier-free world is constrained by the natural environment, is at times economically impractical and is not possible for all people with disability. Moreover, by accommodating some impairments, barriers may then be put in place for other bodies. For example, wheelchair users can find curb cuts useful to enable their movement through the built environment. However, blind people might find that the same curb cuts make it difficult for them to differentiate pavement from road, and leave them walking into the path of a vehicle. Wheeling their way to play in a local tennis tournament, wheelchair users might have difficulties with tactile paving, which gives locational cues to people with visual impairments.

Third, in separating impairment from disability, critics have been keen to stress that the social

model overlooks the importance of the body and impairments in people's lives. Indeed, impairment is not simply biological but profoundly psychological and social too. Consider again the example of SCI. By providing ramps, accessible washrooms, adapted sport and so forth, social barriers are removed and thus, in theory, so should the experience of *disability*. Yet after injury, bodily changes can lead to a host of new issues previously unknown to the individual, including the experience of pain, chronic health conditions, a *new* body and a new method of mobility. Thus to reduce disability to solely sociostructural barriers is overly simplistic. In light of these critiques, other models of disability have very recently emerged but have not yet garnered much attention within the sport and exercise psychology literature.

The Social Relational Model

One such model is an extension of the social model: the *social relational model*. It defines *disability* as a form of social oppression that involves the social imposition of restrictions of activity on people with impairments *as well as* the socially engendered undermining of their psychoemotional well-being. Like the original social model, the social relational model takes account of the sociostructural barriers and restrictions that exclude and oppress persons with disabilities. However, the social relational model also accounts for the social processes and practices that place limits on the psychoemotional well-being of people with impairments. In addition, the social relational model underscores the importance of impairment. It sees impairment and disability as linked and interactive. For example, an able-bodied person may tell a spinal-injured wheelchair athlete that they cannot be a coach at their tennis club *because* they are impaired: The athlete cannot run around the court or stand to serve, so they should not be a tennis coach. In this case, damage may occur to the psychoemotional well-being, and concomitantly the identity as an athlete or coach, of the person with disabilities. The damage is not solely the result of structural barriers. Rather, any damage caused, and the oppression and restrictions of activity that go with this, is an effect of impairment that operates in and through social interaction. Accordingly, the social relational model brings the body and impairment into focus, recognizing the impaired body as a biological, experienced, and psychosocial entity while

maintaining that people with impairments can still be oppressed. It is an approach to understanding disability that keeps social oppression at the heart of matters and yet considers disability in a far more layered, complex, embodied way than was detailed in the social model. Consider once again the example of SCI and athletic identity. Research from the social relational model of disability could explore how participation in adapted sport at the elite level reduces felt stigma and embodied concerns in certain contexts and therefore increases the individual's perceived quality of life. Or, in relation to impression management, it might examine able-bodied people's attitudes toward both people with disabilities who are physically active and people with disabilities who are not.

The Interactional Model

In the *interactional model*, disability and how it is experienced is understood as a consequence of the complex relationship between factors intrinsic to individuals, and extrinsic factors arising from the wider context in which these individuals find themselves. Intrinsic factors include the nature and severity of individuals' impairments, their own attitudes to these impairments, their personal qualities and abilities, and personality. By contrast, extrinsic or contextual factors include the attitudes and reactions of other people; social support systems; the extent to which the environment is enabling or disabling; and wider social, cultural, and economic issues pertinent to disability in society. The interactional model is similar to the social relational model such that impaired bodies are brought back into the understanding of disability. Both models also consider the relational aspect of disability. Therefore, as with the social relational model, a researcher interested in SCI and athletic identity would explore how interactions and impairment intersect to create an identity. In particular, a project framed from this perspective could examine how interactions in specific contexts (like dialogue or conversations in the locker room) either produce or detract from certain identities, such as that of the athlete and that of the person with disabilities.

Relational and Interactional Models Compared

Although similar, there are differences between these two models. First, *relational* in the context of the interactional model refers to the relationship

between intrinsic factors and extrinsic factors that produces *disability*. By contrast, *relational* in the context of the social relational model refers to the relationship of those socially constructed as problematically different, or disabled, as the result of bodily or cognitive variations from the normal and cultural criteria for normal. Furthermore in the interactional model, rather than reserving the word *disability* for impairment effects, oppression, or barriers, the term is used more broadly to describe the whole interplay of different factors that make up the experience of people with impairments. By contrast, impairment is a necessary though not sufficient element in a disability relationship within the interactional model. It is always the combination of a certain set of mental or physical attributes, in a particular environment, within a specified relationship, played out in broader cultural, historical, and political context that, when combined with impairment, create the experience for any individual. Whereas the social relational model places emphasis on oppression and doing things to challenge or eradicate it, the interactional model would suggest there are many things that could be addressed to improve quality of life. This might include physical activity to improve self-esteem, exercise to enhance body-self compassion, or sport to create meaningful friendships and opportunities for travel.

Narrative Inquiry and Narrative Analysis

Understandings of disability are shaped by the model a sport and exercise psychologist adopts. But in addition, the type of theoretical approach adopted also underpins and informs work within sport and exercise psychology on disability. One approach that is garnering interest within sport and exercise psychology disability research is *narrative inquiry*. The core premise of this approach is that a person is essentially a storytelling animal; stories structure our experiences. We make meaning or sense of things through storytelling, and stories act on and in us, often working to shape and inform human behavior. Such a narrative approach is, therefore, of some relevance to sport and exercise psychologists who as professionals are in the business of dealing with *experience*, *meaning*, and *human behavior*. In both applied practice and for research purposes, we often ask athletes with disabilities to share with us their personal accounts of key moments or phases in their career.

In so doing, we are inviting stories. These stories are not passive or uninformative. They are needed in order for people to represent experiences that remain inchoate until they can be given a narrative form. Stories not only offer and impose form to experience, they express experiences. They are one of the most powerful means we have for communicating to others events that have happened, along with our emotions, attitudes, beliefs, and identities. When people tell stories, these stories have the capacity to affect what we do or do not do, shaping who we are and might be. As such, there is much to be gained from inviting stories and then analyzing them.

Narrative analysis is an umbrella term for an eclectic mix of methods for making sense of, interpreting, and representing data that take the form of a story. It takes stories or storytelling as its primary source of data and examines the content, structure, performance, or context of the story or storytelling as a whole. The analytical interest is not simply on what is said in a story in terms of content. The language and telling itself is also examined along with the environments that give shape to narrative content, structure, and performance. That is, in a narrative analysis the interest moves between *what* is being said, *how*, and *why* a person or group tells and performs the story as they do, in certain places and under specific conditions. For example, the narrative analyst is interested in how a story is put together to convey meaning, namely, to make particular points to an audience. For whom was this story constructed and for what purpose? What particular capacities of a story does the storyteller seek to utilize? Why is the sequence of events structured that way and not another? What narrative resources from the cultural menu does the storyteller draw on, take for granted, or ignore? Where do these resources derive from and under what circumstances and conditions? Are there gaps and inconsistencies in storytelling that might suggest preferred, alternative, or counternarratives? What does the story say and do on, for, and with people? How do listeners or readers respond to a story, with what affects, and on whom?

A narrative analysis of disability and sport might therefore highlight performance stories in which winning at all costs is the dominant theme. It could illuminate stories of anxiety and choking or stories of moments when everything comes together and the athlete experiences the sensation

of flow. An analysis of the stories of athletes with disabilities who have just retired could reveal that the type of story several of them get caught up in is one that structures and shapes their retirement experiences as meaningless and devoid of purpose. The story is one that enacts a past full of glory and excitement, but now a present that is empty and a future that is perceived as desolate. Such a story, therefore, can be seen as acting on these athletes in dangerous and negative ways. Alternatively, a narrative analysis might reveal a type of story that following retirement calls on metaphors associated with a journey of self-discovery, notions of being changed for the better following retiring, and time tenses that link the person to living fully and happily in the immediate present. In this story, resilience is a dominant theme. It is used as a resource that works for the athletes to positively adapt to retirement and the adversity that can ensue. In such ways, therefore, sports and exercise psychologists can generate a compelling account of how stories affect human lives and put in place practical resources for people with disabilities to live differently.

Conclusion

As we have seen, disability is a multidimensional construct that can be modeled in several different ways. Moreover, these models lead to different ways of operationalizing disability and thus will have important methodological and practical implications for sport and exercise psychologists. When considering the literature and prior to beginning a research project, it is essential for researchers to reflect on which model of disability frames their understanding, given the impact this understanding will have on the ensuing research approach and method. To develop humanistic, complex, and rich understandings of the lives of people with disabilities, researchers might also consider using narrative inquiry. Stories, after all, affect human lives.

Brett Smith and Marie-Josée Perrier

See also Disability and Exercise; Disability and Sport; Disability Coaching

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DISABILITY AND EXERCISE

Appropriate physical activity engagement promotes a host of psychosocial benefits. These benefits are especially valuable for individuals with disabilities because they have high rates of overweight and obesity. Additionally, because people with disabilities often have associated secondary

conditions (e.g., pressure sores, diabetes) physical activity is particularly important as a mechanism to prevent or attenuate secondary conditions. Unfortunately, most individuals with disabilities are sedentary or get limited physical activity. Low levels of physical activity among people with disabilities have been documented in North America, Europe, Asia, and Africa. Minimal physical activity is evident irrespective of age, gender, the setting (physical education vs. leisure), disability type (spinal cord injury vs. visual impairment), or assessment method (pedometers vs. self-report). Older females with severe disabilities from low socioeconomic-status groups are most at risk for low physical activity engagement. In this entry, the various psychosocial benefits of physical activity are highlighted, followed by a discussion of the barriers to physical activity.

Benefits

Increases in strength and endurance typically contribute to enhanced perceptions of competence. Mastery experiences and social influences both serve to contribute to enhanced competence perceptions. Competence benefits have been derived from activities as varied as lifestyle leisure activity, martial arts, youth sport, fitness training, and horseback riding. Most people also derive social benefits such as self-esteem enhancement stemming from being physically active with friends. Children in adapted sport programs find that they can more easily be themselves. Participation in physical activity also leads to increased social integration and reduces feelings of loneliness and isolation.

However, being physically active does not automatically confer social benefits upon participants. For example, it is not unusual for children to be excluded from physical education classes, ignored when sport teams are selected, and teased at recess, for looking different. Adults with disabilities (e.g., spina bifida) participating in inclusive recreation programs report both positive and negative reactions and experiences. In one study, participants with disabilities reported having difficulty ascertaining whether able-bodied participants disliked them or were simply ambivalent toward them. Other times, adult participants perceived that able-bodied exercisers treated them like children. Despite these reports, physical activity involvement helps minimize the negative influence of a

disability on self-perceptions. A participant in one study reported that she felt “normal” when being physically active. Other individuals with disabilities have experienced less stigmatization which they attributed to their physical activity involvement. Researchers have also found that able-bodied people tend to evaluate people with physical disabilities more positively if they perceive them as being physically active. Two other benefits expressed by people with disabilities are greater feelings of independence from being physically active, and more opportunities to travel away from home on a sport team.

In addition to competence and social benefits, engaging in physical activity is simply enjoyable, and such experiences are valuable for quality-of-life reasons. In addition to the momentary value of increased positive affect, physical activity can also help in mood management when individuals do not have good days. For example, individuals with spinal cord injury (SCI) reported increased positive affect and decreased negative affect after exercising irrespective of whether participants had positive or negative life events that day. There is also preliminary evidence that individuals with neurotic tendencies benefit the most from the mood enhancing benefits of physical activity. In a recent study, virtually half the respondents noted enhanced emotional functioning that they viewed as stemming from their physical activity involvement. Many male adults with SCI have also experienced reduced stress, depression, pain, increased life satisfaction, and subjective well-being as a result of exercise. In brief, much research has supported physical activity as an important vehicle for quality-of-life enhancement by reducing negative emotional states and increasing positive affective states.

A growing body of research has also documented the relational benefits experienced by families who are physically active. Researchers examining family recreation and leisure experiences show that such involvement enhances family quality of life. A small body of research pertained to the family as a unit and to how physical activity experiences impact children, siblings, and parents. Adolescent participants were asked to describe how their adaptive skiing or horseback riding experiences influenced the quality of their family life. Virtually 70% agreed or strongly agreed that their experiences in these two activities enhanced their family life. Additionally, almost 80% of the

participants agreed, or strongly agreed, that skiing or riding with family members contributed to the meaning of the activity. This finding has also been supported by scientists who have found that parents, particularly mothers, in families that have children with disabilities often engaged in activities like bike riding with their children in order to enhance family relationships.

Unique experiences, such as outdoor skill training programs and outdoor adventure trips including both parents and children with disabilities, are promising. Parents, for instance, have reported that their experience helped them overcome or eliminate constraints to being physically active. Both qualitative and quantitative results indicated that parents believed the experience enhanced family interactions, and promoted greater cohesion. Swedish families involved in orienteering, golf, and archery found that their children believed their participation helped the family experience “a feeling of togetherness.” In brief, physical activity-oriented experiences are vehicles that enhance the psychosocial functioning of families.

Parents also benefit from their children’s involvement in physical activities. Fifteen families with children with disabilities, who participated in adapted baseball, found parents derived mutual support from each. In particular, the shared social reality nature of the support was particularly meaningful. In summary, physical activity engagement results in a plethora of benefits to participants although sometimes participants report negative experiences. Individuals participating with or supporting the physical activity experiences of people with disabilities can detract from or enhance the physical activity experience.

Barriers

The benefits of being physically active are often difficult to obtain because of a multitude of barriers to physical activity. Individual based barriers include disability type. For instance, leg amputees cannot of course run or walk, although they can use crutches and wheels. Pain or discomfort is also a major barrier to being physically active. People with cerebral palsy (CP) and SCI list pain as frequently interrupting and preventing physical activity engagement. The physical discomfort of exercise is also viewed as a barrier by some individuals. In addition to pain, people with disabilities also note that fatigue, a lack of energy, disease,

injury, and poor health prevent them from being physically active.

Not knowing where exercise facilities are located or how to start and develop an exercise program also inhibits physical activity. A lack of time for physical activity is also reported by adults because of work and family responsibilities. Limited financial resources reduce or prevent travelling to exercise clubs, buying exercise equipment for the home, or purchasing exercise club memberships. A fear of developing tight muscles and joints from exercising has also been reported. Finally, inner city residents are often afraid to leave their homes for fear of crime or fear of falling.

Barriers that are more social in nature are also common. For instance, most children need their parents to transport, pay for, and facilitate their physical activity involvement. At the same time, parents are often fearful that their children might get teased by their peers for looking different. Parents also fear for their children’s physical health if they engage in sports viewed as dangerous by their parents (e.g., basketball for a blind child). Parents, in turn, criticize community recreation personnel for not knowing enough about adapted sport programming principles and various disability conditions. Physical education teachers are also seen as lacking appropriate training in disability conditions and adapted physical education. Physical education teachers themselves confirm this view by indicating that they often lack adequate professional preparation in their teacher education programs. It is not unusual for physical education to be replaced by therapy or mobility training or canceled because of blanket medical excuses from doctors. Caregivers of adults with CP have asserted that they did not believe an exercise program was of value for their residents, and about a third of them believed exercise would not help their client’s CP. Some caregivers even believed, contrary to research, that exercise would worsen CP.

In addition to individual and social barriers, the environment can also hamper physical activity engagement. For instance, children with disabilities often note that there are very few places to be active and rarely are facilities conveniently located. As children get older, their limited physical activity opportunities tend to diminish. Opportunities can also be illusionary, as wheelchair basketball league officials sometimes ban motorized wheelchairs and swimming pools are viewed as too cold by some

swimmers with disabilities. Built environmental barriers, such as a lack of a curb cut or crosswalks without auditory signals, can prevent individuals with disabilities from crossing the street. Ramps that are ostensibly designed to facilitate access are often built too steep and cannot be used.

Barriers are also subtle. For example, although wheelchairs help people with disabilities to move, many individuals with SCI view their wheelchair as the number one barrier to physical activity. In one study, participants rated their wheelchairs as bigger barriers to physical activity than their disability. Most medical personnel and clinicians have only limited training in prescribing wheelchairs, which might explain why some individuals with SCI find manual wheelchairs uncomfortable, too wide and heavy, and therefore hard to move.

Researchers such as James Rimmer, Barth Riley, Edward Wang, Amy Rauworth, and Janine Jurkowski have examined the accessibility of health clubs and found that many (over 50%) clubs do not have curb cuts for easy access or clear paths to lockers. Most weight rooms do not have enough room for exercisers to move their wheelchairs around the exercise floor. Inadequately lighted biking and running paths and wooded walking trails with exposed tree roots can be barriers to individuals with vision loss.

In summary, physical activities ranging from formal exercise and sport opportunities to recreation and leisure lifestyle activities can all provide a wide ranging set of psychosocial benefits to participants. However, many physical activity barriers organized across individual, social, and environmental categories limit how physically active people with disabilities are, and thus, the benefits they accrue.

Jeffrey J. Martin and Laurel Whalen

See also Adapted Physical Education; Disability; Disability and Sport; Disability Coaching

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DISABILITY AND SPORT

This entry presents an overview of the psychosocial research conducted at both ends of the spectrum of sport engagement. First, we examine the psychosocial benefits of recreational youth sport involvement for adolescents with disabilities. The next section discusses the psychological challenges and mental preparation at the highest levels of sport involvement for athletes with disabilities: the Paralympics.

Youth Sport Participation

This section will report on two areas within the area of youth sport. First, the research on common sport self-perceptions, such as physical self-concept, thought to be influenced by sport, is examined. Second, the sport as a setting where children can develop social connectedness and reduce loneliness is discussed. It is commonly thought that individuals with disabilities may experience low self-esteem because their disability may limit their ability to experience sport success. At the same time, sport can be a vehicle that increases competence if children participate and improve their abilities. Unfortunately, researchers have shown that poorly coordinated children are less active at play, spending more time watching other children play than better coordinated children do, thus limiting their opportunities to improve. Additionally, researchers have shown that children with movement difficulties may fail more in sport relative to children without movement difficulties and as a

result have lower perceptions of athletic competence compared to children without developmental coordination disorder (DCD).

In addition to reduced feelings of pride and satisfaction from limited mastery experiences, significant others may withhold positive feedback and be critical. Children with DCD have reported feeling anxious and humiliated because they were teased. Although accomplished youth athletes with disabilities view themselves as serious athletes, they often perceive that others do not.

In contrast to the above findings, other researchers examining the self-esteem of children with disabilities have found self-esteem scores comparable to that of able-bodied athletes. In a rare intervention study, children participating in a 4-week wheelchair tennis program enhanced their general perceived and tennis competence but did not increase overall self-esteem. In a second intervention study, the impact of a 12-week sport intervention on motor skill development was examined. Students with and without disabilities were randomized into intervention and comparison groups and the intervention group participants showed increases in object control and locomotor skills compared to the comparison group. Other researchers examining both wheelchair sport and wheelchair dance have reported favorable increases in self-perceptions ranging from global self-esteem to self-confidence and control.

The motivational climate that youth sport athletes perceive can also have an important influence on perceived competence. For example, children with disabilities who perceived a strong mastery climate in physical education had higher levels of competence compared to children who viewed the climate as less mastery oriented. Social support by significant others is also important as adolescent athletes' self-efficacy is positively related to three different types of social support: listening, emotional challenge, and technical challenge support provided by friends, parents, and coaches. These findings suggest that sport self-efficacy may partly be the result of encouragement and social support coming from diverse relationships.

The social nature of sport makes it a potentially important vehicle for psychosocial development. For example, adolescents perceive that their involvement in sport increased the favorable impressions able-bodied others had of them. Given that children with disabilities have fewer friends and are lonelier than children without disabilities,

sport offers a tremendous opportunity to promote social connectedness while simultaneously enhancing health outcomes. Jeffrey Martin and colleagues have found that youth athletes with disabilities derived a variety of social benefits from having a best friend on their sport team. Gender differences were apparent with girls, compared to boys, indicating that their best friend looked out more for them, they had more common interests, stuck up for each other more, and were more encouraging after mistakes. Girls involved in formal disability sport receive more social support for their sport engagement compared to girls participating in more informal sport. Girls engaged in more serious high-level formal sport had role models who provided shared social reality support, emotional support and challenge, and technical support and challenge.

Parental encouragement has also been related to physical ability perceptions and sport commitment among international level adolescent athletes with disabilities. Athletes reporting more encouragement from their parents reported stronger perceived physical ability and a greater commitment to sport, compared with athletes reporting less parental encouragement. In contrast, children in a therapeutic martial arts program received lots of parental encouragement, but it was unrelated to their perceived physical ability. Because mean levels of support were high, the researchers interpreted their findings positively as parents did not limit their support to higher skilled children and ignore less-skilled children.

Youth with disabilities who played in an after-school disability sport program mentioned the importance of connecting with other youth who had disabilities. This sense of connectedness helped them feel more authentic. The researchers conducting the study speculated that the specific adapted nature of the sport program versus a nonadapted sport program was instrumental in creating a climate that allowed the participants to be personally expressive. Other sport psychology researchers have reported on enhanced social integration during sport because of increased opportunities for children to interact with their teammates. A benefit of this increased interaction is greater social bonding and a broadening of children's social networks.

Interviewers of 20 Swedish children participating in disability sports programs have also concluded that sport provides a means for the development of new friendships. The 36 adolescents participating

in an adapted sport program who had the highest positive affect also reported the strongest peer relationships. Thus, a potential outcome of positive peer relations in a sport context is enhanced quality of life (positive mood states).

Some research suggests, however, that the experiences of children with disabilities in sport are sometimes less than positive. For instance, some individuals feel as if their physical abilities are denigrated and their sport participation trivialized. Even good intentions can go awry. For example, children with disabilities sometimes see offers of assistance in sport settings from able-bodied children as a negative assessment of their ability. Their self-esteem is threatened when they perceive that help is offered based on a negative assessment of their ability. Sport participants also report feelings of frustration and inadequacy when attempting to accomplish activities that they are inexperienced in or lack the skills to be successful in.

In summary, it should be clear that the sport setting is not inherently a context that promotes well-being among all participants all the time. As such, sport settings cannot be labeled as exclusively positive or negative. Adults, youth without disabilities, and youth with disabilities themselves, exert influence on the outcomes they experience in sport. The potential for positive experiences is greater when adults with emotional attachments (parents) or with sport expertise like coaches are actively engaged with the children in ways that are supportive and promote skill development. The potential for negative outcomes is reduced if peers are not allowed to engage in discriminatory and unkind behavior.

Elite Sport Participation

The Paralympics, like the Olympics, are considered the highest level of competition that an athlete with a disability can aspire to. The increased attention that the Paralympics get and the importance that governments, coaches, and support staff attach to Paralympic performance have the potential to make the Paralympic experience both exciting and stressful. Research with U.S. Paralympic coaches reveals that they believe athletes encounter pressure to perform well and win. Coaches also believe that the increased media and family attention increases stress.

Many researchers have substantiated the value of athlete's feeling confident, developing optimal

mood states, and employing psychological skills. Researchers examining elite amputee soccer players, Paralympic wheelchair basketball players, international-level wheelchair road racers, and swimmers have all supported the value of psychological skills. Well-developed psychological skills help athletes manage and cope with stress. Sources of stress among athletes with disabilities reflect common sport sources of worry (e.g., lack of fitness), sources of stress unique to disability sport (e.g., wheelchair or prosthetic concerns), disability (e.g., pressure sores), and the nature of the Paralympic experience. We next focus on the unique challenges presented by the Paralympics.

British elite Paralympians have reported that traveling to major international competitions is stressful and noted specific factors such as having to get onto the plane first and off last as adding to their stress. A sport psychologist working with one Paralympian reported that his client's biggest source of stress was managing the 26-hour flight to the Paralympics. Getting on and off the plane, transfers from seat to wheelchair to toilet and back as well as personal care in general during the flight were all worrisome. Skiers in the 2014 Winter Paralympics will compete 60 kilometers from the Paralympic village. As a result, athletes will have extensive travel arrangements or have to stay off site and miss part of the Paralympic experience. Para-equestrian dressage riders at the 2008 Beijing Paralympics competed off site in Hong Kong. All the riders reported that they missed the main Paralympic experience. Hence, it seems wise that Paralympians psychologically prepare for either extensive travel, depending on their event, or prepare to not spend all of their time in the Paralympic village. According to British Paralympians, being away from home for a long time period is stressful. Athletes miss their significant others and the social support they receive from them.

Paralympians spend a substantial amount of time in the Paralympic village. For athletes who compete at the end of the games, they must spend a critical amount of time in an unfamiliar setting and sleep in a bed they are not accustomed to. In a study of Brazilian Paralympic athletes participating in the Beijing Paralympics, a majority of them had poor sleep quality during the games. Anxiety was linked to poor sleep as 72% of the athletes exhibiting poor sleep quality had medium levels of anxiety, whereas only 28% of the athletes who had good sleep were anxious. British National

team members have also reported that they were concerned with whom they would share a room and how well they would be able to compete after a poor night of sleep.

Paralympic support staff need credentials to access the Paralympic village but these credentials are limited. As a result, compared to non-Paralympic competition, athletes may have less access to their sport psychologists and coaches. When available, coaches and support staff are often overworked and tired. The ability to see doctors, therapists, masseurs, and athletic trainers is similarly constrained.

Athletes with disabilities are functionally classified, which determines who they compete against. Being classified, which occurs prior to competition, can be stressful because athletes may anticipate being reclassified at a different level than they were previously. If athletes are reclassified, they may have to compete against better athletes. Clearly such a scenario can reduce confidence and increase anxiety at a crucial time.

Drug testing occurs at the Paralympics and may be a new, and therefore stressful, experience. Testing at the Paralympics can potentially be more involved and complicated compared with the testing done by able-bodied Olympians. For instance, many athletes urinate via catheters. Other Paralympians may have to file exemptions because of the medications they consume related to their disability. Some Paralympians endure chronic pain and may manage it with medical marijuana, which, although legal at home, is a banned substance at the Paralympics.

In summary, most elite athletes are affected by the events and conditions of their sport world. However, Paralympians have to deal with many unique conditions and challenges that can be barriers to optimal performance.

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See also Adapted Physical Education; Disability; Disability and Exercise; Disability Coaching

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DISABILITY COACHING

Optimal athletic development and sport success is almost always the product of multiple factors. Genetics, opportunity, effort, and consistent training over many years are critical. However, quality coaching is also recognized as an important influence on athletic success. This entry discusses the history and current status of coaching in disability sport, the importance of quality coaching from a psychological perspective, and unique disability sport coaching challenges.

Historically, many disability sport athletes have had to coach themselves. For example, in a survey conducted almost 15 years ago by Michael Ferrara and William E. Buckley, it was reported that only 58% of 319 elite adult athletes from the United States and only 33% of a diverse group of international athletes from Australia, Japan, and the Netherlands had coaches. Hence, it is understandable that many disability sport athletes have been known to overtrain, train inconsistently, train in nonsport-specific ways, fail to taper for major competitions, and fail to rest after major performance efforts. Consequently, wheelchair road-racing athletes often develop upper respiratory illnesses after marathons, and experience inadequate postrace rest, both of which exacerbate their difficulties. For athletes who self-coach, appropriate disability sport specific information is often hard to locate.

In recent years, athletes from wealthy countries have been supported by high-quality coaching.

Canadian Paralympic swimmers, for instance, have access to the same high-level coaches as able-bodied swimmers. However, Paralympians from other countries are deprived of professional coaching support. Many athletes are still self-coached or receive less than optimal coaching as coaches may not have the appropriate sport science or adapted physical activity education. However, the need for increased coaching quality has been recognized by national organizations. For instance, the American Association of Adapted Sports Programs offers an 8-hour coaching certification course that covers diverse topics such as sport psychology, physiology, management, and philosophy. It should be acknowledged that some athletes prefer not to have coaches.

Although the research on disability sport and coaching is limited, sufficient research allows sketching out a portrayal of the value of good coaching in disability sport. Disability sport athletes who view their coaches as supporting their autonomy express a strong sense of control over their sport involvement and have positive relationships with their teammates. Also, athletes who view their coaches as supporting their desire to be independent have higher levels of intrinsic motivation relative to athletes who perceive their coaches as less supportive. Disability sport coaches influence critical competitive psychological states in their athletes, in particular, confidence and anxiety, which play an important role in athlete's sport performances. For example, coaches who simultaneously support and challenge their athletes to become better are more likely to develop confident athletes. Coach influence on team cohesion is particularly important especially for sports where strategic and set plays are common, as in wheelchair rugby and basketball, and need to be learned in a short time span. National team athletes often live in different parts of the country, making the development of team cohesion difficult. Hence, coach influence on team task and social cohesion at national team camps is critical.

Coaching disability sport is challenging. Coach's prior experiences are usually with able-bodied athletes, heightening the importance of finding good disability sport specific coaching literature. Historically, coaches have reported having a hard time finding disability sport specific coaching material. However, a recent increase in research and disability sport literature, such as Vicky Goosey-Tolfrey's *Wheelchair Sport: A Complete*

Guide for Athletes, Coaches, and Teachers, has made finding coaching material easier.

In addition to understanding their athlete's sport, coaches must gain knowledge on their athlete's disability condition. Because most coaches lack the life experience of living with a disability, specific knowledge must be learned. For coaches of youth athletes, developing sound relationships with the athlete's parents and their physical therapists helps capture their individual athlete's unique needs. In the case of athletes who are deaf, coaches without hearing impairments have additional communication challenges. Coaches also must consider the competition facility's accessibility for all of their athletes. Finally, coaches of high performance sport athletes (e.g., Paralympians) must support their athletes with issues related to traveling. Having to get on the plane first and off last can result in stress and pressure sores. Flying can promote dehydration, which necessitates the access to safe water (bottled water). Prior to travelling, and depending on the country, issues such as travel insurance, immunization, passports, and visas must be considered.

Unique to disability sport is the classification process where athletes are classified for competition depending on the severity of their disabilities. Some athletes have been known to *sandbag*, which means purposely underperforming at classification in order to be slotted into a lower classification where they will encounter inferior competition. Certainly coaches should be cognizant of this practice and insure their athletes do not engage in it. In summary, athletes with disabilities often lack coaching. Research on disability sport indicates that coaches often face challenges that are quite unique compared to the challenges of coaching able-bodied athletes. However, the limited research on disability sport coaching practices suggests that coaches can be quite influential on their athlete's motivation and confidence.

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See also Coaching Efficacy; Disability; Disability and Sport

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DISTRACTION THEORY

See Attention–Performance Relationships

DIVERSITY

The concept of diversity encompasses a broad range of qualities and characteristics that distinguish people from one another. Diversity is used broadly to refer to demographic characteristics including, but not limited to, *sex, race, ethnicity, sexual orientation, class, ability status, age, national origin, religious beliefs, and education*. Diversity is important for a number of reasons. It allows a person to view issues and problems from multiple standpoints, drawing from different experiences, perspectives, knowledge, and connections. Rather than viewing the world from a single-focus lens, a person is able to expand views and consider multiple options. Diversity moves people beyond their ethnocentric and egocentric viewpoints, allowing them to not only learn about others' experiences and backgrounds but also more about themselves.

Sport is a diverse environment that includes individuals from different cultural and racial backgrounds. Sport and exercise psychology professionals work with and study these diverse groups of athletes, exercisers, and coaches. The Association for Applied Sport Psychology (AASP), an international, multidisciplinary, professional organization that offers certification to qualified professionals

in the field of sport, exercise, and health psychology, outlines seven ways in which AASP actively promotes the respect and value of human diversity within its members and through professional actions. AASP members (1) do not discriminate; (2) do not tolerate remarks that reflect disrespect for individuals based on physical or cultural bias; (3) promote human diversity in research; (4) promote individuals with different backgrounds to join and participate in the organization; (5) seek education and training in multicultural methods to best serve the diverse clientele in sport and exercise psychology consulting; (6) promote equity and multicultural representation in AASP publications, conference presentations, and professional activities; and (7) demonstrate attitudes of respect and positive regard toward all colleagues, students, and clients.

Given the diversity among sport and exercise participants, it is surprising that until recently, diversity and multicultural perspectives have received only minimal consideration in the field of sport and exercise psychology. Much of the scholarly attention has focused on race or ethnicity and gender, with little attention devoted to other identities or the intersection of the varied identities. Sport and exercise psychology has traditionally been practiced from an ethnocentric, white male perspective. It is only recently that researchers have begun to seriously address diversity issues. For professionals and students in the field, it is imperative that there is an understanding of the histories and social experiences of the various cultural and identity groups with which we work or will work with in the future; the delivery, implementation and monitoring of sport and exercise psychology skills and interventions requires an understanding of cultural diversity.

Some research has addressed information about different cultural groups in relation to sport psychology consulting. Much of the earlier work in this area, however, relied on stereotypes of racial and ethnic groups and offered rudimentary recommendations of how to work with athletes of color. The early work in this area also treated athletes of color as exotic *others*, placing whiteness as the normative, privileged position. This work, however, served as the foundation for much of the recent discussion and work in this area. Recent research in this area, and in particular the work in cultural sport psychology, has begun to explore the issues pertinent to cultural and racial or ethnic diversity

among athletes and multicultural approaches to sport psychology interventions. Research has also examined white racial identity and privilege in sport psychology consulting. Several researchers have noted the need for consultants to confront their personal feelings, experiences, and attitudes toward race by critically examine how their biases and assumptions influence their interactions with athletes of color.

In line with traditional psychology, gender research in sport and exercise psychology has historically focused on biologically based sex differences and neglected the study of complex gender issues and relations. More recently, researchers have stressed the need to consider athletes, exercisers, and coaches in the social context and work to understand the complexities of gender in sport and exercise.

In order to provide appropriate services to all individuals, as recommended by AASP and the American Psychological Association, Division 47, students and professionals in sport and exercise psychology should have training in issues that relate to diversity, culture, and identity. This training typically occurs in a student's graduate program in sport and exercise psychology, and requires that students take at least one course related to multiculturalism or social basis of behavior. Many have argued that multicultural training will enhance sport and exercise psychology effectiveness. Sport and exercise psychology consultants would benefit from an understanding of people of different cultures from their own. However, multicultural competency requires a heightened awareness of cultural influences in society, an ability to work with individuals from different cultural and racial backgrounds, content knowledge and intervention skills that are relevant to cultural groups, and skills to communicate across cultures. Such competencies will most likely not occur from one graduate course related to the social basis of behavior. Rather, an integrated model of multicultural training is recommended which involves integration and application of multicultural issues within each course in the curriculum. There is a need to move beyond delineating diversity as a special topic or special population toward integration of the social aspects of sport into all areas of sport and exercise psychology, including anxiety, motivation, personality, injury, and group dynamics.

Emily A. Roper

See also Cultural Competence; Gender; Multiculturalism; Race; Racism/Whiteness

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DROP-OUT

The term *drop-out* has two meanings. In elite sport, drop-out refers to a premature termination of a sport career before the athlete could reach individual peak performance level. Drop-out is a typical phenomenon among athletes in childhood and adolescence. Therefore, the (young) age of the athlete may be regarded as an important evidence for premature drop-out. In contrast, career termination after reaching peak performance is called retirement. Typically retired athletes are older than drop-outs in the same sport.

In recreation and sport, drop-out means ending one's participation, for example in a club or a fitness center. In health-related sport, drop-out means leaving an exercise or rehab program before the end of that program—for whatever reason. In this sense, drop-out may happen at every age and is characterized by concluding any supervised physical activity. Different from elite sport, participants are not pursuing a sport career with the goal of peak performance. In this entry, the focus will be on the first meaning.

Drop-out can be considered from the viewpoint of the sport system and the athlete. From

the viewpoint of the sport system, drop-out may be regarded as a nonnormative transition that could be avoided if only the athlete had been more motivated and had been better supported by the environment. Therefore, drop-out is considered a loss of talents and, economically speaking, a lost investment. From the viewpoint of the athlete, drop-out may be a source of regret and negative feelings, which would then accompany the transition to a career outside elite sport. Therefore, research studies and applied sport psychology alike are concerned with the reasons for drop-out, its prevention, and athletes' coping efforts. Drop-out is regarded as a complex phenomenon with a multicausal history. It may result from a deliberate decision of the athlete, for example, after seeing no future in sport due to no performance increases. But on the other hand, it may result from a forced decision because of a career-ending injury. Research shows that it makes a difference for an athlete if the decision for drop-out happens to be voluntary or involuntary. Very often, athletes make an easier transition to the postcareer if they see it as the result of a voluntary retreat that they had planned for.

Which are the most often mentioned reasons for drop-out in elite sport? As noted earlier, there is always a multitude of reasons. Most often, young athletes feel no longer able to combine school education with the high demands of sport training and competitions. They therefore finish their career in order to give priority to their education. Also, young athletes may realize that they lack the potential to make it to the top and perceive any further investment into their career as a waste of time. This feeling may be heightened by performance slumps—particularly during or after puberty—and by motivational crises, particularly after injuries. In addition, coaches may be regarded as being no longer supportive, and athletes would feel forced to leave the training group on the whole. And last, but not least, financial support—particularly from the sport system—may be withdrawn.

There exists not only a multitude of reasons for drop-out but also a multitude of reactions. Athletes who plan their postcareer tend to make a smooth transition and to feel happy with their new life. They see the end of their sport career as a necessary and even positive consequence for their further development. In these cases, drop-out can lead to positive feelings like relief and happiness, which are helpful for leading a new life. But

if athletes perceive their drop-out as a critical life event this may cause problems, which the athletes have to cope with. This feeling results more often from involuntary than voluntary drop-out. In these cases, drop-out can lead to negative feelings and even a crisis.

Apart from the reasons for drop-out mentioned above, former athletes who terminated their career prematurely seem to lack volitional and motivational qualities and they often perceive less social support or too much pressure from parents and coaches than active athletes. Some authors even speculate that coaches play a decisive role in the drop-out decision process.

Athletes thus could be better prepared against drop-out by their being helped to cope with the dual demands of sport and school or education. Special sport schools can be a solution, but also any other systematic support system. Also, coaches should have a positive attitude toward the athlete's pursuing a dual career. If athletes have problems in handling the consequences of drop-out, it would be helpful to have sport psychologists and mentors—former elite athletes in particular—teach the drop-outs how to overcome sadness and regrets and how to cope with the new demands of the postcareer.

Dorothee Alfermann

See also Burnout; Career Transitions; Talent Development

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DRUG USE AND CONTROL

Drugs are used for performance enhancement purposes in elite, competitive, and amateur sports. Unlike heroin, barbiturates, hallucinogens, and substances used for recreational purposes in other subcultures, athletes use and abuse drugs presumed to enhance athletic ability and performance, known as *ergogenic substances*. The most common performance-enhancing drugs include amphetamines and a range of substances thought to increase strength and muscle mass like anabolic steroids. The use of performance enhancing drugs is considered doping practice (see note at end of entry) and is therefore prohibited by the World Anti-Doping Agency (WADA), National Anti-Doping Organizations (NADOs), and sports-governing organizations. However, because athletes may face illnesses or health conditions that require medication, the use of certain prohibited substances is allowed for therapeutic purposes (known as therapeutic use exemptions). This process is regulated by the WADA's International Standard for Therapeutic Use Exemptions (ISTUE).

Health Consequences of Drug Use

Small or moderate dosages of performance-enhancing drugs are unlikely to result in noticeable changes in athletic performance. Therefore, athletes aiming to significantly improve their performance tend to engage in systematic and excessive use of drugs, especially during periods of intensive training. In fact, athletes typically self-administer drugs at much greater dosages than those prescribed for the treatment of medical conditions. However, abusing large doses of performance-enhancing drugs is associated with a wide range of side effects, including possible death. One notable example is the death of cyclist Tom Simpson in the Tour de France in 1967 as a result of amphetamine abuse. Furthermore, the abuse of anabolic steroids has been associated with various physical and psychological side effects, including but not limited to, liver toxicity, hormonal imbalance, irritability, increased aggressiveness (known also as steroid or roid rage), mania, major depression, and psychotic symptoms. Some studies have also noted that anabolic steroids can be potentially addictive, leading to withdrawal and dependence symptoms.

The Ethics of Drug Use

The spirit of sports is characterized by a range of ethical principles, such as fair play, ethics, honesty, health, character and education, respect for the self and other participants, to name just a few. Using prohibited drugs to increase athletic performance defies those principles, and, consequently hurts the image and integrity of sports. Furthermore, drug use gives an unfair competitive advantage to athletes who have the resources to secure drug supplementation, as compared to athletes without access to drugs, or who lack the financial resources needed to support drug supplementation.

Availability and Drug Use Patterns

Drugs can be supplied through drug trafficking networks either face-to-face or online. In many countries, the supplementation, possession, and trafficking of performance-enhancement drugs, and even encouraging someone to purchase prohibited ergogenic drugs, is penalized via legal sanctions. Common patterns of drug use include cycling, stacking, array, and pyramiding. *Cycling* refers to the use of drugs for periods between 8 and 12 weeks, or even longer. *Stacking* describes the concurrent use of several types of drugs to achieve greater performance effects, whereas *array* refers to using other drugs to counter the side effects of steroids, including estrogen blockers, masking agents, and diuretics. Finally, *pyramiding* is the gradual increase of dosages, from relatively low doses of drugs at the beginning of the cycling period, to higher doses toward the end of the cycle. Some athletes may also start with lower doses, move on to higher doses, and then return to lower doses of drug use at the end of the cycle. In many cases, the patterns of cycling, stacking, or pyramiding are based on arbitrary criteria and are not accompanied by the strict medical supervision that oversees therapeutic drug use. This is even more so in amateur sports and fitness, where users base their drug use patterns on word-of-mouth, or on information provided by informal channels such as drug-related websites.

Social and Psychological Predictors of Drug Use

There have been several attempts to identify the psychological drivers of drug use in sports. Relevant studies were initiated as early as the

1980s and centered on athletes' KAB (knowledge-attitudes-behavior). Over the last decade, research on drug use expanded considerably and has moved from mere KAB surveys to studying deeper psychological processes involving the interplay between personal traits and dispositions, motivational variables, cognitions and normative pressures, and behavioral intentions. Some scientists also focus on novel ways to assess social desirability, or capture the implicit beliefs and attitudes of athletes and exercisers toward performance enhancing drugs. Such research is based on the premise that athletes may be reluctant to disclose their preferences and beliefs toward prohibited drugs; therefore, studying their mental representations about the issue with unobtrusive and indirect methods can yield more reliable findings. Other researchers argue that a narrow focus on decision-making processes and cognitive functions may undermine broader cultural and social influences on drug use, such as the effects of use-reduction or harm-minimization policies, and macrolevel normative influences. It is noteworthy that social science research on performance-enhancing drugs has been supported by official antidoping organizations like WADA's Social Science Research Program, which was launched in 2005, and stimulated research on the psychological, behavioral, and social factors associated with drug use. The available studies suggest that doping is a deliberate process that requires decision making and intention formation. Doping intentions are likely to be influenced by doping-related attitudes, goal orientations, motivational variables, and social influences.

Drug Control

A commonplace drug control method is urine testing. This approach is based on a detect-and-punish premise and assumes that drug traces may be present in an athlete's urine during drug use and remain for some time after. Nevertheless, athletes can evade drug controls or tamper with their results by refusing to participate in drug testing, failing to declare whereabouts (a requirement for elite athletes to state their location to official sporting associations and NADOs in order to facilitate unexpected drug controls), discontinuing drug use before competition so that the drug is cleared from the body, drinking excessive amounts of water, or taking masking agents in order to tamper with

or invalidate urine tests. Another problem with existing doping control methods is that drug testing cannot be applied to the entire population of professional and junior athletes, mainly because of the high administrative and financial costs involved. Also, drug controls often fail to detect new substances and are even less effective in capturing gene doping. Finally, the detect-and-punish approach does not prevent the onset of drug use among young people and early-career professional athletes.

Athletes' Beliefs Toward Drug Control

Research on athletes' beliefs toward drug control and testing is still limited. Some of the available studies have shown that collegiate athletes in the United States were in favor of mandatory drug control if it was followed by systematic efforts and education programs aiming to increase awareness about the side effects of drug use. Other studies have shown that collegiate athletes lacked knowledge and awareness of the standard procedures involved in drug controls and displayed little awareness of the side effects of prohibited performance enhancement drugs. On the other hand, evidence from elite athletes is controversial; some athletes appear supportive of rigorous drug testing, whereas others are less in favor of drug controls, and even believe that existing procedures (e.g., urinating in public to supply a sample for analysis) hurts their dignity and personal privacy.

Education and Prevention

An alternative approach to drug control in sports that has gained momentum in the recent years is concerned with the investigation of psychological and social risk factors. This approach suggests that drug use can be explained by the interplay between a person's motivations, cognitions and beliefs, and normative influences. Knowledge about these processes can be used to inform subsequent, evidence-based preventive strategies that focus on education, awareness, and behavior modification interventions.

The ATLAS (Adolescents Training and Learning to Avoid Steroids) and ATHENA (Athletes Targeting Healthy Exercise and Nutrition Alternatives) projects are based on these principles, and are used to train adolescent and junior athletes, trainers and coaches, and team leaders about the effects of drug use. ATLAS and ATHENA provide

knowledge about the side effects of drug use, build refusal efficacy skills, and inform athletes about alternative and safer performance enhancement methods. Empirical studies have shown that both ATLAS and ATHENA are effective in significantly reducing future use of steroids and related performance enhancement drugs among young athletes in both the short and long term.

Conclusion

Drug use is an ongoing issue in both competitive and elite sports and in amateur, nonprofessional sports and leisure time exercise. Anabolic steroids, amphetamines and stimulants, and other anabolic agents are used by athletes to enhance performance. Users tend to self-administer high dosages of drugs and engage in unregulated and unsupervised drug use patterns, thus endangering their physical and mental health. Research on drug use in sports has shown that athletes engage in this practice for various reasons, including motivational and dispositional variables, as well as normative influences. Recent research trends focus on unobtrusive methods to capture athletes' beliefs toward drugs, whereas other researchers call for studies that address broader sociocultural influences such as policy making. So far, drug control approaches have focused largely on punitive methods, but more recent trends support education and awareness-raising campaigns that tackle the psychosocial processes underlying drug use in sports and exercise and contexts and prevent future use by young athletes. Still, it appears that the road to drug-free sports is still long, and much work has to be done in order to eradicate the seemingly well-established drug-based performance enhancement mentality.

Lambros Lazura and Vassilis Barkoukis

Note: The World Anti-Doping Agency prohibits the use of anabolic steroid agents and other ergogenic substances, such as stimulants and amphetamines. Nevertheless, as described in the entry "Performance-Enhancing Drugs" in this encyclopedia, athletes may use nutritional supplements and herbal products to enhance their performance. These substances are considered legal and are not prohibited by WADA or related organizations, and their use is still unregulated. Nevertheless, for reasons of clarity and to avoid overlap with other entries in this encyclopedia, the present entry discusses only prohibited performance-enhancing drugs.

See also Diet Drugs; Narcotic Analgesics; Performance-Enhancing Drugs; Psychopharmacology

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DUAL-PROCESS THEORY

Although dual-process theories have become popular over the last few decades, ideas about mental division have existed for centuries. Significant philosophers and psychologists, such as Plato and

Sigmund Freud, believed that the mind was partitioned, and the early work in this area has contributed much to modern dual-process theories. These theories, which have recently become popular in a variety of psychologies, share the notion that humans possess two distinct modes of information processing. One, which is commonly called System 1, the *impulsive system*, or the *automatic system*, is often characterized as fast, effortless, automatic, nonconscious, and it places little demand on working memory. It is a form of universal cognition that humans share with animals. System 2, sometimes referred to as the *reflective system* or *controlled system*, is commonly described as slow, effortful, controlled, conscious, and it is demanding on working memory. It permits abstract reasoning and hypothetical thought, and it is uniquely human. Early evidence for these two information processing systems was obtained from research in which participants were asked to respond to vignettes from three perspectives: how they believed most people would behave in the situation described, how they themselves would behave, and how a logical person would behave. Participants tended to indicate that people, including themselves, would often not act in accordance with logic.

Many dual-process theorists contend that processing in System 1 relates to the automatic access of knowledge or affective reactions that have become associated with a cue. The repeated pairing of a stereotype (slow) to a social group (elderly), for example, can lead to automatic access to the stereotype after perception of the group. Behavior and judgments can then align (or, in some cases, conflict) with the primed stimulus. Although this associative system operates fast, it is built via a slow conditioning process. Processing in System 2 is often discussed as analytical and rule based, and one which can draw from both slow-learning and fast-learning memory systems. The actual operation of System 2 is significantly slower than that of System 1.

Factors That Influence People's Reliance on Systems 1 or 2

Motivation

System 1 and System 2 compete for control of our inferences and actions. System 1 is generally more influential under conditions of low motivation, whereas System 2 can override the impulses of System 1 when motivation is available. The

rule-based system that System 2 operates under is subjectively effortful and requires attentional resources. Thus, if people are not motivated to engage in this form of processing, responses will be governed by the effortless operation of System 1.

Cognitive Capacity

Cognitive capacity, as well as motivation, is important in determining the contributions of System 1 and System 2. Capacity refers to available processing time as well as attentional resources. Thus, responses that are made quickly or when the perceiver is busy or distracted will be governed by the associative processing of System 1. Alternatively, if an individual is given adequate time and is not distracted, System 2 may override the operations of System 1.

Type of Judgment

Motivation and cognitive capacity are perhaps the two most widely recognized factors that impact the utility of Systems 1 and 2. Nevertheless, other factors, including type of judgment, have also been proposed to influence people's reliance on the two systems. Those judgments that are more intuitive and affective, for example, "How would you feel if an opponent injured you behind play?" seem to be more influenced by System 1 than more rational judgments that are governed primarily by System 2 like "Why did the player injure you behind play?"

Generality of Stimuli or Judgment

The average of a collection of attitudes toward individual members of a team might be quite disparate from an overall attitude of the team. One reason for this is that the associative processing of System 1 seems to prefer specific, concrete stimuli, whereas general and abstract stimuli are better processed by symbolic rules (System 2). Thus, the systems are likely to contribute differently to questions or responses that vary in generality.

Mood

The associative system (System 1) is more often relied upon when positive mood is experienced, whereas negative mood promotes rule-based processing (System 2). Evolutionary reasons have been proposed to cause this effect, such as that consciousness is employed to direct individuals away from the aversive stimuli causing the negative mood.

A positive mood implicitly indicates that all is well, and the effortless System 1 is more likely to be allowed control in these circumstances.

An Example of the Effects of Systems 1 and 2 in Exercise

The motivation of an individual to attend an exercise class might be primarily a function of System 1, System 2, or a combination of both. A person for whom exercise is habitual will rely on System 1; motivation (and behavior) for the class will be activated automatically, and they will begin the session without much conscious effort. Alternatively, a new exerciser might contemplate attendance at an exercise class, and will therefore engage System 2. As a final alternative, both systems might be operational. A person's conscious thoughts about the benefits of an exercise class (System 2), for example, might be biased according to the operation of System 1.

James Dimmock

See also Automaticity; Automaticity: Evaluative Priming; Automaticity: Implicit Attitudes; Decision Making; Habit; Information Processing; Mindfulness; Priming

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DUAL-TASK PARADIGM

It has been known for a long time that there are significant limitations in the human capability to attend to and perform two or more tasks concurrently. Observations of the performance errors arising from the simultaneous performance of multiple tasks date back at least to the late 19th century and form the basis for cognitive theories that regard attention as a limited capacity or resource. When attention is divided between tasks in a way that exceeds the available capacity or resources,

suboptimal performance in the form of increased errors or delayed responding becomes evident on one or more of the tasks. The dual-task paradigm is an established methodology to measure performance capability under these conditions where attention must be divided between two concurrently performed tasks. This entry first outlines the basic rationale and assumptions underlying the dual-task paradigm before describing some of the key factors that affect dual-task performance and some of the existing and prospective uses of the paradigm within the sport domain.

Basics of the Dual-Task Paradigm

As the name implies, the dual-task paradigm involves the simultaneous performance of two tasks—a primary task and a secondary task. In the simplest application of the method, participants are required to assign priority to the primary task (so that performance on this task remains at a comparable level in the dual situation as when the task is performed alone), and then changes in performance of the secondary task are used to provide a measure of the attentional requirements of the primary task. The primary task is generally the task that the researcher is most interested in understanding, whereas the secondary task is simply a tool used to help assess the attentional demands of primary task performance. For example, in research examining the attentional demands of walking control in the elderly, the walking task is the primary task while there is a wide range of possible secondary tasks available, including continuous tasks (such as counting backwards from 100 in 3s) or discrete tasks (such as a reaction time task with an auditory stimulus and a vocal response).

The extent of the deterioration in secondary task performance between the condition in which it is performed concurrently with the primary task compared to when it is performed alone provides a measure of the attentional demands of the primary task. Poor secondary task performance accompanies primary tasks that are extremely difficult and require many of the participant's information processing resources, whereas secondary task performance that remains unchanged from the solitary performance condition to the dual performance condition suggests a primary task that can be performed essentially automatically, requiring no processing resources.

An alternative instructional set that is sometimes used is to ask participants to assign equal priority to both tasks. This particular approach is used if the interest is less in quantifying the attentional requirements of one of the tasks and more in ascertaining the general time-sharing and attention-switching capabilities of the participants. In all applications of the dual-task paradigm, the selection of the secondary task, the monitoring of adherence to the instructional set, and the careful measurement of the performance on both tasks in isolation as well as contemporaneously is pivotal to the generation of interpretable data.

Factors Affecting Dual-Task Performance

Although there is some indication that the total available attentional resources (or processing capacity) available at any given instant may vary with factors such as arousal—being theoretically maximal when arousal is optimal—by far the biggest factor influencing dual-task performance is the complexity and attentional demand of the primary task. The proportion of the available processing capacity that is consumed by the primary task depends on the extent to which the primary task can be controlled by automatic processes and this, in turn, is dependent on the extent to which the skill has been learned. Primary tasks that are intrinsically simple or that are effectively made simple as a consequence of extensive amounts of practice and skill learning require relatively little central processing capacity and can be performed concurrently with only limited impact on secondary task performance. A large number of studies from different domains including aviation, transport, rehabilitation, and sport have consistently demonstrated that the ability to perform two (or even more) tasks simultaneously improves with practice of the primary task and is superior for experts than novices in the task domain. The improvements that occur with skill development are likely attributable to both a transition of at least some of the control of the primary task from deliberate processing to automatic processing and the development of effective time-sharing and attention-switching strategies that help minimize interference between the concurrently performed tasks.

Uses of the Dual-Task Paradigm in Sport

Because many sports activities necessitate the dividing of attention between two or more

concurrent tasks (e.g., the basketball player dribbling the ball while scanning around for a suitable teammate to whom to pass), the dual-task paradigm is ready made for the study of skill in sport. The main use of the paradigm to date has been in assessing the skill level of individual athletes and in this respect the dual-task method is advantageous in that it offers a means of assessing the extent to which primary skills are automated and a means of revealing individual differences that are not apparent from simply observing the primary skill performed by itself. A fundamental premise is that athletes whose primary skills are well automated are more likely to be able to cope with competitive pressure and the time constraints of high-level competition than are athletes who need to devote a considerable proportion of their available attention to their basic skills in order to perform these to an acceptable level. In a recent study in rugby league, it was demonstrated that performance on a dual-task test of 2-on-1 passing skills was predictive of match performance of the same skill. Other potential, but as yet largely untapped, uses of the dual-task paradigm in sport include (1) using comparisons of the secondary task performance under match and simulation conditions to ascertain the effectiveness of simulation drills, (2) making objective assessment of the relative demands of different skill learning drills in order to introduce them in appropriate order of increasing complexity, and (3) exploiting the attentional overload available through dual task methods to attempt to further stimulate the automation of key skills or encourage the learning of skills via a form of control that is below the level of consciousness.

Conclusion

The dual-task paradigm offers a simple but potentially powerful approach for the assessment of the attentional demands of skills and tasks that are fundamental to successful performance in many domains. Although the paradigm has a long history of use in cognitive psychology, especially in the testing of different theories of attention, it has thus far been underutilized in the understanding and enhancement of the skills for sport.

Bruce Abernethy

See also Attention Theory; Attentional Focus; Automaticity; Expertise; Motor Control

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DYNAMICAL SYSTEMS

Every biological system—microbe, athlete, or team—can be described formally in terms of its time evolution or dynamics. This becomes feasible if one capitalizes on a crucial characteristic of biological systems, namely that they exchange energy and matter, and in some cases also information, with their surroundings. In statistical physics, such systems are called open systems, as opposed to closed systems without environmental contact. Certain classes of open systems may self-organize and form coherent macroscopic patterns due to interactions between their constituent subsystems. When applied to sports behavior, the term *subsystems* may refer to the neurons, muscles, organs, and limbs of an individual athlete, but also to the individual athletes themselves, for instance when in combat with an opponent, be it one on one (boxing), one against many (cycling), or in teams (football). Furthermore, the term *system* may refer to virtually any aspect of sports behavior, including action, perception, emotion, and cognition, as well as their physiological underpinnings. Finally, each of these behaviors may change as a function of development and learning, which introduces an extra layer of dynamics. All of this implies that the opportunities for applying the concepts and methods of dynamical systems theory in sports science

are virtually boundless and that any attempt at a priori demarcation is futile. This entry first provides an intuitive, yet necessarily abstract, introduction to the main concepts of dynamical systems theory, which are then brought alive by a collection of examples that highlight their relevance to sports science.

Concepts: A Fling With Physics and Mathematics

An important feature of open systems consisting of many interacting subsystems is that they form spatial, temporal, or functional behavioral patterns. Such coherent macroscopic patterns may be described by a small number of collective variables, called *order parameters*. Spontaneous switches between macroscopic patterns are termed *non-equilibrium phase transitions*¹ in equivalence with similar qualitative, structural changes studied in statistical physics. Phase transitions may be induced by continuous, gradual changes in relevant system parameters that do not specify the macroscopic patterns and are called control parameters. In what follows, it will be argued that, in the vicinity of these phase transitions, any system can, to a good approximation, be described as a low-dimensional dynamical system. The art of dynamical systems modeling is to find appropriate low-dimensional descriptors—that is, the order parameters—and to derive dynamical equations that describe their time-evolution and nonlinear dependence on control parameters. Such equations provide formal analogies of the macroscopic patterns and phase transitions exhibited by the system under study.

Steady States: Fixed Points, Limit Cycles, and More

Like macroscopic behaviors, dynamical equations often show steady-state behavior. That is, as time goes to infinity, they will settle asymptotically on a steady-state solution, which is often an attractor (a subspace of the state space to which trajectories are attracted). Dynamical systems theory may display four types of attractors: fixed points, limit cycles, limit tori, and chaotic attractors. Fixed points and limit cycles are the simplest attractors: a single point or a series of connected points that are visited at certain intervals, respectively. Some attractors are governed by more than one frequency. If these frequencies

stand in rational relation to one another (are commensurate), the resulting behavior will still be a limit cycle. However, if these frequencies stand in irrational relation to one another (are incommensurate), the resulting trajectory is no longer closed, and the limit cycle becomes a limit torus. In this case, the resulting behavior is called *quasi-periodic*, implying that the trajectory passes arbitrarily close to every point on the torus without ever revisiting precisely the same point. A chaotic attractor is no longer a simple geometrical object like a point, cycle, or torus and thus defies precise definition. The outstanding property of chaotic systems is that they are sensitively dependent on initial conditions: trajectories that emanate from two arbitrarily close starting positions diverge at a characteristic rate. Hence, chaotic behavior, although entirely deterministic, is inherently unpredictable.²

Instabilities and Phase Transitions: Isolated Switches in Behavior

Phase transitions are accompanied by a colossal separation of time scales between different system components. Consider the case in which a certain macroscopic pattern is being replaced by another one. In dynamics terms, the first one becomes unstable and the second one becomes stable. Being close to an instability, however, implies that it takes a long time for a system to return to its steady state after a perturbation.³ The order parameters hence evolve arbitrarily slowly, whereas the underlying (real) subsystems maintain their individual, finite time scales. From the viewpoint of the order parameters, all the subsystems become arbitrarily quick so that they can adapt instantaneously to changes in the order parameters. The system dynamics thus amount to that of the order parameters, implying the ordered states can always be described by a very few variables if in the vicinity of behavioral changes. Put differently, the state of the originally high-dimensional system can be summarized by a few variables or even a single collective variable, the order parameter(s). The relationship between the subsystems and the macroscopic structure, in which the subsystems generate the macroscopic structure and the macroscopic structures enslave the subsystems, implies a circular causality. This effectively allows for a low-dimensional description of the dynamical properties of the system of interest.

Complexity: Ongoing Switches in Behavior

The merger of dynamical systems with concepts of statistical physics has provided a thorough understanding of nonequilibrium phase transitions and hence of self-organized pattern formation. One can go a step further and investigate the case in which the system under study remains in a critical state, a phenomenon called *self-organized criticality*. Self-organized critical systems have a critical point as an attractor. Consequently, their macroscopic behavior resembles the temporal scale-invariance characteristic of the critical point, which often includes fractal dynamics (as in chaotic systems) and power laws for, for example, the system's temporal correlations or spectral distribution (long-term correlations and $1/f$ -spectra).

Complex Networks

Macroscopic, scale-free behavior of complex systems may not only emerge as a result of self-organized criticality. More recent studies on complex networks⁴ have revealed that certain forms of network growth yield scale-free networks; that is, the distribution of connections per node in the networks obeys a power law. The mechanism is that of preferential attachment: The likelihood that a new connection will be formed with a node depends on the number of connections of this node. Thus, nodes that already have a large number of connections are more likely to get even more connections ("the rich getting richer"). Many real networks such as the World Wide Web, collaboration networks of scientists, and brain networks are probably scale-free.

Examples: From Physiological Rhythms to Match Play

Entrainment of Respiration and Locomotion: A Matter of Efficiency?

A scenario germane to dynamical systems theory is the spontaneous entrainment of respiration and locomotion cycles in mammals, also known as locomotion–respiration coupling (LRC). During walking or running, particular integer frequency ratios are adopted between cadence and respiration (e.g., 2:1, 3:1, 3:2, and so on). Recall the multi-frequency attractors with commensurate frequencies explained above. Top athletes explicitly train their optimal frequency ratio and learn to flexibly switch between different ratios as a function of

demand and strategy. Why would LRC be beneficial? For years, it has been hypothesized that it serves to enhance performance efficiency. More recently, this notion has been linked to optimization of the effective oxygen volume in the lungs. In brief, the cyclical abdominal pressure modulates self-sustained breathing and causes maximum oxygen concentrations at integer frequency ratios between cadence and respiration.

Over the years, it has become evident that LRC is not simply a result of (bio-)mechanical constraints, such as the vertical impulse arising from the footfalls, as entrainment of breathing has been reported for a wide variety of daily activities with markedly different mechanical constraints, including cycling, wheelchair propulsion, and rowing. In general, the degree of coupling and the resulting frequency ratios depend on a variety of factors associated with task, environment, and athlete. Theoretically, this implies that respiration-locomotion entrainment is a generic phenomenon that is instantiated by, but cannot be reduced to, specific mechanisms or processes. Practically, the finding that greater expertise is associated with a broader range of solutions renders this dynamic phenomenon intrinsically relevant to training and performance enhancement in all cyclical endurance sports.

Dimensionality Reduction: A Way to Understand Coordination and Perception

An expedient tool for studying complex patterns of coordinated behavior is principal component analysis (PCA), a statistical technique for identifying reduced dimensionality in multivariate time series. The dimensions or principal components that account for the most variance in the original data set may be interpreted as the degrees of freedom, or alternatively as the order parameters of the system under study. PCA has found wide application in the study of movement coordination, also in relation to sports, and appealing examples may be found throughout the literature.

PCA revealed how (bio-)mechanical constraints lead to a considerable reduction in the number of components required to describe human walking. Only four components proved to be sufficient for this purpose. Coordination patterns were found to change markedly from walking to running, and PCA revealed a profound reduction of dimensionality during speed-induced gait transitions,

as discussed earlier. Importantly, the evidence for phase transitions was prominent in the coordination patterns but not in the stride parameters. This approach may be readily extended to gait patterns in competitive running, as well as other sports behaviors like catching, throwing, and hitting, as exemplified by several recent studies.

Quite a different application of PCA can be found in the study of visual anticipation skills in tennis, in particular shots in different directions (left, right) and to different distances (long, short). A few components appeared sufficient to capture most of the variance in the shots, but they differed across shots, and tennis players could predict shot direction based on these components alone. This suggests that visual anticipation skill in tennis, and most likely also in other sports, involves the extraction of low-dimensional dynamic information from high-dimensional displays.

This series of examples highlights the fact that (physical theories of) emerging patterns (order or disorder) can be utilized for decomposing complex movement patterns. Ultimately, a better understanding of the principles that govern the complementary processes—pattern formation and pattern decomposition—will help advance perceptual and motor skill training in sports.

Temporal and Spatial Correlations: From Individuals to Teams

Heart rate variability is a measure that is often used in sports practice to individualize training intensity and recovery. But how should it be understood from a dynamical systems perspective? Considering the complexity of the mechanisms regulating heart rate, it is reasonable to assume that its dynamics are nonlinear, although no firm evidence for this could be found. Instead, it appeared that the heart rate time series of athletes is characterized by long-term correlations with a fractal signature. In particular, scaling analysis (detrended fluctuation analysis or DFA) of heart rate time series revealed scale invariance in distinct regions, corresponding to well-known frequency bands in the power spectra of heart rate variability. Furthermore, during training, marked changes were found in the scaling exponents of the scale-invariant regions in question, as well as during recovery after heavy exercise. These findings suggest that (also) the scaling characteristics of heart rate variability might be used to monitor the training status of athletes.

Dynamics of Team Sports

Matches among sports teams represent highly complex dynamical events. The positions of the players on the field evolve over time and may be described in terms of their spatial, directional, and temporal characteristics. Although this is already a rather laborious exercise, it is not sufficient for an adequate understanding. The reason is that the actions of the players on the field only make sense in light of specific, yet not necessarily known, individual and collective objectives and strategies. In any situation, players have a multitude of solutions at their disposal to pursue their strategic objectives, while actions unfold in parallel. As a result, the dynamics of match play represent a notoriously difficult area of investigation. Nevertheless, although still in its infancy, the analysis of match play is on its rise due to both technological and conceptual advances. Notable among the latter is the advent of network theory, which concerns itself with the study of graphs as a representation of connections between discrete objects (nodes).

As a case in point, a network analysis of passes among the players of the Spanish team during the FIFA World Cup 2010 was performed, where the team was considered a network with players as nodes and passes as (directed) connections. This revealed the effectiveness of the Spanish game in terms of several network measures over time, most importantly the clustering coefficient and passing length and speed. Likewise, in searching for the *greatest* team in cricket, matches played from the late 19th century onward were analyzed as networks using a page-rank algorithm to assess the importance of the wins and the rank of teams and captains; the same approach is used, for example, by Google to rank websites by keywords.

Team sports do not only form a network as players interact during team play but also sociologically in that players of the same club are linked. The latter relations have been analyzed in soccer using bipartite networks with nodes being players and clubs. The probability that a player has worked at n clubs or played m games shows an exponential decay, whereas the probability that he has scored g goals represents a power law. If two players who have played simultaneously for the same club are connected by an edge, then a new network arises with an exponentially decaying degree distribution. Of course, performance

(scoring) and social interaction (club sharing) influence each other.

Conclusion

The study of a system's dynamics comes with a rich and sophisticated conceptual framework to address the complexity of sports behaviors. This is true for both the research questions that may be addressed and the methods used for that purpose. Absolute guidelines do not exist for this, only reasonable assumptions. Theoretical concepts may serve as sources of inspiration when linked to real-life phenomena and pertinent questions; conversely, the specifics of those phenomena and questions require an adequate conceptual and methodological approach. The focus on dynamical systems in sports science is still in its infancy but—given its marvels and intricacies—holds great promise for the future.

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See also Anticipation; Coordination; Freezing; Modeling; Motor Control; Motor Development; Motor Learning; Movement; Perception

Notes

1. The term *phase* refers to a system's state, a term that will be used throughout this article. Switches in macroscopic patterns are seminal for self-organizing systems, as they involve no external agents imposing order from the outside.
2. Chaotic systems exhibit a random-like behavior that is not readily distinguishable from real random behavior originating from a stochastic process based on chance events, that is, the probabilistic counterpart to a deterministic process. In order to do so, special mathematical techniques are required that only work well when specific conditions are met. Unfortunately, when studying dynamical systems in the real world, their behavior will typically result from a combination of both deterministic and stochastic processes that are unknown beforehand. This turns the identification and modeling of the behavior of interest into a formidable problem.
3. At precisely the critical point the dynamics of the order parameter is scale-invariant, that is, it does not change if its time scale is multiplied by a common factor.
4. The theory of networks has its roots in both mathematics and sociology. In 1736, Leonard Euler (1707–1783) solved the “bridges of Königsberg” problem: Is it possible to take a walk crossing exactly

one time each of the seven bridges connecting the two islands in the river Pregel and its shores? Euler proved that this is not possible by representing the problem as an abstract network: a “graph.” This is often considered the first rigorous proof in graph theory.

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E

EATING DISORDERS

The eating disorders of anorexia nervosa and bulimia nervosa are characterized by severe disturbances in body image, eating, and engaging in compensatory behaviors that result in serious medical, psychological, and social problems. For example, eating disorders increase the risk of obesity, nutritional deficiencies, depression and anxiety disorders, chronic pain, osteoporosis, insomnia, neurological symptoms, cardiovascular problems, substance abuse, and death. The criteria for anorexia nervosa include an intense and unrealistic fear of becoming fat, engaging in behaviors intended to produce distinct weight loss, and amenorrhea resulting from the refusal to maintain a healthy weight. The body-image disturbance and consequential denial of the negative health effects of one's low weight are defined as maintaining a weight that is less than 85% of what is considered an ideal body weight for the individual's age, gender, and height. This denial is evident by a physiological criterion of *amenorrhea* that is defined as the absence of at least three consecutive menstrual cycles for women.

Two specific types of anorexia nervosa, restricting type and binge-eating-purging type, are based on how the extreme low weight is reached and maintained. The *restricting type* is defined as the absence of bingeing and purging behaviors. The *binge-eating-purging* type states that during the current episode of anorexia nervosa, the individual also engages in binges (eating inappropriately

massive amounts of food in one set period of time) or purging behavior (self-induced vomiting, misuse of laxatives, diuretics, or enemas). While anorexia nervosa can affect men and women of any age, race, and socioeconomic and cultural background, the occurrence of anorexia nervosa is 10 times higher in the female population than among males.

The criteria for bulimia nervosa are similar to that of anorexia in that they also outline an intense fear of becoming fat, but they differ by including the requirements of powerful urges to overeat and subsequent binges that are followed by engaging in some sort of compensatory behavior in an attempt to avoid the weight gain effects of excessive caloric intake. The fear experienced by an individual with bulimia nervosa is also in regard to body-image disturbance. The paradox is the presence of the uncontrollable urges to overeat, resulting in binges. These binges are defined as occurring within 2 hours and eating an amount of food that is larger than most people would consume in a similar time and setting coupled with a sense of lack of control (inability to stop eating) during the binge. Both the body-image disturbance and binges result in engaging in compensatory behaviors to evade weight gain. Compensatory behaviors are separated into purging and nonpurging types. Purging behaviors include self-induced vomiting, use of laxatives, diuretics, enemas, or medication abuse. In comparison, nonpurging behaviors include fasting or excessive exercising. A qualification for a diagnosis of bulimia nervosa is that the binge eating and inappropriate compensatory behaviors must occur,

on average, at least twice a week for 3 months. In contrast to those with anorexia nervosa, people with bulimia nervosa are able to maintain body weight at or above a minimal normal level.

Bulimia nervosa is considered to be less life threatening than anorexia nervosa; however, the incidence of bulimia nervosa is higher. Bulimia nervosa is nine times more likely to occur in women than men. The vast majority of those with bulimia nervosa are at normal weight. Antidepressants are widely used in the treatment of bulimia nervosa. Patients who have bulimia nervosa are often linked with having impulsive behaviors involving overspending and sexual activity, as well as having family histories of alcohol and substance abuse and mood and eating disorders.

Eating disorders exist and develop on a continuum with health detriment occurring throughout the entire span of the development and maintenance of the disorder. That is, cognitions and behaviors seen in full-blown disorders begin with less frequency or intensity and subsequently increase as the disorder develops. While the antecedents and causes of eating disorders are myriad and complex, excessive exercise has been the focus of much research and clinical attention. Although study design and quality have varied greatly, researchers typically find that a large percentage of individuals with eating disorders engage in excessive exercise. Thus, this entry will primarily focus on the relationship between excessive exercise behavior (exercise dependence) and eating disorders. Note that the entry on the female athlete triad focuses on eating disorders and athletes.

Of importance, according to the *Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision (DSM-IV-TR)*, exercise becomes excessive when “it significantly interferes with important activities, when it occurs at inappropriate times or in inappropriate settings, or when the individual continues to exercise despite injury or other medical complications” (American Psychiatric Association, 2000, pp. 590–591). This definition, however, fails to quantify the amount needed to determine if exercise is excessive. Because excessive exercise and its negative health outcomes have been studied far less than the other diagnostic symptoms, more research is needed to better understand excessive exercise and eating disorders. In an attempt to provide clarity to the excessive exercise construct, researchers advocate for either revising the diagnostic criteria

with regard to excessive exercise, or eliminating excessive exercise as a diagnostic criterion because of lack of empirical support for it. In short, considerable debate exists regarding eating disorder classification in general, and in particular with excessive exercise, regarding how best to define it, or even whether to include excessive exercise as a compensatory behavior for bulimia nervosa.

Relationship Between Excessive Exercise and Eating Disorders

Current exercise guidelines identify the minimum amount of exercise needed to experience health benefits. The guidelines also recommend that an increased amount of exercise is associated with additional benefits. Although increases above the minimum guidelines are encouraged, no consensus exists on *how much is too much*, that is, at what point any further increase in exercise may have a negative effect on one’s health.

By definition, a regimen of exercise that has become detrimental to an individual’s physical and psychological health constitutes *excessive exercise* or *exercise dependence*. Simply stated, exercise dependence is a craving for leisure-time physical activity, resulting in uncontrollably excessive exercise behavior that manifests itself in physiological (e.g., tolerance) or psychological (e.g., withdrawal) symptoms. Characteristics of exercise dependence include exercising despite either injury or illness; experiencing withdrawal effects when an exercise session is missed; and giving up social, occupational, and family obligations to exercise. Exercise dependence may also play a pivotal role in explaining the function of exercise behavior in the development and maintenance of eating disorders.

The relationship between exercise and eating pathology, however, is complex and controversial. Diagnostic criteria, correlational research, and clinical observation show a higher prevalence of exercise in individuals with bulimia nervosa and anorexia nervosa than in non-eating-disordered samples. This is in part because of exercise’s ability to offset caloric intake, resulting in weight loss. For many individuals beginning to experience an eating disorder, diet and compensatory behaviors, such as picky eating, skipping meals, and fasting, may only reduce the number of calories consumed. Consequently, weight loss is slowed and the individual may seek complementary methods to accelerate weight loss. If progress with weight

loss seems slow, compulsive exercise may be added in an attempt to increase weight loss. While this seems reasonable and sufficient to explain the role of exercise in eating disorders, simply examining the amount of exercise does not explain either why or to whom excessive exercise may become problematic. Thus, more recent investigations reveal that psychological factors such as exercise dependence may better explain the role of exercise in eating disorders. Therefore, a closer examination of prevalence rates and psychological factors indicates a much more complicated relationship between exercise and eating disorders.

The belief that exercise is associated with the development and maintenance of eating disorders is based largely on cross-sectional, retrospective, and case study designs that fail to adequately assess and quantify excessive exercise. For example, there is a long standing clinical observation that most hospitalized inpatients receiving treatment for anorexia nervosa engage in excessive amounts of exercise during the development or maintenance of their eating disorder. However, no definition is provided for what is considered excessive exercise. Similarly, recent studies have correlated participation in athletics (populations that engage in large amounts of physical activity) with deleterious eating attitudes related to eating disorders. Thus, researchers have focused on exercise amount contributing to the development and maintenance of eating disorders. However, focusing on exercise amount may be misleading because much of the research examining *excessive* exercise has relied on biased sampling methods using unvalidated self-report exercise measures that lack a clear, concise, and consistent definition of how much exercise is excessive. Furthermore, many of the operational definitions used for excessive exercise fail to meet the minimum amount of exercise needed to achieve the health-related benefits of physical activity.

Primary Versus Secondary Dependence and Eating Disorders

Most of the research examining the relationship between excessive exercise and eating disorders has focused on the amount of exercise contributing to the development of eating disorders but has overlooked psychological variables that may mediate such a relationship. Understanding the psychological antecedents of exercise may help clarify the

relationship between eating disorders and excessive exercise by offering insight into the distinction between primary versus secondary exercise dependence. *Primary exercise dependence* occurs when the individual meets criteria for exercise dependence and continually exercises solely for the psychological gratification resulting from the exercise behavior. *Secondary exercise dependence* occurs when an exercise dependent individual uses increased amounts of exercise to accomplish some other end, such as weight management or body composition manipulation.

Simply stated, it is important to distinguish whether the individual is exercising excessively to satisfy the need to exercise (primary dependence) or if they are engaging in increased amounts of exercise as a compensatory behavior that is secondary to other pathology, such as an eating disorder (secondary dependence). Because exercise can be used as a compensatory behavior to prevent or reverse weight gain, secondary exercise dependence in the context of eating disorders occurs when individuals meet the criteria for exercise dependence and continually exercises to manipulate and control their own body; thus, exercise dependence is secondary to an eating disorder. Recently, researchers have found that exercise dependence symptoms, not exercise behavior, mediate the relationship between exercise and eating pathology. Thus, psychological factors, not the amount of exercise, may better explain why the exercise dependence–eating disorder relationship exists.

The Exercise and Eating Disorders Model

The limitations of biased clinical observations, retrospective research designs, vague operational definitions of excessive exercise, inconclusive animal research, and overlooking potential mediating psychological variables support the need for theoretically driven models that explain the relationship between eating disorders and the psychological motivation as well as the physical effects of exercise.

Previous models have been advanced that postulate how and why obligatory attitudes toward exercise may influence the development and maintenance of eating pathology and disorders. However, these models offer limited insight into why the benefits typically experienced as a result of regular exercise do not occur in eating disorders. For example, exercise may impart positive

improvements on the eating disorder risk factors of anxiety, body image, depression, stress reactivity, and self-esteem in non-eating-disordered populations. Similarly, cardiovascular benefits, such as increased cardiac mass, increased stroke volume and cardiac output at rest and during exercise, lower resting heart rate and blood pressure, and a decreased tendency for blood clotting are pertinent to eating disorder research because cardiac damage can occur early during eating disorder development. Exercise also has the ability to reduce adiposity, thus contributing to a leaner, more fit, and culturally ideal body type. Moreover, socio-cultural pressures to be thin and social comparison are risk factors for the development of eating disorders. Furthermore, the metabolic benefits of exercise include decreased triglycerides and increased high-density cholesterol, increased insulin mediated glucose uptake, and a possible increase in resting metabolism. Finally, exercise increases skeletal muscle mass and bone density in youth and it is related to the retention of bone mineral density in older adults. This has implications in the development of osteoporosis, a common consequence of prolonged eating disorder behaviors. Thus, exercise is an effective intervention for many physical and psychological health issues, and yet recent recommendations for research to reexamine the role of exercise in eating disorders have been largely overlooked.

Heather Hausenblas, Brian Cook, and Nickles Chittester presented a conceptual model examining such aforementioned relationships (see Figure 1). Their Exercise and Eating Disorders Model states that regular exercise is associated with improvements in several physical (cardiovascular, metabolic benefits, decreased adiposity, and increases in bone density), psychological (body image, depression, anxiety, stress reactivity, and self-esteem), and social benefits that are also risk factors, maintenance factors, outcomes, or diagnostic criteria for eating disorders. Hence, this exercise and eating disorders model has consolidated and supported several narrative and meta-analytic reviews that have shown exercise's ability to impart positive improvements on eating disorder risk, development, and maintenance factors. The model also extends our current understanding of the relationship of exercise and health status by including exercise dependence. That is, exercise dependence may explain why the development of eating disorders may supersede the expected benefits of

exercise. Simply stated, this model posits that in the absence of pathological psychological factors such as exercise dependence, the benefits conveyed by regular exercise—improvements in depression, anxiety, stress reactivity, self-esteem, and body composition—may counteract the risk factors for eating disorders like body dissatisfaction, depression, anxiety, and increased body mass.

Recent research by Cook and colleagues, Hausenblas, Daniel Tuccitto, and Peter Giacobbi, has provided initial support for this exercise and eating disorder model, revealing that the psychological health benefits, conveyed by exercise can reduce the risk of eating disorders.

First, university students completed self-report measures of physical and psychological quality of life, exercise behavior, eating disorder risk, and exercise dependence symptoms. Structural equation modeling analysis found support for the mediation effect of exercise dependence on eating disorders as well as the effect of psychological well-being on eating disorders. Together, exercise behavior, psychological well-being, and exercise dependence symptoms predicted 22.9% of the variation in eating disorders. Thus, these results indicated that the psychological health benefits conveyed by exercise reduced eating disorders. These results were replicated in a more diverse sample of college students in another study by Cook and various colleagues in 2011.

Exercise Interventions and Eating Disorders

The initial tests of the Exercise and Eating Disorders Model suggest that the model may synthesize two divergent lines of research. That is, exercise may play a role in the development of eating disorders when exercise dependence is simultaneously present. Similarly, the psychological health benefits of exercise may also reduce eating disorder risk for individuals without exercise dependence.

Conclusion

Preliminary research has found that mild to moderate exercise appears to attenuate eating disorder symptoms in patients suffering with anorexia or bulimia nervosa. However, before exercise interventions are considered mainstream, future randomized controlled trials are needed to establish a dose-response relationship for exercise and identify the conditions (e.g., type or severity of eating

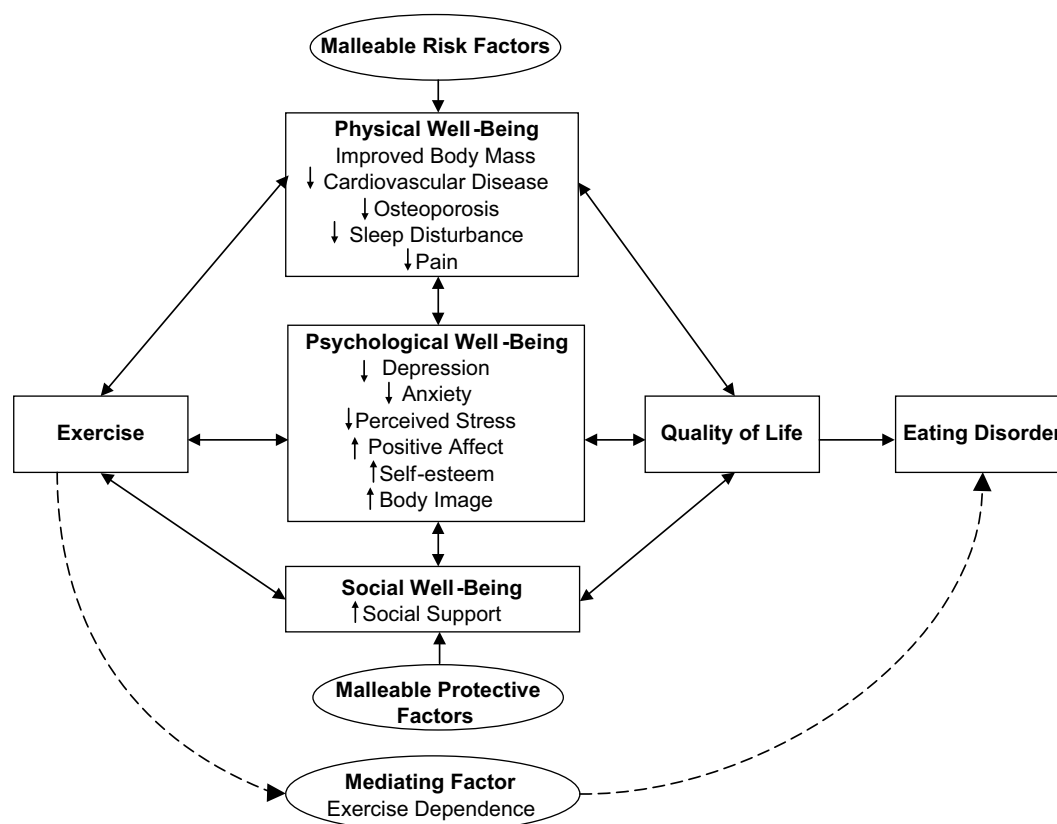


Figure 1 The Exercise and Eating Disorder Model

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disorder, minimum weight level, and exercise environment) under which an exercise intervention may be undertaken. Interest in exercise dependence and eating disorders is recent but provides a context as to why the relationship between exercise and other pathologies, such as body-image disturbance and eating disorders, exists.

Heather Hausenblas and Brian Cook

See also Body Dissatisfaction; Body Image; Exercise Dependence; Female Athlete Triad; Psychological Well-Being

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ECOLOGICAL THEORY

Ecological theory is a global perspective or metatheory, because the broad heading represents several scientific approaches that view human behavior as the result of the relationship between individuals and their environments. Ernest Haeckel, German zoologist and evolutionist, coined the term *oekology* in 1866 to define a field of study that examined organisms in their living environments. Ellen Swallow Richards, chemist and founder of the field of home economics, brought the term to English-language use in 1907. She described human ecology as the study of the influence of the surroundings of human beings on their lives. Swallow Richards provided an example of the application of ecological theory to physical activity with several observations about the interactions between individuals and their environments to promote healthy lifestyle behaviors. For example, in her 1904 publication *The Art of Right Living*, she noted that there is no better way to raise a child to perform healthy habits than to expose a child to environments that utilize the natural desire for effective movements. She recommended giving a child a garden because the care of a garden bed combines exercise, enjoyment, and adds indirect instruction on attaining satisfaction through work. Since the early 1900s, the study of human ecology has been fragmented, with work occurring in a range of disciplines, such as anthropology, geography, sociology, psychology, and kinesiology. The focus here is on describing several ecological perspectives that fall under the heading of ecological theory and illustrating their influence on sport and

exercise psychology. By illuminating the influence of ecological perspectives on sport and exercise psychology, this entry offers a description of an ecological metatheoretical approach. The ecological perspectives described include Kurt Lewin's ecological psychology, Roger Barker's ecological psychology, James Gibson's ecological approach to human perception, Uri Brofenbrenner's ecological systems theory, and recent application of these ecological perspectives to inform physical activity promotion.

Kurt Lewin's Psychological Ecology

In 1943, Kurt Lewin used the term *psychological ecology* to propose that, for an understanding of individual and group behavior, it was necessary to study the opportunities and constraints of environments. But Lewin did not focus on studying environmental opportunities and constraints. Rather his major influence on sport and exercise psychology was his well-known equation $B = f(P, E)$, which stated that behavior is a function of the person and one's environment. While the person–environment interaction concept is adopted by social cognitive approaches, it is a central focus of ecological theory. Lewin viewed individual behavior as a function of the immediate social environment and believed that the social group to which individuals belong was the basis for individual psychological processes. The ecological perspective of Lewin's work departs somewhat from most contemporary ecological theories in that he considered the environment to only have an indirect influence on people's behavior and it was people's subjective perception of the environment that should be the central focus of study and intervention. In contrast, ecological theories place greater emphasis on the study of and intervention on actual, rather than perceived, features of the environment.

James Gibson's Ecological Approach to Perception

James Gibson rejected the premise that perception is part subjective and part objective. He argued that there is a subjective aspect and an objective aspect to every experience, but this does not mean that there is subjective determination of perception. Gibson's approach differed from Lewin's because the target of study and intervention began with environmental features. Because perception was

described as a direct process of extracting information from the environment, the differences among individuals in perception were largely attributed to attention to environmental features. Gibson introduced the concept of affordances. An *affordance* was defined as a latent feature of the environment that provides action possibilities. An affordance was viewed as ecological (person–environment interaction), because an object’s use is always dependent on individual capabilities. For example, a 7-foot-tall individual may not perceive a small chair as affording the opportunity to sit, while a 5-foot-tall individual perceives the same objective features and sits down. Much of the influence of Gibson’s ecological approach within sport and exercise psychology has been in the area of motor skill learning and performance.

Roger Barker’s Ecological Psychology

The study of objective features of the environment in the daily lives of individuals was the center of the work of Roger Barker from the late 1940s to the early 1970s. Although Barker spent 2 years as a postdoctoral fellow with Kurt Lewin and was strongly influenced by him, he shifted his path of research from a focus on the subjective perception of the environment to conduct studies of human behavior within natural environments. He set up the Midwest Psychological Field Station in the small town of Oskaloosa, Kansas, developed empirical methods for observing the activities of individuals in their everyday settings, and did copious measurements of daily life.

Barker’s *ecological psychology* focused on person–environment interaction within the immediate environmental experience, which he labeled a *behavior setting*. Behavior settings were defined as the social and physical environments where behavior takes place. Behavior settings are a natural phenomenon that is bounded by time and place, a specific measurable unit, and recognized by people going about their daily lives. A school, for example, is not a behavior setting. But a basketball game at a school could be a behavior setting. It has a standing pattern of behavior, is not only a characteristic of the individuals involved but can function with different individuals, and is tied to a time and place.

Barker established that the strongest influence on behavior was not personality or global social inputs but rather the places in which people live

their lives. For example, when a young girl entered a worship service, she behaved “worship service.” If she left the worship service, and then went to a basketball game, she behaved “basketball game.” After identifying the behavior setting as the unit of study, Barker shifted his focus from studying individual behavior to studying the environment. Barker and his colleagues provided evidence that knowledge about the place of behavior was more important than individual differences in predicting individual behavior.

Behavior settings were shown to influence physical activity. Studies of small schools and large schools were shown to have a similar number and types of behavior settings, such as basketball games and practices. But small schools offered more behavior settings relative to the population (understaffed). For example, a basketball game requires five players plus substitutes on each team. A basketball practice requires more players, but the number involved in practice games is limited to 10 plus substitutes. Although basketball game or practice behavior settings are similar at both large and small schools, at large schools behavior setting forces eliminate individuals from participation in physical activity through a selection process because of being “overstaffed.” At small schools, however, the behavior setting may foster recruitment of individuals to participation in physical activity because more players are needed for a game or practice than may initially turn out. While Barker’s theoretical perspective has not had a large impact on sport and exercise psychology, many contemporary ecological perspectives have adopted the term *behavior setting* to represent a social and physical environment for physical activity.

Uri Bronfenbrenner’s Ecological Systems Theory

Uri Bronfenbrenner outlined his theory of the ecology of human development beginning in 1958 and continued to develop his ecological perspective over 5 decades. Bronfenbrenner’s theory was strongly influenced by Kurt Lewin, under whom he worked as a young PhD in the Office of Strategic Services of the Army during World War II. Lewin constantly reminded Bronfenbrenner that space is not physical but psychological. This focus on the perceived environment contributed to distinguish Bronfenbrenner’s approach from Roger Barker’s.

In addition, Bronfenbrenner's ecological systems model focused on the impact of the environment on the individual. He viewed individual development as the result of being exposed to many different environments over time.

Bronfenbrenner's largest contribution to contemporary sport and exercise psychology was the concept that the environment is a set of nested systems or multiple levels of influence, each inside the next like a set of Russian dolls. The most proximal of these systems was termed the *microsystem*. This is the immediate situation where person–environment interaction occurs. Key to the nested-system concept is the idea that the immediate microsystem environments that people encounter, while the key to behavior and development, are the outcomes of the larger systems in which they are housed. The next-level (larger) system, called the *mesosystem*, was defined as a system of microsystems. For example, a child may encounter the school bus, classroom, school lunch, recess, and home environments throughout the school day. These could all be considered microsystems. The mesosystem is the combination of exposure to all these microsystems for the developing individual. The next level of the environment is the *exosystem*, which is a microsystem that does not contain the developing person but impacts on the child's micro- and mesosystem. For the child, the exosystem may include the parent's workplace or a school district's administration. Decisions made within each of these microsystems have an influence on the microsystems that the child encounters. The next level in the multilevel system is the *macrosystem*, which was defined as the larger sociocultural context. Bronfenbrenner proposed that cultural influences filtered down to the microsystem level to influence the daily practices of individuals that create the environments with which individuals interact. Later in his work, Bronfenbrenner added the *chronosystem* to his theory to recognize the impact of the patterning of environmental events over time. The last development of Bronfenbrenner's ecological systems theory was to imbed it within a bioecological model that addressed the genetic influences on behavior and development within the ecological environment.

While there has been little true integration across the ecological perspectives described earlier, they share many principles that can inform sport and exercise psychology research and practice. A summary of the principles of ecological theory

suggests (a) the focus of study represents person–environment interaction; (b) study and intervention begins with examining characteristics of the environment; (c) person–environment interaction leads to behavioral outcomes and proximal developmental processes; (d) behaviors and proximal developmental processes occur in environments that, when studied directly, exhibit characteristics of self-regulating behavior settings; and (e) multiple levels of influence operate to impact the quality of the person–environment interaction over time to determine physical activity and development.

Application of Ecological Theory to Physical Activity Promotion

Perhaps one of the strongest influences of ecological theory within sport and exercise psychology has been in the study of physical activity promotion. Physical activity promotion researchers have embraced the principle of multiple levels of influence, as well as the focus on environmental characteristics. The public health focus of physical activity promotion researchers directs their attention to the primary target of population health. Individually targeted behavior change interventions that do not address environmental factors have not been successful in changing the health behaviors of communities. To solve this problem, in 1988, Kenneth R. McLeroy proposed that interventions be targeted at multiple levels of influence, defined as *interpersonal*, *organizational*, *community*, and *public-policy* factors. Corresponding with physical activity being identified by the U.S. Surgeon General as a major public health problem in the late 1990s, first mention of the adoption of an ecological approach within the sport and exercise literature occurred. A few public health scientists, including David Dzewaltowski, James Sallis, and Neville Owen charged researchers to embrace an ecological perspective. While several years and significant work has been conducted, there is still no consensus on an ecological theory for physical activity promotion or on how to define the levels of influence.

Ecological principles have been applied in the design of some interventions. Physical activity interventions for diabetes management have included, in addition to individual health behavior change components, multilevel strategies targeting healthy system change and community resources for regular physical activity. Interventions targeting settings

such as worksites and schools have included broad environmental and policy components. In schools, for example, students may receive curriculum on self-regulation skills, school policy change interventions to increase the amount and quality of physical education, and community outreach to increase physical activity opportunities. Comprehensive community multilevel programs may include legislation for physical environmental changes like parks and walking and bicycle paths; mass media campaigns; and targeted school, worksite, and other delivery setting programs that include building social and physical environments for physical activity and curriculum-based education on behavior-change skills.

Many community-based physical activity approaches also use participatory strategies, whereby individuals in the community are empowered to lead change. These social and physical environment change leaders become champions of physical activity initiatives. Consistent with the ecological approach, interventions are tailored to the natural environments where people live and it is the residents of these environments that are best equipped to understand these unique local variables. Since ecological-psychology perspectives focus on the multiple systems that filter down to create local environmental variables, this multiplicity presents substantial logistical challenges for study and practice.

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See also Health Promotion; History of Exercise Psychology; Leadership in Sport: Social Cognitive Approaches; Motor Learning; Perception; Psychological Skills Training

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EFFORT

Individuals possess a remarkable ability to detect and interpret sensations arising from the body during physical work. As noted by William P. Morgan, terms such as *perceived exertion*, *perception of effort*, and *effort sense* have been used to describe this psychophysiological phenomenon. Interest in this aspect of performance was initiated by the pioneering work of the Swedish psychophysicist Gunnar Borg in the early 1960s. He introduced the concept of perceived exertion and proposed ways by which overall exertion, breathlessness, and localized sensations of fatigue could be numerically and verbally rated. This led to the development of rating of perceived exertion (RPE) scales, the most common of which is the Borg 6–20 RPE Scale. This entry describes Borg's concept of perceived exertion and outlines its underpinning factors, methods of measurement, and how it has been used.

Effort Continua

Borg originally proposed that subjective responses to an exercise stimulus involve three main effort continua: perceptual, physiological and performance. Although a number of more complex models of perceived exertion have been developed, they are founded on the same three effort continua. Borg used the *perceptual continuum* as the initial basis from which to explore the derivations of the ratings of perceived exertion, on the premise that perception plays a fundamental role in

human behavior and in how one adapts to a situation. He stressed that the perceptual continuum will be influenced by a person's subjective experience and that these experiences will be directly affected by psychological traits. The physiological continuum includes a wide variety of variables, such as heart rate, blood lactate, oxygen uptake, and ventilation, which may be characterized by different growth curves as a function of exercise intensity. For example, heart rate and oxygen uptake are characterized by a linear growth in relation to increases in intensity as measured by power output (watts), whereas blood lactate concentration and ventilatory volume are characterized by a nonlinear (positively accelerating) growth function.

The influence of the perceptual and physiological continua on RPE will also be moderated by the situational characteristics of the performance (the third continuum). In this regard, one has to take into account the nature of the performance and the social and physical environment in which it takes place. Performance may involve timed (short or long) incremental stages to exhaustion, the highest workload that can be sustained for a specific period of time, the greatest distance one can cover in a given time period or the fastest time in which one can cover a given distance. Submaximal performances may involve monitoring the time to exhaustion at a given exercise intensity (for example, at a given percentage of maximal oxygen uptake or at an intensity corresponding to a ventilatory threshold reference point). Knowledge of the duration of the task or distance to be completed is also a critical factor in gauging effort. Social environmental factors may include whether or not performance is alone or with others, whether it is competitive, the presence or absence of an audience, and how supportive the audience is. Physical environmental factors include location, ambient conditions of temperature, humidity, and altitude, and any external distractions such as music and visual stimuli.

So, effort perception involves the collective integration of afferent feedback from cardiorespiratory, metabolic and thermal stimuli and feed-forward mechanisms to enable an individual to evaluate how *hard* or *easy* an exercise task feels at any point in time. It is moderated by psychological factors, such as personality traits, cognition, memory, previous experience and understanding of the task; situational factors, such as knowledge of the

end-point, duration, and temporal characteristics of the task; and social and physical environmental factors. The extent to which these factors moderate the perception of exertion has been shown to be exercise intensity dependent.

Measurement and Calibration of Effort

The most common method of measuring perceived exertion is the Borg 6–20 Category Scale followed by the Borg Category Ratio 10 (CR10) Scale. The development and correct use of these scales is described in detail by Borg. The scales are designed to assess sensations of exertion in relation to physiological markers, such as heart rate and oxygen uptake, which rise commensurately with increments in exercise intensity. In both scales, numbers are anchored to verbal expressions. For example, in the 6–20 Scale the numbers 6, 9, 11, 13, 15, 17, 19 and 20 are anchored to verbal expressions of “no exertion at all, very light, light, somewhat hard, hard, very hard, extremely hard and maximal exertion,” respectively. As research questions and applications involving perceptions of exertion have developed, a number of additional scales for adults have been developed. These include Foster's 0–10 Session RPE Scale, used in the calculation of an athlete's training load, and Garcin's 1–20 Estimated Time Limit Scale, which is used in the estimation of the time remaining until volitional exhaustion.

As the antecedents of RPE include the memory of physical work experiences and the level of cognition and understanding, a number of simplified RPE scales for children have also been developed. For a child to perceive effort accurately, and then reliably produce a given intensity at a given RPE, learning must occur. Implicit in the process of learning is practice and the cognitive ability of the child. According to Jean Piaget's stages of development, children around the ages of 7 to 10 years can understand categorisation but find it easier to understand and interpret pictures and symbols rather than words and numbers. For this reason, more recent pediatric RPE scales include pictures to portray the degree of effort and acute fatigue experienced and understood by the child. These developments have also recognized the need for verbal descriptors and terminology that are more pertinent to a child's cognitive development, age, and reading ability. These scales therefore use a limited number range based around 0 to 10 or

1 to 10, pictorial descriptors, and wording that is more familiar to children (e.g., Roger Eston and colleagues' CERT, BABE, CALER, E-P Scales and Robertson's OMNI Scales).

Central and Localized RPE

Sensations of effort can be used to assess overall respiratory-metabolic (central) perceptions of exertion or they can be used to differentiate between central and peripheral (local) signals of exertion. For example, differentiated ratings of perceived exertion may be used to segregate the sensations arising from the upper body and the lower body during cycling exercise or during rowing, running, or stepping. For cycling, localized perception of exertion in the leg muscles tends to dominate the overall perceived exertion response. Consequently, the strength of the perceptual signal of exertion is greater for a given work rate for cycling compared to treadmill exercise.

Estimation and Production of Effort

The sensation of effort has been applied in a variety of ways to assess and understand performance. It is generally observed that RPE measured during an exercise bout increases as exercise intensity increases in adults and children, particularly when the exercise stimulus is presented in an incremental fashion. Such relationships have been most frequently observed using the so-called passive *estimation* paradigm. In this way, a rating of perceived exertion is given in response to a request from the exercise scientist or clinician to indicate how "hard" the exercise feels. The information can be used to assess changes in fitness using standardized submaximal exercise test procedures, such as in the Lamberts and Lambert Submaximal Cycling Test. It may also be used to assist the clinician or coach in prescribing exercise intensities. For example, an exercise intensity (e.g., heart rate, work rate or oxygen uptake), which coincides with a given RPE, may be prescribed by the coach or clinician.

Given the robust relationship between RPE and exercise intensity, particularly if known for an individual, the RPE can be used as a subjective guide to gauge exercise intensity during cardiorespiratory and resistance exercise. Thus, an active *production* paradigm can be employed whereby the individual is requested to regulate exercise intensity to match specified RPE values. A number

of studies show support for the use of the RPE in this way for both aerobic fitness training and for estimating aerobic power and fitness.

Assessment of Training Load

As optimization of training load is a key factor for peak performance, the quantification of effort is considered to be important. Recently, the session RPE method, calculated by multiplying the relative perceived exertion of the session (scale of 0–10) by the duration of the exercise (in minutes), or the number of repetitions for resistance training, has become a popular method of assessing acute and chronic training load in athletes.

Prediction of Maximal Exercise Levels

The RPE elicited from submaximal work rates can be used to provide acceptable predictions of maximal aerobic power ($\text{VO}_{2\text{max}}$) that are as good as or better than heart rate. This is true for healthy active and sedentary groups and able-bodied and paraplegic athletes. The RPE also predicts 1-RM (one repetition maximum) in adults and children and maximal intermittent vertical jump performance.

It has been noted that RPE 20 is infrequently reported at volitional exhaustion during maximal incremental exercise tests or constant-load tests to maximal volitional exhaustion. As the subjective limit of fatigue normally occurs around RPE 19 (extremely hard) on the Borg 6–20 scale, studies have shown that RPE 19 is a better predictor of $\text{VO}_{2\text{max}}$ than the theoretical maximal RPE 20.

Perceptually Regulated Exercise Testing

On the basis that RPE alone may be used to regulate exercise intensity, perceptually regulated exercise testing is an alternative method of estimating maximal exercise capacity and training status. The standard procedure involves a series of short incremental stages (2, 3 or 4 min.) that are clamped to RPEs of 9, 11, 13, 15 (and sometimes 17). Extrapolation of the individual RPE: intensity relationship to a maximal RPE (19 or 20) enables $\text{VO}_{2\text{max}}$ or maximal work rate to be estimated with reasonable accuracy. This method has the advantage of allowing subjects the autonomy to set the intensity of exercise to a given RPE, through changes in pace, work rate, or gradient. First applied in cardiac patients, the

efficacy of this method has been confirmed across a broad range of ages, fitness levels, and levels of physical ability.

Relationship With Time or Distance Remaining to Exhaustion

When the RPE is expressed against the proportion (%) of the time or distance completed, regardless of the length of an effort, the RPE rises similarly relative to the percentage of distance or duration completed or yet to be completed. This has been observed in open-loop, fixed intensity exercise to exhaustion (in which the distance or time is unknown) and during closed-loop tasks (the duration or distance to the end point is known) despite the effects of changed environmental conditions and competitive distances. Certainty about the exact duration and end point has been shown to affect both the RPE strategy and performance, as the rate of increase in perceived exertion is not always constant in all conditions, but changes in relation to the degree of certainty about the endpoint of exercise as well as exercise duration. Disruption in the rate of RPE increase occurs when uncertainty about the anticipated end point is invoked by deception during fixed intensity cycling and treadmill exercise.

The practical implication of this knowledge is that it is theoretically possible to use the rate of increase in the RPE to estimate the exercise duration or time remaining to exhaustion at a given work rate or pace. It is postulated that athletes continually compare their *momentary* or *conscious* RPE with an expected RPE (the “template RPE”) through a process of internal negotiation at a particular portion of a race, and adjust pace to match the anticipated and experienced values for RPE.

Murielle Garcin and colleagues introduced the 1–20 Estimate Time Limit (ETL) scale to provide a direct measure of how the effort at any given point during exercise can be used to provide a subjective estimation of the time remaining to exhaustion. The validity and applications of the scale have been reviewed by Coquart et al. (2012). In this scale, the numbers 17, 13, 9 and 4 relate to an anticipated end time to exhaustion of 4, 15 and 60 minutes and 2 hours, as listed on the ETL Scale, respectively. The ETL scale provides further information on the psychological load (intensity and duration) of exercise and allows for a direct subjective estimation of time that can be maintained

at any intensity and at any given instant. Use of the ETL, in conjunction with the RPE, provides a further method of understanding the relationship between perceived effort, exercise intensity, and the duration that remains until physical exhaustion.

Conclusion

The concept of perceived exertion is a key variable of interest in sport and exercise science. It has applications in children and adults, from sedentary through elite athletic status. Practitioners have used it successfully with paraplegic, partially sighted, obese, cardiac and other clinical populations. It can be used to predict the limits of exercise, to regulate exercise intensity, to assess training load and to compare training status. However, its utility has to be considered within the dynamic context of the fundamental continua originally identified by Borg.

Roger Eston and Gaynor Parfitt

See also Brain; Fatigue; History of Exercise Psychology; Mindfulness; Personality Traits and Exercise; Psychophysiology; Timing

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ELECTROENCEPHALOGRAPH (EEG)

One approach to understanding the effects of exercise on the brain and the cortical processes underlying peak performance is to measure brain activity using electroencephalography. Electroencephalography is a noninvasive technique that uses highly conductive silver or silver chloride (Ag/AgCl) electrodes to record brain activity, which is also referred to as electroencephalographic (EEG) activity.

The EEG recording is actually a measure of electrical signals that are produced by neural cells in the brain and are measurable at the scalp. Because the electrical signals must pass through the dura mater, cerebrospinal fluid, skull, and skin before reaching the electrodes, these signals are recorded in a small unit of measurement called microvolts. EEG can be measured using either single electrodes (flat metal disks) that are attached at particular locations on the head or an electrode cap that has fixed electrodes sewn into the cap. The electrodes are adhered to the scalp using a specific type of gel or paste that maximizes conductance. The electrodes are connected by wires to an amplifier and a computer that records the brain activity. The internationally standardized 10–20 system developed by Herbert Jasper in 1958 is frequently used to communicate particular scalp locations used to record EEG. The locations of the sites are judged relative to landmarks on each individual's scalp, including the bridge of the nose, the bony protuberance at the base of the skull, and the midpoints of the ears. This original system included 21 electrodes that reflect measurement at frontal (F), temporal (T), parietal (P), occipital (O), and central (C) regions of the brain (Figure 1). In this system, the odd number indicates that the locations are on the left side of the brain and the even numbers indicate that the locations are on the right side of the brain. More modern EEG systems can include electrode placements that record information from 32, 64, 128, or up to 256 sites. In addition to the electrodes that are used to record EEG from sites of interest, an additional electrode called the *reference electrode* is used to subtract out basal activity so that the resultant recording from the sites of interest is reflective only of activity at those sites. This electrode is typically placed on the bridge of the nose or on an earlobe.

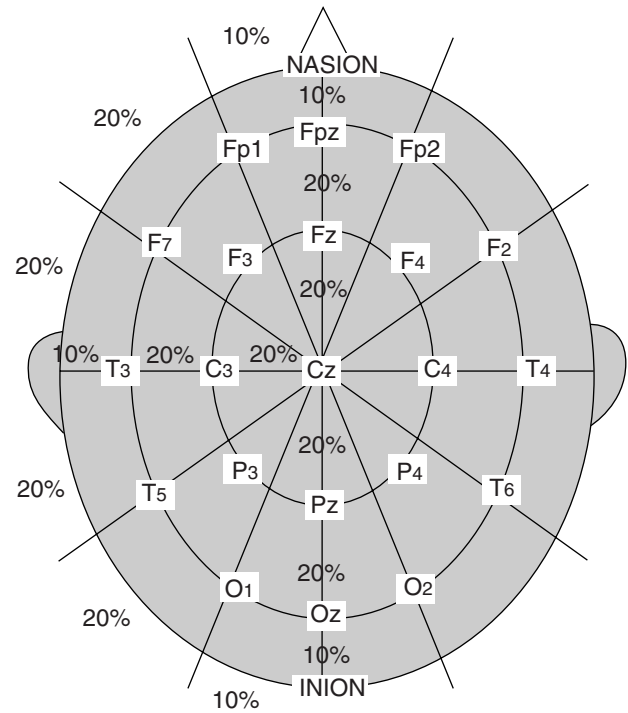


Figure 1 Jasper 10–20 System for Electrode Identification

Source: Adapted from Teplan, M. (2002). Fundamentals of EEG measurement. *Measurement Science Review*, 2(2), 1–11. Used with permission.

There are several computer software programs that can be used to record and analyze EEG data. In addition to recording EEG activity, some software programs can also be used for creating and presenting different types of visual and auditory stimuli so that EEG activity can be recorded relative to behavioral responses to those stimuli. More advanced software programs integrate EEG data with measures using other neuroimaging techniques like magnetic resonance imaging (MRI), positron emission topography (PET), and single photon emission computed tomography (SPECT) data. When collecting EEG data, there are several important things to remind participants of when preparing for the test. Participants should avoid taking certain medications that may alter the brain's electric activity, such as sedatives, muscle relaxants, and sleeping aids. Participants should also avoid caffeine and exercise within 12 hours of the test. The use of hair spray, gels, oils, or other hair chemicals should be avoided on testing day. The participant should be instructed to be as still as possible and remain quiet when assessing resting EEG. However, current EEG technology does

permit EEG measures to be taken during exercise and sport performance. When the EEG data is collected, analog filters are used to limit the recording of low and high frequency signals outside the range of interest, such as low frequency signals from breathing. Nonetheless, prior to analyzing the EEG data, several steps must be taken to further ensure that the collected data is clean. Clean data most purely reflects the EEG data itself and does not include data that reflect muscle movement. Thus, in addition to the EEG electrodes, participants typically wear two electrodes placed around the outside of the eye to record eye blinks. Eye blinks and other muscle movement (e.g., from a cough, tensing muscles in the jaw) alter the EEG signal in a readily observable fashion and portions of the recording that contain these artifacts can be excluded prior to analysis using either manual or automated techniques. Once the data has been cleaned, it is then digitally filtered, which decreases extraneous data in the signal while maintaining the integrity of the EEG data. At this point, the data is ready for analysis.

Spontaneous EEG

There are four main wave forms measured by EEG: alpha, beta, theta, and delta. *Alpha activity* consists of waveforms that occur 8 to 12 times per second (Hz) and is interpreted as reflecting a relaxed state. Alpha activity tends to be low during mentally challenging situations. In sport psychology research, alpha activity has been associated with being relaxed and focused and mentally prepared for performance. EEG data has been used to distinguish mental states between novice and expert performers prior to task execution like a golf putt or free throw. Imagery training has also been linked to increased alpha activity. Biofeedback training has been used to teach participants to increase alpha activity with the expectation that this will result in better performance.

Beta waves (13–30Hz) are the most common type of EEG activity during wakefulness and are present during mental thought and activity, particularly during decision-making processes. Theta waves (4–8Hz) appear during drowsiness and light sleep. Delta waves (.5–3.5Hz) are found during periods of deep sleep and are characterized by very irregular and slow wave patterns. Delta waveforms are of particular interest to researchers exploring the effects of exercise on sleep quality.

In addition to interest in activity at a given site, there is also interest in examining hemispheric asymmetry in EEG responses. In the sport performance literature, expert performers typically display a *quieting* of left hemisphere activation, as inferred by greater EEG alpha power. This has been interpreted as being indicative of expert performers' ability to block out distractions, unwanted emotions, and negative thoughts prior to motor responses and suggests that there is a causal link between this ability and performance. Exercise psychology literature has focused on EEG asymmetry as a marker of affective changes often associated with exercise. The cerebral lateralization hypothesis suggests that anxiety reductions and enhanced affect caused by exercise are due to a decrease in right, relative to left, hemisphere activation.

Event-Related Potentials

Event-related potentials (ERPs) are time-locked waveforms that are identified after averaging the EEG response recorded relative to a particular event. The event-related portion of the name expresses that the timing of the EEG activity is directly related (time-locked) to an event; this is typically a stimulus presentation (e.g., seeing or hearing a stimulus) or a voluntary motor response that is made (e.g., pushing a button). The EEG signal is averaged using the presentation of the stimulus as the anchor so that the patterns of activity are synchronized to the same event. The potential portion of the name expresses that ERPs reflect cumulative electrical potentials generated by neurons in the brain.

There are several different kinds of ERPs. ERPs that occur in response to a stimulus are called *sensory-evoked potentials*; ERPs that occur relative to a motor response are called *motor potentials*. A common experimental paradigm that is used to assess ERPs is the *oddball paradigm*. In this paradigm, the participant is asked to watch a computer monitor on which frequent stimuli (e.g., an X) and infrequent stimuli (e.g., an O) are displayed. The participant is instructed to respond as quickly as possible to one stimulus (e.g., the X) by pressing a button but is asked not to respond when the other stimulus (the O) is displayed. The EEG signal is time-locked to the presentation of the stimulus, which allows sensory-evoked potentials to be observed. Interpretations of ERP data

are based on the amplitude of the potential (the vertical distance from the baseline activity to the peak or trough of the component) and the latency of the potential (the elapsed time to the peak amplitude of the component from the time-locked event). The amplitude and latency of the sensory-evoked potential differ depending on whether the stimulus was the frequently occurring stimulus or the rarely occurring stimulus and also depending upon whether the participant responded correctly, for example, pressing the button when X was presented or incorrectly by pressing the button when O was presented to the stimulus. The motor potential is observed when the EEG signal is time-locked to the initiation of the motor response to press the button.

Two additional types of ERPs are referred to as *slow* ERPs because they occur over a relatively longer period of time than do the sensory-evoked and motor potential. Contingent negative variation (CNV) is observed when a participant is given a warning stimulus prior to presentation of a stimulus that is to be responded to, and its amplitude is increased by attention and decreased by distraction. A readiness potential is evident in the period prior to a voluntary movement and its amplitude has been shown to be related to motivation and to movement speed.

Once the data have been cleaned and averaged, the resultant waveforms are examined to identify the components of interest. The names of the components of the ERPs reflect the direction of the waveform (positive, *P*, or negative, *N*) relative to the baseline activity prior to the presentation of the stimulus and the approximate timing of the component relative to the stimulus. As previously described, components of the ERP are quantified based upon their amplitude and latency.

One ERP component that is of interest to sport and exercise psychology is the P300. The P300 is a positive waveform that occurs between 250 and 500 ms after the stimulus presentation. This component has been shown to be linked to the allocation of attention. The amplitude of the P300 is thought to be indicative of increased attention while the latency provides evidence of the time necessary to evaluate the stimulus (speed of cognitive processing). When considered simultaneously, the amplitude of the P300 is expected to be larger and the latency shorter when participants are using relatively few attentional resources to evaluate a stimulus and perform a task. The CNV (slow ERP)

occurs approximately 260 to 470 ms after a warning stimulus and is a sustained negative component of the waveform. The CNV is thought to be indicative of a participant's expectancy or anticipation of a stimulus. In the sport and exercise psychology literature, higher amplitude CNV has been linked to quicker reaction time.

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See also Brain; Neuroscience, Exercise, and Cognitive Function; Psychophysiology

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ELECTROMYOGRAPHY (EMG)

Electromyography (EMG) is an electrical record of muscle activation. It is a measure that is recorded by placing sensors on the skin of a participant and monitoring changes in the electrical activity of the underlying musculature relative to movement. Greater levels of activation occur as motor unit recruitment increases in order to perform movements successfully. EMG provides a useful window into mind–body relationships, both within exercise and sport domains. While the use of EMG in sport

and exercise psychology has been fairly sporadic, there are pockets of research in both exercise and sport where the use of EMG has provided valuable insights into the mind–body relationship.

One area in which EMG has been used with interesting results is in the study of the *tranquilizer effect* of exercise. More specifically, the study of the anxiolytic (anxiety-reducing) properties of exercise has been a popular topic for decades. Anxiety is a psychophysiological construct, having both psychological (e.g., feelings of worry, fear, apprehension) and physiological (e.g., increased heart rate and respiration, increased muscle tension) manifestations. As such, if one operationalizes anxiety as being reflected as increased muscle tension, EMG can be used to examine changes in that tension before and after exercise. One scientist, who made significant advances in this area, indeed coining the phrase *tranquilizer effect of exercise*, was Herbert deVries. In a series of studies beginning in 1968, deVries demonstrated that muscle tension was significantly reduced following aerobic exercise. This effect proved fairly robust, occurring for relatively brief bouts of exercise in college-age students, in older adults recruited for having elevated levels of resting muscle tension and being superior to tranquilizer medication, and at different levels of the skeletal muscular signal (innervation of muscle fiber, innervation of muscle spindle). This effect has been shown by others as well, with some finding the tension-reducing effect of leg cycling exercise to be specific to the soleus muscle and not to other muscle groupings like flexor carpi radialis. Furthermore, the tension-reducing effect has been shown to occur following both active and passive leg cycling exercise.

Relatively little work has tried to examine the relationship between the physiological reduction in muscle tension and self-reported tension, namely state anxiety (assessed via questionnaire). In one of the few studies, Robert Motl and Rod Dishman examined the effects of moderate intensity leg cycling on muscle tension in the soleus, reflected through the Hoffmann reflex (H-reflex; index of motor neuron excitability), and self-reported state anxiety. Further, using an interesting approach, they manipulated anxiety by having participants ingest a relatively large dose of caffeine (based on body weight) prior to exercise. Muscle tension was reduced following exercise regardless of whether caffeine or placebo was used, but self-reported anxiety was reduced only following the exercise

+ caffeine condition (exercise + placebo did not result in self-reported anxiety reduction). They also showed no significant relationship between changes in muscle tension (H-reflex) and changes in state anxiety. Finally, in an approach using the startle probe paradigm developed by Peter Lang and Margaret Bradley, the relationship between changes in facial muscle tension, specifically the musculature surrounding the eyes (corrugator supercilii), and different intensities of leg cycling exercise has been examined. An overall reduction in EMG activity followed both low- and moderate-intensity exercise compared to seated quiet rest. There was no relationship between the reductions following exercise and self-reported anxiety changes, as anxiety was reduced following both exercise and quiet rest. Self-reported anxiety reductions were largest following quiet rest and smallest following moderate-intensity cycling, but EMG reductions were largest following moderate-intensity cycling and smallest following quiet rest (exactly opposite of what would be expected if the two were related).

Clearly, more work is needed to better understand the relationship between muscle activity and affective responses, particularly with respect to exercise. Most of the work to date has focused only on self-reported anxiety, but other measures of affect related to anxiety, such as tension, tiredness, and calmness, should also be investigated. Furthermore, timing of measurements of both the physiological and psychological events is crucial to uncovering any relationship between the two, should they in fact exist.

The relationship between muscle tension (using EMG) and motor performance has also been a topic of interest in sport psychology. In an initial study in 1976, Robert Weinberg and Valerie Hunt measured self-reported anxiety and EMG activity during a task involving throwing a tennis ball at a target. Participants completed trials without feedback and then completed trials where they were given failure feedback meant to induce pressure. Participants who had higher levels of anxiety had EMG patterns reflecting longer contractions of both the biceps and triceps than those who had lower anxiety. This high-anxious pattern was interpreted as reflecting a reduction in muscular efficiency. Although it did not have negative influence on performance, the low-anxiety group did show performance improvement (greater accuracy) when given failure feedback, as it was thought that

the high-anxiety group could have performed better if muscle tension had not increased under pressure. Subsequent work has essentially replicated these findings. More recent work, using golf putting as the motor task, has examined the effects of performance pressure on anxiety, effort, and EMG activity during the putt swing. Increasing pressure resulted in worsening performance (fewer putts made), increased anxiety, and increased sense of effort. Like the original Weinberg and Hunt study, the patterning of EMG activity was indicative of inefficient use of muscular energy.

Another area of sport and exercise psychology where EMG has shown promise is in the area of imagery, mental practice, or visualization. Whereas imagery has been shown to result in improved performance, the reason for this improvement has been elusive. One explanation for the improvement in physical performance following mental practice has been termed the *inflow explanation*. In essence, the inflow explanation, captured in psychoneuromuscular theory, proposes that EMG activity should be similar in patterning during imagined movements as it is during the actual movement as it activates the same motor structures as the actual movement. Dating back to Edmund Jacobson's pioneering work in the 1930s, researchers have sought to determine whether muscle activation (EMG) during imagery is similar to that obtained during the actual movement. This inflow explanation was supported in work by some, but not by others. More recent studies seem to suggest that the EMG activity seen during imagery is more an outcome of the imagery rather than what causes the imagery or its benefits.

It is actually rather surprising, given the results from studies like those presented, that the use of EMG as a physiological measure has not occurred more frequently. As measurement technology continues to improve, allowing more real-time assessments to take place, it would be worthwhile to see the extent to which some of the findings presented above can be replicated in more natural settings, both in the exercise and sport domains. Although it would certainly require background and training in psychophysiology, the yields of such research could be fruitful indeed.

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See also Affective Responses to Exercise; Effort; Emotional Reactivity; Imagery; Mental Rehearsal; Psychophysiology; Relaxation

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EMOTIONAL REACTIVITY

Participants in sports and other performance settings routinely encounter emotionally salient cues or stimuli that can affect the quality and enjoyment of the activity. Given the emphasis placed on consistently performing at a high level, researchers, coaches, consultants, and practitioners have sought to better understand how athletes and other performers respond to various affective states. A myriad of factors contribute to successful performance, but emotional states directly influence motivation, attention, and movement execution. In some instances, emotional information must be attended to perform optimally. Other times, emotions precipitate internal and external distractions that should be ignored. In either case, the period of emotional reactivity following the onset of a stimulus is critical to performance. Information regarding emotional reactions and the ability to regulate them during competition is ultimately used to develop practice and performance recommendations that ensure peak performance. In this entry the focus is on reactivity; other entries in this encyclopedia address the ability to regulate emotions once they are elicited.

Emotion

Emotions are holistic phenomena, involving both subjective appraisals of affective stimuli and overt physiological changes that prepare the body to interact with the environment. While moods can be considered a general averaging of affective states over a period of time, emotions are comparatively brief reactions. By preparing the body to move, emotions motivate behaviors and actions toward desired goals and away from undesirable situations. In sport, these motivations can be to score a goal, win a championship, or avoid injury, among others. In exercise settings, emotions can motivate individuals to adhere to an exercise program, or to attain new personal records. Emotions can be self-initiated or externally generated. They can also exist in the period preceding a performance or spontaneously erupt during competition. Finally, emotional responses are influenced by situational demands and individual traits. Historically, self-report indices have been relied on to assess emotional reactivity in the sport psychology literature, but a more comprehensive assessment of emotional reactivity can be obtained through a conjunctive evaluation of three primary response systems: *subjective* feelings, *physiological* arousal, and *behavioral* indices.

Reactivity: Self-Reported Feelings

Self-reports assess an athlete's subjective feeling states prior to, during, or after competition. Although a wide array of emotions can be experienced in a performance scenario, anxiety is the most commonly measured emotion among sport and exercise psychologists. Anxiety can be measured at a dispositional (or trait level) and a competition (or state) level. The State-Trait Anxiety Inventory (STAI) is a popular general measure of trait and state anxiety. Sport specific measures of trait like the Sport Anxiety Scale-2 (SAS-2) and state anxiety like the Competitive State Anxiety Inventory-2 Revised (CSAI-2R) are also common. General and sport-specific self-reports differentiate anxiety into both somatic (bodily symptoms) and cognitive (worrying thoughts) components. For example, the CSAI-2R assesses the degree to which an athlete has bodily feelings of tension or *stomach sinking*. Additionally, athletes state how concerned or confident they are about performing well. Such questionnaires provide a more detailed individual profile of emotional reactivity. Sport

psychologists also study specific emotions, such as anger, joy, and multidimensional concepts such as passion. Self-reports are not the only method of determining emotional responses. Questionnaires are often used alongside physiological measures to quantify the relative pleasantness and motivational direction of perceived emotions with physiological changes in arousal.

Reactivity: Physiological Arousal

Emotions are psychophysiological phenomena, resulting from the interaction between environmental stimulation and neurochemical communication between sympathetic and parasympathetic divisions of the body, such as circulatory, respiratory, integumentary, nervous, and muscular. A limitation of self-report measures is that they only provide subjective perceptions of emotional arousal, and these perceptions do not always coincide with actual changes in arousal. Physiological measures of reactivity are aimed at quantifying those changes. Sport and exercise inherently increases physiological arousal, explaining why many athletes cannot accurately describe changes in their arousal due solely to emotional responses. The most common measure of physiological arousal is heart rate (HR), because of the ease of collection, measurement reliability, and being a noninvasive measure. Additional measures of physiological arousal, such as electrodermal activity (EDA), electromyography (EMG), and electroencephalography (EEG), use surface electrodes to assess changes in voltage across or under the skin. EDA and EMG measure changes in voltage related to skin conductivity or motor unit activation respectively, while EEG measures the frequency of cortical activity across the skull. EEG activity can be summed to relate changes in emotion with activation of brain areas known to involve arousal, attention, and cognitive processes. Finally, brain imaging techniques such as functional magnetic resonance imaging (fMRI) are emerging as viable methods to assess brain regions involved in emotional reactivity and regulation.

Reactivity: Behavioral Modification

Self-report and physiological measures can be used in conjunction to establish changes in arousal following an emotional experience, but behavioral responses are the most proximal predictors of overall performance. The most overt index of

behavioral reactivity is how the emotional experience manifests in athletes' movements. From a movement perspective, emotions can impact how quickly we perform a task, the amount of muscle tension or cocontraction in agonist muscles, the smoothness of movement, and error from movement targets. Another behavioral index of emotional reactivity is gaze related behaviors (eye movements), which are linked to changes in attention and effort. In both sport and other performance environments, attention to the right situational information at the right time is crucial to performance. Emotions alter both preferences in the visual field and the length of visual fixations. In sports, where decisions and subsequent movements must be made in a quick and efficient manner, changes in gaze behaviors can have a significant impact on performance. Another important gaze behavior which is affected by emotion is the quiet eye period. Quiet eye is the duration between the last fixation to the target and the onset of movement. Longer quiet eye duration is associated with expertise and improved performance, but emotions have been shown to reduce this period, negatively impacting performance.

Conclusion

Emotions affect what we attend to and the way we move, which affects how well we are capable of playing sports and performing other physical activities. How we move can also impact the emotions we experience and our motivation to continue to participate in sport, exercise, and performance settings. Emotional reactions, therefore, play a critical role in sport performance and are a central focus of sport psychology interventions.

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See also Affect; Affective Responses to Exercise; Basic Emotions in Sport; Emotional Responses; Emotional Schemas; Individual Response Stereotype; Stress Reactivity

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EMOTIONAL RESPONSES

Regular leisure-time physical activity provides a boost to positive emotions: Individuals who exercise are routinely found to have lower levels of depression, anxiety, stress, and hostility. Moreover, programs of exercise have also been shown to be an effective treatment for depression, with similar findings emerging for anxiety disorders. This entry focuses on the nature of the link between exercise and emotion, explicating what is known about the range, limits, and mechanisms behind emotional responses to exercise.

Exercise Intensity and the Time Course of Emotional Responses

The positive emotional effects of exercise do not depend on high-intensity workouts. Positive mood benefits are routinely observed after moderate-intensity exercise, with benefits (reductions in negative emotions and enhancement of feelings of well-being) reported within 5 to 10 minutes of completing exercise. Positive emotional responses also follow high-intensity exercise, but may be delayed by almost a half hour. This delay in positive mood may be a product of the negative moods that are induced by higher intensity or emotionally taxing workouts so that the experience of positive mood benefits from exercise must await recovery from the negative mood induction brought by intense exertion. Indeed, one prominent determinant of negative moods during exercise is exertion beyond the ventilatory threshold—commonly reached when one has difficulty talking during exercise. Also, during exercise, more negative

moods in response to feelings of physical exertion are reported by sedentary individuals, those who are more sensitive exercising in front of others (secondary to concerns that others may judge their appearance negatively), and those who tend to fear sensations of emotional arousal (termed *anxiety sensitivity*). Mood during exercise is of particular importance for the maintenance of leisure-time physical activity; negative mood emerging during exercise appears to be a powerful predictor of exercise adherence 6 and 12 months later. In short, more feelings of pleasure during exercise translate into higher rates of exercise over the longer term.

Changes in Depression

The beneficial mood effects of exercise are reliable enough that programmed exercise has emerged as a treatment for depression. Numerous clinical trials indicate that depressed psychiatric patients, as well as individuals depressed due to medical or other stressors, can successfully use exercise as a treatment. In these treatment trials, moderate exercise (frequently walking for 30 to 40 minutes) is prescribed 4 to 5 times a week. Results of this exercise can rival that for antidepressant medications when responses are evaluated after 3 to 4 months of treatment. Exercise also appears to offer benefit to depressed outpatients who have failed to respond to previous trials of antidepressant medication. As such, programs of regular exercise can be considered an empirically supported treatment for clinical depression, as well as a strategy for nonclinically depressed individuals to boost their mood.

Changes in Anxiety

As indicated by almost 50 studies, programs of moderate exercise can also significantly reduce anxiety and stress. This evidence comes from studies of adults challenged by medical conditions or other specific stressors, studies of the elderly, studies of adults in the general population, and studies of psychiatric samples. Again, in the context of well-controlled clinical trials, results of exercise programs for anxiety rival those found for antidepressant medications. A program of regular exercise has also been shown to extend the benefits offered by cognitive behavioral treatment for anxiety.

Type of Exercise and Mood Change

The question of whether the mood benefits of exercise depend on dose (frequency, intensity, duration)

or exercise type (aerobic training vs. strength training) has been examined only in a handful of studies. These studies show that there is no reason to discourage people from focusing on activities such as weight training or calisthenics for mood improvement instead of pure aerobic activity. That is, while the evidence base is strong for aerobic training, such as walking, running, or biking, the studies that have compared aerobic training to strength training have tended not to find meaningful differences. Whether more is better is also not entirely clear in the case of exercise for mood and anxiety. Results of studies examining the effects of exercise on anxiety, for example, indicate that the amount of change in anxiety reaches a peak at a dose of exercise that is roughly 1.5 times higher than the public health recommendation for exercise, which is 30 minutes of moderate exercise on 5 days of the week. At doses higher than that, the anxiety reduction benefits of exercise tended to decrease.

Exercise, Brain Physiology, and Mood Changes

Exercise brings about a cascade of brain activity, with current research supporting a number of mechanisms behind the positive mood changes with exercise. Animal studies provide reliable evidence that exercise alters some of the same neurochemical systems presumed to underlie pharmacologic treatments for depression and anxiety. Specifically, exercise appears to modulate serotonergic and noradrenergic activity in the brain, with some emerging evidence for γ -aminobutyric acid (GABA) modulation as well.

Acute aerobic exercise has been shown to increase brain-derived neurotrophic factor (BDNF) levels in both healthy individuals as well as those with psychiatric disorders. BDNF is a neurotrophin involved in brain neuroplasticity, differentiation and survival of neurons in both the central and peripheral nervous system; adequate BDNF levels appear to be an important factor in the maintenance of normal cognitive function and mood. Low BDNF has been associated with depression and anxiety, and antidepressant medications tend to enhance BDNF levels. As such, BDNF is considered as having a potential role in antidepressant activity, perhaps including the antidepressant and antianxiety effects of aerobic exercise.

Although endorphin activity has been considered a correlate of the *runner's high* that may occur after prolonged exercise, at the present time

there are several lines of evidence suggesting that endorphin activity does not explain the mood effects brought by moderate exercise.

Finally, atrial natriuretic peptide (ANP) is a peptide hormone that inhibits hypothalamic pituitary adrenocortical activity and may have anxiolytic properties. Importantly, submaximal and maximal exercise bouts significantly increase ANP concentrations, with early work linking these changes to anxiety reductions, raising the possibility that ANP changes may mediate some of the antianxiety effects of exercise.

Exercise to Buffer the Negative Effects of Stress on Mood and Anxiety

Among other biological and psychological processes, attention has been paid to reduced stress reactivity as an additional mental health benefit of exercise. The idea is that exercise may help individuals better modulate the kind of stressors that are so often the trigger for the development of mood and anxiety problems. Several studies have now shown that trained individuals show less physiological (e.g., heart rate, blood pressure, cortisol) and psychological (e.g., anxiety, tension, sadness) reactivity when exposed to psychological stressors as compared with their untrained counterparts. Importantly, research also shows that these protective effects are evident with single exercise sessions. That is, people tend to be less bothered by stressors when these stressors are preceded by exercise. In addition to getting protection against the development of chronic mood and anxiety problems, these observations suggest that people may notice meaningful immediate mood benefits with regular exercise.

Overexercise and Negative Mood

Although there are anecdotal reports of the emergence of depression-like conditions from overexercise in humans (and anxiety-like reactions from overexercise in rodents), there has not been systematic study of these effects. What is more certain is that strenuous exercise may lead to a plummeting mood during the exercise session, and that these mood changes may make adherence to exercise more difficult. Also, given the substantial delay between exercise and improvement in fitness or shape goals, the process of regular exercise in pursuit of weight reduction or shape improvement is a daunting enterprise. Attending to the emotional

benefit brought by exercise offers exercisers a more contingent (tightly time-linked) positive outcome for their exercise efforts.

Conclusion

Evidence from a variety of sources provides justification for prescribing exercise as a mood and anxiety management strategy. The mood and anxiety benefits of exercise are apparent immediately following exercise bouts, particularly among individuals who experience significant distress. For these people in particular, exercise training programs that help them establish a habit of repeated (moderate intensity) exercise sessions every week can provide long-term mood benefits that rival those provided by established pharmacological and psychological treatments.

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See also Affective Disorders; Affective Responses to Exercise; Brain; Psychological Well-Being; Quality of Life; Resilience

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EMOTIONAL SCHEMAS

Emotional feelings are fundamental aspects of human experience. In sport, emotions have powerful influences on athletes' thoughts and actions. The study and explanation of emotions in sport is difficult because there are many ongoing controversies and disagreements. Most theorists would agree, however, that emotions have motivational and regulatory functions, and that there is an interface of emotional feeling with cognitions and behavior in many adolescent and adult experiences. Psychologist Carroll E. Izard has forwarded a theory of emotion, called *differential emotion theory* (DET), which attempts to explain how emotional feelings and cognitions interact to influence the self-regulation of plans and actions. In DET, Izard proposed there are clear distinctions between basic or primary emotions and emotional schema. Basic emotions require only minimal lower order cognitions, whereas emotional schemas involve both emotional feelings and typically higher order cognitions in a dynamic interaction.

To understand emotional schema, which Izard believes involves most emotional experiences, it is necessary to briefly discuss emotional feeling of basic emotions. Basic emotions arise in infancy and are modified throughout childhood, adolescence, and adult development. These basic emotions, such as joy and happiness, sadness, anger, disgust, and fear, are thought to be principally a product of specific neural circuits in "old brain" systems (e.g., amygdala, basal ganglia, anterior cingulate cortex, and insula) that support a distinct feeling. These feeling states have distinct properties that serve to motivate behavior. Basic emotions occur primarily in infancy and are triggered rapidly and automatically by evolutionary meaningful stimuli.

Emotion schemas occur when the motivational and cue-producing emotion feeling interacts with cognitions to influence thought, actions, and sometimes other emotions. Emotion schemas may be triggered by simple or highly complex cognitive evaluation (appraisal) processes and always involve higher order cognitions, such as memories and thoughts connected to self-concept, values, and beliefs. As such, the developments of emotional schemas are highly influenced by the acquisition of language and social cognitive development.

Specific emotional feelings are associated with specific emotional schemas within a person. For example, feelings of fear activate fear-related cognitions and actions but not cognitions and actions related to joy. However, the actual thoughts and actions of an emotional schema in a particular context can vary widely across individuals because of individual differences in personality, culture, learning, and self-concept.

Emotional schemas help explain the great variety of emotional reactions seen within and across sport. Emotional feelings are often triggered by the athlete's appraisal of the personal and social meaningfulness of the situation. Although personal and social meaningfulness may be shared by athletes, there are differences among athletes even within a team because of their different social histories. The emotional feelings triggered within a particular context interact with higher order cognitions to produce plans and actions. The emotional feeling and associated emotional schemas can also trigger a cascade of physiological responses associated with the emotion process. However, the emotion schema can create a number of emotion-specific experiences that have the same core emotion feeling but have different thoughts and actions. Because of individual differences, culture, and learning, the same situation may activate an emotion feeling state in one athlete but not another. Even if the same emotion feeling is activated in two athletes, differences in higher order cognitions, such as evaluations of consequences, values, expectations, and coping options, will result in a qualitatively different emotional schema.

The complexity of emotional schemas is revealed in the example of a fear-related emotional schema in a high-performance gymnast attempting to perform a difficult routine on a balance beam. The routine involves two back flips on the beam followed by a double back somersault dismount. Years of skilled training has allowed the athlete to overcome the initial fear feeling state often associated with performing a potentially dangerous skilled action. High self-efficacy for the specific task allows the athlete to appraise the skill as challenging. However, a bad fall resulting in pain, but no serious injury, results in many of her teammates telling her of their surprise that she was not seriously hurt. She freezes upon mounting the beam the next time she attempts the routine, with the memory of the fall, pain, and comments of her

teammates triggering a very rapid and automatic emotional feeling of fear. The emotional feeling is associated with high arousal and the urge to flee. The emotional feeling is quickly integrated with higher order cognitions, such as the importance of being a high-performance athlete, beliefs of overcoming fear, evaluations of skill, but also the consequences of falling and images of other gymnasts being seriously injured. These conflicting higher order emotions result in continuing experiences of fear and a less than desirable action plan. She begins the routine but bails out after the first back flip and dismounts. Successive attempts produce similar fear emotional schema, resulting in either freezing behavior or incomplete routines.

Sport-specific culture and social learning also help explain why some participants within a sport may share similar emotional schemas to a specific emotion-provoking situation. For example, some professional baseball managers emotionally react to perceived poor calls by umpires in a very sport-specific ritualistic manner, such as standing toe to toe and arguing with the umpire or kicking dirt over bases. These actions would seldom, if ever, be demonstrated by coaches in a sport like volleyball.

It is rare in the sport psychology literature to see any distinction between emotion and emotional schema. Many sport researchers have investigated joy, pride, interest, anger, anxiety, guilt, and shame but seldom examine the integration of emotional feeling and higher order cognitions. From a DET perspective, it is important to distinguish *basic emotions* from *emotion schemas*. Basic emotions typically only occur in infancy and motivate rapid actions that are in response to evolutionary defined survival situations. These emotions are of short duration and highly adaptive. After infancy, emotional experience is increasingly defined by emotion schemas. Schemas may have a brief duration or continue for a longer time course. Schema can continue through a long duration because of the role of higher order processes like reflection and rumination that continue to trigger emotional feelings. The interface of emotional feeling and higher order cognitions is critical to athlete functioning since the cognitive component is intimately part of the emotional experience and is involved in formulating plans and actions. From an applied perspective, understanding the interacting features of emotional feelings and higher order cognitions within an athlete will allow practitioners to

develop effective emotional utilization and emotional self-regulation strategies.

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See also Affect; Basic Emotions in Sport; Self-Conscious Emotions; Stress Management

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ENERGIZING (ACTIVATION) STRATEGIES

Energizing strategies, sometimes called activation strategies, are primarily designed to increase the task-specific level of performer's mental and physical activity. They are of interest to applied sport psychologists, coaches, and athletes alike as on occasion performers require strategies that help stimulate levels of physical and mental activity.

Arousal and Activation

Given their often interchangeable use by applied sport psychologists, it is important to distinguish between the concepts of arousal and activation. The critical distinguishing factor here rests with the unplanned (automatic) versus planned (prepared) nature of the two responses. *Arousal* is the mental and physiological response activity experienced in relation to an unexpected (or unplanned) input into the system like an unexpected shout from the crowd. *Activation* is the mental and physiological activity geared deliberately toward preparing a planned response to an anticipated situation or stimulus like the execution of the long-jump in track and field. Therefore, the focus in this entry is on activation states and, in particular, activation states that facilitate performance on a given sport task. This will differ from sport to sport and from task to task; for example, the

activation state required by a golfer attempting to sink a 4-foot putt differs markedly from that required by an Olympic weight lifter in the clean and jerk. Therefore, in some instances athletes will be required to increase their state of mental and physical activation. The strategies outlined in this entry give some insight into how performers may induce such increased activated or energized states. These strategies are usually introduced to the athlete once they have become proficient at using strategies designed to lower their activation state, such as relaxation strategies.

Breathing Strategies Used to Energize and Increase Activation State

In addition to being used to relax performers, controlled breathing can also be used to increase levels of activation. Once athletes have attained a controlled, rhythmic, relaxed breathing pattern, they should be asked to consciously increase breathing rate. Inhaling, the athlete should purposefully try to imagine being more energized, whereas in exhalation, the performer should imagine that with each expired breath fatigue and wasted effort are removed. As the breathing rate is steadily increased, cue words can be used to supplement the energized breathing pattern. For example, “energy in” can be verbalized with each inhalation and “fatigue out” can be the self-statement used during exhalation.

Verbalization and Self-Talk Strategies Used to Energize and Increase Activation State

The use of self-talk strategies can provide performers with a route to increase their level of activation in a very efficient manner. Traditionally, the generation of self-talk statements involves athletes diarizing the type of words, statements, and feeling states they associate with an energized activated state. Examples of such words or statements include *explode*, *power*, *fast*, *hit*, and *psych up*. Once athletes have produced a bank of statements, they should select those they deem most applicable for use in their sport or for a particular task within their sport. The applied sport psychologist should ensure the self-talk statements adhere to the guidelines surrounding optimal use of self-talk—specifically, ensuring the statements are phonetically simple, positive in form, related to the activation state conducive to the action involved in the forthcoming task, and consistent

with the energized state the performer is attempting to induce.

Imagery and Visualization Strategies Used to Energize and Increase Activation State

In order to create a more activated state, imagery content, such as imagining a particular situation in one’s mind, can be manipulated to create high-impact visualizations that help alter the energized level of the performer. The athlete will require a high level of imagery ability to create these images, owing to the often fast-paced form of the imagery routines. As such, assessment and work on the imagery ability of the performer should be a prerequisite of such strategy use. To increase the activation state of the performer, the image should reflect fast moving, powerful, impactful, and energized content; examples include working machinery, animals, forces within nature, and the execution of explosive sporting movements. Where circumstances allow, the performer can be asked to develop images that show a progressive increase in an energized state in order to steadily increase the activation state—for example, a large boulder slowly gathering momentum as it falls downhill, gaining speed and energy as it begins to fall faster and faster before reaching full speed and power crashing through the obstacles in its path.

Combinations of Strategies Used to Energize and Increase Activation State

A combination of several of the strategies outlined is a useful approach to increasing activation states. Breathing strategies are often combined with self-talk statements to increase the efficacy of the energizing strategy. Imagery routines can often be supplemented with self-talk statements oriented around the content and desired activation state the performer is seeking to create. Further, the nature of the situation may shape the performer’s choice of energizing strategies used. On occasion, the performer may not have time within the fast-paced, real-time environment of the sport to use an imagery routine. Instead, short, sharp, phonetically simple self-talk statements like “fast” might better suit the confines of such a situation. In comparison, an athlete who requires an increased activation state for a training routine might have the time to combine breathing, self-talk, and imagery strategies into one holistic energizing strategy.

Conclusion

The applied sport psychologist, coach, and performer all have a role to play in establishing the applicable activation state for the performer. The strategies outlined in this entry offer insights for increasing activation states. These strategies can either be used individually or combined to form a more holistic energizing program. However, research testing the efficacy of energizing strategies for the performer remains in its infancy within applied sport psychology literature. Very few studies have tested the efficacy of individual or combined activation strategies on performance. Those that have done so recommend that sports, or specific closed skill tasks within a sport that require high activation or energized states, do benefit from some form of energizing strategy. Clearly, further research is required to establish the potential performance enhancing effects of activation strategies and whether one strategy, or a specific combination of strategies, is more effective at raising the energized state of a performer over another. Therefore, the practitioner should teach the performers a range of strategies and allow them to explore which is most effective for them and their needs. As with the use of all new strategies or techniques, the performer needs to become proficient in the execution of these strategies before attempting to use them in competitive settings.

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See also Breathing; Imagery; Relaxation; Self-Talk; Stress Management

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ENERGY, EFFECTS OF EXERCISE ON

For more than 100 years, researchers have noted how exercise enhances the subjective sense of mental and physical energy. *Affect* refers to the quality of a subjective mental state along the dimensions of valence (pleasant or positive vs. unpleasant or negative) and activation (alert or activated vs. sleepy or deactivated). This entry will summarize research related to the effects of exercise on affective states having positive valence and moderately high levels of activation. This category of affective states describes a *sense of energy* and can be termed *positive activated affect* (PAA).

Improvements in postexercise self-reported PAA occur for activities such as walking, jogging, swimming, weight training, and tai chi, and a recent study provides some support for a causal effect of exercise on PAA. Published meta-analyses show medium effect sizes for aerobic exercise (0.47 for acute bouts and 0.57 for regular participation). Taken together, the meta-analytical results of acute and regular aerobic exercise indicate that self-reported PAA increases by about one half (.50) of a standard deviation from pre- to postexercise in participants who performed aerobic exercise (experimental groups) compared to those who did not (control groups). Quantified another way, the meta-analytical results indicate that a randomly chosen person from the population who recently finished an aerobic exercise session would be about 65% to 70% more likely to report higher PAA compared to a randomly selected non-exerciser.

Several variables influence the size of the effect of exercise (in particular aerobic exercise) on postexercise PAA. Participants with lower pre-exercise PAA often report greater postexercise PAA, a finding in line with the idea that exercise can be a self-regulatory strategy to improve energy. Low and moderate intensity exercise appears best for increasing postexercise PAA, although higher intensity exercise can improve postexercise PAA after a short delay, especially in fit individuals. Exercise sessions of 15 to 45 minutes improve PAA, but durations longer than 60 minutes may result in little improvement. PAA typically peaks within 5 minutes after exercise and remains elevated above baseline for 20 to 30 minutes. This pattern of postexercise PAA change applies to typical exercise sessions (low to moderate intensity, 15 to 60 minutes), but not to extreme bouts like

marathon. The motivational state of the exerciser may be another variable that alters the effect of exercise on PAA. For example, participants with a goal-oriented motivational state during exercise will likely report lower postexercise PAA compared to participants in a less goal-oriented, more playful motivational state. A variable that does not appear to change the magnitude of the effect is the length of an exercise program. In other words, successive acute bouts simply restore postexercise PAA from previous sessions without producing an additive effect over time providing further support for exercise as a self-regulatory method to improve energy. Data from college students comprise the majority of the study results and more information is needed on community and clinical samples.

Affect during exercise is related to exercise self-efficacy, exercise history, hydration and blood glucose levels, postexercise affect, and exercise adherence. For example, higher PAA during exercise predicts exercise adherence. In line with pre-to-post exercise data, there is an inverse relationship between exercise intensity and affect during exercise. According to the dual-mode model, cognitive and physiological factors mutually influence affect during exercise, with cognitive factors prominent at lower intensities and physiological factors becoming more salient at high intensities. There is also evidence of a shift in attentional focus toward pleasant stimuli and away from unpleasant stimuli during moderate-intensity exercise. As a result of these interrelationships, PAA during exercise may provide a useful method of monitoring exercise intensity for the development of subjectively based exercise prescriptions. However, although the data show that exercise improves mental well-being including a sense of energy, most people underestimate how much they might enjoy exercising and are therefore physically inactive in part because they anticipate a lack of benefit. This is an interesting inconsistency worthy of further study.

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See also Adherence; Affective Responses to Exercise; Psychological Well-Being; Self-Regulation; Stress Management

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ENJOYMENT

Sport enjoyment, or *fun* in children's terms, is the most important and most studied positive emotion in youth, adolescent, and elite sport. Early youth sport research found that fun was one of the most important reasons given by athletes for choosing to participate in sports, and lack of fun was a prime explanation for dropping out. Consequently, to have fun in sports was one of the rights included in the "Bill of Rights for Young Athletes," written by a special task force established by the National Association for Sport and Physical Education. Subsequent research defines *sport enjoyment* as athletes' positive emotional response to their sport involvement that encompasses generalized feelings, such as fun, pleasure, liking, and love. Detailed in this entry are the research findings addressing two key questions. First, how does enjoyment affect athletes' motivation in sport? Second, what makes sport enjoyable? Importantly, the results presented on both of these issues characteristically generalize or apply to both genders, as well as to diverse sports and competitive levels.

How Does Enjoyment Affect Athletes' Motivation in Sport?

Sport enjoyment has important motivational consequences and is incorporated into many theories dealing with athletes' participation, persistence, and exertion of effort in their sport. Findings show that the higher the enjoyment experienced,

the more athletes elect to participate and persist in their sport, the more they feel the desire to exert greater effort and perceive that they actually have expended more effort, and the higher their *enthusiastic sport commitment*. Enthusiastic sport commitment is a psychological state defined as athletes' enduring determination and desire to continue participating in their sport. Research focused on the sport commitment model, elaborated elsewhere in this encyclopedia, shows that sport enjoyment is the strongest and most consistent predictor of commitment to sport.

What Makes Sport Enjoyable?

Knowing that enjoyment is so crucial to positive motivation makes it important to understand what makes sport enjoyable. Knowledge of the sources of enjoyment not only allows for a more comprehensive understanding of emotion and motivation in sport from a research perspective, but it also tells us the ways to make sport enjoyable for athletes from an applied point of view. In effect, the sources of enjoyment are the buttons that can be pushed to create the enjoyment that kindles positive motivation. Extensive study reveals that there are many potential sources of enjoyment inherent in sport and that they are diverse.

Sources of enjoyment fall under three broad categories: *intrapersonal sources* that emanate from the person; *situational sources* that are inherent to the sport context; and *significant other sources* that are influences from important people to the athlete such as parents, coaches, and teammates. Specific examples of sources of enjoyment will be presented under these three classifications.

Intrapersonal Enjoyment Sources

Athletes' perceptions of high personal competence in their sport, and a number of achievement factors related to these ability perceptions, make sport enjoyable. These sources reflect the positive processes and outcomes of engaging in mastery—including exerting effort, learning, practicing, improving, and perfecting challenging athletic skills—and achieving mastery of these skills.

Athletes' own motivational goal orientation is also a source of enjoyment. Greater enjoyment is experienced by high-task-oriented athletes who define success in a self-referenced manner with criteria like successful skill mastery and high effort expenditure.

Personal movement experiences emanating from the movement and execution of sport skills are enjoyable. These sources include feeling athletic; experiencing movement sensations, such as freedom and exhilaration; and being able to creatively express one's self through movement.

Situational Enjoyment Sources

Both the process and outcomes of competition are sources of enjoyment. In terms of process, enjoyment is experienced from engaging in competition and comparing one's skill against a competitor. Winning a game and having a winning season are enjoyable outcomes.

Receiving recognition from other people for successful performance is also enjoyable. This recognition includes such specific sources as being in the newspaper, yearbook, or record books of one's sport and receiving medals, trophies, and standing ovations.

Significant Other Enjoyment Sources

There are two sets of significant other sources of enjoyment: those that come from adults, and those that stem from teammates. With respect to parental and coach sources, enjoyment is experienced when athletes perceive that these adults are supportive and highly involved in their sport experience, and are satisfied with their performance. Elite athletes find it enjoyable to bring pleasure and pride to these adults through demonstrating their talent. Sources specific to parents involve athlete perceptions of realistic parental performance expectations and lack of pressure to participate in sport. Finally, sources of enjoyment come from being on a team with peers, including positive interactions with teammates, building friendships, and supporting each other. And so we see that there are numerous and many different types of sources or buttons that can be pushed in sport to create the enjoyment that increases and maintains positive motivation.

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See also Affect; Basic Emotions in Sport; Commitment; Emotional Responses; Passion

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ENJOYMENT, AS MEDIATOR OF EXERCISE BEHAVIOR CHANGE

To enjoy means to take pleasure or satisfaction from something; *enjoyment* is the act or condition of enjoying. Thus, enjoyment of physical activity is the act or condition of taking pleasure or satisfaction from physical activity. This entry discusses the conceptualization and assessment of enjoyment, as well as the role of enjoyment as a predictor of physical activity behavior and a mediator of the effects of physical activity promotion interventions on adherence to physical activity programs.

Enjoyment is a broad concept encompassing both pleasure and satisfaction. Pleasure is typically conceptualized as the subjective experience of positive hedonic tone and can occur in the absence of higher cognitive processing regarding the meaning of a stimulus—in this case, physical activity. Satisfaction occurs following the fulfillment of a need or goal and thus, relative to pleasure, typically entails more cognitive mediation to ascertain the meaning of the triggering stimulus. Thus, one may report enjoyment of physical activity like a walk in the park because of the immediate pleasure experienced during the activity or because of thoughts about having accomplished one's daily physical activity or how it might benefit health.

In research on physical activity enjoyment, the concept is most often assessed by the Physical Activity Enjoyment Scale (PACES) developed by Deborah Kendzierski and Kenneth J. DeCarlo. The PACES includes 18 semantic differential items, which require respondents to select a point

along a 7-point continuum between two opposite descriptors relevant to enjoyment of physical activity (e.g., I enjoy it . . . I hate it; I dislike it . . . I like it; I find it pleasurable . . . I find it unpleasurable). While separate measures have been used to independently assess pleasure and satisfaction in response to physical activity, the PACES remains the most popular measure of the global concept of enjoyment.

Enjoyment of physical activity, as measured by the PACES, is a robust correlate and predictor of physical activity behavior in youth, as well as younger and older adults. This correlation between physical activity enjoyment and physical activity behavior simply means that people who enjoy physical activity are more likely to engage in it.

Considerably less research has examined whether enjoyment of physical activity can act as a mediator between physical activity promotion interventions and increases in physical activity behavior. That is, few studies have examined the effects of public health interventions on people's enjoyment of physical activity, and, in turn, increases in physical activity behavior. Thus, it remains unclear as to whether enjoyment of physical activity is malleable through intervention or whether certain people are—because of biological predispositions or early childhood experiences—more likely to enjoy physical activity and more likely to engage in it. More research is needed to investigate whether, how, when, and for whom physical activity enjoyment can be increased through public health interventions or individual efforts.

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See also Affect; Affective Responses to Exercise; Basic Emotions in Sport; Emotional Responses; Hedonic Theory; Intrinsic/Extrinsic Motivation, Hierarchical Model of; Models of Emotion–Performance; Pleasure

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ERRORS

Biological organisms cannot repeat a movement exactly the same way across practice trials or attempts. In other words, there is an inherent amount of imprecision in the motor systems of animals, including humans. This variability can be considered a source of error. Second, in any attempt to solve a new motor problem, whether a young athlete pole vaulting or a dog attempting to catch a tennis ball for the first time, there is inaccuracy—in other words, error—as the organism figures out how to solve these motor problems.

Definition and Measurement

Error is defined as the difference between an observable behavior and a desired behavior. More formally, error is the difference between an observed score and a desired score, called a target:

$$e_i = X_i - T$$

where e is the error, X is the score on trial i , and T is the target score. Traditionally error has been classified as bias, precision, and accuracy.

Bias means that a set of scores tends to produce either a negative or a positive error. Bias is also called *algebraic error*, as it has a magnitude and a direction (positive or negative) on each and every trial. The average algebraic error, or bias, over a series of trials has been called *constant error*.

Precision is a measure of response consistency. In other words, does the individual perform identically every time? *Variability* is the usual measure of imprecision. Variability is considered the average distance a set of scores is from the mean score. Typically, the measure of precision is just the standard deviation of the set of scores. Thus, a smaller standard deviation means an increase in consistency (precision) compared to a larger standard deviation. Historically, *variable error* is the name of this error score.

Accuracy, the most commonly used and important measure of error, is defined as the average *distance* a set of scores is from a target value. Traditionally, *absolute error*, defined as the

average absolute difference (computed on a trial by trial basis, and then averaged) between a score and a target, was the accuracy measure of choice. Absolute error has several statistical issues that have resulted in researchers abandoning this measure. A much better measure of accuracy is root mean squared error (RMSE). This measure uses the one-dimensional distance formula and then takes the average of the set of distances. RMSE is a combination of bias and variability.

So far in this discussion of error, we have focused on measuring performance as the outcome of a single trial and then describing the average of a set of trials. In other words, these measures are *outcome* scores. Error also can be used to capture the *process* of movement. The movement of the effector over time (a trajectory) provides a rich source of information concerning the underlying movement control processes. For example, a movement trajectory can be compared to an ideal movement trajectory and deviations from the ideal would be considered an error. The standard measure of error for a continuous trajectory is RMSE. The error is summed up across samples across a single trajectory. The larger the RMSE, the less *accurate* the trajectory is to the template or standard.

The trajectory RMSE measure can be slightly altered to ask an important theoretical question in motor control. Are trajectories scalable in space and time? In other words, does the production of a skilled action show evidence for an ideal template that might exhibit spatial variability and temporal variability, while the movement appears to be derived from a common invariant trajectory? In speech motor control, Anne Smith, Lisa Goffman, Howard Zelaznik, Goangshuan Ying, and Clare McGillem developed a measure known as the *spatial-temporal index* (STI). First, the trajectory time base is changed to a percentage. Therefore, all movements have a temporal scale from 0 to 100. Second, the displacement values are placed on a normal scale (they usually range from -1.5 to $+1.5$). At each 2-percentage interval, the standard deviation of the normalized displacements is computed. The 50 standard deviations are either summed or averaged. The STI captures how closely a set of trajectories follows the average spatial-temporal trajectory. The average normalized trajectory purportedly represents the invariant template. The STI measure does not make any theoretical statements about the cause of any spatial-temporal invariance. The measure has been shown

to capture development of speech motor patterns and the effects of practice on speech learning. Although the assumptions of linear scaling might not be fully justified, the measure appears to be fairly robust in spite of these issues.

The Value of Errors

The previous section presented the descriptive aspects of error. Now, attention is focused on whether errors are good, bad, or indifferent. Obviously, at high levels of performance errors generally are considered *bad*. Missing the putt to lose the championship is not a good thing. In competition, errors might seem to always be bad. But, that clearly does not have to be the case. A good baseball pitcher deliberately may throw a bad pitch so that the batter might swing and miss, or the batter might be expecting a pitch in the future series of pitches. Is home plate the target on every pitch? No! Thus, errors in pitching are difficult to measure. However, errors in pitching can be measured in practice. The target changes, but during the game, error in pitching can only be measured if we know the goal of the pitcher.

What about errors during practice? For high-level performers errors might be necessary for an expert to figure out the nuances of a difficult task. For example, when concert pianists need to learn a new fingering sequence, they experiment with the best method to get it done. In other words, they are making deliberate errors to help them search for the solution to this new problem. New learners need and use errors to guide the learning process. James A. Adams, one of the first, if not the first, to propose a major role of errors in learning, believed that errors provided motivation and information that learners use to change their attempt on the next trial. Ann Gentile, although not explicit on this matter, believes that learning is a search process. The learner is attempting to find a reasonable solution to the motor learning problem. She calls this *getting the idea* of the movement. After this approximate solution, the learning then works at refining the solution. One can define getting the idea when the learner is consistently producing an acceptable level of error and then moves to a deliberate attempt to reduce error. What would be the consequences of a performer not being able to control and produce error? One would be an inflexible robot. Errors provide the flexibility for individuals to respond to their environment.

Human beings would not be functional if they only possessed fixed action patterns, such as stereotypical reflexes.

Errors also are organized. Robert Sessions Woodworth in 1899, Paul M. Fitts in 1954, and more recently Richard A. Schmidt, Howard Zelaznik, Brian Hawkins, James S. Frank, and John T. Quinn in 1979 described this organization. The organization is called the speed-accuracy trade-off. With some exceptions the cost of increasing movement speed (decreasing duration, increasing distance, or the combination of the two) is a decrease in movement accuracy in the spatial dimension. Home run hitters tend to strike out more often than singles hitters. Why? A home run in baseball requires increased bat speed. The cost of that increase is paid in the currency of an increase in spatial variability. There is a clear exception to this speed-accuracy trade-off in the temporal domain. Temporal variability decreases as the timed interval decreases (in other words, doing the task faster). Therefore, in tasks that require both spatial as well as temporal accuracy performers need to learn the optimal movement speed to minimize spatial error but maximize temporal accuracy.

What is the source of error and variability? One way to view errors and variability in performance is via the classic view of test theory. A performance score is composed of two components, the *true* score and a source of random variations called an *error*. The true score represents the mean of an infinite number of trials. In other words, the true score is a construct, and thus it represents the individual's level of performance.

Recently, scholars in motor behavior have begun to discover that error over a series of trials does have a random component. However, error also has a predictive component. In other words, a performance score in series predicts a performance score in later in the time series. These relations are called *long-term correlations*, and are thought to be the results of the underlying non-linear dynamics of human performance. These long-term correlations are shown in fluctuations of step-cycle duration, reaction time, and cycle duration in repetitive timing tasks. There are some very exciting hints that these long-term fluctuations are crucial to healthy behavior, and the lack of these correlations might be the signature of diseased states. Fluctuations, which are not random, but structured, are a reliable signature of normal biological function.

The observation that errors have a nonrandom component might be the reason for the success of sport psychologists working with high-level performers. Obviously if errors in performance were solely random, the only way to improve performance would be to dramatically increase the amount of deliberate practice. However, most high-level performers already practice many hours per day. If performers acquire techniques to control the nonrandom component of performance fluctuations, performance will improve.

Howard Zelaznik

See also Coordination; Motor Commands; Motor Control; Motor Learning; Sensorimotor Representations; Skill Acquisition

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ESPECIAL SKILLS

An *especial skill* is a unique case where one very specific variation of a motor skill exhibits a markedly superior task performance in comparison to performances on all other variations within the same skill class. This effect has been clearly demonstrated with basketball set shots from the regulation free-throw (foul) line, as well as with baseball pitches over the regulation pitching distance. To

illustrate, as one moves closer to or farther from the basketball goal there is typically a consistent change in the percentage of shots made, with accuracy decreasing linearly as the distance to the target is increased. However, for highly practiced players, the shooting percentage is significantly better than expected at exactly 15 feet (4.57 meters) straight out from the basket—precisely at the foul line. It is reasoned that the especial skill effect emerges due to the massive amount of practice at that one variation of the skill relative to any other variant. The free-throw line is the only distance from which basketball players regularly and consistently practice set shots. This phenomenon is an example of the specificity principle of practice taking precedence over the generality principle of motor skill development. It is of particular interest because no current theories of motor control have been able to satisfactorily explain the intricacies of the especial skill phenomenon.

Specificity and Generality of Motor Skills

The notion of specificity of training in motor skill learning and performance is well established. The more that conditions of the performance task match those that were present during practice, the greater the transfer of benefits from training to test performance. The same shot from the same distance, with the same ball, under the same environmental conditions would be a classic example. The more those conditions are dissimilar, the less transfer is seen from training to test performance. It is reasoned that repeated training of the same skill under the same conditions builds up a strong memory representation for the correct motor action. Such a representation would be at best only partially correct if any condition for performing the skill was changed.

Specificity, therefore, poses theoretical problems because it would be essentially impossible for even an expert to practice every possible variation that they might encounter during motor skill challenges. Under most normal situations, every motor skill execution is subject to differing sensorimotor, contextual, or processing demands. To explain flexibility in performance, theories of generality (most prominently, Schmidt's schema theory of motor skill learning) suggest that performers build up general representations for each class of movement skill. Through practice the skill is increasingly capable of being altered to match

the demands of the task at any particular moment. The performer learns the common coordination pattern for the skill class, and then learns the rules by which forces and timing might be adjusted to create the skilled action for that situation. As the performer adds experience and practice, improvement is seen across the whole skill class. General motor program concepts can be used to explain, for example, how a basketball player can be accurate at jump shots from any one of numerous court positions under continuously changing dynamics of game play.

The most interesting feature of especial skills is that they seem to be an example of specificity emerging within a landscape of generality. All basketball players can shoot set shots (a shot where the feet do not leave the ground) with a general level of accuracy commensurate with their experience and the physical difficulty of the shot. All baseball pitchers can throw a straight fast-ball-style throw to a target of varying distances, again with accuracy varying according to experience and the physical demands of the throw. When the range of those variations is examined for patterns, theories of generality accurately account for performance outcomes. However, in the particular instance of a case that is special to that sport—the one that is practiced far beyond all others—the principle of specificity appears to take precedence.

Explanations

Several theoretical explanations have been offered for the especial skill effect. The *visual context hypothesis* posits that certain consistent visual information, such as the particular angle of viewing, may enhance perception–action coordination under the one special condition. This concept received some support from research showing that 15-foot set shots at angles different than straight out from the basket did not produce the effect. However, other studies where the visual context was significantly altered have nonetheless demonstrated especial skills. The *learned parameters explanation* suggests that repeated practice of one variation of a skill leads to optimization of movement parameters, such as force, speed, or release angle, for that specific instance. This hypothesis has received some research support when weight of the ball was altered, but not supported in other research when the ball and distance were constant but changing the visual angle eliminated the

effect (different viewpoint, but identical movement parameters). Another hypothesis is that experienced performers might develop a special general motor program used solely for the one massively practiced variation of the task. Hypothetically, this would be evidenced by kinematic information (relative angles, velocities, or accelerations of the involved joints and limbs) that shows unique coordination patterns for the especial skill in comparison to those common to the rest of the skill class. Yet, such kinematic analysis has failed to support this explanation with basketball shooting. Lastly, it has been suggested that the phenomenon might be simply a matter of confidence, rather than some difference in motor control per se. However, research has shown measures of performance efficacy to vary independently of the especial skill effect. In fact, such psychological measures call into question any cognitive (conscious processing) explanations. In sum, the phenomenon has yet to receive a fully satisfactory theoretical explanation.

Conclusion

The especial skill effect is interesting because it reflects high levels of practice or expertise, it may serve as a marker in the development of motor skill expertise, and it may highlight noncognitive aspects of human motor control.

Jeffery P. Simons

See also Expertise; Generalized Motor Program; Laws of Movement Learning and Control; Transfer

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ETHICS

Ethics is the investigation of the primary moral assumptions held by individuals, organizations, or professions that are used to help members make sound decisions about what is right and wrong. To expand on this definition, ethics refers to an organization's attempts to protect the welfare of clients by developing, adopting, and enforcing guidelines that regulate member conduct in professional and scientific settings. These ethical guidelines are essentially a set of values that have been agreed upon by the members of an organization or profession. By developing ethical guidelines, the organization or profession is protecting the welfare of those they serve but also communicating their values to society. Ethics in sport psychology have been an area of concern, as the discipline grows in research, education, and practice. While psychologists are encouraged to behave in socially acceptable ways within their personal lives, the ethics codes are only designed to govern their actions in professional settings (research, education, and practice) in their work as psychologists.

Many consistencies exist in the codes of ethics between general psychology and the field of sport and exercise psychology. In fact, as a subset of psychology, many of the sport and exercise psychology codes of ethics are closely based upon those previously written in psychology.

Ethical practice can be considered to be the *coin of the realm* for psychology and sport and exercise psychology practitioners. It is through ethical behavior that professionals take steps to protect their clients from harm. Ethical guidelines of ethical behavior are aimed at helping practitioners make decisions in difficult situations that could prove to be harmful to the client or related person.

Components of Codes of Ethics

It is common for ethical codes to have several sections to them, each serving its own purpose. These sections often include an introduction, preamble, principles, and standards. The purpose of an introduction is to discuss issues such as the intent of the code of ethics, as well as to provide procedural and organizational clarity about the codes and their use. The purpose of a preamble is to outline the value structure of the organization and to encourage

practitioners to meet the highest possible ideals set by the organization and outlined within the code.

Principles are commonly seen as general statements about the codes of ethics that give background context into the rationale for the development of the specific standards and guide practitioners toward the highest ideals for practice. In essence, the principles are aspiration and unenforceable value-driven statements designed to provide guidance to individuals who are faced with ethical decisions. While principles may change from ethics code to ethics code, the most common psychology based principles involve such things as

- Do no harm (nonmaleficence)
- Help others (beneficence)
- Respect and promote autonomy
- Treat others fairly and equitably
- Be trustworthy
- Respect dignity

Ethical standards are the enforceable and more specific portion of the ethics code. These aim to influence the conduct of psychologists in professional settings. While ethical standards are more specific than principles in terms of directing behavior, they are often written somewhat broadly to help psychologists make difficult decisions across a broad array of situations. Because of the many different settings in which psychologists practice, general standards help them make difficult decisions which vary in their working.

Role of Ethics Codes

Students and professionals generally understand and appreciate the usefulness of codes of ethics, but that is not always sufficient to keep individuals behaving ethically. To consistently behave in an ethical manner requires practitioners to possess a number of skills and attributes. While knowledge of and familiarity with the appropriate codes of ethics and a sincere desire to act ethically are a good starting point to help people behave ethically, this step in and of itself is not enough to ensure that professionals will behave ethically when faced with challenging and complicated scenarios. Other factors that help practitioners behave ethically include the following:

- Being able to recognize when challenging ethical situations arise

- Having and using maturity, judgment, discretion, and wisdom in one's decision making
- Understanding the competing influences affecting one's judgment
- Thoroughly considering the consequences of one's actions
- Having a clear understanding of the principles behind the code and using them during the decision-making process
- Understanding and utilizing problem-solving models
- Developing and using a professional consultation network

Important Ethical Issues in Sport and Exercise Psychology

While many codes of ethics for sport and exercise psychology organizations may closely resemble codes of ethics from general psychology organizations, differences do exist. These differences, some stated and some implied, are important to consider, as they are often created or maintained to help deal with the unique field and setting of competitive sport. Further, there are many ethical situations or dilemmas that are unique to specific sport and exercise psychology settings.

Confidentiality

Confidentiality is probably the most commonly cited ethical concern within the field of sport and exercise psychology and is commonly identified as the most important ethical standard. Confidentiality relates to practitioners not allowing others to know their client's identity, as well as not letting others know the client's concerns or related treatment. There are some common limits to the confidentiality that are commonly addressed with a client at the beginning of the service delivery via a consent form.

While confidentiality is important in sport and exercise psychology, the guidelines by which it is judged are different from those in traditional psychology. Because of their high-profile status on campus and in the community, athletes are often more identifiable than other individuals when they enter an office. Further, because of their time commitments, travel schedules, and performance requirements, it is not uncommon for a practitioner to work with a client out in the open where others can see them talking with each other. This may happen on a pool deck, in

a gym, or on the side of a field. Under these circumstances, the goal changes from not allowing others to know that the practitioner is working with a client to not allowing others to know what is being said.

Multiple Roles

A multiple-role relationship can be described as having a professional relationship with a client and at the same time interacting with the client or a close friend or family member of the client in another role. Multiple-role relationships are not necessarily unethical if handled properly. However, practitioners should avoid entering into multiple-role relationships that could cause harm to the consulting process by affecting objectivity or causing exploitation. If a multiple-role relationship is entered into, the consultant needs to be mindful of the potential problems and be willing to stop the consultation process if any are perceived.

Multiple-role relationships may be more common within the field of sport and psychology than in the general psychology. This may be due to the limited number of sport and exercise psychology practitioners. For instance, on many college campuses, sport and exercise psychology professionals often serve in other roles. A practitioner may consult with individuals and teams as well as teach classes and advise students. With such a multitude of responsibilities, practitioners are likely to have contact with clients in more than one setting. Therefore, it is essential that practitioners discuss possible multiple-role relationships with athletes at the outset of services.

Competency

The ethical issue of competency relates to the necessity for practitioners to practice only in areas and with people whom they have the requisite knowledge, skills, and abilities to provide quality services. To be viewed as competent, an individual must understand the domain-specific issues, demonstrate skills for interventions, and be able to assess their outcomes. Common forms of competence include (a) intellectual competence, which is gained from formal education; (b) emotional competence—the ability to emotionally deal with clients and limit personal biases; and (c) experiential competence or the wisdom gained from previous professional experiences.

It is difficult to identify the distinction between competence and incompetence with a given client or presenting concern. For the most part, competence is determined by one's knowledge and experience that is generated through education, training, supervised experience, and professional experience. However, competence does not last forever, and to remain competent in a particular area, practitioners must update their knowledge through continuing education, consultation, study, or additional education.

Individuals who are licensed psychologists, but receive little training in the sport sciences, must take steps to gain the sport science training necessary to understand the athletic context. Sports science trained practitioners who are not licensed need to understand the limitations of their training and their inability to provide therapy to clients. It is essential that practitioners understand their training and only provide services that are in line with their competency.

Teletherapy

Teletherapy can be described as the use of technology to deliver services to clients from a distance. Teletherapy services can be delivered asynchronously via text or e-mail platforms but can also be delivered synchronously via phone and video conferencing platforms. There are both strengths (e.g., service availability for remote, disabled and traveling clients; cost effectiveness; access to self-help resources; client disclosure and anonymity) and limitations (e.g., concerns about confidentiality, relationship development, limited accessibility, credentialing across state and provincial lines) to the use of teletherapy.

Clients are some of the primary driving forces behind the use of teletherapy. Because of their age, comfort level with technology, busy training schedules, and constant travel, athletes often feel comfortable with the use of technology in their consultation sessions. However, practitioners need to be aware of several issues before agreeing to provide teletherapy services: (1) legal issues for those who provide psychological services across state, provincial, or national lines; (2) malpractice insurance for such services; (3) personal competency in the use of technology; (4) maintenance of confidentiality; and (5) appropriateness of teletherapy for client's presenting concern.

Overidentification

Sports are often perceived by people and the public at large as very prestigious, and thus many wish to be associated with it. As such, it is not uncommon for individuals with only a peripheral connection to teams, to identify themselves closely with those teams. This is also true of practitioners who consult with teams. While overidentification with a team does not always cause problems, it has the potential to cause bias and the loss of objectivity in the practitioner.

Identifying the Client

An ethical situation that commonly occurs in sport and exercise psychology is the identification of the client (Who is the client?). In an athletic setting, one may question if the client is the athlete who has been referred for help, the coach, or the athletic department who is paying for the services. While this answer may seem simple, it becomes less clear when the practitioner is employed by the athletic department. In such cases, employees of the athletic department may question how much information they have a right to know. Such situations arise more often when the practitioner is an employee of a referring agency or is receiving payment from a third party.

Ethical Decision Making

The principles and standards identified within an organization's code of ethics are intended to support and guide professionals in the process of making ethical decisions. However, making sound ethical decisions can be complicated since ethical principles and standards oftentimes contain gaps, contradictions, and grey areas because of the multiple considerations that must be addressed. To help combat these concerns, ethical decision-making models can serve as a practical framework that professionals can use to resolve these situations.

Theoretical perspectives serve as the foundation for ethical conduct in each of us and influence how we interpret situations and potential outcomes, as well as how we use the decision-making models.

Teleology

Teleology is an ends-oriented approach that evaluates actions based upon the consequences

of those actions, regardless of the intended consequences or the means used to achieve those ends. From this perspective, actions that lead to the most good for the most number of people are deemed to be ethical, with the opposite being perceived as unethical. Teleology can be viewed from both an act and a rule perspective. *Act-Utilitarianism* is a perspective that focuses on the ends, achieving the greatest good for the most people, with no regard for the means of achieving this result. *Rule-Utilitarianism* is a perspective that encourages focusing on the ends, while following previously established rules to achieve those ends.

Deontology

Deontology can be defined as a means or principled approach to ethical decision making. From this perspective, behaviors are evaluated as ethical or unethical based upon the intentions and quality of the actions rather than the outcomes produced by those actions. From this perspective, ethical behavior should be grounded in principles that would be applicable in different settings.

Models of Ethical Decision Making

While there are similarities and differences to each decision making, there are components to them that are consistent. Ethical decision-making models often include the following suggestions or steps:

- Identify and prioritize the relevant ethical issues, practices, and values.
- Identify the people affected by this situation.
- Identify the relevant ethical principles and cultural issues underlying the situation.
- Consult with other professionals about the situation.
- Identify personal factors that may distort one's perspective of the situation.
- Brainstorm alternative courses of action.
- Analyze the short- and long-term risks and benefits of each course of action.
- Choose a course of action and inform the stakeholders of the decision.
- Act on the situation and assume responsibility for the consequences.
- Evaluate the results and make changes if necessary.
- Document decisions and the decision-making process.

Common Factors That Influence Ethical Decision Making

When making ethical decisions, it is important for practitioners to understand and take into consideration the environmental influences that may affect their decisions. Ethical decisions do not happen in vacuums, and making decisions without taking external influences into consideration may risk these factors unconsciously influencing one's decisions. The primary environmental factors influencing professionals in sport and exercise psychology are likely (1) individual influences, such as moral and cognitive development and world view; (2) situational influences like the culture related to this issue in one's workplace; (3) external influences, such as politics, economy, social issues; and (4) significant other influences, such as how will others perceive one's decisions and how have they influenced one's beliefs about this situation.

Jack C. Watson

See also Character Development; Consultant Effectiveness, Assessment of; Moral Development; Moral Reasoning; Training, Professional

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ETHNICITY

Ethnicity refers to shared cultural traditions and history of a group or population. The collection of people who share an ethnicity is often called an *ethnic group*. An ethnic group shares a common culture that is reflected in language or dialect, religion, customs, clothing, food, and music, literature, or art. Ethnic groups are often associated with nationality, geographic region, or country of origin. For example, in the United States, African Americans, Italian Americans, and Mexican Americans are examples of ethnic groups. Other countries would have different ethnic groups, based on different cultural traditions, such as religion or language.

The term *race* and *ethnicity* are often confused, but these terms are not identical. Race is based on meanings given to biological features, particularly skin color, as well as other physical traits as hair and eyes. Ethnicity is based on a shared culture and claimed identity associated with a sense of belonging and pride. As scholars have noted, ethnicity and race are dynamic, historically derived and institutionalized ideas and practices.

Economic power tends to be unequally distributed among ethnic groups. The dominant ethnic groups have power, also termed *privilege*, whereas nondominant groups often experience oppression or discrimination. Power—or the ability to get what one wants—is held by the majority group when they establish a system based on their own cultural values. Less powerful groups, such as ethnic minorities, occupy a lower status in society because of power relations.

Ethnicity is important in sport and exercise psychology because it impacts sport and physical activity patterns. Sport participation among ethnic groups varies by group, tradition, and rituals. In general, ethnic minorities have experienced a long history of being excluded from participation and leadership from organized sport, competition, and physical activity programs. Today, a few popular sports have a higher percentage of ethnic minorities, but in many sports, ethnic minorities are almost completely absent. Even when participation rates are high, ethnic minorities have been excluded from positions of power in sport, including coaching and management at the youth, collegiate, and professional levels.

Research within sport psychology indicates that negative stereotypes related to ethnicity and other

social categorizations are common in sport and lead to performance decrements. Stereotype threat, which has been found to occur in sport, refers to being at risk for confirming a negative stereotype of one's group. Stereotype threat has been found to lower the performance of African Americans in academic situations because of the negative stereotype that African Americans are less intelligent than other groups. For example, members of a racial or ethnic group believed to be academically inferior score much lower on tests when reminded of their race or ethnicity beforehand.

Scholars have suggested that sport psychology professionals are often oblivious to their own beliefs about athletes from different ethnic groups. In fact, sport psychology professionals' beliefs often are based on a Eurocentric worldview due to the lack of research and discussion related to cultural diversity in sport psychology. Furthermore, cultural diversity training in sport psychology is almost completely absent, which likely contributes to many professionals' lack of understanding of ethnicity. To expand our worldview, sport and exercise psychology professionals need to conduct more research on ethnicity. Those practicing in sport psychology are encouraged to be aware of their own biases related to ethnicity and to develop appropriate skills to work with members of ethnic minorities.

Cindra S. Kamphoff

See also Race

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EXERCISE DEPENDENCE

Many people become physically active to improve their health and to look and feel better. But physical activity may become addictive for a small proportion (3%–5%) of the population. It is important to emphasize that while exercise may represent an addictive behavior for some people who engage in it to an extreme and unhealthy level, habitual exercise itself is not inherently abusive and people should be active to meet guidelines (e.g., 150 minutes of moderate-intensity physical activity per week) to experience its numerous health benefits.

Exercise Dependence Defined

When the desire to exercise becomes an obsession and it negatively impacts a person's physical and psychological health, it is considered a serious problem known as *exercise dependence*. Simply stated, exercise dependence is a craving for leisure-time physical activity that results in uncontrollable and excessive exercise behavior that manifests in physiological (e.g., tolerance) or psychological (e.g., withdrawal) symptoms. Other general terms for exercise dependence include *exercise addiction*, *overexercising*, *obligatory exercise*, *compulsive exercise*, and *exercise fanaticism*. Common characteristics of exercise dependence are continuing to exercise despite having an injury or illness; giving up social, occupational, and family obligations for exercise; and experiencing withdrawal effects, such as anxiety and tension when not able to exercise.

The growing consensus in recent years is that exercise dependence is similar to other substance dependence disorders and should be defined within a cluster of cognitive, behavioral, and physiological symptoms. Researchers have thus defined exercise dependence as a maladaptive pattern of exercise, leading to clinically significant impairment or distress, as manifested by three (or more) of the criteria listed in Table 1, occurring at any time in the same 12-month period.

Exercise dependence may also play a pivotal role in explaining the function of exercise behavior

in the development and maintenance of eating disorders. To this end, it is necessary to distinguish the etiology of the exercise dependence symptoms. Understanding the psychological antecedents of exercise may help clarify the relationship between eating disorders and excessive exercise by offering insight into the distinction between primary versus secondary exercise dependence. *Primary exercise dependence* occurs when the individual meets criteria for exercise dependence and continually exercises solely for the psychological gratification resulting from the exercise behavior—for example, a runner who continually increases distance and speed to facilitate the enjoyment that is experienced from the running itself. *Secondary exercise dependence* occurs when an exercise-dependent individual uses increased amounts of exercise to accomplish some other end, such as weight management or body composition manipulation—for example, a runner who continually increases distance and speed to burn calories and facilitate weight loss. Because exercise can be used as a compensatory behavior to either prevent or reverse weight gain, secondary exercise dependence in the context of eating disorders occurs when an individual meets the criteria for exercise dependence and continually exercises to manipulate and control one's own body; thus, exercise dependence is secondary to an eating disorder. Recently, researchers have found that exercise dependence symptoms, not exercise behavior, mediate the relationship between exercise and eating pathology. That is, psychological factors, rather than the amount of exercise itself, may better explain why the exercise dependence–eating disorder relationship exists.

Prevalence and Societal Importance of Exercise Dependence

Despite the worldwide increase in obesity over the past several decades, there is nevertheless a growing rate of excessive behavioral addictions, such as gambling; Internet use; and exercise dependence that can result in poor psychological and physiological health outcomes, including increased risk for illness, stress, depression, and injury. Recent evidence suggests the estimated prevalence of exercise dependence in the general adult population is 3% to 5%. Among particular subgroups of adult exercisers, for example marathoners and sport science students, the prevalence of exercise dependence is even higher. In general,

Table 1 Exercise Dependence Criteria

<i>Criteria</i>	<i>Description</i>	<i>Example</i>
Tolerance	Need for increased exercise levels to achieve the desired effect, or diminished effects experienced from the same exercise level	Running 5 miles no longer results in improved mood
Withdrawal	Negative symptoms are evidenced with cessation of exercise, or exercise is used to relieve or forestall the onset of these symptoms	Anxiety, depression, or fatigue experienced when unable to exercise
Intention	Exercise is undertaken with greater intensity, frequency, or duration than was intended	Intended to run for 5 miles but ran for 7 miles instead
Lack of Control	Exercise is maintained despite a persistent desire to cut down or control it	Ran during lunch break despite trying to not exercise during work hours
Time	Considerable time is spent in activities essential to exercise maintenance	Vacations are exercise related, such as skiing and hiking trips
Reduction in Other Activities	Social, occupational, or recreational pursuits are reduced or dropped because of exercise	Running rather than going out with friends for dinner
Continuance	Exercise is maintained despite the awareness of a persistent physical or psychological problem	Running despite shin splints

Source: Heather Hausenblas and Danielle Symons Downs.

men present higher rates of exercise dependence symptoms, which parallels the higher rates of physical activity participation among men than women. The prevalence rate is about 10% for at risk for exercise dependence among regular adult exercisers. Even more concerning is the recent evidence that youth may also be at risk for exercise dependence symptoms. One study found that 6% of youth ages 14 to 16 years were classified as at risk for exercise dependence and an additional 65% were classified as nondependent but with some symptoms. Compared to the girls, the boys reported more overall exercise behavior and exercise dependence symptoms. While it has historically been thought that men develop exercise dependence symptoms in early adulthood as they hit their peak fitness level, this recent evidence suggests that boys may be more at risk for developing exercise dependence symptoms at an earlier age in the developmental timeline—possibly due

to their greater involvement in leisure-time play, youth sports, and the types of vigorous activities like running and weight lifting that can perpetuate overexercising. This is problematic because it increases the chances for long-term chronic health problems (injury, illness).

There is also a growing body of research beginning to quantify the exact aspects of health that are affected by exercise dependence. Specifically, compared with nondependent controls, individuals with exercise dependence often experience an increased amount of overuse injuries like tendinitis and muscle injuries, negative affect (particularly during exercise cessation), anxiety about the shape of one's body, neuroticism, hypochondria, and compulsive shopping or buying. Unfortunately, exercise dependence tends to be overlooked because it represents a socially acceptable (or socially tolerated) addiction that appears to be a reasonable form of dependence.

Measurement of Exercise Dependence

Measurement of exercise dependence relies primarily on self-report assessments due to the manifestation of psychological, behavioral, and cognitive symptoms. Assessing the frequency, duration, and intensity of physical activity alone does not provide an accurate estimate of the underlying psychological factors that drive the symptomatology. Although there are several unidimensional measures of excessive exercise (e.g., Negative Addiction Scale, Commitment to Running Scale, and Obligatory Exercise Questionnaire), these measures have been criticized for not providing a complete assessment of the construct of exercise dependence.

The growing consensus among researchers and practitioners is that exercise dependence is better conceptualized as a multidimensional construct that parallels behavioral addiction, and therefore, is more appropriately measured by multidimensional assessments that take into account the theoretical underpinnings of addiction. Researchers have posited that behavioral addictions manifest when a behavior can provide either pleasure or relief from internal discomfort of anxiety or stress, and the behavior is characterized by feelings of powerlessness or loss of control, and is maintained despite serious physical or psychological consequences. It is now accepted that addictions are a part of a biopsychosocial process that share a set of common symptoms, such as tolerance, withdrawal, mood modification, conflict, and relapse.

One validated self-report measure that falls within this categorization of addiction is the Exercise Addiction Inventory (EAI) developed by Annabel Terry, Attila Szabo, and Mark Griffiths. The EAI assesses six components of exercise addiction: salience, mood modification, tolerance, withdrawal, conflict, and relapse. Another psychometrically validated measure to assess exercise dependence is the Exercise Dependence Scale-Revised (EDS-R) developed by Heather Hausenblas and Danielle Symons Downs and is based on the *Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision* (DSM-IV) criteria for substance dependence. This 21-item measure assesses symptoms across seven criteria (tolerance, withdrawal, intention effects, loss of control, time, reduction in other activities, and continuance). The EAI and EDS-R are easy

to administer and offer researchers and practitioners useful tools for assessing exercise dependence symptoms.

Exercise Dependence Diagnosis and Treatment

Currently there are no formal diagnostic criteria for exercise dependence. However, the *DSM-IV* now includes behavioral addictions (although gambling is the only designated behavioral addiction in this category). A key factor in identifying a person with exercise dependence symptoms is distinguishing between the excessive exercise behavior and eating pathology. As noted above, primary exercise dependence occurs in the absence of an eating disorder, whereas secondary exercise dependence co-occurs with eating pathology. Thus, a diagnostic hierarchy is required whereby a person should be first evaluated for an eating disorder such as anorexia nervosa or bulimia nervosa. If the exercise behavior is secondary to the eating pathology, then referral and treatment for an eating disorder should be the priority. If eating pathology is ruled out, the individual may meet the criteria for primary exercise dependence. However, focusing only on the excessive exercise behavior in terms of frequency and intensity of exercise patterns fails to take into account the criteria that are unique to substance dependence, such as tolerance and withdrawal. Thus, determining if a person meets the aforementioned criteria for exercise dependence may provide insight to the excessive exercise. Both the EAI and EDS-R are reliable and useful tools that may assist clinicians in assessing exercise dependence. Also considering other coexisting addictions, such as alcohol or drugs or behavioral addictions like gambling, spending, or Internet use, can provide additional context to whether the person has an addictive personality disorder. Identifying individuals who are at risk for exercise dependence is a major challenge because exercise is considered a positive health behavior. Thus, excessive exercise often goes unnoticed as a negative health behavior until it has reached an extreme form. A key warning sign to distinguish between healthy and dependent exercise is that healthy exercisers organize their exercise around their *lives*, whereas dependents organize their lives around their *exercise*.

Danielle Symons Downs and
Heather Hausenblas

See also Adherence; Body Dissatisfaction; Body Dysmorphic Disorder; Muscle Dysmorphia; Eating Disorders

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EXPECTANCY-VALUE THEORY

Why do some individuals participate intensely in sport activities over many years while others never get actively involved in sport or exercise? What influences initial participation in sport or exercise? What influences continued participation? What influences the intensity of participation?

How do we explain drop-out from sport and exercise engagement? How do we explain both individual and group differences in sport and exercise engagement? Questions such as these have driven work in the area of motivation and sport and exercise for at least the last 50 years. Two main theoretical perspectives have emerged to address these types of questions: expectancy-value theory and self-determination theory. This entry focuses on one particular version of expectancy-value theory: Eccles and colleagues' Expectancy-Value Theory of Achievement-Related Behavioral Choices. This theoretical model has two components: a psychological component, illustrated in Figures 1 and 2, and a socialization component, illustrated in Figure 3. The following sections provide a brief description of each of these components and a summary of related empirical research.

Psychological Influences on Sport and Exercise Participation

Building on the work of Norman T. Feather, Victor Vroom, Albert Bandura, and John W. Atkinson, Jacquelynne Eccles and colleagues developed a comprehensive theoretical model of achievement-related choices, first published in 1983 (see Figure 1) and first directly applied to sport by Eccles and Rena D. Harold in 1991. It is henceforth referred to as the Eccles Expectancy-Value Model (EEVM). Eccles and colleagues hypothesized that both individual and group, for example gender or race or ethnic group, differences would be most directly influenced by individuals' expectations for success and the relative importance or value individuals attached to sport or exercise activities compared to other activity options (see Figure 2). They then hypothesized how these quite domain specific self- and task- related beliefs are influenced by cultural norms and stereotypes, experiences, aptitudes, and more general personal beliefs. In 1993, they specified a model of parental influences on the development of the psychological components of EEVT (see Figure 3).

According to the EEVM, people will be most likely to participate in those activities that they think they can master and that have high subjective task value for them. Expectations for success (domain-specific beliefs about one's personal efficacy to master the task), in turn, depend on both the confidence that individuals have in their various abilities and the individuals' estimations of

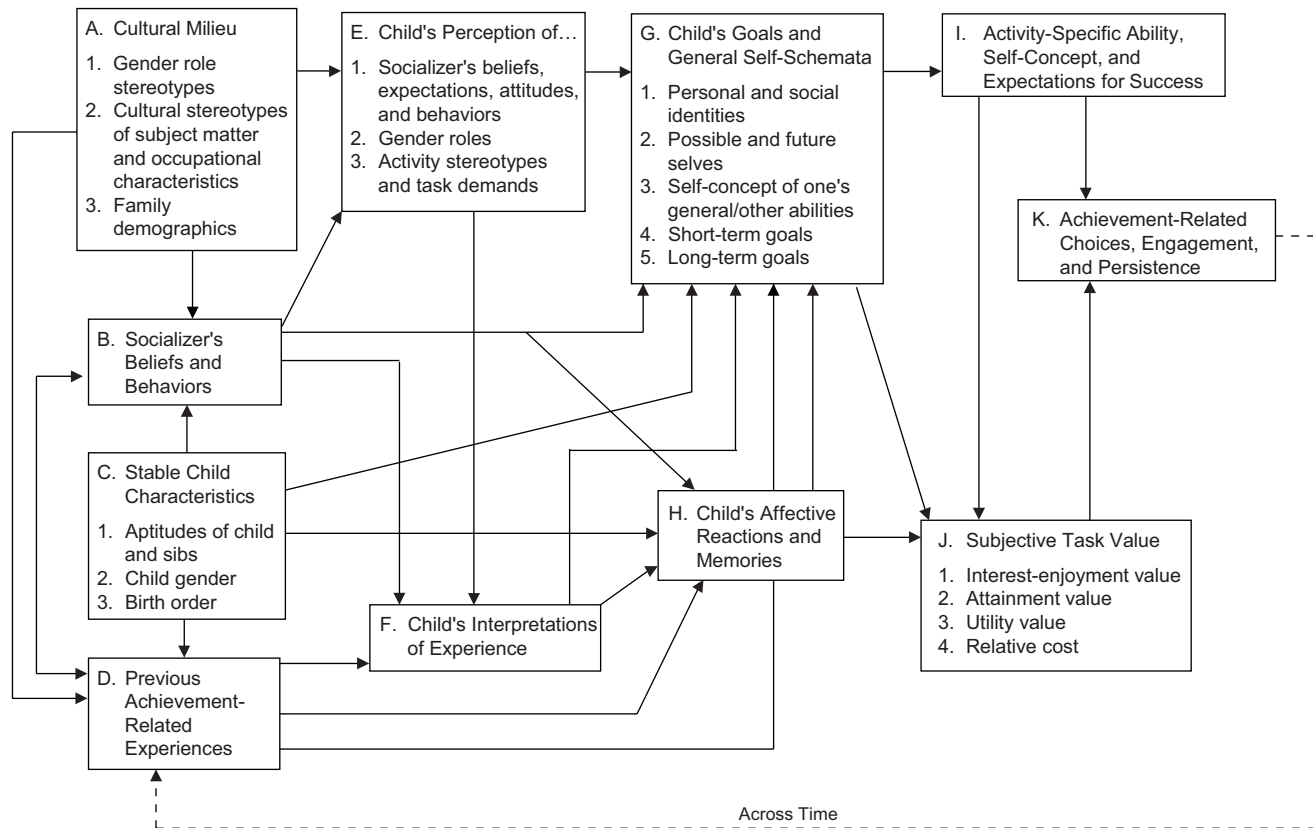


Figure 1 General Eccles Expectancy-Value Model of Activity Choices

Source: Copyright © Jacquelynne S. Eccles.

Note: Since we are not assessing F, we have removed most of the arrows to and from this box.

the difficulty of the various options they are considering. Eccles and associates also predicted that these self- and task-related beliefs were shaped over time both by experiences with the related activities, and by individuals' subjective interpretation of these experiences. For example, does the individual think that personal prior and current successes reflect high ability or lots of hard work? And if the latter, will it take even more work to continue to be successful? Conversely do personal failures or difficulties reflect lack of *natural talent* or deficits in current abilities? Can the current ability level be changed through practice and instruction or does it reflect a stable entity linked to genetic or other stable biological influences—that is does it reflect natural talents? Are these natural talents believed to be linked to gender or other group characteristics? The latter interpretative questions are particularly important for explaining gender and other group differences in participation in sports and exercise. It is likely,

for example, that females in many cultures receive less support for developing a strong sense of their talent for sport from their parents, teachers, and peers than males and are more likely to attribute their difficulties in mastering sports to insufficient natural talent than males. Research has supported this prediction.

Likewise, the EEVM model specifies that the subjective task value of various activities is influenced by several factors. For example, how much does the person enjoy doing sport compared to other activities (referred to as intrinsic interest in the model)? Is participation in sport or exercise seen as instrumental in meeting the individual's long- or short-range goals (referred to as utility value)? Is being good at sport or exercise a critical part of the individual's identity (referred to as attainment value)? Does participating in sport or exercise interfere with other more valued options because of the amount of work needed to be successful either in the major or in the future

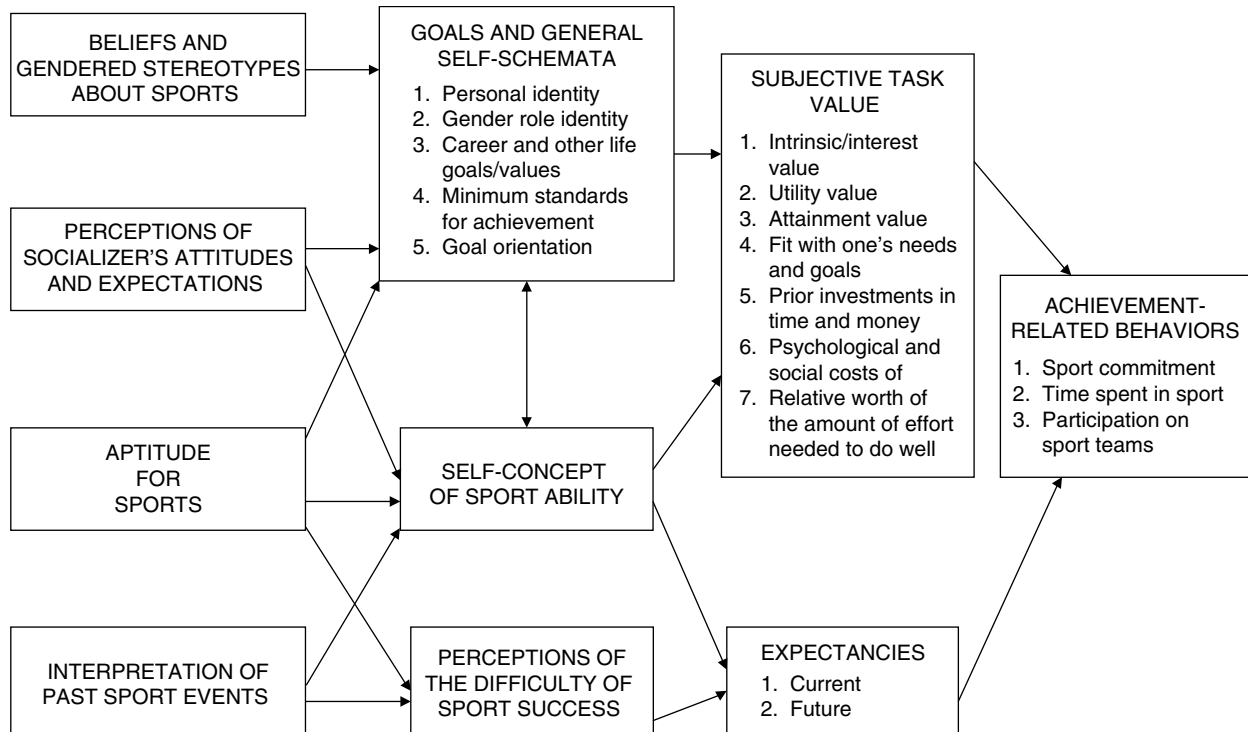


Figure 2 EVM for Sport Participation

Source: Copyright © Jacquelynne S. Eccles.

professions linked to the major (referred to as cost)? The answers to these kinds of questions have proved critical for explaining both individual and groups differences in sport and exercise participation.

As the model was developed in the mid-1970s, it was clear that the theoretical grounding for understanding the nature of subjective task value was much less well developed than the theoretical grounding for understanding the nature of expectations for success. Consequently, Eccles and associates elaborated their notion of subjective task value to help fill this void. Drawing upon work associated with achievement motivation, intrinsic versus extrinsic motivation, self-psychology, identity formation, economics, and organization psychology, they hypothesized that subjective task value was composed of at least four components:

1. Interest value (the enjoyment one gets from engaging in the task or activity)
2. Utility value (the instrumental value of the task or activity for helping fulfill another short- or long-range goal)

3. Attainment value (the link between the task and one's sense of self and identity)
4. Cost (defined in terms of either what may be given up by making a specific choice or the negative experiences associated with a particular choice)

It is the belief of Eccles and colleagues that the last three of these are particularly relevant for understanding gender and other group differences in activity choices. By and large, research has supported these predictions.

Furthermore, Eccles and colleagues believe that both individual and group differences in these aspects of subjective task value are heavily influenced by social forces and socialization, which themselves are influenced by cultural norms and stereotypes. For example, have the individual's parents, counselors, friends, or romantic partners encouraged or discouraged the individual from participating in sport or exercise? More specifically with regard to gender, for example, Eccles and colleagues argued that the socialization processes linked to gender roles are likely to influence

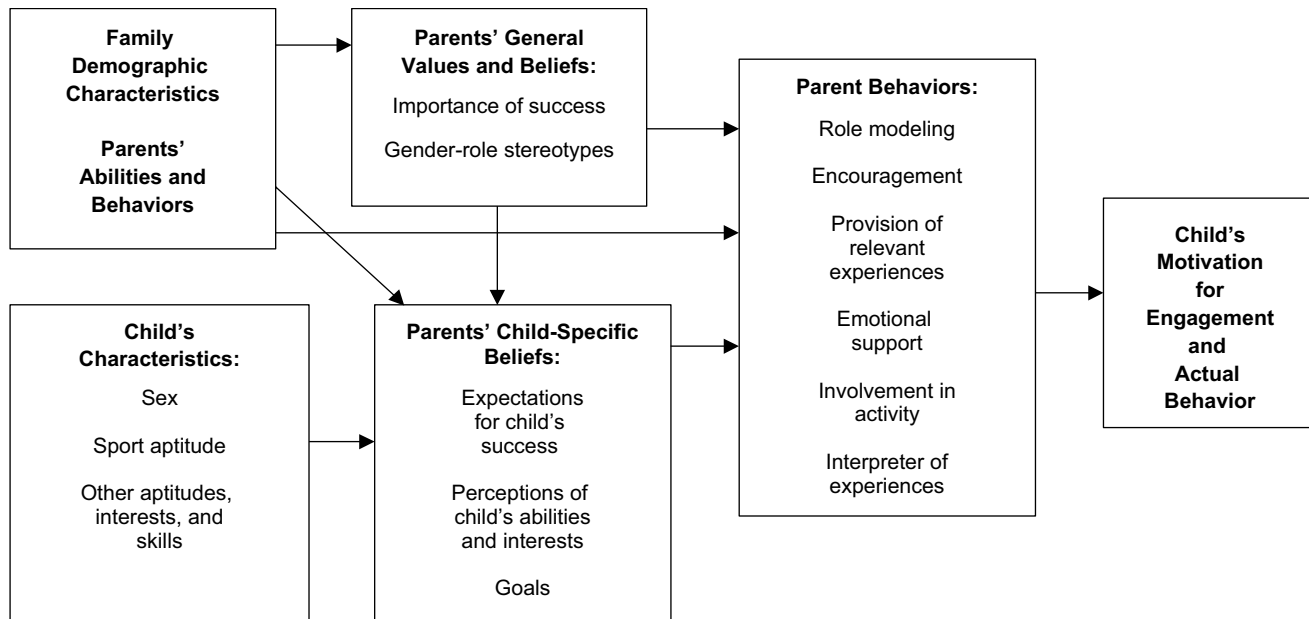


Figure 3 Eccles Parent Socialization Model

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both short- and long-term goals and the characteristics and values most closely linked to core identities. They predicted that males would receive more support for developing a strong interest in sport from their parents, teachers, and peers than females; research has supported this prediction. For example, gender-role socialization has been shown to lead to gender differences in adolescents' sport ability self-concepts, the subjective task value of sport, and actual engagement in sport activities. These outcomes, in turn, have been shown to predict adult involvement in sport and exercise activities.

In summary, Eccles and colleagues assume that activity choices, such as participating in sport and exercise, are guided by (a) individuals' expectations for success (sense of personal efficacy) regarding the various options, as well as their sense of competence for various tasks; (b) the relation of the options to their short- and long-range goals, to their core personal and social identities, and to their basic psychological needs; (c) individuals' culturally-based role schemas linked to gender, social class, and ethnic group; and (d) the potential cost of investing time in one activity rather than another. They further assume that all of these psychological variables are influenced by individuals' histories as mediated by their interpretation of

these experiences, by cultural norms, and by the behaviors and goals of one's socializers and peers.

Eccles and her associates have spent the last 40 years amassing evidence to support each of these hypotheses. The findings related to gender differences participation in sport and exercise are quite robust. Here are just a few examples of both the psychological and socialization models. (All of the survey instruments, details on the studies, and the publications themselves are available at www.rcgd.isr.umich.edu/garp.)

First, the researchers looked at the psychological predictors of participation in sport in high school. They used path analysis to determine whether the gender difference in sport participation was mediated by constructs directly linked to expectations for success (specifically the students' self-concept of their sport ability) and subjective task value (specifically, how much they enjoyed sport), the subjective importance of doing well in sport, and the usefulness of sport while controlling for the students' scores on. As predicted, the gender difference in participating in sport was mediated by the gender differences in these beliefs.

The second part of the research has focused on the role of experiences in the home in shaping both individual and group differences in the

self- and task-related beliefs just discussed, as well as such differences in sport and exercise participation. First, using a variety of longitudinal modeling techniques, Eccles and colleagues were able to confirm most of the basic links summarized in Figure 3. Many parents do endorse gender stereotypic beliefs about general gender differences in abilities and interest in many and parents' gender-role stereotypes predict their perceptions of their own children's math, sport, and English abilities and interests even after independent assessments of their children's actual abilities in these three domains have been controlled, leading them on average to underestimate their daughters' athletic abilities. In turn, parents' estimates of their children's abilities in different skills predict developmental changes in their children's own estimates of their abilities in these skills, as well as the children's expectations for success in these different skill areas, leading to the kinds of gender differences in the children's own beliefs and task values discussed earlier.

Second, on average, parents provide their daughters and sons with gender-role stereotypic types of toys and activity opportunities, particularly in relation to sport. Not surprisingly, these differential patterns of experiences partially mediate the emergence of gender differences in children's confidence in their own abilities, their interests in participating in different activities, particularly, and their actual emerging competencies. Not surprisingly, the extent to which each of these facts holds true is more marked when parents endorse the traditional gender-role stereotypes associated with both ability and interest in these different skills areas.

Conclusion

Research conducted over the past 4 decades by Eccles and colleagues, and by other researchers as well, has provided clear evidence that gender-role related personal and social processes explain, at least in part, the gender differences we see in sport participation. Much less work has been done on exercise. Furthermore, interventions have been created that can help ameliorate these processes, with the result that both girls and women and boys and men are able to make less gender-role stereotypic choices for their own lives.

Jacquelynne S. Eccles

See also Gender; Identity; Parenting; Participation Motives; Self-Determination Theory; Self-Efficacy; Self-Regulation

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EXPERTISE

Expertise refers to the underlying qualities and skills that distinguish highly accomplished people, referred to as experts, from lesser skilled people. Experts are people who are exceptionally skilled in a particular task or domain and their expertise is typically evident across numerous aspects of human performance, including perception, cognition, and motor execution. Interestingly, despite the clarity by which we can identify an expert, definitional inconsistencies of what constitutes an expert are common in the literature. One of the defining features of experts is the amount of practice and effort they have devoted to achieve this status. A number of models detailing the development of expertise have been suggested, creating some confusion for practitioners in the field. Expertise research is conducted in a wide variety of domains, as diverse as sport, medicine, education, the performing arts, and the military.

Why Study Expertise?

There are clear scientific reasons to study expertise. First, discovering the mechanisms underpinning an expert's superior performance in a particular domain, such as sport, provides a broader understanding of the factors that may limit a non-expert's quest to become more skilled. Over time, the collective efforts of expertise researchers have allowed the development of more robust and generalizable theories of expertise and its acquisition (see later section "How Is Expertise Acquired"?). A second key reason to study expertise is to understand what type and volume of practice is critical for the acquisition of skill, and ultimately expertise. Importantly, this information cannot be obtained from the study of non-experts even when trained to perform thousands of practice repetitions in a controlled skill-acquisition experiment.

The scientific study of expertise can assist in providing recommendations to practitioners in the field, such as coaches, teachers, and skill acquisition providers, to assist in the refinement of practice conditions to enhance performance. Similarly, as a higher premium is placed on the identification and subsequent development of talent, testing protocols can be implemented that focus on the

domain-specific elements of performance known to be required to achieve expertise.

A Brief History of Sport Expertise Research

While humankind's fascination with expert sport performance can be traced back to at least the time of the first Olympics, quite surprisingly an identifiable program of research investigating sport expertise is only a few decades old. Seminal work investigating chess masters in the mid- to late 1960s and early 1970s proved to be the stimulus for the current interest in sport expertise research. The common experimental paradigm adopted in most studies has involved an expert–novice comparison where a group of experts or highly skilled performers complete a particular task thought to be representative of that completed in the performance setting and their results are compared to those of a lesser skilled group. It is assumed that the performance variables that reliably distinguish the experts from the lesser skilled are elements important to expertise (see later section "Common Features of Expertise").

In 1965, Adriaan De Groot investigated the perceptual attributes of chess players. He found that after providing chess masters with only a brief view of the chess pieces on a chessboard, they were able to reconstruct the locations of those pieces better than lesser skilled players. These early findings were proven to be robust by subsequent researchers and led to William Chase and Herb Simon's 1973 theory of expertise. A key finding central to this theory was that expertise was only evident when the pieces were structurally arranged such that they were in similar positions to those encountered in normal game play. When the same pieces were arranged in completely random configurations, the chess masters lost their reconstruction advantage over lesser skilled players. This finding suggested that expert chess performance was not the result of an enhanced memory capacity but rather of a capacity to overcome the short-term memory limitation of 7 ± 2 pieces (chunks) of information. While the notion of chunking continues to be revised, the superior performance of experts from many domains is grounded in their capability to process larger amounts of task-relevant perceptual information and then use this information to prepare an appropriate movement response.

In addition to the memory-processing explanation of expertise, domain-specific knowledge has also been considered an important underlying mechanism. Declarative knowledge and procedural knowledge have been proposed as the primary cognitive structures underlying skilled performance. *Declarative knowledge* has been defined as the knowledge of factual information, whereas *procedural knowledge* has been defined as the knowledge of how to do things. A declarative knowledge base has been suggested to provide the foundation for the development of the more complex structures present in procedural knowledge. Compared to non-experts, sports experts possess structured knowledge at a more sophisticated level consisting of highly structured offensive and defensive sport concepts, which in turn are thought to facilitate their decision-making processes. However, particularly in sport, the possession of a strong declarative knowledge base does not guarantee equivalent levels of procedural knowledge. For example, players may possess a substantial declarative knowledge about their sport but possess a lower level of procedural knowledge that acts to limit their overall performance capability.

Cognitive approaches to the study of expert performance have been especially influential in the study of experts in sport. However, sport expertise clearly demands not only enhanced cognitive functioning but also well-timed and accurate motor execution. Accounting for the central role of motor skill within sport expertise has been a consistent challenge for researchers, particularly in relation to creating representative tasks that are sufficiently well controlled to permit the underlying components of expertise to be accurately and reliably recorded. However, technological progress in the measurement tools available to researchers has seen a recent shift from laboratory to field-based measurement with a concomitant increase in the use of more representative tasks; this, in turn, has helped build a more complete understanding of sport expertise.

The expert performance approach was first described by Anders Ericsson and Jacqui Smith in 1991. Three stages were proposed within this approach. The first stage involves capturing expert performance or, put alternatively, identifying the domain-specific situations where the expert excels and then developing representative tasks that allow this superior performance to be re-created in

experimentally controlled conditions. The second stage aims to assess the underlying mechanisms that account for superior performance on the representative tasks. The identification of processing differences between expert and non-expert performers can be obtained through a variety of experimental techniques, such as gaze tracking, verbal reports, and occlusion techniques. The third and final stage involves examining how the identified expertise was developed through experience and practice, again by using a variety of experimental methods. While the expert performance approach has not been adopted universally, it does provide a logical framework from which to consider some of the common features of expertise that have emerged from research.

Common Features of Expertise

The collective efforts of expertise researchers to understand the nature of the expert advantage have revealed a number of common features of expertise that systematically emerge across sport and other domains. These are summarized below.

Expertise Is Specific

Expertise is not transferable between domains (or across sports) and will only appear when the usual context of the sport task of interest is identical or sufficiently representative of the performance setting. For instance, generalized measures that lack sport specificity, such as a reaction time task where the performer is required to react to a flashing light rather than a sport-specific situation requiring a fast reaction (hitting a baseball), consistently fail to distinguish experts from lesser skilled individuals. Recently, the extent of expertise specificity has been challenged. An intuitive view is that talented performers from one sport have some common skills that transfer across sports or even to other domains. While there is some limited empirical evidence to suggest that some elements (anticipation and decision making) transfer from one related sport to another, much more research is needed before a definitive position can be reached.

Experts Recognize and Recall Patterns From Their Domain of Expertise More Effectively

The chunking capabilities of chess masters described earlier are also consistently found in the

team-sport domain with expert players recognizing and recalling structured game patterns more completely than lesser skilled players. It has been suggested this capacity is central to the advanced decision-making skills demanded in time-stressed situations common in such sports as basketball or football. One matter of current debate is whether pattern recall tasks require individuals to use processes they may not otherwise have used, at least explicitly, during a typical game. Accordingly, some researchers have suggested pattern recall is a good example of a task that is not sufficiently representative to elicit true expertise differences in a sport domain despite the demonstration of some expert–novice differences on these types of task.

Experts Possess Knowledge Superior to That of Lesser Skilled Individuals

Not only do experts tend to possess more declarative knowledge or facts about their specific sport, but importantly, experts also appear to know how to navigate an effective course of action in competition (procedural knowledge). Expertise researchers have systematically explored the various connections between different types of knowledge and observable skilled performance to better understand the relative contribution of knowledge to sport expertise. This aspect of sport expertise has been characterized by some innovative research designs where sports experts are compared with expert coaches or spectators who may share similar levels of declarative knowledge but differ in their motor skill proficiency.

Experts Anticipate Their Opponent's Behavior More Effectively Than Non-Experts

In time-stressed tasks, such as returning a tennis serve, an expert player is able to alleviate some of the time stress and in turn allow more definitive response preparation through the reliable anticipation of advance information. The expert performer has been found to pick up two forms of advance information more selectively than lesser skilled performers—situational probability information and opponent movement characteristics. Situational probability information consists of generic information available to a performer before the opponent commences their skill execution. The information typically arises from pre-game knowledge of player strengths, weaknesses, and preferences, as well as court or field position,

and event probability information. The source of the expert performer's attunement to their opponent's movement pattern is argued to stem from a direct link between their perceptual expertise and the pickup of invariant and predictive kinematic information within their opponent's action.

Expert Motor Performance Is More Efficient and Subconsciously Controlled

Expert motor performance has been associated with the ability to produce coordinated movements that are both efficient and effective in terms of meeting the environmental demands of the task or game. The expert's movement coordination superiority over non-experts stems from a range of factors, including the superior recruitment of force, increased fatigue resistance, and the capacity to more effectively control multiple degrees of freedom (limb, joint, and muscle combinations). In contrast to experts, novices tend to approach skills with rigid movements and lock bodily segments at the joints, freezing the degrees of freedom. In a nice illustration of the multidimensional nature of expertise, experts are also able to direct their attention to the external environment to a greater extent than their lesser skilled counterparts as evidenced by their capacity to perform two tasks simultaneously, for example dribbling a basketball while scanning for a passing option). Experts are thought to automate their movement production in that they not have to consciously think about skill execution because of extensive practice relative to lesser skilled individuals. Interestingly, experts can experience skill performance decrements in conditions that force them to process their skills in a more conscious or novice-like manner, leading to what is colloquially referred to as *paralysis by analysis*.

How Is Expertise Acquired?

A long-standing debate within the expertise field concerns determining the most effective pathway to the acquisition of expertise. The most common approach to this question has been to retrospectively examine the practice histories of expert relative to non-expert performers via interviews, training diaries, and questionnaires. Two dominant positions have emerged in regard to the best approach to the development of expertise, that is, a practice-focused approach referred to as the *theory of deliberate practice* and a

play-before-specific-practice approach, referred to as the *developmental model of sports participation* (DMSP). In practical terms, these two approaches can also be aligned to the debate on the relative merits of early (deliberate practice) versus late (DMSP) specialization into a sport.

In 1993, Ericsson, Ralf Krampe, and Clemens Tesch-Romer proposed that expertise is primarily a matter of practice rather than reliance on innate talent, albeit acknowledging that height and body size are clearly important to success in many sports. Deliberate practice is considered to have occurred when a well-defined task, set at an appropriate difficulty level for the learner, is completed with access to feedback and opportunities for practice repetition and correction of errors. Such practice requires effort, generates no immediate rewards, is motivated by the goal of improving performance rather than inherent enjoyment, and consequently occurs over an extended period of time (usually in excess of 10 years, referred to as the 10-year rule). Practicing deliberately is argued to incrementally develop the underlying mechanisms that lead to expertise. The key tenets of the deliberate practice framework were originally formulated based on the practice histories of expert-level musicians and research highlighting the plasticity of cognitive skills to the effects of practice. The importance of deliberate practice has been substantiated in some sport settings, albeit with a number of qualifications suggested by those working within the sport domain. For instance, in contrast to the original definition, it has been found that deliberate practice activities can be both highly enjoyable and high in concentration in the sport setting. In addition, squad or team practice, rather than practice alone (or individually with a teacher), has been identified as being most predictive of skill level in team environments.

In contrast to the notions of deliberate practice, others have advocated a more diversified and playful commencement as desirable for the lengthy journey toward expertise. Jean Côté, Joseph Baker, and Bruce Abernethy building on the work of Benjamin Bloom, formulated the DMSP and suggested three chronological stages of sport participation from early childhood to late adolescence. The sampling years (ages 6–13) emphasize fun and excitement through participation in a large number of sport activities. Voluntary participation of the child in activities that are intrinsically motivating, pleasurable, and

provide immediate gratification are key attributes of this stage. These characteristics are typical of pick-up, backyard, or neighborhood-style games that are self-initiated by children with the sole determinant of having fun (commonly referred to as *deliberate play*). In the specializing years (ages 13–15), involvement in other activities gradually decreases and the focus shifts toward only one or two specific sports. Positive experiences with coaches, encouragement from siblings and friends, and the simple enjoyment of the activity influence the child's decision to gravitate toward a specialized sport or sports. Sport-specific skill development is an important feature of the specializing years and marks a transition toward a more deliberate practice approach. The investment years (15 years and older) suggest the intent of an athlete to pursue expertise in a single sport. The sampling and deliberate play activities reported in the previous stages are replaced by large volumes of intense, sport-specific practice (similar to deliberate practice).

While the DMSP has been supported by a number of subsequent investigations in sport, it has also been challenged. In particular, a key issue is whether expertise requires a more focused approach to deliberate play whereby children are encouraged to engage in playful activity primarily related to their preferred sport of interest rather than diversifying or sampling too many sports. It is argued that such an approach may meet the needs of specialization while simultaneously providing sufficient diversification. It has also been suggested that the relative importance of early versus late specialization may be largely sport- and culture-specific.

Conclusion

Research into the development of expertise and expert performance focuses on the identification and understanding of the mechanisms that distinguish experts from lesser skilled performers. Expertise is domain specific and requires the development of representative tasks if it is to be reliably demonstrated and examined in an experimental setting. A number of common features of expertise have been demonstrated in perceptual, cognitive, and motor components of performance. The development of expertise is perhaps the most debated aspect of this research field with two competing approaches being suggested. This debate

has important practical implications in regard to whether it is advantageous to specialize early in a sporting task or diversify across a number of sports to reach expert status. There are many fertile areas for future research before a complete understanding of expertise will be gained.

Damian Farrow

See also Anticipation; Automaticity; Chunking/Dechunking; Decision Making; Genetics/Nature–Nurture Determinants; Memory; Pattern Recognition and Recall; Practice; Skill Acquisition; Talent Development

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EXTRAVERSION–INTROVERSION

Extraversion–introversion is a personality factor that refers to the degree to which a person's basic orientation is turned inward (toward oneself) or outward (toward the external world). Introverts are said to be shy and prefer to work alone; they tend to withdraw into themselves, especially when they experience emotional stress and conflict. In contrast, extraverts are said to be sociable and prefer occupations that enable them to work directly with other people; when experiencing stress, they will tend to seek company.

Another related personality factor is *emotional instability–stability* (or neuroticism), which is a continuum of emotionality that varies from moody, anxious, temperamental, and maladjusted persons in the *unstable or neurotic* end, as opposed to *stable* people, who are said to be calm and well adjusted.

These two personality factors are strongly associated with the name of Hans Eysenck. In psychology, the question of which primary source accounts for the apparent variance in human behavior has long been debated. The proponents of the trait approach advocated stable, intraorganismic constructs as the main determinants of behavior. One of the most noted of the trait researchers during the 1960s and the 1970s was Eysenck, who conceived traits as relative. This psychologist used psychiatrists' ratings of patients' characteristics to arrive at two superordinate personality factors ranging on continuums, namely *introversion–extraversion* and *emotional instability–stability* (better known as *neuroticism*); these most significant dimensions emerged from many factor-analytic studies of the intercorrelations between numerous and different traits.

Strictly speaking, these two personality factors only provide phenotypic descriptions; therefore, Eysenck also suggested causal explanations on the biological level. He interpreted the introversion–extraversion dimension as being based on the balance between cortical excitation and inhibition. Excitation is related to the facilitation of perceptual, cognitive, and motor responses in the central nervous system, whereas inhibition is related to the exact opposite effect, namely the suppression of such responses. Emotionality or neuroticism is biophysically explained through the instability of the autonomous neural system; it is assumed that an autonomous response is dependent on a

person's constitutional structure, which mediates the strength of his or her responses to incoming stimuli. Thus, it is the autonomous neural system that is most probably the basis of individual differences concerning emotionality.

Trait theorists have typically used objective inventories that involved direct quantification of personality dimensions, without intervening interpretations of participants' responses. Such was the Eysenck Personality Inventory (EPI), which was among the most popular trait inventories taken from general psychology and used during the 1960s and 1970s in early sport personality research. The basic idea here was to try and "match" athletes to sport disciplines; for example, it was argued that an extravert's cortex is less aroused than that of an introvert, and therefore an extravert will need much stronger external stimuli than the introvert, who will in turn suffer from such a strongly stimulating environment; an introvert will clearly prefer a weak environmental stimulation.

According to this line of reasoning, if (a) lower cortical arousal is mostly related to extraversion and higher cortical arousal to introversion, and (b) minor arousal fluctuations are more typical of emotionally stable people and substantial arousal fluctuations are mostly apparent in neurotics, then we could argue that for an elite athlete, although being stable is almost always better than being unstable, the performance quality of introverts versus extraverts will depend on the task (sport discipline) to be performed. Thus, as demonstrated, for example, by J. E. Kane, in a stressful environment when the fulfillment of a complex task is required, as in a basketball game, stable extraverts seem to exhibit motor advantage not only over unstable extraverts, but also over stable introverts (and in particular, of course, over unstable introverts). By contrast, stable introverts will enjoy an advantage in sport disciplines such as marathon running, long-distance swimming, or rifle shooting, which are monotonic and environmentally poorly stimulating.

For the aforementioned reasons, it was particularly the introversion–extraversion dimension that caught the attention of much of the sport-personality research from its earliest days. In general, athletes' personality structures were found to lean toward extraversion, which is not quite surprising given the larger numbers of people actively engaged in extravert sports (team sports such as football or basketball). However, lone athletes like marathon runners were indeed found to be more introverted;

moreover, it was found that top, superior athletes tended to be more introverted in comparison to lower level athletes, even within extravert samples (not to mention introvert ones). Kane contended that superior athletes of all sport disciplines enjoy the advantage of being able to isolate themselves better in order to appropriately concentrate and prepare for task performance, an ability which is clearly related to introverted tendencies.

Despite the fact that Eysenck has been one of the most cited psychologists of his time, and even though his two previously mentioned factors are included among the *Big Five* of personality, it seems that simply knowing someone's personality trait will not really help to predict how that person will perform under specific circumstances. Accordingly, sport-personality research waned toward the 1980s and 1990s, probably because much of this research resulted in what were viewed as statistically significant differences that were often meaningless. However, questions, such as whether extraverts will perform better in team situations and introverts in individual ones, can lead to more accurate predictions of athletes' behavior in comparison to relying only on traits, or alternatively, only on situations.

It may therefore be possible to more usefully apply Eysenck's two factors within an interactionist framework, such as the one suggested by Robin Vealey. Strictly speaking, the basic idea mentioned above, namely that of matching individuals and sports, actually reflects an interactionist perspective. Moreover, transactionist approaches such as Michael Bar-Eli's crisis theory, in which introversion–extraversion and neuroticism were included in a reconceptualization of athletes' prestart states, should be encouraged in order to more fully exploit the potential concealed in these two big personality factors, and further advance our understanding of sport behavior.

Michael Bar-Eli

See also History of Sport Psychology; Models of Emotion–Performance; Personality Tests; Personality Traits and Exercise; Psychological Skills; Psychophysiology

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EYE MOVEMENTS/GAZE

The eyes perceive spatial information with full acuity when light falls on a small region at the back of the retina called the *fovea*. Because of the small area of the fovea, the area over which we are able to see clearly is only about 2° to 3° of visual angle. Visual angle indicates the size of an image on the retina. It is usually determined by extending lines from the edges of the object as viewed in space through the lens to the retina. One way to estimate the size of this area is to hold your thumb up at arm's length on object in space. The width of your thumb subtends 2° to 3° of visual angle projected into space. Gaze control is the process of directing the eyes to objects or events within a scene in real time and in the service of the ongoing perceptual and cognitive processes. The reason the eyes move is to keep objects of interest projected on the high-acuity fovea. Usually the eyes move before the head, localizing a target first. Head movement follows because of the greater size and inertia of the head compared to the eyes. The movement of the eyes and head to the target is normally smooth, with the processing of visual information occurring within 100 ms of the eyes stabilizing on the new location.

Eye movements were first studied with the head fixed as even the slightest movement of the head could cause large errors in the eye data being collected. Today, with the advent of light mobile eye trackers, it is possible to record accurate eye movements as the participant moves naturally throughout the world, making head movements. Eye-tracking methods permit the measurement of the location of the gaze in space or the eye

movements relative to the head. Most eye trackers used in sport are corneal-reflection systems that consist of two video cameras mounted on a headband or glasses. One camera records a video of the eye and the other the scene in front of the performer. The location of the gaze is determined using the center of the pupil and the corneal reflection, which is artificially placed on the eye (using infrared light). Data are collected digitally in *x/y* coordinates, as well as the location of the gaze being shown on the scene video as a cursor or other marker every frame or field of video. The accuracy of most corneal-reflections systems is about 1 degree of visual angle. The most common National Television System Committee (NTSC) data collection is 30 Hz, or 30 frames per second (33.33 ms per frame); or 60 Hz, or 60 fields per second (16.67 per field).

Types of Eye Movements and Gaze

Most eye-tracking systems record fixations, pursuit tracking, and saccadic eye movements. A fixation occurs when the gaze is held on an object or location within 3° of visual angle for 100 ms or longer. The 100 ms threshold is approximately the minimum amount of time the brain needs to recognize or become aware of what is being viewed. Pursuit-tracking eye movements are similar to fixations, except the eyes follow a moving object, such as a ball or a person. The 100-ms threshold still applies for the same reason as it is used for fixations. Saccades occur when the eyes move quickly from one fixated or tracked location to another. Saccades are among the fastest movements that humans can make, exceeding 900 degrees per second. Saccades are ballistic eye movements that bring the point of maximal visual acuity onto the fovea so that an object can be seen with clarity. During saccades, information is suppressed, meaning the information between two fixated locations is not consciously perceived. Instead, the information gained during fixations is maintained in memory thereby ensuring a stable, coherent scene. Saccadic suppression prevents us from seeing a world that is blurry and therefore difficult to comprehend.

The fixated or tracked information falling upon the fovea constitutes focal vision. The focal system is used when fixated or tracked information falls on the fovea and aspects of an object or location are viewed with full acuity or detail.

In contrast, information that falls off the fovea becomes increasingly blurry (of low resolution) and is viewed using ambient or peripheral vision. The ambient or peripheral system is specialized for motion and low light conditions. Both the focal and ambient systems are needed for the visual system to function properly.

Visual Motor Pathways

Visuomotor control is the process whereby visual information is used to direct and control movements. Visual information is registered first on the retina, then passes through the optic nerve, the lateral geniculate nucleus, and the optic or striate radiations to the visual cortex or occipital lobe at the back of the head. Located in the occipital cortex are visual sensors that begin the processing of registering and interpreting what the performer sees, with specific detectors for initial registration (V1), shape (V2), angles (V3), motion (V3a), color (V4), V5 (motion with direction), depth and self-motion (V6), and depth of stereo motion (V7). Once an object, person or location is registered within the visual cortex, visual information moves rapidly forward in the brain, along the dorsal attention network (DAN) and the ventral attention network (VAN). The DAN projects from the occipital lobe to the parietal lobe and forward to the frontal lobe in a journey that goes roughly over the top of the head to the frontal lobe. In contrast, the ventral attention network (VAN) projects forward along the sides of the head through the temporal lobes to the frontal areas. Commands are then sent from the frontal areas to the motor cortex to initiate action. With practice and the development of expertise, it is thought the DAN system sustains attention on critical spatial locations, thereby blocking out competing anxiety-producing stimuli that may intrude from the VAN system while the movement is being performed.

Eye Tracking Methods

Two research methods are widely used in sport, called the *visual-search* and *vision-in-action* paradigms. The visual-search paradigm is the older of the two paradigms, dating back to the beginning of psychology. When the visual-search method is used, eye movements are recorded as the athlete sits or stands and views photographs, videotapes, virtual reality, or computer simulations of events from their sport. Motor responses usually

involve a manual key press, step sensor, or pencil-and-paper response. Using this paradigm, it has been established that expert athletes, compared to novices, are superior at recalling and recognizing sport-specific patterns of plays, are faster in detecting specific cues, are more efficient in recognizing objects, and are better at anticipating upcoming events.

The vision-in-action paradigm differs from the visual-search paradigm in recording the athlete's gaze, in situ, under conditions similar to those found in real world play or competition. Elite and non-elite athletes perform well-known tasks, such as golf putting, basketball free throw, and soccer or ice hockey goaltending, in experiments that couple perception and action. Most studies require the athlete to perform the task until an equal number of successful and unsuccessful trials have been completed thus allowing the detection of gaze and perceptual-motor differences during successful and unsuccessful performance. A major finding resulting from this method is the discovery of the *quiet eye*, which is the final fixation or tracking gaze maintained on a specific location prior to a critical movement in the task. Quiet eye onset occurs prior to the movement and quiet eye offset occurs when the gaze deviates off a specific location for more than 100 ms. Numerous quiet eye studies have shown that elite athletes have earlier and longer quiet eye durations than non-elite; a longer duration quiet eye is also a characteristic of more successful performance. Quiet eye training has proven to be effective in improving performance compared to traditional methods in a number of sports and other areas.

A few studies have also tested the same athletes using both the visual search and vision-in-action methods, with different results reported in terms of the gaze and motor performance. When the athletes were tested using the vision-in-action method, they used fewer fixations to fewer locations in the task environment. Their fixations occurred earlier and were longer in duration than when they were tested using the visual-search paradigm and video simulations of the same task. When the vision-in-action paradigm was used, the athletes' anticipatory actions occurred earlier and both their reaction time and movement times were faster than when the visual-search method was used. These results suggest that the demands of real-world performance create a tight coupling between perception and action that can

only be detected when the athlete is tested under realistic conditions.

Joan N. Vickers

See also Expertise; Perception; Vision

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F

FAIR PLAY

The purpose of fair play is to ensure that every competitor has an equal chance of being successful in any given competition. Fair play is supported by a philosophic belief that every player, team, official, and fan respects and honors (1) the rules of the game (constitutive, regulatory, and sportsmanship), (2) those who play the game (opponents), and (3) those who call the game (officials). This notion of fair play also transcends the actual playing to include all off-field preparation for the game as well as postgame activities for how opponents are treated.

The principle of fair play is ingrained in the Western tradition—fleshed out on the playing fields of English preparatory schools from the late 1700s through the 1800s. In Victorian England specifically, it was argued that young men could learn the important virtues of honesty, fair play, justice, propriety, and decency through the playing of games. Children of the upper classes who attended the elite private schools were taught that sport, played through the virtues of fair play, was a means unto itself. The goal was not about winning but about taking part and being a member of a team. One learned self-control, discipline, and virtue, such as being honest, being fair, and acting properly and decently on the field of play and off the field of play. More importantly, the stated goal was not about victory but about the honorable journey of playing and working together. This was a worthy cause where boys learned to be

gentlemen by following rules and being respectful to opponents. This notion of fair play was seen as imperative to the education of gentlemen. Most of the aristocrats from the ages of 13 to 18 attended the elite boarding schools including Eton or Harrow. Students who attended these schools believed deeply in the importance of learning to play fairly, and they noted it as such in their writings. For example, the end of Napoleon's reign occurred following the Battle of Waterloo when the combined armies of Europe under the command of the Duke of Wellington defeated him. Wellington—a graduate of Eton—is often quoted as having said that the Battle of Waterloo was won on the playing fields of Eton. His implication was that the officers at the battle, most of whom had attended Eton, worked together, followed rules, and had a strong sense of fair play and decency, which helped England win the battle.

This concept of learning to be gentlemen through sport was transported to the British colonies that became the United States—and the philosophy was instituted in the elite schools of the time, including Harvard and Yale. This institutional belief in the importance of developing men, and later women, of character blossomed throughout the United States and became part of a mystical belief that sport builds character—the actual virtues of being honest, trustworthy, and fair. This notion continues today in the five U.S. military service academies. The philosophy guiding the programs stems from General Douglas MacArthur, superintendent of the U.S. Military Academy in 1920, who believed that every cadet

should play and compete in athletics at their highest level. McArthur said, "On the fields of friendly strife are sown the seeds that on other days and other fields will bear the fruits of victory." This same belief in discipline and athletics sits as the ideal philosophy that sports builds character, and hence a heightened sense of being honest, fair, and respectful, which serves as evidence to include sports and athletics as both curricular and extra-curricular activities in U.S. education.

The same fair play philosophy was incorporated by Baron de Coubertin, the founder of the modern Olympic movement, into the overarching purpose of the Olympic Games. Coubertin, classically educated, was highly influenced by Englishman Thomas Arnold (headmaster of Rugby School) that sport was necessary for the balanced education of a person. A balanced education was about developing people of character who knew how to play a game well and fairly. Coubertin spent his life lecturing, writing, and championing the character components of sport. Hence, the Olympic movement worldwide is based in the philosophy of fair play.

The mythical ideal, however, is not always the reality. The institutional and social practices of today support a distorted form of competition—an objectification of the opponents, the game, and the rules in which the one true product valued is the win. One would be hard-pressed to find a fair play standard as noted earlier in most competitive athletic programs in the United States or for that matter in the world today. Instead, sports and athletics are played through a different ethos of competition called *gamesmanship*—getting an advantage using whatever dubious ploys and tactics without getting caught in order to win. Today, it is not the journey that is important, generally, it is the result. "If you ain't cheating, you ain't trying" is a motto today, and often is found pinned to a locker room bulletin board.

In the last year alone, sad unethical practices have occurred at all levels of athletic competition, including youth sport (parents fighting with parents at youth sport events), collegiate sport (famous coaches being fired or censured for not reporting unethical behavior of their players or criminal behavior of their own coaches), professional sport (players placing a bounty on injuring a player), and Olympic sport (athletes being expelled from the games for using illegal performance substances or cheating in relation to the

rules). Few media stories of true fair play exist, and when a story is highlighted that focuses on fair play, it becomes viral through social media because it is so unusual. For example, two softball players carried an injured opposing player around the bases so she could touch the bases and score a run. The opposing player had hit a home run but was injured on rounding first base. She crawled back to first base and asked for a time out. She could not continue, and the rule supposedly stated that she must touch all of the bases. None of her team could aid her to run the bases or take her place. As such, the player would be awarded a one-base hit, and not the home run that she actually earned. Two of her opponents at this point performed an exceedingly unselfish act. They asked if it was legal if they carried her around the bases so she could touch the bases, and thus score the run. It was legal to do so, and so they did. The team of the players who carried the athlete lost the game that day by two runs. They might have won it had they not carried the injured player around the bases.

As noted, fair play does exist, but it can be difficult to find it in the social fabric as defined by the Victorian age. Perhaps a better question is: Can fair play exist and flourish in the highly competitive, gamesmanship world of today? Many observers believe it can, if those individuals who administer and coach athletic programs truly believe in the moral value of fair play. If they do, then missions should be written to support fair play and ethical programs and educational programs should exist for administrators, coaches, athletes, and, in youth sport, parents to support the philosophy of fair play.

In fact, such programs do exist, but they are costly in both resources and time. The World Anti-Doping Agency and U.S. Anti-Doping Agency are presently supporting research to develop fair play intervention programs. Educational programs to recapture the notion of fair play are not a simple process of making rules and expecting people to follow them. Such programs demand reading, writing, and reflection to inspire cognitive change about how we teach, coach, and administer athletic and sport programs. Unless those who are in charge of athletic and sport programs are willing to make the effort, unethical practices are likely to continue while fair play continues to suffer. The problems described here are not unique to the United States. Fair play as a social construct

may be expected in sport and competition but is often violated. At the 2012 Olympic Games, two badminton teams from China were disqualified because they violated the Olympic Oath. They conspired to play poorly in early rounds of competition in order to lose. If they lost, they would be placed in a less competitive bracket and would then have a greater chance of making the finals. Officials realized what they were doing, and disqualified the teams from further competition.

Notwithstanding, the belief that sports build character can be found today at all levels of sport, even in the absence of scientific research to support that notion. There may well be reason to suppose that the philosophy of fair play could again become the overarching purpose of sport and athletics—if those who teach, coach, and administer athletics are willing to make it happen.

Sharon Kay Stoll

See also Character Development; Ethics; Moral Development; Moral Reasoning; Moral Values and Attitudes; Sportspersonship

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FATIGUE

Fatigue is an overwhelming sustained feeling of exhaustion and decreased capacity to complete physical and mental work. It is multidimensional with emotional, behavioral, and cognitive components. Feelings of fatigue are associated not only with disease states but also with healthy functioning. Approximately 20% of adults worldwide report current fatigue. When examined in terms of physical activity, feelings of fatigue are related to issues of physical exertion, inadequate rest, and sedentary life style. This entry reviews the evidence regarding the relationship between physical activity and feelings of fatigue and touches on issues of measurement and biological mechanisms.

Conceptualization and Measurement of Fatigue

Italian physiologist Angelo Mosso (1846–1910) began to shift the focus of research from fatigue of the body to fatigue of the mind during the late 19th century. Influenced by the work of Mosso, clinicians began to draw a sharp distinction between objective and subjective fatigue, thus broadening the conceptualization of fatigue. The concept of subjective fatigue raised measurement issues related to the indirect measure of central nervous system activity.

Contemporary theories on physical activity and feelings of fatigue have changed little over the last century and continue to focus on the multidimensional nature of the construct. In addition, the non-specific, subjective experience of fatigue remains difficult to operationalize through measurement instruments because of the lack of known biological markers. Thus, self-report measures have dominated the clinical and scientific community. These measures range widely in their ability to offer a valid interpretation of fatigue. Despite these limitations, clinicians and researchers recognize exercise as a valuable treatment for fatigue in which the mechanism for the positive effect is likely an interaction among biological, psychological, and psychosocial variables.

Empirical Evidence

Despite the limitations in the measurement of both physical activity and fatigue, the evidence for the effect of physical activity on feelings of fatigue

generally is both positive and consistent. Both epidemiological and experimental research in the areas of exercise adoption, exercise cessation, and overtraining support this association.

Population Studies

Epidemiological evidence suggests that people who are physically active in their leisure time have about 40% reduced risk of experiencing fatigue compared with sedentary individuals. There is general agreement among population-based studies of a strong, consistent, temporally appropriate, dose-response relationship between physical activity and feelings of fatigue. Furthermore, feelings of fatigue appear to be alterable in relation to the adoption or cessation of physical activity. The beneficial effects of physical activity appear to be largest for sedentary individuals who initiate and then maintain an exercise program.

Exercise Adoption

Experimental studies examining the effects of exercise training on feelings of fatigue suggest that exercise programs reduce fatigue in previously sedentary individuals. These effects are consistent across both healthy people and patient groups. Physical activity may be especially beneficial in reducing feelings of fatigue in certain patient groups with chronic diseases such as fibromyalgia, cardiovascular disease, and cancer. Exercise may represent a low-cost, efficacious adjunctive therapy that can address other health outcomes beyond fatigue in these patient groups.

Exercise Cessation

Experimental studies examining the effects of temporary exercise cessation by people who regularly exercise suggest that exercisers experience increased feelings of fatigue during periods of exercise withdrawal. These temporary mood disturbances related to fatigue appear to be alleviated with the reintroduction of the exercise regimen.

Overtraining

Overtraining is characterized by an overload in frequency, intensity, or duration of exercise such that an individual's training regime exceeds the recovery capacity. Overtraining can result in decreased performance and symptoms of under recovery or burnout such as fatigue. Increases in training

volume during overtraining periods are associated with increased feelings of fatigue, whereas decreases in training volume during tapering periods are associated with decreased feelings of fatigue. Despite increases in negative overall mood in response to overtraining, feelings of fatigue often exhibit the earliest and largest changes in mood state.

Biological Mechanisms

Brain functioning is controlled by genes, but social, developmental, and environmental factors can alter gene expression. These alterations in gene expression can induce changes in brain functioning and behavior. This integrative process supports the conceptualization of fatigue as a multidimensional construct influenced by a variety of biological, psychological, and psychosocial factors. The specific brain mechanisms that generate the feelings of fatigue are unknown, but monoamines, histamine-, acetylcholine-, and gamma-aminobutyric acid (GABA)-mediated neurotransmission have been implicated. There is evidence that physical activity can alter these neurotransmitters and neuromodulators in key brain areas associated with fatigue, such as the dorsolateral prefrontal cortex, striatum, cerebellum, and spinal cord.

Conclusion

Physical activity is a healthful behavior that has promise for combating feelings of fatigue. Historically clinicians have recognized the physiological and psychological aspects of fatigue and have consistently recommended exercise as a treatment. The subjective nature of feelings of fatigue has made conceptualizing and measuring the construct difficult. Despite such limitations, epidemiological and experimental evidence suggests physical activity does reduce feelings of fatigue across both healthy groups and patient populations. However, intensive exercise training can produce feelings of fatigue in the event of overtraining. The biological mechanisms associated with physical activity and fatigue should be considered in relation to the nature of the exercise stimulus.

Timothy W. Puetz

See also Burnout; Mental Health; Neuroscience, Exercise, and Cognitive Function; Overtraining Syndrome; Underrecovery Syndrome

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FEEDBACK

Feedback, or response-produced feedback, consists of all the information an individual receives as a result of a practice trial of a motor skill, classically divided into two parts—*intrinsic* and *extrinsic*. Intrinsic feedback is all of the information one receives *naturally*, such as vision, audition, and proprioception. Extrinsic feedback is information provided over and above intrinsic feedback, often by a teacher, coach, or experimenter. In the laboratory, tasks or procedures are used such that the learner typically cannot detect how well one has met the task goal, and then extrinsic feedback is manipulated to access its effects on learning. Using this method, *augmented feedback* has been considered a key variable in the learning process, without which learning does not occur at all. It operates to guide the learner to the correct movement pattern. The learner uses this information to correct errors on subsequent trials, until the desired skill level is achieved. Researchers distinguish between two types of augmented feedback: (1) *knowledge of results* (KR), provided after a trial about the movement outcome in relation to its goal, and (2) *knowledge of performance* (KP), provided during or after the movement about the nature of the movement pattern. Even though KR and KP may, on occasion, have somewhat different functions in the learning process, both seem to

follow the same principles in the way they affect skill learning. Therefore, here we refer to them both as *feedback*.

Research related to the role of feedback in the learning of motor skills has a relatively long history dating from the early 1900s. The view of the role feedback plays in the learning process has changed throughout this time and continues to change. The following sections describe how our understanding of the functions of feedback has developed from the early views (1900–1970) to the second phase (1980–2000) and current research (2000–present).

Early Views: 1900–1970

Toward the beginning of the previous century, psychologists saw feedback as having primarily a reinforcing role. In particular, Edward Thorndike's law of effect stated that actions tended to be repeated if they had pleasant or rewarding consequences and avoided if they were followed by unpleasant or punishing effects; thereby, it provided an account of feedback's role in learning. Much of this early research was guided by animal (occasionally human) research showing, for example, that feedback delays and feedback frequency had negative and positive effects on animal learning, respectively. Starting in about the 1950s, researchers such as Edward A. and Ina M. Bilodeau focused more on the informational role of feedback. Feedback was considered vital for continued improvement with practice, and learning often did not seem to occur without it. Practice without feedback was thought to allow performance to drift away from the goal movement pattern and to weaken the memory representation of the movement. This view is also reflected in Jack A. Adams's (1971) closed-loop theory and Richard A. Schmidt's (1975) schema theory of motor learning. In both theories, the memory representation of a skill (perceptual trace in Adams' theory; recall and recognition schemata in Schmidt's theory) was thought to develop as a function of practice and feedback. That is, the greater the number of trials with feedback a person had performed, the stronger memory representations were assumed to be, and the better the person's ability both to perform and to detect and correct errors without augmented feedback. According to this view, feedback would be most effective if it was provided frequently and

immediately. In the 1980s, these ideas began to be questioned.

Second Phase: 1980–2000

The second major phase of feedback research was initiated and inspired by an influential review of the feedback literature by Alan W. Salmoni, Richard A. Schmidt, and Charles B. Walter in 1984. It became clear that many of the earlier studies had shortcomings, most notably a lack of retention or transfer tests to assess the more permanent, or learning, effects of different feedback manipulations (e.g., temporal delays, frequencies, precision). Inferring learning from the performance of different groups during the practice phase—that is, under the influence of different feedback conditions—became unacceptable. Instead, it was argued, the use of delayed tests under common conditions was crucial to assessing stable learning effects of various experimental variables including augmented feedback. Importantly, Salmoni et al. provided a new conceptual framework for feedback, which was formalized as the *guidance hypothesis*. The term referred to the role of feedback in guiding the performer to the correct movement pattern during the learning process. Aside from this positive function of feedback, several potentially negative effects of frequent feedback were proposed as well. In particular, it was hypothesized that the learner, with very frequent feedback, could become dependent on the augmented feedback, thereby neglecting the processing of intrinsic feedback, which would be necessary for the development of intrinsic error detection and correction mechanisms. Frequent feedback was also assumed to result in excessive variability in performance, as it would prompt learners to constantly correct even small (perhaps acceptable) errors (so-called maladaptive short-term corrections) that perhaps reflected inherent variability in the motor system. The result would be the development of a less stable movement representation. The guidance hypothesis therefore suggested a positive influence of frequent, immediate, or precise feedback during practice while it was present, but it could have a detrimental impact on learning if it were overdone.

Numerous studies examining various feedback manipulations provided support for the guidance idea. These studies typically used feedback manipulations that in some way attempted to reduce the detrimental effects of frequent feedback by

encouraging learners to attend to and utilize their intrinsic feedback. For example, reducing the relative frequency of feedback (reduced percentage of trials after which feedback is provided) was found to enhance learning compared with feedback after every trial (100% feedback). Also, summary or average feedback (feedback for individual trials or as an average, respectively, presented only after a set of trials has been completed) have been shown to be more beneficial for learning than feedback after every trial. Furthermore, bandwidth feedback (feedback provided only when errors exceed a certain predetermined bandwidth) appears to reduce movement variability and enhance learning. Finally, delaying feedback, even by a few seconds, and asking learners to estimate their errors prior to receiving feedback have been demonstrated to yield more effective learning outcomes than providing feedback immediately after the completion of the movement, or concurrently with the execution of a movement.

The guidance hypothesis had an important impact on motor learning research. It has contributed to a better understanding of how feedback influences performance and learning. Yet, even though there has been considerable support for the guidance hypothesis, this support comes primarily from studies using relatively simple laboratory tasks or involving situations in which learners were deprived of intrinsic outcome information and therefore had to rely on the augmented feedback provided by the experimenter. In recent years, it has become clear that the guidance view does not provide a comprehensive description of the various functions of feedback in the process of (complex) motor skill learning. One important factor that influences the effectiveness of feedback, and that qualifies the influence of the feedback frequency, is the attentional focus induced by it. Furthermore, there is converging evidence that feedback has not only an informational role, but that its motivational influence on learning is more important than previously thought.

Current Research: 2000–Present

Feedback and Focus of Attention

Studies have consistently shown that feedback or preperformance instructions promoting an external focus of attention—whereby attention is directed to the movement effect on the environment (e.g., the motion of implement, trajectory of

a ball)—enhances learning compared with those instructions inducing an internal focus by directing one's attention to one's body movements. That is, a simple change in the wording of feedback can elicit an external or internal focus and produce markedly different learning results. For example, references to the movement of a golf club have been shown to be more effective than those related to the performer's arm movements. External focus advantages for learning have been found for many different types of motor skills, different age groups, and levels of expertise.

Feedback that directs performers' attention to their own body movements causes them to use a more conscious mode of control, which constrains the motor system and interferes with automatic control processes (the *constrained-action hypothesis*). In contrast, by adopting an external focus on the intended movement effect, performers use a more automatic type of control that makes use of unconscious, fast, and reflexive processes. Studies have shown associations of external attentional foci and various measures of automaticity—including reduced attentional demands, reduced pre-movement times (more efficient motor planning), high-frequency (reflexive) movement adjustments, and reduced muscular activity. The result of adopting an external focus is typically enhanced motor performance and learning, as seen in increased movement effectiveness and efficiency.

Not only does feedback promoting an external focus of attention lead to more effective learning than feedback referring to body movements, but the previously found benefits of a reduced relative to a high-feedback frequency have been found to be reversed. That is, despite the informational content being the same, feedback inducing an external focus is most effective when it is provided frequently (e.g., 100%), whereas feedback inducing an internal focus is least effective when it is given frequently. The interaction of feedback frequency and attentional focus, as well as the overall benefits of external relative to internal focus feedback, cannot be explained by extant conceptualizations of feedback (e.g., guidance hypothesis).

One possibility is that at least some of the learning benefits of reduced feedback frequencies found in previous studies were not primarily due to learners' becoming dependent on frequent augmented feedback. Rather, the detrimental effects of frequent feedback may have been because of constant internal focus reminders, with those (negative)

effects being attenuated under reduced feedback conditions. The interactive effects of feedback frequency and attentional focus would be consistent with the concept that experiencing less of a detrimental influence (limited use of an impairing thought such as an internal focus of attention) is good, as more of a good thing (frequent reminders to maintain a beneficial external focus) is good.

It has been suggested that the mere mention of body parts like fingers, arms, or feet—within internal focus feedback or instructions—might provoke a focus on the *self* and ensuing self-regulatory activity. Efforts to manage self-related thoughts and emotions may be so demanding that available attentional capacity is exceeded and performance suffers. In contrast, when feedback promotes an external focus, a focus on the self is reduced. The mechanisms underlying the beneficial effects of feedback inducing an external focus may not be dissimilar to those responsible for the learning advantages seen when feedback is given about successful trials, or when it suggests above-average performance relative to a peer group. These types of feedback are discussed next.

Feedback After Successful Trials

Until recently, most researchers were concerned with the *informational function* of feedback, that is, its role in providing information about an individual's performance relative to the task goal. Similarly, practitioners often see performance feedback from this perspective. For instance, a coach might identify deviations from the optimal technique in an athlete's movement patterns and suggest corrections. While such feedback plays an important role in any learning process, a somewhat underappreciated aspect of feedback has been its influence on the performer's motivational state. Recent findings demonstrate that positive (or negative) feedback affects motor learning via its motivational influence.

Providing learners with feedback after good trials, as opposed to poor trials, has consistently been shown to result in more effective learning. In several studies, feedback about task performance (i.e., accuracy of throwing an object at a target) was provided after each of several blocks of practice trials. However, it was provided on only half of those trials. Unbeknownst to the learners, they were given feedback either about their most effective trials or about their least accurate trials.

Participants receiving feedback after their best trials demonstrated more effective learning on a retention test. Thus, feedback emphasizing successful performance, while ignoring less successful attempts, benefited learning. This effect has been linked to participants' enhanced intrinsic motivation. Learners often appear to have a relatively good feel for how they perform, and instructor feedback indicating errors may not only be superfluous, but it can also irritate learners or heighten concerns about the self that may hamper learning (see below).

Social-Comparative Feedback

Effects of *normative feedback*—which involves norms such as a peer group's actual or false average performance or improvement scores that are provided in addition to the learner's own scores—have been examined in a related line of research. Normative information is a potent basis for evaluating one's own performance. Favorable comparisons with others typically result in perceptions of competence, increased self-efficacy, and motivation, while negative comparisons have the opposite effect. Importantly, normative feedback also has differential effects on motor learning, with learning being enhanced by positive relative to negative or no normative feedback. In other studies, bogus feedback about a peer group's average block-to-block improvement resulted in enhanced learning if it conveyed to the learner that one's own improvement was greater than average, compared to less than average. Thus, favorable social-comparative feedback affects the degree to which task skill is retained.

Positive normative feedback not only leads to improved outcome scores but also produces qualitative differences in participants' control of movements, such as greater automaticity in movement control. It is interesting that positive feedback benefits learning compared with both control conditions without comparison information and negative normative feedback conditions, which have similar effects. This suggests that the latter conditions may trigger thoughts about the self (similar to feedback inducing an internal focus), and ensuing self-regulatory activities in attempts to manage thoughts and affective responses, which hamper learning of the primary task. In contrast, when one is purportedly performing well (above average), such self-related concerns and activities

to suppress them might be unnecessary—with the consequence that learning is enhanced.

For ethical reasons, providing false feedback in practical settings would not seem to be appropriate. However, the findings suggest feedback that implies one is an effective performer, or the provision of positive, competence-affirming feedback is critical for learning. While many practitioners may intuitively provide such feedback, others may be more focused on correcting errors—with unintended consequences for motivation and learning.

Conclusion

The view of how feedback functions in the process of motor skill learning has changed over the past few decades. It is now clear that the role of feedback goes far beyond providing reinforcement or guidance to the goal movement. There is mounting evidence for the motivational role of feedback—which not only has an indirect effect by increasing the amount of practice—but a *direct* impact on motor learning. The learning of motor skills not only involves the fine-tuning of motor programs and movement parameters but requires effective self-regulation of cognitive and affective processes, as well as attentional focus demands. A more integrated perspective on motivational and informational aspects of feedback in motor learning research will benefit future theoretical conceptualizations—and should also yield practical motor skill learning insights, as neutral task information is not easily found in the natural social contexts in which movement skills are learned.

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See also Attentional Focus

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FEMALE ATHLETE TRIAD

The female athlete triad (triad) refers to the co-occurrence of three interrelated conditions: low energy availability, menstrual dysfunction, and low bone-mineral density. Factors within sport and exercise environments can increase the risk of developing these conditions. Though the prevalence of the full triad is low, many girls and women will experience one or two of the conditions, which can increase their risk of developing the triad. The triad represents a significant health concern, as it can lead to serious and long-lasting health consequences.

Interrelated Conditions

Originally, the triad included disordered eating, amenorrhea, and osteoporosis. However, in 2007, the American College of Sports Medicine revised their position and redefined the triad as low energy availability, menstrual dysfunction, and low bone-mineral density (BMD). *Low energy availability* refers to the amount of energy after exercise or training that is available for physical functioning and may result from excessive exercise, insufficient caloric intake, or other methods of purging, such as laxatives or vomiting. Eating disorders (EDs), including anorexia nervosa (AN), bulimia nervosa (BN), and eating disorders not otherwise specified (EDNOS) may be present and contribute to low energy availability. However, low energy availability can be present without EDs.

Menstrual dysfunction, the second defining condition of the triad, includes primary amenorrhea (delay of menarche), secondary amenorrhea (absence of menstrual cycle, after menarche, for more than three consecutive months) and oligomenorrhea (more than 35 days between menstrual cycles). Abnormal menstrual functioning often results from low energy availability and can contribute to low BMD. *Low BMD* is defined as a BMD of at least one standard unit lower than

age-group matched controls, lower than normal levels of estrogen, history of deficient nutrition, and previous bone fractures. Low BMD can result from insufficient accumulation of bone mineral during childhood and adolescence or from bone mineral loss during adulthood.

Prevalence

The prevalence of low energy availability has not been well documented. However, estimates of clinical EDs, which may contribute to low energy availability, range between 0% and 13.4% among college and elite athletes. Prevalence of subclinical levels of disordered eating is much higher, though rates of specific pathogenic weight-control behaviors vary widely. For example, few athletes report using laxatives or diuretics or engaging in self-induced vomiting, but many diet and exercise excessively. Menstrual dysfunction, a product of low energy availability, ranges between 16.6% and 54% among high school athletes and up to 63.9% among college athletes. Low BMD has been found in up to 21.8% of high school athletes and 11% of elite athletes.

Prevalence estimates of the co-occurrence of all three conditions are lacking. Several studies using the original, more narrowly defined, criteria report rates of 1.2% (high school), 2.7% (college) and 4.3% (elite). In a study using the current ACSM guidelines (2007), the triad was found among 15.9% of college runners. When considering the presence of two of the three conditions, prevalence rates can range up to 18% among high school athletes and 26.9% among elite athletes. The presence of any of these conditions may increase the risk of developing the full triad.

Risk Factors

Numerous factors can increase female athletes' risk for developing triad conditions. Distorted attitudes toward food, eating, and body weight and shape are associated with unhealthy eating and weight-control behaviors that can create low energy availability. Athletes in sports that emphasize leanness, low body weight, or appearance, such as distance running, gymnastics, or dance, may be more likely to experience pressures from coaches, teammates, family, and other sport personnel (e.g., judges) about body size or shape. Such stressors may lead athletes to restrict caloric intake and exercise excessively.

Evidence also suggests that sport participation at an elite competitive level increases risk of the triad. Pressure and expectations to consistently perform and achieve at a high level often increase among elite-level athletes. Moreover, there may be serious consequences for poor performance for elite athletes. For example, losing a spot on the national team or losing a sponsorship or endorsement may significantly impact an athlete's livelihood. Thus, athletes competing at elite levels are not only likely to engage in extensive physical training but also may be susceptible to disturbed body image and weight-related pressures, placing them at increased risk for low energy availability.

Health Consequences

All three triad conditions put female athletes at risk for a number of negative health consequences. Disordered eating behaviors, which can contribute to low energy availability, are associated with low self-esteem, anxiety, depression, and a myriad of physical health problems, such as cardiovascular problems, constipation, electrolyte imbalance, muscle cramps, or muscle weakness). Menstrual dysfunction can increase the risk of infertility and result in muscular problems and decreases in BMD. Menstrual irregularities and low BMD increases the risk of stress fractures.

Screening and Prevention

The American College of Sports Medicine recommends screening for the triad during preparticipation and annual health screenings. Additionally, female athletes who present with one triad condition should be screened for the other two. The Female Athlete Triad Pre-Participation Evaluation (Triad PPE), available from the Female Athlete Triad Coalition, can be used to assess eating habits and attitudes, weight and menstrual histories, and history of stress fractures. Potential low BMD can be evaluated via dual-energy X-ray absorptiometry (DXA) when there is a history of low estrogen, disordered eating, or history of stress fractures.

To reduce the risk of the triad, coaches, administrators, and sports medicine professionals can create body-healthy environments by avoiding weigh-ins, weight logs, and weight-related joking, as well as pressures to achieve an unrealistic body size or shape. Moreover, sport governing bodies can promote healthy sport environments by providing triad education and training to officials,

coaches, and athletes. Prevention is important because treatment can be challenging; for example, treatment may not result in restoration of normal levels of BMD and ED recovery rates are low. Education, increased awareness, and early interventions are important steps in triad prevention.

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See also Body Dissatisfaction; Body Image; Eating Disorders

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FEMINISM

Feminism is a movement to end oppression, especially as it relates to sexism. Feminism can be taken up in many contexts such as sport and exercise, where theorists and practitioners engage with feminist theory, feminist activism, feminist politics, feminist education, feminist class, race, or gender struggle, and global feminism to varying degrees. In this entry, common myths about feminists are explored. This is followed by a discussion of feminist history, theory, and advocacy and some concluding remarks.

Common Myths Surrounding Feminists

Several myths abound related to those who take up feminism and feminist theory. These include (a) feminists are aggressive; (b) feminists hate men; (c) feminists do not wear make-up, are not feminine, are manly, or are ugly; (d) all feminists

are lesbian; (e) feminists have no sense of humor; (f) feminists want to take over the world; (g) feminists are obsessed with gender; and (h) we live in a postfeminist age now—we don't need feminism anymore.

In terms of the myth that feminists are aggressive, since the late 1800s, feminists—including some men—have been advocating for equal rights for women in a peaceful and nonviolent way. In fact, the feminist movement is one of the few social movements where there is virtually no documented violence related to protests calling for this type of social change. Regarding the myth that feminists hate men, there are plenty of men who stand side-by-side with women who are involved in feminist movements because they are also disturbed by patriarchy's effect on their lives and on the lives of their mothers, sisters, daughters, wives, female best friends, cousins, and so on. When people say that they think feminists do not wear make-up, are not feminine, are manly or are ugly, what they are referring to is traditional notions of *heterosexual beauty* in a society, which includes (for women) applying foundation, eye shadow, eye liner, mascara, rouge, and other cosmetics, in an effort to fit what is defined as heterosexually *beautiful* or *sexy* for women. Those who do not want to wear make-up are seen as ugly; the next leap that most people make is that because they are not interested in wearing make-up or looking *feminine*, they are not heterosexual. The myth that all feminists are lesbian is related to this leap; however, not all feminists are lesbian just like not all women (regardless of sexual orientation) are political. While it is true that many lesbian and bisexual women were a part of the early feminist movement, it was partially because they themselves had already wrestled with pushing against rigid sexual, gender, class, race, and political categories. There is also the belief that feminists have no sense of humor. This may be because those who advocate feminism do not laugh at jokes made at the expense of women. If they are not laughing when others are, they may be perceived as having no sense of humor; in reality, however, they probably do not find it humorous when women are made fun of. The notion that feminists want to take over the world often comes from the fear (by some) that men will be replaced in positions of power, including in the economic, educational, governmental, and political sectors. While it is true that liberal feminists fight for an equal percentage of men and women in these contexts,

according to recent research, it is also true that successful corporations must have at least one third of their management (e.g., those in power) represented by minority workers, including women. The perception that feminists are obsessed with gender appears related to the fact that feminists are concerned with gender oppression. This is because, in most societies, males are valued over females and masculine interests are privileged over feminine interests. Those who are in the marginalized positions in society are acutely aware of the differences that are highlighted in often taken-for-granted daily encounters. Finally, the myth that we live in a post-feminist age now and we don't need feminism anymore *appears* most prevalent in young women who have grown up in a post-Title IX era. Interestingly, these women are usually college age and may not have experienced sexism in the workplace because they have not yet entered the workforce full time.

Brief History of Feminism, Feminist Theory, and Advocacy

Most researchers agree that there are at least three major waves of feminist movement: (1) from the mid-1800s through 1930; (2) during the 1960s and 1970s; and (3) post-1970s. Feminism actually began as a campaign for women to get the vote. During the first wave in mid-1800s, women advocated for the right to vote first in England, then in France, and then in the United States. In the United States in 1837, Mount Holyoke—the first of the *seven sister colleges*—was founded. This was followed in 1850 by the suffragette movement in the United Kingdom. However, neither England nor the United States was one of the first countries to grant women voting rights; it was New Zealand in 1893, Australia in 1902, the United States in 1920, and the United Kingdom in 1928.

There was a lull in feminist movement in the United States between the first two waves (between 1930 and 1960). This was because men were coming home from World War II in the 1940s and taking back the jobs they once held in factories and other work settings. During the war, these jobs were filled by women participating in the war effort (e.g., working in factories which manufactured parts for the war). However, once the men returned, women were expected to go back into the home, become housewives, and start families.

During the second wave (roughly 1960–1980), the focus was on women taking control of their

own bodies, relationships, sexuality, health, and vitality and gaining equality in the workforce. Researchers utilized major feminist theories, including liberal, critical, Marxist, radical, socialist, and ecofeminism, as well as feminist epistemology, empiricism, standpoint theory, and postmodernism. In 1963, Betty Friedan—a wife and mother—wrote a groundbreaking book called *The Feminine Mystique*, which described how “unfulfilled” she was, limited to just those two roles. Friedan and 27 others founded the National Organization for Women (N.O.W.) in 1966. Major theoretical debates were framed around Marxism (e.g., social class issues). In 1970, Shulamith Firestone’s *The Dialectic of Sex* was published at the same time that organized marches, demonstrations, bra burnings, and consciousness-raising sessions began. In 1972, Gloria Steinem started *Ms. Magazine*.

Third-wave feminism began post-1980. The focus during this wave was on redefining the feminist movement, especially to include younger women and direct advocacy. While researchers continued to utilize major feminist theories (e.g., liberal, critical, Marxist, radical, socialist, and ecofeminism), many incorporated a hybrid of theorizing that reflected their beliefs about the ways women know what they know (feminist epistemology), how to include women more fully in the scientific process (feminist empiricism), the fact that each woman—particularly those women who have multiple marginal identities—has a unique story to tell (feminist standpoint theory), and a shattering of what was once taken-for-granted notions about *truth* and the structure of gendered *identity* (feminist postmodernism).

Conclusion

Feminism has had at least three distinct waves and has been described as a movement to end sexist oppression. While myths about feminists abound, those who advocate feminism have succeeded in calling attention to gender inequity in the workplace as well as in other contexts such as sport and exercise.

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FLOW

Flow is a special psychological state of total absorption in a task. When in flow, athletes are fully focused on what they are doing, and this heightened attention is associated with a number of positive factors. Accompanying a focused mind-set are factors such as knowing exactly what one is going to do and how one is doing, having a sense of oneness with the task being performed, and feeling in control of one’s performance. A number of factors have to be in place for flow to occur, and it’s not an easy state for most to attain. However, once experienced, individuals are motivated to re-experience flow, because of how intrinsically rewarding an experience it is. Understanding the flow experience is important because it provides a gateway to optimal subjective experience.

Theoretical Background

Mihaly Csikszentmihalyi developed the flow concept in the 1970s, after investigating the experiences of individuals when everything came together during times of involvement with a chosen activity. The types of activities initially investigated by Csikszentmihalyi were diverse, ranging from surgery to dancing to chess and rock climbing. Despite such diversity in setting, there was considerable consistency of responses regarding what was felt during moments that stood out as being special in some way for the individual.

Since his initial investigations where the term *flow* was chosen to denote these special absorbing experiences, Csikszentmihalyi has continued a research program examining this experience. Flow has been examined across diverse settings, from daily living to a state of mind associated with scientific discoveries. There has been remarkable consistency in how flow has been described by individuals across diverse settings. In addition to the enjoyment that flow brings an individual, the experience of flow is associated with many positive psychological characteristics, and is an optimal performance state. Flow has been identified as a key psychological construct in *positive psychology*, a growing field of interest in psychology, particularly with regard to positive subjective experience.

The Experience of Flow

When in flow, one feels strong and positive, not worried about self or failure. Flow can be defined as an experience that stands out as being better than average in some way, where the individual is totally absorbed in what is being done, and where the experience is very rewarding in and of itself. This definition covers several characteristics of flow, and Csikszentmihalyi has detailed the experience of flow into nine dimensions.

The first and perhaps most critical dimension of flow is the concept of challenge–skill balance. Flow is predicted to occur when the individual moves beyond average experience of challenge and skill. The moving beyond average signifies an investment of mental energy into a task. When the perceived challenges are matched by a belief in having the skills to meet the challenge, the stage is set for flow to occur. The perception of challenge, and of skill, is more important than any objective level of challenges or skills in flow state. That is, prediction of the experience of flow is more accurately based on what individuals perceive the levels of challenge and skills are in situations than by reference to the levels of challenges and skills that may actually exist in those situations.

In flow, one is totally involved in the task at hand. Flow can occur at different levels of complexity but, by definition, flow is intrinsically rewarding, regardless of whether it involves a simple task, or a complicated and dangerous gymnastics routine. Csikszentmihalyi categorized the different levels of flow into micro and macro flow experiences.

Micro flow experiences were proposed to fit the patterns of everyday life, whereas macro flow was reserved for experiences associated with higher levels of complexity and demand on the participant. These latter experiences are often associated with peak performance and peak experience.

The Dimensions of Flow

Csikszentmihalyi conceptualized the flow construct in terms of nine dimensions; the first of these dimensions, challenge–skill balance, has already been described. The other dimensions are action–awareness merging, clear goals, unambiguous feedback, concentration on task, sense of control, loss of self-consciousness, time transformation, and autotelic experience. Together, these nine dimensions represent the optimal psychological state of flow; by themselves, they signify conceptual elements of the flow experience.

The action–awareness merging dimension involves a feeling of being at one with the activity being performed. Often used in descriptions by people asked to discuss what it was like being in flow, perceptions of oneness with an activity bring about a sense of peace and harmony to an active engagement with a task. For some performers, feelings of automaticity are described, with well-learned routines enabling them to process subconsciously and pay full attention to their actions.

Clear goals occur in flow, with individuals describing knowing exactly what it is they are supposed to do. Such clarity of purpose occurs on a moment-by-moment basis, keeping a person fully connected to the task and responsive to relevant cues. Closely associated with clear goals is the processing of how performance is progressing in relation to these goals. This is the dimension of unambiguous feedback. When in flow, feedback is easy to receive and interpret. The performer receives clear, unambiguous information that processes effortlessly, keeping performance heading in the right direction.

The next dimension of flow is total concentration on the task at hand. This is one of the clearest indications of being in flow—one is totally focused in the present on the specific task or activity with which one is engaged. Being totally connected to the task in which one is engaged epitomizes the flow state. This connectedness relies on a present-centered focus; flow resides in being in the present moment.

A present-centered focus leads to the next dimension of flow, known as having a sense of control over what one is doing. Being fully connected to the task or activity in which one is engaged allows a person to perceive a sense of control and confidence in what one is doing. This is an empowering feeling, and one that frees a person from a fear of failure that can creep into performance. The absence of thoughts of failure enables an individual to engage in the challenges at hand.

Loss of self-consciousness is another flow dimension, one characterized by a lack of concern about what others may be thinking of them. A lot of the time people live their lives surrounded by evaluations of how they are doing. This sense of evaluation can prevent a full focus on the task at hand. Because flow is defined by being totally focused on the task at hand, it allow for a loss of self-consciousness during engagement in an activity.

A sense of time passing differently defines the dimension of flow known as time transformation. For some, the experience is that time stops. For others, time seems to slow. Or it may be that time seems to pass more quickly than expected. These sensations come about through the intensity of involvement in flow. Because awareness is tightly focused during the intense concentration of flow, people can lose track of time, and afterwards can be surprised by the actual passing of time that has occurred during their task involvement.

Autotelic experience is the final flow dimension, and is the end result of the coming together of the previous eight dimensions. Autotelic experience describes the intrinsically rewarding nature of flow. As described by Csikszentmihalyi, the word is derived from two Greek words that describe doing something for its own sake (*auto*, which means “self” and *telos*, which means “goal”). The idea is that experience of flow is sufficiently enjoyable as to be, in itself, a much sought-after and self-rewarding state. The nine dimensions of flow provide a conceptually coherent framework for understanding optimal experience.

Flow in Sport

Research on sport involvement was a part of Csikszentmihalyi’s landmark 1975 book, *Beyond Boredom and Anxiety*, wherein the flow construct was initially conceptualized. Investigation of flow and related concepts (e.g., peak experience, peak performance) in sport started to become evident

in the literature the 1980s. In the 1990s, Susan Jackson’s systematic qualitative and quantitative efforts to understand the athletic flow experience led to the 1999 publication of the Jackson and Csikszentmihalyi book *Flow in Sports: The Keys to Optimal Experiences and Performances*. Jackson’s in-depth qualitative examination of athletes’ flow experiences, for example, have demonstrated strong support for Csikszentmihalyi’s nine-dimensional model. Across various quantitative studies, positive associations between flow and several important constructs in sport psychology, including task-focused motivation, perceptions of ability, self-determined forms of motivation, hypnotic susceptibility, and use of psychological skills have been reported as well as consistent negative relationships between flow and anxiety. Factors perceived by athletes as influencing flow were identified in Jackson’s research that served to provide useful understandings of antecedent and disruptive factors. Examples of factors perceived by athletes to influence whether or not flow occurred during their performance included level of motivation toward the performance, physical preparation and readiness, confidence, and focus. Considerable interest in the flow construct led to development of different ways of empirically studying flow, and this research effort has made flow a more accessible psychological concept for the applied researcher and also to practitioners interested in flow in their performance settings.

Measuring Flow

Interest in the flow concept has led to a range of approaches to tap into this experience. Both qualitative and quantitative approaches have been used to assess flow. In-depth information about the experience of flow has been obtained through interviews. Sampling of experience as it occurs through the use of pagers programmed to sound at random times has been another approach used to assess flow. Self-report questionnaires are part of this approach, known as the experience sampling method. The development of self-report questionnaires that can be administered after performance, as well as more generally, has facilitated assessment of flow along with other psychological constructs, to understand factors associated with the flow experience. The set of Flow Scales developed by Susan Jackson and colleagues provide researchers and practitioners with a range of

measurement options for assessing flow. These include scales based on the nine-dimensional flow model, assessed at state and dispositional levels.

Conclusion

Flow is an optimal psychological state that occurs when challenges and skills are balanced and extending an individual. The total focus of flow and the associated positive experiential characteristics of this state provide an opportunity for individuals to move their experience from average to optimal. One outcome of this heightened level of experience is that peak performance is often achieved. However, flow is important not so much for any performance outcomes that may ensue, but primarily for the opportunity it provides to experience full engagement in the present moment.

Susan Jackson and Robert C. Eklund

See also Achievement Goal Theory; Attention–Performance Relationships; Enjoyment; Passion; Self-Determination Theory

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and interact with each other, exemplified by a sand pile, a weather system, and social collectives such as animal colonies and sports teams. An athlete can also be studied in this way. In the complexity sciences, the term *degrees of freedom* typically refers to the independent components of a system that can be reorganized in many different ways as surrounding constraints change. When considering the human body as a complex system, an important challenge is to understand how coordination emerges among the large number of motor system degrees of freedom (e.g., the muscles, joints, limb segments). In motor learning, this challenge is known as Nikolai Bernstein's *degrees of freedom problem*: How can humans organize the large number of motor system degrees of freedom to consistently produce functional actions such as catching a ball with one hand? Even a simple movement of reaching and grasping an object with the hand and arm could require a catcher to regulate 7 degrees of freedom (dfs) of the arm, involving flexion–extension, medial–lateral movement, and rotation of joints (3 at the shoulder, 1 at the elbow, and 3 at the wrist). Of course, more degrees of freedom need to be regulated in coordinating more complex actions such as performing a triple somersault in gymnastics.

Bernstein proposed that performers initially cope with the large number of motor system degrees of freedom by rigidly fixing or *freezing* a small number into a basic motor pattern to achieve a task goal. This strategy leads to the characteristic stiffness that many individuals portray early in learning. The freezing of motor system degrees of freedom is a completely understandable coping mechanism when anyone is placed in an unfamiliar performance context and shows how an individual's intentions, perception, and action interact to constrain the movement pattern that emerges. For example, when novices learn to swim their main intention is to remain afloat and maintain stability in the water in order to breathe and not sink. This intention contrasts with those of an Olympic-level swimmer seeking to move rapidly and efficiently through the water to reach a race endpoint in the shortest time possible. An initial coordination mode in the breaststroke corresponds to an iso-contraction of the nonhomologous limbs: the in-phase muscle contraction of arms and legs together. System stability is enhanced by synchronizing the flexion and extension of both arms and legs together, rather like the directional movements

FREEZING

Complex systems in nature are defined as having many individual components that are free to vary

of an accordion. The accordion mode of coordination corresponds to a juxtapositioning of two contradictory actions: leg propulsion during arm recovery and arm propulsion during leg recovery. It is not mechanically effective and does not provide high swim speed because each propulsive action is thwarted by a recovery action. However, this freezing coordination strategy is functional for novice swimmers because it is the most stable and easiest to perform early in learning.

As learners become more familiar with a task, their intentions change quickly and they can abandon the coping strategy of freezing degrees of freedom by reorganizing them into specific functional muscle–joint linkages or synergies. Bernstein advocated that these more functional groupings help learners compress the numerous physical components of the movement system to make the relevant dfs for an action become mutually dependent. Synergies between motor system components help make the body more manageable for learners when they discover and assemble strongly coupled limb relations to cope with the huge number of movement system degrees of freedom.

Synergies are functional, being designed for a specific purpose or activity, such as when groups of muscles are temporarily assembled into coherent units to achieve specific task goals, like throwing a ball or performing a triple salchow in ice skating. Good quality perceptual information is necessary in assembling coordinative structures because the details of their specific form or organization are not completely predetermined and emerge under the constraints of each performance situation. Assembling a synergy is a dynamical process dependent on relevant sources of perceptual information related to key properties of the performer (e.g., haptic information from muscles and joints) and the environment (e.g., vision of a target or surface). Synergies emerge from the rigidly fixed configurations that learners use early on to manage the multitude of motor system dfs and become dynamic and flexible as learners use information to tune their functional organization.

Bernstein's ideas were a precursor to recognition of the human body as a complex system and were instrumental for movement scientists seeking to understand how coordination can emerge in human movement systems with their huge number of degrees of freedom, such as muscles, joints, and limb segments. It has been suggested that the degrees-of-freedom problem can be resolved in

a human movement system if the human movement system is conceptualized as a complex, dynamical system in which cooperation between subsystem components can lead to a reduction in system dimensionality through the emergence of synergies or more compact movement patterns. Some research on how skilled and unskilled individuals kick a football has supported these ideas. D. I. Anderson and Ben Sidaway's detailed analysis of kicking confirmed the different ways that motor system degrees of freedom are reorganized during learning. They demonstrated that novice kickers did not display the same coordination patterns as skilled individuals. The rigidity of novice movement patterns and the flexible nature of skilled kicking patterns were clearly depicted in their work. Before practice, the joint range of motion (ROM) for knee flexion and extension during kicking by unskilled participants was smaller in magnitude than the values observed in skilled kickers. Smaller ranges of joint ROM tend to signify greater rigidity of movement patterns. After practicing for 10 weeks at 15 minutes per week, the novice group's coordination pattern began to lose its rigidly fixed characteristic and tended to resemble the more flexible pattern of skilled kickers.

Keith Davids

See also Coordination; Motor Commands; Motor Control; Motor Development; Motor Learning; Movement; Task Constraints

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FRIENDSHIPS/PEER RELATIONS

Peers have a particularly powerful social influence on youth development, particularly during adolescence. Positive peer interactions can help

adolescents acquire a range of skills, attitudes, and behaviors. In sport settings, high levels of peer support and quality friendships have been associated with higher ratings of sport enjoyment, commitment, intrinsic motivation, and perceived competence.

Definitions

Alan L. Smith and Meghan McDonough have generated some important understandings of peer interactions in youth sport contexts. But, as they observed, the term *peer* has rarely been explicitly defined in sport and exercise psychology research. Peer groups are often context specific, which means that individuals can engage with different peer groups in different contexts, such as school, sport teams, or groups people hang out with informally. Peer groups can also change over time as adolescents' interests and habits evolve. This issue makes it difficult to specifically define who are one's peers, but in a general sense, *peers* can be defined as individuals of similar age and with similar interests. For example, in sport teams or schools, teammates or classmates can be considered as peers.

Peer networks reflect the structure of peer connections. For example, individuals on a sport team can identify a network of people they interact with most frequently (e.g., *cliques*). *Peer support* is a concept that stems from the peer network. Support can be viewed in terms of structural, functional, and perceptual dimensions. *Structural* dimensions are represented by the composition of a peer network. *Functional* dimensions refer to specific tangible functions of support provided by others (received support). *Perceptual* dimensions refer to the extent to which individuals feel supported by others (perceived support).

Theoretical Frameworks

The role of peers has been recognized in various theoretical frameworks used in sport and exercise psychology. For example, Susan Harter's *competence motivation theory* considers ways in which social agents influence motivation in achievement domains. Reinforcement, modeling, and approval of mastery attempts can enhance individuals' perceptions of competence and intrinsic motivation to participate in a task. In Albert Bandura's *Social Cognitive Theory*, personal (individual) factors, environmental (social) factors, and behavior are

proposed to influence each other in a reciprocal manner. Peer support is a means by which peers can socially reinforce certain behaviors, such as providing encouragement for individuals to act in certain ways. For example, peers can have a positive influence on children's and adolescents' physical activity behaviors and enjoyment of sport participation.

Kenneth H. Rubin, William Bukowski, and Jeffrey G. Parker put forward a contemporary framework of peer relationships in the developmental psychology literature that has recently been used by sport psychology researchers. Rubin argued that peer experiences are considered across several nested levels of social complexity: individuals, interactions, relationships, and groups. The framework begins with the *individual*. Individuals bring relatively stable social orientations to groups. Individuals involve each other via *interactions* (the simplest level of social complexity), which are short-term social exchanges. Interactions change in response to fluctuating circumstances of the social situation, such as the characteristics and responses of a partner. During interactions youth may cooperate, compete, fight, resolve conflict, or engage in a range of other behaviors.

Interactions are shaped by *relationships* (a higher order level of social complexity). Relationships are influenced by past events and future anticipated events, can take many forms, and have characteristics that are not necessarily found in lower level interactions. But, the nature of a relationship is defined by the characteristics of its members, the types of interactions that take place, and the individual's history of earlier relationships. Friendships are an example of the relationship level, and are discussed with reference to sport psychology research here. Relationships are further embedded in and influenced by *groups*, which are networks of relationships with relatively clearly defined boundaries like a competitive sport team. Peer groups are more than an aggregate of individuals, interactions, and relationships. Rather, groups have norms (e.g., shared cultural conventions), processes (e.g., cohesion) and properties (e.g., hierarchical organization) that are not necessarily found in children's experiences at lower social levels. Groups help define the type and range of relationships that occur within them. Groups therefore influence interactions and relationships and vice versa.

Peer Acceptance

Peer acceptance (or popularity) is based on how a group views the status of an individual. Adolescents with higher levels of athletic ability tend to enjoy more peer acceptance and be more popular in school than adolescents with lower levels of athletic ability. Even within sport teams, the most highly skilled athletes tend to be most accepted and viewed as leaders among their teammates. Additionally, peer comparison and acceptance are important sources of information for judging physical competence. In fact, higher peer acceptance has also been associated with greater physical self-worth, positive affect, and intrinsic motivation for physical activity. Therefore, skilled athletes tend to enjoy high peer acceptance, which is further associated with other positive outcomes.

Friendships

Friendship is a feature of the *relationships* level of peer experiences. There are three fairly distinct aspects of friendships: whether or not a child has friends, who these friends are, and the quality of these friendships. Some of the most influential work on peer friendships in sport was conducted by Maureen R. Weiss, Alan L. Smith, and Marc Theeboom in the United States. Through interviews with 8- to 16-year-olds they identified that the key aspects of *best sport friendships* were companionship and pleasant play, loyalty and intimacy, self-esteem enhancement and supportiveness, things in common, conflict resolution, and conflict. These findings were used by Weiss et al. to develop a questionnaire (the Sport Friendship Quality Scale; SFQS) to assess friendship quality in sport. Research using the SFQS has revealed that adolescent junior tennis players who reported higher quality of sport friendships rated tennis enjoyment and commitment higher than players with lower quality sport friendships.

Peer Conflict

Playing on a team or training with a squad expands adolescents' social networks and requires them to learn to deal with different types of people. Moreover, different types of people may not otherwise be part of the adolescent's social group if it were not for involvement in sport. A recent study of peer relationships among members of early adolescent girls' soccer teams showed that

players integrated new members into the team and learned to interact with different types of people, but as the season progressed, they had to manage conflicts that arose. For example, teammates attempted to intervene and mediate conflicts among other teammates, and people learned to accept others' point of view. Over time, a structure of leadership emerged and players learned to work together. Although peer interactions may not always appear to be going smoothly, the ways in which peers learn to deal with each other and manage difficult situations seems to have important implications for their social development, both within sport and beyond.

Constellations of Social Relationships

Peer acceptance, friendship, and conflict are important issues, but artificially separating these issues for the purposes of research may produce only limited understandings of peer experiences. In adolescents' day-to-day lives, they likely experience a complex mixture of peer interactions. Smith has encouraged researchers to consider multiple dimensions of social experiences in youth sport and physical activity settings by highlighting the notion that constellations (or combinations) of social relationships influence adolescents' lives. For example, studies looking at such constellations of social relationships have shown that higher perceptions of adaptive peer group profiles (like high peer acceptance *and* friendship) predicted adaptive motivational responses. Subsequent research has considered an even broader range of social relationships and shown that positive perceptions of issues such as peer friendship quality, peer acceptance, and child-parent relationship were associated with more positive motivational outcomes like enjoyment and perceived competence. Combined, these findings highlight the importance of examining combinations of social relationships in youth sport contexts.

Gender Differences

During the early stages of adolescence, gender differences emerge in peer relations, with females usually assigning higher importance to friendships, intimacy, and emotional support than males. One study conducted in the United States showed that for female high school athletes peer acceptance and friendship quality, along with perceived competence, instrumental, and expressive

behaviors, predicted self-ratings of leadership, whereas coach and teammate ratings were primarily related to ability only. For males, the psychosocial variables and ability were related to self, teammates,' and coaches' ratings of leadership. Interestingly, other research has shown girls who play *gender-appropriate* or traditionally feminine sports such as volleyball and gymnastics typically report high social acceptance. Also, physical ability in boys is viewed as a desirable quality associated with popularity and leadership, while in girls it is valued in some groups but not others.

Social Competence and Life Skills

One of the goals of studying peer relationships in sport and physical activity settings is to find ways in which to promote individuals' social competence. Rubin et al. described social competence in peer relationships as adolescents' capacity to engage effectively and successfully with each social level in their framework (interactions, relationships, and groups). Hence, a socially competent adolescent would be able to engage in a peer group structure and effectively participate in group-oriented activities, nurture satisfying relationships based on balanced and reciprocal interactions with others, and fulfill individual goals and needs through these interactions with peers.

Although such dimensions of social competence have yet to be widely explored in youth sport settings, several studies have shown that learning to deal with different types of people and working as a team to achieve common goals are some of the most important life skills people acquire through their involvement in youth sport. In addition, such social life skills seem to readily transfer from sport settings to other areas of adolescents' lives. For example, youth sport participants have reported that learning to interact with people in sport teams has helped them work better in group assignments in academic settings and effectively engage with others in work settings. Hence, the peer relationships and friendships that occur through participation in youth sport have potentially important consequences for youth development.

Nicholas L. Holt

See also Cohesion; Conflict; Fair Play; Group Characteristics; Group Formation; Moral Atmosphere; Moral Behavior; Moral Development;

Sportspersonship; Status; Team Attributions; Team Building; Team Communication

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FUNCTIONAL VARIABILITY

To achieve performance goals in competitive sport there is a need to strike a delicate balance between movement pattern stability and variability because, although athletes need to achieve consistent outcomes, they also need to be able to successfully adapt their movements to changes in the performance environment. To achieve these aims, the theory of ecological dynamics advocates that there is an intertwined relationship between the specific intentions, perceptions, and actions of individual athletes that constrains this relationship between movement pattern stability and variability in each individual performer. This intertwined relation between an individual's intentions, perception, and action processes needs to be carefully understood because of the insights it provides on expert performance in sport.

Traditionally, a high level of expertise in sport has been associated with the capacity to be able to reproduce a specific movement pattern consistently and to reduce attention demands during performance by increasing the automaticity of

movement. It was assumed that the central nervous system (CNS) functioned as an executive organizer and prescriber of motor programs and action plans charged with the task of producing stable movement patterns from an individual's effector system. From that viewpoint, expertise in sport was associated with a reduction in deviations in task performance from an ideal standard or movement template which was represented in the CNS. By harnessing integrated feedback systems, athletes were considered to modify the motor program entry parameters until expert behavior was eventually achieved after many hours of practice. Traditionally, therefore, movement variability was considered as noise in performance and learning which should be minimized or eradicated to enable the production of highly functional movement programs (movement variability was considered to be an artifact limiting an individual system's processing of information from input to output).

However, research in ecological dynamics has shown that movement system variability should not necessarily be construed as noise, detrimental to performance. Nor should it always be viewed as error or a deviation from a putative expert model, which should be constantly corrected in learners. Inspired by insights from the Russian physiologist Nicolai Bernstein, movement system variability instead is now considered to exemplify the functional flexibility of a skilled athlete to respond to changes in dynamic performance constraints. For this and other reasons, the concept of functional variability has gained a significant amount of empirical support in the sport psychology field. A key idea is that movement pattern variability can be viewed as a functional property of skilled performers to help them adapt their movement behaviors to changing task constraints (see entry on "Task Constraints"). Traditional research has typically focused on the amount of movement variability exhibited by an athlete during performance, assessed by statistical measures such as the standard deviation or variance around a distribution mean. According to Karl Newell's (1986) *Constraints on the Development of Coordination*, these statistics indicate the amount of noise in a single measurement—that is, the standard deviation only indicates the *magnitude* of variability recorded during task performance (the amplitude, the spatial aspect of the scores in a performance distribution. It provides little information on the

structure of movement variability exhibited by an athlete, which is needed to understand whether it is functional or not. For this reason, Newell and his co-investigators recommended studying the temporal structure of movement pattern variability by analyzing the spectral range of noise, which provides information on the deterministic (preplanned) or stochastic (emergent) nature of movement variability. Also pointed out was that it would be wrong to consider that deterministic processes specify the invariance of a movement pattern and that stochastic processes specify its variance. Given these theoretical advances in understanding movement pattern variability, ecological dynamicists have argued that there is no ideal motor coordination solution (a classical technique) that all athletes should aspire to during learning. Rather, functional patterns of coordination emerge during practice from the interaction of constraints on each individual athlete (task, environmental, and organismic), leading to intra-individual and interindividual movement pattern variability as consistent performance outcomes are achieved.

Recently, the functional role of movement pattern variability has also been supported by research highlighting the property of *neurobiological system degeneracy*, technically defined by Gerald Edelman and Joseph Gally as the capacity of system components that differ in structure to achieve the same function or performance output. This structural property in humans indicates the availability of an abundance of motor system degrees of freedom—that is, the many components of the movement system (alluded to by Bernstein), which can take on different roles when assembling functional actions during sport performance (captured by system degeneracy).

Allied to these ideas on neurobiological degeneracy, research on sport performance has begun to explain why expert performers often display higher levels of intraindividual movement pattern variability than novices in sport, data traditionally viewed as counterintuitive. The movement variability exhibited by skilled individuals can play a functional role. For instance, it highlights an expert athlete's capacity to perform several types of movement or to adopt one of a number of coexisting modes of coordination, that is, exploit system multistability (the many functional states of system organization) and metastability (the capacity to switch between functional states), in

order to achieve the same functional performance outcomes. In the past years, empirical research on sport performance has clearly exemplified how intraindividual and interindividual movement variability can play a functional role in the performance of team-based and a range of individual physical activities, such as a cyclical movement task in an aquatic environment (breaststroke swimming) and a continuous discrete task in the wilderness (ice climbing) (see studies by Duarte Araújo, Keith Davids, and Ludovic Seifert and coworkers). These key ideas on functional variability have now been integrated into a motor learning theory by Wolfgang Schöllhorn et al., known as Differential Learning, which advocates adding noise to initial performance conditions to provoke learning by forcing the individual to adapt unexpectedly.

Keith Davids and Ludovic Seifert

See also Adaptation; Coordination; Dynamical Systems; Ecological Theory; Expertise; Freezing; Task Constraints

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G

GAZE

See Eye Movements/Gaze

GENDER

Gender has a clear and powerful influence in society, and a particularly powerful and persistent influence in sport and exercise. Indeed, the sport world seems to exaggerate and highlight gender. Sport and physical activities remain largely sex segregated and male dominated. Gender is so embedded that trying to be nonsexist and treating everyone the same is difficult. Moreover, trying to treat everyone the same may well do a disservice to participants.

Understanding Gender

Gender refers to psychological, social, and cultural experiences and characteristics associated with being male or female. The terms *sex* and *gender* often are used interchangeably, but they have different meanings. *Sex* refers to biological aspects of being male or female. *Gender*, often mistakenly assumed to be biologically based, is *socially constructed*, which means that characteristics associated with men and women are not innate biological distinctions but develop through social interactions. Moreover, gender is not a dichotomous category. People vary greatly in the extent to which they hold and convey gendered thoughts,

feelings, and behaviors. *Gender expression* refers to the way people convey their gender through mannerisms, behaviors, or expressions.

Gender often is conflated with sex, and considered a naturally determined, unchangeable *gender binary* with two opposing categories. That is, male and masculine (associated with strength, independence, competitiveness, etc.) are the opposite of female and feminine (associated with emotionalism, weakness, nurturance, gracefulness, etc.). When people do not fall neatly into these two categories, sexual orientation is often questioned. That is, a woman whose gender expression reflects masculine characteristics is assumed to be lesbian, and a man whose gender expression reflects feminine characteristics is assumed gay, which further conflates sex, gender, and sexuality. Again, gender is not biological sex, or sexual orientation, but socially constructed roles, behaviors, activities and attributes. Simple, dichotomous gender categories cannot explain gender.

When considering gender, it is important to recognize that gender involves *power relations*. Socially constructed gender roles and expectations lead to inequities and disparities, which reflect power relations. The dominant group has power or *privilege*, and nondominant groups experience *oppression* or discrimination. In gender relations, men have power in society and in sport.

It is also important to note that gender is only one of many sociocultural identities. Everyone possesses many other social identities, such as nationality, class, race and ethnicity, and those multiple identities *intersect*. The mix of multiple,

intersecting identities and power relations varies with time and context. Gender affects men as well as women, and gender interacts with other cultural identities. Sport is not only considered male but also considered white, young, middle-class, heterosexual male. A more useful sport and exercise psychology must consider the intersections of gender, race, class, and other power relations.

Gender in Sport and Exercise

Gender is deeply embedded in sport and exercise, not only reflecting the gendered cultural boundaries of society, but also emphasizing physical and biological processes. Sports have separate categories for men and women because of the assumption that men are naturally (biologically) *better* than women, which links to gender disparities and power relations.

Although women's and girls' participation in sport and exercise has exploded in the last generation, the numbers of female and male participants are not equal. More important, gender influences thoughts, feelings, and behaviors within physical activity settings. *Citius, Altius, Fortius*—the Olympic motto—translates as “swifter, higher, stronger,” which underscores that sport is competitive and hierarchical (masculine characteristics) as well as physical. Gender disparities reflect power relations. Before 1972 when Title IX was passed, over 90% of U.S. women's athletic teams were coached by women. Today, even though more girls and women participate, less than 50% of their coaches are women.

Sex segregation is not as obvious in exercise, but gender influence is clear. Women and men may exercise in the same fitness center, but the aerobics and yoga classes are predominantly women's spaces, whereas men dominate the free weights area. Public health reports indicate that physical activity is limited by gender, as well as by race, class, and physical attributes. Men engage in more physical activity than women across all age groups and all other categories.

Gender stereotypes are connected to gender disparities. Over 50 years ago, Eleanor Metheny identified gender stereotypes, concluding that it is not socially appropriate for women to engage in contests that involve bodily contact, bodily force, or long distances. Gender stereotypes persist, and media coverage reflects gender bias. Female athletes receive much less coverage, with the emphasis

on athletic accomplishments for men, and on femininity and physical attractiveness for women. Men are expected to be bigger, stronger, and faster. Boys who are not athletic often are teased, as with the common insult, “You throw like a girl.”

Sport studies scholars have described sport as a powerful force that socializes boys and men into a restricted masculine identity. We expect to see men dominate women, and we are uncomfortable with bigger, stronger women who take active, dominant roles expected of athletes. Gender stereotypes may restrict men in sport even more than women. Men who deviate from the masculine norm within the athletic culture often face ridicule, harassment, or physical violence.

Stereotypes are a concern because people act on them, exaggerating minimal gender differences and restricting opportunities. Both girls and boys can participate in figure skating or ice hockey. Yet children see female figure skaters and male ice hockey players as role models; peers gravitate to sex-segregated activities; and parents, teachers, and coaches support gender-appropriate activities.

The gendered context of sport and exercise has changed, particularly for women and girls, but gender stereotypes and disparities persist. Sport and exercise are clearly linked with masculine values and behaviors. Those gender stereotypes restrict opportunities and behaviors for both men and women, and may encourage unhealthy behaviors, such as overtraining or unhealthy eating behaviors. The limited gender research focuses on women, which highlights neglected issues, but sport and exercise psychology scholars have far to go to understand gender in physical activity settings.

Diane L. Gill

See also Diversity; Feminism; Heterosexism, Homonegativism, and Transprejudice; Sex Differences

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GENERALIZED MOTOR PROGRAM

When learning sequential movements, such as those involved in speech production, handwriting, typing, drumming, or sports skills, performers exhibit the ability to modify a learned movement sequence from execution to execution in some ways but not in others. This is thought to occur because a generalized motor program (GMP), which can be used to produce a specific class of movements, has developed and been stored in memory. When the GMP is retrieved, movement parameters must be specified in an effort to scale the program output to meet the specific demands at hand. Because the movement output can be altered by the premovement specification of parameters, the program was termed *generalizable* by Richard “Dick” Schmidt. The notion of a generalizable program is quite different from the more traditional notion of motor program as a prestructured set of central commands that can be used to produce a specific movement.

A generalized motor program is thought to develop over practice and provides the basis for generating movement sequences within a class of movements that share the same invariant features, such as sequence order, relative timing, and relative force. Specific movements are produced by the premovement specification of movement parameters like absolute timing, absolute force, and effectors. For example, when a movement situation requires a learned sequence to be produced either faster or slower than typically practiced, the invariant features remain unchanged, but the absolute timing parameter is changed to accommodate the rate at which the movement is produced. Thus, the relative times used to produce the elements in the sequence remain unchanged, but the absolute time is rescaled to meet the specific demands. Changing the absolute timing parameter results in slower movements that could be considered *stretched out* (in time) copies of faster movements. Likewise, a lower specification of the force parameter would result in a movement sequence generating reduced forces, which could be thought of as a compressed (with respect to force) copy of a more forceful movement.

A common example used to exemplify the notion of a GMP is writing one’s signature. Each of us can write our signature under a variety of conditions. According to the GMP perspective, we do this by specifying the movement parameters needed

to meet the requirements at hand while maintaining the invariant features of the GMP. For example, if I were asked to write my name (Charles Shea) very quickly on a sheet of paper, the signature maintains the relative timing characteristics invariant to the GMP, but the specification of absolute timing would be reduced. This will result in me taking the same proportion of the total time to write the *Ch*, for example, in my first name when writing quickly as when writing under normal time constraints, but the absolute time used would be reduced. Similarly, if I were asked to write my name in a small box on a sheet of paper or much larger on a white board in the classroom, the invariant features would not change, but the actual timing, actual forces, and even the specific effectors used to produce the movement would change. In the smaller situation, one would use primarily finger movements to produce a signature, while in the larger situation one may use shoulder and arm movements with minimal or no movements of the fingers.

The notion of a GMP has a great deal of intuitive appeal. It seems efficient for each motor program in our movement repertoire to be able to generate a class of movement sequences. This reduces not only the potential storage problems that would result if different programs were needed each time the movement requirements changed but also would reduce potential retrieval problems that would be associated with selecting from among a group of similar motor programs. The notion of a GMP is also consistent with our experience with computer programs and electronic video devices. Indeed, the record player analogy often used to describe the invariant and variant features of the GMP feeds this intuitive appeal. In the record player analogy, a phonograph record is used to illustrate the invariant and variant feature of the GMP. For example, a phonograph record could be played at different speeds (33 $\frac{1}{3}$ rpm or 78 rpm), played with different settings on the volume control, or the output directed to different speakers while maintaining the invariant features of the recording. In this analogy, speed is used to indicate absolute time, volume to indicate force, and speakers used to illustrate different effectors.

There is, however, also a good deal of empirical evidence to support the notion of a GMP. Research has demonstrated that learned movement sequences when scaled in time or force exhibit a pervasive tendency toward approximately proportional scaling. Whereas participants are able to rescale

learned movement sequences in time and force, variable practice, in which the learner is exposed to various parameter requirements, appears important for the learner to accurately specify the time or force parameter. There is also a good deal of evidence that participants can execute movement sequences using different effectors. As noted earlier, one's signature can be executed with the dominant limb using a variety of difference muscle groups. This literature is, however, a little cloudy especially in terms of transfer to homologous and nonhomologous limbs. That is, when asked to produce a movement sequence with the left hand after learning with the right hand, for example, transfer is not always very effective. Thus, there do seem to be some limits to parameterizing a GMP.

Charles H. Shea

See also Especial Skills; Laws of Movement Learning and Control; Motor Control; Motor Learning

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GENETICS/NATURE–NURTURE DETERMINANTS

Conceptualizations of the factors affecting skill acquisition and the demonstration of expertise generally reflect qualities associated with biological factors such as genes (*nature*) or those related to environmental or experiential factors such as training and coaching (*nurture*).

Historical Background

Although the conceptualization of the nature versus nurture debate can be traced at least to Platonic and Aristotelian discussions of human nature in ancient Greece, more contemporary discussions usually start with Francis Galton's 1874 *English Men of Science: Their Nature and Their Nurture*; Galton may have taken his title from Shakespeare's *The Tempest* (Act 4.1)—“a

born devil, on whose nature Nurture can never stick.” Galton's approach was very clearly influenced by the work of his cousin, Charles Darwin, and focused on the stable, biological factors that are transferred from one generation to the next (known as *heritability*, the proportion of variance in a population attributed to genetic factors). Alternately, some researchers see human development as starting from a blank slate (*tabula rasa*) with no innate traits or characteristics, and that all forms of learning and behavior result from interactions with our environment. This extreme nurture viewpoint is demonstrated by behaviorist John B. Watson's famous boast:

Give me a dozen healthy infants . . . and my own specified world to bring them up in, and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist . . . regardless of his talents, penchants, tendencies, abilities, vocations and race of his ancestors. (1998, p. 82)

The history of this debate in psychology is marked by radical shifts in opinion, usually driven by social and cultural factors. For example, after World War II, political and intellectual thought changed to reflect the perspective that differences between individuals resulted from opportunities and experience. This perspective was deemed more socially acceptable than ideologies that sought to separate individuals on the basis of biology since this latter approach was so strongly associated with Nazism. A similar social change occurred in 2001, after the publication of the human genome when researchers advanced positions based on the ultimate power of biology for understanding interindividual differences. James Watson, codiscoverer of the structure of DNA and a key developer of the Human Genome Project, said, “We used to think our future was in the stars. Now we know it is in our genes.” More recently, we have seen a change in direction, at least in some fields of psychology and education, as researchers focus on the solitary role of deliberate practice (a nurture-related variable) in creating the expert performer. The role of nature and nurture factors in sport and exercise is reviewed as follows.

Nature Determinants

Nature determinants focus on stable biological factors. Each person is the product of a unique sequence of genetic material, half coming from

one's mother and the other half from one's father in the form of *chromosomes*. This genetic material (DNA, or deoxyribonucleic acid) is made up of a combination of four chemical bases—adenine, thymine, guanine, and cytosine (abbreviated A, T, G, and C), which group together into base pairs (A with T and G with C) and links of these base pairs form discrete sequences of DNA called *genes*. Each gene corresponds to a specific biological outcome; for example, the COL5A1 gene found in humans on chromosome 9, encodes for a specific collagen protein relevant to flexibility of ligaments and tendons. Within the normal population, there are multiple variants of this gene, which produces the variability seen across a population. This variability results from different *alleles* (a variant in the gene sequence) and is known as a *polymorphism* (the occurrence of many forms of a given gene). Your individual collection of genes (estimated to be between 20,000 and 25,000 genes containing about 3 billion base pairs) is referred to as your *genotype*.

There is strong evidence that genotypic features affect health, fitness, and performance outcomes. For instance, using both heritability statistics and individual genetic markers, data from the HERITAGE Family Study have shown that specific genotypes are better suited for aerobic activities. Since the publication of the human genome in 2001, hundreds of genes and polymorphisms related either directly or indirectly to performance and fitness phenotypes have been identified. Some of these candidates include the COL5A1 gene mentioned earlier as well as genes for the cardiovascular catalyst angiotensin-converting enzyme (ACE), and the skeletal muscle protein alpha-actin type 3 (ACTN3), which have had varying degrees of success explaining interindividual differences in performance. Moreover, despite the considerable attention given to the search for individual genes responsible for athletic achievement, the manifestation of any biological or psychological effect is likely the result of a complex interaction of thousands of genes, and as a result, identification of individual genes that account for meaningful variation in sport- or exercise-related outcomes is extremely difficult.

Nurture Determinants

Few researchers dispute the important role of the environment (nurture) in determining health and fitness related outcomes. In the social and

behavioral sciences, few relationships are as robust as the one between practice and achievement. The power law of practice is a mathematical model of this relationship whereby improvements occur quite rapidly at the onset of practice but become more difficult to obtain (requiring greater practice) as the performer becomes more skillful. This law (or derivations of it) has been widely applied and is proposed to describe human learning of perceptual–motor skills ranging from learning to roll cigars and reading text upside down, to scoring a layup in basketball or hitting a bull's-eye in darts.

Studies from the field of human expertise and expert performance have also supported the relationship between training and ultimate achievement. For example, the proposition of deliberate practice, developed by psychologist Anders Ericsson, is based on the notion that proper training (deliberate practice) performed at the appropriate period of biological and cognitive development is a necessity for the acquisition of expertise in any domain, including sports. Ericsson's (1996) extensive work on the acquisition of expert performance has shown that an individual's level of performance in any domain is determined by the amount of time spent performing “a well defined task with an appropriate difficulty level for the particular individual, informative feedback, and opportunities for repetition and correction of errors” (pp. 20–21).

In addition to the primacy of practice or training to explaining human achievement in sport, other environmental factors are important. For example, Joseph Baker and Sean Horton proposed a number of *secondary factors* that facilitate the acquisition of appropriate training and practice, including obvious factors like access to proper coaches and having supportive parents, as well as less obvious factors, such as growing up in talent hotbeds or being born at the right time of year to take advantage of relative age effects.

Interactions of Nature and Nurture

Although discussions in sport and exercise psychology generally focus on nurture-related factors, it is clear that most health, fitness, and performance outcomes result from both nature and nurture. Perhaps the clearest example of nature–nurture interaction (or gene $\times$ environment interaction) comes from medicine and relates to the condition phenylketonuria (PKU). This genetic disorder causes brain damage and progressive

mental retardation in 100% of individuals with the genetic marker for this condition; however, PKU is highly treatable if individuals eliminate phenylalanine from their diet. In this example, a quality with perfect heritability (PKU) is not expressed because of an environmental manipulation (dietary change). Importantly, the DNA that causes this condition remains unchanged; the individual simply does not express the PKU condition. Although examinations of gene $\times$ environment interactions are rarely examined in sport and exercise psychology, it is likely that similar effects occur. For example, the COL5A1 gene, by affecting injury risk, likely constrains the amount and intensity of training individuals in certain sports (e.g., sprinting) can perform throughout their development.

Modern Conceptualizations

Most evolutionary psychologists and geneticists have moved away from the nature–nurture conceptualization on the basis that it is too limited to explain the nuances of the interactions that occur throughout human development. As a result, researchers have developed more comprehensive models of these relationships. For example, *dynamic systems theory* presumes that human behavior is best seen as the interaction of different systems (biologic, social, psychological) with development being emergent, nonlinear and multidetermined. From this theoretical standpoint, genetic diversity is responsible for some variability in the differences between individuals in how they respond to different experiences and beneficial changes like performance adaptations happen when there is a favorable interaction with important environmental constraints. For example, there is growing consensus in the study of human obesity that the contribution of genetic factors is exacerbated in North American environments that are high in caloric availability.

In evolutionary psychology, focus is on the differences between obligate and facultative adaptations. Obligate adaptations occur no matter what environment a person experiences while facultative adaptations are sensitive to environmental variations. The fact that sugar tastes sweet and lemons taste sour (obligate adaptations) does not change if they are experienced in a different environment. Other, more complex qualities reflect facultative adaptations, similar to *if–then* statements. The attachment style one has as an adult

(i.e., how you approach long-term relationships), for example, is affected by the level of trust one had in their caregivers as a child. If a young person often has promises broken by a parent, this may lead to trust issues as an adult. Given that obligate adaptations do not typically vary across development, scientists in this area focus on how different developmental experiences affect the expression of facultative adaptations.

Another important element in this discussion relates to the emerging concept of epigenetics, which shows that changes in gene expression can occur through modifications to the genome that do not involve a change in the underlying DNA sequence. This field of research suggests that experiences provide a stimulus for genes to be turned on or off. For example, a Swedish study showed that a single hard workout resulted in genes involved in energy metabolism being switched on, with harder work resulting in a stronger effect.

Collectively, research on the effects of biology and experience on human behavior and performance indicate that the nature versus nurture dichotomy is ineffective for explaining the dynamics of how these factors interact and the complexity of this process.

Joseph Baker

See also Dynamical Systems; Expertise; Learning; Motor Development; Motor Learning; Practice; Skill Acquisition; Talent Development

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GOAL SETTING

A goal is simply something you are trying to accomplish; it is the object or aim of an action. Although goals can function at an unconscious level, the process of goal setting represents the deliberate establishment and refinement of goals and the evaluation of goal progress. The concept of goals and the practice of goal setting are well known and established within settings where performance enhancement is the objective. It is important to understand goals because they have such a broad function in terms of affecting the thoughts and behaviors of those to whom participation, productivity, and performance are important.

In the broader field of performance psychology where the objective is to enhance productivity in its varying forms, the effectiveness of goal setting as a strategy has consistently been verified across tasks, groups, methods for setting goals, and performance indicators. Although it was assumed that the positive effects of goals would be replicated within sport and exercise settings, research in sport has failed to illustrate unequivocally that goals function as effectively in this domain. The reasons why have been debated widely, with the consensus being that sample and task characteristics were markedly different in sport. Despite this, studies that have described the goal-setting practices in sport performers have confirmed that almost all athletes do set goals and the majority find them to be effective. This entry provides a brief overview of the current state of play with regard to goal setting in sport, and critiques, where appropriate, the transfer of

goal-setting concepts to sport and to performers in that domain. The intention is not to present an exhaustive review, but rather to highlight those aspects of goals and goal setting that are most pertinent to the advancement of knowledge in this area from both a theoretical and an applied perspective. The following sections cover definitions and types of goals, proposed mechanisms of effects, parameters of goals, and dispositional and situational antecedents of goals, and the final section addresses future research potential.

Types of Goals

The definition of *goals* as an aim of action serves to portray goals as the drivers (or cognitive regulators) behind goal-directed behavior. Consequently, within the multilayered domain of sport, where the nature and level of engagement varies so much, these underlying drivers of behavior can take many different forms. For example, antecedents of behaviors (goals) might range from winning a gold medal at the next Olympics, through the bending of an injured leg an extra two degrees during a physiotherapy session, to maintaining form through a high knee lift in sprinting.

The sport psychology literature consistently distinguishes between three broad goal types: outcome goals, performance goals, and process goals. *Outcome goals* describe intentions relative to the performance of others involved in the activity. The key delineator of these to other goal types is the notion of social comparison. The objective of winning represents the predominant outcome goal; however, the objective of placing in a race, reaching a final, or simply beating a teammate in an individual race, also represent examples of outcome goals. Unlike outcome goals, *performance goals* are based on *levels* of personal achievement and are entirely self-referenced (subjective). Typical performance goals are to run a race in a certain time, to jump a certain distance, to lift a specific weight, or to do a number of repetitions in a training situation—perhaps within a certain time; they refer to products of performance. These goals are normally based on numeric criteria (e.g., to jump one meter and sixty five centimeters) and refer to a predetermined subjective performance standard. *Process goals* are similarly self-referenced but are distinguished from performance goals because their focus is on the process of performing rather than a product of performance. The variation in process goals is subsequently far broader than that

of outcome and performance goals. For example, they might range from the breathing techniques designed to regulate heart rate in a pistol shooter, to imaging in the mind's eye the flight of a golf ball before taking a shot, to focusing on maintaining position while executing a half-court press in basketball. In essence, process goals center on the execution of behaviors regarded as contributing to effective performance.

Relatively few studies have explored the specific effects of using different types of goals. Of those that have, some support, albeit limited, has been found for the use of outcome goals, while moderate to strong support has been provided for the use of performance and process goals. Moreover, it is suggested that combinations of goal types may be more effective than any single type alone. The benefit of performance goals (compared to outcome goals at least) lies in the fact that they are more controllable and flexible than outcome goals; they do not rely on the performance of others to be achieved. However, the achievement of performance goals may still be influenced by external factors, such as environmental conditions, luck, officiating, or even natural fluctuations in personal performance levels. Process goals, conversely, are almost entirely under the control of the individual, and so there is no reason why external factors should disrupt their achievement. Somewhat paradoxically, of those studies which have explored the goal-setting practices of athletes, the majority have found that performers report using outcome and performance goals far more than process goals. This may be a consequence of the less tangible nature of process goals (e.g. regulating breathing or creating a visual image of the trajectory of the ball) and the reassurance that many athletes have in being able to observe or monitor levels of performance directly.

Why Are Goals Effective?

Goal setting is widely regarded as the most popular basic sport psychology technique and is an integral part of any mental training program designed to maximize athletic potential. It is arguably the bedrock of athlete and coach education from a psychological perspective and supports or underpins many other strategies, such as confidence building and enhancing motivation.

In reviewing the literature on goal setting in sport, one is left with two inescapable conclusions: First, goals work; and second, the mechanisms

behind their effectiveness are neither well understood nor particularly well documented. The latter has, on occasion, been attributed to early research into goal setting having limited theoretical grounding. Similarly, more recent research has been criticized for not moving beyond confirming performance effects: It has failed to consider factors that might mediate the relationship between goals and actual sport performance. It is to the mechanisms underlying the positive effects of goals that we now turn our attention.

There is an established view that goals facilitate performance through motivational effects. Seminal work into the application of goal setting to sport suggested that goals influence task performance in four main ways; they direct attention to the task, promote increases in effort, encourage persistence in the face of failure, and facilitate the promotion of new task-relevant strategies (In what other way could I achieve this objective?). Anecdotally, this motivational function of goals is supported, yet research in the sport domain is limited. While the motivational effect of goals on performance is largely direct, goals can also exert an indirect effect through changes in confidence. The successful achievement of goals over a period of time is regarded as pivotal in altering individual perceptions of capabilities. As goals are achieved, new goals are identified; as these new goals are achieved in turn, performers develop a stronger and more stable level of confidence. Repeated failure to achieve goals, on the other hand, undermines confidence.

Although it is suggested that the function of different types of goals may vary, there have been few studies exploring the relative effects of different goal types (e.g., performance vs. outcome goals). From the limited work that has been conducted, researchers within sport and exercise psychology have argued that focusing on self-referenced process and performance goals, rather than outcome goals, appears fundamental to the effective use of goals. Somewhat paradoxically, it is also suggested (and strongly supported anecdotally) that outcome goals (e.g., a desire to break into the world's top 10 in a particular sport) appear to facilitate longer term motivation.

Considering the mechanisms of effects, the limited research that has explored the use of process goals suggests that their beneficial effects can be attributed to focusing more on the task, increasing confidence, reducing concerns about

evaluation from others, and developing personal interest in learning about the task. However, the narrow scope of this work to date reinforces the importance of research into understanding how goal setting works and ensuring consequently that clear guidelines are provided to promote effective goal setting.

A Framework for Effective Goal Setting

As alluded to in the previous section, early theorizing in relation to goals provided an immediate stimulus for research in sport. This research did not, as one might have expected, attempt to explain how goals functioned to enhance performance, but rather focused on the specific content of the goal and its effects. Consequently, while not necessarily increasing our understanding of how goals work, this research did offer some clarity into the nature of what might be regarded as effective goals. At the very least, this provided practitioners with a useful framework upon which to base future goal-setting interventions. The following paragraphs provide a brief summary and critique of this work as it pertains to educating athletes and coaches on the nature of appropriate goals.

Research on goal content in sport has provided some support for the positive effect of goals and highlighted the importance of a number of qualities of effective goals. These aspects are often referred to as the moderators of the goal setting and performance relationship because they are considered pivotal to describing the qualities of goals that enable them to be effective. The aspects of goals include *goal difficulty*, *goal specificity*, *goal proximity*, and *goal collectivity*.

One of the earliest conclusions of research in organizational settings on the relationship between conscious goals and task performance was that individuals striving for goals that were both specific and difficult performed better than those who had goals that were specific and easy, those who had goals that were vague (e.g., “I want to do my best”), and those who had no goals. While there are obvious (and widely debated) contextual differences between business and sport settings, research in sport suggests that moderate levels of goal difficulty were most effective in facilitating performance. Furthermore, responses to extremely difficult goals in sport were very different—instead of withdrawing effort, individuals in sports settings, when faced with relatively difficult goals, modified them

to ensure they remained relevant and achievable. Similarly, in terms of specificity, while specific goals are more effective than no goals or vague goals, those instructed to “do their best” in sport settings do not perform any worse. It is argued that this is because one of the fundamental differences in sport participants is that they actively engage in personal goal setting in response to this type of ambiguous suggestion.

Goal proximity refers to the time aspect of goals, and this can range from immediate intentions to future aspirations. *Long-term goals* have been described as those whose attainment is 6 or more weeks away, whereas goals of shorter duration are termed *short-term goals*. Goal proximity research conducted within sport settings has been rather limited; however, researchers do suggest that combinations of long-term and short-term goals are more effective than using either type alone. Arguably, much of the limited research in this area simply illustrates that having goals is better than not having goals. Nevertheless, in terms of effectiveness, the overriding message is that long-term objectives are most likely to facilitate performance and motivation when short-term goals represent flexible and controllable stepping stones to achieving them. In other words, long-term goals provide direction, while shorter term goals appear to provide opportunities to develop confidence and maintain motivation in pursuit of more distal objectives.

The study of *goal collectivity* concerns itself with the effects of team or group goals on collective performance. Early work on team goals in sport suggested that these goals can facilitate group performance and, in addition, promote team satisfaction, cohesion, and motivation. Specifically, team goals are argued to offer direction for the team and help individual members establish appropriate personal goals to support team objectives. Furthermore, it is logical that team goals should be accompanied by individual goals to ensure task focus and effort levels are maintained by individuals within the team. These individual goals should be based on the individual roles that each player needs to fulfill in order to maximize unit (e.g., a defensive group), and in turn, team effectiveness.

Dispositional and Situational Antecedents of Goals

One important consideration when seeking to more fully understand the nature and function of

goals that athletes set is where do the goals come from and what factors lead to individual variations in goal preferences? The concept of the person by situation interaction has been used by many social scientists attempting to explain thoughts and behaviors. Specifically, and of relevance here, it has been applied in the sporting environment to explain variability in goal-setting practices; the personal and situational antecedents of goals are described in the paragraphs that follow.

At a personal level, the goals individuals adopt are argued to give meaning to their behaviors and energize their actions; they reflect the objective of striving and provide a framework through which an individual can interpret performance-related information. According to the predominant goal theory, two categories of achievement goals exist, and these are consistent with personal views of what is required to demonstrate competence in settings where the potential for competition and evaluation occurs. Goals, and the individuals utilizing such goals, concerned with demonstrating capability through personal improvement and learning about a task with reference to personal performance criteria are said to be *task involved*. Conversely, goals that focus on demonstrating competence with respect to others are described as being *ego involved*. The engaged reader may have noted a parallel between what are described here as achievement goals (task- and ego-involved goals) with what were earlier labeled as goal types, such as outcome, performance and process goals. Put simply, ego-involved goals, where the intention is to outperform others, are logically equated to the use of outcome goals. Similarly, task-involved goals are characterized by self-referenced intentions, which might focus on anything ranging from personal performance targets (a performance goal) to successfully executing a skill or action (a process goal). While the goals described reflect moment to moment targets, individuals are also predisposed to pursue goals reflecting their personal theory of what represents achievement. Researchers have labeled these dispositional (or trait-like) tendencies as goal orientations, for example, one predisposed to using ego-involved criteria to judge competence is said to be ego oriented, and those predisposed to using self-referenced criteria are said to be task oriented, and this aspect of achievement goals has stimulated the majority of research in this area.

Broadly speaking, sport-based research in this area has espoused the motivational benefits of a

high level of task orientation, and regards high levels of ego orientation as placing the individual motivationally and behaviorally at risk, especially when accompanied by lower levels of perceived competence. One of the key characteristics of task and ego goal orientations is that they are independent. This means that an individual can be high or low in each or both orientations, and thus at a general level make judgments of their competence using a variety of personal and social comparison-based criteria. Although these goal orientations may be regarded as relatively stable, the specific goals used by individuals are dynamic and can change from moment to moment in response to the ongoing stream of information presented within the context of their sport involvement. In other words, specific goals can change during a task or performance.

The preceding paragraph outlines a case for how dispositional preferences affect goal choice. In line with numerous other motivational theories, situational factors also play a role in shaping the goals one adopts. Specifically, in situations where performance is evaluated (achievement settings), the psychological environment (labeled as the *motivational climate*) created by the coach (or those with similar leadership roles), can emphasize or promote, within the individual, a variety of criteria for judging competence; unsurprisingly, these impact individual goal preferences. Based originally on work conducted in education settings, two types of motivation climate have been described and applied to achievement contexts such as sport. A *mastery* climate operates when, for example, individuals perceive that task mastery and self-referenced goals are promoted, effort is rewarded, groupings for training tasks are not based on ability, mistakes are regarded as a natural part of learning, and success is evaluated with regard to personal improvement. Conversely, a *performance* climate exists when, for example, individuals perceive that time constraints limit mastery opportunities, superior performance compared to others in the training group is rewarded, groupings for training are based on ability, mistakes are punished, and success is evaluated with regard to outperforming others. Although motivationally adaptive behaviors are most likely to be associated with a mastery climate, certain individuals, most notably those with high levels of perceived competence, may also flourish in environments that promote social comparison (performance climate).

There is no research examining the direct relationship between motivational climate and goal choice. Nevertheless, one might still predict that the relationship between motivational climate and goal preferences parallels that which has been proposed to exist between achievement goal perspectives and goal types. Athletes exposed to a strong mastery climate are most likely to consider using self-referenced goals focused on personal improvement and learning (performance and process-based goals), whereas athletes in a strong performance climate would more likely lean toward goals based on social comparison (outcome goals). Adopting an alternative perspective, recent research on motivational climate suggests that a mastery climate may actually be promoted by encouraging athletes to adopt self-referenced goals focused on personal improvement and learning or mastery of tasks, that is, performance and process goals, respectively.

Research specifically examining the predictive effects of dispositional and situational antecedents of goal choices has not been forthcoming to date—this is most likely a result of an absence of effective measure of goal choices. Nonetheless, anecdotal evidence and common sense suggest that high levels of task orientation and a mastery motivational climate are likely to promote self-referenced performance and process goals, whereas high levels of ego orientation and an environment emphasizing social comparison are likely to promote outcome goals.

Conclusion

Goal setting is an established technique to increase motivation and enhance confidence. It is used widely across all levels of sport, and goals provide essential direction at both an immediate and long-term level. Goals can take a variety of forms, and they have both personal and situational antecedents. The consensus of research and anecdotal accounts is that an individual's primary focus should be on personal task-focused objectives, rather than social comparison. Nevertheless, factors such as personality, perceived ability, the psychological environment, and support with goals by others significantly influence the motivational and behavioral consequences of goal pursuit. While there is some conjecture regarding the specific nature of effective goals, it is generally accepted that setting and using challenging, specific goals with subgoals formulated to act as stepping stones

to longer term objectives seems important. Despite undoubted progress, many questions about goals in sport remain unanswered. More clarity is needed on the functionality of goals and which goal types influence different aspects of personal psychology. Perhaps most importantly, practitioners need to develop methods for process-based goals to be integrated into the day-to-day routines of athletes. Finally, there is a need for much higher quality, theoretically grounded research to provide practitioners with clearer, evidence-based guidance for effective goal setting at an individual and group level.

Kieran Kingston

See also Achievement Goal Theory; Coach–Athlete Relations; Mastery and Control Beliefs; Psychological Skills; Self-Efficacy

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GROUP CHARACTERISTICS

A *group* is defined as a social aggregate of two or more people that involves mutual awareness, interaction, and interdependence of its members. The characteristics of the group shape the beliefs and behaviors of its members. In this entry, two categories of group characteristics are examined, namely (1) characteristics of the group and (2) characteristics of group members.

Characteristics of the Group

Albert V. Carron and Mark Eys examined the many definitions of groups and identified five common characteristics: (1) common fate—sharing a common outcome with other members; (2) mutual benefit—an enjoyable, rewarding experience associated with group membership; (3) social structure—a stable organization of relationships among members; (4) interaction and communication among members; and (5) self-categorization—perceiving oneself as a member of the group. Research in exercise and sport settings was aimed at examining these five characteristics in relation to cognitions and behaviors of its members. A considerable body of research was accumulated about the influence of social structure (e.g., roles and norms of group members) on exercise adherence. In general, the findings support the importance of norms like acceptable forms of behavior and a clear understanding and acceptance of roles to promote exercise behavior. Other research findings pertained to the pattern of communication between coaches and athletes. This research has identified effective verbal and nonverbal communication practices for coaches working with athletes. Research has also begun to focus on self-categorization. Researchers have recently examined how self-categorization influences perceptions, preferences and behaviors in exercise groups.

Kevin S. Spink, Kathleen S. Wilson, and Carly S. Priebe (2010) recently examined all five of the common group characteristics together as a multidimensional construct of *groupness* in relation to

adherence in structured exercise settings. Drawing on previous work, Spink et al. viewed groupness as the extent to which the five group characteristics can be perceived by group members. They hypothesized that perceptions of groupness, as captured by the five group characteristics, would be positively associated with individual exercise behavior. Support was found for this hypothesis as enhanced perceptions of groupness were associated with improved frequency and attendance in exercise.

Group Size

In addition to the five common group characteristics conceptualized by Carron and Eys, group size was considered important in this context. Origins of group size research date back to Norman Triplett's first social psychology study in 1898. Triplett examined the influence of others on cycling performance. Since then, research has indicated that as group size increases members report less cohesion, intimacy, satisfaction, and communication, as well as report greater tension, anxiety, competitiveness, being argumentative, feeling more threatened, and displaying more inhibition.

Group Size and Productivity

A topic of considerable interest to researchers has been the relationship between group size and productivity. In 1913, Max Ringelmann investigated the relationship between group size and individual and group performance in a rope-pulling task. Ringelmann's results, coined the *Ringelmann Effect*, revealed that as group size increased, relative productivity of each individual group member decreased. Ivan Steiner later examined the research on group size and productivity and developed a model to illustrate the relationship between the two concepts. Steiner proposed that as the number of members in a group increases, the potential group productivity also increases due to an increase in available resources up a point when the group's productivity plateaus.

Steiner offered two possible explanations for the decrease in individual productivity as group size increases. The first was coordination loss. Steiner proposed that as group size increased, the number of coordination links increased, and it was more difficult to synchronize the efforts of the individuals. The second explanation was the reduction in individual motivation as group size

increases. It was hypothesized that as the size of the group grows, personal accountability is reduced, and it is more difficult to motivate individuals. Later work replicating Ringelmann's classic rope-pulling experiment research found support for both explanations.

Based upon the evidence on group size and productivity, it begs the question of what is the ideal group size to maximize productivity? Researchers in many fields continue to struggle with this question. Research in work settings supports smaller groups (three to six members) being more productive than larger groups (more than seven).

Group Size Research in Sport and Exercise Settings

Research on group size in sport and exercise began in the early 1990s in a series of studies conducted by W. Neil Widmeyer, Lawrence R. Brawley, and Albert V. Carron examining group size and cohesion on sport teams. Smaller teams were found to be more optimal for the development of commitment to group goals (task cohesion), while moderate-size groups were best for building strong relationships and friendships among group members (social cohesion). Subsequent work in exercise settings by Carron and Spink found task and social cohesion to be higher in smaller groups. Carron and Spink also found the group-based intervention of team building to offset negative effects of increased group size on cohesion.

Other researchers have examined the influence of group size on affective outcomes in sport and behavioral outcomes, such as attendance and retention in exercise classes. Retention and attendance were greatest in the smallest ($n = 5-17$) and largest ($n = 32-46$) exercise classes, but poorest in the medium classes ($n = 18-31$). Group members' enjoyment, satisfaction, and belief in team's ability tend to progressively decrease with increased group size.

Characteristics of the Group Members

The characteristics of group members individually and collectively (referred to as group composition) have been a topic of considerable interest to researchers and practitioners alike. Group members can differ in age, sex, ethnicity, education, and social status. Further, members can vary in physical size, attitudes, motives, needs, mental and motor abilities, and personality traits. Researchers in social psychology have been

interested in understanding how the level of diversity of individual characteristics of group members has implications on group functioning and member cognitions, affect, and behavior. Work in this area has prompted researchers to examine the contextual exercise preferences for adults. In an exercise environment, there is growing support for individuals to prefer to be active with others of similar age and gender. Overweight individuals, in particular, report a greater preference for same-gender exercise classes (versus mixed-gender classes) than normal weight individuals. Research was aimed at examining how intragroup perceptual similarity in surface (age, ethnicity, or physical condition) and deep-level (attitude, beliefs, or values) qualities are related to perceptions of cohesion and exercise adherence within exercise classes. Surface level qualities, notably age, were associated with social cohesion and exercise attendance, while deep level qualities were associated with task cohesion. Collectively, these findings highlight how *similar* we perceive ourselves to be in relation to other group members in terms of various individual characteristics like age can have important implications in terms of our exercise preferences, cognitions, and behavior.

Group Composition

Widmeyer and John W. Loy developed a conceptual framework for group composition suggesting that the properties of sport groups can be considered from three perspectives: (1) amount of group resources, such as skills or personal attributes, present among group members; (2) the variability like homogeneity or heterogeneity in the resources of group members; and (3) compatibility between and among group members.

Strong empirical evidence supports the first perspective as higher individual measures of task ability and motivation have been associated with greater team performance. In regard to Widmeyer and Loy's second perspective, *variability* (e.g., homogeneity vs. heterogeneity in group resources like member attributes) can be considered beneficial or detrimental, or even both depending on the nature of the group and its task. For example, in basketball, effective teams ideally have heterogeneity in playing skills—guards who are gifted at ball handling and forwards who can rebound. Variability in sex, age, and race has also been examined. Among the identified variables, the

relationship between race and playing position has been investigated extensively. To date, no conclusive evidence exists to support a relationship between race and playing position.

An area of future consideration in relation to Widmeyer and Loy's first two perspectives is the amount and *variability in team personality composition*. In work groups, researchers have found higher mean levels of extraversion and emotional stability contribute positively to social cohesion.

Researchers have also focused on the variance in team personality composition. This latter approach is based on research suggesting that averaging individual team member personality scores may mask important information, such as the negative effect of a single team member on a group. As such, researchers have used a minimum scores method to examine the personality scores of individual team members. Using this approach, a minimum rather than average score is established for each personality trait for the group. Additional research with work groups revealed minimum levels of conscientiousness and agreeableness to contribute positively to both task cohesion and team performance. Future research in team personality composition in sport and exercise settings is needed.

Widmeyer and Loy's third category is *compatibility in group resources*. Compatibility is described as the ability of an individual to *fit* (e.g., function effectively and in harmony) with other group members. Across social, work, and sport group settings, compatibility contributes to enhanced individual satisfaction and performance. Research on compatibility in sport has predominantly focused on compatibility of coach–athlete relationships (referred to as coach–athlete dyads), member abilities, and group roles. A behavioral approach has frequently been used to examine the compatibility of coach–athlete dyads and teams. This has often involved observations of coach and athlete behaviors. Early work used William Schutz's fundamental interpersonal relations orientation (FIRO) theory to evaluate compatibility of the coach and athlete. Schutz proposed that in interpersonal relationships, individuals need to express three needs—*inclusion*, *control*, and *affection*. More recent work utilized a novel video observation and analysis approach titled *state space analysis* to examine coach–athlete interactions in youth sport teams over time. The state space approach was adapted from work in psychology examining parent–child relationships.

Collectively, behavioral approaches have attempted to shed light on the compatibility of coach–athlete dyads and group members to achieve effective performance.

Conclusion

Understanding the characteristics of a successful team is a topic of interest shared by researchers and practitioners alike. Further research and the continued integration of theory from other fields, such as organizational psychology or group dynamics, is needed in order to develop a better understanding of the role of group characteristics in sport and exercise settings.

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See also Cohesion; Group Formation; Self-Categorization Theory; Social Identity Theory; Status; Team Building

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GROUP FORMATION

Why do groups form and how do groups develop? In this entry, different perspectives on group development are examined. There are a number of reasons that people join groups. William Schutz theorized humans seek out groups in an effort to fulfill one or more of the following fundamental needs: (1) need for inclusion—desire for affiliation, belonging, and acceptance; (2) need for affection—desire for intimacy; and (3) need for control—desire for power and opportunities to dominate others. Once a group is formed, there are different perspectives on how groups develop.

Group Development Perspectives

Group development involves the evolution of structure, cohesion, and maturity of a group over the course of time. Group dynamics scholars have proposed over a hundred theories to explain group development. Holly Arrow, Marshall S. Poole, Kelly B. Henry, Susan Wheelan, and Richard Moreland reviewed the literature and classified the theories into five general categories: (1) linear (sequential stage) perspective, (2) cyclical (repeating cycles) perspective, (3) robust equilibrium model, (4) punctuated equilibrium model, and (5) adaptive response. The following is an overview of the five perspectives with an emphasis on the linear and cyclical perspectives, which have received the greatest attention in the group dynamics literature.

Linear (Sequential) Perspective

In the mid-1960s, Bruce Tuckman reviewed the literature on group development and proposed that

groups progress through four stages of development. The four stages include: *forming*, *storming*, *norming*, and *performing*. Tuckman in collaboration with Mary Jensen later added a fifth stage, *adjourning*. In the first stage, forming, group members become familiar with each other and identify the group's task. This stage is similar to an orientation phase as the primary focus of the group members is on inclusion and dependency. Storming is characterized by tension and conflict among members and with the group's leader. Issues of power and authority are focal points as members exert their preferences for the group's approach to a task and question, or even resist, decisions of the leader. During norming, the group members come together to reach a consensus on the group's goals and objectives, and the group's norms, the accepted standards for behavior within the group. The norming stage is also characterized as a period of cooperation, clarification of individual member roles within the group, and development of group cohesion. In performing, the group becomes more stable as group members display a clear understanding and acceptance of their roles. The focus of group members is on the successful achievement of the group's task rather than conflict among members and the leader. Finally, adjourning is highlighted by a termination of roles or duties, reduction of dependency, and breakup of the group. Collectively, this linear perspective has two defining characteristics: (1) the five stages are sequential, and (2) the duration of each stage is variable.

Cyclical (Repeating Cycles) Perspective

One of the defining features of the linear perspective on group development is that groups move through a sequence of stages that begin with formation and end with termination (forming, storming, norming, performing, adjourning). In contrast, the cyclical perspective proposes that group development occurs through a *repeating* cycle of five stages. In the first stage, *discontent*, group members do not yet feel a part of the group or consider the group as part of their identity. An example would be during tryouts for a sport team when new players are vying for a spot on the team. The second stage of *group identification* occurs when individuals display commitment to the group and identify group members from nongroup members. The third stage, *group productivity*, occurs when members

strive to contribute toward team goals. The successful attainment of goals often leads to the next stage, *individuation*, in which members seek out recognition for their personal contributions. This may be in the form of an athlete seeking out validation from the coach, fellow teammates, or others for performance. As the group moves through the individuation stage, the group progresses to the final stage, *decay*, where the members become less interested and put less energy toward achieving goals of the group. This often leads the group to cycle back to the first stage of *discontent* beginning the cyclical process again. The group may experience a number of cycles while together.

An analogy to differentiate the cyclical perspective from the linear perspective is a sport team progressing through various segments over the course of a season (preseason, regular season, playoffs). For example, during the preseason, a basketball team may display signs of *decay*, the final stage of the cyclical perspective, for example, lack of interest toward the team's goals, and cycle back to the initial stage of *discontent*. However, the start of the regular season may reinvigorate the group and push the group to the stage of *group identification* as members have renewed interest and seek out their roles for the team to be successful. Conversely, a linear perspective would not describe the team as cycling back to early stages in group development during the preseason. Rather, a linear perspective would view the team as progressing sequentially over the season (forming and storming during the preseason, norming and performing during the regular season).

Additional Perspectives on Group Development

Three alternative group development perspectives identified by Arrow and colleagues include the robust equilibrium model, punctuated equilibrium model, and adaptive response. The robust equilibrium perspective is characterized by group development (e.g., delineation of norms and roles) during the early time periods of the group's formation. Then the structure of the group becomes relatively stable over time. The punctuated equilibrium model is a variation of the robust equilibrium perspective where group development is perceived to consist of stable periods that are interrupted by sudden bouts of instability. During bouts of instability, the group is restructured and may involve a turnover in group members or

reassignment of roles within the group. The final perspective, adaptive response, rejects the notion that all groups move through a similar progression of stages. The adaptive response perspective proposes that groups develop in their own unique manner based upon the stressors and challenges each group needs to overcome. As such, each group's pattern of development is idiosyncratic and a response to environmental constraints and opportunities.

Integrated Model of Group Development

Susan Wheelan examined the group development literature and developed an Integrated Model of Group Development that built upon Tuckman's linear model of group development and the early work of Wilfred Bion. Similar to Tuckman's model, there is a linear progression of stages. Conversely, Wheelan's perspective emphasized that groups achieve maturity as the group members work together rather than simply going through stages of activity. Group development in early stages in the model is associated with dependency, counterdependency, and trust while the later, more mature stages are where productivity is a focus.

To reflect the characteristics of each stage in the Integrated Model of Group Development, Wheelan relabeled the stages beginning with the first stage—*dependency and inclusion*. Similar to Tuckman's model, the stage is characterized by the member's concerns for inclusion and dependency on the group's leader for direction. Within the second stage, *counterdependency and fight*, there is conflict among the group members about the group's goals and procedures. If the group can work through the conflict with open communication, the group can foster commitment and trust, and progress to the third stage, *trust and structure*. The structural component of the third stage reflects the negotiation of roles and conformity to norms (acceptable procedures) within the group. This stage also highlights the establishment of positive work relationships. The final or *work* stage is a period of enhanced productivity and effectiveness, whereby previous deleterious issues within the group have been resolved and the focus of group members' energy is on the tasks and achievement of group goals. An emphasis of Wheelan's model is the link between the productivity of the group and the different stages, whereby lower productivity is

evident during Stage 1 and higher productivity is evident during Stage 4.

Group Development Assessment

Several attempts have been undertaken to assess group development. Susan Wheelan and Anthony Verdi developed the Group Development Observation System (GDOS). This approach involves transcribing, coding, and analyzing the patterns of communication of group members based upon seven group-development categories (dependency, counterdependency, fight, flight, pairing, counterpairing, and work). GDOS has been used to examine the patterns of communication between sexes, whereby researchers have compared same-sex (homogeneous) and mixed-sex (heterogeneous) groups during progressive phases of group life. In this work, no significant differences were found to exist between same-sex groups. However, mixed-sex groups differed from same-sex groups but this was attributed to a larger group size rather than sex composition. In sum, sex composition does not appear to influence group developmental patterns.

Susan Wheelan and Judith Hochberger used the integrated model of group development to create the Group Development Questionnaire (GDQ). The GDQ includes four scales that correspond to the four stages of Wheelan's integrated model. The items in each scale are designed to assess the characteristic behaviors of groups in each stage (e.g., Stage 1 scale assesses dependency and inclusion). Calculating group mean scores for the four scales can be used to determine each group's stage of development, productivity, and effectiveness. Higher group mean scores on the higher stages are indicative of greater productivity and group effectiveness.

Wheelan has conducted group development research with the GDQ in a number of settings, such as work and schools. One study in elementary schools found greater staff group development (e.g., when teachers displayed more trust and cooperation) to be positively associated with student learning and performance. More recently, Wheelan used the measure to examine the relationship between group size, development, and productivity in over 300 work groups. Results revealed that group size is an important factor in group development and productivity. Smaller groups of three to eight members were found to be significantly more productive and more developmentally

advanced and with higher mean scores in more mature stages on GDQ than larger groups of nine or more members. Groups consisting of three to four members were found to be most productive and developmentally advanced.

Empirical Evidence in Sport and Exercise

Although there has been an abundance of work on group development in organizational and work settings, research on group formation and development in sport and exercise is relatively limited in scope. The vast majority of research in this area has focused on the relationship between managerial and team member turnover and team effectiveness. In general, longitudinal examinations of team success after a managerial change in a number of professional and intercollegiate sports, such as baseball, basketball, and hockey, have revealed higher managerial turnover to be associated with lower team effectiveness (e.g., poorer win-loss, league standing) or only short-term improvement in team performance, that is, no long-term effects beyond the season.

Similar to managerial turnover, research on the relationship between athlete turnover and team effectiveness has revealed that higher athlete turnover is associated with lower team effectiveness (e.g., poorer win-loss, league standing) in a number of professional sports (baseball, basketball, football). Nevertheless, contradictory evidence also exists, with one review of professional ice-hockey teams, from 1950 to 1966, revealing that both coach and athlete turnover were unrelated to team performance. Indeed, some coaches and researchers have suggested that turnover is not always problematic as too little turnover can lead to complacency and diminished motivation among team members. The ideal time and level of player turnover with regard to team success has been a related area of study and found to vary by sport and position. Given that player turnover can happen for a variety of reasons—injury, retirement, trade—and can potentially impact the group's development and team effectiveness, it is likely to be a topic of future research interest.

Conclusion

Social scientists have highlighted the importance of groups for individuals. While group development has been a topic of considerable interest in

organizational settings, this topic has received much less attention in sport and exercise settings. A fruitful line of research would involve examining the influence of group development on group effectiveness, such as team morale and performance, and on individual members' satisfaction or adherence. in exercise and sport settings.

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See also Cohesion; Group Characteristics; Norms; Roles; Team Building

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H

HABIT

Repetition can make a simple behavior very powerful. Improving your physical health by exercising, increasing body strength by working out, or building up your potting skills in playing snooker can only be achieved by frequently executing these behaviors. However, people often struggle to maintain such regimes; sport schools typically see a decline in attendance a few weeks after the season start, and many training programs fail due to a lack of perseverance. Turning a beneficial behavior or practice into a habit is therefore the key to success. Unhealthy or ineffective habits may form barriers for change and may prevent establishing positive outcomes or improvements.

In the psychological literature, *habit* has always been defined in terms of frequency of behavior, which was inherited from the behaviorist school. However, there are problems with this definition. The first problem is that it is unclear how frequently behavior must be executed in order to qualify as a habit; does a habit exist after 2, 4, or 20 repetitions? Secondly, frequent behavior is not necessarily a habit. An athlete may regularly attempt to break the world record in sprinting, but few would consider this as a habit. We therefore need a better definition.

Such a definition of habit rests on three pillars. The first pillar obviously is *repetition*. Habits are formed by repeating behavior with satisfactory outcomes and without repetition there is no habit. However, repetition is a necessary but not a sufficient condition for behavior to qualify as

habit, hence a second pillar, which is *automaticity*. Habits are typically executed with relatively little mental effort, awareness, and conscious intent. These qualities make habits being experienced as *easy* or *fluent*; there is not much thought involved or needed to carry them out. While habits are forms of automatic behavior, they are functional and goal directed; we develop habits that serve us in one way or another, no matter whether these goals are constructive or unconstructive. If a habit has been developed, the recognition or activation of a goal may thus automatically elicit the accompanying habit. This relates to the third pillar of habit, which is *context stability*. This pillar is not so much a feature of the habit itself, but rather of the conditions and environments where habits occur. Habits are often elicited by specific cues, such as a particular time, location, physiological state, or the presence of an object or person. Habits thrive in stable conditions, where the same cue elicits the habitual response time and again. For instance, a person may have developed the habit of running twice a week after work and before having dinner. Such a habit may flourish if this person has a stable pattern of returning from work and having dinner at the same time and location but would be difficult to maintain in the face of irregular work times or if this person would travel a lot.

Some Caveats

Measurement. If we accept that behavioral frequency is an inadequate definition of habit, the traditional way of using behavioral frequency as a

measure of habit should be replaced by one that is commensurate to the new definition. The Self-Report Habit Index has been developed for that purpose. It is a 12-item scale, which assesses habit as a psychological construct. It incorporates the various facets of habit, such as the experience of repetition and the lack of awareness and conscious intent.

Locus of control. The fact that habits are dependent on and elicited by cues in the environment where the behavior takes place points to a shift in locus of control when behavior becomes habitual. Whereas new or deliberate behaviors are controlled by conscious intentions and *willpower*, which is very much in line with prevalent socio-cognitive models such as the theory of planned behavior, habits are to a much larger degree controlled by the external environment. Habits are thus less likely to change through motivation and attitude change. Although motivation and attitude change may certainly help make changes, managing the environments in which behavior occurs may be paramount to success if strong habits are involved. This implies identifying and changing or removing the cues that trigger old habits and engineering the environment for new habits to establish and flourish. Habits may thus be planned in a particular environment and may be promoted by, for instance, the use of implementation intentions.

Where is the habit? One may wonder whether habits really exist in the domain of sport and exercise. After all, many who practice sport and exercise do that intentionally and mindfully. These activities are characterized by repetition, but seemingly not so much by automaticity. It is therefore important to analyze precisely where exactly habits may reside. Take for instance exercising. While exercising may be executed intentionally and mindfully, it is often the *decision* to exercise where the problems lie. If we would deliberate every time whether or not to exercise, we would be vulnerable to our well-developed skills to rationalize and find excuses to stay home. On the other hand, if we make the decision to go exercising a habit and build this into our daily or weekly routines, we would not need to deliberate whether or not to exercise, but rather which route to take or which clothes to wear. In other words, while the activity

itself may be executed in a mindful, and hopefully joyful fashion, the decision to exercise is the critical habit.

Opportunities to break and create habits. As breaking old habits often requires rearrangements of the environment where habits reside, we may look for opportunities to break habits that sometimes naturally arise. This happens particularly when people undergo life course changes, such as moving to a new house, starting a family, changing a job, or starting retirement. These events provide short windows of opportunity where old habits are temporarily broken, and individuals may be looking for new ways to arrange their lives. In those situations, people may consider making changes that they have contemplated all along. These may also provide opportunities where behavior change interventions may be more effective and thus provide more value for money.

Mental habits. We have not only habits of doing but also of thinking. Mental habits are repetitive patterns of thought, which may possess the crucial features of behavioral habits such as described by the three pillars. Mental habits may involve negative thoughts, which then may get in the way of good performances and may affect outcomes such as one's self-esteem or mental health. Mental habits may also involve positive thoughts, which may then play constructive roles in keeping up one's motivation and believing in oneself.

Bas Verplanken

See also Automaticity; Decision Making; Goal Setting; Interventions for Exercise and Physical Activity; Leadership in Sport: Social Cognitive Approaches; Learning; Mental Health; Mindfulness; Positive Thinking; Thought Stopping

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HEALTH ACTION PROCESS APPROACH

Theories of health behavior change are needed to explain, predict, and improve self-regulation of physical activity. Such theories are being divided into continuum models and stage models. In continuum models, people are positioned along a range that reflects the likelihood of action. Influential predictor variables are identified and combined within one prediction equation. The goal of an intervention is to move the person along this route toward action. Health promotion, then, focuses on increasing all model-inherent variables in all persons, without matching treatments to particular audiences. The *theory of planned behavior* (TPB), developed by Icek Ajzen, is one such *continuum model*.

In contrast, according to *stage models*, health behavior change consists of an ordered set of categories (or stages) into which people are classified. These categories reflect cognitive or behavioral characteristics, such as the intention to perform a behavior. The main purpose of applying stage models lies in the identification of relatively homogeneous target groups for interventions and the design of stage-matched treatments. The most popular stage theory of health behavior change is the *transtheoretical model* (TTM), developed by James Prochaska, that proposes five stages of change.

Both continuum models and stage models have their advantages and disadvantages. Continuum models have been found useful for explanation and prediction, whereas stage models are often preferred to guide interventions. For health promotion, the continuum models are often too general because all variables involved in such a model need to be addressed in interventions, without considering the special needs of particular subgroups of participants. However, it is possible to integrate both approaches when researchers use a continuum model as a theoretical template and,

when it comes to interventions, subdivide the audience into stage groups to allow for stage-matched treatments. The *health action process approach* (HAPA), developed by Ralf Schwarzer, is such a *hybrid model* with a continuum layer as well as a stage layer.

Mediating Mechanisms

The traditional continuum models have been criticized mainly because of the so-called intention–behavior gap (referring to the frequent failure of intention to predict behavior). HAPA explicitly includes postintentional factors to overcome this gap. It suggests a distinction between (1) preintentional motivation processes that lead to a behavioral intention, and (2) postintentional volition processes that lead to the actual health behavior. Within the two phases, different patterns of social cognitive predictors may emerge (see Figure 1). In the initial motivation phase, a person develops an intention to act. In this phase, *risk perception* is seen as a distal antecedent (e.g., “I am at risk for cardiovascular disease”). Risk perception in itself is insufficient to enable a person to form an intention. Rather, it may set the stage for a contemplation process and further elaboration of thoughts about consequences and competencies. Similarly, *positive outcome expectancies* (e.g., “If I exercise five times per week, I will reduce my cardiovascular risk”) are chiefly seen as being important in the motivation phase, when a person balances the pros and cons of certain behavioral outcomes. Further, one needs to believe in one’s capability to perform a desired action, *perceived self-efficacy* (e.g., “I am capable of adhering to my exercise schedule in spite of the temptation to watch TV”). Perceived self-efficacy operates in concert with positive outcome expectancies, both of which contribute to forming an intention. Both beliefs are needed for forming intentions to adopt difficult behaviors such as regular physical exercise.

After forming an intention, the volitional phase is entered. When a person is inclined to adopt a particular health behavior, the *good intention* has to be transformed into detailed instructions on how to perform the desired action. Once an action has been initiated, it has to be maintained. This is not achieved through a single act of will but involves self-regulatory skills and strategies. Thus, the postintentional phase is further broken down

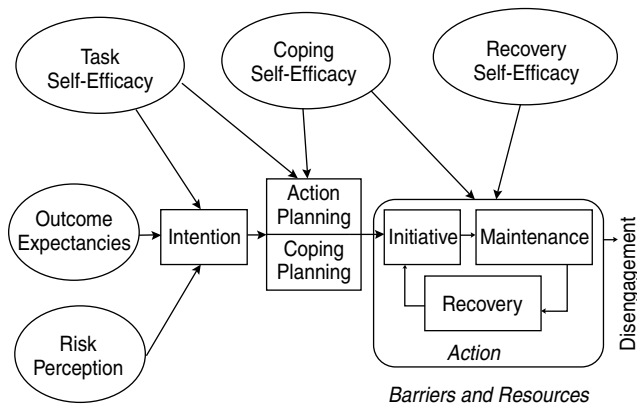


Figure 1 Health Action Process Approach

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into more proximal factors, such as *planning*, *coping self-efficacy*, and *recovery self-efficacy*. Additional volitional constructs that have been included in HAPA research are *action control* and *social support*.

Including planning and self-efficacy as volitional mediators renders the HAPA into an implicit stage model because it implies the existence of at least a motivational and a volitional phase. The purpose of such a model is twofold: It allows a better prediction of behavior, and it reflects the assumed causal mechanism of behavior change. Research that is based on this continuum layer of the model employs path-analytic methods.

Preintenders, Intenders, Actors

When it comes to the design of interventions, one can consider turning the implicit stage model into an explicit one. This is done by identifying individuals who are located either in the motivational stage or in the volitional stage. Then, each group receives a specific treatment that is tailored to this group. Moreover, it is useful and theoretically meaningful to subdivide the volitional group further into persons who perform and those who only intend to perform. In the postintentional-preactional stage, individuals are labeled *intenders*, and in the actional stage *actors*. Thus, a pragmatic subdivision within the health behavior change process yields three groups: preintenders, intenders, and actors.

The term *stage* in this context was chosen to allude to the stage theories, but not in the strict

definition that includes irreversibility and invariance. The terms *phase* or *mindset* may be equally suitable for this distinction. The basic idea is that individuals pass through different phases on their way to behavior change. Thus, interventions may be most efficient when tailored to these particular mindsets.

For example, preintenders are supposed to benefit from confrontation with outcome expectancies and some level of risk communication. They need to learn that the new behavior like becoming physically active has positive outcomes, such as well-being, weight loss, and fun, as opposed to the negative outcomes that accompany the current (sedentary) behavior, such as developing an illness or being overweight. They also need to develop an optimistic belief that they are capable of performing the critical behavior.

In contrast, *intenders* should not benefit much from health messages in the form of outcome expectancies because, after setting a goal, they have already moved beyond this mindset. Rather, they should benefit from planning to translate their intentions into action.

Finally, *actors* should be prepared for particular high-risk situations in which lapses are imminent. Interventions help them if they desire to change their routines by adopting or altering a behavior, for example.

Five Principles

The HAPA is designed as an open architecture that is based on principles rather than on specific testable assumptions. Developed in 1988, it was an attempt to integrate the model of action phases by Heinz Heckhausen and Peter Gollwitzer, with social cognitive theory developed by Albert Bandura, based on five principles.

Principle 1: Motivation and volition. The health behavior change process is divided into two phases. There is a switch of mindsets when people move from deliberation to action. First comes the motivation phase in which people develop their intentions. Afterwards, they enter the volition phase.

Principle 2: Two volitional phases. In the volition phase, there are two groups of people: those who did not yet translate their intentions into action, and those who did. Thus, there are inactive as well

as active persons in this phase. In other words, in the volitional phase one finds intenders as well as actors who are characterized by different psychological states. In addition to health behavior change as a continuous process, one can also create three categories of people with different mind-sets, depending on their current location within the course of behavior change: preintenders, intenders, and actors.

Principle 3: Postintentional planning. Intenders who are in the volitional preactional stage are motivated to change, but they do not act because they might lack the right skills to translate their intention into action. Planning is a key strategy at this point. It serves as an operative mediator between intentions and behavior.

Principle 4: Two kinds of mental simulation. Planning can be divided into action planning and coping planning. Action planning pertains to the when, where, and how of intended action. Coping planning includes the anticipation of barriers and the design of alternative actions that help to attain one's goal in spite of impediments.

Principle 5: Phase-specific self-efficacy. Perceived self-efficacy is required throughout the entire process. However, the nature of self-efficacy differs from phase to phase. This is because there are different challenges as people progress from one phase to the next one. Goal setting, planning, initiative, action, and maintenance all pose challenges that are not of the same nature. Therefore, we distinguish between preactional self-efficacy, coping self-efficacy, and recovery self-efficacy. Sometimes the terms *task self-efficacy* instead of preaction self-efficacy, and *maintenance self-efficacy* instead of coping and recovery self-efficacy are preferred.

Assessment of Constructs

Risk Perception. Risk perception can be measured by items such as "How high do you rate the likelihood that you will ever get one of the following diseases: (a) cardiovascular disease (e.g., heart attack, stroke), (b) diseases of the musculoskeletal system (e.g., osteoarthritis, herniated vertebral disk)?" Any other health risk can be added, if relevant to the research context.

Outcome Expectancies. Positive outcome expectancies (pros) and negative outcome expectancies (cons) can be assessed, for example, with the stem "If I engage in physical activity at least three times per week for 20 minutes . . ." followed by pros such as "then I feel better afterwards," or "then I meet friendly people," and followed by cons such as "then every time would cost me a lot of money," or "then I would be financially depleted."

Self-efficacy. Perceived motivational and volitional self-efficacy can be composed of items such as the following. Motivational self-efficacy (task self-efficacy) refers to the goal-setting phase and can be measured with the stem "I am certain . . ." followed by items like "that I can be physically active on a regular basis, even if I have to mobilize myself," or "that I can be physically active on a regular basis, even if it is difficult." Volitional self-efficacy refers to the goal-pursuit phase. It can be subdivided into coping self-efficacy and recovery self-efficacy. Coping self-efficacy has been measured with the stem "I am capable of strenuous physical exercise on a regular basis . . ." followed by barriers like "even if it takes some time until it becomes routine," or "even if I need several attempts until I will be successful." Items on recovery self-efficacy can be worded "I am confident that I can resume a physically active lifestyle, even if I have relapsed several times," "I am confident that I am able to resume my regular exercises after failures to pull myself together," or "I am confident that I can resume my physical activity, even when feeling weak after an illness."

Intention. Intention to perform a behavior is assessed in correspondence to the behavior itself: for example, "I intend to perform the following activities at least 3 days per week for 20 minutes: strenuous (heart beats rapidly, sweating) physical activities," "moderate (not exhausting, light perspiration) physical activities," and "mild (minimal effort, no perspiration) physical activity." Answers can range, for example, from not at all true (1) to absolutely true (6). Alternatively, intention can be assessed in terms of frequency and duration of more specific behaviors: (a) physiotherapeutic exercises (e.g., back training), (b) fitness activities (e.g., using an exercise bike), (c) resistance training (training muscle strength, e.g., on machines), and (d) physical activity while commuting (e.g., going by bicycle or walking for longer distances).

Planning. Action Planning can be assessed with items addressing the when, where, and how of the activity: for example, “I have already planned . . .” (1) “which physical activity I will perform (e.g., walking),” (2) “where I will be physically active (e.g., in the park),” (3) “on which days of the week I will be physically active,” and (4) “for how long I will be physically active.” Coping planning, on the other hand, can be measured with the item stem “I have made a detailed plan regarding . . .” and the items (1) “what to do if something interferes with my plans,” (2) “how to cope with possible setbacks,” (3) “what to do in difficult situations in order to act according to my intentions,” (4) “which good opportunities for action to take,” and (5) “when I have to pay extra attention to prevent lapses.” Another option is to ask participants to actually generate their plans, which, however, confounds measurement and treatment.

Action Control. While planning is a prospective strategy, that is, behavioral plans are made before the situation is encountered, action control is a concurrent self-regulatory strategy, where the ongoing behavior is continuously evaluated with regard to a behavioral standard. Action control can be assessed with a six-item scale comprising three facets of the action control process: *self-monitoring* (“I consistently monitored myself whether I exercised frequently enough,” or “I consistently monitored myself when, where, and how long I exercise”), *awareness of standards* (“I have always been aware of my prescribed training program,” or “I often had my exercise intention on my mind”), and *self-regulatory effort* (“I really tried hard to exercise regularly,” or “I took care to practice as much as I intended to”).

Staging. When using the continuum model for the prediction of behavior, these variables are being specified as predictors and mediators in a path model (see Figure 1). When employing the stage variant to conduct an intervention study, an assessment of stages is necessary. Stage theories employ algorithms for the staging procedure which can be regarded as a *fast and frugal tree* with satisfactory validity. For a three-stage procedure, one needs two steps. First, behavior is assessed on the basis of a context-specific dichotomous criterion (yes = already active, no = not yet sufficiently active). Those who meet the preselected criterion are defined as *actors*. Those who don’t are subject to

the second step by asking them whether they intend to become active or not. If they do intend, they are defined as *intenders*; if they don’t, they are *nonintenders* (or *preintenders*). Such straightforward diagnostic procedures may be too simple to account for response bias and temporal fluctuation. However, when subdividing large samples to assign stage-matched treatments, such a pragmatic procedure results in more homogeneous subgroups that allow for more effective interventions.

Conclusion

In sum, the HAPA has two layers: a continuum layer and a phase (or stage) layer. Depending on the research question, one might choose one or the other. HAPA is designed as a sequence of two continuous self-regulatory processes, a goal-setting phase (motivation) and a goal-pursuit phase (volition). The second phase is subdivided into a preaction phase and an action phase. One can superimpose these three phases on the continuum model as a second layer and regard phase as a moderator.

This two-layer architecture allows switching between the continuum model and the stage model, depending on the given research question. The stage layer is useful for designing stage-matched interventions. For preintenders, one needs risk and resource communication, for example by addressing the pros and cons of a critical behavior. For intenders, planning treatments are helpful to support those who lack the necessary skills to translate their intentions into behavior. And for actors, one needs to stabilize their newly adopted health behaviors by relapse prevention strategies.

The HAPA allows both the researcher and the practitioner to make a number of choices. Although it was initially inspired by distinguishing between a motivational and a volitional stage, and later expanded to the distinction between preintenders, intenders, and actors, one need not necessarily group individuals according to such stages. If the purpose is to predict behavior change, one would specify a mediator model that includes postintentional constructs (such as planning and volitional self-efficacy) as proximal predictors of performance.

For stage-tailored interventions, however, usually three stage groups would be established. This does not exclude the possibility of generating more than three stages. For example, for some research

questions, one might subdivide the preintenders into precontemplators and contemplators, according to the TTM. Or one might opt for a distinction between preintenders, who are (a) unaware of an issue, (b) aware but unengaged, or (c) still deciding. Thus, HAPA is not a puristic stage model, but a principle-based framework that allows for a variety of approaches. Evidence from intervention studies shows good results, in particular for patients with various chronic illnesses and disabilities. Planning interventions and action control programs proved especially successful.

Ralf Schwarzer

See also Adherence; Control Theory; Interventions for Exercise and Physical Activity; Self-Regulation; Social Cognitive Theory; Theory of Planned Behavior; Transtheoretical Model

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HEALTH BELIEF MODEL

The health belief model, grounded in John Atkinson's expectancy-value theory of achievement motivation, proposes that people are rational decision makers who, during decision making, take into consideration advantages and disadvantages associated with physical activity. The theory also posits that motivation is unidimensional and that the construct of intentions, which represents motivation, is one of the most immediate determinants of physical activity. Therefore, the health belief model is a *motivational theory* that explains intention formation. It does not explain the processes by which people carry out their previously formed intentions.

The health belief model proposes that an individual's readiness (intention) to engage in physical activities is a function of the perceived vulnerability to a health condition and the probable severity of that condition. Consistent with the expectancy-value model, the model posits that readiness is determined by a person's beliefs about the benefits to be gained by a particular behavior such as exercise weighted by one's perceived barriers

associated with physical activity. Finally, the model predicts that readiness to action may not result in physical activity unless some instigating event occurred to set the action process in motion. Irwin M. Rosenstock termed such instigating events as *cues to action*.

Overall, the health belief model predicts that strong intentions emerge when individuals feel vulnerable to an illness, the illness is perceived to be severe, and individuals believe that physical activity will reduce the health threat associated with the illness. For example, individuals may feel susceptible to cardiovascular disease because they have a poor diet and had been told by their physician that they have hypertension. They may also believe that regular exercise may reduce the threat of cardiovascular disease. According to the model, these perceptions are likely to motivate the individual to participate in physical activity.

Thus far research has shown that perceived severity and beliefs about the benefits of physical activity exert strong influences on readiness to engage in that behavior, while perceived severity and barriers have lesser impact. In addition, evidence suggests that the direct effects of perceived vulnerability, severity, susceptibility, benefits, and barriers on physical activity are small and are mediated by readiness. Further, there is evidence to suggest that the health belief model does not sufficiently capture all of the psychological determinants of physical activity and that the model may benefit from considering effects of other constructs such as self-efficacy on intentions and behavior.

One limitation of the health belief model is that it does not define perceived vulnerability clearly, nor does it specify how different variables combine in influencing readiness and physical activity. For example, it is unclear what type of vulnerability to disease should be measured. Shall we measure vulnerability to cardiovascular disease or vulnerability to back pain? In addition, vulnerability to cardiovascular disease may not predict the physical activity behavior of young individuals, given that getting a heart attack is a remote prospect for youth. Moreover, the model does not explicitly state whether perceived vulnerability would facilitate exercise or healthy dieting given that both behaviors would be effective in ameliorating cardiovascular disease risk. Hence, the model does not address behavioral choice. As a result, empirical evidence related to health belief model varies greatly across studies because different studies

have used different operational definitions for psychological constructs or populations.

An important function of research is to provide information about the content of interventions. Generally speaking, the greater the relative importance of a factor in predicting physical activity intentions, the more likely it is that changing that factor will influence intentions and ultimately physical activity behavior. Given that studies adopting the health belief model have shown that appraisals related to perceived vulnerability and perceived severity, and appraisals related to benefits and barriers influence intentions to exercise, it can be suggested that attempts to change exercise behavior should try to manipulate threat appraisals alongside perceived benefits and barriers. An important question, therefore, is how health appraisals can be influenced.

Threat appraisals can be manipulated through fear-arousing communications that highlight (a) the painful and debilitating effects of an illness (perceived severity) and (b) that people who do not exercise regularly are vulnerable to heart disease (perceived vulnerability). Physical activity benefits can be manipulated by exposing people to information that explains the effectiveness of exercise in preventing disease. The negative impact of perceived barriers on exercise can be circumvented by asking people to engage in types of physical activities that are relatively easy to perform or by prompting people to invent coping strategies that help them cope with barriers.

One caveat of interventions based on the health belief model is that although they may be successful in strengthening intentions, they may not always bring substantial changes in exercise behavior. Therefore, it cannot be expected verbatim that application of this model will produce strong effects on exercise behavior. Instead, the effectiveness of the health belief model in changing exercise behavior may be enhanced through the implementation of volitional techniques that can help people translate intentions into actions. Another limitation of the health belief model is that the threatening messages can sometimes undermine rather than enhance intentions. Generally speaking, people have a desire to protect or enhance sense of self. As a consequence, they may not easily accept and endorse health-threatening messages. Therefore, fear-arousing communications should be designed and applied with caution. Health messages should not be too threatening because

otherwise interventions will elicit a maladaptive coping response. For example, telling individuals that exercise reduces the risk of cardiovascular disease may be more easily accepted than telling individuals that they will have a heart attack if they do not exercise on a regular basis. It is therefore always desirable to pilot intervention strategies in a small group of people before conducting large-scale interventions.

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See also Health Action Process Approach; Social Cognitive Theory; Theory of Planned Behavior; Transtheoretical Model

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HEALTH PROMOTION

Health promotion has been defined in a number of different ways and used in reference to a constellation of behaviors, such as eating, exercise, and stress management. For the purposes of this entry, we consider *health promotion* to be defined as strategies that (1) facilitate awareness of the benefits of, and opportunities for, physical activity; (2) enhance exercise motivation and skills; and (3) intervene to make environments supportive for active living. Within this definition is the explicit acknowledgment that both individual- and environmental-level factors are related to health promotion. In this entry, we focus on health promotion strategies to increase exercise or physical activity that focus on individuals, groups, and specific settings.

Individual-Level Interventions

Strategies that are based on underlying theories appear to be more effective than strategies that are not. Social cognitive theories that include outcome expectations, control beliefs, and social influences are the typical underlying theories for individual-level physical activity promotion interventions. Using these models, participants develop goals and plans to achieve these goals. This personal action planning is essential to many of the individual-level interventions.

Personal Action Planning

Personal action planning can be useful in health promotion interventions targeting physical activity. Action plans allow the participants to focus on personally relevant motives as to why they should increase physical activity. For example, one person may list that engaging in an exercise routine will maintain or improve health while another may list the desire to stay young with one's children. These personally relevant motives are directly related to the outcome expectations a person may have

relative to a given behavior. Personal action plans also allow for the behavior changes to be goal directed, meaning a person may cite specific and measurable goals and activities to increase their physical activity. To overcome barriers that may decrease a person's self-efficacy, or the belief in ability or competence to perform a certain action, a person may brainstorm ways to overcome obstacles that may impede personal goals. For example, if one cites not enjoying exercise because it is boring, that person may want to try different activities at the local YMCA until finding something stimulating. A personal action plan process works well to assist participants in resolving these issues. It is also a key tool in a goal setting and feedback loop that can be initiated between a participant and a health professional.

There are a number of effective individual-level interventions that vary in the way they are delivered. It is clear that one-on-one counseling can be an effective method to increase physical activity and that the more frequent and longer duration of the intervention, the more likely it is to lead to behavior changes. One-on-one counseling has been done successfully using face-to-face methods, telephone methods—both live and automated—and, more recently through computer interactive sessions and mobile phones.

The ALIVE! study provides a good example of an individual-level intervention that includes action planning. The study used a computerized program to motivate participants to increase their physical activity as well as a number of other behaviors. The intervention included assessing participants' health behaviors and then encouraging them to set a goal in the area of greatest behavioral need (the behavior that they were currently not doing). Over the next three months, the program facilitated a series of small goals to provide participants with an opportunity to be successful before progressing to more challenging goals. During the program, participants received feedback on their accomplishments. The results of ALIVE! showed that people who participated in the intervention increased their physical activity more than those in a control group.

Group-Based Interventions

Group-based interventions are those that use a group-dynamics approach to promote physical activity. It is important to distinguish between

individual-level interventions that are delivered to an aggregate of people—like a group-education class—and a group-based intervention that includes strategies that consider the positive and negative forces that reside within groups and target those forces to enact behavior change.

For group-based interventions, one of the driving concepts is that group cohesion is key in the process of behavior change and that three general areas of influence can be used to facilitate group cohesion. First, the group environment, which includes the sense of distinctiveness participants hold within the group and the size of the group, is thought to influence group cohesion and adherence. It appears as though the ideal exercise class size is 5 to 17 members, and the most common method used to generate a sense of distinctiveness is to create a group name. Some use games to create member proximity to remove barriers that are typical when people exercise together. Second, the structure of a group, including norms, roles, and even location of members within group meetings, facilitate group cohesion. Group norms are often developed through the public sharing of individual behavior and goals. Roles can be developed formally to aid in participants feeling a stronger responsibility toward the group (e.g., attendance tracker, warm-up leader) and participants can be encouraged to exercise in the same space at each group meeting to give a sense of belonging and fit within the group. Third, specific group processes that included communication, cooperation, and competition in the pursuit of a common group goal can enhance group cohesion. The common goal can be an aggregate of the individual goals of people within the group so that they feel accountable to the group. A group goal is critical and perhaps the most important aspect of group dynamics strategies to promote physical activity.

When a physical activity promotion strategy includes components that focus on group environment, structure, and process, it will typically lead to large changes in behavior. In fact, when compared to individual interventions (with or without social support) group-based interventions lead to significantly larger changes. Interestingly, unlike individual-level interventions, group-based interventions seem to be effective regardless of the intensity and duration of the intervention program. Early research that focused on group-based approaches typically used interventions that included three to five sessions per week over a 3- to

6-month period. These interventions were successful in improving adherence, outcome expectations, and control beliefs as well as increasing exercise participation. These types of interventions also demonstrated improved maintenance of exercise after the sessions were completed.

At the other end of the spectrum, studies like Move More tested the degree to which group-based interventions could increase and maintain physical activity with just two, 2-hour sessions spaced a month apart with the focus of the second session on increasing exercise once the program was completed. Move More targeted insufficiently active adult patients, and the participants were randomly assigned to either a group-based intervention, which used group dynamic strategies, or an enhanced-standard care group, which used social cognitive theory and a self-directed approach (action planning workbook, information on resources in the community, a follow-up telephone call). The results showed that although both interventions succeeded in increasing physical activity over the first 3 months, the participants in the group-dynamics approach were able to sustain and further increase their physical activity 6 months after the intervention ended while those in the enhanced-standard care group did not.

Site-Specific Interventions

The next sections discuss health promotion efforts across a number of different sites. In each case, individual-level strategies, group-based strategies, or both could be applied. As you will note in the examples below, site-specific interventions also regularly use social (e.g., policy changes) and physical (e.g., development of walking paths) environmental strategies to support physical activity.

School-Based Interventions

Children spend much of their time in school, and school provides a great place to offer health promotion activities. As children get older, their time doing physical activity significantly declines. In the Healthy Youth Places Project, an intervention rooted in social cognitive theory, researchers wanted to increase the availability and acceptability of physical activity for middle-school students. The underlying proposition of the Healthy Youth Places Project was that engaging student, staff, and faculty leaders within the intervention would enhance the likelihood that intervention strategies

would be effective. Change teams made up of these key stakeholders received training on effective strategies to promote physical activity and implemented a number of different environmental strategies to improve student perceptions of connection, autonomy, skill building and healthy norms related to physical activity. The results of the Healthy Youth Places project demonstrated that students in the schools with change teams increased their physical activity over 3 years significantly more than those students in the control group.

Worksite Interventions

Like schools for children, worksites are the location that adults spend most of their waking hours. Worksite wellness has a long history because of the role of health in worker productivity and satisfaction. Worksites also provide a wide variety of avenues for intervention strategies (e.g., interpersonal relationships, group settings, organizational structure and policy). The Active for Life work-based physical activity intervention provides a good example of a typical approach for this setting. The intervention included changes to the physical environment in the form of posters and onsite health fairs and individually targeted materials in the form of regular newsletters, goal setting, self-monitoring, and incentives. It also included a group-based approach that was facilitated by site captains and included interoffice friendly competitions. The program reached about a third of the employees and lasted for 10 weeks. By the end of the intervention, the proportion of employees who were meeting the recommended guidelines for physical activity increased from 34% to 48%. However, 6 months following the intervention most of the employees had returned to a less active level. This lack of maintenance of physical activity change is common across health promotion strategies and provides an area in need of considerable research focus.

Health Care-Based Interventions

Medical clinics have the potential for broad reach into the U.S. population. Furthermore, clinicians are considered credible and objective sources of health information for patients. A number of researchers have attempted to take advantage of the credibility of physicians to promote physical activity. Unfortunately, it appears as though a

simple prescription for physical activity is insufficient to initiate physical activity change. Some researchers have attempted to increase the amount of time that physicians spend counseling patients on physical activity and have also used techniques such as motivational interviewing. While there has been some success with these approaches in efficacy trials, there is limited evidence that physicians have the time necessary to even insert a 2- to 3-minute counseling session for each patient that presents with a low level of physical activity. Where the most promise appears to be in regard to clinic-based physical activity promotion is at the intersection of health information technology and community resources. With the growing prevalence of the electronic health record and the push to include exercise as a vital sign, clinics have moved to the forefront as a location to identify people who could benefit from more physical activity. Still other researchers have shown that when physicians refer patients to a proactive community physical activity organization like the YMCA, the results are typically quite good in terms of increased physical activity.

Faith-Based Interventions

The rationale for intervening in faith-based institutions is not very different than some of the other place-based or settings-based interventions mentioned above. A large proportion of the American population attends a weekly faith-based service, and these services provide a great opportunity to promote physical activity. Just as with worksites, there are a number of different levels of potential intervention in a faith-based organization—sermons, socials, and service opportunities. Guide to Health provides a strong example of an effective intervention that was evaluated across 14 churches in southwest Virginia. The intervention prompted congregates to create weekly step-count goals, and when step counts were met, the future goals changed by an additional 500 steps. If a congregant failed to meet a weekly goal, helpful self-regulation strategies were used to help that person meet personal goals. The intervention was successful in initiating physical activity among congregation members.

Paul A. Estabrooks and Erin M. Smith

See also Ecological Theory; Group Characteristics; Group Formation; Social Cognitive Theory; Team Building

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HEDONIC THEORY

Hedonic theory, or theory of psychological hedonism, is the idea that human behavior is motivated by the pursuit of pleasure and the avoidance of *pain* (or, more accurately, displeasure). Its origins can be traced to the beginnings of Western philosophy. Although its prominence within psychology waned during the 20th century, updated versions of hedonic theory have emerged in behavioral economics and neurology. As researchers in exercise psychology have begun searching for postcognitivist explanations for variations in exercise behavior, hedonic theory has attracted attention as a perspective of considerable potential value.

History

Early expressions of hedonism can be found in Aristippus (435–366 BCE) and Epicurus (341–270 BCE), both of whom considered pleasure as the ultimate good. Aristippus emphasized physical pleasures, whereas the Epicureans promoted a holistic view of pleasure that included serenity, a

sense of belonging, and overall well-being. Holistic views of pleasure were also promoted by Plato (428–348 BCE), who believed in the balance of the rational, emotional, and appetitive parts of the soul, and Aristotle (384–322 BCE), who coined the term *eudaimonia* to signify the pleasure derived from a virtuous and fulfilling life.

Hedonism reemerged during the Renaissance, with thinkers struggling to align hedonistic ideas with the stern doctrine of the Church. Erasmus (1466–1536) and Thomas More (1478–1535) argued that the pursuit of pleasure is consistent with religion. René Descartes (1596–1650) accepted that passions, including pleasures, influence human behavior but maintained that the mind must control these passions in its pursuit of higher ideals.

In Britain, hedonism was at the core of debates on the appropriate goal of societies and political systems. Thomas Hobbes (1588–1679) and David Hume (1711–1776) accepted that humans are motivated to pursue pleasure and avoid pain. Hume further claimed that reason, which previous thinkers from Plato to Descartes considered capable of keeping passions under control, is powerless. French philosophers La Mettrie (1709–1751) and Helvétius (1715–1771) also endorsed hedonism, arguing that pursuing pleasure and avoiding pain are and should be the primary human motives.

Hedonistic ideas gained wide exposure in utilitarianism, a movement pioneered by Jeremy Bentham (1748–1832) and continued by James Mill (1773–1836) and his son John Stuart Mill (1806–1873). Utilitarianism was an ethical philosophy, according to which the goal should be the maximization of *utility*. In Bentham's hedonic calculus, the utility (usefulness) of each action is computed as the algebraic sum of the pleasure to be obtained minus the pain to be caused. The *fundamental axiom* of utilitarianism was that “the greatest happiness of the greatest number . . . is the measure of right and wrong.” In the opening lines of his book *Introduction to the Principles of Morals and Legislation*, Bentham famously wrote that “nature has placed mankind under the governance of two sovereign masters, pain and pleasure; it is for them alone to point out what we ought to do, as well as to determine what we shall do.”

The utilitarian pursuit of pleasure was criticized for encouraging unprincipled self-interest and a disregard for costs to others. Critics argued that this emphasis on pleasure abolished any distinction between humans and animals, characterizing

hedonism a doctrine “worthy only of swine.” This attack prompted John Stuart Mill to revise Bentham's earlier undifferentiated view of pleasure by distinguishing between *lower* and *higher* forms (e.g., education, art), the latter being of higher utility.

The ideas of the utilitarians greatly influenced the psychologists of the 19th and early 20th century. Herbert Spencer (1820–1903) emphasized the evolutionary advantage of hedonism, insisting that the only type of behavior that is conducive to life is the behavior that ensures a surplus of pleasure over pain. This evolutionary advantage was also highlighted by Alexander Bain (1818–1903), William James (1842–1910), and William McDougall (1871–1938). Their opinions diverged on the question of whether the pursuit of pleasure and the avoidance of pain are the only (the ultimate) human motives, with Bain supporting and James and McDougall rejecting this notion. The writings of Bentham also influenced Sigmund Freud's (1856–1939) *pleasure principle*.

Fragments of hedonic theory can even be found during behaviorism and cognitivism, periods during which psychology largely ignored subjective states, such as pleasure, focusing instead on overt behavior and cognitive appraisals, respectively. According to Edward L. Thorndike's (1874–1949) law of effect, when a behavior is paired with pleasure, it becomes more likely to be repeated, whereas, if it is paired with displeasure, it becomes more likely to be avoided. Similarly, for Albert Bandura (1925–), pleasant and unpleasant emotional states can have an influence on behavior, albeit not directly but rather by influencing self-efficacy.

Contemporary Iterations

References to hedonic theory in contemporary psychology are rare because, in the prevailing view, behavior and decision making are driven by the rational cognitive analysis of information. Among the exceptions have been emotion theorist Silvan Tomkins (1911–1991) and his students, who have considered affects, including pleasure and displeasure, as the prime human motives. Revivals of psychological hedonism have come mainly from disciplines outside of psychology proper. In behavioral economics, Nobel laureate Daniel Kahneman proposed that, because human rationality and cognitive ability are limited compared to

the complexity of the problems that humans face, decision making is aided by *heuristics*, including the *affect heuristic*, that is, people do what makes them feel better and avoid what makes them feel worse.

In neurology, Antonio Damasio, working with patients with focal brain lesions, showed that when areas involved in affective processing are damaged, one may be able to list the pros and cons of various behavioral options but has difficulty making decisions. According to Damasio, different behavioral options are associated with pleasant or unpleasant configurations of bodily state, called *somatic markers*, which influence the decision-making process.

Application in Exercise Psychology

Prompted by the limited variance in exercise behavior explained by cognitive variables, researchers have begun exploring the potential of hedonic theory. Data show that exercise enjoyment (which partly reflects pleasure from exercise), affective associations (pairing the idea of exercise with positive or negative descriptors), and ratings of pleasure–displeasure obtained during exercise are significant correlates and predictors of exercise behavior.

Panteleimon Ekkekakis

See also Affect; Affective Responses to Exercise; Enjoyment, as Mediator of Exercise Behavior Change; Pleasure

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HETEROSEXISM, HOMONEGATIVISM, AND TRANSPREJUDICE

Heterosexism, homonegativism, and transprejudice are prejudices aimed at lesbian, gay, bisexual, or transgender (LGBT) people. These beliefs and actions are common in sport and negatively impact all participants, regardless of their sexual orientation or gender identity. This entry discusses common types of prejudices faced by LGBT sport participants, defines related terminology, and notes the effects of biased sport climates.

Heterosexism is a belief system in which it is assumed or expected that all people are heterosexual. In a heterosexist environment, heterosexuality is viewed as the natural or normal sexual orientation. This belief creates the assumption that people are inherently *straight* (heterosexual), and that those individuals who do not meet this expectation are deviant. As a result, nonheterosexuality is considered strange or abnormal. Heterosexism forms a climate where heterosexuality is privileged or rewarded with high social status and respect not readily granted to nonheterosexuals.

In sport, heterosexism is common. The prevailing silence about the existence of LGBT athletes or coaches renders them invisible. For example, heterosexual privilege is provided when locker rooms are divided by sex. This separation is based on the assumption that through creating all-male and all-female spaces, all sexual innuendos, harassment, or relationships among teammates will be prevented since everyone is expected to be heterosexual. That people in sport tend to only imagine and recognize male athletes as having girlfriends or wives and female athletes solely in relationships with males shows how extensive heterosexism is. Nonheterosexuals participating in heterosexist sport climates may feel excluded from or invisible to their teammates, coaches, or other sport personnel. Heterosexism provides the foundation for climates perceived as unwelcoming for LGBT sport participants.

Homonegativism refers to purposeful negative stereotypes, prejudice, and discrimination toward nonheterosexuals, whereas *homophobia* is an irrational fear of LGBT people. Unlike homophobia, the term *homonegativism* signifies that the bias against nonheterosexuals often is rooted in personal belief systems and maintains social and political functions. Therefore, the term *homonegativism* more accurately addresses the social issues embedded in the discrimination that takes place in sport and connects it to other “isms” in sport (e.g., sexism, racism). Examples of homonegativism in sport include name calling, stereotyping athletes, physical violence, bullying, and benching or cutting LGBT athletes from teams. Additionally, coaches may be fired or overlooked for jobs if they are perceived as LGBT.

Often such negative treatment is based on stereotypes equating masculinity in females with being lesbian and femininity in males with being gay. The application of these stereotypes has serious implications, including causing some people to avoid or quit sport. LGBT athletes and coaches in heterosexist or homonegative sport settings often experience high levels of stress and monitor themselves to conceal their sexual identity. Other consequences of homonegativism in sport include a decline in self-confidence, increased anxiety, difficulty focusing, dejection, and decreased athletic performance.

Heterosexism and homonegativism can only continue to function as long as the gender binary exists in sport. This binary acknowledges only two categories: *male/masculine* and *female/feminine*. These categories are considered distinct and opposite of one another. This sorting system appears throughout sport. For example, youth sport leagues are divided by gender and some sport uniforms and rules change based on the gender of the participant. Boys are often demeaned through phrases like “You throw like a girl,” which further suggests that boys should throw differently (and better) than girls. Through sport, boys and girls learn how to speak and behave appropriately in relation to their gender. Those who do not conform to these heterosexist rules may encounter homonegative treatment.

One’s gender identity refers to the internal sense of one being a girl or boy, woman or man. For many individuals, gender identity aligns with their outward appearance. However, some people have an identity that is outside of the female–male

binary. For transgender individuals, gender identity is not consistent with their biological sex (anatomy), is fluid (changeable), or is perceived as something other than male or female. Intersex individuals are born with anatomical, hormonal, or genetic characteristics of both female and male bodies. Transsexual people intend to undergo or have completed hormone therapy or sex reassignment surgery so gender identity and sex are compatible. They go through a transition period in which they begin hormone therapy to make physical changes to their body while they live full-time consistent with their gender identity. During this time, their physical body is not consistent with binary gender conforming bodies. After surgery, and with sustained hormone therapy, transsexual athletes can compete in the gender category consistent with their reassigned sex. All of these *trans* people challenge the gender binary because, instead of following the belief that one is born into a prestructured gender classification system, these individuals move across the boundaries. Yet, there is no place for trans people in a heterosexist sport environment and they often face transprejudice (bias based on gender identity, also called *transphobia*). Most sport policies do not include trans and transitioning individuals.

Though the negative effects of heterosexism, homonegativism, and transprejudice persist, they do not describe all sport settings. Some organizations are working to create more inclusive sport climates that do not privilege one particular sexual orientation or gender identity. Many coaches also are working diligently to provide inclusive environments that allow all of their athletes to be successful in and out of the playing arena. More LGBT sportspeople are coming out, or letting others know of their nonconforming sexual orientation or gender identity than in previous decades. That more people are feeling comfortable to do so suggests that, at least in some contexts, the climate of sport is becoming more open and welcoming to diverse people.

Vikki Krane and Mallory Mann

See also Cultural Competence; Diversity; Gender; Multiculturalism

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HIERARCHICAL SELF

Researchers and practitioners have long believed that how people feel about and describe themselves can strongly influence motivated behavior in sport and exercise. Two key constructs studied in the sport and exercise psychology literature are self-esteem and self-concept. Since many people have argued that judgments about the self will influence the selection and maintenance of physical activity behavior, it is critical to distinguish between self-concept and self-esteem. *Self-esteem* or self-worth refers to how a person feels about oneself. It is a self-evaluation that provides a sense of how an individual assesses worth. *Self-concept* refers to self-perceptions of personal attributes such as skills, abilities, and physical characteristics.

Historically, researchers studied both self-esteem and self-concept from a very broad global perspective. Modern work, however, recognizes that self-esteem, and especially self-concept, is best viewed from a multidimensional and hierarchical perspective. Starting with the work of Richard J. Shavelson and colleagues in the 1970s, self-concept has been conceptualized as a multifaceted hierarchy consisting of the global self at the top, with specific domains (e.g. academic, social, and physical) and subdomains nested underneath. For example, one important domain of global self is the physical self, which can be further divided into several subdomains, such as flexibility, endurance or conditioning, sport skills, strength, and appearance. Each subdomain can be further differentiated into more discrete components. For example, physical flexibility can be further separated into facets such as leg flexibility, back flexibility, hip flexibility, and shoulder flexibility. Typically, subdomains of physical self-concept have moderate to strong interrelationships. Aspects of the physical self, especially body appearance, demonstrate

moderate correlations with perceptions of global self across the life span.

Although there is strong support for the multidimensional view of self-concept, support for the hierarchical perspective is less compelling. Work by Herb Marsh and others has found that relationships among some specific domains or subdomains are low or non-existent. Furthermore, global self-concept also has low correlations with specific self-concept domains or subdomains. A hierarchical model also implies either a top-down or bottom-up direction of causal flow. The bottom-up approach proposes that situation-specific and task-oriented experiences influence perceptions in specific subdomains, which in turn will influence domain specific self-concept and overall global self-concept. Researchers conducting interventions using this model would target a specific facet of physical self (e.g. leg strength) and would expect changes in the facet to effect perceptions of a specific physical subdomain (e.g. physical strength). Changes in the specific physical subdomain would then result in expected changes to global physical self, which would effect change in global self-concept.

The top-down model hypothesizes that changes in global self-esteem will influence lower order domains (e.g. physical self), which will impact specific subdomains and in turn would influence specific physical activity behaviors. Therefore, top-down interventions target global self-esteem as it is assumed that individuals with higher self-esteem would be more confident to engage in physical activity behaviors regardless of experience and perceived competence in that domain. A top-down model implies that changes in global self-concept (or self-esteem) should cause changes in lower order domains and influence subsequent situation specific experiences. Unfortunately, longitudinal and experimental research has found little evidence of either top-down or bottom-up causal flow between the domain level and global level.

Measurement of Hierarchical Self

Investigating the causal relationship between self-esteem, self-concept, and physical activity behaviors has been greatly enhanced by multidimensional models and subsequent measurement instruments. Many instruments have been developed, evaluated, and modified for various populations and age groups. Within sport and exercise research, there are a few instruments that have consistently

demonstrated strong measurement properties. Susan Harter and colleagues developed the Self-Perceptions Profiles for various age groups. These instruments were based on hierarchical models with global self-esteem and various domains (e.g., social, athletic, appearance) depending on the age group. Separate work by Kenneth Fox and Herb Marsh sought to capture the physical self in more detail. Kenneth R. Fox and Charles B. Corbin developed the Physical Self-Perception Profile, which consists of four subdomains of physical self-concept and one global domain of physical self-worth. The subdomains include sport competence, attractive body, physical strength, and physical conditioning. Marsh and colleagues developed the Physical Self-Description Questionnaire, which specifies physical self-concept as nine unique physical self-dimensions (strength, body fat, activity, endurance or fitness, sports competence, coordination, health, appearance, and flexibility) nested under general physical self-concept and global self-esteem. Marsh and colleagues also developed the Elite Athlete Self-Description Questionnaire to assess physical self-concept in elite athletes using six different dimensions. Elite athlete self-concept dimensions include skill, body, aerobic fitness, anaerobic fitness, mental competence, and overall performance.

Self-Esteem, Self-Concept, and Physical Activity Behavior

Researchers in sport and exercise psychology recognize the importance of self-esteem and self-concept as not only significant mental health outcomes, but also as potential determinants of physical activity behavior. There is solid evidence, in both youth and adults, that physical activity behavior is correlated with aspects of physical self-concept and sometimes with global self-concept and self-esteem. For example, many researchers have found that conditioning and endurance self-perceptions predict objective physical fitness and physical activity levels. Similarly, self-perceptions of body fat or physical appearance self-concept predict objective measures of body mass index or body composition. Physical self-concept measures are far superior compared to global self-concept or self-esteem measures in predicting physical activity behaviors. However, self-perceptions of body appearance and body fat are often unrelated to physical activity behaviors.

Many physical activity intervention studies in children, youth, and adult populations have primarily focused on global self-esteem or self-concept and physical activity. In studies with children and youth, the majority of these studies have found a modest short-term significant intervention effect. However, many of these studies have been criticized for being poor quality. In adults, John Spence, Kerry McGannon, and Pauline Poon conducted a meta-analysis of over 100 exercise intervention studies and reported that exercise intervention seemed to have a small but significant effect on either global self-concept or self-esteem. Overall, researchers suggest that physical activity interventions can produce small changes in global self. However, there is little consistent evidence about the characteristics of the physical activity intervention (intensity, duration, types) required to produce changes.

Physical activity intervention research targeting changes in the various levels of the hierarchical self is more limited. In one review of intervention studies in various populations, Kenneth Fox concluded that physical activity conferred significant improvements in some components of the physical self. Typically, the intervention effect was stronger for physical self-perceptions compared to global measures. More recent work in adults suggested that interventions have a stronger impact on physical self-concept compared to global self.

Based on a mix of findings, researchers seem to suggest that physical activity can cause changes in both global and physical self. Intervention studies often indicate a bottom up model in that physical activity will cause a change in self-concept or self-esteem. However, recent work by Herbert W. Marsh and Rhonda G. Craven and Marsh and other colleagues strongly suggests there is a reciprocal relationship between achievement behaviors and domain level self-concept. Using longitudinal designs and complex modeling, they found that self-concept and physical activity are both a cause and effect for each other. The reciprocal model suggests that previous physical activity behavior would influence physical self-concept, which would in turn impact physical activity behavior even after controlling for previous physical activity behavior. This reciprocal effect was found in several studies using various physical activity contexts, such as elite swimming, physical education, and extracurricular activities. The work of Marsh and colleagues in various populations and achievement

domains indicates that the reciprocal effect occurs primarily at the domain level. The use of the reciprocal model has implications for research and applied practice. For instance, it implies that interventions should be targeted at both the self-concept level and at specific physical skills or perceptions related to the targeted domain. Also, the level at which self-concept is measured needs to be considered, as reciprocal effects are stronger when the domain in self-concept is congruent with the achievement domain.

Gender and Cultural Differences

Researchers have consistently found that males have more positive global and physical self-perceptions than females in both youth and adults, although there are large variations within each gender. This parallels the findings in the physical activity literature. However, there is no consistent evidence of the relationship between self-concept and physical activity behaviors being different across genders. Some researchers have argued that gender stereotypes, cultural beliefs, and the dominance of traditional boys' sports in school systems and community organizations results in sports being more highly valued for males and thus more prominent in the development of their self-concept and self-esteem. There is, however, no reliable evidence that physical activity interventions are more effective in enhancing self-perceptions for either females or males.

Understanding cultural differences in the global and physical self-concept is challenging. Most studies comparing samples from various countries show similar results in terms of physical self-concepts being meaningfully related to physical activity behaviors and there is strong evidence that males have more positive self-perceptions across groups. However, most studies compared nationality (e.g., Turkey vs. United Kingdom) rather than assessing cultural values, such as filial piety, familism, collectivism, fatalism, and machismo. Overall, there are few quality longitudinal or intervention studies in this area to make any definitive statements.

Conclusion

Self-esteem and self-concept are best understood from a multidimensional perspective. Physical self-concept is an important dimension in understanding motivated behavior in sport and exercise

settings. Males tend to have more positive perceptions of physical self as well as higher levels of physical activity behaviors. Physical activity interventions have a stronger impact on enhancing the physical self-concept compared to the global self. There is growing evidence, however, that there is a reciprocal effect between physical self-concept and physical activity behavior over time. Cultural differences are not well-understood.

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See also Body Self-Esteem; Self-Appraisal/Assessment/Perception; Self-Awareness

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HISTORY OF EXERCISE PSYCHOLOGY

The history of exercise psychology is closely intertwined with the history of sport psychology. Although discussed in separate entries in this encyclopedia, the history of sport and exercise psychology can be considered as one history, with some differing paths and forks, but continual connections. That history has roots going back into classical times, as reflected in the classical ideal of *mens sana in corpore sano* (a sound mind in

a healthy body), highlighting the mind-body connection, which is the essence of sport and exercise psychology. Hippocrates, widely acknowledged as the father of Western medicine, clearly connected mind and body in citing the role of exercise in health. As Janet Buckworth, Rod Dishman, Patrick O'Connor, and Phillip Tomporowski note in their 2013 exercise psychology text, the major religions emphasized healthy lifestyles in the early writings. In the Hebrew Bible, one can find, "She girdeth her loins with strength and strengthened her arms," and 12th-century Jewish philosopher Maimonides cautioned that anyone who leads a sedentary life and does not exercise will have painful days.

We can find research on physical activity and health in the early physical education and psychology work, but as an identifiable academic area, sport and exercise psychology is relatively young, emerging as a subdiscipline within physical education in the 1960s. Sport psychology emerged in the 1970s, and exercise psychology began to develop in the 1980s. Still, from the classical times, through the early roots around 100 years ago, through the development over the last 50 years, to today's expansive academic discipline, sport psychology and exercise psychology have always been connected. This entry focuses on the history of the *exercise* psychology side of the disciplinary area that might better be termed *psychology of physical activity*.

W. Jack Rejeski and Lawrence R. Brawley were first to formally define and delimit exercise psychology in 1988 as "the application of the educational, scientific, and professional contributions of psychology to the promotion, explanation, maintenance, and enhancement of behaviors related to physical work capacity." A few years later, Rejeski and A. Thompson suggested a slightly broader scope by substituting *physical fitness* for physical work capacity. Today, most scholars who might adopt the exercise psychology label would take an even broader approach and substitute *health-related physical activity*. As the following historical review suggests, health-related physical activity could well describe the emphasis in the early roots stage of sport and exercise psychology 100 years ago. As the discipline developed, sport psychology narrowed to focus on competitive sport performance, and subsequently, exercise psychology split off to connect with exercise science and focus on physical fitness. It should be noted that the sport

psychology–exercise psychology separation is a North American development. International sport psychology has maintained the sport psychology title but gradually incorporated more exercise and fitness topics since the 1980s. With more recent rapid growth and expansion of research and practice, the sport psychology–exercise psychology categorization does not capture the rich diversity of topics, populations, outcomes, contexts, perspectives, and approaches within the psychology of physical activity today.

Early Roots: Late 1800s to Early 20th Century

Psychology and kinesiology (then physical education) both began to organize as academic disciplines over 100 years ago, and that early work includes evidence of the psychology of physical activity. Scholars who have described the history of sport and exercise psychology have noted the words of G. Stanley Hall, founding president of the American Psychological Association, who clearly connected physical activity with psychology in his 1908 statement in the *Proceedings of the National Education Association* that physical education makes "the intellect, feelings and will more vigorous, sane, supple and resourceful" (p. 1015). Similarly in 1899, William James wrote about the importance of the "well-trained and vigorous body" for a "well-trained and vigorous mind" and explicitly highlighted the importance for men and women alike. Hall's and James's statements portend some current exercise psychology topics, such as the role of exercise in mental health, and influence of exercise on cognitive performance.

Many sport and exercise psychology texts cite Norman Triplett's 1898 study of social influence and performance as a widely recognized early contribution. Triplett observed that cyclists seemed motivated to perform better with social influence (pacing machine, competition), and devised an experiment to test his ideas. Other early scholars from both psychology and physical education espoused psychological benefits of physical education and conducted isolated studies, including George W. Fitz of Harvard, who conducted experiments on the speed and accuracy of motor responses in the late 1800s.

Sport and Exercise Psychology: 1920s–1940s

The pioneering sport psychology work of the 1920s to the 1940s is discussed in more detail in the entry

“History of Sport Psychology.” Coleman Griffith was a prolific researcher who established a lab at the University of Illinois and published widely. Although Griffith’s applied work with athletes is most often cited, he actually was more concerned with developing the research and knowledge base. In his 1930 article, “A Lab for Research on Athletics,” he noted the abundance of anecdotal reports and explicitly called for a more scientific and experimental approach to psychological issues such as skill acquisition and the effects of emotions on performance. Griffith clearly connected sport and exercise. He closed the 1930 article with a list of 25 specific topics that might be investigated in his lab. Notably, that list included several topics that we might now call *exercise psychology*, although Griffith clearly did not make that distinction. The first topic listed was the relation between physical exercise and learning, and the list included effect of exercise on length of life and resistance to disease, the nature of sleep among athletes, photographic analysis of muscle coordination during fear, sex differences in motor skill tests, and effects of nicotine and other toxins on learning—to name just a few. Not only did Griffith merge topics that many today separate into sport and exercise, he included motor skills, coordination, and development and covered a range of topics that fit into the psychology of physical activity.

Around the same time that Griffith was working in the United States, Robert Werner Schulte in Germany and Avksenty Cezarevich Puni in Russia were developing sport psychology labs and active research programs. Although both clearly identified their pioneering work as sport psychology, like Griffith, their research often included topics such as exercise and memory that could be considered exercise psychology.

C. H. McCloy, one of the early scholars bringing research and a scientific perspective to physical education during Griffith’s time, explicitly called for an end to debates on *of* the physical versus *through* the physical, which reflect the old mind-body dualism. According to McCloy, mind and body cannot be separated. As McCloy understood, mind and body are connected, *of* and *through* the physical are connected, and sport and exercise are connected. The dualism of sport psychology versus exercise psychology is artificial and inaccurate. The full range of physical activities and related issues, including positive health, youth development, life skills, quality of life, and lifestyle

physical activity, belong in psychology of physical activity. In his own pioneering research, McCloy investigated character building through physical education as well as his many studies of motor skills and development, topics that fall within a psychology of physical activity.

From Griffith’s and McCloy’s time through the late 1960s when an identifiable specialization emerged, sustained programs were non-existent. After World War II, several scholars developed research programs in motor behavior that incorporated sport and exercise psychology topics, but research was sporadic.

Emergence of Sport and Exercise Psychology as an Academic Discipline: 1950s–1970s

Despite the innovative work during the first half of the 20th century, sport and exercise psychology did not emerge as an identifiable field until the late 1960s, when several individuals, typically in physical education departments, developed research programs, graduate courses, and eventually, specialized organizations and publications. Notably, these emerging programs and scholars were housed in physical education (now kinesiology).

The International Society of Sport Psychology (ISSP) formed and held the first International Congress of Sport Psychology in Rome in 1965. The ISSP, and international sport psychology, was more closely connected to applied psychology and performance enhancement than in North America, but exercise psychology can be found even in the early development stages. For example, the proceedings of the second ISSP congress in 1968 include several papers on *emotional health*, and another large section of papers on *the child and physical activity*. Specific papers included “Physical Activity Attitudes of Middle-Aged Males” by Dorothy V. Harris, and “Drug Therapy and Physical Performance in Emotionally Disturbed Children” by William P. Morgan, as well as several papers on motor behavior that do not clearly fall into sport or exercise psychology but fit within psychology of physical activity. The inclusion of exercise and physical activity, as well as motor behavior, is not unusual, and around the world, sport psychology is typically understood as including all forms of sport, exercise, and physical activity.

As international sport psychology was organizing, North American scholars also began to

organize, and the North American Society for the Psychology of Sport and Physical Activity (NASPSPA) was officially incorporated in 1967. The organization of NASPSPA reflected the overlapping of sport and exercise psychology and motor behavior in the 1960s and 1970s, with subareas of motor learning, motor development, and social psychology of physical activity (now the sport and exercise psychology area).

As graduate programs and organizations developed, research expanded and sport and exercise psychologists developed specialized publications. The *International Journal of Sport Psychology* began publishing in 1970. The *Journal of Sport Psychology* (JSP) appeared in 1979 (*Journal of Sport & Exercise Psychology*, since 1988) and was immediately recognized, as it is today, as the leading publication outlet for sport and exercise psychology research.

Development of Sport Psychology and Exercise Psychology: 1980s–2000

From the 1970s through the 1990s, sport and exercise psychology gradually became the largest and most diverse of the three areas within NASPSPA. Rainer Martens's 1975 text, *Social Psychology and Physical Activity*, reflects the content and orientation of those early years. Major psychological theories framed the content; most supporting research was from psychology; and the sport psychology work cited seldom involved *sport* (or exercise), but more often involved laboratory experiments with motor tasks. Martens also clearly described physical activity as an inclusive term, not limited to competitive sport, but encompassing varied forms of movement in a wide range of settings.

Separation of Sport Psychology and Exercise Psychology

Sport and exercise psychology from 1975 to 2000 was characterized by *narrowing* and *separating*. As noted, in North America the field began as social psychology and physical activity. Soon, the academic focus shifted, and in the 1970s and 1980s, the field became more sport specific. Sport psychology (exercise was not part of the label or the scope) began to narrow its focus and shifted away from social influence toward psychology of the individual. Before 1980, application largely meant physical education; but with the 1980s, it came to imply psychological skills training with

elite competitive athletes. Following a NASPSPA vote to include only research presentations at the conference, a group split off to form a new organization focused on *applied sport psychology*. October 1985 marked the beginning of the Association for the Advancement of Applied Sport Psychology, shortened to the Association for Applied Sport Psychology (AASP) in 2006. As summarized in the first issue of the *AAASP Newsletter* in 1986, "AAASP provides a forum for individuals interested in research, theory development, and application of psychological principles in sport and exercise." From the beginning, however, AASP meetings and publications have been dominated by the focus on competitive sport with little attention to exercise or physical activity. Meanwhile, NASPSPA maintained its research focus, including research on exercise and physical activity.

International sport psychology, reflected in ISSP and the European Federation of Sport Psychology (FEPSAC) which was formed in 1969, began with an emphasis on sport and competitive athletes, and thus did not experience such a dramatic narrowing and separation. Instead, international sport psychology maintained its emphasis on sport and continued to develop applied programs for athletes, often within psychology.

Emergence of Exercise Psychology

Just as some scholars chose to split off to focus on applied sport psychology, others (a smaller number) shifted to align more closely with exercise science and focus on psychology of exercise and fitness. Several exercise psychology sources refer to the fitness craze of the 1970s and 1980s as a key factor in the growing interest in exercise psychology. Activities such as aerobics, jogging, and weight training gained more popularity, and the fitness industry grew to meet that interest. NASPSPA and the JSP concentrated on strong research and maintained connections with exercise and motor behavior. In 1988, JSP added exercise to the title, becoming the *Journal of Sport & Exercise Psychology*, and more explicitly sought research on health-oriented exercise as well as sport.

The American Psychological Association (APA) organized an interest group, and in 1986, Division 47—on Exercise and Sport Psychology—became a formal division of APA. Although many in APA were interested in applying psychology in

competitive sport, it is notable that the group highlighted *exercise* in the division name. William P. Morgan, a key leader in the emerging exercise psychology research was the first president of Division 47.

Morgan, who spent most of his career at Wisconsin, might well be considered the father of exercise psychology. Certainly Morgan and his many students, particularly Rod Dishman, conducted much of the early research in the 1980s and 1990s and set the standard for continuing exercise psychology research. Like Morgan, most exercise psychology researchers were aligned with kinesiology programs, often collaborated with exercise physiology researchers, and adopted similar research approaches. For example, studies might investigate the influence of type, duration, and intensity of exercise on psychological outcomes such as mood state. On the other side of the exercise–psychology relationship, the role of psychological factors in exercise adherence was an early exercise psychology topic. That early research was connected with the medical model that dominated exercise science at the time. As exercise psychology research expanded, scholars brought in more psychological theories and turned to the *biopsychosocial* model that dominates health behavior research. However, the social aspect was largely ignored at that time, and remains understudied today.

Psychology of Physical Activity in the 21st Century

The psychology of physical activity continued to expand rapidly through the 1990s and into the 21st century, and in many ways became more diverse, for example, including exercisers as well as athletes and a larger settings and psychological perspectives. However, rather than encompassing diverse participants, settings, and issues, sport and exercise psychology researchers, organizations, and programs split into separate, and often intentionally separated, subareas with little connection to each other. Moreover, the separate factions tended to be elite with little connection to the traditional physical education base or wider physical activity contexts. The sport psychology faction focused on elite athletes. Meanwhile, the exercise psychology faction moved to exercise physiology labs with a focus on young, fit exercisers.

The editor who initiated the name change to *Journal of Sport & Exercise Psychology*, Diane

Gill, also coauthor of this entry, fully supports the exercise psychology movement but firmly believes the sport psychology–exercise psychology split is artificial and destructive. *Psychology of physical activity* is more encompassing and more appropriate for this area. Stuart Biddle and Reinhard Fuchs clearly took a psychology of physical activity approach despite the title of their 2009 chapter, “Exercise Psychology: A View From Europe.” After describing the expansion from sport to sport and exercise psychology in the 1980s, and citing the early definition of *exercise psychology* as referring to “behaviors related to physical work capacity,” they clearly defined the scope and major issues in the field as psychology of physical activity. They referred to physical activity throughout their excellent article, and the few times they cited exercise psychology, the term was in quotes.

Again, the separation of sport and exercise is not so dramatic at the international level. Rather than develop a separate exercise psychology, international sport psychology has gradually incorporated more research and issues related to exercise and physical activity, particularly in the 21st century. Notably, ISSP, the first and still dominant international organization, changed its official journal to the *International Journal of Sport and Exercise Psychology* in 2002 explicitly recognizing the expansion of sport psychology into the psychology of exercise and physical activity and health issues. As Gershon Tenenbaum and Dieter Hackfort (2003) noted in their opening editorial statement,

The *IJSEP* encourages researchers and practitioners from all the sub-disciplines of sport and exercise psychology to share their contributions and views through this outlet. It is through integration of the specific knowledge gained by scholars in different fields, and through sharing of different views that our domain will benefit. (p. 8)

Similarly, the European FEPSAC and newer (1989) Asian South Pacific Association of Sport Psychology have maintained the sport psychology title but increasingly incorporated exercise, physical activity, health research, and programs.

Today, the psychology of physical activity has matured and moved in many directions. Despite the multiple facets and factions in the 21st century, there are promising directions and many connections. During the 1980s and 1990s, attention shifted to competitive *sport* and applied work with

athletes. Attention has now shifted back to *exercise*, meeting the public concern for health and fitness with increased research and applied emphasis on physical activity and health promotion. Both sport psychology and exercise psychology have moved toward a broader psychology of physical activity. *Sport psychologists* investigate mental health and eating behaviors of participants, stress and injury, and adherence to rehabilitation exercise as well as sport performance. *Exercise psychologists* have expanded beyond fitness training to consider the relationship of physical activity to quality of life among older adults and clinical populations and the role of reducing sedentary behavior in health. Some researchers focus on exercise parameters and adopt methodologies of exercise science. Others adopt public health models and approaches with a wide range of target populations to investigate lifestyle physical activity and psychological outcomes ranging from cognitive performance to quality of life. In many ways, we have returned to the mind–body connection, albeit with a wider range of measures, models, and methods. Traces of our roots and development are evident, but today’s psychology of physical activity is multifaceted and diverse in research and practice and truly global.

Conclusion

The history of exercise psychology is inextricably intertwined with the history of sport psychology and best understood within a broader psychology of physical activity perspective. Although relatively young as an identifiable area, we can trace our roots back over 100 years with early connections between psychology and physical education. Sport and exercise psychology emerged in the late 1960’s and expanded rapidly during the 1970s, creating a knowledge base and specialized publications. During the 1980s, with sport psychology narrowly focusing on applied research and competitive sport, some scholars turned attention to the relationship between exercise and psychological factors. Exercise psychology has continued to grow and expand into new populations and contexts, and today is more appropriately considered as the psychology of physical activity. Today’s global psychology of physical activity is multifaceted and diverse in research and practice, while continuing to address mind–body connections and the role of physical activity in health and well-being.

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See also Adherence; Affective Responses to Exercise; History of Sport Psychology

Further Readings

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HISTORY OF SPORT PSYCHOLOGY

In many ways, the history of sport psychology mirrors the history of other longstanding disciplines, including psychology, physical education, and other kinesiology-related disciplines, and has been influenced by larger sociocultural trends for decades, for example, the growth of the Olympic movement, professionalization of sport, and women’s liberation. Progressing through a number of eras, sport psychology has grown into a dynamic and continually advancing field. Although there are many detailed written histories of sport psychology, the purpose here is to provide a brief overview. Specifically, this entry begins with a summary of major time periods and key contributors and concludes with general remarks on sport psychology and its development. Importantly, most historical accounts of the field are described from a North American or European white male perspective. However, recent efforts have aimed to recognize and share a more holistic history inclusive of diverse individuals all over the world. Attempts are thus made here to present these global contributions. Finally, the history of sport psychology

cannot truly be divorced from the history of exercise psychology. However, to highlight the historical accounts that uniquely shape these disciplines today, the history of exercise psychology is described in a separate entry of this encyclopedia.

Eras in the History of Sport Psychology

The history of sport psychology has often been organized into six key eras or time periods that mark the field's development. These eras serve as rough guidelines for retrospectively examining events that have shaped sport psychology today. The eras include (1) the prehistory of the field from antiquity to the early 1900s, (2) the development of sport psychology as a specialty in the 1920s and 1930s of the 20th century, (3) preparation for the discipline between the 1940s and 1960s, (4) the establishment of the academic discipline in the late 1960s and 1970s, (5) the science and practice of sport psychology between the late 1970s and 1990s, and (6) contemporary sport psychology over the last decade.

Era 1: Pre-History (Antiquity–Early 1900s)

While most accounts of the history of sport psychology start in the late 1800s, interest in the area can be traced back to ancient times. For example, with the start of the Olympic Games around 776 BCE, the ancient Greeks embraced the mind–body connection and discussed both physical and mental preparation of athletes. In fact, the ancient Greeks are said to be among the first to systematically explore athletic performance with much of the work paralleling modern-day study in the areas of sports medicine and sport psychology. For instance, comparisons have been made between the Greek tetrad system used to prepare athletes for competition (preparation, concentration, moderation, and relaxation) and the contemporary concept of periodization where training occurs in phases.

Fast forwarding to the Victorian era of the late 1800s in the United States and Great Britain, athletes, educators, journalists, and others demonstrated interest in the many topics studied in sport psychology today, such as the psychological characteristics of high-level athletes as well as cultural issues in the sporting world. By 1894, French physician Philippe Tissie and American psychologist Edward Scripture had published some of the first studies in the field. While Tissie studied psychological changes in endurance cyclists, Scripture examined reaction time in fencers and runners. Notably,

Scripture's work reflected efforts to establish a *new psychology* that focused on data collection and experimentation versus subjective opinion as well as an emphasis on applying scientific findings to the real world (e.g., enhancing athletic performance). Although less recognized, Harvard professor G. W. Fitz was also examining reaction time in athletes around the same time. In 1898, American psychologist Norman Triplett conducted the first known experiment to blend the principles of sport and social psychology. Examining the influence of others on cycling performance, Triplett's study contributed to the development of social facilitation theory often studied in contemporary sport and exercise settings. Researchers continued to explore these and related topics throughout the early 1900s; American psychologists Karl Lashley and John B. Watson conducted a series of studies on skill acquisition in archery.

Perhaps the person who demonstrated the most consistent interest in sport psychology during this era was the French founder of the modern Olympic movement, Pierre de Coubertin. A prolific writer, Coubertin wrote numerous articles relevant to topics studied in sport psychology today, such as the reason children participate in athletics, the importance of self-regulation, and the role of psychological factors in performance improvement. He was also the catalyst in organizing several Olympic Congresses, two of which focused on the psychological aspects of sport. Interested in the blend between body, character, and mind, Coubertin's efforts garnered publicity around the critical role of psychology in sporting activity and continued to influence the field's development well into the 1940s.

The contributions of this era involved several noteworthy psychologists, physical educators, and physicians whose work demonstrated interest in what is now referred to as the field of sport psychology. However, while the work of these scholars dabbled in the world of sports, none of these individuals were considered sport psychology specialists who made consistent contributions. Few concerted efforts were made to systematically study the area until the 1920s.

Era 2: The Development of Sport Psychology as a Specialty (1920s–1930s)

In the 1920s and 30s, professionals continued to show interest in the psychological aspects of sport

through periodic writing, research, and exploration. For example, baseball great Babe Ruth was brought to Columbia University in 1921 and psychologically tested to determine the reasons for his exceptional hitting skills. In 1926, American psychologist Walter Miles, his student B. C. Graves, and legendary football coach Pop Warner conducted an interesting study on the influence of signal calling on charging times among the offensive line players.

However, it is also in this era that individuals from around the world began to specialize in the area by developing more systematic lines of research, presentations, and publications that marked a more sustained interest in the psychological aspects of sport. In Charlottenburg, Germany, Robert Werner Schulte started one of the first sport psychology laboratories in 1920 at the Deutsche Hochschule für Leibesübungen where he wrote a book titled *Body and Mind in Sport* and continued his work until his untimely death in 1933. In Russia, notable scholars Piotr Antonovich Roudik and Avksenty Cezarevich Puni began work at the Physical Institutes of Culture in Moscow and Leningrad, respectively. Roudik conducted studies on perception, memory, attention, and imagination while Puni examined psychological preparation and the effects of competition on athletes.

Around the same time, Coleman Griffith was directing the Research in Athletics Laboratory at the University of Illinois. Griffith, who is recognized as the father of North American sport psychology, published approximately 25 studies on topics ranging from motor learning to personality and character. He also published two classic books, *Psychology of Coaching* (1926) and *Psychology of Athletics* (1928), and outlined the key functions of the *sport psychologist*. Griffith's laboratory closed during the Great Depression of the 1930s, but in 1938, he was hired by the Chicago Cubs to assist in improving the team's athletic performance. Unfortunately, Griffith's applied work is said to have been far from successful because of resistance from players and coaches.

The 1920s and 1930s were characterized by not only scholars who dabbled in sport psychology work, but also those across the globe who set up sport psychology laboratories and devoted significant portions of their career to studying the area. Interestingly, with the exception of some Russian scholars whose work continued through the 1960s, the research findings of these early pioneers

had little direct influence on the scientific advancement of the field. Griffith, for example, was ahead of his time but worked in isolation with few students to immediately build upon his work. Still, the emergence of sport psychology as a discipline was ready to begin.

Era 3: Preparation for the Discipline (1940s–1960s)

Scholars who trained future generations of students and professionals set the stage for the development of sport psychology as an academic discipline. As previously mentioned, Puni's work in Russia continued well into the 1960s and 70s and is still recognized today. The same is true in North America where Franklin Henry of the University of California at Berkeley established a psychology of physical activity program and trained physical educators. Upon earning their graduate degrees, these students initiated systematic lines of research across the country. While much of Henry's work focused on what would now be considered motor learning and control, some of his students examined social psychology topics such as athlete personality and the arousal–performance relationship. This era of sport psychology research was also likely influenced by sociocultural events, such as increasing emphasis on performance at the Olympic Games, the Cold War, and the space race, which spurred considerable interest in the development of science in many fields.

On the applied front, the work of David Tracy is noteworthy as he consulted with semipro baseball players as well as the St. Louis Browns' major league team. Tracy helped athletes improve their performance by teaching relaxation skills, using confidence building techniques, autosuggestion, and hypnosis. His work is crucial to this era in that it garnered a great deal of publicity for the practice of sport psychology. It was reported that the Browns' front office believed that if other industries used psychologists, then it would be logical for professional baseball to do so as well. This may have helped initiate a shift in the attitudes of sportspersons and administrators toward the usefulness of sport psychology practitioners.

In addition to Tracy, the work of female pioneer Dorothy Hazeltine Yates is also noteworthy. Best known for her consulting with university boxers, Yates emphasized the use of positive affirmations and relaxation to enhance performance. She also taught a psychology course for athletes

and aviators. Embracing the science–practitioner model, she went on to conduct an assessment of her interventions with boxers. A reflection of her work reveals that Yates made important contributions in an otherwise male-dominated field.

Era 4: Establishment of Academic Sport Psychology (Late 1960s–1970s)

The latter part of the 1960s was marked by a major occasion. The First World Congress of Sport Psychology was held in Rome, Italy, in 1965. While few delegates were considered *sport psychologists* given that the field was only emerging as a discipline, this event marked the beginning of worldwide interest and institutionalization of the field. It was here that the International Society of Sport Psychology (ISSP) was born. With Ferruccio Antonelli of Italy as its first president, the inaugural issue of the *International Journal of Sport Psychology* arrived just five years later in 1970. Serving as a model, ISSP inspired the development of several other professional organizations of sport psychology across the globe, including the North American Society for the Psychology of Sport and Physical Activity in 1966, the British Society of Sports Psychology in 1967, the French Society of Sport Psychology in 1967, the Canadian Society for Psychomotor Learning and Sport Psychology in 1969, and the German Association of Sport Psychology in 1969. Born out of disagreements within ISSP, the European Federation of Sport Psychology (FEPSAC) was founded in 1969 with female pioneer Ema Geron of Bulgaria as its first president. Both ISSP and FEPSAC remain prominent influences on the field today.

In the late 1960s and 70s, physical education as an academic discipline was firmly taking hold in the United States. Professors were asked to begin research programs in all the sport sciences, curriculums were revised to include more academic sport science coursework, and graduate programs were developed. Sport psychology, now considered distinct from motor learning and control, was a part of this change. Increased activity on the applied side of the field was also occurring during this era, albeit not without controversy. In 1966, clinical psychologists Tom Tutko and Bruce Ogilvie wrote a controversial book, *Problem Athletes and How to Handle Them*, and developed the Athletic Motivation Inventory, a personality assessment that was said to predict athletic success. The book was controversial because it suggested that athletes

were problematic and needed to be controlled, while some scholars felt that their personality test was based on questionable science. Despite the controversy, Ogilvie was active in working with elite athletes and teams and was seen as a role model for many young professionals with an interest in applied work. In fact, he has been called the father of applied sport psychology in North America.

Era 5: Bridging Science and Practice in Sport Psychology (Late 1970s–1990s)

As the name of the era implies, it was between the late 1970s and 1990s that sport psychology came of age as both a science and an area of professional practice. This era was characterized by increasing interest in the field with scientists devoting their entire careers to the field, and a growing number of practitioners working directly with athletes and coaches. For example, the U.S. Olympic Committee developed a sport psychology advisory board, hired its first resident sport psychologist, and sent its first sport psychologist to the Olympic Games. New academic journals, including the *Journal of Sport Psychology* (1979; known as the *Journal of Sport & Exercise Psychology* since 1988), *The Sport Psychologist* (1986), the *Journal of Applied Sport Psychology* (1989), and the *Korean Journal of Sport Psychology* (1989), were published. The development of organizations to meet the needs of a growing field continued. For example, the Japanese Society of Sport Psychology was developed in 1973 followed by the Korean Society of Sport Psychology in the 1980s. Following the dissolution of other related organizations, the British Association of Sport and Exercise Sciences was developed in 1984 with *The Sport and Exercise Scientist* becoming its official publication. The Association for Applied Sport Psychology (formerly the Association for the Advancement of Applied Sport Psychology) was established in 1986, established its Certified Consultant designation in 1991, and continues to be the largest applied sport psychology organization in the world. In 1986, the American Psychological Association formed a new division, Division 47, devoted specifically to exercise and sport psychology. Born out of the First National Congress of Sport Psychology in Barcelona, the Spanish Federation of Sport Psychology was founded in 1987. Shortly thereafter, the Australian Psychological Association was developed with sport and exercise psychology as

a specialization in 1988, followed by the creation of the Asian South Pacific Association of Sport Psychology in 1989.

Finally, women of this era were afforded more opportunities and entered the field in greater numbers thanks to the concerted efforts of several female pioneers, including Ema Geron, Dorothy Yates, Dorothy Harris, Jean Williams, Carole Oglesby, Tara Scanlan, Maureen Weiss, and Diane Gill, to name just a few. Recently, determined efforts have been made to write these female pioneers into the history of the field and recognize their many contributions.

Era 6: Contemporary Sport and Exercise Psychology (2000–Present)

Contemporary sport psychology is a firmly established discipline. The popularity of the field was evidenced by over 700 delegates from 70 countries attending the 2009 World Congress of Sport Psychology held in Morocco. A large number of universities now offer specializations at the graduate level with hundreds of research studies being conducted every year. Reflecting a shift in the field's focus toward the inclusion of both sport and exercise contexts, the *International Journal of Sport Psychology* changed its name to the *International Journal of Sport and Exercise Psychology* (IJSEP) in 2002. The British Psychological Society also developed a sport and exercise psychology division in 2004. Greater attention is being paid to the practice of sport psychology with the development of an applied journal, *The Journal of Sport Psychology in Action*, published by the Association for Applied Sport Psychology in 2010. As sport psychology becomes increasingly popular through media outlets and social networking, Olympic and professional athletes continue to work with sport psychology specialists, as do a number of developing and recreational athletes.

With the growth of both the science and practice of sport psychology, a number of major changes are occurring in the field today. For example, fueled by rising obesity rates and a decrease in physical activity in many Western countries, there has been an explosion of interest in health and exercise-related research in the last decade. Exercise motivation and adherence, the role of physical activity in mental health, and the psychology of athletic injuries have all been topics of considerable interest (see also the entry "History of Exercise Psychology"). Life skill development,

cultural issues, and diversity in the sport context have also become popular areas of inquiry. In addition to the topics studied, the methods employed are also expanding. An increase in qualitative research methods in particular is evidenced by the development of new journals solely devoted to this type of research like *Qualitative Research in Sport, Exercise, and Health*.

At the same time sport psychology has seen tremendous growth and advancement, several important challenges face the field. For example, although they are more widely accepted, qualitative research methods have been criticized for their subjective nature and have led to some debate over the best way of gaining knowledge. Economic support for higher education in many Western societies has declined and caused sport psychology laboratories to become more conscious of the need to secure external funding (e.g., grants, fellowships) to support their research efforts. Tension between researchers and practitioners of sport psychology continues, which has led various professional organizations to purposefully attempt to bridge the gap. Other concerns include the job outlook for many being trained in the field. While sport psychology has certainly grown, academic positions are somewhat limited, and few practitioners will land full-time positions working with athletes and teams. Finally, perhaps the greatest challenge is defining the educational training necessary to become a practitioner in sport psychology. Disagreement over the role of kinesiology versus counseling-based training has led to heated debates among professionals. Increasing attention paid to certifications, licensures, and ethical standards are intended to safeguard against those who may unethically practice sport psychology without the appropriate competencies. Addressing these challenges and embracing the aforementioned successes of the field will fuel the growth of sport psychology in the future.

Conclusion

While sport psychology emerged out of other disciplines, the field certainly has a longer and deeper history than has often been described. Recent efforts have unveiled the contributions of individuals who demonstrated interest in the psychological aspects of sport long before Coleman Griffith and brought to light a history characterized by diverse and global influences not previously recognized. Sport psychology is a worldwide phenomenon with

strong research and practice components that parallels the growth of other established disciplines, including psychology and physical education.

Daniel Gould and Dana K. Voelker

See also History of Exercise Psychology

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HOME ADVANTAGE

The association of being at home with feelings of increased physical comfort, safety, and psychological well-being are reflected in a wealth of popular expressions and sayings, such as *Home free*; *Home is where the heart is*; *East–west, home is best*; *Home sweet home*; *There’s no place like home*. Thus, it is hardly surprising that the term *home advantage* has been adopted in sport to represent two related phenomena—both of which are founded on the belief that a team’s own park, stadium, or venue is, indeed, a good place in which to compete.

One phenomenon pertains to (for lack of a better term) *competition location*. In most professional sports in North America, for example, teams play a full schedule of regular season games to determine final league standings. Those teams ranking higher in the final standings are awarded

one extra competition in their home venue in every 3-, 5-, or 7-game playoff series. That extra opportunity to compete at home is referred to as a *home advantage* (e.g., “New York will have the home advantage in its playoff series against Boston”). If the team happens to lose one of its extra competitions at home, it is said to have lost its home advantage. Although competition location can represent a home advantage, it does not always turn out to provide that advantage. For example, in the first round of the National Hockey League (NHL) playoffs in 2010, home teams won 22 games and lost 27 games; in 2011, they won 23 games and lost 26 games. So, certainly from the perspective of competition location, home teams in those 2 years were not able to capitalize on their home advantage.

A second phenomenon, flowing directly from the first, pertains to *probability for a successful outcome*. In almost every instance where large sets of data have been examined—for team and individual sports, for female and male competitors, for international competitions between nations, for athletes and teams from across the age and experience spectrum—the home competitors have had a superior winning percentage that is beyond chance (discussion of these findings in the section that follows).

During the 2011–2012 NHL season, ice-hockey fans closely followed the Detroit Red Wings as they obtained an exceptional 75.6% success rate at home. In their 41 home games, they won an NHL-high of 31, including a league-record 23 straight. This success rate is atypical, of course, but does serve to illustrate an exceptional instance of home advantage.

In this entry, discussion of the home advantage phenomenon is limited to results pertaining to an increased probability for a successful outcome. To this end, discussion is focused on the extent of the home advantage in team and individual sports over a variety of contexts and the explanations that have been advanced by fans, the media, athletes, and sport scientists to help explain its causes. Also, the implications of competing at home for the psychological states and behaviors of athletes and coaches are examined.

The Extent of the Home Advantage

In almost every sport examined, teams have better results when they compete at home. In professional

sport, for example, over a recent 5-year period, the winning percentage was 53.7% in baseball, 61.0% in English football (soccer), 54.6% in ice hockey, 58.2% in American football, and 61.0% in basketball.

In most sports examined, athletes competing individually also have superior results when they compete in their home territory. For example, in World Cup Alpine Skiing, skiers competing in their home country on average improved 16% from where they were seeded going into the race to where they actually finished. Interestingly, professional golf and tennis are the only two individual sports where a home advantage has not been found.

In terms of international competitions, there also seems to be evidence of a home advantage for host countries in both the Summer and Winter Olympic Games, as well as the Fédération Internationale de Football Association (International Federation of Association Football; FIFA) World Cup. In the case of the Winter Olympics, for example, host countries showed an average improvement of about four medals over the previous Olympiad. The only host country in the history of the Winter Olympics that failed to improve was Italy (in Torino in 2006); it won 11 medals compared to 13 in Salt Lake City in 2002.

In the case of the Summer Olympics, host nations show an average improvement of approximately five medals over their previous Olympiad. This respectable improvement, however, is dwarfed by China's performance in the 2008 Summer Olympics held in Beijing; China improved its medal count by 37 (for a total of 100 medals) from Athens in 2004.

The FIFA World Cup has also yielded results that seem to show that host nations benefit from competing at home. The first World Cup took place in Uruguay in 1930 and there have been 19 competitions every 4 years since; the most recent World Cup was in 2010 in South Africa. There were no tournaments in 1942 and 1946. The host nation has reached the semi-finals in 12 of 19 tournaments, the finals in 8 of 19 tournaments, and has won 6 out of 19 times. (Given that FIFA is now making an effort to provide a host opportunity to countries and regions that based on their world rankings have minimal chance of winning a medal [e.g., United States, 1994; South Africa, 2010], the overall results are impressive from a home advantage perspective.) The six host countries that have won the FIFA World Cup

include Uruguay (1930), Italy (1934), England (1966), West Germany (1974), Argentina (1978), and France (1998). The runners-up include Brazil (1950) and Sweden (1958). Finally, Chile (1962), Italy (1990), and Germany (2006) all finished third when they hosted.

Causes of Home Advantage: Popular Beliefs

The benefits that accrue to host teams have given rise to considerable discussion, speculation, and inquiries among fans, athletes, media, and coaches about *why*. What are the principal factors that underlie the home advantage? As might be expected, there is some overlap among the groups in the explanations advanced.

For example, crowd support was the first choice of fans in one survey and one of the top three choices advanced in another conducted with inter-collegiate athletes. Two other choices (endorsed by athletes) were familiarity with the home court and the elimination of the need to travel. The belief that greater familiarity with the nuances of the home venue provides home competitors with the major source of their advantage also was the top reason advanced by coaches.

After working for years as a sabermetrician in major league baseball (*sabermetrics* is the specialized study of baseball through objective statistics), Craig Wright, with the assistance of Tom House, a major league pitching coach, offered his appraisal. Wright and House estimated that 5% of the home advantage is due to a psychological lift from the crowd, 5% results from the advantage of batting last, 10% is due to familiarity with the stadium, 10% is due to the ability of the home team to select and use personnel best suited to its home stadium, 30% is due to a regime regularity, and 40% is due to an umpire bias that favors the home team.

Causes of Home Advantage: Empirical Analyses

Figure 1 is a framework Albert Carron, Todd M. Loughhead, and Steven R. Bray offered to systematically examine the home advantage. As a starting point, they proposed that the location of the competition (home vs. away) differentially influences four main factors—the degree of crowd support (and through crowd support, possible favorable officiating decisions), the need to travel, learned familiarity with the venue, and some rule advantages (e.g., batting last in baseball).

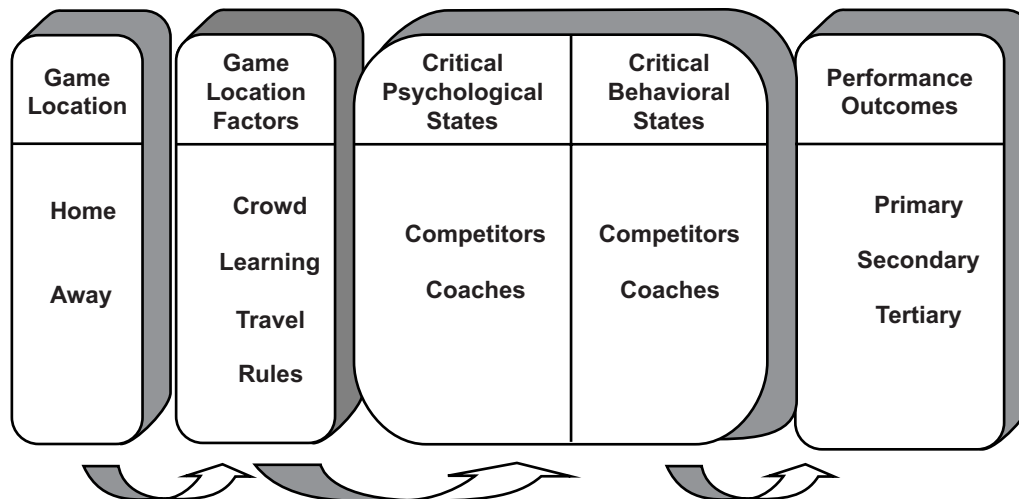


Figure 1 Conceptual Framework of the Home Advantage

Source: Adapted from Carron, A. V., Loughhead, T. M., & Bray, S. R. (2005). The home advantage in sport competitions: Courneya and Carron's (1992) conceptual framework a decade later. *Journal of Sports Sciences*, 23, 396.

Note: The term *officials* was originally presented in the Carron and Courneya (1992) model. Upon revision of the model in 2005, the term was removed from the boxes because, unlike coaches and athletes, officials do not have a home or away status.

Game Location Factors

As is the case with many areas of scientific inquiry, results from studies examining the nuances of each of the game location factors have been consistent in some cases but have shown mixed results in the case of other factors. For example, studies that have examined the influence of the rules factor have been consistent; the home team does not have an advantage from the rules.

Insofar as crowd support is concerned, the results have been mixed. At the risk of oversimplification, the following generalizations seem reasonable:

- Absolute crowd size is generally unrelated to the home advantage.
- Crowd density is consistently positively related to the home advantage.
- The nature of crowd behavior (i.e., booing versus cheering the home team) has no consistent influence on the home advantage.
- Laboratory studies have shown that the home crowd has an influence on officiating decisions (i.e., home teams receive more favorable calls). However, field studies and archival research have not supported these results. In well controlled studies, there is no evidence that crowd support produces more favorable officiating for the home team.

The need for a visiting team to travel to compete has also received a great deal of research attention. Again, the results are mixed. At the risk of oversimplification, the following generalizations seem reasonable:

- Distance traveled (e.g., a 120-mile trip to compete versus a 100-mile trip to compete) does not influence the visiting team's disadvantage (and, of course, the home team's advantage).
- The duration of the road trip does not influence the visiting team's disadvantage in professional basketball and baseball. In professional ice hockey, visiting teams are less successful in the initial games of a road trip.
- Travel across time zones can be a source of disadvantage for visiting teams. The adage "traveling west is best" does seem to have some validity. Professional teams traveling from western to eastern regions of North America are at a greater disadvantage than teams traveling from the east westward.

The final game location factor in the framework illustrated in Figure 1 is the home team's learned familiarity with its own venue. There are a number of elements that fall under this category; these can be categorized as either *stable* or *unstable*. The latter are elements in the home team's environment that can be manipulated to

one's own advantage. For example, anecdotal accounts in the media have reported a professional baseball team competing at home providing excess water to the base paths to reduce the speed advantage possessed by the visiting team. Another reported a visiting professional basketball coach's concern that the home team might overly inflate the balls to facilitate a higher dribble that would favor the preferences of its point guard.

Stable elements are idiosyncratic aspects of the home team's venue. The Green Monster in Boston's Fenway Park would be one example. Presumably Boston outfielders, as a result of their greater opportunities to practice and play in that environment, would be more familiar with caroms. As another example, professional ice-hockey teams competing on an ice surface at home that is smaller or larger than the league average could benefit from increased familiarity.

The following generalizations about the role that familiarity might play in the home advantage seem reasonable:

- Professional soccer teams with a playing surface larger or smaller than the league average have a greater home advantage.
- Professional baseball teams with artificial turf have a greater home advantage than those teams without artificial turf.
- Professional baseball, basketball, and ice-hockey teams moving to a new facility (thereby temporarily losing their superior knowledge of their own venue) experience a reduction in their home advantage. This result is moderated by team quality. Teams with a superior home advantage prior to relocation (a home advantage greater than 50%) experience a temporary significant reduction. Conversely, teams with an inferior home advantage prior to relocation (a home advantage less than 50%) have a temporary significant improvement in their home advantage.

As Figure 1 illustrates, the game location factors are thought to contribute to different critical psychological states for the home versus visiting athletes and coaches.

Critical Psychological and Physiological States

There is relatively consistent evidence that supports the conclusion that coach and athlete

psychological states are superior when playing at home. The generalizations that seem reasonable are as follows:

- Both athletes and coaches have greater personal confidence and confidence in their team prior to competitions at their home venue.
- Athlete emotions and mood states are superior at home. For example, cognitive and somatic anxiety, depression, tension, anger, and confusion are lower prior to a home competition.
- Athletes feel more vulnerable at competitions held away from home because they know they will have to deal with the taunting of away fans (commonly seen in basketball).

Critical Behavioral States

As Figure 1 illustrates, competing at home versus away is also thought to have a differential influence on the behaviors of home versus visiting athletes and coaches. A sense of *territoriality*, which refers to an animal's occupation and defense of a geographical area where it feeds, nests, and mates, has been used to explain the home advantage. Athletes do have higher levels of testosterone prior to home competitions. They are thought to compete more aggressively, expend more effort, and persist longer.

The studies that have been carried out comparing athlete and coach behaviors at home versus away contribute to the following generalizations:

- From a strategy and tactics point of view, coaches adopt more defensive tactics for away games and more aggressive strategies for home games.
- Home versus away teams do not differ in defensive behaviors, such as errors, shots blocked in basketball, or double plays in baseball, but home teams do exhibit more aggressive offensive behaviors like shots taken in ice hockey and basketball.
- Home and visiting teams do not differ in number of aggressive penalties, such as penalties that have intent to injure, as their critical component.

Studies have found trends that away teams seem to be penalized more often and home teams get away with more. In addition, star players seem to get away with more at home.

Conclusion

The aspect of group territoriality known as the *home advantage* has undoubtedly been one of the most examined phenomena in the sport context by coaches, athletes, researchers, administrators, and consultants alike. Generally speaking, the home advantage seems to share some consistency in prevalence across all sport types. Although it would be fair to say that while the home advantage is enjoyed throughout all sport, it is not necessarily enjoyed by all teams within those sports. Certain factors like team quality do moderate the effects of the home advantage. Additionally, it is likely fair to assume that questions going forward surrounding the home advantage would not be to determine *whether* such a phenomenon exists; as outlined in this entry, the pervasive evidence demonstrating a superior winning percentage for the home team goes beyond chance. The fact that the home advantage has also been well documented over the last 100 years should calm any uncertainties. Future directions on investigating the home advantage should maintain a primary focus on why this phenomenon continues.

The conceptual framework presented in this entry is intended as a useful guide to further investigation. No claim is made, however, that it encapsulates everything and all factors pertaining to the home advantage. What it does provide is a simple depiction of what is presumably a dynamic construct (in that it fluctuates) depending on a wide variety of variables and factors specific for each sport and each team.

Albert V. Carron and Kyle F. Paradis

See also Aggression; Competition; Group Characteristics

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HUMAN FACTORS

Human factors (HF) is a multidisciplinary area that aims to understand and support the interactions between a human user and other elements of a sociotechnical system. Because human factors research addresses psychological, social, biological, and other task-related parameters of interactions between humans or between a human and a technical system in the context of work and industrial production, such disciplines as psychology, robotics, industrial design, engineering, anthropometry, biology, and graphic design are incorporated. Sport psychology can contribute to the field as well because of its knowledge concerning performance acquisition and the development of psychological skills like routines, inner speech, or the regulation of stress; all are useful in designing an optimal work environment. The term *ergonomics* is often used interchangeably with human factors.

The goal of HF research is to develop human-oriented technology that adapts to the physical and psychological needs of the human operator. Because humans must interact with technical and production-oriented systems, the research and applied discipline *human factors science* (HFS) exists to optimize and maintain the health and well-being of humans in particular tasks and work environment constellations. Another related goal is to develop and maintain productivity in a particular interaction between the system and the human. Therefore, experts and practitioners in *human factors engineering* (HFE) work to design an optimal fit between the technical production system and the human action system. Dimensions of such an optimal fit between human and technical systems are, for instance, task- and person-related equipment (e.g., body-oriented shape and flexibility of

equipment), the presentation and change of information in accordance with human perceptual systems, and a working structure that considers the biological needs of humans.

From a person-related view, *human factors* are cognitive, motivational, perceptual, physical, or biomechanical properties of individuals and their behavior that may influence the interaction between the human and the technological system. Many approaches in HF research, therefore, explicitly address the structure and cognitive basis of human action and human motion as a main factor in the production process. For instance, research concerning attention in technological systems is based on a close interaction between psychologists and engineers. Psychologists use their knowledge about attention, perception, information processing, and memory to inform engineers, who design products based on these principles, for instance, to improve aviation safety or for constructing new cars.

There are many links between human factors science and sport psychology. For example, performance plays a central role in both areas. Therefore, both disciplines have some potential links to *action theory*. The action theory approach has a number of historical roots; one of which is a book on planning and the structure of behavior, which was written by American psychologists George A. Miller, Eugene Galanter, and Karl H. Pribram in 1960. This book broke away from behaviorist concepts and formulated preliminary ideas about the functional construction of action. Further roots can be found in Russian and German psychology. Thus far, the action theory approach has been formulated most elaborately for human factors by American researchers such as Donald A. Norman and German scientists like Winfried Hacker. Action theory addresses, for instance, the finding that the various elements of a behavior that can be observed externally are based on a deep hierarchical structure and are carried out in order to attain a specific action goal. Hence, activities in production processes or sport environments are always performed relative to a goal and are directed toward this goal. This gives all of the psychological processes and structures (emotions, representations, etc.) within a human-machine interaction or a particular sport setting an action-regulating function. Based on such a common background, it is possible for sport

psychology to inform human factors researchers about well-investigated elements of action and performance and vice versa. While human factors research topics like perception, cognition, attention, and motivation in human-machine interaction are well investigated, important topics often investigated in sport psychology like emotion are mostly ignored. In 2011, David W. Eccles, Paul Ward, Christopher Janelle, and other researchers addressed this discrepancy. They argued that the development of emotional self-regulatory skills in human operators could support system performance, explain risk-taking behavior in human-machine interaction, and may also influence motor planning and motor control in the context of technical systems.

In recent years, the expanding field of cognitive robotics has offered new opportunities to study the construction and functionality of human factors like cognitive representations, attention, and communication with technical platforms, and in doing so has changed some perspectives in human factors research. The long-term goal of newly developed human factors disciplines like *cognitive interaction technology* is to develop robots with unprecedented sensorimotor, emotional, and cognitive intelligence to assist human activities in the industrial production process. Interestingly, current robot technology has matured to the point of being able to approximate a reasonable spectrum of specialized perceptual, cognitive, and motor capabilities, allowing researchers to explore the bigger picture that is the architecture for the integration of these functions into robot action control. This provides the opportunity to fit existing human models of perception, representation, motor control, and decision making together with architectures generated for robot actions. Cognitive interaction technology research labs have produced impressive humanoid robots, robot musicians, dancing robots, robot arms, and brain-machine interfaces, among others, to study the organization and functioning of human action and human-machine interaction in more detail. Based on such platforms, HF researchers are not only addressing the attention of a human user as the guiding factor of human-machine interaction, but furthermore, they wish to create a shared action and a shared attention between humans and robots. This research aims to systematically investigate the principles needed to build artificial

cognitive systems based on the human archetype that can interact with a human in an intuitive way, including the acquisition of new skills by learning.

Thomas Schack

See also Attentional Focus; Biofeedback, Neurofeedback; Task Constraints

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HUMOR

In the year 2000, psychologists Martin Seligman and Mihaly Csikszentmihalyi were authors of an influential article proposing a new focus in the field of psychology: the positive psychology movement. Since that time, researchers have begun to explore the positive aspects or strengths of life such as satisfaction, optimism, happiness, and other positive emotions. Included in this repertoire of constructive attributes is the sense of humor. Humor can simply be described as the quality of being funny, laughable, amusing, or comic. However, how humor is utilized, that is sense of humor, is of main interest in exercise and sport psychology because of humor's ability to serve as a powerful, robust coping mechanism, cohesion enhancer, and communication technique. Humor has also shown to have positive benefits in many other domains, such as therapy, marriage, job satisfaction, education, leadership, and health. People who use humor are usually better able to successfully build relationships and tend to be more popular.

Over the years, several theories of humor have been proposed. Four of these theories are briefly mentioned here. The release theory proposed by

Sigmund Freud stated that humor is used as a coping mechanism to release sexual or aggressive tensions. Many of the other humor theories attempt to explain why humor is found in various circumstances. The incongruity theory states that humor is found in situations where a particular action or outcome is expected, yet an unexpected action or outcome occurs. In other words, the element of surprise plays a role in humorous events. The superiority theory of humor states that persons feel *sudden glory* over another when the other person is the target of a joke. Finally, the benign violation theory states that humor is found when norms or expected behaviors are violated (similar to incongruity theory) without any serious or threatening consequences.

There are numerous forms or types of humor—sarcastic, self-deprecating, hyperbolic, parodic, ironic, and situational—to name just a few. Each may have various effects on the sender and receiver. In other words, although many believe humor has several positive mental and physical benefits, when used differently humor may serve to increase stress, humiliate, or decrease motivation. Used in a positive manner, however, humor may be an effective cognitive strategy (particularly in dealing with stressful situations) and help people to reassess life situations by allowing them to see the problem more realistically—providing psychological distance from the situation. Also, humor may be categorized as reactive or productive. Reactive humor is the ability to accurately appreciate and respond to humorous events (i.e., understand, laugh). Productive humor represents the capacity to devise, create, or use humor in situations not initially seen as humorous.

The sense of humor may be considered to have trait and state components. Measurement instruments have been developed to measure these various aspects of humor, such as the Humor Perceptiveness Test, Coping Humor Scale, Humor Styles Questionnaire, Multidimensional Sense of Humor Scale, Sense of Humor Questionnaire 6, Sense of Humor Scale, and Situational Humor Response Questionnaire. These scales measure facets of the sense of humor: the ability to appreciate various types of humor, humor creativity, use of humor to cope, styles of humor, and the relationship of humor to certain personality characteristics.

As early as the 1960s, studies were conducted on the effects of humor on teaching. Humor has

been found to assist in information recall and is valued by both the teachers and students. Teachers who use appropriate humor usually receive higher teaching evaluations, are seen as more approachable, and are better able to establish rapport with students. Interestingly, although coaching is seen by many as similar to teaching, very few studies have been conducted on the use of humor by coaches. Also, there are very few studies of the use of humor in sport and exercise in general. The few studies found in coaching indicated head coaches who use appropriate humor were more liked by their players—particularly by female athletes. Also, coaches who use humor were found to be perceived as having higher abilities and evaluated more positively.

Future investigations of the effectiveness of humor as a coping mechanism in sport and exercise may provide valuable insight to this often discussed, yet not fully understood behavioral characteristic. How humor may be used by leaders and participants to assist in adhering to exercise programs would significantly impact the fitness and medical industries. Discussions of the use of humor by teammates and coaches to enhance team relationships, cohesion, enjoyment, or satisfaction, as well as for coping with sport circumstances, would be valuable additions to the literature.

Kevin L. Burke

See also Adherence; Coach–Athlete Relations; Coaching Efficacy; Cohesion; Coping; Enjoyment; Friendships/Peer Relations; Optimism

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HYPNOSIS

The term *hypnosis* is often shrouded in misconception, myth, and apprehension because most views about hypnosis are influenced by entertainment stage shows. These shows often highlight participants' engaging in strange and often embarrassing behaviors. However, hypnosis is consistently reported to be an effective and reliable technique in many domains (including medicine, dentistry, and psychology) for the treatment of a number of physical and psychological issues, including pain management and addictive behaviors. Hypnosis is a cognitive behavioral process using the influence of suggestion to bring about changes in thoughts, perceptions, feelings, memory, and behavior. Successful hypnosis is associated with situations in which (hypnotic) suggestions are more readily accepted and acted upon by the participants. Indeed, hypnosis is a psychological technique available to athletes and exercisers to facilitate the mindset for peak performance and regular exercise. The nature of hypnosis, hypnotic procedures, theories of hypnosis, and the applications of hypnosis to sport and exercise are outlined in the following sections.

Nature of Hypnosis

Hypnosis is a process that involves an interaction between the hypnotherapist and participant and typically combines the procedures of visualization, relaxation, and the presentation of hypnotic suggestions. Hypnosis is usually characterized by extreme relaxation, intense concentration, and an increased responsiveness to hypnotic suggestion. Suggestion is an important facet of hypnosis and refers to the issuing of verbal statements—that is, words and metaphors of how a person would like to think, feel, or behave—by the hypnotherapist to a participant. Suggestions are provided during hypnosis to alter perceptions, thoughts, feelings, sensations, and behavior. Further, post-hypnotic suggestions may be presented to participants during hypnosis that are intended to trigger responses affecting behavior during a normal waking state. For example, post-hypnotic suggestions may be presented to help a soccer player feel more calm, composed, and confident during training and competition situations.

Fundamental to hypnosis is determining the degree to which individuals are susceptible to

hypnosis. Therefore, many measures of hypnotic capacity and susceptibility have been developed and evidence demonstrates that from 10% to 15% of all individuals are highly responsive, 10% to 15% are almost completely unresponsive, and the remaining majority of individuals are able to respond to some but not all hypnotic procedures.

Hypnotic suggestions are considered effective when they are presented to participants when they are in or entering a trance. Participants in a trance may look as though they are asleep, but a hypnotic trance is different from sleep. Indeed a person in a trance is aware of their environment, is relaxed, will respond to suggestions, and can usually remember later what transpired in the trance. Trance allows access, via suggestions, to the unconscious mind that holds memories and feelings that are below the level of conscious awareness but exert an influence on behavior, thoughts, and emotions. The properties of trance typically include intense relaxation, time distortion, and an increased tolerance of discomfort and pain.

Hypnotic Procedures

Hypnosis usually includes four phases. Phase 1 is based around preparing the participant for hypnosis by educating them about hypnotic procedures. In Phase 2, a process of induction and deepening occurs that typically involves relaxation procedures to enhance participants' susceptibility to hypnotic suggestions. Phase 3 is where the therapy or change will happen. When a participant reaches a deep level of hypnosis, suggestions and also post-hypnotic suggestions are presented about the way the participant thinks and behaves when alert. Phase 4 of hypnosis is about bringing the participant out of hypnosis and alerting them to the surroundings. In addition, participants may then use self-hypnosis to further increase the acceptance and the effects of suggestions. To illustrate, athletes can use self-hypnosis procedures as part of a preperformance routine, further enabling them to acquire an optimal performance state.

Theories of Hypnosis

Generally, theories of hypnosis fall into one of two camps: state and nonstate. A *state* perspective presents the effects of hypnosis as a result of trance states, altered or divided consciousness, or

dissociation. For example, a trance state is posited to be responsible for increasing a participant's acceptance of hypnotic suggestions. In contrast, a *nonstate* perspective views the effects of hypnosis as a consequence of positive attitudes, motivations, and expectations held by both the hypnotherapist and participant. To illustrate, a hypnotherapist may create expectancy in participants that hypnosis will be effective for them and that they will have certain experiences and responses.

Applications of Hypnosis

The amount and breadth of sport and exercise psychology literature demonstrating the efficacy of hypnosis is currently somewhat scant. However, the research to date has revealed positive effects for hypnosis in increasing optimal performance states like flow and peak performance, augmenting the use of mental imagery, reducing precompetition anxiety, influencing perceptions of effort and physiological responses at rest and during treadmill running exercise, and enhancing athletes' self-confidence. For example, in a recent study using a controlled group design, data revealed the immediate and long-term effects of hypnosis interventions on soccer players' self-confidence and shooting performance. Future researchers should consider evaluating the efficacy of hypnosis on other important psychological factors necessary for optimal sport performance (e.g., concentration, anxiety, and motivation) using controlled group designs.

In sum, hypnosis is a potentially viable and effective strategy available to coaches, athletes, and exercisers for bringing about meaningful psychological and performance gains. To illustrate, during preparation for an important competition, hypnosis could be used to increase relaxation and concentration, therefore reducing muscle tension and distraction. Further, the presentation of positive suggestions prior to and during competition about the feelings and emotions that relate to performing successfully may increase the likelihood of an athlete's experiencing increased motivation and feelings of confidence as well as reducing anxiety.

Jamie Barker

See also Imagery; Interventions for Exercise and Physical Activity; Mindfulness; Psychological Skills; Relaxation; Self-Regulation

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ICEBERG PROFILE

The *iceberg profile* in sport is a visual representation of desirable emotional health status, characterized by low raw scores on the *tension*, *depression*, *anger*, *fatigue*, and *confusion* scales and above norms (the “water line”) on *vigor* as assessed by the Profile of Mood States (POMS). The iceberg profile as a metaphoric image has been employed for better understanding of emotion–performance relationships and well-being in competitive and high-level athletes. This entry examines the utility of the iceberg profile in description of interaction effects of multiple negative and positive mood states affecting preparation and athletic performance.

Iceberg Profile as a Metaphor

An iceberg (*ice mountain*, from Middle Dutch *ijsberg*; Norwegian *isberg*) is a large mass of ice floating in the sea. The iceberg image is often used metaphorically to represent the notion that only a very small amount (the tip) of information about a situation is available or visible whereas the real bulk of data is either unavailable or otherwise hidden. The principle gets its name from the fact that only about one tenth of an iceberg’s mass can be seen above the water’s surface, while about nine tenths of it is submerged and invisible. Another interesting feature of the iceberg profile, which should be emphasized, is its ability to identify the interactive effects of the components of different phenomena. That is why the

iceberg profile as an image, principle, or a model is popular in various contexts. The *iceberg diagram* has also been helpful to identify some of the crucial aspects and influences in management and organizational settings. The iceberg graphically demonstrates the idea of having both visible and invisible structures interact. It is also helpful for understanding of global issues and is often used in systems thinking. In this case, at the tip above the water are events happening in the world; below the waterline there are often patterns or the recurrence of events. Patterns are important to identify because they indicate that an event is not an isolated incident. Like the different levels of an iceberg, deep beneath the patterns are the underlying structures or root causes that create or drive those patterns.

Profile of Mood States-Based Iceberg Profile in Sport

William Morgan introduced the term *iceberg profile*, as a metaphor, in sport in the late 1970s; it was based on his systematic research and monitoring of overtraining and staleness in competitive and elite athletes across different sports. In his assessments, Morgan used the 65-item POMS and noticed that elite athletes and active individuals in general tend to score below the population average on the tension, depression, anger, fatigue, and confusion scales. Moreover, these individuals usually scored about one standard deviation above the population average on *vigor*. This profile has been called the iceberg profile because the resulting configuration resembled an iceberg. All five

negative mood states fell below the population average (*T*-score of 50), and one positive mood state was one standard deviation above the population mean (see Figure 1). (It has been observed that it was good luck that the developers of the POMS placed the vigor subscale *fortuitously* in the middle, or there would be no iceberg profile at all!)

The POMS yields five negative mood states measures, one positive mood state, and a global measure of mood. A *global score* is computed by adding five negative mood states (tension, depression, anger, fatigue, and confusion) and subtracting the one positive mood state (vigor). Since this computational procedure sometimes yields negative values a constant of 100 is added ($Ten + Dep + Ang + Fat + Con + 100 - Vig$). These values are employed as a baseline for individual athletes (and in groups) to estimate the dynamics of staleness during the season. If the athlete suffers from chronic fatigue and is unable complete a workout session, his POMS profile can become *inverted*. The inverse iceberg profile is characterized by a lower level of vigor and higher levels of tension, depression, anger, fatigue, and confusion than the average individual. This type of mood profile is associated with a poor state of physical and mental functioning.

The POMS has several response sets depending on the focus in the assessment of mood states. These sets include *statelike* (*right now, today, or prior to your last competition*) and *traitlike* (*generally, usually, typically during a week or a month*) *foci*. For instance, if the athlete is asked to respond to the question “How have you been feeling *during the past week, including today?*” rather than “How have you been feeling *today?*” then both statelike and traitlike aspects of emotional state are assessed.

Iceberg Profile and Mental Health Model

The POMS-based iceberg profile represents visually the *interaction of multiple mood states*, and it is grounded in Morgan’s mental health model (MHM). Briefly described, the MHM assumes that performance is inversely correlated with psychopathology. Thus, positive mental health is associated with high performance levels whereas mood disturbances are predicted to result in performance decrements. The basic premise of the model is that positive aspects of mental health should be associated with broadly defined success in sport.

There are several concerns with the application of POMS-based iceberg profiles for testing the

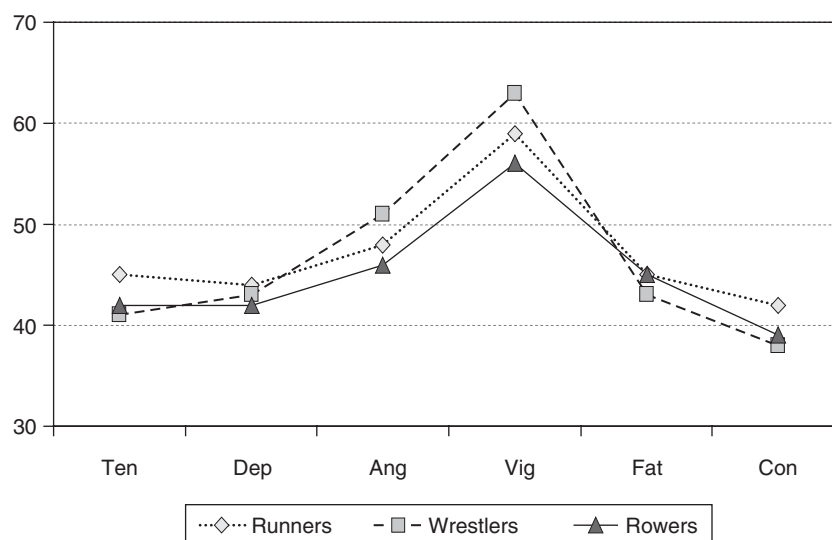


Figure 1 Profile of Mood States Iceberg Profiles in High-Level Runners, Wrestlers, and Rowers

Source: Adapted with permission from Morgan, W. P. (1985). Selected psychological factors limiting performance: A mental health model. In D. H. Clarke & H. M. Eckert (Eds.), *Limits of human performance* (pp. 70–80). Champaign, IL: Human Kinetics.

Note: Ten = tension. Dep = depression. Ang = anger. Vig = vigor. Fat = fatigue. Con = confusion.

validity of the MHM in prediction of athletic performance, especially at the individual level. First, POMS emotion descriptors are predominantly negative (five negative versus one positive mood state). Second, the researcher-generated mood descriptors are “fixed” and thus do not capture the idiosyncratic nature of athletes’ individual emotional experiences. Third, POMS-based iceberg profiles as measures of mood states are not related directly to preperformance and mid-performance situations. Neither are they focused on optimal and dysfunctional impact of mood states on performance. Fourth, population average (norms) rather than individually optimal and dysfunctional profiles serve as criteria to evaluate postperformance emotion impact. Finally, there is a need to examine the effectiveness of deliberately created iceberg profile based on other than POMS assessment measures—for instance, idiosyncratic emotion-centered and action-centered profiles.

Extension of Iceberg Profile Applications

The utility of POMS-based iceberg profiles can be enhanced by adding context-specific and relevant response sets. These response sets include *recalled* iceberg profiles of past emotional experiences (“How did you feel prior to your season best competition?”); *anticipatory* profiles (“How do you think you will feel prior to the upcoming competition?”); *performance related* (pre-, mid-, post-event) profiles; *interpersonal* profiles (“How do you feel while interacting with your coach or a teammate?”), and *intragroup* profiles (“How do you feel in this team and in your previous team?”).

Iceberg profile applications can also be extended by the assessment of emotions other than POMS mood states. For instance, does the iceberg principle work for other multiple emotion measures, such as CSAI-2 (with self-confidence, cognitive and somatic anxiety) or the individualized *emotion-centered profiling*? Some of these concerns were addressed empirically in the individual zone of optimal functioning (IZOF) model that predicts high probability of individually successful performance if the intensity of athlete’s optimal pleasant (P+) and optimal unpleasant (N+) emotions interacts with low intensity of dysfunctional unpleasant (N-) and dysfunctional pleasant (P-) emotions. Figure 2 depicts two individualized iceberg profiles based on the assessment of personally

relevant and performance-related idiosyncratic emotions. These iceberg profiles were constructed by deliberately placing optimal positively toned and negatively toned emotions in the middle and dysfunctional emotions (negative and positive) by their sides (N- > N+ and P+ > P-).

Individualized emotion-centered profiling captures the fact that one athlete may perform quite well when angry or anxious whereas another athlete with the same mood may perform poorly. That is why it is crucial to use task and personally relevant items (idiosyncratic emotion descriptors). In both cases, the predominance of functionally optimal positively toned and negatively toned emotions (P+N+) over the dysfunctional negatively toned and positively toned emotions (N-P-) serves as a predictor of high probability of individually successful performance (well-functioning). On the other hand, predominance of positives (well-being P+P-) over the negatives (ill-being N+N-) characterizes the athlete’s situational mental health status. A distinction between well-being (vs. ill-being) and well-functioning (vs. ill-functioning) is made by identifying both positive and negative emotions that have optimal and dysfunctional impact on mental health and athletic performance.

Variability of optimal intensity zones for each emotion descriptor and their interaction are identified during the repeated (recalled and current) assessments of emotion profiles prior to several successful performances in a single athlete. Prior to unsuccessful performance, the profiles are “flat” or positively skewed or negatively skewed. Consistent results suggest that there are two success-related and two failure-related iceberg profiles based on measures of multiple positively toned and negatively toned emotions. These four profiles are the summary of intraindividual emotion dynamics contrasting individually successful and unsuccessful performances (Figure 3).

Enhancing the Utility of Individualized Emotion Profiling

There are several ways to enhance the utility of the iceberg profile and its applications.

1. “Ideal” emotion iceberg profiles should be identified by multiple intraindividual profiling of five to seven personally successful (self-referenced) performances.

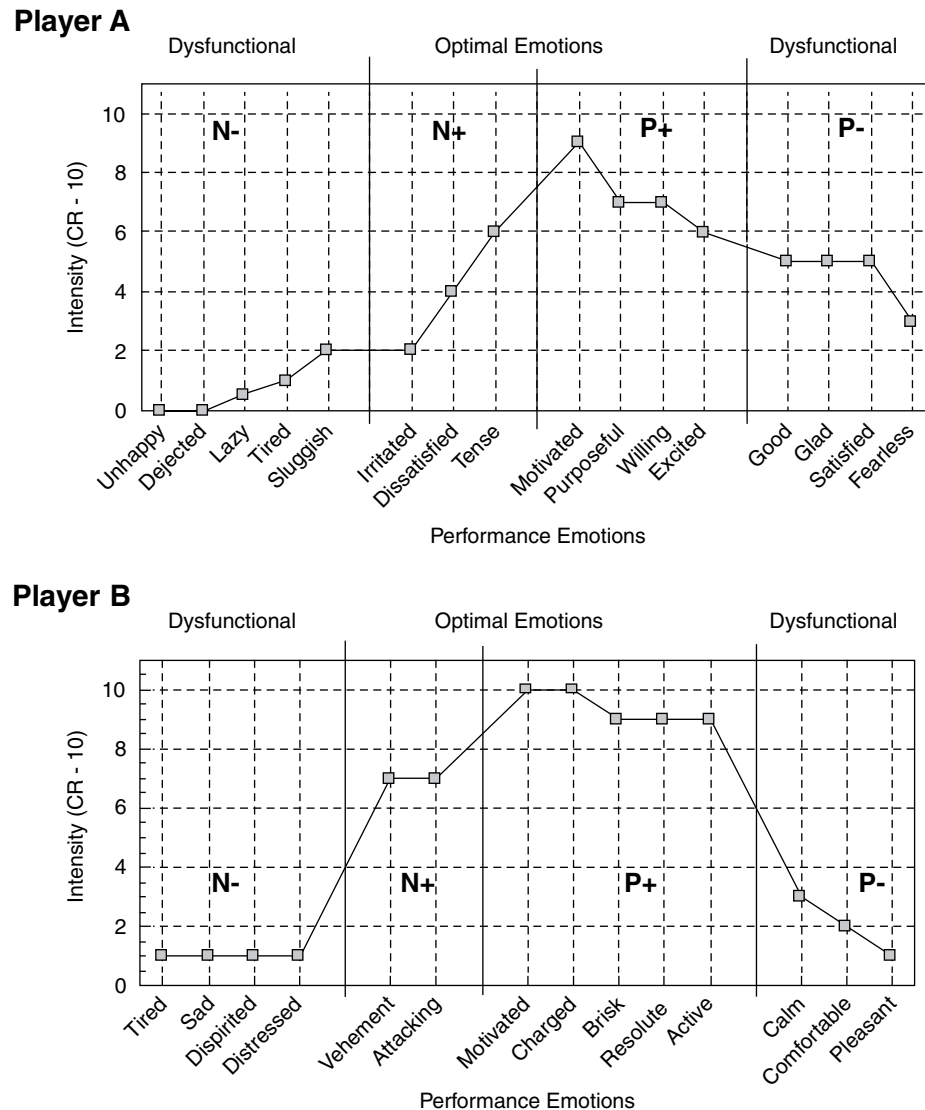


Figure 2 Individualized Pre-Game Zone of Optimal Functioning Emotion Profile of Soccer Players A and B

Source: Adapted from Hanin, Y. L. (2000). Soccer and emotions: Enhancing or impairing performance? In J. Bangsbo (Ed.), *Soccer and science—In an interdisciplinary perspective* (pp. 69–89). Munksgaard, Copenhagen, Denmark: Institute of Exercise and Sport Sciences, Copenhagen University.

2. Emotion descriptors should be idiosyncratic and self-generated by the athlete rather than fixed researcher-generated markers
3. To create iceberg profile, these multiple success-related factors should be placed in the middle; poor performance factors are located by the sides.
4. The main focus in research and applications should be on distinguishing intraindividually between the profiles related to success and poor performance.
5. Action-centered profiling is also recommended in considering successful and unsuccessful performances processes.

Conclusion

The findings suggest that iceberg profile predicting successful performance is possible to create by using idiosyncratic athlete-generated and aggregated researcher-generated emotion descriptors. Successful performances are predicted by two predominant types of profiles (P+) or (P+N+) that are similar for most athletes even if they have different emotion markers. On the other hand, inverse (“flat,” “cavity-shape,” N-skewed, or P-skewed) iceberg profiles predict high probability of individually poor performances. A future challenge in sport psychology (SP) is the development of

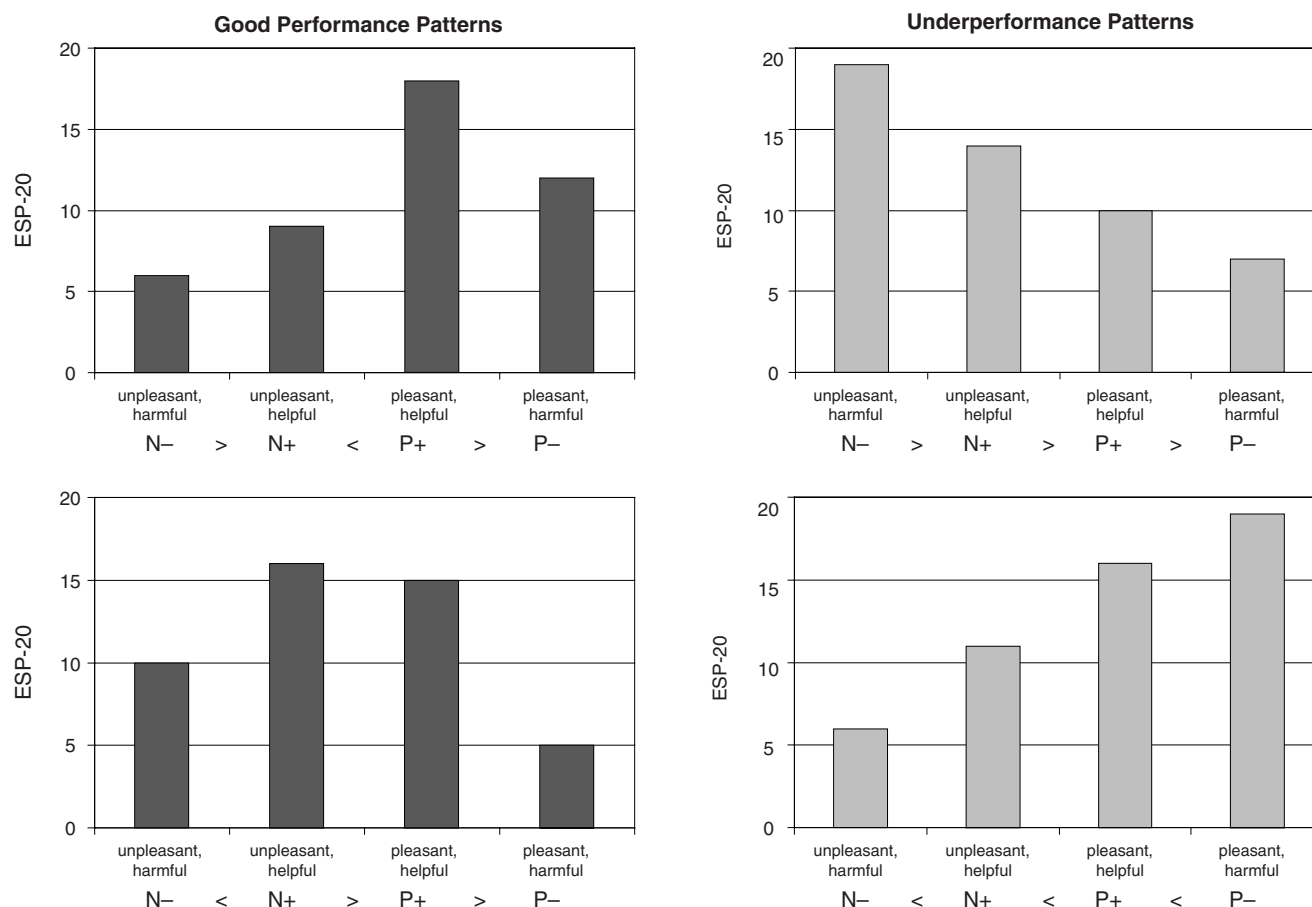


Figure 3 Individual Zone of Optimal Functioning Emotion Profiles Before Successful (“Iceberg”) and Poor (N-Skewed or P-Skewed) Performances

Source: Adapted from Hanin, Y. L. (2011, July 9). *Emotions in sport: Current issues and perspectives*. Paper presented at the invited symposium “Emotions in Sport,” 16th ECSS Congress Liverpool, UK.

individualized emotion scales capable of capturing interactive effects of emotions on athletic performance. Iceberg profile provides a partial solution to this challenging task.

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See also Affect; Basic Emotions in Sport; Emotional Responses; Individual Response Stereotype; Mental Health; Overtraining Syndrome; Psychological Well-Being; Stress Management

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IDENTITY

Exercise identity is a construct that captures the extent to which one sees exercise as a part of one's self-concept, or who one is. This self-perception has been related to exercise behavior and may be of interest to researchers and practitioners who are invested in understanding and promoting exercise adherence. Adhering to recommended levels of exercise has been reliably linked to the procurement of health benefits, yet the majority of the population remain insufficiently active to benefit from exercise. Consequently, researchers and practitioners alike pursue a variety of solutions to the widespread and complex problem of exercise non-adherence. As one avenue to address this problem, researchers have begun to investigate how self-perceptions, or how one views oneself, influence exercise behavior. This approach to understanding exercise behavior is consistent with the position advocated by social psychological theorists that self-views have implications for the motivation and execution of goal-directed behavior. One self-perception that has received increased research attention relative to exercise adherence is *identity*.

Identity and Identity Theory

Identity is a construct appreciated and studied in a wide range of contexts. For example, the term *identity* has been applied relative to, among other categories, cultures, nations, groups, and individuals. When applied with the goal of understanding how identity influences the self-regulation or management of goal-directed behavior, the conceptualization of identities as provided by Peter J. Burke is useful. Burke, an influential proponent of identity theory, views identities as role meanings, or what it means for an individual to hold a particular role in society. Socialization is thought to result in shared conceptualizations among individuals about what it means to hold a particular role in society that, in turn, prescribes acceptable behavior for a given role. Given shared social knowledge, most people are able to describe what it means to hold particular roles in society, such as exerciser (e.g., an exerciser is someone who

engages in regular exercise). Identification involves classifying oneself as holding that societal role and internalizing the meanings associated with that role (e.g., I am an exerciser, and exercisers engage in regular exercise. Engaging in regular exercise is a part of who I am).

Burke has suggested that identities function as self-regulating control systems that encourage identity-consistent behavior. Because the meanings associated with an identity are incorporated into one's sense of self, identities serve as a personally relevant standard for behavior; individuals are motivated to maintain consistency between their identity (e.g., I am an exerciser. Exercisers engage in regular exercise) and their behavior (e.g., I will engage in regular exercise). According to identity theory, when an individual perceives that his or her behavior is at odds with his or her identity, he or she should experience negative affect (e.g., dissatisfaction, guilt) that, in turn, motivates the individual to verify the identity through his or her behavior. As an example, an individual who perceives herself as an exerciser (and therefore someone who exercises regularly) who also perceives that her exercise identity has not been verified through her behavior (she has not exercised for over a week) should experience negative affect and resume exercise as a means of verifying the identity. When no discrepancy between identity and behavior is detected, identity verification should be in place and no adjustment in behavior is necessary. While an identity is theorized to serve as a self-regulatory control system for all individuals who identify with a behavior variations in identity strength or salience are thought to influence the effectiveness of the self-regulatory control system; identity theorists posit that individuals are more likely to behave consistent with an identity when it is strongly endorsed.

Exercise Identity and Exercise Behavior

Researchers and practitioners interested in understanding and promoting exercise adherence recognize the behavioral self-regulatory implications of the identity construct. Consequently, the relationship between exercise identity (or related variations of the construct such as physical activity [PA] identity) and exercise behavior has been examined. It has been well established that exercise identity is positively associated with a variety of exercise-related outcomes. For example, strength of exercise

identity has been associated with how many minutes and how frequently people exercise in a week, how hard people exercise (perceived exertion) and physiological outcomes of exercise including measures of cardiorespiratory fitness (e.g., VO_2 max), and anthropometric variables (e.g., percentage of body fat). While most of the research on exercise identity has been done on university and community samples, support for the ameliorative effect of identity on exercise behavior extends to older adults.

In addition to being associated with exercise behavior, exercise identity has also been associated with variables known to influence the self-regulation of exercise. For example, when compared to individuals with lower scores on exercise identity, individuals with higher scores also report holding their intentions for future exercise more strongly and exhibit higher scores of self-regulatory efficacy—or one's confidence in one's ability to manage (e.g., schedule, plan) exercise behavior. Further, when studied over time, exercise identity appears to serve as a mechanism through which exercise identity exerts its influence on exercise adherence. Specifically, having a strong exercise identity predicts self-regulatory efficacy for exercise that, in turn, predicts frequency of exercise behavior. Planning of exercise also appears to explain in part the relationship between exercise identity and exercise behavior; PA identity has been found to predict planning of exercise that then predicts individual perceptions of progress toward PA goals.

Researchers from the self-determination theory (SDT) literature also suggests that exercise identity may be related to variables that are known to be important in the self-regulation of exercise. SDT, a psychological theory conceptualized by Deci and Ryan, suggests that when individuals engage in a behavior for self-determined reasons, such as enjoyment or because doing so is in line with one's personal goals (as opposed to external reasons such as external pressure or pursuit of reward), positive behavioral outcomes such as adherence result. Stronger endorsement of exercise identity positively relates to engaging in exercise for self-determined reasons. According to SDT, exercise identity may encourage exercise adherence through its association with adaptive motives for exercise.

Further support for the idea that exercise identity promotes the successful self-regulation of exercise comes from research examining how

individuals respond when their exercise identity is challenged. Individuals who report high and those who report moderate levels of exercise identity who are asked to imagine that they have been much less active than usual both respond in a manner that suggests they would self-regulate to bring their exercise back in line with their identity. For example, individuals report that they would experience negative affect such as disappointment and guilt, would intend to increase their exercise, and would employ self-regulatory strategies to help get their exercise back on track if they found themselves in a situation where they had been much less active than usual. However, individuals who report higher levels of exercise identity have reported a *stronger* self-regulatory response to the scenario than individuals who report moderate levels. These findings hold regardless of whether the individual perceives the cause of the identity-inconsistent behavior to be within (e.g., poor time management) or outside (e.g., exercise facility unavailable) his or her personal control. Individuals react similarly to *real-life* perceptions that their recent exercise has not been consistent with their exercise identities. For example, strength of exercise identity moderates affective reactions to one's perceptions that recent behavior is at odds with one's exercise identity; negative affect increases as perceptions of identity-behavior inconsistency increases but this relationship is stronger for individuals with stronger exercise identity scores. It also appears that participants who identify as exercisers exhibit a self-regulatory response to identity-challenging feedback when feedback about exercise identity comes from *others* (e.g., others in a situation perceive that the participant is not an exerciser). Finally, among members of running groups, strength of exercise identity (specifically, runner identity) appears to promote adaptive responses to the possibility of the running group disbanding. For example, strength of runner identity has been associated with self-efficacy for running and less difficulty running in the face of group disbandment. Taken together, these findings are in accordance with identity theory; exercise identity appears to act as a self-regulatory control system that helps people adhere to identity-congruent behavior even in the face of challenges and the effectiveness of this system may be influenced by strength of exercise identity.

Based on what is known about exercise identity to date, exercise identity can be considered a

reliable correlate and predictor of exercise adherence. Further, exercise identity is associated with a variety of variables that are recognized for their ameliorative influence on exercise adherence. In the case of some of these adherence-related variables (e.g., self-efficacy, planning), exercise identity may exert its influence on exercise adherence through these variables. Together these findings support identity theory propositions regarding the role of exercise identity in promoting the successful self-regulation of exercise.

Practical Implications

Given that exercise identity has been established as a reliable correlate and predictor of exercise regulation and adherence, a logical and practical question to ask is how can we promote the strengthening or formation of exercise identities as a means for fostering exercise adherence? A first concern for the enterprise of building and strengthening identities is whether or not this construct is amenable to change. Proponents of identity theory such as Burke describe identities as being relatively stable and, by their very nature, resistant to change. Indeed, previous research has demonstrated that once established, people work hard to protect and confirm their exercise identities. Efforts at changing identities may be best targeted at situations where identity change is most probable such as when individuals are embarking on an exercise program and a few studies have demonstrated exercise identity change in these settings. These preliminary findings suggest that change in exercise identity is a possibility.

As reviewed previously, many correlates of exercise identity have been identified (e.g., self-efficacy, intentions). Deborah Kendzierski and her colleagues offered a formal model, the PA self-definition model, aimed at outlining factors associated with PA self-definition, a construct conceptually similar to exercise identity. This model demonstrates that when studied cross-sectionally, perceived commitment and ability relative to exercise directly relate to PA self-definition while enjoyment of exercise, perceived wanting, and trying to exercise are indirect determinants. This model makes a contribution by imposing some order on some of the many correlates of exercise identity and suggesting how they may work together in association with this construct. Further, this model holds potential as a basis for

intervention efforts aimed at increasing exercise identity.

Importantly, researchers are also beginning to determine which variables are associated with *change* in exercise identity. Preliminary efforts to this end employ both quantitative and qualitative research and suggest a number of variables are associated with change in exercise identity. These variables include perceptions of changes in skill mastery, physical changes in the body, progress toward exercise goals, perceived achievement, control, and belonging. Thus, research is starting to point to factors that may be important in strengthening or building exercise identities. Future intervention efforts that target these variables to bring about change in identity will help determine the best path to identity change. Exercise identity appears to be an important construct related to exercise adherence and further research exploring the role of this construct in the promotion of exercise, including intervention efforts, is likely and warranted.

Shaelyn M. Strachan

See also Self-Appraisal/Assessment/Perception; Self-Categorization Theory; Self-Construal; Self-Discrepancy; Self-Regulation; Self-Schema

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IMAGERY

Imagery involves internally experiencing a situation that mimics a real experience without experiencing the real thing. As a conscious process that is deliberately employed by an athlete or exerciser to serve a specific function, it is distinctly different from daydreaming or just thinking about something. The terms *mental rehearsal* and *visualization* are sometimes used to refer to imagery, but this can be misleading for two reasons. First, although imagery is a popular type of mental rehearsal, this term encompasses a variety of mental techniques athletes and exercisers employ such as observation and self-talk. Therefore, imagery and mental rehearsal are not synonymous, but imagery use does fall within the category of mental rehearsal. Second, the term *visualization* implies that imagery only contains a visual component. However, it is well known that mentally simulating an experience can involve multiple sensory modalities. As well as being able to see the scenario, imagery allows an individual to feel associated movements and bodily sensations, and experience the sounds, smells, and even tastes related to the actual situation. Consequently, imagery is the most appropriate term to describe this cognitive process.

Imagery is deliberately employed by athletes and exercisers to achieve a range of affective, cognitive, and behavioral outcomes. When used effectively, this technique results in better performance, both directly and indirectly via improvements to, among other things, motivation, confidence, and attentional focus. Moreover, the frequency of imagery use is a marker of success in sport as well as level of engagement in physical activity. It is well established that athletes competing at a higher level and more active exercisers report greater use of imagery. Consequently, imagery has emerged as a popular topic within sport and exercise psychology and is extensively researched. This entry summarizes key research findings including (a) the main imagery modalities and perspectives characterizing athletes' and exercisers' imagery use,

(b) the functions and outcomes this imagery use can serve, and (c) how imagery can be used most effectively.

Imagery Modalities and Perspectives

Although imagery can be experienced through different sensory modalities, within movement domains such as sport and exercise, the two most commonly used are visual and kinesthetic. The visual modality refers to what the individual sees in the image and is therefore commonly referred to as the mind's eye. Visual imagery can be performed from either a first-person perspective or a third-person perspective. In a first-person perspective, also referred to as an *internal visual imagery perspective*, the individual views the scenario through their own eyes as if they were performing the movement. An athlete who is imaging herself kicking a ball from this perspective may see the ball down on the ground, her feet running toward the ball, her foot making contact with the ball, and the ball rising up in front of her. In a third-person perspective, also referred to as an *external visual imagery perspective*, the individual views the movement as if they were adopting someone else's point of view to see the scenario. This can be done from different viewpoints or angles, with the most common being in front, behind, the side, and above. Returning to the ball-kicking example, if the athlete was to view herself from a third-person perspective, she may see her entire body performing the kicking movement.

Kinesthetic imagery refers to how it feels when experiencing the situation. Most commonly this internal sensation refers to the muscles associated with performing a movement. A runner may image how his legs feel while performing the running action. However, kinesthetic imagery can also encompass other bodily feelings including body limb positioning, tactile information (e.g., the feet make contact with the ground), physiological responses (e.g., an increase in heart rate, pain, fatigue), and emotions (e.g., excitement, anxiety).

Historically, there has been some confusion related to the concept of imagery perspective and the terminology used. Initially, the first-person-internal visual imagery and third-person-external visual imagery perspectives were known simply as internal imagery and external imagery. However, these terms and how they were defined conveyed

the impression that kinesthetic sensations could only be experienced using internal imagery. By confounding perspective with modality, this led to the assumption that *internal imagery* was the more effective way to image by providing the individual with a realistic and complete sensory experience (i.e., a movement could be both seen and felt). It has now been established that kinesthetic imagery can also accompany visual imagery performed in the third person. Furthermore, adopting the position of an observer appears to be particularly beneficial when imaging tasks with a focus on form or body positioning. This allows the individual to see information that otherwise would not be available from the first person perspective. If a figure skater is trying to improve her performance of a spiral, imaging the scenario from a third-person perspective is likely to help her better see the arch in her back, the height of her leg, and whether her toes are pointed. Alternatively, a first-person perspective is considered more beneficial when perception and timing are important to the skill being performed. A canoeist might view his slalom run from a first-person perspective to determine how to time the turns needed through different gates on the course. In other words, the benefits of imaging from a particular visual perspective will depend partly on the demands of the task being imaged and/or reasons for imaging.

Individual preferences also appear to matter, with some favoring a first-person perspective while others prefer a third-person perspective. It is likely that adopting the preferred visual perspective will make it easier to generate more vivid and controllable images, which in turn, would result in greater benefits from imaging. Individuals able to easily switch between perspectives would be able to maximize the benefits of imagery by appropriately matching the visual perspective to the task being imaged. Moreover, while the visual and kinesthetic senses can be used in isolation, combining these will create more effective images. During such multisensory imagery, individuals can experience both modalities simultaneously or switch their attention between what they are experiencing visually and kinesthetically to focus on a different modality at a particular time.

Imagery Use and Outcomes

Due to imagery's flexible nature, it is used at different times and in various locations. The most

frequent occurrences for athletes are just prior to competing or during training, but they will use imagery throughout the season including during the off-season. Exercisers similarly report using imagery during an exercise session but will commonly use it beforehand. Both types of individuals will typically image within the sport and exercise environment where the benefits of this technique are maximized; for example, it would be more effective for a swimmer to mentally rehearse her race start by adopting the appropriate position on the starting block at the swimming pool, compared with sitting on a chair at home. However, athletes and exercisers should still be encouraged to use imagery in other locations, particularly when injured, ill, or traveling. Indeed, imagery is reportedly used at home, work, and school and in rehabilitation sessions.

Similarly to when and where imagery is used, what is imaged may vary greatly between individuals. Common images range from skills and strategies to those involving thoughts and emotions as well as one's own appearance and health. This content can be very specific and short in duration (e.g., imaging the action of throwing a ball) or be more complex and/or longer by combining different types of content (e.g., imaging an entire gymnastics floor routine with the appropriate attentional focus and emotions associated with best performances). Alongside the duration and complexity of the image, this content can take on different characteristics including the sensory modality, visual perspective, viewing angle, agency (e.g., imaging oneself vs. imaging other individuals), and timing (e.g., slow motion vs. real time). For example, a basketball player might image his teammates and therefore hear them calling for the ball during the scenario.

When deliberately employed, the imagery content is usually intended to serve a particular function or functions. These are most typically categorized as being either cognitive or motivational that, in turn, are classified at specific or general levels resulting in five main functions: (1) cognitive specific, (2) cognitive general, (3) motivational specific, (4) motivational general-arousal, and (5) motivational general-mastery (see Table 1 for definitions and examples).

Although imagery is most frequently used for motivational rather than cognitive functions, individuals typically use imagery for all five functions. Furthermore, reasons for imaging are not only

Table 1 Cognitive and Motivational Imagery Functions

<i>Imagery Function</i>	<i>Description</i>	<i>Athlete Example</i>	<i>Exercise Example</i>
Cognitive Specific	To learn, improve, or maintain performance of specific skills	Improving performance of a tennis serve	Maintaining performance of a dumbbell curl
Cognitive General	To learn, improve, or maintain performance of strategies, game plans, or routines	Learning a new offensive play	Improving a strategy for a 5K run
Motivational Specific	To achieve specific process, performance, and outcome goals	Winning a race	Performing a set of 10 leg extensions
Motivational General-Arousal	To regulate emotions, mood, and arousal and anxiety	Reducing precompetition anxiety	Psyching up for a training session
Motivational General-Mastery	To modify cognitions	Improving confidence	Enhancing motivation to train

Source: Jennifer Cumming.

limited to these functions. Additional functions include those associated with injury rehabilitation (e.g., facilitate healing and pain management) and artistic endeavors (e.g., to choreograph a routine and understand how to interpret movements to music). The different imagery functions are intended to achieve various cognitive, affective, and behavioral outcomes. An exerciser might use imagery to improve his weight lifting technique and enhance his confidence to lift a particular weight for a number of reps. What is imaged to improve this technique and/or raise confidence will depend on both the individual and the situation. It is most common for individuals to closely match the content of their imagery to the function (e.g., image the skill for the purpose of improving that skill), but this is not always the case (e.g., image the skill for the purpose of reducing anxiety). Furthermore, imagery content can mean different things to different individuals. An image of winning a competition might be used by one athlete to motivate herself to train hard, but the same image might be used by another to maintain focus. The same imagery content can also be used in different situations to achieve varying functions. For example, a gymnast might image herself performing a routine correctly at practice to help

improve her performance but image the same scenario immediately prior to competition to improve her confidence.

Effective Imagery

When imagery is combined with physical practice it can lead to greater improvements in the performance of a skill compared with just physical practice. Additionally, imagery can maintain performance levels in the absence of physical practice such as when injured or unable to train. The most contemporary explanation behind imagery's effectiveness is based on the partial overlap of certain neural networks involved with the planning and execution of motor movements during both imagery and execution of a particular skill. More simply put, some areas of the brain that are active when a skill is imaged are also active when the skill is physically performed. Researchers have described this partial overlap as a functional equivalence existing between the two activities. These similarities have also led to the suggestion that imagery might serve to prime performance of a skill by enabling neural networks to activate more accurately or more readily during actual performance.

Other theoretical explanations to imagery's benefits also exist, including the psychoneuro-muscular theory, the symbolic learning theory, the bioinformational theory of emotional imagery, the triple code theory, dual coding theory, the action-language imagination (ALI) model, and the arousal or attentional set theory. Each theory has helped shape our understanding of imagery in different ways, but with the exception of the bioinformational theory, few of these have received empirical support.

Beyond merely improving the physical performance of skills and strategies, imagery also allows someone to rehearse and experience a scenario before it happens for real. This can help an individual mentally prepare for what they are likely to experience in the actual situation. Imagery can therefore allow the individual to anticipate what to expect. For example, a long jumper preparing for his first national championship might try to create a realistic preview of demands unique to this event by including in his imagery the presence of a loud crowd and distractions. By rehearsing how he will perform optimally in this situation, the athlete will likely feel prepared and more confident in his ability to cope when he actually experiences the situation for real.

A factor strongly influencing the effectiveness of imagery in serving its intended function is the individual's ability to image. Although everybody has the capacity to image to some extent, this ability will vary between individuals. For example, elite athletes tend to display a better imagery ability compared with lower level athletes. These differences can be reflected in a number of ways reflective of the imagery process, including how clear and vivid the image is, how realistic it is, and how well it can be modified and maintained once generated. Consequently there are a number of different methods used to assess imagery ability such as self-report questionnaires, computer tasks, and even brain imaging techniques. It is important to include some measure of imagery ability when conducting interventions because research demonstrates that individuals with higher levels of imagery ability experience greater benefits from imagery use compared with their lower-level counterparts who can experience few or sometimes no benefits. Moreover, although termed *ability*, an individual's capacity to create and control vivid and realistic images can be improved or enhanced with invested time and effort, similar to physical skills. It is

therefore thought that while some individuals may inherently find it easier to image compared with others, imagery can be honed with practice.

Making Imagery Effective

When imagery for skill learning and development is used to best effect, it will closely reflect the actual situation where the skill will take place. There are a variety of methods and techniques that can be used to improve imagery ability and maximize imagery's effectiveness. These methods can involve certain triggers or cues such as physical performance, sport or exercise equipment, observations and demonstrations, and imagery scripts to help prompt or guide the imagery scenario.

Physical performance can be done prior to imaging to remind an individual what it looks and feels like to perform a particular movement or action. This can help make the imagery more realistic. A soccer player might physically perform a penalty shot, then use this experience to help her image herself taking a penalty in the shoot-out of a championship tournament. As well, adopting physical characteristics associated with a scenario, actually incorporating certain pieces of equipment can facilitate more effective imagery. A javelin thrower who images himself throwing a personal best while holding a javelin and standing in the stance he would adopt before his run-up would likely prompt certain feelings and sensations associated with the situation thus creating a more vivid and realistic image.

Similarly to physical practice, observation clips and demonstrations can act as a template for the image and provide the individual with specific information or details about what the imaged movement or situation should look like. Unlike physical practice, however, this can be done for movements the athlete is not yet capable of performing. A novice golfer learning to perform a tee shot may use a coach's demonstration, either performed live or played back on video, to remind him of the positioning of his grip or head during the swing and follow-through. Additionally, these prompts may also help the imager work out some of the feelings associated with the image. The novice golfer may watch the smooth swing action of a demonstration to work out the feelings he or she should experience during the swing phase of the image. A few studies also suggest that observations can also be used when trying to image certain

situations. Showing a video of a group of athletes performing a ski run on a particular course may help athletes image themselves performing on the same course.

Imagery scripts are used by athletes and exercisers to help them keep more focused on the scenario they are imaging. These scripts usually provide details of what and how the individual should image and guides them through the experience. Although typically audio recorded, some individuals may prefer to have a written script. Regardless of the format, scripts provide a clear beginning, middle, and end to the scene and prompt the individual to focus on specific modalities (e.g., by referring to your heart beating faster or hearing your teammates call to you). This can be particularly helpful for individuals who find it difficult to control or focus their imagery appropriately.

Finally, personalizing the imagery can make it more emotively meaningful to the athlete or exerciser. Including the relevant emotions in an image allows the individual to be able to draw from memories of their own real experiences more readily. In turn, these memories help make the imagery more vivid, detailed, and realistic. If an athlete or exerciser is unable to relate to the scenario, they are much less likely to be able to image the scenario clearly, making it unlikely to be effective.

Jennifer Cumming and Sarah E. Williams

See also Mental Rehearsal; Mirror Neurons; Modeling; Multimodal Mental Training; Simulation Training

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IMPLICIT/SELF-THEORIES OF ABILITY

Implicit or self-theories of ability refer to individuals' views on the stability and changeability of personal attributes. Two lay theories are argued to exist: an entity theory, whereby individuals consider qualities and attributes of the self or others to be fixed and traitlike, and an incremental theory, whereby qualities and attributes are thought to be dynamic and changeable, and which can be developed. People can thus be described as entity or incremental theorists in relation to their beliefs about specific human qualities. In this entry, the pioneering work of Carol Dweck in the domain of intelligence is discussed. This body of work essentially examines how the two sets of beliefs lead to the adoption of different goals, which in turn, result in different outcomes in educational settings. This section will be followed by a presentation of key issues and empirical findings in sport and school physical education, demonstrating the theoretical and practical relevance of this area of research in sport and exercise psychology.

Research in the Education Domain: Beliefs About Intelligence

Interested in why children of similar ability respond to challenging tasks and achievement setbacks in a very different manner, Carol Dweck proposed that a mastery response (characterized by responding to challenge and setbacks with increased effort and determination to succeed) is underpinned by the pursuit of learning goals, which themselves are undergirded by the belief that intelligence can be improved. In contrast, the helpless response (characterized by responding to challenge and setbacks with negative affect and withdrawal from the task at hand) is underpinned by the pursuit of performance goals, which themselves are undergirded by the belief that one's level of intelligence is fixed. Learning goals refer to striving to develop competence in a self-referenced fashion whereas performance goals refer to striving to prove one's competence to others. Thus, beliefs and goals combine to create a

meaning system or personal framework in achievement settings such as education. Adopting an incremental view of intelligence sets in motion an adaptive (mastery) response in which children, *irrespective* of actual ability, try hard, value the giving of effort, and are more likely to achieve their potential. On the other hand, adopting an entity view of intelligence sets in motion an adaptive response *only* when children are confident in their ability. When doubts about competence exist, a maladaptive (helpless) response is predicted, as children are unlikely to believe in their ability to demonstrate normative competence, and thus reveal to others their lack of (fixed) intelligence. It is thought that individuals do not possess a generalized cognitive style; instead, domain-specific frameworks exist within the person. This idea has led researchers to consider incremental and entity perspectives in the physical domain and, in particular, the notion of implicit beliefs about athletic ability.

Research in the Physical Domain: Beliefs About Athletic Ability

Initial research in the area of implicit beliefs in the physical domain has centered on youth sport and school PE. Essentially, researchers were interested in whether the two theories could be identified with reference to *athletic ability*. Thus, the concept of intelligence was substituted with physical competence in sport and PE activities. In order to discover whether processes of goal adoption and resultant outcomes were evident in the manner found in the classroom setting, it was necessary to devise a reliable measure of self-theories of athletic ability. Early efforts in this regard identified three components underpinning both incremental and entity theories. In relation to the incremental view, children and adolescents responded to items tapping learning, improvement, and specific aspects of athletic ability; in relation to the entity view, participants responded to items assessing gift, stable, and general aspects of athletic ability. Subsequent research resulted in the subscales of general and specific components of ability being dropped. The Conception of the Nature of Athletic Ability Questionnaire–Version 2 (CNAAQ-2) conceptualizes and measures the two theories as higher-order constructs underpinned by notions that ability in PE and sport can be learned and improved upon through hard work (incremental),

and by notions that ability is a genetic gift and immutable (entity). Work in the physical domain implicitly adopts the perspective that individuals' self-systems are domain-specific and not uniform across broad life domains (e.g., relationships, morality, stereotyping).

Correlational Research

Findings from correlational research with young people in sport and PE have generally supported the theoretical and empirical work of Dweck and colleagues in education. First, psychometric work with the CNAAQ-2 has supported the existence of the two implicit theories. Second, these theories have been linked with achievement goals and important outcomes. More specifically, an entity belief has been shown to positively predict amotivation in PE and sport (a maladaptive form of motivation characterized by a lack of purposeful intent and behavior—e.g., “I don’t know why I’m doing this”). In contrast, an incremental belief has been shown to positively predict enjoyment. In both cases, these effects were partially mediated by performance and learning goals respectively. Work has also provided evidence of reliable links with affective responses, self-handicapping, and self-regulated learning in PE classes. For example, among Norwegian PE students, an entity conception of ability has been associated with increased anxiety and reduced satisfaction, regardless of perceived competence. In the same study, an incremental conception of ability predicted increased feelings of satisfaction among students.

Qualitative Research

In addition to correlational research with the CNAAQ-2, investigators in this area have undertaken qualitative forms of inquiry both to substantiate prior findings in regard to links with goals and outcomes but importantly to identify the socialization factors responsible for the development of theories within individuals. One study with low handicap golfers revealed the role of the coach in facilitating the view of golf ability as acquirable, the importance of high-profile role models such as Tiger Woods, learning about the game, and player’s own practice and development. Peers were viewed as potentially endorsing either belief. Exposure to talented and dedicated golfers helped to formulate the view that natural talent could be built upon. Unexpected development in

golf ability, as well as the notion that players with little all-round sporting ability could improve their golf, also underpinned the theme of developing natural attributes. Moreover, this study explored the idea of a ceiling effect (an upper limit at which the individual believes further improvement is not possible) and the potential specificity of golf beliefs (e.g., some aspects or components of the game may be seen as more fixed than others).

A second interview-based investigation has examined elite (Olympic standard) track and field athletes, specifically with respect to sprinters and throwers. As with golfers, the socialization of athletes' beliefs is complex, illustrating the value of conducting both quantitative and qualitative methods of inquiry in this area. In accordance with previous findings, athletes endorsed the importance of key social agents including family members and coaches in shaping beliefs. Interestingly, these studies have also identified psychological attributes, such as confidence and motivation, as components of achievement that are subject to entity and incremental perspectives.

One area of interest in sport, as in education, is how individuals respond to setbacks and the role played by implicit theories of ability in formulating a response. Setbacks have been found to revolve around injury, poor performance, and relationships. An incremental theory may help to buffer against setbacks and lead to a more optimistic view of the future—that is, that injury can be overcome, that performance can improved upon, that relationship issues can be resolved. However, more research is necessary to validate these notions.

Longitudinal Research: Are Young People's Implicit Theories Stable?

There has been some interest in whether and how implicit theories of athletic ability change as children transfer to secondary school. This work, although limited, has shown that both beliefs decline across the transition. Similar work has found evidence for a decline in athletic ability beliefs among older secondary school students. More longitudinal work is required—both to illuminate the direction and extent of changes in implicit beliefs that occur and to link these changes with important outcomes (e.g., goals, behavior) along with the factors that might explain instability in beliefs. One promising line of inquiry is to examine possible change in motivational climate

(the extent to which environmental cues in school PE classes or sports teams endorse either learning or performance goals) as a predictor of instability in incremental and entity theories.

Experimental Research: Can Ability Beliefs in Sport and Exercise Be Changed?

From an applied perspective, it is vitally important to investigate whether implicit theories themselves can be changed—that is, whether an incremental belief can be fostered in individuals. Studies have shown that school and college students' beliefs about athletic ability can be altered, at least temporarily. In experimental studies, beliefs are typically manipulated through participants reading a passage that espouses either an entity or incremental view of the ability concerned, such as golf putting ability or another novel exercise task. After conducting a manipulation check, links with cognitive, affective, and behavioral outcomes are examined. For example, attributions to ability following failure have been found to be higher among school students exposed to an entity theory manipulation than students exposed to the incremental condition. Researchers need to investigate alternative means of manipulating beliefs using video and Internet sources, as well as conduct longer-term interventions with PE classes, sports teams, and exercise groups. Research with adults in this regard is sorely lacking. Changing adults' perspectives on ability may prove more challenging compared with young people.

Conclusion

This entry has sought to discuss the concept of implicit or self-theories of ability in the physical domain. Building on original research in the education domain, it has been shown that incremental and entity perspectives of athletic ability can be identified and have important links with achievement-relevant outcomes. In addition, work in physical settings using multiple methods has sought to identify the antecedents of the beliefs, the role played by beliefs in dealing with setbacks or adversity, temporal characteristics of beliefs among young people, and the susceptibility of beliefs to experimental manipulation. More work in this area is warranted, not least because of the applied implications for coaches, exercise instructors, teachers, and, indeed, parents. Evidence to date suggests that identifying key socialization

influences, and in particular, finding ways to foster and cement incremental views of ability, could have positive consequences for motivation and behavior in the physical domain.

Christopher M. Spray

See also Achievement Goal Theory; Attribution Theory; Personality and Psychological Characteristics of Athletes; Social Cognitive Theory; Youth Sport, Participation Trends in

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IMPRESSION MANAGEMENT

See Self-Presentation

INDIVIDUAL RESPONSE STEREOTYPE

Research and application in sport and exercise psychology (SEP) has relied heavily on psychophysiological applications. Heart rate (HR), blood pressure (BP), skin conductance (SC), and measurement of brain activity through sophisticated techniques such as positron-emission tomography (PET), electroencephalography (EEG) and magnetoencephalography (MEG), and functional magnetic

resonance imaging (fMRI) are only a few of the many psychophysiological measures that have been successfully used to better understand psychological phenomena and clarify mechanisms in exercise and sport settings. For instance, studies examining the relationship between arousal and performance in sport have relied heavily on HR in order to classify individual performance at various levels of physiological activation or arousal. This area of research has led to the idea that individuals perform best when they are at an optimal level of arousal, not too over- and not too under-aroused. Physiological measures have also been employed in exercise psychology to better understand the effects of exercise on a wide variety of psychological and mental health outcomes including mood, anxiety, depression, cognitive function, and stress reactivity. In all of these studies, subtle changes in sensitive physiological measures are used to infer changes in readiness to perform or psychological state. These subtle physiological responses are believed to be a function of the demands of a situation or particular stimuli, an individual’s predisposition to respond in a specific fashion, and the interaction between the two. Individual response stereotypy (IRS), or the tendency of individuals to evidence particular physiological response patterns from one condition or situation to another, is an important consideration in all studies or applications incorporating psychophysiological measures. The construct of IRS, also referred to as autonomic response patterning, has led to a growing recognition that the psychological and health significance of certain psychophysiological states are better reflected by the overall pattern of activity across multiple measures of physiological state, rather than from an acute response or sensitive change across only one or a select group of response domains, such as fluctuations in HR or SC. Complex patterns of physiological responses, which are determined by multiple factors, are thus best studied by incorporating multiple measures of physiological responding.

According to IRS, some individuals will respond to certain situations or conditions (e.g., anticipation of an upcoming event, performance of a difficult cognitive task, drawing of blood from the finger, rest) with the greatest degree of activity in the same physiological measure, regardless of the situation or stressor. This is particularly relevant for studies or applications that rely only on a few or unitary situations and response measures. As an

example, in the area of stress reactivity, researchers have been interested in determining whether exercise or fitness are associated with blunted or altered cardiovascular responses to stress. When assessing BP responses to stress following acute aerobic exercise, it is important to determine not only baseline BP values (in order to assess change in stress-related BP following exercise) but also any individual difference variables that might influence BP responding. Individuals with borderline hypertension or essential hypertension tend to respond to stressful situations with greater increases in BP, particularly when compared to normotensive individuals. Assessing hypertension status would therefore be important. It is also important to note that individuals differ widely in both resting (tonic) and task-related (phasic) physiological responses. Understanding the factors that predispose some individuals to respond in unique ways helps inform the design of research studies as well as understand for whom and under what situations one person may perform optimally or realize the greatest benefits from exercise. This individual, unique pattern of responding has led to IRS being conceptualized as a physiological trait, or an enduring disposition of the body to show specific elicited physiological states or responses.

Although not all studies are in agreement, the notion of IRS points to several important implications. First, it is important to realize that individuals do not vary haphazardly in their physiological reactivity in a given response measure, nor do they vary haphazardly in their pattern of reactivity. Rather, some individuals may respond with maximal physiological activation in the same response measure or to a particular stimulus while another may be considered a low responder or nonresponder. Also, a strong tendency exists for a unique pattern of physiological response to be reproduced from situation to situation. This is so even though the physiological and psychological demands of the different situations may be unrelated. Second, IRS has resulted in a better understanding of the specific situational factors, familiarity with the situation, and range of behavioral choices that contribute to the expression of physiological responses. These factors can be used to strengthen future experimental designs and justify caution in oversimplifying physiological measures. Future studies should aim to elucidate the specific determinants of these unique patterns of response and consider IRS and situational

demands simultaneously. The problem of IRS does not currently appear to be a serious concern as long as careful consideration is given to the number and selection of physiological responses to be measured in psychophysiological studies. Also, a focus on IRS highlights the importance of studying *individual differences* in SEP research.

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See also Autonomic Nervous System; Biofeedback, Neurofeedback; Cardiac Function; Emotional Reactivity; Neuroscience, Exercise, and Cognitive Function; Psychophysiology; Stress Reactivity

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INFORMATION PROCESSING

During the early part of the 20th century, psychology was dominated by the school of thought known as *behaviorism*, which emphasized that psychological processes could only be examined at the level of observable behaviors. This approach assumed that all behaviors could be understood in terms of simple stimulus–response (S–R) relationships and that references to mental processes were neither useful nor valid. To this extent, the behaviorists viewed all mental processes as a *black box* whose internal workings are not directly observable, and are inconsequential to understanding how behaviors are governed by environmental stimuli.

Driven in part by the inability to explain more complex behaviors, the field of psychology underwent a major paradigm shift in the 1950s, which later became known as the *cognitive revolution*. The internal processes of the black box once ignored by the behaviorists now became the primary interest in the emerging field of cognitive psychology. Consequently, understanding what occurs *inside* the “box” (the human mind) was viewed as essential for explaining complex human

behaviors. Coinciding with this new direction, researchers began to observe similarities between human cognition and the early computers of the time. Both humans and computers could be viewed as general symbol manipulators, which can “take in” information, perform mental operations on the data, and finally output the information via behaviors and actions (e.g., images on a monitor). As a result, references to computers became a central metaphor for exploring human cognition, as it could provide direction and useful analogies for how humans process information mentally.

Humans as Information Processors

Based on this perspective, humans are fundamentally considered to be processors of information, with cognition understood as a sequence of computational processes. That is, information in the environment, such as words on a page, are taken in; stored in various memory systems; and processed via mental operations that encode, transform, and give meaning to the information through comparison with previously stored information (i.e., memory). Central to this approach is the suggestion that all information is encoded as internal symbolic representations of the external reality. These mental representations are constructed for all knowledge, information, ideas, and memories and are the subject of mental operations.

Systematic attempts have been made to study the capabilities and limitations of human information processing. Just as computers have certain limitations regarding speed of processing and how much information can be processed at any one time, similar limitations have been investigated with regards to human information processing. Given that mental operations occur within a black box, and thus are not directly observable, researchers have adapted and created a number of experimental methods from which overt behaviors are used to infer knowledge about the underlying cognitive processes. Chief among these approaches has been the use of *chronometric methods*, which emphasize the use of reaction times (RTs) to infer the temporal properties of mental operations. The Dutch physiologist F. C. Donders was the first to use such RT tests to determine the time needed for certain mental operations. Specifically, Donders argued that the duration of mental operations could be determined by subtracting the RTs of various types of tests

(e.g., simple RT, choice RT, go or no go RT) that have different cognitive processing requirements.

Three Stages of Information Processing

Every conscious action by humans, including those of athletes during action execution, is believed to be a consequence of response selection from long-term memory (LTM). LTM consists of a hierarchical structure neural network, which stores information after interacting with the environment. By definition, response selection indicates adaptive behavior based upon the capacity to solve problems. This “behavioral effectiveness” is directed by cognitive processes and mental operation. The effectiveness of these processes consists of the richness and variety of perceptions processed at a given time—that is, the system capacity to encode (store and represent) and access (retrieve) information relevant to the task being performed. From an information-processing perspective, motor behaviors consist of encoding relevant environmental cues through the utilization of attentional strategies, processing the information through an ongoing interaction between working memory and LTM, making an action-related decision, and executing the action while leaving room for refinements and modifications. Under pressure, changes in each of these components are seen. These changes are sequential in nature (i.e., they begin with the perceptual components, continue with the cognitive components, and end with the motor system).

Based on Donder’s *subtractive method*, a number of different processing stages are conceived to exist between the presentation of a stimulus and the initiation of a response (see Schmidt & Lee, 2011). Specifically, at least three information-processing activities must occur during RT. First, the presentation of a stimulus must be detected and identified—the *stimulus-identification stage*. Next, the proper response to the stimulus must be chosen—the *response-selection stage*. Finally, the selected response must be prepared by the motor system and initiated—the *response-programming stage*. In the past 50 years, a number of studies have been conducted to determine the processing limitations of each of these stages as well as the factors that influence processing performance.

The *stimulus-identification stage* consists of the mental operations concerning the sensing and encoding of environmental information. As a

stimulus contacts the body's sensory systems, it is taken in and encoded such that the neurological impulses activate the appropriate mental representation and knowledge associated with the information. A number of stimuli-related characteristics have been found to impact the speed at which the system is able to detect and encode relevant environmental stimuli. The *clarity* of the stimulus (i.e., how distinct and clear the stimulus appears) has been found to significantly impact the speed at which processing occurs during this stage. Specifically, RTs have been found to be slower when the stimulus is less defined compared to when the stimulus is presented with increased clarity (e.g., blurry vs. sharp picture). Similarly, the *intensity* at which the stimulus is presented impacts the speed of identification, with more intense stimuli resulting in faster RTs. Likewise, the modality of stimulus presentation has been found to influence the speed at which a stimulus is detected, as tactile and auditory stimuli have been found to result in faster RTs compared to visually presented stimuli. Once the stimulus is detected, it must also be correctly identified. In real life, the stimuli are often complex, and decisions must be made regarding a complex set of features. The ability to *recognize patterns* or features within the information set is important. Research indicates that task familiarity significantly influences the ability to quickly identify relevant features and patterns within the environment.

Following the detection and identification of the stimulus, a decision must be made regarding how to respond (i.e., what action to take). This stage is referred to as the *response-selection stage*. As with the previous stage, research has identified a number of different factors that influence the speed and selection of responses during this stage of processing. Not surprisingly, the time it takes for a person to decide upon a response has been shown to be influenced by the number of possible response choices they have. As the number of *S-R alternatives* increases, associated increases in choice RTs are observed. The mathematical relationship between RT and the number of response alternatives is given by Hick's law, which states that choice RT is linearly related to the $\log_2$ of the number (N) of S-R alternatives. Additionally, the *compatibility*, or the degree of fit, between the stimulus and response has been shown to influence RTs. For example, when the presentation of a stimulus and the required motor response are

spatially congruent (e.g., left stimulus and left response), RTs are quicker compared to when the stimulus and response are spatially incongruent (e.g., left stimulus and right response).

The final processing stage concerns the programming and initiation of the selected response, known as the *response-programming stage*. During this stage, the selected response must be compiled and transformed into overt muscular activity. F. M. Henry and D. E. Rogers demonstrated that the *complexity* of the movement required during the response is a critical factor in the latency of RTs during this stage. Specifically, responses that require more complex movements show larger latencies before the initiation of the response. This increased latency in responding is suggested to result from the additional time needed to prepare and program the upcoming motor action. A number of key factors relating to movement complexity have been identified which significantly influence this stage of processing. First, as the number of movement parts increases, corresponding increases in RT latencies are observed. For instance, responding to a stimulus with a simple finger movement is initiated faster compared to a response in which the whole arm must be utilized. Additionally, the accuracy requirements of the movement affect RT. As the precision demands of the movement response increase, increases in RT and movement time (MT) latency are likewise observed. Finally, responses with longer movement durations display increased RT and MT to initiate the motor response.

Conclusion

With the emergence of cognitive psychology, the view that humans are processors of information became the dominant framework from which to consider mental operations. In this regard, mental operations are viewed to consist of a series of information processing steps that begin with the taking in of information from the environment, processing that information, and then outputting the information via movement responses. Based on this framework, significant advancements have been made in the areas of intelligence, attention, decision making (DM), linguistics, and memory. Although more recent theoretical perspectives, such as ecological psychology and dynamical systems, provide alternative accounts of cognitive functions, the theoretical roots of information

processing remain a strong influence in modern psychology.

William Land and Gershon Tenenbaum

See also Decision Making; Knowledge Structure; Memory; Motor Commands; Motor Control; Motor Development; Motor Learning

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INJURY, PSYCHOLOGICAL SUSCEPTIBILITY TO

Sport and recreational-related injuries have become a significant public health concern for physically active persons. For example, in 2006 the Centers for Disease Control and Prevention estimated that participation in high school sports will result in approximately 1.4 million injuries reported to medical staff at a rate of 2.4 injuries per 1,000 athlete exposures (i.e., practices or competitions). As sport and other physical activities continue to be promoted as part of maintaining and restoring health, there will continue to be an increase in sport-related injuries. These trends have resulted in multifactorial perspectives of injury prediction and prevention. Advances in safety of equipment,

improvement to the physical environment, and new policies to protect sport participants have been implemented. Of relevance to this resource, but not likely as well known, are those approaches that involve the monitoring and modifying of psychological factors associated with sport injury.

Psychological Stress and Sports-Related Injury

In the 1960s, Thomas Holmes and colleagues suggested that changes in social stress, as measured by assessing accumulated stressful life events, were a precursor to changes in overall health. Essentially, life events often evoked changes in psychological status that required increased efforts to cope. This response to stress, or stress reactivity, was the mechanism underpinning the relationship between stress and health, whereby as stress demands and coping efforts increased, health would be compromised (e.g., onset of illness, progression of disease). In the 1970s, Holmes and later S. T. Bramwell and colleagues were the first to explore this notion in sport. They surveyed American football players and found greater levels of stress were associated with increased likelihood of athletic injury. This prompted further exploration of the relationship between stress and sport injury, yet much of what followed failed to replicate Holmes and Bramwell's findings. The literature lacked a unifying framework until 1988, when Mark B. Andersen and Jean M. Williams's model of stress and athletic injury filled an important gap and provided direction for future efforts.

Model of Stress and Injury

The stress-injury model posits that three categories of psychological risk factors (i.e., personality, history of stress, and coping resources) influence athletes' response to stressful athletic situations that, in turn, influences the likelihood of athletic injury through various stress-response mechanisms (see Figure 1). In addition to proposing psychological risk factors, their model offered specific avenues for psychological interventions and skills training to buffer adverse stress-related consequences for athletes' health and to minimize, or ideally prevent, athletic injury.

Stress Response Mechanisms

Cognitive Appraisal

Central to the stress and injury model is one's cognitive appraisal. Based upon the pioneering

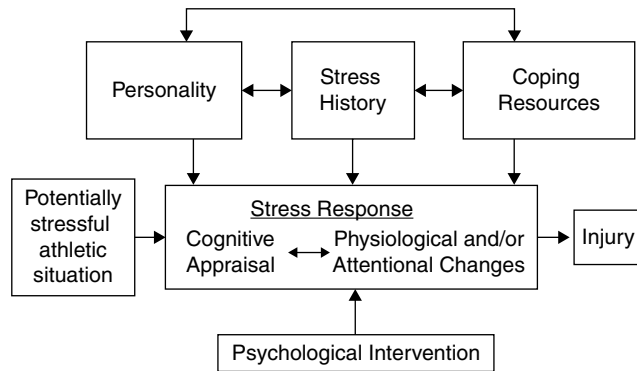


Figure 1 Stress and Injury Model

Source: Williams, J. M., & Andersen, M. B. (1998). Psychosocial antecedents of sport injury: Review and critique of the stress and injury model. *Journal of Applied Sport Psychology*, 10(1), 5–25. Adapted and reprinted with permission.

work of Richard Lazarus and Susan Folkman, the cognitive appraisal involves a balance (or imbalance) between two perceptions. The primary appraisal reflects perceived demands or threat of the stressor whereas the secondary appraisal reflects perceived personal and situational resources to cope with those demands and/or threat. When coping resources are perceived to be adequate to manage threat or demands, stress reactivity is minimal. In contrast, an imbalance occurs where one's perception of stress demands exceeds available coping resources, at which point stress reactivity is heightened.

Physiological Considerations

Very little attention has been given to physiological considerations. Within the model, stress responses that were initially described involved increase in muscle tension, which was thought to potentially impair motor control and slow reaction time (RT) that, in turn, may heighten injury vulnerability when engaged in sport competition and/or training. Over time, three additional pathways emerged associating psychological stress to physiological response. Perturbations in visual attention involving peripheral narrowing of visual field were described among highly stressed athletes and were hypothesized to heighten risk for injury (i.e., being blindsided). The other pathways linking stress to injury involved excess autonomic activity (e.g., increases in stress hormones) that were posited to increase injury risk by impairing immune

function and other cellular processes necessary for muscle repair following strenuous exercise training, or by altering sleep and associated secretion of growth factors also required for muscle anabolism. To date, excess elevation in psychological stress-related autonomic activity remains as the leading mechanism associated with athletic injury, but the specific causal pathway(s) remains unknown.

Attentional Considerations

In contrast to physiological factors, research studies have explored stress-related implications on attentional performance. Measures of visual and attentional indices have included peripheral narrowing and increased distractibility. In 1999, Mark Andersen and Jean Williams first demonstrated that peripheral narrowing mediated the relationship between injury outcomes and high stress in college athletes, and this was later replicated among high school athletes in 2005 by Traci Rogers and Dan Landers. While stress-related effects upon attentional and visual performance have been examined, there are other avenues (e.g., working memory capacity) that may be very relevant yet remain unexplored in the psychological injury vulnerability literature.

Psychological Factors That Influence Stress Response

Athletes who have a history of stress, a personality that amplifies perceptions of stress demands and/or reactivity to stress, and few available or effective coping resources are more likely to have cognitive appraisals of athletic situations that heighten their stress-reactivity. For those athletes, the consequences of heightened stress reactivity experienced within a sporting environment increase their risk of injury.

Personality

Research examining personality has measured patterns of behavior likely to exaggerate perceptions of stress demands and/or responses. In the original 1988 model, six likely personality variables were proposed and those included hardiness, locus of control, sense of coherence, competitive trait anxiety (TA), and achievement motivation. After nearly a decade of research, only four of those had received any attention and other characteristics had emerged that seemed appropriate that were not originally included (e.g., dispositional

optimism). Today, well over 20 different personality factors have been explored as potentially associated with injury vulnerability. While evidence for an injury-prone athlete personality type has not emerged from any of this literature, very few personality variables have been examined in more than one study. Those exceptions that have been explored across multiple studies are anxiety, locus of control, mood states, and anger. Anxiety is the most frequently measured, and while operational definitions vary considerably across studies, those studies examining competitive TA have consistently demonstrated significant associations with injury. In contrast, the research on locus of control, mood states, and anger remains inconclusive.

Associated with personality, other characteristics and sport-related patterns of behavior involving behavioral genetics may also influence the athletes' psychological stress and injury vulnerability. Genome-wide association studies (GWAS) have found particular single nucleotide polymorphisms (SNPs) associated with sport-related ligament injuries. GWAS has not been applied to the study of possible mechanisms linking personality or psychological stress factors to athletic injury. However, behavioral genetic findings associated with mood regulation and conscientiousness identified in the general population may have relevance to athletes' emotional response to injury and behavioral adherence to recovery protocols. As an emerging area of inquiry, further research is necessary.

History of Stress

Stress history was the initial and continues to be the most commonly examined psychological risk factor associated with injury. Athletes' history of stress has been examined through measuring three different variables: major life events, daily hassles or minor life events, and prior injury history. Early studies by Holmes and Richard S. Rahe and colleagues measured accumulated life events (or total life stress) irrespective of the specific nature or impact of those events. Over the years, a focus has shifted away from measuring only accumulative life events and toward measures that examine the impact and valence of stressful events, as well as both life and sport-related stressful events typical of athletic populations. The evidence supporting a relationship between life events' stress and athletic injury is by far the clearest and most consistent.

In contrast to these findings, research examining minor life events or daily hassles has been less clear. By definition, daily hassles occur frequently and therefore contribute to sustained stress activation. Unfortunately, daily hassles have not always been measured in a manner that captures its reoccurring nature; when it has, it has been significantly associated with injury. The third variable, prior injury history, influences athletes' stress reactivity in a few important ways. Athletes who have been injured previously may be physically and/or psychologically vulnerable to re-injury due to returning to sport prematurely either because physical and/or psychological recovery was not yet complete. For example, athletes may be physically recovered but may still have considerable self-doubt and anxiety. Either independently or conjointly, cognitive and physiological symptoms of anxiety further heighten stress reactivity and may have a more pronounced impact upon cognitive, attentional, and/or physiological functioning. In recent reviews of the literature done by Jean Williams as well as our own recent work, approximately 80% to 90% of studies have documented significant relationships between stress history and sport injury. Collectively, the evidence supports the centrality of athletes' stress reactivity in determining stress-related vulnerability to sport injury.

Coping Resources

Compared to stress history and personality, less attention has been given to coping resources in the injury vulnerability literature. Coping resources involve both internal or personal factors as well as external or environmental factors, which collectively reflect the strengths and vulnerabilities in managing demands of stress. Internal coping resources examined have involved measuring athletes' general coping behaviors or self-care (quality of nutritional intake, sleep, etc.) and psychological or sport coping skills (regulation of one's thoughts, energy, attention, emotion, etc.). Researchers have not always been able to demonstrate a significant relationship between internal or personal coping resources and sport injury. When significant links have been reported, they have almost always reflected a protective effect—greater coping was directly or indirectly associated with lower injury risk.

External coping resources have primarily been evaluating the quality, quantity, and/or effectiveness of athletes' social support networks. In

contrast to personal coping resources, results from studies examining social support are quite contradictory. Some studies have reported stress-buffering effects of social support while others have demonstrated increased stress-reactivity and injury risk with greater social support. These contradictory findings are perhaps suggestive of different stress-injury mechanisms across different sports.

Psychological Interventions for Health Promotion in Sport

The most exciting avenue of research is the efficacy of psychological interventions to prevent athletic injury. Intervention studies, based on Donald Meichenbaum's cognitive behavioral stress management (CBSM), involve the provision of education and skills training to athletes aimed to foster adaptive cognitions (e.g., thought stopping, restructuring) and/or manage physiological and attentional functioning (e.g., relaxation, mental rehearsal). To date, intervention studies have yielded medium to very large effects (e.g., 0.67–0.99), and the quality of this research has occurred at the highest level for therapeutic interventions (i.e., a randomized clinical trial). Findings provide strong support for psychological services for athletes to mitigate negative health-related consequences of sport participation (e.g., reduced injury or illness, time loss due to injury).

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See also Autonomic Nervous System; Biopsychosocial Model of Injury; Stress Management; Stress Reactivity

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INJURY, RETURN TO COMPETITION FOLLOWING

Evidence suggests the challenges of injury recovery may not cease at the completion of athletes' physical rehabilitation. Over the past decade, researchers have uncovered a range of psychosocial issues, challenges, and demands associated with the return to competition following injury. In this entry, athlete experiences returning to competition, the motivational issues surrounding return, and options for evaluating athletes' mental readiness to resume competitive activities are examined. Throughout this entry, the phrase *return to competition* refers to injured athletes' transition from injury rehabilitation to sport-specific training and competition.

Athlete Experiences of the Return

In an attempt to examine athlete experiences in returning to competition following injury, researchers have solicited the perspectives of athletes, coaches, and sport medicine practitioners. Issues of competence, autonomy, and relational concerns prominently appear throughout published literature. Competence refers to a sense of being capable or proficient in one's pursuits. Competence concerns comprise the most commonly reported sources of athlete apprehension during the return to competition transition.

Competence issues are salient in athlete expressions of anxiousness over re-injury, concerns about the impact of injury on skill execution or loss of overall physical fitness. Athletes also typically report uncertainties about performing at pre-injury levels. Concerns over re-injury and competing at pre-injury levels are not surprising, given elite athletes' interest in athletic attainment and the inability to pursue goals over a potentially prolonged absence during injury recovery.

Research has also highlighted the salience of autonomy issues among returning athletes. Autonomy pertains to an individual's sense of choice or control over one's actions and behaviors. During the return to competition transition, athletes may experience varying degrees of choice and control over the timing and circumstances of their return. Whereas some athletes may be free to return at a time and manner of their own choosing, evidence suggests that many athletes face external pressures to return from coaches, teammates, or even sport medicine practitioners. Such pressures appear to be largely a function of the temporal proximity of upcoming competitions and the importance of the athlete in attaining desired team outcomes. Further complicating the situation are the internal pressures athletes place upon themselves to return to competition. Internal pressures may stem from intraindividual concerns regarding an inability to perform sport skills, self-generated worry that one is losing considerable fitness, or self-induced distress over "falling behind" fellow competitors during one's sport absence. Unfortunately, indicators of functional capacity (e.g., proprioception, joint range of motion) or consideration of the athlete's long-term health and well-being can sometimes be a secondary concern.

Finally, relational issues pertaining to the lost sense of connection to others and one's sport have been reported in the psychology of sport injury literature. Athlete perceptions of inadequate social support have also been highlighted. In particular, feelings of detachment and isolation, occurring during rehabilitation, may persist until such time as athletes are able to fully resume competitive play and contribute to valued team outcomes. Inadequate levels of social support may also contribute to relational concerns. For example, athletes report some coaches to be distant and insensitive to injury, uninterested in providing desired rehabilitation guidance, or as lacking a belief in their ability to return. Such distancing

by a coach may contribute to the perception that one is not a valued member of the team and that only those who are competing are deserving or worthy of the coach's attention. Confronted with this reality, it is not surprising that athletes all too commonly feel a sense of estrangement from their sport.

A lack of social support may also come in the form of insufficient information about recovery progressions and the requirements for attaining return to competition readiness. Specifically, some athletes have reported a lack of guidance and information from coaches and physiotherapists regarding adequate rehabilitative training to facilitate re-entry to the competitive arena. Such instances are unfortunate, given substantial evidence supporting the benefit of social support for injured athletes. Social support from coaches, family members, and athletic trainers may be essential in enabling the athlete to overcome the demands and uncertainties inherent in the return to competition.

Given the apparent relevance of competence, autonomy, and relational issues, Leslie Podlog and colleagues examined the impact of rehabilitation environments that satisfy athletes' need to feel competent, volitional, and connected to relevant others. Findings reveal that support for athletes in these three areas promotes enhanced well-being and effective return-to-competition outcomes. In particular, athletes who experienced a sense of competence and autonomy during rehabilitation had more positive emotions, which likely fostered a "renewed perspective on sport" following the return to competition (e.g., greater sport appreciation, heightened motivation for success, enhanced mental toughness). Moreover, those who felt a sense of connection to relevant others indicated less negative affect and greater self-esteem—both of which decreased the likelihood of athletes' experiencing "return concerns" (e.g., increased competitive anxiety, re-injury concerns). These findings highlight that the satisfaction of injured athletes' needs may facilitate perceptions of well-being and positive return to competition outcomes. Given these findings, it seems reasonable to suggest that interventions targeting competence, autonomy, and relatedness needs may help mitigate the challenges inherent in the return to competition. Further applied research aimed at minimizing athletes' return to competition concerns is needed, however, to explore the merits of this contention.

Motivational Issues

Motivational researchers Edward Deci and Richard Ryan suggest that the reasons why people engage in a behavior (i.e., their motives for action or involvement) may have important implications for the sustainment of such behaviors and the extent to which individuals experience optimal functioning or well-being. A wealth of evidence across various life domains including education, work, and relationships supports this line of reasoning. Operating from a similar assumption, Leslie Podlog and Robert Eklund recently examined the implications of athlete motives to return to competition on perceptions of post-injury performance. In an investigation with high-level athletes from Canada, Australia, and England, the researchers found that intrinsic motives to return to competition such as a love of the game or the excitement of participation were positively associated with a “renewed perspective on sport.” On the other hand, athletes who were extrinsically motivated—that is, those who returned to competition because they wanted to attain external rewards or to avoid punishments, such as exclusion from participation—experienced greater “return concerns.” Return concerns included, but were not limited to, diminished confidence, unsatisfying performances, and heightened competitive anxiety. Similarly, in an experimental investigation with professional Australian football league players, Podlog and Eklund found that greater intrinsic (versus extrinsic) motivations resulted in more positive thoughts and emotions regarding a return to competition. Interestingly, athletes who returned for intrinsic reasons such as a love of the game reported diminished perceptions of threat, unfairness, and potential damage to one’s ego than their extrinsic counterparts. These findings suggest that the motivations underlining athletes’ return to competition may have important consequences for the quality and nature of their return. In particular, the aforementioned studies by Podlog and Eklund indicate that intrinsic reasons for returning to competition may be associated with positive perceptions of a return to competition and enhanced post-injury performances.

Assessing Psychological Readiness to Return to Competition

Taking into account the abundant challenges and motivational demands placed upon returning

athletes, the need for psychological evaluation is critically important. Several options presently exist for assessing athletes’ psychological status upon return to competition. One option is D. W. Creighton and colleagues’ three-step return to competition decision-making model. In Step 1 of the model, the health status of the athlete is assessed through the evaluation of medical factors (e.g., medical history of the patient, lab tests such as X-rays or MRIs, severity of the injury, functional ability, psychological state). Step 2 involves consideration of the risks associated with participation by assessing variables such as the type of sport played (e.g., collision, noncontact), the position played (e.g., goalie, forward), the competitive level (e.g., recreational, professional), the ability to protect (e.g., bracing, taping, padding), and the limb dominance of the patient. Step 3 in the decision-making process includes consideration of nonmedical factors that can influence return to competition decisions. Relevant considerations here include the timing in the season (e.g., playoffs), pressure from the athlete or others (e.g., coach, athlete’s family), ability to mask the injury (e.g., pain medications), conflict of interest (e.g., potential financial gain or loss to the patient or clinician), and fear of litigation (e.g., if participation is restricted or permitted). The model provides a framework outlining the complex interaction of factors ultimately contributing to return-to-competition decisions. Utilizing the three-step process can help guide clinician decisions regarding athletes’ return to competition.

In an effort to assess athletes’ psychological state mentioned in stage 1 of Creighton and colleagues’ model, researchers and practitioners are encouraged to use Douglas Glazer’s Injury–Psychological Readiness to Return to Sport scale. The inventory is a valid and reliable scale consisting of six questions designed to assess athlete confidence regarding various aspects of their return to competition. For example, athletes are asked to rate their overall confidence to return to competition, their confidence to play without pain, and their confidence to give 100% effort. Given its concise nature, the readiness to return questionnaire can be easily administered by sport psychologists and health practitioners in the rehabilitation setting.

Another possibility in assessing psychological readiness to return to competition is Natalie Walker and colleagues’ Re-Injury Anxiety Inventory. The questionnaire contains 28 statements aimed at

uncovering the extent to which athletes experience uncertainty regarding re-injury. Fifteen of the questions pertain to re-injury anxieties during rehabilitation. These 15 questions require athletes to reflect upon the extent to which they are worried or feel nervous about becoming re-injured during rehabilitation. Another 13 questions assess re-injury worries upon return to competition. These 13 questions ask athletes to indicate how “worried” or “nervous” they are about becoming re-injured during re-entry into competition. Walker and her collaborators made efforts to distinguish fear (a flight-or-fight response to danger) from anxiety (uncertainty, worry, or concern) indicating the latter is a more accurate description of the athlete’s state of mind.

Both the readiness to return and re-injury anxiety questionnaires may facilitate assessment of athletes’ psychological state as they approach a return to competition. Low scores on either scale may alert health practitioners to the fact that athletes may not be psychologically prepared to resume competitive activities. Initial evidence suggests that athletes who experience heightened re-injury anxiety may have an elevated risk for poorer return-to-competition outcomes such as heightened competitive anxiety and diminished confidence in performing sport skills. It is also possible that re-injury anxieties or a lack of confidence in return-to-competition abilities increase athletes’ actual risk of re-injury. This suggestion, however, requires further research.

Conclusion

Returning to competition following injury may entail a range of psychosocial challenges and demands. Issues pertaining to athletes’ sense of proficiency, perceptions of autonomy, and feelings of connection to relevant others appear to predominate during the re-entry period. Evidence also reveals that athlete motivations to return to competition following injury may have important consequences for post-injury performances. Given the plethora of demands and motivational challenges associated with a return to competition, ensuring a holistic assessment of athletes’ physical and psychological readiness is imperative. Creighton and colleagues’ three-step decision making model can be a useful tool in determining an athlete’s ability to safely and successfully compete after injury. Finally, two questionnaires, the Injury–Psychological Readiness to Return to

Sport scale and the Re-Injury Anxiety Inventory, show promise in facilitating evaluation of athletes’ mental preparedness to return to competition.

Leslie Podlog and Robert C. Eklund

See also Injury, Psychological Susceptibility to; Intrinsic/Extrinsic Motivation, Hierarchical Model of; Psychological Well-Being; Self-Determination Theory; Self-Doubt; Self-Efficacy

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INTERDEPENDENCE THEORY AND THE COACH–ATHLETE RELATIONSHIP

In sport, and sport coaching more specifically, the connection between coach and athlete is instrumental for optimal functioning, be it physical, psychological, mental, or social. In fact, there is strong evidence to suggest that success in sport is the product of the combined interrelating between the coach and the athlete. Athletes are unlikely to produce top-level performances without the direction and encouragement of their coaches, and

coaches are unlikely to be successful without their athletes' hard work, commitment, and enthusiasm. Thus, coaches and athletes who are locked into a relationship that is harmonious and stable are also more likely to achieve sport success (e.g., skill development, tactical awareness, trophies, medals) and make the journey to success more pleasurable, satisfying, and personally fulfilling.

Interdependence Theory

Interdependence theory (IT) is an important framework for understanding the processes of personal and social relationships including a central relationship in sport—namely, the coach–athlete relationship. It focuses on how those in relationships, such as coaches and athletes, cause one another to experience positive versus negative outcomes. Positive outcomes are reflected in the *rewards* the relationship and its members experience such as happiness, gratification, and pleasure while negative outcomes are reflected in the *costs* the relationship and its members experience, such as anxiety, conflict, and antagonism. *Interdependence* is a fundamental element because it can affect how, for example, a coach and an athlete influence one another's outcomes through the processes of exchanging, communicating, and interacting. Moreover, these processes allow relationship members to understand themselves and one another.

Interdependence in the Coach–Athlete Relationship

The notion of interdependence in the coach–athlete relationship was recently captured by the constructs of *closeness*, *commitment*, *complementarity*, and *co-orientation*, known in the literature as the 3 + 1 Cs model. Closeness represents coaches' and athletes' affective bond and is reflected in such relational properties as mutual trust, respect, appreciation, interpersonal liking, as well as emotionally caring and supporting one another. Commitment represents coaches' and athletes' thoughts of maintaining an athletic partnership that is close over a long period of time (e.g., the emphasis is on staying near to each other physically or symbolically; to want to coach the athlete or be coached by the coach now and in the future). Complementarity represents coaches and athletes' behavioral exchanges that are (a) affiliative and corresponding, such as members being friendly, responsive, and comfortable in each

other's presence, and (b) organized and reciprocal, such as members assuming their specific and respective roles in a suitably and effective manner (e.g., coach instructs and athlete executes). Finally, co-orientation reflects the degree to which athletes and coaches are perceptually similar (i.e., have a similar understanding as this may relate to the relationship and its outcomes). The 3 + 1 Cs model provides a conceptual and operational model to study the degree to which relationship members are interdependent.

Interdependence Structures

According to IT, the *interdependence structure* in dyadic relationships such as the coach–athlete relationship can be defined by examining the main effects and interaction of each relationship member's behaviors, including (a) actor control (e.g., a main effect of athlete's actions on athlete's outcomes), (b) partner control (e.g., a main effect of coach's actions on athlete's outcomes), and (c) joint control (e.g., an interaction of coach's and athlete's outcomes). Interdependence structure is composed of four properties known as degree of dependence, mutuality of dependence, basis of dependence, and correspondence of interests.

Degree of dependence reflects the extent to which an athlete depends on his or her coach and, thus, the degree to which an athlete's outcomes depend on his or her coach's actions. Dependence is a psychological significant relational characteristic that can exert positive and negative effects on the coach and the athlete as well as their relationship. Research has shown that, for example, coaches' and athletes' dependence on each other in terms of the 3 + 1 Cs can promote open channels of communication and self-disclosure, skill and performance development, physical self-concept, motivation, passion, team cohesion, collective efficacy (CE), and well-being, to name a few. However, coach–athlete dependence can also have negative consequences. For example, low levels of dependence in terms of the 3 + 1 Cs have been found to be detrimental to athletes' morale, performance, satisfaction of basic psychological needs, and well-being. In this case, athletes and coaches would appear to be uncomfortably dependent on one another and more often than not they perceive one-sided dependence (caused by situational factors such as power struggles and/or individual difference factors such as personality, age, maturity).

Mutuality of dependence describes the extent to which a coach and an athlete are mutually rather than separately dependent on one another for generating enjoyable or fulfilling outcomes. Healthy, harmonious, and stable coach–athlete relationships as characterized by the 3 + 1 Cs are expected to involve a considerable degree of mutual dependence (e.g., we support each other) rather than one-sided dependence (e.g., I support him, but he doesn't support me). Mutually dependent relationships and interactions tend to yield rewards. There is evidence in the literature that when coaches and athletes are mutually dependent (both relationship members experience and perceive mutually high levels of 3 + 1 Cs), personal and interpersonal rewards are yielded. However, when coaches and athletes have been found to be nonmutually dependent, upset and discomfort were more readily experienced. In one study, it was found that athletes feel less dependent and satisfied with the relationship (while coaches may continue to feel highly dependent and satisfied), when they perceive that their coaches' resources (e.g., experience, skills, knowledge) are depleting or their self-interests lie elsewhere (e.g., more emphasis is given to other athletes in the team or squad).

Basis of dependence describes the way in which coaches and athletes influence one another's outcomes or whether dependence results from, for example, one's member control (e.g., coach). There are many empirical and anecdotal examples that highlight coaches' power to have control over their athletes, but there are also examples whereby both coaches and athletes have joint control over each other's outcomes. Coordinated forms of interaction whereby, for example, the coach orchestrates the proceedings of training, instructs, and supplies advice while the athlete receives the instructions and advice positively and responsively are indicators of relationship members' comfort with the basis of dependence. If discomfort was experienced due to a member being unwilling to interact in a coordinated fashion (instead interacted in a one-sided or competitive fashion) the likely result would be poor individual and relationship outcomes. Finally, *correspondence of interests* describes the extent to which an athlete's and a coach's actions yield rewards for both members in a corresponding fashion. While interaction is relatively easy when both a coach and an athlete's interests correspond (e.g., we want to win an Olympic title; we want to maintain an

effective and successful partnership), interaction is relatively tricky when their interests conflict (e.g., Coach wants me to go to a summer camp but I don't; athlete never talks, opens up, shares news). Correspondence of interests may range from perfectly correspondent to perfectly noncorrespondent and may be reflected by the degree to which coaches and athletes perceive high or low and corresponding or noncorresponding levels of the 3 + 1 Cs. An open channel of communication has been viewed as a major factor that leads to correspondence of interests and has been found to be a major contributor to enhancing interdependent coach–athlete relationships.

Comparison Level and Comparison Level for Alternatives

IT posits that relationship members' evaluations of their relationships are influenced by two standards: *comparison level* (CL) and *comparison level for alternatives* (CL-alt). CL is a criterion against which coaches and athletes evaluate how satisfactory the relationship is whereas CL-alt is a criterion coaches and athletes are likely to use to decide whether to stay in or leave the relationship. Within IT, satisfaction with a relationship per se is a function of comparing the rewards and costs of that relationship with some type of internal standard (CL). In contrast, deciding whether to stay in a relationship is a function of comparing the rewards and costs of that relationship with the rewards and costs of other relationships that are available (CL-alt). For example, someone may remain in a dissatisfying relationship (below his or her CL) because the satisfaction available in other relationships (CL-alt) is less. Alternatively, someone may leave a satisfying relationship (above his or her CL) because the satisfaction available in other relationships (CL-alt) is greater. Research that focuses on the links between interdependence and satisfaction has found that a coach's and an athlete's interdependence as measured by the 3 + 1 Cs are positively related to satisfaction with training, instruction, and performance accomplishments, as well as satisfaction with personal treatment and athletic relationship. Moreover, it has been found that interdependence (as defined by the 3 + 1 Cs) and satisfaction with training, instruction, and personal treatment were weaker for lower-level (i.e., club) competitors than for higher-level competitors (i.e., regional, national,

and international). Finally, associations between interdependence and satisfaction were stronger for longer relationships and same-gender dyads (all male and all female coach–athlete dyads) than mixed-gender dyads (female coach–male athlete and male coach–female athlete). Collectively the findings of these studies underline that coaches and athletes depend on each other for obtaining valued outcomes; this is reflected in their evaluations of developing a positive relationship and in their beliefs that the other relationship member meets important outcomes.

Conclusion

From an IT point of view, coaches and athletes are likely to be both attracted to a relationship and satisfied with it as the rewards associated with the relationship increase and costs decrease. The research evidence to date suggests that more interdependent relationships (defined by the 3 + 1 C) may be more satisfying because they fulfill basic human needs, heighten positive affect, and enhance self. The application of IT can provide a framework to understand and explain important research questions that relate to complex processes within the phenomenon of the coach–athlete relationship.

Sophia Jowett

See also Attachment Theory and Coaching; Coach–Athlete Relations; Commitment; Conflict; Passion; Relational Efficacy Beliefs in Coach–Athlete Relations; Self-Determination Theory; Team Communication

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INTERFERENCE

The need for consistency in training, motor learning, technical preparation, and competition is well understood by coaches, athletes, and applied sport psychology (SP) consultants. A number of factors, however, can interfere with athletes' ability to perform or learn at an appropriate level during training, technical preparation, and competition. The term *interference* addresses the decrease in performance due to conflicts between different cognitive modules or motor structures within the human action system in a specific environmental constellation.

Interference in Cognitive Learning and Verbal Memory

In the psychological realm, interference is mostly discussed in the context of memory and learning. Interference is addressed as a memory problem that occurs when a learning process is impaired because of an existing, stabilized memory structure or when the activation of consolidated memory representations is affected because of newly learned material. Interference in learning with verbal material has been observed and studied for more than 100 years. Notably, the studies of the German psychologist Georg Elias Müller opened up new perspectives on the topic of memory consolidation. While the studies of Hermann Ebbinghaus (1885) hypothesized forgetting as a function of time, Müller and his student Alfred Pilzecker provided evidence for interference-based memory loss and an interference-based decrease of learning. The term *retroactive interference* was coined by Müller and describes the phenomenon seen in memory in which newly learned material affects the recall performance of previously learned information. In the opposite direction, *proactive interference*

specifies the phenomena that—depending on the task—previously learned information potentially impairs the learning of new information. Proactive interference is commonly explained as competition between two learning processes that are based on the same cognitive resources. From this point of view, in the case of retroactive interference, the learning of new material influences the consolidation of previously learned material in memory. During proactive interference, consolidated memory information is challenged by new information and inhibits the consolidation of new information in memory. Subsequent studies have found that similarity between the information already stored in memory and the new material affects the learning process and causes interference.

Interference in Motor Learning and Motor Memory

In motor control research, interference is mostly discussed in the context of motor learning. The learning of new skills is based on previously learned sensory-motor patterns that are stored in memory. In general, there are two possible types of interactions between previously learned skills and recently acquired skills: (1) a positive and supportive interaction called *transference* or (2) a negative interaction called *motor interference*. Transference was originally addressed by researchers like Nikolai A. Bernstein and is explained as the facilitation of learning as a result of similarity between the required motor pattern and the existing stored pattern, while conversely, motor interference is mostly hypothesized as a negative overlap (i.e., dissimilarity) between the coordination patterns of previously learned motor actions and the structure of a recently acquired skill. From this point of view, learning to crawl could generally facilitate skills like canoeing because of similarities in motor structures (transference), while techniques in sports like tennis and badminton at the same time seem to be related at a surface level but suffer from interference because of significant differences between the relevant sensory-motor structures at a functional level.

Charles H. Shea and other researchers addressed the topic of *contextual interference* and compared performance during practice of a motor task under randomized and blocked conditions. They found that practicing the task in a blocked order, repetitively and under stable conditions (low contextual

interference), resulted in better performance than did practicing the task under randomized conditions (high contextual interference). Interestingly, however, when the performance between random and blocked practice was compared, these and other studies revealed that the random practice group showed superior performance to the blocked practice group. It has been hypothesized that in the short term, reduced interference supports the learning process of simple motor skills, but in the long term, randomized practice supports the establishment of more stable representations in motor memory and leads to better performance compared to the more artificial (low contextual interference) conditions.

Interference Between Motor Control and Verbal Memory

Until now, the phenomenon of interference has been mostly investigated in the area of verbal memory research, isolated from motor learning and vice versa. Interference in the context of motor learning has often been discussed as it relates to memory resources, but it remains unclear whether or not motor planning and verbal working memory share common cognitive resources. In our daily life, verbal memorization and motor action are often performed together—for instance, remembering a list of consumables in a shop while walking along the shop floor and searching for the next product.

In the research area concerning the learning and storage of lists in working memory, the *serial position effect* is a stable finding. This effect is observed when subjects memorize and recall a sequence of items. Subjects are best able to memorize the items presented at the beginning of the list (primacy effect) and the end of the list (recency effect). This recall paradigm has often been used to learn about interference in verbal memory. Combining a motor task and a verbal memory task could lead to two kinds of interference: *cognitive interference* or *motor interference*. In the case of cognitive interference, the memorization of verbal lists would decrease the motor performance; conversely, the opposite would be true in the motor interference scenario. In a variety of tasks such as replanning of motor actions and sequential motor planning in unimanual and bimanual coordination, researchers like Matthias Weigelt, Mark G. Fischman, and Marnie A. Spiegel have investigated in recent years the interference between motor planning and

recall performance. In multiple experiments, they found evidence for the so-called *motor interference hypothesis*—that is, that motor planning reduces memory performance of verbal items because of the challenge between these two entities.

Thomas Schack

See also Attentional Association and Dissociation; Automaticity; Memory

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INTERVENTIONS FOR EXERCISE AND PHYSICAL ACTIVITY

Interventions in the field of exercise or physical activity (PA) psychology focus on issues related to health rather than on issues related to performance in sport. Exercise, by definition, suggests a form of PA that is often structured and undertaken with the aim of improving fitness. However, health benefits can be obtained from more incidental modes of PA that are not primarily done for fitness benefit such as gardening, dog walking, or bicycling to work. Such activity may have different determinants than those that could be defined as exercise and so, to encompass both, the term *PA-exercise psychology* is used in this entry. PA-exercise psychologists employ a wide range of skills, some of which overlap with the skills used by sport performance psychologists, to assist individuals and groups achieve sufficient PA to gain health benefits. PA-exercise psychologists also engage in research to further understand what constitutes effective interventions that may influence population health. When psychologists intervene with individuals, groups,

or communities, they use approaches that have a strong evidence base for successfully increasing PA levels. This health-related aspect of sport and exercise psychology (SEP) is growing in importance because of the global public health risk posed by lack of sufficient PA. This entry summarizes the importance of PA for health, discusses the need to decrease sedentary time, introduces the global call to action in the Toronto Charter for Physical Activity, describes a series of interventions that have global support and represent good investments for countries seeking to increase population levels of PA, and finally makes a recommendation that all interventions should be properly evaluated.

The Importance of Physical Activity for Health

Physical inactivity is the fourth leading cause of mortality from chronic diseases such as heart disease, stroke, diabetes, and cancers, and inactivity contributes to the rising obesity levels noted around the world. For children, the recommended minimum amount of activity shown to be beneficial is 300 minutes of moderate intensity activity over the course of a week, and for adults, the recommended minimum is 150 minutes of moderate intensity activity over the course of a week. Public health physical inactivity is a serious challenge: Inactivity is a risky health behavior and, because the majority of Western populations do not achieve the minimum amount of activity for health, it is also a very prevalent risk. There is also a considerable challenge in increasing the levels of PA within a country's population. Psychologists are part of a team of professionals contributing to determining how best to increase population levels of activity.

The New Issue of Sedentary Behavior

A new aspect of this health-related field is the developing understanding that lengthy periods of sedentary time, such as those experienced in many desk jobs or through TV viewing, may be detrimental to health even if the person achieves the currently recommended minimum amount of activity for health. There is therefore also a need for interventions to focus on how people might decrease the number of minutes they spend in sedentary activities. Exercise and PA psychology has a clear role to play in the design, implementation, and evaluation of interventions that promote PA behavior or decrease sedentary time.

The Toronto Charter for Physical Activity

The Toronto Charter for Physical Activity, which was launched in May 2010 by the Global Advocacy Council for Physical Activity (GAPA), is an important milestone in the history of PA and public health. The aim of the charter is to create sustainable opportunities for physically active lifestyles for everyone. Within the Toronto Charter there are nine guiding principles listed for a population-based approach to PA. Psychologists should use the charter to guide their role in developing interventions to increase PA and decrease sedentary time. The charter's principles identify the importance of evidence-based approaches—of embracing equity by reducing social and health inequalities or removing disparities in access to PA. Importantly, the principles acknowledge the need to move beyond the individual to include environmental and social determinants of physical inactivity. Other principles identify sustainability, a life-course approach to promoting activity, and the need to garner political support and resource commitment at the highest level.

What Kinds of Interventions Are Appropriate?

One common framework for understanding the various levels at which intervention aimed at improving health may take place is known as the socioecological model. This model suggests that there are multiple influences on PA and sedentary behavior, such as individual psychology, social circumstances, the surrounding physical environment, and wider sociopolitical influences such as culture and policies. This suggests that interventions aimed at increasing individuals' activity levels must take into account these wider influences, and psychologists need to work in partnership with other agencies to advocate for environments that encourage activity, such as the creation of foot paths and bicycle lanes. In addition, exercise psychologists must also work at the policy level and advocate for the creation of local and national policies that lay out strategic objectives for raising the level of PA. However, the issue of which interventions to use for greatest effect is not easy to resolve because the evidence base is relatively young, and one cannot say with certainty which approaches are best. Agencies such as the Centers for Disease Control and Prevention in the United States and the National Institute for Health and Care Excellence (NICE) in the United Kingdom

issue guidance from time to time, which is based on thorough and critical reviews of the available evidence, and these resources are useful starting points for psychologists wishing to understand the current evidence. For example, NICE issued a set of guidelines on the best approaches to changing the environment to promote PA.

Which Interventions to Use?

GAPA and leading academics and practitioners from around the world have reviewed evidence for interventions that were effective in increasing PA levels. This review led to the production of a companion document to the Toronto charter, titled "Seven Investments That Work" (www.globalpa.org.uk/investments). These seven approaches are as follows:

1. *Whole-of-school programs.* In these programs, schoolchildren of all ages are encouraged to be active in a variety of ways such as on the journey to and from school, during school break times, and after school and via quality physical education (PE) programs.
2. *Transport policies and systems.* By intervening at the level of transport systems, people might be encouraged to use public transport, which is almost always a more active journey than a car, or they might be encouraged to actively commute from place to place by foot or by bicycle. Transport policies that prioritize walking and cycling and provide safe infrastructure, such as well-lit pavements and off-road cycle paths, could have a substantial impact on population levels of PA. Scandinavian countries provide good examples of this approach.
3. *Urban design regulations and infrastructure.* Towns and cities that provide for equitable and safe access to recreational PA, such as playgrounds and sports facilities, with opportunities for all ages to participate, support lifelong involvement in PA.
4. *Primary care programs.* Every patient in a health service scheme, who is visiting a doctor or nurse for whatever reason, should be asked about his or her current level of activity and given encouragement and opportunities to increase activity to benefit health.
5. *Public education.* Many people simply do not know the risks of inactivity nor the amount of

activity needed to gain health benefits. Public education is therefore needed and might include mass media to raise awareness and change social norms around PA to help create a culture in which being physically active is normal behavior.

6. *Community-wide programs.* Offering opportunities to be active in one's local community and mobilizing community engagement and resources is seen as an effective route in helping people be more active.
7. *Sports systems and programs that promote "sport for all."* When sport is taught within schools and communities, coaches should employ a "sport for all" ethos that encourages participation across the life span in preference to an ethos that only promotes elite-level sport.

How Should Physical Activity–Exercise Psychologists Design and Evaluate Interventions?

When PA–exercise psychologists intervene, perhaps in one of the seven areas listed previously, to promote PA or reduce sedentary behavior, they should base their work on a recognized theory of behavior change and use the principles of that theory to develop appropriate materials. They must also consider a framework to guide the evaluation of their intervention. At a minimum, all interventions must measure PA behavior, ideally with an objective monitor such as a pedometer or accelerometer, to determine if improvements have occurred.

Nanette Mutrie

See also Adherence; Affective Responses to Exercise; Energy, Effects of Exercise on; Intrinsic/Extrinsic Motivation, Hierarchical Model of; Obesity; Quality of Life; Sedentary Behavior; Stress Management

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INTRINSIC/EXTRINSIC MOTIVATION, HIERARCHICAL MODEL OF

The hierarchical model of intrinsic and extrinsic motivation (HMIEM) is a comprehensive theory that seeks to describe human motivation and its determinants and outcomes from a multilevel perspective. Its major premise is that in order to more completely understand the motivation of sport participants (e.g., athletes, coaches, referees, fans), one needs to consider their motivation in various contexts and at various levels of generality. In line with self-determination theory (SDT), the HMIEM posits the existence of three types of motivational constructs: intrinsic motivation (behaving out of pleasure and free choice), extrinsic motivation (behaving to obtain rewards or avoid punishments), and amotivation (the relative absence of motivation). These motivational constructs are organized both vertically and horizontally within the HMIEM. This structural arrangement serves to integrate knowledge on the personality (vertical axis) and social psychological (horizontal axis) determinants of motivation.

First, the vertical organization of elements in the HMIEM specifies that the three types of motivation (intrinsic, extrinsic, and amotivation) exist at three different levels of generality: global, contextual, and situational. The most stable level is situated on top (the global level) and the most ephemeral (the situational level) at the bottom. The global level corresponds to a person's usual way of functioning at the personality level (e.g., Susan is typically intrinsically motivated). The next level is the contextual level. It corresponds to one's usual motivation in a particular context such as sports, education, leisure, or interpersonal relationships. This type of motivational orientation is moderately stable and can vary from one context to another. For instance, Susan may be intrinsically motivated to play sports but extrinsically motivated to go to school. Finally, the

situational level is the most specific and refers to one's motivational state in the present moment, the here and now. For instance, you may observe that Susan is intrinsically motivated to play tennis right now.

Top-Down and Bottom-Up Effects

Importantly, the HMIEM postulates that relationships exist between the levels of the hierarchy such that the type of motivation one has at a given level influences his or her motivation at the other levels. The relationships between motivation at the different levels of the hierarchy take place through the *top-down* and *bottom-up effects*. Top-down effects refer to the influence of higher levels in the hierarchy upon lower levels. Specifically, global motivation influences both contextual motivation and situational motivation whereas contextual motivation influences only situational motivation. Each level has the strongest influence on the level immediately below (i.e., the proximity principle). In other words, global motivation will have a stronger influence on contextual motivation than on situational motivation. For example, if Susan has a high global intrinsic motivation, all else being equal, she is also likely to have a high level of intrinsic motivation at the contextual level (next level below) such as in sports. Research supports this top-down hypothesis showing that the more athletes have a high level of contextual intrinsic motivation for sports the more they have a high level of situational intrinsic motivation during a game. Similar findings have been obtained in physical education (PE) settings.

Second, bottom-up effects represent the influence of lower levels in the hierarchy upon higher levels. This effect explains how repeated motivational changes at a given level instigate changes in motivation at the next higher level. For example, if Susan repeatedly experiences increases in intrinsic motivation (e.g., during many tournaments), she may become more prone to adopt an intrinsically motivated orientation in the context of sports. In turn, this increase in intrinsic motivation at the contextual level in sports may also lead to increases in intrinsic motivation at the personality level (i.e., the global level). The proximity principle applies here as well. Thus, situational motivation will have a stronger effect on contextual motivation than it will on global motivation. Longitudinal research provides support for these hypotheses.

Social Psychological Processes

The HMIEM also describes a horizontal organization of elements that describe the nature of the social psychological processes through which changes in motivation take place and lead to outcomes. The same sequence takes place at all three levels of the hierarchy: "Social Factors→Psychological Needs→Motivation→Outcomes." At the beginning of the sequence, it is postulated that motivation is determined first and foremost by social factors. For instance, global social factors are so pervasive that they are present in most areas of a person's life. One example is parenting, which represents a rather continuous influence on children because it spans many life contexts and situations. The way parents raise their children determines, in part, whether the children will adopt an intrinsic or extrinsic way of interacting with the environment (i.e., a global motivation). Second, contextual social factors are recurrent factors but only in a specific context such as in basketball (i.e., a basketball coach) or in school (i.e., a teacher). Third, situational social factors are present at a single point in time. For example, winning a basketball game on a Sunday afternoon may have an impact on one's situational motivation to join another scrimmage match when the opportunity comes up. Importantly, global social factors determine global motivation, contextual factors determine contextual motivation, and situational factors determine situational motivation.

Next, the HMIEM postulates that the influence of social factors on motivation occurs through basic psychological need satisfaction. Need satisfaction acts as a mediator between social factors and motivation at every level of generality in the model. The more an individual's psychological needs are met in general, in a given context or in a specific situation, the more he or she will engage in activities in an intrinsic way at the global, contextual, or situational level. For example, the more athletes perceive their coach as supporting their needs for autonomy, competence, and relatedness, the more they will develop an intrinsic motivation for sports. Much empirical support exists for the role of need satisfaction as mediators of the effects of social factors (e.g., the coach's behavior) on athletes' motivation. Of additional importance is that motivation in different contexts can interact in different ways. For instance, motivation in a given context (e.g., education) may add to, or even

conflict with, motivation in another context (e.g., sport). In other words, there might be compensation or additive effects among different contexts. Research in sport and education empirically supports the presence of such effects.

Finally, the social psychological process ends with the effects of motivation on outcomes of three types: affective, cognitive, and behavioral. Such effects take place at every level of the hierarchy and are specific to the level of generality. In other words, global motivation produces global outcomes, contextual motivation produces contextual outcomes, and situational motivation produces situational outcomes. For example, Susan may generally experience positive emotions, enhanced concentration, and better performance when playing sports because she has an intrinsic motivation in this context. However, she may experience less desirable consequences such as negative emotions, decreased concentration, and poorer performance in another context (e.g., at school) because she is more extrinsically motivated in this context. At all levels of generality, intrinsic motivation leads to the most positive outcomes whereas extrinsic motivation produces the least positive consequences. Once again, research supports these postulates.

In sum, the HMIEM offers a comprehensive explanation of athletes', coaches', referees', and

fans' motivation from multilevel, personality, and social psychological perspective.

Daniel R. Lalonde and Robert J. Vallerand

See also Passion; Self-Determination Theory

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K

KNOWLEDGE STRUCTURE

Knowledge about tasks and the environment is organized hierarchically in the human cognitive system, involving the diverse long-term memory (LTM) systems and working memory. Knowledge structures underlying task performance are fundamental elements of action control in sports. Experts' skills are often based on the efficient access to relevant task knowledge as well as on enhanced physical abilities. In the history of research in general psychology, motor control, and sport psychology (SP), knowledge structures have been addressed in different ways, focusing on their significance for task performance from different perspectives.

Structured Motor Programs, Schemata, and Events

In the field of motor control, Richard A. Schmidt's generalized motor program (GMP) theory has been one of the most prevalent theories used to explain how we control coordinated movements and how we therefore develop structured plans and knowledge. GMP theory is based on feedback-based models, such as James A. Adams's closed-loop theory of motor learning, and adopts the concept of schema, making it more adaptable and thereby a more realistic explanation of motor learning, especially as it occurs in sporting contexts. Schmidt describes GMPs as templates that serve as abstract plans for basic movements. GMPs can be adapted

to individual situations by modifying variant and invariant parameters. Parameters such as absolute timing, absolute force, and building blocks (individual muscle actions) within the GMP can vary (variant parameters). In contrast, the sequence, relative timing, and relative force of these building blocks are thought to remain invariant.

Schmidt uses the term *schemata* (recall schema, recognition schema) to explain how movements can be learned on the basis of GMPs. Support for a structured organization comes from research investigating the underlying neuronal structures of movement. According to these studies, the basic definition describes organization principles in action and perception that are based on neuronal structures but include functional levels of behavior and cognition up to and including language. According to Michael A. Arbib's theory, motor schemata and perceptual schemata interact in complex ways but are coordinated by control programs that generate and control action performance on a higher level.

More recently, event segmentation theory, developed by Jeffrey M. Zacks, has provided an interesting approach to studying structure and modularity in perception and knowledge memory, showing that the way in which a person spontaneously segments an observed action determines the understanding and later memorizing of this action. This approach can be well applied to sporting contexts. When watching a basketball match, we spontaneously segment the observed stream of movements performed by the different athletes into meaningful actions. Our understanding of the

scenario is based on the way we store an event in memory and segment what we see, the number of segmentation points, the choice of cues, and so on. Naturally, the quality of our understanding is strongly influenced by our own experience and our structured knowledge about the type of action.

Declarative and Procedural Knowledge Structures

A very interesting and often-cited distinction of knowledge quality and structure has been provided in the adaptive components of thought (ACT) theory by John Anderson, a psychologist at Carnegie Mellon University. ACT and later versions such as ACT-R have been inspired by the idea of Allen Newell to develop a cognitive architecture of knowledge. Such a cognitive architecture is computationally implemented in a technical system and is able to simulate a wide range of tasks. ACT is based on production systems. As do other psychological theories, it differentiates between declarative knowledge and procedural knowledge. Knowledge about facts (e.g., a BMW is a German car) has been called *declarative knowledge*. The relevant knowledge elements are facts and events. This knowledge has been characterized as explicit knowledge and as *knowing that*. *Procedural knowledge* about, for instance, bicycle riding is implicit knowledge about how to perform a task. It has been therefore characterized as *knowing how*. Both declarative and procedural knowledge are stored in LTM and are related to each other especially in the learning process. Neurophysiological research has shown that their storage and retrieval are based on different brain structures.

Anderson has described the development of knowledge structures within three phases of motor skill learning. The first phase is characterized as a cognitive phase. In this phase, the learning agents are developing a declarative encoding of the motor skill. They store a lot of facts and explicit elements of the learning process in memory. The second phase is an associative phase. In this phase, single elements of procedural and declarative knowledge become more strongly interconnected, and motor production rules are created. In this phase, it is possible to perform some motor acts step by step and to develop a declarative frame for motor activities. The last phase is the autonomous phase. In this phase, the motor procedures become faster and more and more automated.

Based on perspectives coming mostly from the ACT theory, a number of researchers have addressed the question of how declarative knowledge is structured and networked in sport actions. The major issue of relevance in the SP domain is whether we can confirm that improved performance is accompanied by a higher degree of order formation in the sense of knowledge structuring and hierarchies. Research has therefore been undertaken to confirm expertise-dependent differences in the classification and representation of context-specific problem states in, for instance, springboard divers, judokas, triathletes, and weight lifters. Research on springboard diving has revealed that the nodes of the representation structures in experts possess far more features than those of novices. This result replicates findings in the problem-solving domain. Likewise, expert springboard divers show a greater number of connections between nodes, just as do experts in problem solving. An interesting study about the development of declarative knowledge structures and performance in sports was conducted by Karen E. French and Jerry R. Thomas. They assessed various components of basketball performance (e.g., control of the basketball and cognitive decisions, dribbling and shooting skills) along with declarative knowledge in children ages 8 to 12. Declarative knowledge was measured via a paper-and-pencil test. Results confirmed relationships between knowledge and the decision component of performance, suggesting that knowledge plays an important role in skilled sport performance.

Studies using categorization tasks have shown that experts classify problems according to underlying functional principles, whereas novices operate more strongly with superficial features. Furthermore, questionnaire methods and interviews have revealed the structure and organization of movement knowledge in sports.

Mental Knowledge Representation and Hierarchies in Memory

The idea that actions are mentally represented in functional terms as a combination of action execution and the intended or observed effects is well established in cognitive psychology and has received growing acceptance in the fields of motor control and SP. Perceptual-cognitive approaches, such as the ideomotor approach to action control, propose that motor actions are formed by mental

knowledge representations of target objects, movement characteristics, movement goals, and the anticipation of potential disturbances. David A. Rosenbaum and coworkers demonstrated that movements can be understood as a serial and functional order of goal-related body postures, or goal postures, and their transitional states. The link between movements and perceptual effects is bidirectional and based on information that is typically stored in a hierarchical fashion in LTM. Complex movements can be conceptualized as a network of sensorimotor information. The better the order formation in memory, the more easily information can be accessed and retrieved. This leads to increased motor execution performance, which reduces the amount of attention and concentration required for successful performance. The nodes within this knowledge network contain functional subunits or building blocks that relate to motor actions and associated perceptual (including related semantic) content. These building blocks can be understood as representational units in memory that are functionally connected to perceptual events or as functional units for the control of actions, linking goals to perceptual effects of movements.

Research in complex actions in sports, dance, rehabilitation, and manual action has demonstrated that basic action concepts (BACs) are fundamental building blocks at the cognitive level of representation. BACs are based on the chunking of body postures related to common functions in the realization of action goals and are conceptualized as representational units in LTM that are functionally connected to perceptual events. From this point of view, action control is organized as perceptible event through a structures representation of anticipated characteristic (e.g., sensory) effects, with the corresponding motor activity automatically and flexibly tuned to serve these effects.

The integration of representation units, like for instance BACs, into structures of representation has been studied with a wide range of methods. Based on a new experimental approach, Thomas Schack and Franz Mechsner studied the tennis serve to investigate the nature and role of LTM in skilled athletic performance. In high-level experts, these representational frameworks were organized in a distinctive hierarchical treelike structure, were remarkably similar between individuals, and were well matched with the functional and biomechanical demands of the task. In comparison, action

representations in low-level players and nonplayers were organized less hierarchically, were more variable between persons, and were less well matched with functional and biomechanical demands.

The results of different studies in golf, soccer, windsurfing, volleyball, gymnastics, and dancing have shown that the mental representation structures in place relate clearly to performance. As different studies have shown, these representations are also position- and thereby task-dependent. These representation structures are the outcome of an increasing and effort-reducing formation of order in LTM. This order formation reveals a clear relation to the structure of the movement. With increasing expertise, the representation of the movement corresponds more and more exactly to its topological (spatiotemporal) structure. At this level, the representation has nothing to do with a muscle-oriented effector code. Evidently, the representation structures are formed through the sensory movement effects of distinctive node points (body postures) of the movement. Therefore, the representation structure itself possesses spatiotemporal properties, corresponding with the structure of the movement. Accordingly, movement control becomes possible by representing the anticipated intermediate effects of the movement and comparing them with incoming effects. Importantly, this also means that no special translation mechanism is required between perception, representation, and movement.

Results from another line of experimental research have showed that not only the structure formation of mental representations in LTM but also chunk formation in working memory is built up on BACs and relates systematically to movement structures. These studies have revealed a plausible relation between chunking and priming processes in working memory and the structure of human movements, suggesting a movement-based chunking. Such findings provide experimental evidence that structures in movement and memory mutually overlap.

These results and perspectives are of central meaning for the understanding of cognitive learning and coaching processes. A disadvantage of traditional procedures in mental training (imagery) is that they try to optimize the performance through repeated imagination of the movement without taking the athlete's mental knowledge representation into account (i.e., they are representation-blind). Problems arise if the movement's cognitive

reference structure has structural gaps or errors, as these will tend to be stabilized rather than overcome by repeated practice. The alternative approach is to measure the mental representation of the movement before mental training and then integrate these results into the training. This mental training based on mental representations has now been applied successfully for several years in professional sports such as golf, volleyball, gymnastics, and windsurfing and recently in the rehabilitation of hand functions in patients after stroke.

Thomas Schack and Bettina Bläsing

See also Chunking/Dechunking; Expertise; Information Processing; Memory; Pattern Recognition and Recall

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KÖHLER EFFECT

See Social Processing Effects



LAWS OF MOVEMENT LEARNING AND CONTROL

Various laws of movement learning and control have been proposed on the basis of research. In this entry, the focus is on three of the most firmly established of these laws: the law of practice, Fitts's law, and Hick's law. These laws are of interest to sport and exercise psychologists because they specify relatively simple relationships between different variables related to movement learning and control. The defining feature of these laws is that the relationships they specify apply to many different populations (e.g., males and females, the young and the elderly) and types of movement in sport and exercise. The laws are useful because they afford the identification of practice-related conditions that promote movement learning and predictions about how individuals will perform in various situations requiring movement. As such, the laws provide a basis for the design of practice and training curricula, schedules, and environments and also for the design of equipment for sport and exercise. The laws are described here, in turn.

The Law of Practice

The *law of practice* states that more practice of a motor task will lead to more learning of that task. This definition is usually accompanied by a second statement: Changes in motor task performance that follow practice are generally large and rapid

at first and become gradually smaller with continued practice. As an example, consider a scenario in which 30 novice gymnasts are each provided with 10 practice sessions, in which the gymnast attempts to stand one-footed on the beam, with the other leg bent at the knee, without wobbling or falling off the beam. Each practice session is separated by a 3 days' rest, lasts 2 hours, and begins with one test of how long, in seconds, the gymnast can do the task before they experience an obvious wobble or a fall. Consider then that the average of the 30 gymnasts' test scores is calculated for each of the 10 days' tests and the average values are plotted on a graph against the practice sessions undertaken. The law of practice predicts that the graph would look something like the (hypothetical) example graph shown in Figure 1. Note from the graph that, consistent with the law of practice, the increase in time spent balanced one-footed on the beam is greatest early in practice, between Practice Sessions 1 and 2, and becomes smaller with more practice; the smallest increase occurs between Practice Sessions 9 and 10. One interpretation of this effect is that the rate of improvement at any point in practice is determined by how much learning remains. Early in the practice of a given task, much remains to be learned and thus the rate of learning is very rapid. After extensive practice, there is not much left to be learned, so learning is much slower. At this late stage of learning, it can take months or even years of practice before any noticeable gain in performance is attained. As with all of the laws described here, the law of practice applies to many different populations and types of

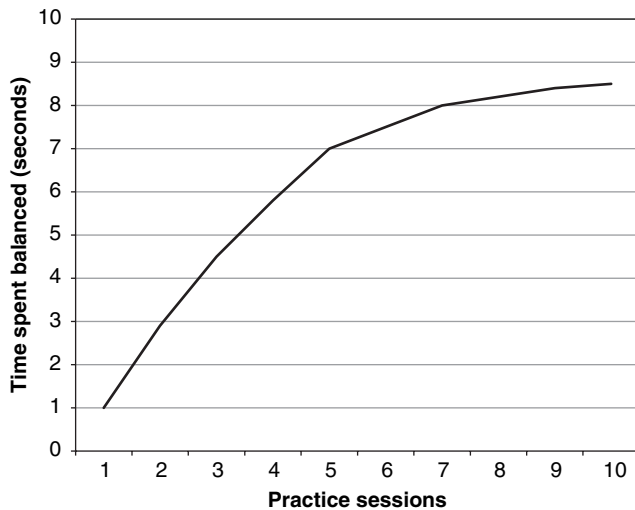


Figure 1 A Hypothetical Example Demonstrating the Relationship Between Task Performance (Time Spent Balanced One-Footed on a Gymnastics Beam) and Amount of Practice as Predicted by the Law of Practice

Source: David W. Eccles.

movement task, from turns in skiing to throws in discus.

It is important to define practice in relation to the law of practice. Practice activities can differ widely, and these differences are not always appreciated, leading some individuals to engage in what they consider to be practice activities only to be disappointed when little or no learning follows. While the exact qualities of practice activities affording optimal learning in sport and exercise are in contention, Anders Ericsson and his colleagues have proposed the term *deliberate practice* to capture some of the qualities that appear associated with learning. Deliberate practice is structured, purposeful practice relevant to improving performance. It requires concentration and/or effort and is often undertaken alone to allow the learner to concentrate; these characteristics make the *process* of engaging in deliberate practice inherently unenjoyable, even if the *result* of deliberate practice is enjoyable, such as the acquisition of a new skill. Deliberate practice activities are often focused on improving weaker components of current performance—for example, a backhand volley in tennis. Coaches are often employed to identify these components and prescribe specific deliberate practice activities to improve them—for example, drills to practice backhand volleys at various distances from the net. These components

are often practiced in isolation from other components of the sport, which affords an increase in the rate of attempts to practice the component; a tennis player might devote an entire practice session to practicing her backhand volley.

Thus, the deliberate practice of a task contrasts with mere engagement in, and experience of, a task. A common example of this difference concerns golf. A golfer seeking to improve his game adds a nine-hole round with friends to his weekly golf schedule, which he considers extra practice but no improvement in his game results. On closer inspection, however, the holes are played in a relaxed way and the “practice” session lacks the qualities of deliberate practice. Thus, this golfer’s addition of nine holes to his weekly golf schedule affords him the opportunity to obtain more experience of the game but does not involve its deliberate practice. Hence, there is no improvement in the golfer’s game.

Fitts’s Law

Fitts’s law describes in more formal terms what is known in everyday terms as the speed-accuracy trade-off; that is, it describes the relationship between the speed of a movement and its accuracy. In the 1950s, researcher Paul Fitts created an experiment in which two targets were laid upon a table-type surface in front of a participant; one target was positioned slightly to the left of the participant and the other slightly to the right. The participant’s task was simply to tap anywhere on the surface of the left target and then the right target using a stylus, which was a penlike rod, held in one hand, and then repeat this (i.e., left target, right target, left target) as fast as possible for a specific time such as 30 seconds. The task was like hitting one drum with a drumstick, and then another drum (with the same drumstick in the same hand), then the first drum, and so on, as fast as possible. In the experiment, the width of the two targets could be varied; for example, they each might be small, at 2 cm in width, or large, at 10 cm in width. Also, the distance between the targets could be varied; for example, they might be closely spaced, with 10 cm between them, or farther apart, with 20 cm between them. The task was scored as the number of taps the participant could make in the time allowed. Accuracy was stressed by asking the participant to make sure that no more than 5% of the taps fell outside of

a target. Fitts calculated the average time taken between two consecutive taps by dividing the time over which the participant was requested to tap (e.g., 30 seconds) by the number of taps made: If 30 taps were made in 30 seconds, the average time between 2 consecutive taps was 1 second. In other words, it took the participant a second on average to tap one target and then move his or her hand to the other target and tap that target. Fitts called this variable *movement time* (MT).

Fitts explored what happened to MT when the width of each target and the distance between the targets was varied. When targets at a fixed distance apart (e.g., 10 cm) each had their width reduced (e.g., from 4 cm to 2 cm), MT increased; participants appeared to need to move slower to be more accurate in the spatial positioning of their taps. Also, when targets of a fixed width (e.g., 4 cm) were moved farther apart (e.g., from 10 cm to 20 cm), MT slowed. Participants appeared to need to take longer to move their hand over the extra distance if they were to maintain the spatial accuracy of their tapping. While these findings were interesting, Fitts's really important finding was that MT increased by a constant amount whenever the distance between the targets was doubled or the size of each target was halved. Furthermore, this finding was basically the same for all the participants tested. This meant that Fitts was able to predict, with reasonable accuracy, the MT of participants he had not yet tested, based simply on the width of the targets and their distance apart. Of course, not every task involving movement is like Fitts's tapping task, but since his research, studies involving different populations and different types of movement task have yielded findings consistent with his own. These consistent findings have led to Fitts's early observations being considered a law of motor control.

One explanation for Fitts's law, in simple terms, is as follows. An increase in the speed with which a given limb (e.g., an arm) is moved requires an increase in the impulse applied by the muscles to that limb; note that an impulse is the amount of force applied multiplied by the time for which that force is applied. A characteristic of the human movement system is that there is a slight and unavoidable difference between the impulse an individual intends to apply to a limb to make it move and the impulse that the individual actually applies to the limb. This difference can be considered error. Research has indicated that the larger

the impulse applied, the greater the error. Now let us put these separate statements together: When humans attempt to move a limb faster, they must apply a larger impulse to the limb: The larger the impulse applied to the limb, the larger the error in the impulse applied; the larger the error in the impulse applied, the greater the inaccuracy in the spatial positioning of the limb.

While tasks involving simple movements like Fitts's tapping task offer the researcher a chance to study motor control under controlled laboratory conditions, these movements might seem at first glance to be of limited relevance to sport and exercise. However, simple movements form the basis for the more complicated movements made in sport and exercise and, as such, Fitts's law is reflected in sport- and exercise-related movements. For example, pole-vaulters take a long time to learn to place the pole tip into the box (i.e., the hole in the ground) correctly at speed; until then, they use a shorter approach, which affords a relatively slow approach run, to achieve an accurate tip placement. In addition, rock climbers who attempt to reach up very quickly to grab the next hold, because they feel their other points of contact with the rock are tenuous, are less likely to accurately place their fingers on the desired hold.

Hick's Law

Hick's law describes the relationship between the time taken to prepare a movement response and the number of possible movement response alternatives. In more everyday terms, the law states that individuals are slower to react when required if it is not clear *before* they are required to react exactly *how* they should react—that is, if it is not clear what kind of movement response is required. An example of Hick's law can be seen in football. Consider a down in which the defense is unclear about what type of offensive play the opponent is likely to run: When the ball is snapped, the defense will be relatively slow to react with an appropriate response. In contrast, if the defense knows exactly what offensive play the opponent is going to run, the defense will be relatively fast to react with an appropriate response when the ball is snapped.

In the 1950s, researchers William Edmund Hick and Ray Hyman studied reaction time (RT) using what has become known as a "choice reaction time" task. In one type of this task, the participant

is seated, with one hand laying palm-down on a “response panel,” and each of the four fingers of the hand resting on a separate button mounted in the panel. In front of the participant, four lights are presented in a row (i.e., horizontally) at head height. Each light is considered a *stimulus*, to which the participant must respond as fast as possible with a movement. Specifically, when a light is lit, the participant must press the button, from the four available, that corresponds to the lit light. Thus, if the leftmost light is lit, she must press the leftmost button with her index finger (if she is right-handed), whereas if the rightmost light is lit, she must press the rightmost button with her little finger, and so on. Usually, participants are unable to predict exactly which light will be lit or when it will be lit and thus are unable to initiate a response in advance of a light being lit. RT is measured as the time taken from a light being lit to the beginning of the participant’s finger movement on the button.

In the choice RT task presented previously, there were four “stimulus–response” (S–R) choices or alternatives; that is, the participant could be presented with one of four stimuli, and each of these stimuli required a unique response. For example, when the second-to-left light was lit, a press of the second-to-left button was required by the middle finger. Hick and Hyman studied the relationship between RT and the number of S–R alternatives by presenting various numbers of lights associated with an equal number of buttons to be pressed. One study condition presented to the participant might feature two S–R alternatives, involving two lights and two buttons, whereas the next study condition presented might involve four S–R alternatives, as in the previous example. The researchers found that RT slowed as the number of S–R alternatives increased. In other words, when it was less clear how they would be required to respond, participants were slower to react.

However, Hick and Hyman’s findings extended beyond identifying that RT slowed under these circumstances. What these researchers discovered was that RT slowed by a constant amount, which was approximately 150 milliseconds on average for the type of choice RT task described here, every time the number of S–R alternatives was doubled. Of course, RT was also quickened by this amount every time the number of S–R alternatives was halved. Thus, when a “one light and one button condition” was changed to a “two lights and two

buttons” condition, RT slowed by 150 milliseconds on average, and when an “eight lights and eight buttons” condition was changed to a “two lights and two buttons” condition, RT quickened by 300 milliseconds on average.

Researchers have found that Hick and Hyman’s findings apply to different populations and different types of movement task, which has led to the finding being considered a law of motor control. One explanation for Hick’s law, in simple terms, is that, when there are more S–R alternatives, the participant must process (i.e., think about) more information to be able to identify and produce the appropriate response. As humans can only process a limited amount of information in a given time, RT is slower when there is more information to be processed.

As with Fitts’s law, the choice RT task described here might seem at first glance to be of limited relevance to movements made in sport and exercise. However, Hick’s law helps explain performance in real sports, especially sports in which responding faster conveys a performance advantage. Consider a defensive team in football that knows the opponent’s offense has two key running plays and two key passing plays and then observes the opponent’s only ball carrier get injured so that running plays are no longer an option. At the line of scrimmage, the defense knows there are now only two possible play options, not four. In terms of Hick’s law, the number of S–R alternatives has been halved, affording the defense a quicker RT (by 150 milliseconds) at the snap. Hick’s law is also reflected in findings from studies of skilled athletes’ preparation for upcoming competitions. For example, David Eccles and his colleagues presented evidence that skilled athletes gather information affording the identification of (and thus a reduction in uncertainty about) the types of stimuli that will be presented during a competition. An example of this process is when athletes and coaches in soccer or football watch game film of an upcoming opponent to identify patterns of play unique to that opponent or when kayakers and orienteers explore the river and forest, respectively, in which they will compete (under time pressure) to identify in advance the demands unique to these environments. According to Hick’s law, these information-gathering activities, which reduce uncertainty about the types of stimuli that will be presented during a competition, lead to quicker RTs during that competition.

Conclusion

In conclusion, there are three well-established laws of movement learning and control: the law of practice, Fitts's law, and Hick's law. The law of practice describes the relationship between practice and learning; Fitts's law describes the relationship between movement speed and accuracy; and Hick's law describes the relationship between the time taken to prepare a movement response and the number of possible movement response alternatives. The relationships described by these laws apply to many different populations and types of movement in sport and exercise. The laws afford the identification of practice-related conditions that promote movement learning and predictions about how individuals will perform in situations requiring movement. As such, the laws are fundamental within the discipline of sport and exercise psychology (SEP).

David W. Eccles

See also Anticipation; Information Processing; Learning; Motor Control; Motor Learning; Practice; Skill Acquisition

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LEADERSHIP IN SPORT: MULTIDIMENSIONAL MODEL

An established model of leadership in sports is Packianathan Chelladurai's multidimensional model of leadership (MML). This model was the substance of a doctoral dissertation in management

science. It represented a synthesis and reconciliation of the models of leadership found in the mainstream management literature. These preexisting models tended to focus more on either the leader, or the member, or the situation. However, as leadership is a concept that encompasses all three factors—the leader; the members; and the organizational context including goals, structures, and processes—it was reasonable to propose the model illustrated in Figure 1.

A unique feature of the model is that it includes three states of leader behaviors. *Required behavior* (Box 4) is the set of prescriptions and proscriptions of the situation in which leadership occurs. Required behavior is mostly defined by the situational characteristics (Box 1) that include the goals of the group, the type of task (e.g., individual vs. team, closed vs. open tasks), and the social and cultural context of the group. The nature of the group defined by gender, age, skill level, and such other factors would also partly define required behavior. *Preferred behavior* (Box 6) refers to the preferences of the followers for specific forms of behavior (such as training, social support, and feedback) from the leader. Members' preferences are a function of their individual difference characteristics (Box 3) such as personality (e.g., need for affiliation, tolerance for ambiguity, attitude toward authority) and their ability relative to the task at hand. Members are also aware of the situational requirements; thus, their preferences are influenced by those requirements. The *actual behavior* (Box 5, i.e., how the leader actually behaves) is largely based on leader characteristics (Box 2) in terms of personality, expertise, and experience. However, the leader would also be constrained to abide by the requirements of the situation (Box 4) and to accommodate member preferences (Box 6) as well.

Another significant feature of the MML is its *congruence hypothesis*. That is, the model specifies that the desired outcomes of individual and team performance, and member satisfaction will be realized if the three states of leader behavior are congruent with each other. Any misalignment among the three states of leader behavior would diminish performance and/or satisfaction. Further, if there is continued discrepancy between actual leader behavior and the other two states of leader behavior, the leader's position within the group would become untenable. The dynamic nature of leadership is highlighted in the model with backward arrows indicating feedback from attained

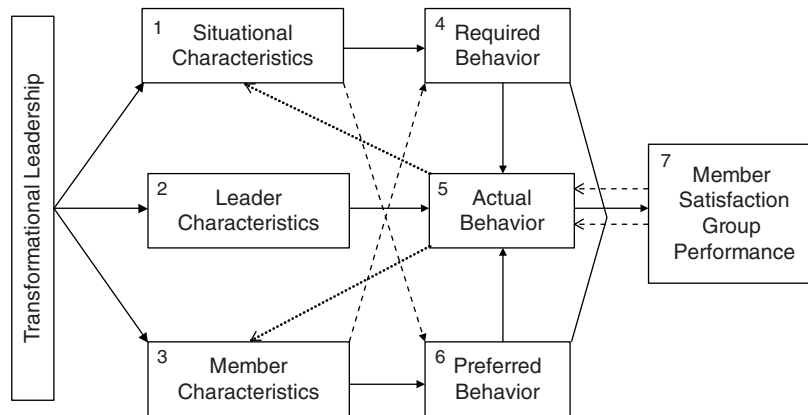


Figure 1 Multidimensional Model of Leadership

Source: Adapted from Chelladurai, P. (2007). Leadership in sports. In G. Tenenbaum & R. C. Eklund (Eds.), *Handbook of sport psychology* (p. 17). New York: Wiley. Used by permission.

performance and/or member satisfaction. That is, the leader may begin to exhibit more of task-oriented behaviors if she or he feels that the performance was below expectations. On the other hand, leader behaviors may begin to be more interpersonally oriented if it is felt that members were low on morale and/or satisfaction.

In 2007, Chelladurai made one significant modification to the model. It was the incorporation of the concept of transformational leadership into the multidimensional model. In his view, leadership exhibited by coaches is largely concerned with pursuit of excellence. In the process of pursuing excellence, the person is transformed from a relatively unaccomplished novice into an expert performer. Thus successful coaches do exhibit transformational leadership and, as such, incorporating the concept into the model was necessary as well as easy. In essence, the coach as the leader transforms member characteristics in terms of aspirations and attitudes and changes the situational requirements by articulating a new mission and convincing the members of the viability of the mission and their capacity to achieve that mission.

The idea (as shown in Figure 1) that transformational leadership influences leader characteristics (Box 2) would be most relevant where a coach has one or more assistant coaches. That is, the chief coach would attempt to transform the assistant coaches in the same way he or she would transform player characteristics. If there is only one coach, it would mean that the coach would change his or her own characteristics to fit the transformational mold and attempt to change the

situational characteristics as well as the characteristics of the members as indicated by the dotted arrows flowing from actual behavior (Box 5) to situational characteristics (Box 1) and to member characteristics (Box 3).

A theory is useful only to the extent that the variables of the study can be measured and the relationships among the variables can be verified. With this in mind, Chelladurai developed the Leadership Scale for Sports (LSS) to measure the three forms of leader behavior contained in the model. It is composed of 40 items to measure these five dimensions of leader behavior: *training and instruction* (13 items), *democratic behavior* (9 items), *autocratic behavior* (5 items), *social support* (8 items), and *positive feedback and rewarding behavior* (5 items). The response format is a 5-point scale ranging from (1) always; (2) often, about 75% of the time; (3) occasionally, about 50% of the time; (4) seldom, about 25% of the time; to (5) never. The scale has been used to measure athletes' preferences, their perceptions of their coaches' behavior, and coaches' perceptions of their own behavior pertaining to those five dimensions of behavior.

The psychometric properties of the LSS have been verified and supported in several studies. However, the subscale of *autocratic behavior* has been shown to be weak in almost all studies. One reason for such low internal consistency estimates is that the items in the subscale relate to three different forms of behavior—being aloof, being authoritative, and making autocratic decisions. Another conceptual issue that plagues the dimension of autocratic behavior is that whether a coach

should be autocratic or democratic is dependent on the attributes of the problem in question. The items in this subscale do not capture the situational contingencies. The entry on “Decision-Making Styles in Coaching” (this volume) deals with contextual differences that indicate the degree of participation by team members in decision making.

There has been an attempt to improve the LSS by James J. Zhang and his colleagues. Their Revised LSS includes the five dimensions, the instructions, and the response format of the original LSS. It also includes a new dimension titled *situational consideration* behaviors. However, the Revised LSS has not been subjected to confirmatory analyses and the new dimension is subsumed by the original five dimensions. Therefore, parsimony would dictate the use of the original LSS.

Commentary

Both the MML and the LSS have been used and tested in several studies. Much of the research on the notion of congruence suggested in the model has been restricted to only two states of leader behavior—preferred behavior and perceived behavior used as a proxy for actual behavior. While the notion of congruence between these two states of leader behavior has been largely supported, it is disappointing that the congruence among all three states of leader behavior has not been tested adequately. Researchers could have omitted required behavior from consideration because of the difficulty of measuring required behavior. In his original research, Chelladurai employed the average of coaches’ reporting of their own behavior as a surrogate of required behavior. Future research may consider asking expert coaches specifically about what should be the required behavior in a given situation defined by age, gender, ability level, goals of the program, and so on. The average of these responses may be used as required behavior in the relevant situation.

In developing the LSS, Chelladurai resorted to the leadership scales then popular in the mainstream management literature such as the Leadership Behavior Description Questionnaire (LBDQ). He collated more than 100 items from these scales and reworded them to suite the coaching context. However, he was forced to reduce the scale to 99 items because the computers available at his university at that time could not handle more than that number of items. He administered the initial questionnaire to members of the

university basketball teams in Canada and derived five dimensions of leader behavior through exploratory factor analysis. Subsequent research studies employing the more sophisticated confirmatory factor analysis has supported the robustness of the LSS. While these steps are acceptable, it is necessary to generate sport specific items, group them into meaningful categories, administer the questionnaire to different teams in different sports, and subject the data to confirmatory factor analyses. Furthermore, the LSS does not tap into the dimensions of transformational leadership, which has been recently incorporated into the MML. It is expected that future research will focus on refining the existing subscales and developing new subscales for transformational leadership.

Finally, it must be noted that although the MML was advanced in research related to leadership in athletics, the model itself is applicable to any context (e.g., business, industry, military) where leadership is a critical process. The model, after all, is a synthesis of other models from business and industry. Thus, reversing the process and applying the model to other contexts including business and industry is feasible. While the situational and member characteristics may vary from context to context, the concepts of required, preferred, and actual behavior are meaningful in any context. On a different note, Chelladurai included group performance and member satisfaction as the outcome variables. But other outcomes such as individual performance, individual growth, group cohesion, group solidarity, commitment, identification, and organizational citizenship can easily be accommodated in the model. Further, business outcomes such as profitability, market share, and return on investments can be used as outcome variables.

Packianathan Chelladurai

See also Autonomy-Supportive Coaching; Coach–Athlete Relations; Coaching Efficacy; Decision-Making Styles in Coaching; Group Characteristics; Leadership in Sport: Transactional and Transformational Learning; Leadership in Sport: Trait Perspectives; Team Building

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LEADERSHIP IN SPORT: SITUATIONAL AND CONTINGENCY APPROACHES

Shortly after World War II, Ralph Stogdill published a highly influential review paper in which he concluded that effective leadership is not derived through the expression of some set of personality traits but is invariably dependent on a range of situational factors that include the context in which leaders find themselves, as well as the alignment of various personal characteristics of the leader with the qualities, activities, and goals of those being led. In essence, Stogdill suggested that effective leadership occurs as a result of the interaction that takes place between a person and his or her environment.

Fiedler's Contingency Model

Over the next three decades (1950s to 1970s), a number of theoretical frameworks emerged that sought to examine the various situational characteristics upon which effective leadership is *contingent*. Perhaps the most widely studied situational or contingency model of leadership corresponds to Fred Fiedler's contingency model. Although the majority of research that tested this model was conducted within organizational units, it is

noteworthy that his foundational work in this area was conducted with high school basketball teams. According to Fiedler, leadership effectiveness is contingent upon both the leader's preferred style of interacting with others and the favorableness of the situation (also referred to as *situational control*). Fiedler suggested that leaders are typically either task oriented or relationship oriented, with task-oriented leaders primarily interested in maximizing goal or performance attainment among followers, whereas relationship-oriented leaders are primarily concerned with maximizing interpersonal connections with followers. Fiedler further theorized that the effectiveness of the leader's personal style (i.e., task- vs. relationship-oriented) is dependent on three situational dimensions that include the quality of leader–member relationships, the clarity and structure of the task and goals being pursued, as well as the extent to which the leader possesses formal authority and power. When each of these is maximized, the situation is described as being most favorable—or in other words the leader has the greatest amount of situational control.

The core tenet of Fiedler's model is that under conditions of high *and* low situational control (or favorableness), a task-oriented leader is preferable; however, under conditions of moderate situational control, a relationship-oriented leader is theorized to be most effective. Specifically, Fiedler contended that a more direct, task-oriented approach is beneficial when members have complete clarity in terms of what is expected and the leader has established authority as well as support (i.e., high-quality relationships) from those being led. Conversely, when teams or organizations are beset with instability and volatility (i.e., low situational control), Fiedler suggested that a highly structured and task-oriented approach would also be most appropriate to instill order out of chaos. However, within teams or organizations beset by moderate situational control (e.g., involving some goal or task uncertainty or diminished power available to the leader), Fiedler suggested that a leader who can adeptly manage the interpersonal relationships among group members would be most likely to evoke better individual performances, rather than a leader who takes a more authoritative task-driven approach.

Although meta-analytic evidence has provided some support for Fiedler's model within organizational settings, research in this area has also been the target of considerable debate and criticism

on both conceptual and methodological grounds. One of the most vociferous concerns that has been leveled at work in this area corresponds to the assessment of leadership styles in these studies. Specifically, Fiedler's contingency model was operationalized by measuring a leader's interpersonal style via an *indirect projective assessment*. Specifically, the least preferred coworker (LPC) scale involves the appraisal, by the focal leader, of *another person* (i.e., his or her LPC) to make an inference about whether the leader's interpersonal style is task- or relationship-oriented. In sum, the very nature of what a LPC score actually represents has been markedly questioned, and with it the validity of the model itself. In the context of sport, other than Fiedler's early work with youth basketball teams, only a few studies have directly sought to test the theoretical tenets of this model, and these have generally demonstrated inconclusive and mixed results. For example, in one study by Anne Marie Bird, members of winning (i.e., theorized to reflect high situational favorableness) Division II U.S. university volleyball teams perceived their coaches to be more task-oriented, and losing (i.e., low situational favorableness) teams perceived their coaches to be more socioemotional. However, the opposite effect was found for Division I volleyball players, where winning teams perceived their coaches to be more socioemotional, and losing teams perceived their coaches to be more task-oriented.

House's Path-Goal Theory

Another prominent situational and contingency model of leadership corresponds to Robert House's path-goal theory. However, unlike Fiedler's model, which focused on the effectiveness of personality factors that were contingent on the situation, House's model focused on the effectiveness of various leadership *behaviors* that were contingent on various situational constraints. Specifically, the extent to which leaders engage in four types of leadership behavior (directive path-goal clarifying behavior, supportive behavior, participative leader behavior, and achievement oriented behavior) was theorized to translate into effective leadership (as indicated by improved affective states, motivation, and performance among followers) but that these leadership behavior-subordinate outcomes were theorized to be dependent on various task-related situational constraints (e.g., clarity or ambiguity of

task demands) as well as the attributes and characteristics of those being led (i.e., team member's preferences for independence). For example, path-goal clarifying behavior utilized by leaders was theorized to be particularly effective when followers experience role ambiguity but was expected to be a relatively redundant form of leadership behavior when followers are clear about their role responsibilities and task demands.

Chelladurai's Multidimensional Model of Leadership

Although research that directly sought to test the tenets of House's path-goal theory in sport settings is somewhat limited, early work by Packianathan Chelladurai and his colleagues found that different coaching behaviors are preferred in different sport contexts (e.g., closed vs. open sport) by athletes with different characteristics (e.g., males vs. females). Interestingly, Chelladurai built upon his initial tests of the path-goal model to develop his multidimensional model of leadership (MML). Specifically, the MML recognized that leadership effectiveness is contingent upon both the behaviors displayed by the leader *and* the preferences of the athletes being coached as well as the demands of the situation. Parenthetically, in addition to drawing from House's path-goal theory, Chelladurai also drew from elements of Fiedler's contingency model in developing his MML. Specifically, Chelladurai recognized that the behaviors utilized by coaches, and the effectiveness of these behaviors are, to some extent, shaped by the interaction of the coach's personal characteristics (e.g., personality traits) with the situational context in which the coach operates. In sum, the contingency and path-goal approaches developed by Fiedler and House, respectively, provided an important basis for Chelladurai's model, which to date has been one of the most extensively used models of leadership in sport.

Hersey and Blanchard's Situational Leadership Theory

The final contingency model described here corresponds to Paul Hersey and Ken Blanchard's situational leadership theory (also referred to as a *life cycle theory*). The core tenet of this theory is that the effectiveness of task- and relationship-oriented leadership behaviors is contingent upon the relative *maturity* levels of those being led.

According to Hersey and Blanchard, the extent to which task-oriented behaviors should be used decreases as the maturity of followers increase. Additionally, the degree to which relationship-oriented behaviors should be used forms an inverted U pattern, whereby at low and high maturity levels relationship-oriented behaviors should be used minimally, while at moderate levels of maturity they should be used more frequently. In workplace settings, empirical support for the life cycle theory has been mixed. In sport settings, the findings have been inconclusive at best and on a few occasions have been directly contrary to the theoretical postulates of this framework. For this reason, the life cycle theory has not generated much traction within the field of sport psychology (SP), although researchers and practitioners have generally recognized that coaches should adapt their coaching behaviors to the developmental levels (vis-à-vis maturity) of their athletes.

Conclusion

When taken together, the contingency approaches to leadership that have been developed over the past 60 years have provided an invaluable basis for recognizing that the utility of various leadership behaviors in sport is dependent, to some extent, on the situation in which leaders (i.e., coaches and captains) and followers find themselves (i.e., a *person × situation* interaction). Collectively, these frameworks have also played a major role in informing more recent (and sport-specific) models of leadership such as Chelladurai's MML.

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See also Leadership in Sport: Multidimensional Model;
Leadership in Sport: Social Cognitive Approaches;
Leadership in Sport: Trait Perspectives

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LEADERSHIP IN SPORT: SOCIAL COGNITIVE APPROACHES

The core tenets of social cognitive theory (SCT) focus on the interrelationship among three sets of factors—namely personal, environmental, and behavioral. These factors are often described as being part of a reciprocal causal network whereby environmental, personal, and behavioral factors interact to determine a range of attitudinal and behavioral consequences. One of the key underpinning factors of the SCT approach is that it recognizes that human interactions do not occur in a vacuum; rather, they occur in an ever-changing environmental context, and this context influences the nature and outcomes of these interactions. For example, an identical set of interactions may produce one set of outcomes in a particular context (e.g., elite level sport) but different outcomes in another context (e.g., a youth physical education [PE] setting). Another key underpinning assumption of SCT is that humans are all different and that these differences will influence the nature and outcomes of their interactions with others. That is, an identical set of interactions will have

a different impact depending on the personality of the individuals involved. For example, individuals high in self-esteem may respond differently to coach criticism than individuals low in self-esteem. SCT models of leadership that have been developed in the context of sport delineate the complex *social* and *cognitive* processes that influence coach behavior as well as the effectiveness of those behaviors on athlete and team outcomes. The major SCT models included in this entry are mediational model of leadership (MML), the coaching effectiveness model (CEM) of leadership, motivational model of the coach–athlete relationship (MMCAR), and coach-created motivational climate (CCMC).

Mediational Model of Leadership

The MML—developed by Ronald Smith, Frank Smoll, and their colleagues—delineates the complex interaction between person, environment, and behavior that typifies an SCT approach. The core of the model specifies that coach behaviors influence athlete perceptions and recall, which, in turn, influences athlete's evaluative reactions. One of the underlying principles of the MML is that the effects of coach behaviors is mediated through the *meaning* that athlete's give to their coaches' behaviors. That is, an athlete's appraisal of his or her coach's behaviors serves as the mechanism through which a coach influences that athlete's attitudes toward the coach and their sporting experience in general. For example, if an athlete perceives a coach to be supportive and helpful, he or she will likely have a positive attitude toward their coach and their sporting experience in general. These positive experiences will then likely lead to enhanced psychological well-being (PWB) (e.g., self-esteem), enhanced performance, and continued participation in their sport. Conversely, if an athlete views his or her coach as being overly critical and unsupportive, they are likely to form negative attitudes toward their coach and their sporting experience in general. This is likely to lead to a reduction in self-esteem, performance and reduced participation in their sport.

As a core tenet of this model, an understanding of the variables that can influence coach behaviors, athletes' perceptions and recall of these behaviors, and athlete evaluative reactions to these behaviors is critical. To this end, the MML identifies factors that are theorized to influence coach behaviors,

athlete perception and recall of coach behaviors, and also factors that are theorized to influence the athlete's evaluative reactions to coach behaviors. Coach behavior is hypothesized to be determined by three sets of factors including coach *individual difference variables*, *situational factors*, and *coach perceptions of athletes' attitudes*. For example, the MML hypothesizes that a coach's normative beliefs impact the way in which they behave. Coach normative beliefs refer to what the coach believes is his or her primary role. For example, one coach may believe that her primary role is to win, whilst another coach may believe her primary role is to promote personal growth and enjoyment. The MML hypothesizes that coaches with a normative belief of winning will display more punitive, authoritarian behaviors, and will be more regimented in their approach than those coaches who see their role as promoting athlete growth. The second broad factor that is theorized to impact coach behavior is situational factors. For example, current game status (i.e., winning, losing or tied) is theorized to impact coach behaviors such that coaches will tend to display more punitive behaviors when losing. The third broad factor that is proposed to influence coach behavior corresponds to coaches' perceptions of player attitudes. For example, a coach will likely behave differently toward an athlete who he believes has a contentious maladaptive attitude compared to an athlete who they believe has a positive attitude.

Athletes' perception and recall of coach behaviors are primarily determined by how coaches actually behave but also by a range of athlete individual difference variables and situational factors. For example, trait anxiety (TA) experienced by an athlete is an individual difference variable that reflects a dispositional tendency to selectively attend to threat-related information in the environment. Because of this attentional bias toward threatening stimuli it is likely that athletes with high TA will interpret a coach's behavior as being more threatening in nature (e.g., more punitive and less reinforcing behaviors) than those athletes with low TA. An example of a situational variable that might impact athlete perception and recall of coach behaviors is the pressure of the situation (e.g., practice vs. competition vs. championship finals). In highly pressured situations, athletes may become more aware of pressure-relevant cues in the environment, such as critical and controlling

coach behaviors. As such, athletes are likely to report experiencing more of these types of behaviors in highly pressured situations.

Athletes' evaluative reactions of their perceptions and recall of coach behaviors are also proposed to be influenced by athlete individual difference variables and situational factors. For example, following on from the previous rationale regarding the influence of TA on perception and recall of coach behavior (i.e., perceive that her coach is displaying more threatening behaviors), TA is also theorized to impact evaluative reactions to these coach behaviors. That is, athletes with high TA perceive punitive coach behaviors to be more threatening than those with low TA. Thus, not only will TA influence the amount of punitive coach behavior that they perceive themselves as receiving but will also influence the magnitude of the effect of these behaviors.

Studies that have tested the tenets of the MML have found that coaches who display more reinforcement of desirable behaviors and effort combined with encouragement and technical instruction toward mistakes are liked by their athletes; the athletes have more fun, and athletes like their teammates more. Coaches' self-ratings of their own behavior tend to be uncorrelated with athletes' ratings of those same behaviors. However, athletes' ratings of their coaches' behavior are often correlated with expert observers' ratings of coach behaviors. Furthermore, supportive coach behaviors are generally associated with positive attitudes toward the coach, and punitive behaviors are generally related to negative attitudes toward the coach. Finally, team win-loss records have been found to be unrelated to athletes' liking of the coach. However, athletes' perceptions of their parents liking of the coach are related to win-loss records (athletes on winning teams perceived that their parents liked their coach more than athletes on losing teams).

The MML formed the basis for the development of a prominent coaching effectiveness training program by Smith and Smoll that was designed to enable coaches to relate more effectively to their athletes. This training program involves targeting (and modifying) coaches' behaviors in order to enhance the sport experience of their athletes. Five guidelines underpinning this coaching effectiveness training program are to (1) focus on effort and enjoyment rather than winning at all costs, (2) emphasize a positive approach to

coaching, (3) establish norms around interpersonal helping behaviors, (4) involve athletes in decision making (DM), and (5) enhance self-awareness of coaches. This program has been shown to be effective in modifying coach behaviors, developing athlete self-esteem and enjoyment, liking of the coach and teammates, and reducing attrition rates in sport.

Coaching Effectiveness Model

The CEM, which was developed by Thelma S. Horn, is one of the most recent models of coaching that has emerged from a social cognitive perspective. The CEM, in some respects, can be considered an extension of the MML. Many of the predictions and pathways specified in the CEM are consistent with those specified in the MML. For example, both models highlight the importance of coach behaviors and athletes' evaluative reactions of these coach behaviors. However, the CEM also extended the MML by identifying and adding more explicit mediational pathways to the coaching process. These mediational pathways have been added to both the antecedents of coach behavior and the effects of coach behavior. For example, where the MML specifies that coach individual difference variables and situational factors can combine to impact coach behavior the CEM posits that the effects of these factors on coach behaviors is mediated by coaches' expectancies, values, beliefs, and goals. Furthermore, the CEM identifies that coach behavior directly impacting athlete performance has an indirect effect on athlete performance via athlete self-perceptions, beliefs, and attitudes. These in turn impact athlete motivation and performance.

The various linkages and relationships specified in the CEM are based on a wide range of theoretical models. For example, the link between coach expectations and coach behaviors is supported by principles of self-fulfilling prophecy theory. Achievement goal theory (AGT) is also used to provide a rationale for identifying factors that impact coach behaviors. Specifically, the CEM hypothesizes that coaches who have more of a task orientation display behaviors that are consistent with that task orientation and, for example, emphasize self-referent goals rather than goals that involve comparisons with others. Given that the core principles of the CEM are largely similar to the MML, the research that

supports the main tenets of the MML also support the CEM.

Motivational Model of the Coach–Athlete Relationship

Genevieve Mageau and Robert J. Vallerand's MMCAR is premised upon self-determination theory (SDT) contentions. Similar to the other models discussed in this entry, the MMCAR specifies antecedents and consequences of coach behavior. The main difference of the MMCAR from the other models is that the MMCAR is almost exclusively grounded in SDT. In line with the principles of SDT, the MMCAR hypothesizes that the impact of coach behaviors on athlete motivation is mediated by the satisfaction of the three basic needs for *autonomy*, *relatedness* and *competence*. That is, the more coaches satisfy these three basic needs, the greater the athlete's self-determined motivation will be.

The MMCAR also identifies the coach's personal orientation (either autonomy supportive or controlling), coaching context, coach perception of athlete behavior, and motivation as determinants of coach behavior. For example, coaches' expectations of their athletes' ability are likely to influence the behavior that they display toward athletes. Specifically, according to the MMCAR, a coach who has low performance expectations of an athlete is likely to send messages of mistrust, emphasize mistakes, and ignore success in that athlete (i.e., exhibit behaviors that are likely to thwart need satisfaction). The MMCAR also identifies structure that a coach provides as being important in forming athlete perceptions of competence. Namely, structure in the form of specific instructions from the coach provides athletes with information by which to gauge their progress. Without sufficient feedback, athletes cannot experience progress, thus thwarting their need for competence. Lastly, coaches' involvement with the athlete (i.e., demonstrating care for the athlete's welfare) is theorized to make the athlete feel more related to the coach and thus satisfy their need for relatedness.

The MMCAR posits eight key indicators that are theorized to be consistent with the satisfaction of the three basic needs in athletes: provide a rationale for tasks, acknowledge athletes' feelings and perspectives, provide athletes with opportunities for initiative taking and independent work, provide

noncontrolling competence feedback, avoid overt control, avoid criticisms and controlling statements, avoid tangible rewards for interesting tasks, and prevent ego involvement in athletes. There is a large body of evidence that supports the core principle of the MMCAR that satisfaction of the basic needs for autonomy, relatedness, and competence is positively associated with the internalization of motivation. There is also research evidence that supports the notion that coaches who display an autonomy supportive style of leadership have positive effects on athletes, whereas coaches that display more controlling behaviors tend to have negative effects on their athletes.

Coach-Created Motivational Climate

Developed by Joan Duda and Isabel Balaguer, the CCMC approach to examining leadership in sport is grounded in the principles articulated in AGT. Consistent with the other models of leadership described in this entry the CCMC specifies antecedents of coach behavior (e.g., coach individual difference variables, situational characteristics, and coaches' perceptions of athletes' attitudes), and it also delineates the consequences of coach behavior via specific mediating mechanisms. However, the CCMC diverges from the other models in that all the effects of coach behavior are proposed to operate predominately through the creation of task-involving and ego-involving climates. Based on the work of John Nicholls, AGT posits that there are two fundamental higher-order goals—namely, task and ego goals. Task goals refer to goals that are primarily articulated around self-referent criteria and involve factors such as effort and improving skill level, whereas ego goals are primarily articulated around the demonstration of superior competence in relation to others. The CCMC model posits that coaches create climates that promote either task-involvement or ego-involvement, which in turn impact the types of goals that athletes set. The types of goals that athletes set (or the reference by which they evaluate their performance) have an impact on their own attitudes, motivation, and behavior.

A substantial body of research has generally supported the predictions of the CCMC. For example, task-oriented climates have been found to be related to a large number of positive outcomes, such as self-determined motivation, satisfaction, effort, enjoyment, persistence, team

cohesion, collective efficacy (CE), and sportsmanship behavior among athletes. Furthermore, ego-oriented climates have been found to be related to maladaptive coping strategies, anxiety, worry, tension, and amoral sport behaviors.

Conclusion

The social cognitive approaches to leadership that have been developed within the field of sport psychology provide strong evidence that coaches play a significant role in shaping athletes' experiences. The models and theories of coach leadership that have emerged from the social cognitive perspective have identified in excess of 20 factors that are theorized to influence coach behaviors. These factors can be subsumed under the broad headings of coach individual difference variables, situational variables, and contextual variables. In excess of 15 coach behaviors have been identified with three different methodologies of measurement (observer rating, coach self-report, and athlete rating). The main findings to emerge from the social cognitive paradigm of sport leadership can be summarized as follows:

- a. The coach plays a central role in influencing the athlete experience.
- b. Athlete cognitive evaluations of coach behaviors are the primary mediator of the impact of coaches' behavior.
- c. Coaches who provide their athletes with autonomy supportive environments provide positive reinforcement and structure, demonstrate care for their athletes, create task-oriented environments, and avoid controlling behaviors tend to engender positive outcomes in their athletes.
- d. Coaches' individual differences, situation, and context play a major role in determining coach behavior.
- e. Athletes' individual difference variables influence their perceptions and evaluations of their coaches' behaviors.
- f. Athletes' individual difference variables moderate the effectiveness of coach behaviors.
- g. Coaches' self-perceptions of their behaviors often appear to be incongruent with athlete perceptions and observer ratings of those same coach behaviors.

In sum, it is apparent that the social cognitive approach has been very influential in helping

researchers and coaches understand the process of influence in a sport context. It is also apparent that the different models that have been proposed share similar features at a general level (e.g., they all specify antecedents of coach behavior and propose outcomes of coach behaviors) but differ substantively at a specific level (e.g., the specific antecedents of coach behavior that are identified, the specific variables involved in the coach behavior influence process, and the primary theoretical grounding employed). Finally, it appears that the coach has an incredibly difficult yet very important job in developing his or her athletes toward both optimal performance and PWB.

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See also Achievement Goal Theory; Coach–Athlete Relations; Coaching Efficacy; Expectancy–Value Theory; Leadership in Sport: Trait Perspectives; Relational Efficacy Beliefs in Coach–Athlete Relations; Self-Determination Theory; Transformational and Transactional Leadership

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LEADERSHIP IN SPORT: TRAIT PERSPECTIVES

An enduring question within the field of sport and performance psychology concerns the origins of effective leadership (as displayed by both coaches and athletes) and, in particular, whether displays of leadership can be attributed to the emergence of any underlying set of personality traits. In the early 20th century, much research interest, especially within the field of organizational psychology, focused on the role of personality as a predictor of leadership. Following World War II, this interest receded considerably, largely in response to a prominent review paper by Ralph Stogdill. In this, he concluded that the development of leadership does not occur as a result of a set of personality traits but rather through the complex interactions that exist between leaders and followers, as well as the varied situations in which leaders find themselves. In short, he emphasized that situational factors are more salient than personality traits in determining leadership effectiveness and that although people might emerge as leaders in one context they may not in another.

From the 1960s through the 1980s (which coincided with the pronounced development of sport psychology as a recognizable field of study), a number of alternative paradigms to the trait leadership perspective gained prominence. These reflected a growing recognition of the role that social, situational, and self-regulatory processes play in the development of leadership. Indeed, during this time period, and as a reflection of research especially within the field of organizational psychology, interest in the study of personality traits in relation to leadership in sport began to wane. To a large extent, the diminished interest in trait approaches to leadership within sport can be traced to the highly variable and inconclusive nature of the research findings. For example, although some studies reported evidence that some personality traits were associated with improved ratings of leadership ability among athletes, very different patterns of

findings were observed across many other replication investigations. It has been suggested that the failure to identify a clear pattern of associations between personality traits and leadership behaviors in sport was due, to a large extent, to the widespread use of conceptually limited and atheoretical approaches alongside deficiencies in measurement, which failed to explain how the extensive number of traits being measured might be related to the criterion variables of interest. As a prominent example of this, an instrument called the Athletic Motivation Inventory (AMI) was developed in the late 1960s, which was purported (by the authors of that instrument) to assess a range of personality traits associated with, and predictive of, athlete success. Specifically, the AMI was designed to assess traits that included “drive,” “leadership,” “responsibility,” and “mental toughness.” Following publication of the AMI, several scholars raised concerns with the use of this instrument (which was widely marketed for use in professional sports), highlighting that no published evidence for the reliability or predictive validity of measures derived from this instrument was ever provided.

Balanced against this backdrop has been a resurgence of interest in the study of personality traits and leadership in recent years, especially within the field of organizational psychology (interestingly, this interest has not been matched within SP). The impetus for this renewed focus can be ascribed to the general recognition of the five-factor model (FFM) of personality, which provided a unifying framework to organize the most salient dimensions of the personality construct. The five personality traits incorporated within the FFM include extraversion, neuroticism, conscientiousness, agreeableness, and openness to experience. In a prominent meta-analytic review by Timothy Judge and his colleagues, the personality traits subsumed within the FFM were not only implicated in the prediction of which people tended to emerge within leadership positions (i.e., leader emergence) but also predicted indicators of leadership effectiveness. Specifically, four out of the five traits (namely, extraversion, conscientiousness, neuroticism, and openness to experience) were found to be related to measures of both leader emergence and leader effectiveness, with extraversion demonstrating the strongest correlation with both criterion measures. When taken together, the five traits displayed noteworthy multiple correlations with both leader emergence ($R = .53$) and leader effectiveness ($R = .39$).

Unfortunately, only four studies included within that meta-analysis were conducted with athletes and coaches. As such, any inferences regarding the extent to which these effects might be evident in sport settings are limited. Nevertheless, it is entirely conceivable, for example, that those who “emerge” into positions of leadership within sport teams would be athletes who are more vocal and outgoing (i.e., score higher on extraversion). In short, it would be interesting to examine whether the same patterns of findings reported by Judge and his colleagues are also evident within the context of leadership in sport (e.g., team captains).

With the emergence of the FFM as a viable framework to understand leadership processes in sport, there is some recent evidence that the *concordance* of personality traits between coaches and their athletes is related to improved indicators of relationship quality. Specifically, in a study by Ben Jackson and his colleagues, when athletes were found to display a high degree of concordance (i.e., similarity) with their coaches in terms of extraversion and openness to experience this was found to be related to higher levels of commitment and relatedness among both coaches and athletes. Conversely, when athletes and coaches were found to be dissimilar on these personality traits, this was found to be related to lower levels of commitment and relatedness between both parties. In summary, and in spite of the recent dearth of research on trait leadership perspectives in sport, advances in personality science, including the widespread acceptance of the FFM, may have the potential to advance our understanding of leadership among athletes and coaches alike.

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See also Extraversion–Introversion; Leadership in Sport: Situational and Contingency Approaches; Leadership in Sport: Social Cognitive Approaches; Personality and Psychological Characteristics of Athletes; Personality Tests; Personality Traits and Exercise

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LEADERSHIP IN SPORT: TRANSACTIONAL AND TRANSFORMATIONAL

Over the past 25 years, there has been considerable interest in the application of the *transactional* and *transformational* leadership paradigm to understanding the effects of leadership behaviors in relation to various psychological (e.g., motivation, self-confidence) and behavioral (e.g., individual and team performance) outcomes among those being led. Originally conceived within political and organizational settings, the transactional and transformational leadership paradigm (also referred to as transformational leadership theory) has been studied across numerous domains of human achievement including (but not limited to) financial services, multinational project teams, military combat units, sport teams, and education. The basis for sustained academic inquiry can largely be traced to the seminal work of Bernard Bass, who differentiated between transactional and transformational leadership, as well as the subcomponents of those leadership dimensions. Broadly conceived, the essence of transactional leadership involves a series of exchanges (or transactions) between leader and follower, whereby leaders make use of rewards and reinforcement to foster compliance and encourage followers to meet previously agreed-upon standards. Transformational leadership, on the other hand, takes place when leaders go beyond their own self-interests and inspire, encourage, and stimulate others to exceed minimally expected standards.

Components of Transactional and Transformational Leadership

Transactional leadership is conceptualized as including both *management-by-exception* and *contingent reward* subcomponents. Management-by-exception

includes both active and passive forms. Active management-by-exception involves actively monitoring the behaviors of those being led and taking corrective action when necessary. Passive management-by-exception involves waiting until problems become serious before intervening and reprimanding those being led. Contingent reward involves the clarification of expectations and providing rewards and recognition that is contingent on successful goal attainment.

In the context of sport, the active monitoring of athletes' competencies and actions and taking corrective action when necessary is seen as the foundation of "good coaching." Similarly, coaches providing praise and recognition when athletes perform well is seen as an important basis for bolstering their self-confidence and acts as an invaluable barometer, letting athletes know whether they are meeting their coaches' expectations.

From the perspective of transformational leadership theory, however, these transactional dimensions are theorized to be insufficient to maximize salient follower outcomes. This is not to suggest that these leadership behaviors are problematic. They are far from it. It is rather that the more active forms of transactional leadership (i.e., active-management by exception and contingent reward) provide a basis (or a bare minimum) for effective leadership, and in order to get the best out of those being led (i.e., exceed minimally expected standards), these transactional leadership behaviors need to be supplemented with transformational practices. Bass referred to this process as an *augmentation effect*, whereby the effects of transformational leadership build upon (and indeed supersede) the effects of transactional leadership.

Although a few different conceptual models have been advanced, it is generally recognized that transformational leadership is composed of four behavioral subcomponents. These include *idealized influence*, *inspirational motivation*, *intellectual stimulation*, and *individualized consideration*. Idealized influence involves—to use the old adage—"doing the right thing because it's the right thing to do." In essence, it involves acting as a role model, engendering the trust and respect of others, and articulating and acting on the leader's personally held value system. In sport settings, this might involve ensuring the coach practices what she preaches and following through on her words, even if it involves following the path of most (not

least) resistance. Inspirational motivation involves articulating a compelling vision of what is possible, setting high expectations of others, as well as displaying considerable optimism and enthusiasm about what others can accomplish. In sport, this might involve a coach conveying to his athletes, his philosophy for working effectively as a cohesive team, and also raising team members' expectations about what they can accomplish personally and collectively. Intellectual stimulation involves encouraging others to look at various challenges, assumptions, and problems from new and alternative perspectives and actively involving them in decision-making processes. From a coaching perspective, this might involve relinquishing some power and responsibility to one's athletes and having them become active agents in shaping adaptive team norms, strategizing, and developing group goals. Finally, individualized consideration involves paying attention to, and acting to satisfy, individuals' personal and psychological needs. In sport, this might involve coaches tailoring their training programs to align with each athlete's individual strengths and improve upon their weaknesses, ensuring that sufficient time is allocated to listening to their specific concerns or challenges.

Empirical Evidence From the Sport Domain

Although considerable research has been devoted to examining the transactional and transformational leadership paradigm in organizational settings, its application in the field of sport psychology (SP) is relatively recent. In one of the first studies to examine the external validity of the transformational leadership construct in the context of sport, Danielle Charbonneau and her colleagues investigated the extent to which perceptions of transformational leadership predicted motivation and sport performance among youth athletes. In this study, transformational leadership predicted elevated levels of athlete performance, and this relationship was mediated via elevated levels of intrinsic motivation.

Consistent with a core tenet of transformational leadership theory, research has also provided some support for the augmentation hypothesis specified earlier. Specifically, in the context of martial arts classes, displays of transactional leadership utilized by sensei (i.e., coaches) were found to be significantly related to three indicators of coaching effectiveness. These included measures of student

satisfaction with the coach, self-report measures of “extra” effort, as well as student perceptions of coaching effectiveness. Most notably, however, the results of this study revealed that the effects of transactional leadership in relation to these three criterion measures were significantly augmented by the effects of transformational leadership in predicting higher levels of those same outcomes.

In addition to predicting enhanced individual-level outcomes, a growing number of studies have sought to identify the potential effects of transformational leadership behaviors in relation to group-level outcomes. Consistent with work conducted within occupational settings, elevated levels of transformational leadership have been found to be associated with improved task and social cohesion. In other work, conducted within the context of youth (ice-) hockey teams, when coaches were found to make use of transformational leadership behaviors, this was associated with a lower likelihood that athletes would engage in aggressive behavior. In this study, the relationship between transformational leadership and athlete aggression was mediated by team aggression levels. Taken together, this suggests that coaches have the ability to shape the culture and climate within a given team (in this case in relation to maladaptive or deleterious behaviors) and that this climate, in turn, can play a substantive role in shaping individual-level athlete outcomes.

One area of inquiry that has received considerable attention outside of the field of SP concerns the extent to which transformational leadership behaviors are shaped by heritable factors, early childhood experiences, or through intervention later in life. Some estimates, based on recent twin studies (involving comparisons of monozygotic and dizygotic twins), place the level of heritability in transformational leadership behaviors at just under 50%, which suggests that both nature *and* nurture play a substantive role in bringing about these behavioral characteristics. Although research has yet to examine the genetic bases of leadership behaviors among coaches and athlete leaders in sport, there is some evidence to suggest that the family environment can play an important role in developing transformational leadership behaviors among young athletes in sports. In a prominent study by Anthea Zacharatos and her colleagues, researchers examined the relationship between the leadership behaviors displayed by youth athletes

and the leadership behaviors provided by their parents. In this work, parents’ displays of transformational leadership were found to be associated with their adolescent children’s displays of transformational leadership (based on self-ratings as well as ratings provided by their teammates and their coach). Furthermore, when adolescents made use of transformational leadership behaviors in the sport team context they were rated by both their teammates and coach as being more effective. This suggests that the home environment created by parents, and the manner in which parents interact with their children, can shape, to some extent, the development of transformational leadership among youth. In terms of the extent to which transformational leadership can be trained through intervention, this remains a question that has yet to be empirically tested within the context of sport. However, in organizational, military, and more recently within educational settings, a growing number of intervention studies have been conducted, which support both the viability and efficacy of transformational leadership training programs. In these studies, transformational leadership has been found to be amenable to change (i.e., trainable) and thus enhanced through intervention, resulting in improved cognitive (e.g., self-efficacy, self-determined motivation) and behavioral (e.g., achievement, performance) outcomes among those being led. In light of the moderate to large training effects derived within these studies, this would suggest that such intervention initiatives might have considerable value within the sporting domain.

Transformational Leadership and Physical Activity Engagement

Recently, researchers have begun to examine the transformational leadership behaviors utilized by teachers in relation to the behavioral engagement of students within physical education (PE) classes. While work in sport settings has primarily been concerned with the extent to which transformational leadership behaviors utilized by coaches might be related to performance-related outcomes among athletes, research in educational settings has primarily centered on the extent to which transformational leadership behaviors used by PE teachers might be related to health-enhancing cognitions and physical activity (PA) outcomes among adolescents. Recent studies indicate that when PE

teachers make use of transformational leadership behaviors in their interactions with their students, their students tend to respond with improved self-efficacy beliefs, more self-determined motivation, and greater engagement in PE classes. Furthermore, there is evidence that when PE teachers consistently display transformational leadership behaviors in their interactions with their students this not only predicts effortful behaviors by adolescents within class time but, perhaps more importantly, also predicts greater engagement in PA behaviors during their leisure time. This finding points to substantive cross-domain effects, with PE teachers influencing adolescent behavioral engagement beyond the confines of the school.

Consistent with studies conducted within industrial and organizational settings, there is also evidence that transformational leadership behaviors used by PE teachers can be developed through short-term (1-day) training programs, resulting in elevated motivational responses among students. Transformational leadership training initiatives have typically taken a social-learning approach, whereby leaders are provided with exemplars of transformational leadership, opportunities to practice and receive feedback on their leadership behaviors, as well as make use of self-regulatory processes (e.g., goal setting) to maximize their sustained use of transformational leadership over time. In light of the omnipresence of professional development workshops for teachers, such training programs have the potential to maximize the quality of interactions between teachers and their students.

Conclusion

The transactional and transformational leadership paradigm has received a growing amount of empirical attention within both the sport performance literature and the field of educational psychology. Studies to date point to the predictive utility of the transformational leadership construct in relation to a range of adaptive athlete and student outcomes. Future work, however, is clearly warranted to test the effectiveness of interventions guided by transformational leadership theory, especially in sport settings in relation to enhancing coach behavior, as well as salient athlete–team outcomes (e.g., motivation, group cohesion, individual and team achievement).

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See also Athlete Leadership; Cohesion; Team Building; Transformational Parenting

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LEARNING

The ability to learn defines much that is unique about human behavior and underlies many aspects of sport and exercise psychology (SEP). Attempts to develop sweeping laws of learning have generally been unsuccessful, and it is unlikely that a universal theory of learning can be developed. Learning is often described as a process during which lasting changes occur in the potential that an individual has for a specific behavior. Such changes are a consequence of experience within a particular environment, rather than attributes

of growth or development or temporary changes caused by fatigue, boredom, injury or even drugs or aging. Some authors define learning as a “biological device” that facilitates primarily adaptive changes that extend an individual’s capability to survive.

This entry provides a much-condensed summary of learning as it has been understood over the past 120 years of study. Influential concepts and theories of learning are discussed in a relatively chronological sequence, and an effort is made to show how the theories culminate in recent approaches to learning in sport and exercise. Behaviorist theories regard learning to be an observable effect of the environment on an organism’s behavior, whereas cognitive theories regard learning to be relatively permanent storage of knowledge as processes or representations in the brain. Constructivist theories consider learning to be the active construction of knowledge about the world. Jerome Bruner (1915–), for example, argued that learning is primarily driven by an active process of discovery about the meaning of information.

Learning by Association

A forerunner to any of these approaches, and probably the firstborn theory of learning, held that learning was a consequence of the formation of associations through experience. A person may feel pain on the first occasion that he or she goes running and thereafter associate exercise with pain as a consequence of learning the stimulus–response (S–R) relationship. Learning by association primarily is a function of conditioning. Ivan Pavlov (1849–1936) first demonstrated this as classical conditioning. By presenting food at the same time that he sounded a bell, Pavlov conditioned dogs to associate the idea of food with the sound of a bell. The dogs thereafter salivated when they heard a bell, regardless of whether food was present. Pavlov’s work paved the way for behaviorism, as conceptualized by John Watson (1878–1959), who rejected subjective inferences about the influence of cognition on behavior in favor of objective measurement of external, overt actions. Watson argued that conditioning is the key mechanism that underlies learning in animals and humans. He argued that all are born *tabula rasa* (a blank slate), and the environment governs how each learns to behave. Watson (1930) famously claimed that the aim of psychology is “to predict, given the stimulus, what

reaction will take place; or, given the reaction, state what the situation or stimulus is that has caused the reaction” (p. 11).

Shaping to Learn

An important building block of behaviorism was provided by Edward Thorndike’s (1874–1949) “Law of Effect,” which states that behaviors are likely to be repeated if they are followed by favorable (e.g., pleasant) consequences but will eventually cease if they are followed by unfavorable (e.g., disagreeable) consequences. Thorndike showed that cats trying to reach a food outside a puzzle box eventually pressed a lever that gave them access to the food and that over trials they became faster at pressing the lever because of the favorable outcome associated with that response. In essence, the cats had learned from the consequences of their behaviors. Thorndike’s work gave rise to operant (instrumental) conditioning, as demonstrated by B. F. Skinner (1904–1990). Skinner showed that behaviors could be modified by the type of consequence that followed a desired response. That is, a behavior would occur with greater frequency if it was reinforced and with less frequency if it was punished or even be extinguished if there was no consequence. Skinner also showed that behaviors could be gradually modified (shaped) by reinforcing iterations that approximated the desired response. Coaches often use shaping to modify an inappropriate technique in sport, or chaining to link appropriate responses together. A coach might use verbal praise to reinforce gradual increases in the height of the ball toss when a child serves at tennis, for example, and chaining might involve linking the ball toss to knee bend, followed by the swing of the tennis racquet, the snap of the wrist at contact, and finally the follow-through.

Many criticisms have been directed at behaviorism. For example, even when a behavior has been learned through conditioning, individuals, and animals, clearly can change the behavior if new information becomes available. Gestalt psychology abandoned the step-by-step S–R approach to learning in favor of an approach in which behaviors were seen to be driven by dynamic patterns of information available in the environment as a whole. But the great criticism of behaviorism, as embodied in Sigmund Freud’s (1836–1939) psychodynamic approach, was that by considering

only objective, observable behaviors, behaviorism ignored the influence of cognition, internal mental states of mind, on behavior.

Thinking to Learn

Cognitive approaches to learning differ from the behaviorist approach in that they define learning in terms of relatively permanent changes in organization and storage of information as a consequence of experience, rather than relatively permanent changes in behavior itself. Consequently, internal mental processes, such as information encoding and processing, perception and memory, and insight or intuition are seen to be key factors in learning, which mediate the relationship between a stimulus and a response.

Cognitive theories therefore try to account for the influence of internal thought processes on learning. Albert Bandura (1925–) proposed that internal psychological factors (the person), external observational factors (behavior), and the situation all interact to influence social-interpersonal forms of learning. *Social learning theory*, now called *social cognitive theory* (SCT), proposed that most learning occurred observationally, via modeling, and the tendency for a person to persist at it was governed by factors such as the person's sense of their own capability to carry out the required behavior effectively (i.e., self-efficacy).

Multistore models of memory that propose separate sensory, short-term, and long-term stores for information and multicomponent models of memory that explain how task-relevant information is temporarily stored and manipulated have provided a popular framework for examining the role of internal mental processes in learning. Evidence suggests that long-term memories (LTMs) present as rewired patterns of activation that require a process of consolidation to be laid down permanently, whereas short-term memories present as patterns of neural activation that are somehow prolonged by working-memory mechanisms such as subvocal rehearsal. Working memory facilitates a crucial component in most cognitive approaches to learning, which is verbal hypothesis testing about contingencies associated with actions, especially in terms of reasoning and problem solving. B. F. Skinner even delineated between rule-governed behavior and contingency-shaped behavior, because behaviorist approaches have difficulty accommodating the

tendency for humans to verbally mediate behavior. Skinner argued that behaviors that solve a problem can arise from direct shaping by contingencies (operant condition) or from rules that are hypothesized by the person solving the problem or from instructions provided by an agent with prior experience of the problem. In sport, a particular behavior may be shaped gradually by its consequences or by verbal rules that the athlete acquires by hypothesis testing or by instructions from a coach who has previously acquired the relevant information.

Conscious and Unconscious Awareness of Learning

An important distinction that arises from the cognitive approach to learning is between conscious and unconscious aspects of behavior, most recently approached within the context of implicit and explicit learning. Much of our interaction with the environment is implicit, resulting in accrual of knowledge without conscious awareness and sometimes without even intent to learn. Experimental studies of implicit learning began in the 1960s, when Arthur Reber used Markovian grammar chains to study the way in which participants learned knowledge underlying complex tasks. When participants memorized lists of exemplars (letter strings) created using the artificial grammars, they could distinguish between grammatically correct and incorrect exemplars that they had not seen previously, even though they were unable to consciously express knowledge of the grammatical rules that supported their decisions.

The double dissociation between performance of a task and the ability to express knowledge that guides performance of that task has been demonstrated using other paradigms. For example, in the serial reaction time task (SRTT), participants are required to rapidly depress keys that match positions indicated on a monitor. When the same sequence of positions (usually about 12–15) is repeated on trials, participants learn to anticipate each position in the sequence and thus respond by depressing the matching keys very rapidly. Few participants become consciously aware that they are responding to a specific sequence of key presses and fewer still can report the sequence, suggesting that the sequence may have been learned implicitly.

Implicit and Explicit Motor Learning

The conscious and unconscious dichotomy has been applied to learning in SEP in the context of implicit and explicit *motor* learning. Left to their own devices, humans display a pervasive tendency to acquire declarative knowledge explicitly when they learn motor skills. Usually, this knowledge is accrued by instructions from an agent (such as a teacher or coach) and conscious hypothesis testing during a trial-and-error process in which the learner makes attempts to move in a way that solves the motor problem. In particular, visual feedback about the outcome of each attempt is used to confirm or refute the hypotheses that are tested. Take, for example, a father and son at a golf driving range. The father may instruct his son to hold the golf club in a particular way. The son may use this grip but see that the ball travels to the left. Consequently, the son may try a different grip and watch closely to see if the ball travels in the desired direction. If the grip works, it is likely that the information will be stored as declarative knowledge for further use.

The ability to test hypotheses and store and manipulate information that can be used to make motor responses is made possible by the information-processing capabilities of working memory. Implicit motor learning tries to discourage hypothesis testing about motor responses or disrupt working memory storage of information that can be used for hypothesis testing, thereby limiting the amount of declarative knowledge that is accumulated during learning. For example, when motor learners carry out a secondary working memory task, such as tone counting or random letter generation, they tend to be unable to test hypotheses about the primary motor task that they are practicing. Consequently, they learn the primary motor task implicitly. Other methods devised in the context of SEP cause implicit motor learning by reducing the commission of errors during practice or providing reduced feedback about the outcomes of the movement. These methods prevent working memory involvement in learning by removing the necessity or the ability, respectively, to test hypotheses. If a performer does not make an error when executing a movement, there is little point in testing a hypothesis; if a performer does not become aware of the outcome of each movement, it is not possible to test the outcome of a hypothesis. Another method

that has been used to facilitate implicit motor learning entails presentation of a movement analogy (e.g., “kick like a dolphin”), which describes an appropriate technique by which to achieve the desired movement response, without the need to present explicit instructions. Analogy learning is only effective if the similar concept upon which it is based (a dolphin’s tail movement) is understood by the learner. While it is unlikely that any form of human motor learning is purely implicit or explicit, implicit motor learning techniques appear to reduce conscious access to task-relevant knowledge and thus reduce potential destabilization of automatic movement by conscious thought processes.

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See also Adaptation; Feedback; Motor Learning; Movement; Practice; Skill Acquisition

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LIMBIC SYSTEM

The limbic system is composed of a group of brain structures associated with various functions, most notably emotion, cognition, fear, and motivation. There is considerable variation in what structures researchers consider to constitute the limbic system, though the two primary structures of the system are consistently noted to be the amygdala and hippocampus. It should be acknowledged that the concept of the limbic system as the predominant view of the limbic system as the locus of emotional regulation of emotional regulation is considered by many to be flawed because it is overly simplistic. Some researchers in mainstream psychology have suggested that the concept of the limbic system be abandoned altogether. Nonetheless, the key structures of the limbic system (i.e., the hippocampus and amygdala) play important regulatory roles in behavior and may factor into some of the psychological effects seen with exercise. Both the amygdala and hippocampus have demonstrated important links with emotion, cognition, and the adaptation to stress.

The Limbic System and the Stress Response

Perhaps one of the main reasons that the limbic system is associated with the stress response and behavior is because of the direct links that it shares with the hypothalamic-pituitary-adrenal (HPA) axis. The HPA axis, hippocampus, and amygdala (along with the autonomic nervous system [ANS] and dorsal raphe nuclei) all respond to stressful stimuli. Furthermore, these areas also influence corticotropin-releasing hormone (CRH), serotonin, and other endocrine responses associated with stressful stimuli. The stress responses appear to be mediated through mineralocorticoid and glucocorticoid receptors in target organs as well as in the limbic system itself. In conjunction with serotonergic input from the dorsal raphe nuclei, the amygdala and hippocampus help integrate cognitive and behavioral responses associated with the HPA axis during exposure to stressors, including exercise. Neuroanatomic pathways

between the hippocampus and hypothalamus provide for integration of metabolic, emotional, and cognitive information.

As further evidence of the link between the limbic system and the HPA axis, glucocorticoids have structural and functional impacts on the hippocampus. Marked impairment of cognition has been noted with chronic exposure to stress. Additionally, chronic stress induces direct changes in CRH and CRH gene expression in the hippocampus as well as a variety of morphological changes associated with cognitive impairments. It appears that exercise, along with other positive stimuli such as an enriched environment, can counteract these deleterious effects and promote neurogenesis. Optimal challenge or stimulation positively influences limbic system development. An inverse-U relationship appears to exist for the effects of stress on hippocampal function and structure. Stressors of a moderate intensity, such as exercise, effectively enhance hippocampal-related cognitive function. On the other hand, excessive or chronic stressors impair cognition and, as noted previously, produce negative changes in the hippocampus and related gene expression.

The hippocampus plays a central role in regulating cognitive and endocrine responses and adaptations to stressors, including exercise. Part of this regulatory role is derived from the inhibition that the hippocampus exerts over HPA axis activation. It is also important for terminating HPA axis activity following stressor exposure in order to promote systemic recovery. The hippocampus exerts negative feedback on the paraventricular nucleus under situations of high glucocorticoid secretion in order to decrease CRH secretion. Additionally, there is a counterregulatory influence by the glucocorticoids on hippocampal activity that results from acute stress. Under these conditions, the link between stress, the hippocampus, and the amygdala should also be recognized as stressors that impair the hippocampus yet enhance amygdala activation. This enhanced amygdala activation is particularly pronounced in the bed nucleus of the stria terminalis (BNST), which is a key projection site that has been associated with anxiety. Anxiety may manifest itself as such things as excessive worry and apprehension, physical tension, heightened cardiovascular tone, and, in animal models, freezing behaviors and decreased free-roaming.

Hippocampal Influences on Depression and Cognition

Dysfunction in the limbic system has been implicated in the development of depression and stress-related disorders. One of the mechanisms by which certain antidepressants appear to exert effects is through upregulation of neurotrophic factors, particularly brain-derived neurotrophic factor (BDNF), in the limbic structures. Further support for the link between the limbic system and depressive etiology is the fact that a number of morphological and metabolic changes occur in the hippocampus of individuals suffering from depression. It may partly be through this pathway that exercise exerts its effects on depression and anxiety. BDNF increases in the hippocampus in response to exercise and facilitates use-dependent neuronal growth. Despite the BDNF downregulation that is common during intense or prolonged stress, secretion is upregulated during exercise. Furthermore, there appear to be at least 33 exercise-regulated hippocampal genes, many of which are involved in growth factor and neurotrophic factor signaling and production. One growth factor in particular that is stimulated by exercise, VGF, appears to be involved in energy balance, synaptic plasticity, and has demonstrated antidepressant actions.

There is some evidence that brain uptake of insulin-like growth factor-1 (IGF-1) from endocrine or paracrine sources is necessary for exercise-induced hippocampal neurogenesis. Additionally, exercise helps halt the stress-induced efflux of glutamate from the hippocampus that can be detrimental to hippocampal structure and function. Activity-related increases in norepinephrine and serotonin may also have mediating roles in neurogenesis and cell proliferation, respectively.

The Amygdala: Emotion and Adaptation

The amygdala attributes emotional valence and arousal to external stimuli and integrates adaptive

responses to stressors. The basolateral region of the amygdala integrates with the hippocampus to derive contextual information from stimuli. The centromedial area has projections to the lateral hypothalamus to help control blood pressure and projections to the BNST to regulate HPA activation. Because of these anatomical connections, amygdalar activation directly stimulates HPA axis responses and allows the limbic system to play a key role in regulating psychological and physiological adaptations to stressors, including exercise.

The amygdala plays a key role in emotional behaviors, fear conditioning, reward, and nociception. Much like the hippocampus, chronic stress can affect neuronal morphology and synaptic plasticity in the amygdala. This may be impacted significantly with exercise, though limited data currently exist to support this. The amygdalar CRH systems appear to be activated by stressors of a predominantly psychological nature, which may be related to cognitive interpretations of the nature of the exercise stimulus.

Shawn M. Arent

See also Brain; Cognitive Function; Emotional Responses; Mental Health; Stress Reactivity

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MASTERY AND CONTROL BELIEFS

Control has been conceptualized in many different ways in psychological literature, but the prototype for control is the belief that an agent acting through some means can affect an outcome. Exercise and sport have been examined both as outcomes of personal control beliefs and as means to achieve health and fitness outcomes. However, the effects of personal control beliefs on exercise and sport can differ with the definition of control that is used.

In a comprehensive review of the control beliefs literature, Ellen Skinner argued that control has three main components: agents–ends beliefs (i.e., personal control beliefs) refer to the belief that an agent causes an outcome (e.g., I can win a sport competition); agents–means beliefs (i.e., self-efficacy beliefs) refer to the belief that an agent has access to a behavior (e.g., I can attend practice regularly over the course of the sport’s season); and means–ends beliefs (i.e., outcome expectations) refer to the belief that an action causes an outcome (e.g., if I attend practice regularly over the course of the sport’s season, I will win the competition). These beliefs can be conceptualized along a continuum from global and pervasive to proximal and situation-specific. Importantly, it is the *perception* of the ability to exert control, rather than the objective ability to exert control, that has meaningful consequences for outcomes. In other words, people will only attempt to exert control if they believe they have control; they will not enact

control strategies if they do not perceive personal control over the situation.

Self-efficacy, defined as confidence in an agent’s ability to perform a specific behavior, is a situation-specific agents–means belief. For example, exercise self-efficacy refers to a belief about an agent’s ability to perform exercise, while sport self-efficacy refers to a belief about an agent’s ability to participate in a sport. A wealth of literature examining multiple domains of functioning and diverse populations demonstrates that self-efficacy beliefs are associated with concurrent behavior and also predict future behavior. For example, research examining behavioral interventions reveals that participants who enhance their exercise self-efficacy beliefs over the course of the treatment protocol are more physically active during the intervention and also maintain their exercise behavior after the intervention has ended.

In order for self-efficacy beliefs to be effective at predicting future behavior, the agent also needs to believe in the efficacy of the means to produce the desired effect or end goal (i.e., outcome expectation and means–ends belief). In other words, the agent should be confident in the efficacy of the means to produce the desired outcome. For example, people might believe that they are capable of exercising but might not believe that exercise will improve their health. These people are less likely to exercise than people with high outcome expectancies (i.e., those who *do* believe that exercise improves health). Thus, self-efficacy (an agent–means belief) is only effective at producing outcomes when

outcome expectancies (a means–ends belief) are also high.

In contrast to self-efficacy and outcome expectancies, personal control is best conceptualized as a global agents–ends belief. In other words, control beliefs refer to more general beliefs that an agent can produce an outcome. Control has been examined in exercise and physical activity (PA) literature and studies reveal that this more general sense of control is correlated with exercise, such that those participants with a greater sense of personal control are also more likely to exercise. However, the effect sizes (ES) in these studies are typically small, indicating that the relationships reported among general control beliefs and exercise behavior are usually weak.

Because general control reflects a personality trait, it may manifest to different behaviors for different people via specific self-efficacy beliefs. For example, some people high in general control will also have high sport self-efficacy, while others might have high music self-efficacy or culinary self-efficacy. In studies simultaneously examining self-efficacy and general control beliefs for their relationships to outcomes, self-efficacy is the stronger predictor of behavior. General control, on the other hand, is not consistently associated with sport and exercise outcomes. Thus, specific control beliefs and self-efficacy is associated with specific behaviors and should be bolstered in individuals who want to enhance sport and exercise practices.

While self-efficacy beliefs impact behavior, behavior also impacts self-efficacy. Experiencing a success in a particular skill domain is considered a mastery experience, which boosts people's confidence in their abilities in that domain. On the other hand, experiencing a failure in a particular skill domain undermines confidence. In accordance with Albert Bandura's social cognitive theory (SCT), research on sport and exercise shows that specific mastery experiences build self-efficacy and that self-efficacy predicts concurrent and future successful behavior. Thus, mastery experiences and self-efficacy share a reciprocal relationship.

On the basis of research and theory describing control beliefs and mastery, the most effective way to bolster sport and exercise behavior is to elicit successful mastery experiences by setting achievable and efficacious goals. Achievable goals are developed by identifying and targeting behaviors that are amenable to change and setting small, realistic goals. Furthermore, behavior should be

monitored to ensure that the agent's actions actually cause the outcomes, thus aligning subjective control with objective control and ensuring that goals are efficacious. These mastery experiences build self-efficacy beliefs, which then make future success more likely. Furthermore, if sport and exercise are areas of functioning that are particularly meaningful to the individual attempting to change his or her behavior, these mastery experiences and self-efficacy beliefs will bolster their global sense of control. Thus, sport and exercise behavior have potential implications for enhancing multiple domains of functioning.

Kelly A. Cotter

See also Self-Efficacy; Social Cognitive Theory; Theory of Planned Behavior

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MEMORY

Memory is a cognitive module in action organization in which information about objects, movements, events, environmental elements, and the action-related constellations between these entities are stored. Memory could be described as well as a process by which such information about the aforementioned elements are encoded, consolidated, stored, and recalled for use in attaining action goals. The structural organization of memory is based on units, categories, and expertise-dependent order formation and is therefore strongly related to learning processes.

The study of human memory has a long history, making it one of the oldest and most investigated topics in psychology. The initial scientific studies of memory are usually ascribed to Hermann Ebbinghaus's work in 1885. He investigated the serial learning and forgetting of new information (nonsense syllables) and described the speed of forgetting by formulating the so-called *forgetting curve*. William James was the first scientist to address the topic of structure in memory with the distinction between *primary* and *secondary*

memory, around the last decade of the 19th century. Over the years, the wealth of research assessing memory has greatly increased. The methods used have also evolved over time, spanning from experimental approaches in healthy subjects, to neurobiological and biophysical studies on neuronal cells, to neuropsychological studies on patients with brain damage, and in recent decades to neuroimaging research.

Memory Systems

The first memory studies, and over the years the bulk of memory studies, have been conducted to learn about verbal memory using methods like *serial learning*, *associative learning*, *free recall*, and *recognition tests*. Through studying such learning processes, researchers have found that memory performance is constrained by both *capacity* and *time-related storage* (duration). Halfway through the 20th century, the idea of a distinctive *short-term memory* (STM) came forward, mostly supported by the investigations and theoretical work of Donald Broadbent in the 1950s and Richard Atkinson and Richard Shiffrin in the 1960s. From their points of view, STM temporarily allows a limited storage of information that can be conveniently retrieved without performing control processes like rehearsal. The capacity of information storage is limited as well. As a famous example for such *capacity limits*, in 1956 George A. Miller described the magical number 7 (± 2) as the limited number of items that may be stored in STM. He was also able to show that this memory capacity could be increased through processes like *chunking*—that is, grouping items (e.g., digits) into reasonable frames or subgroups, such as three-digit-chunks. Based on various experiments in the 1960s using a *partial report paradigm*, George Sperling defined *sensory memory*, which refers to holding an iconic mental picture of the relevant items for approximately 100 to 500 ms after an item is presented. Based on such research lines, Atkinson and Shiffrin were the first to describe in 1968 a multistore model of memory featuring both (a) *structure* and (b) *control processes*. Concerning structure, the further developed *Atkinson–Shiffrin memory model* distinguished between three separate stores: sensory memory, STM, and long-term memory (LTM). Control processes are used for encoding, consolidating, and rehearsing information in memory; improving memory capacity; supporting the

transfer from STM into LTM; and vice versa. This model became an influential framework in cognitive psychology and sport psychology (SP) because it provided an integrative theoretical perspective and a testable background for experimental studies.

In the years since the appearance of the original Atkinson–Shiffrin model, many alternative models have been developed, some of which hypothesize only one memory structure. In such models, sensory registers or short-term stores are hypothesized as temporary states of activation in a unique memory network. These perspectives are similar to the idea of a unified cognitive memory model like Adaptive Character of Thought–Rational (ACT–R), which was developed over two decades (from the 1970s to the 1990s) by John Robert Anderson. This model does not define memory structures by their storage duration. Rather, it distinguishes two types of information that may be stored in memory: *declarative information*, which consists of facts about our world, and *procedural information*, which is the knowledge about how to perform actions, for instance writing a letter. ACT–R theory also includes a computational implementation to study principles of human memory through simulation.

Much effort has been made in the field of memory research to define and describe *working memory*. While there are commonalities between working memory and STM, STM researchers emphasize *time limitations*, whereas working memory researchers address *storage limitations* from a functional point of view. The term *working memory* was originally used by George A. Miller, Eugene Galanter, and Karl H. Pribram in 1960 to describe the function of memory in the planning, implementation, and cognitive control of behavior. Because various working memory studies demonstrated that stimuli are differentially processed with respect to the sensory nature of information, Alan D. Baddeley proposed in 1986 a working memory model with three active components. The model distinguishes between two active slave systems: a *phonological loop*, responsible for the storage of auditory information, and a *visuo-spatial scratch pad*, which stores visual and spatial information. Furthermore, it has been assumed that these different types of representations are controlled by an attention-based *central executive*. With the help of this model, it is possible to explain why it is easier to perform a dual task with two different

sensory channels, with for instance auditory and visual information, compared to a dual task with the same kind of information, such as two types of visual information. This model of working memory remains influential even today.

Memory and Performance

Research over the past 40 years has revealed the close relationship between memory and performance. For example, the chess studies of Adriaan de Groot and later William G. Chase and Herbert A. Simon have highlighted differences in STM performance among expert and novice performers. These authors used different methods to learn about the cognitive mechanisms underlying expert performance. De Groot used think-aloud protocols, in which participants were instructed to verbalize their thoughts during the task, while Chase and Simon used a 5-second recall task to learn about how experts “chunk” meaningful game constellations. These chess studies revealed that experts are better than novices at storing task-relevant information in STM. However, their superiority is limited to meaningful game constellations. The experts’ advantage is no longer evident when players must reproduce meaningless constellations of chess pieces. Chase and Simon developed a chunking theory and proposed that experts are better able to memorize perceptual information because they adopt an organized pattern of information and a large number of chunks in LTM through practice. However, later research has shown that it is speculative to make direct statements as to how far chunked-representations in LTM mediate the formation of a chunk pattern in STM. As a consequence, William G. Chase and K. Anders Ericsson developed at the beginning of the 1980s the *skilled memory theory*, which argues that experts’ LTM structures, as well as their encoding and retrieval skills, spill over into working memory, supporting its capacity. This research perspective was later generalized to studies on the relationship between STM and performance in sport. New studies employing a variety of tasks such as menu orders, medical expertise, and text comprehension highlighted the limits of skilled memory theory. As a consequence, in 1995 Ericsson and Walter Kintsch presented their concept of *long-term working memory*. From this point of view, experts do not only use encoding and retrieval skills but also develop a retrieval

structure to address relevant information patterns in a particular task domain. Skilled performers are able to use long-term working memory to anticipate future retrieval demands and identify the task-relevant information in the environment and in memory. This work has made major contributions to the study of the functional links between retrieval processes in LTM and the coding and chunking processes in working memory, and it shows that the capacity for task-related information storage also increases as a function of performance in domains like high performance sport.

Further research has addressed the storage of knowledge components in LTM. As opposed to the focus on information storage capacity in working or STM, these studies have been more concerned with how knowledge is structured and networked in LTM. Hence, a major issue in this domain is whether we can confirm that improving performance is also accompanied by a higher degree of hierarchy in the knowledge structure. A wide range of methods and populations have been used to study expertise-dependent differences in the classification and memory representation of context-specific problem states, for instance, among springboard divers, judokas, triathletes, and weight lifters. Such research has revealed that the nodes of experts’ representation structures in memory possess far more features than those of novices.

Such nodes of representation in motor memory might involve formats such as *propositions*, *relational structures of many kinds*, and *concepts*. Researchers from various fields such as cognitive psychology, cognitive robotics, and SP have provided evidence in recent years of so-called *basic action concepts* (BACs) in the control of human movements. BACs are based on the cognitive chunking of body postures and movement events concerning common functions in the realization of action goals. Based on this definition of representation units in motor memory, Thomas Schack, Franz Mechsner, Bettina Bläsing, and other researchers studied the link between memory and motor skills in various kinds of sport and dance to investigate the nature and role of LTM in skilled athletic performance. In high-level experts, these representational frameworks were organized in a distinctive hierarchical treelike structure, were remarkably similar between individuals, and were well matched with the functional and biomechanical demands of the task. In comparison, action

representations in low-level players and non-players were organized less hierarchically, were more variable between persons, and were less well matched with functional and biomechanical demands. The results from a number of different studies in domains such as golf, soccer, windsurfing, volleyball, gymnastics, and dancing have demonstrated that mental representation structures in memory are functionally related to motor performance.

Thomas Schack

See also Chunking/Dechunking; Expertise; Interference; Pattern Recognition and Recall

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MENTAL BLOCKS

A mental block is the inability to cognitively process thoughts or recall information. The effect can potentially interfere with performance. Sometimes confusion, lack of action, or indecision can occur with mental blocks. Performers of all ages, backgrounds, and activities can experience a mental block in varying degrees of severity. The block may be caused by repression of painful information; a traumatic physical or emotional event; or lack

of focus that results from fatigue, fear, or a poor previous performance. Mental blocks are believed to be temporary and surmountable if properly addressed. This entry discusses the harmful effects of mental blocks and several psychological skills that can help in overcoming mental blocks.

Harmful Effects

Athletes have described mental blocks in phrases like “my brain is stuck” or “my body won’t move.” Harmful thoughts seem to overtake their cognitive processing and physical movements. As a result, several negative effects can stem from mental blocks. Athletes may have a decrease in performance or in severe cases the inability to perform their sport or a specific sport skill. The lack of mental readiness to perform can produce high stress and anxiety. Over time, if the mental block is not resolved, an athlete can experience avoidance behavior or more severe symptoms of psychological distress.

Overcoming Mental Blocks

The following psychological skills can help in overcoming mental blocks.

Imagery

Imagery is the process of rehearsing a performance of a physical task while using all the senses to create or recreate the experience in the mind. Imagery should employ a variety of senses and should be practiced with an internal perspective or visualizing through one’s own eyes. This skill enhances performance and focus, and regulates anxiety. Imagery is a powerful tool because brain activity during imagery (e.g., when imaging a perfect golf putt) parallels brain activity during actual physical movement. Imagery should be relevant and specific to the present task and should be part of a regular routine, used before, during, and after practices and performances. Mentally rehearsing a perfect response can alleviate mental blocks by increasing success expectations and therefore increasing successful performances. For example, when a golfer is experiencing a mental block in putting, she can use imagery to visualize herself completing a smooth, automatic stroke and sinking the putt in the hole. Visualizing successes can increase her confidence and help her break free of the mental block.

Goal Setting

Goal setting is a systematic approach to acting and thinking in purposeful ways to achieve specific accomplishments. Goal setting affects behavior by increasing focus, effort, and motivation and affects cognitions by influencing levels of anxiety, motivation, and confidence. Goal setting can help athletes avoid mental blocks through breaking down complex skills into more manageable tasks. For example, if an athlete is experiencing a mental block pole-vaulting, he can break the skill down into smaller, more manageable goals, such as improving his approach and clearing the bar one foot lower than his desired height. As he experiences success with these smaller process goals, the athlete may begin to have confidence that he can complete the task; he can conquer his mental block. As a result, his effort and motivation to clear his goal height should increase, leading to further performance success.

Relaxation

Relaxation is an important skill to master because athletic performance and mental blocks can be highly stressful. Athletes respond to stress cognitively (i.e., thoughts) and/or somatically (i.e., physiological reactions). Physical relaxation is a way to reduce reactions to stress and purposefully control the body to stimulate a more relaxed state. Athletes can benefit from physical relaxation because when employed properly, relaxation reduces muscle tension and can alleviate somatic symptoms of anxiety (i.e., a racing heart or shallow breathing). Physical relaxation techniques can be used as a momentary response during competition (e.g., before a foul shot), to conserve energy before games or practices, and as a way to rest and recover between or after performances. Techniques include power breathing and the tension-release method. Power breathing is using the diaphragm to consciously slow one's breathing to reduce tension and symptoms of somatic anxiety. An example of using power breathing is taking a long, steady inhale for 4 seconds, holding the breath for 2 seconds, and exhaling the breath for 6 seconds. The tension-release method requires the athlete to systematically tighten and relax muscles throughout the body. Not only does this help the athlete feel loosened up but also it helps the athlete become more aware of tension throughout the body. For example, when a gymnast is experiencing a mental block with a back tuck, utilizing relaxation

techniques can help ease any muscle tension that is inhibiting her from performing the skill successfully. Using power breathing before practices and before performing the skill itself can help relieve any existing muscle tension and set the gymnast up for smooth execution of the skill.

Preventing and overcoming mental blocks are an important aspect of successful performance. While there is not a lot of research specifically examining mental blocks, we do know that using psychological skills, such as imagery, goal setting, and relaxation, helps to maintain confidence, lower anxiety, and ensure that athletes are mentally ready to cognitively process thoughts and perform their best.

Melissa A. Chase and Emily Galvin

See also Attention Training; Cognitive Function; Perception; Self-Efficacy

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MENTAL HEALTH

Mental health has been described as the emotional and spiritual resilience that enables people to enjoy life and to survive pain, disappointment, and sadness. It is fundamental to health and well-being and is more than the absence of symptoms of mental illness. Mental illnesses are health conditions that are characterized by alterations in thinking, mood, or behavior associated with distress and/or impaired functioning. Mental health and mental illness should be considered distinct. An individual

may have a diagnosis of a mental illness, such as schizophrenia, yet still attain positive mental health. Mental health problems, such as sub-clinical levels of depression, can affect individuals without necessarily becoming a diagnosed mental illness. Consequently, mental health concerns everyone, and strategies to promote mental health should be implemented for the benefit of all. No single factor causes poor mental health or mental illness. Rather, these are influenced by a range and combination of factors such as genetic inheritance, biochemical and psychological factors, life events, and socioeconomic circumstances. Similarly, when mental health improves, it is likely that a combination of factors is at work.

The physical health benefits of exercise and physical activity (PA) are well established. PA, as a general term, refers to any movement of the body that results in energy expenditure above that of resting level. Exercise refers to a subset of PA in which the activity is structured and often supervised and undertaken with the aim of maintaining or improving physical fitness or health. Research is rapidly increasing concerning the impact of both on mental health. A focus on this relationship is very relevant as mental health outcomes may motivate people to stay physically active while also having a potentially positive impact on mental health. Without regular participation, both mental and physical benefits will not accrue.

This entry describes four main functions of PA in impacting mental health. First, PA may prevent mental health problems. Second, exercise has been examined as a treatment or therapy for existing mental illness. Third, exercise may improve the quality of life (QOL) for people with mental illness. Fourth, exercise and PA can improve the mental health of the general public in a broad number of ways.

Prevention

There is some evidence that participation in PA prevents mental health problems from occurring at a later time. Given our aging society, senile dementia and Alzheimer's disease represent significant public health burdens. Some evidence suggests a reduced risk of dementia for individuals who are physically active. More extensive evidence is available regarding depression. In 2010, the World Health Organization predicted that depression would be second only to ischemic

heart disease in global overall disease burden by the year 2020. This prediction is alarming when considering depression's detriment to physical and mental health. Depression is commonly associated with high morbidity and mortality rates; increased risk of suicide; major suffering; and a decrease in physical, cognitive, and social functioning. To look at the role PA has on the prevention of depression, epidemiological, prospective longitudinal studies are required to identify a temporal relationship. Such studies involve large numbers of people and measure levels of PA and depression at two or more time periods. For example, a study conducted by William Strawbridge and colleagues in California collected measures of PA and depression from 1,947 community-dwelling adults ages 50 to 94 years at baseline in 1994 and after 5 years of follow-up. After excluding individuals with depression in 1994, an increase in PA was associated with a significant (20%) reduction in the likelihood of being depressed in 1999. Perhaps this finding could be explained by bias in some way—for example, individuals with a physical disability may have both a higher risk of depression and lower levels of PA. Most studies like this one, however, continue to demonstrate a relationship between PA and a decreased risk of depression while accounting for a wide range of possible confounding factors, such as disability, smoking, alcohol, and social status, in the statistical modeling. Overall, these types of studies suggest PA has a protective effect against developing depression.

Treatment

Exercise might be used as a therapeutic strategy to alleviate mental health problems or mental illness. As with the preventive function, the most compelling evidence comes from studies in the area of clinical depression. As depression is the most common mental health problem, it has received more attention by researchers than other conditions. Meta-analyses quantify results from a number of studies and allow statistical techniques to be used as a means of analysis. The basis of a meta-analysis is the effect size (ES), which transforms differences between an experimental and control group to a common metric. As a guideline, ES, usually determined through the calculation of differences between an exercise and control group or from pre- to post-exercise intervention, of $\geq .2$, $\geq .5$, and $\geq .8$, is generally described as reflecting a small,

moderate, and large effect, respectively. In one meta-analysis of 28 studies examining exercise and depression, comparing exercise with no treatment or a control intervention, the ES was -0.67 indicating a moderate clinical effect. However, when only the four methodologically strongest trials were analyzed, the ES was reduced and was small at -0.31 . Notably, this ES is comparable to those seen in meta-analyses of the effects of medication on depression. These findings suggest exercise might be an additional consideration for treatment for those who are interested in trying it. Researchers have started to examine how exercise might be an adjunctive treatment for other illnesses such as bipolar disorder, anxiety, and drug dependence.

Physical Activity for Quality of Life and Coping With Mental Illness

For people with severe and enduring mental health problems, improvement in QOL tends to enhance the individuals' ability to cope with and manage their disorder. In this case, exercise might be useful in improving the mental health of individuals with a mental illness. A good example is schizophrenia—a serious mental illness that is characterized by psychotic symptoms (e.g., hallucinations, delusions), disorganized speech and behavior, negative symptoms (e.g., apathy, blunted or flattened affect), and neurocognitive and social cognitive deficits. Epidemiological surveys demonstrate an excess of physical morbidity and premature mortality in individuals with schizophrenia with the largest causes of death being related to cardiovascular disease, obesity, and diabetes. Potential causes of this excess mortality and morbidity are varied but can be broadly categorized in terms of changes in treatment (metabolic side effects of medication); greater prevalence of engagement in unhealthy behaviors such as smoking, physical inactivity, and poor nutritional habits; and limited access to health care. Consequently, PA promotion is well warranted in this population irrespective of mental health gains. One meta-analysis suggests small effects in terms of improvements in some of the negative symptoms of schizophrenia such as depression, low self-esteem, and social withdrawal. There is less evidence that exercise may be a useful coping strategy for dealing with positive symptoms such as auditory hallucinations. There are limitations to this research literature, but the evidence suggests exercise interventions are feasible in this

population and can have healthful effects on both the physical and mental health of individuals with schizophrenia.

Mental Health Promotion for All

There are many components of mental health such as self-esteem, cognitive functioning, stress, and sleep. Although research has been conducted on all of these topics, its extent and degree of methodological rigor vary. The majority of existing research examining exercise and mental health is included in this category. Self-esteem has received much attention because many consider it a central aspect of mental health, as it reflects the overall evaluation or appraisal of one's own worth. In a meta-analysis, there was a small relationship between exercise and self-esteem ($+0.23$). One well-accepted approach to self-esteem considers the construct to be multidimensional and hierarchical in nature. That is, self-esteem can be broken down to many constituent parts comprising an extensive number of life domains. For example, one's self-esteem could be based on how well he or she performs at work, in relationships with others, or how one looks and feels physically. Theoretically, one might expect exercise to have a stronger relationship with components of self-esteem related to the physical self. Research confirms a stronger effect is seen when there are significant changes in physical fitness, for example, and this can extend to more generalized improvements in self-esteem. Overall, there is evidence that exercise can change people's perceptions of their physical self and identity in a positive way.

How Does Exercise Benefit Mental Health?

Understanding how exercise improves mental health may lead to the development of exercise prescriptions that more effectively and reliably improve the mental health of individuals. Researchers have investigated a range of mechanisms that can be broadly described as being biochemical, physiological, or psychosocial in nature. A commonly suggested biochemical mechanism is that endorphins (endogenous opioid peptides) are responsible for euphoria or other changes in mood during or after acute exercise. While such a mechanism is plausible, there is still no compelling evidence to support such claims. Changes in fitness, a physiological mechanism, may explain improvements in mental health. However, some studies demonstrate mental health

benefits even when there are no changes in fitness. Psychosocial mechanisms have also been explored, including how exercise may enhance perceptions of mastery or competence, or provide opportunities for distraction and/or social interaction. Current consensus is that there is unlikely to be one central mechanism that explains mental health benefits for all people. Because mental health is influenced by many factors, it is just as likely that many factors positively influence mental health and that exercise could be helpful through a number of pathways. Several mechanisms most likely operate in concert, with the precise combination being highly individual-specific and varying through exercise characteristics such as intensity and type, preferences of the individual, and even when and where the exercise takes place. Current guidelines for PA aimed at adults (e.g., accumulating 150 minutes of moderate-to-vigorous physical activity [MVPA] per week) are supported in terms of their potential for enhancing mental health.

Conclusion

The field of exercise and mental health is a rapidly growing focus for researchers and practitioners. As with most fields of study, more research is required—particularly, rigorously designed experimental studies comparing the effects of exercise to other therapeutic approaches. Furthermore, there is also scope for exploring issues such as dose-response (e.g., how the type, intensity, frequency, and duration of exercise is related to mental health outcomes), participant characteristics (e.g., who responds most positively to a “dose” of exercise), and the mechanisms that explain mental health change. Because PA is an effective method for improving important aspects of physical health, the promotion of PA for mental health can be seen as a win-win situation—the physical health benefits are indisputable, and evidence suggests that there can be important mental health benefits as well.

Guy Faulkner and George Mammen

See also Adherence; Affective Disorders; Affective Responses to Exercise; Psychological Well-Being; Quality of Life

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MENTAL REHEARSAL

Mental rehearsal is an umbrella term that covers several techniques used by athletes and exercisers to improve performance. It happens covertly and without any actual movement and typically involves the representation of an action or behavior using nonverbal (e.g., imagery, observation) or verbal processes (e.g., self-talk). For example, an athlete may think (in the “mind’s eye”) about a skill to be performed using imagery or repeating key words associated with successful execution of that skill using self-talk. Moreover, mental rehearsal may involve a combination of techniques. Described here are the characteristics of mental rehearsal and three common types of techniques employed by athletes and exercisers to mentally rehearse their activity: imagery, observation, and self-talk.

Characteristics of Mental Rehearsal

Mental rehearsal is deliberately employed by the individual with the intention of achieving specific cognitive or motivational outcomes. Cognitive outcomes are usually associated with the learning and performance of skills, strategies, and routines. A figure skater might observe someone successfully perform an axel to learn how it should be correctly executed. Motivational outcomes typically include managing thoughts and emotions as well as goal setting. Someone attempting to become more active to lose weight might imagine himself or herself looking thinner for motivation to be more physically active. Mental rehearsal is not necessarily limited to a singular outcome. Rather,

it is possible to obtain multiple outcomes—for example, observing a particular skill to obtain improvements to both technique and confidence of performing the skill.

Systematic use of mental rehearsal is one of the qualities that distinguishes elite athletes from those who do not excel in their field. The benefits of mental rehearsal are gaining recognition within an exercise context, with more active individuals reporting greater use of the quality to self-regulate their thoughts, feelings, and behaviors.

Imagery

Imagery involves the experiencing or re-experiencing of a situation through multiple sensory modalities (e.g., visual, kinesthetic). It is well known that when combined with physical practice, imagery leads to greater improvements of a motor skill compared to physical practice alone. The proposed mechanism underpinning these improvements is the activation of some common neural networks during imagery and actual execution of the same skill. This has resulted in imagery being viewed as an effective mental rehearsal technique that supplements and improves training and can even stand in or be substituted for some amount of actual practice. Beyond these cognitive outcomes, imagery is also well established as a confidence-enhancing technique that enables individuals to manage symptoms associated with anxiety.

When imagery is combined with relaxation, this subtype of mental rehearsal is known as *visual motor behavior rehearsal*. The two-step process begins with relaxation (e.g., take a deep breath) followed by imagery to fully re-experience an event or situation (e.g., you are standing on the green again, holding the putter). It can be used to strengthen desirable responses (e.g., you are confident as you take the shot) and/or eliminate undesirable ones (e.g., reducing or reappraising symptoms associated with anxiety). While this standardized training method might be useful for modifying thoughts and feelings, it is not always appropriate to relax individuals before they engage in imagery. This is because activation levels might fall below those typically experienced in the real-life situation, which can make imagery less effective.

Observation

Another common form of mental rehearsal is observation. When used for the purposes of

acquiring a new skill or behavior, it is referred to as observational learning or learning by demonstration. Sharing many characteristics with imagery including the activation of similar neural and cognitive processes, observation involves watching oneself or another perform a behavior; these demonstrations can be live or videotaped. For example, a dancer might observe a video recording of herself to feel more confident in performance. Alternatively, a basketball player new to a team might observe the team run a set play to learn how it should be performed. Consequently, as with imagery observation can facilitate learning and improve or alter performance and behaviors.

Self-Talk

Self-talk involves having a dialogue with oneself, either out loud or in one's head. This mental technique is used by individuals to instruct and motivate performance of skills and exhibition of certain behaviors. Within the context of mental rehearsal, statements to the self would be used to give instructions on a skill or strategy to be performed in the form of specific cues or longer phrases. The athlete rehearses the skill or strategy by talking to herself or himself, which is also known as instructional self-talk. This form of mental rehearsal can also be done in conjunction with imagery and/or observing. For example, a field hockey player might image performing a block tackle and say specific instructions such as “step left” and “stay low” while imaging those specific parts of the skill.

Jennifer Cumming and Sarah E. Williams

See also Imagery; Modeling; Multimodal Mental Training; Relaxation; Self-Talk

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MENTAL TOUGHNESS

Athletes are confronted with a variety of stressors, challenges, and adversities, external (e.g., hostile crowds, referee errors, challenged by an opponent, sport and life balance) and internal (e.g., fatigue, self-doubt, emotional instability), which are characteristic of the training and competition contexts of sport. Some athletes manage these demands or challenges positively, either having a smooth progression through the performance cycle or successfully negotiating these challenges in constructive ways. However, for other athletes, such demands or challenges can overwhelm their coping resources, creating major distress and negatively influencing their performance and goal attainment. What accounts for these individual differences in athletes' ability to manage both negatively (e.g., injury, deselection) and positively (e.g., winning streak, taking the lead in a match) construed challenges and demands? Most scholars, practitioners, and the general public suggest that the answer lies in an athlete's mental toughness. Although mental toughness remains a contested concept, common themes among most contemporary conceptualizations reveal that mental toughness encapsulates a reservoir of personal resources that enable individuals to produce consistently high levels of performance or goal attainment despite everyday challenges and significant adversities.

Historical Perspective

The scientific study of mental toughness, which has its evolutionary roots in personality research on "tough-mindedness" dating back to the 1950s, has garnered a modest amount of empirical attention in recent years. Prior to this influx of research and theory development, much of what was known about mental toughness was based on

anecdotal reports and disseminated predominantly through the popular media. The result was a diverse selection of definitions and explanations of mental toughness that largely included an assortment of beneficial psychological characteristics (e.g., resilience, insensitivity to criticism) and mental skills (e.g., arousal regulation, visualization). The conceptual confusion created from this lack of consistency and understanding resulted in numerous beneficial psychological characteristics being incorrectly labeled as mental toughness, particularly as they were based on authors' opinions rather than empirical research. Encouragingly, empirical research on mental toughness has blossomed over the last decade, with independent but related streams of research focused on its conceptualization, measurement, and development.

Conceptualizing Mental Toughness

Mental toughness is typically employed by scholars, practitioners, and the general public as an umbrella term to describe a multitude of facets, including cognitive, affective, and behavioral concepts that are pertinent to the attainment and sustainment of high performance despite stress, challenge, or adversity. Much of the past decade of scholarly activity has been concerned with elucidating an understanding of the core personal resources associated with mental toughness. Owing to the lack of prior theory and empirical research, most researchers have approached the systematic task of generating insights into this elusive concept using an inductive process in which key stakeholders' (e.g., athletes, coaches, sport psychologists) perceptions of mental toughness were generated and explored. With an emphasis on the rich, real-world context in which the concept occurs, these scholars have sought to identify and understand patterns of relationships within and across sources of either sport-general (i.e., informants from a variety of sports) or sport-specific samples (e.g., informants from a specific sport such as cricket, soccer or football). Other researchers have sought to integrate these inductively derived data with established theory and empirical research on hardiness, which is conceptualized as a cognitive personality variable consisting of a sense of control, commitment, and challenge.

Research on such a broad psychological concept has led to several different ideas about what mental toughness encapsulates. For example, as depicted

in one of the most widely employed conceptualizations, mental toughness is said to comprise four distinct components of confidence, control, commitment, and challenge. Other conceptualizations of mental toughness encompass a much wider net of personal resources to include key attributes such as sport knowledge, attentional regulation, emotional awareness and regulation, and coping with failure and success. Nevertheless, despite these differing viewpoints on the content and breadth of its key facets, several core components have emerged across these studies and appear to capture the essence of the current state of affairs. These core components include *general self-efficacy* (i.e., generalized belief that one is capable of achieving his or her goals or producing high levels of performance), *optimism* (i.e., tendency to expect positive events in the future, and attribute positive causes and outcomes to different events), *success mind-set* (i.e., desire to achieve success and ability to act upon this motive), *sport knowledge* (i.e., knowledge of the performance context and its application for goal attainment or high performance), *self-regulation* (i.e., capacity to optimally manage one's thoughts, focus, and emotions), *resilience* (i.e., perceived capacity to successfully cope with or bounce back from potentially significant risks or adversities), and *buoyancy* (i.e., ability to effectively execute the required skills and processes in response to the challenges and pressures of everyday life). Interestingly, each of these key facets has a rich history of theory and empirical research that have not yet been adequately drawn upon for the conceptual evolution of mental toughness. For example, Charles Snyder's theory of hope offers a useful backdrop upon which to conceptually integrate the success mind-set (i.e., agency thinking) and sport knowledge (i.e., pathways thinking) facets of mental toughness. Theoretical integration and development remains an important avenue of future work for this concept.

Measuring Mental Toughness

The debate over a common definition and conceptualization of mental toughness has not slowed the amount of research on its measurement. Indeed, the methodological development of measurement tools for key facets of optimal human functioning can greatly accelerate scientific advancements in theory (e.g., basis for rigorous empirical tests of construct conceptualization) and practice

(e.g., monitoring and evaluation). As with efforts designed to generate insights into its key components, scholars have developed self-report measures of mental toughness that are designed to generalize across all sports (e.g., Mental Toughness Questionnaire 48 [MTQ48]) or capture conceptualizations that are unique and specific to individual sports such as cricket (Cricket Mental Toughness Inventory) and Australian football (Australian Football Mental Toughness Inventory).

Conceptual (e.g., clearly articulated model supported by empirical evidence), statistical (e.g., analyses performed to develop and substantiate reliability and validity), and practical (e.g., predict or explain behavior or performance) issues are inextricably linked to the construct validation process and fundamental to ascertaining the usefulness of a psychometric tool for theory and practice. An assessment of the currently available sport-general and sport-specific measures against these three criteria reveals that there is currently no "gold standard" tool for assessing mental toughness. For example, although the conceptual model underpinning the Cricket Mental Toughness Inventory has received support via rigorous, hypothesis-testing statistical procedures (i.e., confirmatory factor analysis), it suffers from a major concern related to its ability to generalize to other sports and across most athletes (e.g., males and females). In contrast, the MTQ48, which is the most widely employed measure of mental toughness, has been the subject of debate with regard to its conceptual rationale and factor structure. These conceptual, statistical, and practical concerns remain the primary focus of future work on its measurement. In the meantime, an alternative approach to the measurement of mental toughness in future research might involve the use of existing measures to capture the key facets (e.g., Generalized Self-Efficacy scale, Dispositional Hope Scale).

Developing Mental Toughness

Attempts to understand mental toughness often lead to discussions or questions about its development. These questions on mental toughness development raise some of the most interesting issues facing this scholarly field of inquiry. Among the most heated discussions are those that concern the "nature" versus "nurture" debate. Although aspects of mental toughness may be considered heritable, as with related research on personality,

one cannot deny that a significant amount of variance remains unexplained by nature, thereby implicating the important role of environments (e.g., home environment, sport environment) and social agents (e.g., parents, coaches) for its development or enhancement. Indeed, the centrality of both nature and nurture is evident among key stakeholders' retrospective perceptions of mental toughness.

Several important findings can be gleaned from the existing research on mental toughness development. First, individuals seem to acquire some degree of mental toughness from their family network (e.g., offering social support, providing reinforcement and encouragement, having realistic beliefs in their child's abilities, serving as a positive role model) during their formative years prior to participating in sport. This "generalized form" of mental toughness is subsequently refined into a "sport-specific form" and maintained through one's interactions with key social agents such as coaches (e.g., coach-athlete relationship, player development vs. coaching success, motivational climate) and teammates (e.g., mentoring, competitive rivalry, role models). Second, there is evidence to suggest that some mechanisms of influence facilitate (e.g., simulation training, social support) the developmental process, whereas others (e.g., controlling coaching, low and unrealistic expectations) thwart or undermine efforts to develop mental toughness. In particular, these underlying mechanisms are said to operate in a combined, rather than an independent manner, to facilitate the development of mental toughness. Third, the most beneficial learning and motivational climates appear to differ throughout this long-term developmental process as athletes gain experience and transition into higher levels or different types of competitions. For example, enjoyment and skill mastery are considered key to the early years or the period in which one initially becomes involved in sport, whereas competitive rivalries and performance setbacks become prominent features in the middle years when sport demands greater discipline from individuals owing to structured training regimes and sport specialization. Finally, it appears that mental toughness can be both *caught* indirectly through experience and *taught* directly via coaching, education, psychological skills training (PST), or strengths-based approaches. In other words, mental skills training alone should not be considered "mental toughness development."

Conclusion

Mental toughness is the umbrella term often employed to describe one's capacity to deal effectively with both major assaults on one's normal level of functioning and "everyday" challenges in the pursuit of high performance. Despite entering its second decade of empirical research activity, what constitutes mental toughness and therefore how it is assessed remains a complicated area of study. Some consistencies across the available research have emerged with regard to its core facets and important development influences, yet debate still exists as to a precise or commonly accepted definition and conceptualization of mental toughness. Although recent advancements in its conceptualization, measurement, and development are important, mental toughness, as a scholarly field of inquiry, is still in its infancy when compared with the literatures on its key components (e.g., self-efficacy, optimism). Thus, future work on mental toughness could do well to leverage off such established theory and research. Mental toughness has the potential to become an influential concept in the field of sport and exercise psychology (SEP), yet much scholarly work remains to be done to better understand its makeup and influence on behavior.

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See also Competition; Coping; Optimism; Resilience; Self-Efficacy; Stress Management

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MENTORING

Mentoring is a process in which a mentor, who is typically more experienced or older, helps a mentee or protégé grow and develop in some way. As such, a mentor may be thought of as a guide, tutor, counselor, or adviser. In sport and exercise

psychology (SEP), mentoring is commonly thought of in terms of the supervision of graduate students or professionals becoming involved in the field. In such settings, mentors help these individuals develop their consulting skills, as well as develop professional attitudes and values about important issues such as ethical practices. However, mentoring has many other applications for SEP professionals to consider. In many cases, the individuals that professionals in the field of SEP work with or teach are involved in mentoring efforts. For example, physical education (PE) teachers mentor undergraduate students, as do certified athletic trainers, exercise professionals, and coaches. In addition, many sport-based programs focused on youth development have included mentoring of young people as a key process in achieving their goals.

Given the widespread application of mentoring in sport and exercise settings, as well as in SEP, it is important that professionals in the field have an understanding of mentoring and the mentoring process. This entry is designed to provide an overview of mentoring and its application to SEP.

Defining Mentoring

Unfortunately, there is no universally accepted definition of mentoring. Dictionaries define mentors as trusted counsels, teachers, coaches, or loyal advisers. When describing mentors, those in psychology and business have used terms such as *role models*, *individuals who help others achieve life goals*, or *career guides*. Mentoring, then, seems to be equated with and used interchangeably with a number of other terms. These include teaching, coaching, advising, counseling, educating, tutoring, guiding, training, developing, molding, supporting, and shaping.

While no universal definition of mentoring can be found, looking across various definitions reveals many common characteristics. These include (a) a situation usually involving two individuals where a relationship built upon mutual trust and respect occurs; (b) a mentor who possesses more power, knowledge, and/or experience than the mentee; (c) a flow of information from mentor to mentee; (d) a process that involves the transfer of knowledge, experience, or power from the mentor to the mentee; and (e) benefits for both the mentors and mentees. Therefore, for the purpose of this entry mentoring is defined as the process by which an

older or more experienced individual formally or informally works with and counsels another young or less experienced individual(s) for the purpose of enhancing their growth and development whether that involve the development of skills, attitudes, values, and/or dispositions.

Types of Mentoring and Mentors

There are a number of types of mentoring that can occur. These include traditional one-on-one mentoring, peer mentoring, or group or team mentoring. It should be noted, however, that in group mentoring situations, it is recommended that the ratio not exceed one mentor to four mentees at any one time. There is also e-mentoring, which is undertaken using various technologies like Skype, videoconferencing, or the Internet. Mentoring can also occur in formal structured programs such as a supervisory experience in a counseling or clinical SEP or informally where a new member of an athletic team is mentored by a veteran without being formally asked to do so. Whatever the context or structure, however, mentoring has been found to be effective in both formal and informal settings.

Mentoring Research in Sport and Exercise Psychology

When the research on mentoring in sport and exercise settings is reviewed, several conclusions are clear. First, there is a lack of research on mentoring. More often than not, one finds articles calling for the use of mentoring with sport and exercise participants, articles discussing how mentoring might be used or reviews of the literature often relying on research and studies conducted in other fields like business and education. Second, there are almost no systematic lines of research. Third, most of the studies are descriptive in nature and few attempts have been made to propose and test possible theoretical explanations for mentoring and mentoring effects.

While research on mentoring in sport and exercise is certainly limited, studies are beginning to emerge. For example, in a 2009 study of mentoring of sport psychology (SP) graduate students, Watson, Clement, Blom, and Grindley reported students had positive perceptions of their mentors and the mentoring process whether it was formal or informal. This study also found that peer mentoring was evident at much higher amounts than

in other fields. Other studies have looked at how coaches and athletes are mentored. For instance, in a 2010 study, Carter and Hart interviewed 38 black female college athletes to understand their views of mentors in the lives. Results revealed they viewed mentors similar to the ways in which mentors are viewed in the broader research literature (e.g., as guides, role models, and supporters). These athletes also reported that different individuals fulfilled different mentor roles (athletic support, career or academic support, and psychosocial support). Finally, some intervention and evaluation studies have begun to be conducted investigating the effect of mentoring programs within SEP settings. These studies typically compare the effect of mentoring on one group of participants against a group that did not receive mentoring. For instance, in a 2012 study, Armour and Duncombe examined the effectiveness of a school-based mentoring program that used successful athletes to deliver motivational activities to disengaged youth. Results revealed that both the young people targeted and their teachers reported positive reactions to the program, but little evidence was found for its broader impact on factors such as self-esteem and attendance at school.

More research has been carried out to examine the role of mentoring in the development of coaches and physical education (PE) teachers. For example, studies have examined how coaches perceive their roles as mentors, the working relationships between supervising physical educators, student teachers, and university instructors coordinating the student teaching mentoring experience. All of these studies merely identify and describe mentoring as a positive tool and process.

In summary, more research is needed on mentoring in sport and exercise settings. The existing research parallels that in other fields. However, confusion remains regarding inconsistent and imprecise definitions of mentoring. There is a need to move beyond descriptive studies to ones that test the effectiveness of mentoring interventions or relationships between factors influencing mentoring and mentoring outcomes. Tests of theoretical explanations for mentoring are also needed.

Key Mentoring Findings

Looking across the research in SEP, kinesiology, and in other fields such as business, adolescent development, and education, a number of key

mentoring findings can be identified. Some of the more important ones are summarized next.

Mentoring Works

Research conducted across diverse settings and individuals has shown that mentoring works and leads to such positive outcomes such as higher performance, faster career advancement, positive emotional states, and psychological growth and development. Mentoring effects are robust, having been shown to be effective in a variety of educational, sport, and business settings as well as in the field of positive youth development.

Mentoring Effectiveness Is Influenced by a Variety of Individual and Environmental Factors

While mentoring works, it has been found to be influenced by both individual specific, cultural, and community factors (e.g., mentor experience, mentee readiness to change, context). For example, mentoring effectiveness has been shown to depend on individual characteristics related to the mentor and the mentee such as levels of motivation, cooperation, and involvement. Mentoring effectiveness is also influenced by broader environmental factors such as the community and culture the mentor and mentee are situated in.

The Length of Time Mentored Is Highly Related to Mentoring Success

Research has shown that that longer one is mentored, the more effective mentoring is, with 1 year to 18 months being the minimum length needed to show effects. It has also been shown that across this time period mentoring should occur a minimum of 4 hours per month.

The Quality of Mentor–Mentee Relationship Is Critical for Success

The quality of the relationship between the mentor and mentee has been found to be one of the most important factors influencing mentoring success. The closer and more trusting the mentor–mentee relationship, the more effective mentoring is. Little trust and poor relationships are associated with unsuccessful mentoring efforts.

Mentor–Mentee Match Is Critical

Because the mentor–mentee relationship is so important for mentoring success it is no surprise

that the mentor–mentee match is considered critical for program success. In most cases, the mentor should have more knowledge, experience, and power than the mentee. However, more successful mentoring efforts consider mentor–mentee age, mentee needs, and common interests between mentors and mentees as factors influencing mentor effectiveness. Natural mentors such as teachers, older relatives, other athletes, and coaches can be particularly powerful mentors; in many cases, these relationships are the most effective.

Mentor Training Is Important

While there is evidence that natural mentors, who often are untrained, can have an important influence on mentees, research with formal mentoring programs has shown that mentor training is important. Mentoring experts suggest that mentors should have at least 6 hours of training. This training should emphasize that mentors should only mentor in areas in which they are competent, and should include ways to build relationships based on trust and respect, the importance of maintaining a consistent presence in the mentee's life, the need to focus on the needs of the mentee, how to set and establish boundaries and specific expectations, and ways to make the experience enjoyable. In youth mentoring programs, mentors should be made aware of the importance of efforts to get to know the mentee's family without getting overly involved. Finally, mentors need to realize that authoritarian styles do not succeed in mentoring. Mentoring should be a dialogue between the mentor and mentee. Emphasis needs to be placed on changing the mentee's behavior by building a trusting relationship with him or her, not forcing him or her to do things.

Formal Mentoring Unfolds in Stages

Mentoring experts have found that the mentoring process unfolds in typical stages, although the exact stages identified vary to some degree across authors. Often-mentioned stages include the following:

1. The development of program goals and focus
2. An initiation stage that involves mentor screening and entry
3. Mentor training
4. The cultivation of the mentor–mentee relationship
5. Maintenance
6. Separation, termination, or closure of the mentor–mentee relationship
7. Redefinition of the mentor–mentee relationship if continued contact is expected.

How Mentoring Works

Research on how mentoring works is underdeveloped, but initial findings suggest that mentors help set expectations in their mentees. They also convey various skills and attitudes. For example, in SEP supervisory situations, the mentor and mentee may discuss consulting techniques as well as ethical issues. Finally, mentors are thought to influence mentees by providing support and encouragement.

Steps to Designing Formal Mentoring Programs

In the United States, the National Mentoring Partnership has identified steps to designing formal mentoring programs. These include the following:

1. Define the individuals that the mentoring program is designed to serve (What is the need? What are their ages, gender, and common characteristics?).
2. Identify the types of individuals that would best serve as mentors.
3. Determine the type of mentoring program that will be offered (e.g., one on one, group mentoring).
4. Determine if the program will be structured as part of an existing organization or a stand-alone program.
5. Determine the nature of the mentoring program (e.g., youth development, career assistance, improved academic performance, character development).
6. Identify what outcomes will be associated with what the program is hoping to accomplish.
7. Decide when the mentoring sessions will take place.
8. Determine the length of the program and how often mentors and mentees will meet.
9. Decide where the mentoring will take place (e.g., school, faith based, work, e-mentoring).
10. Define program stakeholders (e.g., advisory board, parents, mentors, mentees), and identify how they can promote the program.
11. Determine ways the program can be evaluated for its effectiveness.

12. Establish administrative procedures and protocols to ensure that they monitor and support the extent and quality of all mentoring efforts.

These general steps can be applied to almost any formal mentoring program and across cultures. Sport and exercise psychologists involved in developing mentoring programs would be well advised to employ them.

Conclusion

Mentoring is an important process that has the potential to influence a wide variety of individuals and behaviors in sport and exercise settings and in the field of SEP itself. While there is a broad base of mentoring research in other areas, few studies have been conducted in SEP. Moreover, studies from related fields like coaching and PE tend to be descriptive and devoid of any theory. In spite of this, these research findings parallel the general mentoring research, clearly indicating the steps that sport and exercise psychologists interested in mentoring need to take.

Daniel Gould and Andrew MacIntosh

See also Career Paths; Certifications; Consultant Effectiveness, Assessment of; Consulting; Credentials

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MINDFULNESS

Mindfulness is a state of nonjudging awareness and acceptance of internal experiences. This state has been achieved through evidence-driven mindfulness-based interventions that are useful for the treatment of psychological concerns and for the enhancement of performance among athletes and other performers. This entry describes the basic processes associated with mindfulness, the intervention components that enhance these processes, and the usefulness of mindfulness interventions for the enhancement of athletic performance.

Exploring Mindfulness

From the general public to sport psychologists, most would agree that optimal performance seems to require athletes to maintain a fully engaged and absorbed focus on the athletic task, a state often referred to as “flow.” Yet there are many obstacles to maintaining that focus, such as crowd noise, pressure to succeed, physical pain, and thoughts about possible failure. These obstacles are inevitable, and data indicate that efforts to control these variables (such as trying to ignore pain, crowd noise, and thoughts and emotions that seem to be getting in the way) are largely ineffective. In contrast, mindfulness interventions help athletes develop a nonjudging awareness (the ability to be aware of one’s experiences without judging them as good or bad, right or wrong, etc.) of internal and external experiences; an acceptance of their internal cognitive, emotional, and physiological (i.e., bodily sensations) states; and a willingness to experience internal and external states while continuing to engage in behaviors that are consistent with what really matters to them in athletics. Importantly, the goal of mindfulness practice is not to reduce, avoid, or change athletes’ subjective distress or increase characteristics such as confidence and positive thinking. Rather, the goal is to develop the skills necessary to engage in present-moment attention without becoming entangled in internal or external

experiences, without judging these experiences as right or wrong or good or bad, and without trying to alter these experiences. Instead of engaging in behaviors in order to reduce uncomfortable experiences (known as experiential avoidance), mindfulness encourages experiential acceptance, which is defined as a willingness to pursue (rather than avoid) one's goals (e.g., practicing hard during each practice) and values (e.g., maximizing athletic skill) regardless of how or what one may think or feel at any moment.

Along this line, mindfulness can be described as *being* with one's internal processes (i.e., thoughts, feelings, bodily sensations) regardless of what they are, rather than *doing* something with those processes in an impossible attempt to create the perfect internal state or situation within which to perform. For instance, this approach helps athletes learn that they can perform well while at the same time experiencing a variety of internal states (e.g., frustration, tiredness, worry) or dealing with an array of external distracters (e.g., opponents, poor officiating, coaching behaviors). Athletes can therefore shift from a perspective of "I want to perform well, *but* I don't feel confident today and I'm irritated" (a perspective that encourages experiential avoidance strategies) to the perspective of "I want to perform well, *and* I'm not feeling confident today and I'm irritated." This shift enhances the athlete's ability to non-judgmentally accept internal experiences (even if they do not feel good) and by doing so helps the athlete maintain awareness and attention to the immediate task at hand. By developing the capacity to perform well while experiencing a wider range of internal states and among shifting environmental situations, athletes can function more effectively by maintaining an increased capacity to remain task-focused as circumstances inevitably change.

Mindfulness Techniques

A number of mindfulness exercises are used to convey both the impossibility of eradicating or controlling the array of normal processes that athletes experience and the perspective that successful performance does not require an ideal physical, emotional, or cognitive state. Surely, if athletes had to feel physically sound, think only positive thoughts, and feel no negative emotions in order to perform well, few athletes would be able to perform on any

given day. The same reality applies for all humans within any life context.

Mindfulness techniques include a variety of meditative-type exercises that develop the ability to (a) become aware of one's internal and external experiences and recognize the inevitable and transient nature of these experiences and (b) recognize the costs of attempting to control, avoid, or modify experiences judged as "negative" versus the benefits of a nonjudging awareness and acceptance of internal experiences. A primary mechanism by which mindfulness exercises work is by developing a decentered (i.e., distanced) perspective about one's own thoughts and emotions and recognizing that thoughts and emotions are internal events that inevitably come and go, do not have to be evaluated or altered, and do not have to direct one's behavioral choices. Mindfulness techniques do not alter the content of athletes' internal experiences at all. Instead, they alter the relationship athletes have to their internal experiences. So, mindfulness does not attempt to change the frequency of thoughts and emotions, the type of thoughts and emotions, or the intensity of thoughts and emotions in order to promote optimal performance but instead advocates that athletes can function optimally while having these experiences. This leads to both on- and off-field behaviors that promote more functional athletic performance, from disciplined practice behaviors to committed competitive behaviors.

Mindfulness practices include various exercises that can range in length from 5 to 45 minutes and often involve sitting in place and allowing attention to move from aspects of one's body (e.g., breathing, parts of the anatomy) to the various thoughts, sensations, and emotions that naturally come and go. Ultimately, mindfulness practices create a climate within which athletes become aware of their cognitive, emotional, and physiological states; accept and allow for the presence of these internal experiences; maintain focus on task-relevant stimuli and contingencies; develop the capacity to self-regulate attention as desired (i.e., direct attention as needed to perform better); and embrace behaviors that are consistent with their goals.

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See also Centering; Flow; Psychological Well-Being; Self-Awareness

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MIRROR NEURONS

The term *mirror neurons* refers to neurons that show spike activity in response to the performance of an action and to the perception of the same type of action in an observed subject. These neurons, first found in the brain of macaque monkeys, are regarded as first evidence for a functional overlap between action and perception on a neuronal level.

Mirror neurons were first described in the 1990s by Giacomo Rizzolatti's research group at the University of Parma in Italy. Anecdotal evidence given by members of the group indicates that the discovery of mirror neurons was due to chance. Neurophysiologists were investigating neurons expected to respond to hand and mouth actions, such as manipulating objects or picking up food items for eating. Neuronal activity was measured from electrodes placed in the ventral premotor cortex of macaque monkeys. When the experimenter took a bite from his sandwich during the experiment, neurons from the macaque responded unexpectedly, indicating activity owing to the mere observation of the investigated action, this time performed by the human.

The first mirror neurons were identified in the F5 area of the inferior frontal gyrus of the macaque brain; later studies also identified mirror neurons in the inferior parietal lobule. Following their findings, a multitude of research groups engaged in the study of mirror neurons. Neurons with mirror properties were found in the premotor cortex, classically ascribed to the motor

system, and in parietal areas involved in the processing of perceived information. Following the observation of neurons responding to the perception and performance of manual actions, neurons responding to actions involving the mouth, face, and body were investigated. Today, various types of mirror neurons have been described, including neurons that respond selectively to actions observed in extra- or intrapersonal space, neurons that respond not to visual but to acoustic stimuli, and neurons that link actions to related acts of communication. Mirror neurons vary in their level of congruency; highly specialized neurons respond only to one type of action, whereas others generalize over actions that resemble each other, have the same goal, or follow each other in typical sequences.

Results from studies on mirror neurons provided strong support for theoretical perspectives that had so far lacked neuronal evidence. The overlap between action and perception processing in the brain had been addressed over decades by Wolfgang Prinz and colleagues through the common coding principle and the ideomotor theory; the concepts of embodiment and situatedness of cognitive processes are related to this work. These perspectives, together with the discovery of mirror neurons in monkeys, provided a strong motivation for the quest for neural structures with similar properties in the human brain. Studies investigating human brain areas with mirrorlike properties chiefly used neuroimaging techniques (i.e., functional magnetic resonance imaging [fMRI], positron-emission tomography [PET]), electroencephalography (EEG), transcranial magnetic stimulation (TMS), as well as behavioral paradigms. Areas with mirror properties were found in the ventral and dorsal premotor cortex and in parts of the parietal cortex, including the inferior parietal lobe, the superior parietal lobe, and the superior parietal sulcus. Numerous studies have corroborated that these brain regions are involved in and modulated by a large number of cognitive tasks, as well as social functions. Their activity has been claimed to underlie the planning and understanding of actions, imitation, language comprehension, empathy, social learning, and cultural transmission. Malfunctioning in these brain regions has been claimed to play a role in the emergence of disorders like autism. Because of their broad functional involvement, it has been argued that the term *mirror system* was misleading, and other

terms have been suggested instead, such as *action observation network* (AON).

Neurocognitive research, often applying sports- and dance-related paradigms, has shown that the relevant brain regions are more strongly activated during the observation of biological and especially human agents compared to artificial ones (e.g., robots) and that the degree of activation is modulated by the observer's capability and experience of performing the observed actions. Furthermore, there is evidence that individual regions within the network are specialized on different aspects of action perception and performance. Beatriz Calvo-Merino and colleagues studied brain responses to movement observation in capoeira and ballet experts. They found that the activation of relevant brain areas was more distinctive while the experts watched movements from their own discipline, and specifically movements belonging to their own active movement repertoire, compared to movements they had frequently watched but not performed themselves. Salvatore Aglioti and colleagues showed that expert basketball players performed better than experienced observers in predicting the success of a free throw during very early stages of the movement, and that their motor system and hand muscles were activated in a movement-specific way. Using a dance-training paradigm, Emily Cross and colleagues showed that the superior temporal cortex preferentially responded to the presence of a human model, whereas the ventral premotor cortex responded specifically to motor familiarity of the observed movement.

In recent years, critical views have been expressed regarding the interpretation of mirror neuron studies, specifically the existence and function of mirror neurons in the human brain. It has been argued that neurons with mirror properties might not represent a distinct class of cells but rather neurons in the motor system that have developed mirror properties as an artifact, additional to their original and more relevant functions. Based on empirical findings, Cecilia Heyes proposed that human mirror neurons might emerge as a by-product of associative learning and social interaction. Gregory Hickok strongly argued against the claim that mirror neurons were crucial for action understanding, emphasizing the lack of evidence of this function in monkeys in which the neurons had been described. Arthur Glenberg and Patricia Churchland both pointed out that most of the phenomena the human mirror neuron system (MNS)

has been associated with (including understanding intentions and actions, language comprehension, imitation, and disorders like autism) are not yet fully understood and that more research is needed to substantiate such claims.

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See also Body Image; Mental Rehearsal; Simulation Training

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MODELING

Along with the operationally defined concepts of dynamical systems theory comes a rich arsenal of mathematical methods and modeling tools that may be usefully employed in the study of complex sports behaviors. This entry highlights just a handful of those, which are all centered on the “problem” of dimensionality reduction. Before going into modeling, the entry first outlines some useful approaches to data analysis aimed at system identification, starting with more heuristic

approaches that capitalize on an educated guess about which state variable is (most) relevant, followed by more unbiased, statistical approaches.

System Identification

Parameter Extraction: Stability and Regularity

When seeking to identify the main characteristics of a system's dynamics, one typically applies conventional statistics in combination with stability-related measures like Lyapunov or Floquet exponents. Lyapunov exponents are the method of choice in estimating the stability properties of a dynamical system involving the emergence and disappearance of macroscopic patterns, such as the transitions from walking to running, and from ankle to hip strategy in upright posture.

The Lyapunov exponent measures the exponential divergence of nearby trajectories and thus the robustness of a (movement) pattern against (small) perturbations. Only if the exponential divergence is negative, a perturbation will be compensated and the pattern is (locally) stable; otherwise, a transition is likely to occur or is permanently present (as in chaotic systems or self-organized criticality). In the study of movement coordination, Lyapunov exponents and other stability-related measures have proved rather useful in addressing movement flexibility, stability, and adaptability.

Of similar importance for the identification of dynamics are entropic measures. In physics, entropy is used to quantify the degree of disorder in a statistical ensemble such as a gas. When looking at movement trajectories, in contrast, entropy may serve to quantify the degree of regularity. A frequently used measure in the analysis of noisy time series, such as movement data, is sample entropy. Entropic measures have been used, among others, to assess balance in gymnastics and dance, and the effects of sports-related anxiety and brain injury on postural control. Entropy can be readily estimated, irrespective of the type and dimensionality of the data in question. For instance, predictability of competitive balance in sporting contests has been assessed through entropy using the example of the 2006–2007 English Premier League results.

Dimensionality Reduction: Principal Component Analysis and More

Advances in data acquisition have allowed for simultaneous recordings of multiple signals for

considerable time spans, resulting in huge data sets. This development has increased the need of a priori data reduction. Principal component analysis (PCA) suits this purpose.

When analyzing movement, an estimate of the covariance matrix between trajectories (or time series) is central to PCA. Eigenvalues of this covariance matrix rank the degree to which a so-called principal component contributes to the entire data variance (which provides an index of its strength). The corresponding eigenvectors are the principal components, also referred to as factor loadings. When projected onto these components, the movement trajectories yield factor scores (here the time series of the components). PCA has found wide application in sports, and many examples may be found in the literature.

Alternative but related approaches include, for example, clustering techniques and artificial neural networks, which all pursue the same goal: reducing dimensionality. However, an even more principled and rigorous approach to pursue that goal is to focus on phase transitions because theory dictates that in the immediate vicinity of the critical point the dynamics of the complex system under study is reduced to a (small) set of order parameters. Whenever that scenario occurs, mathematical modeling in terms of dynamical systems becomes feasible.

Identifying Dynamics: Capitalizing on Noise

This section highlights a recently established method for identifying stochastic dynamics. Originally, this analysis had been developed to study stochastic processes in the context of turbulent flows, but of late, it has also been applied successfully in the analysis of a broad variety of experimental data including human movement. The method produces numerical values of dynamical equations as a function of phase space coordinates. From these numerical values, specific model terms can be estimated, but they can also be analyzed directly by reconstructing vector fields and comparing those across experimental conditions; see Figure 1.

Modeling

To model a complex dynamical system, consider x_k as state variable dependent on time t . Further, let the system be high-dimensional (e.g., of dimension n) such that the entire system can be cast in

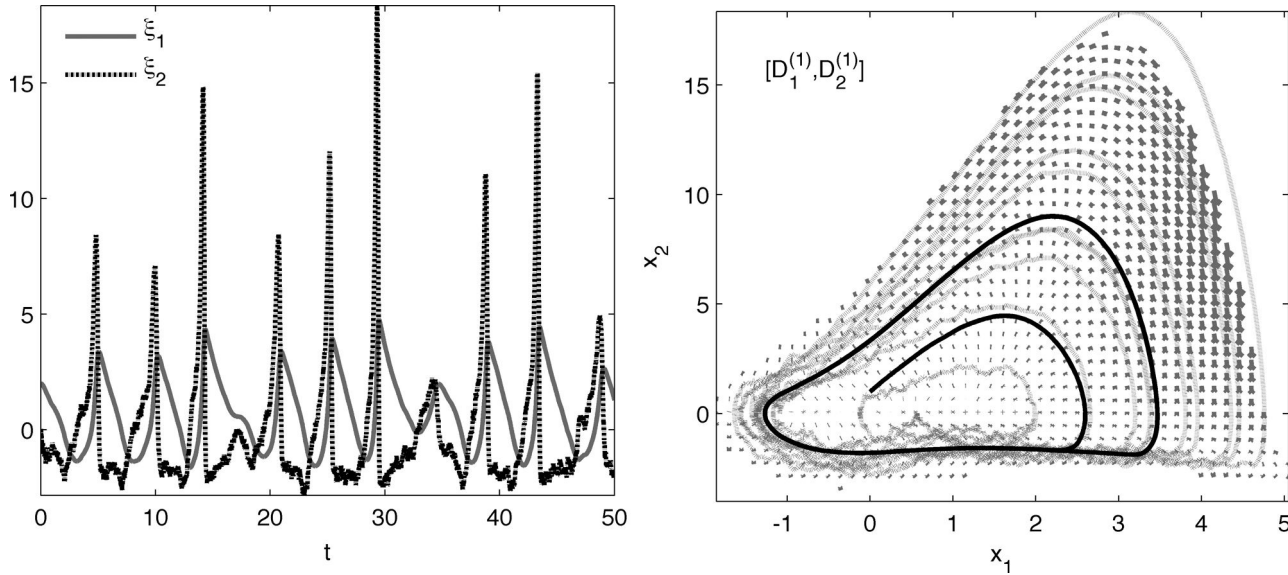


Figure 1 Identifying Dynamics

Source: Andreas Daffertshofer.

Note: From the recorded signals (left panel), the so-called drift coefficients $D^{(1)}$ are determined. With those drift coefficients, the deterministic part of the dynamics can be reconstructed (right panel). This procedure can also be applied in the n-dimensional case, although numerical implementations still need to be improved.

the vector-form $X = (x_1, x_2, \dots, x_n)$. Its stochastic dynamics may then read

$$\dot{X} = F(X, t) + \text{noise}, \quad (1)$$

where the dot-notation refers to the derivative with respect to time. In the presence of phase transitions one can split the state variables into those that become unstable (u , representing the switch in macroscopic patterns) and those that stay stable (s). One may formally write $X = (u, s)$. The previously explained separation of time scales in the vicinity of the phase transition implies that the stable parts can be eliminated adiabatically,¹ which somewhat counterintuitively has the mathematical form $\dot{s} = 0$. This adiabatic elimination allows for expressing the stable components as mere functions of the unstable ones, i.e., $s = M(u)$, which known as the center manifold theorem according to which the stable components are “enslaved” by the unstable ones. The latter are hence the so-called order parameters whose dynamics read

$$\dot{u} = N(u, t) + \text{noise}. \quad (2)$$

Importantly, the dimension of u is always much smaller than that of X by which one can the rigorously analyze the system X through well-established mathematical techniques.

As an example, consider the case in which the order parameter dynamics can be written using the gradient of a potential V —that is, $\dot{u} = -V'(u)$. If V is depicted as a landscape, then the systems stability can be readily determined by means of the hills and valleys of the landscape: the former represent the unstable states and the latter the stable states of the dynamics.

Movement Coordination: Low-Dimensional Dynamics

The previously given approach had a major impact on the study of the coordination of rhythmic movement through the seminal work of J. A. Scott Kelso and colleagues on frequency-induced phase transitions in rhythmic finger movements. This transition was modeled by means of potential for the relative phase between the finger movements, which changed shape as a function of movement frequency, as shown in Figure 2, and an underlying system of two nonlinearly coupled nonlinear oscillators. The model motivated a flurry of research on rhythmic bimanual coordination, with ramifications to learning and development, the accompanying brain activity, and to interpersonal coordination in dyads and teams. Through the intimate connection between the study of motor

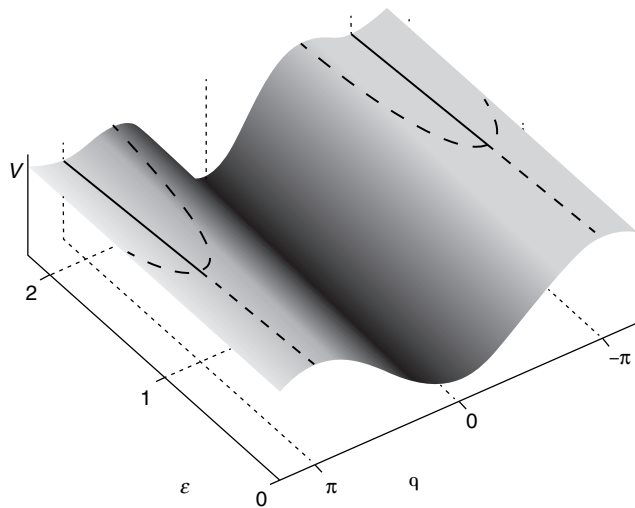


Figure 2 Potential Underlying the Dynamics of the Relative Phase

Source: Andreas Daffertshofer and Peter J. Beek.

Note: The order parameter is here given as the relative phase between two oscillatory units—that is, $u = \varphi = \phi_1 - \phi_2$. Order parameter dynamics are given through a potential depicted here as $V(\varphi) = -\cos(\varphi) - \frac{\varepsilon}{4}\cos(2\varphi)$. Dependent on the control parameter ε different states are stable. Solid lines display the valleys and dashed lines display the hills. A phase transition occurs if ε is increased beyond the critical point $\varepsilon = 1$. There $\varphi = \pm\pi$ loses stability.

control and learning and sports behavior, it also penetrated the domain of sport science.

Entrainment of Locomotion and Respiration—Also Low-Dimensional

A closely related approach revealed how the optimization of oxygen extraction in animal respiration can explain the multifrequency synchronization of breathing and locomotion observed in competitive rowing. In this case, the to-be-analyzed potential is formed by the negative effective value of lung volume (modulated by the cyclic muscle contraction due to locomotion). The corresponding variational principles (searches for hills and valleys) do not only predict steady—that is, optimal—states but also (points of) transitions between those states whenever control parameters—again, movement speed but also fatigue—are being altered; see Figure 3.

Graph Theory

The analysis of complex networks as a discretized and distributed version of complex

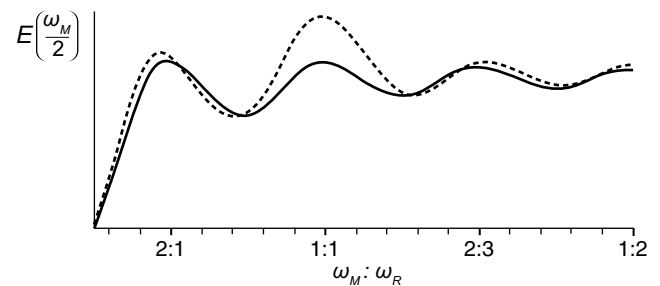


Figure 3 Effective Value of the Lung Volume as a Function of Locomotion and Respiratory Frequency Ratio

Source: Andreas Daffertshofer and Peter J. Beek.

Note: The effective value of lung volume E (symbolizing the aerobic capacity) is shown as a function of frequency ratio $\omega_M:\omega_R$ between movement (stroke) and respiration (inhaling). Maxima of E are stable solutions ($\sim$ minima of valleys of the potential).

systems has found many applications in recent years. Complex networks link to dynamical systems by virtue of the phase transitions that occur when a network grows and connections are formed; in random networks, transitions occur toward (almost) fully connected networks dependent on growth rate and the probability to connect. It is duly recognized in the literature that match play in (ball) sports are amenable in terms of an analysis in terms of network theory. And indeed all ingredients for this are available: The players represent the nodes, and they are connected to each other via ball passes and via perception-based collective motion. These connections represent the edges. The number of passes between players provides a strength of their connection and so does—although more difficult to quantify—coherent motion. An analysis of a sports match in those terms is already arguably more revealing than an analysis in more conventional terms like average speed, distance covered, or areas visited. Using a variety of so-called centrality measures, it may provide information about the importance of nodes (hubs) and edges in the network and thus reveal strategic behaviors. Various attempts have been made in this regard, but a crucial element has so far been missing—namely the fact that the connections between players serve the purpose of making and preventing goals, implying that goal effectiveness and directionality should be incorporated into the analysis. In other words, network theory needs to be appropriated to match analysis in order to make it transcend the current divide

between match play analyses based on temporal correlations (e.g., relative phase) and spatial correlations (e.g., Voronoi diagrams).

Conclusion

Concepts of dynamical systems build on a potent arsenal of mathematical tools. Exploiting this arsenal yields a highly sophisticated assessment of complex sports performance. Models in motor control are seminal for the predictive capacity of this approach. In particular, the macroscopic switches in behavior can be analyzed with great precision. Like the general focus on dynamical systems in sports science, also modeling efforts are still in their infancies. The rigor of the employed mathematical tools, however, renders the approach comprehensive and appealing.

Andreas Daffertshofer and Peter J. Beek

See also Anticipation; Coordination; Dynamical Systems; Freezing; Modeling; Motor Control; Motor Development; Motor Learning; Movement; Perception

Note

1. This is a term lent from statistical physics where it refers to the fact that a process does not show any gain or loss of heat with its surrounding. Here it implies that the stable subsystems are so quick that they can instantly adjust to changes of the unstable ones—that is, s is too quick to affect u .

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MODELS OF EMOTION– PERFORMANCE

Emotion is an integral part of human functioning and can enhance or hinder individual and team performance. This entry defines emotion and then provides an overview of theories and models that have been used to explain and describe the relationship between emotion and performance. These include multidimensional anxiety theory (MAT), the individual zones of optimal functioning (IZOF) hypothesis, and catastrophe models, as well as some of the associated elements of the emotion–performance relationship, including interpretation of emotion and the cognitive–motivational–relational theory (CMRT) of emotions.

Richard Lazarus proposed that emotions are organized psychophysiological reactions to news about ongoing relationships with the environment. Emotions arise when people attend to situations that are deemed important to their goals (e.g., competing, supporting a team). Although the terms *emotion* and *feeling* are often used interchangeably, they can be differentiated on the grounds that emotions have both “feeling” and “doing” aspects; feelings do not necessarily involve action or the impulse to act. Most researchers agree that it is the action or the impulse to act that leads to changes in the autonomic nervous system (ANS) and the neuroendocrine system. The ANS is responsible for the control of involuntary or unconscious bodily functions, and the neuroendocrine system is the system that involves both nervous stimulation and endocrine secretion. Both of these systems anticipate a behavioral response to the emotion experienced, and humans can develop an awareness of their emotions and the associated changes. It is this awareness that led William James, a philosopher and psychologist considered by many to be the father of American psychology, to view emotions

as response tendencies that may be controlled in a number of ways. Within sport and exercise it is this control feature of emotions, or the ability to regulate emotion, that has received most attention.

Researchers have typically conceptualized emotions in one of two ways. The first is to conceptualize emotions as lying along two dimensions: hedonic tone (pleasure or displeasure) and activation (relaxation or tension)—often referred to as *Russell's circumplex model of affect*. In this model, each emotion (or *affect*) is considered as a combination of high or low hedonic tone and high or low activation such that all emotions can be categorized in one of four quadrants: high hedonic tone and high activation (e.g., excited), high hedonic tone and low activation (e.g., happy), low hedonic tone and high activation (e.g., anxious), and low hedonic tone and low activation (e.g., sad). Although appealing, this model does not allow one to distinguish between some emotions that might otherwise be considered distinct. For example, anger and anxiety are thought to be distinct emotions but both are classified as low in hedonic tone and high in intensity in the circumplex model.

The second way that researchers have conceptualized emotions is to consider them as primary discrete states that vary in important ways from each other. The ways in which these discrete emotions vary is influenced by individuals' different appraisals of a situation. An appraisal is the cognitive evaluation a person makes regarding the nature and significance of an event. The idea that people can respond with different emotions to the same situation depending on how they appraise (interpret) the situation is one of the central features of cognitive appraisal theories of emotions.

Within such an appraisal framework, the modal model of emotion has been utilized to understand emotion and its behavioral responses. The modal model of emotion is a sequential model, the starting point of which is an individual in a situation that is important to him or her. The interaction between the individual and this situation forces the individual to attend to what is happening. The individual then interprets what is happening and forms an appraisal of the situation. This appraisal results in a multisystem response to the individual-situation interaction (e.g., fight or flight). Thus, in this model, it is the cognitive appraisal that underlies all emotional states and different individuals can display different responses to the same

situation. The view that individuals can display different emotional responses to the same situation led researchers to begin to identify specific discrete emotions.

Paul Ekman and colleagues pioneered research that examines emotions in relation to facial expression and identified seven basic emotions on that basis: anger, disgust or contempt, fear, happiness, interest, sadness, and surprise. Other researchers have extended this list and proposed other discrete emotions. For example, Carroll Izard proposed that guilt, shame, and distress are also discrete emotions; Richard Lazarus further proposed shame, envy, pride, jealousy, anxiety, relief, hope, love, and compassion as discrete emotions. Within sport and exercise, a range of emotions have been observed, such as anxiety, frustration, disappointment, happiness, hope, and anger. It is anxiety that has received by far the most attention especially in relation to the emotion–performance relationship.

Anxiety

Anxiety is considered an unpleasant emotion that can be situation-specific (state anxiety [SA]) or dispositional (trait anxiety [TA]). Charles Spielberger suggested that SA is subjective with conscious feelings of tension and apprehension associated with arousal of the ANS. SA is normally relatively transitory, as it is the individual's response to a specific threatening situation or event. Conversely, TA is a general disposition to respond to a variety of situations with high SA. Early theoretical accounts of anxiety were from a one-dimensional perspective. It is now accepted, however, that anxiety has at least two distinguishable components: a mental component normally termed *cognitive anxiety* or *worry* (e.g., I am worried about not performing well); and a physiological component normally termed *somatic anxiety* or *physiological arousal* (e.g., an increase in skin conductance [SC], often manifest in sweaty hands). Outlined next are a number of models and theories that have been proposed for the anxiety–performance relationship.

Multidimensional Anxiety Theory

Although it is referred to as a theory, Matthew Martens and colleagues' MAT is largely a model that describes the relationships between anxiety and performance. MAT posits that cognitive anxiety and somatic anxiety are related to performance in two independent ways. According

to MAT, *cognitive anxiety* has a linear negative relationship with performance. This relationship is proposed on the basis that worrying thoughts (cognitive anxiety) take up some of an athlete's cognitive resources. As the anxious athlete has fewer cognitive resources, he or she is no longer able to meet the cognitive demands of the task and as a result his or her performance deteriorates. In other words, the more athletes allocate cognitive resources to worry, the fewer resources they have for their performance and poorer performance results. According to MAT, *somatic anxiety* has an inverted-U relationship with performance whereby performance is optimal at moderate levels of somatic anxiety. The somatic anxiety–performance relationship is based on the Yerkes–Dodson inverted-U arousal–performance model. According to this adaptation of the inverted-U model, performance improves as an athlete becomes more physiologically aroused (e.g., they become faster, more efficient, more accurate, more alert, more focused), but this performance improvement is only up to a point; this point is termed the *point of optimal arousal*, as it is the point at which best performance is achieved. Beyond that point, if the athlete continues to become more anxious then performance will gradually start to decline (e.g., become slower, less efficient, less accurate, less alert, less focused).

Interpretation of Emotions

Most emotions are conceptualized as being either negative or positive. For example, anxiety is considered a negative emotion and happiness is considered a positive emotion. However, a negative emotion does not simply lead to a negative performance, and athletes also do not typically interpret their emotions in this simplistic fashion. For example, an athlete can have uncomfortable anxious thoughts and feelings about an upcoming sporting event and feel that these “negative” feelings are desirable, even necessary, for optimal performance. Studies by Hardy and colleagues with basketball players and of crown green bowlers in the United Kingdom have shown that a negative emotion (e.g., cognitive anxiety) can indeed sometimes have a positive effect on performance. In the specific context of anxiety interpretation, Mahoney and Avenier were the first to report that performers could interpret their anxiety in different ways: The more successful gymnasts tended to use their anxiety as a stimulant to better performance, whereas

less successful athletes seemed to reach a state of near panic. Furthermore, Graham Jones and colleagues have reported that successful athletes tend to view their anxiety as helpful to performance.

Individualized Zones of Optimal Functioning

In the 1970s and 1980s, Yuri Hanin proposed the IZOF hypothesis. The central tenet of the IZOF hypothesis is that each athlete has his or her own optimal zone of preperformance emotion within which he or she is more likely to attain optimal performance. If the level of an emotion lies outside of this zone of optimal functioning, performance will be impaired. To illustrate, a person's optimal level of anxiety is specific to that particular individual and if the athlete is in his or her “zone,” peak performance will follow. Athletes may thus perform differently from each other even if they are experiencing the same degree of anxiety. Investigations including emotions such as anger, disappointment, frustration, excitement, and joy to derive IZOFs have supported the applicability of the IZOF concept in the emotion–performance relationship. While literature on the IZOF has predominantly focused on anxiety, researchers have contended that a recipe of emotions should be used for optimal performance. For example, some athletes may flourish under high levels of anxiety with high levels of hope (or other positive emotions), while other players will do better with lower anxiety in combination with higher or lower levels of other emotions.

Catastrophe Models

In the early 1990s, Lew Hardy and his colleagues developed the cusp catastrophe model as a description of the relationship between anxiety and performance. The cusp catastrophe model (Figure 1) predicts that an athlete's performance will depend on a complex interaction between cognitive anxiety and physiological arousal. The cusp catastrophe model is a three-dimensional relationship among cognitive anxiety, physiological arousal, and performance with the following predictions: (a) when physiological arousal is low, increases in cognitive anxiety will be beneficial to performance; (b) when physiological arousal is high, increases in cognitive anxiety will be detrimental to performance; (c) when cognitive anxiety is low, changes in physiological arousal will result in small and continuous changes in performance

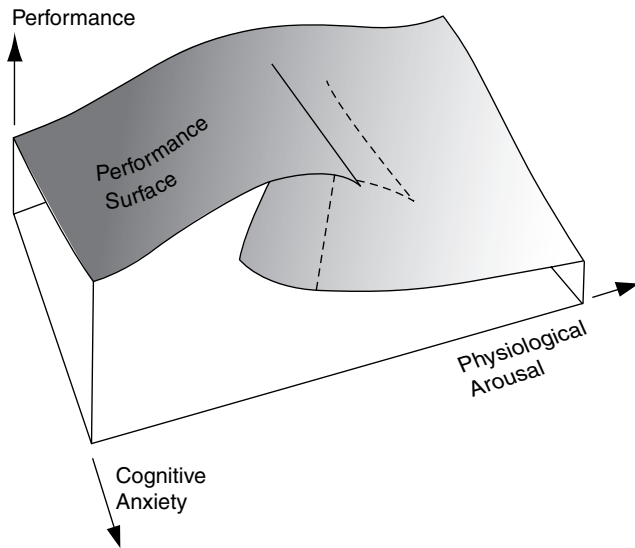


Figure 1 Two-Surface Catastrophe Model

Source: From L. Hardy, G. Jones, & D. Gould (1996). *Understanding psychological preparation in sport* (p. 151). Chichester, UK: Wiley. Copyright © John Wiley & Sons Ltd. Reprinted by permission.

in the form of a mild inverted-U; and (d) when cognitive anxiety is high, physiological arousal will either help or hinder performance, depending on the level of physiological arousal that is experienced. Specifically, when cognitive anxiety is high, performance will increase as a function of physiological arousal, up to a point. If physiological arousal increases beyond this point, the athlete suffers a dramatic drop in performance: a catastrophe. This large and dramatic drop in performance is reversible only if the athlete's physiological arousal reduces to a point that was lower than the point at which the performance catastrophe occurred. Thus, a catastrophe can only be reversed by a substantial reduction in physiological arousal. When cognitive anxiety is high, the small changes in physiological arousal that can result in large and discontinuous changes in performance is termed *hysteresis* and changes in performance are different depending upon whether physiological arousal is increasing or decreasing.

Hardy also developed the butterfly catastrophe model, which is an extension of the cusp catastrophe model. The butterfly catastrophe model incorporates self-confidence as a fourth factor, which is called the bias factor. The bias factor (self-confidence) moderates the interaction between cognitive anxiety and physiological

arousal on performance such that the front face of the model (see Figure 1) swings to the right (high self-confidence) or to the left (low self-confidence). Under high cognitive anxiety, highly self-confident performers are able to withstand higher physiological arousal before suffering a sudden drop in performance (i.e., the front face of the model swings to the right) than their less self-confident counterparts. Hardy and colleagues' studies with golf and basketball players have shown support for the catastrophe model. In addition, Stuart Beattie and colleagues have provided further evidence for the catastrophe model and hysteresis in a rugby context. Operationalizing a performance catastrophe as disengagement from the task, these researchers found that an increase in physiological arousal led to sudden disengagement from the task when cognitive anxiety was high.

Cognitive–Motivational–Relational Theory of Emotions

Richard Lazarus proposed the cognitive–motivational–relational theory (CMRT) of emotions in sport. According to Lazarus, *cognitive* refers to knowledge and appraisal of what is happening in an encounter: Knowledge is both situational and generalized beliefs about how things work; appraisal is an evaluation of the personal significance of an encounter. *Motivational* refers to two interrelated components. First, it is a characteristic of a person: a dispositional variable that one brings to every encounter. Second, the disposition to attain a goal has to be activated by the demands, constraints, and resources that the environment provides (i.e., a suitable environment to achieve that goal must be present). *Relational* suggests that emotions are always about person–environment relationships involving harms or benefits. Emotions are created by person–environment relationships that change over time and circumstances. A central premise of the theory is that each emotion involves a distinct core relational theme that describes the interaction between the individual and the environment. The core relational theme is a summary of the appraisals that individuals make in assessing the risk and reward involved in a particular situation. For example, the core relational theme of anger involves the perceived commitment of an offence. Each core relational theme has an associated action tendency that directly represents the manifestation of the

person's appraisal of the stimulus in relation to the self. For example, according to Lazarus, the associated action tendency for anger is the impulse to retaliate with a view to restoring self-esteem.

In CMRT, it is proposed that the core relational theme and the associated action tendency will influence performance depending on the complex relationship between the athlete and the situation. For example, anger may negatively impact performance if it draws resources away from the primary task at hand. However, if the task requires a "lashing out" motion toward an aggressor or opponent, performance may be facilitated due to its close association with anger's action tendency. Tim Woodman and colleagues have offered support for CMR in a "kicking out" task where anger was primed via an imagery script. Performance on the physical task was significantly greater in the anger condition compared with happiness and emotion-neutral conditions. In contrast, on a cognitive task, which does not match anger's action tendency, anger did not have a facilitative effect. From this research, it appears that in order for anger to be an effective performance-enhancing emotion, the task needs to be aligned with anger's lashing out action tendency. The perspective that action tendencies have matching themes has also been applied to other emotions (e.g., hope, pride, happiness).

Conclusion

Researchers of emotion and sport performance have proposed a number of models and theories examining how emotions impact performance. While there has been a predominant focus on anxiety, more recently researchers have begun to focus their attention on the role of other emotions. Understanding the complexity of emotions continues to be of value to researchers, athletes and applied practitioners with a view to optimal performance. Although the early models of emotion–performance conceptualized the emotion–performance relationship as a series of two-dimensional relationships, it is the interaction between different components of emotions or between emotions themselves that stands to further our understanding in this fascinating and complex research domain.

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See also Attention–Performance Relationships; Attention Theory; Choking; Emotional Responses

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MORAL ATMOSPHERE

The concept of moral atmosphere was originally described by Lawrence Kohlberg and his associates, who investigated school and prison environments to determine the influence of the

group norms of these settings on moral reasoning and behavior. They proposed that, over time, groups develop their own culture and a shared understanding of what is appropriate behavior as a result of interaction among group members. These shared group norms define the moral atmosphere of a group. Thus, moral atmosphere involves a set of collective norms regarding moral action on the part of group members. In this entry, the relevance of moral atmosphere to sport is explained, research examining this construct in sport is discussed, and the link between moral atmosphere and other relevant variables is highlighted. The entry ends with study limitations and conclusions regarding the importance of moral atmosphere in sport.

Moral atmosphere is particularly relevant to sport settings, especially team sport. For example, in a soccer team certain philosophies are developed over time regarding appropriate behavior in that context. These philosophies are partly the outcome of characteristics of the coach and team members and develop as players interact with each other and with their coach. Teammates' choices in situations that give rise to moral conflict are also part of the moral atmosphere. Moral atmosphere (which is also referred to as team norms) is assumed to affect moral decision making (DM) and behavior in sport.

D. E. Stephens and Brenda J. L. Bredemeier were the first to examine moral atmosphere in relation to self-reported likelihood to aggress against an opponent in young female soccer players. They presented participants with an aggression scenario featuring a hypothetical protagonist who was faced with the decision of tackling an opponent from behind, thereby risking injuring her. Athletes were asked to imagine themselves in this situation and indicate how likely they would be to tackle the hypothetical opponent from behind. Moral atmosphere was assessed via players' perceptions of the number of teammates willing to engage in the behavior and the degree of importance the coach placed on normative success. When athletes perceived that a large number of their teammates would engage in the described behavior and that being the best team was important to their coach, they indicated greater likelihood to behave aggressively.

In another line of research by Maria Kavussanu, Steve Miller, and their colleagues, moral atmosphere was examined in relation to multiple components of morality—namely, moral judgment, intention,

and behavior. These researchers have typically presented athletes with scenarios describing inappropriate behaviors—for example, faking an injury, pushing an opposing player when the referee is not looking, and risking injuring an opposing player. Players' perceptions of the number of teammates who would engage in these behaviors, as well as their perceptions of their coach as encouraging the behaviors in question if it was necessary for the team to win, were measured as the two dimensions of moral atmosphere. Soccer and basketball players who perceived their coach as encouraging the described behaviors and a large number of teammates as likely to engage in the behaviors, were more likely to judge the behaviors as appropriate and to report the intention to engage and greater frequency of engagement in the behaviors.

An interesting finding revealed in previous research is the strong relationship between moral atmosphere and performance motivational climate. This refers to a goal structure that values normative success and rewards only the best players and is typically created by important others in the achievement context such as coaches. In some studies, players who perceived that their coach created a performance motivational climate in their team also perceived their coach as encouraging them to push another player, fake an injury, or risk injury to their opponents, if such behaviors were necessary for the team to win. This finding suggests that coaches who create a performance motivational climate in their team value winning over the players' welfare.

In all sport studies that have examined moral atmosphere, the relationship between this construct and moral variables is very strong. It is important to note that in all studies to date, moral atmosphere has been measured via athlete reports in response to the same scenario that is also used to measure athletes' behavior. It is possible that, to a certain degree, athletes may project their own thoughts on others. That is, athletes may think that their teammates would behave in a certain manner because they are likely to behave in this way. This issue is important to keep in mind when interpreting the findings of previous research. However, overall, there is strong evidence for the relationship between moral atmosphere and moral behavior in sport, and it makes sense that the prevailing team norms should affect players' behavior.

Maria Kavussanu

See also Aggression

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MORAL BEHAVIOR

The term *moral behavior* is used with at least two different meanings in the literature. The first is when researchers explicitly define moral behavior and refer to the conditions in which an act is right, or ethical; when these conditions are met, the behavior can be called moral. For example, Augusto Blasi defined moral behavior as behavior that is intentional and a response to some sense of obligation, with the latter being a response to an ideal. James Rest stated that without knowing what produced the behavior we cannot call it moral. David Shields and Brenda Bredemeier have emphasized the importance of assessing a person's motives in the study of moral behavior. Albert Bandura highlighted the consequences of the act as important considerations for the labeling of behavior. When the term *moral behavior* is used in this manner, researchers do not call measured behavior "moral" because from this perspective we cannot call behavior moral unless we know one's motives for the behavior.

The second meaning of the term *moral behavior* is behavior that is the subject of the moral domain. Elliot Turiel and his colleagues found that children and adolescents perceive behavior as a moral transgression when it is intentional and has consequences for the victim. Behaviors that are classified in the moral domain are these that involve physical harm (e.g., pushing, shoving, and hitting), psychological harm (e.g., teasing, hurting

feelings, ridiculing, or name calling), fairness and rights (e.g., breaking a promise or destroying others' property), and positive behaviors (e.g., helping another in need, and sharing). Although researchers do not always explicitly refer to the moral domain, the meaning of the term *moral behavior* is implied by the content of reviews or articles that refer to moral action but discuss research on poor sporting behavior, temptation to play unfairly, prosocial behavior, and aggression. This entry introduces prosocial and antisocial behaviors, which have been referred to collectively in past research as moral behavior. First, prosocial and antisocial behaviors are defined providing examples from the sport context. Then their measurement is discussed followed by a description of the antecedents of these behaviors. The entry ends with some concluding remarks.

Definitions

In sport, as in any interpersonal context, intentional behaviors that typically have positive or negative consequences for others occur. For example, athletes encourage and support their teammates after a mistake and also help other players off the floor or when they are injured but also cheat, break the rules, or try to hurt other players. These two sets of behavior correspond to what Bandura has called proactive and inhibitive morality, respectively—that is people doing good things but also refraining from doing bad things. The terms *prosocial* and *antisocial* behavior have been linked to proactive and inhibitive morality, respectively, and are defined next.

Prosocial behavior is voluntary behavior intended to help or benefit another individual or group of individuals. Examples in sport are lending equipment to an opponent; helping another player off the floor; and encouraging, supporting, or congratulating a teammate. Prosocial behavior can be performed for a variety of reasons. For instance, when a basketball player helps his teammate off the floor, he intends to help or benefit the recipient. However, he could do this because he truly cares about his teammate, or he wants to impress his coach, or he expects to receive some other benefit. Thus, although the behavior is voluntary (i.e., non-accidental) the reasons for the behavior can vary.

Antisocial behavior is behavior intended to harm or disadvantage another individual or group of individuals. Examples of antisocial behavior in

sport are trying to injure an opponent, verbally abusing a teammate, retaliating after a bad foul, faking an injury, and breaking the rules of the game. Antisocial behavior overlaps with aggression, which is verbal or physical behavior that is purposeful, intended to cause injury, and has the capacity to cause psychological or physical harm to another person. However, antisocial behavior is broader than aggression. For example, breaking the rules of the game or faking an injury are behaviors intended to disadvantage the opponent, but they do not involve physical or psychological harm. Thus, these behaviors can be classified as antisocial but not aggressive. In contrast, verbally abusing or physically intimidating another player can harm the player; thus, they are both antisocial and aggressive. The main difference between antisocial and aggressive behaviors is that the term antisocial refers to a broader class of acts, some of which can be classified as aggressive.

Prosocial and antisocial behaviors typically have positive and negative consequences, respectively, for others. Thus, helping an injured player should alleviate his or her distress, and kicking a player should cause him or her to experience pain. However, if the player does not feel any pain (e.g., because he or she is distracted or has taken analgesic medication), the behaviors will not have the expected consequences. Similarly, encouraging a teammate may not benefit the athlete who does not appreciate such encouragement; verbally abusing a player may not cause psychological harm to the individual who ignores the abuse; and cheating may not negatively affect others if the offender is caught and the situation is rectified. Thus, prosocial and antisocial behaviors have the *potential* to affect others.

Many studies have examined prosocial and antisocial behaviors in sport. A consistent finding in these studies is that these behaviors are either inversely associated or unrelated to each other. When they are inversely associated, the correlation between them is small or medium. This means that they are relatively independent. Knowing that someone has acted prosocially toward another player does not tell us much about his or her antisocial behavior. Thus, it is important to examine both types of behavior.

Measurement

Prosocial and antisocial behaviors in sport have been measured using questionnaires and behavioral

observation. Questionnaires consist of several items referring to specific behaviors (e.g., helping an opponent off the floor, congratulating a teammate, physically intimidating an opponent, verbally abusing a teammate). Participants are asked to indicate how often they engaged in these behaviors in a specified period of time. Responses to items are averaged to form separate scores for prosocial and antisocial behaviors. Behavioral observation involves videotaping games and coding behavior to predetermined categories based on operational definitions. For example, helping a player off the floor is operationally defined as pulling a player off the floor, and every time this behavior is observed, a frequency is recorded in the respective category. Frequencies for prosocial and antisocial behavior are computed separately to provide a total score for each behavior type. Studies using these two methods have identified several antecedents of moral behavior, which are discussed next.

Antecedents

A number of individual difference and social environmental factors have been associated with prosocial and antisocial behaviors. Maria Kavussanu has provided a comprehensive overview of these variables. Three of the main antecedents identified in past research are *goal orientation*, *moral disengagement*, and *motivational climate*. Goal orientation refers to the subjective criteria of success people use in achievement contexts such as sport. Two major goal orientations operate in sport: task and ego orientation. Athletes high in task orientation feel successful when they try hard, do their best, improve their performance, and master new skills. These players are more likely to engage in prosocial behavior while playing sport. In contrast, players high in ego orientation feel successful only when they do better than others; they are preoccupied with winning and showing that they are the best. These players are more likely to engage in antisocial behavior and think that it is appropriate to cheat and injure other players.

The second variable, moral disengagement, refers to a set of cognitive mechanisms that people use to justify their antisocial behavior. Through these justifications, athletes manage to engage in antisocial behavior without experiencing negative feelings, such as guilt, that normally controls this behavior. For example, players may displace responsibility for their actions to their coach,

blame their victim for their own behavior, claim that they cheated to help their team, or downplay the consequences of their actions for others. Moral disengagement is associated with higher frequency of antisocial behavior and lower frequency of prosocial behavior.

Finally, the motivational climate of the team refers to the criteria of success that are dominant in the social environment and is created by parents, teachers, and coaches. Through the feedback they provide, the rewards they give, and, in general, the way they interact with the players, coaches make clear what are the criteria of success in that achievement context. For instance, when coaches provide feedback about how good a player is relative to others and reward only the best players, they create a performance motivational climate sending a clear message to athletes that only high ability matters. Players who perceive a performance climate in their team are more likely to display antisocial behavior. In contrast, a mastery climate is one that focuses on individual player improvement, rewards effort and personal progress, and has been associated with prosocial behavior.

Conclusion

In conclusion, moral behavior is a term used to refer to behaviors that are included in the moral domain and it is often used to refer collectively to prosocial and antisocial behaviors. These behaviors are morally relevant because they can have consequences for others. Both dispositional and situational factors are associated with prosocial and antisocial behaviors in sport. Individual difference variables such as goal orientation and moral disengagement as well as social environmental factors such as the motivational climate of the team are important determinants of prosocial and antisocial behaviors. By focusing on personal progress, rewarding effort and improvement, and avoiding explicit social comparison, we could promote prosocial and deter antisocial behavior from the sport context.

Maria Kavussanu

See also Aggression; Moral Disengagement; Moral Values and Attitudes; Sportpersonship

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MORAL DEVELOPMENT

Moral development refers to the age-related developmental processes that individuals undergo as they become more knowledgeable, consistent, and responsive in their ethical functioning. Moral development includes cognitive, affective, volitional, and behavioral elements, and developmental lags in any of these areas can result in ethical violations. For example, an athlete who exhibits sophisticated thinking about ethical issues yet has poor emotional control may still impulsively cheat when there is little chance of detection. The following sections introduce major theories of moral development and how they have been used in sport psychology (SP) research.

Jean Piaget's Approach to Moral Development

The famed Swiss developmental psychologist Jean Piaget was the first to conduct a careful examination of developmental trends in children's thinking about moral issues. Piaget's classic book, *The Moral Judgment of the Child*, published in 1932, begins with nuanced descriptions of children playing marbles in the street. Piaget interviewed the children about the rules, their source, and their openness to revision. Drawing from his extensive interviews and observations, Piaget went on to describe children's thoughts not only about game rules but also about stealing and lying, harm, and justice.

Piaget identified two broad stages in children's moral development. Initially, children embrace a heteronomous approach to morality. At this stage, morality is viewed as an external set of

requirements handed down by authority figures, like parents. Rules are unchangeable. Eventually, as children gain experience in collaborative activity with peers, heteronomous morality gives way to autonomous morality. Children come to see moral regulation as something that is socially constructed and open to revision.

When considering Piaget's contribution to moral development research, it is helpful to keep in mind his broader contribution to psychology. Piaget was the most influential architect of an approach to psychology known as *constructivism*. In contrast to behaviorists, who argue that the child is simply the product of environmental determinants, Piaget argued that the child is an active agent. According to Piaget, the mind doesn't simply passively receive information; it actively seeks to interpret experience. Metaphorically, every child is a budding philosopher who tries to make sense of the world and construct meaning through interactions with physical and social reality.

Lawrence Kohlberg's Approach to Moral Development

With Piaget having pioneered the scientific study of moral development, it was Lawrence Kohlberg who brought it into the mainstream of psychology. Kohlberg began his work in the late 1950s, and his theory and research dominated the field for nearly a half century.

Kohlberg began with an initial sample of 50 males, ages 10 to 28, whom he interviewed every 3 years for 18 years. Kohlberg claimed that each of his subjects went through the same sequence of changes in the structure of their moral thinking. Based on his interviews, Kohlberg proposed a six-stage model of moral development. Over the years, the precise descriptions of the stages evolved, and Kohlberg eventually dropped Stage 6. Before his death in 1987, Kohlberg and his colleagues had interviewed thousands of subjects from a diverse range of cultures and found general support for the universality and invariant order of his stage sequence. The next sections describe his theoretical orientation, his methodology, and the stage sequence that he proposed.

Kohlberg's Theoretical Orientation

Kohlberg's theory synthesized philosophical and psychological commitments. He was heavily influenced by Kantian philosophers, such as John

Rawls, who argue that there are universal principles of morality that can be discovered through reason. But reason doesn't come fully formed in the young child, and Kohlberg found inspiration in Piaget's studies of the development of children's cognitive reasoning.

Kohlberg offered a cognitive developmental theory of moral development. According to Kohlberg, cognition is key to morality. A moral act is one motivated by commitment to moral principles. People may not always act consistently with their best thinking, but sophisticated moral thinking is a necessary, if insufficient, precondition for acting morally. The defining characteristic of mature moral reasoning, according to Kohlberg, is an ability to weigh everyone's claims in a fair, unbiased, impartial manner. Stated differently, the core of morality is the principle of justice.

Kohlberg's Methodology

To investigate moral thinking, Kohlberg distinguished between the *form* and the *content* of moral thought. The content of moral thinking refers to the specific thoughts that a person has about a moral situation—the specific beliefs that they hold. The form or structure of moral reasoning refers to a deeper pattern to the way people think. Consider, for example, an athlete who argues that a baseball pitcher should hurl a fastball close to a batter's head in retaliation for the opposing pitcher hitting a teammate with a pitch the previous inning. The athlete's specific belief about what the pitcher should do is part of the content of their moral reasoning. Underlying that belief is a patterned way of thinking about issues of rights, responsibilities, fairness, and human welfare. That pattern of reasoning is the structural level of moral thinking and development.

Kohlberg presented his research subjects with brief hypothetical dilemmas and asked them what the protagonist should do. For example, in his most famous story, people are told about "Heinz," who must choose whether or not to steal a drug to save his dying wife. Whether the person says that Heinz should steal the drug or not is irrelevant to scoring their stage of development. That decision reflects the content of their moral judgment. Kohlberg was interested in the pattern of reasoning that the person used to reach their conclusion. By asking why a person thinks the way they do and probing their responses, Kohlberg sought to isolate the "moral grammar" beneath their judgments. Thus,

moral stages are defined in terms of the structure of moral thinking.

Kohlberg's research methodology is time-consuming and expensive. An alternative approach, based on a similar developmental model, was developed by James Rest. His Defining Issues Test (DIT) was developed to provide a paper-and-pencil alternative to Kohlberg's interview method. The DIT presents some of the same moral dilemmas as Kohlberg uses but then provides a set of possible responses to the dilemmas. Respondents rate the adequacy of the responses and then select the responses that most appeal to them. Rest's assessment methodology has been used in a number of sport investigations. In addition, his later elaboration of moral functioning processes, which extend beyond developmental issues, has laid the foundation for significant sport research (see the entry *Moral Judgment*, this encyclopedia).

Kohlberg's Moral Levels

According to Kohlberg, children move through a regular, age-related sequence of stages. While the stages of development are universal, children's progress through the stages can occur at varying rates, and many people plateau in their development without reaching the highest stages. Growth from one stage to the next is facilitated by experiences that lead them to question the adequacy of their existing stage of reasoning.

Each stage in Kohlberg's model is said to be better than the previous stage. By better, Kohlberg meant that each stage is an advance both philosophically and psychologically. From a philosophical standpoint, each stage improves upon earlier stages because it is more comprehensive in the kinds of moral issues that it considers and more internally coherent in the way that it addresses them. Psychologically, each stage is said to be better because it is more stable, comprehensive, and coherent than the previous stage (for a brief description of the stages, see the entry *Moral Reasoning*).

Post-Kohlbergian Perspectives

Though Kohlberg's theory has been the most influential, there have been many critics and alternative proposals. Carol Gilligan, noting that his original sample was all male, thought that Kohlberg neglected women's voices. If he had listened equally to women, she argued, he would have realized that

his focus on justice needed to be complemented by an equal concern for care and interpersonal responsiveness.

Others have critiqued Kohlberg's philosophical commitments and how they shaped various aspects of his theory. For example, Norma Haan developed an alternative moral development model based less on Kantian-style reasoning and more on how people create informal norms of reciprocity in interpersonal relationships. Though Haan did not hold to all of Kohlberg's strong claims about stage development, she did present a five-level developmental sequence that characterizes increasingly adequate forms of moral thinking.

A number of theorists have also focused on "prosocial behavior," which is behavior intended to benefit others (regardless of motive), rejecting Kohlberg's more stringent requirement that moral action arise from moral reasoning. The behavioral focus also characterizes those who divide moral action into antisocial and prosocial behaviors, both of which are defined independent of underlying reasons or motives.

Since Kohlberg's pioneering work, the field of moral development research has evolved to focus on a number of other major dimensions of morality in addition to cognition. Augusto Blasi, for example, has written extensively on the concept of moral identity, which refers to both how salient moral concerns are to one's sense of self and what specific dimensions of morality one elevates to highest concern and how moral identity serves as a bridge between moral thought and action. Jonathan Haidt, among others, has highlighted the development and role of unconscious processes in moral functioning. Some researchers have also sought to elucidate the biological and evolutionary underpinning of moral development, though this line of investigation has not yet had much of an impact on sport morality research.

Moral Development in Sport Research

In the 1980s, Brenda Bredemeier and David Shields pioneered a line of sport morality research drawing from both Kohlberg's and Haan's theoretical paradigms. They found that the maturity of an athlete's moral reasoning has important correlates. Research, for example, suggests that college basketball players with lower stage moral reasoning are more likely to be rated by their coaches as more aggressive; they are also more likely to

accept potentially injurious behavior as legitimate. Similarly, children who are less mature in their moral reasoning are more likely to use aggressive tactics than assertive or submissive ones.

Some researchers have sought to determine whether participation in sports has an impact on moral stage growth. To date, research on this question has been cross-sectional, so no cause-effect conclusions can be reached. Moreover, the existent research is inconclusive. A few studies have found that participation in some sports, particularly at higher levels, is associated with lower moral reasoning maturity, but these findings have not been robust. It is unlikely that any conclusions about “sports” in general will hold up since sport experiences are so varied. Other influences, such as education and diverse role-taking opportunities, are likely more influential.

There have been a small number of intervention studies that have sought to increase moral reasoning maturity through physical education (PE) or sport programs. The primary intervention strategy is to provide participants with structured opportunities for role-taking, dialogue, and negotiation about moral issues. The results of these pilot investigations have demonstrated that leaders, equipped with training, can promote moral development.

Recent trends in SP have been away from an explicit focus on moral development in favor of other dimensions of moral functioning, such as moral reasoning and behavior and how these are influenced by other variables of interest. One reason for this trend may be the difficulty of assessing moral development; existent measures, such as the interview methods of Kohlberg and Haan, require considerable time to learn, administer, and score. Even the paper-and-pencil DIT is lengthy. Still, moral development has been documented to be an important influence on moral action in sport, and future researchers are likely to shed additional light on this important dimension of the sport experience.

*Brenda Light Bredemeier and
David Light Shields*

See also Character Development; Cheating; Moral Behavior; Moral Judgment; Moral Reasoning

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MORAL DISENGAGEMENT

Moral disengagement is the conditional endorsement of transgressive behavior through the use of any of eight psychosocial mechanisms and is apparent in sport when players rationalize harmful behaviors such as injuring or deliberately fouling opponents. The eight mechanisms of moral disengagement operate by minimizing or eliminating unpleasant emotional reactions (e.g., guilt, shame) normally associated with transgressive acts, therefore promoting such behavior. The following section of this entry defines the eight mechanisms of moral disengagement and provides sport examples. The discussion then turns to how mechanisms can also be used together and how progressive increases in moral disengagement may lead to increases in the gratuitousness of harmful behavior.

Mechanisms of Moral Disengagement

Within his social cognitive theory (SCT) of moral thought and action, Albert Bandura outlined eight mechanisms of moral disengagement. The eight mechanisms are *moral justification*, *euphemistic labeling*, *advantageous comparison*, *displacement of responsibility*, *diffusion of responsibility*,

distortion of consequences, dehumanization, and attribution of blame. All mechanisms have been evidenced in sport, and definitions and examples from sport are provided in the subsections that follow.

Moral Justification

This mechanism operates when culpable behavior is cognitively reinterpreted so that it is considered justifiable both personally and socially. This cognitive reconstrual centers on the purpose of the behavior, portraying it as achieving a valued social or moral purpose. When acting under moral justification, people follow social or moral imperatives based upon valued purposes. In nonsport contexts, detrimental conduct is morally justified by serving both social (e.g., honor and reputation) and moral (e.g., religious beliefs) functions. However, in sport, moral justification largely occurs when harmful behavior is seen to facilitate achievement of valued social outcomes. An example of this is when a soccer player breaks the rules of the game to score and considers this acceptable because it helps the team win.

Euphemistic Labeling

Capitalizing on people's ability to use language to reshape their thought patterns and associated emotions, euphemistic labeling uses selective language to cognitively disguise the harmfulness of culpable activities or bestow a respectable status on them. In doing so, one avoids the negative emotions (e.g., shame, guilt) that are normally associated with detrimental conduct. In sport, euphemistic labeling can involve the use of softening language to make behaviors appear less harmful or obscuring linguistics to hide the true meaning of an act. An example of using softening language in sport is when an athlete using performance-enhancing drugs refers to them as "juice" and of obscuring linguistics is when players talk of "bending the rules" when they really break them or "letting off steam" when in fact they are acting aggressively.

Advantageous Comparison

Advantageous comparison occurs when a harmful act is compared to another behavior considered more harmful. In doing so, it is possible to diminish the perceived seriousness of the less harmful act through exploitation of the contrast principle.

The contrast principle describes how seeing two things in sequence that are different from one another leads to the tendency to see the second one as more different from the first than it is in reality. Thus, by comparing a particular detrimental act to a more harmful behavior, one can lessen the perceived harm caused by the perpetrated act. Importantly, the more divergent the characteristics of the two behaviors compared, the greater the reduction in the perceived harm. An example of advantageous comparison in sport is when use of abusive language is compared to physically injuring an opponent.

Displacement of Responsibility

Displacement of responsibility occurs when people committing harmful acts consider their actions to originate either from the directives of another person or to be imposed by the situation. In such circumstances, people are unlikely to experience deterrent emotions because personal accountability for action and/or its detrimental consequences is diminished or eliminated. An example of displacement of responsibility to another person in sport is when players say they have been told to break the rules by their coach. The legitimacy of the person authorizing detrimental acts is a key social factor determining the ease with which people displace responsibility to others. As such, authority figures such as coaches are ideally placed to facilitate transgressive behavior in sport through this mechanism. Displacement of responsibility to the situation is illustrated by a player's saying she had to take the law into her own hands because the officials were failing to do their jobs correctly.

Diffusion of Responsibility

Personal accountability for harmful actions and their consequences can also be minimized through diffusion of responsibility. There are three primary means through which this mechanism can operate: division of labor, group decision making (DM), and collective action. Division of labor occurs when individual members of a group carry out separate acts that are harmless in isolation, but when combined have detrimental consequences for others. This form of diffusion of responsibility is more often seen in organization settings in which harmful processes can more easily be separated into apparently harmless subcomponents. In contrast, group DM is more suited to sport and

occurs when a group collectively takes the decision to engage in injurious conduct. For example, a team may jointly decide to use illegal tactics in an upcoming match during a team meeting. In such situations, no individual takes responsibility for the decisions made by the group, instead externalizing the decision to others in the group. Similarly, diffusion of responsibility can also operate in sport through collective action, whereby people engage in harmful behavior as a group, and thereby obfuscate personal responsibility for any harm caused. This is seen in sport when teams collectively use aggressive tactics to gain competitive advantage.

Distortion of Consequences

This mechanism occurs when the detrimental consequences of harmful conduct are ignored, minimized, distorted, or disbelieved. By either avoiding or minimizing the harm caused by one's actions, it is possible to prevent vicariously aroused emotions that are normally experienced when harm is inflicted on others. Distortion of consequences can occur through selective inattention, cognitive distortion, or by actively discrediting evidence of any harm caused. Selective inattention is seen in sport when a player actively avoids finding out the extent of injuries to another player resulting from his actions. Alternatively, if it was not possible to avoid such information, the player could convince himself the harm caused was not as bad as it appeared (i.e., cognitive distortion) or actively discredit the evidence presented by describing situations in which players recovered very quickly from similar injuries.

Dehumanization

The emotional reactions experienced as a result of detrimental conduct are dependent in part on how perpetrators view their victims. Similarity between perpetrator and victim heightens empathic or vicarious emotional responses and therefore deters future detrimental conduct due to associated unpleasant emotions. Dehumanization prevents such reactions by either divesting victims of human qualities (e.g., describing an opponent as a savage) or by attributing animalistic qualities to them (e.g., calling an opponent a dog); both forms of dehumanization decrease the perceived similarity between perpetrators and victims, thereby decreasing feelings of empathy toward victims. Practices that separate people into ingroup

and outgroup members facilitate dehumanization through social separation. Team sports are particularly susceptible to this, as depersonalized opponents (i.e., outgroup members) are more easily deprived of human qualities.

Attribution of Blame

Attribution of blame occurs when a person feels caused to transgress against another due to forceful provocation by the victim or by compelling circumstances. As a result, perpetrators consider themselves victims of circumstance rather than being culpable for their actions. This mechanism is often used when an isolated incident from a series of reciprocal acts is cited as the reason for a provoked defensive reaction. Thus, a player who has had a series of altercations with an opponent may suggest one particular foul by her opponent forced her to foul her opponent as a defensive reaction.

Combining Mechanisms of Moral Disengagement

The interrelatedness of the eight mechanisms determines that the practical application of moral disengagement can involve the combined use of more than one mechanism. Importantly, the combined use of mechanisms can increase the overall effect in comparison to that of the sum of the effects when the mechanisms are used in isolation. This suggests emotional self-regulation may be most effectively obfuscated by combining different mechanisms. For example, two mechanisms that can be used in conjunction with one another are moral justification and attribution of blame. An example of this in sport is when a player fouls an opponent and declares she acted in retaliation for an earlier foul by her victim on one of her teammates. In this scenario, the perpetrator could attribute blame to her victim by declaring the foul was a response to the earlier foul by her victim. She could also morally justify the act by suggesting she acted to defend her teammate's honor, thereby serving a valued social purpose.

Progressive Moral Disengagement

People's use of moral disengagement is thought to develop gradually, resulting in a gradual shift in the harmfulness of behavior, rather than imparting rapid changes in its gratuitousness. Initially, people engage in relatively minor transgressions that they can easily endorse through use of one or more of

the eight mechanisms. As an individual's adeptness in using the eight mechanisms increases, conditional endorsement of more harmful acts occurs. This gradual development of moral disengagement is thought to continue for as long as the harmful acts serve one's self-interests or are considered profitable. This process is subtle in that people are not necessarily aware of the changes occurring to their moral cognitions and behavior, eventually resulting in people engaging in reprehensible acts that at one time they would have considered repugnant. Such a process has been seen in doping scandals in professional cycling, where riders join a new team only to find subsequently that participation in a doping program is expected of them. Such athletes often report initially resisting or agreeing only to use certain substances at low dosages. However, over time they gradually develop the ability to justify engagement in the full doping program, ultimately taking substances and dosages that at one time they would never have considered. This gradual development of moral disengagement in parallel with the level of engagement in the transgressive act allows athletes in such situations to progress their use of performance enhancing drugs without experiencing emotions such as guilt or shame that may deter progressive engagement in doping practices.

Conclusion

Moral disengagement involves the conditional endorsement of transgressive behavior using any of eight psychosocial mechanisms. Given that the social nature of sport provides the necessary conditions for all eight mechanisms, it is important to understand the etiology and definitions for the eight mechanisms so their use can be identified and actively discouraged. As well as understanding the nature of the eight individual mechanisms, it is also important to recognize their combined use may be particularly harmful, and how over time progressive moral disengagement may lead to engagement in more reprehensible acts.

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See also Moral Atmosphere; Moral Behavior; Moral Development; Moral Judgment; Moral Reasoning

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MORAL JUDGMENT

Moral judgment refers to the determination a person makes about an action (or inaction), motive, situation, or person in relation to standards of goodness or rightness. People articulate a moral judgment, for example, when they say that an action is right or wrong, that a person is good or bad, or that a situation is just or unjust. Athletes frequently make moral judgments about moral issues that arise in sports, and such judgments have been investigated by sport psychologists. This entry distinguishes moral judgment from moral reasoning, reviews methods of investigating moral

judgments, and summarizes key findings from the sport-related moral judgment literature.

Moral Judgment and Moral Reasoning

The terms *moral judgment* and *moral reasoning* have sometimes been used interchangeably. For example, the influential moral psychologist Lawrence Kohlberg called his scoring manual *The Measurement of Moral Judgment*. In reality, the manual was written as a guide to scoring the developmental stage of a person's moral reasoning. As the field of moral research evolved, however, a more refined vocabulary also developed and moral judgment was often distinguished from *moral reasoning*, typically representing its conclusion. Thus, many researchers use the term *moral judgment* when referring to the outcome of a process of moral reasoning. A moral judgment is the conclusion that a person reaches about the ethical quality of something or someone. A moral judgment is a moral evaluation.

The term *moral judgment*, however, remains controversial. Reflecting a broader debate in cognitive psychology, there is dispute over the role of explicit reasoning versus more implicit and unconscious processes. The central question is this: When people say that an action is right or wrong, good or bad, is their judgment the result of conscious, deliberative processes (i.e., moral reasoning), or is it a result of unconscious motives and intuitions? Certainly people can offer moral reasons for their beliefs, but some theorists see those reasons as genuinely motivating the person's beliefs, while others see those reasons as simply post-hoc rationalizations for moral judgments that have their actual origins in unconscious processes.

James Rest has provided one of the most influential accounts of moral action processes, which are instructive to how these relate moral reasoning and judgment. For Rest, before a person can behave morally in any situation, a set of psychological processes, divided into four categories, must activate and function optimally. First, a person must interpret the situation. Second, a person must engage in moral reasoning; that is, he or she must weigh any competing moral issues and form a mature and responsible judgment about what should be done. Third, the moral judgment must be prioritized over other values and action choices. Finally, the person must put their moral intention into action.

It is Rest's second set of processes that are relevant here. In the second component of his model, the potential moral agent needs to arrive at a moral judgment. The agent needs to consider all of the morally relevant information, apply their moral reasoning to it, and make a decision about what *should* be done. Rest acknowledges that not all moral reasoning is conscious and deliberate; some of it is rather automatic and habitual. Moreover, moral reasoning itself has both structural and content dimensions to it. In other words, it reflects both a person's mode of thinking, which is influenced by their level of cognitive sophistication, and their specific beliefs, attitudes, and values, which reflect a range of individual and cultural influences. The key point here is that this complex process of moral reasoning—both conscious and unconscious, influenced both by stage of development and specific moral beliefs—leads to a situationally specific judgment about what morality requires the actor to do.

The Assessment of Moral Judgment

Moral judgments are typically assessed in a straightforward fashion. The researcher, for example, may present study participants with pictures or stories of legal, illegal, and/or ambiguous actions in a sport setting and ask respondents to make judgments about whether the depicted actions are okay. These determinations are often referred to as *legitimacy judgments*. Judgments about cheating, aggression, and doping have been the most frequently investigated, along with specific sets of actions designated as antisocial or prosocial behavior.

Brenda Bredemeier developed a moral judgment measure called the Continuum of Injurious Acts (CIA). The CIA consists of a set of six cards describing aggressive acts in basketball that have intended consequences that become increasingly more serious. The CIA cards are presented in random sequence to respondents, who sort them into those deemed legitimate and those judged as illegitimate. Variants of the CIA, adapted to different sport settings, have been used by a number of other sport researchers.

Another moral judgment measure is the Hahm-Beller Values Choice Inventory (HBVCI). Though it has sometimes been said to measure moral reasoning maturity, it is more accurately described as an assessment of moral judgment. Respondents

are asked to indicate how much they agree or disagree with a set of scenarios, each of which briefly describes a moral belief. The HBVCI items are designed to determine the extent of respondents' agreement with deontic ethics (i.e., a philosophical approach that states actions have inherent moral qualities that are independent of the consequences of the actions).

A number of instruments have been developed by sport psychologists that combine moral judgments with other related constructs. For example, Dawn Stephens and colleagues developed the Judgments About Moral Behavior in Sport Questionnaire (JAMBYSQ). The instrument consists of three soccer scenarios, each depicting hypothetical protagonists faced with choices about whether to engage in unfair but advantageous behaviors (specifically, lie to an official, hurt an opponent, and violate a game rule). Piloted with upper elementary female soccer players, the instrument was designed to assess respondents' (a) fair play action tendencies; (b) moral judgments—that is, what they thought the protagonist should do; (c) moral motives; and (d) perceptions of team norms pertaining to unfair play behavior. Both Stephens and other researchers have adapted the measure to fit a variety of other ages and sports. Similarly, Sandra Gibbons and colleagues developed a hybrid measure that asked respondents to answer three questions about each of 10 behaviors. The three questions were designed to tap moral judgment, moral reason, and moral intention.

Maria Kavussanu and her colleagues have developed a number of measures that tap moral judgment. In some studies, similar to the Bredemeier methodology, she has asked respondents to rate the moral legitimacy of a list of acts. In other studies, she has used hybrid measures, similar to those of Stephens and Gibbons, which assess moral judgments, along with other components of Rest's moral functioning model.

Moral Judgments in Sport Research

Why does one athlete believe that doping is okay, while another believes it is cheating and immoral? Moral judgments (e.g., moral beliefs about what is right) do not have a single biographical, psychological, psychosocial, or cultural source. Such judgments reflect a broad and diverse range of influences. For example, various studies have shown that such demographic characteristics as

one's age, gender, race, and economic status often influence moral judgments. Male athletes, for example, tend to be more lenient in their moral judgments of aggression than female athletes.

In addition to demographic characteristics, there is a range of other influences on moral judgments. A person's level or stage of moral reasoning, while not determinative of moral judgments, is one developmental variable that influences judgment. Other sources have also been investigated, including moral identity, affective self-regulatory efficacy, self-esteem, and, most especially, motivation.

Researchers in the area of achievement motivation (or goal orientation) have shown consistently that there is a significant empirical link between motivation and morality. Those individuals who participate in achievement contexts like sport because they are motivated to pursue learning, growth, and mastery are much more likely to make prosocial, positive moral judgments, such as endorsing fair play and condemning cheating and aggression, than those whose motivation is primarily to demonstrate superiority over others.

A variety of contextual influences also affect moral judgments. Not surprisingly, moral judgments are influenced by the sport and competitive level. For example, recreational golfers may, on average, make different moral judgments about the legitimacy of steroid use than professional baseball players.

One major influence on moral judgments that has received considerable attention is the moral atmosphere. The term *moral atmosphere* refers to the shared, collective moral norms of a group. David Shields and colleagues opened this line of research in the sport realm by developing the Team Norm Questionnaire, a simple six-item assessment that asked athletes to estimate the number of their teammates who would engage in ethically inappropriate behavior (i.e., cheating and aggression) if it would help their team win and, second, whether the coach would want them to do it. Other researchers have developed similar measures of moral atmosphere and have documented that shared norms have a significant influence on individual judgments and behavior.

Like the moral atmosphere, the motivational climate has also been shown to influence moral judgment. The motivational climate refers to social cues that tend to elicit either a task or mastery motivational orientation or an ego or performance orientation. For example, when a coach emphasizes

winning or normative ability, it tends to encourage an ego orientation. In contrast, when a coach emphasizes doing one's best and praises effort more than outcome, a mastery climate is fostered that supports task motivation. Generally, a performance climate—one that augments an ego goal orientation focused on outperforming others—tends to result in more antisocial judgments.

Another important question is the relationship between judgment and action. From research in both general psychology and sport psychology (SP), it is clear that moral judgments are neither irrelevant nor determinative when it comes to predicting or understanding moral action. People sometimes act consistently with their moral beliefs, but situational factors may override moral judgments. For example, when there are high expected benefits associated with acting in antisocial ways, athletes may act inconsistently with their more "abstract" moral judgments. Similarly, when intense emotions are activated, moral judgments may be overridden.

Moral judgments also can be suspended through processes of moral disengagement. In other words, an athlete who normally judges aggressive acts as wrong may, nonetheless, engage in an aggressive act and excuse her behavior by blaming the other party—"she started it." Drawing from the work of Albert Bandura, sport researchers such as Maria Kavussanu have investigated a number of psychosocial moral disengagement processes that function to suspend a person's normal moral judgment.

In conclusion, a moral judgment reflects a person's assessment of the moral quality of something (e.g., an action, person, policy, institution). Athletes and coaches make moral judgments frequently as they consider behavioral options ranging from the obviously deplorable to the clearly praiseworthy. To increase the likelihood that moral judgments will guide behavior, athletes and coaches can seek to prioritize ethics, promote task motivation, foster a mastery motivational climate, and implement strategies to counter the temptation to use processes of moral disengagement.

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See also Achievement Goal Theory; Achievement Motive Theory; Character Development; Cheating; Moral Atmosphere; Moral Behavior; Moral Development; Moral Disengagement; Moral Reasoning

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MORAL REASONING

Moral reasoning is focused on what, from a moral standpoint, a person *should* do in a given situation. Moral reasoning addresses issues of right and wrong, good and bad, virtue and vice. Moral reasoning seeks to answer these questions: What ought to be done? What is the right thing to do? Athletes use moral reasoning to make practical decisions about such topics as cheating and fair play, aggression and altruism, disrespectfulness and sportsmanship, doping and honesty.

Moral reasoning, while often focused on the practical resolution of dilemmas or problems, is also a form of philosophical reasoning. Moral reasoning appeals to decision-making criteria, formal or informal, that reflect claims about what is true,

right, and/or good. When a young athlete asks herself, Should I risk my health by severely restricting my diet so that I can compete at a target weight? she is asking fundamental questions about human welfare, personal and social responsibilities to self and others, and basic issues of rights and fairness. Her reflections are practical, in that they are oriented to decision making (DM) and philosophical consideration, in that she is evaluating the moral principles that might inform her choice.

Next, we discuss the use of moral principles to inform moral reasoning, the development of moral reasoning, and moral reasoning in sport research.

Reasoning From Moral Principle

Not all moral action originates from or is guided by explicit moral reasoning. If the right thing to do is unambiguous and clear, there may be very little need for deliberative moral reasoning. Similarly, sometimes a person may act in morally charged situations from habit or intuition. But some situations lead a person to pause and reflect. This may happen, for example, when a situation presents a person with a conflict among values. For example, an athlete may think about whether to use performance-enhancing drugs (PEDs) because, on the one hand, she values optimal physical performance and, on the other hand, she values health, as well as fairness and honesty. Sifting through and prioritizing these values necessitate moral reasoning.

When people engage in moral reasoning, they make use of moral rules or principles. *Moral rules* are specific, prescriptive guides to action: don't cheat, don't lie, and play fair. In contrast, a *moral principle* is a generalized guide to behavior, such as the Golden Rule: Treat others as you would want to be treated.

There is a long tradition of philosophical debate over moral principles. While there are many variants, the two main philosophical camps are the *utilitarians* and the *deontologists*. Utilitarians determine the morality of an action by examining its consequences. The most widely cited utilitarian moral principle is to act so as to bring about the greatest good for the greatest number. An athlete may decide cheating is wrong, for example, if that athlete believes that sports will benefit more people if contests are fair. In contrast, deontologists define moral principles in terms of the quality of the action itself. According to deontologists, an unjust act cannot be justified by appeal to the

act's consequences. For the athlete who embraces deontology, cheating is wrong not because of the consequences of cheating but because of its inherent qualities. According to Kant's famous version of deontology, an act is moral if, and only if, you want all people to act that same way in similar circumstances. Since an athlete does not want everyone to cheat (especially opponents), cheating is inherently wrong regardless of its consequences.

Developmental Stages of Moral Reasoning

In the mid-20th century, the Harvard psychologist Lawrence Kohlberg combined a neo-Kantian view of moral principles with the constructivist psychology of Jean Piaget, and outlined a six-stage sequence in the development of moral reasoning. Kohlberg's pioneering work represented a fundamental break with the behaviorist approach to morality and psychology that was then dominant.

Drawing from the work of the Swiss developmental psychologist Jean Piaget, Kohlberg claimed that children were not passively molded by the environment; rather, children actively seek to interpret their experience; they are meaning makers. Children seek to form coherent understandings of their world. As part of their ongoing effort to construct increasingly complex and comprehensive understandings of the world, children gradually develop more adequate forms of moral reasoning.

A stage of moral reasoning is simply an organized pattern of thinking about what makes something right or wrong. To understand what Kohlberg means by a moral stage, it is important to distinguish between the *content* of moral reasoning and its *structure*. The content of moral thinking refers to the specific thoughts that a person has about a moral situation. If a coach believes that an athlete should play through injury, that belief is part of the content of the coach's reasoning. The structure of moral reasoning, in contrast, refers to a deeper organized pattern of DM and prioritizing values. The structure of moral reasoning, to use an analogy, is like the "grammar" that gives coherence to a wide range of moral thoughts and decisions. To continue our example, the coach who believes an athlete should play through injury may hold that belief because he systematically, across numerous situations and contents, prioritizes the value of winning over other values, such as concerns for

welfare. Typically, people engaged in moral reasoning are aware of their specific thoughts about whatever moral issues they are facing. But moral structure operates below the surface; people are no more conscious of the structure (or stage) of their moral reasoning than they are of the grammatical rules that give linguistic structure to their speaking.

Moral stages are defined in terms of the structure of moral reasoning. According to Kohlberg, children progress through a regular, age-related sequence of stages. Although the stages of development are universal, children do not always progress at the same rate, nor do they all stabilize at the same level. Children's moral reasoning development may plateau at any stage.

Growth from one stage to the next is dependent on experience. Kohlberg identifies what he calls *cognitive disequilibrium* as the primary motivator that leads a child to gradually abandon one stage and slowly construct a more adequate one. For example, imagine an athlete thinks that "the good" is equivalent to "what's good for me." The athlete's moral perspective, which is quite immature, is limited to considering only one viewpoint: that of the self. However, the athlete's teammates are likely to be impatient with that perspective and may argue or retaliate. The contrast between the athlete's stage of reasoning (which assumes that the world revolves around the self) and the negative reaction of others is likely to create a sense of mental incoherence (i.e., cognitive disequilibrium). With sufficient experience, the young athlete's self-centered perspective becomes unstable, initiating a search for a more comprehensive perspective. The result is the child's development of a new stage of moral reasoning.

The six stages of Kohlberg's model of moral reasoning development are divided into three levels: the *preconventional*, *conventional*, and *postconventional* levels. The preconventional level is egocentric in the sense that the child, like the athlete mentioned previously, cannot think beyond his or her own needs and desires. The conventional level is reached when the child embraces a social perspective that takes account of others and group norms. For example, an athlete may start to think in terms of the team and its behavioral norms. Finally, the postconventional level is reached when the person orients to universal values and moral principles. At this level, an athlete's moral reasoning may be guided by norms

of fairness and welfare that are prioritized over whatever expectations may characterize a particular team.

The stages can be briefly summarized in the following way:

Preconventional Stages

Stage 1: Morality is equivalent to whatever is rewarded; wrong is equivalent to whatever is punished. Moral reasoning is guided by considerations of authority and power. "Whatever the coach says is right" would be an example.

Stage 2: Morality is equivalent to what is good for the self. Moral reasoning is guided by a desire to obtain benefits. Simple reciprocity is embraced: "I'll scratch your back if you'll scratch mine." In sports, it may also take the form of "cheating is okay when opponents cheat."

Conventional Stages

Stage 3: What is good or right is equivalent to stereotypic notions of what a "good" person does. This stage involves a superficial embrace of group norms. Moral reasoning is guided by ideas about what is nice, polite, or conforming. If the person identifies with an antisocial group, this conformist mode of thinking may take on a more rebellious tone. In sports, conformity to superficial conventions of good sportsmanship may reflect this stage of thinking.

Stage 4: Social maintenance is the core value of this stage. This can vary from a "law and order" orientation to a strong but uncritical embrace of one's cultural beliefs and practices. Moral reasoning is guided by rules, formal or informal, that serve group maintenance functions. The athlete who defines morality by rigid appeal to rules is likely reasoning at this stage.

Postconventional Stages

Stage 5: This stage is characterized by a humanistic perspective that seeks to uphold universal values. Moral reasoning is guided by allegiance to chosen values that the person believes should be upheld everywhere by everyone. The athlete who thinks about the "purpose of the rules," rather than just the literal rules themselves, may be reflecting Stage 5 reasoning. For example, consider a skier who shares a pole with an opponent who breaks hers. The skier may do so, even though not required to

by the rules, to uphold the “spirit of fairness” that the rules seek to express.

Stage 6: Justice is the core decision-making principle at this stage. Moral reasoning is guided by a concern to apply the principle of justice impartially to all parties. It should be noted, however, that Kohlberg dropped Stage 6 from later versions of his scoring manual because too few individuals were found who actually demonstrated unambiguous Stage 6 reasoning. He retained the sixth stage only as a hypothetical terminal point to the stage sequence.

Post-Kohlbergian Perspectives

Kohlberg’s pioneering work is without parallel in breadth and scope and continues to have lasting significance. Still, even before his death in 1987, numerous criticisms of his theory had been advanced. Carol Gilligan, noting that his original sample was all male, critiqued Kohlberg’s focus on justice to the relative neglect of the ethical requirement to care for others. John Gibbs supported Kohlberg’s model up through Stage 4 but suggested that the postconventional stages were based on philosophical arguments that went beyond the empirical research. Elliot Turiel and Larry Nucci conducted research that suggested Kohlberg’s model confused and blended moral concerns with social conventions.

Norma Haan, whose approach has been used in a number of sport psychology (SP) research projects, studied moral reasoning in the context of moral action rather than following Kohlberg’s preferred methodology of relying on responses to hypothetical dilemmas. Kohlberg, for example, had asked respondents whether a hypothetical “Heinz” should steal a drug to save his dying wife. Haan, by contrast, recorded people’s actions and utterances as they participated in simulations games that required moral negotiations to occur. She coded how people balanced their own interests and perspective with those of others as they sought to navigate toward shared solutions.

Moral Reasoning in Sport Research

Research conducted in sport settings has demonstrated that the maturity of an athlete’s moral reasoning has important correlates. Some research, for example, suggests that athletes with lower stage moral reasoning are more likely to cheat or try to injure an opponent. Since preconventional

moral reasoning is guided by self-interest, this is not surprising.

Some researchers have sought to determine whether participation in sports has a facilitative or inhibitive effect on moral stage growth. Do athletes, compared to their same-age peers, exhibit more or less mature moral reasoning? The research on this question has been inconclusive. There have been a few studies that suggest that participation in sports, particularly at a higher level, is associated with lower moral reasoning maturity, but these findings have not been robust. It is unlikely that any conclusions about “sports” in general will hold up. People’s experience in sports is highly variable depending on the sport, the competitive level, the culture of the team, the coach, and numerous other factors.

Another line of research has focused on differences between moral reasoning about issues occurring in sports and moral reasoning about similar issues in everyday life. Brenda Bredemeier and David Shields proposed a theory of “game reasoning” that suggests many athletes use moral reasoning that resembles lower stage reasoning when they respond to issues in sport but shift to more mature reasoning when considering moral issues outside of sport. Sometimes this shift toward a more egocentric form of reasoning simply reflects a playful and nonserious deviation, but game reasoning can also degenerate into serious moral lapses.

There have been a small number of intervention studies that have sought to increase moral reasoning maturity through physical education (PE) or sport programs. While the number of studies has been small, the results were generally encouraging. When coaches foster opportunities for moral dialogue and negotiation, when they seek to build a culture of fairness, it is likely that they can have a positive impact on the moral reasoning of their athletes.

In conclusion, it is clear that moral reasoning is an important process that informs many decisions and actions within sport settings. The adequacy of moral reasoning that an athlete or coach exhibits will depend of a variety of individual and environmental factors. Certainly, the developmental level of a person’s moral reasoning is one factor, but it is not the only one. The content of a person’s moral reasoning is also influenced by socialization and contextual factors. Explicit efforts to improve moral reasoning in

sport settings will need to pay attention to all of these dynamics.

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See also Character Development; Ethics; Moral Development; Moral Judgment

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MORAL VALUES AND ATTITUDES

The study of morality in sport has attracted the interest of many sport psychologists, partly because of the pervasive and long-held belief across many scholars and lay people that sport builds character. Research in this area of sport psychology (SP) has looked at whether sport participation is indeed linked to moral behavior by examining issues pertaining to moral development, moral reasoning, and an array of prosocial and antisocial behaviors. This entry presents work on values in sport, pioneered by Martin Lee, which was based on research in social psychology conducted first by Milton Rokeach, and subsequently

by Shalom H. Schwartz. This entry also outlines parallel work by Martin Lee on ethical attitudes.

Values in Sport

Rokeach defined values as a set of beliefs individuals have concerning desirable modes of behavior or goals (e.g., kindness, success, independence, security). Values represent the standards that individuals and societies set for themselves. Values motivate behavior by guiding action and choice of activities. Individuals vary in the importance they place on different values. Thus, individuals and societies rank values hierarchically in terms of importance by developing value systems. Values differ from attitudes; the latter have been defined as predispositions to respond in a favorable or unfavorable manner with respect to a given object. Values provide standards and transcend actions and situations, whereas attitudes are not hierarchically valued and usually refer to specific actions or objects. Thus, values are fewer than attitudes and are usually considered as their antecedents. For example, a positive attitude toward cheating in sport might reflect an underlying value system in which achievement is highly ranked. Further, attitudes can be positive or negative, whereas values are considered only as expressions of desirable ends.

Schwartz presented an influential model of human values that distinguishes 10 universal values according to the motivational goal they express. Schwartz proposed the values of *self-direction* (independence), *stimulation*, *hedonism*, *achievement*, *power*, *security*, *conformity*, *tradition* (respect for customs), *benevolence* (preserving and enhancing in-group welfare), and *universalism* (preserving and enhancing the welfare of all people). These values reflect the satisfaction of biological needs, demands of coordinated social interaction, and survival and welfare needs of groups. Schwartz mapped these 10 values in a circular structure that has two dimensions. The first ranges on a continuum from *self-enhancement* (power, achievement) to *self-transcendence* (universalism, benevolence). The second dimension ranges on a continuum from *openness to change* (stimulation, self-direction) to *conservation* (security, conformity, and tradition). The closer the values are in this structure, the more compatible they are (e.g., power and achievement). For example, pursuing achievement in sport is congruent

with the pursuit of power. In contrast, values far apart from each other are conflicting; pursuing achievement in sport often conflicts with the value of protecting the welfare of opponents (i.e., universalism).

Research on values in sport has been conducted chiefly by Martin Lee, who distinguished values from attitudes, followed Rokeach's lead and operationalized values as guiding principles and conceptions of desirable end states and attempted to measure these values in ways compatible to Schwartz's work. In initial research, Lee posed sport-specific moral dilemmas to athletes on the assumption that a moral dilemma provides the basis for a discussion that can elicit athletes' values. Possible and desirable reactions to these dilemmas were explored via semistructured interviews. This line of work eventually identified 18 values: *enjoyment*, *personal achievement*, *sportsmanship*, *contract maintenance* (play properly), *being fair*, *compassion*, *tolerance*, *showing skills*, *obedience*, *team cohesion*, *conscientiousness*, *excitement*, *health and fitness*, *self-actualization* (feel good), *public image* (show a good image to others), *companionship*, *conformity*, and *winning*. In subsequent work, Lee and colleagues developed a questionnaire of values in youth sport that taps five moral values (*obedience*, *fairness*, *sportspersonship*, *helpfulness*, and *contract maintenance*), three competence values (*achievement*, *showing skill*, and *self-direction*), and three status values (*winning*, *public image*, and *leadership*).

Ethical Attitudes

As explained earlier, attitudes differ from values in that they are bipolar, specific to a particular object, and have no hierarchy of importance. An attitude could reflect one or more underlying value. Lee investigated attitudes to moral decision making (DM) in youth sport in parallel with his work on values. Lee was particularly interested in the endorsement of attitudes toward unethical behaviors. Based on the youth sport literature, interviews with British young athletes, coaches, and sport administrators, Lee and his coworkers developed and validated a questionnaire that measures three key ethical attitudes in sport: *endorsement of cheating*, *endorsement of gamesmanship*, and *keeping winning in proportion*. The first two factors capture attitudes toward antisocial behaviors. Endorsement of cheating refers to attitudes

toward violating the written rules of the game (e.g., using hands to control the ball in soccer) or the employment of deception in order to gain an unfair advantage. Endorsement of gamesmanship refers to attitudes toward unwritten rules of the game in an effort to upset the opponents but without actually violating an official rule. For example, in a soccer game, an opponent might try to psychologically unsettle a player just before he or she takes a penalty kick by making a reference to unfavorable weather conditions. Last, keeping winning in proportion represents a prosocial attitude to restrain the pursuit of winning when this compromises ethical principles. An example of this would be of an athlete who feels that winning is not valuable if it is achieved in a dishonest manner; as a result of this attitude he or she might reveal to a game official that a call to his or her advantage was incorrect.

Research on values and ethical attitudes has examined the links between these constructs and motivation, given that values can guide choice of activities and that attitudes toward prosocial and antisocial behaviors reflect different motivational influences. In a study involving British youth sport athletes, Lee and coworkers explored the links between values, ethical attitudes, and achievement goals proposed by the classic achievement goal theory (AGT) (i.e., goals that individuals have in achievement situations and which reflect different notions of how ability is construed and success is achieved). This study tapped not only attitudes toward negative behaviors (namely, endorsement of cheating and endorsement of gamesmanship) but also attitudes toward positive behaviors. For the latter, two factors from Robert J. Vallerand's work on sportspersonship were chosen: commitment to participation and respect for social convention. The results of that study showed that competence and moral values positively predicted what the authors labeled as prosocial attitudes. In contrast, antisocial attitudes were positively predicted by status values and negatively by moral values. Furthermore, competence and status values predicted task achievement goals (i.e., emphasis on individual improvement and hard work) and ego achievement goals (i.e., emphasis on showing superiority over others), respectively. In further analyses, the authors were interested in the mediated or indirect effects of values on achievement goals via attitudes. The results indicated that task and ego goals partially mediated the effect of competence

values on prosocial attitudes and of status values on antisocial attitudes, respectively.

In a different study linking ethical attitudes and another theoretical framework of motivation—self-determination theory (SDT)—it was shown that autonomous motivation for sport participation (i.e., motivation based on enjoyment and valuing of sport) was positively linked to prosocial attitudes and negatively related to antisocial attitudes. The opposite pattern of results was observed between controlled motivation (i.e., motivation based on feelings of guilt, external rewards or pressure) and prosocial and antisocial attitudes. Thus, there is evidence in the literature to suggest that values are important precursors of achievement motivation and that ethical attitudes can be meaningfully predicted by the achievement goals and motivation of athletes.

The empirical evidence regarding values and ethical attitudes in sport is relatively limited in breadth and applications on various accounts. First, it has concentrated on athletes in youth sport. Second, some of the questionnaire items that purportedly tap ethical attitudes seem to be capturing self-reported behavior. Third, there have been no studies employing objective outcomes of values and ethical attitudes (e.g., disciplinary records of athletes, recorded cases of cheating), nor any experimental work that draws from the values and ethical attitudes identified in Lee and coworker's research. Despite these limitations, the work by Lee and his coworkers represents a promising initial step in mapping the value system of young athletes and understanding how this might be implicated in observed variations in their ethical attitudes and motivation in sport.

Nikos Ntoumanis

See also Achievement Goal Theory; Aggression; Cheating; Moral Behavior; Moral Disengagement; Moral Judgment; Moral Reasoning

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MOTOR COMMANDS

There are two causes of body movement. First, an external force can act on the person or animal; a strong gust of wind, for example, may cause the movement of living organisms. Second, through biological machinery such as muscles, force is produced internally, leading to movement. The truly interesting aspect of our movement within a physical environment is that without exception, our movement is a consequence of both muscles generating force internally and the environment producing forces on our body externally.

How then does the nervous system deal with the generation of instructions to the muscles to control movement? In other words, what is the “language” for motor commands? Let’s begin with a straw person model, which we know cannot be correct but is nonetheless useful to enhance our understanding. Perhaps the motor commands specify an exact sequence of muscular forces along with the actual temporal pattern of these forces. This type of idea is seen in the *impulse-variability theory* for the speed–accuracy trade-off. This model is based upon the assumption that motor commands specify the intensity and duration of muscular contractions.

Nikolai Bernstein argued that the central nervous could not directly specify muscular forces. His argument is quite simple. Imagine flexing and extending your elbow joint in rhythm to a metronome. In one case you are moving in the horizontal (transverse) plane so that gravity is neutral. In the second case, you move in the vertical (sagittal) plane. In this latter case, the effects of gravity are dependent upon whether you are extending the elbow (moving down) or flexing

the elbow (moving up). In these two conditions of planar motion, the movement trajectories are the same, but the actual internally generated muscular forces are different. In other words, motor commands cannot be specific because we cannot know (predict) the exact nature of external forces acting upon us. Motor commands must be more general.

Delos Wickens, in a classic experiment published in 1938, showed that motor commands are not linked directly to muscle commands. Human research subjects placed their index finger on a button, with the palm of the hand facing down. A stimulus was presented, and a short time later, the index finger received a mild electrical shock. In a short time, the subjects were conditioned to avoid the shock. What did these people learn? Did they learn to contract a muscle, or did they learn to move away? Wickens then had these subjects put their index finger on the button but this time with the palm up. If in fact the individuals had developed motor commands to contract their flexor muscle of the index finger, they would not avoid the shock. If the individual's motor commands were written to "move away"—if, in other words, the motor commands were a spatial code—then the participant would easily avoid the shock. For 9 out of 10 subjects, the movement was immediate and away. In other words, at the highest levels of the central nervous system (CNS), the motor commands were set up in reference to what is called *allocentric space* (with reference to the external spatial world) compared to *muscular space*.

Are specific muscular commands unimportant? The answer is negative. One can design a very easy experiment to capture this point. Take your favorite pen, and sign your name on a piece of paper. Now, use your nondominant hand and sign your name. Research clearly has shown that these two signatures look very similar but at the same time are different. One's nondominant hand signature is not as clear as that of one's dominant hand. In other words, motor commands have *generality* (the spatial commands can be transferred to different effectors) and also have *specificity*. Specificity means that one's nervous system has specialized commands for well-practiced muscular patterns. These commands might be much more motor-specific. We do not know much about the nature of this type of specificity.

Another way to examine the nature of motor commands is to make the assumption that human beings strive to minimize a particular characteristic

of movement. For example, to minimize movement time (MT), muscular contractions would produce force as quickly as possible and then decelerate ("jam on the brakes") at the last possible instant. This is the driving strategy of a typical 16-year-old. However, unless there is a sense of urgency, it seems as though human beings attempt to move to minimize some aspect of the energy expended. The way to do this is to move smoothly. Smooth is defined as lacking changes in acceleration. Point-to-point movement trajectories look as though the *minimization principle* is related to reducing changes in acceleration. Movement trajectories tend to show a bell-shaped velocity profile so that there is a period of positive acceleration and then a negative acceleration. Scientists are not in full agreement concerning the actual muscular and/or neural quantity that is minimized, but it is clear that biological motion is geared to reduce energy expenditure.

Finally, motor commands must possess a high degree of generality. Handwriting is the most popular example. People can sign their name on a whiteboard, a check, a letter, and using a mouse with a computer screen. Signatures are large or small, written fast or slow, but a signature maintains the spatial-temporal pattern that makes it "yours." What type of motor commands allows humans to do this? Perhaps it makes sense to think of a set of motor commands like a set of rules, or equations, that define the trajectory in space and time. These rules, or equations, are scalable so that the essence of our signature or our tennis forehand exists, but the particular muscle fibers, or muscle units, are organized independently of these rules or commands. Thus, our motor commands are not in muscle language, which allows for flexibility as well as specificity.

Howard Zelaznik

See also Control Theory; Coordination; Dynamical Systems; Errors; Laws of Movement Learning and Control

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MOTOR CONTROL

Motor control, in reference to movements of an organism or motions of a robot, is often conceived of as a computational problem. How is something or someone able to move to achieve various environmental goals? For human movement, in particular, the question of how individuals are able to organize the motor system at multiple levels (e.g., joints, muscles, neurons) defines the study of motor control. In this way, motor control is a solution that is arrived at by the individual, which satisfies numerous and sometimes competing goals (e.g., to remain balanced, to reach a hot cup without getting burned, to avoid obstacles, to move quickly and accurately, to minimize energy expenditure, to avoid injury or uncomfortable positioning). The problem of control is typically conceived at multiple levels and is often distinguished with respect to the voluntary or intentional nature of control versus a more automatic or reflexive level. In the former case, movements are goal-directed and at least part of an unfolding movement designed to reach the goal is under voluntary control of the actor. In the latter case, movements are not under intentional control and take place without perceived control or conscious attention. This issue concerning what is controlled, prepared, or programmed in advance of a movement and what is controlled as a movement unfolds dominates much of the discussion in the motor control literature. For seemingly very simple movements like balance and walking, to more complex movements that are considered to define many sporting accomplishments, the problem of motor control is enigmatic.

Hierarchical Control

Many of the processes underlying human movement take place without explicit awareness on the part of the actor, but many movements are still voluntary. That is, an actor makes a conscious decision to act and this desire ultimately leads to movement. Yet the exact parameters of the movement are usually unknown and not directly controlled by the actor. Therefore, motor control is the problem of which transformations intervene between the thought of attaining a goal and the

muscle activations that result in movement. Models and theories of motor control differ, but most if not all of the models are hierarchical in nature, suggesting that (a) the goal of the task is loosely set at a high level (sometimes referred to as the task level); (b) a sequence of movements is specified at a lower level (also known as response programming); and (c) at the lowest level, a specific pattern of muscle activation allows for execution (execution level). Although for clarity we treat these various levels as independent, it will be clear from the following discussion that these processes are neither serial nor mutually exclusive and our distinctions across levels serve primarily to aid thinking about the various problems involved in controlling movement.

Intending and Planning to Act (Response Identification and Selection)

At the task level, the goal of the movement is specified consciously by the actor, although there is evidence that actions are substantially influenced by purely perceptual stimuli without conscious intention on the part of the mover (e.g., different types of handles afford different grasping postures). At the task level, the issue of movement preparation or planning is important, and considerable research has been conducted to explore how movements are planned in advance of execution. We have learned a lot about movement preparation through simple measures of reaction time (RT) (i.e., the time between an environmental event and the actual action). Defining when an action actually begins is not always a simple procedure. For some people, action onset is defined with respect to a change in displacement of a limb in comparison to a baseline resting value, whereas movement onset has also been defined with respect to muscle activation as measured by electromyography (EMG). One of the most robust findings is that RT increases in a predictable, linear fashion as the number of stimulus–response (S–R) alternatives increases, so termed Hick's law. Hick's law is a problem of response selection. As choice increases, the time needed to make an appropriate movement response will also increase.

Further inferences about movement planning based on measures of RT have been verified with more sophisticated recording techniques that identify brain areas involved in movement preparation. For instance, areas in the prefrontal cortex

(PFC) are particularly involved in the task-level processes of motor control. Human neuroimaging studies show decreased activity in the PFC as a result of motor learning, suggesting that as a skill is practiced there is less reliance on explicit (attention demanding) control of the movement. When people learn a sequence of movements implicitly (i.e., without awareness of learning and sequence regularities), activity in PFC decreases relative to explicit learning conditions (i.e., with verbalizable knowledge of the sequence). Similarly, when implicitly learned regularities in the sequence are explicitly revealed, PFC activity increases to baseline levels. Damage to PFC has also been shown to reduce conscious control of movement, whereby brain-lesioned patients use or respond to objects in a stereotyped manner that is inappropriate for a given context.

Significant attention has also been paid to the role of vision in the planning and execution of actions. One neurophysiologically supported theory is that there are two main processing routes in the brain that are activated, with selection depending on whether actions are required toward an object. When individuals interact with objects or move to targets, processing of visual information is assumed to be primarily governed by a dorsal route in the brain (i.e., a route that involves areas of the brain near the top of the cortex, particularly the posterior parietal cortex). In contrast, when individuals make decisions about a target's position or identity, ventral route processing in the brain is involved (involving lower-level cortical areas, primarily occipital-temporal cortex). Patients with temporal cortex lesions are severely impaired when visually identifying objects but show unimpaired tool use and object manipulation. Conversely, patients with parietal cortex lesions have unimpaired recognition abilities but significantly impaired reaching and pointing accuracy (suggesting a deficit in calculating spatial locations). This double dissociation between recognition in the temporal lobe and perceptual integration in the parietal lobe suggests that the ventral and dorsal streams of vision might be differentially involved in response selection and response programming, respectively.

Response Programming

The history of response programming in motor control dates back to the late 19th and early

20th century. Programming is thought to be the process preceding voluntary actions whereby action plans are organized and potentially stored in cortical or subcortical structures in the brain, ready to be released when a response is required. Evidence for programming is that people are unable to inhibit actions when they are close to their release point, studies of RT show that as complexity of a movement increases so does RT, stereotyped movement features are observed when an action is repeatedly executed, suggesting that at least part of the response is programmed in advance and is not influenced by feedback (termed *open-loop control*), and features of a movement remain when the movement is unexpectedly blocked or is initiated early due to a startling acoustic stimulus (i.e., the start-react effect).

The term *motor program* has had varied usage in theories of motor control, although there is little consensus as to what this motor program might be and even whether such terminology helps in our understanding of motor control. Some of the discussion has been with respect to the specific or general nature of motor programs (i.e., are programs for throwing specific to certain distances, ball sizes, limbs), the neuroanatomical location of such a motor program (cortical, subcortical, spinal), as well as the necessity of motor programs, if one considers actions to be assembled on the fly in response to real-time constraints in the environment (i.e., the principle of self-organization in dynamical systems theory). More recently, a language of control has emerged that is based on the idea of internal models, rather than programs. This language and associated ideas were primarily a result of computational motor control theorists who emphasized predictive processes occurring as an action unfolds (i.e., forward, anticipatory models) as well as the idea of control processes that arguably function like motor programs (i.e., responsible for programming motor commands). Despite the debate about the nature or even existence of motor programs, it is evident that actions are at least in part assembled prior to action execution. The nature of the programming that is required is discussed next.

One of the first processes that must occur at the programming level is the translation of objects in allocentric space (the relation of objects in space to each other) into egocentric space (the relation of objects in space to the actor). This translation is tremendously important for sport. Consider a

soccer player who is trying to connect their body to volley an incoming ball. Making contact with the ball depends on an accurate calculation of where the ball is in space, where the player's foot is in space, and calculation of the relationship between the moving ball and the moving leg. Neurological evidence suggests that multiple egocentric spatial frames exist centered around the head, the arm, or the leg depending on the effector that is being used in the action. Other physiologically driven research has led to the suggestion that rather than the whole trajectory of a moving limb being computed in advance, that only the endpoint of the movement is programmed. Rather than individual joints or muscles being represented in the cortex, many features of a movement arise from the peripheral nervous system (rather than the brain) and intrinsic properties of muscles and joints (e.g., equilibrium point theory). However, there is evidence that the sequence of programming is hierarchical in nature, based on such parameters as the extent and direction of a movement as well as the choice of limb required. It is not possible to program movement amplitude if one does not know the limb required to make the movement.

Programming time is affected by the complexity of the response required. Movements that involve more components or that are of a longer duration than others take longer to assemble or program. This programming time is however influenced by the skill level of the actor. After extended practice, action-response sequences become grouped or unitized (requiring less time to program), whereas for novices these responses exist as small(er) movement "chunks" requiring more time to prepare or assemble. A skilled tennis player is therefore expected to take less time to prepare a return lob in tennis than a more novice player, if for the novice player the lob is conceived as a sequence of components potentially involving a backswing and forward swing phase.

Also at the programming level, the correct sequence of movements needs to be specified in order to generate the correct endpoint and movement effect. As motor sequences are learned, the movement becomes more accurate, faster, and more automatic. Neurophysiologic data show that the PFC and presupplementary motor area (pre-SMA) are important for learning new sequences. As these sequences are learned, however, there is a decrease in these areas and a corresponding increase in activity in the supplementary motor

area (SMA) and primary motor cortex. The connection of these cortical structures to lower levels of the brain, (i.e., the basal ganglia and the thalamus) creates feedback loops that are important for the programming of movement sequences in a goal-directed fashion (i.e., learning to calculate the appropriate "next-step" given a current task goal). Interestingly, activity in the SMA is found when individuals physically perform an action and when they mentally imagine performing an action. This suggests that mental imagery may be an effective way to practice certain aspects of a motor skill because mental imagery, to a limited extent, requires programming of the motor sequence.

Making the Movement

At the execution level, there is little or no access to awareness and conscious control as this level consists of detailed specification of motor unit recruitment in both time and space. The type of representation used at this level is not clear, but there is research to suggest that rather than programming individual muscle contractions or joint torques, the execution level relies on stereotyped relationships and patterns of muscle activity referred to as synergies or "motor primitives." For simple movements, much of this execution-level programming appears to occur at the level of the spinal cord. Electrical stimulation of the spinal interneurons in animals leads to relatively smooth movements of the whole effector toward a consistent endpoint. In contrast, stimulation of the alpha-motor neurons, the next level below interneurons, leads to motor unit contractions within a single muscle at a constant force dependent on the magnitude of stimulation.

Reflex loops and hardwired neural pathways exist at the execution level that differ depending on the type of control being examined. That is, there are specific neural pathways that generate relatively consistent patterns of muscle activity across individuals that maintain balance and posture, stabilize patterns of locomotion, or coordinate movements of the eyes relative to the head in order to maintain gaze. These fast, low-level motor control processes operate outside of awareness and can be very difficult to override if they conflict with voluntary motor actions. However, there is evidence that these pathways can be modified through experience and can operate in a goal-directed manner (i.e., the result of a reflex loop

can change depending on the current goal of the motor system).

Sensory feedback, as well as the prediction of what sensory feedback will look, feel, and sound like, is critical at the execution level so that actions are controlled appropriately to achieve specific functions and goals. The motor system has its own internal error correction mechanisms that facilitate success in movement and learn from experience to progressively fine-tune goal-directed actions. The exact process by which these modifications are made to the motor system is not clear, but several variables must be calculated in order to detect errors and correct the transformations from task level to execution level. These transformations might be conceived of as internal models or dynamic biological processes involving both predictive and control processes. In order to tune these processes, the motor system must represent the actual state of the system, the desired state of the system, and the predicted “next state” of the system. By comparing these different states, error signals can be generated that tune both the predictors and the controllers. Predictors are tuned through the comparison of the predicted state and the actual state (i.e., Did that do what I thought it would?). Controllers (or motor commands) are tuned through comparison of the desired state with the actual state.

It appears that these processes associated with internal models take place at a mostly subconscious level. Neuroimaging data show that when a movement sequence is explicitly attended to, there is an increase in the activity of the PFC and a decrease in the SMA (i.e., a shift to the neural structures that were involved early in the learning process) and a corresponding reduction in the speed at which a movement or sequence of movements is produced. Behavioral evidence also suggests that attending to the details of movement, compared to attending to the desired goal, leads to worse performance and less efficient motor control.

One final consideration for movement execution is critical task variables that appear to affect how the movement is executed. For example, it is well established that movement time (MT) is (predictably) dependent on the size and extent of a movement required (so called index of difficulty). Large movements to small objects are slower to execute than small movements to large objects, arguably because more time is needed to process visual feedback to accurately hone in on

the object. Measurement of movement kinematics (or movement process measures), such as displacement, velocity, and acceleration of a limb, has allowed researchers to infer how movements are controlled, based on sensory feedback, as an action is executed.

“Offline” Success Evaluation

This general, hierarchical framework provides a basis for understanding how a command makes the transformation from thought to action. However, in order to achieve the desired goal accurately and reliably, the motor system needs to be able to modify (“tune”) these transformation processes based on sensory feedback errors about the success of an action. This evaluation can happen after the movement in what is termed an “offline” fashion. Correcting subsequent movements can be an onerous task for the motor system because there is considerable ambiguity in what sensory errors mean from a motor control standpoint. Errors convey information about the success of a movement (e.g., hits vs. misses) and some feedback about what needs to be corrected in the movement (e.g. a miss to the left versus the right), but the error does not specify which aspect of the movement needs to be modified (the credit assignment problem) or when in the movement a modification needs to occur (temporal credit assignment). This is where a coach can play an important role, using his or her personal experience and expertise to identify where and when an error occurs and then provide additional (augmented) feedback to an athlete to fix the error.

Keith R. Lohse and Nicola Jane Hodges

See also Motor Learning; Movement; Skill Acquisition

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MOTOR DEVELOPMENT

Motor development is the movement science discipline concerned with age-related changes in movement patterns and the underlying processes that drive those progressions. The changes in movement pattern tend to be sequential, occurring in a series of steps. In the early portion of the life span, change is from execution of simple (even though coordinated) movement units to highly organized and complex movements. In the later portions of the life span, change reflects adaptations to physical aging and disease onset.

Early in the establishment of motor development as a field of study, researchers tended to study the young, especially infants. While many came to associate motor development only with the study of young performers, current interest is in change throughout the life span. Study of the entire life span has facilitated understanding of the underlying processes that drive development, as researchers can observe how a change with growth compares to a change with aging. For example, the stride length of a newly walking toddler is short, compared to that of a young adult, as is the stride length of some senior walkers. Motor developmentalists are interested in whether this reflects underlying processes that are similar or different, with the answers enriching the understanding of developmental processes.

The study of motor development includes the description of change and documentation of *when* changes occur in the life span. Yet, the field goes far beyond description to focus on underlying causes of change or the *why* and *how* of change. This has not always been apparent to researchers in other movement sciences. A number of early studies describing progression in movement patterns demonstrated a consistent pattern of emerging new skills across human infants. At the same time, the famous “nature–nurture” debate was ongoing in a

number of fields concerned with development. The consistent pattern of skill emergence was often taken as support for the view that “nature” was the predominant influence on development. As a result, the perception of scholars outside the field of motor development was of a field dominated by a maturational perspective and using only methods of description as a research tool.

Despite the perspective of others that motor development emphasized influences of nature (genetics), many early researchers in motor development had adopted an interactionist perspective—one that acknowledged that nature *and* nurture (environment) interact and both influence development with neither dominating. They sought understanding of all the processes driving developmental change. Nevertheless, it was not until the later portions of the 20th and early 21st century that perception of the field changed to acknowledge this great interest in how genetically driven development and the environment interact in the developmental change in motor patterns.

While motor development far exceeds just the description of change in movement patterns, the developmental descriptions produced by early scholars introduced a rich body of information that is used even today. Those rich descriptions identified sequences of new skill acquisition in infancy (interskill sequences), such as the sequence of rolling over, holding the head up, pushing the chest up off the surface, and pushing up to hands and knees. Also identified were sequences of change within a skill (intratask sequences). One brief example for the skill of throwing is for trunk movement pattern: no rotation, simultaneous rotation of the upper and lower trunk, and then lower trunk rotation followed by upper trunk (differentiated) rotation.

Motor development has its roots in biology and psychology. The roles of physical growth, physiological maturation, and physical aging in driving changes in movement are obviously rooted in biology. Yet, the discipline of psychology, with its emphasis on human behavioral development, was a greater influence on motor development as a field of study, particularly by proposing models of development and means of studying the human organism.

Focus on longer periods of change contrasts motor development with the field of motor learning that focuses on the change occurring in shorter time periods, for example, as a result of weeks of

practice. The fields are further distinguished by an emphasis on the movement product in motor learning, but an emphasis on the movement pattern in motor development.

The current field, then, is concerned with development over the life span, processes underlying age-related change in skill performance, and the relationship of genetic and environmental factors as they drive motor development. It is important to understand how contemporary perspectives are shaping both research and practice in the field. Two are particularly important: *modeling movement* as arising from the relationship of constraining factors and *modeling developing systems* as dynamic systems.

The Model of Constraints

The current emphasis on the interaction of multiple influences on development can be seen in the common use of Karl Newell's model of constraints by both researchers and practitioners in the field. The model uses a triangle to represent movement as arising from the interaction of constraints that fall into three categories: the *organism* (here, the human performer), the *environment*, and the *task* to be executed. The constraining, or shaping, factors associated with the performer can be further divided into structural and functional constraints. Examples of structural constraints include the body's physical structure and its systems, such as the muscular system, the nervous system, and so on. Examples of functional constraints are motivation, experience, and fear. Environmental constraints include characteristics of the environment, such as gravity or altitude as well as the surfaces and dimensions of the setting for skill execution. Task constraints include the intended goal for the movement, equipment, and any rules established for achieving the goal.

A particular movement is shaped by the interaction of all relevant constraints. As any even single constraint changes, the relationship among the constraints differs, potentially resulting in a different movement pattern. Usefulness of the model to motor developmentalists is obvious when one considers how the structural constraints change with physical growth and maturation or aging. As an individual grows, for example, the relationship of the structural constraints with the environmental and task constraints changes, giving rise to differing movements. Hence, the model provides

a vehicle for relating changes in these relationships with growth and development to qualitative changes in movement patterns.

This framework is useful both to researchers who focus on specific relationships among the constraints and to practitioners. While growth or aging typically cannot be manipulated in the short term, the environment and task constraints can be manipulated in conjunction with changes in structural constraints. Practitioners can manipulate environment and task constraints to design developmentally appropriate tasks for those at any point on a developmental continuum of change in the structural constraints. As a simple example, if the movement desired is to shoot a basketball through a hoop, the height of the hoop from the ground and/or the weight of the basketball can be incrementally changed as the height and strength of the performer change to achieve the desired goal over the period of growth and maturation. Another example is how practitioners have designed a series of adapted bicycles for children with disabilities—for example, replacing the rear wheel with two rollers, to facilitate development to riding conventional bicycles.

Modeling Body Systems as Dynamic Systems

How qualitative change in movement patterns emerges is one of the most interesting questions in motor development yet to be answered. Motor developmentalists have modeled the developing body systems involved in movement as dynamic systems to address this question. The study of dynamic systems has its roots in applied mathematics. Dynamic systems are time-evolving systems; hence, their usefulness to the study of motor development becomes obvious. Body systems evolve, or grow and mature and age, over time. The characteristics of dynamic systems have much to offer when addressing a number of fundamental issues in motor development today.

Consider that new movement patterns are not "more" of a previously executed pattern but rather different patterns. There are qualitative changes. The study of dynamic systems has demonstrated that elements of a system can self-organize, to place themselves in new relationships with each other. This could explain how new movement patterns appear: the body limbs and joints that are elements of a movement pattern could self-organize into a different movement pattern.

Dynamic systems are nonlinear, and their development is nonlinear. Hence, dynamic systems offer another advantage for modeling skill performance. Dynamic systems alternate between *periods of stability* and *instability*, and what prompts the change from one state to another might suggest how new behaviors emerge in development. Dynamic systems are pushed out of a stable state by a change in one or more components of a subsystem. Likely the change must reach a critical value to push the overall system into a period of instability from which a new stable state eventually emerges. In motor development, changes such as an increase in bone length, an increase in muscle strength, or formation of neural connections might be the types of changes that, once reaching a critical value, stimulate appearance of a new movement behavior. Dynamic systems theory predicts that a performer would enter a period of variability in a movement pattern before a new pattern eventually emerges and becomes stable. Whether this actually occurs is a question for future research.

One of the multiple systems involved in a movement might develop more slowly than the other systems. Hence, that system discourages the appearance of a new skill until a critical value is reached, changing the relationship among systems and facilitating appearance of a new pattern. For example, the leg strength and body weight ratio might have to reach a critical level before an infant can stand. In such a case, the slowly developing system is described as the *rate limiting* system for a particular movement skill. Motor developmentalists are interested in identifying rate limiting systems for various movements.

Another issue in the movement sciences in general concerns the mechanism for controlling so many body parts into a coordinated and smoothly controlled movement, and obviously, this too is an issue in the development of motor control. Since body parts have a number of ways of moving—that is, multiple degrees of freedom (df)—a challenge for movement scientists is to explain how those df are harnessed to execute goal-directed movements, especially when the starting position of limbs and joints can vary from time to time the movement is executed. For motor developmentalists, the challenge is extended to explain how control is initially achieved. Dynamic systems can be constrained into functional units, and this suggests that a human body comprised of dynamic systems can spontaneously organize into functional units,

thus minimizing the number of individual elements that must be controlled by the developing nervous system. These functional units have been called *muscle linkages* or *coordinative structures*. An example would be how the muscles of the ankle, knee, and hip act as a unit in swinging the leg forward in walking. Motor developmentalists are interested in exploring how these coordinative structures develop, how they might reorganize into new movement patterns, and whether they change over time or come to be incorporated into new movement patterns.

As mentioned earlier, a particular challenge for modeling the control of movement, especially in developing individuals, is the probability that the initial conditions for beginning a movement are almost always different, yet movement patterns are successfully completed. Sensitive dependence on initial conditions has become widely known as the *butterfly effect*, so named for the notion that a butterfly stirring the air might influence storm systems halfway around the world a month later. The challenge specifically for motor developmentalists is explaining how movement patterns are acquired, including how they sometimes can be executed correctly on the first attempt, despite this *context-conditioned variability*. As a simple example, developmentalists must account for how even infants can reach to an object despite an almost infinite number of starting positions of the arm.

Dynamic nonlinear systems also are modeled as *oscillators*, including pendula. This has permitted an interesting analogy for walking, since we might think of each arm, swinging back and forth with each walking stride, as a pendulum. The arms maintain a relationship, systematically staying in opposition, and they maintain a relationship with the legs, moving forward and backward, opposite in direction to the same-side leg. This *synchronization* is called *entrainment*, and oscillators have been shown to entrain. Entrainment might be a means for the nervous system to more easily control what would otherwise be a set of complicated movements. If each limb, with its joints and muscle units, moved as an independent pendulum, control of walking would be a daunting task. Entrainment might be an example of how dynamic systems constrain or harness df to simplify control of some movements. Very early walkers tend to hold their arms up and not swing them but with walking experience begin to swing the arms at their sides like pendula, moving with the opposite

leg. This suggests that early walkers might begin to use entrainment as a means of more efficiently controlling the movement of multiple limbs.

Motor developmentalists, then, have used the features of dynamic systems to address some of the fundamental issues surrounding the emergence and control of movement patterns in developing movers. Current research into the development of control and coordination in movement is examining many of these features as contributions to our understanding of the underlying processes driving developmental change.

The Sequential Nature of Development

As mentioned earlier, motor development is considered to be sequential. Longitudinal study of the development of basic skills typically reveals a series of qualitative and quantitative changes that eventually result in the execution of a skill at an advanced level. For example, very young children are typically observed throwing without taking a step and therefore with little weight shift. The distance the thrown ball travels is consequently short. At a higher level, children might be observed stepping with a throw but with the foot on the same side of the body as the throwing arm, hence limiting the contribution of trunk rotation to the throw. Then at even more advanced levels throwers can be seen stepping with the foot opposite the throwing arm. The length of the step might also increase with development so that a throw for distance can be maximized. Hence, development of leg action in overarm throwing is a series of qualitative changes and occurs by progressing from level to level.

Motor developmentalists seek to identify those processes that drive change to the next level. The goal of assisting performers to progress from level to level (with developmentally appropriate tasks) is a key feature of a developmental perspective. Rather than seeing each early level as an error because it is not the most advanced level, developmentalists give credit for advancement to the next qualitative level, or for a quantitative increase within a level, all as progress toward the goal, the most advanced level.

Consider that performers move through developmental levels in different ways. Take the development of the leg action in throwing described previously. There also are developmental levels for the action of the trunk and of the arm. One performer might advance a level of leg action before a level of arm action but another vice versa. Even as

researchers explore what underlying processes are involved in these variations, performers advance sequentially. Perhaps this is related to sequential change in the component body systems (serving as constraints for a task). Practitioners who present developmentally appropriate tasks take developing movers from level to level over time. This too characterizes a developmental perspective on complex skill acquisition.

As motor development concerns age-related, sequential change, the ideal research design for those studying motor development is a *longitudinal* design. The same individuals are observed over time. This design permits researchers to actually observe the phenomenon of interest as it changes in individuals. An obvious challenge for researchers is the length of time individuals must be observed to see the change in some phenomenon, as well as repeated measurement. Considering this, other designs are often used in motor development research. Cross-sectional designs allow researchers to observe a phenomenon in those of different ages to *imply* developmental change. So too are designs that combine longitudinal and cross-sectional designs. These allow developmental research to be conducted in a shorter time period and with fewer repetitions of a measurement. Nevertheless, the longitudinal design remains the preferred method for confirming true developmental change.

Other Areas of Interest

Many motor development researchers are concerned with the sequential changes in movement patterns, the development of postural control, and the development of coordination and control of movements. Yet others study simultaneous change in other constraining, interacting factors that can influence movement patterns. The study of growth and maturation and also aging is important to an understanding of motor development. Physiologically, children are not miniature adults, and there are a number of unique aspects in how children respond to exercise loads. Some researchers, then, have specialized in age-related changes to exercise or in the interaction of exercise with the structural constraints, especially body composition to address aspects of general health and wellness in young and old populations.

Motor developmentalists are also interested in the functional constraints and their age-related change. Changes in self-efficacy regarding skill performance are studied, as are motivational

factors in participation in physical activities. Mastery learning climates have been studied as a means to provide developmentally appropriate tasks. Societal expectations, culture, and values also have been examined, particularly as they might influence gender differences in performance or selection of particular physical activities

Conclusion

Motor development as a field of study has become an important component of the movement sciences, both for practitioners and for researchers. Practitioners deal with movers who change over time and attempt to facilitate that change by assisting movers to attain complex advanced movement patterns. Researchers have found that models of the control and coordination of movement are only robust if they account for developmental change. Hence, the motor development field is integral to the important issues taken up by movement scientists. The fundamental questions taken up by motor developmentalists remain on the forefront of efforts to achieve a better understanding of the movement sciences.

Kathleen Haywood

See also Coordination; Dynamical Systems; Ecological Theory; Genetics/Nature–Nurture Determinants; Mastery and Control Beliefs; Motor Control; Motor Learning; Practice; Self-Efficacy; Skill Acquisition; Task Constraints

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MOTOR LEARNING

Motor learning is the shaping of individual sensorimotor capabilities by the physical and social environment. It is based on changes of neural networks of the brain that enable a relatively permanent improvement of performance, even though this may not always be manifest. Motor learning is induced by experience or practice and can take place with the intention to learn but also without; indeed, even without conscious awareness that something has been learned. In this entry, three issues are addressed: (1) the behavioral changes that give evidence of motor learning, (2) the internal representations and mechanisms involved, and (3) the practice conditions that affect motor learning.

Behavioral Changes

Motor learning is associated with a number of behavioral changes. Prominent among them are improvements of speed and accuracy, increasing movement consistency, economy, and automatization.

For many motor skills, higher speed means better performance. Exceptions are skills for which correct temporal patterns are defined, as in playing a musical instrument, or where movements have to be synchronized with environmental events, as in catching balls. The improvement of speed is captured by the *power law of practice*, which is a good approximation for a variety of skills. According to this empirical law, the time T needed to perform a particular action declines with the number of repetitions N in a way that can be described by a power function, $T = kN^{-\alpha}$. Across the first repetitions the decline of movement duration is rapid. Later on it becomes progressively smaller, and finally, it can hardly be detected against the background of random variations of performance. Diminishing returns in terms of improvements are a fairly universal characteristic of practice curves not only with temporal measures of performance

but also with many other dependent variables. As a consequence, the early stages of practice are generally more rewarding than later stages.

The increase of accuracy in space and/or time is a second major behavioral change. For many tasks, there is a trade-off between speed and accuracy—for example, in movement aimed at a target, known as Fitts's law. However, motor learning overcomes this trade-off. Typically accuracy increases in concert with speed. Nevertheless, there is an important difference between learning-related changes of these two performance characteristics. Accuracy is always defined with respect to a specific target, but speed is mostly “the faster the better”—that is, the goal is maximization (of course, maximization can also be a goal for spatial characteristics in tasks such as throwing a ball as far as possible).

With increasing number of repetitions of a motor skill, performance tends to become more consistent—that is, less variable. This can also represent an increase of accuracy. As an example, consider the task of throwing a ball over a distance of 10 m. The throws will have a mean deviation from the target (called *constant error*), and they will also vary around the mean (called *variable error*). A reduction of the constant error typically requires that the learner has information on the error, whereas a reduction of the variable error can result from simple repetitions.

The example of throwing a ball over a certain distance does also illustrate that movement strategies can be adapted to increase accuracy of the outcome, the constant and variable error of throwing distance. The distance covered by a thrown object depends on its initial velocity and the angle of its initial flight path with the horizontal plane. For a certain distance, the required initial velocity is smallest for an initial angle of 45 degrees and for deviations from this angle it increases. Deviations from the strict relation between the initial flight angle and velocity required for the target distance result in different throwing errors depending on the range of flight angles and velocities where they occur. Thus, performance improvement can also result from a proper choice of strategy. More generally, for many motor skills there may be strategies for which variability in movement production has comparatively small effects on the outcome.

During practice of many motor skills, increasing economy of movement production can be experienced. A good example, which unfortunately is

somewhat outdated, is the operation of a scythe. This is an exhausting exercise for a beginner, whereas an expert can do it smoothly for a long time. Operation of the scythe requires exact guidance of the instrument. In particular, skills that require high accuracy are initially accompanied by cocontractions. These are concurrent activations of opposing muscles. They do not result in net torques but serve to increase the impedance of the limb and thus make it more resistant against perturbations. In many motor skills, cocontractions decline in the course of practice so that movements become more economical in terms of muscle activity involved and thus in terms of energy consumption.

Finally, motor skills tend to become automatic after sufficient amounts of practice. The “autonomous” phase represents the final stage of a quite popular conceptualization of skill acquisition by Paul Fitts. It follows the initial “cognitive” phase, in which motor patterns are produced with strong cognitive involvement, and the subsequent “associative” phase, in which the components are gradually bound together or associated. Subjectively automaticity comes close to the motor skill running off by itself once it has been started. Functionally, automaticity is inferred from the absence of interference by concurrent cognitive activity. Absence of dual-task interference, however, is not a universal phenomenon with highly practiced motor skills. First, the perturbation of a practiced motor skill by a concurrent cognitive activity depends on the nature of the activity. Second, in the literature, one can find examples that even apparently quite separated cognitive activities such as mental arithmetic interfere with motor skills such as standing (in older persons in particular) or filing of skilled precision mechanics—that is, with skills that seem fully automatic at first glance.

Representations and Mechanisms

In the course of motor learning, the control of movements is changed, which results in better performance. There are underlying changes of internal representations, and there are mechanisms that bring these changes about.

Types of representation, which are acquired or modified during motor learning, are representations of correct movements, of environmental characteristics, and of transformations the motor system has to master. Representations of correct

movements have been posited by a number of theories. For example, a core construct in the influential theory of Jack Adams is the *perceptual trace*. This is a representation of the reafference of the correct movement that is claimed to serve as a reference for *closed-loop control*. A number of theorists have posited (*generalized*) *motor programs*, which are representations of the motor outflow. These have been conceived as prestructured motor commands but also as prototypical force-time profiles that can be varied in certain ways so that different variants of movements of a certain type can be produced.

Representations of environmental characteristics have received only little attention in the motor-learning literature. For example, in playing a musical instrument, one can learn which notes are to be played next, or in playing soccer, one can learn what a certain opponent player is likely to do next in a certain situation. Such representations are involved in anticipation, and they can contribute to pretty extensive changes of perception. They also allow to prepare movement sequences in advance and to form chunks of the elements of a sequence, such as in typing, which can then be produced in rapid succession. Sometimes representations of environmental regularities are hard to distinguish from representations of movements. For example, skilled typing involves chunks of finger movements that are produced in rapid succession, but at the same time, these movements reflect the environmental regularity of the letter sequence.

Representations of transformations are generally referred to as internal models. To clarify what is meant by transformations, consider again the example of throwing a ball. Muscles are activated, forces of various muscles are generated and combined, joints are rotated, the hand is moved along a certain path with a certain velocity, and finally the ball flies a certain distance. To produce the proper hand movement and to release the ball at the right time requires a fairly complex internal model of the transformations on the way from muscle activation to ball flight. More precisely, what is required is an *inverse model* that specifies the proper input to the transformation (muscle activity) that results in the desired output (flight of the ball over 10 m). However, the *forward model* is also useful in that it allows rapid predictions of the outcome of a movement. In the example, it allows prediction of the success of the throw before the ball has even left the hand.

A popular variant of the notion of internal models is provided by Richard Schmidt's *schema theory*, which is actually several years older than the notion of an *internal model*. According to this theory, a representation called *motor schema* establishes a link between desired outcomes of an action (e.g., the desired amplitude of a throw) and the parameters of a generalized motor program (GMP) (e.g., the overall force level of a throwing movement or its total duration). One of the major predictions of schema theory is that variable practice should be more beneficial than practice under constant conditions. The reason for this prediction is basically that to learn a relation between variables one has to encounter a range of these variables and not just single values.

Sufficiently accurate (inverse) internal models of the transformations involved in motor control are a prerequisite of *open-loop control*. In fact, motor learning has been characterized as a progression from closed-loop to open-loop control and also as a progression from the use of visual feedback to the use of proprioceptive feedback. Such progressions may indeed be found for certain motor skills, but there are also motor skills that remain dependent on visual feedback (as well as proprioceptive feedback) even after prolonged practice. Balancing a vertical rod on the tip of the index finger is an example.

Regarding the mechanisms involved in motor learning, repetition effects, error-based corrections, reinforcement, motor resonance, and consolidation are among the important ones. Historically, the existence of repetition-based (or use-dependent) learning without any feedback about the outcome of the action has been doubted, but this is clearly unjustified. Pure repetition can serve to organize visual input (unsupervised learning), for example. Also movements will become faster and less variable. A basis for such changes might be facilitation within the neural networks that are involved in the production of a certain movement and in the perception of the relevant environmental conditions.

Error-based learning and *reinforcement learning* are sometimes hard to distinguish, and in the early 20th century, the distinction was uncommon. A concept that combines these two types of learning is knowledge of results (KR), which is typically provided after the end of a movement. It can take various formats. For example, it can be evaluative ("good," "poor"), or it can be informative by way of indicating the precise error ("20 cm too short").

In the first case there is likely reinforcement, an associated neural event that basically strengthens what has been rewarded (“good”). What has not been reinforced is likely to be changed in the next attempt, and without further information, the change will be more or less random. In the case of informative KR, there is error-based learning in that the next attempt can be modified in a way that compensates the error indicated. In spite of their similarity, these two mechanisms of learning invoke different neural structures.

A currently quite popular mechanism of motor learning is *motor resonance*. This mechanism is crucial for observational learning. Humans can observe movements produced by other people, and often they can reproduce these movements immediately. The popularity of this kind of learning mechanism has been boosted by the discovery of *mirror neurons* in monkeys. These neurons are active not only when a certain movement is produced but also when the same movement is observed. In humans, overlapping patterns of activated brain areas have been observed in producing and observing movements and also in producing and imaging them. Thus, observation of a certain movement is likely to activate at least a subset of those neurons that are also involved in its production—the motor system “resonates” in response to the visual input.

A final mechanism is *consolidation*. Consolidation refers to neural changes that serve to stabilize or even improve what has been practiced before. It is invoked when it comes to an improvement of motor performance after a break or after a night of sleep. It is also invoked when the question is addressed whether the acquisition of a new internal model of a transformation overrides an older internal model or is added to it. In any case, consolidation is the most comfortable mechanism of motor learning in that it does not require any activity of the learner.

Practice Conditions

From the various representations and mechanisms involved in motor learning, a number of principles for the design of practice conditions are obvious. Motor learning requires information to enable error-based learning and evaluative feedback to enable reinforcement learning, variability should facilitate the acquisition of internal models of motor transformations, and sufficient spacing can

bring in consolidation and serve to prevent muscle fatigue as well as mental fatigue. Among the less obvious design principles are the avoidance of over-optimization, the use of imagery and observation of the motor skill, and the proper direction of the focus of attention.

In order to optimize practice conditions and to obtain a rapid improvement of performance several measures can be taken. Some of them can produce immediate unintended (and perhaps counterintuitive) effects. For example, error-based learning can be facilitated by sufficiently accurate error information. When errors are presented visually, for example, they can be amplified; when they are presented numerically, they can be given in smaller rather than larger units. The risk of such measures is that the scale of error information exceeds the precision of movement production. In the extreme case, when movements are basically accurate and vary only randomly, the random errors are fed back and learners try to correct them, which is doomed to failure. It only has the effect of increasing variability.

Unintended effects of optimized practice conditions can also become visible after a delay; for example, when the motor skill has to be performed under real-life conditions after augmented feedback has been removed. Performance can become dependent on augmented feedback and break down when augmented feedback is no longer available. In addition, optimization of continuous visual feedback for closed-loop control can impede the acquisition of internal models that are needed for open-loop control. In recent years more and more robots have been designed to support motor (re-)learning, in particular in neuro-rehabilitation. Haptic guidance provided by them results in high levels of performance. It also demonstrates how the correct movement feels. But it prevents active generation and shaping of motor commands as well as active error corrections. Therefore, it can impede rather than facilitate motor learning.

Motor imagery and movement observation can serve to improve motor performance. The reason is that both imagery and observation share neural structures with actual movement production. In a certain way, they are similar—and functionally equivalent—to central processes of overt motor behavior. In general, observational and mental practice are less efficient than physical practice, but combinations of the different types of practice can be superior to physical practice alone.

Observation can offer the opportunity to notice aspects of performance that remain unnoticed when one moves oneself. Imagery offers possibilities of slow motion and the exploration of skill variants that might even be physically impossible. Alternation of periods of observational or mental practice with periods of physical practice allows the combination of the respective advantages of the different procedures.

Finally, during the last several years, a critical role of the distribution of attention for motor learning has been documented. Basically an external focus of attention results in superior learning as compared with an internal focus. The external focus is on movement outcomes—for example, the swing of the golf club—whereas the internal focus is on the moving limbs; that is, the movements of the arms. Most likely the benefits of an external focus of attention are related to the fact that motor control typically starts with a desired outcome (e.g., a throw of a certain distance), and motor commands are selected according to an internal model that has been acquired during motor learning. In the performance of many motor skills awareness of the details of the movements is quite limited, and an internal focus of attention (and thus the attempt to voluntarily control details of muscular contractions) may actually interfere with the required delicate timing of motor commands or other aspects of proper motor outflow.

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See also Anticipation; Attentional Focus; Automaticity; Feedback; Generalized Motor Program; Imagery; Reinforcement and Punishment

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MOVEMENT

Riding the bicycle to work, walking up the stairs to the apartment, taking a book from the shelf, and playing the violin are all examples of motor activities, which are signified by the planning of future goal states, the coordination of different limbs and various whole-body postures, and the regulation of muscle forces during the dynamic control of the different movements. The skillful coordination of such movement actions is one of the major challenges in mastering the various tasks of our daily life. To do so, people do not only passively take in the information from the environment with their senses when they interact with objects or with other people but also plan their movements in advance to change the environment in a purposeful way. Most of the time, people's movements are therefore voluntary, goal-directed, and intentional, relying mainly on higher-level perceptual–cognitive control processes. This is markedly different from motor reflexes, which are hardwired, involuntary, and unintentional, relying on lower-level sensorimotor control processes and providing the basic building blocks of human behavior (e.g., stepping reflex, grasping reflex). This entry focuses on the performance of complex movements that can be readily observed in various sport disciplines and settings. Of chief interest are three aspects: (1) conceptualizing movements within different taxonomies, (2) practicing movements within different schedules, and (3) mastering movements at different levels of expertise.

Conceptualizing Movements Within Different Taxonomies

There are at least three ways to differentiate the various kinds of movements into different

taxonomies. The first is to consider the number of different movement elements, components, or actions that need to be generated and executed. In this context, three different categories can be identified: *discrete movements*, *serial movements*, and *continuous movements*. Discrete movements are characterized by a well-defined beginning and endpoint. The motor action is usually executed within a short time window. Examples of discrete movement skills are kicking a soccer ball, throwing punches, or dancing a pirouette. Serial movements are of longer duration, as they consist of two or more discrete motor actions, which are executed in an ordered sequence of events. A gymnastic routine, high-jumping, or the turnaround in swimming are examples of serial movement skills. Continuous movements have no recognizable beginning and end, as the different elements are repeated in a way that the movement's final phase merges into the new initial phase, starting the whole series of motor action all over again. Examples of continuous movement skills are running, swimming, and cycling.

The second way to classify movement skills into different categories is to look at the relative importance of perceptual, motor, and cognitive elements and components for task performance. *Perceptual skills* rely mainly on the processing of visual, auditory, and/or tactile information, whereas the execution of corresponding motor action is of less importance. For a catcher in baseball, the demands on the motor elements (and on decision making [DM]) of the catching action are (considerably) simple compared to the demands on visual anticipation of the fastball pitch. *Motor skills* depend primarily on the quality of movement performance. A weight lifter, for example, must be able to maximize his lift-up and push-out actions, in order to realize a heavy weight. *Cognitive skills* require complex DM and the use of strategies. The strong safety in American football must recognize different playing patterns of the attacking team and come up with quick decisions for his defensive maneuvers. When looking at these three previous examples, however, it should be noted that task performance for each of these different types of movement skills does not solely rely on a single element or component. Rather, the focus is on the movement element or component, which is mainly "responsible" for successful task performance.

The third way of skill categorization is to address the context-specific environmental

demands during task performance. On one end of the continuum, these may be always stable and predictable, whereas on the other end, task demands may be constantly changing in a most unpredictable way. Accordingly, movements are either *closed skills* or *open skill*, or they are in between these two poles somewhere on the continuum. The extent to which changes in the environment can be predicted may be further affected by time constraints and/or the interaction with teammates and opponents. Playing darts is an example of a closed skill, where the task demands are always similar and the environmental context does not change, while the discrete movement can be executed without much time pressure and no physical interaction with the opponent. The same is true for high-jumping (indoors), but the serial movement has to be adapted to changes in the task context, as the bar is raised higher. Backcountry skiing is more complex, because serial and continuous movements have to be adapted to the (natural) environment of the skiing resort, making this a slightly more open skill. Still more complex and open is the Olympic discipline of freestyle skiing, because there is high time pressure, as the skiers race against one opponent (although without coming into physical contact). Playing rugby is an example of an open skill, where performance must be adapted to varying task demands and a constantly changing environment, under high time pressure and against a number of different opponents.

Practicing Movements Within Different Schedules

Movements become skills when they are practiced deliberately. K. Anders Ericsson, Ralf Krampe, and Clemens Tesch-Römer defined deliberate practice as "activities that have been specially designed to improve the current level of performance" (p. 368). Thereby, distinct *changes of performance* are the direct result of the activities experienced during the training session. Most all of the time, such activities (i.e., physical practice) lead to effects of performance improvement, which can be readily observed in an increase of some critical variables (e.g., score points, reaction time [RT], movement time [MT], performance errors). But under specific circumstances, physical practice can also produce effects of performance decrement. Take the implementation of a new training method

or drill as an example. After this new method or drill has been introduced into training, athletes may not be able to perform a certain skill with the efficiency as before. Such changes of performance, however, may only be of short duration, when the movement is not practiced over a longer amount of time. More permanent changes of performance are thought to reflect effects of *motor learning*, which results from extended amounts of physical (and mental) practice. Characteristic effects associated with motor learning are the higher consistency of performance and better performance results, the optimization of movement economy and the (quasi) automatic execution of skills, as well as the ability to flexibly adapt any movement to fast changes in the environment. Also, with extended practice, the execution of even the most complex movement skills will demand less attentional resources, which can be used to direct attention to other aspects in the environment. Whether or not an athlete has actually learned from practice can only be assessed with a retention test (i.e., testing the level of performance after a period without practice) and/or a transfer test (i.e., testing the level of performance in a new task or under a new task context).

Deliberate practice can be scheduled in many different ways. However, the most important variable of motor learning—with all other factors being equal—is practice. That is, only a large amount of deliberate practice will improve the level of performance in the long term. Many studies therefore focused on how to schedule physical training and to distribute the amount of practice for a particular skill. Consider the following example: A basketball player would like to improve her performance for the jump shot. She can go ahead and perform 100 jump shots, without a break and always from the same spot on the court. Such a training schedule is referred to as *massed practice*. Alternatively, she may perform 10 shots in a row, take a short break, and then continue with the next 10 shots, take a break, continue for 10 shots, and so on, for a total amount of (again) 100 jump shots from the same spot. This is called *distributed practice*. As research shows, athletes will learn a certain skill better when the activity is distributed over several blocks of practice, as seen in better retention and transfer test results. Some of the benefits of distributed practice schedules may be explained with the recovery of perceptual, cognitive, and motor functions during practice breaks.

In this basketball example, the player was practicing the jump shots always from the same spot and thus shooting the ball to the basket over the same distance in all of the practice trials. This is referred to as *constant practice*. A way to alter this practice schedule would be to introduce some variability into the task conditions. To this end, the player could move up closer to the basket on some trials and move farther away from it on other trials, shooting either from shorter or longer distances. This is an example of *variable practice*. Hence, the amount of variability may be another factor to consider when scheduling practice session. Much research shows that practicing movements under constant conditions increases the level of performance for a short period of time, whereas performance improvements are more permanent under variable conditions. If variable practice is more beneficial than constant practice, then the question that needs to be answered in this regard is how to schedule the variable conditions. Assume that the basketball player would now like to practice the 100 jump shots from four different distances (e.g., 2, 3, 4, and 5 m). She could perform 25 shots in a row from each distance, before moving on for another 25 consecutive shots from the second distance, and so on. Alternatively, she could take only a single shot from one distance and then move to another distance in a random sequence, without attempting to shoot from any distance twice in a row. One schedule is organized as *blocked practice* and the other schedule as *random practice*. Again, more variability within a particular schedule seems to benefit motor learning and improve movement skills in the long term. This has been explained either with a deeper conceptual processing of the movement during practice or with ongoing forgetting and reconstruction processes, which benefit the retention and transfer of the movement.

Mastering Movements at Different Levels of Expertise

The deliberate practice approach suggests that extensive amounts of domain-specific practice are necessary to reach an expert level of performance. It has been suggested that athletes need at least 10,000 hours of deliberate practice or more than 10 years of training within a particular field, such as music, typing, chess, and sports. Most interestingly, it seems that everyone can become an expert performer, for as long as he or she will put in the

effort of such prolonged and extensive practice for at least 4 to 5 hours a day over the number of years. If large amounts of practice are dedicated to train a single skill, the level of performance may become exceptionally high. For example, the basketball player referred to before most likely spends much time to practice his or her shots from the foul line. As a result, the likelihood of scoring from the foul line will be much higher than what would be expected when only considering the foul-line shooting distance relative to the success rate from locations nearby. Such an exceptional performance in sports relies on so-called *especial skills*, which are an important factor of domain-specific expertise.

Before an athlete becomes an expert for any movement or sports skill, however, he or she moves through different stages of proficiency during the learning process. For the acquisition of new movement skills, Paul Fitts and Michael Posner proposed a *three-stage learning model*: In the first stage, the cognitive stage, the *novice performer* tries to get an idea of what to do and how to solve a particular movement problem. He or she learns the basic sequence of events constituting a particular movement. Movement execution is aided by verbal cues. At this early point in time, movements are performed slowly, inaccurately, and with the generation of too much force, resulting in more or less stiff-looking motor actions. Much attention is drawn to the organization and execution of the movement pattern. In the second stage, the associative stage, the *advanced performer* intentionally varies different task components and associates these variations with success or failure. Therefore, the quality of feedback that the learner receives plays a major role on skill acquisition during this phase. Overall, the level of performance is high as long as the task context remains stable and the movement can be carried out without much interference by an opponent and time pressure. Movement execution relies less on conscious (verbal) control and more on automatic processes. In the third stage, the autonomous stage, the *expert performer* produces his or her movements virtually without spending much thought on it. Movement execution is realized quickly, with high precision, and almost effortlessly. Skilled performance is achieved by automatic processes, allowing the athlete to focus on other important aspects during competition.

Motor learning is associated with (relatively) permanent changes of the *procedural memory*

system. Based on extended deliberate practice, movement experts can rely on distinct memory structures during DM in sports, in which so-called *basic action concepts* (BACs) provide the representational basis for the voluntary control of complex movements. In high-level experts, these representational frameworks are organized in a distinctive hierarchical treelike structure, are remarkably similar between individuals, and are well matched with the functional and biomechanical demands of the task. In comparison, movement representations in novices are organized less hierarchically, are more variable between persons, and are less matched with the task's functional and biomechanical demands. Thus, the quality of any movement performance may strongly depend upon the quality of movement representations in the procedural memory system.

Expertise in a number of game sports, however, may also be reflected by the degree of the athlete's individual *bilateral competence*, which signifies his or her ability to perform particular skills well on both sides of the body. The basketball player who is able to handle the ball with equal proficiency with the dominant and the nondominant hand has an advantage, because he or she can readily adopt the execution of skills to fast changes of play (e.g., while switching from the dominant to the nondominant hand), enabling him or her to flexibly adjust movements to new situations. In contrast, the athlete who is not able to handle the ball with equal proficiency on both sides may not be capable to adjust his or her play sufficiently to new situations and is (therefore) constrained to perform a particular skill with the dominant hand, even if the situation requires use of the nondominant hand. Thus, the generation of optimal solutions for different game situations in basketball (and other game sports) highly depends on the ability to perform specific movement skills with either side of the body, without much decrement in performance when using the nondominant hand. A lack of bilateral competence may hinder the progression to higher levels of competitive play in a number of game sports, harming the professional development of athletes.

Matthias Weigelt and Tino Stöckel

See also Coordination; Errors; Expertise; Motor Commands; Motor Control; Motor Development; Motor Learning; Practice; Timing

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MULTICULTURALISM

Culture refers to the context in which people develop their perspective on, and approach to, life. Numerous cultures exist, representing the multiple geographic regions and racial, class, ethnic, and gender groups across the world. Multiculturalism refers to an approach that integrates these varied perspectives into society and an awareness of how an individual's cultural background may impact personal and professional interactions with others.

In sport and exercise psychology (SEP), until recently, the idea of multiculturalism has received little attention in the literature. However, pioneers in the fields of sport sociology, counseling psychology, and community psychology have highlighted

the importance of cultural variables in research and practice settings. Early work involving multiculturalism and sport focused exclusively on racial or ethnic variables as the indicators of a person's cultural background. These efforts focused on identifying and discussing the relevant social issues within each racial or ethnic subgroup, such as lack of economic resources, limited access to training or employment, restricted opportunities for sport participation, and lower power equality relative to the dominant group.

Much of the multicultural literature in the counseling field has focused on training professionals to become culturally competent providers of services. These localized trainings, often within the context of graduate programs in psychology, provide educational modules or practical experiences aimed at exposing trainees to the cultural practices of specific client populations. This narrow approach to training can be contrasted against the broader goals of multiculturalism within psychology, which instead focuses on awareness of one's own cultural heritage and the diversity of social and political factors within society that can contribute to a person's cultural and personal identity. Many graduate programs have evolved to include both localized and broader approaches in their curricula and applied experiences. Within sport psychology (SP), professionals using this lens in their practice will understand the biases and limitations of their own cultural identity as well as the potential issues affecting their work with clients or within particular settings. Thus, multicultural practices de-emphasize the biological determinants of behavior, while prioritizing environmental factors. Using these guidelines, practitioners may seek out (through interview, assessment, or observation) how each client's cultural identity will impact the therapeutic relationship.

Multicultural competencies are becoming more important every year in a world where information and perspectives can be shared through various technologies with relative ease. The Internet age has dramatically increased our exposure to multiple cultures through access to images and videos, language translation, and news reports from around the world. The many factors that determine one's cultural reference point have contributed to the rise in popularity of phenomenology and other qualitative approaches in SEP research. The expansion in our exposure to, and appreciation of, multiple cultures has led sport psychologists to consider

many additional aspects beyond racial and ethnic background including socioeconomic status (SES), religion, sexual orientation, and gender, among others. The inclusion of different aspects of culture in research methodologies may help the field move forward toward an approach that does not use Western models as the default frame of reference but instead chooses a perspective and lens appropriate to the question at hand.

Culture has become an important frame of reference for designing research projects and applied interventions. From single-case designs to larger-scale randomized trials, evidence has been accumulated in support of the idea that culturally targeted approaches are more effective than standardized interventions. These cultural adaptations can range from simple adjustments in language or terminology to more radical reconceptualizations of popular concepts or methodologies. A simple example of this adaptation is when academic professionals collaborate across borders to translate and reverse translate a research instrument for use in cross-cultural research. Another methodology growing in popularity in health and exercise research is called *community-based participatory research*. This approach to data collection, intervention, and evaluation allows flexibility in the methods and means of data collection so that researchers can adapt to contextual differences present in various cultural regions or among subgroups once the project has started. This flexibility may include changing who delivers the assessments and intervention to include local partners; the language used; or the timing, length, and depth of contact.

Each new generation of students and professionals in SEP is more exposed to, and more educated about, cultures beyond their own. Professionals who desire to conduct ethical research and practice must learn and account for the biases of their own worldview while continuing to ask good questions about the importance of someone else's views. These developments in our field are but one small part of the growing multicultural nature of our global society, but they are reflective of the whole.

Sam J. Zizzi and Matthew P. Martens

See also Cultural Competence; Cultural Safety; Diversity; Ethnicity; Gender; Race; Stereotyping, Cultural/Ethnic/Racial

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MULTIMODAL MENTAL TRAINING

Multimodal mental training, also known as mental skills training or psychological skills training (PST), involves educating athletes, coaches, and exercisers on the effective use and implementation of psychological techniques and skills that are associated with sporting and exercising excellence. Discussed here are the goals of such interventions; common components of multimodal mental training; and phases commonly involved with multimodal mental training's planning, delivery, and evaluation.

Goals of Multimodal Mental Training

Multimodal mental training typically adopts an approach that helps athletes and/or coaches to self-regulate their thoughts, feelings, and behaviors. The normal focus of multimodal mental training is on improving performance and personal

well-being. However, there is growing recognition that multimodal mental training can also improve the development of the person as a whole. This more recent, holistic approach is rooted in the positive youth development movement and posits that the mental techniques and skills learned in sport settings also can be applied to other life domains (e.g., school or work environments). Multimodal mental training therefore allows those responsible for the development of young people to fulfill the moral obligations of this role and equips athletes with the skills to successfully navigate various life transitions that are considered a prerequisite for long-lasting sporting success. Indeed, multimodal mental training can likely play a major role in talent development, whether it is to accelerate learning, sustain intrinsic motivation, or cope with difficulties such as injury or deselection.

Components of Multimodal Mental Training Programs

A number of psychological qualities distinguish more successful performers from their less successful counterparts. These include a high level of self-confidence and positive expectations for success, experiencing appropriate emotions and arousal levels, being totally concentrated and focused on the task at hand, feeling in control and viewing difficult situations as a challenge rather than a threat, having positive thoughts and images about performance, and being determined and committed. These psychological qualities can be regulated as appropriate through mental skills (e.g., building or maintaining confidence, attentional and/or arousal control) and mental techniques (e.g., goal setting, imagery, self-talk, and routines) employed to develop those skills.

Different mental techniques have separately been shown to enhance the mental skills associated with greater performance. Implementing a combination of these within the same program helps ensure that the individual needs and preferences of the targeted group are met. It is likely that athletes and coaches will favor certain techniques over others, and this preference may vary within groups of individuals receiving the same intervention. Within a rugby team, for example, the forwards who participate in the scrum might prefer to focus more on techniques to increase their physical activation, whereas the backs might wish to focus on improving their passing and kicking

techniques by imagining themselves performing these skills well.

For this reason, multimodal mental training components are usually designed to meet the needs of individuals. Therefore, when implementing multimodal mental training, it is important to consider the physical, technical, logistical, and psychological demands of a specific sport and/or player position, as well as the age, competition level, and previous experiences of the athletes. Athletes and coaches with negative attitudes toward sport psychology (SP) and/or who have had poor previous experiences may be less accepting of a multimodal mental training program delivered by a sport psychologist. However, research shows if athletes believe the program has been tailored to their personal needs and preferences, it is more likely they will buy into what is being taught and adhere to recommendations for its use and implementation.

Phases of Multimodal Mental Training Programs

Multimodal mental training programs are typically divided into assessment, education, acquisition, practice, integration, and review phases. In the assessment phase, the aim is to obtain sufficient information about the intervention athlete or group of athletes to determine (a) what are the aims of the program; (b) which techniques and strategies should be taught; (c) how should the techniques and strategies be taught (e.g., length and frequency of sessions) and by whom (e.g., sport psychologist or coach); (d) when should they be taught (e.g., off-season, competitive season); and (e) in what sequence should they be taught.

Next, individuals are taught the techniques and skills (education). This phase involves learning how to use techniques and skills as well as understanding why they influence performance. Resources are often developed to aid this process such as diaries or worksheets, video clips, websites, and online discussion blogs. Also used are reports of well-known athletes and coaches describing their personal use and experience with the techniques and skills being learned.

Opportunities for athletes to become more proficient in the use of these techniques and skills are then provided by employing structured training methods in a nonevaluative environment (acquisition). Once a certain level of competence and comfort with the technique or skill use has been

reached, these are systematically integrated into practice routines and customized to meet individual needs (practice). The next step is to integrate techniques or skills into competitive situations. Finally, it is important to evaluate and update the program to prevent boredom and ensure it continues to be relevant to the group thereby promoting adherence. The review process, although usually ongoing, can be informal, just gaining the impressions of the individuals involved, or formal, whereby objective performance tests, questionnaires, and semistructured interviews are used.

Jennifer Cumming and Sarah E. Williams

See also Goal Setting; Imagery; Modeling; Psychological Skills Training; Self-Talk

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MULTIPLE BEHAVIOR CHANGE

Traditional approaches to helping individuals change health behaviors focus on reflective processes. In other words, in these approaches, the first thing a practitioner might do is identify what a client thinks about a behavior. Thus, initial counseling may focus on increasing positive beliefs and perceived benefits associated with that behavior and challenging less positive beliefs. Support may

turn to building confidence to initiate and then maintain the behavioral change.

Less directive approaches, such as those involved in motivational interviewing, may give the individual greater freedom to identify which behavior to work on, but again, the approach usually focuses on separate behaviors with an emphasis on working through cognitions about that behavior. Given that people often have a cluster of poor lifestyle behaviors, the challenge emerges to identify how best to help clients with changing multiple health behaviors, as shown in Figure 1.

Debate may follow in epidemiology about the risk associated with each behavior, and thus which to focus on first. Also, specialist health promotion counselors may focus on physical activity (PA) while others focus on smoking cessation or diet. Besides being inefficient, trying to change multiple behaviors using multiple but parallel or distinct counseling approaches can lead to cognitive overload and perhaps multiple failure. Yet, so far, theory and practice remain primarily oriented toward understanding and changing single health behaviors.

There is, however, increasing interest in developing new theories and guidance for practitioners to help clients with multiple behavior change, as discussed by James O. Prochaska. Not only does Figure 1 show how different lifestyles may mediate

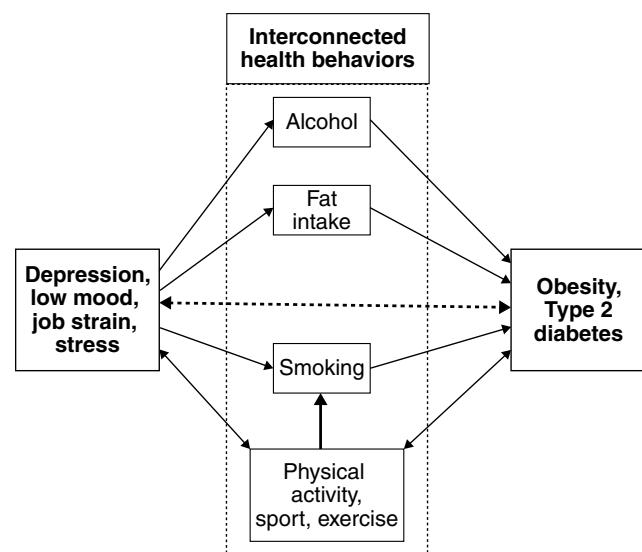


Figure 1 The Mediating Role of Health Behaviors in the Reciprocal Relationship Between Mental and Physical Well-Being

Source: Adrian H. Taylor.

the reciprocal relationship between mental and physical health but it also highlights that there are complex interrelationships between these different behaviors. For example, surveys suggest that more active people tend to also drink less alcohol, eat fewer snacks, and smoke less. The relationships are not always so straightforward: Some types of sports participation are associated with fewer healthy lifestyles than others.

The next question is whether engagement in PA prevents the reduction of healthy lifestyles, and, if it does, how? What changes as a result of playing sport or regularly exercising that discourages smoking, excessive alcohol use, or poor diet? Likewise, do sedentary lifestyles, common in Westernized society, lead to other less healthy lifestyles and why? If we know the answers to these questions, this may help us develop PA interventions that impact on multiple health behaviors. These questions are relevant not only in the context of engaging young people in sport and PA but also for adults who already have clusters of poor lifestyles.

There is evidence that more physically active children and adolescents are less likely to progress to smoking, taking illicit drugs, drinking alcohol excessively, and engaging in antisocial behavior.

However, it is difficult to produce strong causal evidence, as there may be a number of confounding variables. Nevertheless, there may be plausible psychological mechanisms, such as enhanced self-efficacy, self-perceptions, and self-esteem, and enhanced mood from PA. Excessive sedentary behavior in adolescence may lead to deactivation and fatigue, which may lead to turning to high-energy food and other substances to enhance mood.

Much more is known about whether PA (of various doses) causally influences engagement in other behaviors among adults, and especially smoking and snacking. Neuroscience and cognitive psychology have greatly advanced our understanding of addictive processes involved in both these behaviors in particular. Common areas of the brain are activated in anticipation of a reward or reinforcement, when faced with cues or images associated with the behavior. As Figure 2 shows, with the circumplex model of affect at the center, when people are deprived of a substance (regularly used to regulate mood), or feel under pressure (with low mood) or view cues associated with the behavior (e.g., a cigarette pack, a picture of someone else smoking), this generates a state

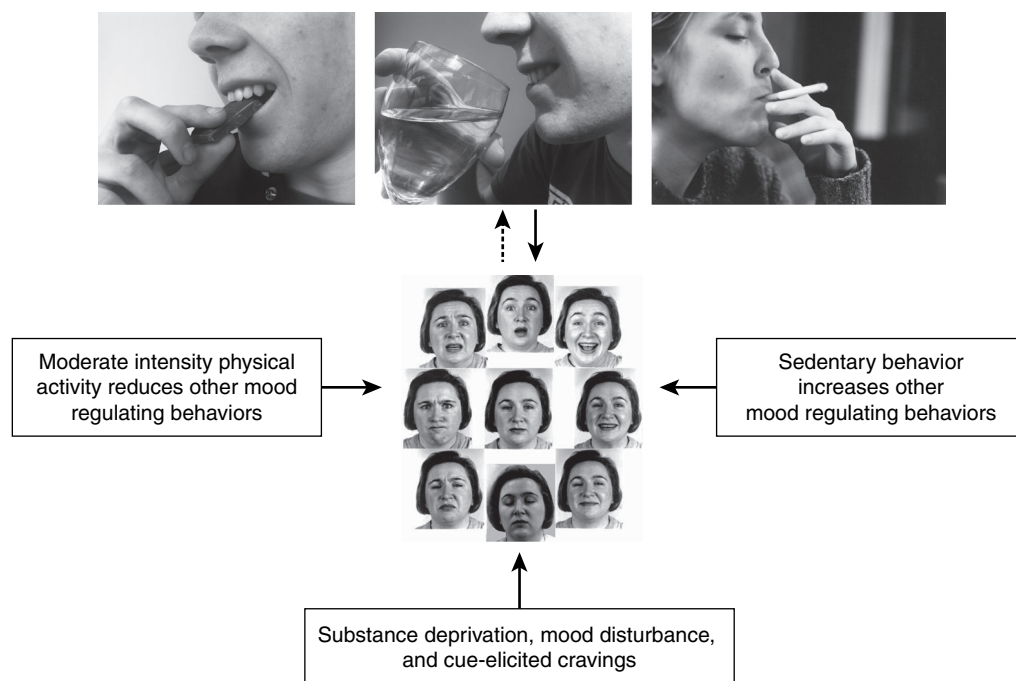


Figure 2 The Potential Role of Physical Activity and Sedentary Behavior in Regulating Hedonic Behaviors Such as Snacking, Consuming Alcohol, and Smoking

Sources: Adrian H. Taylor; center composite is reproduced courtesy of James Russell (<https://www2.bc.edu/james-russell>).

of wanting. By smoking or snacking, a reward is received and a sense of pleasure (on the horizontal axis of the model) or increased activation (on the vertical axis of the model) is experienced.

A great deal of evidence has been accumulating over the past 30 years on the acute and chronic effects of exercise on mood and affect. Briefly, moderate-intensity exercise increases affective activation and pleasure. Recently, work has turned to also understanding the effects on urges, desire, and cravings for various substances such as nicotine and high-energy snack food. Researchers have discovered that even a 15-minute brisk walk (that most people can achieve throughout a typical day) reduces substance cravings, attentional bias to salient images, ad libitum smoking and snacking, during temporary abstinence, under stress and in the presence of salient cues. Brain-imaging studies have shown reduced activation in areas of the brain associated with viewing smoking-related images, after exercise compared with rest. With these studies in mind, new counseling interventions have been developed to encourage smokers to use PA as an aid to reducing and quitting smoking. Given that self-regulation of snacking and smoking and other health behaviors is a challenge, and these behaviors are often driven by automated rather than reflective mental processes, promoting PA appears to have a role in enhancing the self-regulation of multiple behaviors, irrespective of whether there is an intention not to smoke or eat high-energy snack food.

Adrian H. Taylor

See also Addiction, Effects of Exercise on; Affect; Affective Responses to Exercise; Pleasure

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MUSIC-BASED INTERVENTIONS

The effects of music in sport and exercise contexts have been of interest to researchers for over 100 years. Recent advances in digital music technology and the facility for users to formulate their own playlists have dramatically increased the prevalence of music in both the sports training and health club environments. Even at the high altar of international competition, the use of music has become commonplace as athletes seek ever-more effective means of optimizing their mental and physical states. Interventions assume one of three types depending on when the music is delivered in relation to the task: pre-task, in-task, or post-task. Very little research attention has been given to the post-task application of music as a recuperative tool; therefore, this entry will focus on pre-task and in-task music, which represent the mainstay of its use in the sport and exercise domain. The personal and situational factors that influence music selection will also be introduced.

Pre-Task Music

Research findings indicate that music used before a physical task can serve as a potent stimulant. This is especially so when the task itself requires a high degree of psychological and physical arousal; activities such as weight lifting or sprinting that depend on the use of every major muscle group and a short, intense effort are particularly good examples. In concert with the available research evidence, many athletes have used music as a pre-task intervention prior to winning Olympic medals, an example being the swimmer Michael Phelps. Pre-task music may also serve to relax a performer prior to competition as was demonstrated by Olympic gold medalist runner Kelly Holmes at the 2004 Summer Olympics in Athens. Further functions include the use of music as a *trigger* to invoke mental images that are relevant to the task or to emphasize a particular aspect of technique. The

suggestions borne by the lyrical content of music can be especially potent in this regard.

In-Task Music

There is now a wealth of research evidence documenting the effects of music used in accompaniment of various physical tasks. Its principal psychological effects are the enhancement of positive feeling states and arousal and distraction from the unpleasant sensations associated with physical effort and fatigue. It is thought that this distraction effect takes place at a neural level: the nerve signals bearing auditory information (music) taking precedence over those relating to exertion. However, as the lactate threshold is approached, which is, broadly speaking, the intensity at which one can no longer continue to exercise and keep one's breath, the signals relating to effort become stronger than those pertaining to the music and the distraction effect is reduced.

While music may not divert attention at high exercise intensities, it retains the capacity to elevate feeling states, which may in turn alter our interpretation of effort and fatigue. Music also exerts behavioral consequences on exercisers—namely, an increase in self-selected exercise intensity or an extension of voluntary endurance. Research has shown that music is particularly effective as a work enhancer when movement is synchronized to its rhythmic qualities. This may enhance movement efficiency by establishing a more regular or *metronomic* work rate. This *synchronous* use of music is especially applicable to repetitive endurance activities such as running or swimming.

Music Selection

The benefits of music depend on its appropriateness for both the task at hand and the listener in question. In both sport and exercise, the goal generally is to enhance feelings of pleasure and arousal. However, in target sports such as shooting, golf, or archery the aim is typically to sedate rather than to stimulate; therefore, soft or slow music (less than 80 beats per minute [bpm]) is recommended. In terms of heightening arousal, studies have shown that loud music with a fast tempo (over 120 bpm) and pronounced rhythmical features is particularly effective. Indeed, music is most preferred when its tempo is selected with the intensity of the exercise in mind. A range of 125 to 140 bpm would

appear to describe the preference for music tempo across the entire range of exercise intensities (e.g., from walking to fast running). In simple terms, as one works harder, one prefers faster music until the preference ceiling of ~140 bpm is reached. The relationship between preferred tempo and heart rate (HR) as plotted on a graph is not linear but curvilinear in nature (see Figure 1).

Music qualities thought to promote pleasure include the melody (tune) and the series of harmonies used. Harmony refers to the simultaneous sounding of notes, which lends music its emotional “color.” Music may prove particularly effective if it bears associations that the listener finds personally motivating. Such associations may be transferred through popular culture, examples being found in the theme music linked with sporting films or events. To be maximally effective, music should be selected with the age, gender, personality (e.g., introversion vs. extraversion), and sociocultural background of the listener in mind.

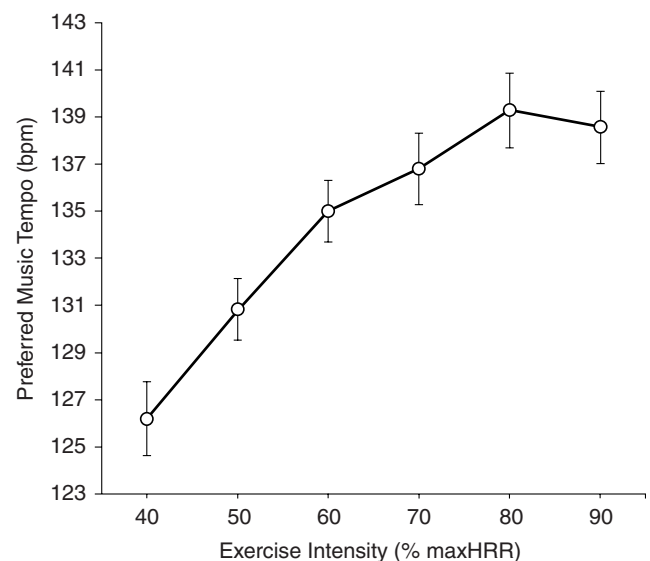


Figure 1 Observed Relationship Between Exercise Heart Rate and Music-Tempo Preference

Source: Reprinted with permission from *Research Quarterly for Exercise and Sport*, 82, 274–284, Copyright © 2011, by the American Alliance for Health, Physical Education, Recreation and Dance, 1900 Association Drive, Reston, VA 20191.

Note: The plotted scores are aggregates of participants' most-preferred tempi at each exercise intensity with standard error bars.

There are now in advance of 100 published studies demonstrating the psychological and work-enhancing effects of music in various sport and exercise contexts. It is entirely possible that carefully selected music may exert longer-term effects such as increased adherence to exercise. Music may exert a compound effect when used in a social environment because it can influence *esprit de corps* in addition to altering the behavior of supporting crowds or group leaders. In the sporting sphere, music has become part and parcel of many athletes' pre-event routines and is a potentially powerful tool for the forward-thinking sport psychologist.

Costas I. Karageorghis

See also Adherence; Affect; Attention Theory; Attentional Association and Dissociation; Preperformance Routines; Psychological Skills

Further Readings

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N

NARCOTIC ANALGESICS

The term *narcotics* is commonly used to refer loosely to a broad range of drugs, from marijuana to cocaine. More accurately, the narcotic drugs consist of opium and its derivatives, also known as opiate drugs. The most common opiate drugs include heroin, morphine, and codeine, and their use leads to a sense of numbness resulting from the suppression of the central and autonomic nervous systems. Because of these properties, narcotics are used as effective painkillers or analgesics. In fact, morphine and heroin were commonly used as over-the-counter pain relievers in the early 1900s, but their widespread use was regulated as soon as the addictive properties of narcotic analgesics were revealed. Today narcotic analgesics are used in the treatment of chronic pain, and their local application is thought to minimize the side effects, such as sedation, respiratory depression, and nausea, produced by other local anesthetics. Systematic and long-term use of narcotic analgesics, however, is not free of undesirable short- and long-term side effects (e.g., cardiovascular disease, opioid-induced hyperalgesia, cognitive dysfunction, immunosuppression, drug addiction) that may offset the benefits of pain relief.

Narcotic Analgesics in Sport

Although there is no scientific consensus about the ergogenic properties of narcotic analgesics, athletes still use these substances for performance enhancement purposes. Narcotic analgesics are

principally used to alleviate pain resulting from musculoskeletal injury and to induce euphoric states. Through pain reduction before or during the competition, and by lowering the pain associated with injury or fatigue, narcotic analgesics may enable sports participation and provide a competitive advantage to overtrained or injured athletes. For instance, athletes may resort to narcotic analgesics in order to recover quickly and cope with the pain caused by injury or fatigue in tournaments requiring continuous participation and effort within a limited time frame, such as the Olympic Games and world championships. Furthermore, given their mood-altering properties, narcotic analgesics can be used to alleviate negative emotions, psychological stress, and performance anxiety; thus, they elicit an indirect performance enhancement effect by improving the athlete's morale. Due to their mood-altering effects, narcotic analgesics may be more common in combat sports and sports requiring fine motor control.

Regulating Narcotic Analgesics

Because narcotic analgesics are highly addictive, harmful to health, and can be used to provide an unfair competitive advantage, their use has been officially regulated by several sporting organizations and authorities, including the International Olympic Committee (IOC), the World Anti-Doping Agency (WADA), and national anti-doping organizations (NADOs). According to the WADA 2012 list of prohibited substances, the following narcotic analgesics are banned:

- Buprenorphine
- Dextromoramide
- Diamorphine (heroin)
- Fentanyl and its derivatives
- Hydromorphone
- Methadone and morphine
- Oxycodone and oxymorphone
- Pentazocine
- Pethidine

Under specific circumstances, WADA may allow the use of prohibited narcotic analgesics for therapeutic use in athletes, but this requires strict implementation of the Therapeutic Use Exemptions (TUEs) scheme and proper medical supervision.

Epidemiology of Narcotic Analgesics

The use of narcotic analgesics is not widespread compared to other pain relievers, such as nonsteroidal anti-inflammatory drugs (NSAIDs), corticosteroids, and beta-agonists. The overall prevalence of narcotic analgesics for therapeutic uses in sports is less than 1%. This is probably due to the fact that their use may yield a positive doping test and because other drugs (NSAIDs, cyclooxygenase-1 [COX-1], cyclooxygenase-2 [COX-2] and related enzymes) can be equally or more effective and elicit analgesia without the reported side effects of narcotics. However, the use of narcotic agents outside competition is an issue of major importance. There is vast anecdotal evidence in daily press about athletes who become addicted to narcotics, and as a result, they cannot train or compete at high level and accordingly withdraw from sports.

Preventing Abuse of Narcotic Analgesics: The Role of Education

Preventing abuse of narcotic analgesics mainly rests with the physician's knowledge and perspective on therapy and treatment of injury and pain. Thus, strategies to prevent the abuse of narcotic analgesics by athletes should target medical and paramedical professions involved in sports (e.g., sports medicine physicians, physiotherapists, athletes, and coaches). Furthermore, strategies should incorporate specialized training courses in medication and nonmedication strategies for pain management, as well as for raising awareness about the risk for drug addiction. The available evidence shows that such educational programs have increased sports physicians' capacity to treat

acute and chronic pain conditions by minimizing the addictive effects of narcotic analgesics, or even without using any medication at all. In addition, the implementation of prescription monitoring and surveillance systems for dispensing and administration of narcotic analgesics would further prevent the abuse and trafficking of narcotic analgesics in sports. Finally, athletes should be also educated on the harmful health and performance side effects of narcotic analgesics.

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See also Alcohol Abuse; Drug Use and Control; Performance-Enhancing Drugs

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NEUROLOGIC DISORDERS

Among the most frequently occurring neurologic diseases in Europe and North America are multiple sclerosis, Parkinson's disease, cerebrovascular diseases, brain and spinal cord trauma, and chronic headache. These neurologic disorders are also the most relevant for treatment with sport therapy, a motion-therapeutic method that compensates and regenerates disturbed bodily, mental, and social functions; prevents secondary damage; and promotes health-oriented behavior. Sport therapy is rooted in a biopsychological approach to the human organism. The methods of sport and movement therapy are based on medical, exercise, educational, psychological, and sociotherapeutic principles.

Historically, the therapy of neurologic disorders was limited by the common view that brain development is confined to childhood and that the adult brain has lost the capacity to develop. Recently, however, the observation of plasticity in the adult brain has shed new light on the potential for sport and exercise in the therapy of neurologic disorders. It is now recognized that not only the strength of synaptic transmission but also the size and activation pattern of cortical networks are activity dependent. These findings provide a scientific basis to elaborate the therapeutic use of sports and exercise in the treatment of neurologic disorders.

This entry provides an overview of the most relevant neurologic disorders that are currently treated with sport and movement therapy. For each disorder, first the medical basics are described, and then the specific methods used in sport and movement therapy.

Multiple Sclerosis

Multiple sclerosis is a demyelinating disease characterized by an autoimmune destruction of the myelin sheaths of nerve fibers. Genetic, environmental, and psychosocial factors have been identified as etiologically relevant. The time course and location(s) of the demyelination process are highly variable both inter- and intraindividually. Thus, patients display a variety of symptoms such as blurred vision or loss of sight, motor weakness, numbness and tingling, vertigo, facial pain, coordination deficits including the speech and ocular muscles, fatigue, impaired control of the bladder, and depression. The course of the disease is unpredictable and can be relapsing–remitting or primarily progressive. Women are affected three times more often than men. The onset of the disease is typically between 20 and 40 years of age.

The fatigue symptom is an important factor to be considered when planning sports therapy for multiple sclerosis patients. The lacking economy of movement in this patient group may have different causes. Important factors are muscular hypertonus, coordination deficits, and fatigue. All three factors may lead to a decreased quality of coordination and thus lead to an additional expenditure of energy and strength. Sensory loss can lead to qualitative and quantitative changes in the subjective body image and this should be considered in the context of sports and exercise therapy. The fact that mental health can be influenced as well by

physical activity (PA), especially in groups, makes this an important approach in sports therapy with multiple sclerosis patients. Since the localization of inflammatory foci in the central nervous system and the course and severity of the disease differ, in practice heterogeneous groups are built. This has to be considered when organizing a group. The division of a wheelchair group and also a walking group could be useful. In addition, the intensity of training should be dosed individually and be modified sufficiently for patients with varying levels of performance. An acute episode is a contraindication for participation in a sports therapy group.

Parkinson's Disease

Parkinson's disease is a degenerative illness characterized by a loss of neurons and a depletion of dopamine. The lack of this neurotransmitter results in alterations of posture and movement. Core symptoms are a resting tremor, a reduction and retardation of body and facial movement, stiffness, postural instability with gait disturbance, loss of dexterity, and a soft and monotonous voice. Further symptoms are depression and vegetative symptoms, such as the dysregulation of blood pressure (BP), body temperature, and bladder control. The prevalence of this disease increases with age from 1% to 2% of the population over the age of 65 years to 3% over the age of 80.

The early use of therapeutic movement measures in Parkinson's disease is an important supplement to medical therapy. To overcome the reduction of body movement (freezing phenomenon), optical, acoustic (music), and tactile stimuli may help initiate the movement. Repetitive training of corrective support reactions can bring a positive effect on stability and contribute to prevent falling. A dosed run treadmill training (duration method) can influence the course rhythm and stride length positively. If patients suffer from orthostatic hypotension, this can be improved by targeted cardiac training. Important elements of sports and exercise therapy are, for example, practicing an upright gait with long steps and contralateral movement of the arms and practicing the changing of direction. These measures are supported for example by a functional everyday life-oriented training.

Cerebrovascular Diseases

Stroke, which is a sudden focal neurologic syndrome, is the most frequent manifestation of a

cerebrovascular disease. Stroke is mainly caused by a lack of blood supply (ischemia) or by cerebral hemorrhage. The occlusion of a major cerebral artery often results from an embolus, typically from the heart. Small vessels are affected by diabetes mellitus, high BP, or cigarette smoking. The symptoms strongly depend on which artery is occluded. Frequent symptoms are a paresis and numbness of one half of the body; impairments in language, praxis (conceptualization and execution of movements), spatial cognition or attention; a visual field defect, vertigo, or diplopia. Similar symptoms are found in cerebral hemorrhage. The latter, however, is often accompanied by an acute and severe headache.

In sports and exercise therapy with stroke patients, improving of strength and coordination, promoting local and general dynamic endurance, and training of sensorimotor function and perception build the focus in addition to the general objective and content of movement therapy. If spasticity (muscular hypertonus) develops in paralyzed limbs, a regular PA can help to reduce it temporarily and to develop strategies to counteract spastic patterns. Beyond a well-dosed motor skill training, coping with cognitive tasks is also part of the individual sports and exercise therapy. In summary, suitable exercises for stroke patients are disciplines with manageable coordinative requirements that promote aerobic endurance. These objectives can be realized by the active stay in the water, provided that the patient's ability was tested and secondary diseases are excluded. In the water, a commodification of cardiovascular function can be achieved and the water density facilitates the movements. A practical example is aqua-jogging or aqua-walking.

Brain and Spinal Cord Trauma

About half of brain trauma cases are due to traffic accidents and about one quarter each to falls and accidents during sports, leisure, and work. If present, the duration of the loss of consciousness is prognostically relevant. A wide range of symptoms may occur and persist depending on the location of the lesion, including more subtle symptoms such as nervous instability. The incidence of brain trauma is highest between the ages of 15 and 24, as well as above the age of 75. Men are twice more likely to suffer brain trauma than women are.

About half of all spinal cord trauma cases are due to traffic accidents; one fifth are due to accidents at work. The higher the lesion is in the spinal cord, the more extensive are the resulting impairments. Lesions at the level of the neck result in a paresis and a loss of sensibility of the arms and legs, an impairment in the control of bladder, as well as disturbances in the regulation of vegetative functions such as BP, heart rate (HR), and body temperature regulation. Lesions at lower levels result in a paresis and sensory loss of the legs and an impairment in the control of bladder. Sixty-one percent of spinal cord traumas occur between the ages of 16 and 30, with a male-to-female ratio of 5:1.

The intention of sports and exercise therapy is to stabilize, compensate, and improve the functions affected by the brain damage. This resource-oriented therapy involves the improvement of coordination, since a common symptom in brain trauma is residual ataxia. The stand-gait coordination and eye-limb coordination can be trained by simple and complex motor actions. The development of compensation strategies, increasing overall capacity and improving the strength of the locomotor apparatus should be further goals in exercise therapy with brain injury patients. Improving the quality of the motor task can be achieved by regular practice.

The aim of exercise therapy concerning motor skills of paraplegic patients includes the compensation of muscle function in particular of the upper extremities. Sports therapy with tetraparetic patients often uses hand biking to prevent cardiovascular diseases. The maximum oxygen uptake and aerobic capacity can be improved by circuit training (interval method), for example, to economize the cardiovascular capacity. Other focal points of sports therapy with paraplegic patients are the training of sitting balance, weight-bearing exercises, and the rehearsal of everyday movements in order to obtain and provide the greatest possible independence of auxiliary persons and material. In addition to its therapeutic function, sports therapy has a major psychosocial significance for paraplegic patients, bringing into play the elementary skills and abilities that demonstrate to wheelchair users the activities necessary for practice and provide them with sufficient competence to deal with everyday situations. As a sport, for example, wheelchair basketball, wheelchair rugby, or table tennis could be considered. The

changing game situations require quick decisions and accurate movements. This leads to increased concentration of mental and physical abilities and contributes to the improvement of balance. In addition to the previously mentioned examples, there are also many opportunities for athletes with disabilities to participate with people without disabilities (integrative sport).

Headache

Of the various types of headache, tension headache and migraine are epidemiologically most relevant.

Chronic tension headache has no structural correlate. Pain from chronic tension headaches is usually bilateral, sometimes with a frontal or occipital accent. The pain is most often described as dull, with tightness or pressure. The intensity of the pain is mild to moderate and reportedly does not interfere with daily activities. Sports and exercise may diminish an ongoing tension headache. Approximately 2% to 3% of the population suffers from chronic tension headache, with men and women being equally affected.

Migraine headache is thought to be the consequence of disturbances of the trigeminal system. The genetic disposition is well established. Migraine headaches take the form of attacks that may last from 4 to 72 hours. In most of the patients, the headache is unilateral and pulsatile in character. The intensity is moderate to severe and increases with exercise. The headache is accompanied by nausea; vomiting; and sensitivity to light, noise, and smell. The headache attack may be preceded by a temporary focal neurologic deficit, the aura, which can manifest as a disturbance of vision consisting of flashes of light, formations of zigzag lines, or an enlarging blind spot; focal numbness or weakness; or mild disturbances of language production. The aura lasts from 5 to 20 minutes and ends with the onset of the headache. Incidence is twice as high in women as it is in men, with respective population-based rates of 14% and 7%. Migraine episodes typically begin in adolescence or in early adult life.

The aim of sports therapy in headache is to improve the subjective well-being of patients. Cardiovascular endurance training can reduce migraine and tension headache symptoms. It is required to increase the cardiopulmonary performance—that is, the exercise intensity should be in the range of 60% to 75% of maximum HR

and be regularly (two to three times per week) performed with an effective training time of 30 minutes per session. In line with sports therapy, also endurance sports, especially jogging, has a relaxing effect on headache patients (reduction of stress level).

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See also Motor Development; Music-Based Interventions; Neuroscience, Exercise, and Cognitive Function

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NEUROSCIENCE, EXERCISE, AND COGNITIVE FUNCTION

In the fields of neuroscience and cognitive science, human cognition is broadly defined as a component of brain function that includes information processing, memory, attention, perception, language, and executive function related to decision making (DM) and the initiation or inhibition of behavior. In the context of sport and exercise psychology, researchers have been interested in the possible benefits of increased leisure-time physical activity (PA) and the performance of acute and chronic exercise on various aspects of cognitive function, from infants to older adults. The focus of this entry is human older adult cognitive function. The effects of exercise training and leisure-time PA on magnetic resonance imaging (MRI) derived measures of brain structure and function related to cognitive performance among healthy older adults and those who are at increased risk for cognitive decline and dementia are also discussed.

Acute and Chronic Exercise in Healthy Adults

The performance of a single session of exercise results in improved cognitive performance in healthy younger adults. However, the type, duration, and intensity of the exercise are important, as well as the timing of the cognitive performance during the exercise and after the exercise has ended. In

addition, some types of cognitive function improve more than others. During exercise, impairments in cognitive function can occur for complex cognitive tasks, especially during exercise that also may be more cognitively demanding, for example, running compared to stationary cycling. However, evidence for cognitive improvement during exercise may occur for simple tasks that involve rapid DM or require a fast reaction time (RT), and in tasks that are very well learned and, thus, can be performed without much thought or planning. After exercise has ended, improvements in cognitive performance, especially tasks that involve information processing, memory encoding, and memory retrieval, occur over the first 15 minutes and then dissipate thereafter. It is presumed that the heightened physiological arousal during the recovery period contributes to these effects. It is unknown if these effects of acute exercise also occur in healthy older adults or those with cognitive impairments.

Exercise, Physical Activity, and Older Adult Cognition

Among healthy older adults, greater levels of cardiorespiratory fitness and periods of exercise training (other than acute bouts) are associated with better cognitive function. These effects are largest for cognitive tasks that involve information processing and executive control, such as attention and performance during a dual task, efficient switching between different types of tasks, DM, and response inhibition.

There is some evidence that beneficial effects of exercise on cognitive function are stronger in those who are at genetic risk for Alzheimer's disease, the most common cause of dementia. Apolipoprotein E (APOE) allele status is related to risk for Alzheimer's disease (as well as cardiovascular disease) through its handling of cholesterol. Possession of one or two copies of the apolipoprotein-epsilon4 (APOE-e4) allele increases the risk of future cognitive decline up to 10 times compared to noncarriers of the APOE-e4 allele. However, engaging in moderate levels of leisure time PA substantially reduces the risk of future cognitive decline in APOE-e4 allele carriers, equal to the risk for noncarriers (the other variants being the most common e3 allele, and the protective and less common e2 allele).

Very little information exists regarding whether these beneficial effects extend to those with existing

cognitive impairment. Individuals diagnosed with a very early stage of Alzheimer's disease, termed *mild cognitive impairment*, may benefit from exercise training in their ability to perform a semantic fluency task on a day they did not exercise (e.g., naming as many animals or fruits as possible in 30 seconds) but not in tasks that involve episodic memory (e.g., learning a list of words and then being able to later, after doing other tasks, recall that list without any reminders). However, these effects have not been replicated or shown in large samples.

Magnetic Resonance Imaging to Measure Brain Function and Brain Structure

MRI is a tool that can be used in research to assess brain function and brain structure. There are multiple modalities of MRI that can be used (termed *multimodal MR imaging*) to assess differences between groups of individuals or the effects of interventions, such as exercise, on brain function. Functional magnetic resonance imaging (fMRI) is the most commonly used modality and depends on the blood-oxygen-level-dependent (BOLD) signal—derived from differences in oxygenated and deoxygenated hemoglobin presumed to reflect oxidative metabolism in nerve cells. The BOLD signal is an indirect, but validated, estimate of relative neuronal activation. The absolute rate of cerebral blood flow can be measured using a MRI technique called arterial spin labeling, and using radioactive contrast agents (e.g., gadolinium) cerebral blood volume can be estimated. Structural information about the brain, such as the volume of gray matter, white matter, and cerebrospinal fluid compartments of the cerebrum, can also be obtained using MRI. The structural integrity of brain white matter fiber tracts can be measured using a MRI technique called diffusion tensor imaging (also diffusion-weighted imaging), which assesses the diffusion characteristics of water molecules, which in a healthy person are constrained to diffuse along the boundaries of the intact myelinated white matter fiber bundle. Finally, MR spectroscopy can be used to measure the concentrations of certain metabolites or markers of neurotransmitter function in a single voxel (small three-dimensional cubes of brain tissue). The use of MRI in the context of sport and exercise psychology is appealing; however, caution is warranted as very little is known about the physiological effects of exercise

on fundamental MRI signals that may occur independently from, but could appear as, alterations in neuronal firing or cerebral blood flow.

Effects of Physical Activity and Exercise on Multimodal Magnetic Resonance Imaging Outcomes

After robust growth in synaptic connections and brain volume during maturational development, the volume of the brain gradually decreases from roughly the age of 30 years until death. This decline in brain volume contributes to normal age-related cognitive decline, but brain atrophy is accelerated in neurodegenerative diseases such as Alzheimer's disease, Parkinson's disease, and Huntington's disease. There is accumulating evidence that modifiable lifestyle behaviors, such as PA, may help to preserve brain tissue and promote a neural or cognitive reserve, and perhaps the maintenance of cognitive abilities, into old age.

Using MRI to examine the structural volume of the brain, several studies have demonstrated the benefits of PA and cardiorespiratory fitness as a method to preserve brain volume. The greatest effects of exercise on brain volume have been shown in the hippocampus, a medial temporal lobe brain region critical to all learning and memory processes; a region that is an early first target of Alzheimer's disease neuropathology. People who self-report being more physically active, as well as people who possess greater cardiorespiratory fitness, tend to have greater brain volume in several additional brain regions, including the parietal, frontal, prefrontal, and subgenual cortices. Importantly, exercise training may result in an increase in the volume of the hippocampus in healthy older adults. This exercise training effect appears to be stronger in the anterior portion of the hippocampus, which is known to show more severe atrophy in Alzheimer's disease. Among healthy, cognitively intact, older adults who are at increased genetic risk for Alzheimer's disease, greater levels of self-reported PA reduces the risk of future cognitive decline and helps to preserve (but not increase) hippocampal volume over 18 months. However, it is not known if leisure time PA will lead to reduced rates of Alzheimer's disease diagnosis. In patients previously diagnosed with early stage Alzheimer's disease, greater cardiorespiratory fitness is associated with greater brain volume. However, it is currently unknown

if exercise training will help preserve brain tissue volume over time in older adults diagnosed with Alzheimer's disease, or if these effects translate into a slowing of disease progression.

Evidence from fMRI experiments suggests that PA and cardiorespiratory fitness are associated with enhanced patterns of neural activation during executive control and semantic memory tasks. For example, in one study, healthy older adults completed a flanker task during the scan, which consisted of indicating the direction a central target arrow was pointed among flanking arrows that were congruent ($> > > >$) or incongruent ($> > < >$) with the direction of the target. This task involves attention and visual information processing as well as inhibition of motor responses during the more difficult incongruent condition. Older adults who had greater cardiorespiratory fitness and others who had completed a 6-month walking exercise intervention (compared to the less fit and the stretching exercise controls groups, respectively) showed greater activation in areas involved in executive control, including the right middle frontal gyrus and superior parietal lobule, and lesser activation in the anterior cingulate cortex, a region activated in response to unexpected conflict and adaptations to attentional control processes. In another study, older adults completed a famous name discrimination task. In this task, the participant makes a right index finger button press to indicate the name is famous (e.g., Frank Sinatra) and a right middle finger button press to indicate the name is not famous (e.g., Rebecca Hall). Older adults, even those with cognitive impairment, perform the task very well with about 90% accuracy. Only correct trials are included in the analysis in order to remove activation related to errors in memory performance. In the analysis of the brain activation response, a "famous" minus "unfamiliar" metric is calculated in order to remove brain activation related to the common sensory and motor aspects of the two name conditions, thus providing a measure of activation related to semantic memory retrieval. Greater levels of self-reported PA were associated with a greater spatial extent and a greater intensity of neural activation in several brain regions involved in semantic memory. Furthermore, these effects were much greater in the more physically active participants who possessed a genetic risk for Alzheimer's disease with the APOE-e4 allele. Larger effects of PA on brain

amyloid levels, measured using positron emission tomography, have also been reported in APOE-e4 allele carriers. Accumulation of amyloid plaque in the brain is a hallmark feature of Alzheimer's disease, and the early accumulation of brain amyloid is greater in APOE-e4 allele carriers, even prior to any symptoms of memory loss. Physically active APOE-e4 allele carriers showed substantially lower brain amyloid than those who were less physically active, levels that were similar to those who did not possess the genetic risk factor. Thus, exercise and PA may help preserve brain volume in regions involved in memory and executive function and may help preserve the ability to activate these regions to perform cognitive tasks. These effects are hypothesized to provide a cognitive or neural reserve that may provide protection against potential neural insults or neuropathological processes. Despite a genetic disposition to develop Alzheimer's disease in APOE-e4 allele carriers, PA may promote cognitive resilience and the ability to maintain intact cognitive function and functional independence with age.

Exercise-Induced Angiogenesis and Neurogenesis

The possible neurophysiological mechanism(s) for the effects of PA on brain function have been well characterized using animal models. The most well-known finding is that exercise induces neurogenesis, the birth and growth of new nerve cells, in the hippocampus. Exercise in rodents produces increases in brain-derived neurotrophic factor (BDNF) and BDNF messenger RNA in the hippocampus and dentate gyrus. These exercise-induced neurotrophic effects in the hippocampus are hypothesized to contribute to the mnemonic benefits of exercise on memory. An important concomitant of neurogenesis is angiogenesis, the birth and growth of new blood vessels or capillaries, and exercise has been shown to induce angiogenesis in rodent motor cortex. The neurogenic and angiogenic effects of exercise are coupled and likely combine to affect cognitive function. In younger healthy adults, exercise training led increased cerebral blood volume in the dentate gyrus, an effect also observed in mice along with markers of hippocampal neurogenesis. Importantly, these effects are related to improved memory performance.

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See also Brain; Brain Imaging; Cognitive Function; Memory; Neurologic Disorders

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NORMS

In a broad sense, *norms* reflect inferences about accepted, appropriate, valued, and/or desirable behavior. Norms differ from rules and laws in that they are implicit (as opposed to explicit) in nature, and normative influences have been studied since at least the late 19th century in the fields

of social psychology and sociology. More recently, these processes have been integrated as fundamental components within prominent group dynamics and health behavior models (see, e.g., the entry Theory of Planned Behavior, this volume), and in the contemporary literature, norms are often categorized on the basis of whether they are injunctive or descriptive in nature. *Injunctive* norms refer to one's appraisals about what significant others think one should do (e.g., my parents think that I should exercise), whereas *descriptive* norms relate to one's appraisals about what significant others actually do (e.g., my parents exercise a lot). As a method for guiding our intentions and behavior across a variety of settings, context-specific norms exist in relation to various independent (e.g., exercise, diet) and interpersonal (e.g., sports teams, coach-athlete relationships) activities. We might, for instance, feel that our parents place great importance on us making healthy dietary choices, that it is necessary to be punctual for meetings with our coach, or that members of our sports team are expected to exert maximum effort during training drills. At their most global level, injunctive and descriptive cultural norms (e.g., regarding eye contact, greeting rituals) also provide a framework for what is considered acceptable (or not) for diverse groups of people across the globe.

In sport and exercise, norms have been investigated in individual, relationship, and team-based contexts. With respect to individual pursuits, perhaps the most widely studied normative concept is that of *subjective norms*. Within the theory of planned behavior, subjective norms reflect an injunctive perception that develops as a function of (a) individuals' inferences about the extent to which others think they should perform a behavior (e.g., my friends think it's really important to exercise) and (b) the degree to which they are motivated to comply with those inferred views (e.g., I don't care much about what my friends think I should do). In conjunction with attitudinal and control perceptions, subjective norms have been shown to be important in predicting individuals' intentions to perform a given behavior. A substantial body of literature has demonstrated that individuals may hold elevated exercise intentions when they believe that their friends and family think they should exercise regularly and when they are motivated to conform to those opinions. Alongside subjective norms, there is also evidence that descriptive norms support behavioral intentions in exercise

and health settings. For example, it is not uncommon for adolescents to seek to engage in desirable (e.g., exercise) or undesirable (e.g., smoking) health behaviors due to the notion that "everyone else seems to be doing it."

In relational and team contexts, norms are common in dictating interpersonal behavior and team harmony. For example, in order to limit conflict and promote relationship maintenance in one-to-one interactions, coaches and athletes may develop norms regarding what they consider to be desirable communication strategies, rewards or incentives, and practice behavior, among other things. Similarly, in larger interdependent networks, such as sports teams, *group norms* emerge that help provide consensus about the behaviors that are (and are not) considered appropriate and desirable. Group norms develop through the interactions and reinforcement processes that exist within a team, and they serve an important function in terms of informing group members about behavioral standards, as well as unifying the group around a common set of principles. Research has shown that group norms develop regarding both task (e.g., at competition and practice) and social (e.g., at social events) aspects of behavior. For instance, group members have often been shown to endorse productivity norms, which state that all players are expected to display a strong work ethic during practice and competition, and to maintain their fitness and physical condition during the off-season. Importantly, these perceived group productivity norms can facilitate team member effort and bolster overall team effectiveness. It has also been shown that teams develop implicit standards around various other issues that include, but are not limited to, appearance (e.g., we all stick to the dress code), punctuality (e.g., players must be on time for all team meetings), attendance at practice and social events (e.g., it is expected that all team members attend all team activities), and attitudes toward social events (e.g., we don't embarrass our team during outings). Group dynamics research in sport has documented that adherence to group norms is at its greatest when individuals feel they are part of a highly cohesive (i.e., united) team. However, when an individual transgresses and violates a group norm, sanctions are likely to be placed on that person that help further reinforce the nature and importance of these consensus standards. For example, if a player violates a norm for fair play by committing an aggressive foul and

harming a teammate during practice, she or he may receive criticism from the rest of the team and in extreme cases may be dropped (temporarily or permanently) from the team's roster.

Another type of normative belief that has been studied in relation to competitive behavior within team sports is the concept of *moral norms*. Moral norms reflect players' perceptions of the behaviors that others, such as coaches and teammates, view as either moral or immoral and have been shown to align with important aggression-related outcomes. For instance, when coaches place a strong emphasis on winning (rather than on personal or team mastery), players are more likely to endorse a norm that supports the acceptability of aggression and seek to injure an opponent, as well as show reduced respect for rules and officials. In sum, injunctive and descriptive norms not only help shape our intentions and behavior in sport

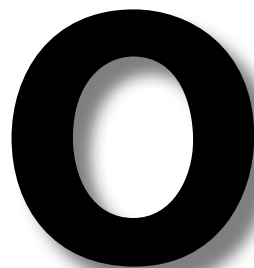
and exercise but they also provide an important implicit scaffold that guides our interactions in relationship and team settings.

Ben Jackson

See also Conformity; Group Characteristics; Roles; Team Building; Theory of Planned Behavior

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OBESITY

Obesity is commonly defined as a body mass index of 30 kg/m² or higher. Increasing rates of obesity in many countries represent a great challenge for public health. In the United States, for example, the rate of adult obesity exceeds 35%. Obesity is associated with increased risk of premature mortality resulting from chronic diseases (e.g., diabetes, cardiovascular and cerebrovascular disease, certain cancers), increased disability (e.g., knee osteoarthritis, chronic back pain, fatigue), and increased health care costs.

The etiology of obesity is multifactorial, comprising both genetic and epigenetic factors. While an increasing number of putative risk factors continually emerge (including such novel contributors as sleep deprivation and stress), the “big two” causal factors are diet and physical inactivity.

Obesity is of special interest to exercise professionals because obese individuals exhibit lower rates of physical activity (PA) participation and higher rates of nonadherence and drop-out than their normal-weight counterparts. However, the reasons behind this phenomenon remain unexplored, severely limiting the intervention options that are available to practitioners.

The Challenge

While most scientific organizations and public health agencies currently recommend a minimum of 30 minutes of moderate-intensity PA on 5 days per

week for health maintenance in the general population, the recommended dose for weight management is more demanding. Specifically, most organizations recommend at least 60 minutes of activity per day to prevent weight gain and at least 60 to 90 minutes per day for the avoidance of weight regain in individuals who were previously overweight or obese.

Nearly all obese individuals report that they would like to weigh less (90% of men, 96% of women). Approximately two thirds report doing something in an effort to lose weight. Even of those obese individuals who say that they are doing something to lose weight, however, the percentage of them who report engaging in PA (approximately 61% of men, 56% of women) is lower than that in the general population.

The magnitude of the problem becomes more clearly evident when one examines how much PA obese individuals do. The percentage of those who do not meet the minimum recommended amount of PA for health approaches 80%. In 2000, according to Behavioral Risk Factor Surveillance System data, 21.3% of obese men and 15.9% of obese women trying to lose weight reported combining eating fewer calories with at least 150 minutes of PA per week. Only 6% of men and 3% of women reported combining eating fewer calories with at least 420 minutes of PA per week (i.e., 60 minutes per day on average). When activity is measured objectively via accelerometers, the percentage is even lower. In a 2009 study of 1,297 obese participants, which was conducted by Deborah Rohm Young and collaborators, only 1.7% were found to be active at least at a moderate intensity for

at least 60 minutes per day. On average, the participants in this sample did only 12.8 minutes of moderate and 2.0 minutes of vigorous PA per day.

Reverse Causation

When considering the relationship between physical inactivity and obesity, most researchers and practitioners assume that causation is unidirectional: People who do less activity become obese. However, longitudinal studies from several countries have shown that, whereas the inactivity–obesity link is usually weak or nonexistent, the obesity–inactivity link is always robust.

This finding has several implications. First, it underscores the importance of efforts to address the growing problem of childhood obesity because it suggests that, once obesity develops, it becomes an important barrier to PA participation. Second, this finding shows that obese individuals face unique challenges in their efforts to become and remain physically active, beyond those faced by their normal-weight and even overweight counterparts.

Psychosocial Impact

Extensive evidence indicates that obesity is associated with significant declines in quality of life (QOL) and a range of mental health problems. Among obese youth, self-reported QOL has been found to be comparable to that of children and adolescents diagnosed with cancer. Shame and social isolation are presumed to be significant mediators for this effect. Among adults, epidemiological data indicate that obesity is associated with a 25% increase in the risk for mood and anxiety disorders. Longitudinal studies, such as the British Whitehall II study, have also shown that, while obesity predicts mental health problems, the case is at least as strong for the reverse order: Individuals with a higher number of mental health problems have a higher risk of becoming obese over time.

Possible Reasons for Physical Activity Avoidance

The reasons for the extremely low rates of PA participation among obese individuals are not fully understood. Based on preliminary evidence, the hypotheses that have emerged include the following. First, possibly due to media messages, many obese individuals begin activity regimens with unrealistically high weight loss expectations. When these expectations are violated, frustration ensues.

Second, a high number of obese individuals experience self-presentational concerns in the context of PA, resulting in social physique anxiety (SPA). Under the influence of this anxiety, several environmental cues, such as the presence of others or wall-to-wall mirrors, are interpreted as threatening. Third, perhaps as a result of prior negative experiences, a high percentage of obese individuals report negative or dysfunctional attitudes toward PA. Approximately 25% report feeling “too fat to exercise.” In turn, this perceived barrier is associated with feeling too shy or embarrassed and having a poor exercise identity. Fourth, several studies have found evidence of both explicit and implicit antifat bias among exercise, nutrition, and medical professionals, as well as obesity specialists. For example, exercise science students, on average, endorse statements such as “Fat people are physically unattractive” and “There is no excuse for being fat.” Fifth, a growing number of studies show that physiological and biomechanical problems associated with obesity contribute to making the exercise experience less pleasant. It has been found, for instance, that obese individuals experience more pain (possibly due to heavier impacts on knees and lower back), higher perceived exertion, more dyspnea, and more uncomfortable thermal sensations during exercise.

Panteleimon Ekkekakis

See also Affective Responses to Exercise; Body Dissatisfaction; Body Self-Esteem; Social Physique Anxiety

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OPTIMISM

Optimism is an expectation for positive or desirable outcomes to occur. Viewed by some as an inherent and evolutionarily adaptive aspect of human biology, it has been examined by psychologists both as a relatively stable dispositional trait ("big optimism") and as a less stable, situation-specific state of mind ("little optimism"). In its dispositional form, optimism has numerous similarities with the psychological constructs of explanatory style and hope. As a situation-specific variable, it has similarities with the constructs of confidence and self-efficacy. In both forms, optimism is believed to be closely linked to positive emotions and approach behavior.

Optimism and Health

Numerous studies have examined optimism as a correlate of mental and physical health, and there is strong evidence for its wide-ranging benefits. From a mental health perspective, optimism is associated with lower daily hassle scores and lower levels of perceived stress. Studies across a wide range of populations (e.g., students, working professionals, victims of violence, medical patients, caregivers) have shown that optimism is also associated with lower levels of anxiety and depression. In addition, greater perception of control, fewer mood disturbances, less loneliness, higher levels of life satisfaction, and higher levels of self-esteem have been observed as correlates of optimism. Finally, optimism also appears to be associated with lower levels of chronic anger and less suppression of anger when it does occur.

From a physical health perspective, optimism shows positive associations with both self-reported and objectively assessed outcomes, and the strength of these associations is similar for healthy cohorts and patient cohorts. For example, healthier lifestyle habits, fewer physical symptoms, better physical functioning, and lower levels of pain have been documented among individuals high in optimism. Physiological markers of health (e.g., blood pressure [BP], blood glucose level) and immune

system functioning also appear to be better among individuals high in optimism. Among cardiac patients, optimism has been associated with fewer postsurgical complications, reductions in postsurgical angina, better ratings of physical recovery by hospital staff, decreases in coronary risk scores over time, and fewer readmissions to the hospital. Positive consequences of optimism have also been observed among cancer patients for outcomes such as physical symptomatology, physical functioning, and cancer antigens. Studies that have examined optimism in relation to survival or mortality have generally shown small but significant relationships between the two constructs.

Optimism in Sport and Exercise

Studies in the domains of sport and exercise have also documented a number of benefits associated with optimism. Interviews with Olympic athletes, their coaches, and family members suggest that these individuals are characterized by high levels of dispositional optimism and hope. Optimism has also been shown to correlate positively with measures of mental toughness among athletes. Similar to findings in other domains, athletes high in optimism report greater use of approach-oriented coping strategies (e.g., logical analysis, increased effort) and less use of avoidance-oriented coping strategies (e.g., disengagement) when facing difficulties. They have also been shown to exhibit greater persistence following unsuccessful performances and to experience higher levels of positive affect following successful performances. In addition, optimistic athletes appear to have lower levels of perceived stress as well as lower burnout risk scores.

Exercise-related studies have supported the distinction between big (dispositional) optimism and little (activity-specific) optimism, and they have shown that these constructs are differentially related to exercise cognitions. Specifically, big optimism appears to correlate positively with longer-term thinking about exercise (e.g., planning and intentions), while little optimism appears to correlate positively with shorter-term thinking (e.g., decision making [DM]). At a more global level, there is evidence that high levels of optimism are associated with high levels of physical activity (PA), while lower levels of optimism are associated with lower levels of PA. This positive relationship between optimism and PA levels has been observed in samples of young adults as well as samples of elderly adults.

Conclusion

Optimism is defined by positive outcome expectancies, and it is reliably linked to positive affect and approach behavior. Numerous mental health correlates of optimism have been identified, including lower levels of perceived stress, anxiety, and depression. Physical health correlates have also been identified, and they include healthier lifestyle habits, better immune system functioning, and more positive reactions to trauma and disease. Among athletes, optimism is associated with approach-oriented coping, mental toughness, persistence following failure, and reduced burnout risk. Within the wider population, it is associated with exercise-related planning, exercise-related DM, and higher levels of PA. Together, these associations suggest that optimism is an important correlate of cognitions, emotions, behavior, and performance.

J. Robert Grove

See also Attribution Theory; Positive Thinking; Resilience; Self-Doubt; Self-Efficacy

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performances. K. Anders Ericsson suggested that athletes must practice for 10,000 hours or 10 years to become experts in an activity. However, extensive training holds many pitfalls if athletes practice too much, without appropriate recovery. Recovery demands increase with intensified training. Consequently, the overtraining syndrome can be judged as the result of too much training and stress with insufficient recovery periods. The overtraining syndrome develops over time and occurs when athletes are unable to refill their energy stores adequately and continue to practice in a tired state. Being in a recovered condition is necessary to show top performances and to achieve training effects. The risk of overtraining is particularly salient for athletes who compete in numerous competitions per season. Feelings of tiredness and being drained may arise, which finally may result in the overtraining syndrome.

There are many examples of overtraining syndrome in which elite performers prepare intensively for important challenges but leave the competition with disappointing results. Aside from other factors, underperformances may be due to an imbalance between physical and/or mental demands and the lack of adequate recovery. For example, the number of tournaments in tennis has increased in recent years, as has the pressure to play tournaments to win points for the annual ranking. Consequently, during the 2011 US Open, favorites appeared to be in poor shape, seeming to be underrecovered. “Out of the 14 players, 11 [had] musculo-skeletal problems, and all [appeared] to be because of overload” (Bricker, 2011).

The relevance, impact, and consequences of overtraining have been addressed in many publications. The most frequent causes of overtraining cited by athletes are (a) too much stress and pressure, (b) too much practice and physical training, (c) physical exhaustion and allover soreness, (d) boredom because of too much repetition, and (e) poor rest or lack of proper sleep. Underrecovery (insufficient and/or lack of recovery time) between training sessions is the main cause of the overtraining syndrome and can be characterized as a precondition. Factors such as nutrition, sleep deficit, sickness, travel, and competitions increase the negative effect of insufficient recovery.

In 2013, a joint consensus statement of the European College of Sport Science and the American College of Sports Medicine regarding

OVERTRAINING SYNDROME

The statement “No sweet without sweat!” is well known by athletes of any age. It expresses the time and cost athletes have to invest to achieve top-class

the prevention, diagnosis, and treatment of the overtraining syndrome was published, summarizing the challenges of training to increase performance. The publication differentiated among training, functional and nonfunctional overreaching, and the overtraining syndrome. The reward of balanced training and regeneration is an increase in performance. Training may include functional overreaching, which is characterized by an intense period of exercise with a short-term recovery. Negative psychological symptoms should not arise when athletes experience short-term performance decrements as a result of functional overreaching. However, problems may arise when overreaching is driven inappropriately or is extended for too long, which may lead to harmful, nonfunctional overreaching. Achieving overload for successful training necessitates being aware of the nonfunctional combination of excessive overload plus inadequate recovery.

The difficulty of distinguishing nonfunctional overreaching and the overtraining syndrome is caused by similar physical and mental symptoms of affected athletes. Additional adaptations of several biological, neurochemical, and hormonal regulation mechanisms may be used to identify the overtraining syndrome and the prolonged maladaptation of the athlete. Typical symptoms of the overtraining syndrome, such as fatigue, performance decline, and mood disturbances, are more severe than those of nonfunctional overreaching. Therefore, training load (e.g., stress–recovery relationship, training volume, intensity, methods, technique training, frequency of competitions) must be carefully controlled to increase performance without incurring nonfunctional overreaching and overtraining syndrome.

In addition to training load, nontraining stress factors have a huge impact on the athlete as well. Performance abilities are influenced by many factors, such as state of health (e.g., cold, fever, gastric and intestinal diseases, infections), lifestyle (e.g., sleep, daily schedule, nutrition, alcohol consumption and smoking, housing conditions, leisure activities), and environment (e.g., family, roommates, teammates, social contacts, job or school, coach). Emotional worries or social fights inside and outside of the training environment (e.g., illness, quarrels with friends or partners, parents' divorce) can strongly affect athletes. Problems and obligations at school, difficulties in time management (e.g., practice, school, friends),

and other responsibilities can increase stress. Often, individuals can cope with these situations, but when an additional heavy training load is added to an already high stress load, the athlete's coping resources may be inadequate for the combination.

In light of this, young athletes may be especially susceptible to overtraining overload symptoms. Due to the incomplete developmental process, the variability of psychological and psychological characteristics is higher than that of older athletes. For example, young athletes may lack sufficient coping strategies. Young motivated athletes may practice additional units at home, which coaches are typically unable to monitor. Adding more hours of training outside normal practice can contribute to excessive training load, ultimately contributing to overtraining syndrome. Often young athletes show earlier signs related to overtraining, but the symptoms may not immediately be identified as stemming from the training process due to other developmental processes experienced by young athletes.

Consequences

Consequences of overtraining, or rather the overload of practice and noncompliance of recovery, can be differentiated in short- and long-term consequences. Besides the short-term consequences of tiredness and exhaustion, the long-term consequences can be harmful. Stagnation or decreases in performances can be observed in competitions. In addition, feelings of depression, general apathy, decreased self-esteem, or emotional instability can occur. Behavioral and physical reactions include restlessness, a lack of sleep, disturbed sleep, weight loss, increased resting heart rate, hormonal changes, and an increased vulnerability to injuries. Overtraining may also increase the risk for infections or respiratory diseases.

The consequences of poorly structured and/or monitored training are particularly serious for young athletes. Frequently, these athletes drop out of the sport before having achieved top levels. Unfortunately, when athletes observe performance stagnation or decline, they commonly tend to increase training efforts, leading to even worse stages of nonfunctional overtraining. In such cases, common statements from coaches like "You need to do more!" are counterproductive and miss the target.

How Coaches and Athletes Can Prevent Overtraining

"I am better this year because I train less; in other years I was already tired before the race," a professional cyclist commented on his late-season fitness levels. This statement underlines the importance of well-structured practice and emphasized the need for balance between training and recovery periods. During an intensive season with high amounts of training during the off-season, a complete physical and psychological recovery may seem difficult to achieve. However, coaches and athletes need to be aware of the costs of inadequate regeneration periods. Coaches need to plan breaks and changes in strains to prevent an overload. A break could include a different sports activity (e.g., swimming, cycling) or the reduction of intensity. Such variability positively affects athletes' performances. Breaks should take place without irritations, disturbances, and annoyances. For example, having a nap between two practice units sounds relaxing but only if this nap is not disturbed by a neighbor's loud music. Breaks must be times in which athletes actually relax and replenish their resources.

Active and proactive recovery represents the best overtraining syndrome prevention strategy. Self-initiated active processes to reestablish psychological and physiological resources help to prevent symptoms of overtraining and support performance ability. Strategies like muscle relaxation, yoga, or a night out with friends as a kind of social recovery may refill empty stores. Athletes should learn how to relax themselves and establish their own personal way of recovering. Recovery is an individual phenomenon so the effectiveness of different strategies will vary across athletes. Accordingly, coaches must be aware that athletes should choose their own effective strategy. A trusting relationship between athletes and coaches is critical to guarantee an open exchange of ideas, problems, and advice. When athletes realize that the load of training is too high, a talk with their coach is indispensable. Communication about private circumstances (e.g., changes in job, school) is important to reduce stress, optimize recovery, and prevent overtraining.

Unfortunately, when symptoms of overtraining become obvious, it is often too late for prevention, and the athlete should visit a medical doctor specialized in high-performance sport. A commonly accepted view in the literature suggests that

rest and recovery phases from weeks to months, depending on the seriousness of nonfunctional overreaching or overtraining syndrome, represents the best treatment for overtraining.

Conclusion

The concept that "sometimes less is more" may be serious advice when it comes to the interrelation of overtraining, underrecovery, and performance ability. When performance plateaus occur, athletes and coaches often increase their efforts and increase the training load. High loads of intense practice need adequate periods of rest and regeneration, but the latter tend to be neglected. The consequences of imbalanced exercise and recovery not only concern performance but also motivation, infections, injuries, and early drop-outs. The risk of the overtraining syndrome can be reduced, if not eliminated, through careful periodization of training, the reduction of intensity, changes in activities, or the mere implementation of longer or more efficient recovery periods. Young athletes should be discouraged from doing additional workouts without informing the coach. Unfortunately, rest is often just a four-letter word for athletes, so they should be taught and, if necessary, compelled to recover.

Michael Kellmann and Sebastian Altfeld

See also Mental Health; Practice; Relaxation; Stress Management; Underrecovery Syndrome

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P

PARENTING

The developmental psychologist Jacquelynne Eccles suggested that parents influence their children's involvement in sport in three ways: as providers, role models, and interpreters. Parents *provide* children with opportunities to participate in sport by signing them up for programs, transporting them to practices and matches, paying registration fees, and so on. Parents can act as *role models* by taking part in sport themselves, and they can also influence their children's attitudes and behaviors by modeling appropriate and inappropriate behaviors in sport contexts. Parents help children *interpret* sporting experiences by communicating beliefs and values regarding sport, performance, and success. Parents' beliefs can influence how their children evaluate their own performances, the value of winning, and the importance of participating in sport.

Parental Support

Parental support has been defined as children's perceptions of their parents' behavior aimed at facilitating their involvement and participation in sport. Parents can provide children with tangible support (e.g., financial assistance), informational support (e.g., feedback after games), and emotional support (e.g., comfort after a loss). Such parental support has been associated with positive outcomes for children. For example, parental support is one of the main sources of enjoyment

for child-athletes and can positively influence children's perceived competence, confidence, coping skills, and intrinsic motivation to remain involved in sport.

Parental Pressure

Parental pressure can be defined as parental behaviors that are perceived by children as indicating high, unlikely, or possibly even unattainable expectations. Parental pressure can include parents overemphasizing winning, providing excessive criticism following competitions, and engaging in coercive behaviors. Parental pressure has been associated with children's reports of reduced enjoyment of sport, lowered self-confidence and self-esteem, and increased anxiety. Ultimately, excessive parental pressure can cause children to drop out of sport.

Parental Stressors

Parents themselves can experience stressors as they attempt to support their children's involvement in sport. A variety of studies have shown parents report numerous stressors relating to youth sport, including factors associated with sport competition, their child's sporting and overall life development, and a range of logistical and organizational issues (e.g., financial concerns, relationships with coaches, family stress, and sibling rivalries). Therefore, when attempting to understand parental involvement in sport it is important to remember that being a "sport parent" can be a difficult and challenging task for parents.

The Parent–Coach

It is common for parents to take on a coaching role in their children's sport. When parents fulfill this dual parent–coach role, it can have positive and negative outcomes for children *and* parents themselves. From a child's perspective, the positive aspects of having a parent–coach include increased amounts of praise, additional technical instruction, inclusion in team discussions, and an opportunity to spend quality time with their parent. On the other hand, when their parent is also their coach, children are often exposed to increased expectations for performance and behavior and receive additional criticism regarding mistakes. Children may also experience conflict with their parent and perceive they are being unfairly treated.

Having the opportunity to coach their children can provide parents with a chance to spend quality time with their children and have an opportunity to teach children skills and values that are important to them. Parents also have the opportunity to meet other parents and children and witness and take pride in their children's achievements. However, parents can struggle to separate their parenting and coaching role, which can reduce the support they provide to their children and increase the pressure they place on them. Parents can also struggle to cope with the additional time demands associated with the coaching role and might encounter rebellious behaviors from their children.

Parents' Behaviors During Competitions

Attending competitions can be a stressful experience for parents. As the emotional intensity of competition increases, well-intentioned parents may become pressuring, outcome-oriented, and make more negative comments to their children. Whereas sensationalized media stories often depict cases of "nightmare parents," research actually shows that only a minority of parents are inappropriately involved during competitions. Self-report surveys administered to parents in the United States have shown that approximately 20% to 25% of parents acknowledge having displayed negative behaviors or made negative comments to children during competitions. Observational studies conducted in Canada and New Zealand have shown that parents make positive verbal comments during competitions far more frequently than they make negative comments. But parents tend to make more negative comments as the emotional intensity of

games increases. The emotional intensity of games can be triggered by issues such as the quality of the officials, unequal distribution of playing time, and seeing children struggle during games. These studies point toward the importance of understanding more about parents' experiences in the social context of competitions to ensure steps can be taken to change or improve the competitive experience for parents.

Children's Preferences for Parental Involvement

Researchers have asked children how they would prefer their parents to be involved at competitions. Children indicated that prior to competitions they wanted their parents to help them with physical preparation (e.g., equipment, water) and understand the different ways in which they prepared mentally for games. During competitions, children wanted parents to show respect to opponents, officials, and coaches; display positive body language; and focus on effort rather than outcomes. Children did not want parents to be overly loud or embarrassing, attempt to coach, become involved with refereeing decisions, or make negative comments. After competitions, children preferred parents to provide positive but realistic feedback about their performance. Children may have different preferences for parental involvement in certain situations, so constant communication between parents and their children regarding parental involvement is likely to be beneficial.

Parenting Styles and Practices

The ways parents approach their parenting responsibilities (in general and within sport) can be thought of in two compatible, but different, ways. At a general level, there is a *parenting style*. Parenting style broadly refers to the range of attitudes parents have toward their children and the emotional climate parents create. Within their general style, parents also display more specific *parenting practices*. Parenting practices reflect parents' goals for their children, which have a direct influence on their children's behaviors in specific contexts. Parenting styles and parenting practices appear to have an influence on children's involvement and enjoyment in youth sport.

Although there is not a great deal of sport psychology (SP) research examining parenting styles, the handful of studies published to date suggests

that authoritative or autonomy-supportive styles of parenting are most beneficial. Authoritative or autonomy-supportive parenting styles are characterized by parents allowing their children to feel they have control over their behaviors (rather than being authoritarian or controlling). In sport contexts, these parenting styles have been associated with children having adaptive perfectionist tendencies, adhering to rules, enjoying greater satisfaction with their sport, and parents having open modes of communication and being able to “read” their children’s moods.

Educational psychology researcher Wendy Grolnick has provided a clear framework of autonomy-supportive parenting styles. Autonomy-supportive parenting is the extent to which parents value and use techniques that allow children to feel they have control over their behaviors (rather than being authoritarian or controlling). There are three elements to autonomy-supportive parenting: *autonomy-support* versus *control*, *structure*, and *involvement*. Autonomy-support is the degree to which children feel they initiate their own actions. Autonomy-supportive parents provide children with the options to choose, solve problems on their own, and exert minimal pressure to act in a certain way. Structure refers to parents providing clear and consistent guidelines, expectations, and rules for their children’s behaviors. Children can then make decisions within the limits set by their parents. Involvement is the extent to which parents are involved in their children’s lives. More involvement is generally better when parents provide children a sense of independence and appropriate structure. Excessive parental involvement that undermines children’s autonomy should be avoided.

Conclusion

Parents play vital roles in youth sport. When they perceive their parents are involved in supportive ways, children report a range of positive psychosocial outcomes. Conversely, perceptions of parental pressure are associated with a range of negative psychosocial outcomes. Being a sport parent can be stressful for parents themselves, and although few parents make inappropriate comments during competitions, the increased emotional intensity of games has been associated with parents making more negative comments. Parents should seek to understand their children’s preferences for parental involvement during competitions and the

specific behaviors in which parents should engage. An autonomy-supportive parenting style appears to be the most appropriate type of parental involvement and should be matched with appropriate behaviors.

Nicholas L. Holt and Camilla J. Knight

See also Stress Management; Talent Development; Youth Sport, Participation Trends in

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PARTICIPATION MOTIVES

The most commonly and consistently cited motives for participating in sport are developing and displaying competence (from learning new skills), experiencing challenges and success, acquiring social benefits that arise from affiliation to a group or team, improving fitness, and having fun. On the other hand, reasons for sport withdrawal include the attraction of other activities, lack of success, reduced playing time, overemphasis on winning, and not meeting new friends. Hence, it appears there is a greater likelihood children and adolescents will drop out of sport when their reasons (i.e., motives) for participation are no longer being fulfilled.

Developmental Differences in Participation Motives

There are developmental differences in sport participation motives. A study conducted by Daniel Gould and his colleagues in the 1980s showed 8- to 11-year-old swimmers cited encouragement from parents and friends and liking the coach as important motives for participation. Alternatively, older swimmers (12- to 19-year-olds) cited fitness, skills, excitement, and challenge. Another study from 1990 showed younger children (ages 6–9) rated competition-related motives, liking the coach, and pleasing family and friends as more important motives for participation than older age groups. The older children (ages 10–14) and adolescents (ages 15–18) rated social status motives higher than the younger group. Interestingly, the reasons participants drop out of sport also seem to vary with age. For example, one study showed swimmers ages 15 to 19 who dropped out cited not being good enough more than swimmers ages 10 to 14 who dropped out.

Gender Differences in Participation Motives

Girls are generally less active than boys, especially during adolescence. One reason for the gender difference in sport participation may be the different societal expectations that have existed for boys and girls. Traditionally, sport has been perceived as a male domain that provides boys with opportunities to display strength, skill, and other physical attributes. Although these views seem to be changing and girls are increasingly participating in sport, research continues to indicate that boys receive more financial and logistical support to participate in sport than girls. It has also been identified that girls and boys value different motives for participating in sport. For example, boys tend to value the competitive aspects of sport more than girls. In contrast, girls appear to place a higher premium on the social aspects of sport than boys. Girls also appear to consider the benefits of sport relating to fitness, fun, weight control, and physical appearance more important than boys.

Theories of Motivation

Various theoretical perspectives help reveal more information about motives for youth sport participation. Developed to explain motivation in achievement settings, Susan Harter's *competence*

motivation theory proposes that children are motivated to feel competent in achievement settings and engage in mastery attempts so they can display competence. That is, if children are successful in their mastery endeavors they feel more competent and continue to seek out opportunities to display mastery and competence. With regards to sport, this theory predicts that youth who perceive higher levels of physical competence are likely to be motivated to remain in sport. Harter's theory also highlights a number of antecedents that underpin perceptions of competence and motivation for sport participation. Specifically, feedback from social agents (e.g., parents, coaches), previous experiences (e.g., success or failure in sport), and perceived control over the outcome (e.g., the extent to which success is under the athletes' control) are proposed to influence children's perceptions of competence. Children's perceptions of competence also influence affective outcomes (e.g., anxiety, enjoyment) associated with sport participation.

Achievement goal theory (AGT) posits that to understand the motivation of young athletes it is necessary to understand the function and meaning of their goal directed actions. Individuals' goals (e.g., to demonstrate competence or ability or avoid demonstrating low ability in achievement domains) influence behaviors. Individuals' perceptions of their ability can be construed in two different ways. Ability may be perceived in relation to individuals' own task mastery (i.e., a self-referenced perspective) or using normative reference standards (i.e., an other-referenced perspective). Individuals who seek to demonstrate their ability in relation to others are labeled as ego-involved. Such individuals tend to define success through the demonstration of superior abilities or performances in relation to others. In contrast, task involvement refers to when individuals seek to demonstrate ability in relation to their own task mastery. For task-involved individuals, ability can be evaluated through learning new skills, improving skills, mastering a task, or giving their best effort. Research shows high task-involved individuals select more challenging tasks and expend greater effort as they strive to improve their own performance. On the other hand, highly ego-involved individuals select tasks they are confident of achieving to display superior ability compared to others. The value significant others (i.e., teachers, coaches, and parents) place upon

self-improvement versus succeeding against others has been associated with the development of children's task- or ego-involvement. For example, coaches who provide positive feedback when their athletes win and negative feedback when they lose may encourage the development of ego goals in their athletes (due to the emphasis they place on performance relative to others), whereas coaches who provide feedback to their athletes regarding the development of specific skills and the effort they display are likely to encourage athletes to develop task orientations. As such, striving to foster task-involvement in youth may enhance their motivation to participate in sport.

Edward L. Deci and Richard M. Ryan's *self-determination theory* (SDT) is based upon the premise that humans have an innate drive to maximize their positive feelings and effectively master challenging tasks. According to the SDT, people have three universal needs that must be fulfilled. These needs are *competence* (the need to feel behaviors and interactions are effective), *autonomy* (the need to perceive behaviors and thoughts are freely chosen), and *relatedness* (the need to feel connected to people around us). SDT also includes four "mini-theories" that explain how the interaction between humans and social contextual factors influence motivation and well-being. These are *cognitive evaluation theory* (CET), *organismic integration theory* (OIT), *causality orientations theory* (COT), and *basic needs theory*. The extent to which intrapersonal and interpersonal contexts support individuals' basic needs is proposed to dictate their enjoyment of an activity and their feelings of self-determination regarding behavior. Behavior driven by intrinsic motivation (engaging in an activity because it is innately enjoyable) is the most self-determined. Amotivation (a lack of motivation) results in the least self-determined behavior. Extrinsic motivation (engage in a behavior to achieve a separate outcome) sits between intrinsic motivation and amotivation. Thus, striving to fulfill youths' basic needs and promoting intrinsic motivation are important for maintaining children's participation in sport.

The *sport commitment model* (SCM) was developed by Tara Scanlan and colleagues to illustrate factors associated with an individual's desire to continue his or her sport participation. Antecedents of sport commitment are *enjoyment*, *involvement opportunities* (anticipated or expected

benefits afforded from continued participation), *involvement alternatives* (attraction to other activities as opposed to sport), *personal investments* (how much an individual has invested in sport), and *social constraints* (expectations from others to continue in sport). However, research indicated that the role or importance of these antecedents with regards to sport commitment may not be equal. Rather, it appears that sport enjoyment is particularly important and mediates the influence the other antecedents have upon individuals' sport commitment.

Implications

The theories just described provide several implications for fulfilling the motives children and adolescents have for participating in sport. It is important to create an environment that enhances competence, which can be achieved when individuals received praise from their parents, coaches, and peers. Praise and feedback related to individual improvement and effort, as opposed to their success in relation to others, is likely to enhance perceptions of competence and intrinsic motivation more consistently. Youth also need to be reminded that their success is a result of their own actions. Youth involved in sport also need to have opportunities to develop friendships and relationships with their peers, coaches, and other social agents. One way to do this is to provide opportunities for participations to gain the experience of playing the sport without being overly constrained by the technical and tactical aspects of training or competition. By creating a sporting environment that maximizes the potential for youth to experience success, affiliation, and enjoyment, there may be more chance of youth remaining in sport.

Conclusion

The most commonly cited motives for participating in sport are developing and displaying competence (from learning new skills), experiencing challenges and success, acquiring social benefits that arise from affiliation to a group or team, improving fitness, and having fun. When these motives are not fulfilled, youth may drop out of sport. Approximately 30% of youth drop out of a given sport program annually, and highest attrition occurs during adolescence. Several theoretical frameworks provided useful strategies for facilitating involvement and enjoyment in youth sport.

These frameworks can be used to examine and facilitate participation in youth sport.

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See also Achievement Goal Theory; Achievement Motive Theory; Competence Motivation Theory; Drop-Out; Effort; Enjoyment; Friendships/Peer Relations; Parenting; Self-Determination Theory

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PASSION

The dualistic model of passion (DMP) describes two types of passion, namely harmonious and obsessive passion. This model allows for a better understanding of the passion of people involved in sport (athletes, coaches, referees, and fans) as well as the outcomes they experience.

The DMP defines *passion* as a strong inclination toward a self-defining activity (person or object) that one likes (or even loves), finds important,

and in which one invests time and energy. In fact, passionate activities become part of one's identity because they are so important and valued. The process through which the activity comes to define someone is called the internalization process. In other words, with the internalization process, what was once external to the self becomes a part of it. More specifically, the internalization of a passionate activity can take place in two different ways, which lead to two different types of passion.

First, *harmonious passion* results from an autonomous internalization, which occurs when individuals freely accept the activity as important for them and choose to engage in their passionate activity, without any contingencies attached to it. With harmonious passion, there is a smooth integration between the passionate activity and our identity. Thus, the activity occupies a significant, but not overpowering, space in the person's identity and remains in harmony with other aspects of the person's life. Consequently, people with a harmonious passion are able to fully focus on the task at hand and experience positive outcomes both during and after task engagement. Furthermore, when prevented from engaging in their passionate activity, people with a harmonious passion are able to adapt well to the situation and focus their attention and energy on other tasks that need to be done. Thus, there is little or no conflict between the person's passionate activity and other life domains. Finally, with harmonious passion, the person can decide when to and when not to engage in the activity. This is because, with harmonious passion, the person is in control of the activity.

In contrast, *obsessive passion* is the product of a controlled internalization. This type of internalization originates from intra- and/or interpersonal pressure such as of self-esteem or social acceptance contingencies that are attached to the activity or because the feeling of excitement derived from activity engagement is uncontrollable. With obsessive passion, the activity becomes a part of one's identity because it is loved but also because it brings other, more extrinsic benefits such as a boost of self-esteem or a reward. Therefore, people with an obsessive passion find themselves in the position of experiencing an uncontrollable urge to partake in the activity they view as important and enjoyable. This is because, with obsessive passion, the person is not in full control of the activity. Rather, the activity controls the person.

On the foregoing basis, the DMP posits that harmonious and obsessive passion for sport ought to lead to different affective, cognitive, and behavioral consequences. Specifically, harmonious passion is conducive to more positive outcomes compared with obsessive passion.

Passion, Affective Consequences, and Well-Being

The type of passion one holds toward an activity plays a significant role in the affective consequences that individuals are likely to experience both during and after activity engagement, as well as when prevented from engaging in the passionate activity. With harmonious passion, people volitionally engage in the passionate activity with an openness and a mindfulness that allow them to fully partake in the activity and thus to experience positive affective experiences more fully. This is not the case with obsessive passion, where one's engagement is more defensive and is less conducive to full engagement in the activity. Specifically, research has shown that harmonious passion is positively related to positive affect in sport as well as in life in general, while it is unrelated to negative affect. In contrast, obsessive passion is conducive to negative affect and is unrelated or only slightly related to positive affect in sport as well as in life in general.

Along the same line, harmonious passion is associated with higher levels of psychological well-being (PWB) compared to obsessive passion. In line with Barbara Frederickson's *broaden-and-build theory*, the positive relation between harmonious passion and PWB is mediated by the effects of positive emotions experienced with harmonious passion. This is because positive emotions allow one to have access to the self by broadening the attention and thought-action repertoires made available to the person, and subsequently facilitating the use of adaptive processes, such as coping with adversity and stress. This is not the case for obsessive passion, as it is typically unrelated to positive emotions.

Passion and Cognitive Processes

Harmonious passion entails an open and mindful form of task engagement, whereas obsessive passion facilitates a more defensive form of activity engagement. Therefore, harmonious passion facilitates adaptive cognitive processes (such as concentration, flow, and better decision making

[DM]), whereas obsessive passion leads to less positive cognitive processes. In fact, empirical evidence shows that soccer referees with an obsessive passion make worse decisions during a soccer match than those with a harmonious passion. Another study shows that obsessive passion from fans prevents full concentration in other life activities taking place on the same day as the passionate activity (i.e., game). In contrast, this is not the case for harmonious passion.

Passion and Physical Health

Physical health can be affected by passion in a number of ways. For example, regularly engaging in a passionate activity such as sport may positively contribute to health by leading people to experience increases in physical well-being over time. However, obsessive passion can also put people's health at risk by leading them to engage in risky sport or because of ill-advised rigid persistence in the activity, such as cycling on icy roads. In addition, obsessive passion can constitute such a risk factor for injuries. This is because with obsessive passion, people continue to engage in the activity when they should not and thus risk creating or aggravating an injury. This is not the case for harmonious passion. In fact, research demonstrates that harmonious passion leads individuals to derive positive psychological and health benefits from regular engagement in the activity. On the other hand, research conducted with dancers reveals that obsessive passion is a risk factor for the development of *chronic* injuries.

Passion and Performance

Passion leads athletes to engage in deliberate practice, a specific form of practice that focuses on improving one's skills. In fact, one must love an activity dearly and have the desire to keep on practicing if one is to engage in the activity for long hours over several years (sometimes a lifetime). Therefore, both forms of passion lead athletes to engage in deliberate practice that, over time, leads to improved performance. Research has shown that achievement goals (i.e., competence-based aims that an individual seeks to accomplish in an achievement setting) serves as mediator in the relationship between both types of passion and performance. Specifically, people with a harmonious passion pursue mastery goals (i.e., a focus on the development of personal competence and

task mastery) that lead to deliberate practice that, in turn, leads to performance. In contrast, people with an obsessive passion also pursue mastery goals (that lead to performance through deliberate practice), but they mostly pursue performance-approach goals (i.e., a focus on the attainment of personal competence relative to others) and especially performance-avoidance goals (i.e., a focus on avoiding incompetence relative to others) that negatively and directly influences performance. In addition, only individuals with a harmonious passion experience PWB while attempting to reach high levels of performance in sport.

In sum, both types of passion may facilitate performance. However, it appears possible that the highest levels of performance can be reached through a painless, even happy, passionate engagement in the sport to the extent that it is harmonious in nature.

Passion and Interpersonal Outcomes

A better understanding of the coach–athlete relationship is important if we are to help athletes reach their goals in sport while feeling good about their activity engagement. The type of passion one holds for sport represents an important factor that allows both athletes and coaches to experience high-quality relationships. Specifically, athletes' harmonious passion toward their sport is positively related to various indices of relationship satisfaction with their coach. In contrast, athlete's obsessive passion is mostly unrelated to those relationship indices. This is because harmonious passion leads people to fully immerse themselves in the activity and to experience positive emotions that can be shared with others. Furthermore, positive affect, generally experienced by coaches, serves as a mediator in the relation between harmonious passion toward coaching and coaches' perceived relationship quality with their players. Conversely, obsessive passion for coaching is unrelated to positive affect or relationship with the athletes. In addition, obsessive passion for soccer can conflict with the romantic life of soccer fans. It is not the case for harmonious passion.

On the Development of Passion

There are at least three processes through which an interesting activity such as sport can transform into a passionate activity. These processes are (a) activity selection, (b) activity valuation, and (c) the type of

internalization process that takes place. First, activity selection refers to the person's preference for the activity over other activities. To the extent that the person feels that such selection reflects true choice and interests and is consonant with her or his identity, it should promote the development of passion toward that activity. Activity valuation (or the subjective importance given to the activity by the person) is expected to play an important role in the internalization of the identity. The more valued the activity is, the more this activity will be internalized in the person's identity and the more passionate the person will be toward this activity. The type of passion that will develop also depends on the type of internalization that takes place. A harmonious passion will develop if the internalization process is carried in an autonomous fashion. In contrast, an obsessive passion will develop if the internalization takes place in a controlled fashion.

The internalization process is influenced by the social environment as well as by personal factors. More precisely, social environment (i.e., parents, coaches, peers) and personal factors (i.e., individual differences and personality processes) that promote a person's autonomy will lead to the development of a harmonious passion by facilitating the autonomous internalization process. In fact, coupled with high-activity valuation, both high autonomy support from close adults (parents, coaches, etc.) and autonomous personality style predict the development of a harmonious passion over time. In contrast, controlling social environment and personal factors will lead to the development of an obsessive passion by facilitating the controlled internalization process. Specifically, coupled with strong activity valuation, both the lack of autonomy support and controlled personality style predict the development of an obsessive passion.

In conclusion, the two types of passion, namely harmonious and obsessive, matter greatly for sport participants because they lead to a host of important consequences (affect, cognitions, subjective well-being [SWB], performance, physical health, etc.) that are typically adaptive and maladaptive, respectively.

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See also Affect; Autonomy-Supportive Coaching; Coach–Athlete Relations; Commitment; Intrinsic/Extrinsic Motivation; Hierarchical Model of; Psychological Well-Being; Self-Determination Theory

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PATTERN RECOGNITION AND RECALL

In the sporting domain, pattern recognition and recall refer to the capability of athletes to recognize or recall the patterns formed by the configuration of key elements (such as teammates and opposing players) that exist within the playing environment. An enhanced capability to quickly and accurately extract information from these unique pattern structures has been suggested to contribute to other important sports skills such as anticipation (i.e., predicting future events such as the movements of other players) and decision making (DM) (i.e., deciding what to do next). This entry briefly outlines the basic methodological approaches used to measure pattern recognition and recall as well as summarizes the results from empirical research in this area.

Pattern recognition performance is typically measured in two phases. In the first phase, participants are shown a series of patterns presented sequentially. The task in this initial phase generally requires participants to either memorize each of the presented patterns or make some form of a decision such as determining what they think will happen next in each particular scenario. In both cases, the principal objective is to encourage participants

to closely attend to the patterns. Shortly after the first phase, participants are shown another series of patterns but on this occasion, the series contains some patterns that were displayed in the first phase (usually 50%) and some patterns that are entirely new. When each pattern is presented, participants are asked to determine whether it is a previously seen pattern or a new pattern. Recognition performance is then measured by the number of correct responses.

The method used to measure pattern recall performance in team sports typically involves presenting experienced and lesser-skilled participants with a series of stimulus patterns extracted from a specific sport. Each pattern is displayed to the participants for a few seconds before it is removed from view. Once it is removed, participants are required to recall the final locations of the players in the pattern by placing or drawing icons of the players on a scaled diagram of the playing area. The patterns themselves can be displayed in a number of ways with the more common approaches using schematic images (where icons are used to show the locations of players), static slides (still images such as photographs that have been taken from an actual game), or videos (video footage of an actual game). Each pattern is carefully selected to ensure it is either highly structured (i.e., shows the locations of players as they would appear in a typical game) or relatively unstructured (i.e., shows an atypical pattern where players are located in random positions that do not reflect game structure, such as when the game has been disrupted by a time-out). Recall performance is assessed by comparing the location of each player in the actual pattern to those indicated by the participant and, from this, generating an overall recall accuracy (or recall error) score.

Studies employing the pattern recognition and recall methodologies have generally shown a significant advantage for experts over lesser-skilled participants when the patterns depict gamelike information, but this advantage typically declines when the patterns become more unstructured. This demonstrates that expert performance on recognition and recall is moderated by the amount of structure contained within the patterns themselves. For example, A. Mark Williams and his colleagues have shown that experts extract information from the interrelationships that exist between the elements in the patterns, and this perceptual information is utilized by experts more so than some of the

finer details, such as the specific postural orientation of the players in the image. Their research has also revealed that this structural information may in fact transfer between sports, particularly those that have similar rules, tactics, and playing structures. This may help explain how experts from one team sport are sometimes able to perform quite well in other, similarly structured team sports.

It is also believed that the structural information may play an important functional role in anticipation and DM in the normal competitive setting. Research by A. Mark Williams and Keith Davids in 1995, for example, suggested that an enhanced capability to recall the configurations of players in an evolving team sport pattern may assist experts to anticipate the next most likely outcome to occur in the playing sequence. Further support for the use of pattern information in anticipation was provided by André Didierjean and Evelyn Marmèche, who used a recognition task to show that the encoding of structured patterns includes anticipatory information where the future locations of the players in the pattern are incorporated into the memory representation. However, K. Anders Ericsson and his colleagues have argued that memory-based tasks, such as recognizing and recalling patterns of play, are not a requirement of normal participation and so the extent to which pattern recognition and recall skill actually contributes to performance in the competitive environment remains an area of scientific debate.

Tests examining pattern recognition and recall skill have helped researchers to better understand the nature of expert performance and the underlying mechanisms that contribute to perceptual skill in team sports and other domains. An understanding of the key characteristics that contribute to expert performance can help practitioners design training sessions that specifically target the underlying variables that significantly enhance the development of skilled performance.

Adam D. Gorman

See also Anticipation; Decision Making; Expertise; Memory; Recognition and Recall Paradigms

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PERCEPTION

Perception brings us into contact with people, objects, and places within the environment. Perception relies on sensory systems that enable humans to see, hear, feel, smell, and taste. It is estimated that one third of the human brain is dedicated to perception. Early Greek philosophers were among the first to study the science of perception. Their strong bias toward sight continues to this day, and sport psychology (SP) is no exception. For example, in addition to the wide body of research that has studied the motor dexterity of elite athletes, a growing number of researchers have studied the visual perceptual skills of these athletes. Visual perception is critical for decision making (DM); for example, to recognize an opponent's pattern of play to assume a strategically apt position for intercepting the ball and for the guidance of action, when for instance, controlling running speed and direction in order to successfully catch a fly ball. This entry addresses perception by providing an overview of the physiology of the visual system and the two dominant theoretical approaches used to study perception: *information processing* and *ecological psychology*. In line with the typically predominant interest in visual perception, this entry uses the term *perception* to refer to visual perception, except where stated otherwise.

Physiology of the Visual System

Visual perception starts with rays of light, or stimulus information, falling upon the *retina* at the back of the eye. The retina consists of different types of light sensitive photoreceptors colloquially named *rods* and *cones*. The rods greatly outnumber the cones and are distributed across the entire retina except at the *fovea*, the location of highest visual resolution, where the cones are highly concentrated. The rods are more sensitive to light than cones and are involved in the perception of location and motion. Rods only provide coarse perception of the environment (i.e., low acuity). The cones function best in bright light; they are important for perceiving detail (i.e., high acuity) and color. The stimuli at the retina are relayed to the brain along the *optic nerve*, which consists of different populations of ganglion cells that project onto the *lateral geniculate nucleus* and from there to the *primary visual cortex* (V1) at the rear of the brain. *Magnocellular* ganglion cells are fast and transport stimulus information important for the perception of motion and location or distance. *Parvocellular* ganglion cells are much slower and transfer stimulus information about color and edges (i.e., shape). A third population of ganglion cells project directly to the subcortical areas of the visual system (i.e., the *superior colliculus*) to facilitate the control of eye movements. From the primary visual cortex, stimulus information is further distributed to higher cortical areas, each of which selectively processes different visual properties. For example, V2 (and V1) process orientation, V3 supports shape perception, V4 is involved in color perception, and V5 or MT is dedicated to motion perception. Importantly, beyond V1 the visual brain is organized into two distinct pathways, one toward the parietal lobe (the *dorsal stream*) and the other toward the temporal lobe (the *ventral stream*). There is fierce debate over the respective functions of the two visual systems. Traditionally, the systems have been distinguished in terms of their input: The dorsal stream is referred to as the *where* stream reflecting its primary function for spatial perception (i.e., the perception of motion, distance, and location), whereas the ventral stream is named the *what* stream consistent with the supposition that it is responsible for object perception (i.e., the perception of shape, size, and color). However, a relatively recent reinterpretation emphasizes the distinctive *functions* of the two

streams rather than their input. Hence, the dorsal stream is proposed to be the *action* stream, underpinning the fast, unconscious, and online control of movement, while the ventral stream is the *perception* stream, a relatively slow but mostly conscious system surmising knowledge about the environment. This distinction between action and perception is particularly relevant when considering perception in sport.

Information-Processing Approach to Perception

In SP, as in most other domains of psychology, information-processing approaches have undoubtedly been the dominant theory for understanding perception. Information-processing approaches consider human perception to act in a manner similar to a digital computer. In this metaphor, the visual system is considered to be the “hardware” that underlies perception. Yet, the most distinctive feature of these approaches for understanding perception is the “software” conceived to be necessary for the different stages of information processing. In one of the earliest encompassing accounts of the information processing approach in SP, H. T. A. Whiting distinguished a series of successive and separate stages of information processing: the *input*, *decision-making*, and *output* stages. In the *input* stage, the eyes register stimulus information from the environment and transmit it to the brain. Importantly, the transmission of information is selective; because of the vast amount of stimuli, it is not possible to attend to all the stimulus information that is available at any one time.

The stimulus information in itself is inherently ambiguous. For example, the retinal size of an object varies according to the distance between the object and observer, and retinal shape varies as function of the orientation of the object relative to the observer. A single object can therefore generate an infinite number of different stimulus patterns on the retina and hence is considered to be insufficient as a basis for perception. The stimulus information needs to be interpreted or enriched. Consequently, in the *decision-making* stage of processing, central mechanisms in the brain must interpret and process the stimulus information to form a representation of the environment. These internal representations then provide the basis for other central mechanisms that make decisions about a new action,

or adjustments in an ongoing action, to program appropriate movement responses. In the final *output* stage, the programmed movement responses are performed by the musculoskeletal system. Notably, for information-processing approaches, perception ends the instant that an internal perceptual representation of the environment has been created. For this reason, within the information processing approach, perception and action are conceived as largely independent processes that can be studied in isolation from one another.

Information-Processing Approach in Sport Research

In sport research, the information-processing approaches have above all been used to better understand perception and perceptual expertise in DM and anticipation. In this respect, a common feature for perception in sport is that it is performed under restrictive time constraints—for example, when attempting to interrupt the opponents' attacking play and passing the intercepted ball to a teammate. Hence, successful performance requires perception or *anticipation* of the outcome of others' actions before it is fully manifested in ball flight: a player is often too late if he or she waits until the outcome information is available. Early investigations, predating the information processing approach, assumed that expert performers must possess visual hardware that is better than in lesser-skilled performers (i.e., better visual acuity [VA], color perception, or peripheral vision). On the whole, this research failed to demonstrate a strong relationship between sport skill and the low-level physiological properties. Yet, given the recent dramatic improvement in our understanding of the visual brain, it may be premature to completely discard this proposal. For instance, measures like motion and speed perception, which purportedly reflect processing of higher cortical areas, may or may not eventually prove to distinguish athletes with different levels of skill.

More recent evidence has shown that experts possess a superior sport-specific *knowledge base* in addition to a superior ability to *process* stimulus information. These capacities speak to the very heart of the information-processing approach, which holds that the inherently insufficient information in the stimulus needs to be enriched for perception to become meaningful—that is to

say, to form a high-fidelity representation of the environment. This enrichment comes about by relating the stimulus to task-specific knowledge built from past experiences via the processes of *encoding* (i.e., transforming stimulus information into a neural code that can be stored in the knowledge base) and *retrieval* (i.e., the reverse process whereby stored perception information is accessed). Perception is more truthful and rapid when the knowledge base becomes wider and when encoding and retrieval become more efficient. Indeed, experts are shown to recognize and recall patterns of play more accurately and much quicker than lesser-skilled players. In a seminal study, Adriaan D. de Groot briefly presented images of chess positions to players of different skill levels and asked them to reconstruct the positions. The expert chess players were able to recall the positions with relatively few mistakes, whereas players below master level made a large number of errors. When players in a subsequent study viewed both structured (i.e., real) and unstructured (sometimes impossible) chess positions, the recall of the experts was found only for the real configurations. This demonstrates that skilled players possess more efficient game-specific encoding and/or retrieval of chess positions rather than exceptional visual memory or speed of processing. The expert players reproduce the positions by grouping together or *chunking* small groups of individual pieces into larger meaningful units. This chunking allows the experts to process more information while using the same (limited) memory capacity.

The superior encoding and retrieval shown by experts in a static game like chess have been shown to also exist in more dynamic sports like basketball, field hockey, and soccer. Typically, this work presents high- and low-skilled players with pictures or video clips of structured and unstructured patterns of play. Superior recognition and recall is commonly observed among high-skilled players, suggesting that expertise entails an enhanced ability to perceptually chunk the observation of individual players into larger (meaningful) units. Presumably, this allows these athletes to anticipate the outcome of patterns of play earlier, and more accurately, than lower-skilled players.

Indeed, studies employing the *temporal occlusion paradigm* have provided ample evidence that high-skilled players anticipate action outcomes by extracting early, pre-ball flight stimulus

information (or advance cues). In the temporal occlusion paradigm, participants view video clips simulating an athlete's view of an evolving pattern of play (e.g., an offensive move by a team) or an opponent's action (e.g., a badminton serve). The clips are selectively occluded at predetermined times relative to a critical moment in the action sequence (e.g., ball contact). Participants predict the outcome of the action, usually by producing a verbal or a simplified movement response. High-skilled players are typically better at reliably anticipating the outcome of the occluded pattern of play or opponent's action. Within the information-processing approach, this is taken to mean that high-skilled players distinguish themselves by their ability to use their extensive knowledge base to process early visual cues to infer the forthcoming event.

Some have argued that the expert perceptual advantage in sport is not limited to the processing of stimulus information but that it also entails the ability to better attend to relevant stimuli or cues. High-skilled athletes appear to be able to visually search the environment in a more efficient manner by selectively fixating longer on those areas that are relevant for successful task performance. Yet, *visual search* has also proven to be highly sport- and task-specific, rendering it hazardous to make inferences about general skill-related characteristics for visual search. Nonetheless, it is fair to conclude that athletes' visual search is not only driven by environmental events, but is also directed by their internal knowledge base. Experts know what are the most informative aspects of an evolving pattern of play or an opponent's action, and what aspects are less relevant for further processing.

The Ecological Approach to Perception

The central notion of the information-processing approach is that the environment is perceived indirectly via an internal representation constructed by cognitive processes that add meaning to inherently ambiguous stimuli. In contrast, the ecological approach holds that visual perception is direct. It claims that all the information that is needed for perception (and action) is "out there" in the environment. James J. Gibson argued that the *optic array*, which refers to the ambient light that is lawfully patterned by the surfaces of the environment, carries information that is rich and sufficient for perception. Because information

in the optic array is *specific* to the properties of the environment (or the observer-environment system), internal processes of enrichment are redundant. Visual perception therefore entails the pickup or extraction of specifying information that resides in the optic array. As a rule, information is granted by virtue of the observer's movements. For instance, a moving observer generates a global expansion pattern in the optic array, the focus of which is specific to the direction of locomotion. Importantly, these movement-induced patterns in the optic array (denoted *invariants*) also form the basis for the visual guidance of action. Accordingly, when steering, the only thing a cyclist needs to do is to move or steer in such a manner that the focus of expansion remains at a position farther down the road. Within the ecological approach this is referred to as *perception-action coupling* (or sometimes, and perhaps more appropriately, *information-movement coupling*). A critical corollary of this intricate coupling of perception with action is that the available information, and hence perception, is grounded in action: Perception and action cannot—and should not—be understood and studied as independent processes. Accordingly, the primary objects of perception are thought to be so-called *affordances*, or opportunities to act.

Within the ecological approach, perceptual learning is described as *convergence* or *education of attention*, a process along which perceivers learn to pick up specifying information. Early in the learning process, perceivers can be expected to attend to less useful nonspecifying information (i.e., information that is not specific to the environmental property of interest), resulting in misperception. During learning, however, the observer comes to attend to the specifying information that will enhance perceptual accuracy. Moreover, accurate perception (or action) not only requires the pickup of specifying information but also an appropriate scaling or *calibration* of perception or action to the exploited information. Hence, proponents of the ecological approach posit that in addition to the education of attention, perceptual learning entails a process of calibration.

The Ecological Approach in Sport Research

The concept of perception-action coupling in particular has had, and continues to have, a powerful influence on research in perception in sport.

The baseball outfielder trying to catch a fly ball is a paradigmatic example. Rather than inferring the location that the ball will land based on the perception of its current distance and velocity and internal knowledge of ball flight characteristics, the outfielder should—according to proponents of the ecological approach—simply run in a manner that generates systematic changes in the optic array. During early ball flight, the optical height of the ball changes according to the location the ball will land relative to the outfielder. The ball will land behind the outfielder when there is vertical optical acceleration. Conversely, in the case of vertical optical deceleration it will land in front of them. This invariant not only specifies what the situation affords for action (i.e., running forward or backward) but also how to run to get in the right place at the right time to intercept the ball. That is, catchers have been shown to run in a manner that zeroes out or negates optical acceleration or deceleration and keeps optical velocity constant.

The outfielder catching a ball underlines that perception and perceptual expertise cannot be investigated separate from action (i.e., without action, optical acceleration cannot be negated). Accordingly, the ecological approach has criticized information-processing approaches for doing just that: studying perceptual expertise in sport using experimental designs that isolate perception from action. Soccer goalkeepers, for instance, demonstrate superior anticipation (and interception) when diving to save real penalty kicks (in situ) when compared to making verbal responses in response to video clips (as typically required within the temporal occlusion paradigm). Importantly, these differences in anticipation are reflected by critically different gaze patterns. In addition, faster and more agile goalkeepers are found to wait longer before moving to block the ball when tested in situ. In doing so, the pickup of information supporting anticipation of the kick can be based on (potentially less deceptive) information that is only available in the latter part the penalty taker's actions. Consequently, perceptual expertise can only manifest itself relative to dexterity in action. Neurophysiologically, this would suggest that skilled athletes rely on interacting and complementary contributions from the ventral (perception) and dorsal (action) visual streams. Therefore ecological psychologists consider it impossible, or perhaps even misleading, to try capturing (or

achieving) perceptual expertise in sport by examining (or practicing) perception on its own, decoupled from action.

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See also Anticipation; Dynamical Systems; Expertise; Eye Movements/Gaze; Information Processing; Pattern Recognition and Recall; Vision

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PERFECTIONISM

Perfectionism is a personality disposition characterized by striving for flawlessness and setting exceedingly high standards for performance, accompanied by tendencies for overly critical evaluations. It is a disposition that may pervade all areas of life, particularly areas in which performance plays a major role (e.g., work, school). Therefore, it comes as no surprise that perfectionism is a common characteristic of competitive athletes.

Perfectionistic Strivings and Perfectionistic Concerns

Traditionally, perfectionism has been regarded as a sign of psychological maladjustment and mental disorder, because people seeking help for depression and anxiety often report elevated levels of perfectionism. Traditional conceptions regarded perfectionism as a one-dimensional personality disposition. In the 1990s, however, a more differentiated view emerged that conceptualized perfectionism as a multidimensional and multifaceted characteristic. From this view, a consensus has emerged that perfectionism should be differentiated into two main dimensions: *perfectionistic concerns* and *perfectionistic strivings*. Whereas the dimension representing perfectionistic concerns captures those aspects associated with concern over mistakes, fear of negative evaluation by others, feelings of discrepancy between one's expectations and performance, and negative reactions to imperfection, the dimension representing perfectionistic strivings captures those aspects of perfectionism associated with self-oriented striving for perfection and very high personal standards of performance.

The differentiation between these two dimensions is central to the understanding of perfectionism. Whereas the two dimensions are often highly correlated—most people who show elevated levels of perfectionistic concerns also show elevated levels of perfectionistic strivings; they show differential, and often contrasting, patterns of relationships. Perfectionistic concerns show strong and consistent negative relationships—that is, positive associations with negative characteristics, processes, and outcomes (e.g., neuroticism, maladaptive coping, negative affect) and with indicators of psychological

maladjustment and mental disorder (e.g., depression). In contrast, perfectionistic strivings often show positive relationships—that is, positive associations with positive characteristics, processes, and outcomes (e.g., conscientiousness, adaptive coping, positive affect) and with indicators of subjective well-being (SWB) and good psychological adjustment (e.g., satisfaction with life).

The differentiation between the two dimensions is of central importance also in sport and exercise psychology (SEP), where perfectionistic concerns and perfectionistic strivings show different—sometimes opposite—relationships with emotion, motivation, and performance. As in the general perfectionism literature, however, the positive associations of perfectionistic strivings are often “masked” because of the overlap with perfectionistic concerns and may be revealed only when the negative influence of perfectionistic concerns is controlled for. Consequently, it is important for studies in SEP investigating perfectionism to use statistical analyses that control for the overlap between the two dimensions in order to uncover the often-marked differences that the two dimensions show.

Emotion

In the general perfectionism literature, a frequent and well-replicated finding is that perfectionistic concerns are associated with positive affect, whereas perfectionistic strivings are associated with negative affect. The same holds for perfectionism in sports regarding general affect and specific affective experiences. For example, research using imagined scenarios to investigate how athletes respond to success and failure in important competitions found that when the overlap between the two dimensions was controlled for, perfectionistic concerns were associated with negative affect after failure, whereas perfectionistic strivings were associated with positive affect after success. Differential relationships of the two perfectionism dimensions have also been found for competitive anxiety, an important emotion frequently experienced by competitive athletes that has received much attention in sport psychology (SP). In the literature, three aspects of competitive anxiety are usually differentiated: cognitive anxiety, somatic anxiety, and self-confidence. Cognitive anxiety represents the side of competitive anxiety involving negative cognitions about

possible failure, and somatic anxiety represents the side involving perceptions of bodily symptoms and heightened negative arousal. In contrast, self-confidence involves positive cognitions and feelings indicating that one is up to the task and capable of giving one's best possible performance. Consequently, self-confidence in competitions is a sign of low competitive anxiety and has shown positive relationships with performance. A number of studies have examined the relationships between perfectionism and competitive anxiety in athletes. Overall findings show that only perfectionistic concerns are associated with higher levels of cognitive and somatic anxiety and lower levels of self-confidence. In contrast, perfectionistic strivings (once the overlap with perfectionistic concerns is controlled for) are associated with low levels of cognitive and somatic anxiety and high levels of self-confidence, suggesting that athletes who strive for perfection (and are not overly concerned about not achieving perfection) approach competitions with a positive mind-set that may help them achieve a higher performance.

Motivation

Motivation is another key variable in SP, particularly achievement motivation. An important predictor of athletes' achievement motivation is achievement motives. Achievement motives are stable individual differences in learned, affectively charged anticipatory responses to achievement situations that energize and direct people's behaviors. Research traditionally differentiates between two achievement motives: *hope of success* (motivating people to achieve success) and *fear of failure* (FOF) (motivating people to avoid failure). Mirroring the findings from the competitive anxiety literature, studies controlling for the overlap between the two perfectionism dimensions have found that only perfectionistic concerns were positively associated with FOF. In contrast, perfectionistic strivings were positively associated with hope of success (and negatively with FOF), suggesting that perfectionistic strivings are associated with a positive mind-set regarding achievement situations that may help athletes attain success, whereas perfectionistic concerns are associated with a negative mind-set focusing on possible failure.

Another variable closely associated with achievement motivation is achievement goal orientation. As with achievement motives, perfectionistic

strivings and concerns show a clear differential pattern with achievement goal orientations, particularly when the 2×2 model of achievement goals is considered. The 2×2 model differentiates two dimensions of goal orientation—definition (performance vs. mastery) and valence (approach vs. avoidance)—resulting in four different goals: performance-approach, performance-avoidance, mastery-approach, and mastery-avoidance. Performance-approach goals represent the motivation to demonstrate normative competence (e.g., striving to do better than others), performance-avoidance goals represent the motivation to avoid demonstrating normative incompetence (e.g., striving to avoid doing worse than others), mastery-approach goals represent the motivation to achieve absolute or intrapersonal competence (e.g., striving to master a task), and mastery-avoidance goals represent the motivation to avoid absolute or intrapersonal incompetence (e.g., striving to avoid doing worse than one has done previously). A number of studies have investigated the relationships between the two dimensions of perfectionism and the 2×2 goals. Across-studies results showed that when partial correlations were computed controlling for the overlap between the two dimensions, perfectionistic concerns showed positive correlations with performance-avoidance and mastery-avoidance goals, indicating that perfectionistic concerns are associated with avoidance motivation in competitive achievement situations. In contrast, perfectionistic strivings showed positive correlations with performance-approach and mastery-approach goals, indicating that perfectionistic strivings are associated with an approach motivation.

Performance

The findings that perfectionistic concerns are associated with a negative mind-set and avoidance motivation in competitive situations, whereas perfectionistic strivings are associated with a positive mind-set and approach motivation, suggest that the two dimensions of perfectionism also show different relationships with performance. Unfortunately, only few studies so far have investigated perfectionism and sport performance, and the findings they have produced are not as clear-cut as those of the studies regarding emotion and motivation. On the one hand, there are findings that corroborate the many findings from

the general perfectionism literature claiming that perfectionistic strivings are associated with higher performance: In a study investigating basketball training performance, perfectionistic strivings were associated with higher performance across a series of trials in a new, nonstandard training task, and in two studies investigating racing performance of triathletes, perfectionistic strivings predicted higher race performance beyond what was expected from the athletes' previous performance (seasonal or personal best). On the other hand, there are findings indicating that both perfectionistic concerns and perfectionistic strivings are detrimental to performance. In a study investigating athletes' psychomotor performance in a balancing task (stabilometer performance), which took place under laboratory conditions where false performance feedback was given (athletes were told they were performing below expectations), both dimensions of perfectionism were associated with lower performance. Finally, there are findings indicating that both dimensions of perfectionism can enhance performance and lead to performance improvements. In the study investigating basketball training performance, athletes who had high levels of perfectionistic strivings and high levels of perfectionistic concerns—which is a combination of the two perfectionism dimensions that most theories and research in perfectionism would consider unhealthy, maladaptive, or dysfunctional—showed the greatest performance improvements across the trials. It appears that if we disregard the laboratory study with false performance feedback, perfectionistic strivings are not detrimental to sport performance. On the contrary, perfectionistic strivings may be associated with higher performance in training and competitions.

Open Questions and Future Research

Perfectionism is a characteristic that can have important implications for athletes' well-being, their emotional experiences and motivational orientations, and their performance in training and competitions. Yet compared to research in general perfectionism, research on perfectionism in sport is still underdeveloped, and many open questions remain for future research. First, little is known about how individual differences in perfectionism develop. Second, we do not know anything about the long-term consequences of perfectionism in sport. In particular, we must learn more about the

long-term consequences of perfectionistic strivings on performance, as some researchers have suggested that perfectionistic strivings, while boosting performance in the short run, are detrimental to sustained performance and athletic development because they may have negative consequences in the long run (e.g., athlete burnout). Third, we must know more about how perfectionism affects the team (e.g., team cohesion, team performance). While a number of studies have investigated perfectionism in athletes engaged in team sports, these studies have focused on how individual athletes' perfectionism affects the individual athletes but not on how it affects the team. Finally, until now most of the research on perfectionism in SEP has focused on sport, and only very few studies have looked into exercise. More studies are needed to investigate what role perfectionism plays in physical activity (PA) and exercise behavior.

Conclusion

Although some researchers suggest that perfectionism in sports is a negative characteristic because it undermines athletes' performance and stifles athletic development, empirical evidence shows that perfectionism has a dual nature with both negative and positive sides. This dual nature is represented in the two main dimensions of perfectionism—perfectionistic concerns and perfectionistic strivings—that show different and unique relationships with athletes' emotion, motivation, and performance. The pattern of findings suggests that only perfectionistic concerns are clearly maladaptive. In contrast, perfectionistic strivings are associated with emotional and motivational qualities that may give athletes an additional “boost” to make an extra effort and achieve the best possible result. Even some clinical psychologists are beginning to recognize that perfectionistic strivings may form part of a healthy pursuit of excellence and, as such, are not unhealthy, dysfunctional, or maladaptive. However, this may be the case only when perfectionistic strivings are not accompanied by perfectionistic concerns because perfectionistic concerns are clearly unhealthy and maladaptive. While these concerns may have no direct negative effects on sport performance, they represent a serious risk to athletes' psychological well-being (PWB) as well as their motivation, self-esteem, and health.

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See also Extraversion–Introversion; Iceberg Profile; Personality Tests; Type A/B Personality

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PERFORMANCE-ENHANCING DRUGS

Performance enhancement is of major importance in competitive and elite sports. Among the many methods to enhance performance in sports, such as specialized training and diet regimes, some athletes seek resort to performance-enhancing drugs (PEDs). Following the drug-related death of British cyclist Tom Simpson during the 1967 Tour de France, a controversy erupted concerning the extent of drug use in sports and the health consequences of PEDs. Today, PEDs are broadly categorized into licit and illicit substances.

Illicit Performance-Enhancing Drugs

Since 1999, the World Anti-Doping Agency (WADA) has been regulating the use of illicit PEDs in sports. WADA annually issues an official list of prohibited PEDs and regulates drug testing and doping control procedures in collaboration with national anti-doping organizations (NADOs) across the world. The WADA list of prohibited substances distinguishes between substances that are banned at all times (both inside and outside

competition), substances prohibited only in-competition, as well as substances prohibited only in particular sports.

The first class of substances (i.e., those that are banned at all times) includes mainly anabolic-androgenic steroids (AAS), such as human hormone testosterone and its synthetic or natural derivatives (e.g., androstenediol, androstenedione, DHEA, and 7-keto DHEA). Other anabolic agents in this category include clenbuterol, selective androgen receptor modulators (SARMs), peptide hormones, growth factors, and related substances (e.g., growth hormone, insulin-like growth factor-1 or IGF-1, and erythropoiesis-stimulating agents), beta-2 agonists, diuretics, and masking agents.

Prohibited in-competition substances include stimulants, narcotics, cannabinoids, and glucocorticosteroids. Additionally, some sporting associations prohibit the use of alcohol and beta-blockers. Some of the prohibited stimulants are considered legal if they are found in relatively low concentrations in the athlete's urine during drug testing. Also, certain prohibited PEDs are allowed to be used for therapeutic reasons (Therapeutic Use Exemptions [TUEs]) and under medical supervision. This related mostly to narcotic analgesics and other drugs that can have a therapeutic value.

Health Effects

The use and abuse of illicit PEDs, such as amphetamines and anabolic steroids cause various psychological and physical health symptoms, such as irritability, aggressiveness, violent outbursts, depression, suicidal thoughts, acne, baldness, infertility and hypogonadism (in males using AAS), facial hair and deepening of the voice (in females using AAS), heart disease, and liver toxicity. Prolonged or excessive use of growth hormone has also been linked to the development of cancer, disfigurement of facial bones, and sudden death.

Epidemiology of Illicit Performance-Enhancing Drugs in Professional and Amateur Sports

There is a plethora of known cases of elite athletes using or abusing illegal PEDs. However, establishing reliable and accurate prevalence estimates of illicit PEDs in sports represents a difficult task for numerous reasons. First, the use of biomedical testing has proven problematic in relation to the high financial and administrative costs involved. Second, current testing procedures often fail to

detect newly introduced, prohibited PEDs. Third, biomedical testing can only be applied to a small subsample of the total population of elite athletes, thus some athletes may be regular users of prohibited PEDs but escape testing. Finally, self-reports of drug use are also problematic, as elite athletes are reluctant to disclose drug use patterns. Therefore, while anecdotal evidence suggests that doping in elite sports may be more than 50%, official reports only support that the prevalence of illicit PEDs range from 5% to 10%. Clearly, there is still a great deal of work to be done concerning the epidemiology of illicit PEDs in sports.

Notably, several studies indicate that the use of illicit PEDs is not limited to competitive sports. Rather, illicit PEDs are also used in nonprofessional sports. Amateur athletes and adolescents engaged in recreational sports and leisure time physical activity (PA) self-report use of anabolic steroids and other banned substances. Some studies show that around 6% of adolescent athletes involved in organized sports in the United States have used AAS at least once in their lifetime. Other studies show that at least 10% of amateur, youth athletes, or leisure time exercisers attending fitness centers engage in illicit PED use. These findings are alarming because they show that the use of illicit PEDs is not a problem limited to the elite sports community. The reported prevalence rates among larger populations of leisure time exercisers and noncompetitive athletes indicate that illicit PED use is a growing and emerging public health risk. In fact, it can be argued that the health benefits of exercise and PA can be offset by the use of illicit PEDs, especially among chronic users and younger people.

Licit Performance-Enhancing Drugs

Nutritional supplements and licit performance-enhancement substances provide an alternative to illicit drug use in sports with no to minimal side effects on users. These products include a wide variety of substances, such as proteins, herbal-based products (e.g., *Tribulus Terrestris*), creatine, amino acids, stimulants (e.g., caffeine), multivitamins, and diet drugs. There is evidence that most of these substances yield benefits to physical health (e.g., the antioxidant properties of vitamins and herbal products support immune functioning), and increase athletic performance in diverse ways. Given their “legal” nature, nutritional

supplements are widely used in both professional and amateur sports, across age groups, and even among people who engage in exercise for recreational purposes.

Nevertheless, there are several reasons to question the apparent legality and safety of licit performance-enhancement substances. First, nutritional and dietary supplements are not subject to strict testing and scrutiny by official regulatory authorities, such as the Food and Drug Administration (FDA). Consequently, many supplements may be contaminated with illegal PEDs, commonly anabolic steroids, or with other toxic substances, such as heavy metals. Such contamination can pose serious threats to consumer health and safety. Second, official reports reveal that elite athletes tend to abuse legal nutritional supplements by concurrently using up to 20 different substances, a practice known as “stacking.” The lack of proper regulation and labeling of the ingredients and the uncontrollable use of dietary supplements raise questions about the safety of licit PEDs. Specifically, unknown interactions between substances, and liver toxicity resulting from prolonged use or abuse of PEDs, may prove health-threatening. Finally, several studies suggest that the systematic use of licit PEDs may act as a gateway to the use of illicit substances, especially among junior and early-career athletes. Thus, while legal PEDs can yield benefits to the user, there are major concerns surrounding safety of use. Also, the use of licit PEDs may enforce a mentality of drug-assisted performance enhancement, and this may increase the risk to use illicit PEDs.

Determinants and Consequences of Performance-Enhancing Drug Use

The main reasons for PED use include weight management, improvement of physical stamina, building leaner muscle mass and burning fat, and/or gaining other health benefits, such as a healthier skin or countering the effects of aging. In this respect, body image disturbances (e.g., body checking, investment in appearance) and social physique anxiety (SPA) can be related to the use of PEDs. Further, nutritional supplement users, particularly adolescents, are willing to engage in excessive use of supplements to achieve a better physique or improve performance, and tend to overlook the health risks involved in excessive use of nutritional supplements.

Prevention and Education

Drug control mainly applies to the use of illicit PEDs in sports, as the use of legal performance-enhancement substances is still unregulated by official international sporting associations. The ATLAS (Adolescents Training and Learning to Avoid Steroids) and ATHENA (Athletes Targeting Healthy Exercise and Nutrition Alternatives) projects are based on education, awareness-raising, and training of adolescent athletes to increase abstinence from licit and illicit PEDs.

Conclusion

PEDs can be either licit or illicit, and they are used mainly to increase athletic performance and improve physical appearance. The use of illicit PEDs is related to adverse health problems and is banned from international sporting organizations. Athletes using such substances face legal sanctions and career termination. The potential negative outcomes of illicit PEDs may explain why the use of licit PEDs, typically known as nutritional or dietary supplements, has expanded considerably in both athletes and nonathletes. Licit PEDs may bring several health benefits to the user and assist performance enhancement. Nevertheless, careless and excessive use of nutritional supplements and lack of proper regulation of the supplements' actual ingredients pose threats to the health of users. Most importantly, using nutritional supplements from an early age may reinforce polypharmacy trends and accordingly facilitate transition to illicit PEDs (the gateway hypothesis). Preventive efforts and educational approaches for illicit PED abstinence, as well as the safe use of licit PEDs are needed. Action should be also taken to regulate the ingredients and labeling of legal PEDs, so as to reduce the risk of contaminated products.

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See also Diet Drugs; Drug Use and Control; Fair Play; Narcotic Analgesics; Recreational Drugs

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PERSONALITY AND PSYCHOLOGICAL CHARACTERISTICS OF ATHLETES

Personality has been defined as the relatively stable organization of an individual's character, temperament, intellect, and physique, which shapes the individual's behavior and his or her actions in a given situation. For each individual, there are core personality components that are quite stable and unchanging. However, when one observes patterns of behavior in an individual not only his or her base psychological core should be considered but also the dynamic organization within the individual that determines his or her unique adjustment to the environment. In essence, the interaction between the core trait and the peripheral (e.g., state) aspects of the individual's behavior defines his or her personality.

This interaction between the core traits and peripheral aspects in the individual is expanded in a current view on personality. This view, called the "New Big Five," is composed of five different aspects associated with the concept of personality:

(1) an individual's unique variation on the general evolutionary design for human nature, expressed as a developing pattern of (2) dispositional traits, (3) characteristic adaptations, and (4) integrative life stories complexly and differentially situated (5) in culture. According to this view, the psychology of personality should provide an integrative framework for understanding how each person is like all other persons (understanding species-typical characteristics of human nature), like some other persons (understanding individual differences in common characteristics), and like no other person (understanding unique patterns of an individual's life).

Based on the New Big Five approach, a *psychological profile* is often used to describe the type of personality that is exhibited by the individual (e.g., an athlete). For example, elite performers in sport are expected to be highly motivated toward practicing hard, aimed at enhancing their abilities and skills, focusing on their performance, and coping effectively with physical (e.g., illness, injury) and mental (e.g., failure, lack of playing time) barriers. A unique psychological profile (i.e., a "sport personality") consisting of sport-relevant psychological characteristics is required to attain high performance standards. In addition, individuals act in given settings that also have the potential to influence their personality. In sport, athletes are typically required to execute acts in particular settings (e.g., practice sessions, competitions, games), where they are exposed to specific task and environmental demands. Under such conditions, observed behavior depends largely on the interaction between the athlete's core personality and the environmental demands. Therefore, *personality* is composed not only of core psychological characteristics but also of an arsenal of plans, strategies, and skills that individuals implement to achieve their goals. The integrated approach of the core psychological characteristics (i.e., "hard" aspect) and the skills and strategies (i.e., "soft" aspect) that are used by the individual is adopted in our discussion on personality and psychological characteristics of elite athletes.

Since the early 1960s, researchers in sport psychology (SP) have used introspective tools to measure the personality profile of athletes. Two of the earliest tools for measuring personality were the Minnesota Multiphasic Personality Inventory (MMPI) and the Cattell 16PF Questionnaire. These personality tests were typically given to

adult athletes who had already attained the highest level of performance in their sport. The personality scores of these high achievers were compared with those of nonathletes or with general population norms.

Personality tests failed to associate specific traits with sport performance and ignored the unique psychological characteristics related to sport performers. Consequently, sport-related psychological skills questionnaires were developed. An example of one of these questionnaires was the Psychological Skills Inventory for Sports (PSIS), developed by Michael J. Mahoney, T. J. Gabriel, and T. S. Perkins, which measured the psychological skills athletes use in practice and competition. This inventory and others, such as the Self-Analysis of Mental Skills (SAMS), developed by J. R. Grove and S. J. Hanrahan, and the Ottawa Mental Skills Assessment Tool (OMSAT), developed by Jordache D. Bota, were administered to elite and novice athletes. Specifically, the psychological strengths and weaknesses of the novice and elite athletes were assessed. In some of the studies using questionnaires related to sport, a number of methodological (e.g., sample size or sample availability) or conceptual (e.g., the difficulty in defining the term *elite athlete*) concerns were noted, and therefore, determining the psychological characteristics associated with skill levels was limited and still remains inconclusive.

To overcome the methodological and conceptual barriers associated with personality-testing instruments, such as the MMPI or PSIS, in-depth interviews and narrative stories have been among the alternative methodologies used for determining sport-related psychological characteristics. In the in-depth interviews, athletes are asked about the psychological attributes and characteristics that most helped them to attain their athletic performance. From the narrative stories (e.g., autobiographies, biographies told by athletes), researchers obtained unique information on the athletes' personal journey of "climbing to the top," which helped to provide insight into the psychological characteristics or skills central to the athletes' achievements.

Psychological Characteristics and Skills of Adult Elite Athletes

A number of psychological characteristics and skills of elite athletes have been identified from

the extant research using questionnaires and interviews. These are outlined next.

Psychological Characteristics

Among the psychological characteristics of elite athletes are (a) *commitment* (the dedication to devote oneself to all activities associated with the selected sport from early phases of talent development until achieving sport expertise); (b) *internal motivation* (the internal drive of the athlete to direct his or her effort toward achieving high-standard goals, such as the willingness to be involved in a multiphased task-pertinent training program for an extended period of time, the desire to excel in competition, and the ability to cope with feelings of discomfort and failure); (c) *learning capability* (the ability to learn complex sport skills); (d) *control* (the command of the ongoing events on and off the court, field, or gym); (e) *competitiveness* (the ability to compete against others and strive for success); (f) *self-confidence* (the belief in one's ability to achieve the best results while overcoming physical and psychological barriers); (g) *adaptability* (the ability to perform well under unfavorable conditions); and (h) *mental toughness* (the ability to control emotions and thoughts, and to remain focused under challenging and stressful conditions, particularly under negative circumstances).

The aforementioned psychological characteristics have been observed in athletes in both team and individual sports. However, in some early studies comparing the personalities of athletes of various sports with nonathletes, it was evident that athletes who played team sports demonstrated less abstract reasoning, more extraversion, more dependency, and less ego strength than nonathletes. Athletes who competed in individual sports exhibited higher levels of objectivity and dependency, and less abstract thinking, compared to nonathletes. In regards to the female athlete, early research showed that female athletes were more achievement-oriented, independent, aggressive, emotionally stable, and assertive than female nonathletes.

Psychological Skills

Elite athletes have been found to possess various mental skills that are at significantly higher levels than those of less-than-elite athletes. Specifically, the following psychological skills have been found to be used by elite athletes and associated with

their efforts to achieve high levels of proficiency in their sports: (a) *goal setting* (the ability to set challenging but realistic goals for short- and long-term periods); (b) *imagery* (the ability to imagine oneself successfully performing a sporting act); (c) *focusing attention* (the ability to focus effectively, ignore external distractions, and avoid negative internal thoughts); (d) *emotional control* (the ability to stay calm when feeling anxious or nervous or to become "psyched-up" when not appropriately aroused or excited); and (e) *psychological routines* (the use of pre-, during-, and postperformance routines to self-regulate behavior and emotional states before, during, and after performing the sporting act).

Psychological Characteristics and Skills of Young Athletes

Not only have psychological characteristics of elite adult athletes been studied but also those of young individuals in earlier phases of talent development (e.g., children up to the age of 14). Researchers interested in talent detection and early development in sport identify age 14 as the final year of *early involvement* in sport and the beginning of the *specialization phase* (i.e., focusing solely on one sport activity). Based on data obtained by researchers such as Craig R. Hall and Tara Scanlan, who administered questionnaires to young athletes involved in competitive sport programs in basketball, gymnastics, hockey, soccer, and wrestling, as well as conducted interviews with these athletes, it was found that young athletes showed more persistence than nonathletes. In terms of psychological characteristics and skills, it was observed that (a) the level of trait confidence in the young elite athletes was negatively related to the level of state competitive concerns; (b) motivation to participate in sport was more intrinsic than extrinsic in the young elite athletes—more specifically, young elite athletes more often reported skill development, team affiliation, fun, excitement of the game, liking to compete, and liking to do something they are good at as important motives for participating in sport; and (c) using imagery as a psychological skill is common among young athletes, but the type of imagery and the reasons for using imagery vary. For example, the types of imagery used by the young athlete were visual, auditory, kinesthetic, or tactile. Some of the athletes used imagery for developing game strategy while others used it only to serve individual goals. In addition, it was found

that gender differences exist in some, but not all, psychological characteristics. Girls reported professional or international ambitions less often than boys and reported enjoyment as a motive for playing sports more often than boys. More boys than girls attributed their success to physical factors.

Can Sport Build Character?

A question that is often asked by researchers and practitioners deals with the contribution of sport to the development of psychological characteristics. Specifically, “Can sports help athletes build character?” Although the belief that sport enhances psychological development is widespread, research in SP has yet to provide clear-cut evidence supporting this assumption. For example, studies have examined the influence of sport competition on the prosocial behavior of cooperation and altruism in children. Data from these studies showed that sport competition had a negative effect on prosocial behavior and that children who gained more experience in competitive sports were significantly less altruistic than those who accumulated less experience. In other studies aimed at examining the contribution of organized sport programs to the moral behavior of young children, it was observed that some aspects of moral behavior did improve (e.g., attitudes concerning sportsmanship, and moral reasoning). However, some of the sport activities that were performed in these studies failed to reflect actual competition or game situations, and therefore the ecological validity of the studied activities was relatively low.

The question of whether elite athletes are born with sport-relevant psychological characteristics remains unsolved. Additional studies incorporating personality and genetics may advance our knowledge on this issue. At present, more effort should be devoted to research tools and methodological aspects deemed appropriate for the study of psychological characteristics of athletes.

Ronnie Lidor

See also Extraversion–Introversion; Iceberg Profile; Optimism; Perfectionism; Personality Tests; Type A/B Personality

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PERSONALITY TESTS

Personality is typically defined as a person's distinctive and enduring (i.e., cross-situational) thoughts, emotions, and behaviors that characterize the person's reactions to life situations. Personality traits are also defined as specific properties that predispose a person to react in certain ways in given classes of situations—sport, for example. Some personality traits are genetically determined and formed in early childhood. The terms *traits* and *dispositions* are often used interchangeably (e.g., personality dispositions). Traits are not the same thing as orientations or styles, however.

Whereas traits are permanent, enduring, and not susceptible to change, *orientations* or *styles* reflect a person's preference, or tendency, to think or act in a predictable manner given a specific set of circumstances. An athlete's competitive orientation, for example, reflects his or her tendency to feel more motivated and demonstrate higher quality performance under competitive situations, as opposed to practice settings or other low competitive environments (e.g., recreational sport).

Behavioral tendencies, which are similar to and often expressed as *routines*, reflect the athlete's observable actions that are performed regularly, often leading to one or more routines. A baseball player's tendency to unwrap, then rewrap his or her batting gloves between pitches, the chewing and spitting of tobacco, eating a particular pregame meal at a specific time, or engaging in any behavior pattern that is predictable and observable would be classified as behavioral tendencies.

For many years, researchers, coaches, athletes, and sport psychology (SP) practitioners have wanted to know if it is possible to explain or predict the behavior of high-quality (elite level) sports competitors based on a "paper-and-pencil" personality inventory. Can we predict a person's—especially a child's—future success in sport? The ability to accurately determine psychological characteristics of athletes that both describe and predict future sport success has two important implications. First, ideally, an athlete's responses to a questionnaire would allow coaches, parents, and sport psychologists to invest time, equipment, coaching, and financial resources on promoting a person's future participation in sport, particularly at advanced levels. Second, determining strong evidence for possessing certain psychological characteristics as a successful athlete would allow sport psychologists to provide interventions that promote these desirable psychological features. For instance, knowing how elite-level athletes deal with anxiety, cope with stressful events, maintain concentration, regulate their arousal level, and develop confidence would suggest that lower-skilled athletes can learn particular mental skills and apply them under the right circumstances.

An example of one of these characteristics is mental toughness, which has gained great interest by SP researchers in recent years. Describing and identifying mental toughness requires validating an inventory that accurately measures it. Then, researchers would have to conclude that mental

toughness is a characteristic that is desirable and unique to elite-level athletes. Predictably, coaches would want their athletes to possess high mental toughness scores, and sport psychologists would provide interventions that improve mental toughness. However, is mental toughness, not unlike other dispositions such as confidence and coping style, a personality trait? Or is it a thinking tendency, called a disposition or orientation? The ability to differentiate personality, which is not subject to change through training, from related concepts such as styles, orientations, dispositions, and behavioral tendencies is a limitation in the sport personality literature.

Pros, Cons, and Guidelines for Using Personality Inventories

Researchers and practitioners have examined whether competitive athletes have unique personality traits, as compared to nonathletes, or if athletes' personality traits differ as a function of skill level, gender, or sport type. Results of selected studies measuring psychological characteristics suggest that elite athletes score high for dispositions such as competitiveness, confidence, optimism, proper coping styles, and self-control, and low in state (situational) anxiety and helplessness. Elite athletes of both sexes are particularly adept at managing—not eliminating—stress, anxiety, and arousal. These characteristics have been strongly linked to performance success. Nevertheless, sport personality testing has several limitations.

Poor Prediction Rates

For instance, researchers have shown that inventories can predict athletic behavior and success only 8% to 10% of the time.

Attempts to Measure Changes in Traits

Personality scales have been used inappropriately to examine changes in personality traits over time—for instance, before versus after the season, or the failure to establish baseline scores. This is incompatible with the design of any instrument that examines personality, because by definition, personality traits are stable and enduring. Consequently, results cannot be interpreted to measure personality change. An example would be researchers who measure changes in athletes'

anxiety at various points before sport competition using a trait anxiety (TA) inventory; changes in TA should not be expected.

Inconsistent Use of Terms and Traits

Sometimes certain terms and factors used in personality scales are not universally defined so that the results of studies lack generalizability (i.e., external validity). For example, determining who is an athlete has differed among studies (e.g., participants in recreational competition as opposed to competitors on a school team or at more elite levels). A related limitation occurs when personality traits (e.g., sociability, ego strength, shrewdness, dominance) are defined differently, depending on the inventory being used, limiting the ability to accurately measure and generalize findings beyond the present sample.

Inappropriate Use of Inventories

The traditional personality inventories used in sport personality research are often not created for sport participants. For example, the Minnesota Multiphasic Personality Inventory (MMPI) was originally meant to diagnose mental illness. The Cattell 16PF Questionnaire and the California Personality Inventory (CPI) inherently assume that each of many factors is interpretable in sport situations. The CPI requires a reading comprehension level equal to about the 10th grade, making younger athletes or persons with poor reading skills ineligible for this assessment tool. These inventories do not include a single item related to thoughts, emotions, or behaviors in competitive sport situations.

Using the Cattell 16PF Questionnaire has shown that athletes tend to be more stable, extraverted, tough-minded, and highly efficient, as compared to nonathletes. In other (European) studies, the MMPI has shown male soccer players to be aggressive, intelligent, and dominant. American football athletes differ from competitors in other sports on traits called tough mindedness, extraversion, and self-control. Critics of these findings, however, question the relevance and real-world application of these traits.

In support of using psychological profiles, highly successful sport performers do differ from less successful competitors. For example, elite female athletes, as compared to non-elite athletes, score higher for dominance, aggressiveness,

adventurousness, sensitivity, independence, self-sufficiency, and introversion. In addition, female sports competitors in general tended to be more assertive, dominant, self-sufficient, reserved, achievement oriented, and intelligent, while lower on emotionality, than female nonathletes. Practitioners question, however, whether these findings have relevance in SP consulting or in any other aspect of sport.

Poor Sampling Techniques

Past attempts at examining sport personality have included examining several variables in the same study (e.g., team vs. individual sports) without controlling for skill level, age, gender, and cultural differences. This flaw makes it impossible to draw conclusions about identifying a set of personality traits associated within a specific group or cohort.

Despite flaws in research tools and other limitations, sport psychologists generally acknowledge that highly skilled athletes score relatively low in neuroticism, tension, depression, anger, fatigue, and confusion. They tend to score very high in self-confidence, self-concept, self-esteem, vigor, need achievement, dominance, aggression, intelligence, self-sufficiency, mental toughness, independence (autonomy), sociability, creativity, stability, and extraversion. Whether these results are meaningful in identifying and consulting sports competitors remains questionable. Here are a few sample sport personality traits that have been studied over the years.

Risk Taking

One characteristic of highly successful competitors is risk taking. Risk taking is a function of narrowing the margin of safety, both physically in terms of bodily harm and psychologically in terms of the probability of success or failure. Elite athletes seem to thrive on and to prefer the excitement of engaging in risk-taking behaviors.

Stimulus Seeking

A psychological disposition of elites that is similar to risk taking is called stimulus seeking, or sensation seeking. Athletes prefer situations that foster tactile and other forms of sensory stimulation as high stimulus-seekers. Stimulus seeking is a motivational factor to participate in sport and to engage in the risk-taking behaviors.

Competitiveness

The extent of the desire to win consists of three dimensions: competitiveness (desire to strive for success in competition), win orientation (focus on winning and avoiding losing), and goal orientation (focus on personal goals). Athletes score higher on all three dimensions, with competitiveness being the major discriminator.

Expectations for Success

Highly skilled athletes have high expectations of success: they expect to win, which improves their concentration and attentional focusing, and fewer distracting anxious thoughts.

Mental Toughness

This orientation concerns an athlete's propensity to reach and sustain under pressure high performance—the athlete's ideal performance state. Mental toughness is learned, not inherited. Mentally tough competitors are self-motivated and self-directed (their energy comes from internal sources; it is not forced from the outside), in control of their emotions, calm and relaxed under fire, and highly energetic and determined.

Fear of Failure

Fear of failure (FOF) is a motivational disposition to avoid failure: a core belief that attaining desired goals is not likely. Persons who fear failure are likely to avoid challenging tasks. Better athletes successfully manage their anxiety and have low FOF.

Fear of Success

Some athletes fear the social and emotional isolation that accompanies success (i.e., fear of success, or FOS). An athlete may feel that performing at a level far superior to that of teammates or peers will result in social discomfort, even ostracism. In addition, some athletes prefer not to deal with the pressure to constantly match or exceed their previous best performance. Trying to live up to the expectations of spectators, coaches, parents, and the media can place an extensive amount of pressure on athletes to reach or exceed their previous best performance.

Can Personality Tests Predict Future Talent?

There is considerable scientific evidence that each of us has an optimal level of performance quality

called *motor capacity* that is genetically determined. Our genetic capacity to perform sport skills, which cannot be altered, is an important reason psychological inventories do not predict a world-class champion. Predicting an athlete's future success in sport through psychological testing is at the heart of a process called *talent identification* (TID).

The primary purpose of TID is to identify a child's sport in which they are most likely to succeed and to encourage them to participate in that sport or those sports. TID is also used to screen young athletes to determine which individuals are most likely to succeed in sport and then to direct them toward the sport(s) to which they are most suited. TID is the process of recognizing individuals currently involved in sport who have the potential to become elite athletes. To date, the scientific literature does not support using personality profiles or any other type of psychological inventory to select or eliminate those individuals based on test score.

Conclusion

The following conclusions about sport personality appear to reflect the current literature:

1. Elite athletes are self-confident, have a high need to achieve, and maintain a relatively high self-image, self-esteem, and confidence in the sport environment as opposed to their non-elite counterparts.
2. Highly skilled athletes feel anxiety, stress, and other negative emotions but manage better than their less successful counterparts.
3. Highly skilled athletes tend to be mentally tough, resilient, and tenacious in dealing with the array of "storms" that competitive sport regularly presents.
4. Elite-level athletes, particularly from contact sports, are more tolerant of acute pain than nonathletes.
5. Personality scales can predict an athlete's future success in sport only 8% to 10% of the time. This is cause to avoid personality scales for prediction purposes. They should also be avoided as part of TID programs.
6. Finally, differences exist between personality traits, dispositions, styles, orientations, and behavioral tendencies, and some are more susceptible to change than others.

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See also Extraversion–Introversion; Iceberg Profile; Optimism; Perfectionism; Personality and Psychological Characteristics of Athletes; Type A/B Personality

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PERSONALITY TRAITS AND EXERCISE

In the past 30 years, personality researchers have amassed a considerable body of evidence to support the importance of traits as crucial determinants of behavior. This research has shown that personality is structured similarly across over 50 cultures, shows evidence of genetic heritability, has high stability across time, and does not relate strongly to parental rearing style. Although several pathways for how personality interacts with health have been postulated, most physical activity (PA) research has been based on the hypothesis that personality affects the quality of one's health practices. More specifically, personality is hypothesized to affect social cognitions (e.g., perceptions, attitudes, norms, self-efficacy) toward a behavior, which in turn influence the health behavior itself.

Personality Trait Structure

Although several models of trait personality have been researched and postulated, the dominant two frameworks are Hans Eysenck's three-factor model and the five-factor model (FFM). Eysenck originally developed his theory with two factors: extraversion–introversion (e.g., tendency to be sociable, assertive, energetic, seek excitement, and experience positive affect) and neuroticism–emotional

stability (e.g., tendency to be emotionally unstable, anxious, self-conscious, and vulnerable). He later added a third trait called psychoticism–superego (e.g., risk taking, impulsiveness, irresponsibility, manipulateness, sensation seeking, tough-mindedness, nonpragmatism) to his model. The most popular personality model at present is a five-factor taxonomy. This model suggests that neuroticism, extraversion, openness to experience and intellect (e.g., tendency to be perceptive, creative, reflective and appreciate fantasy, and aesthetics), agreeableness (e.g., tendency to be kind, cooperative, altruistic, trustworthy, and generous), and conscientiousness (e.g., tendency to be ordered, dutiful, self-disciplined, and achievement oriented) are the basic factors of personality structure.

Both the three- and five-factor taxonomies are thought to represent the basic building blocks of personality and subsequently cause the expression of more specific subtraits. Thus, individuals high in extraversion may express this higher-order trait through excitement seeking, sociability, a positive outlook, energetic activity, or another possible outlet that is feasible and socially conditioned from one's environment. Mixtures of these five factors of personality may also produce traits of interest. For example, type A behavior may be a combination of high extraversion, high neuroticism, low agreeableness, and high conscientiousness. While it is important to continue with a higher-order super-trait understanding of personality and behavior, these facet or specific traits may also help define the relationship between personality and specific behaviors like PA.

Personality and Physical Activity

In 2006, Ryan E. Rhodes and N. E. I. Smith reviewed and systematically appraised the relationship between personality and PA for the three- and five-factor models of personality. Overall, neuroticism showed a small negative relationship with PA, while extraversion and conscientiousness had positive correlations. The most compelling evidence for these relationships is from a 40-year longitudinal trial conducted by Margaret L. Kern and colleagues in which neuroticism and extraversion were found to be predictive of PA changes across time. Agreeableness and openness to experience did not have a relationship with PA. Eysenck's third construct of psychoticism also had no evidence of a relationship with PA.

Extraversion concerns the differences in preference for social interaction and lively activity; the seeking of PA behaviors appears a logical extension for people high in this trait while the disinterest in PA seems likely for those low in extraversion. Individuals with high neuroticism are those with less emotional stability and more distress, anxiety, and depression than those with lower neuroticism. Avoidance of PA or cancellation of PA plans is a logical extension of this trait. High scores of conscientiousness represent a purposeful, self-disciplined individual, suggesting that this factor may be important in terms of adherence behavior. The predisposition to maintain PA behavior appears to be logical for individuals who possess higher conscientiousness than their low conscientiousness counterparts.

Rhodes and Smith and a follow-up review also examined whether other variables interacted with the relationship between personality and PA. The stability of personality suggests that it should be relatively invariant to most factors across demographics and cultures and this is generally what was found. They concluded that no reliable differences by gender, age, and study design or instrumentation were evident. They did notice a potential trend by country, where extraversion had a larger correlation with PA in Canada and the United States than in the United Kingdom. They also found that strenuous and moderate intensity modalities of PA may be associated with personality factors of neuroticism, extraversion, and conscientiousness more than lower intensity activities.

Rhodes and Leila A. Pfaeffli reviewed the evidence for several of the more specific facet and lower-order traits and their relationship with PA. Sociability (preference for the company of others and gravitation to social situations) and dispositional optimism (generalized expectations of positive outcomes) were not related to PA, but type A, activity, and industriousness-ambition showed some evidence as correlates.

Type A personality gained popularity from its association with coronary heart disease and it is marked by a blend of competitiveness and hostility with agitated behavior and continual movement patterns; thus, PA could conceivably be a natural extension of type A individuals. Rhodes and Pfaeffli reviewed six studies that have appraised the relationship between type A and PA. Overall, five of the six studies in nonclinical populations showed some significant positive association

between type A and PA in the small to medium effect size (ES) range.

Extraversion's activity facet represents a disposition toward a fast lifestyle, being high energy, fast talking, and keeping busy as opposed to a more *laissez faire* disposition. These properties could conceivably make regular PA a behavior of choice given its energy demands. Rhodes and Pfaeffli's literature review identified six studies that have examined the association of the activity facet with PA. In all cases, the activity facet showed a medium to large correlation with behavior. More compelling, the three studies directly comparing the predictive capacity of activity against the superordinate dimension of extraversion showed the superiority of the activity facet. Overall, the results suggest that extraversion's activity facet is a reliable and strong predictor of PA.

Conscientiousness's facet of industriousness-ambition has also received attention in four studies. The facet comprises aspects of achievement-striving and self-discipline and a natural extension of this type of disposition could be regular exercise given its challenge, impact on health and appearance, and self-regulatory barriers. Three of the four studies found a significant relationship between this facet and behavior, suggesting it may be the critical link between conscientiousness and PA.

Mediation and Moderation Tests of Physical Activity and Social Cognition

Personality theorists and social psychologists generally agree that behavioral action is unlikely to arise directly from personality. Instead, personality is thought to influence behavioral perceptions, expectations, and cognitions. The theory of planned behavior (TPB) has been the leading model to test this assumption in the PA domain with 16 of these tests. According to Icek Ajzen, personality should affect behavior through the constructs of attitude, subjective norm, and perceived behavioral control. Thus, the TPB should mediate personality and PA relations. This test, however, was only supported in 3 of the 16 studies. The remaining tests showed evidence for only partial mediation. Overall, the results suggest that attitudes, social norms and perceived control may not always fully mediate the relationship between personality and PA.

Another way that personality may affect behavior is by interacting with PA motivation. A review

of factors that interact with the intention-behavior relationship found convincing evidence that conscientiousness may alter this relationship. The basic premise for this finding is that people who are more conscientious are more likely to act on their good intentions than their less conscientious counterparts. It may be that the disposition toward achievement keeps high-conscientiousness individuals from slipping in their original PA plans. There is also some evidence that extraversion may interact with intention. The rationale behind this proposed relationship is that individuals high on extraversion may facilitate their intentions through gravitating toward active environments than more introverted individuals.

Personality and Intervention

Based on the enduring and stable nature of traits, it seems it would be a difficult enterprise to plan an intervention to change personality. Personality traits have been shown to change, particularly in adolescence, but change among adults generally unfolds very slowly across the life cycle. Personality should resemble other intractable correlates of PA such as age, disability, or gender. Introverts or less conscientious individuals may need targeted interventions to help them increase PA despite a natural inclination to be less active.

The proposal for personality-matched interventions has appeal, but very limited research has been conducted on this approach. Rhodes and D. H. Matheson examined whether a planning intervention among low-conscientiousness individuals could help improve PA over a control group. The effects were null, but it may have been from an ineffective intervention as most of the participants reported that they did not even complete the planning worksheet. By contrast, Yong Peng Why and colleagues examined the effects of a walking intervention and found that messages were more effective in increasing walking behavior among conscientious individuals than their less conscientious counterparts. The results underscore that personality traits may need targeting to help less conscientious individuals.

Conclusion

Personality psychology has a rich and spirited history, but contemporary research has supported the premise that people have enduring and stable individual differences in how they express thoughts,

feelings, and actions. A review by Rhodes and Smith identified extraversion ($r = .23$), neuroticism ($r = -.11$), and conscientiousness ($r = .20$) as the key personality dimensions related to PA. A subsequent review of the more specific facets suggested that extraversion's facet of activity and adventurousness and conscientiousness' facet of achievement striving and ambition may be the critical components that are related to PA. Type A personality, which includes components of both the activity and achievement striving traits (as well as hostility), was also shown as a reliable predictor of PA.

There is some evidence that these traits may affect PA through social cognitive constructs such as attitudes, norms, and perceived behavioral control. Personality may also affect the success of holding to positive PA intentions. Research has been quite convincing in demonstrating that people low on conscientiousness have more difficulty holding to their initial PA intentions than people who are high on conscientiousness. The findings suggest that understanding "at risk" personalities may be important when creating interventions. Interventions that consider one's introversion, for example, and target behaviors accordingly (e.g., home-based, solo or with one friend, low intensity) may have utility. The topic has not received very much research at present. Future research is needed to examine this personality matching strategy.

Ryan E. Rhodes

See also Adherence; Extraversion-Introversion; Genetics/Nature-Nurture Determinants; Leadership in Sport: Trait Perspectives; Personality Tests; Theory of Planned Behavior; Type A/B Personality

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PLEASURE

Pleasure and displeasure comprise a bipolar dimension that is the most important ingredient of core affect. As such, pleasure and displeasure provide texture to conscious experiences and form the foundation of emotions and moods. Furthermore, pleasure and displeasure have long been considered by many philosophers and psychologists as powerful motives. Pleasure is of particular interest to exercise psychologists because, unlike many other health behaviors, exercise can induce substantial changes along the pleasure–displeasure dimension, and these may, in turn, influence subsequent exercise behavior.

Pleasure–Displeasure as the Core Affective Dimension

In 1896, Wilhelm Wundt wrote that, among the “manifold of feelings,” one can distinguish certain “chief affective dimensions,” the first of which consists of “pleasurable and unpleasurable feelings.” In the 1950s, Charles Osgood found that the first dimension to emerge from factor analyses of semantic differential items comprised such adjective pairs as pleasant–unpleasant, good–bad, and positive–negative. This factor explained well over half of the accounted variance. Subsequent analyses of affective adjectives, facial expressions, and emotional ratings of environments have confirmed that the first dimension, accounting for the majority of the variance, refers to pleasure–displeasure.

An interesting question is whether one can feel both pleasure and displeasure at the same time. For example, some runners say that running “hurts so good.” While empirical analyses consistently show

that pleasure and displeasure are polar opposites (i.e., reciprocally related) and cannot co-occur at the exact same time, cognitive appraisals may be independent. Thus, someone can feel “pleased” about the fact that exercise induced a 10-pound weight loss and “displeased” that exercise takes so much time. In these cases, the “pleasure” and “displeasure” reflect conflicting cognitive appraisals. Likewise, a runner may experience “hurt” (a direct, unpleasant sensation from the body) while running but also feel “good” when appraising running as the means to enhanced fitness and better running performance.

Adaptational Properties

In the early 1900s, Herbert Spencer proposed that pleasure and displeasure represent evolutionary adaptations. Their functions make sense when considered within the context of what John Bowlby later called the “environment of evolutionary adaptedness,” the environment that shaped hominid evolution. From this perspective, pleasure and displeasure spread through the population because they conferred a crucial adaptational advantage. Pleasure signifies usefulness and provides an inducement to approach stimuli that promote Darwinian fitness (i.e., survival and reproduction). Displeasure signifies danger and acts as a deterrent.

These ideas also apply to exercise-induced changes in pleasure and displeasure. Exercise can be “useful” in a Darwinian sense, since it provided the only means of subsistence for hunters and gatherers and it can maintain health and vitality. Therefore, it would make adaptational sense to link exercise to pleasure as a means of facilitating this useful activity. On the other hand, exercise can also increase the risk of injury (e.g., skeletal, muscular, cardiovascular) and death (e.g., from heat exhaustion or cardiac arrest). So it would also make adaptational sense to link exercise to displeasure as a means of maintaining exertion within safe parameters.

Alliesthesia

Physiologist Michel Cabanac coined the term *alliesthesia* to describe a phenomenon whereby the same stimulus may induce pleasure or displeasure, depending on the condition of the organism. For example, a certain room temperature may feel comfortable in the summer but cold in the winter. Similarly, a level of exercise intensity may be pleasant under normal circumstances but aversive

or intolerable under conditions of homeostatic perturbation (e.g., sleep deprivation, fever, dehydration, hypoglycemia, high ambient temperature and humidity).

Pleasure as “Common Currency”

Ethologists have long grappled with the problem of the forces that determine the behavior of an animal. Animals are subjected to numerous internal and external factors that can influence their behavior. Some of these may be in conflict (e.g., “go” and “stop” or “approach” and “avoid” signals). However, since an animal can only do one thing at a time (move in one direction), there must be a “behavioral final common path,” a mechanism that coverts these factors to a common scale (a “common currency”) so that they can be compared.

Michel Cabanac proposed that pleasure is this common currency. Pleasure is an integral component of all experiences. Furthermore, evidence indicates that events register in memory not solely as factual information (i.e., what, where, when) but also in terms of the pleasure or displeasure that accompanies them. Pleasure is, therefore, ideally positioned for this role.

Cabanac offered evidence for his hypothesis by investigating behavior under conditions of conflict. In one study involving use of a treadmill, he kept the speed of the treadmill fixed and allowed the participants to choose the slope or kept the slope fixed and allowed the participants to choose the speed. He hypothesized that the choices the participants would make would be intended to maximize pleasure and/or minimize displeasure. Indeed, the choices the participants made were reciprocal, resulting in approximately constant power and constant ratings of pleasure. Moreover, their choices could be predicted from ratings of pleasure–displeasure.

Varieties of Pleasure and Their Implications

Research by Daniel Kahneman has shown that the pleasure–displeasure experienced at different phases of an episode (e.g., a medical procedure) have different implications for subsequent behavioral decisions. Specifically, the peak pleasure or displeasure (called the “peak rule”) and the pleasure or displeasure at the end of the episode (called the “end rule”) are weighed more heavily. On the other hand, how long the episode lasts and, therefore, the total amount of pleasure or

displeasure experienced appears inconsequential (called “duration neglect”).

These principles have interesting implications for exercise. For example, an implication would be that, as long as exercise practitioners ensure that (a) pleasure is not reduced during exercise, so as to induce a negative “peak” and (b) a cooldown is provided, to create a pleasant “end,” the duration of the activity can be extended without adverse consequences for motivation. A study by Britton Brewer and collaborators showed that, in fact, participants prefer a longer and more energetically costly bout as long as intensity is reduced at the end to increase pleasure.

Panteleimon Ekkekakis

See also Addiction, Effects of Exercise on; Affect; Affective Responses to Exercise; Emotional Responses; Hedonic Theory

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POSITIVE THINKING

Traditionally, sport psychologists have placed great value on athletes thinking positively about upcoming and imminent performances. Compared with practicing sport psychologists who have demonstrated a keen interest in positive thinking, theorists and researchers have not. As a result, a systematic and thorough knowledge base regarding what positive thinking is and why it is so sought does not currently exist. However, it would be reasonable to state that positive thinking is a broad term

encapsulating self-affirmations (e.g., I feel strong, I stay focused under pressure), images of successful goal attainment (e.g., winning an important competition), as well as thoughts reflecting optimism and an unshakable self-belief. Although positive thinking can include recollection of previous sporting successes, the essence of positive thinking is a focus on the present goals at hand (e.g., to successfully “sink” the imminent golf putt) and the likely attainment of these. Both personal (e.g., personality traits such as anxiety) and social (e.g., coach feedback about progress concerning the learning of a skill) factors can influence positive thinking. The current research and theories pertaining to positive thinking are discussed here.

As well as sport psychologists, coaches around the world clearly value positive thinking. For instance, we know that North American and Australian tennis coaches strongly encourage their players to think positively and say positive things to themselves in order to bolster their confidence levels. However, despite its perceived importance, it is only relatively recently that researchers have identified four types of sports-oriented positive thinking: psyching up, anxiety control, confidence enhancing, and instructional. Although instructions are not necessarily inherently positive, it is likely that the mere presence of directive thought assists with a positive mind-set by indicating to the athlete that she or he has some control over the situation. Researchers who have examined how athletes might utilize positive thinking have developed a self-talk questionnaire, the use of which has revealed beneficial relationships between athletes' positive thinking and their mood states (e.g., anxiety, vigor, tension, and boredom).

Perhaps unsurprisingly, positive thinking has been shown to be most strongly and positively related to state self-confidence. Indeed, studies involving athletes being trained in the use of positive thinking have revealed that athletes experience reduced precompetition anxiety and elevated levels of confidence, as well as more facilitative interpretations of the symptoms associated with anxiety. That is, athletes are more likely to view characteristics associated with anxiety (e.g., “butterflies in the stomach,” concern with the task at hand) as being helpful rather than a hindrance for their subsequent performance.

Although the association between positive thinking and performance has not been extensively investigated, based on the limited available

literature, researchers have identified a consistent positive effect of positive statements addressed to oneself on sporting performance. One of the proposed reasons for this effect involves self-confidence; more specifically, positive thinking leads to enhanced confidence, which, in turn, assists skill execution and performance. Drawing on the sources of self-efficacy proposed by Albert Bandura offers some guidance about how to develop a positive frame of mind and increase sporting confidence. For instance, athletes can be trained to think more positively through the use of specific positive self-administered verbally persuasive cues (e.g., I can). Research has indicated that use of these cues by moderately skilled tennis players prior to task execution improves both their confidence and ability to execute ground strokes.

Although specific positive thoughts seem to be particularly beneficial for enhancing confidence (and performance), there are a number of more general positive thinking techniques worthy of mention, not least because they are frequently employed by sport psychologists when working with athletes. As well as monitoring one's mind-set (e.g., keeping a diary or log of mental aspects connected with training and competing), a number of methods for modifying thoughts have been proposed. These include thought stopping using physical or verbal triggers (e.g., saying stop), replacing negative statements with positive ones, or more thought restructuring-based approaches such as reframing, which emphasize positive aspects or perspectives within a seemingly unfavorable situations (e.g., an injured soccer player unable to practice enables her to realize how much she enjoys playing the sport and being part of the team).

Practitioners have been advised to assist athletes by challenging and countering irrational or distorted thinking and by helping develop affirmation statements or more extensive positive scripts. In essence, these methods aim to help athletes develop an optimistic and empowered explanatory style of events they experience; that is, athletes learn to avoid catastrophizing about errors or failure, view themselves as having more control over events, and are able to extract positive information from potentially negative experiences (e.g., a loss). It is notable that positive thinking seems to be particularly useful when athletes need to cope with adversity (e.g., injury, being under pressure). In this setting, positive thinking may help the performer make a more favorable appraisal

of the situation and/or his or her ability to cope with it. Thus, positive thinking may have both a direct and an indirect effect (via stress buffering) on performance. Caution needs to be used due to the relatively underresearched nature of many positive thinking techniques advocated in the sport psychology (SP) literature; although theoretically viable, the techniques and reasoning behind them have yet to be tested with athletes.

A final important consideration is that, to date, a “more is better” approach to positive thinking has been taken within the sport psychology literature; indeed, some research evidence supports this perspective. However, alternative perspectives exist that predict that problems arise with excessive positive thinking, reflecting unrealistic expectations and self-perceptions. Consequently, it is possible that one can have “too much of a good thing” and that there is a need to optimize the balance between positive and negative thoughts.

James Hardy and Emily J. Oliver

See also Cognitive Restructuring; Psychological Skills Training; Self-Efficacy; Self-Talk; Thought Stopping

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POSSIBLE SELVES

As a child, whom did you want to be when you grew up? A firefighter? A professional athlete? Where did this idea come from? Something you saw on television? Or perhaps you were emulating

a parent or other significant adult. As you entered high school, it's likely you began to think more seriously about your future. You met with your guidance counselor and created a plan for your future—you took specific courses, applied to particular colleges, or volunteered to gain firsthand experience for that hoped-for career. As a young adult, you continue to have thoughts about the future. You may wish for that ideal job or even fantasize about winning the lottery. In all, our lives are filled with hopes, dreams, and desires for the future. These future-oriented self-perceptions, or *possible selves*, are the subject of this entry.

Definition and Characterization of Possible Selves

Possible selves, a term coined by Hazel Markus and Paula Nurius, are described as how an individual thinks about his or her potential and future. An individual's possible selves are thought to be the cognitive link between past experiences and future hopes, desires, fears, and fantasies. They represent what we hope to become as well as what we hope to avoid becoming; we call these *hoped for* and *feared* selves, respectively. Possible selves that are more likely to be achieved focus on who we want to *be* (e.g., a mountain climber) versus what we might like to *do* (climb a mountain). Both hoped-for and feared selves have been shown to motivate us to act; we develop strategies to achieve our hoped-for selves and avoid feared selves. Importantly, even selves that have little likelihood of being achieved (for example, the lottery winner self-described in the first paragraph) can carry significant emotional weight. Thus, a young athlete could yearn for the hoped-for self of winning an Olympic medal, something that may be unlikely but at least remotely possible, or a sedentary, overweight adult could be able to clearly envision a possible self as thin and strong, even though he or she may be doing little to achieve that possible self. As Hazel Markus and Paula Nurius noted, an individual is free to create any type of possible self he or she wants as long as these are important to the individual and consistent with the sociocultural and historical world in which he or she belongs.

Factors Related to the Endorsement of Possible Selves

People seem to have no trouble coming up with and describing their possible selves, regardless of

age, gender, race, or other individual differences. In a study examining possible selves in people between the ages of 18 and 86, researchers found that no one had trouble generating possible selves. The possible selves ranged from very general (to be happy, to be healthy) to very specific (to play the piano well for my own enjoyment). Other studies support the ease with which people can think of and verbalize possible selves. Even older adults can relate to the possible self-question, Who do you want to be when you grow up? For an older person, this is likely to focus on issues related to health and independence, such as “not being a burden on my family” or “able to enjoy retirement.”

In fact, researchers have found age differences in possible selves. Younger adults tend to emphasize hoped-for selves related to family (becoming a mother) or occupations (having a job I enjoy), and feared selves focus on family and relationships (never finding the “right” person) or are physical in nature (being in poor health). While family hoped-for selves of middle-aged and older adults are also endorsed (seeing my children graduate from college), possible selves in the physical category (be healthy and active, be in good health) become much more important with age. Feared selves are commonly reported in the physical category as well, focusing on avoiding ill health or becoming dependent on others. The fact that possible selves differ across age groups is an indication that this component of the self continues to evolve as we age, with some possible selves staying important and others fading into the background. However, possible selves have also been shown to be resistant to change. In one of the few studies examining possible selves over time, older adults’ hoped-for and feared selves showed much more stability than change over a 5-year span. The changes that did occur involved health-related and physical hoped-for selves, which were added to existing possible selves.

Stability of Possible Selves

Although it might be difficult to see how possible selves can be stable and changeable at the same time, the concept of stability and change is actually a core principle in human development. For example, we can think of our personality as a very stable entity, but if we look at components that make up one’s personality, we see considerable change over time. The principle of stability and change

allows us to plan for the future (e.g., knowing how we generally react to situations) while allowing for more subtle changes that fit the situation we find ourselves in. In a similar way, we may have possible selves that remain very important over our lives (e.g., being an active person) but add possible selves that fit a particular context or time frame (e.g., being able to play with my grandchildren).

Planning Achievement of Hoped-for Self and Avoidance of Feared Self

Possible selves have been linked to long-term goals and identity. Like goals, possible selves require an action plan to be achieved. Since they are associated with our identity, possible selves act as a strong incentive for action and change. You can think of possible selves as “supercharged” goals; they are not just something we want to *do* (like run a marathon) but someone we want to *be* (a healthy adult, able to achieve a goal and persist over time). Building a bridge between who we are now (our current selves) with who we want to be (our future or possible selves) provides an opportunity for behavior change. The following example highlights the general principle of comparing one’s current self with a future (possible) self to initiate action. This process utilizes well-studied techniques from sport and exercise psychology (SEP), including goal setting and imagery. The unique aspect is the specific focus on the self and one’s identity.

Imagine, if you will, two young athletes with a desire to achieve a high level of performance. Athlete A is a young man who wants to earn a college scholarship to play baseball. Athlete B is a young woman who wants to join the professional golf tour and compete with the best in the world. In both cases, they share the possible self of being an elite athlete. Their current selves are also somewhat similar; they see themselves as accomplished in their sport, have aspirations to reach an even higher level, and believe they have the mental toughness needed to be successful. Developing an action plan that links their current self to their future self can help them achieve their goal. Research shows that well-conceived and complex possible selves are most likely to be realized, so the more vivid the plan, the better. These detailed plans facilitate the development of self-regulatory strategies, defined as the ability to work toward one’s short and long-term goals. To begin this

planning process, both athletes would do well to examine their strengths and weaknesses. Having a clear possible self facilitates self-monitoring, a key step in developing self-regulatory skills. Athlete A might then create a highlight video of college players he admires, so as to “see” his possible self more clearly. He would continue on from there, developing specific technical or psychological strategies to help him move from his current self to his hoped-for self. Similarly, Athlete B might create benchmarks toward attainment of her ultimate possible self (e.g., compete successfully in regional tournaments, play in a mini-tour, attend Q-school). By developing a plan and a process for evaluating movement toward their hoped-for self, these athletes will be more likely to achieve this self over time.

Intervention Research

Although little research in the sport domain has explicitly tested the use of possible selves as an intervention strategy, research in the exercise realm has. For example, Judith Ouellette and her colleagues asked students to describe themselves 10 to 20 years in the future as either an exerciser (hoped-for self) or a non-exerciser (feared self), then were asked to provide specific details of those possible selves (thereby increasing the specificity of those selves). Both groups given this brief intervention showed increased exercise behavior 4 weeks later. It may seem surprising that the feared selves were equally motivational as hoped-for selves. However, there is evidence that pairing a feared self with a hoped-for self is motivational, since the discrepancy is more likely to lead to action.

In a recent follow-up to that study, Elisa Murru and Kathleen Martin Ginis examined the effects of a possible selves intervention on young adults. They were interested in *how* possible selves impact behavior—that is, the mechanism. Not only was the intervention successful in promoting exercise (the intervention group had higher levels of activity than a control at 4 and 8 weeks) but they also found that, as predicted by theory, one’s confidence in utilizing self-regulatory strategies (specifically planning) in part allowed the possible self to impact actual behavior. In sum, their research provided further evidence that possible selves impact exercise behavior by facilitating the development of self-regulatory strategies, which in turn makes you more confident about your ability to achieve that

possible self. Although research in this area is just beginning, these studies provide a compelling starting point, suggesting that possible selves can be an effective strategy to elicit health behavior change.

Conclusion

The concept of possible selves is highly relevant to sport and exercise contexts. Most of us have ideas about our future we can verbalize, and these future-oriented selves have been shown to impact our behaviors. For adults, a large percentage of these possible selves relate to health, well-being, and our physical selves, so it seems quite reasonable that physical activity (PA) can be the strategy that links current and future selves. Although possible selves have the potential to be an intervention strategy for sport contexts as well, more research is needed. Most of us contemplate our hoped-for and feared selves but may not have taken the time to examine what we need to do to realize these dreams and desires. With a little work, possible selves can become actual (achieved) selves; it is a concept ripe for future research and practice in sport and exercise contexts.

Diane E. Whaley

See also Identity; Implicit/Self-Theories of Ability; Self-Monitoring; Self-Schema

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PRACTICE

Practice typically comprises activities that are designed to help a person acquire a new skill, improve in an already acquired skill, or maintain a skill. Practice can be deliberate and engaged in for a specific purpose (such as attaining a speed or accuracy goal) or it can be more incidental in nature, potentially engaged in for fun or for the enhancement of some other goal (such as practicing a swimming stroke to help improve fitness). Practice is usually considered with respect to two characteristics: how much practice (i.e., practice quantity) and what type of practice (i.e., practice quality).

Quantity of Practice

Practice quantity is typically thought to be the most important variable for motor skill acquisition. In most cases, more practice is better than less practice, and although there might be diminishing returns associated with practice, there are rarely performance costs associated with increased practice (cf. automaticity). In the past, researchers have attempted to fit performance curves to practice data in order to describe how an increase in practice amount is related to performance. Most commonly, performance data as a function of practice have best been described in a logarithmic fashion, known as the *power law of practice* (although exponential curves, similar to power functions, also fit these types of data well). The logarithm of speed-related variables (such as reaction times [RTs] or movement times [MTs]) or of errors has been shown to decrease linearly with the logarithm of practice attempts. Significant performance gains are seen early in practice, but as practice proceeds, the performance benefits start to decrease and appear to plateau. Some researchers have argued that this continuous, linear pattern of behavior is an artifact of averaging across different individuals. Indeed, there is considerable variation in the learning curves observed between individuals and across tasks, suggesting that the power law of practice is an oversimplification of the learning process. It might be the case that learning is a nonlinear process, with moments of success and failure and/or long periods of practice that show no observable change in performance. It is clear that merely looking at practice amount to

make predictions about learning or to intervene to improve performance is by itself insufficient.

Practice quantity as a variable of interest has received renewed interest with respect to attainment of high levels of motor skill and expertise. This has been spurred by suggestions that expert performance in sports and other domains requires approximately 10,000 hours (or 10 years) of practice. Researchers have argued that high quantities of sustained, “deliberate” practice are necessary (and potentially sufficient) for elite levels of performance in sport. It has been demonstrated that performance-related markers of expertise in sports and other domains are monotonically related to practice amounts (i.e., a specific increase in practice translates to a specific increase in performance quality). Importantly, it is not only the amount of practice hours that is important but also the quality of practice. High amounts of effortful practice, designed to improve current levels of performance, are deemed necessary for success. There is considerable evidence, across a wide range of sports, that elite-level performers have consistently acquired many more hours of practice that fits this definition than their less elite, age-matched peers. Indeed, one of the most important contributions from this work is the emphasis on the “deliberate” nature of practice necessary to see performance gains (despite the fact that there has been a wealth of studies in sport where the focus has just been on quantifying practice amounts). It is likely that variations in performance curves as described earlier are related to different types or quality of practice undertaken by different individuals that will therefore have differential impacts on performance.

Quality of Practice

Attempts to study quality of practice in sport and movement skills have mostly been conducted by researchers working with nonathletes, practicing mostly artificial, laboratory-based tasks, over relatively short time periods. Exceptions have been noted whereby sport experts are studied in the field (such as on the ice rink) and comparisons are made across differently skilled individuals with respect to the types of activities that are engaged in during practice; the amount of actual physical practice; and the time spent watching, listening, and/or resting. Questionnaire data have also been collected from skilled athletes to ascertain their perceptions of various practice type activities in terms of the

amount of physical effort, concentration, or enjoyment. This has allowed for inferences concerning what types of practice typify sporting excellence and are most related to success. Although there have been manipulations to practice conditions engaged by skilled performers, such that comparisons can be made across performers matched for skill level, research of this nature has been limited. This is probably a function of the time needed to see improvements in a skill for high-achieving athletes, difficulties in intervening and controlling the practice conditions of elite athletes, and gaining access to elite populations.

The advantages of studying practice in the laboratory is that it is possible to create controlled conditions, whereby variables related to practice or individual differences can be manipulated, minimized, or controlled. By studying novel tasks, it is possible to eliminate, or at least minimize, the effects of prior experience on learning and performance and hence isolate what specific practice conditions (when controlling for practice amount) best promote learning. Despite this laboratory emphasis, this has not precluded studies of novice athletes practicing more real-world sporting skills under laboratory-type conditions (e.g., putting, throwing, kicking, serving, batting skills). One of the advantages of studying practice for relatively simple motor skills in the laboratory is that only a short amount of practice is needed to see practice improvements and skill attainment. Although this is obviously a drawback when trying to make comparisons to long-term practice of more complex motor skills, the trade-off is usually one that researchers are willing to make in order to control the practice environment and make conclusions about causality.

Practice Quality Variables

There are a considerable number of practice quality variables that have been shown to influence motor skill acquisition, including performance feedback about outcomes and movement execution; demonstrations and instructions; variability in how the skills are practiced (such as practicing a putting skill across different distances or with different putters); variability in the order of practice conditions (such as practicing three types of shots in basketball in a mostly random order in comparison to performing the same shot in a repeated blocked order); rest periods between practice

trials; whether the practice is determined by oneself or someone else; whether practice is conducted in pairs or alone, in small subcomponents, or as a whole; and the similarity between the practice conditions and those during testing or competition. What has been shown to be critical in finding the optimal or the best conditions of practice is matching the demands of the practice to the skill level of the individual (and the difficulty of the task or tasks) and making sure that the practice is demanding enough to promote long-term learning (retention) and/or transfer to new, yet similar; practice or performance environments.

In motor learning research, it has been necessary to separate short-term performance gains and rate of motor learning from longer-term measures of learning and performance. There have been noted dissociations between conditions that best promote gains during practice and those that best serve retention (and prevent forgetting). One of the best examples in motor learning research is the so termed *contextual interference effect*, whereby interference encouraged by variability in the order that different motor skills are practiced, while hindering rate of motor skill acquisition during practice, has positive effects on retention. The conditions that are hardest, promoting more variability during practice, are retained better than those that are easiest. As noted, however, it is important that the interference or challenge is appropriate to the skills of the individual, as too much interference in practice for a beginner can be as harmful as too little challenge for a more competent performer.

Motivation has always been considered an important variable with respect to practice optimization and the acquisition and attainment of skill in sport. In the early 1900s, theories of learning were primarily based on reinforcement learning and ideas that practice behaviors are repeated in order to gain some sort of behavioral or physiological reward. In recent times, there has been an acknowledgment that motivation-related variables interact with practice-related variables in quite significant ways. For example, merely telling someone the scores of a peer (who either does worse or better on the skill) will affect how that person learns and what they retain. Ideas concerning how motivation acts on practice behaviors and learning are still relatively undeveloped. However, it is thought that manipulations, which increase a person's motivation to learn (such as enhancing feelings of

competency or control over the situation), impact their subsequent processing of task-related information and cognitive effort. It is unclear whether these types of manipulations work in the same way as incentives or rewards.

Conclusion

In summary, methods for optimizing practice time, both with respect to efficiency (i.e., time to acquire a skill) as well as retention (i.e., how well and for how long a skill is retained), have received considerable attention from researchers working within the motor learning and the motor skill acquisition or skilled performance fields. Despite the advances in these fields, there are notable areas where best practice behaviors remain relatively unexplored or unclear. In particular, there has been little study of practice behaviors of skilled individuals and what might be considered maintenance practice. We are also somewhat unclear about what practice behaviors are needed to overcome slumps or plateaus in performance and transition between success and failure at a skill, or the acquisition of new skills on the backdrop of existing skills. In addition to manipulations to actual practice behaviors, there have been manipulations to what have been referred to as covert behaviors, specifically observational practice and imagery practice. These types of practice have received considerable attention of late due in no small part to discoveries of shared cortical areas in the brain between covert and overt practice. However, what is still somewhat unknown are the mechanisms behind their successful or unsuccessful use in teaching new skills and under what conditions observational or imagery practice will work to enhance learning in comparison to more traditional physical practice methods.

Nicola Jane Hodges

See also Automaticity; Developmental Histories; Especial Skills; Expertise; Feedback; Learning; Motor Learning; Skill Acquisition

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PREPERFORMANCE ROUTINES

Preperformance routines refer to the consistent sequence of thoughts and actions in which a performer engages before executing a skill. Typically, they are associated with performing self-paced skills in which the performer decides when to initiate the action. Examples include sports such as golf, snooker, archery, and static target shooting, as well as individual elements of team sports, such as the basketball free throw, goal kicks in rugby, and soccer penalty kicks. Research in the area has addressed both physical (i.e., actions such as taking deep breaths and looking toward the target) and mental (i.e., thoughts or cue words such as “focus”) components of routines. Researchers have also looked to make inferences about mental states from observing physical features such as routine consistency or from other physiological or behavioral measures. Routines may contain ritualized

components or superstitious thoughts and behaviors that, arguably, have a different origin.

Routine Consistency

When observing preperformance routines, a defining feature is in behavioral consistency. For example, in studies examining routine times, high-skilled basketball free throw shooters, golf putters, and rugby goal kickers have a dominant behavioral pattern that is followed before each attempt. This means that their routine times and the behaviors exhibited are more consistent relative to low-skilled or less-experienced performers. Some researchers have hypothesized that routine consistency facilitates higher levels of performance, but the evidence for this is somewhat mixed. On the one hand, there is some evidence that routine consistency is positively correlated with performance. For example, high-skilled golfers have been found to use their dominant pre-putt routine more frequently than beginner golfers and routine consistency is correlated with golfers' handicap. In college basketball, routine time consistency has been found to correlate with free throw success, with more successful players again displaying more consistent times. Further, professional basketball players have been found to be less successful when deviating from their dominant behavioral sequence, most notably when performers added something to their routine (e.g., taking a deep breath). Related to this, it has also been shown that disrupting the rhythm of the routine harms performance. Specifically, speeding up or slowing down basketball players' free throw routines did not impair performance, whereas disrupting the relative timing of different components (e.g., speeding up one aspect, slowing down another) led to lower shooting accuracy.

A problem with inferring from this that routine consistency causes good (or improved) performance is that showing two things are related to each other does not allow us to say that one causes the other. In addition, it is notable that there is often considerable variability between different performers both in terms of the time and content of their routines. Looking at routine time, there has been no advantage found for longer or shorter routines and large individual differences are apparent in some skills. For example, in one study of World Cup rugby union goal kicks, goal kickers were shown to have concentration times ranging from 4 seconds to more than 20 seconds. Similarly, research into

basketball free throws in the National Basketball Association play-offs has shown that there is no difference in success rates between routines of brief, regular, and long durations. Crucially, in the few studies where researchers have attempted to improve routine consistency, they have failed to bring about meaningful or lasting improvements in performance. As some researchers have noted, the behavioral consistency we see in expert performers might be a by-product of the hundreds or thousands of hours of repetitive practice.

Content and Function of Mental Routines

Researchers have hypothesized that the mental and physical components of routines have at least three possible functions. First, they can focus performers' attention, helping them to concentrate on the relevant aspects of the task and block out distractions. The nature of the distractions might be external to the performer, such as crowds making noise or waving their arms, or might be internal in the form of distracting thoughts such as, what if I miss? Second (and closely related to internal distraction), the mental aspects of a routine can help performers regulate their thoughts, behavior, and emotions. Researchers have suggested this is a key function of routines because a defining feature of self-paced skills is that performers have much more time to think and reflect than in externally paced, reactive skills. For example, tennis players have a split second to react to a passing shot during a rally but have up to 25 seconds between points in which to consider the situation, the score, or their emotional state. In a broader sense, then, routines are the vehicle through which performers try to deploy tools to help regulate their emotional, behavioral, and cognitive responses to competitive situations. Last, a consistent sequence of thoughts and behaviors can help trigger or cue automatic, fluent performance and prevent performers from trying to consciously control their actions. This links to the large body of research indicating that that anxiety can lead skilled performers to try to consciously control their actions, resulting in "choking."

The specific function of any individual routine will be reflected in the nature of the content and there is evidence that performers use mental routines for different purposes. For example, research has shown that high-skilled golfers collectively employ a range of psychological "tools" in their pre-shot routines, including distraction techniques,

relaxation strategies, cue words, positive self-talk, and imagery. These psychological tools align with the aforementioned functions and are consistent with the “5-step strategy” proposed by Robert Singer and colleagues, a preparation strategy for performing self-paced skills (e.g., basketball free throw) that has received good empirical support. The steps consist of *readying* (preparing for the act), *imaging* (visualizing the movement), *focusing* (on a meaningful external cue, e.g., the rim of the basketball hoop), *executing* (with a quiet mind, allowing the movement to flow automatically) and *evaluating* (a brief assessment of the effectiveness of each of the previous steps once the skill has been performed). The readying and imaging steps are directed toward self-regulation and enhancing self-efficacy. The focusing and executing steps are clearly directed toward attentional processes. The focusing stage encourages a task-relevant, external focus, while the executing step is an attempt to prompt automatic, effortless performance that is free of conscious interference (e.g., overanalyzing movement technique).

Most experimental research into the mental function of routines has been focused on what performers attend to just before executing the skill. Researchers have used various techniques, from directly manipulating this state through verbal instructions to drawing inferences from physiological data, such as heart rate (HR) or measures of brain activity. Using direct instructions, there is a significant body of research supporting the efficacy of an external, compared to an internal, focus of attention, which involves focusing on the effects of movements rather than the movements themselves. The benefits are evident both in learning new skills and executing well-learned skills. For example, in one study skilled golfers performed better when attending to the motion of the club head than when attending to the motion of their arms as they performed “chip” shots to a target. In interpreting findings from this type of study, researchers have pointed to the self-organizing nature of the motor system, arguing that an external focus facilitates this, whereas an internal focus constrains the system, making it less efficient. Consistent with this work, many performers incorporate a specific external visual focus as part of their mental routine. For example, rugby player Jonny Wilkinson uses the word *spot* to prompt him to focus on the very specific point on the ball he wants to strike when taking kicks at goal.

Psychophysiological studies mostly have been focused on HR, electroencephalogram (EEG), and visual gaze data to compare what expert and less-skilled performers are attending to during their routines. EEG studies record different types of brain activity (e.g., alpha, beta) by placing surface electrodes on different parts of the scalp. Analysis of activity in the few seconds before skill execution has commonly been interpreted as reflecting an acquired ability to suppress conscious control processes. This is evidenced by increased left hemisphere alpha activity, a change in the ratio of activity in the left and right hemisphere, a more widespread decrease in cerebral activity, or lower coherence between left hemisphere and frontal midline activity in the few seconds prior to skill execution. Results from research using brain-imaging techniques such as functional magnetic resonance imaging (fMRI) to compare activity in expert and novice golfers as they imagine their pre-shot routines are consistent with this interpretation. Thus, overall brain activation is lower for experts and appears more efficiently organized compared to diffuse brain activity seen in novices.

Research using HR has focused on deceleration in the few seconds before skill execution. HR deceleration has been shown to be associated with an external focus of attention and decreased cortical activity, whereas greater cortical activity and an internal focus of attention is associated with HR acceleration. This research has corroborated findings using other methods (e.g., verbal instructions to manipulate attention). For example, there is evidence that HR deceleration is more pronounced in high-skilled than less-skilled golfers. Adding further weight to these findings, researchers have also monitored where performers look during their routines, notably on the “quiet eye” period—that is, the final fixation (or tracking of a moving target) the performer makes on a location or object before initiating movement. Among other things, this research has shown that experts tend to have longer quiet eye duration and, what is more, that quiet eye training improves performance. The quiet eye period has also been shown to relate to electrocortical activity indicating a motor programming or preparatory function.

Conclusion

Preperformance routines contain behavioral and mental components through which performers

attempt to regulate their thoughts, emotions, and actions. In so doing, routines direct performers' attention, help control psychological and physiological responses to stress, and allow motor processes to run off with minimal conscious interferences. The weight of research indicates that it is these functions, rather than routine time or consistency per se, that facilitate performance.

Robin C. Jackson

See also Attention Training; Attention–Performance Relationships; Centering; Concentration Skills; Electroencephalograph (EEG); Self-Talk

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PRIMING

Priming refers to the process of temporarily activating an individual's mental constructs (i.e., trait concepts, stereotypes, contexts, goals) and observing the subsequent effect of this activation on psychological, social, and/or motor behavior phenomena. Every individual possesses a set of mental representations (that are constantly being added to or developed) about themselves and the world around them that influence the way in which they perceive their environment and govern how they function within it. These representations or constructs are stored at a subconscious level, and their accessibility is dependent on how frequently they are activated. These differences in activation can, in turn, lead to individual differences in perception and action. The aim of priming is, therefore, to activate a particular representation in one situation so that it is readily accessible, and hence, the probability of its use in a subsequent task or situation is enhanced.

Research has consistently demonstrated that a behavioral response “to an implicit or subconscious cue can be elicited despite [the individual being] unaware of the nature of the cue or even that the cue was presented” (Hull et al., 2002, p. 406). For example, many studies have observed that participant perceptions of an individual displaying an ambiguous expression or behavior are synonymous with the personality construct (i.e., happy or sad, hostile or kind, rudeness or politeness) that has been activated in a previous priming task. In all cases, the participants are unaware of the link between the memory-based, grammatical, or pattern recognition task that they have completed (which contains the primes) and the perception or behavior they are exhibiting. With regard to the motor behavior–sport context, priming has been utilized successfully to change or enhance both motor skill execution and exercise intensity. For example, researchers have demonstrated that exposing participants to elderly stereotype primes via a grammatical task reduced their walking speed on departure from the laboratory compared with their control counterparts.

Prime Exposure

Subliminal Priming

Subliminal priming involves the brief presentation of a priming stimulus (the prime), which is

immediately masked via the presentation of a second stimulus (e.g., a pattern or letters). The duration for which the prime is presented falls below a consciously perceptible threshold; John A. Bargh and colleagues noted that this is ~15 ms for primes located within central vision and up to ~90 ms as they move toward peripheral vision. Yet despite the prime being of insufficient length to enter conscious awareness, a cognitive behavioral response is still elicited.

Supraliminal Priming

Supraliminal priming is also known as conscious priming because the priming stimuli that are presented are detectable to the individual; they are presented explicitly within the task being completed. However, while the stimuli might be fully perceptible, the individual is not aware either of the association between the prime and target behavior or, in many cases, that something other than what is perceived to be the primary task is being assessed. The most commonly adopted supraliminal priming technique is the “scrambled sentence task.” This task requires participants to reorder a series of words to form a grammatically correct sentence. Participants are therefore exposed to the prime concept via the words that are embedded within the task.

In examining subliminal and supraliminal priming, however subtle or implicit researchers perceive the priming stimuli to be, it is imperative that participant perceptions of the experiment and the link between the prime and target behavior are assessed in order to determine whether they were consciously aware of the prime stimuli. While researchers strive to disguise prime presentation, it should also be noted that conscious recognition of primed items might not necessarily have an impact on test performance.

Priming Effects

Behaviors elicited as a result of priming are directly associated with their content, although depending on the mental constructs that are activated and the judgment formation or encoding that takes place, researchers may observe assimilation or contrasting behavior or performance outcomes.

Assimilation

Assimilation refers to a response that corresponds to the mental construct that has been

activated by the prime. For example, primes may be presented that evoke an automated or external focus of attention in skilled performers; thus, in a pressurized condition, automated, fluent performance should ensue rather than performance disruption occurring as a result of explicit monitoring.

Contrast

Contrast, on the other hand, refers to responses that are displaced from the mental constructs that were activated. This often occurs when extreme exemplars are utilized as prime stimuli, which elicit a comparative judgment and subsequent divergent response. For example, with a prime that identifies a well-known and highly skilled footballer (e.g., Lionel Messi) as a standard of comparison, the participant’s perception of their own football skills or that of an “ambiguous” fictitious football player may be deemed particularly poor and decrements in skill or actual performance may be observed. A contrast effect can also occur when a participant becomes aware of the incidence of the prime and/or its potential effect. So, in instances where a researcher may anticipate assimilation, they may actually observe a contrast effect as the participant adjusts their behavior to counteract the perceived influence of the prime.

Kelly J. Ashford

See also Attentional Focus; Automaticity; Evaluative Priming; Self-Awareness; Self-Awareness Theory; Self-Monitoring; Self-Regulation

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PROBABILITIES

In sport, some events, occurrences, and outcomes are more probable than others, and the potential exists to use information about probabilities to aid performance. This entry discusses two levels on which probabilities are relevant to sport performance. The first is the individual, immediate performance level. On this level, probabilities are used implicitly during performance; performers may not even be aware that they are using probabilities, having built up this feature of anticipation and decision making (DM) over time as they acquired skill. The second level is a broader, game-analysis and planning level, on which information on probabilities is actively sought to drive strategic decisions. This second level seeks to explicitly exploit probabilities for planning and competitive advantage. It uses a more comprehensive approach, often accessing large pools of data to identify meaningful patterns that exist on the individual, implicit level. Thus a strong link exists between the two levels of probabilities, with one influencing the other, as researchers use probability information to drive interventions and strategy. The sections that follow discuss the individual and the tactical planning levels of probabilities by presenting examples that illustrate

how probabilities are used in sport and exercise psychology (SEP) research.

Individual Level

Though there is relatively little direct exploration of the explicit use of probabilities in DM for individual action choices, it is well known that performers make informal “calculations” and use probabilities based on the situation. In competition, athletes implicitly assign weightings, or probabilities, to the likelihood of an event occurring and use this information to guide their decisions. These implicit, subjective probabilities influence the ability to respond. For instance, a tennis player’s position on the court can influence the type of shot he or she is able to play. Knowing this, a defender can assign probabilities to each possible shot to anticipate the opponent’s actions. If a defender thinks the probability of a particular shot is high, the defender will show better anticipation of this shot. A shot the defender thinks is less probable will not be as well anticipated if the opponent makes this choice. The attacking player can exploit this information by playing a low-probability shot when another outcome is highly probable and thus catch the defender ill-prepared or off guard.

The effective use of probabilities can also increase performance based on availability of specific situational information. This has been explored by manipulating the situational information accompanying video clips when performers are asked to decide on their next action. For example, a baseball pitcher is more likely to try to throw a strike pitch rather than a pitch outside the strike zone when the pitch count contains three balls than when it contains none. This situational information allows a batter to make an informed decision regarding the pitch location and, by reducing uncertainty, make a quicker decision about whether or not to swing.

The use of probabilities also distinguishes between more and less skilled performers. The use of verbal reports, in which athletes report their thoughts during play, have shown that elite athletes, compared with less skilled athletes, are able to create more detailed and sophisticated event profiles using past and current information to “diagnose” what is currently taking place, as well as predict and anticipate future play. This probabilistic information can be used to effectively manage energy expenditure during the course of

an event. In squash, for instance, the probability of winning a game is based on the probability of winning a rally and the relative importance of each rally. Athletes naturally use this information to regulate energy expended on a particular point and energy trade-offs in the context of competition as a whole and its likely outcome.

Relying on probabilities is not always beneficial, however; athletes who believe in the “hot hand” in basketball, for instance, think there is a greater probability of a player making a successful shot after they have made one or two previous shots. Teammates are thus more likely to allocate or pass the ball to a “hot” player even though the evidence to support the hot-hand belief is equivocal. Sports officials are also prone to using situational probabilities when making decisions; they are more likely to penalize teams in an alternating fashion rather than penalizing the same team twice in a row. Cognitive biases and decision errors such as these provide a place for the use of probabilities in sport on a more tactical level.

Tactical Planning Level

Coaches and athletes often use intuitive methods to derive tactics. Work in the area of calculation, statistics, and machine learning methods are used to support these intuitive processes and avoid errors. Using archived or accumulated performance data, data analytical methods focus on maximizing performance by more fully exploiting information for tactical planning and are especially useful in sports, such as rowing and cycling, where pacing is critical. For example, historical performance data can be used to both create race profiles that typify competitors from one team or country and to determine optimal race patterns in sports like rowing, which have different race portions (e.g., 500-m splits of a 2,000-m race). This information is used to improve planning and the chances of achieving success.

The tactical, planning level of probabilities is also useful in multievent sports, where the final standings are determined after more than one event (e.g., triathlon, decathlon). In some multievent sports, such as the cycling omnium, pacing is considered between events over more than 1 day. The omnium is comprised of six track cycling events that take place over 2 days. The introduction of a new event (the elimination race) to create a six-event omnium, before the 2012 Olympics, is

a case where the tactical, planning level of probabilities is useful to investigate the best strategy to secure a medal. Machine learning analyses have been used to identify the minimum result required in each event that will lead to the highest probability of success, as well as whether the addition of the new elimination race event benefits endurance riders or sprinters. Not only does this use of probabilities help planning before and during the event (i.e., between events, between days) but it can also be useful information for athlete selection.

In general, SEP seeks to increase overall probabilities of successful performance. Individual DM “in the moment” and structured methodical planning based on the likelihoods of different events are influential in reaching this goal.

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See also Anticipation; Decision Making; Expertise

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PROTECTION MOTIVATION THEORY

The protection motivation theory (PMT) originally aimed at explaining why people develop protection motivation and what role fear-appeals play in this process. A protection motivation might be an intention to adopt or adhere to a fitness program. Athletes might fear to perform not at their

best form if they do not attend training. Exercisers could fear that not regularly training and participating in an exercise program diminishes their fitness, worsens their attractiveness, and increases their risks of diseases such as diabetes. The PMT explains motivation by fear, risk appraisal (severity and vulnerability) and coping appraisal (response efficacy/response costs/self-efficacy).

Fear results from and interrelates with the perceived *severity* of a health condition: Some individuals might comprehend more increased severity than others. Also, some incidents are typically perceived as more severe (such as cardiac infarction) and others as less serious (a pound too much body weight for an intermediate time). The more severe a health risk, the more likely a motivation develops and a recommended behavior will be shown. However, this is not always the case: Vulnerability also determines whether individuals maintain or change their behavior.

Vulnerability refers to whether a person feels likely to develop a specific condition. This can vary between individuals and different incidents. If a person feels very vulnerable to develop an adverse effect (e.g., worsen performance level, cardiac infarction), she or he might be more likely to obtain such a condition than a person who believes to be less vulnerable. In addition to differences in individual perceptions, some conditions are also more likely than others. For example, the flu is typically very likely to develop during the year (high likelihood and vulnerability), but the intensity and duration of it is rather limited (low severity). In contrary, a cardiac infarction is less likely to occur (low likelihood and vulnerability), but if one develops it, it is life-threatening and persisting (high severity).

Together, severity and vulnerability provide a basis for *risk appraisal*. Many studies have investigated these effects and found that high-risk appraisals do not guarantee a protective motivation. Additionally, *coping appraisal* is of importance: Response efficacy, response costs, and self-efficacy determine whether humans change current behavior or maintain recommended behavior.

Self-efficacy refers to the belief in one's own competence to perform a behavior even in the face of barriers. Athletes require high self-efficacy beliefs in terms of training even if they have many other duties or when feeling tired. Also, exercisers need self-efficacy to overcome comparable barriers and inner temptations.

Another component of coping appraisals is *response efficacy*: Only if the individual is convinced that a behavior leads to the desired outcome will she or he be more likely to intend to perform the behavior. Athletes need to find an effective training for their performance goals. If a soccer player trains only the muscles, she or he might question how she or he might develop the needed competencies—besides strength—for successfully playing a soccer game. Thus, a mixed training of muscles, flexibility, and coordination would be desired. Alternatively, an exerciser with the aim to mainly prevent pain and to perform muscle training might feel this fits much better his or her needs than a mixed training.

On the other side, perceived *response costs* come into play: If a person perceives much higher costs than benefits (response efficacy), she or he might not perform the behavior over a longer period of time. Only if positive aspects (response efficacy) outperform negative aspects (response costs) will the individual maintain the motivation to perform the recommended behavior. If costs outweigh the benefits, people will be more likely to discontinue.

The PMT has been successfully applied to different behavioral domains. Figure 1 shows results from the 2000 meta-analysis by Milne and colleagues. The indicators (r_+) show that self-efficacy and response-costs appear to be the main predictors of motivation and behavior, while response efficacy is also important. However, threat appraisal appears to be less imperative.

The results of meta-analyses and newer studies demonstrate clearly that fear can hardly change behavior. Also, fear and/or threat appraisal cannot guarantee maintenance of a recommended behavior: *Coping appraisal* is needed to adopt or maintain a behavior and with that to prevent and overcome injuries and slumps. Coping appraisal is also essential for overcoming fears and mental blocks. Only by means of resources like self-efficacy and sufficient knowledge (i.e., response efficacy) inner or external temptations can be overcome.

These resources can be improved in interventions. When promoting the performance of a recommended behavior, the adoption of the behavior (or increasing the motivation to change) is typically differentiated from the maintenance of the behavior (the behavior itself). In recent studies, it was found that threat appraisal and response efficacy are imperative for preparing behavior

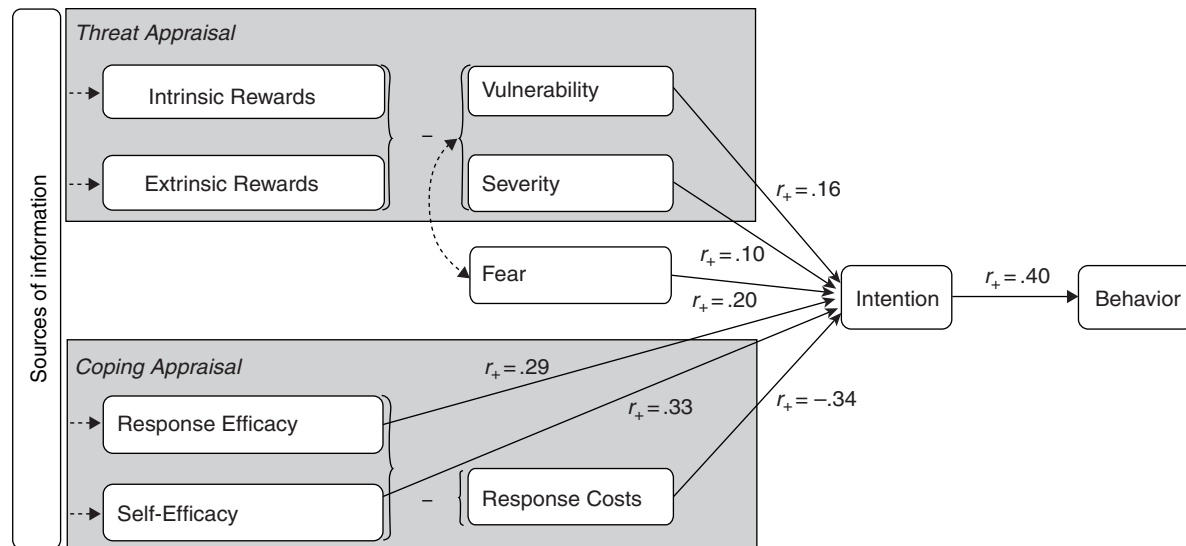


Figure 1 Protection Motivation Theory With Aggregated Correlations (r_+)

Source: Lippke, S., & Renneberg, B. (2006). *Theorien und Modelle des Gesundheitsverhaltens* [Theories and models of health behavior]. In B. Renneberg & P. Hammelstein (Hrsg.), *Gesundheitspsychologie* (pp. 35–60). Berlin: Springer. Reprinted by permission of Springer Verlag.

Notes: Components in gray and dotted lines were not investigated systematically. Data for r_+ from S. Milne et al. (2000). Prediction and intervention in health-related behavior: A meta-analytic review of protection motivation theory. *Journal of Applied Social Psychology*, 30, 126f.

change. Changing behavior and maintaining it over time requires self-efficacy and other volitional variables.

Sonia Lippke

See also Health Action Process Approach; Health Belief Model; Mastery and Control Beliefs; Transtheoretical Model

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PSYCHOLOGICAL CONSEQUENCES OF SPORT INJURY

Injury is a common occurrence in association with sport participation. Most sport injuries are minor, require minimal medical intervention, and have limited impact on the sport involvement of athletes. However, a substantial number of injuries are of sufficient severity to require more extensive medical treatment (e.g., immobilization, surgery), rehabilitation, and restriction of sport involvement. In addition to the physical effects, there can also be psychological consequences of sport injury. Numerous models have been proposed and evaluated to document and predict psychological responses to sport injury. Based on these models, interventions have been developed to facilitate psychological adjustment to sport injury.

Models of Psychological Response to Sport Injury

Beginning in the late 1960s, empirical studies of the psychological consequences of sport injury began to appear in the scientific literature. By the mid-1980s, scholars had adapted theoretical models from psychiatry and general psychology to describe and explain how athletes respond to sport injury. Stage models, which were derived from research on terminally ill patients, were also used to attempt to predict psychological responses to sport injury. In stage models, athletes were posited to experience a sequential series of stages of adjustment (e.g., denial, bargaining, anger, depression, acceptance) following injury. Despite their popular appeal, these models fell out of favor as research findings indicated that athletes' psychological responses to injury are highly variable and follow no universal pattern.

Scholars borrowed another theoretical perspective from research on stress and coping in the form of cognitive appraisal models. According to cognitive appraisal models, an athlete's cognitive, emotional, and behavioral responses to injury are interrelated. Athletes' responses are influenced primarily by their appraisals or interpretations of the injury and their ability to cope with the injury. Characteristics of the athlete (e.g., personality), the situation (e.g., life stress, social support), and the injury (e.g., severity, duration) are thought to affect athletes' cognitive appraisals. Cognitive appraisal models have accommodated the results of numerous studies in which individual differences in psychological responses to sport injury have been found. With consistent empirical support for their basic tenets, cognitive appraisal models provide a useful framework for examining psychological consequences of sport injury.

Psychological Responses to Sport Injury

Researchers have documented a wide variety of psychological responses to sport injury. In accord with cognitive appraisal models, the psychological responses can be grouped into cognitive, emotional, and behavioral responses.

Cognitive Responses

Causal attributions, cognitive coping strategies, injury-related cognition, cognitive performance, self-related cognition, and perceived benefits of injury are among the aspects of cognition associated

with the occurrence of sport injury. Athletes who have been injured readily generate causal explanations for why they experienced their injuries. Athletes may also experience intrusive injury-related thoughts and images. They often report applying cognitive strategies to cope with their injuries, such as accepting their injuries, focusing on healing, thinking positive thoughts, using imagery, and avoiding negative thoughts. Even with the use of coping strategies, athletes who have experienced a recent injury tend to perceive greater risk of sustaining an injury, greater worry about injuries, and less confidence in the ability to prevent injury than athletes without a recent injury. With regard to cognitive performance, neurocognitive functioning (e.g., attention, memory, processing speed, reaction time [RT]) can be impaired not only by concussions but by musculoskeletal injuries as well.

Thoughts about the self appear to be particularly affected by sport injury. Changes in self-related cognitions over the course of sport injury rehabilitation have been documented. Post-injury levels of self-esteem tend to be lower than pre-injury levels of self-esteem. Self-confidence and self-efficacy tend to increase as athletes recover from their injuries. For serious injuries with a long and arduous rehabilitation, self-identification with the athlete role (i.e., athletic identity) may decline over the recovery period, especially for those who are not recovering quickly or well.

Not all cognitive responses to sport injury are negative. Apparently, some athletes derive benefits from their injury experiences. Perceived benefits of injury tend to involve themes associated with growing as a person, enhancing mental skills for sport performance, and developing physical and/or technical skills. Experiencing benefits from injury likely requires athletes to expend effort in acquiring skills or viewing their situation from a new perspective. Full recognition of the benefits of injury may not occur until well after the recovery process has concluded.

Emotional Responses

Given that injuries often disrupt athletes' pursuit of sport-related goals, it is not surprising that athletes report experiencing a variety of negative emotions following injury (e.g., anger, anxiety, bitterness, confusion, depression, disappointment, devastation, fear, frustration, helplessness, resentment, shock). Evidence indicates that

compared with their pre-injury emotional states, athletes experience post-injury emotional disturbance. Injured athletes report experiencing greater mood disturbance than athletes without injuries. Estimates suggest that approximately 5% to 27% of athletes with injuries have clinically meaningful levels of emotional distress. Negative emotions tend to dissipate and positive emotions tend to become more prominent over the course of rehabilitation. Although the general pattern of post-injury emotional adjustment is similar to that what might be predicted by stage models, athletes vary tremendously in the emotions they report experiencing and the order in which they experience them. With such variability in responses across individuals, the concept of stages does not apply and is, therefore, not useful. Cognitive appraisal models provide a more parsimonious explanation for the emotional responses to injury exhibited by athletes. In accord with cognitive appraisal models, numerous personal, situational, and cognitive factors are associated with emotional responses to sport injury.

Personal factors refer to relatively stable characteristics of athletes and, for the purposes of this discussion, characteristics of the injuries that athletes sustain. In general, higher levels of post-injury emotional disturbance have been documented in athletes who are (a) young, (b) in pain, (c) high on neuroticism, (d) less recovered from their injury, (e) injured more acutely, (f) injured more severely, (g) more impaired in their ability to perform everyday tasks, (h) more strongly identified with the role of athlete, and (i) more heavily invested in playing sport professionally.

Situational factors refer to characteristics of the social and physical environments inhabited by athletes. Levels of post-injury emotional disturbance appear to be highest among athletes who (a) commit a large amount of time to their sport activities, (b) experience a high amount of life stress, (c) participate at a low level of competition, and (d) perceive themselves as having little social support for their rehabilitation and lives in general. High perceptions of social support are not only associated with lower levels of emotional disturbance but may also help to reduce the adverse effects of life stress on emotional functioning.

Cognitive appraisals, cognitive coping strategies, cognitive structures, self-related cognition, and causal attributions are among the aspects of cognition for which associations with emotional

adjustment to sport injury have been documented. Athletes who appraise themselves as being unable to cope with their injuries tend to show greater emotional disturbance than those without such appraisals. Similarly, athletes who report that they use avoidant cognitive coping strategies, possess dysfunctional schemas that are used to process self- and world-related information, and/or rate themselves as low in physical self-esteem are at increased risk for experiencing post-injury emotional disturbance. Significant relationships between attributions regarding the cause of the injury and emotional adjustment to injury have also been obtained. The direction of the attribution-adjustment relationship is unclear, however, as internal attributions for injury were associated with better emotional adjustment in one study and worse emotional adjustment in another.

Behavioral Responses

As indicated by cognitive appraisal models, sport injury can produce behavioral as well as cognitive and emotional reactions. Some behavioral reactions are maladaptive, such as those involving suicidal behavior, disordered eating, alcohol abuse, and consumption of banned substances (e.g., amphetamines, anabolic steroids, ephedrine, marijuana). Other observed behavioral reactions are far more benign, including pursuing academic and occupational goals (with the increase in free time), visiting friends and family members, and building networks for social support.

To help deal with the physical, cognitive, and emotional consequences of injury, some athletes with injuries initiate behavioral coping strategies. Common behavioral responses of athletes with injuries include (a) trying to maintain as normal a life as possible while working hard at completing a prescribed rehabilitation regimen ("driving through"), (b) keeping their mind off the situation by staying busy (i.e., distraction), (c) attempting to use social resources as means of support, (d) staying away from others (i.e., isolation), (e) adopting an aggressive stance toward rehabilitation activities, (f) learning about their injuries, (g) trying unconventional treatments (e.g., alternative therapies), and (h) becoming physically stronger. Some of the behavioral coping strategies are instrumental or problem-focused in that they reflect attempts to address the athletes' concerns directly. Other behavioral coping responses are emotion-focused,

as they involve attempts to manage the stress associated with sport injury.

Adherence to rehabilitation programs represents the most researched behavioral response to sport injury. Adherence to prescribed treatment regimens is an area of concern across the health care spectrum, and sports health care is no exception. Adherence to treatment is considered important because of the presumed positive association between adherence and treatment outcomes. Specifically, better adherence is thought to contribute to more favorable outcomes. Actual dose-response relationships between sport injury rehabilitation activities and functional treatment outcomes are known for only a few treatments. Positive adherence-outcome relationships have been demonstrated for relatively few sport injury rehabilitation programs. In the absence of evidence to the contrary, however, adherence to sport injury rehabilitation programs is typically viewed as a worthy aim.

Rates of adherence to sport injury rehabilitation vary considerably depending on the nature of the rehabilitation program under consideration. Generally, high levels of adherence are obtained for clinic-based rehabilitation activities, which typically occur under the supervision of rehabilitation professionals. In clinical settings, adherence is commonly defined operationally in terms of the extent to which athletes attend rehabilitation sessions, expend effort during the sessions, follow the instructions of rehabilitation professionals, and complete prescribed rehabilitation exercises during the sessions they attend. Adherence rates are considerably more variable for home-based aspects of rehabilitation programs, ranging from complete nonadherence (in which athletes complete none of the prescribed regimen or even engage in contraindicated behavior) to gross overadherence (in which athletes do substantially more of the regimen than prescribed). Home-based rehabilitation programs may include elements such as (a) appropriate restriction of physical activities, (b) completion of rehabilitation exercises, (c) consumption of medications, and (d) use of therapeutic devices (e.g., splints, orthotics) and modalities (e.g., ice, transcutaneous electrical nerve stimulation [TENS]).

Consistent with hypotheses generated from cognitive appraisal models, a variety of personal, situational, cognitive, and emotional factors have been found to predict levels of adherence to sport injury rehabilitation programs. Athletes high on

self-motivation, self-identification with the athlete role, pain tolerance, and tough-mindedness tend to adhere better to sport injury rehabilitation than athletes low on these personal factors. With respect to situational factors, adherence levels tend to be high when athletes perceive (a) others as supporting their rehabilitation, (b) their rehabilitation professionals as expecting them to adhere, (c) the clinical setting as comfortable, and (d) the scheduling of their rehabilitation sessions as convenient. In terms of cognitive factors, adherence levels tend to be higher when athletes report believing that (a) their injuries are severe, (b) they are susceptible to additional health problems if they do not complete rehabilitation activities, (c) they can exercise control over their health in general and their injury status in particular, (d) their rehabilitation program is effective, and (e) they are capable of completing the tasks associated with rehabilitation. Athletes experiencing high levels of mood disturbance and fear of re-injury tend to adhere more poorly to sport injury rehabilitation programs than those without such distress.

Interventions to Affect Psychological Consequences of Sport Injury

Many of the cognitive, emotional, and behavioral consequences of sport injury are aversive and can have an adverse impact on athletes' quality of life (QOL) and recovery from injury. Consequently, researchers have developed, implemented, and evaluated interventions to prevent or reduce these circumstances. A small body of controlled experimental studies has documented the effects of the interventions on psychological outcomes. These studies indicate that goal setting, modeling, and several multimodal interventions are effective relative to a control condition not only for certain functional or physical rehabilitation outcomes (e.g., strength, endurance, recovery time) but also for one or more psychological response to sport injury.

Interventions in which goal setting is the primary focus have a beneficial effect on adherence to sport injury rehabilitation programs. A modeling intervention was found useful in promoting self-efficacy for the use of crutches in rehabilitation. Multimodal interventions with components such as relaxation and guided imagery, or stress management, cognitive control, goal setting, relaxation, and guided imagery, have produced

improvements in tension, extraversion, social orientation, security, rehabilitation attitude, readiness for physical activity (PA), and reinjury anxiety. Cognitive behavioral interventions are still in the infancy of their application in the context of sport injury rehabilitation. Nevertheless, informed by research findings and guided by cognitive appraisal models, they offer considerable promise in enhancing psychological responses to sport injury and, potentially, recovery and return-to-sport outcomes.

Britton W. Brewer

See also Adherence; Affect; Biopsychosocial Model of Injury; Injury, Psychological Susceptibility to; Injury, Return to Competition Following

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PSYCHOLOGICAL SKILLS

Psychological skills comprise the deliberate use of pre-prepared and structured sequences of specific thoughts and behaviors by athletes and exercisers to regulate their psychological state (e.g., feelings of confidence). Psychological skills are of interest within sport and exercise psychology (SEP) because psychological states are thought to affect performance in sport and exercise settings and, thus, if performers are able to regulate their psychological states, they may be able to enhance

their performance. In this entry, the theory underlying the role of psychological skills in sport and exercise performance is presented first. Next, four psychological skills that have received attention from researchers are described. The entry ends with a discussion of research indicating that increased psychological skill use is associated with enhanced performance.

Within current theory, it is proposed that a performer's psychological state (e.g., self-confidence, motivation, anxiety, and attention) is considered to affect his or her performance on tasks of interest such as throwing the javelin or completing an exhausting workout. Certain psychological states are considered more facilitative to performance than others. The nature of the relationship between a performer's psychological state and his or her performance depends on three factors: the individual performer, the specific task at hand, and the wider task environment. Regarding the individual performer, Anna might feel her tackling performance in soccer is better when she feels aggressive, but Beth might feel her performance on this task is better when she is calm. Regarding the task, while Anna feels her tackles are better when she feels aggressive, she might feel her corner kicks require calmness. Regarding the wider task environment, Anna might find it easier to feel the anger she requires for tackles during a high-stakes competition than a "friendly" scrimmage.

Thus, a challenge for an individual performer is to attain and maintain a psychological state that facilitates performance, given his or her personality, the specific task, and the wider task environment. The ability to attain such a state is thought to depend, in part, on the use of psychological skills. Such skills are thought to be learnable and learned via instruction (e.g., from coaches) and natural learning experiences such as competitions. The skills take two forms, termed *advanced* and *basic psychological skills*. Advanced psychological skills include being able to regulate one's self-confidence, motivation, anxiety, and attention. A gymnast who makes an error during a routine but is able to refocus quickly on the routine, instead of dwelling on the error, would be considered as having the advanced psychological skill of being able to regulate his attention. Basic psychological skills are considered to serve advanced psychological skills. An example of the use of self-talk (a basic psychological skill) to regulate attention (an advanced psychological skill) is the

aforementioned gymnast saying to himself to stop and refocus after making the error to refocus on the routine and avoid dwelling on the error.

Most research has involved four basic psychological skills: goal setting, mental imagery, relaxation and activation, and self-talk skills. Goal setting involves deliberately establishing, refining, and evaluating progress toward a goal. For example, a weight room exerciser might set a goal to improve her squat technique. Mental imagery describes mental activity that resembles a real experience but occurs in the absence of such an experience. For example, on rest days, a kicker in football might mentally image taking field goals to prepare to do this for real during upcoming games. Relaxation skills involve thoughts (e.g., “relax”) and actions (e.g., breaths) that alter levels or interpretations of anxiety symptoms such as worries and butterflies. For example, a rock climber might take a few slow, controlled breaths in an attempt to feel more in control of her body before making a difficult move on a steep cliff. Self-talk involves talking to oneself in one’s mind or externally, as in the previously given example of the gymnast’s use of the skill.

While more research is required on psychological skills to better establish their effects on performance, to date, two kinds of research have indicated that increased psychological skill use is associated with enhanced performance. First, survey-based research has indicated that (a) increased use of advanced and basic psychological skills is associated with more, versus less, successful competitive performances and (b) athletes competing at higher, versus lower, competitive levels (e.g., international vs. regional) report more frequent use of these skills during practice and competition. For example, the ability to effectively focus attention on one’s performance in the presence of distractions (an advanced psychological skill) is associated with more successful competitive performances. In addition, when compared to non-elite athletes, elite athletes have been shown to spend more time in a typical training week engaged in the basic psychological skill of relaxation, perceive relaxation skills as more relevant to performance, and report greater use of relaxation skills to cope with anxiety during competition.

Second, controlled experiments indicate that psychological skill use benefits performance. For example, reviews of studies of mental imagery show that the use of imagery to mentally practice

a motor skill such as a tennis serve, when compared to no practice (mental or actual) of the skill, enhances learning and performance. Also, a recent review of the use of self-talk indicates that positive self-talk that is motivational (I can do this!) or instructional (e.g., keep my weight over my feet) in nature has beneficial effects on performance.

In conclusion, athletes and exercisers are thought to be able to enhance performance on tasks in their domains by regulating their psychological state through the use of learned psychological skills. These skills include advanced psychological skills, such as the ability to regulate anxiety, and basic psychological skills, such as self-talk, which underpin advanced psychological skills. Survey-based research indicates that the use of these skills is associated with better competitive performances, and controlled experiments indicate that performers who use these skills learn faster and perform better. Athletes and exercisers who learn and practice using psychological skills can expect to capitalize on their positive effects on performance.

David W. Eccles and Kaitlin Riley

See also Burnout; Fatigue; Memory; Motor Learning; Overtraining Syndrome; Practice; Skill Acquisition

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PSYCHOLOGICAL SKILLS TRAINING

Psychological skills training (PST) involves training athletes and exercisers to learn psychological skills (e.g., relaxation skills) that help these performers regulate their psychological state (e.g., their feelings of confidence). PST is of interest within sport and exercise psychology (SEP) because psychological states can affect sport and exercise performance; if performers can regulate

their psychological states via learned psychological skills, they may be able to enhance their performance. In this entry, the theory underlying the role of psychological skills in sport and exercise performance is presented first, PST programs are described next, and evidence demonstrating PST improves sport and exercise performance is presented last.

Within current theory, performers' psychological states (e.g., feelings of motivation) are considered to affect their performance on sport- and exercise-related tasks. Certain psychological states are considered more facilitative to performance than others, depending on the individual performer, the task, and the wider task environment. For example, the psychological state conducive to lifting weights during training might be different from the state conducive to taking a penalty kick during a soccer game; and the state required for each of these tasks may differ across individuals (e.g., introverts vs. extraverts). Thus, a challenge for performers is to attain a psychological state that facilitates performance. The ability to attain such a state is thought to depend, in part, on the use of advanced and basic psychological skills. Advanced psychological skills include being able to regulate one's self-confidence, motivation, anxiety, and attention. A soccer player who misses a penalty kick during a game but refocuses quickly would be considered as having the advanced psychological skill of being able to regulate her attention. Basic psychological skills, which include goal setting, imagery, relaxation, and self-talk, are considered to serve advanced psychological skills. An example of the use of self-talk (a basic psychological skill) to regulate attention (an advanced psychological skill) is the soccer player saying to herself, "Focus on now" when she finds herself thinking about the missed shot during the remainder of the game.

One way psychological skills are learned is by training. Training programs can vary widely, but some typical program characteristics, as well as "best practice" principles for program design, are as follows. Programs can be implemented by the athlete or exerciser but are more likely to be beneficial when delivered by knowledgeable instructors (e.g., sport psychology [SP] consultants). Programs often have three phases: an education phase, in which the potential benefits of psychological skills are explained; an acquisition phase, in which specific skills (e.g., imagery) are introduced; and a practice phase, in which performers practice

applying the skills. Programs are more effective when: tailored to performers (e.g., junior elite-level) and their sport (e.g., rugby) and situation (e.g., recovering from injury); needs-led, so that the program is based on an assessment of facets of the performer's mental game requiring improvement (e.g., anxiety control); and performer-led, so that the performer has as much of a role in identifying their psychological needs as the instructor. Finally, programs will benefit when there is ongoing evaluation of their effectiveness.

While further research is required to better establish the effects of PST on performance, the research to date has provided evidence that such training can improve performance. This evidence has been provided by what we term *experimental studies* and *education studies*. The emphasis within experimental studies is on identifying the effect of the use of a particular psychological skill on the performance of given task. The typical training provided is limited: It involves a single contact with the participant, lasts minutes, and includes simple instructions directing the participant to apply the skill while completing a given task. In a study of self-talk, Antonis Hatzigeorgiadis and his colleagues assigned tennis players to an experimental or control group. The groups completed five tennis practice sessions. In the first session, forehand drives were practiced. In the next three sessions, participants practiced backhand drives. The experimental group was asked to execute each backhand in these three sessions while saying various words (e.g., *low* or *strong*) to themselves; the control group did not use these words. The fifth session, like the first session, involved forehand drives for both groups. In this session, the experimental group chose one self-talk word from the previous three sessions to use as they executed each forehand; again, the control group did not use these words. Forehand drive performance was assessed in the first and fifth sessions for both groups. Performance improved for the experimental group but not the control group.

The aim within education studies has been to identify the effect of a *program* of PST on an athlete's or team's overall performance in their sport. The typical training provided involves multiple contacts with the participant(s) over days or weeks, includes education about using various psychological skills in various situations on and off the "field," and is at least partly classroom-based. Richard Thelwell and his colleagues studied the

effects of a midfielder-specific training program in relaxation, imagery, and self-talk skills on the performance of five soccer midfielders. A trained sport psychologist consultant taught a different skill each day over 3 days and assigned “homework” involving workbook exercises. Training also involved practicing the skills: For example, players practiced using relaxation skills to recover from “bad” first touches on the ball. Players were rated on three midfielder-specific aspects of play performance over nine real soccer games. PST effects on performance were assessed using a multiple baseline research design, which involved players receiving PST individually and at different times within the nine-game period: Player 1 received the training after the third game, Player 2 after the fourth game, and so on. Following PST, all players experienced at least small improvements in each aspect of play performance and four of five players experienced more pronounced improvements in these aspects.

In conclusion, athletes and exercisers are thought to be able to enhance performance on tasks in their domains by regulating their psychological state through the use of psychological skills. PST involves attempts to teach performers psychological skills so that they might obtain these performance benefits. Experimental and education studies indicate PST enhances performance in sport and exercise settings, although further research on this topic is required.

David W. Eccles and Kaitlin Riley

See also Breathing Exercises; Coping; Goal Setting; Imagery; Interventions for Exercise and Physical Activity; Relaxation; Self-Regulation; Self-Talk; Stress Management

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PSYCHOLOGICAL WELL-BEING

Psychological well-being (PWB) is defined as one's level of psychological happiness/health, encompassing life satisfaction, and feelings of accomplishment. At the risk of being dualistic and separating physical well-being from PWB, it is helpful to note that physical well-being encompasses physical health, including disease states, fitness level, and ability to perform activities of daily living (ADL). PWB encompasses the person's perspective on life, including not only perceptions of physical health but also of self-esteem, self-efficacy, relationships with others, and satisfaction with life. A monistic perspective, wherein it is recognized that physical well-being and PWB are integrally interrelated, is preferable.

Subjective well-being (SWB) can be addressed from a primarily psychological perspective. Ed Diener suggested that SWB is comprised of the presence of positive affect, the absence of negative affect, and high levels of life satisfaction. However, research indicates physical well-being should be considered as well in evaluating PWB. The Roman poet Juvenal was correct in that *mens sana in corpore sano* (“a sound mind in a healthy body”) is most desirable. PWB can range on a continuum from the absence of well-being, identified as ill health or even mental illness in some cases, to optimal well-being, perhaps having attained the state of self-actualization described by Abraham Maslow. Many individuals will identify their PWB as being on the wellness side of the continuum. Striving toward one's psychological potential and toward balance in one's life and happiness (perhaps noting a positive psychology approach) might bring one closer to optimal PWB. Brian C. Focht also noted the connections of PWB to health-related quality of life (HRQOL). A “sound mind” can be examined from a number of perspectives within an exercise and sport psychology (SP) framework: assessment of physical capabilities, including fitness; social relationships; level of athletic or exercise identity; the balance in life across a variety of domains, including social, work or academic, and spiritual; and feelings of accomplishment and progress toward attainment of an individual's potential.

The first perspective involves assessment of physical capabilities, including fitness. PWB is certainly affected by fitness level, both in terms

of exercise and sport participation but also in an individual's ability to perform ADL. Ability to perform ADL (e.g., self-care or personal hygiene, mobility, housework) will likely affect one's PWB. Impairments in ability to perform ADL may decrease PWB due to decreased independence and subsequent reliance on others and/or assistive devices to help in accomplishing ADL. PWB would be expected to be high to the degree that one is fit enough (fitness encompassing cardiovascular capacity, muscular strength, muscular endurance, and flexibility) to perform in preferred exercise and sport modalities. PWB level may be affected by goals. If a person is striving to improve exercise or sport performance, then increases in PWB would accompany increases in fitness (perhaps also tied into performance, which may encompass skill development in aspects of a given exercise or sport, such as the backhand stroke in tennis). Bonnie G. Berger and David A. Tobar addressed physical activity (PA) and quality of life (QOL) more extensively.

In terms of social relationships, most individuals desire some level of social interaction (e.g., No man is an island). PWB would be expected to be higher if a person has the level of social interaction desired. If an individual has fewer close friends than desired, however, then PWB would be expected to be lower. Social relationships and social interactions are often found in exercise and sport settings; examples include friendly get-togethers for a Sunday basketball game or a bicycle ride in one's area, or attending a local sporting event. Exercise or sport can be an excellent vehicle for meeting people, such as at a local health club or fitness center or at participatory events such as running road races or triathlons.

Athletic or exercise identity refers to the degree to which one identifies as an exerciser or athlete. The greater an individual's identity is tied to her or his exercise or athletic existence, the greater PWB is likely to be affected by exercise or athletic performance as well as injuries or illnesses that might make participation challenging. Research indicates the desirability of individuals having other coping strategies for times when the exercise or athletic component of their lives is not going as well as desired.

Of particular importance is the balance in one's life across a variety of domains, including social, work or academic, and spiritual. Life is a balance of the roles we play—work, family, religious or

spiritual, exercise or sport, to name primary ones. Concerns might be raised if exercise–athletic identity is primary and other important areas of a person's existence are excluded. It is important to attempt to be nonjudgmental and know the individual before deciding if someone's life appears out of balance. If someone's PWB is high due to the high priority given to one area of their existence (such as exercise or sport), then this should be acknowledged, assuming this is a conscious choice on the part of the individual. Sometimes, however, individuals find themselves drifting into one area of primacy, while neglecting other areas unintentionally. This can lead to exercise addiction in some cases, where a person has lost control of the activity and her or his life is seemingly governed by exercise or sport. A conscious decision-making process on allocation of time and energy to the various important areas of one's life, including the role of exercise and sport participation, is critical to PWB.

Feelings of accomplishment and potential attainment of a person's potential is also related to PWB. PWB may be tied to satisfaction with life—that is, has the person accomplished or attained the goals that were set? Has the person reached her or his potential in one or more of life's domains, as noted previously? To the degree that the person has, then her or his PWB would be expected to be higher. This can encompass exercise or sport as well—attaining one's goals and reaching one's potential in preferred exercise or sport pursuit(s) contributes to enhanced PWB.

In summary, PWB is affected by an array of factors associated with physical and mental health. Exercise and PA and sport play key roles in both physical and mental health, and therefore have profound effects on PWB.

Michael L. Sachs

See also Health Promotion; Mental Health; Quality of Life

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PSYCHOPHARMACOLOGY

Psychopharmacology refers to the study of the actions of drugs on behavior. Drugs and substances with the potential to modify or completely change human behavior are best known as *psychoactive*. It is noteworthy that psychopharmacology has developed along with advances in neuroscience research. These two fields of scientific inquiry revealed the mechanisms of brain function—and more especially how chemical substances naturally occurring in the human brain (i.e., neurotransmitters) influence human behavior. The different types of psychoactive drugs can be broadly categorized as follows:

- Opiates and narcotics
- Neuroleptics or antipsychotics
- Stimulants
- Anti-anxiety agents
- Antidepressants
- Psychedelics
- Sedative-hypnotics

Nicotine, caffeine, alcohol, as well as certain herbal products (e.g., ginkgo biloba and ephedrine) also exert psychoactive effects and change aspects of behavior by inducing alertness and increasing energy levels.

Function and Uses of Psychoactive Drugs

Psychoactive drugs exert their effects on brain cells (neurons) mainly by mimicking or increasing the function of neurotransmitters (agonists) or decreasing and countering the effects of neurotransmitters (antagonists). Some psychoactive drugs are used as sedatives and induce hypnotic states, whereas others are used to counter the negative effects of

anxiety disorders or depression. In principal, psychoactive drugs are used for the treatment of a wide range of psychiatric symptoms and psychological disorders. In addition to their clinical uses, psychoactive substances are consumed for recreational purposes in a variety of settings and social contexts. Alcohol, nicotine, marijuana, and psychedelic drugs (e.g., MDMA, or ecstasy) are commonly used, especially among young people, during social interactions, such as parties. Studies indicate that psychoactive substances may also be used to increase academic performance during exams and increase productivity in specific professions.

Psychoactive Drug Use in Sports

Psychoactive drugs are used in sports to increase athletic performance in numerous ways. Several studies report that psychoactive substances are effective in countering the effects of insomnia and sleep disturbances followed by excessive training in elite athletes. Athletes may also use psychoactive drugs to counter stress and anxiety associated with intensive training and work-life balance issues. Research also indicates that psychoactive substances are used to increase athletic performance. For instance, stimulants like ephedrine-based supplements induce thermogenesis and elevate energy levels. This is assumed to help athletes increase physical stamina and burn body fat more effectively. However, evidence of the actual performance-enhancing properties of psychoactive drugs is unequivocal. The presumed ergogenic effects may come at the expense of physical and mental well-being. Specifically, serious health problems and death may result from prolonged and excessive use of psychoactive substances, such as amphetamines.

Therapeutic Uses and Doping Prevention

International sporting organizations and agencies try to regulate psychoactive drug use among athletes. The list of prohibited substances issued annually by the World Anti-Doping Agency (WADA) includes several types of psychoactive drugs that are considered doping agents, such as narcotics, stimulants, amphetamines, and alcohol (prohibited in-competition in specific sports). Nevertheless, some prohibited psychoactive drugs may be used under the Therapeutic Use Exemptions (TUE) regime. This involves the authorization to use medication for the treatment of specific conditions or

illnesses in athletes and necessitates strict medical supervision of the TUE process.

Psychoactive Drug Use Inside and Outside Sports

Psychoactive drugs are used in different situations, and the use of specific substances (e.g., cannabis, alcohol, nicotine) can be initiated, as previously noted, in nonsporting contexts such as parties and other socializing activities. Thus, unlike other doping substances, psychoactive drugs entail a unique socialization element, and their use may be the result of peer influences or other processes that do not necessarily relate to performance enhancement. By this token, preventing psychoactive drug use in sports should not only focus on a strict detection-and-punishment approach. Rather it could be equally important to address the broader societal influences and trends that facilitate and normalize psychoactive drug use outside sports. For instance, collegiate athletes are familiarized with excessive alcohol drinking and the use of other psychoactive substances in college campus and outside sporting events. This socialization process may facilitate the use of psychoactive drugs in the sporting context—the drugs once used for recreation can be used to facilitate performance enhancement.

Lambros Lazuras and Vassilis Barkoukis

See also Alcohol Abuse; Drug Use and Control; Performance-Enhancing Drugs; Recreational Drugs

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PSYCHOPHYSIOLOGY

Psychophysiology is the scientific study of the connection between the mind (psychology) and the body (physiology). Psychophysiology is based on the premise that there is two-way communication between the mind and the body. Thus, physical changes in the body can affect psychological responses in the mind and psychological experiences may be reflected in physiological measures.

Psychophysiology has been used extensively in the field of sport and exercise psychology (SEP). In sport psychology (SP), psychophysiological measures have been used with athletes to understand physiological variables that are predictive of performance and to use this understanding to improve performance. In exercise psychology, scientists have used psychophysiological measures to provide physical evidence of the effects of exercise on a variety of psychological experiences including cognitive function and brain health, stress and anxiety, affect, arousal, and sleep.

Psychophysiological Measures

The psychophysiological measures relevant to SEP are constantly evolving as technological advances allow for the use of noninvasive measures and theory advances our understanding of potential mechanisms of relevant outcomes. The measures described here are commonly used in SEP research.

Magnetic Resonance Imaging

By utilizing finely tuned magnetic pulses that disrupt the orientation of atoms in the brain, magnetic resonance imaging (MRI) is used to assess cerebral structure. MRI is capable of making

multiple assessments of the brain at varying depths and in different directional planes (sagittal, coronal, transverse). This makes it possible to assess the cerebral volume of specific areas of the brain. Exercise psychologists have used MRI to assess changes in cerebral volume in relation to physical activity (PA) in clinical (e.g., Alzheimer's disease) and nonclinical populations.

Functional Magnetic Resonance Imaging

The brain requires a supply of oxygenated blood in order to support neuronal activation. In areas of the brain where activation has occurred, blood flow is increased to compensate for the consumption of oxygen and other vital nutrients. The increased blood flow results in a higher concentration of oxygenated hemoglobin (the protein that transports oxygen) in the specific areas of activation relative to less active areas. Functional magnetic resonance imaging (fMRI) is able to detect the different magnetic properties of oxygenated and deoxygenated hemoglobin in blood, and this blood-oxygen-level-dependent (BOLD) response is used to infer neural activation, which is interpreted as indicative of cognitive activity.

Heart Rate and Heart Rate Variability

Heart rate (HR) provides a simple measure of arousal and affect. An increase in HR has been interpreted as indicative of an increase in psychological arousal or a change in mood. Assessing HR can be done manually by palpating the wrist or carotid artery and counting the number of beats per minute (BPM), by using a heart monitor, or by measuring electrocardiographic activity using electrodes.

Heart rate variability (HRV) refers to variations in the time between heartbeats (which is also referred to as the *interbeat interval* [IBI]). HRV has been used to make inferences about the control of HR by the autonomic nervous system (ANS). In particular, in the SEP literature, HRV is interpreted relative to high-frequency (parasympathetic activity) and low-frequency (both sympathetic and parasympathetic activity) components.

Electroencephalography

Electroencephalography (EEG) (also referred to as electroencephalographic activity) is the process of recording electrical activity across the scalp. The supposition is that the EEG activity is

indicative of the neural activity taking place in the brain. EEG is described as being high in temporal resolution because observed activity is related to current cognitive processes but low in spatial resolution because data represents the firing of a large number of neurons in a general area that may or may not be located near the recording location. EEG recordings may be used to assess either spontaneous activity or event-related potentials (ERPs). These constructs are described further in the entry titled *Electroencephalograph (EEG)*.

Electromyography

Electromyography (EMG) is used to measure skeletal muscle activity (sometimes referred to as tension) by taking measures of electrical activity either at the skin using surface electrodes or directly from the muscle using intramuscular electrodes. EMG measures have been used to identify the timing of muscle activation and to provide an indication of levels of arousal, stress, and affect.

Galvanic Skin Response

Galvanic skin response (GSR) is measured by assessing skin conductance (SC). An electrical stimulus with an extremely small voltage is applied between two electrodes adhered to the skin, and the conductance between them is recorded. GSR is directly reflective of moisture level and is interpreted as being indicative of sympathetic nervous system (SNS) activity and arousal.

Cortisol

Cortisol is also referred to as the "stress hormone" and is released when the hypothalamic-pituitary-adrenal (HPA) gland axis is stimulated. When a person is exposed to a stressful stimulus, the brain (i.e., hypothalamus and pituitary gland) signals the adrenal glands to release cortisol from the adrenal glands to prepare the body for fight or flight. When it senses the end of the stressor, the brain begins the reuptake processes of removing the cortisol from the blood stream, which returns the body to homeostasis. Cortisol levels are indicative of stress and can be assessed in peripheral blood and in saliva.

In addition to the psychophysiological measures described here, researchers in the field of SEP have also used measures of critical flicker fusion, acoustic startle eye blink responses, neurotransmitters (e.g., serotonin, dopamine), hormones (e.g.,

epinephrine), and body temperature to enhance our understanding of mind and body interactions.

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See also Biofeedback, Neurofeedback;
Electroencephalograph (EEG); Electromyography (EMG); Individual Response Stereotype

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PUPILLOMETRY

See Eye Movements/Gaze



QUALITY OF LIFE

Quality of life (QOL) assessments are instrumental in developing a more comprehensive understanding of the efficacy of disease prevention and health promotion interventions. There is growing interest in delineating the role of exercise in enhancing QOL outcomes. Indeed, findings from multiple reviews on the effect of exercise on QOL reveal that exercise consistently results in small to moderate, yet clinically meaningful, improvements in multiple QOL outcomes, ranging from global indices of life satisfaction to disease-specific measures of QOL dimensions. Collectively, these findings provide compelling evidence of the QOL benefits of exercise across a variety of populations.

Given the well-established QOL benefits of exercise, determining factors that influence the exercise–QOL relationship is a focal point of contemporary exercise psychology research. Therefore, among the objectives of this entry are to identify: (a) considerations in the conceptualization and structure of QOL and (b) potential moderators and mediators of the exercise–QOL relationship.

Definition and Structure of Quality of Life

Definitions of QOL vary across studies examining the exercise–QOL relationship. In the psychology literature, QOL is conceptualized as an index of overall life satisfaction. Conversely, medical and behavioral science disciplines have espoused multidimensional perspectives of QOL

outcomes reflecting one's appraisal of key life components including physical health, psychological well-being, social relationships, and level of independence. In select medical disciplines, QOL is conceptualized as health-related quality of life (HRQOL), a multidimensional, umbrella concept that comprises broad categories of well-being and functioning and more discrete, specific aspects reflecting core dimensions of function and well-being. Subjective well-being (SWB), a related yet distinct outcome, is comprised of life satisfaction (the cognitive evaluation of the overall quality of one's life) and positive and negative affect (the frequency and intensity of pleasant and unpleasant feelings experienced in one's life).

Variability in the conceptualization of QOL across disciplines presents a challenge to synthesizing knowledge of the QOL benefits of exercise. Nonetheless, it can be concluded that QOL, HRQOL, and SWB are related, though not isomorphic, constructs and that exercise consistently elicits improvements of similar magnitude across these outcomes. Emerging evidence also suggests that QOL outcomes are *hierarchically organized* and exercise-related improvements in proximal QOL subdomains predict favorable change in more distal global QOL outcomes. Consistent with this proposed hierarchical structure, QOL can be viewed as an umbrella term involving the subjective appraisal of various dimensions of one's life, encompassing global ratings of life satisfaction, broad dimensions of functioning and well-being, and perceptions of specific domains of functioning and well-being.

Moderators and Mediators of the Effects of Exercise on Quality of Life

With regard to moderator variables, baseline QOL values consistently act as independent predictors of improvements in QOL following exercise across multiple population subgroups and QOL outcomes. These findings provide compelling evidence for the position that those who have the most to gain in terms of improvement of QOL (e.g., those presenting with the least favorable QOL profile at baseline) prior to initiating exercise derive the greatest benefit from exercise interventions across an array of QOL outcomes. Although gender has been proposed to moderate the effects of exercise on QOL, evidence of consistent gender differences in the QOL responses to exercise interventions remains equivocal and warrants further inquiry.

Investigations addressing mediators of the QOL benefits of exercise have adopted *top-down* or *bottom-up perspectives*. Top-down perspectives propose that judgments of satisfaction with specific domains of life, functioning, and/or well-being are determined by ratings of overall QOL and well-being. Thus, those reporting favorable ratings of global life satisfaction or well-being evaluate specific dimensions of life in a similarly positive manner. Conversely, bottom-up perspectives propose that judgments of specific life domains influence global ratings of QOL. Although both top-down and bottom-up models have merit, findings from the exercise-QOL literature have yielded consistent support for the bottom-up perspective and little evidence of the utility of the top-down approach.

Consistent with the bottom-up perspective, recent studies in multiple sclerosis patients demonstrated that exercise positively influenced self-efficacy, disease symptoms, and select indicants of psychological well-being. In turn, changes in these intermediate outcomes were linked with more favorable ratings of global QOL. Similar mediational pathways were observed with exercise interventions targeting arthritis patients. For example, mobility-related self-efficacy beliefs and knee pain symptoms, both domain-specific QOL measures, mediated the beneficial effect of exercise upon functional performance and ratings of generic HRQOL.

Additionally, satisfaction with physical function, knee pain ratings, and mobility-related self-efficacy has been identified as independent mediators of the effect an exercise and dietary interventions on QOL in arthritis patients. Studies

also demonstrated that the favorable effects of exercise on global QOL were partially mediated by improvements in affective responses, self-efficacy, and perceptions of social support among older adults at risk for functional decline.

Thus, the effects of exercise on global indices of QOL can be mediated by changes in proximal QOL domains. The indirect effect of exercise upon global QOL suggests that these potential mediators should be targeted within future exercise intervention trials. It also provides support for the hierarchical structure of QOL outcomes whereby improvements in proximal QOL constructs lead to improvements in distal global QOL outcomes. Thus, exercise may influence proximal QOL dimensions assessed by measures of generic core domains of QOL and disease- or population-specific symptoms. In turn, favorable changes in these proximal QOL outcomes subsequently contribute to improvements in overall, global ratings of QOL.

Directions for Research in the Exercise-Quality of Life Relationship

Select areas of future inquiry may be particularly integral to advancing knowledge of the exercise-QOL relationship. It is critical for researchers to acknowledge that QOL, HRQOL, and SWB represent related yet conceptually distinct constructs. Additionally, emerging evidence supports the position that QOL outcomes are hierarchically organized and that a bottom-up model accounts for the direct effects of exercise on QOL domain outcomes and indirect effects upon global QOL.

Implementation of a hierarchical, bottom-up conceptualization of the exercise-QOL relationship has important implications for designing exercise interventions. This approach suggests that multilevel assessments integrating global indices of life satisfaction, generic measures of QOL, and targeted measures of domain-level QOL outcomes should be included as a standard part of theoretically driven assessment batteries. Augmenting this approach through the measurement of conceptually relevant and evidence-based mediators will also expand understanding of the mechanisms underlying the QOL benefits of exercise. Adoption of the hierarchical model could have ancillary benefits in advancing knowledge of the role of exercise in enhancing QOL. For example, exercise-induced increases in specific QOL domains are mistakenly interpreted as reflecting improvement in other

QOL domains or global QOL. In a hierarchical model of QOL, improvements in QOL domains should not be assumed to be indicants of change in global QOL. Consequently, improvement in one QOL measure should not be used as a proxy for inferring change in different QOL outcomes.

In summary, exercise is valuable for enhancing and sustaining QOL. Despite the well-established QOL benefits of exercise, advances in the conceptualization of QOL outcomes and the mechanisms underlying improvements in QOL have been limited. Adopting a hierarchical, bottom-up approach to the study of exercise-related QOL benefits is an important step in augmenting the incremental progress that has been observed in the investigation of the exercise–QOL relationship.

Brian Focht

See also Adherence; Affect; Affective Responses to Exercise; Psychological Well-Being; Self-Efficacy

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R

RACE

The term *race* is not easily defined; it has complicated and contested meanings contingent upon historical and social contexts. In a general sense, scholars within, and outside, sport psychology (SP) have conceptualized *race* as having biological and/or social distinctions. From a biological perspective, *race* refers to individuals who are perceived by others, and perhaps by themselves, as possessing distinctive, inherited, biological traits (e.g., skin color). People subscribing to a biological perspective typically regard race as a fixed, objective, homogenous category. Enduring assumptions from this perspective include the argument that people can be divided into biologically detached “races” and/or static representational categories of particular races and that similarities within, and difference between, these groups can be reduced to appearance, ability, behavior, and psychological characteristics. From a social perspective, the foregoing assumptions are viewed as simplistic and problematic, as race cannot be solely reduced to a scientific, neutral, and monolithic (i.e., all-encompassing representing a single perspective or experience) category. Instead, *race* is conceptualized as a socially constructed phenomenon, the meaning of which is contingent upon changing and complicated interactions between social processes and lived experiences. Thus, from a social constructionist viewpoint, the term *race* is not viewed as a fixed, natural, and biologically given but is a *constructed* term—the meaning of which

is fluid depending upon the contextual moment and one’s interpretations and experiences within that moment. The degree to which biological assumptions concerning race have endured is not confirmation of their existence but rather shows that such explanations have persisted over time as socially agreed upon ways of speaking about *race*.

Despite the acknowledgment of race as a quasi-biological and social construct, in North America the term is still often reduced to an overarching concept equated with skin color and/or physical appearance. Sport studies scholars and, more recently, scholars in SP have called attention to the political effects of such simplistic “race conceptions,” as they perpetuate a Eurocentric white (and often male) hierarchy of power and privilege in sport. In turn, one perspective in SP emerges as “truth” and as the dominant view against which all experiences are measured and interpreted. Perspectives and voices falling outside of a mainstream Eurocentric viewpoint are silenced, particularly the perspectives of those whose identities intersect with class, gender, and sexuality. Another result of such silencing and privileging of primarily one perspective is that the justification of bias, discrimination, prejudice and stereotypes may result.

Ethnicity is similarly misconstrued and often used interchangeably with the term *race*; thus, the term also warrants mention or highlighting in conceptions and discussions of race. Ethnicity has been used to refer to a unique set of cultural, social, and religious characteristics (e.g., shared customs, values, traditions, worldviews) and does not refer to one’s physical appearance or serve as

a proxy for a particular race. As with the aforementioned biological viewpoint concerning race, when used interchangeably with race, the term *ethnicity* takes on the inherent, historical, and social meanings and the (problematic) implications of the biological viewpoint. In-line with a social constructionist conception of race, ethnicity can also be conceptualized as having multiple meanings that are dynamic and changing because they are socially and culturally (re)produced. Scholars outside, and within, SP have highlighted the need to recognize that while there may be a degree of overlap between conceptions and meanings of race and ethnicity, one should be aware of the foregoing dynamic meanings and that each of these terms have very different theoretical approaches to conceptualization.

While the inclusion of race and ethnicity in SP is relatively new and still a void within the field, it is a growing area of interest in both research and application. To date, many of the approaches used have been to assert that race exists and to ask what effect membership in a particular race or ethnic group has on sport involvement, what psychological implications result, and in turn, consider the implications for applied practice. In addition to such approaches, sport studies scholars have called for critical forms of exploration and theorizing race and ethnicity to move beyond looking at race and ethnicity as interchangeable and/or as fixed variables. Critical forms of theorizing allow researchers and practitioners to acknowledge and explore the complexity and multiple meanings of race and ethnicity—particularly in terms of their intersection with class, gender, and sexuality. Following the recommendations and work of these scholars, it has been suggested that a more profound, and perhaps socially transforming approach for SP, is to conceptualize *race* as a socially and culturally produced marker of a particular relationship of power, to see racial identity as contested, and to ask how and why racial relations are (re)produced through sport.

As the field of sport and exercise psychology moves forward in a global and multicultural society, the inclusion of marginalized identities, cultures, and perspectives outside of the mainstream is important. SP practitioners, educators, and researchers may increasingly find themselves working with athletes, coaches, and physical activity (PA) participants from a wide range of cultural backgrounds. An awareness of the foregoing

distinctions and discussions concerning race and ethnicity will be an important gateway toward enhanced mutual understanding between minority participants and those working in SP contexts. Such mutual understanding may open up a deeper appreciation of PA participants from a multicultural perspective, as well as break down racial, social, economic, and political barriers that impede and/or impact PA participation for minority groups in society.

Kerry R. McGannon

See also Cultural Competence; Cultural Safety; Feminism; Multiculturalism; Racism/Whiteness; Stereotyping, Cultural/Ethnic/Racial

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RACISM/WHITENESS

Race can be understood as a concept that signifies meanings and struggles over power in reference to skin color. Within the social sciences, most consider race to be not a biologically valid concept but rather a social construction. The issue of race in sport and exercise psychology is important because while the majority of professionals in the field are White, the clientele is diverse in terms of racial and ethnic backgrounds. This is particularly true in sports such as baseball, basketball, and football. Scholars have noted that in the conference presentations of the Association for Applied Sport Psychology (AASP), there have been few meaningful attempts to incorporate issues related to racial and ethnic identities. Others have pointed out the infrequency with which race or ethnic identity are meaningfully discussed in the methods and results

sections of articles in the major sport and exercise psychology journals. In this sense, race has been conspicuous by its absence. In comparison, fields such as counseling psychology, nursing, and others have explicitly developed a body of academic work dealing with how race may play into interpersonal interactions with clients and patients.

Over the past several years, sport and exercise psychology scholars have called upon members of the field to move toward a greater sensitivity and engagement with cultural issues. This “cultural turn” has involved an increased attention to issues of race and ethnic identities in the field and their implications for practice. In addition to several well-respected journal issues, two anthologies, *The Cultural Turn in Sport Psychology* and *Cultural Sport Psychology*, signaled a new direction for the field and an attempt to foreground issues such as race in research and application. The “cultural turn” is also related to the more general infusion of critical race theory into academia. Critical race theory involves a praxis-based attempt to study and transform the relationship between racial identity, racism, and power. Thus, researchers should not simply increase the diversity of their samples but actually use research to identify, critique, and dismantle racist structures in sport and society. Further evidence that the cultural turn in sport and exercise psychology has gained momentum is found in popular sport and exercise psychology textbooks that now include sections on racial and ethnic identity.

As one of the major social institutions in the United States and elsewhere, sport has much to do with racial formation, racial ideology, and whiteness. Since the late 1990s, the academic work on racial identities and racism has prompted scholars to address the place of *whiteness* in sport. The term *whiteness* refers to the evolving discourse on white racial identities and privileges associated with having, and identifying with, a White ethnic identity. As opposed to the problematic practice of examining the experiences of minority groups, or the “racial other,” from a supposedly color-blind perspective where “sameness” is prioritized over equal treatment and the recognition of the meaningfulness of racial identity, whiteness studies switches the focus of research onto race by confronting the socially constructed nature of white identity and the impact of whiteness on social relationships. Whiteness studies thus acknowledges that what we mean by “White” is contingent on

sociohistorical and political contexts, and recognizes the need to critique whiteness as a key factor in sociocultural spheres. For example, scholars have examined how African American college athletes perceive their interactions with White sport psychology (SP) professionals.

Scholars in whiteness studies originally conceived of whiteness as being akin to an invisible knapsack that was filled with white and male privileges and that were often invisible to White individuals. It has been argued that white men in particular carry with them unexamined, unearned privileges across many social spheres because they are considered “White.” Listing these privileges is one method to begin to confront, denounce, and eliminate these otherwise unnoticed inequities. However, scholars have recently questioned the effectiveness of this seemingly simple task. Authors of this scholarship in sport have focused less on the examination of individual white privileges and more on the ways that racial inequality and racism are reproduced and experienced in what scholars have labeled the “new racism.” Some aspects of new racism involve a more covert discourse on race, an avoidance of racial terminology (e.g., “post-racial America”), claims of reverse racism by whites, and the incorporation of “safe minorities” as proof that race is less important than it was previously.

Several authors have provided suggestions to address issues of racial identity in sport and exercise psychology. The universalistic approach suggests that professionals be taught cultural competence, in the process allowing them to be more sensitive to racial and ethnic differences. Cultural competence involves the professional recognizing the client’s racial identity, his or her own racial identity, and how race and ethnicity may play a role in their interactions. The role of introspective, self-reflexive practice is crucial to this process. Three areas of cultural competence have been proposed, including an awareness of one’s own values and biases, an understanding of others’ perspectives and worldviews, and culturally appropriate intervention strategies.

The cultural compatibility model, in contrast, recommends matching the background of the service provider with that of the client. Given the lack of racial diversity in applied sport and exercise psychology, this model is less practical. Further, it does little to change the field’s current level of multicultural training. Finally, those involved in the

cultural turn movement advocate for a centralization of issues of racial and ethnic identity, as well as other lines of social identity (e.g., gender, sexual orientation, disability). Addressing race, then, would become one way of understanding the complexity of the experiences of professionals and clients, rather than an “add-on” module in a course. None of these approaches to cultural competence have been fully embraced in sport and exercise psychology, and there is currently no consensus as to how issues of racial and ethnic identity should be addressed.

Ted M. Butryn

See also Cultural Competence; Diversity; Ethnicity; Stereotype Threat; Stereotyping, Cultural/Ethnic/Racial

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RECOGNITION AND RECALL PARADIGMS

Pattern recognition and recall paradigms are the concepts, theories, and methods that are typically used to examine and explain the underlying mechanisms contributing to the capability of performers to recognize and/or recall information from their domain. A vast amount of evidence has demonstrated that experienced performers are better than novices at recalling the locations of

“elements” (i.e., the individual items that comprise a pattern) from a pattern that is presented for only a relatively brief period of time. Similarly, experienced performers can also recognize a previously seen pattern from among previously unseen patterns with accuracy that surpasses lesser-skilled individuals. The theoretical explanations surrounding these attributes are important for helping better understand the methods in which experts (including expert sportspeople) from a range of domains utilize perceptual information as well as assisting in the design and implementation of practical applications for enhancing the perceptual-cognitive skills of lesser-skilled individuals. This entry provides information on the basic theoretical principles used to explain experts’ superiority in pattern recognition and recall tasks. Much of this theorizing has arisen from outside the classical sport domain in other domains such as chess.

The results from research conducted by Adriaan de Groot in the mid-1900s on chess experts are frequently cited as some of the more influential findings in the area. He showed that experienced chess players could recall the locations of chess pieces from only brief viewings of chess boards with extraordinary accuracy compared to that achieved by lower-level players. While this proved to be an interesting finding in itself, subsequent work by William Chase and Herbert Simon in 1973 showed that the superiority of the experienced players was apparent only when the pieces were arranged in configurations that were typical of actual game play. They showed that when the pieces were placed in more random configurations that were less typical of a normal game, the recall performance of the highly skilled players deteriorated to a point where it was largely indistinguishable from that of the lower-level players. The latter finding was of primary importance because it suggested that skilled recall performance is not necessarily based upon a generalized memory advantage. Chase and Simon suggested that the experienced players had acquired the capability to link particular chess piece locations together so they could be encoded into short-term memory (STM) as small clusters of information. This method of encoding essentially lightened the load on STM by reducing the large number of individual chess pieces that had to be memorized into a smaller subset of pieces. (See the entry “Chunking/Dechunking,” this volume.)

Since the development of this theoretical framework, further research has shown that experts can

continue to recall large amounts of information under conditions where STM capacity is likely to be overwhelmed. This prompted a paradigm shift with contemporary memory-based theories such as Fernand Gobet and Herbert Simon's *template theory* in 1996 and the *long-term working memory theory* of K. Anders Ericsson and Walter Kintsch in 1995, emphasizing a more prominent role for long-term memory (LTM). The exact nature of the underlying mechanisms that are proposed varies across the two theories, but the general consensus suggests that after extensive experience in their domain, experts acquire unique and task-specific structures in LTM that allow patterns to be rapidly stored and retrieved without exceeding the capacity of STM. In short, information can be directly encoded into LTM with links and associations created between similar sources of information. Consequently, when a domain-specific pattern is observed (e.g., a defensive setup in basketball), the configurations of elements within the pattern (i.e., player locations) generate cues that rapidly associate the observed pattern with other patterns in LTM (e.g., recognizing the pattern as a zone defense), allowing the necessary information to be quickly stored and then accessed as required.

An alternative theoretical approach, termed the *constraint attunement hypothesis*, was developed by Kim Vicente and JoAnne Wang in 1998. This approach downplayed the role of memory as the primary mediator of expert performance in recognition and recall tasks and instead highlighted the importance of the environment. They proposed that skilled performers do not necessarily rely upon pattern representations stored in memory but rather experts learn to closely attend to the naturally occurring relationships that exist within the typical performance environment. After extensive exposure to the critical environmental features (e.g., rules and tactics) that influence the movements of elements in a typical pattern, experts learn to accurately identify the pattern structures that are common to their domain. However, when the patterns are less structured and more randomly arranged, the relationships between the pattern elements are altered. As a consequence, the pattern information becomes less familiar and so the capability of the experts to accurately recognize and/or recall the patterns under such circumstances is also reduced.

Historically, the majority of the extant research investigating pattern recognition and recall has primarily focused upon memory-based paradigms

although the precise mechanisms underlying these and other theoretical frameworks, such as the constraint attunement hypothesis, are a source of continued debate. Despite this, a common theme across all theories is that skilled performers develop the unique capability to extract information from sport-specific patterns and play configurations, allowing them to accurately and rapidly recognize and/or recall these patterns when the situation demands.

Adam D. Gorman

See also Chunking/Dechunking; Ecological Theory; Expertise; Information Processing; Memory; Pattern Recognition and Recall

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RECREATIONAL DRUGS

Recreational drugs refer to substances that are not used by athletes for the purpose of enhancing athletic performance. The health-related risks of such drugs are well known, and concern about such risks, fear of a failed drug test, and performance-related concerns are reasons many athletes choose not to use them. Nonetheless, issues associated with recreational drug use among athletes are often expressed in the scientific and popular press.

Prevalence Rates

Several studies have compared athletes against non-athletes regarding prevalence rates of recreational

drug use, with athletes usually reporting somewhat less use. For example, rates of past-year and past-30-day marijuana use among U.S. college athletes are approximately 23% and 10%, respectively, whereas rates among the general U.S. college student population are approximately 33% and 18%, respectively. Similarly, approximately 4% of the general college student population report past-year cocaine use, compared to approximately 2% of college athletes. Studies among adolescent athletes across several countries have revealed a similar pattern of findings, with those participating in sport less likely to use drugs like marijuana, cocaine, and psychedelics. Few studies have examined recreational drug use among professional or other elite athletes, but the existing evidence also suggests lower prevalence rates compared with the general population. Researchers have not yet addressed the rates of recreational drug use disorders (i.e., substance abuse or dependence) among athletes.

Reasons for Recreational Drug Use

Even though rates of recreational drug use among athletes tend to be relatively low, it is nonetheless important to understand why it is that some athletes use such substances. A significant number of athletes do use illicit recreational drugs, and it is a virtual certainty that some will eventually experience considerable negative consequences as a result of their drug use. Further, many sport governing bodies provide severe sanctions for documented recreational drug use, so athletes who engage in any recreational drug use may be putting their athletic careers at risk.

It is likely that much recreational drug use and/or abuse among athletes can be explained by universal risk factors such as poor impulse control, lack of alternative coping strategies, high sensitivity to the rewarding effects of recreational drugs, family history of substance abuse, genetic or biological risk factors, or an environmental context that promotes drug use. Over the years, scholars have speculated about sport-related risk factors for substance use, such as excessive stress associated with balancing athletic and other demands, enhanced social opportunities associated with being an athlete, and individual characteristics that promote both an attraction to sport and risk for substance use (e.g., high sensation seeking). It may also be the case that some athletes perceive that recreational drugs provide a performance-enhancing function

(e.g., believing marijuana use will result in relaxation prior to a competition), which could increase the likelihood of use. To date, though, research on sport-specific risk factors for substance use is lacking, so the degree to which such factors do in fact impact an athlete's decision to use recreational drugs is unknown.

Preventing and Treating Recreational Drug Use

Over the past 20 years, a number of intervention strategies have been shown to be effective at either preventing or reducing substance use. These strategies represent a diverse array of approaches, including the following:

- Treatments focusing on facilitating the 12-step process
- Cognitive behavioral treatments that seek to modify thoughts and behaviors that promote substance use
- Motivational interviewing approaches that seek to resolve ambivalence about changing substance use behaviors
- Contingency management treatments that provide incentives for avoiding substance use
- Family-based therapy and prevention programs
- Life-skills prevention programs that focus on assertiveness training and healthy coping strategies

Studies have not examined the efficacy of these types of programs at reducing recreational drug use specifically among athletes. Researchers have shown that some of these programs are efficacious at reducing alcohol use among athletes and may therefore be promising strategies for impacting drug use as well. There is also evidence to suggest that regular drug testing with subsequent consequences for positive screens can have some impact at limiting substance use.

Education-only approaches, whose primary goal involves enhancing the recipients' knowledge regarding the dangers of drug use, is a strategy shown to be ineffective at preventing substance use. Unfortunately, this is the most commonly used "prevention" strategy in most sporting environments. For example, most U.S. college athletes are mandated to receive educational information about alcohol and other drugs, and such programs are also commonly implemented at the youth sport

level. It is unlikely that these education-only programs impact the substance use habits of the athletes who participate in them.

Conclusion

Although athletes tend to be less likely than others to use recreational illicit drugs, such use does occur among athletes and can result in significant personal and athletics-related consequences. A number of intervention and prevention strategies exist for those who experience problems with recreational drugs, although the effects of such strategies have rarely been assessed specifically among athletes.

Matthew P. Martens

See also Alcohol Abuse; Diet Drugs; Drug Use and Control; Narcotic Analgesics; Performance-Enhancing Drugs

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REINFORCEMENT AND PUNISHMENT

Reinforcement and punishment are common verbal and nonverbal responses to successes and failures in sport, exercise, and rehabilitation contexts. These practices may be best understood in the context of operant conditioning. This entry defines

reinforcement and punishment, reviews evidence of their frequency in sport, identifies their motivational implications, and reviews mechanisms for those effects.

Response Consequences in Operant Conditioning

Operant conditioning is a process in behaviorist theory by which behaviors are influenced by their consequences. Over time, people learn to associate certain behaviors with specific consequences and, to the extent that the consequences of certain behaviors are satisfying, those behaviors are more likely to be repeated. Correspondingly, to the extent that the consequences of certain behaviors are unsatisfying, those behaviors become less likely to be repeated. Table 1 illustrates four general response consequences in operant conditioning as a function of (a) whether the consequence involves adding or removing a response and (b) whether the response is intended to increase or decrease the future likelihood of a particular behavior.

The two response consequences that aim to increase the likelihood that a desired behavior will be repeated are positive and negative reinforcement. *Positive reinforcement* refers to the process of introducing a satisfying consequence following a desirable behavior that will increase the likelihood of that behavior being repeated. For example, in sport, coaches often use verbal praise as a reinforcement following a desirable play by an athlete. *Negative reinforcement* refers to the process of removing an unsatisfying consequence following a desirable behavior to increase the likelihood of that behavior being repeated. For example, a basketball coach who constantly criticizes his players about their technique while shooting free throws

Table 1 Four Response Consequences

	<i>Adding a Consequence</i>	<i>Removing a Consequence</i>
<i>Increases the likelihood of future behavior</i>	Positive reinforcement	Negative reinforcement
<i>Decreases the likelihood of future behavior</i>	Punishment	Extinction

Source: David E. Conroy.

but then quiets down after seeing a player use proper technique is using negative reinforcement. Again, both positive and negative reinforcement serve to increase the likelihood that a behavior will be repeated. In the latter example, the coach is reinforcing the correct technique by eliminating his verbal criticism following an attempt with the correct technique.

The two response consequences that aim to decrease the likelihood that an undesired behavior will be repeated are punishment and extinction. *Punishment* refers to the process of introducing an unsatisfying consequence following an undesired behavior to decrease the likelihood of that behavior being repeated. For example, a coach who criticizes an athlete for making an error is implementing punishment in an effort to reduce the likelihood that the athlete will commit that error again. *Extinction* refers to the process of removing a satisfying consequence following an undesired behavior to decrease the likelihood of that behavior being repeated. A coach who intentionally ignores an athlete's undesirable attention-seeking behaviors could be looking to extinguish those behaviors. The possibility also exists, however, that a coach who ignores a player's extra hustle during practice may be (inadvertently) extinguishing that desired behavior. Both punishment and extinction serve to decrease the likelihood that a behavior will be repeated.

These examples focused on response consequences in sport, but these response consequences can be observed in many other contexts of human movement. For example, in exercise settings, a personal trainer might try to increase a client's exercise behavior by praising her adherence to an exercise prescription (i.e., positive reinforcement) or by eliminating comments about a person's excess weight after workouts (i.e., negative reinforcement). Alternatively, a physical therapist could reduce a patient's compliance with an at-home rehabilitation prescription by ceasing to recognize the patient's progress or effort (i.e., extinction) or by making an ill-advised criticism of the person's ability during a rehabilitation appointment (i.e., punishment).

The Natural Frequency of Response Consequences in Sport

Little is known about the frequency of different response consequences in exercise, but the frequency of response consequences in sport has been examined. One classic study of coaching observed

legendary UCLA basketball coach John Wooden over the course of 15 practices. These practices totaled 30 hours of coaching and 2,326 codable coaching behaviors. The coding scheme used in this study was derived from educational research and modified after some preliminary observations of Coach Wooden but it was not designed specifically to study response consequences. Surprisingly, positive reinforcement was relatively rare in this study, with approximately 8% of the coded behaviors classified as "praises" (6.9%) and "nonverbal rewards" (1.2%). Punishment was, fortunately, also relatively rare with only 6.6% of the coded behaviors classified as "scolds" (or "reproofs" as Coach Wooden preferred to call them). Another 8% of the coded behaviors were classified as "Woodens"—a sequence in which a scold or reproof was followed immediately by a demonstration of how to perform a behavior correctly, a demonstration of how the player performed a behavior, and a repeated demonstration of how the player should perform a behavior. This combination was consistent with Coach Wooden's tendency to emphasize instructional comments (50.3%) and demonstrations (4.4%) in his coaching. Only trace amounts of "nonverbal punishment" were observed. None of the coding categories captured negative reinforcement or extinction, but that finding likely reflects the difficulty of capturing the absence of behavior using behavior observation methods. The behaviors exhibited by Coach Wooden while coaching an elite college basketball team may not generalize to all sport contexts.

A subsequent line of studies on youth sport developed a system for coding the observed behaviors of youth sport coaches that sheds light on operant conditioning processes in the context of youth sport. The Coaching Behavior Assessment System (CBAS) divides behaviors into (a) reactions to athlete behavior and (b) spontaneous actions (e.g., organizational activity, general communication). The responses to athlete actions are the most relevant to operant conditioning processes and include *responses to desirable athlete behaviors*, *responses to athlete mistakes*, and *responses to athlete misbehaviors*. The responses to desirable athlete behaviors include positive reinforcement and nonreinforcement, which can be linked to extinction processes in an operant conditioning framework. The responses to athlete mistakes include punishment and punitive technical instruction as well as mistake-contingent technical instruction,

mistake-contingent encouragement, and ignoring mistakes. The first two of these responses might be described as punishment in the operant conditioning sense. Responses to athlete misbehaviors involve keeping control and may involve an element of punishment depending on the eliciting event.

Across studies of youth baseball, basketball, and swimming, positive reinforcement is consistently one of the most common behaviors observed by trained coders and perceived by young athletes. In contrast, punishment and punitive technical instruction are among the most infrequent behaviors observed by trained coders and perceived by young athletes. Unfortunately, even trace amounts of punishment can have devastating motivational consequences for young athletes' motivation, anxiety, and self-perceptions. Another significant finding from this work was that a brief educational training program with a self-monitoring component can lead coaches to use more reinforcement and less punishment.

In sum, the frequency of response consequences in sport varies as a function of the competitive level of the participants. Youth sport coaches tend to use high levels of positive reinforcement and only infrequent punishment, but those proportions may change at different competitive levels. In the youth sport context, even trace amounts of punishment can have detrimental effects on youth development.

Motivational Implications of External Rewards

An interesting debate early in the 20th century centered on the question of whether external rewards might actually harm intrinsic motivation and reduce the likelihood that a behavior will be repeated. From an operant conditioning perspective, external rewards for a behavior should increase the odds that it will be repeated (i.e., positive reinforcement); however, considerable evidence that challenged this view had accumulated by the end of the 1990s. In a now-classic meta-analysis, researchers summarized studies that assessed intrinsic motivation as either people's free-choice behavior (i.e., did they select the behavior absent any external pressure?) or people's self-reported interest in a behavior. On average, results were consistent with the proposition that external rewards reduce future intrinsic motivation regardless of which measure was used; however, closer inspection of the results revealed some important nuances in the data.

The inhibiting effects of external rewards were limited to certain types of external rewards and specifically those rewards that were expected and tangible with one of three task-related contingency structures. First, external rewards that were contingent on a person's mere engagement in the activity reduced intrinsic motivation for desired behaviors. This reward structure is relatively common in sport. For example, the custom of giving out "free" T-shirts to runners who register for a race is an engagement-contingent expected tangible reward. Second, external rewards that are contingent on a person completing an activity reduced intrinsic motivation for that activity. Many youth sport programs implement this type of reward by distributing participation awards, such as trophies or certificates, at the end of a sport season. Finally, external rewards that were contingent on a person's level of performance reduced free choice of the desired behaviors, although these rewards appear to have no effect on people's interest in the behavior. These performance-contingent rewards are pervasive in sport (e.g., medals to the top three finishers in the Olympics, trophies to tournament champions and runners-up).

This meta-analysis also revealed two important exceptions to the general conclusion that external rewards reduce intrinsic motivation for a behavior. First, unexpected external tangible rewards have no effect on intrinsic motivation. Second, one particular type of external reward, verbal praise, actually has the opposite effect and enhances intrinsic motivation for a stimulus behavior. This finding helps to reconcile two competing views on the effects of a popular response consequence. When the reward is verbal praise, behaviorists are correct that positive reinforcement should increase the likelihood that a behavior will be repeated. When the reward is an expected tangible reward that is somehow contingent on the task, the reward will actually serve to reduce intrinsic motivation (what behaviorists might describe as extinction).

Mechanisms Underlying the Effects of External Rewards

Two general explanations have emerged to account for the differential effects of rewards on intrinsic motivation and behavior. Cognitive evaluation theory (CET) posits that the effects of rewards depend on whether the rewards assume informational or controlling properties. When people

construe rewards as providing feedback on the level of their performance, the informational properties of the reward should help to satisfy people's need for competence and therefore boost intrinsic motivation. On the other hand, when the expectation of an external reward becomes the reason for one's behavior, the reward is said to have assumed controlling properties (i.e., the rewards become the reason for the behavior). These controlling properties are thought to interfere with or thwart people's need to feel autonomous (i.e., like the origins of their behavior) and therefore reduce intrinsic motivation.

A complementary explanation posits that expected tangible rewards undermine the intrinsic value of activities. Recent research has shown that, after training with a task under a performance-contingent expected tangible reward structure, people display less free choice behavior when the reward is withdrawn. This reduction in intrinsic motivation corresponds with reduced activation in the striatum and midbrain—regions associated with task valuation. These findings indicate that performance-contingent expected tangible rewards undermine intrinsic motivation by reducing the value that people associate with the task.

Conclusion

In sum, reinforcement and punishment are simple behaviors with complex consequences. They may be used effectively to modify behavior but should be selected carefully to avoid unwanted long-term consequences. Considerably less is known about negative reinforcement and extinction processes in sport, exercise, and rehabilitation contexts.

David E. Conroy

See also Autonomy-Supportive Coaching; Coach–Athlete Relations; Feedback; Leadership in Sport: Social Cognitive Approaches; Self-Determination Theory

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RELATIONAL EFFICACY BELIEFS IN COACH–ATHLETE RELATIONS

Despite the proliferation of self-efficacy research that has occurred over the past 40 years, only relatively recently (i.e., over the last decade or so) have investigators turned their attention to the additional efficacy beliefs that are formed specifically within relational contexts. The tripartite efficacy model provides a conceptual framework for the investigation of this issue and maintains that when individuals occupy a position within an instructional (e.g., coach–athlete, teacher–student) or cooperative (e.g., athlete–athlete) relationship, they develop an interrelated network of confidence beliefs. This network comprises their confidence in their own capabilities (see the entry “Self-Efficacy”), as well as two distinct *relational efficacy* perceptions regarding the person with whom they are interacting. The first of these relational beliefs, termed *other-efficacy*, refers to how confident one person is in the other person's ability (e.g., an athlete's confidence in his or her coach's ability). Athletes and coaches often allude to this concept when discussing how much they “believe” in their significant other's capabilities prior to competition. The other relational construct, namely *relation-inferred self-efficacy* (RISE), is formed as individuals estimate how confident they think the other person is in their ability. In coach–athlete exchanges, for example, athletes form RISE appraisals regarding how highly they believe their coach rates their ability as an athlete (e.g., I think my coach really believes in me). Due to the vagaries of interpersonal impression formation,

our RISE beliefs may or may not align accurately with the other person's actual confidence in our ability, and in that sense, this construct is classified as a metaperception insofar as it reflects an inference about another person's thoughts. This entry presents an overview of the interrelationships between self-efficacy, other-efficacy, and RISE and integrates information on the theorized antecedents and consequences of individuals' relational efficacy beliefs with existing research findings. The final section provides a summary of conceptual and empirical evidence for the existence of an additional relational efficacy construct (*estimations of the other person's self-efficacy*, or EOSE), which has been largely unexplored to date.

Interrelationships Between the Tripartite Constructs

Sport-based investigations with members of coach–athlete and athlete–athlete dyads have typically demonstrated positive interrelationships between the tripartite efficacy constructs, whereby individuals' other-efficacy and RISE beliefs have been found to be associated with higher levels of self-efficacy perceptions. For instance, athletes have been shown to report greater confidence in their own ability when they believe that they are working under a highly capable coach (i.e., other-efficacy) and/or when they feel that their coach believes strongly in their ability (i.e., RISE). However, this does not always appear to be the case, and counterproductive relational outcomes have been documented when discrepancies exist between one's self-efficacy and relational efficacy beliefs. Empirical evidence indicates that, within coach–athlete interactions, a minority of athletes may report strong self-efficacy beliefs *despite* holding relatively low other-efficacy and RISE appraisals. In this situation, when relatively self-efficacious athletes doubt their coach's capabilities and their coach's confidence in their ability, this appears to accompany heightened perceptions of relational conflict, and relatively lower levels of commitment to and satisfaction, with their coach.

Formation and Consequences of Other-Efficacy

Other-efficacy beliefs are theorized to originate out of one's experiences with significant others (e.g., athletes' interactions with their coaches) and one's perception of the other's accomplishments, as well

as through comparison processes and stereotyping (e.g., comparisons with previous coaches, perceiver tendencies). Interviews with members of international-level coach–athlete and athlete–athlete dyads have provided support for these proposed antecedents. For example, coaches and athletes have reported that their confidence in the other person is strengthened when they feel that person boasts a successful track record, compares favorably with their previous partners, appears to be highly motivated, and is held in high regard by third parties. In cooperative settings, experimental procedures have also demonstrated that individuals' other-efficacy beliefs may be substantively influenced by the provision of feedback about the partner's performance effectiveness (or ineffectiveness).

In turn, when individuals believe strongly in the capabilities of their partner, coach, or athlete, this promotes a series of beneficial intrapersonal and relational outcomes (particularly when the “perceiver” occupies a subordinate position in the relationship). From an intrapersonal perspective, one's other-efficacy perceptions have been shown to be positively associated with one's own performance and motivation levels, as well as positive affective states (e.g., enjoyment). In exercise contexts, related research regarding the notion of *proxy efficacy* (i.e., one's confidence in the abilities of a third party to function effectively on one's behalf) has also demonstrated that individuals display more adaptive exercise intentions and engagement when they are highly confident in their instructor's or practitioner's capabilities.

In terms of relational outcomes, those who are highly confident in the other person's ability also tend to display a stronger desire to maintain their relationship with that person, higher levels of trust and rapport, and greater enjoyment regarding their interactions. Due to the interdependent nature of sporting interactions, research has also shown that one person's other-efficacy appraisals may shape outcomes for the other person in the relationship. For instance, when coaches believe strongly in an athlete's capabilities, this has been shown to promote greater relationship commitment and effort on the part of the athlete. This effect is likely transmitted through a causal chain whereby the coach displays encouraging and supportive interpersonal behavior as a result of his or her confidence in the athlete, and this behavior is subsequently detected by the athlete, leading to favorable outcomes for the athlete.

Formation and Consequences of Relation-Inferred Self-Efficacy

RISE appraisals are shaped not only by the behavior of the target individual but also as a result of relevant perceiver attributes. For example, athletes and coaches appear to infer that the other person has confidence in their ability when she or he provides positive performance feedback, displays inclusive and supportive body language, and gives the impression that she or he enjoys their interactions. In addition, when individuals believe strongly in their own ability and are highly self-motivated, they also tend to believe that those around them are confident in their ability (e.g., I know I can do it, so why wouldn't she be confident in me?). As with other-efficacy, RISE beliefs are theorized to account for a range of positive consequences. Most notably, as well as reinforcing one's confidence in one's own ability, individuals are more likely to feel comforted and reassured when they sense that the other person believes in their ability and as a result should report unifying relational perceptions (e.g., greater relationship satisfaction, perceived support, persistence intentions). These relational effects have been supported in existing qualitative sport-based research, as have a number of additional desirable outcomes in terms of elevated performance (e.g., if I feel that she believes in me, then that reassurance helps me to make decisions and do well when I'm competing) and motivation (e.g., when I think he's confident in me, it makes me try even harder for him). Similarly, in high school physical education (PE) contexts, students have been shown to endorse relatively more self-determined motives for PE (i.e., participating primarily due to fun, interest, and enjoyment) when they believe that their teacher rates them as a highly capable student.

In coach–athlete settings, however, there is also contradictory evidence to indicate that, in some instances, RISE beliefs may align with maladaptive relational consequences. Specifically, when considering how the predictive functions of RISE may differ according to one's role (and relative status) in coach–athlete dyads, moderator analyses have revealed a divergent pattern of effects for athletes and coaches. When coaches gauge that their athletes are highly confident in their coaching capabilities, it has been shown to promote adaptive relational perceptions on the

part of those coaches (e.g., greater relationship satisfaction and commitment). However, athlete RISE estimations have been found, at times, to display inverse relations with these outcomes, insofar as athletes may report lowered commitment and satisfaction when they infer that their coach believes strongly in their ability. It is possible that, for these athletes, strong RISE appraisals may encourage a degree of complacency, which is manifest in reduced relationship commitment and satisfaction (e.g., if she's not going to push me any further, then what's the point?). Alternatively, it is also plausible that strong RISE appraisals about a coach might strengthen athletes' confidence in their own ability to such an extent that they strive to find a new coach who can help them progress further and compete at a higher level. While these findings appear somewhat counterintuitive, they are noteworthy nonetheless and will undoubtedly be elucidated further as the relational efficacy literature continues to grow.

Estimations of the Other Person's Self-Efficacy

Although qualitative, observational, and experimental work has begun to map out the implications associated with coaches', athletes', exercisers', and students' other-efficacy and RISE beliefs, another potential relational efficacy construct has yet to receive systematic attention. This construct, EOSE, represents one person's appraisal of his or her partner's level of self-efficacy. To illustrate the nature of this type of metaperception, we might consider a coach who feels that his or her athlete is beset by self-doubt. In this instance, the coach is not directly questioning the athlete's capabilities (i.e., other-efficacy), nor is she or he concerned about the athlete's confidence in his or her ability as a coach (i.e., RISE); rather, she or he feels that the athlete appears to lack confidence in his or her own ability (e.g., my athlete doesn't look very confident about what she's got to do in this competition). Whereas the accurate appraisal of another person's self-efficacy may rely on the detection and interpretation of relatively subtle behavioral cues, it appears that individuals in coach–athlete and athlete–athlete relations do make assessments of this kind about one another. In fact, research has demonstrated that athletes and coaches draw from many of the factors that contribute to their own self-efficacy beliefs when estimating how confident the other person is in

his or her own ability (e.g., gauging the other person's recent performances, verbal and nonverbal behavior, affective and physiological state). Interviews with members of elite sporting dyads have also revealed that individuals may feel more positive about their relationship, as well as more motivated and more confident in their own ability, when they believe that the other person is highly self-efficacious. On the other hand, when athletes and coaches appraise (rightly or wrongly) that the other person in their relationship is experiencing self-doubt, then this may result in elevated anxiety levels, interpersonal difficulties, and in some cases, relationship termination.

Despite the potential for EOSE to shape personal outcomes and relational dynamics in sport, to date there is no single study that has adopted a fully integrated approach to the study of relational efficacy beliefs in sport (i.e., by measuring other-efficacy, RISE, and EOSE simultaneously). Nevertheless, the emerging literature indicates that the tripartite framework has relevance in terms of understanding the interpersonal cognitions that drive behavior and relationship outcomes within sport- and exercise-based interactions.

Ben Jackson

See also Coach–Athlete Relations; Collective Efficacy; Self-Efficacy; Social Cognitive Theory

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RELAXATION

Relaxation has been defined as a psychological strategy used by sports performers to help manage or reduce stress-related emotions (e.g., anxiety and anger) and physical symptoms (e.g., physical tension and increased heart rate [HR]) during high-pressurized situations. Several different types of physical and mental relaxation strategies will be discussed in this entry, all of which can be used to relax the performer and, potentially, benefit athletic performance.

Types of Relaxation Strategies

Different types of relaxation strategies have been advocated within the sport psychology (SP) literature and have been categorized as *physical relaxation strategies* or *mental relaxation strategies*. The rationale for using either type of strategy often has been dependent on the symptoms described by the athlete. Specifically, researchers have advocated matching the treatment (i.e., relaxation type) to the dominant set symptoms experienced by the athlete. Ian Maynard and colleagues termed this treatment approach the *matching hypothesis*, whereby symptoms of somatic anxiety are primarily treated with a form of physical relaxation and symptoms of cognitive anxiety with a form of mental relaxation. The notion also can be applicable to the experience and implications of other emotions such as anger and excitement.

Physical Relaxation Strategies

Physical relaxation strategies can be employed to reduce muscular tension and improve coordination during performance. Examples of such strategies taught by sport psychologists include *breathing exercises*, *progressive muscular relaxation* (PMR), and *biofeedback* (BFB).

Breathing Exercises

Breathing correctly is a simple form of relaxation and has the benefits of increasing oxygen in the blood, improving mood, and reducing muscular tension. The process of breathing properly involves *diaphragmatic breathing*, where the performer is directed to breathe into the abdomen and then the chest. Specifically, when breathing in deeply, the performer should concentrate on

filling the lungs by first pushing the diaphragm down and the abdomen outward then by expanding the chest and raising the chest and shoulders. To promote this breathing in a controlled manner, so that it is of benefit during competitive performances, athletes can be encouraged to be *rhythmic* in their breathing by inhaling, holding, and exhaling to a count of a predetermined number. In addition, making the exhalation audible (e.g., with a “hheerr” sound) could be of benefit to performance by helping to reduce muscular tension during key movements, such as releasing the javelin or striking a tennis ball. Some support for the benefits of breathing exercises have been provided by Adam Nicholls and associates in SP research.

Progressive Muscular Relaxation

Derived from the work of Edmund Jacobson in the 1930s, PMR strategies require an individual to focus on progressively tensing and then relaxing specific muscle groups, one at a time. Through this progressive technique, Jacobson’s premise was that the individual would learn the difference between tension and less tension. Consequently, the individual would become aware when tension occurred and begin reducing it by relaxing the relevant muscles. Jacobson also proposed that this form of physical relaxation would also decrease mental tension. Critically, Jacobson’s program was quite long and, consequently, inappropriate for the regular athlete. To overcome this issue, many sport psychologists such as Graham Jones have advocated a variant of Jacobson’s approach, where the objective is to teach a performer to relax within 20 to 30 seconds. To achieve this, a performer will undergo several relaxation training phases, typically over a period of 10 to 12 weeks, that progress toward much quicker relaxation. The first training phase involves twice daily 15-minute relaxation sessions where the muscle groups are progressively tensed (for 5 to 7 seconds) and relaxed. During this phase, it is commonplace for the athlete to be provided with an audio track that helps systematically work them through a full muscular tension or relaxation program. Once practiced, so that the performer is proficient in using this technique, a 5- to 7-minute *release only* training phase is instigated. Here, the performer is guided (by an audio track or by the sport psychologist) only to relax (release) any tension in the muscles. The next progression is a 2- to 3-minute *cue-controlled* phase where

the focus is still on release only, but the release is instigated by the performer through words such as *relax*. Sometimes rather than associate the relaxed state with a cue word, sport psychologists link the relaxed state to a natural “trigger” cue within the athlete’s environment (e.g., gripping the racquet in tennis, holding a basketball prior to a free throw). This phase is, therefore, only a few seconds long and involves the performer recognizing tension in specific muscles and focusing solely on reducing that tension. Preliminary research by Ian Maynard and colleagues has shown that PMR can help to reduce the intensity of reported bodily symptoms associated with the experience of anxiety (e.g., muscular tension).

Biofeedback

Similar to PMR, in which performers are taught to become more aware of muscular tension, BFB is a method that helps performers become familiar with such autonomic nervous system (ANS) responses as muscular activity, HR, and respiration (R) rate. By becoming more aware of these and other physiological responses, performers can then attempt to control them for the benefit of sporting performance. Biofeedback training (BFBT) involves the use of electronic instruments to provide visual or auditory feedback about selected physiological responses and the requirement of the performer to then use such strategies as relaxation to reduce the level of these responses. For example, if using an electromyograph to measure the electrical activity of the muscles, a high amount of electrical activity may mean muscle tension. Consequently, visual feedback showing high levels could help the performer become aware that muscular relaxation is needed when experiencing sensations associated with that level of activity. This method does require some training in relaxation techniques such as PMR. Limited research has been conducted on the combination of BFB and relaxation, but some evidence has been provided by Tammy Evetovich and her associates that suggests BFB and relaxation can help reduce muscular tension.

Mental Relaxation Strategies

The mental relaxation strategies that have been promoted within the SP literature have been focused primarily on reducing anxiety, which is a negative emotion caused by situational appraisals

of threat or harm. Nevertheless, they also can be used to reduce the intensity of other emotions experienced such as anger or excitement, as these and other emotions can be distracting if too high in intensity. Examples of mental relaxation techniques include *transcendental meditation*, *mindfulness meditation*, and *autogenic training*.

Transcendental Meditation

Meditation generally involves the individual's focusing attention on a single thought, sound (often called *mantra*), or object. Transcendental meditation is an approach in which the individual repeats a mantra, which is a sound (e.g., the syllable *om*) or a key word or phrase that has personal meaning—such as “relax.” This technique has been suggested to reduce the focus on negative thoughts and also lower HR, blood pressure (BP), and R, all cognitive and physiological changes that could be beneficial in fine-motor-skilled performances such as rifle shooting, archery, or golf. When practicing transcendental meditation, the performer is required to sit in a quiet environment, adopt a comfortable position, and repeat the mantra aloud. As with PMR, the challenge here for a sport psychologist working with a performer is to progress from prolonged training sessions in quiet environments to sessions that help the performer mentally relax within seconds in competitive environments. At present, few studies in the SP literature have documented a successful transfer of meditation from peaceful surroundings to the pressurized sporting arena.

Mindfulness Meditation

The practice of mindfulness, which originated within the Buddhist tradition, can be loosely defined as a state of awareness achieved through purposely and *nonjudgmentally* paying attention to the present and ongoing experiences of yourself and others—that is, attempting to put aside judgments of current situations, thoughts, or feelings as “good” or “bad.” Mindfulness meditation is an approach that helps develop this nonjudgmental awareness and that promotes calm and focus in potentially stressful situations. Other documented benefits of mindfulness meditation include reduced reporting of depression, anxiety, and chronic pain. Different forms of mindfulness training exist, but the general principle is to guide the individual to develop concentration by focusing their attention

to the sensation of breathing. Then, when thoughts, emotions, or body sensations distract the focus from the breathing task, the individual is directed to nonjudgmentally acknowledge the distraction and return to the breathing exercise. A progression is to then focus on body sensations during the breathing task. If such labels as “good” or “bad” are used to describe sensations, the individual is guided to again nonjudgmentally acknowledge the label and return to focusing on the body sensations while breathing. Such approaches help individuals to become more aware of the stressors in the present situation (e.g., work demands, family issues, coach expectations) in a nonjudgmental way by reducing the common appraisals that something experienced is good or bad. This nonjudgmental approach then influences more constructive solutions to problems, as they are approached in a more impartial manner. Although the benefits of mindfulness are well documented in other areas of psychology, they have only recently been explored within sport and exercise psychology. Early findings from Rachel Thompson and colleagues support the positive cognitive (appraisal) and performance outcomes of mindfulness training.

Autogenic Training

Autogenic training involves a series of exercises designed to produce warmth and heaviness sensations—feelings that are typically associated with relaxation. Within this form of self-hypnosis, attention is focused upon the sensations the individual is attempting to produce. Developed by Johan Schultz in the 1930s, autogenic training has been associated with reduced anxiety, fatigue, HR, and an increased sense of control, and better focus and sleep. The process involves six sequential training stages where verbal self-statements are used to direct the focus to specific bodily sensations. In Stage 1, the individual is guided to focus on achieving heaviness in the arms and legs, starting with the dominant arm or leg. Here, self-statements such as “My right arm is heavy” are used repeatedly to achieve heaviness, before the sensation is “cancelled out” by the individual—often through bending the arm, breathing deeply, and/or by opening their eyes. Once trained, the next stage is to achieve warmth in the arms and legs. Similarly, the self-statement “my right arm is warm” may be used repeatedly during training sessions to help facilitate warmth in the right

arm and then other extremities. The third stage involves HR regulation, with the self-statement of “my heartbeat is slow, relaxed, and calm.” Stage 4 focuses on regulating breathing rate (e.g., breathing is slow, relaxed, and calm); in Stage 5, the sport psychologist aims to promote sensations of abdominal warmth (i.e., with hand on abdominal area, the self-statement is “my abdomen is warm”), and stage six the cooling sensation of the forehead (i.e., the self-statement is “my forehead is cool”). Once the individual is competent enough to control the sensations of heaviness and warmth of their extremities, their sensations of heart and breathing rates, and the perceived temperature of their abdominal area and forehead, then the potential to achieve a relaxed state increases.

Crossover Benefits of Relaxation

Even though the relaxation approaches identified here have been categorized as either physical or mental in nature, it is documented within the SP literature that a physical relaxation strategy focused on reducing muscular tension also may have mental effects, such as reducing the incidence of negative thoughts. Similar *crossover* effects have also been reported for mental relaxation strategies primarily focused on reducing negative thoughts and emotions; these strategies also have been found to reduce the incidence of negative physical symptoms associated with anxiety.

Relaxation Used With Other Psychological Strategies

Alongside the physical and mental benefits of relaxation, a further benefit is that the reduced incidents of negative thoughts associated with relaxation allows for other psychological strategies such as self-talk and visualization to be used. For example, consider a situation in which a sport psychologist attempts to guide a performer to use self-talk, to talk to himself or herself more effectively to help change negative thoughts experienced. If the negative thoughts were causing extremely high levels of anxiety, then the thought changing exercise would prove fruitless. Consequently, the sport psychologist could first reduce the experience of anxiety by helping the performer to learn to use mental relaxation strategies. Then, with the performer able to achieve a more even-tempered state through the use of these strategies, the commitment to, and understanding of, appropriate self-talk can

be improved. A similar example can be provided when developing a performer's ability to visualize himself or herself performing a certain skill effectively. If the performer is not sufficiently relaxed, the ability to image effectively may be compromised by experiences of intense negative thoughts, emotions, and images.

Centering

Centering is a strategy in which the performer directs their thoughts toward adjusting their body weight so that the weight feels like it is about their center of mass. This allows the performer to feel in control and comfortable so that he or she can consciously modify such physiological symptoms as HR and R, along with their focus of attention. Indeed, focusing internally (in this case focusing on adjusting their body weight) helps the performer to ignore unwanted negative thoughts and then focus on performance relevant information. Within this approach, performers are guided on becoming aware of their center of mass, so to promote recall of where their center of mass should be during stressful situations. This awareness gives the performer a point of focus to switch attention to. Then, the performer is guided to concentrate on breathing appropriately to reduce arousal and tension, which helps the performer focus on the task at hand. This three-step approach of centering, breathing, and task focus is trained and then encouraged just before the performance action is to occur; this way, attention is directed to the appropriate information at the right time in competition (e.g., the behaviors needed to complete a free throw in basketball).

Autogenic Training and Imagery

The use of autogenic training followed by imagery (when trained to visualize effectively too) has numerous benefits. First, when the performer reaches the calm state at the end of autogenic relaxation, he or she can imagine relaxing scenes, colors, or thoughts that help translate the physical relaxation reached through autogenic relaxation into the mind. Second, the relaxed state promoted by autogenic training will allow the performer to imagine proficient execution of performance related skills that can be used to increase confidence and performance, as no unwanted thoughts or images would be present during such relaxed states to disrupt such constructive imaging. The

benefit of combining autogenic training and imagery training for sporting performance has been documented by SP researchers such as Alain Gros Lambert and colleagues.

Conclusion

Given the proposed benefits of physical and mental relaxation strategies to sport performers, sport psychologists will continue to train performers in the use of these strategies to help performers reduce or control their cognitive and/or physical state. However, as with any strategy, the effectiveness of the use of the strategy during competition depends on the extent to which the strategies have been practiced. Once learned, these strategies can be used by performers to function better within competition and everyday life, and to allow other strategies to be learnt more effectively.

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See also Attentional Focus; Autonomic Nervous System; Breathing Exercises; Centering; Imagery; Mindfulness; Response; Self-Talk

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RESILIENCE

Most athletes will encounter one or more major setbacks or adversities during their sporting career. An elite athlete may, for example, experience a career-threatening injury, garner demotion from a top-tier team because of poor performance, or need to relocate to another country to continue competing in their sport. Despite the potential for setbacks and adversities to negatively influence one's developmental trajectories, in some cases, and for some athletes, exposure to major assaults on one's typical level of functioning or performance does not always result in negative outcomes. Why is it that some athletes bounce back from adversity or experience minimal disruption when faced with these major assaults? The concept of *resilience* is central to coping with such demands and challenges. Although there remains considerable debate regarding a formal definition, common themes among most contemporary conceptualizations reveal that resilience encapsulates one's capacity to regain or sustain relatively stable, healthy levels of psychological and physical functioning.

Historical Perspectives

The scientific study of resilience emerged about 40 years ago from research on children “at risk” (i.e., increased probability of a negative outcome) for psychopathology, often due to environmental

factors such as psychiatric history in the family or poverty. Within this line of investigation, researchers observed a significant degree of variability in impairment to normal levels of functioning ranging from maladaptive outcomes to instances of positive adjustment or adaptation. Intrigued by cases of positive outcomes despite risk or adversity, researchers subsequently advanced the study of resilience in four major waves: (1) identifying protective factors within individuals (e.g., optimism, self-efficacy), the family (e.g., socioeconomic status, parenting style), and communities (e.g., neighborhood qualities, public health system) that contribute to positive adaptation in the face of risk or adversity; (2) understanding the processes by which positive adaptation occurs across the life span; (3) evaluating the effectiveness and efficacy of interventions designed to protect or promote positive functioning among individuals, groups, or communities; and (4) integrating the first three waves of research across multiple levels of analysis and across disciplines including genetics, neurobiology, sociology, and cultures.

Resilience in Sport

The scientific study of resilience in sport has typically lacked the empirical and theoretical momentum evidenced within other areas of the parent discipline such as developmental and gerontological psychology. Only a handful of studies have systematically investigated resilience in sport via the simultaneous integration of the two core components of contemporary conceptualizations, namely (1) exposure to significant risk or adversity *and* (2) returning to, sustaining, or attaining positive adjustment or functioning. Nonetheless, a considerable body of knowledge has been generated through separate but related lines of inquiry examining the resilience phenomenon.

Risk, Stressors, and Adversity

Exposure to both acute and chronic stressful events can represent significant sources of risk to the development and functioning of athletes. Recent research on the types of stressors inherent within sport contexts has revealed that elite athletes experience and recall more demands associated with the sport organization (e.g., qualities of the coach and coaching style, poor support networks, financial concerns) than with the competitive environment (e.g., injury, pressure to perform

well). Injury and the processes involved in returning to competition represent a major agenda within the sport psychology (SP) literature, not surprisingly because the phenomenon entails a range of physical and psychological traumas. Aligned with current conceptualizations of resilience, empirical evidence has revealed that some athletes can and do withstand or recover from these stressors or adversities that threaten their normal levels of functioning or performance.

Protective Factors

Researchers have revealed that personal resources are important constructs for understanding how individuals effectively approach and respond to various stressors and adversities. Self-efficacy (i.e., one's belief in his or her ability to perform a specific task), optimism (i.e., tendency to expect good things to happen), hope (i.e., motivational state encompassing the "will" and the "way" to attain personally important goals), and coping (e.g., ways in which people respond when confronted with stress) are among the most important and widely studied personal resources. Most recently, researchers have explored such influential personal resources through the conceptual lens of mental toughness, which is considered a reservoir of personal resources that enable individuals to produce consistently high levels of performance or goal attainment despite everyday challenges and significant adversities. However, unlike mental toughness, resilience is not solely a fixed collection of personal attributes; thus, one must also consider the influence of a multitude of external assets for understanding person–stressor and adversity transactions. Social support, which involves the exchange of resources between two individuals (e.g., parents, coaches, teammates), is one of the frequently studied external assets in sport contexts with regard to managing stressors and adversities. Collectively, research has revealed that these protective factors and external assets can have both direct and indirect (i.e., moderates the negative effects of stress on psychosocial functioning and performance) relationships with stress, adversity, and important outcomes.

Conclusion

Research on the types of stressors and adversities (e.g., injury) experienced in sport contexts, and the various protective factors characteristic of elite

performers and their psychosocial environment, has provided insights into the ways that athletes and coaches respond to and cope with assaults on their development or normal levels of functioning. Nevertheless, the conceptual integration of empirical evidence and theoretical discussions on these independent streams of research within a resilience framework represents an important next step to guide future work. For example, exposure to different types of risk or adversity may trigger different assets and protective factors and, as a result, contribute to different outcomes.

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See also Coach–Athlete Relations; Competition; Coping; Mental Toughness; Optimism; Parenting; Self-Efficacy; Stress Management

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RESISTANCE TRAINING

Resistance training (RT), also commonly referred to as strength training or weight training, is a form of anaerobic exercise that utilizes external resistance of varying loads to improve musculoskeletal fitness. Compared with aerobic exercise, there are significantly more variables to consider when structuring an appropriate RT bout or program. Some of these prescription considerations include repetitions, sets, load, rest intervals, exercise order,

speed of movement, body part training split, and frequency. Also, unlike aerobic exercise, RT volume is not time-dependent but rather a function of repetitions $\times$ sets. Dumbbells, machines, resistance bands, free weights, and body weight can be successfully employed to provide resistance and overload. When applied correctly, RT causes increases in muscular strength, hypertrophy, and endurance. RT has also been used clinically to positively influence such conditions as arthritis, Type II diabetes, and musculoskeletal dysfunction and injury. RT is also particularly appropriate for older adults, as it reverses the typical age-related sarcopenia and consequent loss of strength. It has also become increasingly apparent that RT can impact psychological outcomes in addition to physiological outcomes. For example, RT has been linked with improvements in depression, anxiety, positive and negative affect, self-efficacy, cognition, and quality of life (QOL).

Resistance Training and Affective Responses

Much of the early work in this area found that acute RT resulted in either elevations or little to no change in state anxiety (SA) responses, suggesting that this was an inferior form of exercise for improving mental health in comparison to aerobic exercise. However, a number of studies have since observed improvements in SA and other relevant psychological outcomes. It appears that factors such as the type of RT routine performed, training load, and intensity influence the psychological responses accompanying acute RT.

Altering the RT load can significantly affect the acute metabolic, autonomic, and hormonal responses to training. Consistent with these responses, a growing number of investigations have documented improvements in psychological states following acute bouts of RT incorporating light to moderate training loads. Unfortunately, most of the prior investigations had notable differences in load determination and assignment, total volume, repetition ranges, and rest intervals between sets. This has made drawing conclusions regarding the dose–response relationship between acute RT and psychological responses difficult. Additionally, most of these studies have focused primarily on anxiolytic effects of exercise. However, a recent well-designed dose–response study has demonstrated that a moderate-intensity RT bout (70% of 10 repetition maximum [RM]) produces

improvement in SA, positive and negative affect, perceived energy, and calmness. Similar to what has been seen for moderate-intensity aerobic exercise, these responses appeared immediately following the moderate-intensity RT bout and persisted for at least an hour post-exercise. It is important to note that high-intensity RT (100% of 10 RM taken to momentary muscular failure) can produce unfavorable psychological responses, including increases in SA, negative affect, and tension. When properly defining intensity and controlling for RT volume, acute moderate-intensity RT results in more favorable psychological responses relative to either low- or high-intensity bouts of RT. Many aspects of the RT stimulus other than load (i.e., inter-set rest intervals, volume, set termination criteria: volitional fatigue versus momentary muscular failure) directly influence RT intensity. A modification of these programmatic factors likely has a meaningful impact upon the psychological responses to acute RT.

To adequately understand the expected outcomes of a RT bout and provide an effective prescription, it is imperative that the mechanisms underlying the acute RT-affect relationship be identified. Although several biological and psychological hypotheses have been proposed to explain the mental health outcomes associated with acute exercise, empirical support has been equivocal. There is, however, increasing evidence that the hypothalamic-pituitary-adrenal (HPA) axis may play a pivotal role in influencing responses to different exercise intensities. For example, cortisol and corticotropin-releasing hormone (CRH) have been found to produce anxiogenic responses with central administration. Activation of the stress response could very well explain many of the early findings in this area that suggested that RT may produce psychological disturbances. Increased CRH and cortisol levels have been associated with negative affect.

Resistance Training and Well-Being

RT programs have been found to positively impact QOL in various populations, including older adults as well as individuals who are depressed, sedentary, chronically ill, or overweight. It is likely that the greater physiological gains that would result from longer RT programs emphasizing strength or hypertrophy could translate to improved activities of daily living (ADL), thus producing greater improvements in QOL.

The psychological benefits of a RT program also exist for normal and overweight adolescents. Compared to a traditional physical education (PE) program, 12 weeks of moderate- to high-intensity RT produced significant improvements in total physical self-perception, physical condition, body attractiveness, and global self-worth. These psychological outcomes were related to the significant strength gains and improvements in body composition seen with RT.

There is increasing evidence that RT may also favorably impact cognitive function. However, it appears that program length is a critical factor, with protocols longer than 6 months generally needed to produce significant improvements in cognition. This effect is seen for both moderate- and high-intensity RT programs. This is clearly an area that warrants further investigation.

Conclusion

RT prescription for optimal mental health benefits may not be the same prescription for optimal physical benefits. However, these outcomes are not necessarily mutually exclusive, particularly when applied to training programs rather than acute exercise. Changes in musculoskeletal fitness appear to be related to positive changes in well-being and self-perception. Moderate-intensity RT also appears to effectively improve acute affective states. Furthermore, there is good preliminary support for the role of the stress response as a potential mechanism for affective changes. The efficacy of both acute and chronic RT is seen across many populations but appears to be particularly pronounced for individuals that potentially have the most to gain from a functional standpoint. Despite some inconsistencies in the current literature, it is apparent that this modality of exercise has the potential to positively impact a number of psychological constructs and is a useful tool for improving both physical and mental health.

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See also Affective Responses to Exercise; Cognitive Function; Mental Health; Self-Efficacy

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RESPONSE

All animals must act in their environment in order to survive. They must also react to changes in the environment, for example, when threats or beneficial opportunities arise. Responses are defined as any reaction of the organism to external or internal events. Responses can pertain to one or more levels of body function ranging from cognition, behavior, and physiology to endocrine and biochemical reactions. The overall purpose of human and animal responses is optimal adaptation to environmental demands.

In sports, humans develop and hone sophisticated, explicit systems of action and reaction, often with the sole purpose of competing with one another. At first glance, athletes' responses seem to be simple. A runner waits for the starter's gun, and upon perception of the signal, she sprints as fast as possible toward the finish line. However, humans in general, and athletes in particular, respond in multiple distinct ways to action-relevant events such as a starter's gun. In addition to a behavioral response, humans may respond physiologically (e.g., heart rate [HR] acceleration), emotionally (e.g., stage fright), and cognitively. An example of a purely cognitive response can be seen in athlete's self-talk as a form of attention regulation. Moreover, humans respond not only to external events like a starter's gun but also to internal events like psychological anticipatory stimuli. The latter can be seen in athletes' false starts, where the signal is perceived in illusion possibly as a result of athletes' internal rehearsal prior to the gun. This entry will first introduce a fundamental human

response, the so-called *orienting response*, before describing more generally response characteristics and basic aspects of responses. Then, various response dimensions and response systems of the organism will be introduced before discussing the distinction between overt and covert responses. Measures are then presented that can be used to evaluate or quantify overt and covert responses. Finally, two specific aspects of responses, interference with performance and the value of covert responses, will be illustrated.

The simplest form of a behavioral response is the reflex. The Russian physiologist Ivan Pavlov was one of the pioneers in the investigation of involuntary reflex actions and classical conditioning, which is a form of learning. He believed that the involuntary reflex action is an unconditioned response—that is, it was seen as an inborn behavioral pattern. This reaction of the organism toward some relevant change in the environment could be described as the expression of curiosity and, accordingly, it was also termed the “What is this?” response by Ivan Pavlov. He defined three functions for this reflex: to activate the organism, to draw its attention toward the environmental change, and to prepare for the exploration of the associated section of the environment. Later, Evgeny Sokolov researched this response more systematically and gave it the name that is still used today, the *orienting reflex*.

Six different aspects of environmental events are known to elicit an orienting reflex or response, some of which may be interrelated: (1) novelty, (2) reaching a certain intensity threshold, (3) relevance of the event, (4) its associated surprise, (5) the event's incongruity with the organism's experience, and (6) an induced action conflict (i.e., if two actions are required with the same priority). The orienting response is known to *habituate* (i.e., with repeated presentation of the relevant stimulus, the orienting response becomes weaker). This is also known as the familiarization of the organism with the stimulus.

Responses are conceptualized as the individual inclination to react to external or internal stimuli. External stimuli represent events in the environment that can be perceived with the sensory organs. Internal stimuli are events that originate from within the body and can be either somatic or psychological. A full description of responses includes a description of the elicited change and also how long it takes until the reacting system

comes back to the pre-response state. An individual's inclination to react can be temporally variable or stable and may thus be conceptualized as state- or trait-dependent.

Responses are seen as interactions of the organism with the environment—that is, responses are a basic way for organisms to cope with their surroundings. Responses describe the way individuals react to external and internal events. At the same time, responses also encompass reactions that influence the stimulus situation or environment. That is, responses reflect the influence of the environment on the individual but also the influence the individual has on the environment. Under such a transactional view, individuals can, of course, predict upcoming stimuli and the effects of their own reactions.

Four basic aspects of responses can be distinguished. One is the sensitivity of the organism—for example, the threshold at which a reaction is elicited, or the strength, latency, speed, or duration of the reaction. The second aspect is the adaptability of the organism to repetitive stimulation (habituation), or even constant stimulation (adaptation). The third aspect is the *resilience* of the organism. Resilience refers to the manner, strength, duration, or continuity of the stimulation. Finally, the organism's influence on the stimulus or the whole situation refers to changes in, for example, the source of stimulation, psychological processes (e.g., attention), or the whole situation in which the event occurred. These aspects also make evident that responses can refer to short-term or long-term reactions of the organism.

Responses can be related to different dimensions and to different systems of the organism. Stimulus-related dimensions of responses refer to physical (e.g., loudness or brightness) and psychological aspects (e.g., threat or attractiveness). Responses can also be described by the dimensions of the reaction itself. One can distinguish, for example, the quality, the intensity, or the course of reacting (e.g., latency, duration, or habituation). Two additional important dimensions in the characterization of responses are temporal and situational aspects.

Here, we focus on a major division among different response systems of the organism. The major response systems are the somatic, biochemical and neurochemical, endocrine, immunological, and psychological systems. The *somatic response system* refers to the central and peripheral nervous system and to the muscle system, whereas the

psychological system refers to perception, cognition, memory, emotion, action control, and so on. One major division that cuts across these systems is whether a response is overt or covert.

Overt Versus Covert Responses

Overt responses are characterized by the involvement of movements. That is, they can at least partially be evaluated by an outside observer. Overt responses include gross body movements like starting to run in a competition, small movements, such as shivering, and minor or subtle movements. Movement changes may occur inside the body—for example, by increased activity of the cardiovascular system (resulting in an increased blood pressure [BP]) and may be observed by, for example, changes in skin coloration (e.g., blushing). Covert responses, in contrast, reflect internal processes of information processing. By internal processes, we restrict ourselves here to the psychological system: processes of perception, recognition, memory, thought, action, and emotions. Movements do not constitute an integral part of covert responses. The dichotomy of overt versus covert responses corresponds roughly to bodily or psychological reactions and may serve as an approximation. For other response systems such as the neurochemical or endocrine systems, other classifications may be more appropriate.

Such a conceptual distinction calls for empirical support—that is, the question of how overt and covert responses can be measured and how such measurements differ arises. In particular, parameters are needed that can capture responses regarding their quality, latency, duration, amplitude, course of reaction, and their return to the pre-response state.

Overt responses involve some form of movement. Hence, they are prone to observation—that is, they can be evaluated by an outside observer. Such a form of measurement may require a subsequent, reliable form of movement classification. Depending on the research topic, one must define relevant movement categories such as, for example, *running away*, *approaching an object*, or *preparing for action*. More specific or “smaller” bodily movements can also be directly measured. The measurement of kinematic, or temporal and dynamic (i.e., force), aspects can provide detailed information about physical parameters such as the aforementioned latency, duration, amplitude of

reactions, and so on, by measuring changes in joint angles or muscularity, for example. Such technical measurements provide objective diagnostics of overt responses.

In contrast, covert responses defined as psychological responses cannot be directly measured. To obtain information about covert responses, one can use the method of self-report, whereby the subject describes verbally their responses. Again, the description can relate to different dimensions of the response, such as stimulus- or reaction-related or different response systems (e.g., somatic or psychological). Another, more objective measurement of covert responses can be acquired with psychological questionnaires. Information about the quality of the data, such as reliability and validity, are usually available with this measurement tool. Psychological questionnaires can be used to evaluate trait variables such as hand preference or achievement motivation or state variables such as state anxiety (SA) or emotional arousal. The biological basis of all such covert responses, at least the great majority, is seen in the central nervous system (CNS). As a consequence, the measurement of the activity of the CNS provides a powerful and objective, although less direct way to evaluate covert responses. Of course, the voluntary control of overt responses—action control—also involves the CNS. To measure activity in the CNS, one can record central or peripheral physiological parameters, as these are the closest correlates of psychological responses available to date.

Central Versus Peripheral Measurements

Measures of CNS activity can reflect the reactions of both specific and unspecific nervous subsystems. One example of an unspecific system is the *ascending reticular activating system* (ARAS), a neural network originating in the brain stem that is involved in the regulation of sleep–wake transitions. Its function is unspecific as it regulates the overall arousal level. *Attention* is another unspecific system that is supported by a frontoparietal network in the brain's cortex because external as well as internal events from all sensory domains can attract attention. Examples for specific nervous systems are the (neuronal part of the) *stress response system*, which prepares the organism for strong bodily activity, the so-called *fight-or-flight reaction*; the *motor system*, with its neural circuits to both the basal ganglia (main function: selection

and initiation of voluntary movements) and the cerebellum (main function: control and fine adjustment of movements), is another example of a specific nervous response system.

Depending on the method of investigation, one can measure various parameters of responses within the different functional systems of the CNS—for example, changes in an electroencephalograph (EEG) frequency band that is related to attention. However, it appears to be more difficult to relate recorded parameters—that is, EEG changes in our example to specific dimensions, whereby the dimensions could be either stimulus- or response-related. If a particular EEG response with a given latency is recorded, such as after perceiving a starter's gun, the response may refer to the stimulus (its loudness) to the emotional response (a startle reaction), or to the response preparation—that is, to run as fast as possible.

Today, CNS activity can be measured with a number of invasive and noninvasive methods that differ in their temporal and spatial measurement precision. Noninvasive techniques include the electroencephalogram (EEG) and the magnetoencephalogram (MEG), which have a high temporal resolution. These methods measure the summed electrical (EEG) or magnetic (MEG) fields, respectively, of neural activity in the brain's cortex. Functional magnetic resonance imaging (fMRI) and near-infrared spectroscopy (NIRS) have a high spatial resolution and measure indirectly through changes in blood flow nervous activity; fMRI can do so throughout the brain. Finally, an invasive technique that can only be applied in a patient or animal setting is the intracranial recording of electric fields or action potentials—that is, using recording electrodes implanted into the brain tissue during an operation. This method has a superior temporal and spatial resolution.

Another way to evaluate covert responses is to measure peripheral activity. Some measures of peripheral activity are very good indicators of central nervous processes—for example, the orienting response—even though the peripheral systems have specific peripheral functions. Furthermore, recordings are usually performed with techniques that are easier to apply than the aforementioned techniques for recording CNS activity. Useful systems for the evaluation of covert responses are the cardiovascular system, the electrodermal system—that is, by and large sweat gland activity—and reactions of the eye—for example, pupil dilation or eye

fixations. The cardiovascular system—for example, HR—is a peripheral indicator of the activity in the sympathetic and parasympathetic nervous system (PNS). These important parts of the autonomous nervous system are implicated in the stress response. Both parts, the sympathetic and the PNS, normally show complementary reactions, and it is difficult to isolate one from the other. In order to measure sympathetic activity, electrodermal data is helpful. Sweat glands are solely innervated by the sympathetic part of the autonomous nervous system, and can thus serve as an indicator of the CNS functions associated with sympathetic activity. Eye movements, finally, are a reliable measure for various cognitive processes. They are informative for processes such as visual search—that is, the scan path of the eyes, reading, and attention. Visual search is an important process, for example, to capture the distribution of teammates over the playing field. Of particular interest is the pupil response—that is, a change in diameter that is associated with cognitive workload and affective processing. These peripheral measurements, especially eye movements, can be taken while the participant is fully mobile, which permits an investigation of covert responses in natural settings.

Illustrations in Sport Settings

In sport settings, humans aim for optimal control over their responses. This includes not only the production of wanted behaviors but also the inhibition and suppression of some unwanted responses. Specifically, responses ranging from involuntary muscle activity to psychological distractions due to stress can severely interfere with performance, particularly in competitions. The latter is known as *choking under pressure*. This phenomenon refers to acute performance decrements due to attentional malfunctions or perceived pressure to achieve. Another example of interfering responses are the so-called yips in golf. “Yips” are involuntary hand and arm movements—that is, jerks, tremors, or freezing that adversely affect the movement trajectory possibly involving a deterioration of the basal ganglia. Such unwanted responses can interfere with sport performance, and athletes must train to control them.

Covert responses also play an important role in sports, specifically in mental training. Mental training refers to the imagined reproduction of a movement sequence without any movement execution; it

is a cognitive reproduction of the movement. This form of training can lead to performance improvements, especially for cognitive task aspects and if combined with physical training. It is thought that the activation of movement representations in memory below the motor threshold—that is, a covert response—is responsible for later improvements in movement performance. Thus, mental training is a form of covert response to internal events, which plays an important role in sport settings.

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See also Adaptation; Automaticity; Habit; Movement

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REST

Rest is the cessation of activity. Rest is of interest within sport and exercise psychology because

performance and learning in sport and exercise domains are affected positively by appropriate use of rest and negatively by inappropriate use of rest.

The specific effects of rest depend on the length of the rest period and the length and type of event within which rest is provided. When the event is a practice or exercise session or competition, rest periods are usually taken or provided within the event and last for seconds or minutes. These periods have three functions.

First, they reduce physical fatigue induced by the activity. Physical systems engaged during activity consume energy and produce waste products. Rest affords energy replenishment and removal of these products. Failing to rest prevents these processes, which degrades physical functioning, leading to poor performance. Despite common beliefs that physical fatigue negatively affects psychological functioning (e.g., decision making [DM]), there is little evidence of this currently. Second, rest periods within an event reduce cognitive fatigue induced by the activity. Sport tasks often require full attention, especially if tasks are not well learned or the goal is to improve rather than maintain task performance. Providing full attention is effortful and leads to cognitive fatigue. Consequently, practice sessions requiring full attention are often limited to around one hour before fatigue is experienced and rest is required. Rest replenishes attentional capacity. Failing to rest prevents this process, which degrades the ability to attend, leading to poor performance. Third, rest during practice sessions specifically can enhance the learning of movement skills. The provision of rest within a practice session is termed *distributed practice*; providing no rest is termed *massed practice*. Distributed, versus massed, practice enhances long-term learning of movement skills. Theoretical explanations for this effect can be found in Further Readings at the end of this entry.

Rest periods lasting hours or days are usually taken or provided between events (e.g., competitions). These periods have four functions. First, they allow greater energy replenishment than is afforded by the smaller rest periods typically taken or provided within events. Second, an event placing a greater-than-normal physical demand on the performer, known as an overload, causes a breakdown in the overloaded physical system. Breakdowns stimulate physical adaptations within the system that lead to a new, higher level of system functioning than existed before the overload

was applied, a process termed *supercompensation*. However, days of rest are required for complete adaptations to occur. Thus, rest days between events allow for supercompensation following overload. Third, rest periods lasting hours or days enhance memory consolidation, particularly if they include naps or overnight sleep, which facilitates learning (e.g., of a new movement skill). Memory consolidation involves a gradual stabilizing of newly acquired memories following practice. Consolidation increases the resistance of memories to forgetting and interference from other similar memories, resulting in better learning. Fourth, a high level of motivation must be maintained for the performer to attend fully during an event as well as spend many hours per day within the sport environment (e.g., in the gym). Rest periods lasting hours or days help reverse decrements in motivation that result from extended practice and competition and immersion in the sport environment.

Practice and training regimens spanning a competitive season or longer time frame are often divided into periods, a process known as *periodization*, to allow athletes to obtain a high level of performance when they are required to compete. One common approach to periodization involves three major periods. The first period involves preparation for competition. The second period comprises the target competition(s). As such, the performer undertakes much activity in these two periods. The third period involves a transition between a completed competition period and the next preparation period and is characterized by weeks of rest. Rest in the transition period involves the cessation of activity but often also includes what is termed *active rest*. Active rest involves engagement in activities designed to help the performer maintain a base of physical fitness and movement skill while imposing a low training load.

The weeks of rest provided in the transition period afford physical energy replenishment, physical adaptation, and healing following the demands of the preparation and competition periods. Also, a high level of motivation must be maintained for the performer to (a) provide the physical and cognitive effort required during the preparation and competition periods and (b) spend many hours per day during the months spanning these periods within the sport environment. The transition period, involving weeks of rest and active rest, helps reverse decrements in motivation that may follow the preparation and competition periods.

Chronic failures to rest within and/or between practice or exercise sessions or competitions, and/or between demanding periods of larger-scale training regimens (e.g., preparation and competition periods) can lead to physical and psychological problems that impede performance and are slow to reverse. Among these are the overtraining and burnout syndromes. Overtraining syndrome is characterized primarily by a long-term decrement in performance capacity restored only following weeks or months of rest. The key characteristic of burnout syndrome is a long-lasting experience of emotional and physical exhaustion, sport devaluation, and reduced sense of accomplishment.

In conclusion, rest influences performance and learning in sport and exercise domains. Athletes and coaches as well as exercisers and fitness instructors who include appropriate rest periods within and between their practice and exercise sessions and also competitions can expect to capitalize on the positive effects of rest.

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See also Burnout; Fatigue; Memory; Motor Learning; Overtraining Syndrome; Practice; Skill Acquisition; Underrecovery Syndrome

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likely depends on an individual's memory of the material or skill in question. Not surprisingly, most definitions of memory sound similar to those of retention. For example, one definition of motor memory (i.e., memory of how to produce a motor skill) characterizes memory as the *persistence* of the acquired capability for performance. The notion of persistence implies that a person's memory for a particular skill is capable of being assessed (i.e., with some form of retention test) at a later time (i.e., from days to years). Retention tests are a common phenomenon in most formal educational settings. Teachers assume, rightly or wrongly, that students' performance on tests represents a valid estimate of students' memory of the material or skill in question. In the motor learning and control literature characterizations of memory have ranged from an internal representation of the rules for producing the skill to the persistence of an intrinsic coordination pattern. Regardless of how one conceptualizes memory, however, some form of retention test must be given to obtain an inference of its persistence.

Retention and Transfer

Another concept that is quite similar to retention is that of transfer. While retention pertains to the persistence of performance under the same conditions as those experienced during practice, transfer deals with performance persistence demonstrated under conditions that differ in some respects from practice. Practically speaking, transfer can be considered a special case of retention and, in some sport settings, is the more valid test of the persistence of athletes' skills. For example, open sports requiring rapid and accurate responses to challenges presented by the opponent (e.g., basketball, football, soccer, tennis, volleyball) would likely involve competition experiences that are different in some fashion from those encountered during practice. The extent of the differences between practice and competition experiences would, however, be minimal in closed sports (e.g., gymnastics, swimming, golf, track and field) where the environmental demands are relatively similar. For simplicity sake, the term *retention* will be used throughout the remainder of this entry to refer to any performance assessment that occurs following some period of practice or that is interspersed between organized practice sessions.

RETENTION

Classical definitions of retention emphasize the degree to which people are able to remember (or perform) some previously practiced material (or skill) after an elapsed period of time. Thus, retention

Assessing the Retention of Sport Skills

There are several categories of sport skills that represent candidates for retention assessments. These include *technical skills*, *tactical skills*, and *mental skills*. Technical skills refer to the physical tasks and movement patterns athletes must be able to produce in order to perform effectively in competition. Examples would include the serve and volley in tennis, blocking and tackling in American football, and the floor exercise in gymnastics. Tactical skills pertain to the strategies and decisions athletes must make in order to enhance the prospects of achieving desired performance outcomes. Examples would be maintaining a balanced positioning of players on the field or ice in soccer and ice hockey, respectively; throwing to the correct base in baseball or softball; and varying the target location of serves in tennis or volleyball. Mental skills are important for handling the cognitive and emotional demands of performance, particularly when they occur under pressure. Two mental skills that are essential for optimal performance are focus (i.e., attention control) and composure (i.e., emotional control).

In order to assess athletes' retention of each of the preceding skill categories a coach must be able to identify the desired target behaviors and then come up with some way of measuring them. Typical indicators of performance proficiency include the consistency, accuracy, or frequency with which a desired behavior occurs. One technique many coaches use to assess athletes' capabilities is video analysis. With this approach, the coach obtains a video recording of the athlete's behavior or performance and then reviews the recording in order to determine the extent to which the desired behavior or performance is executed effectively. For example, a basketball coach might evaluate a player's execution of proper rebounding technique (i.e., technical skill) by dividing the number of times the player performed the skill of correctly blocking out the opponent by the total number of rebound attempts. In the case of tactical or mental skills, the coach might record the relative accuracy of an athlete's decisions during a particular phase of competition (e.g., deciding when to maintain possession of the ball and when to pass it to a teammate during the last two minutes of a close game) or the frequency of negative emotional outbursts in response to adversity (e.g., vigorously disputing an official's decision), respectively. By obtaining

periodic measures of athletes' behavior and performance during practices and competitions, coaches can evaluate the extent to which the skills athletes are developing and refining in practice are being retained (i.e., produced effectively) in competition.

When conducting retention assessments, coaches also need to be aware of spurious factors that can influence athletes' behavior or performance. Such factors would include improper warm-up, fatigue, illness, medication, weather conditions, or a loss of motivation. For example, an athlete who is recovering from a minor injury might demonstrate less effective skill execution during a competition than she would if she had not been injured. In this instance, the athlete's retention performance would likely not be an accurate reflection of the skill she had developed in practice. Thus, it is important for coaches to "consider the circumstances" when assessing retention and resist the conclusion that athletes' performance in games is always a valid indicator of what they have learned in practices.

Variables Shown to Influence Retention

Historically, laboratory studies have consistently revealed several variables that can have an impact on participants' retention of simple motor skills. These variables include the amount of original practice, the length of time elapsing between skill acquisition and the retention test, and the existence of interfering material or experiences between acquisition and retention. Put simply, research findings have shown that the greater the amount of skill practice, the shorter the time between acquisition and retention, and the fewer the interfering items, the higher retention performance tends to be. It should be noted however that these results have been obtained under highly controlled conditions using very simple motor skills (e.g., blindfolded limb positioning movements). Thus, it is important to consider the potential relevance of these findings to real world sport situations.

Clearly, the amount of practice would be expected to contribute to athletes' retention regardless of the skill being learned. Indeed, some scholars have suggested that at least 10,000 hours of practice may be necessary to ensure expertise for any given skill. However, it is less likely that the length of the retention interval or other activities and experiences intervening between practice and retention would exert much influence on retention performance—at least with respect to skills

that have been practiced for a considerable length of time. Another variable more likely to exert an affect on skill retention is the similarity between the conditions of practice and those of retention, sometimes referred to as contextual similarity or practice-test similarity. Several studies have shown that the more similar the circumstances and demands of practice to those of the retention test the higher retention performance tends to be. These results suggest that coaches should try to make practices as much like competition as possible if they want their athletes to perform as well as competition as they do in practice. Consistent with this notion, the author of a recent book on sport coaching strongly advocates a “games approach” to practice. The logic for this approach is that by designing practice sessions that place a strong emphasis on developing skills under game-like conditions, coaches are more likely to see their athletes produce those skills in the desired way in competition (i.e., on the “retention test”).

A growing body of recent research in motor learning has revealed yet another variable that can have a positive impact on skill retention. Referred to as “self-control” or “self-regulation,” this variable concerns the degree to which learners are given the opportunity to control some aspect of the practice environment. A number of studies have shown superior retention performance for participants given the freedom to control how often they receive extrinsic feedback or observe a model of correct performance compared with that of participants given no such control. Plausible explanations for enhanced retention by self-control participants include a more active involvement in the learning process (rather than passively responding to the dictates of the researcher) and a more challenging learning experience (due to the learner’s own sense of the optimal challenge at any point in time). Suggested outcomes of self-controlled learning that might be expected to produce superior retention performance include a greater attention to relevant environmental information, a heightened sensitivity to sensory feedback emanating from practice attempts, and an enhanced ability to evaluate and correct one’s own mistakes.

The findings of self-control studies would appear to have important implications for enhancing skill retention in sport competition. If nothing else, they suggest that giving athletes the opportunity to take some control of their skill development and decision making (DM) during practices would

enable them to execute their skills more effectively and make decisions more spontaneously during competition. Encouraging such freedom would appear to be particularly important in competition when the coach is prevented by the rules from exerting complete control. Put another way, by allowing athletes to take some ownership of their technical, tactical, and mental skill development during practices (e.g., allowing athletes to do more problem solving on their own and request feedback only when they need it), coaches would be decreasing athletes’ dependency on them in competition. In some sense, the self-control variable is similar to the contextual similarity variable in that it speaks to the importance of making the conditions of practice as similar to those of competition in order to enhance retention (i.e., performance in games).

Potential Benefits of Retention Assessments for Skill Practitioners

Taken together, the existing research and scholarship suggests several variables that are particularly important for enhancing skill retention. Thus, it would appear that practitioners who consider these variables when designing practice experiences could possibly facilitate participants’ skill development and, as importantly, equip participants for effective skill execution in the retention context—competition. Perhaps the most important questions coaches and other skill practitioners might ask themselves when attempting to maximize the retention benefits of practice are What skills do my athletes or participants need to learn? and Under what conditions (i.e., retention) do they need to be able to perform those skills? After achieving possible answers to those questions, practitioners are in a position to create practice experiences that lead to enhanced retention. By periodically conducting retention assessments, coaches can then evaluate the effectiveness of their practices. For example, evaluations of athletes’ performance in competition could help the coach discern whether drills currently being used in practices for adequately developing athletes’ technical, tactical, and/or mental skills. If retention performance is found to be consistently lower than practice performance, the coach might consider revising or perhaps eliminating some of the drills and replacing them with others that simulate more closely the demands and expectations of competition.

Consistent with the notion of practicing for retention is the teaching philosophy of former Princeton University basketball coach Pete Carril. In his book, *The Smart Take From the Strong*, Carril stated the following:

One of my most fundamental points is that we (his team) will not do one single thing in practice that doesn't show up in a game. Everything we do in practice must show itself somewhere in the game, or else we don't do it.

By practicing the technical, tactical, and mental skills they needed under the conditions they were required to perform the skills in competition (i.e., retention), Carril's teams amassed an impressive record over the course of his coaching career.

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See also Attentional Focus; Cue Utilization; Memory; Practice; Recognition and Recall Paradigms; Self-Regulation; Simulation Training; Transfer

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ROLES

Roles are important structural components of all groups and represent the expectations for behaviors of individuals within a particular social situation.

This entry briefly highlights the history, types, and emergence of roles and presents a number of cognitive (e.g., role clarity), affective (e.g., role satisfaction), and behavioral (e.g., role performance) elements to role involvement.

History

The term *role* is common to many disciplines. For example, the term is employed within the entertainment industry to indicate the part that is played by an actor. Furthermore, roles exist in organizations, sport teams, marriages, and so forth. Essentially, where there are groups, there are roles being enacted by group members. The examination of roles as an academic area of study was facilitated by the work of Bruce Biddle and Edwin Thomas in 1966. Their book, *Role Theory: Concepts and Research*, was an influential publication that communicated the knowledge base pertaining to the concept of roles. Around the same time frame, Robert Kahn and his colleagues published a summary of studies examining two role perceptions in the book *Organizational Stress: Studies in Role Conflict and Ambiguity*. Both of these contributions have served as valuable resources in the development of role-oriented research in sport that began in earnest in the 1990s.

Types of Roles

There are a number of ways in which to describe the types of roles within sport groups. In a very specific manner, roles and the associated responsibilities can be identified on a sport-by-sport basis. For example, roles occupied by basketball players can be identified by the position they hold (e.g., point guard, shooting guard, center) and the specific behaviors expected of them (e.g., rebounding, scoring, defending). Obviously, these roles would differ greatly in comparison to other sports (e.g., curling). However, two categorization schemes have been utilized to more generally describe the types of roles that occur in groups. The first categorization is based on the primary objective of the role and is differentiated based on task versus social orientations. *Task roles* pertain to those that focus on the accomplishment of the group's objectives. In contrast, *social roles* pertain to those that facilitate interaction, integration, and harmony within the group.

The second categorization that can be used to classify roles pertains to the degree of formalization

of responsibilities. *Formal roles* are those that are established by the group, organization, or team and are directly prescribed to group members. Examples of formal roles in a sport context include the positional responsibilities held by athletes and the leadership role of captain. The majority of sport research to date has focused on the formal role responsibilities held by athletes.

Conversely, *informal roles* are those that arise through the interactions of team members but are not directly prescribed to individuals. In essence, they emerge naturally in the group environment. In 2010, Cassandra Cope and colleagues highlighted 12 descriptions of informal sport roles including positive examples such as the mentor (one-on-one counselor), spark plug (energy booster), comedian, and informal leaders. In addition, they noted that negative informal roles such as a distracter and a cancer are also discussed in sport literature.

Overall, research across contexts suggests that informal roles can act to supplement or resist the existing formal roles and can be viewed as positive or negative. Furthermore, it can be stated that some informal roles appear to be present across contexts (e.g., a social organizer may be present in sport, work, church groups, and so forth) while others may be quite unique to a specific sport (e.g., the enforcer in ice hockey).

Transmission of Role Responsibilities

Discussion pertaining to the emergence of roles (i.e., formally vs. informally) leads directly to consideration of how role responsibilities are transmitted. Toward this end, a major contribution of the aforementioned work by Kahn and colleagues in 1964 was the Role Episode Model, which highlights a communication process between two actors. The first actor is called the *role sender* and is the individual who develops role expectations for the second actor, the *focal person*. Within the context of sport, it is typical that the role sender is the coach while the focal person is the athlete. The Role Episode Model includes five events in the transmission of role responsibilities: (1) the role sender must develop expectations for the focal person, (2) the role sender will exert pressure on the focal person to fulfill these expectations, (3) the focal person will experience the role pressure and expectations, and (4) respond accordingly. The fifth event is that the responses of the focal person will be interpreted by the role sender. In addition

to this ongoing cycle of events, a number of organizational, personal, and interpersonal factors are proposed to influence (and be influenced by) this role transmission process.

Elements of Role Involvement

As noted at the beginning of this entry, there are several cognitive, affective, and behavioral elements to an individual's experience with his or her role. In the following section, these elements are described.

Role Performance

This represents the prime outcome of the role episode and is the behavioral element of role involvement. Clearly, the aim of developing expectations for a focal person is to have him or her successfully execute the role responsibilities. However, a number of factors may influence the congruence between the role performance of the focal person and the expectations that were derived by role senders.

Role Clarity or Ambiguity

The degree of understanding one has about his or her role responsibilities is referred to as role clarity. Alternatively, researchers also use the term *role ambiguity* to reflect a similar idea (i.e., a *lack* of understanding regarding role responsibilities). Of all the role elements described in this entry, this concept has received the most research attention within a sport context. In 2002, Mark Beauchamp, Mark Eys, Bert Carron, and Steven Bray proposed a conceptual model of role ambiguity (concurrent with an assessment tool: the Role Ambiguity Scale), suggesting the need to consider athletes' understanding of (a) the general scope of their responsibilities, (b) the specific behaviors necessary to fulfill their responsibilities, (c) how they will be evaluated with respect to role performance, and (d) the consequences of not fulfilling their responsibilities. Research conducted since the development of this conceptual model and assessment tool has provided support for the importance of providing clear and consistent information to athletes regarding their responsibilities on the team.

Role Efficacy

This role perception refers to athletes' beliefs in their capabilities to carry out their role

responsibilities in an interdependent fashion (i.e., in conjunction with other team members). Early work on this concept by Steven Bray demonstrated that athletes with stronger perceptions of role efficacy were rated more positively in terms of role performance by their coaches.

Role Acceptance

The willingness of the athlete to execute his or her role responsibilities is generally referred to as role acceptance. From a sport perspective, limited research has been conducted on this topic despite a great deal of anecdotal evidence suggesting its importance. Previous literature devoted to the topic of conformity and persuasion offers some perspective on how role acceptance develops in the focal person including the person's perceptions of the credibility of the role sender and the power held by the role sender. Recent qualitative research by Alex Benson in 2012 supported this suggestion and also highlighted that the significance of the role, team cohesion, and experience on the team were among several potential antecedents of role acceptance for athletes.

Role Conflict

This cognitive role element refers to the presence of conflicting expectations for the focal person. For example, athletes may receive role-related information from a head coach while simultaneously receiving conflicting messages from assistant coaches. This represents just one type of role conflict, and Kahn and colleagues noted several ways in which role conflict could occur within groups. First, *intersender conflict* refers to the reception of inconsistent expectations from different individuals within the same social context (illustrated by the previous example of conflicting information from different coaches). Second, *intrasender conflict* occurs when the inconsistent information is received from the same person (e.g., the head coach provides role information on one day but provides conflicting information the next day). Third, role requirements may not be consistent with who the athlete is as a person (e.g., a nonaggressive individual who is expected to antagonize opponents). This is referred to as *person-role conflict*. The final type of role conflict is *inter-role conflict*, which refers to incongruent expectations arising from different roles in one or more social contexts. For example, the role responsibilities held by a

competitive youth soccer coach (e.g., ensure that individual outcomes for all players on the team are positive; make decisions that benefit the group as a whole) who is also a father of one of the team members (i.e., with specific role responsibilities to ensure the well-being of his child) may at times come into conflict for that coach–father.

A separate concept, *role overload*, shares some overlap with role conflict. Role overload has not received any attention within sport literature and is not particularly well defined due to a number of perspectives that can be taken on this concept. However, generally, overload refers to issues pertaining to the volume of role responsibilities. For example, role overload could manifest itself in several ways within sport including (a) having one role responsibility that exceeds the capability of the athlete and (b) holding several roles that, in combination, exceed the capability of the athlete (although no one role is unmanageable on its own).

Role Satisfaction

This affective component of role involvement refers to the degree to which one finds his or her role fulfilling. In 1987, Genevieve Rail conducted a study with volunteer sport organization executives to gauge how they derive satisfaction from their role. The information she derived from her interviews suggested that role satisfaction was influenced by the degree of autonomy the individuals held over their roles, the amount of significance ascribed to their responsibilities, the degree to which their specific abilities were used, and how much feedback or recognition they received.

Conclusion

Overall, roles represent important structural aspects of all groups. There is long-standing interest in the topic of roles within organizational psychology. More recently, sport psychology (SP) researchers have recognized the necessity to investigate the previously given concepts in an athletic context. Future research can continue to add to our understanding with respect to the conceptualization, assessment, and practical applications pertaining to role involvement.

Mark Eys

See also Cohesion; Feedback; Group Characteristics; Group Formation; Interdependence Theory and the

Coach–Athlete Relationship; Norms; Shared Mental Models; Status; Team Building

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RUNNER'S HIGH

The “runner’s high” is a euphoric sensation reportedly experienced during running, usually unexpectedly, in which the runner feels a heightened sense of well-being, enhanced appreciation of nature, and transcendence of barriers of time and space. There are many terms in the research and popular literature that have been used to describe the runner’s high. These include euphoria, speed, strength, power, gracefulness, spirituality, and sudden realization of one’s potential, and they also include glimpsing perfection, moving without effort, and spinning out.

There are variations in the percentage of runners who experience the runner’s high across different surveys, but many percentages tend toward the lower end of the spectrum, perhaps 10% or so.

However, this percentage really depends on how one defines the runner’s high, and each individual may have her or his own definition or perception of what the runner’s high means. The runner’s high may be experienced on a continuum from what one might call an enhanced sense of well-being on one end to peak experiences on the other end.

This enhanced sense of well-being may be related to the feel-better phenomenon described by some researchers as being experienced during and after exercise participation. Assuming adequate physical conditioning, running generally feels good, both while one is engaged in the activity and afterward, and this sense of well-being may be interpreted as the runner’s high. While one would not want to suggest that this is not what the runner is experiencing (this experience may be important for a given runner in promoting adherence to running on a regular basis), many researchers have looked for more “significant” experiences than just enhanced well-being or what might be seen as a flow state. Instead, researchers have focused more on exalted and euphoric experiences, closer to peak experiences. However, peak experiences, by definition, only occur occasionally and are sometimes accompanied by peak performances. This is not necessarily what the runner’s high is, although it can certainly be felt during peak experiences (and probably is, most of the time). Rather, the runner’s high is a more frequently experienced, but still generally unanticipated, phenomenon. One cannot start out running and say, “I’m going to experience the runner’s high today,” although some runners do talk about being able to get into a meditative state and experience the runner’s high on a regular basis. In the final analysis, the runner’s high is indeed an individual experience.

What does it take to be able to experience the runner’s high? One starts with a high degree of physiological preparedness—being able to run a fairly long distance. Ideally, the runner should be able to run at least 3 to 4 miles comfortably in order to more readily experience the runner’s high. There is a saying about running long distances that the first half hour is for the body, the second half hour is for the mind, and the third half hour is for the soul. This is exactly where the 3 to 4 miles comes in—the suggestion is that one has to run 3 to 4 miles to get into a physiological “groove” wherein a shift to the mind takes over (for those who can run this comfortably) and one can “escape” into the sensation of the runner’s high on occasion.

What types of settings are more conducive to experiencing the runner's high? Research suggests this is fairly individual, but general suggestions include cool, calm weather with low humidity and few distractions. An absence of hills is recommended along with running in familiar areas. An absence of problems is also desirable, with little anxiety about day-to-day affairs, nor pressure about speed and distance during the run. Rather, one is looking for a nonstressful environmental experience along with few cognitive distractions so that one can "tune out" and enjoy the running experience and then potentially experience the runner's high. However, there are exceptions, with some individuals having said they experience the runner's high in new settings (running through a new city). The runner's high is truly an individual experience.

What is the derivation of the runner's high? Is it related to psychological factors such as entering a flow state and being able to tap into right brain consciousness? Perhaps there are physiological explanations that encompass biochemical elements (e.g., the endorphins) or enhanced physiological preparedness to run smoothly and efficiently or effortlessly over a time or distance that is challenging but within one's capabilities. Unfortunately, no definitive answers are presently available, but the search for such an answer is an exciting one.

The runner's high is a challenging phenomenon to study. By definition, it is unlikely to be experienced in laboratory conditions (such as running on a treadmill), where one could control a variety of variables such as weather, terrain, speed, and so on. Once one begins working with runners in the real world, one has challenging logistical

issues—one could easily communicate with runners via cell phones or other such devices but then this would be a distraction likely to detract from the possibility of experiencing the runner's high in the first place or disturb it if it is being experienced.

In the final analysis, whether what the runner is experiencing is an enhanced sense of well-being, a peak experience, or somewhere in between, the runner's high appears to be a factor in keeping runners coming back for more. Anything that facilitates adherence to exercise is desirable. However, researchers always desire to know what really underlies these types of phenomena, especially one as mysterious and euphoric as the runner's high. The runner's high is likely to remain, for the moment, an elusive but highly desirable and wondrous experience.

Michael L. Sachs

See also Addiction, Effects of Exercise on; Affective Responses to Exercise; Flow; Hedonic Theory

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S

SATISFACTION

Satisfaction is recognized as an important determinant of motivation and commitment in sport and exercise. Although researchers and practitioners have acknowledged that satisfaction influences motivated behavior and also represents a desirable cognitive emotional end state in itself, satisfaction is not recognized as a distinct emotion or mood state. Satisfaction has been used interchangeably in the sport and exercise literature with various terms such as enjoyment, happiness, commitment, liking, and well-being. Although sharing the positive affective experiential aspects of these various terms, satisfaction should be treated as a distinct term. Satisfaction represents a positive cognitive affective state resulting from a cognitive judgment process that what is received or experienced meets or exceeds a personal standard.

Researchers suggest that standards can include psychological needs, values, personal and external expectations, preferred behaviors from others, social comparison (being better than others), self-comparison, and goals. Indeed, the types of specific standards can range markedly across people, contexts, and time. Since the reflective cognitive appraisal process involves comparing what is received or experienced against a “standard,” it is not surprising to find that satisfaction can be conceptualized from very broad levels (e.g., life satisfaction), to specific domains (e.g., athlete satisfaction, exercise satisfaction), to specific subdomains (e.g., body shape satisfaction,

coach–leader satisfaction) to specific facets (e.g., achieving specific goals, strategy decisions, training and instructions).

Life satisfaction represents a cognitive evaluation of the quality of an individual’s life as a whole. Researchers within exercise and sport psychology (SP) who are interested in subjective well-being (SWB) often study life satisfaction. Life satisfaction is often equated with or used as a synonym for general happiness or well-being. Some investigations using younger participants such as university students or athletes have found positive associations between physical activity (PA) and life satisfaction. In older adults, review studies indicate that PA has very little association with life satisfaction. This is not surprising for a few reasons. Life satisfaction can involve not only evaluating one’s life achievements and experiences but also involve future expectations and hopes. Life satisfaction in older adults involves evaluating multiple standards such as health, family, parenthood, intellectual functioning, physical functioning, and economics, to name a few. For many older adults, physical functioning might be more important than actual PA engagement.

There is far greater understanding of satisfaction in specific domains such as sport and exercise. Athlete satisfaction has been linked to team cohesion, motivation, the learning and teaching setting, coach–player compatibility and relationships, and injury rehabilitation. Research by Harold A. Riemer and Packianathan Chelladurai demonstrated that athlete satisfaction is very complex and can involve evaluating numerous standards related to

(a) individual and team performance, (b) leadership, (c) the team, (d) the organization, and (e) the individual. Although standards associated with the achievement of the team and the individual seem obvious, there are other unique aspects associated with the team and the athlete. Team aspects can involve team integration, how the team treats the athlete, and the ethical behavior of teammates. Individual standards include personal dedication, as well as task and social contribution to the team. Leadership factors are also important to athlete satisfaction. How the coach conducts training and instruction, utilizes the athlete in training and competition, develops strategy, and treats athletes are all distinct features of athlete satisfaction. Many of these features influence the quality of coach–athlete relationships. Last, there are several features associated with the sport organization including financial support, quality of medical support, and academic support. Standards associated with the sport organization are likely to vary widely depending on the level of competition and the organizational structure (e.g., community groups, educational institutions, professional sport). Given that there are many distinct features associated with athlete satisfaction, the athlete's level of satisfaction can be mixed, with satisfaction with some features and dissatisfaction with other aspects.

Satisfaction in exercise will be related to meeting or progressing toward personally valued standards. The literature suggests satisfaction in exercise is related to several of the same general features as identified in the sport literature. These features including the instructor's behavior, quality of training or instruction, quality of the exercise training environment, ability to develop competence and achieve exercise goals, and ability to have meaningful social interactions and acceptance. Since exercise settings potentially allow for greater individual freedom and choice, there can be great diversity in the type and level of personal "standards." Generally, satisfaction is related to participation motivation in exercise. However, there is evidence that body dissatisfaction is also a motive for exercise engagement in some individuals.

Body satisfaction, and more commonly body dissatisfaction, is a central feature of body image. Although body image is discussed in more detail elsewhere in this encyclopedia, body satisfaction and dissatisfaction involve comparing one's body or parts thereof to some personal standard. Body satisfaction is typically related to meeting societal

expectations of idealized body shape and composition (muscle and fat) relative to age and gender. Body satisfaction is moderately associated with both physical and global self-esteem. Researchers have found that body satisfaction and dissatisfaction can be factors in motivating the initiation and continued engagement in exercise programs. There is evidence that suggests that exercise is negatively related to body satisfaction in younger women but positively related in older women and men of all ages. Researchers are also interested in body dissatisfaction because it is a risk factor for eating and exercising pathology.

A growing area of study in sport and exercise psychology (SEP) is examining satisfaction related to psychological needs. These needs could include competence (feeling effective), autonomy (feeling as though you make your choices), and relatedness (sharing a meaningful connection with important others). Guided by Edward L. Deci and Richard M. Ryan's self-determination theory (SDT), researchers often assess the degree to which psychological needs are satisfied in sport and exercise contexts. The sport and exercise environment can facilitate psychological need satisfaction. For example, the coach or exercise leader could create environments that are optimal for need satisfaction through encouraging choice, acknowledging personal perspectives and minimizing pressures. Satisfying psychological needs in sport or exercise is associated with optimal outcomes such as increased well-being, intrinsic motivation, exercise behavior, and intrinsic interest and interest in sport. Researchers are also interested in psychological need thwarting (frustration of needs) because it is associated with athlete burnout, negative affect, and perfectionistic concerns.

Peter R. E. Crocker and Katie E. Gunnell

See also Achievement Goal Theory; Attribution Theory; Basic Emotions in Sport; Body Image; Enjoyment; Hedonic Theory; Self-Determination Theory

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SEDENTARY BEHAVIOR

Exercise psychology has evolved from sport and health psychologies and has involved the study of psychological antecedents (e.g., motivational and other influences) and consequences (e.g., mental health effects) of participation or nonparticipation in moderate-to-vigorous physical activity (MVPA). It is now widely recognized that a lack of MVPA (inactivity) is a major public health issue. It is also believed that similar health problems can arise from too much sitting (sedentary behavior).

Sedentary is a term often used to describe individuals with low PA or not meeting a criterion level of PA. However, it is recommended not to define sedentary behavior in this way but instead to define it as sitting or lying behaviors undertaken in waking time. This encompasses behaviors associated with the use of technology (e.g., TV, computer), socializing (e.g., sitting and talking), travel (e.g., motorized transportation, especially car use), and other common sitting behaviors (e.g., reading). Some of these are changing rapidly, such as computer gaming, and may pose significant health risks if sedentary time is excessive. Such risks can occur even when PA levels meet guidelines. One important issue for psychologists is how to effect behavior change—that is, to help people sit less. This is different from “moving more” at the level of MVPA.

Little research has taken place on the psychology of sedentary behavior. It has been found that

sociodemographic (e.g., social class), behavioral (e.g., smoking) and environmental (e.g., having TV in bedroom) variables are all associated with screen time (TV and computer use), but no consistent psychological correlates consistently emerge from such analyses. Psychology may operate indirectly, such as through the negotiation of use of electronic equipment in the home or parents discussing rules about screen use for their children, but few data exist showing consistent psychological correlates of sedentary behavior in young people or adults.

Typical theories used in exercise psychology to help explain participation in PA are *social cognitive theory* (SCT), *theory of planned behavior* (TPB), and the *transtheoretical model* (TTM). These may not necessarily be appropriate for the study of sedentary behavior. *Behavioral choice theory* has been used in sedentary behavior research and suggests that behavior is a function of the ease or difficulty of access and availability of sedentary pursuits and the reinforcement value (e.g., enjoyment) of such behaviors. Access and availability are likely to be high for many sedentary behaviors. The enjoyment of some sedentary behaviors, such as computer games, could also be high. This might make sedentary behavior reduction a significant challenge.

Anecdotally, sedentary behaviors seem to be as much about “habit” than conscious or deliberate decisions. Health behavior change can be achieved in one of two ways. There is the “deliberative route,” where people use thoughts and conscious processing to arrive at decisions and behaviors. However, there is also the “automatic route,” whereby people are influenced in their behavior by environmentally cued responses, habit, and little or no conscious processing. For example, when entering a room with chairs, little or no conscious processing is involved in sitting. No “decisions” are made; most people just sit. Habit will be clearly associated with a behavior if everyday contexts remain stable and are regularly encountered. Clearly, this is the case for sitting. On the other hand, there may be contexts, particularly if people become more aware of the health consequences of prolonged sitting, where more conscious thoughts are used to stand and move. Promoting such behaviors may be beneficial. In short, psychologists may wish to account for more habitual processes in sedentary behavior than has typically been the case for PA. This might involve promoting less sitting with signs and reminders.

Interventions to reduce sedentary behavior have been reported almost exclusively on young people. The overall effect is small but meaningful. The small effect may reflect sedentary behavior being a challenging behavior to change, driven by strong habitual tendencies and social norms.

Approaches to change sedentary behaviors in young people have generally included educational and family interventions, with the former aiming to teach children about sedentary behaviors (e.g., TV viewing) and how to seek alternatives. This is best illustrated by the Planet Health program in the United States, where children study a “power down” TV reduction element alongside other aspects of health. Other intervention approaches have included the use of TV and computer time budget monitors. With adults, studies are emerging that tackle sitting in the workplace, such as through the introduction of standing desks or goal setting.

Sedentary behavior has become a popular area of research although the number of psychological studies is currently small. However, as we move to greater use of interventions, the role of psychology should develop. But much more needs to be known about how we might change such behaviors. This is at a time of an environmental onslaught that encourages and reinforces such behaviors. Psychologists will need to explore what is realistic for sedentary behavior change, how best to prompt and reinforce sedentary behavior change, and to consider what theories or perspectives work best. Moreover, more knowledge on how to communicate this “new” health message is also needed.

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See also Affective Responses to Exercise; CALO-RE Taxonomy of Behavior Change Techniques; Ecological Theory; Habit; Mental Health; Obesity; Psychological Well-Being; Quality of Life; Self-Monitoring

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SELF-ACCEPTANCE

According to psychologist Carol Ryff, the term *self-acceptance* refers to positive evaluations of oneself and one’s past life. Someone with high self-acceptance experiences psychological well-being (PWB) in the form of a positive attitude toward oneself—being able to recognize and accept the good and bad qualities of multiple aspects of oneself—and looks positively upon past life. Alternatively, someone low in self-acceptance is dissatisfied with herself or himself, wishes to be different, is troubled by certain personal qualities, and is disappointed with past life. The term *self-acceptance* has its origins in research on positive PWB and has been identified as one of the central components of mental health, optimal functioning, and the realization of one’s true potential.

Physical Self-Acceptance

Aspects of self-acceptance are featured prominently in research and theory in sport and exercise psychology (SEP), more specifically research on self-esteem and the physical self. For example, in R. Sonstroem and W. Morgan’s model of how exercise might influence self-esteem and the physical self, physical acceptance (i.e., the degree to which an individual accepts both physical strengths and weaknesses) is identified as one of the two main dimensions of the physical self (physical competence is the other dimension). Physical acceptance is described as similar to satisfaction with various parts of one’s body; and body satisfaction itself is often associated with global self-esteem. The model makes a case that it is possible to alter perceptions of the physical self, including physical acceptance, through exercise. However, in comparison to the physical competence dimension of the model, much less is known about the physical acceptance dimension.

Links to Well-Being

Indicative of its role in psychological adjustment, self-acceptance has been proposed as a dimension of human potential consistent with

Ryff's *eudaimonic perspective of well-being*. Self-acceptance is associated with hedonic forms of well-being, such as satisfaction with life and positive affect, and is positively related to the other dimensions of eudaimonic well-being (e.g., environmental mastery, positive relatedness). Along with additional indicators of optimal psychological functioning, such as personal growth and purpose in life, self-acceptance is associated with participation in physical activity (PA), as well as satisfaction of the basic psychological needs of competence, autonomy, and relatedness through engagement in PA. By using the abilities and functions of one's body, PA promotes self-acceptance and therefore striving for the fulfillment of human potential. Being physically active contributes to an able body that meets the demands of everyday life and encourages contentment with one's body despite weaknesses and deficiencies. This allows one to be self-accepting, particularly of the physical self.

Self-Accepting Self-Regard

A sport-specific form of self-acceptance has also been discussed in the literature, consisting of self-accepting self-regard and independence of self-regard. Self-accepting self-regard represents a belief in the inherent, unique, and complex worth of all human beings, whereas independence of self-regard refers to valuing and feeling good about oneself despite one's shortcomings. Together, self-accepting self-regard and independence of self-regard form a sport-specific self-acceptance, which reflects athletes' valuing and feeling good about themselves despite their shortcomings and failures in sport. Sport-specific self-acceptance is related to general self-acceptance, stability of self-concept, and self-esteem. Although sport-specific self-acceptance and self-esteem both involve a positive sense of self-regard, important distinctions exist between the two constructs. In particular, the independence of self-regard dimension of sport-specific self-acceptance means that athletes value and feel good about themselves regardless of whether or not they perceive themselves as competent in sport-related domains. In contrast, high self-esteem is conditional on a sense of competence or adequacy, whereby athletes need to feel good about their strengths and accomplishments. Researchers suggest that sport-specific self-acceptance might be as beneficial as self-esteem

development in helping athletes feel good about themselves.

Self-Acceptance and Self-Compassion

Self-acceptance has also been described as an important aspect of self-compassion, which shows promise as a personal resource for athletes and exercisers. Self-compassion is particularly useful in times of pain and failure, because it generates a desire to heal oneself with kindness rather than berate oneself with harsh criticism. As exemplified in exercise settings, people who are self-compassionate seem to be more accepting of things that they are unable to change and instead focus on things that they can change. Specifically, women exercisers who are less self-compassionate tend to feel a stronger obligation to exercise, even when that exercise might be harmful to them. Women exercisers have also described that acceptance of one's own body limitations can be an important part of developing self-compassion toward the body. In sport, self-compassion has also been proposed as potentially beneficial to women athletes on the journey to self-acceptance of their own muscularity.

Conditional and Unconditional Self-Acceptance

The distinction between unconditional self-acceptance and conditional self-acceptance has also been recognized in the SEP literature, each of which might have important consequences when looking at the outcomes of participation in sport and exercise. Unconditional self-acceptance can be considered an adaptive acceptance of one's self that does not depend on the approval, acceptance, or love from others; rather, it involves accepting oneself fully and unconditionally. Among athletes, unconditional self-acceptance is negatively related to maladaptive forms of perfectionism, such that those with high unconditional self-acceptance are less likely to experience perfectionism based on either the pursuit of exceedingly high standards set by the self or high standards perceived as imposed by others. Conditional self-acceptance, on the other hand, is highly dependent on the perceived evaluations of others and predisposes perfectionists to experience burnout.

Kent C. Kowalski and Leah J. Ferguson

See also Psychological Well-Being; Self-Compassion; Self-Conscious Emotions; Self-Criticism

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SELF-APPRAISAL/ASSESSMENT/ PERCEPTION

It is widely accepted that the way people view and evaluate themselves has important implications for how they feel, think, and behave. For this reason, researchers have come to see self-evaluations as a central and important topic of study. However, despite there being a great deal of literature on self-appraisal, self-assessment, and self-perception, a noteworthy gap is the lack of consistent operational definitions used for each of these terms. Consequently, there has been a repeated failure to make a distinction between these terms, and many researchers have used these terms interchangeably given their conceptual similarities. Because conceptual distinctions cannot be made between these terms, *self-appraisal* will be used in this entry to infer the description and evaluations that an individual makes of his or her attributes and capabilities.

Types of Self-Appraisals

Similar to research with other constructs related to the self, such as self-esteem and self-concept, self-appraisals can be global (i.e., descriptions and evaluations of oneself in general) or can exist within particular domains (i.e., descriptions and evaluations of hockey abilities), which raises the question of whether global or domain-specific self-appraisals are more predictive of certain outcomes in sport and exercise psychology (SEP).

In circumstances where domain-specific self-appraisals have been the focus, the different types of self-appraisals can be classified as (a) *performance-oriented* self-appraisals or (b) *self-oriented* self-appraisals. A main distinction between performance-oriented and self-oriented self-appraisals is that the former is concerned with an individual's evaluation of his or her own capabilities to perform a particular task, whereas the latter is concerned with an individual's evaluation of himself or herself that may include personal qualities, personality traits, physical appearance, and self-worth. From this perspective, self-appraisals can have a broad or narrow perspective. Generally speaking, people who have high self-appraisals of himself or herself or some aspect of the self will think positively of themselves and be confident. In contrast, people with low self-appraisals will think negatively of themselves and lack confidence. In both cases, however, self-appraisals reflect *evaluations* as opposed to *descriptions* and can affect people's thoughts and attitudes and also can play a role in determining people's behavior by influencing their goal commitment, goal pursuit, and goal-related experiences.

Accuracy of Self-Appraisals

Self-appraisals may or may not include objective ratings. In some cases, people might self-appraise based on a number of different criteria such as social norms, social comparisons, past experiences, and expectations. In addition, just as others can affect how people define their self-concept, others can also affect people's self-appraisals generally or within specific domains. Ultimately, the lack of objectivity and presence of others might lead to biased self-appraisals. Indeed, although people's self-appraisals have been shown to become more accurate as they age, researchers have found evidence that self-appraisals do not always correspond with objective measures, such that people either overestimate or underestimate themselves or some aspect of the self. This can be particularly problematic when people perceive there is an advantage to positively appraising oneself. For example, when people are asked to rate their ability for a certain team sport, they might evaluate themselves positively if they believe this could help them achieve some desired goal, such as making a team's final roster. This raises concerns about people's objectivity and ability to view themselves accurately because there can be

consequences if self-appraisals are unrealistic and exaggerated, regardless of the direction. People who overestimate their abilities might try tasks they cannot accomplish and experience unnecessary failure. These failures could then undermine their self-efficacy and result in behavioral withdrawal. Alternatively, people who underestimate their abilities might become self-limiting and restrict their activity choices. Therefore, the degree to which people are accurate in their self-appraisal is important information that could affect motivation, activity preferences, effort exerted, persistence, and subsequent self-efficacy beliefs.

Sport and Exercise Psychology Research on Self-Appraisals

There are a variety of constructs that have been shown to relate to self-appraisals, such as cognitive (e.g., satisfaction), emotional (e.g., affectivity, anxiety, depression), motivational (e.g., goal setting and commitment, persistence), and behavioral (e.g., performance disruptions, effort) variables. For example, researchers have documented that motivation is related to self-appraisals, such that people with negative self-appraisals see challenging tasks as difficult and feel that it is beyond their abilities and/or their control, which in turn lowers their motivation. In contrast, people with positive self-appraisals are highly motivated to complete challenging tasks because they feel they have the ability and control to successfully complete the task. Another key finding in the literature is that people who have high self-appraisals persist longer at tasks in the face of difficulties or setbacks. Finally, researchers have also shown that self-appraisals can affect attributions, whereby individuals who have positive self-appraisals make internal attributions (i.e., ability, effort) for their self-worth, attributes, and capabilities (e.g., I am fit because I exercise regularly), whereas those who have negative self-appraisals make external attributions (i.e., luck, task difficulty; e.g., I am fat because of my genes).

Jennifer Brunet

See also Hierarchical Self; Self-Criticism; Self-Doubt

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SELF-AWARENESS

Self-awareness involves expert knowledge of oneself, independent of others. As such, self-awareness is central to one's personal and social perception and meaning of self.

Development Perspectives

It has been argued that there are at least two inter-related levels of the self that psychologist William James (1842–1910) labeled the “I” self and the “me” self. The former is considered an implicit level whereby the self is a subject of experience, whereas the latter is described as an explicit level where the self is an object of knowledge for oneself. This me self is a conscious representation of the self and involves a complex ability to allow one to be the subject of one's own attention through self-identification and self-recognition.

Implicit and explicit self-awareness are contingent on developmental perspectives, whereby the I self concerns the first 2 years of life and the me self is developed in at least three stages thereafter. Most of the assessment and theoretical testing for self-awareness has been conducted by observing infants' and children's reactions to themselves presented in mirrors. Infants begin with no self-awareness (oblivious) and develop to experience self-differentiation (how the body feels and looks) and situational differentiation (how the body relates to the environment). Around age 2, individuals will manifest recognition of themselves, and this is a landmark in the emergence of a conceptual self. At this initial explicit stage, there is an

identified self that the image in the mirror is “me” and what is experienced from within is related to what is displayed in the mirror. Following this identification, individuals develop a sense of permanence whereby their self-identification has no boundaries (i.e., they can recognize themselves in contexts beyond the mirror, such as movies or photos). The final developmental level of self-awareness has been proposed to include displays of self-consciousness resulting from an awareness that the self is seen from within—what the self is—as well how the self is perceived by others.

High Versus Low Self-Awareness

Self-awareness ranges on a continuum from low levels of awareness (i.e., narrow, temporally limited awareness of movement and sensation in the immediate present) to high levels of awareness (i.e., invoking comparisons of events and self against standards and norms and involve broader time spans). These distinctions of low and high self-awareness have been linked to divergent thoughts, feelings and affect, and behaviors. For example, researchers have found that individuals who are highly self-aware perceive their experiences more intensely, have more accurate self-evaluation and are able to self-regulate more adaptively, react more strongly to social comparison, and generally know themselves better compared with individuals who are low on self-awareness.

Theoretical Framework: Escape Theory

The principle of escape theory is that individuals may find it burdensome and aversive to be aware of himself or herself and therefore find ways to escape. Since it is difficult to completely “turn off” self-awareness, a common strategy is to narrow one’s attention to the present and immediate environment. By narrowing attention, there is limited capacity to engage in meaningful thought about identity and oneself. This deconstructive process, or moving one’s awareness from high to low levels, is a way to escape threats and pressures. The motivation to escape self-awareness hinges on a comparison of the self against standards, norms, and expectations whereby the higher the standards, the greater likelihood of failure, and the more impetus for escape. As it pertains to body shape and appearance, for example, there are high standards of “ideal” physiques for men and women that are perpetuated in the media

and internalized as normative discontent. Based on these high standards, the self is deficient, and actions are carried out to manage the deficiency. Within the escape theory framework, behavioral actions such as alcohol use, cigarette smoking, and binge eating are associated with a reduction of the undesirable state. While not examined empirically, it is intuitive that sedentary behaviors such as playing video games or watching television and physical activity (PA) engagement are also behavioral strategies to manage (escape) the deficiency brought about by high self-awareness. It could be argued that excessive exercise, for example, follows a similar trajectory as binge eating in that it is an escape from immediate or pending threats and pressures.

Manipulations of Self-Awareness

Most researchers involved in experimental studies of self-awareness have manipulated attention and cognition through strategies of distraction (e.g., intense, absorbing movies; challenging math problems; complex motor behaviors) and heightened focus on the self (e.g., mirrors, self-relevant prompts, concentration and focus).

In sport psychology (SP), self-talk is a strategy that can be used to heighten self-awareness. It is proposed that self-talk may be the link between the I self and the me self and helps in the development of motivational thoughts, feelings, affect, and self-monitoring. In this way, self-talk may mediate self-awareness and related strategies such as self-monitoring and regulation.

Too Much Self-Awareness?

There is some evidence that manipulating self-awareness can heighten negative emotion and can foster susceptibility for anxiety and depression and lower self-esteem. More research is needed to better understand the potential maladaptive role of self-awareness.

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See also Body Awareness; Self-Monitoring; Self-Regulation

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SELF-AWARENESS THEORY

Self-awareness refers to the ability to make oneself the object of one's own attention. This self-reflexive quality of consciousness has been the focus of considerable research within the cognitive, social, and sport psychology (SP) domains over the past 50 years. Originating from this research, a theory of self-awareness emerged from the work of Thomas Shelley Duval and Robert Wicklund, which posited a dichotomy between external and internal objects of conscious attention. At any one time, attention may be turned outward toward the environment or directed inward toward the self but may not be focused on both at the same time. Self-awareness theory (SAT) further proposes that self-focused individuals will automatically engage in a comparison of the self against a set of standards. These standards are considered to represent a "criterion of correctness" consisting of beliefs about correct behavior, attitudes, and traits. In this regard, self-awareness plays an active role in the process of self-evaluation.

SAT assumes that the preferred state of the individual is consistency between the self and a particular ideal or standard. Inevitably, however, the comparison between self and standards leads to the recognition of discrepancies, which result in negative affect if standards are not met. In order to reduce this discomfort, SAT predicts that individuals are motivated to seek behavioral responses in order to regain equilibrium between self and standards. Initially, one of two behavioral responses is predicted to occur as a result of these discrepancies: First, individuals may actively change their

actions, attitudes, or traits to be more in line with their standards. Such behavioral responses might result in greater task effort, more concentration, or other steps leading to self-improvement. Alternatively, the individual could elect to avoid the self-focusing situation altogether, thus ending the comparison between the self and standards. More recently, however, research has suggested a third possibility in which individuals reduce discrepancies by altering their standards to be more congruent with the self. Thus, discrepancy reduction can occur either through means of changing the self or one's standards of right behavior, attitude, or traits.

Increased self-awareness has been shown to lead to a number of predictable consequences. Individuals who are made self-aware have been found to be more likely to conform to standards of correct behavior than those who are less self-aware. For instance, heightened self-awareness has been found to increase the willingness of individuals to help others and to decrease the urge to cheat. Additionally, self-awareness is an important determinant of the degree to which individuals are cognizant of bodily states, attitudes, and feelings. To this extent, the accuracy of self-reports are better when obtained under conditions of increased self-focused attention. Self-aware individuals are also more likely to attribute causal responsibility to themselves compared to less self-focused individuals. Finally, self-awareness, especially when induced by the attention of others, facilitates the motivation to make behavioral attempts to reduce discrepancies between one's self and one's values and standards.

Factors That Influence Self-Awareness

Any stimulus or situation that reminds an individual of himself or herself can lead to a heightened salience of self. For example, seeing oneself in a mirror, photograph, or video can lead to increased awareness of particular self-dimensions that are most relevant at that time. Dependent on the situational factors, the relevant dimensions may be related to task performance, attitudes, beliefs, or physical appearance. While any number of situations could potentially lead to heightened self-awareness, four situational factors have been demonstrated to consistently lead to increased self-directed attention, arousal, and increased pressure: (1) *competition*, (2) *reward and punishment*

contingency, (3) *ego relevance*, and (4) *the presence of an audience*.

Competition results from the comparison of one's skills or abilities to those of another. Competition may be either explicit or implicit in nature. *Explicit competition* refers to competitive situations when the performer is clearly aware that his or her performance will be compared to that of another. *Implicit competition* occurs when the performer compares his or her performance to that of another, even though no explicit competitive arrangements have been made—comparing oneself to the performance of a teammate during practice, for example. For both explicit and implicit competition, motivation and self-awareness are heightened when the outcome of the competition is made public or when one's performance is slightly below that of the competitor.

When rewards or punishments are made contingent upon performance, the saliency of the individual's task performance is increased. Often, these situational factors cause the performer to direct attention inward in order to attempt to ensure correct behaviors and responses. However, contingent rewards have been shown to interfere with attentional mechanisms underlying both cognitive and motor tasks. Similarly, contingent punishment often leads to decreased performance on cognitive tasks, likely resulting from shifts of attention to self-dimensions that are not directly relevant for task performance (e.g., thoughts of worry or failure).

Presenting a task as ego relevant can increase one's self-awareness while performing the task. To this extent, performance on an ego-relevant task is considered as a reflection upon a particular aspect of one's self. For example, diagnostic or evaluative tasks such as an IQ test would result in heightened self-awareness, as the individual is concerned with performing in a manner that reflects his or her belief about personal abilities. Ego-relevant tasks also result in heightened performance pressure and motivation as one attempts to perform in accordance with one's standards and beliefs.

Performance in front of an audience can also create heightened self-awareness. This self-directed attention is prompted by concerns for how the audience will view the performance and performer. In this regard, performance in front of an audience has the potential to define the self as either successful or unsuccessful. Furthermore, performance in front of an audience makes these

self-definitions into a publically confirmed reality. Consideration of the influence of outside observers has led to the distinction between public and private self-awareness. Public self-awareness refers to the awareness of oneself as imagined from an outside perspective, whereas a private self-awareness refers to the awareness of oneself from one's own perspective. Increased awareness of the public self has been found to result in increased conformity to societal norms and expectations, whereas heightened private self-awareness has been shown to result in behaviors more consistent with personal standards.

Self-Awareness and Choking Under Pressure

Increased self-awareness has been viewed as one of the main contributing factors to the breakdown of motor performance under pressure. Under situations of heightened performance pressure (e.g., performing in front of an audience), the performer's attention is shifted inward in order to monitor and control behavioral responses in an attempt to ensure correctness of execution (i.e., a shift from automatic to controlled processing). However, attempting to monitor and control the step-by-step components of movement execution leads to the disruption and breakdown of highly learned motor skills. Specifically, skills that have become largely automated, requiring little attentional resources and running largely outside of working memory, are *dechunked* into smaller submovements that must be run and activated separately. This dechunking leads to reduced movement efficiency and, ultimately, reduced performance.

The extent to which an individual chokes under pressure has been found to be moderated by the individual's predisposition to be self-attentive. Individuals with a habitually high level of self-awareness are naturally more cognizant of their own actions, thoughts, and feelings compared to those who are habitually low in self-attention. To this extent, research has demonstrated that those with high dispositional self-awareness are less likely to choke under heightened pressure. Specifically, individuals with high dispositional self-awareness are more accustomed to performing under heightened self-focus, and thus the impact of increased self-focus resulting from pressure is less severe compared to individuals with low self-awareness.

Self-Awareness Training

Based on findings from SAT, self-awareness training strategies have been developed in order to reduce the negative influence of performance pressure. Specifically, self-awareness training strategies work off the assumption that individuals who train under heightened levels of self-focus learn to adapt and acclimate to the increased levels of self-awareness brought about by pressure. For example, research has shown that performers who train while being videotaped were more likely to become accustomed to increased self-focus and did not suffer performance decrements under high pressure situations compared to those who had not been trained to adapt to self-awareness. Self-awareness training strategies seek to replicate the types of situational factors (e.g., punishment and reward contingencies, ego relevance, audience presence, and competition) found within the individual's relevant performance environment that lead to increased self-awareness.

Conclusion

Heightened self-awareness has been shown to lead to a number of behavioral responses, whether it be increased motivation, conformity, or awareness of bodily states, beliefs, or feelings. In this regard, SAT has motivated a substantial amount of research within social, cognitive, and SP, on topics such as choking under pressure, marketing and consumer behaviors, attribution, prosocial behavior, and self-assessment.

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See also Attentional Focus; Choking; Chunking/Dechunking

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SELF-CATEGORIZATION THEORY

As a conceptual extension of social identity theory, John Turner and his colleagues developed self-categorization theory. Self-categorization theory seeks to understand and explain the processes by which people form cognitive representations of themselves and others in relation to different social groups. The underlying premise behind this theory is that people place themselves and others into social categories on the basis of the underlying attributes that are particularly salient, and this process of social categorization shapes a range of attitudes, emotions, and behaviors.

As the precursor to self-categorization theory, social identity theory purported that people not only develop a sense of personal (i.e., individualized) identity through reliance upon factors that make them unique but also possess multiple social identities, based on their membership in social groups. When these social identities are made salient, individuals tend to favor persons who share membership in the applicable group (i.e., in-group members) over those from other social groups (i.e., out-group members). Although social identity theory included recognition of the fact that social identities will carry implications for both within- and between-group behavior, the predominant focus of this framework centered on *between-group* (i.e., intergroup) processes. To explicate the cognitive processes by which people categorize themselves and others, and define themselves in terms of membership *within* different social groups, Turner and his colleagues developed self-categorization theory. Self-categorization theory focuses to a much greater extent on within-group (i.e., intragroup) processes than social identity theory. That said, self-categorization theory and social identity theory do share considerable conceptual overlap, so much so that both theories are often collectively referred to as the *social identity approach*.

Core Tenets

Depersonalization; Conceptions of Fit

One of the core tenets of self-categorization theory corresponds to the extent to which *depersonalization* takes place. By depersonalization, Turner and colleagues referred to the process whereby people categorize the self in terms of a particular

social group (e.g., being a member of a given sports team) rather than as an individual. When this happens, people see themselves less in terms of their unique characteristics and more in terms of the social groups to which they believe that they belong (this involves a shift from “I” to “we”). It is important to note that self-categorization theory is not restricted to self-perceptions as the categorization of others is also possible. When categorizing others, the perceiver interprets an individual as a member of a larger social category rather than a unique, or individualized, entity. Self-categorization and other-categorization occurs through two primary mechanisms, namely, *perceived fit* and *perceiver readiness*.

Perceived fit corresponds to the extent to which a particular categorization aligns with how the perceiver understands and interprets his or her social world. Turner and colleagues differentiated between comparative and normative components of perceived fit. *Comparative fit* corresponds to the potential differences that exist within one particular social category relative to differences that might exist across different categories. For example, if a number of male and female athletes are at a party along with those (both male and female) who self-identify as being inactive or sedentary, those athletes may be more likely to identify with the social categorization of “athlete” than other categorizations that might exist, such as “gender.” Alternatively, if the differences between males and females at that same party are perceived to be more pronounced than the differences between athletes and nonathletes, people at that party may be more likely to self-categorize, and hold as being salient, identities related to being male or female. Turner referred to this as a process of *meta-contrast*.

Normative fit, in contrast, corresponds to the extent to which the behaviors displayed by people in a given social category align with what one would expect (i.e., what is stereotypical) of people in that social category. For example, if some of the athletes at the social gathering just described engage in behaviors that would be inconsistent with what one might expect of “an athlete” (e.g., start smoking), this may inhibit the extent to which the category athlete is salient. Conversely, if people do behave in ways that align with the expectations of that social category, and one believes himself or herself to engage in similar behaviors to those displayed by group members, this can facilitate greater alignment with that social group.

Perceived Readiness

Beyond personal conceptions of fit, a secondary mechanism by which people engage in social categorization corresponds to the *perceived readiness* (also referred to as *accessibility*) of social categories. Although certain categories (e.g., age, ethnicity, and gender) tend to be very identifiable and thus accessible in most people, the perceived readiness of categories also varies between people. Specifically, this perceived readiness is heavily dependent on previous experiences, expectations, and a person’s current goals or objectives. As an example, if a person has been around athletics all of her life, as an adult she may be more likely to categorize people on the basis of activity status and engage in various behavioral processes that align with the identity of “physically active adult” (e.g., seek out and interact with other adults who are active). In a similar manner, if this woman has accrued more positive experiences in the past from being physically active in the company of other women, rather than from exercising in mixed-gender contexts, she might be much more likely to seek out similar same-gendered-exercise contexts in the future, and make salient her social identity as a “physically active woman.”

Crossed Categories

The previously given example highlights an additional complexity of social categorization. Under certain circumstances, categories can become *crossed* such that an emergent subcategory develops. In the case of the woman just described, the categories of *physically active people* and *women* have been crossed to result in the creation of a new salient category, namely that of physically active women. The use of crossed categories offers some promise for mitigating intragroup hostilities within sport and exercise settings. As an example, if an ethnic minority baseball player is introduced into a baseball team composed of members from the ethnic majority, the fact that this minority member shares category membership with his teammates along at least one dimension (i.e., all are baseball players) may contribute to his integration within the team (assuming that this shared membership is recognized). Cross categories, however, also suggest the possibility of contributing to further stereotyping and discrimination by allowing for the successes of subcategory members to be discarded or downplayed (such as when one

is perceived as playing well *for a girl*) as well as tokenism (whereby a single cross categorized group member is used as evidence, or a token, for a lack of prejudice within a sports club or organization).

Why Do We Engage in Social Categorization?

We are predisposed to engage in processes of social categorization primarily for two reasons. First, engaging in social categorization reduces the cognitive demands placed upon the perceiver. It is relatively taxing to interpret each person encountered as an individual rather than a member of one or more social categories. Such individuation does happen. It, however, requires one to be motivated and able (i.e., not cognitively depleted) to engage in such effortful processing. A second reason that we engage in categorization is to try to reduce uncertainty regarding our place in the social world. Specifically, through reliance upon social categorization, a substantial amount of information can be ascertained about others, which in turn can inform our expectations (i.e., reduce uncertainty) for any future interactions.

A principle common to social identity theory and self-categorization theory is that individuals interpret the social groups they are a part of positively, and they do so in order to promote a sense of self-worth. In a related manner, individuals are generally attracted to others with whom they share membership in a given category and repelled by those with whom they do not share category membership. There is a growing body of evidence supporting the notion that the extent to which people self-identify as being similar to, or different from, others within physical activity (PA) contexts (i.e., groups), on the basis of various social categories, influences their attraction to, and level of involvement within, that group. For example, when people perceive themselves to be similar to others within a particular PA group, in terms of age, they report higher levels of adherence to the group. Consistent with this relationship between age similarity and program involvement, across the adult age spectrum people report a relative dislike of being physically active in groups where the other group members are either much older or younger than themselves (i.e., other members are “unlike me”) and a positive preference for exercising in groups comprised of others their own age (i.e., other members are “like me”). Taken together, these findings suggest that the categorization of

age is particularly salient within PA groups, as it not only corresponds to their general exercise preferences but also predicts their PA adherence behaviors within group settings.

As an extension of these studies, there is also evidence to suggest that, among people who like to exercise within group settings, there exists a general preference for exercising within gender-segregated (i.e., same-gender) groups rather than within gender-integrated (i.e., mixed-gender) groups. This preference is particularly pronounced among people who are overweight (i.e., body mass index ≥ 25). From the perspective of self-categorization theory, the results of this research suggest that within mixed-gender classes the social category of weight status becomes particularly heightened such that comparative fit is notably poor for overweight individuals. When offered the opportunity to exercise within same-gender classes, however, weight status becomes a less debilitating and pronounced personal categorization. People can instead benefit from the unity that appears to develop when they self-categorize primarily on the basis of gender.

Conclusion

Self-categorization theory specifies the process by which the self and others are placed into broader social categories. Such categorization is sensitive to both the immediate context (e.g., comparative fit) and personal history (e.g., perceived readiness). It has recently been applied to exercise group settings through examination of the relation between the demographic composition of the group and program involvement. It should be noted, however, that although it is likely equally applicable to sport teams, it has rarely been considered within this context.

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See also Cohesion; Hierarchical Self; Identity; Social Identity Theory

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SELF-COMPASSION

The term *self-compassion* refers to a healthy self-attitude in which one acts in a compassionate way toward oneself, similar to having compassion for others. The term has its origins in Buddhist philosophy, but is a relatively new concept to Western psychology and research. Most self-compassion research to date has been in the general psychology literature, with only a few studies specific to the context of sport and exercise psychology. Despite limited research in sport and exercise, self-compassion has begun to show much potential as a way for athletes and exercisers to manage difficult emotional experiences, particularly experiences related to evaluation and failure, in effective and healthy ways.

Components of Self-Compassion

Being self-compassionate involves taking a non-judgmental attitude toward one's perceived

imperfections, limitations, and failures. Kristin Neff described self-compassion as having three components, including (1) self-kindness, (2) common humanity, and (3) mindfulness. Self-kindness entails offering oneself warmth and nonjudgmental understanding, rather than harsh self-criticism, when confronted with suffering, inadequacy, or failure. Common humanity includes recognizing that being imperfect, making mistakes, and encountering life difficulties are part of the shared human experience. Mindfulness represents a balance between rumination and thought suppression in which painful feelings are neither suppressed nor exaggerated.

Distinguishing Self-Compassion and Self-Esteem

While self-esteem and self-compassion share many of the same potential benefits (e.g., less anxiety and depression), the main distinction is that self-compassion is not based on positive self-evaluations or on favorable comparisons with others. Instead, self-compassionate people recognize that being limited and imperfect is part of the human experience. In addition, the development of self-compassion might help avoid some of the dangers of inflated self-esteem, such as arrogance and narcissism.

Exercise and Sport Settings

Although there are numerous psychological and physical benefits to exercise, the exercise setting can also present many unique challenges related to social evaluations and comparisons that can influence women's exercise behaviors. Hence, self-compassion research in the exercise domain has focused primarily on women's experiences and has shown self-compassion to be related to a variety of motives to exercise and exercise-related outcomes. More specifically, self-compassion is positively related to intrinsic motivations to exercise (e.g., exercising for fun and enjoyment) and negatively related to extrinsic forms of motivation to exercise (e.g., exercising because of pressure from other people; or feeling like a failure for not exercising for a while). In addition, self-compassion is negatively related to various exercise-related outcomes that rely on self-evaluative processes, including ego goal orientation (i.e., striving for exercise goals that depend on an evaluation of the self in relation to others, such as proving that one

is better at exercise than others), social physique anxiety (SPA) (i.e., concerns over other people's negative evaluations of one's physique, figure, and body), and obligatory exercise (i.e., feeling obligated to exercise, even if that exercise results in harm and diminishes well-being). Interviews with women exercisers have also revealed that there might be a self-compassion specific to the body, requiring an appreciation of the uniqueness of one's body, taking ownership and care of one's body, and engaging in less social comparison. Women exercisers have also spoken about the role that other people (e.g., family, friends, and training partners) can play in helping them develop body self-compassion, particularly those who are nonjudgmental and accepting.

Similar to exercise, sport settings can present many evaluative experiences for which self-compassion might be a useful personal resource. Research with undergraduate students shows that self-compassion is an important predictor of anticipated thoughts, emotions, and behavioral reactions when people are asked to respond to a hypothetical scenario in which they are responsible for losing an athletic competition for their team. For example, those with more self-compassion are more likely to have thoughts like "everybody goofs up now and then" and react more calmly to being responsible for losing compared to people with less self-compassion. In addition, more self-compassionate people are less likely to experience negative feelings and less likely to have thoughts like "I am such a loser" when responsible for losing a game for their team.

Research with young women athletes has also supported self-compassion as a potential resource that acts as a buffer against negative emotions, thoughts, and behaviors that rely on self-evaluation processes. More specifically, athletes high in self-compassion are less likely to experience a tendency toward shame yet are also able to experience the shame-free component of guilt. This is important because shame is a particularly devastating self-conscious emotion that arises from a negative evaluation of the entire self, whereas shame-free guilt allows one to engage in prosocial and reparative behaviors when one's actions have not lived up to a personal standard (e.g., making excuses for missing team training sessions). Among women athletes, high self-compassion is also related to lower levels of SPA, as well as less body surveillance (e.g., thinking about one's looks

many times per day), body shame (e.g., feeling ashamed about not having one's desired body), fear of failure, and fear of negative evaluation.

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See also Self-Acceptance; Self-Conscious Emotions; Self-Criticism; Self-Doubt

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SELF-CONSCIOUS EMOTIONS

Shame, guilt, pride, and embarrassment are considered self-conscious emotions that are evoked by self-reflection and self-evaluation. They are founded in social relationships whereby people interact, evaluate, and judge themselves and others. As such, it not surprising that self-conscious emotional experiences are prevalent in sport and exercise settings. In fact, it has been argued that self-conscious emotions are central to motivating and regulating most of people's thoughts, feelings, and behaviors.

Self-Conscious Emotions Versus Basic Emotions

Basic or primary emotions (e.g., happiness, sadness, anger, fear, disgust, surprise) have been studied more frequently in sport and exercise psychology (SEP) research compared to self-conscious emotions. This imbalance of research focus may be due to the difficulty of assessing self-conscious emotions, the complexity of understanding these emotions, or the limited understanding of the distinction between basic and self-conscious emotions. The primary distinguishing feature of self-conscious emotions from basic emotions is that their elicitation requires self-awareness and self-representations. A second distinctive feature of self-conscious emotions is that they emerge later in childhood than basic emotions since they require the formation of stable representations and an awareness of socially appropriate standards that are used for evaluation. A third distinction between basic and self-conscious emotions is that the former is believed to promote survival goals (e.g., fear may cause an individual to avoid a large, angry opponent after a competitive match), whereas self-conscious emotions are thought to promote behaviors that increase the stability of social hierarchies and affirm status roles (e.g., team sport cooperation, dominance). Additionally, self-conscious emotions guide behaviors that are socially valued and acceptable and foster avoidance of behaviors that may lead to disapproval. A fourth distinguishing feature of self-conscious emotions is that they do not have unique, universally recognized facial expressions. Nonetheless, a growing body of evidence suggests that a few of the self-conscious emotions can be identified when specific body postures and head movements accompany the facial features of each expression (e.g., one look at the opponent's downcast, blushing facial expression and slouched body posture after missing the open net would likely portray embarrassment). Finally, self-conscious emotions are more cognitively complex compared with basic emotions. Although basic emotions involve relatively simple appraisals, the elicitation of self-conscious emotions requires individuals to form stable representations about who they are and to direct attention toward those representations. These capacities allow individuals to engage in various complex self-evaluative processes that elicit self-conscious emotions.

Given that the sport environment tends to exemplify conditions for the experience of self-conscious

emotions (i.e., socially constructed, evaluative, public forum where the body and performances are on display and may be judged by oneself and others), it is likely that self-conscious emotions of shame, guilt, embarrassment, and pride are tied to sport participation experiences that foster sustainability or dropping out in sport.

Shame

Shame is an acutely painful emotion that individuals experience when they fail to meet internalized social standards. Shame implies the perceived or feared loss of social status and a failure to live up to one's own standards of excellence, with a focus on deeply rooted global causes (e.g., I am not good at sports). Thus, shame involves negative feelings about the *self*. The phenomenological experiences that accompany shame include a sense of worthlessness, humility, and shrinking, and the desire to hide, or disappear. Generalized forms of this highly painful state are consistently linked to a host of maladaptive psychological, physical, and behavioral outcomes including increased depressive symptoms, anxiety, anger, cortisol dysregulation, cardiovascular reactivity, substance abuse, self-injurious behavior, decreased self-esteem, and adaptive personality characteristics like agreeableness and conscientiousness. Researchers targeting body-related shame also report unique correlates compared with generalized forms of shame including increased body monitoring, weight rumination, body concern, body image importance, internalization of the thin ideal, social physique anxiety (SPA) and also decreased global self-worth, physical self-perceptions, body self-acceptance, self-determined regulations, and physical activity (PA).

Guilt

Guilt is a negative emotion that involves a sense of tension, remorse, and regret over a "bad thing" done with a focus on a specific *behavior* that caused the experience (e.g., I didn't practice enough and missed the winning goal). Feelings of guilt are also frequently accompanied by a preoccupation with the transgression, rumination about the behavior, and thoughts of changing the current situation. As such, guilt motivates an act to repair the transgression instead of avoiding the situation in an attempt to protect the self. For example, an individual who feels guilty about eating junk food

might engage in PA in an attempt to alleviate the transgression of eating poorly.

There is some question in the literature as to whether guilt is an adaptive or maladaptive emotion. The traditional view is that guilt plays a significant role in psychological and physical pathology. Frequent references are made to the maladaptive nature of guilt characterized by chronic self-blame and obsessive rumination over the transgressions. Feelings of guilt are also consistently and positively linked to depressive symptoms, anxiety, and low self-esteem. Body-related guilt is further associated with increased body monitoring, weight rumination, body concern, body image importance, internalization of the thin ideal, SPA, negative affect and decreased self-esteem, physical self-perceptions, positive affect, and body self-acceptance. However, more recent research has stressed the adaptive functions of guilt, particularly for prosocial behaviors such as motivation and PA. For example, an athlete who feels guilty for missing a practice may do extra training on his own time to make up for the lost practice.

While often confounded conceptually and statistically, recognizing that shame and guilt are distinct emotions may help clarify issues regarding the adaptability of guilt. Such a distinction is necessary given that shame and guilt emotional experiences often overlap, and the co-occurrence of guilt and shame may mask the adaptive functions. That is, an individual may start with off with a guilt experience (e.g., I can't believe I didn't exercise today) but then intensify and generalize the failure to the self (e.g., . . . I'm such a bad person). This sequence of appraisals may lead to a loss of the advantages associated with guilt. In cases like this, not only is the exerciser presented with regret over a specific behavior that needs to be fixed but she is also burdened with feelings of disgrace for being an inadequate person. Thus, it is guilt accompanied with shame that most likely leads to the more negative outcomes (obsessive rumination and self-blame) often associated with guilt experiences. One way to address this issue in SEP is to use measures that appropriately distinguish between shame and guilt and/or use statistical corrections to develop guilt-free shame and shame-free guilt scores.

Embarrassment

Embarrassment is a negative emotion that requires attributions to internal causes yet the experience

of this emotion does not depend on cognitive appraisals (i.e., stability, globality, and controllability). That is, embarrassment can occur from events caused by stable, uncontrollable, and global aspects (e.g., being publicly exposed as an incompetent athlete) or by events caused by unstable, controllable, and specific attributions (e.g., tripping during a soccer game in front of spectators). Moreover, embarrassment occurs only when focus is placed on the *public self* (the self as perceived by real or imagined others) and activates corresponding public self-representations.

The phenomenological experiences that accompany embarrassment include a sense of blushing and wanting to hide. Embarrassment is also associated with several physiological reactions including steady increases in both systolic and diastolic blood pressure (BP), heart rate (HR), cortisol, and reddening of the face. When embarrassment is proportional to the eliciting event, it is a productive emotion that stimulates appeasing behavior, elicits favorable evaluations from others, and usually resolves social situations. On the contrary, excessive or unnecessary feelings of embarrassment over extended periods of time can lead to maladaptive behaviors including timidity, passivity, and antisociality.

Pride

One of the least studied self-conscious emotions in SEP is pride, which is a positive emotion that results from an individual engaging in valued behaviors or presenting with positive characteristics (e.g., exercising, being physically fit, participating in sport). Two facets of pride have been consistently identified in the literature: *hubristic* (or alpha) and *authentic* (or beta) pride. Hubristic pride is experienced as uncontrollable and global aspects of the self (e.g., I *have* a great body) typically involving feelings of personal grandiosity and superiority to others, whereas authentic pride is focused on specific, controllable achievements and behaviors (e.g., I *finished* the marathon I *trained* for). Thus, hubristic pride is more closely linked to stable attributions, abilities, and global positive traits such as inherent muscularity or athletic skill, whereas authentic pride is more closely linked to attributions of effort and specific goal attainment and accomplishments like successfully losing weight after following an exercise program, or performing well at an athletic event. Furthermore,

hubristic pride is associated with phenomenological descriptions such as arrogant, snobbish, egotistical, and conceited, whereas authentic pride is associated with phenomenological ratings of being accomplished, achieving, confident, and successful. Thus, authentic pride is related to adaptive, prosocial, and achievement-motivated personality profiles (e.g., high self-esteem, who are extraverted, agreeable, and conscientious) compared to the maladaptive personality profiles (e.g., narcissistic, self-aggrandizing, disagreeable, and nonconscientious) related to hubristic pride.

Although hubristic pride is highly positive and emotionally rewarding, it is difficult to sustain since it is focused on the self and not a specific behavior. Furthermore, those high in hubristic pride may have problems with interpersonal relations, since placing focus on themselves is likely to interfere with the needs and goals of other people. This may be problematic in team sports where collective goals are necessary for long-term success. In sport and exercise contexts, authentic pride is associated with feelings of achievement and motivation to engage in goal-directed behaviors such as PA and the accomplishment of sport-specific goals.

Sex Differences

Researchers have found that there are sex (and likely gender) differences in both the establishment of standards, rules, and goals and the evaluation process that is required for the elicitation self-conscious emotions. Women are more likely than men to make a global appraisal following failure, which leads to the elicitation of shame. Contrarily, men tend to endorse hubristic pride more than women do. Equivocal responses for men and women have been reported for authentic pride, guilt, and embarrassment. Differences in appraisals are most likely due to early socialization (in the classroom, home, on the field or court), yet additional research is needed to explore the reasons fostering the sex differences in the experience of self-conscious emotions and to explore such differences in various sport and PA participants.

Cultural Differences

Culture has an impact on the way individuals conceptualize the self. Individuals from collectivistic cultures (e.g., East Asian cultures) typically conceptualize the self as being interdependent with others in a larger social context. On the other hand,

individuals from individualistic cultures (Western cultures) typically view the self as separate from others. Researchers indicate that self-conscious emotions that focus on the public self and others, such as shame and embarrassment, are more commonly experienced in individuals from collectivistic cultures. Such emotions tend to be valued and even viewed as an appropriate emotion in collectivist cultures because they reaffirm the individual's place and sense of belonging within the social group. However, since individuals do not view themselves as separate from their relationships with others, shame and guilt may be less distinct in collectivistic cultures. In contrast, those in individualistic cultures more commonly experience authentic pride. To date, cultural samples have been primarily restricted to North America and East Asia with a focus on generalized shame, guilt, and pride, limiting cross-cultural comparisons across a wide range of self-conscious emotions and within various sport and PA contexts.

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See also Basic Emotions in Sport; Self-Compassion; Social Comparative Emotions; Social Physique Anxiety

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SELF-CONSTRUAL

Construals are the way in which individuals perceive, understand, and interpret their worlds.

When these construals are focused on the perceptions of the self rather than the social environment, they are defined as self-construals.

Distinct Self-Construals

Self-construals have been distinguished as independent (also referred to as personal) and interdependent (also referred to as social). Independent self-construal involves a focus on the individual as being separate from others and on the individual's unique traits, abilities, preferences, interests, goals, and experiences. Independent self-construals serve to maintain autonomy, and the focus is on the self. Furthermore, the motivation of independent self-construal is in distinguishing oneself from others, whereas the social self-construal involves an inclusion or integration mind-set. In this way, social comparisons may be activated and emphasize differentiation or differences (in the case of independents) or assimilation and similarity (for interdependents). Interdependent self-construal serves to focus on the social environment, group memberships, and interpersonal relationships. Individuals holding interdependent self-construals think and behave in ways that emphasize their connectedness to others. Their sense of self is dependent on the stability, importance, and usefulness of their relationships compared to individuals with independent self-construals whose sense of self is dependent on the extent to which they experience a sense of agency.

Interdependent self-construal has been further differentiated as relational self-construal, whereby individuals predominantly define themselves by their roles in social relationships (e.g., being a member of their sport team), and collective self-construal, in which individuals define themselves within a broader collective (e.g., being a basketball player).

While originally conceptualized as a dichotomous self-representation, researchers have identified that individuals tend to have both independent and interdependent self-construals, with one dominating more than the other.

Gender and Cultural Differences in Self-Construal

Women are more likely to develop interdependent self-construals compared with males. However, gender-related differences in self-construal and related cognition, affect, and motivation may become less pronounced with the changing

dynamics around roles and responsibilities, opportunities, power, and competition.

Most of the studies on self-construal have been focused on cultural differences. Broadly, collectivist cultures such as those endorsed in Asian countries tend to be more interdependent compared with those of European and North American countries. Independent selves value being unique, promoting their own goals, and overtly expressing themselves compared to interdependent selves who highlight the importance of belonging, connectedness, and promoting others' goals. This being said, interdependent selves are focused on the goals and needs of others with whom they relate or desire involvement with (i.e., "in-group" members). Based on these more general cultural differences in self-construals, researchers are beginning to examine the role of acculturation and multiple cultural identities in the formation and maintenance of independent and interdependent self-construals.

Cognitive, Affective, and Behavioral Outcomes of Self-Construal

An individual's type or level of self-construal affects his or her thoughts and perceptions, sense of self, affective experiences, and motivation. Interdependents are more sensitive and attentive to others and have heightened cognitive representations of the self in relation to others. They are also less likely to display several of the basic emotions (i.e., joy and anger) and generally report fewer episodes of positive affect compared with independents.

It has been suggested that individuals who endorse interdependent self-construals are less likely to achieve personal health goals such as weight loss or reducing sedentary time since their personal goals are not always in line with, or perceived as important as, the goals of the collective. This said, it is also suggested that interdependents may be less likely to be impulsive and more likely to engage in behaviors considered socially desirable or aligning with social norms. Independents are more likely to engage in behavior that aligns with their personal goals or emotions.

The study of self-construal has been limited in sport and exercise contexts, in spite of the obvious links to pertinent topics such as physical self-concept, group cohesion, collective self-efficacy, exercise and sport motivation, and sport type. Some studies of self-construal use sport-specific analogies to make the independent (e.g., individual

sport such as boxing) versus interdependent (e.g., team sport like football) distinctions. In one study, there was some evidence that adolescent athletes' preferences for their sport teams were consistent with their preferences for family and friends. Also, some qualitative studies focused on understanding sport practices and preferences have alluded to self-construals as defining cultural distinctions.

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See also Cohesion; Identity; Self-Categorization Theory; Social Comparison; Social Identity Theory

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SELF-CRITICISM

Contemporary definitions of *self-criticism* are typically characterized by negative evaluations of the self, often as a result of failing to meet expectations and standards. Individuals who are self-critical typically demonstrate greater depressive symptoms. However, historically self-criticism was not attached to a notion of vice and negative functioning but rather conceptualized as adaptive and essential to truly knowing oneself. Therefore, the path through history can view self-criticism as both good and bad. Modern views of self-criticism in sport and exercise psychology (SEP) tend to see it as maladaptive.

Goal Pursuit in Sport

There is evidence that self-criticism is negatively associated with goal pursuit among athletes. It

has been proposed that there tends to be hypersensitivity toward potential judgment and criticism for those high in self-criticism; therefore, for self-critics the focus in a performance-related setting becomes one to avoid failure and loss of approval from others, both of which detract from goal pursuit. Because of a tendency to focus on both external and internal pressures in the pursuit of goals, those high in self-criticism might be less motivated by intrinsic regulations (e.g., for fun and enjoyment) than by extrinsic regulations (e.g., to please others or to avoid feelings of guilt). In addition, because of harsh self-criticism, the response to performance failures and not meeting expected performance standards might be associated with more negative emotions for self-critics compared to what is experienced by those who engage in less self-criticism.

Perfectionism in Sport

Based on the link between self-criticism and goal pursuit, it is likely no surprise that self-criticism is often linked to perfectionism. As evidence of its importance to the conceptualization of perfectionism in sport specifically, self-criticism has been identified as a dimension on the Sport Perfectionism Inventory (SPI), which is operationalized by items such as "If I win the competition, I still tend to criticize myself" and "Even when I perform well I think about something I could have done better." While perfectionism can lead to a healthy pursuit of excellence, debilitating forms of perfectionism (e.g., self-oriented perfectionism) are often characterized by the compulsive pursuit of extremely high personal standards accompanied by harsh self-criticism. When self-criticism is associated with perfectionism there tends to be excessive rumination and a focus on personal and interpersonal inadequacies, resulting in outcomes such as social physique anxiety (SPA), anger, precompetitive anxiety, and debilitating performance anxiety among athletes. Furthermore, it has been proposed that when athletes who experience self-oriented perfectionism engage in harsh self-criticism in response to failing to meet excessively high self-set standards, the pattern of self-criticism can become chronic and contribute to burnout in sport.

In addition, there is evidence to suggest that perfectionism and high levels of self-criticism are associated with heightened eating psychopathology in athletes who strive for sporting perfection. The transdiagnostic cognitive behavioral model of

eating disorders provides a framework for understanding how eating disorders may arise. From a theoretical point of view, athletes who experience more conflict in relationships with an influential parent or coach are often more likely to criticize themselves, which in turn results in low self-esteem and increased depressive symptoms. Both low self-esteem and increased depressive symptoms have been linked to disturbed eating behaviors. Furthermore, self-critical perfectionism has been shown to play a more pivotal role in eating psychopathology than standard personal perfectionism. Thus, an athlete who possesses high self-imposed standards might not necessarily experience eating psychopathology; rather, what may be maladaptive is how the athlete evaluates these standards when they are not met. Overly self-critical evaluation thus substantially accounts for the relationship between perfectionism and eating disorder, leading to an overevaluation of one's eating and shape and weight control.

Managing Self-Criticism

The development of self-compassion, which involves treating oneself with the same type of compassion that we extend to others, might be a particularly useful strategy to help people more effectively manage self-criticism. One aspect of self-compassion—self-kindness—specifically requires the avoidance of harsh judgment and self-criticism. Highly self-compassionate individuals might have more accurate self-evaluations than those low in self-compassion because their self-judgments are less tainted by self-criticism. In addition, researchers focused on body image have shown that self-compassion is uniquely, and negatively, related to eating guilt (e.g., feeling bad for eating unhealthy foods) and that women who are self-critical are more likely to experience body preoccupation (e.g., worry about specific body parts).

Paul Gilbert and his colleagues developed compassionate mind training, which is specifically aimed at fostering self-compassion for people who experience high shame and self-criticism. Both shame and self-criticism rely heavily on self-evaluative processes, including self-directed hostility, contempt, self-loathing, and a lack of self-directed warmth and reassurance. Compassionate mind training focuses on changing someone's historical pattern of responding to setbacks and failures with a self-attacking style to a new response of care and compassion. Compassionate mind

training focuses on learning to be compassionate to the self through activities such as compassion-focused imagery and compassionate letter writing. It has been shown to have potential to reduce depression, anxiety, self-criticism, and shame with clinical samples; however, the efficacy of compassionate mind training for sport and exercise samples has yet to be established.

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See also Eating Disorders; Perfectionism; Self-Compassion; Self-Conscious Emotions

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SELF-DETERMINATION THEORY

Self-determination theory (SDT) is a metatheory of human motivation and personality that addresses autonomous behaviors and the conditions and processes that support such high-quality forms of volitional engagement. In the past decade, there has been a burgeoning growth of basic, experimental, and applied research within sport and exercise settings designed to test the theoretical tenets advanced within SDT. Within this entry, the central components and motivational phenomena proposed within SDT are outlined.

Theoretical Framework

SDT starts with the assumption that humans are growth-oriented organisms who actively seek optimal challenges to actualize and new learning experiences to master. This inherent tendency for growth is manifested within SDT through the concept of *intrinsic motivation*, a construct that is held to be inseparably intertwined with the notion of active and spontaneous activity. Intrinsic motivation is defined as partaking in activities for the enjoyment and interest that is inherent within the activity itself (i.e., doing the activity is the reward in and of itself). Yet, SDT also stresses that individuals do not automatically experience growth and healthy human functioning. Rather, social environments serve to shape the way that people act, think, and develop. In this regard, SDT specifies three basic and universal psychological needs that provide the essential nutrients for people to experience growth, development, and well-being. Specifically, these needs are (1) *the need for autonomy* (i.e., the need to experience activities as self-endorsed and choicefully enacted), (2) *the need for competence* (i.e., the need to interact effectively within the environment), and (3) *the need for relatedness* (i.e., the need to feel close, connected, and cared for with important others). Within SDT it is held that social environments that offer supports for these basic psychological needs provide the basis for people to engage more volitionally in activities and experience greater effective functioning and enhanced psychological wellness.

Based on the seminal work of Edward L. Deci and Richard M. Ryan, SDT has evolved through a comprehensive and systematic program of inductive research spanning the past five decades. In its current constellation, the overall SDT framework is represented by five interrelated mini-theories, namely *cognitive evaluation theory* (CET), *organismic integration theory* (OIT), *causality orientations theory* (COT), *basic psychological needs theory* (BPNT), and *goal contents theory* (GCT). Each of the separate SDT mini-theories was developed via lab and field research to explain and empirically test a set of specific motivational phenomena.

Five Mini-Theories

The first of the five SDT mini-theories to be developed was labeled CET. This mini-theory provides a theoretical perspective and organizing structure to identify and synthesize empirical findings pertaining

to how various social factors such as rewards, provision of choice, optimal challenge, feedback, and deadlines affect a person's intrinsic motivation. Within CET, it is posited that (a) events that satisfy the basic psychological needs for competence and autonomy will enhance a person's level of intrinsic motivation and (b) social factors that are not supportive or undermining of competence and/or autonomy (i.e., are deemed controlling) undermine or frustrate an individuals' level of intrinsic motivation.

A central tenet within SDT is that intentional action can be intrinsically driven (i.e., for the inherent interests of an activity) or extrinsically driven (i.e., for a separable consequence). As opposed to pitting extrinsic motivation against intrinsic motivation, SDT conceptualizes extrinsic motivation as multidimensional. Such reasoning provided the basis for the development of a second mini-theory, termed OIT. Within OIT, a continuum of internalization is specified with four distinct types of extrinsic motivation being conjectured. These motivations, which can be experienced simultaneously, are anchored between amotivation (viz., a state of lacking intention to act that can arise due to a lack of competence, a belief that the activity is not important, or a lack of contingency between the behavior and desired outcomes) and intrinsic motivation. These motivational styles encompass regulations characterized by external contingencies (i.e., reflected by a high level of control) through to forms that reflect self-endorsement and personal value (i.e., autonomous forms of extrinsic motivation). Specifically, and from most to least autonomous, the extrinsic forms of motivation described within OIT are termed *integrated regulation* (i.e., when a person partakes in an activity or behavior as it reflects their identity and is akin within their other values, goals, and needs), *identified regulation* (i.e., when an individual volitionally engages in an activity or behavior because they identify and value the purpose and benefit accrued from taking part), *introjected regulation* (i.e., when behavior is underpinned and directed by intrapersonal sanctions such as shame, guilt, and pride), and *external regulation* (i.e., when behaviors are controlled by external contingencies such as tangible rewards and punishments).

As a mini-theory, OIT provides a theoretical lens through which to conceptualize, understand, and define motivation from a quality perspective. That is, within OIT it is proposed that there are

manifold benefits of acting through more autonomous forms of motivation (e.g., enhanced well-being and health, better performance, improved learning, and better activity experiences). The continuum within OIT is not a developmental structure but rather an organizational representation of the motivational regulations. This means that an individual can adopt a regulation at any stage of the continuum depending on the social context. To this end, OIT stresses that contextual supports for the basic needs for autonomy, competence, and relatedness are conducive to supporting internalization and integration.

In terms of social context, within SDT it is hypothesized that autonomy-supportive environments (viz., interpersonal contexts that support choice, initiation, and understanding, while minimizing the need to perform and act in a prescribed manner) facilitate autonomous engagement, better internalization and integration, effective functioning, and enhanced psychological well-being (PWB). The manifold benefits offered to individuals of interacting with autonomy-supportive others (often an authority figure such as a teacher, a manager, or a coach) is that such environments provide necessary supports for the satisfaction of people's basic psychological needs. People also experience increased well-being from providing autonomy support to others as well as receiving (i.e., mutual autonomy support). A number of other social contexts have also been examined within the extant literature in terms of supporting the basic needs for autonomy, competence, and relatedness (e.g., structure, involvement, cooperation, and task-involving climates).

The third mini-theory within SDT is labeled COT and describes variations in tendencies of how people orient toward environments and regulate their behavior. It is held within COT that everyone, to some extent, varies in three orientations: *autonomy orientation* (an orientation toward intrinsic motivation and well-integrated extrinsic motivation), *controlled orientation* (an orientation toward being motivated by reward contingencies, constraints, and directives), and *impersonal orientation* (i.e., a tendency for people to act without intentionality and consider themselves as incompetent). Causality orientations operate at a global level (or life domain) and reflect stable and consistent patterns of thinking. Within COT, holding an autonomous orientation is hypothesized to positively predict effective

functioning, adaptive behavior, and psychological health. Experimental manipulations have also shown that causality orientations can be primed to yield effects parallel to the relations found in research that has been conducted in field settings using self-report measures.

The fourth mini-theory within SDT is termed BPNT. BPNT is built on the assumption that three basic psychological needs are essential nutriments for ongoing wellness and optimal functioning. As referred to previously, the needs specified within BPNT are for *autonomy*, *competence*, and *relatedness*. As basic needs, these nutriments to wellness are posited to be universal as opposed to being learned or accrued via value systems (i.e., they have a functional impact irrespective of culture, gender, developmental stages). At the crux of BPNT is the assumption that a person's development, growth, integrity, and wellness are supported to the extent that these basic needs are afforded by the social context. In a similar vein, and as essential nutriments, BPNT stresses that in the event of these needs being frustrated ill-being, alienated functioning, and passivity will follow.

The fifth mini-theory within the broader SDT framework is GCT. This mini-theory is concerned with explaining why the different goals that individuals pursue differentially predict a person's motivation and well-being. The distinction used to categorize and compare goals varying in foci within GCT is that of *intrinsic* and *extrinsic*. Intrinsic goals are those with internal foci such as personal growth, affiliation, and community contribution. In view of their focus on developing one's personal interests, values, and potentials, intrinsic goals are directly supportive of a person's basic psychological need satisfaction. Extrinsic goals are those with an outward foci such as financial success, image, and popularity. Within GCT, extrinsic goals are held to be less supportive, or even undermining, of basic psychological need satisfaction and are therefore considered to hinder optimal human development (i.e., they are associated with greater ill-being and reduced wellness). Research has also started to examine goal contents within specific contexts such as exercise wherein goals for health management, skill development and social affiliation are classified as intrinsic whereas exercise goals for image and social recognition are categorized as extrinsic. Central to GCT, at both global and contextual levels, is that intrinsic goals are positively

associated with enhanced well-being as a result of such goal pursuits being consistent with the satisfaction of a person's basic psychological need satisfaction.

Conclusion

The five mini-theories described in this entry collectively constitute the broader SDT framework. Research within sport and exercise has been central to developments and applications pertinent to SDT, since early work on CET through to recent applications of GCT. Within the contexts of sport and exercise, research has focused on the differential effects of differing types of motivation and the important implications that result from acting through these reasons for a range of outcome variables such as well-being, behavioral persistence, effortful engagement, and performance. A similar line of inquiry has recently emerged with regards to exercise-based goal contents. Work has also been conducted to better understand the social contexts and goal inducements that facilitate versus undermine autonomous engagement, detailing how communications, feedback, rewards, and relational supports contribute to or derail volition and wellness in these domains. This work has become increasingly couched within BPNT as researchers attempt to better delineate the supports (and thwarts) of basic psychological needs. In better understanding what social supports and goal inducements are supportive of basic psychological needs, there has been an emergence of more applied research and intervention work using SDT.

Although not as yet formalized within the mini-theories of SDT, a number of processes and phenomena that hold relevance to sport and exercise are addressed from the SDT perspective. These processes include the study of vitality (i.e., a marker of mental and physical energy), the positive effects to well-being of being within the natural environment, the study of close relationships, and the role of mindfulness (or awareness) in autonomous functioning.

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See also Achievement Goal Theory; Autonomy-Supportive Coaching; Intrinsic/Extrinsic Motivation; Hierarchical Model of; Self-Regulation

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SELF-DISCREPANCY

Self-discrepancy is incongruence (i.e., mismatch, lack of agreement) in one's perception of his or her actual attributes and one's internalized standards or ideals. In sport and exercise psychology (SEP), self-discrepancies are often studied in the realm of body image, whereby actual weight is compared to an ideal that is either established from societal norms or desired based on conceptions of what is healthy, attractive, and/or socially desirable. Discrepancies are also studied within sport as incongruence in actual athletic abilities and desirable skills for a position or team, and as mismatches between what athletes need and want

(ideal state) compared to what they perceive they get from their coaches (actual state).

Self-Discrepancy Theory

Based on the self-discrepancy theory, proposed by Edward Tory Higgins, people who hold conflicting or incompatible beliefs about themselves will likely experience emotional discomfort. These conflicting or incompatible beliefs arise when people's beliefs of themselves do not match the ideals or standards they have set for themselves or internalized from others (called self-guides). The conflicting or incompatible beliefs of the self can be experienced in any domain of the self, including but not limited to social relationships, moral responsibilities, financial achievement, and physical self.

Domains of the Self

The self is defined as consisting of three domains: the *actual* self (i.e., the representation of the attributes one is believed to possess, either by oneself or others), the *ideal* self (i.e., the representation of the attributes one hopes, wishes, or aspires to possess), and the *ought* self (i.e., the representation of the attributes one should, by virtue of one's sense of duty, or responsibility, possess).

Standpoints of the Self

The "self" domains are also distinguished as two points of view from which one can be judged: one's *own* personal standpoint and the standpoint of some significant *other* (e.g., a parent, sibling, partner, friend). Self-state representations can be distinct for various significant others. Although these distinctions of self-state representations are unique to self-discrepancy theory, most researchers have focused on one's own standpoint.

There are six basic types of self-state representations: *actual-own*, *actual-other*, *ideal-own*, *ideal-other*, *ought-own*, and *ought-other*. Actual-own and actual-other are generally considered one's *self-concept*, whereas the ideal and ought self-states from one's own or other standpoint are considered *self-guides* (i.e., internalized standards). People differ on which self-guide they are motivated to meet, yet they are motivated to reach a condition in which their self-concept (actual self) matches their personally relevant self-guide (ideal

and/or ought selves) from either their own standpoint or from the perceived standpoint of some significant other.

Emotional Outcomes

Based on self-discrepancy theory, a mismatch in actual-ideal or actual-ought selves results in specific emotional and motivational difficulties. Discrepancies between one's actual and ideal self, from his or her own standpoint, result in the absence of a positive outcome (failure to meet hopes and desires), and predicts dejection-related emotions (e.g., dissatisfaction, sadness, disappointment, depression). If the actual-ideal discrepancy is perceived from the standpoint of another, with inherent failure to have met someone's expectations, shame and embarrassment may also be associated with dejection-related emotions.

Self-discrepancies between actual and ought self-domains result in a presence of negative outcomes, and in agitation-related emotions of resentment, fear, and frustration when the ought self is from an other's standpoint (i.e., a person's perceived actual attributes do not match the attributes the person believes some significant other considers to be his or her duty and obligation). Agitation emotions such as guilt, self-contempt, and uneasiness are outcomes when the ought self is from one's standpoint (since people believe they have not met a personally accepted moral standard). Furthermore, researchers suggest that the greater the discrepancy, the more pronounced the emotional outcome. While the self-discrepancy theory is a template to help understand how certain discrepancies relate to emotional outcomes, the actual-ideal self-state, from one's own standpoint, has been the predominant focus in research and practice.

Physical Self-Discrepancies

In SEP, a large proportion of the self-discrepancy research is focused on the physical self. Self-discrepancies related to actual and ideal (or ought) weight status, body shape, and attractiveness are foundational to a number of body image frameworks and theories of motivation and emotion. For example, it has been argued that Westernized cultures are so pervasively driven by unrealistic ideals of thin and toned for females and muscular and tall for males that there is a normative discontent brought about by comparisons between

one's actual self and these often unattainable ideals. It is estimated that a majority of women hold thin body ideals that are unrealistically thinner than their actual bodies, resulting in a mismatch between their *actual-own* and *ideal-own* perceptions. These particular body-related discrepancies have been linked to body dissatisfaction, eating disorder symptomatology, shame and negative emotions, depression, and maladaptive exercise and eating patterns. Compared to men, women tend to report a greater number and magnitude of discrepancies in their actual and ideal self-states pertaining to the physical self. Research examining self-discrepancies in men is limited.

Measures and Analytical Considerations

There is no single measure that has been considered the gold standard in measuring self-discrepancies. Some of the more prominent scales include the Selves Questionnaire (participants identify traits and attributes for different self-states and standpoints, and discrepancies are calculated as the number of matches and mismatches between self-state attributes), self-lines questionnaire (a graphical representation of self-states and the distance between the identified self-states is used to calculate discrepancies), Likert-scale type tools such as the physical self-discrepancy scale (attributes of the physical self are identified and participants identify to what extent their self-states match the descriptors), and index measures such as the Integrated Self-Discrepancy Index that combine idiographic and nomothetic methods. Other commonly used measures for body-related self-discrepancies have involved identifying body shapes and sizes that represent one's actual, ideal, and ought physiques and calculating a discrepancy between the body shapes and sizes.

Commonly, discrepancy scores have been used in analyses predicting emotional and behavioral outcomes. However, discrepancy scores present methodological problems such as reduced reliability, ambiguity, confounded effects, and dimension reduction. Drawing from congruence work in business and commerce domains, it may be more useful and appropriate to use component measures in analyses (i.e., treat the actual, ideal, and ought self-states as independent predictors). The use of polynomial regression techniques, combined with response surface methodology, has been advocated

to best understand the associations among self-discrepancies and affective, cognitive, and behavioral outcomes.

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See also Body Dissatisfaction; Self-Conscious Emotions; Social Comparison

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SELF-DOUBT

Self-doubt has been defined as uncertainty about one's abilities, potential for success, or competence in performance situations. As self-doubt concerning personal abilities increases, global self-esteem tends to decrease because self-doubt presents the threat to global evaluations of oneself. Hence, self-doubt can lead to both self-handicapping (i.e., creating or claiming obstacles that reduce the probability of success while at the same time providing an excuse for failure) and overachievement (i.e., striving to perform beyond one's capabilities) in order to protect the self from the implications of

failure. As such, self-doubt is often studied in the context of performance.

Self-doubt includes thoughts and feelings such as wondering whether or not one has the ability to succeed at important activities, having thoughts that focus on the bad things that might occur, feeling unsure of one's abilities more often than not, and experiencing greater emotional impact as a result of avoiding failure than achieving success. Among university students, self-doubt is negatively related to variables such as achievement motivation, self-esteem, and narcissism and positively related to variables such as self-handicapping, social anxiety, and "impostor" feelings in which success is perceived as undeserved.

It has been proposed that people who experience high levels of self-doubt tend to reduce their level of effort or quickly settle for mediocre solutions when faced with difficulties, challenges, or setbacks in performance situations. Thus, while self-doubt might be a more or less natural reaction to failure, it might be the resilience and ability to regain confidence that is critically important to successful performance when self-doubt occurs. However, subjective overachievement (i.e., the psychological approach to and process of a performance) can occur when self-doubt is combined with a concern over performance. While overachievers tend to perform well on tasks, the challenge to being an overachiever is that the motive to perform well might be driven more by a desire to gain social approval than by intrinsic motives such as the inherent satisfaction in being able to perform a task.

Despite the potential risks of the overachievement relation to self-doubt, a common message appearing in recent sport and exercise psychology (SEP) literature is that some level of self-doubt might actually be beneficial to performance in sport, at least in some circumstances. For instance, there is an important distinction between performance efficacy (i.e., perceived capabilities to be successful during the practice phase of competition) and preparatory efficacy (i.e., perceived capabilities to be successful immediately before the start of competition) and the role that self-doubt might play in each. Self-doubt at game time can act as a barrier to athletes' effectively using their skills because as performance efficacy decreases, performance level tends to decrease; therefore, efficacy beliefs should be as high as possible when competition begins. Alternatively, when athletes

already have at least a minimal level of efficacy, some self-doubt during the preparatory phase of competition can be beneficial because it might help protect against overconfidence and complacency, as well as motivate athletes to expend increased effort in practice as a way to continue to enhance personal growth, skill development, and skill execution capabilities. While theoretical arguments supporting the potential benefits to self-doubt are compelling, there is little research to date directly exploring the complexities of the relationships among self-doubt, preparatory efficacy, and performance efficacy specifically in the sport and exercise context.

The benefits of self-doubt to performance in sport and exercise were observed in a recent experimental study conducted by Tim Woodman and colleagues with participants who were skilled in their ability to skip rope. The experimental group received information that the rope they were going to use in a competition would be more difficult than the rope used in practice trials. This manipulation was a way to decrease self-confidence and induce self-doubt among the participants in the experimental group. Results showed that those in the experimental group, but not the control group, improved performance on the 1-minute skipping competition task. The reason for the improvement was not clear, as there was little support for increased effort being the reason for the increase in performance. Despite study limitations, this research provides evidence that the relationship between self-confidence and performance might be more complex than generally thought and that a little self-doubt might actually be helpful to performance in sport and exercise.

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See also Achievement Motive Theory; Psychological Skills Training; Self-Efficacy; Self-Handicapping

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SELF-EFFICACY

Self-efficacy is a construct introduced by Albert Bandura, arguably among the most eminent cognitive psychologists of the 20th century. Based on his clinical work (mostly dealing with helping people overcome their anxiety and phobias to such things as snakes), Bandura proposed a social cognitive theory (SCT) to explain and predict psychological changes achieved by different modes of treatment. Within this SCT, Bandura believed that people are capable of intentionally making things happen by their actions (human agency) and that making things happen is more likely when people believe in their abilities to do so (self-efficacy). Specifically, self-efficacy is the part of the theory that refers to people's beliefs that they can do something successfully. Self-efficacy beliefs are judgments about what people think they can do with the skills they have (e.g., I think I can . . .). Bandura considered self-efficacy to be the foundation of human agency and the most important part of behavior change. In sport and exercise psychology (SEP), self-efficacy is considered to be the most important psychological state that affects performance. Researchers in SEP have been studying self-efficacy since the 1980s, and there are several hundred articles and at least one book dedicated to self-efficacy in sport. Its popularity stems from its research-tested theoretical assertions and ease of practical application for intervention purposes.

Self-Efficacy Theory

Within the larger framework of SCT, Bandura proposed the self-efficacy theory as a way of identifying modifiable antecedents of self-efficacy and related cognitive, behavioral, and affective outcomes. The theory has self-efficacy acting as the common cognitive mediator between sources and outcomes.

Sources of Self-Efficacy Beliefs

Bandura proposed that four principal sources of information are used to build, maintain, and regain self-efficacy beliefs. They are *past performance accomplishments* (also called mastery or performance experiences or enactive mastery experience and attainment and refer to clear successes and failures), *vicarious experiences* (also called social comparisons, observational learning, or modeling), *verbal persuasion* (defined as encouragement and persuasive techniques, provided by a range of social influences including coaches, teammates, parents, teachers, and even oneself) and *physiological* (e.g., heartbeat, breathing rate) and *emotional* (e.g., mood, emotions, affect) states. Although Bandura operationalized physiological and emotional states as a combined influence on self-efficacy, some researchers have separated physiological states from emotional states to examine these antecedents of self-efficacy separately. Also, imaginal experiences (e.g., imagining oneself or others behaving effectively or ineffectively) have been operationalized as a form of vicarious experience or as a separate unique antecedent of imagery. The key to improving self-efficacy beliefs using imagery is that the athlete or exerciser must see himself or herself demonstrating mastery (rather than failure).

Self-efficacy beliefs, whether accurate or faulty, are a product of a complex process of self-appraisal and self-persuasion that relies on the cognitive processing of information from these sources. For example, past performance accomplishments are considered to be the most influential and dependable source of efficacy information because they are based on a person's own experiences. Success raises efficacy beliefs and failures lower them. The influence of past performance on self-efficacy beliefs depends on other factors such as the perceived difficulty of the task, the amount of guidance received, the pattern of success and failure, the effort expended, and the person's appraisal of locus of control (i.e., ability as skill rather than inherent aptitude). Successful accomplishments on difficult tasks, without external assistance and with only occasional failures, have the greatest positive efficacy value. Based on this type of information, instructional strategies (such as skill progressions, performance aids, physical guidance, simulations, and equipment modifications) have been used to build self-efficacy beliefs. For example, coaches can

help arrange efficacy-enhancing situations by not putting their athletes into situations that are likely to bring repeated failure. Efficacy information can also be gained by observing and comparing oneself to others (i.e., modeling and social comparison, which is part of the vicarious experiences source), by verbal persuasion (i.e., self-talk, feedback framing, coaches' pregame speeches, attribution training), by appraisal and interpretation of physiological and emotional states (e.g., interpreting "butterflies" as being ready for competition instead of being nervous), and by using imagery (i.e., visualizing oneself being successful). Because efficacy beliefs can be changed (e.g., I can't do this to I can do it), several intervention strategies have been developed based on these sources. It is these strong ties among theory, research, and application through intervention that make self-efficacy theory one of the most influential in SEP.

Outcomes of Self-Efficacy Beliefs

Understanding the outcomes of self-efficacy beliefs highlights why so much attention has been paid to self-efficacy beliefs and their sources. Sport psychologists have investigated psychological differences between successful and less successful athletes for years. For example, comparisons between athletes who made an Olympic team compared to those who were cut from the team showed that the more successful athletes had higher confidence. Confidence has often been used as a synonym for self-efficacy because it is more familiar to people like athletes, coaches, parents, and the media. Bandura stated that confidence and self-efficacy differ in the sense that confidence is a nondescript term that refers to strength of belief but does not specify what the certainty is about. For example, someone can be extremely confident that he or she will fail a test. Self-efficacy is theoretically positive in nature, involving affirmation and accomplishment. Many sport psychologists use the word *confidence* in professional practice, but in research writing, especially when referring to the theory, self-efficacy is typically used.

Bandura proposed that efficacy beliefs are the primary determinant of people's level of motivation to accomplish a specific goal. More specifically, self-efficacy beliefs affect behavior (i.e., what people choose to do, how much effort they put into it, and their persistence in the face of difficulties), goals and self-regulation, thought patterns

(i.e., attributions, decision making [DM], style of thinking), and emotional reactions (e.g., anxiety, worry, fear). Much of the focus of sport research has been on the relationship between self-efficacy and performance. A consistent finding is that self-efficacy has a positive relationship with performance. With the requisite skills and motivation, people who are more efficacious outperform those who have lower self-efficacy.

Measurement of Self-Efficacy Beliefs

Self-efficacy beliefs are measured along three dimensions: *level*, *strength*, and *generality*. Level refers to people's expected performance accomplishments at increasing levels of difficulty. For example, a golfer may rate his or her self-efficacy using a 0 (*no confidence*) to 10 (*high confidence*) scale that he or she could putt into a hole at varying distances. Strength refers to the certainty of people's beliefs that they can attain these different levels of performances. For example, two golfers may believe they can make a putt from 30 feet away (level), but one may be more certain than the other (strength). Generality refers to the number of areas that people judge themselves to be efficacious in and the transferability of efficacy judgments across these areas. For example, a golfer may generalize his or her exceptional skills and high efficacy beliefs for putting relate to driving as well. Self-efficacy measures are usually situation-specific, created for the skills or tasks being investigated.

Different Types of Self-Efficacy

There are many different types of efficacy beliefs, with the most commonly studied in SEP being *task efficacy*, *coping efficacy*, *self-regulatory efficacy*, *barrier efficacy*, *scheduling efficacy*, and *collective efficacy* (CE). Task efficacy is used to describe beliefs in capabilities to perform a certain task (e.g., athletes' beliefs regarding their ability to score a goal). Coping efficacy (also called ameliorative efficacy) refers to beliefs regarding one's ability to cope with diverse threats such as stress, unwanted and negative thoughts, and pain. An example of coping efficacy would be athletes' beliefs regarding their abilities to control negative thoughts about their performances. Self-regulatory efficacy describes beliefs about one's ability to complete a task while overcoming difficulties to successful behavioral performance. This type of

efficacy considers a person's motivation, thought processes, emotional states, and patterns of behavior. For exercisers, a common conceptualization of self-regulatory efficacy is called barrier efficacy. An example would be being able to stick with an exercise program when faced with competing time demands, bad weather, or low social support, among others. Scheduling efficacy is a similar type of efficacy belief with particular relevance to exercise settings. When assessing scheduling efficacy, researchers and practitioners are interested in measuring whether one can regularly schedule exercise sessions into daily living activities.

CE is also known as team efficacy. It is considered to be an extension of self-efficacy to the team level although there are important differences because of the group dynamic. Theoretically, the efficacy of the team is based on many of the same sources of self-efficacy but there are unique sources such as group size, the length of time the team has been together, and outside sources like spectator and media influences. The outcome of CE most often studied has been performance, although there is some interest in group dynamic variables like cohesion. There are measures developed to assess CE in sport, and the same results have generally been found: the more confident teams are the most successful. The type of sport or degree of interdependence has implications for these relationships. In more interdependent or interactive sports (e.g., hockey, basketball), CE is a better predictor of performance than self-efficacy. In less interdependent or coactive sports (e.g., golf, bowling), self-efficacy is the better predictor of performance. These results tell us that interventions should focus on CE for interdependent sports and self-efficacy for coactive sports. Another interesting program of research has been to track self-efficacy and team efficacy across a season. Researchers have shown that after a win, both self-efficacy and team efficacy ratings increase. But after a loss, team efficacy decreases but self-efficacy either goes up or remains the same. These findings are interesting because they show that athletes may have a tendency attribute losses externally, which allows them to keep a strong and resilient self-efficacy.

Sandra Short

See also Coaching Efficacy; Collective Efficacy; Relational Efficacy Beliefs in Coach–Athlete Relations; Self-Regulation

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SELF-FULFILLING PROPHECY

A self-fulfilling prophecy can be defined as a prediction that sets in motion a series of events that ultimately causes the original prediction to become true. Within the sport setting, the self-fulfilling prophecy theory has been used to describe how and why the expectations that individuals (e.g., coaches) form about others (e.g., athletes) can serve as prophecies that subsequently dictate or determine the level of achievement that these others (athletes) will reach.

History of the Self-Fulfilling Prophecy Phenomenon

In 1949, Robert Merton, a sociologist, used the phrase *self-fulfilling prophecy* to describe a social phenomenon in which a false belief or expectation is developed and which initiates a series of events that causes the original false belief to become true. Although the term originated in the sociological literature, and the concept itself was popularized in the entertainment world (e.g., plays and movies based on the concept), the self-fulfilling prophecy phenomenon gained particularly wide recognition in the academic realm. In 1968, Robert Rosenthal and Lenore Jacobson published the results of a study they had conducted to determine whether students' academic progress could be influenced by their teachers' expectations of them. These researchers began by notifying teachers at the beginning of the academic year that some of the children in their classes had been identified, via achievement tests,

as “late bloomers” and that these students could be expected to show large gains in academic achievement. What the teachers did not know was that the students who were identified as late bloomers had actually been selected by the researchers at random. However, at the end of the school year, the targeted children, as a group, had made greater gains academically than had their classmates who had not been identified as late bloomers. In discussing these results, Rosenthal and Jacobson concluded that the teachers used the false information provided by the researchers to develop higher expectations for the children identified as late bloomers and then to act in ways that would stimulate higher performance in them (e.g., more individualized instruction, higher rates of positive feedback). Thus, the hypothesized chain of events conformed to the self-fulfilling prophecy phenomenon in that the expectations formed by the teacher (based on false information) were believed to set into motion a series of events (e.g., enhanced teacher-student interactions in the classroom) that caused the original false expectation to become true.

Following the publication of this study, other researchers in the academic domain explored the self-fulfilling prophecy phenomenon. This body of research generally resulted in support for the theory but also revealed that expectancy effects are not evident in *all* classrooms and may exert relatively low effect sizes (ES) when compared to other factors that also impact children’s academic progress. Nevertheless, reviewers of this research do point out that some children (e.g., those from low socioeconomic backgrounds, racial and ethnic minorities) may be particularly vulnerable to the self-fulfilling prophecy phenomenon in educational settings.

Self-Fulfilling Prophecy in Sport and Physical Activity Contexts

Based on the research conducted within academic classrooms, Thomas Martinek and Jack Rejeski began to explore expectancy effects in physical education (PE) and youth sport settings. Their results, along with that of others, have provided evidence that the self-fulfilling prophecy phenomenon can occur in a number of different contexts within the physical activity (PA) setting.

Expectations Formed by Coaches

Much of the early work on coaches’ expectancy effects was based on a four-step model drawn

from the academic literature that describes how coaches’ expectations for individual athletes can serve as prophecies that dictate the ultimate level of performance and type of behaviors that these individual athletes will exhibit. Step 1 begins when coaches form expectations regarding the capabilities of each of their players. These initial coach judgments can be based on person-type cues (i.e., gender, dress, socioeconomic status [SES], racial and ethnic group, family background), performance-based cues (e.g., athletes’ previous performances or their current performance in drills, fitness tests), and/or on perceived psychological characteristics (e.g., coaches’ perceptions of athletes’ self-confidence, maturity, self-discipline). Although all of these sources of information may be available in any sport setting, coaches likely differ in the particular sources they use and in the accuracy and flexibility of their initial expectations. If a coach, for example, initially forms a negative impression of a particular athlete’s capabilities (based, perhaps, on that athlete’s style of dress the first day of practice) but that initially false expectation is quickly revised by the coach as more information is obtained (e.g., athlete performs very well in drills on second day of practice) then the initially false expectation that the coach developed on the first day may not interfere with the athlete’s ability to progress in that sport setting. Conversely, if a coach forms an initially false (and low) expectation for an athlete (based, perhaps, on person-type cues such as gender or family background) and does *not* revise that expectation based on subsequent more performance-oriented information (athlete’s skill test scores), then that coach’s initial expectation was both inaccurate and inflexible. This is the type of situation that can lead to the second step in the self-fulfilling prophecy phenomenon.

At the second step, the coach’s initially false expectation or judgment that an individual athlete on her or his team has low abilities may lead the coach to exhibit behaviors toward and with this athlete that reflect the coach’s low expectation. Specifically, the coach may (a) interact less with this athlete; (b) provide him or her with less time in drills and practice activities; (c) establish a lower standard of performance for him or her; (d) deliver less frequent and less beneficial praise after a successful performance and less informational or corrective feedback after a performance error; and (e) attribute the athlete’s successful performances

to luck or task ease and, correspondingly, attribute the athlete's unsuccessful performances to lack of ability.

Not surprisingly, the differential behavior provided by the coach in the previous example would lead to the third stage of the self-fulfilling prophecy in which the athlete's performance and behavior conform to the coach's original expectation for him or her. Specifically, players who consistently receive less frequent and less effective instruction will not show the same skill improvement as other individuals on that team who are given optimal learning opportunities. Thus, by the end of the season, the low-expectancy athlete may indeed be a low-performing team member. However, this lack of progress is not really due to the player's low athletic ability but rather to the initially false expectation by the coach that caused a series of events (e.g., differential practice time, ineffective feedback) that resulted in the player's inability to improve. Furthermore, the negative coach behaviors toward this particular athlete likely also affected her or his psychosocial state. Thus, by the end of the season, he or she may be exhibiting low perceptions of competence, low enjoyment, low motivation, high anxiety, and possibly even low teammate support and friendship.

At the final step in the self-fulfilling prophecy sequence, the coach sees that this particular athlete's performance and behavior were consistent with her or his (coach's) original expectation. This conformity reinforces the coach's belief that she or he made the right judgment at the beginning of the season. Thus, the coach perceives himself or herself to be especially astute at judging the initial talent levels of athletes, and the cycle continues.

Although research studies have provided support for the four-step self-fulfilling prophecy model, it is important to point out that not *all* coaches exhibit tendencies to be expectancy-biased (i.e., to be quick to make inaccurate and inflexible judgments of individual athletes at the beginning of the season and then to act in ways that are biased by this initial expectation). Furthermore, not all athletes are susceptible to the self-fulfilling prophecy phenomenon (i.e., some are resistant to the negative effects of a biased coach). But there are others who may be particularly vulnerable to such coaches.

In the youth and adolescent sport settings, athletes who could be more vulnerable to expectancy-biased coaches might include children that

are early or late maturers (i.e., those who are proceeding toward full physical maturation at either a faster or a slower pace than are their average and early maturing peers). Late maturing boys, for example, who are typically smaller in physical size and stature than their same chronologically aged peers may be perceived by expectancy-biased coaches to be low-expectancy athletes. Thus, these temporarily smaller, less physically developed boys may not get the optimal instruction and feedback that they need and/or may even be cut from competitive youth programs. For girls, however, it actually may be the early maturing ones who are most vulnerable to expectancy-biased coaching. That is, early maturing girls (i.e., those who are entering the pubertal development stage at an earlier chronological age than their peers) may be seen by their coaches as less physically competent (due to early pubescent changes in body shape, composition, and functioning) and less interested in sport participation, especially as compared to their late-maturing female peers who still have a more linear body build and less body fat.

Another important issue with regard to the self-fulfilling prophecy phenomenon is that of the broader societal stereotypes and biases that can affect coaches' expectations for individual athletes. Research in the sport literature has shown, for example, that coaches' behaviors toward individual athletes can be biased on the basis of their (coaches') sexist, racist, and/or homophobic attitudes, beliefs, and expectations. Thus, coaches who do hold such negative and biased stereotypes may judge individual athletes at the beginning of the season based on person-type cues (e.g., gender, race, ethnicity, sexual orientation, body build) and then exhibit the kind of biased instructional behavior that interferes with individual athletes' ability to develop and/or exhibit their talents and abilities. Again, then, the self-fulfilling prophecy phenomenon may be most evident under such social conditions.

Expectations Formed by Referees

Within the sport setting, referees constitute another group of individuals who may exhibit expectancy effects. Specifically, referees' decisions regarding fouls and other game infractions appear to be influenced by the initial expectations they develop based on such things as team uniform color, team reputation, and even players' body

size. Furthermore, gender stereotypes regarding the appropriateness (or inappropriateness) of females exhibiting aggressive behaviors may exist among referees as female players have been found to be penalized more often than are male athletes for aggressive play. These results are consistent with the idea that the expectations that referees might form either as a result of general stereotyped attitudes and/or during pregame activities would affect their actual behaviors within the game context.

Expectations Formed by Athletes

Some studies have examined expectations formed by athletes and, correspondingly, how those expectations might affect athletes' beliefs and behaviors in relation to their coaches, other individuals in the sport setting, and even their opponents. First, in regard to athletes' expectations about coaches, it appears that athletes do form initial judgments when first meeting (or reading profiles of) a new coach and that these initial expectations are primarily based on dynamic cues (e.g., coach's eye contact, body language) and on third-party reports (e.g., the coaches' previous performance results) and less on static cues (e.g., gender, race, ethnicity, age). At this point, little is known about the effects of such initial judgments on subsequent coach-athlete interactions. However, some studies have found athletes' perceptions of their coaches' leadership styles and behaviors to vary as a function of their coaches' gender. That is, female coaches who exhibit certain types of behaviors and leadership styles (e.g., a controlling or autocratic style) may be perceived more negatively than are male coaches who exhibit the same types of behaviors. Again, such differential evaluations are likely due to gender stereotypes that may be prevalent in the broader sociocultural context within which that sport exists.

Second, athletes also form initial expectations when judging the effectiveness of other types of coaches with whom they may have contact (e.g., sport psychology [SP] consultants, strength and conditioning coaches, personal trainers). In particular, athletes form initial impressions of SP consultants based on such things as gender, dress (e.g., business or athletic wear), and body build (i.e., appearance of physical fitness). For strength and conditioning coaches as well as for personal

trainers, athletes and students may generally indicate a higher preference for, and more positive evaluation of, male coaches or trainers, with male athletes and students exhibiting stronger gender preferences than do female raters. Sexual orientation, as well, has been demonstrated to impact the preferences of athletes and their parents for coaches and the rating of job applicants for personal training positions.

Third, athletes have been found to develop expectations about their opponents based on such person-type cues as the opponents' dress, body language, body size, and uniform color. Furthermore, these initial expectations or judgments formed about an opponent do seem to affect the athletes' own level of confidence and expectancy for success. Based on the four-step self-fulfilling prophecy model, it would be further hypothesized that the expectations that individual athletes or teams form during precompetition events (e.g., watching game films, observing warm-up activities) about the competencies of their opponents (e.g., that athlete or that team is really good) might affect the athletes' own behavior within the competition (e.g., amount of effort exerted, persistence after a mistake) and thus, actually, cause the initially formed expectation (that athlete or team will beat me or us) to become true. As such, then, the self-fulfilling prophecy is fulfilled.

Conclusion

The self-fulfilling prophecy describes a social phenomenon in which an initial prediction sets in motion a series of events that ultimately cause the original prediction to become true. Within the PA setting, research studies have found support for this theory, most particularly with regard to the expectations that coaches develop for individual athletes on their team and that, ultimately, can determine the level of performance that each athlete reaches. More recent work has demonstrated that the self-fulfilling prophecy phenomenon may also apply to other PA situations (e.g., referees' behavior, hiring practices in sport, and PA settings). From a practitioner standpoint, the self-fulfilling prophecy phenomenon can have significant effects on the progress that individual athletes, coaches, and other individuals can make in any particular PA context.

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See also Coach–Athlete Relations; Implicit/Self Theories of Ability; Stereotype Threat; Stereotyping, Cultural/Ethnic/Racial

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SELF-HANDICAPPING

Self-handicapping is a future-oriented, self-protection strategy used to (a) maintain personal perceptions of competence, control, self-worth, and self-esteem and/or (b) protect or enhance one's public image in the eyes of coactors or observers. It consists of thoughts, statements, and behaviors that take place in advance of performance, and that increase the likelihood of situational factors being blamed for poor performance but personal factors being credited for good performance.

Variations and Principles

Two forms of self-handicapping have been identified based on the way in which the process unfolds. When cognitions or verbalizations are used to

proactively defend or enhance self-worth and self-esteem, they are considered to be *self-reported*, or *claimed*, *handicaps*. Examples of this form of self-handicapping include a focus on (or claims of) temporary illness or injury, situation-specific anxiety, mood fluctuations, or recent exposure to uncontrollable negative events. When proactive defense or enhancement of self-worth and self-esteem involves deliberate, observable acts that could conceivably interfere with performance, they are regarded as *behavioral handicaps*. Examples of this form of self-handicapping include ingestion of drugs or alcohol, withholding of effort, choosing to perform under less-than-optimal conditions, or providing assistance to competitors.

From a theoretical perspective, the use of self-handicapping strategies invokes the attributional principles of augmentation and discounting. More specifically, proactively established handicaps permit the individual to augment the role of personal attributes such as ability if performance is good or, alternatively, to discount the importance of personal attributes such as ability if performance is bad. From an empirical standpoint, studies of attributions made for one's own performance in the presence of potential handicaps provide consistent support for discounting (i.e., self-protection) following failure and occasional support for augmentation (i.e., self-enhancement) following success. Some researchers have suggested that self-enhancement effects might be most evident among individuals with high self-esteem.

Measures of the Construct

Individual differences in self-handicapping tendencies have been assessed in several ways within the sport and exercise domains. The most widely used measure of the construct in these settings is a 14-item version of the Self-Handicapping Scale (SHS) that assesses propensities for *excuse making* and *withholding of effort*. Despite its widespread use, this measure has been criticized by some sport researchers on psychometric grounds and because of its generic frame-of-reference, which undermines domain-specific relevance. As an alternative approach, other investigators have utilized an open-ended, listing-of-impediments procedure to examine self-handicapping tendencies among athletes. Overall, the findings from these studies reveal that commitments outside of sport, injury or illness, and training disruptions are the most frequently cited potential handicaps prior

to performance and that some athletes do indeed have a greater propensity than others to cite such impediments. What cannot be determined, of course, is the extent to which these obstacles are actual or perceived for any given individual. As such, there is likely to be some measurement error inherent to the listing-of-impediments approach.

A less-frequently used but potentially informative approach to assessing self-handicapping tendencies involves the construction of study-specific inventories containing contextually relevant descriptions of proactive impression-management strategies that are then rated on a “like me” or “not like me” basis. In the youth sport context, these items might include statements such as the following: “Some players fool around at practice and before games.” Then, if they don’t play well, they say that was the reason. or “Some players get involved in a lot of activities outside of sport.” Then, if they don’t play well, they say it was because they were involved in other things. Such statements are clearly anticipatory, reflective of self-handicapping tactics, and explicitly self-presentational in the orientation. Research in physical education (PE) classes has demonstrated that this approach can be informative, and it deserves

broader consideration within competitive sport. It could also prove useful for advancing knowledge about context-specific self-handicapping in relation to exercise behavior. At present, the only measure of the construct specific to the exercise domain is the Self-Handicapping Exercise Questionnaire, which assesses exercise impediments related to incorporating exercise into one’s routine, training in an exercise facility, and physical health.

Research in Sport and Exercise

Self-handicapping occurs most often in situations that involve public behaviors that are considered important to the individual and, at the same time, are characterized by uncertainty about the likelihood of a good performance. In such circumstances, poor performance is potentially threatening to private or public self-worth and self-esteem. Because sport and exercise environments often contain most of these elements, it is not surprising that psychologists have taken a keen interest in the correlates and consequences of self-handicapping processes within sport and exercise settings. Table 1 summarizes research findings related to self-handicapping in the areas of motor

Table 1 Summary of Findings for Studies of Self-Handicapping (SH) in the Motor Skills, Sport, Physical Education, and Exercise Domains

<i>Behavioral Domain</i>	<i>Constructs Examined</i>	<i>Findings in Relation to Self-Handicapping</i>
Motor Skills	Achievement goals	performance-avoidance goals associated with increases in claimed SH and behavioral SH (reduced practice effort) prior to performance; performance decrements associated with performance-avoidance goals may be mediated by SH processes
	Self-esteem; Self-confidence	low self-esteem associated with increases in claimed SH and increases in behavioral SH (reduced practice effort); self-confidence negatively predicted claimed self-handicapping;
	Motivational climate	performance-oriented climate positively related to claimed SH; behavioral SH (reduced practice effort) greater in performance-oriented climate than in mastery-oriented climate
	Stereotype threat	behavioral SH (reduced practice effort) increased when poor performance might confirm negative stereotype
Sport	Self-esteem; Self-efficacy	SH tendencies negatively correlated with general self-esteem and physical self-esteem; self-efficacy negatively related to claimed SH

(Continued)

Table 1 (Continued)

<i>Behavioral Domain</i>	<i>Constructs Examined</i>	<i>Findings in Relation to Self-Handicapping</i>
Physical Education	Motivational climate	task-oriented motivational climate negatively related to claimed SH
	Group cohesion	SH tendencies positively related to self-reported handicaps when perceived group cohesion high
	Precompetitive anxiety	precompetitive cognitive anxiety positively related to use of SH; precompetitive somatic anxiety positively related to use of SH
	Implicit ability conceptions; Dispositional goal orientations	Static ability conceptions and ego orientation positively related to SH for both team and individual sport athletes
	Perceived competence; Attributions for failure	SH positively related to external attributions for failure and maintenance of ability perceptions following failure
	Coping strategies	use of emotion-focused coping positively associated with SH tendencies
	Flow experiences	SH tendencies positively related self-reports of flow during competition
	Public self-consciousness	certain types of claimed SH positively related to public self-consciousness
	Dispositional goal orientations; Motivational climate	ego-involving climate positively related to SH; weak but potentially meaningful negative relationship between task orientation and SH
	Achievement goals	task goals and performance-approach goals negatively related to SH; performance-avoidance goals positively related to SH
Exercise	Implicit conceptions of ability	stable ability beliefs positively related to claimed SH (excuse-making) and behavioral SH (reduced effort); incremental ability beliefs negatively related to behavioral SH (reduced effort); relationships
	Perceived competence; Fear of failure	perceived competence negatively related to SH tendencies; fear of failure positively related to SH tendencies
	Self-efficacy; Running performance	SH had positive impact on performance for students low in self-efficacy; SH had no impact on performance for students high in self-efficacy
	Perceived enjoyment; Intrinsic motivation	enjoyment positively related to SH for students high in intrinsic motivation; no relationship between enjoyment and SH for students low in intrinsic motivation
	Exercise-specific SH	identification of 3 factors related to SH claims in exercise environments (making exercise a routine, training in an exercise facility, healthy physical functioning); factors were consistent across genders, ages, and exercise settings
	Self-esteem; Persistence	self-esteem negatively correlated with SH tendencies; combination of low self-esteem and strong SH tendencies predicted voluntary withdrawal from a demanding physical training regime

Table 1 (Continued)

<i>Behavioral Domain</i>	<i>Constructs Examined</i>	<i>Findings in Relation to Self-Handicapping</i>
	Performance self-efficacy; Self-presentational efficacy	self-reported handicaps negatively related to performance self-efficacy and to self-presentational efficacy on a graded physical fitness test
	Level of performance	weak negative relationship between SH and performance on a short-duration (1.5 min) strength task

Source: J. Robert Grove.

Note: SH = self-handicapping.

skills, sport, PE, and exercise. The information is necessarily selective, but it provides an overview of how self-handicapping has been examined in these domains and outlines relationships that have been observed with other psychological constructs.

Several noteworthy observations arise from an inspection of Table 1. First, self-efficacy, self-confidence, and self-esteem have been studied extensively in connection with self-handicapping, and the findings have been consistent across behavioral domains. In general, lower levels of self-efficacy, self-confidence, and self-esteem are associated with higher levels of self-handicapping. The interrelated constructs of achievement goals, dispositional goal orientations, implicit ability conceptions, and motivational climate have also been widely studied in relation to self-handicapping. Regardless of behavioral domain, it appears that a mastery (learning) orientation, which is associated with incremental ability conceptions, results in higher levels of self-handicapping than a performance (outcome) orientation, which is associated with fixed ability conceptions.

Most of the studies within these domains have operationalized behavioral self-handicapping in terms of reduced practice effort. While expected relationships have been observed between various psychological constructs and this index of behavioral self-handicapping, some researchers have expressed concern about heavy reliance on this measure. More specifically, it has proven difficult to assess reliably via questionnaire in sport and exercise settings, and direct, observational data related to practice effort (e.g., number of practice attempts) can be influenced by numerous factors other than self-handicapping. Some consideration may therefore need to be given to other indices of behavioral self-handicapping. Additional consideration should also be given to the correlates

and consequences of self-handicapping in relation to real-world exercise behaviors outside of sport and PE settings. Despite some interesting preliminary findings (e.g., an association between self-handicapping tendencies and voluntary withdrawal from demanding physical training programs), it is clear from the summary in Table 1 that relatively little self-handicapping research has been conducted in this domain. Given the importance of real-world exercise behavior to personal health, its social desirability, and the self-presentational issues that surround it, there would seem to be abundant opportunity for more detailed examination of the associated self-handicapping processes.

Conclusion

Self-handicapping is a proactive self-protection and self-enhancement strategy. Classic forms of self-handicapping involve self-protective thought processes or statements that occur in advance of performance (*self-reported or claimed handicaps*) and pre-event behaviors that might interfere with performance (*behavioral handicaps*). Numerous psychological constructs have been examined in connection with these forms of self-handicapping in the motor skill, sport, PE, and exercise domains. These constructs have included personal (e.g., anxiety), group (e.g., cohesion), and environmental (e.g., climate) variables. In general, positive self-perceptions, incremental ability beliefs, and mastery-oriented environments are associated with low levels of self-handicapping, while negative self-perceptions, fixed ability beliefs, and performance-oriented environments are associated with higher levels of self-handicapping. Self-handicapping processes within real-world exercise settings deserve closer scrutiny in future research.

J. Robert Grove

See also Achievement Goal Theory; Implicit/Self-Theories of Ability; Mastery and Control Beliefs; Self-Efficacy; Self-Presentation

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SELF-MONITORING

Self-monitoring reflects the regulation of individuals' expressive behavior, self-presentation, and nonverbal displays of affect in social situations. With a grounding in theories of self, Mark Snyder advanced the study of self-monitoring to heighten understanding of dispositional (i.e., stable) versus situational influences on behavior.

Self-Monitoring Orientations

At its core, the construct of self-monitoring is embedded in the belief that people's behavior is guided by internal (e.g., dispositions, attitudes) and external (e.g., characteristics of the situation) cues in the environment. Based on scores from Snyder and Steve Gangestad's revised 18-item Self-Monitoring Scale, individuals with greater and lower propensity to self-monitor are classified on the basis of a dichotomous (true or false) format. Individuals higher in self-monitoring are responsive to situational contexts and readily engage in self-presentational strategies to fit a given situation or role. As such, these individuals are likely to be adaptive in social situations and able to tailor

their behavior to fit situational and interpersonal demands. In contrast, individuals with a lower tendency to self-monitor are more likely to behave in a manner consistent with "who they are" as opposed to being swayed by situational cues. For these individuals, the expression of behavior is closely linked with their inner attitudes, emotions, and dispositions.

Consequences of Self-Monitoring

Self-monitoring has been related to a wide range of behavioral domains spanning (but not limited to) leader emergence, status, and perceptions of social cues. In exercise contexts, research delineating the influence of self-monitoring has generally been guided by three propositions, each of which is considered next.

Proposition 1. Self-Monitoring Orientation Has Implications for Self-Presentational Processes

In an attempt to unpack self-presentational processes and the conditions that elicit socio-evaluative threat, exercise psychology researchers often manipulate the physical environment (e.g., scenario-based research; wearing revealing clothing). A recent study by Steve Bray and colleagues found that individuals with more propensity to self-monitor reported greater efficacy in their ability to present themselves in a manner consistent with an "exerciser" (e.g., fit, strong) regardless of the manipulation when compared to those lower in self-monitoring. When physique anxiety was evaluated in response to different exercise scenarios, little variation was noted for higher self-monitors. Conversely, individuals with a lower tendency to self-monitor were more likely to report greater physique anxiety in the presence of specific environmental threats (e.g., exercising in a mixed as opposed to same-sex environment). These findings support suppositions of self-monitoring. That is, individuals who are more prone to adapt to the demands of social situations develop greater efficacy in their ability to adjust to environmental cues and demonstrate less affective reactivity (i.e., less variability) in unfamiliar social situations.

Proposition 2. Individuals With a Lower Propensity to Self-Monitor Will Demonstrate Greater Attitude–Behavior Consistency

Attitudes reflect the tendency to evaluate an object, person, or situation as favorable or

unfavorable. Behavioral responses, either through intentions or overt acts, are often deemed expressions of one's attitude. For example, an individual who holds a favorable attitude toward physical activity (PA) should report greater intentions to active and/or behavioral engagement. However, the nature of the attitude-behavior relationship may be more complex than originally believed as individual difference variables or situational factors may attenuate the relationship. Self-monitoring is one personal-related variable posited to moderate the attitude-behavior relationship. More specifically, individuals who are less likely to self-monitor should demonstrate greater attitude-behavior consistency given they are less likely to react to situational cues. The unique influence of self-monitoring on the attitude-exercise intention relationship has been supported in university students. The extent to which individuals lower in self-monitoring intended to exercise was consistent with their favorable versus unfavorable attitudes toward the activity. Conversely, higher self-monitors demonstrated no relationship between attitudes and intention to exercise.

Proposition 3: The Propensity to Self-Monitor Is Associated With Attitudes Toward Messaging

Health promotion specialists have often relied on the use of messaging strategies in an effort to communicate the benefits of PA. Two common forms of messaging are health-based (e.g., cardiovascular health) or appearance-based (i.e., focusing on outcomes linked to attractiveness). Consumer behavior research has suggested that those higher in self-monitoring are more persuaded by appearance-based messages given individuals' desire to create a specific image. Conversely, messaging focusing on more stable dispositions (e.g., personal values) is evaluated more favorably by those lower in self-monitoring. This finding has been supported in an exercise context among university students and refuted among pregnant women such that there were no differences between health- and appearance-based messages.

Conclusion

Self-monitoring has been a widely studied construct to develop understanding of social behavior. Its adoption in exercise contexts has been slower to emerge. Preliminary results demonstrate utility

for the role of self-monitoring in understanding self-presentational and attitudinal variation in exercise contexts.

Diane Mack

See also Personality Traits and Exercise; Self-Efficacy; Self-Presentation; Theory of Planned Behavior

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SELF-OBJECTIFICATION

In their seminal article, Barbara Fredrickson and Tomi-Ann Roberts posited that gender socialization and contextual experiences predispose individuals to internalize cultural standards shaping how meaning is ascribed to one's body. Rooted in socio-cultural approaches to the psychology of women, self-objectification is the tendency to introject an external observer's perspective on one's body, evaluating it in terms of its value and attractiveness to others rather than its value and function (i.e., what it can do). With the adoption of an external perspective, the individual's body is reduced to instrumental status, which serves as one's primary view of the physical self. Most prevalent in adolescence and young adulthood, self-objectification and its theorized consequences may compete with the experience of sport and exercise with implications for psychological health and behavior.

Measuring Self-Objectification

Measurement issues are at the forefront of advancing our understanding of the process of, and consequences from, perceptions of self-objectification. While believed to be a chronic (i.e., trait-like) propensity to adopt an observer's view of the self, certain contexts or stimuli (e.g., viewing fitness models, training in form-fitting clothing) may prime perceptions and behaviors consistent with an objectified state of awareness. When measured at a trait-level, two instruments have dominated the literature. First, Stephanie Noll and Barbara Fredrickson developed the Self-Objectification Questionnaire to assess the extent to which respondents value observable appearance-based body attributes (e.g., physical attractiveness, firm musculature) to be more important than nonobservable competence-based features (e.g., health, strength) for how the physical self is evaluated. Second, the Objectified Body Consciousness Scale, developed by Nita McKinley and Janet Hyde, assesses three components (i.e., body shame, body control, and body surveillance) believed to represent the degree to which perceptions of external observers are internalized. With all three subscales relevant, the eight-item body-surveillance subscale of the Objectified Body Consciousness Scale has most frequently served as a proxy measure of self-objectification. The body surveillance subscale was designed to assess the extent to which an individual views his or her body in terms of its appearance.

Self-objectification can also be induced through experimental manipulation. Situational self-objectification may be triggered through exposure to objectified visual images, language (e.g., beauty advice columns), interactions with strangers, and being in situations where physique salience is emphasized such as exercising in a mirrored environment. Following the chosen manipulation, participants are generally asked to complete survey instruments or perform a task to assess the impact of the triggering manipulation. Feelings elicited through exposure to objectifying situations have been found to persist beyond the manipulation, attesting to its somewhat enduring nature.

Self-Objectification in Sport and Exercise Contexts

Engagement in physical activity (PA) has been advocated as one strategy to offset the deleterious effects of objectification given its focus on

body competence. The inherent challenges embedded in the sport and exercise environment may assist active individuals to view their bodies for how they feel and function as opposed to how they look. Though intuitively appealing, research addressing this issue is relatively nascent, with non-experimental research equivocal in young adult female exercisers or adolescent athletes when compared with nonactive cohorts. Behavioral implications from cross-sectional research have been noted: for example, women scoring higher in trait self-objectification report engaging in less PA. Specifically, individuals who tend to objectify their bodies endorse reasons for exercise aligned with appearance more strongly than motives linked to enjoyment or health, thus reducing sustained PA engagement. Extending to intervention studies, Emily Impett and colleagues found that engagement in a 2-month yoga program was associated with reductions in self-objectification for adult women exercisers. While findings were based on a single group design and small sample, the focus on internal awareness that embodies yoga participation may be one strategy to prevent the development of a sense of self that is object-oriented.

Characteristics of the PA environment have been considered in an attempt to further enhance understanding of the contextual influences underpinning self-objectification. Participation in certain activities (i.e., aesthetic sports, aerobic activities) may predispose individuals to experience greater self-objectification than engagement in activities where physical appearance is less salient to performance. Features of the exercise environment including preferences for positioning in an aerobics studio, exercising outdoors, and attire of physically active women have also been associated with self-objectification. With the understanding that the cues in the PA environment may trigger feelings of self-objectification, practitioners should examine ways of tailoring the exercise environment to minimize these concerns. These environmental manipulations may be particularly important for the individual who is predisposed to view his or her body in an objectified manner.

Consequences of Self-Objectification in Physical Activity Contexts

Emotional Consequences

Beyond the social contextual processes leading to feelings of objectification, researchers have

investigated the consequences of viewing oneself as an object. The internalization of an observer's perspective is associated with increased self-consciousness, habitual body monitoring and lowered personal agency. Emotional experiences most commonly linked with self-objectification are perceptions of appearance anxiety stemming from fear about when and how one's body will be evaluated. Body-related shame has also been identified as a salient consequence that occurs when the body is evaluated as inadequate when compared against the internalized standard. Shrouded with implications including a lack of self-control and failure to meet social expectations, the individual who experiences body-related shame attributes shortcomings to the core self. Body-related guilt resulting from the appraisal that one's actions or behaviors are wrong has recently been advanced as a consequence of self-objectification. For example, restrained eating or excessive exercise engaged in to improve appearance may render an individual feeling body-related guilt as a consequence of these actions. The postulated relationships between self-objectification and appearance anxiety and body-related shame have been consistently demonstrated in athletes and exercisers. The extent to which other self-conscious emotions such as body-related guilt is associated with self-objectification needs to be explored in physically active samples.

While originally grounded in the experiences of women, the emotional consequences of self-objectification in physically active men should not be ignored. Findings linking self-objectification to measures of body-related emotions in males are less consistent than those demonstrated in females. The noted variation between self-objectification and body-related emotions underscores the inherent complexity of examining self-objectification in males. Disproportionate exposure to self-objectifying situations in comparison to females may account for these findings. Measurement issues may also be implicated as existing instruments and manipulations may be less powerful triggers of body objectification in males than females.

Mental Health Consequences

The experience of self-objectification, whether it be acculturated or primed by contextual cues, is posited to play a role in an array of mental health concerns that disproportionately affect women (e.g., eating disorders, depression) and diminished

well-being. In physically active samples, greater self-objectification has been directly linked with increased eating disorder symptomatology, with body-related shame identified as one mechanism underpinning the relationship. Consideration of the mental health risks of self-objectification in athletes and exercisers beyond eating-disordered symptomatology has received minimal attention. This is despite consistent associations noted between self-objectification and increased depressive symptomatology in nonactive samples. As PA is often advanced as adjuvant treatment for mental health concerns including depression, the role of self-objectification in inhibiting effective treatment serves as one area for future investigation. Self-esteem has been identified as one plausible buffering agent to reduce the deleterious effects of self-objectification on measures of mental health. Consequently, greater self-esteem appears to serve as one protective mechanism minimizing the negative effects of self-objectifying experiences in comparison with those lower in self-esteem.

Motivational and Behavioral Consequences

Adopting a third-person perspective as one's sense of self has been theorized to limit mental resources as attention is divided between appearance and performance. As a consequence, self-objectification has been implicated in the reduction of peak motivational states or flow (i.e., a state of being fully absorbed) when engaged in demanding activities. Cognitive disruptions linked to perceptions of self-objectification hold implications for the fulfillment of one's potential in sport and exercise contexts. Given the salience of motivation to sustained engagement and psychological health, it is surprising that the motivational implications of self-objectification have received little attention when compared to emotional and mental health outcomes. The few investigations examining the theorized relationship between self-objectification and flow has demonstrated trends suggestive of the implied in samples of physical active females or former dancers.

In addition to interfering with cognitive performance, self-objectification may also serve to hinder physical performance as a consequence of the fractionation of mental capacity. In the one study examining the previously given contention, Barbara Fredrickson and Kristen Harrison examined the effect of self-objectification on physical

performance in a throwing task in adolescent females. Girls who self-objectified to a greater degree demonstrated lower throwing performance than those with fewer tendencies to objectify. This held even after controlling for age, ethnicity, and prior throwing experience, which speaks to the potential implications for performance of individuals with a propensity to internalize others' perceptions of their bodies.

Conclusion

Since its introduction to the literature, self-objectification and its postulated consequences have received considerable empirical attention with implications for emotional, psychological, and physical development generally supported. While engaging in sport and exercise has been advanced as one strategy to combat the deleterious effects of self-objectification, such conclusions cannot be firmly advanced given the current state of knowledge. In an effort to more fully understand self-objectification in sport and exercise, it may be prudent to move beyond cross-sectional research to consider the processes through which individuals internalize others' (e.g., coaches) standards for appearance and potential strategies to combat their influence. Although speculative, these strategies may include self-regulatory strategies, a focus on the physical self of how the body performs as opposed to looks, and strategies to increase self-esteem.

Diane Mack

See also Body Awareness; Body Image; Self-Conscious Emotions; Social Comparison

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SELF-PRESENTATION

Self-presentation, also referred to as impression motivation, is the process whereby individuals attempt to monitor and control how other people perceive them—that is, they attempt to portray a specific image (or impression) to others. In general, people try to present truthful images of themselves to others. However, in any specific situation, they may choose to highlight certain aspects of themselves and downplay others. For example, people may choose to emphasize their positive qualities (such as organizational skills, previous win–loss record), and avoid drawing attention to their less positive qualities (e.g., poor time management, poor athlete retention) in a job interview for a new coach position. There are also situations in which people may actually decide to make a negative impression—for example, a student pretending she or he has not done readings in class (seeming unprepared) to avoid having to answer questions in front of the whole class.

People self-present for many reasons. First, when they make positive impressions, individuals generally maximize the rewards they get from others (e.g., getting a job, making friends) and minimize the costs (e.g., avoiding public speaking). Second, making positive impressions on others is associated with more positive moods and higher self-esteem, while negative impressions tends to lead to more negative moods and lower self-esteem. Third, if other people have a specific impression of an individual (e.g., a good athlete), that helps reinforce how that individual sees himself or herself (i.e., self-concept). In general, some level of concern for what others think is thought to be adaptive and functional.

Two processes are proposed to underlie self-presentation. The first is impression motivation—the desire to create a certain impression in others. The second is impression construction—how individuals decide which images to create and how they go about creating them. Several factors can influence an individual's level of impression motivation. Some people are generally more concerned with what others think of them in most situations—they have high trait self-presentational concerns. However, certain situational factors can also heighten self-presentational concerns as well. For example, if the goal (what people are trying to achieve by making a specific impression) is important, they will be more motivated to create an impression. Similarly, if someone thinks that how others currently see them (current image) is very different from how they want others to see them (desired image), impression motivation will also be higher. Impression construction can also be influenced by both traitlike and situational factors. For instance, an individual's self-concept influences the images he or she may wish to project. Situational factors include the values of the target (who people are trying to make an impression upon) or how people are currently perceived by others. For instance, people may try to create an image consistent with the values of the target and similar to their current images.

Self-presentation can occur in almost any aspect of one's life, including sport and exercise settings. In these settings, self-presentational concerns can impact how, or even whether, people participate in sport and exercise. In addition, sport and exercise participation can impact self-presentational responses.

Motivation for Exercise and Sport Participation

For many people, self-presentational concerns can motivate them to participate in physical activity (PA). One specific concern relates to physical appearance. In particular, in North American society, there are well-defined ideal physical appearance standards for women (thin, toned body) and men (muscular, lean body). Participation in sport and exercise can help people maintain or improve their physical appearance and achieve these ideals. Thus, for some, the desire to appear more physically attractive to others motivates them to be active. In addition to physical appearance, the wish to be perceived as an exerciser or an athlete may

also encourage people to be more physically active. One reason is that people described as exercisers are perceived more positively than non-exercisers on a variety of physical and personality characteristics. Similarly, being seen as athletic (e.g., strong, coordinated, quick) can also lead to more positive outcomes. For instance, many individuals identify professional athletes as their heroes. Thus, people may participate in PA to achieve these positive social images.

Alternatively, self-presentational concerns can also demotivate people from being active. Often, if people believe they will make a negative impression during PA, they may avoid being active. These negative impressions can also relate to physical appearance and social identity. For instance, since PA by definition places a focus on the body, and often involves wearing more revealing clothing (e.g., shorts and a T-shirt), people may worry about others thinking they are physically unattractive. Similarly, individuals may worry that they may appear weak or unfit if they can't lift as much weight or cycle as long as others. In sport settings, they may worry about looking unskilled if they make a mistake. Thus, if individuals believe that they will create a negative image (e.g., physically unattractive, unfit, or unskilled) then they may avoid PA altogether.

Physical Activity Context

Even if people choose to be physically active, the context of their activities can be influenced by self-presentational concerns. For example, if individuals believe they may make a negative impression, they may choose to be active alone—such as exercising at home. Another aspect of the PA context that is influenced by self-presentational concerns is the nature of the social environment. For example, many women indicate preferring to exercise in all-female rather than coeducational settings because they feel more comfortable with respect to their bodies.

It is also evident that some people may choose to avoid participating in certain activities because of the stereotypes that are endorsed about participants of certain activities. For example, some sports are considered more “feminine” (such as figure skating or gymnastics), while others are considered more masculine (such as football or boxing). Because of these stereotypes, some men may decide not to participate in certain sports to avoid

being seen as too feminine, and some women may avoid certain sports so they are not seen as too masculine.

Effort and Exertion

People's level of effort during PA can also be impacted by self-presentational concerns. In some cases, people may tend to work harder when others are around. If people want to appear fit and hardworking to others, they may put more effort into their workout or their sport. For instance, they may lift more weight at the gym if someone is working out beside them, or they may run faster to chase a ball during a soccer game if there are others watching. On the other hand, some people may report working less hard to make the same impression. For instance, if during a challenging fitness activity someone says it feels easy, she or he may be trying to create the impression of being highly fit.

One particular self-presentational phenomenon that is relevant in sport is self-handicapping, which occurs when people set up impediments (either behavioral, such as not practicing, or claimed, such as saying they feel sick or anxious) in advance of their performance. Thus, if people fail, they can attribute that failure to an external factor (e.g., sickness, lack of practice, highly skilled competition). If they succeed, they manage to look even more talented or skilled, as they were able to overcome the impediment in front of them. In general, self-handicapping is more likely to occur in situations that are considered very important, where the outcome is uncertain, or that are highly public.

Self-Presentational Responses to Physical Activity

While self-presentational concerns can affect choices regarding PA, participation in PA can also lead to certain self-presentational responses. Specifically, participating in sport and PA can increase, or decrease, the experience of certain self-presentational concerns. In general, these concerns are related to social anxiety—the worry or concern that arises when people want to make a specific impression (impression motivation) but are not sure they will be successful.

In sport settings, one of the most common self-presentational responses is competitive anxiety, which is the anxiety that results specifically in competitive sport situations. Although anxiety can originate from many sources, some of the concerns

are self-presentational in nature. For example, athletes may worry about embarrassing themselves in front of fans, teammates, and coaches if they make a mistake or look unprepared or unskilled. They may also worry about evaluations of their body, particularly in sports where physical appearance is judged as part of the outcome (e.g., gymnastics and figure skating), where there are weight restrictions (e.g., rowing or wrestling), or where clothing is particularly revealing (e.g., swimming and diving). In extreme cases, this anxiety may lead to choking under pressure—failing in situations where people are trying particularly hard to succeed. People may place so much pressure on themselves to succeed (and make a positive impression) that they are unable to perform their best.

A second self-presentational response that is common in sport and exercise contexts is social physique anxiety (SPA), which is the anxiety that occurs when one's body is evaluated by others. Interestingly, the level of SPA has been associated with both higher and lower levels of PA. For some people, high concern over others' evaluations of the body leads them to be more physically active, while for others, it is associated with less PA, perhaps to avoid the possibility of their body being evaluated by others. For athletes, participation at higher competitive levels is generally associated with lower levels of SPA—perhaps they become used to their body being evaluated by others or perhaps those very high in SPA drop out of sport early on. In addition, certain factors in the PA environment may make SPA more likely. For example, the presence of mirrors, the presence of men (for women), leadership style, and the wearing of revealing clothing (by oneself or others) are all associated with higher SPA levels.

A third self-presentational response is self-presentational efficacy, which is the confidence people have in their ability to create a desired impression in others. In exercise and sport settings, these images most often revolve around being seen as fit, skilled, and physically attractive. Generally, individuals higher in self-presentational efficacy are more active than those lower in self-presentational efficacy. They also report lower SPA.

Physical Activity Interventions: Their Impact on Self-Presentational Concerns

Generally, PA participation can help reduce self-presentational concerns. Aerobic and strength

training programs have been associated with reductions in SPA and increases in self-presentational efficacy. In addition, there is some evidence that even a single exercise session can lead to these same improvements in self-presentational concerns. Further, these improvements have been found in college men and women, as well as middle-aged and older adults. Interestingly, it appears that these changes are not necessarily linked with actual changes in the body—rather, changes in self-efficacy and perceptions of fitness improvements may be equally influential.

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See also Body Image; Choking; Self-Conscious Emotions; Self-Efficacy; Self-Handicapping; Social Physique Anxiety

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SELF-REGULATION

Participation in sport and exercise requires self-regulation. Examples of self-regulation include an individual fighting the urge to stop during an intense exercise class, a golfer trying to ignore feelings of excitement and focus on a simple putt to win a tournament, and a soccer player resisting the desire to strike an opponent who has repeatedly fouled him. The process by which individuals consciously attempt to constrain unwanted thoughts, feelings, and behaviors and bring these in line with ideals or goals is termed *self-regulation*, or *self-control*. The ability to self-regulate has been

shown to contribute positively to performance and behavior in a number of domains, including sport and exercise.

Athletes and exercisers use numerous self-regulation strategies. For example, in an effort to control anxiety before an important game, a hockey player may engage in the regulation of his thoughts by using self-talk to reappraise the importance of the game, imagine performing well, and distract himself by listening to music. An exerciser fighting the impulse to stay in and watch television instead of working out may do so by using motivational self-statements (e.g., I will feel much better after exercising), imagining achieving her fitness goals, or seeking the support of a fellow exerciser. The aim in using these strategies is to make achieving an important goal (e.g., play well in the hockey match or maintain fitness) more likely by regulating thoughts, feelings, and behaviors that can detract from the goal (e.g., worrying about the outcome, watching the TV). Overriding the impulses that negate our goals is not easy, and to understand why, the core aspects of self-regulation are first outlined here. Following this, a discussion is provided about when self-regulation may be impaired. The entry concludes with a discussion about how self-regulation can be developed.

Core Aspects of Self-Regulation

Self-regulation comprises four core aspects: *standards*, *monitoring*, *strength*, and *motivation*. Regulation means a change to bring into line with some standard, which is not possible without keeping track or monitoring current states, and so the first two aspects are inextricably linked. For example, if a tennis player is to reduce her angry outbursts on the court in line with the umpire's standards, she must monitor her thoughts, feelings, and behaviors to curb displays of frustration following unfavorable line calls. This "curbing" is difficult and, much like any effortful task, requires energy. Therefore, self-regulatory strength will determine whether the tennis player is capable of avoiding displays of frustration. Motivation to change is also necessary; in this example, if the outbursts are leading to point penalties, this may increase the tennis player's motivation to change her responses. The core aspect of self-regulation that has generated most research is the notion that self-regulation requires energy, and as such can be

depleted. This proposition is central to the strength model of self-regulation, which is now discussed.

The Strength Model

The strength model was proposed by Roy Baumeister and colleagues in the early 1990s, and its central premise is that self-regulation deteriorates over time from repeated exertions, as if reliant on a finite energy source. A consistent body of research has supported this view. This research has typically used a dual-task paradigm in which one group of participants completes a task requiring self-regulation and one group completes a task that does not. Both groups then complete a second task that does require self-regulation. The key finding is that participants are more likely to perform worse at later tasks requiring self-regulation if they have already done a task requiring self-regulation. To illustrate, participants in one study who first completed a thought-suppression task requiring self-regulation (try not to think of a white bear) were less able to resist the temptation of a beer before completing a driving simulator task compared with participants who had first completed a five-minute mental arithmetic task. Further, the energy source required for self-regulation is depleted regardless of the type of self-regulation. Of particular interest to sport and exercise participants is that physical performance can be affected by the regulation of thoughts or emotions. To illustrate, self-regulation on a cognitive task (e.g., Stroop test) has been shown to deplete performance on muscular endurance tasks (e.g., handgrip test), with participants having to work harder to achieve the same level of physical performance. The findings suggest that self-regulatory depletion may cause central fatigue, which can be thought of as the failure of the central nervous system (CNS) to drive the muscles. Thus, athletes and exercisers striving to control their thoughts and emotions in competition may be more prone to suffer disruptions in endurance performance.

In sum, research consistently shows that after an initial bout of self-regulation, subsequent self-regulation is disrupted, regardless of the sphere. In other words, it matters little whether the initial bout of self-regulation task involves the control of thoughts, feelings, or behaviors; subsequent self-regulation in the same, or different, sphere is disrupted. This clearly has implications for sport and exercise participants. It could explain why after

a difficult day at work in which self-regulatory resources are depleted, an individual is less able to resist the lure of a TV dinner on the couch and misses his exercise class, while a netball player who depletes her self-regulatory resources in regulating anxiety before a crucial game is less able to resist feelings of tiredness than normal and get up and down the court as efficiently as usual. However, to date we do not have any research specifically in sport and exercise settings exploring of depletion of self-regulatory resources may affect behavior and performance.

The Importance of Glucose for Self-Regulation

To this point, self-regulation has been described as a limited resource, with no explanation provided for *what* is depleted. Most researchers agree that glucose is the crucial energy source for self-regulation and, specifically, that lower levels and the inability to transport glucose effectively to the brain is associated with impaired self-regulation. This does not mean that the more glucose one has the better one can self-regulate. It simply means that self-regulation consumes glucose, and if enough glucose is replaced, self-regulation will be maintained, but if glucose is not replaced subsequent self-regulation efforts are more likely to fail.

Glucose is considered as the finite energy source because the availability and effective use of glucose is paramount for effective brain function, and if the flow of glucose to the brain is inadequate, cerebral functioning is impaired. So it is not surprising that some research has found that self-regulation is associated with changes in glucose levels. For example, controlling attention, regulating emotions, and coping with stress have been associated with a reduction in blood glucose levels compared to tasks that require no self-regulation. However, the data on which these conclusions are based have been questioned. Furthermore, the amount of additional glucose consumed by the brain during the self-regulation tasks employed in these studies, which typically last a few minutes, has been proposed to be minimal at 0.2 calories. So, at the present time, there is uncertainty as to the effect of self-regulation tasks on glucose levels.

There is, however, other research that points to the role of glucose. Low glucose, and inefficient glucose use, has also been shown to be associated with poor self-regulation. For example, low glucose is linked to less vigilance, more aggression,

greater distractibility, higher incidence of emotional disorders, increased emotional outbursts, and impaired performance under stress, while inefficient glucose use is linked to impaired attentional control, poor emotion regulation, and greater impulsivity. The administration of glucose has been shown to counteract the effects of prior exertions of self-regulation. That is, performance on a second task requiring self-regulation is not impaired even after completing a prior task requiring self-regulation provided glucose levels are replenished. In sum, although there is debate about some of the currently available evidence, proponents of the strength model propose that self-regulation appears to be susceptible to changes in glucose levels, and self-regulation failure is more likely when glucose levels are low or when glucose is metabolized inefficiently.

To complement the research supporting the central role of glucose in self-regulation, and the conflicting research suggesting it has a much more limited role, a recent proposal outlines an alternative way in which glucose plays a role in self-regulation. This proposal is that the distribution of glucose, rather than glucose level, determines effective self-regulation. To explain, with ample blood glucose available and motivation maintained to persist in a task, self-regulation is likely to be successful. In the event of low blood glucose availability, as long as motivation to persist is maintained and the task is deemed important for the individual, self-regulation can still be successful, because glucose is redirected to the areas of the brain needed for self-regulation. This could explain why even after a difficult day at work, in which self-regulatory resources are depleted, individuals sufficiently motivated to exercise will be able to resist the lure of a TV dinner, and a netball player who depletes her self-regulatory resources in regulating anxiety before a crucial game, but is highly motivated to win, is still able to work as hard as normal on the court. Again, we do not however have any research specifically in sport and exercise settings exploring the interaction between motivation and levels of glucose depletion.

Improving Self-Regulation

Apart from supplementing glucose, there are four other proposals for strengthening self-regulation: *conservation hypothesis*, *motivation*, *recovery hypothesis*, and *self-regulation exercises*. The

conservation hypothesis holds that individuals can reserve some self-regulatory energy if they are made aware of a subsequent self-regulation task. For example, in one study, after an initial task requiring self-regulation, participants performed a second task with half informed that there would be a third task. The depletion effects in the second task were most pronounced among participants who expected a third task. People do seem able to conserve self-regulation resources for an expected task.

The power of motivation to overcome depletion also suggests that self-regulation is never completely exhausted but temporarily depleted until sufficient motivation is generated to encourage the expenditure of further resources. Laboratory studies indicate that participants can self-regulate successfully if they are offered incentives to do so—even after first being depleted of self-regulation resources. This shows how self-regulation can be maintained in an important competition or meaningful exercise activity.

Sleep also has an important part to play in self-regulation. Insufficient sleep is linked to poorer self-regulatory capacity, and sleep replenishes self-regulatory resources. This idea is similar to the conservation hypothesis but is more akin to a recovery hypothesis. Studies have shown that a rest period between self-regulation tasks results in replenishment of resources and that rest and relaxation allay depletion effects.

Repeated self-regulation exercises can also enhance self-regulatory capacity, much like building a muscle through physical training. Studies have shown that, when compared with a control group, participants who practice self-regulation, such as avoiding sweet foods for 2 weeks, improve performance on other self-regulation tasks. In other words, in the short term, self-regulation depletes resources, but in the long term, repeated self-regulation makes a person's ability to self-regulate stronger.

Conclusion

Given that sport and exercise environments frequently give rise to intense emotions and can test a person's ability to regulate behavior (e.g., effort, decision making [DM]), it is not surprising that self-regulation should be a crucial aspect of participation in sport and exercise. Failures to self-regulate may be reflected in emotional outbursts, poor decisions, and a reduction in physical performance.

While the ability to self-regulate may be depleted, under what circumstances this happens and the interaction between motivation and levels of glucose is still to be determined by researchers. Further exploration of how self-regulation can be developed over time is also warranted given its clear applications to sport and exercise settings.

Marc V. Jones and Martin J. Turner

See also Aggression; Effort; Emotional Responses; Imagery; Psychological Skills; Relaxation; Self-Talk

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SELF-SCHEMA

A self-schema is a cognitive structure involving some aspect of the self that is considered important to the individual, such as exercising or participating in a sport. It is derived from generalizations based on past experience. Self-schemas are not only an important part of our overall self-concept (i.e., how we think of ourselves) but they also serve to facilitate the processing of incoming information and to guide behavior. For example, self-schemas

are thought to influence what we notice, how quickly we process schema-related information, what we remember, and what we do.

Background Information

The term *self-schema* was introduced in 1977 by Hazel Markus, who based self-schema theory on cognitive psychological theory and research on schemas (or schemata). Schemas contain highly organized knowledge in a particular domain. Consider, for example, a schema for a physical object such as a chair. We cannot remember everything about every chair we have ever encountered. Instead, we form a concept of a chair that contains information about different types of chairs, the function of chairs, how you sit on a chair, and some experiences with chairs. Having a chair schema allows us to quickly recognize a chair when we see one and to know what to do with it.

Just as there are schemas for physical objects such as a chair, there are self-schemas for a variety of aspects of the self including, but not limited to, personality traits (e.g., independence–dependence, assertiveness), appearance, gender, race, body weight, sexuality, fashion, and mathematical or scientific self-schemas. There are also self-schemas involving health-related behaviors such as drinking, dieting, eating healthy, and exercising.

Exercise and Sport-Specific Self-Schemas

Within the physical activity (PA) literature, research has focused on exercise self-schemas; sport-specific (baseball and softball) self-schemas have also been studied. Although the overall content of self-schemas has not been examined, an exercise self-schema, for example, would likely contain knowledge about what constitutes exercise (e.g., running, weight lifting), how to do the exercise activity, exercise-related clothing and equipment, exercise outcomes (e.g., increased energy, weight control), the physical sensations involved in exercise (e.g., the feeling of pedaling a bicycle or of sweating), as well as how one feels about doing different forms of exercise, one's past exercise experiences, and so on.

Exercise Self-Schema Measurement and Terminology

To determine whether a person has a self-schema, researchers can use reaction time (RT) measures

because incoming information that is consistent with highly organized knowledge is processed quickly. This strategy is seldom employed because it is impractical for most field studies and for studies with large numbers of participants. Instead, such methodology has been used to validate a paper-and-pencil self-report measure of exercise self-schema status.

With the self-report method, ratings are made of both the self-descriptiveness and the importance to the individual's self-image of three exercise-related phrases (*someone who exercises regularly, physically active, someone who keeps in shape*). Individuals then are classified into groups based on these ratings. *Schematics* are individuals who have clear images of themselves in regard to exercise (as evidenced by either extremely high or extremely low self-descriptiveness ratings) and who consider the exercise domain as extremely important to their self-image (as evidenced by extremely high importance ratings). Two types of schematics can be identified: *exerciser schematics* and *non-exerciser schematics*. Individuals in both groups view exercise as extremely important to their self-images. However, individuals in the former group clearly see themselves as exercisers; those in the latter group clearly see themselves as non-exercisers. Only a very small percentage of participants studied so far have been non-exerciser schematics, making it difficult to gain an understanding of them. The non-exerciser schematic group may actually consist of two subgroups of people: those who want to be exercisers (thus making it an important domain for them) but who do not yet see themselves as such and those for whom *not* being an exerciser is an important part of their sense of self (i.e., proud couch potatoes).

In contrast, *aschematics* are individuals who do not have a clear self-image in regard to exercise (as evidenced by moderate self-descriptiveness ratings) and who do not consider exercise highly important to their self-image (as evidenced by low to moderate importance ratings). Last, individuals who do not fit the criteria for classification as exerciser schematics, non-exerciser schematics, or aschematics are termed *nonclassifiabiles*. When there are theoretical reasons to contrast those who do versus do not have an *exerciser* self-schema, the latter three groups (non-exerciser schematics, aschematics, and nonclassifiabiles) may be combined into a *non-exerciser self-schema* group, which can then be contrasted with the exerciser schematic group.

Terminology differs across research labs; individuals termed *non-exerciser schematics* in one lab are sometimes termed *nonschematics* in another, and the no exerciser self-schema group has sometimes been termed *unschematics*.

Although alternative self-rating methods for identifying self-schema status have been used involving only self-descriptiveness ratings, only importance ratings, or the addition or multiplication of self-descriptiveness and importance ratings (or self-descriptiveness, importance, and certainty ratings), these methods are incompatible with Markus's conceptualization of self-schema. The first two methods equate self-schema with either past experience or importance to the individual and these challenges are discussed in the "conceptual issues" section. The multiplication method, on the other hand, can result in the same numerical score being assigned to people from different self-schema categories, as well as in non-exerciser schematics and nonclassifiabiles receiving either higher or lower scores than aschematics.

Correlates of Exercise and Sport-Specific Self-Schemas

Researchers have found numerous cognitive, affective, motivational, and behavioral correlates of self-schema status. For example, consistent with self-schema theory, baseball or softball schematics possess more baseball or softball knowledge (e.g., rules, strategy) than do aschematics; they also have more trait sport confidence, enjoy the sport more, participate in it more frequently, and expect to participate in it more in the future.

There are many theoretically meaningful correlates of exercise self-schema status. Exerciser schematics not only endorse as self-descriptive more words and phrases related to exercising and fewer related to not exercising, they take less time to make schema-consistent judgments (e.g., indicating that an exercise word is *me*), predict that they will be more likely to engage in future pro-exercise behavior (e.g., to use the stairs instead of taking the escalator at the mall), and recall more specific instances of past exerciser behavior and fewer specific instances of non-exerciser behavior. Interestingly, exerciser schematics show attentional bias for exercise-related words; that is, their attention is unconsciously drawn more to exercise-related words. Likewise, non-exerciser schematics show attentional bias for sedentary

words. In addition, exerciser schematics perceive themselves as more physically fit, have greater exercise self-efficacy, have more positive attitudes toward exercise and fitness, are more interested in and committed to exercising, and have stronger exercise intentions. They also report having more autonomous reasons for their goal of exercising although not less controlled reasons; these reasons include both identified reasons (i.e., exercising because they feel it is important) and intrinsic reasons (e.g., exercising because they enjoy it). Exerciser schematics make more plans to help themselves to exercise regularly and have more tricks or strategies for getting themselves to exercise on days when they do not feel like exercising. They also view the causes of their own (but not others') exercise lapses as less stable than do individuals who lack an exerciser self-schema, even when the causes they mention are similar (e.g., being lazy, stressed, or overloaded); this may help schematics recover better from an exercise lapse.

Given these findings, it is not surprising that undergraduate exerciser schematics have reported exercising more frequently over various time periods than aschematics and non-exerciser schematics, as well as that they do more activities for exercise—and their exercise caloric expenditure over the past week has been found to be higher. In addition, a higher percentage report having exercised at least three times per week in the past semester, having adopted an exercise program during the semester, and having begun to exercise regularly after returning to college from an extended holiday. These behavioral data should be viewed with caution because they are based on self-reports. Fortunately, an unobtrusive objective measure of exercise behavior (attendance at a university fitness facility) provides converging evidence that exerciser schematics engage in more exercise activity. Interestingly, this behavioral difference seems to be due to exerciser schematics' greater tendency to act on their exercise intentions, a finding obtained in studies involving both self-reported and objective measures of exercise.

Conceptual Issues

Self-schema status has never been, and probably cannot be, experimentally manipulated. Consequently, if a relationship is found between self-schema status and some variable, that relationship is correlational in nature. When people

who have different self-schema statuses differ on some other variable, it is therefore inappropriate to conclude that self-schema status actually caused the difference. For example, we know that exerciser schematics act on their exercise intentions more, but we cannot conclude that this is due to their having a self-schema. That *may* be the case but other explanations are logically possible as well. For example, the differences might have been caused by some other variable on which exerciser schematics and the group with which they are being compared (e.g., aschematics) differ—such as self-efficacy or past behavior. So, although self-schema theory predicts that having a self-schema will affect (i.e., cause) information processing and behavioral effects, all we can logically conclude when we find differences along these lines is that the data are consistent with the theory.

A second issue involves how the concept of self-schema, a stable cognitive structure, can be reconciled with the growing body of social psychological research that supports the idea that the self is not stable but instead changes in response to the situational context. The answer lies in the current conceptualization of the self as a working self-concept. The working self-concept includes only a subset of the individual's self-knowledge—namely, those elements of self-knowledge that are activated at the moment. Self-schemas are highly accessible (i.e., easy to activate in memory), so they are likely to be activated across situations. Consequently, self-schemas can be seen as providing some stability within the working self-concept.

Finally, the fact that exercise self-schema status is typically assessed using ratings on two dimensions raises the issue of whether the exercise self-schema findings might really be due just to either past experience (as assessed by self-descriptiveness ratings) or the importance of exercise to the individual (as assessed by importance ratings). There are both conceptual and empirical arguments against these alternative explanations of self-schema findings. From a conceptual standpoint, exercise experience and considering exercise important to the self are each necessary but not sufficient for the development of a self-schema. For example, even if someone has a lot of experience with exercise, he or she will not necessarily develop an exercise self-schema; exercise must also be considered important to his or her self-image. From an empirical standpoint, if self-schema simply reflects past experience, then

we would not expect to obtain the same types of information processing effects for exerciser and non-exerciser schematics because they presumably differ in exercise experience. However, we do obtain them—for example, both exerciser and non-exerciser schematics show attentional bias for schema-consistent words. Likewise, if it were the case that self-schema simply reflects how important exercise is to the individual's self-image, then exerciser schematics and non-exerciser schematics should not differ in their exercise behavior because they both report that exercise is highly important to their self-image—but they do.

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See also Adherence; Identity; Self-Efficacy; Self-Regulation

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reflect positive (e.g., I can do this) or negative (e.g., don't screw it up) verbalizations. However, there is also an interpretative element associated with self-talk, which is idiosyncratic and potentially more important than the content of self-statements per se. For instance, while two exercisers might say the same phrase to themselves when fatigued (e.g., this is tough going), one may view the statement as an indication to give up, whereas the other might interpret it as a sign that the intensity she is working at is the appropriate level and to keep going. Self-talk is sometimes referred to in the research literature as private speech, verbal rehearsal, or inner dialogue.

Although encouraging athletes to use particular types of self-talk is commonplace within the sports setting, when compared with most other mental skills, self-talk remains relatively under-researched despite researchers' having adopted a more systematic approach to the study of self-talk over the past decade. Some early research examined the effects of training athletes in the use of self-talk as part of larger mental skills training that involved training in skills such as mental imagery, relaxation, and goal setting. Such studies provided evidence supporting the use of mental skills packages but did not allow researchers to identify the effect of each individual mental skill. However, more recent research has been focused on self-talk alone. Systematic reviews of the research on self-talk have confirmed that the skill can be effective at enhancing performance and that these benefits hold across various sports or tasks and skill levels. That said, there is a relative dearth of research on the effectiveness of training skilled performers in the use of self-talk; most studies have involved unskilled university students as participants. There is also little research on the effects of self-talk on performance in real competitive settings, as opposed to on laboratory-based tasks or in practice settings.

Structures and Forms of Self-Talk

The structure of self-talk can range from single “cue” words (e.g., *head*), to specific phrases (e.g., *get there*), to full intact sentences; regardless, most self-talk is abbreviated in form. Also, abbreviated cue words or short phrases are usually taught in studies of self-talk training. It is suggested within the research literature that the content of self-talk interventions (i.e., programs of self-talk

SELF-TALK

Self-talk refers to statements that athletes and exercisers address to themselves; these might represent automatic verbalizations or more deliberate forms of speech. Although such statements can be said aloud, most self-talk is said covertly as a silent voice in one's mind. The nature of self-talk can also

training) should be limited to a few, phonetically simple terms, logically associated with movement phases integral to successful task execution. When employing these recommendations, research has generated data to support these claims; for example, saying instructional self-talk words such as *split* and *turn*, representing the parting of the feet to create a firm base and turning of the shoulders to control the racquet head, enhances the accuracy of a tennis net volley. There is also a research base supporting the use of motivationally oriented self-talk with tasks more reliant upon strength and power (e.g., a defensive clearance in soccer) than precision. Even though most of this research has used the rather bland motivational phrase *I can*, there is consistent evidence that the use of this benefits performance. This suggests that the uses of self-talk extend beyond the use of movement-based cue words to organize and prompt the execution of technical movement patterns.

Recognizing that more instructional as opposed to motivational forms of self-talk might influence task execution differently depending on the characteristics of the task at hand, a matching hypothesis has been presented within the research literature on self-talk. It states that, because instructional self-talk helps athletes focus on task relevant cues, it should be more effective than motivational self-talk for tasks dependent on technique and precision. Conversely, motivational self-talk ought to be more effective than instructional self-talk for the execution of gross, strength-based tasks because it helps the performer achieve a more appropriate mind-set reflecting confidence and a positive mood state. While there is some evidence supporting this hypothesis, at present the available literature suggests that the consistency or robustness of the different beneficial effects is questionable. For example, most studies find benefits for both types of self-talk but with no clear difference between the “matched” and “unmatched” self-talk for the task type. For instance, instructional self-talk (e.g., straight and clean, referring to the backswing and contact of a golf putt) is not significantly more beneficial than motivational self-talk (e.g., you can do this) for aiding execution of an accuracy-based task (e.g., 6-foot golf putt).

The categorization of self-talk as either instructional or motivational in terms of the function of self-talk is a relatively new within the research on self-talk concerned with sport. A traditionally held view among sport psychologists, still currently

prevalent, is that positive self-talk is to be encouraged over negative self-talk. To this end, a number of mental techniques (e.g., thought stopping, cognitive restructuring) have been espoused within the applied literature. Given the apparent interest in this aspect of self-talk, the accompanying lack of experimental examination is notable. Nevertheless, the available research does support the belief that positive self-talk can lead to enhanced performance although the opposite is not necessarily true for negative self-talk.

One key area to consider, therefore, is that performers’ interpretation of their self-talk may be of greater importance than its content. Early research exploring athletes’ self-talk identified that some athletes reported negative self-talk to be motivating. While the motivating effects of negative thinking may be only realized by certain athletes under certain circumstances, these findings emphasize the importance of discussing how an athlete views and responds to self-talk as an integral part of working with him or her. For example, if an extremely resilient athlete uses negative and self-critical self-talk to increase his or her own effort or refocus attention following a lapse in performance, this may be an entirely functional use of self-talk and not something a sport psychologist would necessarily want to change.

Recent theory-based studies of self-talk have examined its interpretation in greater depth. Research drawing from Edward Deci and Richard Ryan’s self-determination theory (SDT) has suggested that whether self-talk is interpreted as self-pressurizing or self-supportive may be an important determinant of subsequent motivation, emotion, and behavior. Specifically, self-talk that emphasizes the perspective of the athlete, provides the athlete with information and feedback about his or her competence, and fosters a sense of empowerment is likely to result in more positive forms of motivation, positive emotions, and ongoing task engagement and application of effort. Conversely, self-talk that is pressurizing, critical, and undermines personal empowerment is likely to result in a lack of task engagement and more negative emotional effects. Thus, a positively phrased self-instruction (e.g., keep your head still) perceived by the individual as controlling and pressurizing may in fact have negative consequences. Equally, a stern self-administered “talking to” may emphasize that the athlete has the ability to alter his or her situation, with adaptive consequences.

This aforementioned work goes some way toward helping address the general limitation associated with research on self-talk—namely, a lack of theory-based research and the absence of an actual theory of self-talk to date. In an effort to systematically make sense of the existing data, frameworks summarizing the effects of self-talk are currently being developed and refined. A sport-specific model, suggested by James Hardy and his colleagues, centers around the relationship of self-talk to performance, with theoretically grounded causes of self-talk and potential mechanisms helping to explain the performance effect identified. Specifically, the model emphasizes that both individual difference factors and situational variables can influence athletes' use of self-talk. Individual difference factors may include the athletes' preferences for processing information, their belief in the efficacy of self-talk, and also more global personality characteristics such as optimism, trait anxiety (TA), and neuroticism, for example. Situational variables include task difficulty, game circumstances (e.g., having lost an important point in a tennis game), and the influence of significant others (e.g., coaches). There is some evidence to suggest that athletes model their self-talk from coaches' comments and feedback, consistent with social learning-based models of behavior.

Pathways to Influencing Performance

In terms of the mechanisms explaining how self-talk might influence performance, four main pathways are highlighted: cognitive, motivational, behavioral, and affective. Although conceptualized as separate pathways, it is likely that the underpinning explanations actually work in combination. First, the category of cognitive mechanisms refers to processes such as information processing, concentration, attention control, and attentional foci. Athletes report using self-talk to aid concentration and to direct and redirect attention to selective and important aspects of the skills being executed. Specific cue words have been implicated in the deliberate changing from one attentional focus to another (e.g., prior to the start of a race a sprinter pulling her attention away from the cheering crowd and on to the immediate task at hand—driving as quickly as possible out the blocks after the gun blasts). There is also some evidence that self-talk can reduce the occurrence of more internally oriented distractions such as interfering

thoughts (e.g., task-irrelevant thoughts, such as What am I going to have for dinner?) while performing sport skills.

In terms of motivational mechanisms, self-talk may improve performance by triggering enhanced effort and/or greater long-term persistence. For example, self-talk may act as a form of verbal persuasion, improving an athlete's self-confidence, which in turn causes them to invest greater effort, for longer. However, to date, controlled experiments have found equivocal support for the role of confidence in the self-talk to performance relationship. Nonetheless, the use of specific motivational self-talk phrases (e.g., I can) has resulted in increases in athletes' confidence levels. Alternatively, motivation and, in turn, performance might be influenced by the interpretation of self-talk such that self-talk viewed as reinforcing ability and choice ought to be beneficial and phrases which are self-critical, increasing pressure will likely have detrimental effects.

Behavioral or biomechanical mechanisms underlying the effect of self-talk on performance have perhaps greater evidential support. Changes in athletes' form and movement patterns have been shown to result from the use of either cue words (e.g., "knee" referring to keeping one's knee over the ball when executing a low driven shot in soccer) or longer instructional phrases. Typically, these types of self-talk focus on segmented parts of a movement or action (e.g., phases of a tennis forehand or golf swing); however, some movement changes have been noted following the use of more generic instructional commands—for example, the use of the phrase *drive up* as an attempt is made at a vertical jump.

Last, self-talk may influence performance through a variety of mechanisms concerning the regulation of affective states (e.g., positive and negative moods) and arousal (e.g., being "psyched up"). Different patterns of self-talk are associated with a number of different mood states including depression, anger, anxiety, and so on, and counseling techniques often emphasize changing the nature of self-directed statements as a way of enhancing mood state. Although athletes frequently report using self-talk as a psyching-up strategy to increase levels of arousal, the effectiveness of self-talk for this function has not been experimentally determined. There is, however, some evidence linking the use of self-talk (e.g., cue word *calmly*) to the effective control of anxiety levels.

Conclusion

Although the development of the literature regarding self-talk has greatly advanced in the past decade, there remain many unanswered problems regarding how best to use self-talk, the way in which self-talk enables athletes to maximize performance, and, crucially, exactly why this might be. Contemporary models associated with self-talk have begun to provide some guidance regarding these questions; however, the role of key moderators, such as the athlete's skill level and the type of task being completed, has yet to be fully examined.

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See also Cognitive Restructuring; Positive Thinking; Thought Stopping

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SENSORIMOTOR REPRESENTATIONS

In anatomical terms, sensation and movement are each represented across distributed regions in the human brain. Somatosensory information—that is, information pertaining to sensation of the body—is gathered from the environment and transmitted to the brain via the spinal cord and via the thalamus, from where it is distributed to three regions of the parietal cortex: the primary somatosensory cortex (located in the postcentral

gyrus), the secondary somatosensory area, and the posterior parietal area. These areas are primarily involved in the representation of touch and of limb position, particularly as it occurs in the absence of visual feedback. The motor regions of the brain include the primary motor cortex, located in the precentral gyrus, the premotor cortex, and the supplementary motor cortex. These areas act in concert to contribute to the organization and production of movement and are collectively referred to as the *sensorimotor cortex*.

First illustrated by Wilder Penfield and Edwin Boldrey in 1937, the sensorimotor cortex, in particular the primary somatosensory and motor cortices, is topographically organized, meaning that particular areas of the body map onto particular areas of the cortex. Furthermore, the size of the representation is proportional to the sensitivity (primary somatosensory cortex) and dexterity (primary motor cortex) of the represented area, rather than its physical size. For example, the somatosensory representation of the thumb and the lips are each approximately as large as that of the entire leg. The reason is that the skin of the thumb and the lips are densely populated by cutaneous receptors, giving these areas a high degree of sensitivity and therefore large cortical representations, while the skin on the leg is far less densely populated by these receptors and therefore less sensitive. A similar topographical map exists in the primary motor cortex, in which parts of the body that perform highly dexterous movements, for example the fingers and tongue, occupy a larger volume of the primary motor cortex with projections to the muscles of these areas, compared with other parts of the body such as the trunk and pelvis.

On a cellular basis, neurons in the primary motor cortex are tuned to the direction in which a movement is performed. These cells are activated strongly when movements are performed in one direction in space and less so for movements in another. It is unclear, however, whether the preferred direction for particular cell populations is represented in an intrinsic coordinate system—for example, flexion and extension of a joint—or an extrinsic coordinate system—for example, movement to the left or the right.

Plasticity of Sensorimotor Representations

For many years, it was thought that there was little capacity for sensorimotor representations to

be altered in the adult brain. More recently, it has been demonstrated extensively that sensorimotor representations in the adult brain can undergo functional reorganization, a process known as *plasticity*, in response to damage to peripheral or central structures, or as a consequence of experience. For example, imaging studies have shown that arm amputees activate a larger than normal area of sensorimotor cortex when performing shoulder movements of the corresponding amputated arm, suggesting that the remaining arm muscles have an expanded representation. Similarly, transcranial magnetic stimulation (TMS), which can be used to map the area of sensorimotor cortex projecting to specific muscles, has demonstrated that motor practice results in an increase in the area of sensorimotor representation of the muscles involved in the practice. These observations confirm that sensorimotor representations are dynamic and malleable, depending on circumstances.

Motor Learning as a Tool to Understanding Sensorimotor Representations

Motor learning can be considered as a form of perturbation in which behavior and neuronal activity is examined before, during, and after a period of adaptation. The pattern of behavior during these three important periods of adaptation can reveal different properties of the movement representation in the brain. In particular, the brain's ability to generalize motor learning and apply what was learned to different contexts gives important insights into sensorimotor representation in the brain. For example, learning to hit a tennis forehand does not require exposure to all possible ball trajectories, only a subset. Once learned, the challenge is to optimize the generalization of the skill in the appropriate context. Too little generalization results in improvements that are limited to very specific conditions, whereas overgeneralization results in inappropriate actions for a given condition. For instance, it is no use trying to hit a forehand when a backhand stroke is required. The key point here is that patterns and constraints of generalization can reveal how movements are encoded in the brain.

Levels of Sensorimotor Representation

Complex sensorimotor skills can be represented in a number of different frames of reference. In the example of hitting a tennis ball, the brain must

first represent the purpose of the movement (hitting the ball), the required end-point result (racquet meeting the ball), the required joint torques to position and orientation of the arm, and finally the required muscle activation pattern to achieve the movement outcome. At each stage, there is a different sensorimotor representation that guides the building of the movement. During motor learning, the exact pattern of errors in the movements can give clues about the neural representation in the brain and the frame of reference that was used during the adaptation phase.

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See also Brain; Learning; Motor Control; Neuroscience, Exercise, and Cognitive Function; Transcranial Magnetic Stimulation (TMS)

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SENSORY SYSTEMS: AUDITORY, TACTILE, PROPRIOCEPTIVE

Sensory systems are the peripheral parts of the nervous system responsible for the transformation of physical stimuli into a neural code. Receptors of each sensory system are sensitive to a distinct kind of energy, like the hair cells of the inner ear to sound energy and the mechanoreceptors of the tactile system to mechanical energy or the visual receptors to electromagnetic energy. Each system encodes four essential features of a stimulus. Besides stimulus modality, these are intensity, duration, and spatial location. Vision, hearing, kinesthesia, and touch serve as initial functions for

generating information within the central nervous system (CNS), which in turn contributes sensations and bottom-up input to perception and up to higher cognition. Vision is discussed more in depth in a separate entry, whereas the roles of hearing, kinesthesia, and touch are discussed more fully in this entry.

Nearly all kinds of goal-directed actions in sports rely on currently updated information about the environment as well as about one's own posture and kinesthesia. Though vision is often understood as the most important sensory modality concerning the regulation of actions in sports, it is also crucial to realize that vision only can work in relation to the perceived self. Information about the physical self from different modalities—tactile, proprioceptive and vestibular information—is integrated within a multisensory supramodal coherent representation, the *body scheme*. This body representation is broadly involved into spatial sensorimotor processing of skilled action and it is continuously updated with ongoing movements. Patrick Haggard and Daniel Wolpert distinguished between a body scheme and a *body image* much more dependent on conscious visual representations.

The organization of the somatosensory representation—as a main element of the body scheme—is related to the localization of a distinct area on the body surface. The region of skin from which a tactile stimulus will alter the firing rate of a receptor neuron is called the *receptive field*. Receptive fields have been found for neurons of the somatosensory, the visual, and the auditory systems. Neighboring cutaneous receptive fields from the human skin are represented also in adjacent cortical fields, leading to a *somatotopic* organization of the somatosensory cortex in the form of a *homunculus*. A *topographic* organization of the cortical projection pattern can also be found for the visual system in the primary visual cortex as well as a *tonotopy* pattern in the primary auditory cortex, altogether known as *sensory maps*. Nevertheless visual input is integrated into the body scheme in a mostly unconscious manner, as shown by Michael Graziano and coauthors at the end of the 20th and beginning of the 21st century, and is further modulated by auditory input. Multisensory integration is particularly crucial for the emergence of a coherent body scheme, and some empirical findings illustrate the continuous impact of intersensory adjustments. Examples include the rubber

hand illusion, reported by Matthew Botvinick and Jonathan Cohen, and an altering of perceived size of a single body segment evoked by the vibration of related tendons, reported by Jim Lackner.

Sensation Versus Perception

In contrast to perception, which explicitly includes the level of awareness, sensory sensations do not usually reach the level of conscious awareness. Sensation is the detection of basic features of a stimulus. It is understood as the first stage of an activation of a sensory system, as a peripheral internal representation of distal stimuli, clearly dominated by bottom-up processes. Nevertheless sensory information can directly influence motor behavior and ascending sensory pathways have several linkages to descending motor functions. Besides such physiological indications for sensory motor circuits, there are supplemental references from the field of applied kinesiology. Here, special training procedures as well as scientific approaches in fields like sensorimotor control, sensorimotor training, or sensorimotor adaptation can be found; they involve certain subdomains of motor behavior with a particular low impact of conscious attention such as the regulation of balance or posture, as well as speech articulation. On the other hand, perception is based on higher brain functions and the bottom-up fractions are supplemented much more by top-down fractions like experiences and expectations when interpreting objects and events in the world. However, as the title of James Jerome Gibson's book, *The Senses Considered as Perceptual Systems*, suggests, it is utterly impossible to definitively separate levels and functions of the sensory system from that of the perceptual system.

Proprioception: Kinesthesia, Touch, and the Vestibular System

Proprioception can be subdivided into two or three subsystems describing the tactile system as part of proprioception.

1. Kinesthesia (literally, *sense of movement*) generates information about the positions and movements of the limbs and other body parts including strength sensation. Based on kinesthesia, humans possess information about the rate and the direction of limb movements. This information can be self-generated or externally imposed and

can be encoded by different kinds of mechanoreceptors in joints and capsules (slowly adapting *Ruffini endings*), muscles (*muscle spindle receptors*), tendons (*Golgi tendon organs*) and ligaments (slowly adapting Golgi receptors). Muscle spindles signal limb position and movement by encoding muscle length; Golgi tendon organs encode muscle tension. Subtlety of kinesthesia can be determined by measuring the *just notable difference* or *detection threshold* for a joint, the *threshold displacement in degrees* as well as the *smallest angular excursions* or the *smallest rotation angle* of a joint. Reported values vary between joints and can be related to the initial joint angle, muscle loading, active versus passive effectuation, and age of the participants.

2. The tactile system, as well as parts of kinesthesia, is based on mechanoreceptors in the skin. Besides detection and specification of mechanical deformation of the skin (force, pressure), about 12 different kinds of cutaneous mechanoreceptors play a significant role in kinesthesia. Cutaneous mechanoreceptors can be subdivided into rapidly responding (A) vs. slowly adapting receptors (B).

- A. Receptors of the first subcategory are well suited for the specification of rapid movements and fine motor skills. The loss of skin sensation of the hand—as an example of a body part with a high degree of tactile resolution and a large receptive field in the cerebral cortex—causes severe deficits in fine motor hand skills.
- B. Slowly adapting receptors signal static position information as well as information about slow displacements. Several functions of the tactile system rely on such slowly adaptive proprioceptors, as shown by Type 1 and Type 2 receptors, which are associated with Merkel cell complexes and Ruffini endings and continue discharge with maintained deformation of the skin.

Kinesthetic information and tactile information are essential parts of the somatosensory system. Tactile information is entangled closely to kinesthetic information, as demonstrated by the bending of any joint that stretches a related region of skin around the joint and relaxes another area proportionately.

3. The vestibular system is located in the inner ear as three semicircular converging canals—the

superior, the posterior, and the horizontal canal—which indicate rotational accelerations; they are supplemented by the otolith organs in the saccule and utricle, which indicate linear accelerations. Besides balance control, the vestibular system contributes to spatial orientation and, via mediating structures, posture regulation, as well as control of eye movements. Balance regulation, as a well-known example of sensorimotor regulation, is not solely governed by the vestibular system; kinesthetic information and visual information are integrated seamlessly.

Hearing and the Emergence of Acoustic Events

Although it is obvious that vision is the dominating sense for many sports as well as for motor learning, which is frequently based on visual models, it might be surprising that hearing is a sense that is *exclusively sensitive to the perception of motion*. Sound is an acoustic consequence of a kinetic event—the existence of a kinetic event is essential for generating a sound event. If there is no movement in the surroundings, there is nothing to hear. The air or other media such as water and ground must be set into vibration by motion to generate and to transmit audible sounds within the hearing range of the human ear. Only sound events within the frequency range of about 20 hertz (Hz) and up to 20,000 Hz can be transformed within the hair cells located in the cochlea. Additionally, a minimal pressure of the sound wave—also dependent on frequency—is necessary to elicit an auditory sensation. As noted by Bertram Scharf and Søren Buus, a sound of 2 kilohertz (kHz) is audible at a sound pressure level (SPL) of 0 decibels (dB), a sound of 3.8 kHz at about –5 dB, as has been measured on young adults; both indicate the extreme sensitivity of the human ear. In comparison to the optic nerve, consisting of about 1,000,000 nerve fibers, the auditory nerve subsumes only about 30,000 fibers; though the number and interpretation is arguable, this may indicate that the ear is much more designed to analyze temporal features contrary to vision, which is more sophisticated in analyzing spatial features.

When setting the frequency range of human movements in relation to the hearing range, it is evident that humans are unable to hear their own movements directly. Highest motion frequencies have not been observed in sportsmen but in world-class pianists performing trills at

about 16 Hz and thereby below the threshold of the hearing range at 20 Hz. Only the impact of human movements on ambient surfaces, such as hitting the tennis ball with the racket or stemming the skis into an icy snow, induces audible sounds. But once a kinetic event generates an audible sound, there is much kinetic information coded acoustically. On a wooden surface, a bouncing Ping-Pong ball sounds different from a tennis ball. Kinetic features as well as properties of involved materials (balls, wood) specify the sound parameters systematically, which can be subdivided into two essential categories: material-induced sound features versus kinetically induced sound features.

The material category specifies sound features such as *spectral composition* and *sound envelope*, which includes the components *attack*, *sustain*, and *decay*. These parameters are specified by the physical parameters of the related materials and the related medium, the air. When hitting the tennis ball, the density of the racket as well as the tension of the racket string specify the sound in interaction with the special features of the tennis ball (material, pressure). Further features of the sound are assigned to the kinetic category, like amplitude and duration of sound, which are specified by kinematic and dynamic parameters. The kinetic energy of the approaching tennis ball as well as direction and spin specify *loudness*, *hardness*, *timbre*, and sound duration; the selected technique also influences these characteristics: A slice sound exhibits a longer duration and is of less hardness, so it sounds smoother than a straight backhand. There is much information about the kinetics encoded in sound, and motor control and motor learning are indeed supported by auditory information. Only a few experts are aware of the auditory impact on motor behavior because auditory information processing takes place widely in the background of conscious awareness.

Sensory Auditory Information in Sports

The impact of music and sound on motor behavior is multifaceted. Football players adjust with team members to fans' acclamation; they respond to the acoustical atmosphere within the arena, and they react to the rhythm and the accents of the referee's whistles. Sport gymnasts and dancers synchronize their movements to accompanying

music or arrange certain elements related to the melody individually, together with a partner or within an ensemble. Runners synchronize the step cadence to perceived music and even pace themselves over a race by arranging a special sequence of different music pieces with varying tempi for tuning their running speed over the course. Music can be used to modulate the *heart rate* (HR), the *perceived exertion*, or the *general arousal* as well as the *basic activation* of distinct brain regions. An overview on effects of music on sports has recently been published by Costas Karageorghis and David-Lee Priest.

In most of these correlations between music and motor behavior perceptions, higher cognitive functions are broadly involved: Music is audible only after auditory stimuli are processed by a widespread cortical network. However, a more direct, sensory-based impact of sound on motor behavior occurs as acoustic information from earlier stages of auditory processing is integrated into motor control. Natural motion attendant sounds are used for controlling and optimizing motor behavior: The performance of a tennis player decreases if auditory perception is occluded, and similar findings have been reported on table tennis. Synchronization in team rowing is partially realized by analyzing sounds of the boat and of the swivels. In a diversity of sports, the "all-around sense" of hearing directs the spatially restricted visual attention. For instance, in team sports the orientation of gaze is partially affected by surrounding sounds induced by the steps of teammates or opponents. Auditory input affects gaze behavior as well as orientation behavior fundamentally straight on the sensory level, when integrated with visual and proprioceptive input in the colliculi superiores. These structures located in the brain stem are on the other hand emitting motor efferences, which are directly specified by the special formation of the current multisensory input. The multisensory-motor characteristics of the colliculi superiores have been studied intensely by Barry E. Stein and M. Alex Meredith. This region can be understood as an outstanding example for a straight but flexible responding sensorimotor interface. It indicates how seamlessly multisensory information is merged together already on the subcortical level—and even down to the level of a single neuron: A multisensory convergence neuron integrates auditory, visual, and proprioceptive input,

and the same neuron responds with descending motor output.

Further Perspectives for Sports and Exercise

There is no doubt that skilled action in sports relies on supramodal representations. It has become evident that the integration of information from different sensory modalities in terms of an *early multisensory integration* is crucial for the emergence of a body scheme as well as for spatial representations, where visual, auditory, and body segment-related sensory maps are continuously integrated into supramodal representations. These representations act as a reference framework for the regulation of skilled action. For developing new methods of training and exercises prospectively, mechanisms of multisensory integration can be focused. Information from different modalities is integrated according to certain rules (spatial convergence, temporal coherence). If one of the engaged sensory streams can be shaped in a certain form or an additional perceptual stream can be created in a way that is integrated into the supramodal representation, a diversity of applications can be addressed in the future, to include enhancement of mental training, recalibration of the body scheme, or a directed adaptation of sensations of a distinct modality. The approach of *realtime movement sonification*, developed by Alfred Effenberg and currently neurophysiologically supported by Gerd Schmitz and colleagues, could be a method prospectively used in many different applications in sports and motor rehabilitation.

Alfred Effenberg

See also Coordination; Eye Movements/Gaze; Information Processing; Movement; Motor Commands; Motor Control; Motor Learning; Perception; Vision

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SERVICES

Sport and exercise psychology (SEP) services (e.g., team interventions, one-on-one interventions, and consultation with organizations) aim at performance enhancement with athletic populations and exercise adherence with physically active individuals. In addition, SEP practitioners design their interventions to promote healthy lifestyles and to contribute to their clients’ personal growth, psychological well-being (PWB), and quality of life (QOL). This entry deals with the foundations, recipients, goals, approaches, and delivery considerations involved in SEP services.

Knowledge Base

SEP services are based on an interdisciplinary integration of basic and applied knowledge sampled from (a) SEP, (b) psychology (e.g., counseling psychology, health psychology, clinical psychology), (c) sport and exercise science (e.g., exercise physiology, motor control, sociology of sport), and (d) consulting.

Recipients of Services

SEP services can be offered to (a) organizations (e.g., fitness club franchises, single fitness clubs, sport academies), (b) groups (e.g., sport teams, fitness classes), (c) individuals responsible for helping others achieve goals (e.g., coaches, fitness instructors), and (d) individuals concerned with their own athletic or fitness goals (e.g., athletes, exercisers). Thus, SEP services can, for example, help to formulate systematic mental training services at a sport academy or implement new customer services procedures aiming at increasing intrinsic motivation of fitness club members (organizational level). Teams and fitness groups can use SEP services to increase a sense of cohesion to promote time-on-task productivity and adherence. Further, SEP services can aim at enhancing day-to-day operations of individuals (e.g., improved way of providing feedback for coaches, personal trainers, parents) who further support their trainees, clients, or children. Finally, SEP services assist individuals directly (e.g., an athlete or exerciser can learn and apply positive self-talk to manage distractions while engaged in their respective athletic or exercise activities).

Service Goals

Areas of an individual's psychological functioning that can be improved through SEP services include, but are not limited to, motivation, self-confidence, stress management, resilience, mental toughness, and attention management. Improved psychological functioning aims to enhance the efficacy of practice and exercise drills improve competition outcomes, enhance rest and recovery, and overcome setbacks and failures (e.g., injury, relapse to sedentary lifestyle). Additionally, the goals of the clients' personal growth, PWB, and quality of life (QOL) are pursued in an integrated fashion. SEP services can also target group dynamics and aim at team building (for both athletes and exercisers)

through establishing productive norms, roles (e.g., leadership), and communication to achieve high levels of cohesion, including task (goal advancement) and social (interpersonal relationships) dimensions.

A long-term goal of SEP services is the recipient's self-regulation in relation to the target behavior change (e.g., stress management while performing highly demanding motor tasks or exercise adherence during a major life transition). Thus, the client's independence and self-reliance are promoted.

Service Approaches

The numerous goals of SEP intervention programs can be accomplished through educational and psychological methods (or interventions), both sensitive to multicultural issues. Educational interventions commonly include psychological skills training (PST) and team building and utilize a single method or a combination of methods (i.e., multimodal mental training). For example, PST methods for performance enhancement or exercise adherence are based on a number of cardinal psychological skills such as, for example, activation management (i.e., relaxing and energizing), imagery, self-talk, concentration, goal setting, and mental routines. Alternatively, psychological (as contrasted with educational) interventions can involve cognitive restructuring; mindfulness, acceptance, and commitment approaches; management of interpersonal relations; and other strategies that are grounded in a SEP consultant's theoretical paradigm in psychology (e.g., cognitive behavioral, humanistic, integrative).

Some psychological barriers (or mental blocks) to actualizing one's athletic potential or maximizing one's engagement in a physically active lifestyle can arise from clinical problems (e.g., eating disorders, affective disorders, exercise addiction) or major life adjustments (e.g., developmental transitions, injury, retirement, migration or immigration). To address these complex issues, clinical or counseling services (interventions) are needed.

Service Delivery Considerations

SEP service delivery is a multifaceted professional activity. Assisting a person, group, or organization in behavior change is a very complex process and requires adequate professional qualifications and competencies. In short, SEP service delivery

involves a number of important elements: (a) education, training, and professional experience (including supervised practice); (b) professional ethics; (c) professional philosophy (the practitioner's core personal values and beliefs, well articulated theoretical paradigm, and clearly defined model of practice along with unambiguous intervention goals and methods); (d) making contact (e.g., referral, advertising); (e) the working interpersonal alliance with the client; (f) assessment (e.g., through structured interviews, psychometric tests, observations, or combination of these); (g) conceptualizing athletes' concerns and potential interventions; (h) program implementation (quite often in the practice or exercise setting, competitive venue, or both); and (i) program and consultant evaluation.

Currently in the United States there are two educational paths required for competent delivery of SEP services: (1) sport and exercise science (kinesiology) and (2) psychology. These paths need not be mutually exclusive. To obtain a license in psychology, one needs to earn a doctoral degree in a psychology program and meet other requirements (regulated by state governments). To obtain certification by the Association for Applied Sport Psychology (AASP), a master's degree in either sport and exercise sciences or psychology that includes specific course work in SEP complemented with supervised practice requirements is the minimum criteria.

Professionals and students who seek professionalization of SEP services are organizationally grouped in AASP. AASP attempts to ensure high-quality SEP services through its certification program termed *Certified Consultant of the AASP* (CC-AASP). Although primarily concerned with the advancement of applied sport psychology (SP) in the United States, AASP includes international membership from over 30 countries.

Artur Poczwadowski

See also Adherence; Certifications; Consulting; Ethics; Interventions for Exercise and Physical Activity; Mental Blocks; Mental Toughness

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SEX DIFFERENCES

Sex differences are often referred to interchangeably as gender differences, often to offset taboos associated with the word *sex*. Use of the word *gender* originated as a social construct to include biological sex, social roles, and gender identity and is conventionally used in biological delineations. The significance of physical differences between and within sexes lies in perceptions formed from social comparisons and stereotypes generated and perpetuated within the sociocultural environment by significant others, parents, coaches, peers (social agents), and the popular media. Differences in biological or in physical and physiological characteristics are referred to as sex differences in the literature. However, when perceptions of the self or those of social agents are involved, the term *gender differences* is more common. Because discussions of sport and exercise participation and performance inherently involve interactions among physical, psychological, task and environmental factors, the term *gender differences* is most appropriate unless the discussion focuses solely on biological characteristics. Regardless of semantics, an understanding of male and female differences in sport and exercise psychology (SEP) is necessary to inform research designs, talent development, and practical interventions.

Conceptual models from general psychology and motor behavior help categorize gender differences and promote the notion that athletic and exercise performance quality are functions of interacting individual and environmental variables. Motor behavior models, however, also account for task characteristics. More specifically, dynamical systems and biocultural perspectives applied to motor behavior postulate that athletic performance outcomes (positive or negative) are functions of interacting individual, environmental, and task characteristics. Individual characteristics

can be structural (i.e., physical) or functional (i.e., psychological) and these, too, interact affecting outcomes positively or negatively. For example, if an athlete's structural (e.g., leanness and linearity) and functional (e.g., mental toughness, fearlessness) characteristics match the demands of the task (e.g., motor performance skills involving aerial rotation) and the athlete has access to a comprehensive coaching team (e.g., elite level, certified coaches, mental training advocates), the likelihood of success is enhanced. Alternatively, an example illustrating a possible negative impact of interacting structural and functional characteristics would be a late maturing athlete who is not strong enough to enjoy the successful accomplishments of early-maturing, stronger, and seemingly more skilled peers within his or her same gender group. Consequently, low self-esteem during critical growth periods, which can span five years, may be evident among late maturers and impact behavioral choices. Unfortunately, interdisciplinary studies of development that measure biological and psychological variables concurrently are sparse. The following sections consider gender differences within individual, environmental, and task components to help elucidate other interactions informative to practitioners working in athletic and exercise contexts.

Physical Domain

Sex differences refer to the biological characteristics linked to chromosomes where females (x,x) and males (x,y) vary in reproductive characteristics and functions as well as other physical characteristics that underlie performance differences. Differences in physical size are minimal until about 9 years of age when females enter their pre-adolescent growth spurt (age at takeoff); males follow about 2 years later at 11. Peak height velocity (PHV, the largest amount of growth in height within a 1-year period) occurs about 18 to 24 months earlier in females ($M = 10.7$ years, early, to 13.5 years, late) than in males ($M = 12.5$ years, early to 15.7 years, late). Growth during the year of PHV varies from 5.4 cm to 11.2 cm for females and 5.8 cm to 13.1 cm for males. On average, females will grow 1 cm per year less than males, averaging about 9 cm per year compared to males at 10.3 cm per year. Pubertal growth spurts last 24 to 36 months, and the growth during this time accounts for about 20% of final adult height

where females grow between 23 and 28 cm and males 26 to 28 cm. Peak weight velocity (PWV) coincides with PHV in males but in females, PWV occurs 6 to 9 months after PHV. Males and females also differ in somatotype, a three-component physique indicator characterizing the shape of the body as relatively pear-shaped (endomorphism), muscular and V-shaped (mesomorphism), or lean and linear (ectomorphic). Boys are more ectomorphic than girls early (<12 years) and late (17 years) in adolescence. Differences in mesomorphism are minimal in early adolescence but are marked, favoring boys in later adolescence, whereas girls have higher endomorphism scores than boys at both points in development.

While overall fatness differs between boys and girls, between- and within-gender differences can vary significantly because of factors such as nutritional intake, which is further linked to socioeconomic status (SES). After puberty, the ratio of muscle to fat among boys doubles. Boys also have more lean body mass and greater arm and calf circumference than girls. Because of metabolic differences corresponding with tissue differences, the average daily energy requirements are higher for boys than girls, and this is consistent at each chronological age particularly postpuberty. The magnitude of difference is much greater at late adolescence (male = 2,870 kcal; female = 2,140 kcal) than in early childhood (ages 3 to 7; 1,440 vs. 1,560, respectively) with the gap widening between 14 and 16 years of age.

Human development is age related, not age specific; biological age represents changes in body structure, function, and tolerance to environmental stress and does not equal chronological (calendar) age. Thus, the timing of growth and maturation within the context of social norms may underlie many self-perception variations and behavioral choices within and between sexes. At adulthood, within-sex variability in physical characteristics (i.e., leg length, height, fat mass, physique) are markers of developmental timing pathways traveled by boys and girls because these characteristics vary systematically with the timing of PHV and sexual maturation. Early maturers tend to be shorter and stockier (mesomorphic in physique) than late maturers who are typically more linear (ectomorphic in physique) with longer legs. Longer relative leg length is attributed to the growth plates of the long bones (i.e., legs) remaining open for a longer period, resulting in

taller stature. Accordingly, boys who mature later than girls are typically taller with longer legs than torso (i.e., sitting height–stature ratio) and this phenomenon is also true within each gender; later maturers are typically leaner with longer legs than early maturers.

Sexual development varies significantly between males and females and is hardly comparable with the exception of pubic hair transition, which is about 9 months later in boys than girls (male $M = 11.8$ years old, female $M = 10.2$ years old). Genital development among boys begins around 9.5 years of age where as breast development in girls begins around 10 years. Additionally, girls experience menarche (initial menses) at about 11.8 ± 1.0 years, approximately 1 year after age at PHV. The timing of menarche, typically assessed through retrospective interviews, can be an informative biological characteristic especially in reference to tracking development in areas such as psychological and behavioral characteristics, injuries, and motor performance. To date, accounting for physical or biological characteristics reflecting maturational timing is an exception rather than a norm in psychology research designs. Consequently, the complex mechanisms underlying gender differences in psychological phenomenon such as those comprising the functional domain are limited.

Functional Domain

Functional, or psychological, gender differences span a variety of domains including cognitive abilities, personality, self-perceptions, social behavior, and memory. While not all are directly relevant to athletic and exercise contexts, some of these characteristics may be embedded within athletic domain specific-self perceptions. For the most part, studies show that sport participation has positive influences on psychological characteristics such as self-confidence, self-concept, and self-worth in both males and females but results of other studies suggest this finding is equivocal among females. Some studies show that women particularly with athletic bodies report conflict between their physique perceptions and cultural ideals of leanness and linearity. The female athlete triad, linking body dissatisfaction, disordered eating, and osteoporosis, is an example of how negative self-perceptions often stemming from sport-context pressures to lose weight can negatively influence behavior and deteriorate health.

Body image dissatisfaction is also evident among male athletes who in contrast ideally desire to be bigger, or more muscular.

In general psychology, studies show that gender differences in functional characteristics are often contingent on developmental stage. Among adolescents, there is evidence that males are better than females in the following cognitive variables: mechanical reasoning, spatial perception, and mental rotation. For communication variables of facial expression processing and emotional expression (smiling), females outperform males in both adolescent and adult samples. Males consistently outscore females in a variety of aggression, competitiveness, computer efficacy, sexual behavior, and confidence measures. Developmental trends based on chronological age stratification show males outperform females in complex problem solving but only after the age of 15, and this is often confounded by some of the aforementioned differences in aptitude. Additionally, boys report higher self-esteem than girls but only between 11 and 18 years of age, with small effect sizes ($ES < .35$).

Noting age periods where there are similarities between boys and girls are just as important for informing practice, particularly if scores are low or imply negative behaviors and perceptions. Interestingly, gender difference claims are exceedingly overinflated in this domain. Historically, highlighting male and female differences has dominated popular media, but meta-analytic reviews do not support many of the stereotypes ($ES > .36$). Examples include superior mathematical performance and self-esteem among men and superior nurturing by women.

Understanding gender differences, and lack thereof, is not only important for helping athletes and exercisers but also for understanding perceptions and influences of significant others such as coaches, parents, and athletic trainers in respective social networks. Stereotype threat studies show expectations of gender differences can influence performance; in conditions where females expected gender differences in performance, they underperformed men. Continued expectations for, and communications about, gender differences perpetuated in athletic and exercise environments can undermine confidence, motivation, and perhaps even healthy eating and exercise behaviors. Consider the early-maturing female who may engage in dietary restrictions and excessive exercise

in efforts to control physique characteristics that are genetically linked and not malleable.

Task Characteristics

The overwhelming influence of social environment factors (practice frequency, instruction, and feedback quality) on performance is highlighted by data showing gender differences favoring boys in throwing velocity and distance prior to age at PHV takeoff where size and strength differences between sexes are minimal; throwing for distance ES are from 1.5 to 2.0 for children 3 to 8 years old, and not surprisingly, the difference is stable, increasing to > 3.5 after for adolescents 16 to 18 years old. However, for other variables, gender differences are generally not stable; there is no difference until after puberty for balance, pursuit rotor tracking, tapping and vertical jump. Gender differences favoring boys are evident prior to puberty for short distance running, sit-ups, long jump, grip strength, shuttle run, and catching. There are no age-related trends in gender difference that favor boys for agility, anticipation timing, arm hang, reaction time (RT) (low ES), throwing accuracy, and wall volley (high ES); a low difference favoring girls is evident for eye motor coordination and flexibility tasks. In childhood, studies have shown that girls also develop certain fundamental motor skills earlier than boys including skipping, hopping, and catching by 6 months to 1 year.

Environmental Characteristics

Influences from parents, teachers, and coaches among other in the form of gender role stereotypes and biologically based expectations shape self-perceptions and consequently performance capabilities. Perception and performance differences prior to elementary school are largely attributed to parents. Sons receive more sport opportunities and encouragement and engage in more gross motor behaviors and rough play than daughters. Teachers and coaches tend to treat differences as naturally evolving biological factors formulating expectations conforming to stereotypes. Consider gender specific norms in fitness testing which rightfully account for biological differences yet may permeate areas where differences do not exist and are irrelevant.

The magnitude and even direction of gender differences depend on the context and perhaps to a larger extent than is currently supported by

data, the interactions among individual, structural, and functional characteristics. With the exception of physical characteristics, gender differences are few, small, and unstable. Although socialization toward specific activities and development of competitive orientations begin much earlier than the growth spurt, biologically there are minimal physical differences between boys and girls prior to the growth spurt.

Attending to gender similarities and within-gender variability in maturational timing and its associated physical characteristics is particularly important to counteract gender stereotypes and make the most informed decisions. For example, the 2- to 3-year window when pubertal girls are generally taller and stronger than many boys can be a critical period for girls' confidence development in athletic contexts. But this depends on how social agents shape sport experiences relative to cultural ideals that still favor the lean, linear feminine physique. In contrast, boys' self-esteem is as equally problematic as it is for girls particularly for boys who mature late during early adolescence. Consequently, parents, teachers, and other professionals may overlook boys' self-esteem problems, precluding interventions that they need. The costs of inflated claims of gender differences are also notable and may undermine many areas including parenting, relationships, and work, as well as opportunities in athletic and exercise settings. One historic example is the preclusion of women from marathon running until 1984.

While Title IX of the Education Amendments of 1972 helped improve participation opportunities for women in the United States, gender stereotypes still exist and undermine the importance of being informed about sex and gender differences. Decisions about athletic participation, specialization, training, and fostering positive sport and exercise experiences rely on attending to age-related differences within and between sexes. Equally important is sensitivity to the myriad of self-perceptions cultivated in athletic contexts. A broad sampling of sports throughout childhood and into adolescence is advocated for both genders as a mechanism for reducing overuse injuries and burnout and also maximizing skill development for physical activities that can be enjoyed later in life. It will always be important to attend to gender difference stereotypes perpetuated in the media because regardless of data supported differences, or lack thereof, stereotypes will permeate

individual perceptions, constraining athletic and exercise potential. Practitioners are encouraged to help athletes focus on their personal potential rather than on social comparisons between or within genders.

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See also Body Image; Developmental Considerations; Eating Disorders; Female Athlete Triad; Gender; Motor Development; Talent Development

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SHARED MENTAL MODELS

The *shared mental model* is a term used in industrial and occupational psychology. Within the discipline of sport psychology (SP), a shared mental model is commonly referred to as a *shared knowledge state*. This is a state in which the knowledge held by each member of a sports team about the upcoming actions of the team is at least similar to other team members' knowledge of these actions. In everyday language, the state can be described as team members being "on the same page" about their actions on the field. This topic is of interest to sport psychologists because the ability of a team to coordinate the actions of its members is thought to depend, in part, on the attainment and maintenance of this state. In this entry, the concept of a shared knowledge state and its role in achieving team coordination is outlined first. The processes by which a shared knowledge state is established

prior to a given game, and established and updated during a given game, are then described. Note that there have been few studies to date on team coordination and the role of shared knowledge in achieving team coordination, within the sport domain. As such, while reference to such studies is made in the sections that follow, the content of these sections is largely theoretical.

Shared Knowledge States and Their Role in Achieving Team Coordination

A challenge for teams is to achieve coordination. Team coordination is the process of arranging team members' actions so that, when they are combined, they are in suitable relation for the most effective result. Team members' actions must be arranged so they are correctly related on three dimensions of action: type, timing, and location. First, achieving a team action often requires that each team member undertakes a specific type of action. If a point guard in basketball lobs the ball in an alley-oop pass through the air toward her offensive basket, her teammate might prepare to jump to catch the ball in midair and dunk it in the basket. If at the last moment the point guard makes a bounce pass on the ground, her teammate might not have time to adapt to the bounce pass because she has already begun to jump. This change in action type would likely result in an incomplete pass. Second, achieving a team action often requires each team member to undertake an action at a specific time. In the basketball example, if the point guard makes the alley-oop pass too early, her teammate might be unable to jump fast enough to catch the ball and dunk it. Third, achieving a team action often requires each member of the team to undertake an action at a specific location. If the point guard does not throw the alley-oop pass close enough to the rim of the basket, her teammate will not be in position to catch the ball to dunk it.

While the basketball examples are hypothetical, research has provided a real example of the importance of action timing. Studies of multi-person rowing show that rowers' strokes must be synchronized, requiring that they deliberately coordinate their actions during races. In a recent study of elite-level two-person rowing, Carole Sève and colleagues, a rower (Marion) was shown a film of her and her teammate's rowing performance and asked to comment on their performance. Within her comments, Marion highlighted

a problem of coordinating her strokes with those of her teammate (Lucy), feeling that she (Marion) was being “pushed” by Lucy. This report suggests a problem with action timing; Lucy is stroking too fast for Marion. A subsequent analysis of the rowers’ movements confirmed this: The length of the stroke phase (i.e., the oar in the water) was shown to be shorter for Lucy than Marion, which meant that Marion needed to “catch up” to Lucy during the recovery phase (i.e., the oar out of the water). Having proposed that achieving coordination can be challenging for teams, we now consider the reasons why attaining this outcome can be challenging.

When one individual performs a task, the task being undertaken is controlled by only one executive (i.e., brain). However, when a team attempts a task, there are as many executives as there are team members. Each person possesses unique knowledge about how to perform a given task, so people placed into teams tend to select the type, timing, and location of their actions at their own discretion. The performance of a team suffers when team members select actions at their own discretion because, as proposed previously, team members’ actions must be related in terms of type, timing, and location to actions being undertaken by other team members. Establishing a shared knowledge state, in which team members share (i.e., hold in common) similar knowledge about how to undertake their team task, means that team members can draw on that similar knowledge to undertake this task. As a consequence, team members are able to select the type, timing, and location of their actions in a way that affords effective team coordination.

The Process by Which a Shared Knowledge State Is Established and Updated

In this section, we will distinguish between shared knowledge states established prior to a given game and those established and updated during a given game, as both types of shared knowledge state underpin team coordination. Prior to a given game, teams acquire a shared knowledge state in two ways: through actual play, in both practice and competition settings, and via explicit planning.

Shared knowledge acquired through play is often considered knowledge of “situational probabilities”—that is, knowledge of what the team and its individual members are likely to do

in response to a given game situation. This type of shared knowledge can be acquired simply through experience of playing the sport. Players learn from practicing and competing within their sport what teams and individual team members are likely to do in general in a given situation. Consider a newly formed soccer team where, previously, every team member has played the sport but no team member has played on this particular team. As a result of their experience playing the sport, all the players on this team know that when a midfield player accidentally turns the ball over to the opposition, defenders on his or her team are likely to respond to the turnover by moving back toward their own goal into a defending position.

Shared knowledge of situational probabilities also arises from experience of playing alongside members of one’s own team. Players come to know through playing on a specific team what their team and its individual team members are likely to do in a given situation. Consider how team members playing together on the same ice hockey team for several years would learn that the right winger on their team likes to feint with his stick twice before passing the puck across the goalmouth for an assist. Coaches often design practice sessions and drills, such as “2 vs. 1” drills in soccer, as well as scrimmages with the goal of accelerating players’ learning of the situational probabilities related to their team and individual teammates. In other words, coaches place players in specific game (or gamelike) situations involving coordination so they will get to experience, and in turn learn, how those situations “play out.”

Regarding shared knowledge acquired through explicit planning, coaches or other team members often provide the same information about the team’s intended actions to all team members by communicating plans of action (e.g., approaches, strategies, tactics, and plays) to those members. Coaches usually communicate plans initially via verbal communication and often demonstrate plans using visual aids such as a whiteboard. Subsequently, the coach asks the team to practice executing the plan (e.g., running a play) on the field. Ideally, all team members acquire similar knowledge of the plan from the coach’s communications and demonstrations and/or by practicing the execution of the requisite plan.

The result of (a) playing the sport generally, (b) playing in a particular team within the sport, and (c) creating and practicing the execution of

team plans is a relatively stable form of shared knowledge state. This state is established prior to a given game and forms a cognitive resource upon which team members can draw during a given game to achieve coordination. Consider an example of a play in football. The play is planned by a coach and then discussed with, and demonstrated using a whiteboard to, players prior to a practice session. Next, the play is practiced on the field, and feedback on it is provided by the coach until players feel comfortable executing it. At this point, each player involved in the play knows (a) when to use the play (or what signal from the coaching staff or teammate denotes that the play should be used), (b) what actions those involved in the play are expected to perform, and (c) how his own actions fit with the actions of his teammates. Consequently, the play can be used during actual games.

In addition to the relatively stable shared knowledge state achieved prior to the game, it is usual that a relatively dynamic version of a shared knowledge state is established and updated by a team during actual games. This dynamic “in-game” state is required because games in most team sports are characterized by rapidly unfolding and difficult-to-predict situations within which team members must adapt (e.g., a faked punt by the opposing team in football). Nonetheless, the establishment of this in-game state is often based in part on the shared knowledge state achieved prior to the game. Consider a soccer team that has developed a strategy of attacking up the wings of the field; in other words, the team has acquired a shared knowledge state in which this is the team’s go-to offensive strategy. Of course, the team members would likely also know, and share knowledge of, how to attack up the middle as a team. In other words, although attacking up the middle is not the team’s current go-to strategy, the team has used this strategy and thus knows how to implement it. Then consider that the team begins a game but soon realizes that their opponent is defensively strong on the wings but weaker in the middle. The team would likely switch from attacking up the wings to attacking up the middle. Thus, the in-game shared knowledge state of the team would need to be updated accordingly for the team to achieve the coordination required to effectively execute this new strategy—that is, a strategy that is new to this game, if not to this team.

The updating of this in-game state occurs via incidental and deliberate means. Regarding incidental means, when a given change in the game situation occurs, one or two team members may begin to adapt to this situation by switching to a different strategy. In the previous example, as the soccer team experiences a strong defense against its attacks up the wings, an attacking player (e.g., Emma) may begin to pass the ball to a teammate in the middle of the field rather than dribble it up the outside of the field. When other team members see the change in Emma’s attacking behavior, they infer (on the basis of their experience of attacking up the middle) that Emma is switching from the up-the-wing strategy initially prioritized to an up-the-middle strategy. In turn, these team members infer that they are required to integrate their own actions with the up-the-middle strategy to achieve effective coordination.

Team members also often deliberately update the team’s in-game shared knowledge state by communicating intended changes to upcoming actions during games to other team members. They typically communicate this information using bodily movements (e.g., pointing) and verbal communication. In the soccer example that was just given, consider that when Emma realizes that the strategy of attacking up the wings is not working, she might communicate verbally to her teammates that they should attempt to attack up the middle. Even subtle facial expressions can be used for this purpose. In a recent study of table tennis doubles teams, Germain Poizat and colleagues reported that a particular stare by one table tennis player was enough to communicate intended changes to tactics to the other player.

Conclusion

In conclusion, team coordination is thought to depend in part on the team’s ability to establish a state in which knowledge of the team’s intended actions is shared by team members before a given game and establish and update an “in-game” version of this state during actual games. Establishing a shared knowledge state before a game is achieved by (a) general experience of practicing and competing in the sport, (b) experience of practicing and competing in a particular team, and (c) communication of team plans by coaches to players and practice executing these plans. Establishing and updating the in-game version of this state occur

on the basis of observed changes in teammates' actions, which signal a change of team plan, and deliberate communication by coaches or team members to other team members about intended changes to the team plan.

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See also Cohesion; Decision Making; Knowledge Structure; Practice; Skill Acquisition; Team Building; Team Communication

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SIMULATION TRAINING

Simulation training is a popular technique used in many domains, including aviation, the military, medicine, music and theatre, and sport. It can be described as training or practicing under conditions that are reflective of performing under pressure. Whether this is to practice performing lifesaving surgery, rehearse a new dance piece, or preview the atmosphere of the Olympic Games, the main aim is to prepare individuals to effectively

handle the pressures of performing. Discussed here are two major types of simulation training used in sport and their theoretical underpinnings and main psychological benefits.

Types of Simulation Training

It is well known that the best athletes make extensive use of simulation training by approaching routines, plays, or scrimmages in practice as if they were at competition. This is an active form of simulation and typically involves the athlete wearing the clothes worn in competition and following the same preparation routines. Soccer players will be dressed in their home uniform before hosting a match and rehearse certain plays with the strengths and weaknesses of specific opponents in mind. Other aspects of the competition environment also may be replicated to make the situation as realistic and relevant as possible. Swimmers may preview an important event by holding a mock race in practice that is started by an official and watched by friends and family to create the usual noise and commotion that would be present. Mimicking details that are usually unique to competition helps athletes to become desensitized to potential distracters and stay focused. As well as becoming more comfortable with performing under these conditions, it will also give the athletes a sense of confidence that they can be successful when it really matters.

Studies show that simulation training also can be done more passively by imagining scenarios or using video clips. These passive forms of simulation training may be particularly useful when it is not possible to either create or participate in the real life situation, such as when preparing for certain weather conditions or when an athlete is injured. It is also an effective way of systematically desensitizing athletes to situations that they fear or worry about. When imagining scenarios, the individual will often generate relevant details of the situation in their mind (i.e., the stimulus), including all of the usual thoughts and feelings associated with it (i.e., the response). To weaken the connection made between the situation and any anxiety typically experienced, two different approaches can be taken. The more traditional method is to first relax the individual, using deep breathing or muscle relaxation strategies, before imaging the stress-evoking scene. By starting in a relaxed state, the individual is better able to

endure the imagery and, with time and repeated practice, will come to associate the stimulus (i.e., competition) with a different response (i.e., feeling relaxed).

However, a low level of activation may not always be appropriate. Depending on the activation level an athlete performs best in, an alternative approach is to change the meaning of the response. Also referred to as cognitive reappraisal, the guided imagery may direct the athlete to reinterpret the thoughts and feelings experienced in the scene as helpful toward performance. By doing so, the athlete will feel more confidence and control over symptoms normally associated with anxiety. Rather than view the stress-evoking situation as a threat, the athlete will then come to associate the situation and the responses experienced as a challenge, which will increase the probability of performance success in real life.

Theoretical Underpinnings

The use of simulation training is based on the *encoding specificity principle of learning* (also known as the *theory of identical elements*). This theory is that practice is most effective when it closely approximates competition. By engaging in simulation training, there is a greater likelihood of automatic transfer from the situation in which a skill or routine is learned (i.e., practice) to the situation in which it is performed (i.e., competition). Within research on memory, it has been established that contextual information (e.g., details within the physical environment) affects what is remembered and recalled. Remembering how to perform a skill or routine is better when the information available when storing the skill or routine in the memory is also present when this information is being retrieved. Thus, by recreating unique competition conditions within practice, the learning environment will be optimal and quality training should ensue.

Psychological Benefits

In addition to improving learning and performance, there are many presumed benefits of simulation training including improvements in self-confidence, attentional and emotional control, decision making (DM), and communication and teamwork. By recreating the conditions of competition in practice, simulation training provides opportunities for athletes to succeed in certain situations. According

to Albert Bandura's social cognitive theory (SCT), providing authentic experiences of performing well under conditions that mimic competition will instill a sense of performance accomplishment in athletes. This is considered to be the most valuable form of self-efficacy—a more specific more of self-confidence. Consequently, performing in these conditions should lead the athlete to being more confident.

Better attentional and emotional control can also result from simulation training. By identifying details that are present during competition and incorporating these into training, the athlete will have an opportunity to practice selectively attending to relevant things and disregarding distracters. Similarly, the simulation could involve aspects of competition that typically trigger unhelpful emotions, such as hearing a bad call made by an official, being taunted by an opponent, or taking a penalty kick after a teammate misses. The athlete can then rehearse how they will stay positive and cope more effectively if such a situation ever arises again in real competition.

Finally, and particularly for teams, simulating how plays might work under certain game conditions can lead to greater situational awareness (SA) and better DM. It also provides an opportunity to rehearse communication strategies and develop better teamwork. Research shows that national squads will identify unique aspects of an upcoming championship or competition and try to re-create these in their practice preparation. It is thought that practicing in these conditions can be very beneficial to performance in the competition.

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See also Attentional Focus; Imagery; Mental Rehearsal; Self-Efficacy; Situational Awareness

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SITUATIONAL AWARENESS

Expertise in sport, or the growth of specialist athletic knowledge and skills as a result of effortful experience, has attracted considerable attention in recent years from researchers in cognitive psychology, cognitive neuroscience, and sport psychology (SP). An important impetus for this surge of interest is the fact that the study of athletic expertise can shed light on the relationship between knowledge and skilled action in complex dynamic environments that are characterized by uncertainty and time constraints. Specifically, studies in this field can help to identify the cognitive processes and neural mechanisms that underlie expert and novice differences in pattern recognition, decision making (DM), and skilled performance. For example, research shows that expert athletes are generally superior to novices in recognizing and recalling precise details of patterns of play in their specialist sport—a skill that enhances their “situational awareness” (SA) (or their general understanding of what is going on around them) and efficiency of performance. But what are the components of SA in sport? What do we really know about expert and novice differences in SA? Finally, can SA be trained, and if so, how? The purpose of this entry is to answer these questions.

Components of Situational Awareness

The term *situation(al) awareness* (SA) originated in human factor (HF) research in the aerospace industry and subsequently spread to cognitive task performance in other domains, such as military command operations, surgery, and sport. In general, it describes people’s knowledge of the importance of what is happening in their immediate surroundings. More precisely, SA denotes the accuracy with which performers can perceive relevant events in their environment, establish their significance, and use this knowledge to anticipate future outcomes. Unfortunately, the theoretical foundations of SA have not been well established. To illustrate, there is considerable semantic confusion in this field as few theorists have attempted

to clarify exactly how SA differs from apparently related, but more established, concepts in cognitive psychology such as “mental models” (i.e., the knowledge structures that people create in their minds to understand and explain specific phenomena and experiences). In addition, the logical status of SA is unclear as theorists are divided on the issue of whether it is an individual characteristic (e.g., residing in the mind of the perceiver) or a shared one (e.g., emerging from the constant interaction between the perceiver and his or her colleagues and their shared environment). In view of these problems, it is not surprising that some theoretical models of SA appear to be based more on intuition than on systematic empirical findings.

Typically, SA is postulated to contain three cognitive components—*selective attention* (or the ability to focus on task-relevant information while disregarding distractions), *pattern recognition* (e.g., the capacity to recognize situational characteristics and promising strategic options based on stored knowledge), and *anticipation* (e.g., the ability to make rapid and accurate predictions about future outcomes). Interestingly, the claim that SA is facilitated by rapid access to memorized information highlights an intriguing paradox of expertise. Although expert athletes have a larger database of knowledge to search through than have novices (e.g., the size of a chess master’s database has been estimated at between 50,000 and 100,000 chunks or units of meaningful information), they can retrieve game-specific information in their specialist sport quicker than novices. The most plausible reason for this expert–novice difference in speed of search and retrieval concerns the way in which this knowledge is stored. Specifically, it seems that whereas expert athletes’ knowledge is largely associative and extensively cross-referenced in long-term memory (LTM), novices’ knowledge is more discrete and compartmentalized.

Situational Awareness in Sport

At present, little consensus exists as to the best way in which to assess SA in sport. In some early studies on this topic, athletes’ knowledge of event probabilities was explored using generic choice reaction time (RT) tasks in laboratory settings. As these measures failed to elicit athletes’ game-specific knowledge, however, they were abandoned in favor of more ecologically valid methods. For example, SA in soccer has been investigated

by showing expert and novice players various film sequences of actual match action and requiring them to extrapolate from such information. In these studies when the final frame of action was frozen, the expert players were more adept than the novices at identifying the best strategic options for the players viewed (as assessed by an independent panel of judges).

Arising from such research, several methods are now available to investigate expert–novice differences in the anticipation component of SA in athletes. Typically, these methods include *sport-specific prediction tasks* (e.g., athletes must guess what will happen next in a sport-specific film sequence), *pattern recognition tasks* (e.g., athletes are shown several film sequences and asked to judge whether a particular one had been shown earlier or not), and *immediate retrospective verbal reports* (e.g., athletes are required to describe what they had been thinking while viewing dynamic simulations of a sport-specific task). To illustrate, a recent study combined these three methods in exploring expert–novice differences in anticipation skills in soccer players. Results showed that compared with novices, skilled soccer players demonstrated superior anticipation accuracy and recognition performance (i.e., they were better able to predict future events and more adept at distinguishing previously seen from novel film sequences). Furthermore, analysis of these athletes' verbal reports suggested that the expert performers used more complex memory structures than did their novice counterparts in generating their predictions. In summary, many studies have shown that expert athletes are superior to relative novices in their ability to use SA skills in order to anticipate correctly how a given scenario will unfold in a dynamic sport situation.

Training Situational Awareness Skills

The question of whether or not perceptual–cognitive skills in sport can be enhanced using systematic training interventions has attracted increasing research attention in recent years. Unfortunately, the majority of these studies have used simulation training techniques to improve specific anticipation skills (e.g., the ability of athletes to identify advance visual or postural cues from opponents) rather than general SA. Also, many of these studies have not used appropriate control groups. Nevertheless, focusing only on methodologically sound studies in

this field, there is evidence that novice and developing athletes can benefit significantly from programs that train people to detect informative cues in their specialist sport and to execute appropriate actions accordingly. However, the effects of these programs on expert and/or older athletes are largely unknown. Consequently, there is currently inadequate evidence available to date to demonstrate that SA skills can be improved significantly in athletes. Clearly, the collection and evaluation of such evidence is an important priority for future researchers in this field.

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See also Anticipation; Attention Theory; Cue Utilization; Expertise; Information Processing; Pattern Recognition and Recall

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SKILL ACQUISITION

Acquisition of skill is a type of learning in which repetition results in enduring changes in an individual's capability to perform a specific task. With

enough repetition, performance of the task eventually may become automatic, with little need for conscious oversight. Any behavior that needs to be learned and that is improved by practice can be considered to be a skill. Making a cup of tea can be considered a skill, just as is drinking it. Skill acquisition in sport generally can be thought of as either learning to coordinate the body appropriately to achieve an intended movement outcome or as learning any of the myriad mental aspects associated with effective movement, such as where to move and when. A highly skilled goalkeeper who saves a penalty kick has learned not only how to fling his body most effectively around the goalmouth to keep the ball out but also when and in which direction to fling his body. In most sports, these different facets of skill cannot be viewed in isolation, as it is the combined synchrony of perceptual, cognitive, and motor processes that culminates in highest levels of performance. Performance of the skill will improve over time as a consequence of repeated engagement in the task (i.e., perhaps deliberate, perhaps not) and will be accompanied by increasingly consistent attainment of the desired outcome, which eventually can be achieved in a variety of conditions, contexts, or environments. For example, a skilled golfer may consistently hit the green from 150 yards and have a higher likelihood of doing so in the pouring rain or when it really matters than an unskilled golfer. A skilled performer can thus be defined as a person with the ability to achieve an intended outcome repeatedly in a variety of conditions and usually with less physical or cognitive effort than a less skilled performer.

The study of skill acquisition dates to work that investigated how field operators acquired telegraphic language skills. At the turn of the 20th century, the effectiveness of telegraphy as a form of rapid long-distance communication relied on operators to be highly skilled at the perceptual, cognitive, and motor aspects involved in transmitting, receiving, and translating Morse code. The work, which examined skill level as a consequence of experience, showed that variability of performance decreased as operators became more skilled, and that their performance was less easily disrupted by outside distractions. Although the work showed that it generally took between 2 and 3 years for operators to become skilled, the progression of learning was not continuous. Plateaus occurred during which performance barely

changed, only progressing if the operator did not become despondent and discontinue practice. The researchers argued that performance could not progress to a higher level until attention was freed by the automatization of lower level responses within a *hierarchy of habits*.

Although plateaus in performance level are not always observed over the course of skill acquisition, one characteristic that is observed more often than not is a high rate of change in performance level early in practice, with gradual reductions in rate of change as practice continues. This power law of practice has the function $\log C = \log B + n \log x$, where C refers to performance level, x refers to amount of practice, and B and n are constants.

There is little doubt that the level of performance that can be achieved when acquiring skills is closely related to the extent to which a person practices the skill and the way in which the person practices. Researchers often use retention or transfer tests to determine whether the changes in performance caused by practice are temporary or permanent. By assessing performance sometime after practice has been completed, perhaps a day or a week, it is possible to observe true performance level without temporary practice effects related to factors such as motivation, fatigue, or boredom.

Researchers have examined many methods designed to maximize the enduring changes in behavior that practice must produce for skill acquisition to occur. For example, early work examined whether it was better to practice the different parts of a task independently or together as a whole task. Findings were mixed, and it is probably fair to say that no consensus emerged. The effects of massed and distributed practice have been examined at length. Massed practice involves continuous practice, whereas distributed practice involves the spacing of practice over time. One meta-analysis of 116 studies suggested that distributed practice is preferable for continuous tasks but massed practice is preferable for discrete tasks. A well-known phenomenon is the contextual interference effect, in which interference appears to cause poor performance during practice but good performance during retention and transfer. In particular, the effect is seen when interference is caused by switching randomly between skills when practicing, as opposed to practicing each skill one after the other in a blocked pattern. Two explanations have been proposed for the phenomenon. The *elaboration hypothesis* suggests that random

practice of different skills requires performers to actively retrieve movement solutions each time that they perform, which results in a more distinct representation of each skill in memory. The *reconstruction hypothesis* suggests that switching from one skill to another during random learning causes performers to forget the previous movement pattern. Consequently, performers must continually reconstruct the movement as they practice. A fascinating adjunct to the contextual interference effect has been added in recent years by the concept of *differential learning*, which suggests that a significant learning advantage is gained from maximizing interference during practice by introducing stochastic perturbations of performance. In essence, it is argued that a person learns more by trying to achieve the same skill outcome with maximum differences in the manner of execution.

Many factors influence skill acquisition. The difficulty of the skill to be acquired, and the availability of information or the form in which the information is available, play a significant role in the process. Information about potential movement solutions can be acquired by instructions or guidance from an external agent or by observation or modeling. Much research has examined the effects of feedback on skill acquisition. Intrinsic feedback provides sensory information that arises from executing a skill. Visual feedback often seems the most salient, but feedback from the other senses also plays a part in skill acquisition. Extrinsic feedback provides augmented information about performance of a skill, often supplementing the information that is available via intrinsic feedback. In particular, *knowledge of results* and *knowledge of performance* are crucial types of extrinsic feedback during skill acquisition. Knowledge of results is extrinsic information about the movement outcome, whereas knowledge of performance is extrinsic information about the movements themselves. The information in both instances is usually provided to the performer verbally (e.g., you missed left or keep your head still) or perhaps by video replay or even diagrams. Most people accrue much information by a process of hypothesis testing in which they devise and test strategies to correct performance errors. There are few cases in which learning takes place in the absence of at least some intrinsic or extrinsic feedback. The challenge point framework proposes that what is learned is a function of the information that becomes available during performance and that the crucial factor is

the interaction between the difficulty of the skill to be learned and the level of ability of the performer.

The Cognitive Approach to Skill Acquisition

Cognitive characterizations of the human brain as a processor of information argue that information received by the senses contributes to the development of centralized representations in memory, which are used to control performance using preprogrammed commands that run without the need for modification from feedback (i.e., open-loop control) or from continuous comparison of ongoing feedback with representations generated by previous movements (i.e., closed-loop control). A significant problem for these theories was that a representation needed to be stored for each different movement, creating a storage problem. Subsequently, *schema theory* was proposed as a solution to the storage problem. Schema theory proposed that a general motor program represents a class of movements (e.g., throwing), but that different submovements are produced by specifying the necessary parameters at the time (e.g., speed or force).

The cognitive approach suggests that during practice movement representations can be updated and improved—using knowledge of results or knowledge of performance, for instance. As a consequence, performance efficiency improves. An outcome of this process is that skill learning progresses through a sequence of stages characterized by fairly distinct cognitive characteristics. At first, when the skill has been little practiced, performance tends to be *cognitively controlled* as the learner consciously tries to select strategies to use and the relevant information to attend to. Performance tends, therefore, to be explicit, inefficient, and variable. A second stage follows in which the intense need for conscious, cognitive involvement begins to ease. Strategies are cemented and stimulus–response (S–R) associations begin to strengthen. A final stage is characterized by performance that requires very little cognitive engagement. Performance is consistent, efficient, and can occur automatically without conscious intervention. These characteristics are common in many psychological theories of skill acquisition, although sometimes the theories differ with respect to why practice improves performance.

Some theories (e.g., *instance theory*) are predicated on the belief that practice increases the

number of available memory representations that can be accessed to support performance, whereas other theories (e.g., *adaptive components of thought* [ACT] theory) are predicated on the belief that practice refines strategies and procedures that support performance. These theories often make a distinction between declarative and procedural knowledge that is associated with performance. Declarative knowledge can be thought of as explicit facts or propositions that describe information related to performance and that can be described by the performer, whereas procedural knowledge can be thought of as implicit representations of how to perform, which cannot be described by the performer. Use of declarative knowledge to support skill performance tends to require processing resources and is effortful, but eventually practice allows skills to be executed as procedures, with little or no reference to declarative knowledge. In such cases, far fewer processing resources are used, and performance tends to be more effortless than effortful. Researchers who have used *implicit* motor learning techniques to inhibit the accumulation of declarative knowledge during skill acquisition claim that this promotes the benefits associated with use of procedural knowledge to support performance.

An Ecological-Dynamical Approach to Skill Acquisition

The ecological approach proposes that information that resides in the environment specifies the options that are available for behavior by an organism (i.e., affordances). Actions and perceptions are coupled, and movements can be initiated without reference to memory representations or motor programs, using only information that specifies the constraints on movement. Practice is seen as a process of educating attention to pick up the specifying information rather than the nonspecifying information in the environment. In tandem with the ecological approach, a dynamical systems theory of movement coordination has evolved in which biological organisms are seen as systems that can self-organize in response to existing constraints or parameters, and that are attracted to stable patterns of coordination. This view emerged as a way to understand the complexities presented by a human motor apparatus that has evolved with countless ways in which to move. Nicolai Bernstein called this the *degrees of freedom* (df) problem and

proposed that skill acquisition is a matter of solving the df problem. He argued that initially movement control is simplified by locking various joints in order to reduce many of the df. Gradually the df are released during practice and the performer becomes able to control or exploit energetic forces, such as momentum or inertia, during movement. This approach provides a perspective to skill acquisition that has yet to be fully reconciled with traditional cognitive approaches in sport.

Rich Masters

See also Adaptation; Automaticity; Chunking/Dechunking; Errors; Feedback; Generalized Motor Program; Retention; Transfer

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SLEEP AND EXERCISE

Sleep is critical to normal physiological and psychological function, and people spend nearly one

third of their lives asleep. Usual sleep duration is considered to be between 7 and 9 hours per night for healthy adults. The different stages of sleep have historically been defined according to the 1968 classification rules of Alan Rechtschaffen and Anthony Kales (R&K) as wakefulness, one of four stages of non-rapid eye movement (NREM) sleep (S1, S2, S3, and S4), or rapid eye movement (REM) sleep. The American Academy of Sleep Medicine (AASM) modified these standard guidelines and recommended that NREM sleep be referred to as N1, N2, and N3, with N3 reflecting R&K stages S3 and S4, and Stage REM to be referred to as Stage R. A normal sleep cycle (consisting of all the sleep stages) lasts for approximately 90 minutes and is repeated throughout the night.

Sleep can be measured by several different methods. Self-reports such as sleep logs or diaries are helpful to determine trends over time. However, these are subjective and rely on accurate recollection by the individual. Actigraphy is a method of measuring body movement. For the purpose of measuring sleep, these devices are typically worn on the wrist and periods of low activity scored as “sleep” and periods of high activity scored as “awake.” Due to its noninvasive nature, actigraphy has been used widely as a tool to measure sleep quantity and quality over lengthy periods of time and in the normal (home) sleeping environment, although the method cannot determine the different stages of sleep or transient arousals from sleep. Polysomnography is acknowledged as the gold standard method of measuring sleep, as it allows assessment of all sleep stages and is used widely to assess the presence and severity of sleep disorders. Polysomnography requires many electrodes to be attached to the scalp and face in order to measure brain, eye, and chin muscle activity, and sensors on the chest, finger, and face to monitor cardiorespiratory parameters.

Effects of Acute Exercise on Sleep

Normally, core body temperature peaks in the early evening, starts decreasing prior to bedtime, and reaches its minimum in the early hours of the morning. Exercise in the evening is known to raise core temperature, which could potentially oppose the normal pre-bedtime decrease and interfere with sleep onset. This is one factor underlying the current recommendations that exercise should only be performed 5 to 6 hours before bedtime and

no closer than 3 hours prior to sleep. However, data supporting this widely accepted recommendation are scarce. Indeed, a study reported that high-intensity exercise performed in the evening has little effect on sleep in healthy young people. Similarly, a meta-analysis of the literature on the effect of exercise on sleep, accounting for time of day, showed that acute exercise increased Stage 2 and slow wave sleep, decreased REM sleep, and increased total sleep time.

Effects of Chronic Exercise on Sleep

There is little research investigating the effect of chronic exercise among athletes on sleep, although it has been reported that those athletes who become overtrained report disrupted sleep patterns. In the general population, regular physical exercise is recommended as a way to reduce the effects of other contributing factors to poor sleep such as smoking, ingesting too much caffeine, overeating, and consuming alcohol.

Effects of Acute Sleep Loss on Exercise

Good sleep is considered essential for optimum athletic performance and is regarded as the primary method of recovery from physical activity (PA). Acute sleep deprivation has been shown to negatively impact sport performance. However, the magnitude of this detrimental effect appears to depend on the specific athletic activity. A literature review by Thomas Reilly and Ben Edwards concluded that the effects of acute sleep deprivation or an altered sleep pattern had a marginal impact on predominantly aerobic or anaerobic sports (e.g., sprints, power events, running 3,000 meters, swimming 400 meters) but a negative impact on sports that required high concentration, consisted of a mixture of anaerobic and aerobic activities, or required multiple anaerobic efforts (e.g., field sports, sailing, road cycling, aiming sports, combat sports, swimming, middle distance running, jumping events, weight training). The specific performance decrements included poorer decision making (DM), increased errors, decreased power output, and increased fatigue.

Effect of Chronic Sleep Loss on Exercise

Very little is known about the effects of chronic sleep loss on athletic performance. Case studies of ultraendurance cycling events, which require

sustained periods of wakefulness, report that, in order to maintain optimal athletic performance, at least 2 hours of sleep per night are required. Similarly, a cluster napping technique (multiple short naps) is reportedly adopted in ultra-endurance sailing to achieve an average total sleep time of 5.5 hours per day, presumably as a method to maintain optimal vigilance.

Jet Lag and Exercise

One of the most common transient sleep disorders experienced by athletes is jet lag due to travel across multiple time zones. Symptoms of jet lag include disorientation, light-headedness, impatience, and lack of energy. Westward travel tends to be accompanied by less severe jet lag than eastward travel. Athletic performance can be compromised due to jet lag as the body's circadian rhythms adjust to the new time zone.

Undiagnosed Sleep Disorders and Exercise

Undiagnosed sleep disorders have the potential to reduce athletic performance. One such prevalent disorder among the general population is obstructive sleep apnea (OSA), which is characterized by repetitive narrowing and/or collapse of the pharyngeal airway during sleep, resulting in reduced blood oxygen levels and multiple arousals from sleep. The repetitive arousals fragment sleep and have been associated with decreased motor function and increased fatigue. While usually associated with a sedentary lifestyle and obesity, anatomically predisposed athletes (e.g., high body mass index and large neck circumference) can also be affected by OSA. One example of this is the increased prevalence of OSA among American football players, particularly offensive and defensive linemen.

Conclusion

Sleep is critical to both physiological and psychological function. Sleep loss has a negative impact on athletic performance; hence, education of athletes and coaches on sleep hygiene principles is needed to minimize sleep loss-related performance decrements. Sleep disorders are surprisingly common among athletes and, if suspected, should be investigated and treatment initiated in order to optimize athletic performance.

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See also Competition; Electroencephalograph (EEG); Fatigue; Rest; Underrecovery Syndrome

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SOCIAL COGNITIVE THEORY

Social cognitive theory (SCT), promulgated by psychologist Albert Bandura, has been used widely to explain health behaviors across different populations. Lifestyle physical activity (PA), resistance training (RT), and sport performance are examples of these types of behavior. Interventions to improve sport performance or increase PA participation have been developed using the SCT framework and shown to be successful at changing these behaviors. This entry describes the constructs that are central to the theory; discusses how they work to predict, explain, and change PA and sport behavior; and reviews key literature surrounding this theory.

SCT emerged in the 1980s during a period when social learning theories had been gaining wide recognition. Bandura differentiated this theory by emphasizing the importance of human agency—the notion that a person can influence his or her actions, with one's thoughts and cognitions playing a central role in determining one's behavior. Human beings are seen as capable of self-reflection, forethought, and self-regulation. Accordingly, human function is the result of an interaction between behavior, personal factors (including cognitions), and the environment. This interaction is also known as *triadic reciprocation* and was important in distinguishing SCT from other social learning theories of the same period. Each component in this triadic determination can influence the other two. Thus, a change in behavior

may result in a change in cognition in the same way as a change in cognition can impact behavior.

Main Constructs of Social Cognitive Theory

Bandura laid out specific theoretical constructs that would determine, predict, and explain behavior. He also gave details on how these constructs work together and how they can be influenced, changed, or enhanced to change behavior. The main personal construct of SCT is *self-efficacy* and is defined as the belief in one's ability to complete a specific task. *Barriers to self-efficacy* refers to the confidence a person has to overcome obstacles associated with a certain behavior. This type of efficacy is often used in PA promotion and explanation. Typical barriers include the environmental (e.g., weather, venue, travel), personal (e.g., schedule, feeling self-conscious), and social (e.g., no exercise partner, not receiving encouragement). Self-efficacy is hypothesized to influence how individuals act, what activities they choose to participate in, the amount of energy they put into behavioral attempts, how they combat obstacles, and the level of achievement they attain. Because of these functions, self-efficacy has been put forth as the most central and influential construct in Bandura's SCT theory. In fact, Bandura's SCT is sometimes referred to as *self-efficacy theory*. Self-efficacy also is theorized to influence every other construct, making it a key construct in behavior explanation and prediction.

Primary sources of efficacy information include mastery experience, vicarious observation, social persuasion, and interpretation of affective and physiological stimuli. Successful past performance of the behavior would be classified as mastery experience. People with prior success with the task would have higher levels of self-efficacy than those with prior failure or lack of experience. Another way to change self-efficacy is through vicarious experience. Watching another individual, thought to be similar to oneself, successfully complete the behavior would increase the observer's own self-efficacy. Social input, or persuasion, can also alter self-efficacy. If people receive positive feedback from an important other, their self-efficacy should increase. The final source of self-efficacy is the evaluation of affective and physiological states. If the state is perceived as a normal or positive response to the behavior then it will translate into stable or increased self-efficacy. If that symptom

is perceived as negative or abnormal, however, self-efficacy may decrease. The effectiveness of these strategies in exercise interventions has been investigated and vicarious experience, feedback on past or other's performance, action planning, instruction, and reinforcement of effort all bring about positive effects on self-efficacy. However, relapse prevention strategies, graded mastery, and persuasion have all been found to be ineffective for enhancing self-efficacy in exercise interventions.

Self-efficacy is often compared with perceived behavioral control, an element of the theory of planned behavior (TPB). Although similar in nature, they are two distinct constructs. Self-efficacy is focused on individuals' confidence in their ability, whereas perceived behavioral control is concerned with the amount of control people feel that they have regarding their participation in a certain behavior. Many times, perceived behavioral control is measured using some efficacy items, but these constructs are, in fact, different.

Another SCT construct posited to influence behavior directly and indirectly is the *outcome expectation*—beliefs about the consequences of an action or behavioral outcome. Positive outcome expectations are seen as motives for behavioral engagement, whereas negative outcome expectations are usually detrimental to performance of the behavior in question. For example, someone who believes that PA will result in pain (or exacerbation of current pain) may be less likely to engage in regular exercise than an individual who believes PA will decrease painful symptoms. Age may moderate the effect of positive outcome expectations on PA behavior, with the effect being stronger for older adults than for young adults. Outcome value or proximity also may play a role in this interaction. Typically, positive outcome expectations are measured with outcomes such as chronic illness prevention, which may not have great perceived value for younger adults. These types of outcomes are seen as less relevant (or distal) to their life stage. Thus, the predictive ability of positive outcome expectations, as typically measured, may be less for young as opposed to older adults.

Outcome expectations can be positive or negative but also fall into three categories: physical, social, and self-evaluative. *Physical* outcome expectations are the positive and negative feelings that are associated with the behavior. This category also includes any tangible gains or losses. *Social* outcome expectations refer to the impact

on the person's acceptance by important others. *Self-evaluative* outcome expectations are the positive and negative assessments made relative to one's participation in the behavior. If participation is viewed as improving one's rating of oneself or that it will heighten their self-perceptions, they are more likely to engage in that behavior.

Another integral construct in the social cognitive framework is *self-regulation*, which involves the use of self-directing strategies to help change or maintain behaviors. Various components of self-regulation, including goal setting and self-monitoring, have been found to be related to health behaviors.

There are three main components of self-regulatory systems. *Self-observation* involves monitoring one's behavior and performance. This component includes a self-diagnostic function and a self-motivating function. People can diagnose themselves in terms of the behavioral patterns that they observe over time. They can determine what conditions within their environment are present when they behave in a certain way. This information can be meaningful when attempting to learn a skill or change a behavior. The self-motivating function comes into play when a person sets goals based on his or her previous performance or behavior. The *judgment process* is the second component of self-regulation and involves developing personal standards or goals for performance and behavior. Performance can be compared against a person's previous behavior, a certain group of important others, or in a collective. The collective comparison is an evaluation of a group's performance, rather than an individual's, and is particularly important in the sport domain. The third component of self-regulation is *self-response*. Based on the observed performance and its evaluation, a person will react favorably or unfavorably. The type of reaction will depend on several factors, including the perceived importance of the task and the locus of control. If individuals feel that the task was personally important but they did not have personal control over their performance (e.g., the venue was inhospitable), then the reaction will be very different from if they had felt they had control over their performance. A person trying to elicit behavior change may develop incentives for positive performance of this behavior. For example, if a person met their PA goal for the month, they might treat themselves to a new workout outfit.

Also important in Bandura's SCT are *impediments and facilitators* of behavior. Individuals may evaluate their own ability to overcome barriers to health behaviors, and if the barriers are perceived as too great, the individual may fail to engage in the behaviors. Alternatively, if one is surrounded by an environment that encourages health behaviors, this individual may see this activity as one that is easily adopted. Typical facilitators and impediments include the social and physical environment. Social support is an important facilitator. If an individual learning a new skill or changing a behavior has strong support from valuable others, the person will tend to be more successful. The physical environment, either actual or perceived, can also facilitate or impede behavior change. This includes the built environment (e.g., neighborhood structures, walking trails, essential sports equipment), as well as policies and rules that may be in place to help people be active or engage in sport, or keep them from participation. Changing the environment, either physical or social, can be difficult as these components are out of the individual's control. However, change in the perceptions of these environmental components can help people change their behavior.

All of the aforementioned constructs work in concert with each other to explain current behavior as well as behavior change. Self-efficacy is fundamental to the process of behavior change in that confidence in one's abilities can provide the motivation necessary to initiate a change in behavior. Additionally, self-efficacy is important because it influences several other SCT variables. Individuals who are more efficacious are more likely to believe that the behavior will bring about positive consequences. That is, they have positive outcome expectations regarding the behavior of interest. If individuals have higher self-efficacy, they believe that with personal effort they can overcome the barriers to certain behaviors. Those who have high self-efficacy also set their personal goals higher than those who have lower levels of self-efficacy and are more motivated to achieve these goals. Not only does self-efficacy directly influence behavior, but also it has an indirect influence on a behavior by way of other social cognitive variables. Both outcome expectations and impediments and facilitators also indirectly predict behavior through self-regulation. People with more positive outcome expectations who view their physical and social environment as

supportive of their behavior will have better self-regulatory capacity, which should lead to positive behaviors. In summary, self-efficacy, outcome expectations, and self-regulation have direct effects on behavior. Impediments and facilitators are theorized to indirectly affect behavior through self-regulation.

Evidence of Behavior Prediction and Behavior Change

PA participation has been explained using SCT in several different populations. Evidence has linked self-efficacy with PA behavior in adolescents, emerging adults, older adults, and people with chronic conditions or disabilities. High levels of self-efficacy have also been shown to be related to better adherence to a PA intervention.

The effect of outcome expectation is often muted by the effect of self-efficacy on behavior, thus downplaying the importance of this construct. However, there has been evidence to show that self-efficacy, outcome expectations, social support, and self-regulatory behaviors all have independent predictive effects on PA behavior and play a role in behavior change.

Self-efficacy has been found to be the best predictor of change in behavior and maintenance of that change. People who participate in interventions and increase their self-efficacy for PA tend to also increase their levels of PA. Furthermore, if self-efficacy levels remain higher than at baseline, PA participation also remains higher than initial levels.

Evidence has shown that interventions that include action planning, or detailed planning processes regarding how, when, and where to perform activity, are more successful at increasing self-efficacy for PA as well as actual PA behavior than interventions that do not include action planning. The mechanism for this success is possibly through the successful use of goal setting and implementations. Further, successful completion of goals could increase a person's sense of mastery—thereby increasing their self-efficacy. Another successful intervention strategy is instruction on the resources available to help participants to engage in PA. Receiving specific instruction on what is available can help people to plan their goal implementation in such a way that enables successful completion of goals. This success may then increase self-efficacy and PA behavior.

Self-efficacy has been shown to be the mediating construct between intervention programs and increased PA in a number of different populations. Post-intervention work has shown that people who have higher levels of self-efficacy after an intervention tend to report engaging in greater amounts of PA than those who have lower levels of self-efficacy.

Social Cognitive Theory in Sport

SCT is not only useful for understanding PA behavior, but it is also important when examining sport behavior. There are many similarities in the sources of self-efficacy between PA and sport. Mastery experience, achievement of goals, social support (verbal persuasion), physical and mental preparation (interpretation of physical and emotional states), and vicarious experiences are the same in both domains. Some sport-specific sources of self-efficacy include coaches' leadership, environmental comfort, physical self-presentation, and situational favorableness. Another source of self-efficacy that has been primarily studied in sport is imagery. Athletes can use this technique to envision themselves performing successfully. The use of imagery has been shown to improve self-efficacy as people can see themselves successfully completing the task, beating an opponent, or simply feeling confident in their abilities. Moreover, imagery has been related to improved performance (an effect that may be mediated by changes in self-efficacy beliefs).

Efficacy beliefs help to determine people's behavior in sporting contexts. Certainly, their efficacy level will influence which activity they choose to participate in and what kind of goals they will set. People with healthy levels of self-efficacy will set challenging, but realistic goals. However, it has been suggested that efficacy in the sporting domain may have an inverted-U effect. Those with low self-efficacy will not do well in the activities they participate in because they will not put forth the effort and time needed to be successful. Those with overly high levels of efficacy may set their goals too high and lack the skills to meet them. This ultimately sets the individual up for failure and discouragement that may keep them from developing their skills. Those with optimal levels of self-efficacy tend to strive to improve their skills and continue to work hard even when faced with adversity.

Performance in the sport setting (i.e., competition, practice) is related to self-efficacy. People with healthy levels of self-efficacy tend to perform at higher levels than those with either very low or very high levels of efficacy. Sport performances provide very tangible evidence of goal achievement. Successes and failures are very visible and can be compared with goals, thus creating a context of goal-directed behavior in which SCT constructs become very relevant to performance. For example, those with high efficacy beliefs, when faced with internal thoughts and emotions, are typically able to focus on the challenge of the event and their goals. Those with low efficacy beliefs will typically focus on their worries (e.g., about losing, getting hurt, stress and pressure). This loss of focus can be detrimental to performance in sport. In addition to impaired concentration, sport performance can suffer from low self-efficacy due to increased anxiety or reduced level of self-confidence or self-esteem. Individuals faced with a challenging task but doubting they possess the skills to meet the challenge will have low self-efficacy and likely experience anxiety or concentration problems, leading to poor performance.

Beyond the Individual: Collective Efficacy

There are several different types of efficacy in sporting contexts. One such belief that is very different from the PA domain is the construct of *collective efficacy* (CE). This is the group's confidence in its ability to achieve a certain goal or perform a certain task. CE affects how much effort team members exert in team activities. It also plays a role in determining how long a member is willing to continue belonging to a group or team. CE has been related to team performance, even after controlling for past performance. Thus, building CE in a team may be a way to improve performance. CE can be enhanced using the same methods that are used to enhance individual efficacy, as long as they pertain to the team's performance and goals.

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See also CALO-RE Taxonomy of Behavior Change Techniques; Control Theory; Health Action Process Approach; Mastery and Control Beliefs; Protection Motivation Theory; Strength Model of Self-Control; Theory of Planned Behavior

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SOCIAL COMPARATIVE EMOTIONS

Social comparative emotions are elicited when there is a reflection and comparison of one's personal attributes with an individual or group of others who may pose a threat to the self. Social comparative emotions are evoked when there appears to be large a discrepancy between the self and another, in domains that are important and valued by the individual. Social comparative emotions such as jealousy and envy are developmental in nature and cognitively complex. Jealousy and envy are grouped with emotions of guilt and shame as problematic negative self-conscious emotions since these emotions arise from comparisons with a standard in which the individual comes up short and is left in a state of wanting, each emotion involves a threat to the self, and all emotions are fundamentally interpersonal (with jealousy and envy being explicitly interpersonal). Jealousy and envy are inherent emotions in sport and exercise contexts that are laden with performance accomplishments,

competition and cooperation, and physical appearance and function comparisons.

Characteristics of Envy and Jealousy

The distinction between the emotions of jealousy and envy has been blurred and confounded, in part since given the similar characterizations involving social comparison processes and an innate motivation to maintain positive self-evaluations. Targets of envy and jealousy tend to be individuals who are perceived to be similar to the individual on attributes such as sex, age, social class, sport position, or ability (e.g., a quarterback might feel envious of another team's quarterback). Given that feelings of envy and jealousy arise when a comparison is made in a domain that is important to the self, individuals who place importance on perceptions of physical appearance and performance (arguably most exercisers and athletes) may be prone to heightened envy and jealousy emotions.

Jealousy

Jealousy is experienced when the individual already possesses a desired tangible object or attribute yet feels threatened that he or she may lose this possession to another individual. Specifically, jealousy results when the superiority of one person threatens the self-worth of another person and diminishes the individual's status. The threat involves a perceived loss of a relationship to a rival. For example, a soccer player who has a strong rapport and friendship with the head coach may experience jealousy when another teammate receives praise or extra attention after making a successful play. Jealousy may also be experienced as a fitness partner continues to improve while the other partner does not—the inferior partner may fear the loss of the partnership to an individual of equal or superior fitness qualities. Affective components of jealousy include fear of loss, distrust, anger over betrayal, and uncertainty about the circumstances.

Envy

Envy is experienced when an individual wishes he or she possessed the objects or personal attributes of another that the individual himself or herself lacks. Envy is a negative emotion that is characterized by feelings of inferiority, hostility, and resentment. Generally categorized as an undesirable emotion, envy is associated with detrimental

behaviors such as diminishing the envied person's advantage or feeling pleased when the envied person suffers or loses their advantage (e.g., when an envied fitness partner or teammate gets injured and loses some strength or performance).

There are two distinctions in the semantics of envious experiences. Benign envy is described as longing for or a coveting of what another possesses but lacking feelings of hostility. This type of envy results from an upward comparison where an individual perceives himself or herself as inferior to another with regard to a possession or attribute but without a sense of injustice in the envied individual's superiority. This type of envy places the emphasis on the possession rather than the individual and consists of an appreciation and admiration of a possession—for example, a golfer who compares herself to a Ladies Professional Golf Association (LPGA) player and is envious of the professional athlete's putting skill. Since benign envy is not associated with hostile feelings, it is generally admitted more openly and honestly. Benign envy can be a motivating force to achieve the envied target; however, this depends on an individual perceiving behavioral control (i.e., he or she believes that the goal is within reach).

In contrast, malicious envy is described as an unjust feeling and desire for the envied individual to lack a desired trait. Malicious envy can manifest itself as others' perceived superiority in certain dispositional attributes such as intelligence, athletic ability, and physical attributes that may cause feelings of injustice for the inherent nature of (i.e., not being born with) these traits. For example, competitors may envy a lean and petite gymnast who they believe spends less time practicing due to her ideal sport-specific body type and feel ill will toward her (e.g., hoping the gymnast gains weight). Malicious envy places emphasis on the person being envied, rather than the possession. Due to social stigma associated with these feelings of malicious envy, it is highly unlikely to be openly admitted or discussed. This type of envy has been linked with predisposing individuals to feel pleasure at the downfall of the envied person, an emotion termed *schadenfreude*. Therefore, envy can manifest itself in various degrees of severity, ranging from resentment of another's successes, lessening importance of personal goals to diminish other's advantage, to pleasure in watching an envied other fail.

Psychological and Behavioral Outcomes

A high disposition to experience envy and jealousy has been associated with lower self-esteem, anger, irritability, anxiety, obsessive-compulsive tendencies, neuroticism, and depression. Furthermore, characteristics of envy, such as hostility and resentment, have been reliably linked with increased blood pressure (BP), physiological stress response, and cardiovascular disease.

Catherine M. Sabiston and Eva Pila

See also Self-Conscious Emotions; Self-Discrepancy

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SOCIAL COMPARISON

Social comparison is a process in which self-appraisals are formed and involves evaluating one's skills, attributes, belongings, and so forth compared with those of others.

Social Comparison Theory

Social comparison theory, proposed by Leon Festinger in 1954, states that individuals have an innate drive to maintain stable and accurate appraisals of themselves and do so by searching out comparative standards. There are three fundamental influences related to the drive to engage in social comparisons. First, social comparisons are used to understand one's own standing in relation to others in terms of *attributes*, *skills*, and *social expectations*. Self-evaluations tend to be more stable and accurate when comparisons are made in reference to someone who is similar, rather than when there are large discrepancies between one-self and other. In this regard, social comparisons help to develop *accurate self-evaluations*. Second,

social comparisons help one to maintain a positive self-image by comparing with someone perceived to be in lower standing in a certain domain. In this way, social comparisons help to *self-enhance*. Third, individuals compare themselves to people who are better (e.g., thinner, faster, stronger, more successful) in order to advance or self-improve. Therefore, the third motivation of social comparison is related to *self-improvement*.

Based on the *theory propositions*, individuals harbor a unidirectional drive upward in which they strive to achieve or self-improve. The preferred source of comparison is a person similar in ability; opinion; or attribute in question; or who presents with realistic characteristics, skills, or attributes that the individual can achieve. Maximal information can be extracted for evaluation with an individual who is similar, resulting in a more precise comparison. For example, an aspiring collegiate athlete may compare himself or herself to another rival athlete who is similar in age, physical fitness, and personal performance. This gives him or her a basis for evaluation on the probability of being selected for the available position.

Upward Social Comparisons

In cases where individuals are seeking to self-improve, the comparisons are often made with others who are better off, or superior, and this is termed an *upward social comparison*. Individuals who are competitive, high achievers, or highly motivated to achieve a goal are prone to make more upward social comparisons and do so based on achievable standards. Individuals may also use social comparisons with superior others to be inspired and learn skills from the other to achieve a goal.

However, upward social comparisons may be discouraging, since they increase awareness of one's own inferiority. Theorists claim that making upward comparisons with those who are similar may be difficult to accept when the superior other is highly similar in other dimensions. These upward social comparisons are particularly hurtful to one's self-esteem and foster negative affect and discontent. Also, some upward comparisons can be too difficult to accept (i.e., unachievable), and the individual making the comparison may use cognitive coping strategies, such as self-protection or ego defense. For example, individuals who compare unfavorably may selectively focus on other real or imagined domains that the target falls short on to maintain self-esteem.

Downward Social Comparisons

Downward comparisons occur with individuals who are perceived to be inferior or less advantaged. The primary motive behind these comparisons is self-enhancement. While not as common as upward comparisons, individuals tend to make more downward comparisons on dimensions that are desirable. For example, individuals tend to believe that if they are engaging in a desirable behavior, few others are doing the same. Yet when they engage in an undesirable behavior, they believe many others would do the same. Downward comparisons are used to maintain self-esteem and for ego protection. Strategies used for downward social comparison include purposefully selecting inferior targets, focusing on domains where the self is superior, or avoiding comparisons entirely.

Social Comparison and the Physical Self

Individuals can make social comparisons in many self-domains, yet social comparisons regarding the physical self are particularly pertinent in sport and exercise psychology (SEP). For example, it is argued that social comparison is a process throughout which self-appraisals of physical appearance and attractiveness are formed. Appearance and athletic or fitness ability-focused social comparisons are prominent pertaining to the physical self. Most research on this topic has focused on upward comparisons around the body and suggests that these comparisons are interpersonal processes that shape body image. Many researchers use social comparison theory to describe body dissatisfaction and body image concerns more generally, since the social (and media driven) ideals of attractiveness and fitness (e.g., thin and toned for women and muscular and tall for men) are unachievable for most yet are the target of most upward social comparison. While social comparisons perpetuate body image concerns, there is also some suggestion that the propensity to make such comparisons (i.e., general tendency to compare oneself to others) is a key factor in body dissatisfaction and negative affective and behavioral outcomes such as depression, anxiety, disordered eating, physical inactivity and/or exercise dependence.

The limited research on downward appearance-focused social comparisons suggests that individuals with motives to maintain or improve self-esteem and self-worth make more downward comparisons. Downward social comparison may

also be related to feelings of body-related hubristic pride. Generally, there is limited focus on downward social comparison and the association with body image or physical self-perceptions.

Individual Differences in Social Comparisons

Individual differences in social comparison experiences, such as sex, culture, and ethnicity, have only been minimally assessed. For example, some researchers contend that since females are more likely to define themselves by their interpersonal relationships compared to males, they may be more likely to experience social comparisons. However, most researchers suggest that males and females are prone to making similar comparisons. Also, cultural differences on social comparisons are often explained based on differences in the way the self is identified and internalized with respect to others. In general, cultures that are highly interdependent (e.g., East Asian) are more likely to engage in social comparisons than independent cultures (e.g., North American, European). Furthermore, individuals from independent cultures tend to make social comparisons with self-enhancement motives compared with individuals from interdependent cultures who are likely to display self-improvement motives. However, researchers focused on social comparisons targeting the physical self (e.g., appearance, body shape and size) suggest opposite findings. Independent, Westernized cultures tend to place higher importance on the physique, and therefore, individuals are more likely to make comparisons related to the body form and function and appearance compared to Eastern, more interdependent cultures. Additional individual differences that may predispose individuals to make upward social comparisons include low self-esteem, unstable self-concept, depression, neuroticism, and those high in reactivity to stress.

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See also Body Dissatisfaction; Hierarchical Self; Self-Acceptance; Self-Appraisal/Assessment/Perception; Self-Awareness; Self-Awareness Theory; Self-Discrepancy; Self-Presentation

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SOCIAL FACILITATION

See Social Processing Effects

SOCIAL IDENTITY THEORY

Identity exists along a spectrum that ranges from the personal to the social. The personal end of this spectrum is constituted by the distinctive combination of attributes that an individual deems relevant to his or her self-concept (e.g., being shy or tall). The social end of this spectrum, in contrast, is constituted by the extent to which a person aligns himself or herself with being a member of particular social groups (e.g., my sports team, being Canadian). One's position along this *identity spectrum* varies across contexts—certain situations make salient personal attributes whereas others highlight membership in specific social groups. Developed by Henri Tajfel and John Turner, two European psychologists based at the University of Bristol, social identity theory explains the mechanisms by which people come to function at the social end of the identity spectrum as well as the personal and collective implications that derive from identification with social groups.

Research Findings and Their Implications

In their early work in this area, Tajfel and Turner found that group membership—indeed even group membership established on the basis of quite arbitrary and trivial criteria (e.g., a coin flip)—can substantively shape and mold individual behavioral patterns. For instance, through use of a *minimal groups paradigm* whereby individuals are assigned

to one of two groups at random, participants not only tend to rate members of their own group (referred to as *in-group members*) more favorably than members of other groups (referred to as *out-groups*) but also demonstrate behavioral responses that reflect a sense of in-group bias (e.g., allocating more resources to in-group members relative to out-group members).

When a person's membership in a social group is made salient, he or she becomes more motivated by factors that reflect the interests of this group. Under some circumstances, this may even involve sacrificing self-interest for the betterment of the group. In the context of sport teams, highlighting the common social identity of the team can provide an invaluable basis for enhancing the degree to which members work collaboratively in pursuit of a common set of team-level objectives. For example, when the social identity of the team is underscored, players may become less concerned with their individual performance statistics (e.g., personal points per game) and more concerned with the *team's* success as a whole (e.g., winning the game). From a leadership perspective, an enhanced common social identity can allow coaches and others in position of authority to harness what Turner referred to as “power through” the group rather than “power over” the group. That is, if members are highly united, and motivated, in pursuit of the group's instrumental objectives, then leaders do not need to operate through coercion and instead can rely on players' team orientation to help maximize group performance.

Although highlighting the social identity of a team holds the potential to increase the performance of this group, it also carries with it the possibility of severely limiting its decision-making ability, which in turn may debilitate its capacity to function effectively. Individuals who function at the social end of the identity spectrum may create an environment in which *groupthink* flourishes, whereby alternatives and counterproposals are not voiced due to an increased desire for group unity. If members of this team are tasked with making decisions about the best course of action during a sporting event (such as when a golfer and caddy confer), coaches and players are responsible for deriving a strategy during a pivotal moment in a game (such as when a basketball coach attempts to draw up an appropriate play with his or her point guard) or managers must devise a plan for an upcoming game (such as when the managers of

a baseball team meet to discuss the team they will face later in the week), groupthink may restrict the number of options considered and, ultimately, the team-level outcome achieved.

Most, if not all, people possess membership in multiple social groups. These groups can range from the very specific (or subordinate; e.g., fans of the local basketball team) to the very broad (or superordinate; e.g., fans of basketball), and these differing levels of inclusiveness can have important implications for intergroup relations. A great example of this comes from a couple of experimental studies conducted by Mark Levine and his colleagues. In this research, they assessed the extent to which soccer fans helped an injured stranger on the basis of the shirt this stranger was wearing. In one study, Manchester United Football Club (MUFC) supporters observed the stranger (actually a confederate) slip and fall, ostensibly writhing in pain. Dependent on the condition being operationalized this stranger was wearing either a MUFC shirt, a Liverpool United Football Club (LUFC; historical rivals of MUFC) shirt, or a nonbranded sports shirt. Participants were found to display significantly greater helping behavior toward the stranger when he was wearing an in-group (i.e., a MUFC) shirt relative to when he was wearing an out-group (i.e., a LUFC) shirt, or a plain sports shirt.

In a subsequent study, the same research design was utilized but with one noteworthy exception. Specifically, a more inclusive (superordinate) social group was made salient by emphasizing participants' social identity as soccer fans in general rather than as Manchester United fans in particular. By getting participants to think of themselves as soccer fans rather than MUFC fans, they were found to help this stranger more if he was wearing a soccer jersey—irrespective of whether it was a MUFC or LUFC shirt—relative to those wearing a plain sports shirt. From the perspective of informing theory-driven interventions, the results of this study suggest a potential mechanism to reduce intergroup conflict and discrimination. That is, by emphasizing superordinate social identities, it may be possible to get bitter rivals to see each other as human beings, rather than as despised members of separate subordinate groups.

A key tenet of social identity theory is that, when people are either allocated to, or self-select membership in, specific social groups they not only look to differentiate themselves from other groups but also identify and create ways of demonstrating the

superiority of their own group. Tajfel and Turner referred to this as a motivation to display *positive distinctiveness*. The underlying intent in displaying positive distinctiveness is to bolster feelings of self-worth. In the context of sport, such distinctiveness arises when teams not only distinguish themselves through their physical appearances (e.g., team uniforms and logos) but also through developing intrateam rituals (i.e., team songs) and a unique language or dialogue. Parenthetically, the utilization of strategies designed to reflect positive distinctiveness is theorized to bolster team cohesion and has been frequently used as a practical team building strategy to maximize team functioning. More recently, those concerned with increasing adherence behaviors within the context of exercise classes have sought to make use of distinctiveness strategies (e.g., allocation of a group name, class T-shirts) to foster a sense of group cohesion and thereafter sustained involvement in those classes among group members.

In sport, the drive to display positive distinctiveness is not only evident among athletes on teams but also by supporters (i.e., fans) of those teams. An interesting phenomenon that illustrates this process is referred to as *BIRGing*, or “basking in reflected glory.” In one of the first studies to examine BIRGing, university students were found to have a much greater tendency to wear clothing that reflected the university's colors and name after the university's (gridiron) football team had won, rather than lost, a game. By associating themselves with the success of their university's football team, students sought to positively distinguish themselves from those students at other universities (which, in this case, represented the relevant out-groups). Conversely, if sports fans have developed a particularly ingrained social identity with a given team, and this team experiences a lack of success, this increases the likelihood of those fans attributing failures to external (e.g., biased officiating), rather than internal (e.g., a lack of talent) factors, in the interest of maintaining a sense of self-worth. If, on the other hand, a person's social identity is only weakly aligned with that of a failing team, he or she may well engage in a process of *CORFing*, or “cutting off reflected failure.” That is, this person may seek to distance himself or herself from the underperforming team, again in the interest of protecting a sense of self-worth.

Across diverse life contexts, social identity theory has also been utilized to understand and

ameliorate prejudice between social groups. Within the domain of sport, this is particularly pertinent in relation to race and gender disparities. From the perspective of social identity theory, when the boundaries between social groups are perceived to be impermeable (i.e., one cannot easily shift between social groups, as is the case with race and gender), group members may engage in a process of *social creativity*, whereby they make use of strategies that emphasize the positive qualities of the in-group. For example, in sport this might involve emphasizing the unique athletic, aesthetic, and competitive qualities of women, who in the past have been deprived access to participation in various elite sporting endeavors. Another strategy that is sometimes implemented, when people perceive the basis for status differences between groups to be illegitimate, is *social competition*. This involves low-status group members directly challenging high-status social groups in order to redress the balance of power and redistribute available resources. This is evident, for example, when women have challenged the lack of equity in prize money allocations within professional sports (especially golf and tennis) or when athletes from racial minority groups have challenged the legitimacy of being selected only for marginalized (rather than focal) playing positions.

Conclusion

In summary, the groups with which we identify significantly impact our thoughts, emotions, and behavior. Within the context of sport and exercise, identifying with a team or physical activity group fosters a sense of cohesion and holds the potential to enhance group-level performance and individual-level exercise adherence, respectively. A consideration of social identity theory allows for a better appreciation of why this is so. Furthermore, a consideration of this theory offers insights as to why fans often behave the way they do, coaches sometimes make the wrong call, and members of marginalized groups come to focus on their strengths instead of their weaknesses and/or demand equality. Thus, social identity theory has proven valuable in explaining the *groupness* of our existence.

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See also Cohesion; Hierarchical Self; Self-Categorization Theory; Self-Presentation; Team Building

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SOCIAL LOAFING

See Social Processing Effects

SOCIAL MARKETING AND MESSAGE FRAMING

Social marketing employs commercial marketing strategies to try to solve social problems and to effect voluntary behavior change. An important aspect of social marketing is *message framing*,

which is the tone or valence in which the information related to the behavior is conveyed. This entry discusses social marketing and message framing in the context of physical activity (PA) promotion and exercise adherence, provides examples of successful marketing campaigns, and identifies factors that influence the effectiveness of such campaigns.

PA is unquestionably important for health, yet the majority of people in developed countries are not active enough for health benefits. Part of the difficulty may be that, although there is abundant information available regarding the benefits of being active, many people do not fully understand what it means to be active. Further, there are myriad conflicting messages about “exercise” put forward by commercial companies that can result in confusion. Messages about health compete with messages about achieving a certain look and may cause misunderstandings regarding how much PA is needed, of what type, and how often. Thus, although consistent messaging is needed, “too much confusion” is often cited as a reason for not adhering to positive health behaviors. It is therefore important that health promoters have a single, consistent, clear public health message to counter trends in physical inactivity.

Promoting Physical Activity Through Social Marketing

Social marketers use the “four Ps” of marketing to effect change: a *product* (e.g., the benefits of PA) is offered at a certain *price* (e.g., discomfort, costs of running shoes), and is *promoted* (e.g., using advertisements) in specific *places* (e.g., through television). Social marketers also often work in partnerships with other agencies to try to influence policy. Because it is recognized that a “cookie-cutter” approach to PA messaging is ineffective, a key to social marketing is targeted marketing, where the audience is segmented into groups who share characteristics. As such, a message targeted at older adults will likely be very different from a message targeted at teenagers or working mothers. However, regardless of the audience, it has been advocated that the messages contained within social marketing and health promotion efforts should be framed positively and highlight the benefits of PA. This recommendation is made based on a body of research that examines the effects gain-framed messages (i.e., highlight benefits: PA will help prevent heart disease) compared to those

that are loss framed (i.e., highlight risks: you are at greater risk of heart disease if you are not active). Essentially, the same information is conveyed but in different ways. Reviews of research that examine message framing and PA conclude that using gain-framed messages is more effective than loss-framed messages in changing PA.

Why Gain-Framed Messages Are Effective

Several reasons have been proposed as to why gain-framed messages may be optimal for promoting PA. One suggestion is that gain-framed messages match how people think about PA. That is, for the most part, people consider PA a behavior that will afford sure health benefits. Gain-framed messages reinforce this notion of benefits and have been suggested as the most persuasive message type when people are considering behaviors such as PA that have sure outcomes. There is also some emerging evidence that gain-framed messages may better capture attention and have a stronger impact on peoples’ beliefs about, and attitudes toward, PA than loss-framed messages.

VERB is an example of a successful social marketing campaign that employed gain-framed messages to encourage PA among “tweens” (children ages 9–13) in the United States between 2002 and 2006. Gain-framed messages included “A steady regimen of physical activity doesn’t just keep children fit, it helps build social skills and confidence” and “Kids who are active are a step ahead when it comes to their health.” Children who saw the advertisements had more positive opinions regarding PA and were more active. Parents were also positively influenced by this campaign.

A broader example that has included multiple campaigns targeted at different market segments is ParticipACTION, a national not-for-profit Canadian organization dedicated to increasing PA in the Canadian population. ParticipACTION has received much international recognition and is identified as one of the most successful campaigns ever and contributed to creating societal norms for activity in Canada. Many of ParticipACTION’s campaigns are targeted at parents with the goal of informing them about the importance of PA for children and youth (e.g., the Think Again campaign). Over the years, the content of the messages in the campaigns have varied. Gain-framed messages used by ParticipACTION included “Regular physical activity is an essential part of early

childhood growth and development” and “Being active teaches young kids healthy habits that will be more likely to stick with them for life.”

Message Awareness and the Hierarchy of Effects

Despite these examples of how gain-framed messages can be incorporated into social marketing, PA participation is still low in North America, and the question remains as to why information does not necessarily translate into behavior change. Though there are many possible reasons, we note and discuss the following three: (1) good marketing and message campaigns may have many positive outcomes other than behavior change; (2) sociodemographic differences exist in how quickly information is disseminated into society; and (3) the competing, often conflicting, messages that are pervasive in the wide media context may influence behavior.

The first of these, that behavior change is not the only possible outcome of PA messaging, boils down to a key marketing principle: The crux of a successful campaign is basic awareness. Because increased information is a goal of many campaigns, the hierarchy of effects framework is sometimes used to evaluate the success of a PA campaign. In this framework, awareness of the message is a key effect, followed by changes in variables such as beliefs, attitudes, or intentions, and, ultimately, a change in behavior. The point is that behavior change may not happen because of a messaging campaign, but increased awareness of a problem and perhaps positive changes in attitudes or beliefs could occur. Indeed, evidence supports that gain-framed PA messages are more attention grabbing than loss-framed messages. For instance, researchers used eye tracking technology to measure individuals' eye movements as they read gain- or loss-framed PA messages. With eye movements being an indicator of attention, it was determined that participants attended to gain-framed messages more than loss-framed messages. However, the messages did not affect PA behavior. As exemplified in this study, gain-framed messages may be a good starting point for behavior change but they evidently need to be supplemented with additional behavior change techniques and interventions to produce lasting change.

The Think Again campaign of ParticipACTION (2011–2012) is also a good example of a campaign that had awareness as its main goal. The

problem being addressed was that less than 10% of Canadian children meet national guidelines for PA (i.e., 60 minutes of daily moderate-to-vigorous physical activity [MVPA]), but over 80% of parents think their children are sufficiently active. ParticipACTION, therefore, created a campaign targeting mothers with children ages 5 to 11 to try to address this discrepancy and to raise awareness of how much activity children actually need. Though it is possible other effects will occur because of this campaign (e.g., parents support their children in being more active), the message first needed to be disseminated and understood by parents. For instance, if parents believe that swimming three times a week is enough activity for their child, it is unlikely they will try to help their child be active in other ways. Subsequent campaigns may then target changing beliefs about PA or actual PA.

The Knowledge-Gap Hypothesis

Second, health promoters should be aware that PA-related information does not necessarily reach everyone in society at an equal rate and, when it does reach them, it may not be considered relevant by all message recipients. This problem has been considered through the knowledge-gap hypothesis, which states that as mass media are introduced into society, those with more education will acquire knowledge faster than those with less education. This occurs because people with more education are often better able to manage communication, have more prior knowledge about a topic, have broader social networks, and pay more attention to public health education campaigns. A variety of contextual and personal factors have been shown to influence health information seeking, including demographic, socioeconomic, information environment, health status, and motivation. For example, researchers have argued the “digital divide” of Internet use that exists between seniors and younger adults is due to generational differences in uptake of new technologies rather than differences in cognitive abilities or “stage of life.” A meta-analysis indicated that knowledge gaps in health promotion have persisted and have not decreased over time. The authors of this meta-analysis argue that such health knowledge gaps may account for the differences in prevention behaviors found between high and low socioeconomic status (SES) groups. Thus, ways to overcome the knowledge

gap are needed. Moreover, it is important to consider that once the information reaches recipients its uptake may vary from person to person. There are a number of personality characteristics that have been found to affect whether an individual is receptive to gain-framed information. For example, people who focus on achieving positive outcomes are especially receptive to gain-framed messages, whereas people who are concerned about preventing negative outcomes are less persuaded by gain-framed messages.

Social Marketing in a Complex Media Environment

A final difficulty in translating PA messages into increased behavior is the conflicting messages in the wider media milieu. Several authors have questioned how well health promotion campaigns fare within today's increasingly complex media environment. For example, little is known about how commercial advertisements may compete with public service health promotion campaigns. This is a concern because the number of commercial marketers of exercise products and services is far greater than noncommercial marketing efforts. One manifestation of this competition is the attention afforded commercial advertisements may draw attention away from gain-framed health promotion advertisements. However, whether the attention paid to commercial advertisements is necessarily detrimental to population health remains to be determined. Though limited, research increasingly points to negative effects of promoting PA for appearance reasons. Another difficulty is that we do not fully understand what information is available to consumers. For example, the Internet is a popular source of information, yet the challenges of understanding what PA information is available and who accesses it have not been overcome. At the time of this writing, typing the word *exercise* into the Google search engine resulted in 579,000,000 hits and the specific phrase *physical activity* got 87,900,000 hits. With such an overwhelming amount of information available, those interested in promoting PA are strongly advised to not ignore where people get information regarding PA and how such information is interpreted.

Conclusion

Marketing campaigns are often used to counter trends toward physical inactivity, with messages

promoting PA for health disseminated not only via traditional media such as television and magazines but also through the Internet and social media. Whatever the medium, the general goal of mass-market health promotion campaigns is to increase the amount of information available on a particular health topic. Targeted social marketing campaigns that promote the benefits of being active through gain-framed messages can successfully (albeit modestly) influence societal norms. However, if behavior change is looked to for evidence of success, one may be disappointed. Rather, evaluation efforts should examine awareness of the basic message followed by changes in attitudes and beliefs prior to looking for increased PA. Further, PA messages may reach some people faster than others due to socioeconomic, education, or other barriers. The plethora of information available from multiple sources may also impede uptake of PA. Thus, a consistent public health message is needed to counter trends in physical inactivity. The message can be tailored to meet the needs of different segments of society, but the unfailing basic message should be that PA is unquestionably good for you.

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See also Health Belief Model; Health Promotion; Self-Regulation; Social Cognitive Theory; Theory of Planned Behavior

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SOCIAL NEUROSCIENCE

Exercise scientist William P. Morgan often contrasted how exercise physiologists and exercise psychologists study the exercising human. The physiologists were seen as unconcerned with what happened above the neck, choosing instead to focus only on physiological processes that changed during and after exercise and making the erroneous assumption that affect, cognitions, and perceptions were unimportant in understanding these processes. Likewise, their psychologist counterparts were seen as unconcerned with what happened below the neck, choosing only to focus on the psychological and/or social psychological processes that took place during and after exercise and ignoring the contributions that physiological processes might yield. The reality is, of course, that the body has a head and the head has a body—physiological changes can influence psychological processes and vice versa. Within exercise and sport psychology (SP), the biological plausibility of various social psychological or cognitive theories that dominate the field can be examined. Questions such as the following can be asked: How do all of these social psychological constructs relate to biological function? Where does the “energizing” capacity of goals come from? How might level of self-efficacy be manifested in the body? To grow meaningfully as a field of study, theoretical development needs to be

framed within the context of neurophysiological mechanisms. Social neuroscience provides such a framework.

Social neuroscience is a multidisciplinary field of study that seeks to better understand how various physiological systems of the human body (e.g., immune, autonomic, central nervous system [CNS]) are involved in, and influenced by, social (e.g., social relationships, social support networks) and social psychological (e.g., attitudes, motivation, confidence, intentions, self-regulation) factors. At its essence, social neuroscience seeks to move beyond “simple” biological explanations for behavior by integrating influences across the physiological, psychological, and social levels of interactions between the person with his or her environment. It is worth noting that although the term *neuroscience* is currently the term of choice, and seems to impart immediate implicit respect, it is simply an evolution of already existing fields of study (e.g., psychophysiology, psychobiology) with a more explicit focus on the CNS. Understanding these multilevel interactions (e.g., physiological, psychological, social) can aid in better organizing theories of how the mind and body work. In essence, social neuroscience is a multilevel analysis approach to studying the problems and questions in exercise and SP. Such multilevel approaches are not easy to apply, but the informational yield is far greater than can ever be achieved by continually examining phenomena from a single level of analysis.

A social neuroscientific approach is certainly not impossible. Bradley Hatfield has been engaged in a programmatic line of research aimed at understanding the nature of superior cognitive motor performance and how that relationship is influenced by stress. Emerging from his early work, Hatfield described a model of efficiency that operates on multiple levels (i.e., neural, psychological, physiological, biomechanical). He argued that an environmental challenge can change brain activity, in turn, influencing the emergent psychological state (e.g., self-efficacy, focus, mood). The resulting psychological state then influences the motor control processes, autonomic and endocrine system functions and ultimately results in qualitative change in physiological state and the movement outcome. Hatfield’s empirical work demonstrated how SP, motor control, cardiovascular psychophysiology, and exercise physiology point toward “efficiency” as a main feature

of superior performance. Hatfield's research is a beautiful illustration of what John Cacioppo and his colleagues referred to as multiple levels of analysis (i.e., a social neuroscience approach). Such a model not only informs the findings of the research that they have carried out but also serves to guide subsequent investigations.

As another example, Art Kramer and Edward McAuley have been involved in a systematic line of research examining the influence of physical activity (PA), or exercise on cognitive functioning in older adults. The beauty of this work is in the approach. Instead of examining tasks that simply reflect cognitive functioning (which is a broad-ranging construct), as so many others have done, their approach was to determine (a) what regions of the human brain are compromised with aging; (b) what cognitive tasks are thought to be "controlled" by such brain regions; and (c) whether exercise protocols that increase blood flow and neuronal density (determined from previous animal studies) can influence performance on such cognitive tasks. Given such an approach, it is not surprising that this group found more positive effects than are typically reported in this very diverse literature. As this line of research using what can be classified as a social neuroscience approach has continued, the investigations have yielded not only behavioral findings showing positive effects of exercise in previously sedentary older adults but corresponding changes in brain structure and function (via magnetic resonance imaging [MRI] and functional magnetic resonance imaging [fMRI]).

Many other examples could be cited to illustrate how "reintegrating" mind and body meaningfully in models and theories provides a much greater understanding of behavior. There is far more to gain by examining human behavior as a mind-body phenomenon from multiple levels (i.e., social neuroscience, psychophysiological, social-environmental) than can be gained by separating mind and body and/or staying within single levels of analysis. This knowledge yield is no less important in exercise and SP.

Steven J. Petruzzello

See also Autonomic Nervous System; Brain; Cognitive Function; Decision Making; Electroencephalograph (EEG); Information Processing; Neuroscience, Exercise, and Cognitive Function; Psychophysiology

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SOCIAL PHYSIQUE ANXIETY

Researchers have been studying the measurement properties, correlates, and outcomes of social physique anxiety (SPA) for more than two decades. With roots in social anxiety, self-presentation, and body image affect, SPA is defined as the anxiety that an individual experiences when he or she perceives that others could be negatively evaluating his or her physique. Sport and exercise environments are inherently social and evaluative, and it has been argued that sport and exercise may foster preoccupation with self-presentation and anxiety about how the body appears to others.

Measurement of Social Physique Anxiety

Early research exploring the measurement properties of the Social Physique Anxiety Scale (SPAS) supported the use of the original 12-item scale and offered insight into a two-factor scale comprised of physique presentation comfort and expectations of negative physique evaluation. More recent research argued for a reduced 9-item scale deleting

three of the five positively worded items and maintaining a unidimensional structure. Further testing of the measurement scale has provided evidence for the 9-item or an 8-item unidimensional scale. For younger adolescents, researchers have also advocated for a 7-item scale, essentially removing all positively worded items from the scale. The majority of researchers report using the 9- or 12-item scale.

Further support for the measurement of SPA has been presented with high internal consistency coefficients for the SPAS items (e.g., Cronbach's alpha coefficients are consistently reported to be higher than 0.85) across samples ranging the age life span, gender, ethnicity, sport type, and exercisers. As evidence of construct validity, SPA has been moderately to highly correlated with a range of personality traits, anxiety, self-presentation processes, and body image dimensions.

Physique Anxious Always, or Sometimes? The Trait-or-State Debate

The original conceptualization of SPA was as a stable disposition or trait. In addition to linking SPAS scores with relevant personality traits such as perfectionism, Crocker and colleagues conducted a prospective longitudinal study with adolescents and demonstrated that scores on the SPAS were quite stable over three time waves 1 year apart. While the SPAS was developed with trait-like features in mind, several researchers have provided evidence of statelike properties both using the original scale and in the modification of the scale to target a more malleable physique anxiety. For example, SPAS scores were higher among college students reading hypothetical scenarios placing them in a fitness center when compared to a dining hall and library and reportedly higher among older adolescents when they were in the company of peers compared to being in an evaluative environment with their parents. Further reports suggest physique anxiety is higher among individuals when confronted with an opposite-sex exercise partner or fitness leader compared with a same-sex environment. With the development of a state SPAS and research evidence supporting both trait and state features of SPA, it is best that sport and exercise psychology (SEP) researchers employ the conceptualization and complementary measurement scale that best matches their research questions and study goals.

Age, Gender, and Weight Status Effects

There is cross-sectional evidence that SPA is highest among adolescents and young adults and lowest among older adults. Further research is needed to better understand the developmental features of the experience of SPA. Overweight men and women tend to report higher SPAS scores compared with age- and gender-matched healthy weight counterparts. Furthermore, there is consistent evidence that females score higher on the SPAS than males. These differences are deemed to be true mean differences since multiple studies have demonstrated gender invariance of the SPAS, suggesting that there are no gender differences in the way the scale items are interpreted and completed.

Sport Type

There are no clear divides in SPAS scores among athletes compared with exercisers, nor among competitive versus recreational athletes. Nonetheless, consistent evidence suggests sport type is important in SPA experiences. Athletes in competitive aesthetic and subjectively rated sports such as figure skating, gymnastics, and some martial arts tend to report higher SPAS scores compared to athletes in non-aesthetic sports such as basketball and soccer. There is mixed evidence that the type of uniform worn in sport may also affect SPA experiences, with more revealing uniforms worn in the aesthetic sports and sports such as track and field, swimming, cross-country, and women's volleyball. Therefore, it is important to tease out the relative (and cumulative) effects of sport type or sport constraint contexts on fostering or protecting from SPA experiences. Mandatory uniforms in physical education (PE) classes are also described as highly relevant sources of SPA among adolescents.

Social Triggers

Peer group influences may perpetuate feelings of SPA. In a study by Diane E. Mack and colleagues, male and female adolescents who received encouragement to alter their physiques from friends in their peer groups, and who identified themselves as less attractive than their peers, reported higher SPAS scores. Cox and colleagues have focused on PE settings, finding that general feelings of acceptance and belonging within a peer group have been shown to help to buffer against SPA.

Other sources of SPA may include parents, teachers, coaches, and fitness leaders. For example, mothers of adolescent girls may trigger SPA through indirect commentary and controlling actions such as guiding physical activity (PA) and managing their daughters' diets. Teachers and coaches who place an emphasis on appearance and weight are perceived to foster a more threatening physique-salient and evaluative environment. Additionally, the media may heighten experiences of SPA. In particular, there is evidence that SPAS scores increase when individuals (predominantly studies involving women) are exposed to appearance-salient advertising and decrease when individuals are exposed to health-based advertising. These findings are consistent when the exposure includes fitness videos, television commercials, or print media such as photos and posters.

Self-Perceptions and Social Physique Anxiety Experiences

High SPAS scores have been associated with low physical self-perceptions related to body weight and appearance; low perceptions of competence for PA, healthy eating, and PE skills; low perceptions of autonomy; and lack of enjoyment of sport, exercise, and PE. Body dissatisfaction, body surveillance and monitoring, and social comparison have been consistent correlates of SPA experiences.

Coping With Social Physique Anxiety: Physical Activity Avoidance or Engagement?

Given the self-presentation and social anxiety foundations of SPA, it is not surprising that a predominant strategy for managing physique concerns is behavioral avoidance (i.e., avoiding the situation or event that is likely to elicit the feelings of anxiety). In sport, behavioral avoidance may result in an athlete dropping out of his or her sport. There is evidence that adolescents who experience high SPA in PE classes are more likely to avoid classes by skipping PE altogether or seeking parental notes excusing them from class. Exercisers who feel that the fitness center is a threatening environment for physique evaluation may avoid exercising, or engage in less effective fitness routines to avoid certain locations within the gym (e.g., avoid the free weight area, or the aerobics floor). In fact, there is consistent research that mirrored exercise environments perpetuate SPA, and individuals report

avoiding these areas—perhaps at the expense of proper form and exercise function.

When avoidance of the sport and/or exercise situation is not feasible, there are reports that individuals high in SPA will manage the threatening context by covering up their physiques to avoid the possible evaluations by others. For example, adolescent girls reported wearing loose-fitting clothing in PE classes and at the gym, and a common strategy at the beach or swimming pool is to cover a bathing suit and avoid exposing the body unless necessary. Women who have had mastectomies for breast cancer report using prosthetics to hide the surgery and exercise or participate in sport in non-formfitting clothing.

There is qualitative and quantitative evidence that individuals reporting high SPA are likely to avoid PA situations that provide opportunities for physique evaluation. There is also evidence that SPA may motivate some individuals to engage in PA. This dichotomy of avoidance and engagement has led to an overall finding of ambiguous associations between SPA and PA outcomes. Some reports suggest a negative association (as SPA increases, PA decreases, as in an avoidance model), others a positive association (i.e., an active engagement model), and there are also findings of no significant associations. In an attempt to explain these ambiguous findings, researchers have linked SPA experiences to reasons for exercise and specific behavioral regulations. Higher scores on the SPAS have been consistently associated with more externally regulated reasons for exercise, such as exercising to gain some reward (e.g., weight loss, social affiliation) or to avoid self-conscious emotions such as feeling guilty or ashamed. These findings have been supported in PE contexts among youth. Furthermore, profiles have been identified whereby youth who report relatively high SPAS scores combined with high external regulation for PE reported the lowest leisure time PA behavior and the least enjoyment of PE compared to those reporting low SPAS scores and high autonomous regulations.

Other Health Outcomes

In addition to the association between SPA and PA, researchers have also linked SPAS scores to disturbed eating attitudes and reports of disordered eating behaviors among college women and various elite sport athletes. Athletes high in SPAS

scores have reported higher competitive anxiety in their sports. There is also limited evidence that SPA is associated with depression in exercisers and non-exercisers.

Some recent reports have also associated physique evaluative threat, which may be analogous to SPA, to higher physiological markers of stress. Specifically, individuals in contexts manipulating physique threat have had high post-manipulation levels of cortisol, which is a hormone released by the hypothalamic-pituitary-adrenal (HPA) axis. These studies have demonstrated initial evidence that SPA may have long-term health implications since cortisol and heightened chronic stress are associated with a host of poor health outcomes (e.g., cardiovascular disease, immune system dysregulation, depression). Research studies focused on the long-term effects of SPA are needed.

The Future of Studying Social Physique Anxiety in Sport and Exercise Psychology

SPA experiences have been well defined and documented in SEP literatures. One of the challenges moving forward will be to determine the unique role of SPA within a comprehensive operationalization of body image affect, or body-related emotions. High correlations among SPAS scores and body shame have been reported, and it is likely that SPA experiences are highly tied to embarrassment and guilt. Using qualitative research findings as a guide, feelings of SPA may not be experienced frequently—a finding that is also supported by the consistent low scores on the SPAS scale (the mean values across studies are often below the midpoint of the Likert-type scale, suggesting *slightly to moderately*). It may be valuable to include measures of relevant and related self-conscious emotions in studies focused on physique salience, evaluation, self-presentation, and body image.

Furthermore, one of the biggest limitations in the SPA literature to date is the lack of longitudinal studies. Prospective studies are needed to provide insight on the developmental trends in SPA experiences, health outcomes, performance outcomes, and an understanding of the mechanisms that may help explain the association between SPA and behavioral outcomes among exercisers, non-exercisers, and athletes.

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See also Body Image; Self-Conscious Emotions; Self-Presentation

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SOCIAL PROCESSING EFFECTS

Social processing effects are grounded within individuals' assessments and interpretations of social contextual information. Because sport and exercise psychology (SEP) is the scientific study of how people and their behaviors affect and are affected by the environmental contexts in which they operate, social processing effects can best be viewed through a social-psychological lens. The social context in sport and exercise settings is shaped by members of the audience, teammates, and opponents. This entry is focused on social processing effects by exploring the topics of *social facilitation*, *social loafing*, *the Köhler effect*, and *audience effects*.

Social Facilitation

The influence that others have on a performance, social facilitation, is a notion that can be traced back to the work of Norman Triplett in 1898. Through Triplett's studies of bike race performance and children winding a fishing reel, he noted that individuals tended to perform better in the presence of others when compared to performing the same task in isolation. Triplett attributed this phenomenon to what he termed *dynamogeny* (i.e., the presence of a coacting performer resulting in the expenditure of additional, otherwise dormant, energy). Floyd Allport was the first to use the term *social facilitation* as he found that individuals generally performed better on a variety of tasks when they were grouped with others, as opposed to sitting alone. Both Triplett and Allport were aware, however, that the presence of others did not universally result in performance improvement, as some participants had been observed to perform worse when being observed or coacting with other individuals.

Developing an explanation for inconsistencies that had been observed when performing in the presence of others took until the 1960s and the work of Robert Zajonc. Zajonc argued that when individuals engage in coactive tasks (i.e., similar tasks but independent of each other) or performed in the presence of others (i.e., audience member and groups of two or more), behavior and performance can best be explained utilizing *drive theory*—postulated by Clark Hull in 1943. Specifically, Zajonc posited that the presence of others elicits an arousal response, which increases the chance that the dominant response will occur. Therefore, during the initial stages of skill acquisition, the presence of others would be more debilitating than facilitating; whereas, once a skill is mastered, the company of others would be more likely to positively affect performance. Zajonc highlighted how this relationship held true in a multitude of studies regardless of whether the behavior was focused on trying to achieve a goal or centered on trying to avoid punishment. Other researchers have since noted that a two-factor model (i.e., arousal $\times$ skill mastery) explaining social facilitation may be too simplistic. For instance, people are often more distracted from the important tenets of a task when an audience is present compared to when alone. Thus, the explanations of social facilitation must also include the

element of distraction or attentional changes to fully understand and operationalize this concept. Finally, while Zajonc's model did make significant strides in explaining how social facilitation operates, it did suffer from the limitation of social facilitation only being defined as the mere presence of others with no interaction between participants and observers—something that rarely occurs in sport. Nickolas Cottrell highlighted the difficulty with this limitation and asserted that an additional factor of *evaluation apprehension* needed to be included when examining how arousal levels affect performance. In particular, Cottrell and colleagues found that when skilled individuals participated in a verbal habits recollection task—with group members blindfolded—individuals did not perform as well as when individuals were being observed by others unknown to the participant. This finding helped to spur on other research related to the effect audience members can have on individuals if the participants believe their performance is being evaluated.

Social Loafing

At the opposite end of the spectrum from social facilitation (i.e., the presence of others facilitating an individual's performance) is the phenomenon of social loafing (i.e., the reduction in an individual's efforts when working in a group setting). In 1913, Max Ringelmann had reported that individuals had the tendency to become less productive as the group size they were working in increased (i.e., the "Ringelmann Effect") after measuring performances of individuals in pushing and pulling a horizontal load in different size groups. The social loafing construct was subsequently formalized in the 1970s in independent research efforts lead by Alan Ingham and Bibb Latané, who presented evidence indicating that motivational losses were primarily responsible for maladaptive group productivity behavior. Specifically, four main explanations have been proposed to explicate why social loafing occurs: *the allocation strategy*, *the minimizing strategy*, *the free rider proposal*, and *the sucker effect*. The allocation strategy contends that people are motivated to work with other individuals; however, they also save their best effort for solo performances that have direct reflections upon themselves. The minimizing strategy suggests that people seek to complete necessary tasks but with the least amount of effort required; thus,

because of increased potential and lack of accountability in groups, social loafing occurs quite frequently. The free rider proposal asserts that group members reduce their efforts because of the belief that others will compensate to complete the necessary task(s). Finally, the sucker effect is the result of individuals reducing their own efforts in a group setting because they do not want to give a free ride to other, less productive members.

One mediating effect of social loafing is task cohesion—as specifically, swimmers who engaged in a relay race with teammates did not socially loaf when they displayed high levels of task cohesion, regardless of whether individual times were reported to the group or kept private. In contrast, swimmers who held low task cohesion socially loafed when individual relay times were kept private from other participants. Through a comprehensive meta-analysis, Steven Karau and Kipling Williams demonstrated that social loafing exists for physical skills, cognitive skills, and perceptual skills and is not confined to a specific segment of the population (i.e., Western cultures). Additionally, to correct the negative implications of social loafing, situations should be structured so that people perceive their contributions as unique, have a standard to which to compare their group's performance, work on a task(s) they find intrinsically motivating, and view collective outcomes as important.

Köhler Effect

The Köhler effect is another social processing effect phenomenon that occurs when individuals work together on tasks. A key element in the manifestation of the Köhler effect is that the tasks to be completed must be conjunctive in nature (i.e., tasks are not complete until all members contribute to the final product; thus, a group's performance is roughly equal to that of the least effective group member). Under these conditions, supporters of the Köhler effect assert that weaker individuals in the group will expend more effort in a group setting than they would when working in isolation.

Otto Köhler first demonstrated the Köhler effect in the 1920s through motivational gains afforded in groups of two to three individuals over performing individually. Since that initial work, studies have confirmed the presence of the Köhler effect with a variety of physical tasks in laboratory

settings and computer-based cognitive tasks. Furthermore, in a 2007 meta-analysis focused on the performance of the most inferior group member, Bernhard Weber and Guido Hertel found that the Köhler effect exists regardless of the demands of the task (i.e., physical or cognitive) and occurs with both men and women.

The two most accepted explanations for the Köhler effect involve the notions of social comparison and social indispensability. In the *social comparison explanation*, it is argued that individuals have an innate need to determine their level of ability; thus, individuals seek out others as a baseline to compare requisite skills. However, it is important to note that the comparison process is not arbitrary, as individuals tend to compare themselves to others who are similar in some important aspect, yet performing better. Once this upward comparison occurs, individuals will learn about their discrepancies and begin a conscious process to reach or maintain an improved level of performance. The *social indispensability explanation* for the Köhler effect is premised on the idea that individuals tend to be especially cognizant of their own contributions toward the group's overall success or failure. In a conjunctive task setting, the most inferior group member is highly motivated because of the belief that his or her contributions to group success are more valuable than those of the higher performing members of the group. Regardless of the mechanism operating to produce the Köhler effect, benefits of this phenomenon include increased group motivation and performance, both of which are adaptive behavior traits for sport teams where all individuals share a role in a group's impending success or failure. Furthermore, Deborah Feltz and colleagues found that the Köhler effect has also been shown to exist in exercise settings regardless of whether participants exercise in real or virtual environments.

Audience Effects

A final concept to examine is the effects that audience members can have on performers. Research in this area developed from social facilitation research—and specifically, the attempt to understand the interplay of arousal and evaluation apprehension, previously noted through Zajonc's and Cottrell's work. In the realm of sport, research has indicated that the presence of an audience can have cognitive and a physiological effects.

Exploring cognitive aspects first, when a performer exhibits a behavior that conflicts with his or her perception of performance, a change in self-concept may result. This process—which Dianne Tice termed internalization—is more pronounced when behaviors are viewed by an audience compared to a performance that takes place in isolation. Turning attention to physiological effects, it has been demonstrated that the presence of an audience can increase arousal; however, the performer must also experience evaluation apprehension, attentional conflict, and an increased presence of social monitoring for the greatest arousal changes to occur. Additionally, performers may also respond differently to increased arousal based on their skill level and situational assessment. For instance, if an individual believes he or she has the skills to meet an impending challenge, it more likely that this person will view the situation as a challenge, as compared to someone who questions his or her competence and is more likely to view an identical situation as a threat.

The distinction between perception of challenge versus threat may help to explain the home court advantage effect from a psychological perspective. Specifically, it has been noted that home teams display more assertive skills and visiting teams commit more fouls and penalties than home teams. It is hypothesized that this relationship exists because while trying to successfully navigate the increased assertive play of home teams, the visiting teams experience more frustration; therefore, they engage in more maladaptive behavior (i.e., the home court advantage may actually be an away court disadvantage). Of course, it also possible that playing in front of a home crowd could impair performance by participants experiencing additional pressure to succeed. This heightened self-awareness can cause individuals to direct attention to elements of the task that are automatic or already well-learned skills; thus, the individual begins to “press” to succeed and may end up “choking” under the pressure. While it is known that home teams in a variety of sports (e.g., basketball, football, and ice hockey) win a majority of their games, a study by Steven Bray, including 20 years of National Hockey League games using archival data, demonstrated that while playing at home is an advantage for any team, high-quality teams experience a much greater home ice advantage than low-quality teams (70% vs. 52% success rate). Thus, it appears that a moderating variable in audience

effects—related to home court advantage—is the quality of the home team.

Conclusion

In conclusion, social processing effects are central in SEP because they often involve social interaction in some form or another. Whether this social interaction is competitive or noncompetitive, participants routinely learn from watching and emulating others, receive directions and corrections from coaches and teachers, and are subject to observation in the process. As a consequence, the effects and constructs discussed in this entry do not exhaust the array of potential social influences but they serve to highlight the power of social processing effects on individuals engaging in sport and exercise behavior.

Todd A. Gilson

See also Basic Emotions in Sport; Cohesion; Feedback; Home Advantage; Self-Presentation

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SPORT PSYCHIATRY

Like sport psychology (SP), the field of sport psychiatry may involve utilization of psychological approaches to enhance athletic performance. However, a primary focus of sport psychiatry is diagnosis and treatment of mental illness in athletes. Sport psychiatry is a relatively new field, and as such, the research base is still developing. This entry reviews the diagnostic epidemiology of, and treatment options for, mental illness in athletes.

Epidemiology of Mental Illness in Athletes

Little systematic research has been conducted of the relative frequency of psychiatric disorders in athletes as compared with the general population. However, a recent review by Claudia L. Reardon and Robert M. Factor showed eating disorders and substance use disorders among athletes might be relatively more common than other psychiatric disorders in this population. Up to 60% of female athletes in certain sport-specific populations suffer from eating disorders, and the rates among male athletes are increasing. Additionally, mental health providers must consider “anorexia athletica” in athletes, in which athletes’ muscle mass masks their low body fat, but they exhibit many of the other signs of an eating disorder.

Substance use disorders, especially anabolic steroid and stimulant abuse, also appear to be diagnoses of relative concern in athletes. Athletes use steroids and stimulants to enhance performance, but these substances can also adversely impact performance. Specifically, steroids can cause mood changes, aggression, psychosis, and cognitive impairment. Stimulants can cause weight loss, tremors, and insomnia. As in the general population, athletes abuse alcohol more than any other substance.

Beyond eating disorders and substance use disorders, any psychiatric disorder could strike an athlete. However, others particularly important to consider include major depressive disorder

(MDD). Some evidence suggests that MDD might be no more common in athletes than the general population, but it can be precipitated by the unique factors of overtraining, injury, poor performance, retirement from sport, and concussion.

Clinicians should also consider compulsive disorders, including *muscle dysmorphia*, in which muscular athletes feel they are too small, and *exercise addiction*, in which athletes experience “withdrawal” anxiety and depression when unable to exercise, and thus continue to exercise despite injury or dysfunction. Importantly, clinicians should not mistake superstitious rituals, common in sport, for obsessive-compulsive disorder.

Treatment of Psychiatric Illness in Athletes

Medications and psychotherapy are the two primary modalities of treatment of psychiatric illness in any population, including athletes. When prescribing medications, clinicians should consider three factors: effects on performance, anti-doping guidelines, and safety. No large, randomized, controlled trials of psychiatric medications in athletes have been published to date. However, small studies of their use in athletes have been reported and can guide medication choices. In considering anti-doping guidelines, prescribers should know the guidelines of the relevant sport federation prior to prescribing any medication for a high-level athlete. Regarding safety concerns, clinicians should especially consider whether medications might become toxic in athletes who are excessively sweating and if the medications can have adverse cardiac effects in athletes pushing themselves to extremes.

There have been small studies of some antidepressants, specifically selective serotonin reuptake inhibitors (SSRIs) and bupropion. Preliminary data show that the SSRI fluoxetine may not impact athletic performance. One study shows the SSRI paroxetine to limit performance, while another shows no such impact. Early data show that bupropion might be performance-enhancing in hot temperatures and especially when used acutely as compared to over longer time periods (the latter representing how athletes generally take the medication if used for psychiatric reasons). Other antidepressants have not been systematically studied in athletes. A survey of sport psychiatrists has revealed that fluoxetine would be their antidepressant of choice for use in athletes needing medication treatment for depression.

Sedative-hypnotics and anxiolytics are another important category of medications to consider, as athletes may suffer from insomnia and performance anxiety. Melatonin is the most widely studied sedative-hypnotic used for insomnia in athletes, and it appears to have no impact on performance. Some studies show that benzodiazepines, when used for insomnia, may have a negative hangover effect on performance. This is particularly true of longer-acting benzodiazepines. Non-benzodiazepine agonists such as zolpidem and zopiclone appear to have less of a hangover effect on next-day performance. Other medications that may be used for anxiety include beta-blockers, which may be performance-enhancing in some sports such as archery, and thus are banned by that sport's international federation. In contrast, they are probably performance-inhibiting in endurance sports. Likewise, the anxiolytic buspirone has been shown in one small study to be performance-inhibiting.

Medications to treat ADHD include stimulants and atomoxetine. The former have been shown in several studies to be performance-enhancing, and thus have been banned for use by athletes at high levels of competition unless the athlete has a Therapeutic Use Exemption (TUE). Atomoxetine is a selective norepinephrine reuptake inhibitor and has not been studied for impact on performance.

Mood stabilizers have been minimally studied in athletes and antipsychotics not at all. The mood stabilizer lithium may be of concern when used by athletes because of the risk of toxicity while sweating. However, even that apparently well-known risk has been called into question, as some evidence suggests that lithium levels can actually be reduced during intense exercise.

Psychotherapy alone may be used for mild cases of depression, anxiety, or sports-related adjustment issues. It may also be used as an adjunct to medications. As with prescription of medications for athletes, psychotherapy with this population presents unique challenges. Denial of psychological problems is common among successful athletes. Additionally, athletes may be used to being the "VIP" and in charge, which can interfere with therapy. Among the different types of psychotherapy, cognitive behavioral therapy may be a particularly reasonable choice for athletes given their comfort with structure, practice, and goal setting.

Conclusion

Clinicians, coaches, and other staff who work with athletes should be aware that mental illness can strike this population, despite athletes' typical presentation as strong and physically and mentally sound. Once a mental illness is diagnosed, treatment may consist of medication and/or psychotherapy.

Claudia L. Reardon

See also Affective Disorders; Alcohol Abuse; Eating Disorders; Performance-Enhancing Drugs; Psychopharmacology

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World Anti-Doping Agency: <http://www.wada-ama.org/en>

SPORTSPERSONSHIP

Sportspersonship has long been a topic of interest in physical education (PE) and sport environments and is one of the most frequently mentioned virtues promoted by sports. Sportspersonship represents a wide-ranging concept that has important consequences for the physical and psychological welfare of all sport participants. From a psychological perspective, the concept of sportspersonship is closely related to that of moral behavior and development. However, theory in this area has not systematically addressed the role of both person and contextual variables in explaining moral behavior and moral development in sport. Therefore, Robert J. Vallerand and colleagues have proposed a social-psychological approach to sportspersonship.

First, this approach posits that both individual differences and social forces influence sportspersonship behavior. Second, the behavior of others plays an important role in helping individuals define the content of sportspersonship. It is through their interaction with their peers, parents, coaches, and other sport participants that children

come to learn what sportspersonship is and what it is not. In line with these assumptions, Vallerand and colleagues proposed and empirically validated a multidimensional definition of sportspersonship that incorporates five facets:

1. A full commitment toward sport participation by working hard and showing up during practices and games, acknowledging one's mistakes, and trying to improve
2. Respect for social conventions to be found in sports, such as shaking hands after the game or recognizing the good performance of the opponent
3. Respect and concern for the rules and officials
4. Respect and concern for the opponent (i.e., fair play), such as refusing to take advantage of an injured opponent or agreeing to wait for a late opponent instead of winning by default
5. Absence of a negative approach toward sportspersonship, which can manifest itself in the adoption of a win-at-all-costs mentality or by losing one's temper after making a mistake.

In sum, sportspersonship can be defined as the concern and respect of one's full commitment for the sport, the rules and officials, social conventions, and the opponent as well as the relative absence of a negative approach toward participation.

The social psychological approach further proposes that it is important to distinguish between three key elements: *sportspersonship behavior*, *sportspersonship orientations*, and the *development of sportspersonship orientations*. Sportspersonship behavior refers to the display of behavior at one given point in time. Sportspersonship orientations pertain to individual differences in the propensity to act in a sportspersonlike fashion. Finally, the development of sportspersonship orientations refers to the process through which sportspersonship orientations develop. In line with the social psychological approach, sportspersonship orientations refer to individual differences in the propensity to behave in line with each of the sportspersonship dimensions outlined previously. For instance, athletes with a high orientation toward concern for the rules and officials would be less likely to cheat than athletes who score low on this orientation. Consequently, Vallerand and colleagues developed a scale in order to capture individual differences

in sportspersonship orientations. This scale, labeled the Multidimensional Sportspersonship Orientation Scale (MSOS), is based on the sportspersonship definition outlined previously and serves to measure athletes' orientation on the five sportspersonship dimensions described previously. Each dimension is measured using five items. Support for this multidimensional conceptualization of sportspersonship has been provided by extensive validation work. Finally, the development of sportspersonship orientations refers to the process through which sportspersonship orientations develop.

Determinants of Sportspersonship

As stated earlier, both personal factors and the social environment influence the determinants of sportspersonship orientations. As it pertains to personal factors, research has shown gender differences in sportspersonship orientations. Specifically, girls have been found to exhibit higher levels of moral reasoning and moral functioning, as well as less aggressive behaviors, than boys in sport. Additionally, motivation toward one's sport has also been shown to influence one's sportspersonship orientation. More precisely, it has been shown that intrinsic motivation was positively, while extrinsic motivation was negatively, related to sportspersonship. Moreover, the motivational climate of the team has been related to sportspersonship orientations. Specifically, athletes who play in a strong task- (or mastery) oriented climate where one focuses on learning and self-referenced achievement and progress are more likely to develop and display higher sportspersonship. Conversely, athletes perceiving a high ego- (or performance) oriented climate where the focus is on normative achievement and criteria of success and failure are more likely to display low levels of sportspersonship. The same line of reasoning applies to individual achievement goal orientation. More precisely, task-oriented goals in athletes are typically associated to high sportspersonship levels, whereas ego-oriented goals are related to low sportspersonship levels.

Consequences of Sportspersonship

Sportspersonship orientations have been shown to be associated to various consequences by influencing the way individuals approach their sport. In fact, individuals who display higher levels of

sportspersonship are less likely to display unethical behaviors, such as cheating or using performance-enhancing substances. Furthermore, stronger sportspersonship orientations have been shown to lead to less reactive aggression in sport. In other words, sportspersonship orientations seem to reduce athletes' tendency to use malicious or injurious behaviors within a competitive setting. Conversely, individuals with high-level sportspersonship orientations are more likely to use instrumental aggression, an unemotional and task-oriented way of interfering with an opponent. Simply put, they will respect the rules and social conventions of their sport when determining what is acceptable while competing.

In sum, a social-psychological and multidimensional approach to sportspersonship provides a broader, yet integrative, perspective of the personal and contextual factors influencing the development and display of sportspersonship orientations. Furthermore, it leads to a better understanding of important consequences of these orientations for moral as well as immoral behavior in sports.

Robert J. Vallerand and Jérémie Verner-Filion

See also Cheating; Fair Play; Intrinsic/Extrinsic Motivation; Hierarchical Model of; Moral Judgment; Moral Reasoning; Participation Motives

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STATUS

Status represents an individual's social standing in relation to others. Attributes of status can be based upon physical characteristics (e.g., age) or intangible attributes (e.g., prestige, prominence). Individuals with higher status are often afforded more social benefits including respect, more unsolicited help from others, greater praise for performance, disproportionate credit for successful collaborative efforts, and more opportunities to share their opinions. Higher status individuals perceive the world and themselves differently and more favorably than do those of lower status. Status is an important concept within group settings as perceptions of status influence thoughts, attitudes, and behaviors of individuals toward other group members. While status and its influence on group members has been a topic of considerable interest in the fields of social and organizational psychology, research on status is relatively recent in sport and exercise settings. In this entry, we investigate attributes of status and empirical evidence on the topic in sport and physical activity (PA) settings.

Attributes of Status in Sport

Shanthi Jacob and Albert Carron conducted a series of studies examining attributes of status in sport. In an initial review of literature in psychology, sociology, and sport sciences, they identified 17 attributes of status. The attributes were further classified into two categories: (1) *achieved status attributes*—attributes such as education or experience that are products of personal effort—and (2) *ascribed status attributes*—attributes such as race or parental education that are independent of personal effort. The most salient status attributes included competitive experience, performance, and team role (captain or cocaptain)—all of which are achieved status attributes.

In a follow-up study, Jacob and Carron examined the association between perceived status and cohesion in sport teams. The sample included intercollegiate Canadian and Indian athletes from a wide cross section of sports (e.g., volleyball, basketball, soccer, hockey), and they found that athletes who perceived greater task cohesion on their respective teams (i.e., higher personal involvement with group's task, productivity, and goals) placed less importance on status within the team.

Recently, Jacob Johnson (née Jacob) undertook a qualitative approach to further understand attributes of status in sport teams. Her analysis revealed four main categories of status attributes: *physical* (e.g., performance, experience, appearance, role, and position), *psychological* (e.g., individual psychological attributes—positive attitude, group psychological attributes—team spirit), *demographic* (e.g., age, income, family status), and *relationships with external others* (e.g., family support). Psychological rather than physical or demographic attributes were deemed most important and most frequently reported by Canadian and Indian athletes.

Empirical Evidence on Status in Sport

Starting status, often dichotomized into *starters* (i.e., athletes who begin competition on the playing surface and typically receive regular playing time) versus *nonstarters* (i.e., athletes that do not begin competition on the playing surface and typically receive irregular playing time), has been a focus of sport psychology (SP) research. Starting (high-status) athletes hold greater perceptions of dedication, social cohesion (attractions toward team social activities), and satisfaction for ability utilization. In contrast, nonstarters have greater role ambiguity (i.e., lack of clarity in roles) and less role efficacy (i.e., belief in their capability to successfully perform role) than starters.

Status Inequality in Sport

Researchers in organizational psychology have highlighted the benefits of equal status within groups. Equal status teams make decisions on the basis of quality of information regardless of the contributor's status and provide lower status individuals with greater opportunities to enhance their self-efficacy and performance. Amy Christie and Julian Barling examined how status inequality on sport teams is related to individual performance and physical health among 30 National Basketball Association (NBA) teams over six seasons. Status was operationalized through player salaries, number of games started, organizational tenure, player awards and recognitions, and star power. Physical health was evaluated based upon absenteeism. Key findings revealed that status inequality impeded performance across time for lower status players in particular. Mixed evidence was found for the status equality–physical health relationship.

The inconsistent findings were attributed by the researchers to the lower status players in both equal and unequal settings feeling pressured to play whether ill or injured due to job insecurity. This explanation is consistent with previous studies on work place sickness and absenteeism.

Status and Physical Activity Participation

Status research in physical activity (PA) has predominantly been descriptive and focused on sociodemographic factors such as age, sex, race, ethnicity, and socioeconomic status (SES). Other status classification categories include education, marital status, parental employment, family poverty, and self-reported health. Among sociodemographic factors, SES and PA have received considerable attention. Higher SES youth and adults consistently report higher PA compared to lower SES individuals. While much of the status research on PA and exercise has been descriptive and quantitative in nature, some limited work has qualitatively investigated the influence of status on PA behavior. Louise Humbert and colleagues asked high and low SES youth to share their perspectives on barriers to PA and how to get youth their age to be active. Findings indicated that environmental factors of proximity, cost, access to facilities, and safety were important participation factors for youth living in low SES areas. Intrapersonal factors (e.g., perceived skill and competence to be physical active) and social factors (e.g., friends, adult support) were identified as key intervention components to improve participation among both high and low SES youth.

Mark W. Bruner

See also Diversity; Gender; Race; Roles

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STEREOTYPE THREAT

Stereotype threat is the perceived risk of confirming, as self-characteristic, a negative stereotype about one's group. Over 300 studies on academic testing show that the threat of confirming a negative stereotype about one's group sets into motion a sequence of psychological processes that inhibit cognitive capacity and exacerbate performance monitoring. As a result, stigmatized individuals underperform relative to nonstigmatized individuals and relative to members of the stigmatized group not under stereotype threat. Several studies indicate that stereotype threat processes can reduce achievement in sports when athletes perceive that their performance might be evaluated through the lens of racial or gender stereotypes.

Evidence for Stereotype Threat in Sports

Jeff Stone and colleagues conducted the first study of stereotype threat in sports to test whether the salience of negative racial stereotypes about Black athletes (i.e., low sports intelligence) and White athletes (i.e., low natural athletic ability) could cause each group to perform more poorly on a golf putting task. The participants were Black and White former high school athletes, and some participants were told that the test measured their sports intelligence, while others were told that the test measured their natural athletic ability. Participants in a control condition were told that the test measured their sport psychology, a stereotype-irrelevant attribute. As predicted by stereotype threat, the results showed that (a) both Blacks and Whites performed equally well on the putting task when it was framed as a measure of sport psychology, (b) Whites performed significantly worse than Blacks when performance was framed as a measure of natural athletic ability, and (c) Blacks performed significantly worse than Whites when performance was framed as measuring sports intelligence. Thus, simply framing the task in terms of a negative

racial stereotype about each group was sufficient to activate the threat processes that reduced their sports performance.

Female athletes suffer stereotype threat when their natural athletic ability is linked to gender differences in sports. For example, Jeff Stone and Chad McWhinnie had White females complete a golf-putting task. To explicitly induce stereotype threat, participants were told that the test measured gender differences in athletic ability. In addition, to create an implicit cue for stereotype threat, a male experimenter ran the testing session. The results were that female athletes required significantly more strokes to finish the putting task when it was explicitly framed as measuring gender differences in athletic ability compared to control conditions. In addition, women performed less accurately on their final putt when run by a male compared to a female experimenter. Thus, both explicit (e.g., the test frame) and implicit (i.e., experimenter gender) reminders of the negative stereotype about female athleticism impaired different aspects of their performance. Similar effects are found for female athletes on the soccer field and in competitive chess tournaments.

Elite-level athletes are not immune to the stereotype threat. Sian Beilock and colleagues recruited expert male golfers who had 2 or more years of high school or college varsity experience or a Professional Golfers' Association (PGA) of America handicap of at least eight. After a baseline round of putting, expert golfers in the stereotype threat condition were significantly less accurate during the second round. Aina Chalabaev and colleagues reported a similar finding with elite female club soccer players in France, who were unable to match their baseline time on a dribble test when it was framed as measuring their natural athletic ability.

The anticipation of being negatively stereotyped in a sport can engage defensive responses, like self-handicapping, that have a negative effect on how athletes prepare for an important competition. For example, in one study, White athletes were told that a putting task measures their natural athletic ability and were then told they could practice on the first hole of the course for as long as they wanted before the test began. The results showed that when under stereotype threat, White athletes self-handicapped by practicing less compared to White athletes in control conditions. These results suggest that stereotype threat processes can start

to impair athletic performance well before a competition begins.

Stereotype threat primarily impacts the performance of athletes who maintain their self-worth through their achievements in sports. Consistent with this reasoning, several studies show that the salience of negative stereotypes primarily reduces the practice and performance of psychologically “engaged” athletes. Psychologically “disengaged” athletes, in contrast, are less affected when a negative group stereotype is salient. However, over time, athletes may respond to stereotype threat by psychologically disengaging from competitive outcomes and eventually by misidentifying and withdrawing from the performance domain. These coping responses may not only cause some talented individuals to quit playing a sport, but to avoid being the target of threat in the future, they may give up participation in recreational sport, which has important consequences for their health.

Reducing Stereotype Threat in Sports

One way to reduce stereotype threat effects in sports is to emphasize the racial or gender neutrality of a competitive performance context. When negative stereotypes are salient during competition, athletes can deflect stereotype threat if they learn to view characteristics like athletic ability or sports intelligence as malleable attributes they can improve with practice. They can also strengthen their sense of efficacy to overcome a stereotype threat through exposure to in-group role models who exhibit strong counterstereotypic attributes. Finally, teaching athletes to recognize the causes and consequences of stereotype threat can empower them to defend against its detrimental effects when they prepare for and perform in their sport.

Jeff Stone

See also Self-Handicapping; Stereotyping, Cultural/Ethnic/Racial

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STEREOTYPING, CULTURAL/ ETHNIC/RACIAL

Stereotyping represents a category-driven, formulaic, and oversimplified process of making broad-brush suppositions and generalizations about a group of people to whom specified sets of characteristics are attributed. The characteristics ascribed to identified groups can be positive, neutral, or negative and can be originated and sustained relatively easily when there are clearly visible and discernibly observable features. Characteristics upon which stereotypes are formed and promulgated are innumerable, and a non-exhaustive categorical list includes cultural, ethnic, racial, age, gender, sexual orientation, socioeconomic status (SES), physical and mental disabilities, personality, careers (e.g., athletes, white-collar jobs, blue-collar jobs), sensory (e.g., hearing, sight, smell, touch, taste), regional (e.g., southern, northern, western, eastern), generational (“the sixties,” “old school,” “the *me* generation”), appearances (e.g., height, weight, skin color), and relational status (e.g., single, married, divorced, partnered). On an individual level, stereotypes imply that a person possesses the characteristics alleged to be part of the group being stereotyped. Accurate or not, these inferences shape beliefs and perceptions about the targeted individual. Last, stereotyping is a within- and across-group social categorization process in which all communities around the world participate. As such, no one individual or group is immune from being stereotyped or from experiencing the relative impact of this always-present and conventional phenomenon.

Stereotypes and stereotyping are learned directly and vicariously, usually emanating from childhood and fostered actively and cumulatively by parents, family, community, school, religious institutions and the media in all of its forms (e.g., visual, print, auditory). A related issue is that a lack of familiarity with other groups (e.g., cultural, ethnic, racial) increases vulnerability to socially promoted images that fuel stereotyping. Intergroup conflicts and competitions minimize similarities and magnify

differences. The utility of stereotypes and stereotyping across time, circumstance and situations suggests its sustainable permanence among our global citizenry.

Stereotyping serves multiple purposes. It positions persons seeking to simplify a complex world with degrees of freedom (df) to (a) organize and synthesize information, (b) control and at some level predict likely outcomes, and (c) decrease the amount of time it takes to absorb and make sense of myriad interpersonal and intrapersonal stimuli radiating from other individuals or groups. The challenges associated with this purpose are expressed in the human tendency to resist new information and input, fearing loss of the hitherto organized data and feeling forced to manage the need to construct new images and perceptions.

For persons holding meritocratic beliefs rooted in notions of superiority stereotyping allows persons in this type of in-group to justify ill-founded prejudices, engage in scapegoating, and to hold on to judgmental attitudes toward alleged “less than” and “inferior” out-group members. Finally, and at its negative extreme, stereotyping positions oppressors and persons bent on exploitation to evade and escape thinking of themselves as wrongdoers and justify continued exploitation. Abundant historical and contemporary examples of the latter purpose, seen across national (e.g., African American, American Indian, Asian-Pacific Islander, and Latino or Latina) and international groups throughout world history are replete in scholarship across disciplines (e.g., psychology, sociology, anthropology, and history).

The impact of stereotyping varies by individuals and groups and can be understood and appreciated within the contexts of circumstances; situations; the conscious and unconscious motivation and intent of the perpetrator; and the receiver’s unconscious and conscious awareness of, sensitivity to, and focus on the stereotype being expressed. Stereotypes and stereotyping distort perceptions and advance social judgments. Stereotypes and stereotyping stimulate overestimation of the occurrence of negative beliefs and perceptions of ethnic minority and other marginalized populations and thus, preclude really seeing the uniqueness of individuals and the groups to which they belong. Caricatures of athletes as “dumb jocks,” portrayals showing that “white men can’t jump,” images of female athletes participating in sports seen traditionally as male (e.g., basketball, football,

baseball, soccer) as “tomboy” and “butch,” and perceptions of Asian athletes as “intelligent” and athletes of African descent as “natural athletes” and “less intelligent” represent just a few examples of impressions that fuel believable misperceptions. Information that persons notice, attend to, and process is impacted by this process of social categorization. Once learned, stereotyping is a difficult habit to break. What’s more, stereotyping becomes self-perpetuating for both the person engaging in stereotyping and the individual or group on the receiving end. Related, stereotyping influences self-perceptions particularly around competence and confidence, and these markers of self-doubt represent cornerstones to experiences of stereotype threat. Stereotype threat, essentially, is an anxiety-based response to an appraisal situation that carries with it the potential for the individual to confirm negative stereotypes about the group (i.e., cultural, ethnic, racial) to which he or she is a member.

Stereotyping relative to cultural, ethnic, racial, and other traditionally marginalized groups continues to be the focus of abundant scholarship, across disciplines. Scholars have examined stereotyping from historical and contemporary perspectives, and much has been learned about how stereotypes begin, persist, and are transmitted to individuals and groups irrespective of the region of the world where it continues to manifest.

Stereotyping is an ever-present social process whose impact worldwide is undeniable. The degree to which stereotyping impacts individuals and groups, however, varies based on the confluence of numerous factors. Also, the ways in which individuals and groups respond to stereotypes and stereotyping differs across myriad dimensions. The institutionalized and systemic nature of stereotyping precludes its eradication. That being said, engaging in difficult dialogues about this complex social force represents a reasonable approach to managing this pervasive social reality.

William D. Parham

See also Cultural Competence; Diversity; Ethnicity; Multiculturalism; Race; Racism/Whiteness; Stereotype Threat

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STRENGTH MODEL OF SELF-CONTROL

The strength model of self-control was proposed by Roy Baumeister, an eminent social psychologist, to describe how individuals can control their behavior, automatic tendencies, and natural desires in order to achieve long-term goals and conform to socially prescribed codes of behavior and norms. In this model, the terms *self-control* and *self-regulation* are used interchangeably to describe the capacity to consciously guide behavior to a desired direction and resist short-term temptations that lure individuals away from their focal goals. The strength model of self-control has achieved a great degree of attention in the psychology literature given that failures to self-control can lead to personal and societal problems (e.g., underachievement in school or work, violence against others).

The ability to self-control is particularly important in terms of engaging in important health behaviors (e.g., regular exercise, healthy diet) and tackling obesity as well as alcohol and substance abuse. Hence, it is not surprising that research based on the strength model of self-control is becoming increasingly popular in the exercise and health psychology literature. In a recent survey of American adults, conducted on behalf of the American Psychological Association, it was revealed that the most frequently reported goals the survey participants had set for 2012 were health-related (e.g., 41% reported a goal to start exercising regularly). However, in the same survey

27% of the respondents reported that poor self-control prevented them from meeting their goals. On a positive note, 71% of them believed that self-control can be improved through practice, which is what research also suggests.

This entry first provides an overview of the strength model of self-control and explains why self-control is a limited resource. It goes on to discuss why self-control is relevant to the promotion of health-related behaviors, and concludes with some practical for strengthening self-control.

Self-Regulatory Strength and Depletion

Following through a New Year's resolution to engage in regular physical activity (PA) requires not only the motivation to change but also the self-regulatory strength to monitor any discrepancies in meeting this goal and fight against competing interests, priorities, or temptations. This self-regulatory strength is colloquially called *willpower*. Willpower helps individuals overcome temptations and achieve valued academic, health, financial, interpersonal, and societal goals. Research indicates that the benefits of having high levels of self-control transcend over time; children with high self-control (assessed by both self-reports and adult reports) not only do better at school but grow up to be more successful adults (in their professional and personal lives) than children with lower self-control. These findings hold up if even when differences in socioeconomic status, intelligence, and other potentially confounding variables are taken into account.

Despite the benefits of having high self-control, individuals are often unable to resist temptations and impulses. The playwright Oscar Wilde famously claimed that he could resist anything but temptation. The strength model of self-control explains self-regulatory failures by arguing that self-control is a limited resource resembling a muscle. The more self-control is exercised (irrespective of the nature of the task), the more it gets tired and eventually depleted. This depletion, called ego depletion in the literature (aligned with the Freudian discussion of the self in terms of energy), makes individuals vulnerable to future temptations and can impair their performance in subsequent tasks that require self-control. Research testing the strength model of self-control has found this depleting effect to be robust across a variety of situations, such as when asking individuals to regulate their emotions, make difficult choices, engage in

problem solving and logical reasoning, suppress thoughts or stereotypes, and resist temptations. For example, in a study by Kathleen Vohs and Todd Heatherton, chronic dieters and nondieters were asked to sit close to or away from a candy bowl and were told to either help themselves to the candy or not to touch it. For chronic dieters, sitting close to a candy bowl and being told to help themselves to the candy requires the exertion of self-control. The results of the Vohs and Heatherton study showed that the chronic dieters demonstrated less persistence in a subsequent cognitive task and ate more candy compared with the dieters who sat away from the candy bowl. For nondieters, sitting near the bowl did not impair their subsequent self-control performance. This finding indicates that the undermining effects of ego depletion occur only in individuals who have the motivation to achieve a goal. Also, in the same study there was not self-regulatory failure when dieters were told not to eat from the candy bowl. This finding illustrates that willpower depletion does not occur when behavior is constrained by external factors and is not under volitional control.

The depleting effects of self-control might vary as a function of a number of personal or situation factors. For example, individuals high in autonomous motivation to exert self-control, or perceiving others to support their autonomy, are less depleted than those with low autonomous motivation or those pressured by others to self-control. Also, increased positive mood and individuals' belief that willpower is not a limited resource may reduce levels of ego depletion. In individuals with high reserves of self-control (i.e., high levels of trait self-control), the undermining effects of situational ego depletion on task performance are less pronounced than for individuals who have low trait self-control. Additionally, participants who anticipated a future self-control task showed larger ego depletion than participants who did not expect a future task. Perhaps in the latter case individuals are more strategic in terms of their resource allocation, choosing to conserve resources for the subsequent self-control task. This conservation hypothesis is aligned with Baumeister's argument that acts of self-control result in partial depletion only, rather than complete exhaustion, because individuals conserve their energy resources for future situations. In fact, research by Vohs, Baumeister, and Brandon Schmeichel has shown that moderately (but not severely) depleted motivated individuals

can utilize some of their remaining resources to maintain their performance levels.

The findings from the strength model of self-control literature have implications for exercise and health promotion. First, individuals whose jobs require a great deal of self-regulatory strength and information processing are likely to feel depleted by the end of their working day. If these individuals are on a weight loss program, they might be less able to exert self-control later on in the day in terms of choosing a healthy meal option or engaging in PA when faced with tempting high calorie or sedentary alternatives (unless these individuals are highly motivated to lose weight). Another implication from the findings in the literature is that, since self-control is a limited resource, individuals should try to consolidate a change in a single health behavior first (i.e., make it more habitual so that it requires less self-control) and then move on to another behavior, as opposed to trying to alter multiple behaviors simultaneously. The latter would be more depleting; in fact, exerting self-control toward one behavior might undermine self-control efforts toward the other behavior. This was well illustrated in a study by Ditka Shmueli and Judith Prochaska in which smokers who were asked to resist a highly tempting food were more likely to subsequently smoke compared with smokers who were asked to resist food that was not tempting. The longitudinal evidence suggesting that children with high self-control are more likely to retain their high self-control in adolescence and adulthood, coupled with evidence indicating a negative relation between self-control and overweight or obesity, imply that self-control training programs that aim to promote healthy behaviors should be implemented from a relatively young age. Obviously, the influence of the obesogenic environment and advertising for unhealthy products cannot be discounted. Expecting individuals to exert self-control when unhealthy temptations are readily available throughout the day might not be entirely realistic. However, self-control training could be used in conjunction with changes in policy and legislation to better facilitate health behavior change. The next section discusses some ideas for strengthening self-control.

Boosting Self-Control

Odysseus tied himself to the mast of his ship and put wax in his shipmates' ears to avoid the fatally

tempting song of the Sirens. Unlike this Homeric hero, however, many individuals overestimate their ability to resist temptation. People who are perceived to have a high capacity to control their impulses expose themselves to more tempting situations and ultimately engage in more impulsive behavior (perhaps as a result of depletion) than those who perceive a lower capacity for self-control. Thus, an effective way of maintaining self-control, particularly when depleted, is to avoid highly tempting situations. For example, new exercisers who plan to exercise in the evening should avoid watching television first “for a little while” before they start their workout. Self-control can also be boosted by inducing positive mood, promoting autonomous motivation for task engagement, reaffirming long-term goals and higher-order values, and taking breaks or engaging in relaxation between self-control tasks. Further, self-control can be enhanced by making it more automatic. For instance, it has been found that forming implementation intentions (“if-then” plans) can overcome the performance decrements associated with ego depletion. Using again the example of new exercisers who plan to exercise in the evening, these individuals should form a plan so that they are prepared when at the end of the working day they are invited out by their colleagues for “a few drinks.”

Another interesting suggestion that has been made in the literature for boosting self-control is to maintain steady blood glucose levels. This is because self-control might deplete blood glucose, which is important for mental activities. It has been shown that depleted individuals have lower blood glucose levels than nondepleted ones and that by consuming a glucose drink, the effects of depletion can be counteracted. Thus, keeping blood glucose levels steady via consuming glucose-based drinks or regular meals might help individuals’ self-control in tempting situations (e.g., being offered a high calorie cake) when they are in a “hot state” (i.e., hungry). However, such findings have been challenged recently as it has been shown that similar bolstering effects on self-control can be found by simply rinsing the mouth with carbohydrate solutions without actual digestion. Thus, it is possible that the glucose effects are motivational (as opposed to metabolic) in nature.

The muscle metaphor of the strength model implies that self-control, similar to a muscle, can be trained regularly to become stronger. For example, a study by Megan Oaten and Ken Cheng showed

that participants enrolled in a 2-month self-regulation program based on regular exercise (which requires self-control) were less depleted in a laboratory task they performed at the end of the program compared to those not enrolled in this program. The self-control training group also self-reported gains in self-control in other health-related behaviors (e.g., smoking, alcohol, and caffeine use).

In conclusion, the strength model of self-control has potential as a conceptual framework for explaining how to protect health behavior change from temptations, in particular before the new behavior becomes habitual. Its utility should be further tested outside laboratory settings by recruiting community and clinical samples (e.g., obese individuals) to self-control training programs with objectively assessed outcomes.

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See also Control Theory; Dual-Process Theory; Fatigue; Health Promotion; Obesity; Self-Determination Theory; Self-Regulation

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STRESS MANAGEMENT

Stress management refers to the environmental, physiological, cognitive, and behavioral techniques employed by an individual to manage the factors and components that underlie the stress process or experience of stress. A primary goal of stress management in sport is to allow the athlete to effectively regulate competition related demands to facilitate optimal performance as well as to enhance psychological well-being (PWB). There are numerous stress management techniques that can be classified into various heuristic categories. Many of these are covered in this entry. However, to understand why these techniques are effective under specific conditions, it is important first to understand the stress and emotion process.

Contemporary thinking in sport psychology (SP) conceptualizes stress as a complex dynamic transaction between environmental demands, such as those associated with high-level competition, and the athlete. Stress occurs when the demands tax or exceed the resources, such as skills or support, that the athlete has at his or her disposal. Since competitive sport is by nature demanding, how athletes evaluate and cope with the demands they encounter has a large impact on the stress process. The environmental demands, as well as internally generated demands from personal expectations and goals, are typically called *stressors*. Stressors can be acute, chronic, or intermittent, and they can also be expected or unexpected.

The stress process is highly influenced by how athletes evaluate the personal and social meaning of stressors. Such evaluation, typically called an *appraisal process*, can be rapid and automatic or reflective and is shaped by social learning, culture, and memories. In many cases, emotional feelings and patterns of thought and behaviors are activated, with corresponding physiological and neurological activation, action impulses, cognitive plans, and actions. Thus, the stress response can include changes in emotion, feelings, cognitions, behavior, and autonomic physiological systems. Stress responses differ from athlete to athlete, and, for any given athlete, stress responses can take different forms in varying situations. Thus, effective stress management can target the actual demands and/or enhance the athlete's ability to regulate the factors that are associated with the appraisal, emotion, and cognitive behavioral response.

Stress management techniques in sport typically target somatic, behavioral, and/or cognitive affective symptoms of stress. Somatic responses involve the athlete's physiological reactions, such as changes in heart rate (HR), respiration (R), sweating, gastrointestinal functioning, muscular tension and control, pupil dilation, urinary system, and salivation. Behavioral responses are the direct actions taken because of the stress, including engagement or disengagement in certain strategies or activities, as well as distraction. Finally, cognitive affective responses include the thoughts associated with the stress, including worries, beliefs, apprehensions, and negative expectations about performance as well as action plans to manage stress. Distinguishing between and being aware of each of these aspects is important for the athlete, coach, and SP consultant, as this knowledge helps to ensure the appropriate stress management skills are applied.

Effective stress management also needs to recognize the temporal aspect of the stress process. Stressful transactions in sport often involve anticipation, confrontation (engagement), and post-engagement stages and can result in an athlete feeling overwhelmed. Stress management techniques can target specific stages or combination of stages.

Types of Stress Management Programs and Techniques

There are a number of stress management approaches in sport to deal with various

components of the stress process. Some practitioners advocate a multimodal approach, which involves using different tactics thought to be more effective in combination. Others suggest focusing on the dominant stressor with a unimodal approach, which uses a singular, focused intervention strategy. Multimodal approaches tend to be favored because of their effectiveness on a wide range of factors related to different elements of the stress process (i.e., actual stressor, emotional feeling, cognition, behavior, and physiological responses). However, there is evidence that situations dominated by one particular stressor may be more efficiently treated with a unimodal approach. The effectiveness of any type of stress management ranges depends on variables such as the athlete's situation, his or her coping resources, and the appropriateness of the approach for the stressor. It is best to create individualized stress management skills programs designed to meet each athlete's specific needs. Common stress management interventions are briefly outlined next, in alphabetical order. These approaches can be seen as an application of theoretical and clinical knowledge to produce a more practical approach, and each of the approaches has varied levels of empirical support, depending on important factors such as context and person variables.

Anxiety Management Training

Anxiety management training involves an athlete's learning to employ relaxation strategies under stressful or arousing situations, including those producing emotions such as anger and anxiety. During anxiety management training, the athlete visualizes the stressful situation and allows the accompanying physiological arousal to be generated within himself or herself. Relaxation techniques, such as applied relaxation, progressive muscle relaxation, breath control or deep breathing, or meditation (outlined later), are then used by the athlete to reduce the symptoms of physiological arousal, such as increased HR, R, and blood pressure (BP). This may also promote management of behavioral responses such as loss of coordination, acts of aggression or frustration, "choking," or withdrawing from sport.

Applied Relaxation

The aim of applied relaxation is to learn the skill of relaxation and develop the ability to apply it

rapidly where needed, in any situation. Connected to this approach are six stages. The first stage is progressive muscle relaxation, a technique where muscles are contracted or tensed and subsequently relaxed, which is used to help facilitate relaxation and help the athlete reduce somatic anxiety symptoms. As the athlete becomes proficient in this skill and moves to stage two, muscle relaxation is promoted by relaxing the muscles without tensing them first. In stage three, the term *relax* is conditioned to bring on a relaxed state when spoken or thought by the athlete. A focus on breathing is also promoted in this stage, as well as a focus on passive concentration, which is an effortless, automatic, yet focused state of mind, similar to mindfulness. Stage four requires the athlete to learn to use the skill in real-life settings, relaxing appropriate muscles while engaging ones needed for activity. Stage five focuses on having an athlete relax while in a naturally occurring, nonstressful situation. Breathing is the trigger of relaxation in this stage and is practiced 15 to 20 times per day. The sixth and final stage is called application training. The relaxation technique is implemented in a practice or training session and then in a low-stakes competition. The more frequently and completely it is implemented, the easier it will be for the athlete to use the strategy in a higher level of competition.

Arousal or Energizing Techniques

Some research suggests that athletes differ on the level of activation needed to produce optimal performance. Various levels of arousal are often conducive to high performance, and it is paramount that the athlete perceives the arousal as beneficial (see Cognitive Control later in this section). While many stress management approaches take an arousal reduction focus, strategies to increase arousal include imagery, self-talk, goal setting, and cognitions or thoughts focused on heightening stimulation.

Autogenic Training

Autogenic training, first introduced in psychiatry by Johannes Heinrich Schultz, involves a series of exercises designed to produce sensations such as warmth or heaviness, to help promote relaxation. The program is based on six stages, each with a separate goal. The stages are learned and practiced in the following order: heaviness in the

extremities, warmth in the extremities, regulation of cardiac activity, regulation of breathing, abdominal warmth, and cooling of the forehead. Verbal cues to the athlete can be used to aid in prompting the sensations.

Biofeedback

Biofeedback training (BFBT) can help control autonomic physiological stress responses, such as increased HR and BP. It also has been used to control anxiety disorders as well as anxiety connected to particular environments or contexts. The premise behind biofeedback (BFB) is for the athlete to become aware of how stress is manifested physiologically, such as changes in BP, HR, breathing, or muscle tightness, using different modes of objective feedback and monitoring. With this increased awareness, athletes are better equipped to control their actions. With training, athletes become less reliant on the feedback, learning to control their physiological responses on their own.

Breath Control and Deep Breathing

Breath control is a relaxation technique using the physical strategy of breathing. It is an effective and relatively easy stress management technique to apply. Irregularities in breathing, such as holding one's breath, hyperventilating, or random shallow breaths, can affect performance, potentially influencing coordination, focus, or rhythm, or can cause the athlete to feel unsettled, causing further stress. Breath control can be practiced by taking a slow, complete breath. Often, the lungs are conceptualized in three parts to aid in proper instruction of a slow, complete breath. The lower lungs are filled by pushing the diaphragm down and forcing the abdomen out. The middle portion of the lungs is then filled by expanding the chest cavity, expanding the rib cage. The upper lungs are then filled by raising the chest and rib cage. The breath is held for several seconds, and then a slow exhalation is made, taking approximately double the time taken for the inhalation process. Breath control is commonly used before a competition or during a natural break during the competition, as it is most practically applied during nonactive times.

Cognitive Affective Stress Management Training

Cognitive affective stress management training is one of the most comprehensive multimodal

stress management programs used in sport. Originally designed by Ronald Smith, the program is designed to teach the athlete relaxation and cognitive skills that can aid in controlling physiological reactions and cognitive thought patterns. Intervention consists of both cognitive and physiological strategies, including relaxation skills, cognitive restructuring, and training that is self-instructed and targets the physical and mental reactions to stress. The premise behind the combination of physical and mental coping strategies is the development of an integrated coping response. The program, which has some empirical support, is educational rather than psychotherapeutic in nature and is designed to help athletes increase their self-control.

The cognitive affective stress management program consists of four distinct phases. In the first phase, the *pretreatment assessment*, the consultant uses an interview approach as well as questionnaires to assess the athlete's issues with stress—namely, what situations tend to produce stress, how the athlete responds to stress, and how the resultant stress affects performance and other behaviors. The athlete's cognitive and behavioral skills are assessed to determine existing resources. This stage is integral in understanding the unique aspects and situation of the particular athlete in question, allowing for a personalized program to be tailored for the athlete. The next phase is the *treatment rationale phase*, the aim of which is to help the athlete better understand his or her stress responses through analysis of personal stress reactions and experiences. Next, in the *skill acquisition phase*, athletes receive training in muscular relaxation, cognitive restructuring, and self-instruction. Muscular relaxation is taught under the guidelines of progressive relaxation, described earlier in this section. *Cognitive restructuring*, as described in more detail later in this section, involves the identification of irrational and destructive thoughts and the subsequent refocusing into more positive thoughts. Self-instruction training aims to teach athletes to provide themselves with specific instructions designed to improve concentration and promote problem solving. The final stage is *skill rehearsal*. In this stage, different levels of stress are induced by the consultant using mediums such as videos or imagery. The athlete is required to apply, and thus practice, the coping skills he or she has learned in the program.

Cognitive Control

Cognitive control involves changes to cognitions that trigger, maintain, exacerbate, or reduce the stress and emotion response process. Many cognitive control strategies were developed for cognitive therapy and help athletes understand how thought processes are involved in the experience of stress. Strategies to control unwanted or maladaptive thoughts include cognitive restructuring, positive thought control, and attentional refocusing. Cognitive restructuring involves helping an athlete to recognize and challenge irrational thoughts and to change these thoughts so that they become more adaptive. There are several steps in cognitive restructuring including identifying automatic thoughts or beliefs that are irrational and negative, challenging or debating the rationality of these thoughts, and then replacing these automatic thoughts with more positive and rational thoughts. *Positive thought control* involves self-awareness to identify negative thoughts and replace them with more adaptive ones. Positive thought control involves three elements: using negative thoughts in a positive way, controlling negative thoughts, and training positive thoughts. The aim is to have the athlete take a more positive orientation regarding the situation. *Attentional refocusing* involves shifting attention or focus from a stressful issue to one with fewer negative connotations attached to it. Some athletes may become too focused on their thoughts and stress reactions, causing them to become more anxious. To a large extent, attention refocusing attempts to shift attention from a self-focus to more of a focus on the features of the sporting environment.

Hypnosis

Hypnosis involves getting the athlete to an altered state of consciousness in which he or she is relaxed and where perceptions, feelings, thoughts, or actions can be changed through suggestion. Although still somewhat controversial and misunderstood, hypnosis has been employed with athletes to help reduce anxiety and manage stress, as well as enhance other mental skills, focus attention, and increase confidence. Other stress management techniques such as relaxation and imagery or visualization are often used in conjunction with hypnosis, but the athlete is in a hypnotic state before they are applied. Typically,

hypnosis is applied in four phases. The *induction phase* involves putting the athlete in a relaxed state and then inducing hypnosis using imagery and/or attention-focusing techniques. In the *hypnotic phase*, athletes are given suggestions designed to target the issue at hand, most of which will be carried out once out of hypnosis. The *waking phase* consists of the athlete coming back to a conscious state, and the *posthypnotic phase* involves the athlete carrying out the suggestions given to him or her while in a hypnotized state. Athletes will benefit from hypnosis only to the extent to which they are able to be influenced on a subconscious level.

Meditation

Meditation is another method of raising self-awareness, allowing an athlete to better manage stress. Through meditation, the athlete becomes more attuned to physical sensations and builds an understanding of the connection between physiological functions (e.g., increased HR, nausea) and psychological state (e.g., anxiety, confidence). There are a variety of approaches to meditation, all directed toward increasing awareness of internal physical and psychological triggers that have potential to prompt certain outcomes. This knowledge can help to promote relaxation or direct other stress management approaches, depending on the situation.

Performance and Competition Planning

Preperformance and competition as well as performance and competition plans can help the athlete manage the stress that is inherent in competition. Such plans allow the athlete to take a proactive stance on stress, identifying ahead of time triggers of stress, and formulating a plan to counteract those issues. Planning allows many athletes to feel more in control of the situation and the self, thereby often decreasing further experiences of stress. It also provides a structure for them to incorporate other stress management and psychological skills into their preperformance and performance routines. Preperformance and performance plans have been suggested to promote proper focus and attention toward task relevant issues and help to attain the proper level of activation for performance, promoting both physical and mental readiness to perform.

Self-Compassion

Self-compassion interventions can help prevent athletes from becoming overly self-critical. Based on the work of psychologist Kristin Neff, self-compassion has three key components. Self-kindness involves being understanding and accepting toward oneself in instances of adversity as opposed to being overly self-critical. Common humanity is the acknowledgment that one's experiences are not isolating, as others also have these experiences. Finally, mindfulness involves a balanced perspective, keeping thoughts and feelings in a state of equilibrium, as opposed to overidentifying with them. Strategies to promote self-compassion include writing, imagery, and psychoeducational components. Interventions are currently being adapted for sport.

Stress Inoculation Training

Stress inoculation training (SIT), developed by Donald Meichenbaum, is based on the idea that if an athlete is exposed to stress and learns to cope or deal with that stress in amounts that increase incrementally, an increased tolerance to stress will be obtained. It is a multimodal approach using coping skills that include creating productive and adaptive thoughts, images, and self-statements designed to benefit the athlete's psychological state, as well as performance. It has been found to be effective in reducing anxiety and enhancing sport performance. SIT involves three stages. The *conceptualization stage* aims to raise the athlete's awareness on the effects of positive and negative thoughts, self-talk, and imagery. The *rehearsal stage* involves the athlete's learning to use a number of specific coping skills such as arousal control, imagery, and self-talk, which creates coping resources. The actual skills will depend on the specific needs of the athlete. Finally, the *application stage* involves the athlete's practicing the skills in increasingly stressful situations. A key feature of SIT is the gradual exposure to stress such that the athlete becomes "inoculated" and is less affected. The application begins with low-stress situations and gradually builds toward higher stress situations as coping skills become more advanced. Specific application procedures involve imagery, role-playing, and simulations of increasing perceived stressfulness.

Other Associated Psychological Skills

There are a number of other psychological skills, such as imagery, identifying strengths, and

goal setting, that can be incorporated into stress management programs. Calming imagery, such as visualizing oneself in a safe, relaxing place, can be used to help reduce cognitive anxiety and arousal and to bring on physical relaxation. Conversely, imagery can be used to energize and motivate by visualizing more stimulating, exciting places or scenarios. Imagery is often incorporated into athletes' preperformance and performance plans and routines. Identifying strengths can help refocus athletes' thought processes toward what they can do rather than what they cannot do and assist in developing competition plans that maximize assets. Goal setting can help the athlete stay focused on the task at hand and keep attention on relevant issues. Setting reasonable goals—ones that are measureable and challenging, yet attainable—can also help keep stress from becoming overwhelming. This is most commonly incorporated into preperformance and performance plans and routines.

Conclusion

Stress management techniques can include any intervention that can modify one or more components of the stress process. Stress management techniques need to be directed at individual needs and the issue at hand, as well as take into account the coping resources the athlete has available. As with the acquisition of any skill, application of stress management techniques requires training, time, and practice. Knowledge is not sufficient, as it does not guarantee an athlete can apply the necessary skills or program to his or her specific issue. Application and practice are necessary, and effort is needed on the part of the athlete to make gains in stress management ability.

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See also Basic Emotions in Sport; Coping; Emotional Reactivity; Emotional Responses; Emotional Schemas; Stress Reactivity

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STRESS REACTIVITY

Although ubiquitous in life, stress is rather difficult to define. Stress reactivity is a dynamic process involving context-dependent, interactive factors subject to personal interpretation that dictate subsequent individual response patterns. Stress is an extremely influential element of any engagement in sport or exercise participation. The present entry provides an overview of stress. It discusses each individual stage of the stress process (while giving particular emphasis to the critical role of cognition and appraisal) and explores the evolution of key theories addressing the effects of stress on all performance with an especial eye to sporting pursuits.

Overview

Despite the universal nature of stress and its evident effect on all of human performance, a fully comprehensive understanding of its fundamental nature still remains elusive. Decades of research in both psychology in general and sport psychology (SP) in particular fail to fully explain why stress provides the impetus for outstanding physical achievement in some individuals, while debilitating others to the extent of spectacular failure regardless their level of preparedness. Notwithstanding the continued debate regarding an appropriate operational definition, much headway has been made to answer why humans vary so widely in their responses to stress, particularly in the realm of sport. One of the clearest explanations of this phenomenon has been provided by Charles Spielberger.

Spielberger envisaged a three-stage concept regarding the stress process. The first stage is constituted by the presence of a stressor; the second, the appraisal to what degree said stressor represents a threat; and, finally, the third comprises the level of state anxiety (SA) prompted by the extent

to which the stressor was then judged to be threatening. Spielberger subsequently proposed that athletes' (re)action patterns were dependent on this induced level of anxiety. Each of these stages is herein examined in greater detail.

Stressors

Stressors are conditions that constitute an objective physical or psychological threat. As stress is a biopsychosocial phenomenon, stressors can emerge from any of these biological, psychological, or social factors. Biological stressors are those that test the physiological adaptability of the athlete. The key physiological stressors are the adaptations the human body experiences as the results of undergoing the physical activity (PA) associated with sport and exercise participation (i.e., pulmonary and cardiac efficiency, skeletal muscle development). Examples of biological stressors outside of the body include extreme temperatures, intense noise, high altitudes, and so on. Psychological stressors, on the other hand, exercise their most profound effect on the performer's cognitive capacities. Mental fatigue (exhaustion), worry (suffering from disturbing thoughts), and competitive anxiety (the predilection to interpret competitive situations as threatening) comprise typically experienced forms of psychological stress. Social stressors are those pressures that athletes perceive to be placed upon them by the sport community or society as a whole. Athletes subject to social stress can suffer from body image issues, pressure to return to training or competition after injury (with insufficient recovery time), as well as unrealistic expectations (such as the pursuit of perfection) from coaches, trainers, teammates, supporters, or their country (at the elite international level). There are stressors that impact the athlete via all three channels such as overtraining—a phenomenon whereby the volume and intensity of an individual's exercise surpass the person's ability to recover (physically and psychologically) that then consequently hinders their progress (causing great concern in the athlete and support staff).

It is critical to remember, however, that each of these so-called stressors merely possesses the potential to induce change in the athlete. With the exception of the very extremes of environmental stress, a stressor, almost independent of its origin, only exercises an effect if the athlete *perceives* it

as stressful or threatening. The following section addresses this crucial role of appraisal in stress reactivity.

The Importance of Appraisal

A stress reaction generally does not take place unless appraisal has first occurred. There are any number of potential stressors continually present within both the environment and the athlete himself or herself. A stress reaction is triggered only when one such stressor or several are interpreted as threatening to an athlete's goal fulfillment.

Not only is such cognitive assessment critical because it initiates the stress process, but appraisal is also crucial because it regulates the severity of the stress reaction. Therefore, the judgment of whether or not (as well as to what degree) a stressor is "stressing" dictates the reactivity of several objective and subjective states within the athlete that forthwith influence performance.

Such changes are referred to as SA, the emotional reaction to an immediate, specific perceived threat (thus subject to fluctuations over time). SA is not to be confused with *trait anxiety* (TA), the more stable, personality-based tendency (or frequency) to perceive stimuli as dangerous. TA does, however, mediate SA levels, as higher TA leads to elevated SA, which some athletes view as a hindrance to successful performance. SA is the final product of the stress reaction and its effects on the objective and subjective states of the athlete are explained next.

Induced Anxiety (Experience of Threat)

Objective and Subjective Changes

Significant objective changes take place throughout the athlete's physiological system during stressful events. Typical physical reactions include increases in perspiration, respiration (R) rate, muscle tension, pupil dilation, heart rate (HR) and blood pressure (BP) elevations as well as accompanying decreases in food consumption and digestive rate.

The subjective reactions of SA include emotions (or feelings) and cognitions. Conventional emotional reactions include nervousness, apprehension, fearfulness, and tension. Each of these emotional reactions is based on the person's subjective evaluation of their own preparedness to cope with the stress of the situation and the knowledge that failure to perform will lead to aversive outcomes.

The thoughts that result from a stress reaction are characteristically adverse or unpleasant in nature and most often include uneasiness and worry that such undesirable outcomes will come to pass. The following section presents a number of theories of how these physical and psychological changes collectively influence performance, with special reference to sport activity.

Theories of Stress and Performance

Several sport-specific theories regarding stress and performance have evolved from more general psychological theories; here we will discuss each of the latter major perspectives in turn. Each theory, however, defines stress in its own way, which is somewhat understandable as they were established some decades apart. Acknowledging this compilation of decades of research, we begin each with an examination of the stress process as described by Richard Lazarus.

Lazarus's explicitly stated prerequisites necessary for an accurate model (that agree with Spielberger's in many respects) are that stress is (a) a dynamic process that is (b) iterative and interdependent in nature, (c) composed of unique interactions between an individual athlete and the environment, and is (d) dependent on the athlete's appraisal of said environment and his or her ability to cope with the potential demands imposed and perceived. To explain these effects on performance, three explicit theories are considered.

Yerkes–Dodson Inverted-U Hypothesis

In the earliest of the three theories, Robert Yerkes and John Dodson proposed in 1908 that superior performance occurs when stress levels are moderate, but as arousal levels reached extremes (i.e., too high—hyperexcitement or too low—boredom), learning capacity deteriorates in a symmetrical and increasingly drastic fashion. Graphically, such a relationship resembles an inverted-U form (hence the theory's enduring name). However, inspection of Yerkes and Dodson's original data reveals many inconsistencies, indicating that the range of arousal deemed "moderate" and therefore optimal is fairly small.

The inverted-U hypothesis, as originally presented, leaves much to be desired, most critically when trying to predict performance outcome. Additionally, attempting to explain past performance using Yerkes and Dodson's theory leads

one to deduce that when performance is poor, the performer is either too aroused or insufficiently aroused, but this may not actually be the case.

Individual Zones of Optimal Functioning Model

Yuri Hanin attempted to take the idea of optimal functioning a step further by stating that the most advantageous state of arousal is not necessarily “moderate” but in fact differs across individuals. Yuri Hanin’s individual zones of optimal functioning (IZOF) model holds that performance is successful when initial stress levels are within an optimal zone, which is specific to that individual athlete. If the athlete’s stress level falls outside his or her personal range, performance degrades. In contrast to the Yerkes–Dodson inverted-U hypothesis, which implies that stress levels should be ubiquitously moderate for optimal performance, Hanin’s model does not specify what level is ideal. The optimal level (whether high, moderate, or low) depends solely on the individual performer.

The Dynamic Adaptability Model

Peter A. Hancock and J. S. Warm’s dynamic adaptability model integrates and synthesizes aspects of the aforementioned theories. Much like Spielberger’s conceptualization, the theory is founded on a trinity of stress, namely *input*, *adaptation*, and *output*. Input encapsulates the physical sources of stress in the environment. Adaptation consists of physiological and psychological adaptations to this input and so reflects the appraisal process. Output denotes the change in goal-directed response efficiency. In sport, this latter selection would represent efficiency on any particular task-related performance. When represented graphically, the dynamic adaptability model resembles an extended-U shape, which retains the inverted-U notion that performance deteriorates quickly at the extremes of stress. However, here there is also an optimum range between the extremes, which yields a successful plateau of stable performance. The extended-U also incorporates the IZOF model’s view that the optimum range is based on the individualistic nature of each performer. Hancock and Warm’s theory further contends that the extent of the optimum range not only is subject to the athlete’s physical and psychological capacities for adaptation to stress but is also crucially contingent upon the recognition that the task itself is the major proximal source of stress. Collectively,

these conceptions serve to describe stress reactivity and provide a theoretical and quantitative basis to understand how athletes respond to the stresses that their sports impose upon them.

G. M. Hancock and P. A. Hancock

See also Adaptation; Basic Emotions in Sport; Burnout; Coping; Emotional Reactivity; Psychological Skills; Psychological Well-Being; Psychophysiology; Resilience; Stress Management

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SUPERVISION

Supervision is a central component of professional training and development, providing opportunities for sport psychology (SP) practitioners at all levels to use the experiences and knowledge of others to examine themselves. Supervision is also required for certification or licensure (e.g., registration, chartering) in most of the psychology and helping professions (e.g., psychology, counseling, social work), and in the exercise and sport science fields (e.g., athletic training, strength and conditioning). Those seeking supervision to fulfill degree requirements, or to become eligible for certification or licensure, must know what requirements are expected by their institutions, and the licensing bodies in that area (e.g., American Counseling Association, American Psychological Association [APA], Association for Applied Sport Psychology

[AASP], British Association of Sport and Exercise Sciences [BASES]). Ideally, receiving and providing supervision are career-long endeavors.

Definition and Purposes of Supervision

Supervision is a relatively long-term interpersonal relationship aimed at developing self-knowledge and comprehension of services and therapeutic processes through feedback and communication with another skilled professional. The primary goal of supervision is the welfare of clients in the supervisees' care. SP supervision also serves several purposes for trainees, experienced practitioners, and the general public. Supervision helps trainees, and even seasoned practitioners, learn to work ethically and effectively, develop new skills, gain confidence, and enhance self-awareness—all of which are necessary for effective practice. For skilled practitioners, supervision helps ensure ongoing feedback and opportunities for growth as service providers. The general public benefits from supervision because supervisors assume high degrees of legal and ethical responsibilities for supervisees and their clients, serving gatekeeping and quality control roles that protect the public from inept, or even dangerous and exploitative, practitioners.

A Brief History of Supervision

In the fields of clinical and counseling psychology, supervision is considered a central feature of professional practice. Thousands of research articles, books, and chapters on supervision models, processes, and training in these areas of practice have been published. With regard to training, licensed psychologists and professional counselors in the United States are required to complete 2,000 to 4,000 hours of supervised training and, in many cases, to take coursework in supervision. The field of physical education (PE) also has a long history of supervision with an emphasis on overseeing of student teachers' internships. As in psychology, supervision is an integral component of training and development. SP is closely related to both psychology and PE, and it is somewhat surprising that supervision has received limited attention in the SP research literature and in terms of requirements for certification (licensure, registration). For example, the BASES-accredited sport psychologists are required to have completed only directed studies and consulting under the guidance of a supervisor who is accredited. Similarly, a practitioner

who becomes a Certified Consultant of the AASP (CC-AASP) must complete coursework but need to have only 400 hours of supervised practicum experience under the guidance of a supervisor who holds the CC-AASP credential. In Australia, attention to supervision of sport psychologists is greater than many other countries but that is because SP practitioners are trained as psychologists, and to become registered (i.e., licensed) psychologists, students must complete 1,000 hours of supervised experience in a master's degree or 1,500 hours in a professional doctoral program.

The limited attention in the SP literature to supervision in terms of research and practice may both reflect and contribute to the low levels of SP supervision provided and received. A study of SP practitioners in the United Kingdom (see the last entry in the Further Readings section) showed that 31% of those surveyed had received no supervision of their practice in the past 12 months, and 50% had participated in only one to five supervision sessions in the past year. Further, when asked about the need for supervision post accreditation (after obtaining licensure, registration, chartering), 71% reported that supervision was important only when needed. One might assume that the 29% who indicated that supervision was important and should occur monthly or fortnightly received regular supervision, but only 5% of the sample had been supervised 11 or more times in the past year. It appears that applied SP supervision is advocated as a best-practice guideline but has yet to be fully integrated into training and practice.

The Focus of Supervision

The focus of supervision varies greatly depending on the theoretical orientations and models of supervisors and supervisees (e.g., psychological skills training [PST], cognitive behavioral approaches, narrative therapy models). In general, supervision involves client-case conceptualization (e.g., formulations of what is happening for the client), discussions of centered around professional growth (e.g., supervisee's self-reflections, knowledge gained, knowledge needed), and the dynamics of the working alliance between supervisors and supervisees.

The "hows," "whys," and "wheres" of conducting supervision also vary widely and may include *in vivo* observations of a supervisee's work with an athlete, coach, or team, and reviews, critiques, and discussions of digital audio and video recordings

of client–practitioner interactions, or indirect observations through supervisees recounting their experiences in narrative forms or engaging in role plays with their supervisors to rehearse upcoming encounters with clients. Also, part of supervision may look like advanced training where supervisors instruct and model new techniques to help supervisees expand their helping repertoires. There are no specific guidelines for the frequency of supervision. AASP recommends 1 hour of supervision for every 10 hours of service delivery, but the duration and frequency of supervision vary depending on the developmental needs of the supervisee. For example, students starting their first SP practicum need much more attention than a seasoned professional taking part in peer supervision.

Models of Supervision

As there are many effective approaches to psychotherapy, coaching, and teaching, so are there many different approaches to supervision. The best model of supervision is the one that fits the supervisor and supervisee in the particular circumstances in which they are working. For many SP consultants, the model of supervision is a reflection of, or runs parallel to, the type of treatment model the supervisee is employing in service delivery. For example, the supervisor of a practitioner who is addressing performance enhancement issues via a PST model might focus on the skills used in teaching those interventions to athletes. Supervisors might also work with supervisees on their own professional concerns (e.g., anxiety, irrational thinking) regarding SP practice, and may support them in using PST interventions to address their development and practice (e.g., mental rehearsal of sensitive client interactions, examination of supervisee self-talk). A supervisor using a psychodynamic approach might examine supervisee–client transference and countertransference processes, as well as the supervisor–supervisee transference and countertransference dynamics.

Most models of supervision include commonalities—supervisees’ expectations, desires, theoretical backgrounds, strengths, weaknesses, skills, knowledge, and anxieties—that are discussed and used to determine the course and focus of supervision. Supervisors and supervisees usually identify mutually agreed-upon goals, monitor progress toward goals, and evaluate goal attainment. These processes typically occur in regular

face-to-face encounters (or distance interactions through Internet or phone connections) between supervisors and supervisees or in small groups (or both). In the relatively long-term intimate working relationship in supervision, effective communications between supervisors and supervisees play central roles. To enhance communication and the quality of the supervision experience, it has been suggested that supervisors be mindfully present in supervision encounters. This mindful stance positions the supervisor in the here-and-now moment and nonjudgmentally open to anything that arises. The supervisory space becomes a caring and holding environment where supervisees’ strengths and frailties are both embraced in loving care. Often supervisees need to disclose to their supervisors difficult or embarrassing material (e.g., countertransference responses). A mindful supervisor with many Rogerian characteristics (e.g., genuine, empathic, nonjudgmental, and having unconditional positive regard) may help reduce the potential perception of threat and allow supervisees a safe and secure space to examine (with curiosity, not dread) their human qualities, their strengths, and their weaknesses.

Supervision as an Assisted Form of Self-Reflective Practice

Self-reflection is an important component of effective practice that can enhance self-awareness, help individuals identify their strengths and weaknesses, and make it more likely that practitioners will approach challenging situations in thoughtful ways. Self-reflection is an essential component of effective practice, but self-reflections cannot take the place of supervision. Supervision involves an interpersonal relationship that is focused on the development of the supervisee’s insight, professional growth, and development. Practitioners who rely on self-reflection without supervision lack an outside voice that might provide corroboration, doubt, oversight for the athlete’s welfare, and gatekeeping to protect the public and the field. Self-reflection is important, but it is probably most effective when combined with supervision.

Clarity on the part of the supervisor can make it more likely that self-reflection will be a part of the supervision process. Effective supervisors encourage self-reflection, the expression of feelings and opinions, and follow all ethical guidelines including those with regard to confidentiality and

modeling professional behavior. Supervisees are likely to benefit from opportunities to self-reflect as part of supervision if they are prepared for supervision sessions, have kept up-to-date progress notes, have acted ethically, have been willing to critically examine themselves as SP practitioners, have not concealed information about their athlete sessions, and have provided feedback about supervision.

The Future of Supervision

SP lags behind psychology and PE in terms of the attention paid to supervision, but there are many opportunities with regard to supervision for SP professionals. The history of supervision in sport and performance psychology has not been particularly bright, but the future has the potential to be all that is needed to best serve the athletes with whom sport psychologists work. For example, in Australia all registered (licensed, chartered) sport and exercise psychologists have to complete at least 10 hours (every year) of continuing professional development that involves speaking to peers or mentors, in a supervision context, about their work with clients (or students). Studies have shown that post-registration supervision is often seen as ad hoc rather than as something essential to practice. A possible path for the future of supervision may be that policies and standards, such as those found in Australia, will become widespread across the globe.

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See also Certifications; Cultural Competence; Ethics; Goal Setting; Mentoring; Mindfulness; Self-Awareness

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SUPPORT GROUP

The term *support group* refers to all those supportive and potentially supportive people in an athlete's or exerciser's environment (e.g., parents, siblings, friends, peers, teammates, coaches, exercise leaders, athletic trainers, physiotherapists, physiologists, psychologists). This topic has been implicated widely within sport and exercise psychology (SEP) and has been noted to explain nearly a quarter of the variance in sports performance. In addition, many of the world's greatest athletes attest to the importance of their support group in helping them to achieve their goals. Provided here are definitions of support, its multidimensional nature, theory and mechanisms, and when support should be effective.

Definitions

Support comprises three major subconstructs. *Social integration* reflects the number of different types of relationships in a person's support group. *Perceived support* reflects a person's potential access to support from the group, and is a subjective judgment that the support group would provide assistance if needed. *Enacted support* reflects

the actual provision of help by the support group within a specific time frame.

Multidimensional Support

Perceived and enacted support are generally considered to be multidimensional. This means that support may be further broken down into specific dimensions, such as emotional, esteem, informational, and tangible support. Emotional support refers to other people being there for comfort and security, leading to a person feeling loved and cared for; esteem support refers to bolstering a person's sense of competence or self-esteem—for example, expressing a belief that an athlete is capable of coming through under pressure; informational support refers to providing a person with advice or guidance; and tangible support refers to providing a person with assistance—for example, acting as a training partner to help an exerciser through a tough weights session. In order to better understand and assess the influence of these specific dimensions, sport and exercise researchers have developed measures such as the PASS-Q (the Perceived Available Support in Sport Questionnaire).

Theory and Mechanisms

The key theoretical perspective used to help understand support is the *stress and coping* perspective. According to this perspective, perceived and enacted support operate through *appraisal* and *coping* mechanisms. Athletes who perceive themselves as having high levels of support are less likely to consider challenging situations as stressful compared with athletes with low levels of perceived support. When athletes do experience stress, however, it is the enactment of support that will then help them to cope with the situation and maintain their performance.

The preceding lines refer to the idea that support can reduce or “buffer” the potentially detrimental impact of stress on outcomes, such as performance in a competition. This is known as the *stress buffer* effect. For example, a stress buffer effect would be demonstrated as follows: At low levels of stress, there would be no difference in performance between those with high and low levels of support; at high levels of stress, however, those with low levels of support would perform worse compared with those with high levels of support. Alternatively, when support impacts

performance independently of stress, this is termed a *main* effect; put simply, the more the support, the better the performance.

Research to date has demonstrated that support is associated with a host of important variables in SEP. For example, support and encouragement enhance the confidence of exercisers in their ability to maintain an exercise program, and they also enhance the confidence of athletes to perform well in competitions. Furthermore, those with good support cope better with stress and injury, perform better, and are less likely to burn out.

What Makes Support Effective?

Although research has demonstrated that social integration and perceived support are associated with many positive outcomes, the evidence for effects of enacted support is mixed—enacted support is often unimportant, or worse still, has detrimental consequences; this is known as the *paradox* of support. The effectiveness of enacted support may be influenced by a number of different factors. Support may be most effective when it matches the needs arising from the situation. According to the *optimal matching* hypothesis, relatively uncontrollable situations (e.g., the pressure of competition) lead to a need for emotional and esteem support, such as someone giving moral support and instilling the player with the confidence to deal with the pressure. In contrast, relatively controllable situations (e.g., technical problems in training) lead to a need for informational and tangible support, such as someone giving technical advice and planning, setting, and organizing training sessions to deal with the problems.

The context in which support is provided may influence its effectiveness. In exercise settings, emotional and esteem support (e.g., encouragement and moral support) may be considered particularly nurturing and beneficial. In contrast, informational and tangible support (e.g., direct advice and being pushed by an athletic trainer) may be considered obtrusive, controlling, and more likely to undermine a person's confidence to complete an exercise program on his or her own.

The timing of support may also be important. Support may be particularly effective when an athlete has already had the chance to consider the demands of the situation and decided to request help. However, when support is provided before it has been requested, there may be a greater risk

that the unsolicited support will be interpreted negatively. In the latter case, support may still have its intended effect if it is delivered in such a way that the athlete does not notice it or interpret it as support; this is termed *invisible support*.

Finally, some members of a person's support group may simply be better than others at providing specific types of support. Coaches and exercise leaders may be particularly important providers of informational support, whereas friends and family may be important providers of emotional and esteem support. Furthermore, the quality of relationship shared between the support group and the athlete or exerciser may well determine how support is viewed and whether it will be interpreted as intended.

Tim Rees

See also Burnout; Coach–Athlete Relations; Parenting; Self-Efficacy; Stress Management

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TALENT DEVELOPMENT

How do talented children become elite adult athletes? Many young people start on the road toward becoming professional athletes, but few achieve this level of performance. The development of talent across a range of achievement domains, such as music, art, science, and sport, is a classic area of psychological research. Over the past 20 years or so, a robust body of talent development research has been published in the sport psychology literature.

Talent development reflects the classic nature–nurture debate in that there are some strong genetic components of talent development (e.g., height, body composition) and also strong environmental aspects (e.g., training, social support). From this perspective, the term *talent* can be used to refer to innate abilities, whereas the word *development* refers to how abilities are nurtured and enhanced. Given neither extreme environmental nor extreme genetic approaches may ever be conclusively supported, many researchers adopt the middle ground and, rather than view talent development as a nature *versus* nurture issue, consider talent development as the interaction of genetic and environmental factors.

It is generally accepted that athletes move through a number of stages of talent development, with each stage being characterized by distinct types of activities and different types of involvement and support from parents and coaches. Two important models are reviewed here. Although the

terminology varies between these different models, the general principles are quite similar.

Bloom's Stages of Talent Development

In 1985, psychologist Benjamin Bloom published a book based on interviews with 120 talented musicians, artists, scientists, and athletes (of whom 21 were Olympic swimmers and 18 tennis players who had been ranked among the world's top 10). Bloom proposed talent development occurred through three stages. The first stage, the *stage of initiation*, was characterized by children engaging in fun and playful activities. They relied heavily on their teacher or coach for guidance and support, and at some point parents, teachers, or coaches noticed the children were apparently talented in some way. Parents played a key role, and they were often responsible for stimulating their child's interest in their own personal areas of activity.

During the *stage of development*, children became hooked on a particular activity. Their pursuits became more serious, and their teachers and coaches were more technically skilled than at the previous level. Coaches took a strong personal interest in their prodigies and expected results through discipline and hard work. Practice time increased significantly and competition was used as a measure of progress. Crucially, parents provided both moral and financial support and helped to restrict their child's engagement in distracting activities such as paid employment and social outings with friends.

The *stage of perfection* represents the time when performers become experts in their chosen activity, which now tended to dominate their lives. Performers were willing to invest the time and effort required to meet their performance goals. Responsibility for training and competition shifted from coaches to the individual. Simultaneously, performers were required to be autonomous and be able to deal with enormous demands from their coaches. Parents played a lesser role as individuals became completely absorbed in their actions and assumed total responsibility for them. Bloom emphasized that precocious children need a long-term commitment and increasing passion for their field if their talent is to develop in later life.

Developmental Model of Sport Participation

One of the most well-known models of talent development in sport was introduced by Canadian researcher Jean Côté, originally based on his 1999 study of elite adolescent athletes and their family members. Similar to Bloom's model, Côté proposed three stages of talent development but used different terminology that reflected a more explicit focus on sport and the involvement of the family.

The *sampling years* (ages 6–13 years) were characterized by parents providing opportunities for their children to enjoy sport with an emphasis on fun rather than intense training. Typically, all the children within a family participated in various extracurricular activities, but at some point, parents recognized that a particular child had a gift for sport. During the *specializing years* (ages 13–15 years), athletes gradually decreased their involvement in extracurricular activities and focused on one or two sport events. Fun and enjoyment remained as central elements of the sporting experience, but sport-specific development emerged as an important characteristic of the participants' athletic involvement. Parents emphasized school and sport achievement, made financial and time commitments to their child-athlete, and developed a personal interest in the child's sport involvement, while other siblings acted as role models of work ethic. The *investment years* (age 15 years and over) reflected increased commitment by the athlete to one sport and parents showed even greater interest in this sport. Play activities were replaced by an enormous amount of practice. At this point, parents helped the athlete

fight setbacks, such as injury, fatigue, or pressure, and demonstrated different behaviors toward each of their children, which sometimes caused younger siblings to show bitterness toward their older sibling's achievement. Côté also suggested this talent development process was characterized by a move from deliberate play to *deliberate practice* (discussed below).

Côté expanded on his original study in a later work, and created the current developmental model of sport participation. Within this model three trajectories (pathways) of development in sport are proposed. The first trajectory, *recreational participation through sampling*, reflects a process whereby children engage in a variety of sports during the sampling years. By engaging in a range of playful games, children gain the building blocks for later participation in recreational (rather than elite) sport. The second trajectory, *elite performance through sampling*, reflects the classic three stages (sampling, specializing, investment) of the original research. In the third trajectory, *elite performance through specialization*, athletes skip the sampling years and are heavily invested in practice and competition activities for a specific sport during childhood (i.e., they specialize in a sport early). While some may achieve elite levels of sport through early specialization, this is a perilous route and early specializers often experience overuse injuries, reduced sport enjoyment, and are more likely to drop out of the sport in which they specialized.

Deliberate Practice

An alternative approach to talent development was provided by K. Anders Ericsson, Ralf Krampe, and Clemens Tesch-Römer. The original work was based on data obtained via diaries of daily activities from talented musicians and has since been replicated and expanded by sport psychology researchers. Ericsson proposed the concept of deliberate practice was crucial in the attainment of elite levels of performance. Deliberate practice activities are highly structured, require high levels of effort, generate no immediate rewards, and are specifically intended to improve performance.

Experts reported higher levels of deliberate practice than amateurs across a range of domains. This research ultimately led to the so-called 10,000-hours rule, whereby the attainment of elite levels of performance is associated with the

accrual of 10,000 hours of deliberate practice over approximately a 10-year span. This does not mean that 10,000 hours of deliberate practice produces elite athletes, but rather it provides an approximation of the number of practice hours in which elite performers have engaged.

Constraints that prevent performers engaging in optimal levels of deliberate practice are related to *resources, motivation, and effort*. Resource constraints, including *time* and *access to coaches and training facilities*, must be negotiated by individuals striving to excel. Because deliberate practice does not lead to immediate social or monetary rewards, individuals must overcome motivational constraints. In fact, individuals who engage in deliberate practice are in part motivated by the belief that deliberate practice ultimately leads to improved performance. Finally, deliberate practice is mentally and physically demanding, so individuals must overcome effort constraints.

Ericsson proposed that deliberate practice is not inherently enjoyable, but athletes have reported that the most relevant and practiced activities in which they engage are highly enjoyable. More recently, Côté put forward a more sport-specific view that refined our understanding of deliberate practice. That is, Côté suggested, children engage in *deliberate play* activities. Deliberate play activities in sport are those designed to maximize enjoyment and are activities with flexible rules that are set up and monitored by children. During deliberate play, children are less concerned with outcomes or improving performance, but rather the focus is enjoyment. Hence, during sampling years children should likely engage in high amounts of deliberate play and low amounts of deliberate practice. In fact, the early specialization trajectory is often associated with high amounts of deliberate practice and low amounts of deliberate play, and this may account for the overuse injuries, lack of enjoyment, and drop-out associated with early specialization. Sport psychologists should encourage a focus on sampling a range of sports during childhood with an emphasis on deliberate play. Increasing levels of deliberate practice comes with later specialization and investment in sport.

Other Factors Associated With Talent Development

Although stage models and deliberate play or practice activities are the most prominent aspects of

talent development research in sport psychology, several other factors have been implicated in the attainment of high levels of performance. Three such factors are reviewed below.

Relative Age Effect

Relative age effect refers to the time of year a person is born with respect to the age-eligibility cut-off for a particular sport. For example, if the age cut-off for a sport is January 1, 2002 (i.e., to play on a team a child must be born before January 1), a child born on December 31, 2001, would be eligible for this team, but so too would a child born on January 2, 2001. The January-born athlete is nearly a full year older than the December-born athlete and has advantages of additional growth, development, and practice. As size, speed, and coordination (valued attributes in many sports) are highly correlated with age, relatively older players often demonstrate superior levels of performance and are selected for elite teams ahead of relatively younger players. Relative age effect is generally a robust finding for elite-level age-group sports during adolescence. That is, on elite age-group teams, there are often a greater proportion of athletes born earlier in the age-eligibility year than those born later. The effect of relative age diminishes over time and is less relevant in sports that do not have strict age-group boundaries.

Birthplace Effects

The location of an athlete's place of birth can influence the likelihood of playing professional sport. Research with male and female athletes from several sports (including golf, baseball, basketball, and ice hockey) has shown that athletes who were born and grew up in smaller cities are more likely to become elite athletes. Areas of lower population may provide conditions more conducive to the development of expertise than larger city environments because in smaller municipalities, children have fewer opportunities to engage in a range of activities and may have quite extensive support for specific sports; the hockey rink in a small town is the only option available, and the sport is widely supported in the community. Comparative analyses have suggested contextual factors associated with place of birth contribute more to the achievement of an elite level of sport performance than relative age effects.

2D:4D Ratio

The ratio of the second digit (index finger) on a hand to the fourth digit (ring finger) is known as 2D:4D. A smaller index finger than ring finger (i.e., a low 2D:4D ratio) has been associated with high levels of attainment in sport. One study with professional soccer players showed lower 2D:4D ratios than controls (a sample of men in the general population). Furthermore, within the sample of professional players, those who played in 1st teams (rather than reserves) and those who had played at international level had lower 2D:4D ratios than their counterparts who had not reached these pinnacles of professional soccer. Although the reasons 2D:4D effects have been observed among elite athletes are largely unproven, it is possible that fetal and adult testosterone may be important in establishing and maintaining abilities associated with physical competitiveness (and therefore the attainment of high levels of performance in sport). These findings reflect the idea that talent development is a combination of genetic and environmental factors.

Nicholas L. Holt

See also Burnout; Career Transitions; Coach–Athlete Relations; Drop-Out; Friendships/Peer Relations; Parenting; Skill Acquisition; Transition; Youth Sport, Participation Trends in;

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TASK CONSTRAINTS

The importance of interacting personal, task, and environmental constraints on the emergent behaviors of individuals, as they assemble functional states of movement organization in achieving task goals, is well established. Personal (or organismic) constraints include factors such as individual anthropometrics (height, weight, and limb lengths), fitness (strength, aerobic capacity, and flexibility), mental skills (concentration, emotional control, and motivation), perceptual and decision-making skills (recognizing patterns of play, anticipation by reading the movements of opponents) and personality factors (risk taking behaviors). As part of his constraints model, Karl M. Newell distinguished between physical environmental constraints, such as gravity, ambient temperature, and altitude and task constraints, which are task related and concerned with the goals of a specific activity. More recently, sociocultural constraints, such as family support, cultural expectations, and access to facilities, have also been considered as environmental constraints. In the study of sport performance, task constraints can include factors such as rules of games, equipment used, boundary playing areas and markings, nets and goals, number of players involved in a practice task, and the information present in specific performance contexts. In sport, task constraints can be most easily manipulated by coaches and teachers to channel the acquisition and performance of specific coordination patterns and decision-making behaviors.

Emergence of ecological dynamics (the fusion of dynamical systems theory and ecological psychology) has developed understanding of how information constrains movement coordination due to the reciprocal link between perception and action. Informational constraints, such as highly structured optical energy arrays formed by light reflecting off objects within the performance environment, represent a most significant task constraint. Indeed, the information available in specific performance contexts can be perceived and used directly by athletes to shape their ongoing behaviors and movement responses.

A key point is that individual, environmental, and task constraints all interact in order to shape the way that a performer achieves a specific task goal. Constraints can be deliberately manipulated by practitioners, such as physical conditioning,

equipment modification, pitch sizes, or “rules” of a practice task, or they can be outside their control like the physical environment including weather patterns, growth and development of performers, or official changes to rules of a sport. During practice, coaches and teachers manipulate task constraints to direct the ongoing search for functional movement solutions by athletes. Manipulating constraints is a strategy that forms the basis of nonlinear pedagogy. An important task is to identify key constraints that impinge on an individual performer, considered as a nonlinear dynamical system, in order to stimulate the emergence of functional behaviors during goal-directed performance. Sport practitioners need to identify key informational constraints that can lead to transitions in behavior patterns to help performers overcome performance rate limiters (factors that might limit current performance levels).

Task goals relate to the specific intentions and aims of individuals during task performance. With few exceptions, such as predetermined movement patterns specified by the rules of a sport, exemplified by the performance criteria in diving, ice skating, or gymnastics, task goals tend to not precisely specify how a task should be achieved. Movement coordination solutions are, therefore, only optimal for individuals due to the unique interactions between individual, environmental, and task constraints, meaning that the search for a putative general *ideal movement pattern* or *classical technique* is a redundant goal in coaching and teaching. Functional movement patterns of an individual performer may vary, even within activities which require high levels of performance outcome consistency, such as a gymnastic vault, a long jump approach run, or a golf swing, because the task, environmental, and individual constraints differ from performance to performance. A good example to demonstrate that individual solutions for outcome goals need not be specified is the penalty save in association football. A functional solution for each goalkeeper is a product of the action capabilities of the goalkeeper. Goalkeepers with a faster dive time are able to leave the initiation of their dive to save the ball until later in the kicking action of the penalty taker. Goalkeepers with slower movement times tend to move earlier and rely more on anticipation processes to start their diving action early in the penalty kick.

Specific rules in sports constrain how sports performers achieve task goals. For example, in the different rugby codes, there are rules common

to all formats (i.e. league and union; 7 aside, 13 aside, and 15 aside versions), such as the specifying ball shape and the need for passes to be made backward. There are also some fundamental differences in rules such as those for team numbers, tackling (what the player is allowed to do with the ball once tackled), and game restart rules when the ball is played off the field. These rules constrain the attributes of players most suited to each game type and result in the emergence of different movement patterns and tactics. Even minor rules changes have led to significant changes in the functional movement strategies for achievement and can also lead to variations in the fit between different individuals and rule-based task constraints. In learning design, it is essential that practice tasks include boundary markings since a performance location on field or on court has been shown to lead to changes in individual decision making and actions. For example, research by Headrick et al. (2012) has demonstrated how proximity-to-goal influenced behaviors in a one-on-one practice task in soccer. Players significantly vary their behaviors depending on the distance to the attacking or defensive goal area. Data provided implications for the design of practice tasks in relation to key visual reference points in the environment, such as goals and line markings.

Changes to equipment design can lead to significant changes in the way that athletes meet their performance goals. For example, the Fosbury Flop was a performance solution that emerged coinciding with the use of new technology in the high jump: softer landing areas in modern times. Directing the body over the bar, head and shoulders first and sliding over on one's back would have likely led to injury without modern technological developments in landing mats. In soccer, the development of waterproof balls led to a 30% increase in kicking distance and led to the development of new kicking techniques in wet climates that had been impossible with heavier nonwaterproof balls. In a well-known performance paradigm shift in soccer, the first players to *bend* the trajectory of balls when shooting (adding swerve and dip to flight) were South Americans in the 1960s rather than European players. Why was this? Because prior to the 1970s, footballs had a leather skin that picked up moisture as a game progressed in wet conditions. Games played during winter in the Northern Hemisphere led to the ball increasing in mass to almost double its original value. In

the generally drier climates of South America, the ball did not pick up as much moisture and mass, with aerodynamic forces having more influence on the lighter ball, enabling skilled players to acquire expertise in bending and curving ball trajectories. With the advent of waterproof balls in the 1970s, Northern European experts were able to re-create the same effects as their South American counterparts. In turn, changes in task constraints like physical characteristics of equipment influenced the tactical strategies available to teams and consequently reemphasized the acquisition of specific shooting and long passing skills in players. These developments showed that practitioners can deliberately manipulate the characteristics of implements to facilitate the reorganization of movement coordination.

Changing task constraints can lead to changes in new individual performance characteristics being best suited to the new task. For example, changes in specification of the javelin in 1986 led to changes in the optimal throwing technique for throwers. Because of the desire to develop javelins that landed point first on every throw, the projectile's center of gravity was moved forward by 4 cm. The change was observed to result in the optimal technique being a higher release angle. It has also been suggested that the changes would benefit more powerful athletes who could generate high release speeds. Sudden interathlete changes in performance levels can be explained by changes in interacting personal and task constraints.

In sport, as expertise is enhanced, informational constraints designed into practice tasks can progressively attune an individual to the specifying information sources that support the organization of actions and enhance the capacity to adapt to changes in a performance environment. James J. Gibson (1979) proposed that invariant (persistent features) and variant information can act as *affordances* for action, through which a performer perceives information from the environment in relation to what it offers or demands in action responses. Over time, performers become attuned to information through experience and practice in different performance environments, creating relationships between movement patterning and specific sources of perceptual information (information–movement coupling). The importance of ensuring the presence of key specifying information in practice tasks is captured by the ecological concept of *representative*

learning design. Traditionally, representativeness has referred to the generality of task constraints in a specific research context to the perceptual variables available in actual performance settings. In sports, practice environments are the equivalent of experimental settings, suggesting that they need to be accurately designed to ensure congruence with a performance environment in which the movements will be implemented. Changing the informational constraints on action might result in less representative practice designs and changes to a performer's acquisition of functional movement control. This idea has been exemplified in cricket batting research. Ross A. Pinder, Keith Davids, Ian Renshaw, and Duarte Araújo demonstrated that batters adapted spatiotemporal characteristics of emergent action when facing a live opponent through the pickup of advanced kinematic information, in contrast to facing balls delivered via a projection machine where movements were delayed through a need to sample early ball flight to determine the bounce point of the ball.

Verbal informational constraints such as instructions or feedback, can constrain the movement patterns adopted by sport performers. For example, an instruction to a tennis player to make sure the first serve goes in will likely elicit a different movement pattern from the server than the instruction is to hit the ball as hard as possible without worrying if it goes out. A more effective pedagogical strategy involves the careful manipulation of task constraints within the context of interacting task, individual, and environmental constraints, facilitating the emergence of functional movement patterns and decision-making behaviors in learners (see Davids, Araújo, Hristovski, Passos, & Chow, 2012).

Ian Renshaw and Keith Davids

See also Anticipation; Coordination; Decision Making; Dynamical Systems; Ecological Theory; Expertise; Functional Variability

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TEACHING PERSONAL AND SOCIAL RESPONSIBILITY MODEL

The term *social responsibility* refers to an ethical ideology whereby organizations and individuals have an obligation to act in manners that benefit society. Models of social responsibility abound in various scientific domains, and in the field of sport and exercise psychology the *Teaching Personal and Social Responsibility* (TPSR) model has been developed to promote social responsibility in youth. In this entry, an overview of the TPSR model and examples of youth programs using TPSR are presented.

The TPSR model was initially developed in the early 1970s by Don Hellison from his work teaching physical education to inner-city youth in Portland, Oregon. More than 40 years of instruction and evaluation have led to the establishment of preferred practices that form the basic components of the model today. The fundamental purpose of the model is to provide a guiding framework for helping youth take responsibility for their own well-being and be sensitive to the well-being of others. TPSR is composed of five levels of responsibility, with each level having its own

set of goals. The responsibilities are depicted as levels in order for instructors to implement them progressively during physical activity lessons.

The first level consists of introducing *respect*, which is viewed as the most important responsibility. Displaying respect, by minimizing abusive actions, solving disagreements calmly, and promoting inclusion, is considered necessary in order to protect the rights of all participants. As participants progress to other levels, respect is regularly revisited for participants to understand that it is the core responsibility. At the second level, the development of *self-motivation* is encouraged to help participants demonstrate sustained effort and persistence, try new things, and redefine how they view success. The third level involves *self-direction*, which is an advanced level of responsibility that builds on respect and motivation. Participants attain this level when they become more independent, implement their own personal physical activity programs, and are able to set and attain personal goals. The fourth level consists of *helping others and exhibiting leadership*. At this level, participants act in manners that are responsive to the well-being of others and do so without extrinsic reward, simply because it is the right thing to do. Achieving this level is difficult because participants must demonstrate leadership by taking away time from their own personal interests to help others have positive experiences. The final level of *transference* is achieved when participants apply responsibilities outside of the program and become role models for younger kids. The goal is to have participants test new approaches at school, work, and home using the competencies learned in the program.

To foster social responsibility, instructors are trained to work toward five goals. First, instructors develop solid instructor–participant relationships by recognizing participants’ individual strengths, listening to their opinions, and believing in their ability to make good decisions. Second, instructors integrate responsibilities into physical activity content for participants to learn through practice how to be socially responsible. The third, fourth, and fifth goals are, respectively, to empower participants (e.g., involvement in decision making and negotiations), promote reflection (e.g., group problem solving), and facilitate transfer (e.g., giving reminders, providing personal examples).

Five implementation strategies have been designed to structure physical activity lessons and

put responsibilities into practice. First, *relational time* is allotted at the beginning of lessons, giving instructors opportunities to acknowledge the presence of participants, foster relationships, and thank participants for their contributions to the program. Second, *awareness talks* are performed to teach participants the five levels of responsibility, and activities are specifically planned to facilitate direct instruction. For example, to foster self-motivation, instructors ask volunteers to tell other participants what the program is about. Third, *physical activity lessons* take place, during which instructors integrate strategies to teach responsibility concepts to the participants in competitive and noncompetitive game options. For example, to promote inclusion, instructors tell participants to make a minimum number of passes to teammates during games like soccer and basketball before taking a shot on goal. Fourth, *group meetings* are organized, allowing participants to voice their opinions, problem solve, and reach decisions. For instance, if a participant has shown disrespect for others during an activity, that individual can negotiate the consequences of this action. Finally, participants are provided with *reflection time* to evaluate their attitudes and behaviors as they relate to the five responsibilities. This activity helps participants develop self-direction by allowing them to define success for themselves, work toward their own goals, and decide if they want to volunteer for teaching roles.

Physical activity programs based on TPSR have been implemented in a wide range of contexts in the United States as well as internationally in countries such as New Zealand, Great Britain, and Spain. Examples of programs designed for elementary school children include Project Effort, the Coaching Club, and the Energizers Club and generally, these programs were developed to foster resilience and social responsibility in underserved populations. Programs based on TPSR have also been developed for underserved middle and high school-age participants to keep them involved in productive activities. The Youth Leader Corps, Cross-Age Leadership Program, and Project Lead are examples of programs that have helped participants develop leadership, confidence, and problem-solving skills by teach younger children the responsibilities of the TPSR model.

Although interests in alternative activities and commitments to “real life” responsibilities have occasionally proven to be hindrances in having participants develop social responsibility, the TPSR model has generally been shown to be

effective at facilitating the positive development of youth. The model serves as a useful framework to provide participants physical activity instruction that teaches them to take responsibility for their well-being and that of others.

Martin Camiré

See also Character Development; Moral Development; Resilience; Transfer; Youth Sport, Participation Trends in

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TEAM ATTRIBUTIONS

In his influential theory of motivation and emotion, Ivan Weiner proposed that attributions, the reasons that people use to explain the causes of behavior, are powerful determinants of emotions and motivation. Although attribution researchers in sport have predominantly focused on self-referent attributions (attributions for one's own behaviors), it has been proposed that *team attributions*, the explanations provided for group behavior, may also be important determinants of individual (e.g., emotions and motivation) and group (e.g., team cohesion and communication) level consequences. Team attributions are sometimes described and studied as the shared belief held by the group (team attributions) and sometimes as the perception held by the individual athletes within the team (team-referent attributions).

Although there are an almost infinite number of possible team attributions, attributions are best understood when classified into underlying dimensions. Historically, team attributions have been considered to vary in terms of *locus of causality* (the extent to which causes are perceived as residing within the team or outside of the team), *stability* (the extent to which causes are seen as variable over time), and *controllability* (the extent to which

causes are perceived to be regulated by the team or other factors). More recently, it has been suggested that a greater number of attribution dimensions should be considered. These include *globality* (the degree to which the cause is perceived as localized or occurring across many situations) and *universality* (the degree to which the cause is perceived as common among other teams).

Consequences of Team Attributions

The fundamental drive for exploring team attributions is that, as with self-referent attributions, they are considered to be important determinants of a range of outcomes that, in turn, are important determinants of achievement and enjoyment in sport. Researchers have proposed that team attributions will influence emotional and cognitive responses to team success and failures. In terms of the emotional responses, evidence suggests that following successful performances, individuals who report controllable and internal team attributions like teamwork and tactics experience greater levels of self-esteem than individuals who report uncontrollable attributions like luck or refereeing decisions. Additionally, it has been proposed that internal and controllable team attributions after successes will promote higher levels of happiness, pride, and satisfaction; whereas, following unsuccessful performances, internal and controllable (and global) team attributions will lead to higher levels of dejection, hopelessness and shame.

In terms of the cognitive responses, research also provides partial support for the effects of attributions on collective efficacy (a group's belief in its ability to achieve specific goals) and expectancies for future team performances. Specifically, research has shown that, following successful performances, greater increases in collective efficacy are seen in individuals or teams who use controllable and stable attributions (team coordination), whereas following failure, greater decreases in collective efficacy are seen in individuals or teams who use uncontrollable and stable attributions (team ability levels). However, such relationships are modest and research has not consistently identified the same associations between attributions and collective efficacy.

In addition to these consequences, it has also been proposed that team attributions influence perceptions of the team's cohesion; following success, attributions that give more credit to the team

for the performance are likely to promote cohesion. Also influenced are psychophysiological reactions, and recent research indicates that hormonal responses, such as changes in testosterone and cortisol levels, to success and failure are moderated by self-referent attributions; and behavioral responses, where again, research conducted with self-referent attributions indicates that persistence following failure is enhanced if individuals engage in internal, controllable, and unstable attributions.

Thus, it appears as if team attributions have the potential to influence some important responses to performances. Overall, it is contended that those people or teams who use team-serving attributions (internal, controllable, and stable attributions following successes and internal, controllable, and unstable attributions following failures) will experience more positive emotions, hold higher expectancies for future performance levels, be more attracted to being part of their team, and devote more effort in the pursuit of team goals than will individuals or teams who report different attribution responses to successes and failures.

Determinants of Team Attributions

Given the potential role of team attributions in determining important team consequences, researchers have also begun to investigate determinants of team attributions and those factors that may promote team-serving attributions. Underlying this interest is the belief that team attributions may be relatively amenable to manipulation and a tendency to engage in team-serving attributions can be trained. Essentially, although it is acknowledged that individuals will vary both in their tendency to use certain attributions following success and failure and in their knowledge of the determinants of success and failure in their sport, the development of attributions will also be determined by the interplay of situational factors such as the results of the sporting encounter; the publicly expressed explanations of coaches, teammates, and spectators; team cohesion; and event importance. At present, many of these potential sources of attributions have yet to be empirically examined, but research does indicate that factors such as match outcome, event importance, and gender may interact to influence the team attributions that people use. Specifically, research has consistently shown that winners or subjectively successful performers use different attributions than losers or subjectively unsuccessful performers, although

the exact nature of these differences is not consistent and the extent to which team-serving attributions are used following success may be moderated by factors such as gender (males tend to be more team-serving than females) and event importance (athletes tend to use team-serving attributions following important encounters than after less important encounters). It is possible that this research will lead to the development of team-attribution retraining programs.

Iain Greenlees

See also Attribution Theory; Basic Emotions in Sport; Cohesion; Collective Efficacy; Emotional Responses; Psychological Skills Training

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TEAM BUILDING

The fact that groups can influence outcomes important to both the individual and the group has been recognized for decades. One group variable that has been proposed to be important in this regard is the member's perception of the group's cohesiveness—staying with a group to achieve goals or for member satisfaction. Numerous studies have emerged illustrating that perceptions of cohesiveness are associated with a multitude of factors that impact individual (e.g., adherence, satisfaction) and group outcomes (e.g., team success, collective efficacy). While recognition of these relationships is important, there is an equally important question for those interested in behavior change. How may the cohesiveness of the group be enhanced? One established group-based intervention used by coaches and exercise leaders to enhance cohesion is team building. Although it has been defined in numerous ways, team building in activity settings typically refers to programs that

use group dynamics principles to increase cohesiveness, which then enables the group to function more effectively.

Team-building programs have been used in both exercise and sport settings but typically look to target different outcomes. In the sport setting, the focus is usually on the process of locomotion occurring within groups, with the flow moving from team building → increased cohesiveness → improved group effectiveness. In contrast, the other key process occurring within groups, maintenance, is often the focus in exercise settings. Research questions typically surround how to keep individuals adhering to their activity regime. The flow is captured as team building → increased cohesiveness → increased group maintenance. Given their different outcomes, team-building programs take different forms across settings, and these differences will be highlighted in this entry. Further, it also is evident that team building transcends the stereotypical notion of having teammates go to a movie or theme park together.

Team Building in the Sport Setting

In the sport setting, with its emphasis on creating a more effective team, the results of a meta-analysis by Luc Martin, Albert Carron, and Shauna M. Burke revealed the use of four different forms of team-building programs. These included, in ascending frequency of reported use in the field setting, *interpersonal relations*, *goal setting*, *adventure*, and *omnibus*. In the interpersonal relations framework, the focus is on building an interpersonally competent group, with the assumption being that this will translate into group cohesiveness and effectiveness. This approach works well in situations when there is a requirement to improve poor personal relationships or poor communication among members. Within the goal-setting framework, the focus is on helping group members develop individual and group goals and the actions required to achieve them. This approach is typically used when the team's goals are not clear or the actions required to achieve the goals are not clearly identified. The development of cohesion derives from the group's focus on its conjoint goals. The adventure framework approach proposes that engaging team members in an adventure or outdoor experience will enhance group cohesion and ultimately team effectiveness. The adventure experience can range from

providing a challenge in an outdoor environment or gym wherein members must work together cooperatively to solve some problem to having team members go on an extended trip where they spend concentrated blocks of time together. The omnibus framework is underpinned by the idea that the enhancement of cohesion and team effectiveness is created when members are exposed to a number of protocols derived from group dynamics principles, such as creating uniqueness, making sacrifices for the group, or clarifying roles.

The results of the meta-analysis revealed that team-building protocols that focus on goal setting are of longer duration and target individual-type sports like track appeared most effective. In terms of the relationship between the overall team-building interventions and specific outcomes, positive relationships with performance effectiveness, adaptive cognitions like self-efficacy or satisfaction, and to a lesser extent social cohesion, have been reported for those exposed to team building than those not. Given these interventions were primarily directed toward enhancing performance effectiveness, ostensibly a task focus, the emergence of social cohesion (orientation toward maintaining social relationships within the group) might be perplexing. One possible explanation might be that many of the identified protocols involved collaboration as in the goal-setting protocol where members had to agree on a common goal, and this need for collaboration was reflected in social cohesion.

Team Building in the Exercise Setting

As noted above, outcomes in most exercise settings that employ team-building protocols typically concern the maintenance of the group, whereby interest is directed toward enhancing the adherence of individuals. Unlike the sport setting, where multiple frameworks have been employed, one model has served as the foundation of numerous interventions conducted in the exercise setting. It is the four-stage, team-building model developed by Bert Carron and Kevin Spink that uses the exercise leader as the agent of delivery. The efficacy of this model has been demonstrated across many exercise settings and age groups ranging from adolescents to the elderly. The stages in the model include *introduction*, *conceptual*, *practical*, and *intervention*. The first three occur in a workshop conducted by a team-building specialist such as an

exercise psychology consultant working with exercise leaders. The final stage involves the leaders going back to their group and implementing the team-building strategies that were formulated during the workshop. The specifics of each stage are presented next.

In the first stage, which focuses on establishing a rationale, exercise class leaders are presented with a brief overview of the benefits of cohesion specific to their setting. For instance, if the leaders run exercise classes for older adults, the positive relationships that have been established between elderly exercisers who perceive their group as more cohesive and as having better exercise class attendance are mentioned.

The second stage provides a frame of reference for the leaders. This is accomplished by introducing a conceptual model that outlines how to use group dynamics principles to enhance cohesion. In the model, cohesion within the group is viewed as an output (or product) of conditions that arise from three different categories of group characteristics. Two categories are the *environment* of the group and the *structure* of the group (inputs), and one category is group *processes* (throughput). It is assumed that specific elements in the environment and the structure of the group will contribute to enhanced group processes, which in turn lead to member's increased perceptions of cohesiveness in the group. Furthermore, within each of the three categories, specific factors are identified that have previously emerged as being associated with enhanced group cohesiveness. These include highlighting *group distinctiveness* (group environment), *fostering group norms and individual positions* (group structure), and *increasing communication or interaction and individual sacrifices* (group processes). By way of illustration, to highlight the factor of distinctiveness, leaders are told that when something in the group's environment is somehow made distinctive (e.g., all group members wear the same T-shirt), members develop a stronger sense of *we*, can more readily distinguish themselves from nonmembers of the group, and ultimately develop stronger perceptions of cohesiveness. This procedure of presenting the leaders with a research-based rationale justifying the inclusion of each factor continues until all the factors in the model presented above are addressed.

The practical stage is the final part of the workshop. The purpose is to have the exercise leaders become active agents in developing practical

strategies that they will use in their own group settings. This is done by having them use the conceptual framework to brainstorm as many specific techniques as possible to be used for team building in their own groups. The leaders are asked to use distinctiveness, norms, positions, sacrifice, and communication or interaction as frames of reference. From the lists of suggestions generated, each leader is free to take the suggestions thought to work best.

In the intervention stage, the leaders take the team-building protocols they have developed at the workshop and introduce them into their groups. Examples are now presented to illustrate the outcomes that have resulted when this intervention has been used in the exercise setting. The purpose of the initial studies into team building using the Carron and Spink model was simple—to determine if cohesion in exercise classes could be enhanced using the principles in the model. The results of numerous studies revealed that use of this team-building intervention significantly enhanced group members' perceptions of task cohesion in exercise classes across the age span. The emergence of task cohesion (orientation toward achieving the group's goals) was consistent with the fact that the team-building protocol used in this model targeted task cohesion.

This positive relationship also has been extended to another key outcome salient to the exercise setting—adherence. Results generally revealed that those exposed to team-building protocols report better attendance, less lateness, and a lower probability of dropping out of the exercise group than those not exposed to such protocols. Cognitive outcomes, such as satisfaction, also have been examined in some of these exercise studies, with those exposed to a team-building condition reporting higher levels of satisfaction with the group experience than those not exposed to the team-building protocol.

Delivery of Team-Building Programs

A closer examination of the team-building programs reveals that the protocol can be delivered to group members either directly or indirectly. In the indirect approach, which is the predominant mode of delivery in the exercise setting, the program is filtered through the exercise leader. Specifically, a team-building specialist works with the exercise leader to develop a team-building protocol based

on group dynamic principles each leader will implement with the particular activity group. While the indirect approach also has been used in the sport setting, most sport interventions have used a more direct form of team building, in which the intervention is presented directly to the team members. In direct approaches, the individual responsible for implementing the intervention (e.g., sport psychology consultant) works directly with the team members to provide them with greater insight and greater independence in applying the appropriate group dynamics principles. It is assumed, through the contact and education, team members will become more intrinsically motivated to develop team bonds as a result of their enhanced competence and self-determination. Of interest, comparisons of the direct versus indirect approaches in the sport setting have revealed that both are equally effective in generating the expected positive outcomes.

While the effects of team building are typically perceived to generate positive outcomes, there is one form of team building that may not be as effective as commonly assumed, and that involves hazing. *Hazing* is the use of harassment, abuse, or humiliation as a way of initiating new members to a group (e.g., having rookies wear odd clothing in public places as a type of team initiation). It is often assumed that team-building behaviors such as hazing serve to build team cohesion. However, there is little to support this conclusion as results have revealed that athletes who report doing or seeing more hazing-type activities also report that their teams were less cohesive.

Kevin S. Spink

See also Adherence; Cohesion; Drop-Out

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TEAM COMMUNICATION

Communication is commonly defined as a transmission of thoughts, feelings, information, knowledge, and ideas by means of written or verbal messages. However, when people communicate face-to-face, they position their bodies in a certain way, vary their stance, control their eye gaze, and move their hands in particular manners. Therefore, there is an additional set of nonverbal behaviors, which is often coordinated with verbal message exchange. Verbal (overt) and nonverbal (covert) are the communication channels athletes use in practice and competition. Team communication consists of a least two individuals who share meaningful exchange of information in which a person attempts to influence the response of another person or the team as a whole. The shared information in verbal communication is open, explicit, unambiguous, precise, and clear. Conversely, in nonverbal communication the information conveyed is hidden, implicit, covert, unexplained, and sometimes, not obvious to the casual observer.

The communication process consists of the following sequence: deciding to make a message, encoding the message by the sender, sending the message, decoding the message by the receiver, and finally, responding to it. Communication among team members can suffer due to the sender's inability to successfully encode the message, noise or interference in the communication channel (e.g., physical noise, psychological barriers), and the receiver's inability to effectively decode the message. In sport, noise can be created by cheering fans or a whistling umpire (physical noise), as well

as the context of the competition, such as game standing and time phase (psychological noise), which can equally disrupt the communication process.

Team Communication in Sport

The contemporary viewpoint is that teams, more than individuals, can find more solutions to problems evoked during competition. To become effective, the members of the team must communicate, particularly when participants wish to convey a message. Communication processes can result in persuasion, evaluation, information transfer, motivation, and problem solving. Besides, communication may carry more than one function at the same time. For example, a playmaker in basketball can motivate and inform a teammate about the next play in order to overcome a defensive strategy set by the opponent team. Thus, communication may serve for the coordination (cognitive function) and for motivating the team members (affective function). Communication patterns (e.g., how do members of the team respond to each other's message) are found to discriminate between more effective and less effective teams. Effective sport teams tend to (a) display more consistent types of communication, and (b) use frequent planning statements.

Many sports are characterized by constantly changing locations and time constraints. In sport such as basketball, soccer, and field hockey, players change locations and the ball moves at speed from one player to another. Under such circumstances, members of the team often rely on both overt and covert communication in an attempt to increase team coordination and performance. In baseball, the pitcher and catcher communicate through a series of hand signals what pitch will be thrown. At the same time, the shortstop and second baseman can communicate who will protect the second base. In basketball, the playmaker communicates to the rest of the teammates which play will be performed, to slow down, or to speed up the tempo of the game. Finally, soccer teammates communicate a particular target to a player who executes a corner kick. It follows that team members in sports communicate with each other to achieve better solutions to challenges posed by the opposing team players, and in so doing are better positioned to coordinate and implement team strategies among themselves.

Team Communication Moderators

Culture moderates verbal and nonverbal communication on a large-scale level. Customs, norms, and display rules are areas in which one culture is differentiated from another. Furthermore, the fact that genetically and culturally determined rudimentary messaging (displaying aggression, play, and association) is universal, there is remarkable inconsistency in how individuals, as well as groups, articulate and understand verbal and nonverbal signals. Gender is an additional variable, which moderates encoding and decoding processes. Evidence in the recent literature supports the conclusions that men and women differ in terms of encoding and decoding verbal and nonverbal behaviors. First, differences exist in relation to nonverbal expressivity. When compared with men, women tend to use more smiling, touching, and eye-contact behaviors. In addition, women have more speech fluency. On the contrary, men tend to use extraverted gesticulation and movements more so than women. The majority of these differences are salient when in public. Second, men tend to be socially less responsive and less attached than women. Moreover, men tend to interrupt and talk more, listen less, and are overall more dominant during conversations. Also, men tend to assume less proximity and signal less friendliness in comparison to women. Furthermore, women tend to have better facial recognition and are more precise in interpreting other people's nonverbal messaging.

Research findings also show that the tendency to receive messages visually is positively associated with receiving messages audibly (different encoding channel abilities correlate one with another). In other words, people who better receive, interpret, and respond to visual messages tend to be better in doing so when receiving auditory messages. Also personal traits, such as extraversion, being talkative and expressive, having high self-esteem and trait self-consciousness, positively relate with encoding skills. Furthermore, if people are outgoing, vigorous, autonomous, mentally accommodating, and have low levels of trait anxiety, they tend to be better predisposed in terms of their nonverbal communication skills. That said, it is important to note that nonverbal communication can improve with practice. In addition, encoding skills are affected by age; they improve as one gets older but deteriorate when one becomes

old. No meaningful relations tend to exist between communication encoding skills and race, education, or IQ.

Team Communication Assessment

The assessment of communication in sport has primarily related to aspects of team cohesion and centered on general communication styles. There is no evidence to date of published studies in which researchers sought to measure communication at a behavioral level (i.e., amid teammates members during actual play) or in which they related such assessments to the performance of teams. Recent research by Domagoj Lausic, Gershon Tenenbaum, David Eccles, Allan Jeong, and Tristan Johnson has assessed verbal communications through video and audio recording technology to elicit the type of messages that tennis players exchange during doubles tennis competitions. In this research, winning teams more than losing teams were found to respond more constructively and confirm actions offered by each player, as well as using motivational statements during the competition.

While research in the area of nonverbal communication has increased during last 25 years, there is a paucity of research on nonverbal communication in sports. Three procedures are typically used to assess nonverbal skills: (1) standardized performance measures, (2) individualized performance measures, and (3) self-report measures. Standardized performance measures consist of audio or video presentations about which the respondent draws conclusions about a perceived emotional message, an interpersonal relationship, or the presence of deception. The respondent is typically tested on accuracy in recognizing emotions that are communicated by another person's face, body, and voice. Individual performance measures include videotaping subjects while partaking in various group activities, where videotaping captures a range of emotions. The tapes are subsequently viewed and analyzed by a group of judges. Self-report measures capture participants' assessments of their own nonverbal skills.

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See also Group Characteristics; Group Formation; Shared Mental Models; Team Building

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THEORY OF PLANNED BEHAVIOR

The theory of planned behavior, developed by Icek Ajzen, is a social cognitive theory that has guided a large majority of theory-based research on physical activity. The theory of planned behavior is an extension of the theory of reasoned action developed by Martin Fishbein and Icek Ajzen in 1975. Since its introduction over 25 years ago, the theory of planned behavior has become one of the most frequently cited and influential models for predicting human behavior.

The theory of planned behavior specifies that some or all of the following four main psychological variables influence our behavior: intention, attitude, subjective norm, and perceived behavioral control. The combination of an individual's expectations about performing a behavior and the value attached to that behavior form the conceptual basis of this theory. Figure 1 displays the main variables of the theory of planned behavior, and each of these main variables is described in more detail below.

Theory of Planned Behavior Variables

Intention to perform a behavior is the central determinant of whether or not an individual engages in that behavior. Intention is reflected in a person's

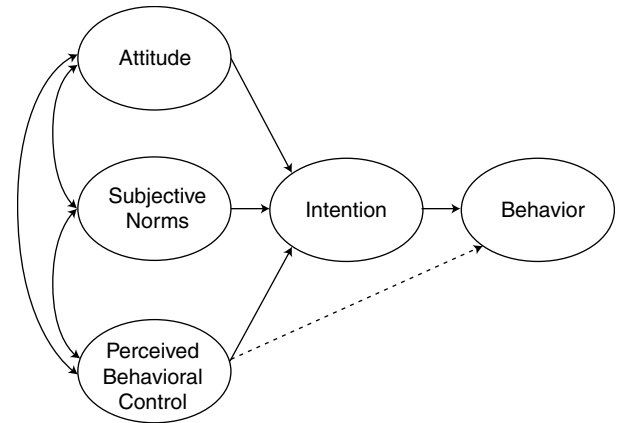


Figure 1 Main Variables of the Theory of Planned Behavior

Source: Heather Hausenblas.

willingness and how much effort that individual is planning to exert to perform the behavior. The stronger a one's intention to perform a behavior, the more likely one will be to engage in that behavior. Thus, if someone has a strong intent to go for a walk this afternoon, that person is likely to go for that walk.

As might be expected, a person's intention can weaken over time. The longer the time between intention and behavior, the greater the likelihood that unforeseen events will produce changes in people's intention. For example, a young adult may intend to be a regular lifetime runner. However, after running for a few years, this person may become bored with the activity and start to swim instead. The runner did not expect boredom to affect the intention to run. A person's behavioral intentions are influenced by personal attitudes about the behavior, the perceived social pressures to perform the behavior (subjective norm), and the amount of perceived control over performing the behavior (perceived behavioral control).

Attitude represents an individual's positive or negative evaluation of performing a behavior. Do you find regular exercise useless or useful, harmful or beneficial, boring or interesting? An older adult may have a negative attitude toward engaging in a vigorous physical activity such as running, but rather have a positive attitude toward walking in the neighborhood. Our attitude toward a specific behavior is a function of our *behavioral beliefs*, which refer to the perceived consequences of carrying out a specific action

and our evaluation of each of these consequences. A college student's beliefs about playing doubles tennis could be represented by both positive expectations (e.g., it will improve my social life because I will meet lots of people) and negative expectations (e.g., it will reduce my time to study). In shaping a physical activity behavior, the person evaluates the consequences attached to each of these beliefs. Common behavioral beliefs for physical activity include that it improves fitness or health, improves physical appearance, is fun and enjoyable, increases social interactions, and improves psychological health.

Subjective norm reflects the perceived social pressure that individuals feel to perform or not perform a behavior. Subjective norm is believed to be a function of *normative beliefs*, which are determined by the perceived expectations of important significant others, such as family, friends, physician, or priest, or groups like classmates, teammates, or church members, and by the individual's motivation to comply with the expectations of these important significant others. For example, an individual may feel that his wife thinks he should exercise three times a week. The husband, however, may not be inclined to act according to these perceived beliefs. Common normative beliefs for physical activity include those of family members, friends, and health care professionals.

Perceived behavioral control represents the perceived ease or difficulty of performing a behavior. Perceived behavioral control influences behavior either directly or indirectly through intention. People may hold positive attitudes toward a behavior and believe that important others would approve of their behavior. However, they are not likely to form a strong intention to perform that

behavior if they believe they do not have either the resources or opportunities to do so. You may have a positive attitude and enjoy swimming; however, if you do not have access to a pool you will not be able to perform this behavior.

Perceived behavioral control is a function of *control beliefs*. Control beliefs represent the perceived presence or absence of required resources and opportunities (e.g., there is a road race this weekend), the anticipated obstacles or impediments to behavior (e.g., probability of rain on the weekend is 90%), and the perceived power of a particular control factor to facilitate or inhibit performance of the behavior (e.g., even if it rains this weekend, I can still participate in the road race). The most common control beliefs for physical activity include lack of time, lack of energy, and lack of motivation. Table 1 contains sample items for measuring the theory of planned behavior constructs in relation to regularly exercising during the winter.

Theory of Planned Behavior and Physical Activity Research

Several meta-analytic reviews have supported the theory of planned behavior for explaining and predicting a wide variety of physical activities in a wide variety of populations, including ethnic minorities, youth, pregnant women, cancer patients, diabetic adults, cancer survivors, and older adults. In general, researchers found that intention is the strongest determinant of behavior followed closely by perceived behavioral control. And our intention to perform a behavior is largely influenced by our attitude and perceived behavioral control followed to a lesser extent by subjective norm. It is important to note that the influence of each of the theory

Table 1 Theory of Planned Behavior Sample Items

For me to exercise regularly during the winter will be: [Attitude]	useless						useful
	1	2	3	4	5	6	7
I intend to exercise regularly during the winter. [Intention]	strongly disagree						strongly agree
	1	2	3	4	5	6	7
Most people who are important to me would like me to exercise regularly during the winter. [Subjective Norm]	strongly disagree						strongly agree
	1	2	3	4	5	6	7
How much control do you have over exercising during the winter? [Perceived Behavioral Control]	very little control						complete control
	1	2	3	4	5	6	7

Source: Heather Hausenblas.

of planned behavior construct can vary from population and context.

Elicitation Studies

A main strength of the theory of planned behavior is that an elicitation study forms the basis for developing questions to assess the theory's variables in a specific population. The elicitation study enables a practitioner to determine the specific beliefs for a specific population. This is very important because we know that beliefs vary by population and even by activity. For example, the main behavioral beliefs for breast cancers survivors are that *exercise gets my mind off cancer and treatment, makes me feel better, improves my well-being, and helps me maintain a normal lifestyle*. In comparison, the main behavioral beliefs for pregnant women are related to pregnancy-specific issues such as having a healthier pregnancy and easier labor and delivery.

Because beliefs vary by population, researchers and practitioners are encouraged to refer to research that has determined the physical activity beliefs of their specific intervention population, for example postpartum women, cancer survivors, or high school students. If physical activity beliefs for a practitioner's population of interest have not been determined, then it is recommended that an elicitation study be conducted to determine the pertinent beliefs concerning a behavior for that specific population. Table 2 provides an example of open-ended items used to assess the beliefs of pregnant women during their first trimester. Women during their first trimester would be asked to list three to five responses for each of the items. Then a content analysis (a simple frequency count) to determine which beliefs are most salient and finally a structured belief questionnaire would be developed from the content analysis (see Table 2).

Structured items that arise from the elicitation study should be specific to the target at which the behavior is directed, the action or specificity of the behavior under study, and the context and time in which the behavior is being performed. For example, when developing a walking intervention for older adults, the practitioner should ask a sample of older adults to list the main advantages of walking briskly three times a week for 30 minutes outside during the summer. This information will help researchers develop an intervention based on the salient behavioral beliefs of these older adults that is specific to the behavior. According to the theory of planned behavior, once beliefs are modified, intention will be altered and the desired behavior change will occur. The relative contribution of the theory of planned behavior constructs may fluctuate from context to context. Thus, before interventions using this framework are implemented, the predictive ability of these constructs with the specific population and specific context should first be tested.

Using Theory for Practice

The theory of planned behavior is useful in identifying psychosocial determinants of physical activity. Therefore, it has been useful for developing community, group, and individual exercise programs. For example, people intend to exercise when they hold a positive evaluation of exercise. Exercise programs that offer a positive experience would likely increase intention to exercise, which would likely positively influence exercise behavior. Positive behavioral beliefs and their evaluation may be enhanced if people are given experiences with enjoyable types of physical activities and then gradually encouraged to increase the intensity, duration, and frequency of those activities. Perceived behavioral control is

Table 2 Assessing the Theory of Planned Behavior Beliefs in Pregnant Women During Their First Trimester

Belief	Item
Behavioral beliefs	List the main advantages of exercising during your first trimester. List the main disadvantages of exercising during your first trimester.
Control beliefs	List the main factors that prevented you from exercising during your first trimester. List the main factors that helped you exercise during your first trimester.
Normative beliefs	List the individuals or groups who were/are most important to you when you thought/think about exercising during your first trimester.

Source: Heather Hausenblas.

an important factor in intention to be physically active. When individuals perceive physical activity as difficult to do, intention is low. Assisting people to overcome barriers, such as time involvement, other obligations, or feelings of inability, should enhance perceptions of control about carrying out exercise. We will now take a closer look at a physical activity intervention called Wheeling Walks that was guided by the theory of planned behavior.

Wheeling Walks was an 8-week mass media walking campaign developed by Bill Reger et al. (2001). The main goal of the intervention was to promote walking among sedentary adults ages 50 to 65 years old in the city of Wheeling, West Virginia. This "communication intervention used the theory of planned behavior . . . constructs to change behavior by promoting 30 min[utes] of daily walking through paid media, public relations, and public health activities." The impact of the campaign "was determined by pre- and postintervention telephone surveys with 719 adults in the intervention community and 753 adults in the comparison community and observations of walkers at 10 community sites" (p. 285).

To develop the messages to be used in the intervention, the theory of planned behavior was used to determine the salient beliefs that should be targeted during the walking intervention. They found that sedentary and irregularly active people shared similar attitude and subjective norm beliefs with regular exercisers but showed very strong differences on perceived control. Sedentary adults believed that they had less control over their time and scheduling of exercise compared to regular exercisers. Wheeling Walks promoted 30 minutes or more of moderate intensity walking on almost every day for better health and feeling more energetic. To address, the I-don't-have-time belief, which was identified as a primary barrier to physical activity, the ads suggested "start walking ten minutes a day at first, then twenty minutes . . ." and also compared the 30-minute time period to that of "one TV program." Each ad ended with the tagline, "Isn't it time you started walking" (p. 287).

Behavioral observation showed a 23% increase in the number of walkers in the intervention community versus no change in the comparison community. They also found that 32% of the baseline sedentary population in the intervention community reported meeting the physical activity guidelines by walking at least 30 minutes at least 5 times per week versus 18% in the comparison community. An important question is how long did the

increased walking last by the adults in Wheeling, and can these results be replicated in other communities. The answer to both questions is yes. A follow-up study revealed that the campaign succeeded in sustaining the increase in walking 12 months after the intervention. As well, the campaign has been successfully replicated in other communities including rural and larger communities.

Theory Limitations

Although the theory of planned behavior has been successful in explaining and predicting physical activity behavior, limitations of the theory exist. Some researchers reject the theory of planned behavior outright as an adequate explanation of human social behavior. Most critics, however, accept the theory's basic assumptions but either question its sufficiency or inquire into its limiting conditions. Some of the theory's potential limitations are briefly discussed below.

First, factors such as personality, affect or mood, demographic variables (e.g., age, gender, socioeconomic status), past exercise behavior, and habit are not directly taken into consideration within the theory of planned behavior. This is a limitation because researchers examining the determinants of physical activity have consistently found, for example, that the percentage of the population reporting no physical activity is higher among female than male populations, among older adults than younger adults, among extraverts than neurotics, and among the less affluent than the more affluent.

Second, there is ambiguity regarding how to define perceived behavioral control, and this creates measurement problems. Ajzen stated that the term *perceived behavioral control* may be misleading and to avoid misunderstanding, he suggested that perceived behavioral control should be read as "perceived control over performance of a behavior." Ajzen further noted that perceived behavioral control is composed of two components: self-efficacy (ease or difficulty of performing the behavior) and controllability (beliefs about the extent to which performing the behavior is up to the actor); and thus, measures of perceived behavioral control should contain items that assess both self-efficacy and controllability.

Third, the longer the time interval between intention and behavior, the less likely the behavior will occur. Research reveals that the predictive power of intention will vary inversely with the time between

the measurement of intention and performance of the behavior. The longer the time interval between intention and behavior, the more likely intention is to change with new available information. As time passes, an increasing number of intervening events can change people's behavioral, normative, or control beliefs and can modify attitudes, subjective norms, or perceptions of control, thus generating revised intentions. Changes of this kind will tend to reduce the predictive validity of intentions that were assessed before the changes took place. This new information would result in a diminished relationship between intention and behavior. Consistent with this argument, shorter intervals between assessment of intentions and behavior are associated with stronger correlations than are longer time intervals.

A fourth potential limitation pertains to the construct of subjective norm. Consistent throughout the physical activity literature, the theory of planned behavior variables of attitude and perceived behavioral control have been significant predictors of intention. Subjective norm, however, is generally a weaker predictor of intention. One reason for the inconsistent usefulness of subjective norm is that the role of significant others may not be important in encouraging participation of physically active individuals. Support for this view comes from the fact that subjective norm is a stronger predictor of intention for other health behaviors, such as contraceptive use, where the role of significant others is deemed to be more important for the decisions made, and thus cannot be ignored.

Conclusion

Changing people's behavior is very difficult to do, especially when dealing with a complex health behavior such as physical activity. To increase the success of predicting, understanding, explaining, and changing physical activity behavior, researchers and practitioners should use a theoretical framework such as the theory of planned behavior as a guide. Researchers have found support for the utility of attitudes, perceived behavioral control, and to a lesser extent, subjective norms in explaining people's intention to becoming physically active. Also, researchers have found a strong relationship between someone's intention to be active and one's engaging in the behavior. Of importance, the size of the intention-behavior effect shrinks significantly when past exercise behavior is controlled. Furthermore, a person's perception of the personal

control over engaging in physical activity can also directly predict behavior. In general, there is strong evidence that the theory of planned behavior can explain and predict people's physical activity intentions and behaviors. In summary, because of the success of the theory of planned behavior to explain and predict exercise behavior, intervention research reveals that it is also a useful framework to guide physical activity interventions.

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See also Control Theory; Habit; Health Action Process Approach; Social Cognitive Theory; Strength Model of Self Control

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THOUGHT STOPPING

Thought stopping has its origins in the late 1950s and is a class of cognitive techniques (involving mental or behavioral aspects) commonly employed by sport psychologists to eliminate athletes' recurring negative, self-defeating, or anxiety-related thoughts. Consequently, an underpinning foundation of thought-stopping techniques is the assumption that such thoughts are detrimental to task performance, to well-being, or to both. While there is an abundance of evidence supporting this perspective within the clinical psychology literature, there are some emerging and credible alternative viewpoints within the sport psychology literature. For example, although there has been support for the theorized negative associations between negative thoughts and performance as well as between negative thinking and confidence, there is evidence within the sports-oriented self-talk research that negative thinking might have a beneficial motivating effect. However, caution is required when interpreting the potential benefit of negative self-statements; the motivating effects of negative thinking may be realized only by certain athletes under certain circumstances, depending upon their view of the content of the negative thinking. For instance, immediately following a basic mistake within a practice session, a competent athlete might self-scoold, which is interpreted as being motivational.

Nevertheless, few practitioners would proactively encourage the use of negative thinking. On the contrary, negative thinking would be discouraged and some practitioners would advocate the use of traditional thought-stopping techniques to achieve this. Thought stopping represents the use of a mental or behavioral cue to prevent the occurrence of, or cease, recurring negative thoughts. Mental cues might involve the use of self-directed verbal cues ("Stop!") or the creation of mental images like a stop sign or a red traffic light immediately upon recognition of a negative thought. Alternatively, behavioral cues such as a slapping of one's thigh or pinching oneself can be utilized, with some sport psychologists reporting best results

when using mental and physical cues in combination. Anecdotal reports also suggest that thought stopping is more effective when the problematic symptom is largely cognitive in nature rather than accompanied or driven by unwanted behaviors, such as an extreme negative thought accompanied by disengagement from the task at hand.

It is believed that thought-stopping techniques are effective because the cue is distracting and can represent a punishment-oriented command; as such, frequently exhibited negative thoughts are consistently punished and reduced. An additional view is that thought-stopping cues are assertive responses that can be followed up with the use of additional mental techniques offering reassurance. In fact, some psychologists report greatest effectiveness of thought stopping when it is accompanied by the redirection of the performer's thoughts to emphasize positives within a seemingly poor situation or to refocus attention back on the task at hand.

Typically, thought stopping is employed to tackle a single persistent thought (e.g., "I'll never get this right"). Deliberately practicing thought stopping over a number of days has been suggested to increase its effectiveness. The process of introducing thought stopping can involve the athlete recalling a personal experience of a common situation where the habitual thought appears with the psychologist shouting "Stop!" upon the presence of this thought. When this brings about the desired effect of disrupting the targeted thought, the client shouts "Stop!" when experiencing the negative thought (instead of the practitioner). Following successful thought stopping because of the client's shouted "Stop!" the client practices reducing the cue from normal talking volume to whispering, with the goal that, eventually, the cue can be used covertly.

Despite reports of successful use in the sporting environment, there are various theoretical complications with the use of thought stopping. For example, the theory of ironic effects developed by Daniel Wegner argues that attempting to influence thoughts and mental control involves the balance between two opposing processes; the effortful intentional operating process and the unconscious monitoring process. Whereas the operating process tries to create the desired state of mind, the monitoring process continuously searches for inconsistency and failure of mental control. When situational conditions reduce available mental

capacity (as when we are under stress), the monitoring process overwhelms the intentional operating process and ironically produces the unwanted effect. These unwanted effects have been shown in word recall tasks, those involving movement errors, and, crucially, for thought suppression. Thus, the very act of an athlete trying to stop or not to think a certain thought may increase the likelihood of the thought occurring. Of further concern, this problem is likely to be exacerbated in athletes who have dispositional issues with anxiety. Mental techniques that involve rationalization of intrusive thoughts rather than suppression may be a realistic alternative.

As well as cognitive restructuring, there is a relatively newer and less well known form of thought stopping termed the *eye movement technique*, which is believed to interrupt negative thoughts by sequentially activating the two sides of the brain. This method requires the client to rapidly move the eyes back and forth between two reference points, for example, two corners of a room or window or hands placed on knees when seated, approximately 25 times. In doing so, it's theorized that clients stop focusing on the stressful event or negative thought. Repeated use of the technique may be needed if the unwanted thought is still experienced after the first application of rapid eye movements. However, the eye movement technique seems to be most effective when the recurring thought is only moderately stress inducing.

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See also Cognitive Restructuring; Positive Thinking; Self-Talk

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TIMING

Timing our actions is something we do thousands of times a day without thinking twice—controlling our eyes to read this text or reaching out to pick up a cup. To understand timing, however, it is first necessary to draw the distinction between the concept of time and timing. *Timing* is a human-made physical means of measuring the passage of time (e.g., seconds, minutes, hours). How we perceive time, *per se*, is very different from how time is measured by a watch or clock. Although the brain has no known dedicated time receptors, we are still very adept at organizing and guiding our actions in space and time (timing our actions). Indeed our survival in the dynamic world we all live in requires that the timing of our actions involve some understanding or perception of the time remaining before something happens. Prospective information provides information about the current future, that is where we are going and how we will get there based on our current course of action. For example, when one is driving a car, the changing patterns of light picked up by the driver's eyes will inform the driver about when to start braking so as to avoid colliding with the car in front. This is an example of how prospective information available in the world we live in can be used to allow us to act ahead of time. In other words, the perceptual information available to us is important in allowing us to successfully control the timing of our interactions with other moving objects or people who live in the same dynamic environment. Before discussing how to control the timing of our actions in sporting contexts, it is necessary to draw a distinction between different types of timing and illustrate how they relate to different types of temporal information.

Different Types of Motor Timing

Catching a ball, moving to a beat, and hitting a golf ball are all examples of actions that require precise timing. More specifically, the outcome of these actions will be determined by how well the unfolding movements are controlled with respect to either intrinsic or extrinsic temporal information specifying how the event is unfolding. For example, when catching a ball, continuous visual perceptual information specifying where the ball is going and when it is going to get there is extrinsically

available in the ball's unfolding trajectory. In this case, the temporal control of the movement is called *extrinsic*. That is to say the movement of the ball that is external to the person determines how and when one should move to get to the right place at the right time to catch it.

Alternatively, when clapping to a beat, the perceptual information available to guide the action is intermittent. The beat demarcates when the clap should take place but does not provide any continuous perceptual information specifying *when* the next beat will sound. As with all movements, the movement needs to start before the beat sounds (prospective control), so the hands come together as the beat sounds. The intermittent acoustic information available when the beat sounds is insufficient to time the movement of the hands. In this case, some other form of temporal information must be provided by the brain to allow one to get their limb to the right place at the right time to clap to the beat.

Finally, when golf putting or taking a penalty kick in rugby or soccer, neither the beginning nor the end of the movement is specified by any form of extrinsic temporal information. Instead, the timing of the putting movement or kicking action, which is crucial for determining the velocity of the club or foot at ball impact, must again be specified by some intrinsic neural mechanisms in the brain. In this case, the action is totally self-paced in that the player determines when to begin the action and how it should unfold. The spatial temporal dynamics of the club or leg movement largely determine the success of the action with the timing of all self-paced actions being provided intrinsically.

This distinction in the different types of action timing is critical when considering skill-based learning in sport (e.g., interceptive type actions vs. self-paced). In the next section, a theory is presented for how the temporal information (intrinsic or extrinsic) provides a solid basis to guide the timing of actions in sport.

Theories for Timing Movement

A plethora of research has studied timing mechanisms in the brain when tapping along to a beat. These studies have used the tapping paradigm, focusing on the size and variability of temporal errors made when synchronizing taps with beats and when continuing to tap when the beat stimulus is removed. Although timing theories have emerged, they tend to be paradigm driven

only explaining movement timing when tapping a finger not when dancing, catching, or striking a ball. For instance, when dancing, synchronization of body and limb movements is patterned (spatially) relative to the temporal structure imposed by the music. In other words, the timing of movements is neither temporal nor spatial but is *spatiotemporal*—how the limb position changes over time. This poses a genuine problem in the application of traditional timing models to the understanding of timing in the control or coordination of goal directed action.

Thus, to understand the concept of timing in the spatiotemporal context the notion of time needs to be reconceptualized. Instead of considering the brain representing time like a clock, it is more appropriate to think of time's passage, or a time interval, as being represented by something that changes over time. That is, time can be considered more of an abstraction that emerges from the way things change over time, which resonates with James J. Gibson's (1975) idea that "events are perceivable, but time is not." It is therefore more prudent to consider that the spatiotemporal control of actions or timing of actions must inevitably be related to the perception of events. The theory presented here, tau-coupling theory, follows from this and hypothesizes that the perception of temporal *information* (either intrinsic or extrinsic) *guides* the timing of our *movements*.

Temporal Information

To understand how temporal information can guide the timing of movement, it is first of all important to describe what is meant by information. In the physical world, the way in which a spatial gap between a ball and a target closes provides robust temporal information that can be used to control the timing of an action. David Lee suggested that the *time-to-closure* (tau) of a motion gap (in this case the distance between the ball and the target) at its current closure rate provides powerful information for timing movement. Not only is tau simple, general, robust information, it is also prospective, offering temporal information about the time remaining until a gap is closed. The appeal of a prospective informational variable in the context of timing is that it allows for the preparation and initiation of an action *ahead of time*, something that is critical in all sporting contexts.

Although tau as described above can explain action timing in the extrinsic temporal control

context like catching a ball, it does not explain how we can time our actions in cases where the temporal control is intrinsic, as in self-paced actions—putting a golf ball. Here no continuous perceptual temporal information is available to guide movements prospectively (e.g., kicking a stationary ball). This informational void has to be, therefore, filled by some internal dynamic temporal representation generated in the player's brain that helps provides a temporal framework to guide the timing of their movement. Previous research has proposed a mathematical model where this temporal information is specified in the brain in a way that is similar to the specification of temporal information in the physical world, namely a changing neural quantity (firing rate or neural power) over time.

To control the timing of our actions, the theory proposes a simple robust solution where the way the movement changes over time (the movement τ - τm) is linked to the perception of the dynamic temporal information (the informational τ - τ -guide [ti]) generated by an event in the environment (extrinsic) or the brain (intrinsic). This dynamic temporal information therefore *prescribes* the temporal course of the movement so that the information and movement are coupled in such a way that $\tau m = k\tau i$ (where k a constant represents the volitional control of movement). This theory explains parsimoniously how we control the timing of our movements through the principles of information movement guidance.

Improving Motor Timing

Learning a new self-paced skill, such as controlling the velocity of the club head when putting a golf ball different distances, is a difficult skill to master. The velocity of the club head can be controlled by varying the amplitude of the back or forward swing, the duration of the swing, or both of these parameters. In other words, the way the action unfolds over space and time to give a particular velocity can arise from a mixture of different space-time-action permutations. Being a self-paced action, as mentioned above, the temporal framework for this type of movement is intrinsic, generated within the brain, and is not directly accessible to the learner or the trainer. One exciting new way of looking at how the timing of these actions can be improved is by seeing how some form of extrinsic temporal information can be provided that specifies how the action should

unfold over time. Recent research has shown that by externalizing the temporal framework within which an action takes place, movement variability can be significantly reduced and the consistency of the timing of the action be improved. Coaches and players seeking to improve the timing of self-paced actions should therefore consider the use of extrinsic temporal information that could help the player prepare and execute the timing of the action in a more consistent manner. Furthermore, the role of preperformance routines, often adopted by elite players to help form the temporal envelope within which a self-paced action unfolds, should also be carefully considered so they include some form of extrinsic temporal information that could significantly improve the outcome of that action. That is, adopting the aforementioned information-movement coupling approach to consider how the brain controls the timing of an action (intrinsic vs. extrinsic information) could lead to new advances in coaching techniques and greatly advance research and practice in the area of motor timing. Finally, for interceptive actions, the information that players use is emerging as critical for determining performance outcomes.

Recent studies debunk previous research that suggests that responding as quickly as possible is of primary importance. Instead, this research has shown that tuning into the right source of external information (e.g., the player's centre of mass) that specifies an action relevant property (e.g., final running direction) and acting upon this information in the appropriate time frame determines expertise. It is acting in the right way at the right time that defines sports performance in handball, soccer, and rugby, not acting as quickly as possible. More research is now required to understand how coaches can educate players' attention to enable them to tune into the right perceptual information that can be used to guide the timing of their actions.

Cathy Craig

See also Decision Making; Ecological Theory; Expertise; Motor Learning; Movement; Perception; Response; Skill Acquisition

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TRAINING, PROFESSIONAL

Individuals preparing for a career in sport and exercise psychology (SEP) have traditionally followed one of two educational paths, training in kinesiology or training in psychology. In keeping with most international credentialing guidelines for certification or licensure of SEP professionals, both educational paths typically feature substantial coursework in the other discipline, reflecting the interdisciplinary nature of the field. Retraining is possible for established professionals in either discipline who wish to specialize in SEP. Continuing education is available to help SEP professionals stay abreast of the latest developments in the field. Taking into account the roles that they wish to fulfill in the field of SEP, prospective professionals can choose the training options that best match their career goals.

Professional Practice in SEP

SEP professionals have been identified as performing research, educational, and clinical or counseling functions. These three functions encompass a wide range of professional activities. The research function is generally performed within colleges, universities, and other academic institutions by doctoral-level professionals who examine such topics as psychological factors affecting sport performance, motivation to engage in sport and exercise activities, coping with stress, group and team dynamics for athletes and exercisers, psychological skills training for sport and exercise, and psychological effects of sport and exercise. In addition to

conducting research, professionals occupying the research role are likely to teach undergraduate and graduate courses.

The educational function involves using educational approaches to enhance the experience, performance, and outcomes of sport and exercise involvement. In this, the most versatile of the three primary functions, professionals with master's and doctoral degrees apply psychological principles and methods to address performance-related issues in a variety of settings, such as academic institutions, military environments, youth sport academies, health maintenance organizations, and private consulting firms.

The clinical function involves providing services to sport populations and exercise participants for subclinical or clinical issues. Professionals who perform the clinical function generally have a graduate degree (a doctorate in some countries but a master's degree in many other countries) in clinical or counseling psychology, licensure or registration (depending on the country) appropriate for the provision of clinical services, and specialized training in SEP. Among the sorts of issues addressed through the clinical function are mental disorders (such as anorexia nervosa and dependence on performance-enhancing substances, which can be directly intertwined with physical activity), and subclinical syndromes (such as adjustment reactions to injury and other life stressors) that can have a profound impact on sport and exercise involvement. Professional activities within the clinical function may include prevention, diagnosis, and treatment of mental disorders in sport populations as well as enhancement activities, such as career development or life skills programming, more traditionally associated with counseling psychology. Some professionals use exercise as a therapeutic technique with their clients.

It is important to note that the three primary professional functions in SEP are neither mutually exclusive nor exhaustive. Professionals may perform multiple functions across clients or with a single client, as in the case of a professional who works with a client on a clinical issue, uses an educational approach for performance enhancement, and documents the work in a case study. Also, there are professional activities in SEP that do not fall neatly within the three primary functions, for example, talking with the media, but are nevertheless within the realm of professional practice in SEP.

Paths to Professional Practice in SEP

Given the diverse array of activities that constitute professional practice in SEP, it is not surprising that there are multiple paths that people aspiring to become SEP professionals can take. Beginning with undergraduate education and proceeding through graduate education, possible retraining, and continuing education, SEP professionals-in-training face numerous choices and options on the way to launching their careers in the field.

Undergraduate Training

Opportunities for specialized training in SEP at the undergraduate level vary considerably around the world. At some Asian universities, students can enroll in undergraduate degree programs in sport psychology. At some universities in Europe and Australia, students can pursue advanced study in SEP as part of their undergraduate honors degree. In much of the world (including North America), however, most colleges and universities do not offer an undergraduate major in SEP, and the bachelor's degree is not generally considered a terminal degree in SEP. Consequently, in regions where there is no specialized undergraduate training in SEP, undergraduate education can be thought of as a means of preparing prospective SEP professionals for graduate school. The undergraduate paths trod most commonly on the way to graduate school in SEP are majors in kinesiology (also known as sport science, physical education, human movement studies, human kinetics, health and human performance, and exercise science) and psychology. Although it is most common for students to select one major or the other, some students double major in kinesiology and psychology or major in one discipline and minor in the other.

The content of undergraduate kinesiology curricula varies widely, with some universities adopting a liberal arts approach in which students are exposed to a broad base of courses across the discipline and other universities maintaining a professional career-based approach in which students receive training in a subdiscipline of kinesiology (e.g., athletic training, cardiac rehabilitation, strength and conditioning) that prepares them for a specific career. In either type of program, students who pursue an undergraduate degree in kinesiology can expect to take courses in general subjects such as biology, chemistry, physics, and psychology, as well as discipline-specific subjects

such as biomechanics, exercise physiology, motor learning, SEP, and sport sociology. Such coursework not only helps prepare students for graduate study in SEP but also can be counted toward requirements for certification as a SEP consultant in some countries.

Students enrolled in undergraduate psychology programs generally sample broadly from major subareas in the discipline. Common course offerings include surveys of introductory psychology, social psychology, experimental psychology, abnormal psychology, physiological psychology, sensation and perception, personality psychology, developmental psychology, applied psychology, health psychology, history of psychology, research methods, and statistics. Aside from the occasional specialty course, SEP does not typically figure prominently in undergraduate psychology education. Students wishing to augment their training in the fundamentals of psychology with coursework in SEP are sometimes able to do so by taking classes offered through departments of kinesiology.

Graduate Training

Because a graduate degree is the terminal degree for SEP in most of the world, it is important for students to align their choice of degree program with the activities they wish to perform as professionals (and the degree requirements corresponding to the certification or licensure criteria for engaging in those activities). Students seeking to be involved primarily in SEP research activity are typically best served by earning a doctorate in kinesiology, as SEP research and coursework do not regularly receive strong support in academic departments of psychology. Students interested in activities associated with the educational function in SEP can benefit from graduate training in either kinesiology or psychology, as long as the graduate program in which they enroll is receptive to the students' applied interests and affords the students opportunities to partake in relevant coursework and supervised practical experiences. Students wishing to perform the clinical function in SEP, almost without exception, choose to pursue graduate study (either a master's degree or a doctoral degree, depending on local professional and legal standards) in clinical or counseling psychology and augment their training with SEP coursework and supervised experience. Although it is not mandatory for students to have completed an undergraduate degree in the same

field as their intended graduate degree, they may be more competitive in the application process if they have done so and may be required to complete undergraduate coursework in the field of their graduate degree if they are admitted into a graduate program in a field not corresponding to their undergraduate degree.

Graduate study in kinesiology with a concentration in SEP is similar to undergraduate study in kinesiology except that the specialization in SEP and the emphasis on research are likely to be more pronounced. Degree requirements typically include coursework in general research methods and statistics; discipline-specific, such as exercise physiology, motor control, motor development, motor control, sport history, and sport sociology; and concentration-specific, such as applied sport psychology, exercise psychology, and sport psychophysiology courses. Supervised and independent research on specific topics in SEP is also a central feature of most SEP master's and doctoral programs in kinesiology. Graduate SEP programs in kinesiology departments vary widely in terms of their acceptance of and emphasis on applied SEP, so prospective students aspiring to do applied work are advised to investigate this aspect prior to enrolling in a given program.

Similar to graduate study in kinesiology, graduate study in psychology is more specialized than its undergraduate counterpart. Among students in graduate psychology departments interested in SEP, most enroll in clinical or counseling psychology programs. Some students with research interests in SEP pursue degrees in developmental, experimental, industrial or organizational, or social psychology, but these students do not receive training to perform the clinical function in SEP. Graduate psychology degree requirements may include courses on biological bases of behavior, cognitive and affective aspects of behavior, social aspects of behavior, history and systems of psychology, psychological measurement and assessment, research methods, techniques of data analysis, and concentration-specific topics. For students in clinical and counseling psychology programs, training and supervised practica in counseling and psychotherapy are prominent components of graduate study. Supervised and independent research also tend to be salient aspects of graduate programs in psychology. Among doctoral programs, the extent to which research is the emphasized aspect of the program is influenced to a large degree by whether

the program awards a doctor of philosophy (PhD, research-oriented) or doctor of psychology (PsyD, practice-oriented) to students who complete degree requirements.

Retraining

Sometimes, people who have completed a terminal degree in kinesiology, psychology, or a related field and worked in their profession for a period of time develop an interest in SEP and wish to shift their professional focus to become more involved in SEP. For these individuals, retraining may be required to enable them to achieve competence in the professional practice of SEP and satisfy credentialing requirements in SEP. Retraining typically involves some combination of independent study, coursework, and supervised practice. In cases where there is a large gulf between the credentials of professionals and the requirements of the certifying or licensing body in SEP, professionals may need to undertake another degree to acquire the requisite knowledge, skills, and academic background.

Continuing Education

Professional training in SEP does not end with the receipt of a degree or completion of an internship. The knowledge base is continually expanding with advancements in the science and practice of SEP. Credentialing bodies in SEP recognize the need for professionals to stay abreast of the latest developments in the field and build continuing education into the process of certification and licensure. It is common for professional organizations to offer continuing education programs, sometimes in workshop format in conjunction with annual meetings and sometimes in the form of home study using text and video materials. Regular enrollment in continuing education offerings provides professionals with the opportunity to keep current and hone their skills while maintaining their career development trajectory.

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See also Career Paths; History of Exercise Psychology; History of Sport Psychology; Supervision

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TRAIT PERSPECTIVES

See Leadership in Sport: Trait Perspectives; Personality and Psychological Characteristics of Athletes; Personality Traits and Exercise

TRANSCRANIAL MAGNETIC SIMULATION (TMS)

Transcranial magnetic stimulation (TMS) is a non-invasive method of activating regions of the brain by small electric currents generated by a magnetic pulse. Over the last 25 years, TMS has become an integral technique for studying brain function in humans in both healthy and diseased states and is a valid and reliable technique for investigating brain mechanisms of popular interventions in sport and exercise psychology, such as action observation and imagery.

Principles of TMS

Based on the principle of electromagnetic induction, a brief, high-intensity current is passed through a coil composed of copper windings encased in plastic, creating a transient magnetic field. The magnetic field has the capacity to induce a small electric current in nearby excitable tissue, including brain cells, peripheral nerves, and muscle. When placed on the surface of the scalp,

the induced current can cause neurons in the outer part of the cortex to fire. Because magnetic fields pass through the body without activating sensory afferents like cutaneous and pain receptors, TMS has been referred to as *ouchless* stimulation.

Types of TMS

Three distinct types of TMS are routinely used. The first, *single pulse TMS*, when applied to the primary motor cortex, produces muscle activity known as a *motor evoked potential* (MEP), which can be recorded in the electromyogram. Single pulse TMS is routinely used in research to monitor changes in motor pathways, for example, in response to exercise or skill acquisition. Clinically, single pulse TMS is used to assess the integrity of corticospinal pathways for diagnostic purposes in conditions such as stroke, spinal cord injury, multiple sclerosis, and motor neuron disease.

In the second type, *paired pulse TMS*, two stimulation pulses are delivered either through the same stimulating coil or through two separate coils. Delivering two stimuli in a conditioning-test format through the same coil allows local inhibitory or excitatory circuits to be assessed, depending on the interstimulus interval. For short intervals (2–4 milliseconds) a subthreshold conditioning stimulus followed by a suprathreshold test stimulus is thought to activate short-interval intracortical inhibition (SICI) networks within the local circuits of the motor cortex. In contrast, longer intervals (8–20 milliseconds) are thought to activate local excitatory circuits and can be used to assess intracortical facilitation (ICF).

Paired pulses delivered by two separate coils can be used to assess the functional connections between regions of the cortex. For example, a conditioning pulse delivered to the dorsal premotor area has an effect on a test pulse delivered to the primary motor cortex 8 milliseconds later. In this manner, the functional connections of various motor association areas can be examined and their role in important processes such as skill acquisition can be determined.

The third type of TMS involves *repetitive stimulation*, known as *rTMS*. The effects of rTMS outlast the period of stimulation by several minutes and, depending on the specific stimulation profile used, can induce long lasting potentiation or depression of cortical circuits. As a result, rTMS

can be used to study such higher brain functions as language, memory, and attention. Furthermore, rTMS can result in behavioral changes and may be effective for the treatment of a number of neurological and psychiatric disorders. For example, in randomized, controlled placebo trials, rTMS treatment of depression has been shown to have a comparable effect to pharmacological treatments.

The rTMS is also being examined for use in relation to a number of movement-related pathologies, including rehabilitation for aphasia (the loss of the ability to understand or produce speech) and motor function following stroke, improvement of motor function in Parkinson's disease, and alleviating the effects of dystonia (abnormal muscle tone). Other clinical applications of rTMS include the management of chronic pain, migraines, bipolar disorder, and tinnitus.

TMS in Sport and Exercise Psychology

In sport psychology-related work, TMS has been used to examine the corticospinal changes that occur in response to action observation and motor imagery, two fundamental techniques used in skill acquisition and sport psychology practice. Observation of an action performed by self or others, in the absence of any recordable overt movement or muscle activity, modulates the excitability of the corticospinal pathway in humans, resulting in increased amplitude of MEPs specific to the muscles involved in the observed action. Similarly, motor imagery has been shown to result in increased MEPs in response to TMS. Interestingly, the increases in excitability of the corticospinal pathway are greatest when the *feel* of the movement is imagined (kinesthetic imagery) compared with when the *look* of the movement is imagined (visual imagery). These findings confirm that both action observation and motor imagery result in functional changes in the central nervous system that may explain their effectiveness. It is also likely that TMS can be used to examine the quality of imagery since the degree of corticospinal modulation is correlated to the vividness of imagery as assessed by self-report questionnaires.

Safety Considerations

Although considered safe, TMS is not without risk. Of primary concern is the risk, although small, of inducing seizures. Of the small number of TMS-induced seizures reported in the literature,

in many incidences there were predisposing factors, including brain lesions and familial history of epilepsy. Other associated risks include discomfort as a result of stimulation of the nerves and muscles of the scalp resulting in localized muscle contraction, headaches, and some mild effects on hearing (as there is a loud click when the coil discharges). TMS should never be applied in conjunction with metal electrodes or in persons with metal implants near the site of stimulation as the magnetic discharge can cause heating in the metal. In 2009, the Safety of TMS Consensus Group issued comprehensive guidelines for the application of TMS in research and clinical applications and considers the overall risk of TMS to be low.

Stephan Riek

See also Brain; Electromyography (EMG); Imagery; Learning; Mental Rehearsal; Motor Commands; Motor Control; Motor Learning; Neurological Disorders; Neuroscience, Exercise, and Cognitive Function

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TRANSFER

In sport, *transfer of learning* is generally defined as the influence of previous experience of performing a skill on the learning of a new skill or on performance of the same skill in a new context. This influence may be positive or negative. Positive transfer occurs when previous experience of performing a skill is beneficial for learning a new skill or for performance of the same skill in a different context. For example, a hockey player might reasonably expect that previous experience of striking a ball with a hockey stick will transfer positively

to learning a golf swing, and most golfers hope that practicing their swing in the car park before a round will transfer positively to the golf course. Negative transfer occurs when previous experience of a skill inhibits learning a new skill or disrupts performance of the same skill in a different context. For example, groundstrokes in tennis, which are best performed with a relatively firm wrist, usually transfer negatively to squash shots, which are best performed with a flexible wrist, or vice versa. In this entry, testing and assessing transfer of learning in sport, theories of transfer, and the significance of transfer in sport are discussed.

Testing for Transfer

Researchers commonly use a simple experimental design to test for the influence of previous experience of a skill on learning a new skill (*intertask transfer*) or on performing the same skill in a new context (*intratask transfer*). One group of participants (the experimental group) practices the first skill and a second group (the control group) does not practice. Both groups are then tested on either a second skill (sometimes called the criterion skill) or in a different context. Any pre- or postdifferences between the groups on the second test may be attributed to positive or negative *transfer*.

Assessing Transfer

There are many ways to quantify the *amount* of transfer from one skill to another skill or from one context to another context. Although no method is infallible, the most common method used by researchers is to compute the percentage of transfer, which is the proportional difference in performance scores of the experimental and control groups on the second skill or context:

$$\text{Percentage of transfer} = \frac{\text{Experimental group} - \text{Control group}}{\text{Experimental group} + \text{Control group}} \times 100$$

To illustrate, a researcher might conduct a transfer experiment to examine whether learning darts for 1 week enhances or inhibits archery performance. The researcher can use a transfer test to compare the archery performance of an experimental group that practiced only their darts during the week immediately before or a control group that was prevented from practicing. Performance during the first ten trials of the archery task might reveal that the experimental group strikes the

bull's-eye of the target on an average of four occasions, whereas the control group strikes the bull's-eye on an average of one occasion. Applying the above formula,

$$\frac{4-1}{4+1} \times 100 = \frac{3}{5} \times 100 = 60\%$$

the experimental group shows 60% transfer, suggesting that they perform 60% better at archery compared with the control group. The amount of transfer is usually estimated on the first few trials of performance because performance by the control group on later trials may be confounded by the practice that group is receiving. If the measure of performance indicates that lower values represent better performance (e.g., error, speed), then the formula should be modified by reversing the numerator.

Theories of Transfer

Research on transfer of learning began more than a century ago, when studies were conducted to examine the *theory of formal discipline*, which proposed that studying formal disciplines, such as Latin, logic, or mathematics, caused positive transfer to *thinking* in general by training the mind's faculties. William James, however, tested the principle on himself by memorizing a new poem each day. His ability to learn new poems was not improved by this self-imposed case study. Later, Edward Thorndike, his student, as well as Robert Woodworth, further refuted the theory of formal discipline, arguing that good thinkers are more likely to study such disciplines so when they are successful at other disciplines, it only appears that they have benefited from their earlier studies of formal disciplines. Nevertheless, some evidence continues to support the concept of formal disciplines transfer. For example, it has been shown that learning probability statistics improves general reasoning skills.

Identical Elements Theory

Thorndike later formulated the *theory of identical elements*, in which he argued that earlier learning is only advantageous if the second task to be learned has elements that are identical to those of the first task. The amount of transfer thus depends on the similarity between the elements of two skills or of two performance contexts. The problem

with this theory was that it did not specify what the elements were or how their similarities were to be assessed. Two possible ways to specify an element in the context of motor learning are that (1) it is any observable movement component of a skill, such as the swing of the leg in a kick; or (2) it is any task-specific coordination dynamic, such as the synergy of the elbow, wrist, and fingers when throwing a Frisbee. Predictions on the basis of common elements between two skills suggest that the amount of transfer between a tennis serve and a golf putt is likely to be less than between a tennis serve and a volleyball serve because a tennis serve has more elements in common with a volleyball serve than a golf putt. Predictions on the basis of common elements between two contexts or domains suggest that learning to anticipate the direction of a tennis serve by reacting in real life will yield more positive transfer to game play than learning to anticipate the direction of a tennis serve by reacting in a video game.

The Theory of Generalization

Subsequent efforts to understand transfer included cognitive factors. For example, Charles Judd argued that psychological factors, such as motivation, combine with learning of general principles to give rise to transfer. Judd's *theory of generalization* suggested that transfer occurs when general principles learned in one situation are applied in a new situation. Children aware of the principles of light refraction, for instance, throw darts more accurately at a target submerged under the water than children unaware of the principles.

Transfer-Appropriate Processing Theory

Transfer-appropriate processing theory suggests that learning is best if practice involves processing activities that are similar to those activities which occur during transfer. To structure an optimal learning environment or encourage positive transfer of learning, a practitioner therefore must match the processing activities in which the learner engages during practice to the cognitive processing activities that are associated with the criterion skill or when performing in a different context. For example, a training protocol for kicking penalties in soccer, which incorporates the processing associated with kicking under psychological pressure, would be expected to produce more positive transfer to a penalty kick in the World Cup final

than a training protocol that does not incorporate such processing. Transfer-appropriate processing theory also offers an explanation of the phenomenon known as *bilateral transfer*; the processing activities involved in performing a skill with one limb are likely to mimic the processing activities involved in performing the same skill with another limb, thus facilitating transfer between the two. This explanation is supported by recent work in which functional magnetic resonance imaging (fMRI) showed that brain activation at the supplementary motor area (SMA) is positively correlated with the amount of transfer between learning a sequential finger tapping skill with the right hand and performing the same skill with the left hand. Such transfer disappeared when SMA activation was inhibited by the use of transcranial magnetic stimulation (TMS).

Despite such findings, neither transfer-appropriate processing theory nor any other theory can fully explain the transfer effect. Much remains unclear about transfer of learning occurrence.

The Importance of Transfer

From both theoretical and practical points of view, however, the principle of transfer is important for understanding the effects of practice on performance. This is perhaps best exemplified by recent advances in our understanding of how expertise develops in sport. Experts often display developmental histories that are characterized by participating in many other sports. Transfer of learning is an obvious explanation for such a phenomenon. Experts, for example, are better than non-experts at recalling patterns of play from not only their own sport but also sports in which they are not expert, suggesting that learning to recognize patterns of play in one sport transfers to another to some extent.

With respect to motor tasks, the consensus probably is that, although transfer is generally positive when it comes to motor tasks, the effects are trivial unless the tasks are almost duplicates. Nevertheless, the principle of transfer implies that the practice of one skill will affect the learning of subsequent skills, which suggests that sequencing skills to be learned in a logical progression is important when designing training protocols or programs. For example, when physical educators, coaches, or therapists teach motor skills in school, at a club, or during rehabilitation, they often

follow the simple-to-complex rule so that learners progress from fundamental skills through to complex skills that require mastery of the basic skills. Additionally, the principle of transfer informs diversity of instructional and training methods. Modern technology is advancing so quickly, for instance, that surgeons are able to train on simulators that mimic more and more of the underlying similarities of real surgery on actual patients. Thus, the costs and the risks associated with training can be largely reduced. Indeed, virtual reality training, which allows practice of a skill in three-dimensional, computer-generated environments, is becoming more accessible and thus more popular, as a training procedure that relies on the principle of transfer. Finally, the principle of transfer provides the best assessment of the effectiveness of a practice routine or instruction method. For sports coaches, the transfer test is whichever game or competition that the team or the athlete was preparing for. For piano teachers, the transfer test may be next week's recital at school. For physical therapists, the transfer test may take place the moment the patient leaves the clinic or perhaps later at home. Performance in the transfer test, whatever the test may be, is the true test of practice.

Rich Masters and Frank Zhu

See also Expertise; Learning; Motor Learning; Movement; Practice; Retention; Skill Acquisition

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TRANSFORMATIONAL AND TRANSACTIONAL LEADERSHIP

See Leadership in Sport: Transactional and Transformational

TRANSFORMATIONAL PARENTING

Just as businesses and organizations require effective leadership behaviors in order to grow and develop successfully, it has been suggested that the same is required within families in order to foster a family climate that promotes healthy child development. Although a range of predictors are associated with child and adolescent health, arguably the family (and especially parents) remains one of the most influential determinants of child and adolescent health behaviors. Although many theories of parenting are utilized in sport and exercise psychology, a contemporary framework of *leadership* has recently been extended to the parenting domain in order to examine the influence of parenting behaviors in relation to physical activity participation (and other important sport and health-related cognitions and behaviors) in youth.

Grounded in transformational and transactional leadership theory, which was initially conceptualized by Bernard Bass, *transformational parenting* comprises four behavioral dimensions; idealized influence, inspirational motivation, intellectual stimulation, and individualized consideration. When parents engage in idealized influence, they behave as role models to their children, lead through the demonstration of personally held values and beliefs, and are consistent and dependable. Inspirational motivation involves parents energizing and inspiring their children to go beyond minimally accepted standards and displaying optimism with regard to what their child can achieve. Intellectual stimulation includes behaviors that get their children to think for themselves, encourages them to contribute to decision making and behaviors that are respectful of their children's ideas and opinions. And finally, when parents display individualized consideration they show a genuine interest in their children's life, display a sense of care and concern, and are sensitive to their children's unique abilities and needs.

The behaviors that underpin transformational leadership in many ways parallel effective parenting strategies. For example, both leaders (within organizations) and parents (within families) aim to encourage those they lead to become increasingly autonomous in their actions and seek to inspire and motivate them to achieve established goals or objectives. Furthermore, successful leadership and parenting both involve maximizing the *quality* of their relationship with others. It is the focus on higher order intrinsic needs of followers and the long-term vision for success (rather than a focus on short-term goals) that differentiates transformational leadership theory from other leadership theories and makes this framework especially pertinent to parenting.

Given the positive associations that have been demonstrated between transformational leadership and important follower outcomes in diverse settings (self-efficacy, intrinsic motivation, leader satisfaction, positive affect), and the significance of parenting in promoting adolescent physical activity and health, researchers are now applying transformational leadership to the parenting domain with a focus on understanding the effects of specific parenting behaviors on child and adolescent physical activity and health-related outcomes.

Although empirical research integrating principles from transformational leadership theory and parenting is at a relatively early stage, a growing number of studies have extended this framework to parenting. In the context of sport, perceptions of transformational parenting behaviors are associated with adolescents' own use of transformational leadership behaviors displayed within their peer interactions in sport settings. Given the positive associations between transformational leadership and important group-level outcomes, such as group cohesion and improved conflict management, fostering transformational leadership in youth sport has important implications for subsequent participation and performance, and it is plausible to suggest that this can be enhanced through parents' own use of transformational behaviors in the home. Furthermore, research suggests that displays of transformational parenting may translate into less aggressive behaviors in youth sport. These outcomes arise as transformational parenting involves behaving in ways that fosters identification with the leader and desire to emulate their (prosocial) behaviors. Furthermore,

transformational parenting involves setting high expectations for (appropriate) behavior and getting children and adolescents to think about alternative ways to overcome problems and frustrations, thus making it less likely that they will choose to enact aggressive behaviors.

Outside of the sport arena and from an exercise and health psychology perspective, research has demonstrated positive associations between parents' use of transformational leadership behaviors (transformational parenting) and adolescents' self-efficacy beliefs to engage in regular physical activity and to sustain a healthy diet, as well as indices of adolescent psychological well-being. Furthermore, adolescents' perceptions of transformational parenting are also associated with their actual physical activity and healthy eating *behaviors*. Interestingly, these studies suggested that *both* mothers' and fathers' use of transformational parenting was associated with these improved health-related outcomes among adolescents. Transformational parenting is postulated to bring about improvements in health-related cognitions by communicating high expectations with regard what adolescents can achieve, thus motivating them to choose to enact voluntary health behaviors such as increasing their physical activity. Furthermore, transformational parenting involves empowering rather than controlling behaviors and also creates conditions where adolescents are encouraged to think for themselves, thus bringing about improved self-regulation to engage in such behaviors.

In light of the predictive effects of transformational parenting behaviors in relation to adaptive adolescent exercise and other health-related outcomes, it seems pertinent in future research to examine the potential of transformational parenting interventions. Given the well documented success of training and developing transformational leadership in other diverse settings through relatively short-term intervention, this framework holds much promise for fostering adolescent health and well-being via improvements in these family leadership processes.

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See also Intrinsic/Extrinsic Motivation, Hierarchical Model of; Leadership in Sport: Transactional and Transformational; Parenting; Self-Efficacy; Psychological Well-Being

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TRANSITION

The term *transition* has been employed in various academic fields to explain a process of changes in a certain phenomenon, such as economic, political, biological, and psychological changes. The concept of *transition* in psychology is closely related to individuals' period of change and life-span development based on one's aging (e.g., retirement), development (e.g., entering to higher education), and significant life events (e.g., getting married, giving birth). Therefore, individuals experience a series of transitions throughout their lives, and research on transitions has been discussed in various terms, such as crisis, adaptation, and stress management. Within the general psychology literature, the term *transition* seems to explain individuals' process of change because it covers broader events, not just crisis but also less observable life changes, and explains individuals' experiences as processes, not just a consequence of the change but also a process of adjustment.

According to Nancy Schlossberg, transitions occur as a result of event (anticipated and predictable event like high school graduation, job entry) or non-event (unanticipated and nonscheduled

event like job loss, divorce), which accompany changes in one's world, including oneself and one's behavior and relationships. The transitions do not necessarily occur in a given sequential order, and individuals' responses to their transitions can vary in terms of adapting to and dealing with their changes. In addition, these changes can be the source of individuals' distress and traumatic experiences or unique opportunity for personal growth and development. Thus, the outcomes of individuals' transitions can be both positive and negative, and their cognitive appraisal is closely related to outcomes of the transition. For example, sport career termination can be considered as a life crisis and cause feelings of loss and isolation for one athlete but not for another; the same event can become an opportunity to challenge new areas or developing new interests and social networks.

As early research on career transitions in sport focused on examining athletes' retirement process, *transition* has been introduced in sport psychology area to explain and understand athletes' sport career termination process. Although a need to approach athletes' sport career termination as a transition had been suggested in 1980, the term *transition* has not been widely used in the sport psychology area until early 1990s, because until then athletes' retirement tended to be considered as either a social death or a singular event. However, based on the early findings, which indicated changes in athletes' identity, lifestyle, and social networks, athletes' sport career termination has been considered as a social rebirth or a process, and researchers have focused more on athletes' overall transition throughout their sport careers via developmental perspectives. Therefore, in order to understand athletes' career transition as a process, researchers have employed transition models in the sport psychology area.

Model of Human Adaptation to Transition

The model of human adaptation to transition is one of the models most employed to investigate the athlete career transition process. The model explains all kinds of transitions that human beings experience over the course of their lives. The model consists of three major factors that influence transition outcomes, including *individuals' perceptions of the particular transition*, *characteristics of pre-transition and posttransition environments*, and *characteristics of the individual*.

Perceptions of the Transition

Perceptions of the particular transition indicate individuals' perceptions of the characteristics of transition. Transitions can happen via *internal* (self-chosen) or *external* (being cut) *forces*, *on time* (predictable, planned) or *off time* (unexpected, abrupt), and *gradual* or *sudden*. The changes can also be *permanent* or *temporary*, and can be either *positive* or *negative*. All aforementioned characteristics of transition can be predictors of individuals' degree of stress and the level of difficulty in adjustment process. For example, individuals who confront gradual, self-chosen, and planned transitions, generally experience less difficulties and smooth adjustment to their posttransition lives than people who experience sudden, forced, and unexpected transitions.

Characteristics of Pretransition and Posttransition Environments

Characteristics of pretransition and posttransition environments contain all kinds of external support that individuals can receive during the transition. These include *interpersonal support systems*, such as intimate relationships, family unit, and network of friends; *institutional supports* like occupational organizations, religious institutions, and community support groups; and *support from their physical setting*, for instance, climate and weather, urban or rural location, and neighborhood. Individuals' degree of support from close others, institutes, and physical setting is closely related to the quality of individuals' adaptation to transition. For example, when individuals confront difficult life situations, such as loss of family members or divorce, close others' and institutional support play important roles in coping with critical life events because those external support help individuals reduce their stress levels and enhance their degree of confidence.

Characteristics of the Individual

The characteristics of the individual reflect individual differences, including demographics and developmental differences, such as one's sex, age, previous experiences, and psychosocial competence. The relationship between sex and the quality of transition is complex, but perhaps individuals' sex-role identification might be related to one's quality of adaptation to their transitions. For example, since women and men generally have

different social roles and attitudes, and experience different kinds of transitions through their lives, they may perceive or react differently while facing similar transitions. In the transition literature, *life stage* seems to be preferred over the term *age* because psychological (e.g., capacity to respond to societal pressure), social (e.g., social roles), and functional (e.g., ability to perform or function in society) ages tend to be more meaningful to explain the adaptation process than chronological age. Individuals' previous experiences influence their posttransition adaptation. Specifically, having experienced transitions in the past and successfully coped with the associated changes, it becomes easier to adapt to a subsequent transition. The reverse is also true. Finally, psychosocial competence, including one's abilities in self-evaluation, optimism, and overall coping skills, has been considered as one of the influential determinants for ones' success or failure in adaptations to transitions. If individuals can maintain their self-esteem through positive self-evaluation, can focus on hope and goals via optimistic views, and can employ active coping strategies, they tend to experience healthier transitions.

Athletes' Career Transitions

The model of human adaptation to transition has been supported by research findings on career transitions in sport and seems to help understand athletes' transition as a process. However, limitations in using the general psychological models for accounting for athletes' career transition are evident because of absence of specific models that relate to sport career transitions. Hence, since the middle of the 1990s, new models, which aimed at explaining athletes' sport career termination, were introduced. These models, mainly those developed by Dana Sinclair and Terry Orlick, and by Sandy Gordon, centered on overall athletic careers through focusing on life span developmental perspectives.

In the modified transition model for athletes, transition characteristics include voluntariness of retirement, retirement trigger, timing of retirement, durations of retirement, role changes involved, and degree of stress. For examples, if athletes terminate their sport careers because of injury or deselection from the team, the transition can be unanticipated and sudden for them; therefore, they may experience more difficulties during the transition process than athletes who retire by free choice.

Environmental characteristics contain *available social support inside and outside sport*, *degree of preparedness*, and *options*. Similar to findings from the general psychology, athletes' social support networks had positive relationship with the quality of career transition. In addition, when athletes felt more preparedness both psychologically and vocationally during the transition process, and had options in post-sport careers or other interests outside sport, then they tended to have less transition difficulties than those who had less support from others, a low degree of preparedness, and less options.

Individual characteristics indicate *demographics*, *degree of athletic identity*, and *degree of life skills development*. Demographic factors have been examined in the study area, but findings have not provided a clear association between demographics and the quality of career transition, apart from competitive level and marital, financial, and educational status. Athletes who competed at the Olympic or international level seemed to experience fewer transition difficulties than did those who competed at a lower level (national), and *marital status* tended to relate to a degree of social support (married athletes had stronger support systems). As *financial difficulties* were one of the concerns for retiring athletes, *financial status* was positively related to the quality of career transition, and *educational status* was related to the athletes' personal development and vocational options.

Athletic identity is one of the most examined factors in athletes' career termination study area. A strong athletic identity can be a source for transition difficulties and identity crisis during the adjustment process. Athletes' identity development is also influenced by their athletic experiences. Many elite athletes tend to spend a large amount of time for training, competing, and considering their sport performance excellence, and do not have other interests or roles outside their sport while actively competing. Therefore, they often fail to develop multiple and balanced identities across inside and outside their sports but have strong athletic identity.

Similarly, athletes' *degree of life skills development* is influenced by their athletic careers and this affects the quality of their career transition. A large number of elite athletes tend to delay their life skills development, including social skills, coping skills, and vocational skills, during their athletic time, in terms of devoting their time, energy,

and effort to sport performance enhancement than developing any other areas of their life skills. As a result, when they confront sport career termination, they seem to have difficulties in finding post-sport careers and other interest outside sport and adapting to new lifestyle and social networks. Through the development of the athlete career transition research, the focal orientations have broaden from athletes' sport career termination to their overall life span development, including entry into elite sport, within sport career transitions, and effectiveness of applying life development intervention (see the following section).

Natalia Stambulova's psychological models of the sport career consists of two different models: the *synthetic description* and the *analytic description models*. The aim is to explain psychological components related to the athletes' development and athletic performance through five concepts: *time*, *space*, *information*, *energy*, and *substratum* for predictable developmental stages.

The synthetic description model consists of four objective characteristics: *length of sport career and age borders*, *generalization-specialization*, *level of achievement*, and *cost of achievement*; and two subjective characteristics: athletes' *satisfaction* and *level of success*. Each of these characteristics is individually important; all of them have the potential to interact with each other. These interactions can have a positive or negative impact on athlete's development. For example, an athlete who has a lengthy career and achieves all personal goals is more likely to experience a positive transition, whereas an athlete who specializes at sport at an early age but is not satisfied with personal success can experience transition difficulties.

The analytic description model contains four different factors: *period of sport training*, *period of any career*, *level of sport*, and *age category in sport*. Period of sport training presents athletes' changing athletic involvement, while period of any career pertains to athletes' sport career with the four activities of *preparation*, *start*, *culmination*, and *finish*. Level of sport refers to individuals' competition levels, and age category refers to individuals' chronological development (children, juniors, and adults). The model indicates that athletes may face crisis during seven periods in their sport careers: (1) the beginning of sports specialization, (2) the transition to special intensive training in the chosen sport, (3) the transition from mass popular sports to high-achievement sports, (4) the transition from junior to adult sports,

(5) the transition from amateur sports to professional sports, (6) the transition from the culmination to the end of the sport career, and (7) the ending of the sport career.

Paul Wylleman and David Lavalley introduced the developmental model based on previous studies in both within-sport career transitions and retirement from sport. The developmental model has a developmental and holistic perspective on athletes' normative (predictable, anticipated) transitions during their athletic careers from the initiation to the discontinuation stages. The model consists of four levels that are related to athletes' life span development: *athletic*, *psychological*, *psychosocial*, and *academic and vocational* levels.

Athletic level refers to factors related to athletes' transitions based on changes in their competitive levels, and the changes are determined by organizational characteristics. The athletic level includes four different stages: *initiation*, *development*, *mastery*, and the *discontinuation*. The initiation stage refers to the period when young athletes enter their sport, the developmental stage refers to the time when athletes dedicate time and effort to their sport to develop their sporting skills, the mastery stage is the period when athletes reach their highest performance, and the discontinuation stage refers to transition out of competitive sport.

The psychological level in the model contains major developmental stages, including childhood, adolescence, and adulthood, which relate to a specific pattern of engaging in competitive sport. In childhood, young athletes exhibit readiness for structured sport competition, including *motivational viewpoints* like degree of interest in and attention on participating in the sport, and *cognitive viewpoints*, such as capacity for understanding rules, responsibilities, relationships, and causes of performance outcomes. Adolescence and adulthood are the periods when athletes are facing various life skills development, such as social skills and self-identity. In this period of time, athletes build new or mature relationships with peers and become emotionally independent from parents. Athletes' self-identity development in the adolescent stage is crucial, and their identities can be both positively and negatively influenced depending on the degree of their commitment to their sport and other life skills development.

The psychosocial level presents athletes' development of social networks, including interaction with their surroundings, and changes in the roles

of significant others through the athletic career. The model postulates that parents, siblings, peers, and coaches are the most influential others for young athletes, and in the latter stages of an athletic career, partners, families, and coaches play important roles in athletes' social networks.

The academic and vocational level describes athletes' educational and occupational development during participation in competitive sport and contains *primary education*, *secondary education*, *higher education*, *vocational training*, and *professional occupation*. The academic transition possibly accompanies changes in their sport teams and social networks so that athletes must cope with these changes. Later, when those athletes confront transition out of a sport career, they must step into the vocational level and acquire vocational training or learn a professional occupation.

Life Development Intervention

Since athletes' career transitions are closely related to their life span development, the life development intervention (LDI) has also been discussed as useful model to understand and assist athletes' career transitions. The LDI refers to a framework for the practice of sport psychology based on individuals' life span development.

The major perspective of the LDI is psychoeducational development that emphasizes continuity in growth and changes in one's life and sees individual development as a life span process. In addition, from the LDI perspective, an individual's life changes are inevitable, and the changes can disrupt one's daily routine or relationships, can be sources of stress, and can be used as opportunities to grow. Each major change that individuals confront in life is called a *critical life event*. The event can occur to individuals either with preparation and expectation, which is called an *on-time event*, or in an unplanned and unpredictable way, which is termed an *off-time event*. If some event happens on time, individuals can get more support during the transition period, whereas by contrast, off-time events are usually accompanied by coping difficulties.

Athletes, during their athletic careers, may confront various critical life events, including within-sport career transitions, annual selection processes, injuries, and retirement from their sport. For some athletes, especially those who put a huge amount

of effort and time into athletic success, deselection from the team or career-ending injury tends to be a critical life event because leaving the sport or team is accompanied by negative experiences, such as *feelings of loss, identity crisis, loss of attention, loss of everyday training, and changes in social networks*. The LDI model suggests several strategies that can be applied to assist athletes' career transition process, including *enhancement* (e.g., helping anticipate life events, transfer skills from one domain to another), *support* (e.g., organizing support groups, personal mentors), and *counseling* (e.g., understanding one's problems, planning future events) strategies.

Conclusion

Applying the term *transition* and the transition models in sport psychology area has contributed to broaden knowledge in athletes' career transition research, both in terms of examining the process of changes in athletes' experiences and identifying correlates of the quality of career transition. In addition, changes in the perspectives of athletic experiences in going from a singular event through the transition process have led researchers to address developmental issues in the transition with holistic approaches.

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See also Career Transitions; Coping; Drop-Out; Stress Management

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TRANSTHEORETICAL MODEL

The Transtheoretical Model of Behavior Change (TTM) was originally developed by James O. Prochaska and Carlo C. DiClemente for understanding the stages and processes of smoking cessation and over the past 3 decades has been broadly applied for understanding change across a variety of behaviors, including exercise and physical activity. The TTM was developed to be comprehensive model of behavior change by including cognitive, behavioral, and temporal components into a single framework for behavior change. The five components within the TTM are *stages of change, processes of change, self-efficacy, decision balance, and temptation*. Stages of change classify one's preparedness and progress toward behavior change, and progress is directed by the other four TTM components. Indeed, processes of change, self-efficacy, decision balance, and temptation differ in level and importance among stages, and can be guiding pillars of interventions for promoting progression and preventing regression of individualized behavior change.

The stages of change characterize the preparedness or readiness dimension into five progressive stages or timelines along which behavior change occurs. The first stage is *precontemplation* and includes persons with no intention of change within the next 6-month period. Essentially, the person has not even considered behavior change. The second stage is *contemplation* wherein the person has considered behavior change within the next 6-month period. Essentially, the person has begun thinking about a change in behavior.

The third stage, *preparation*, describes a person who possesses immediate intentions, plans, and commitment for changing behavior (e.g., within next month) but has not done so yet. This is followed by the *action stage* wherein persons have initiated behavior change, but this change has been sustained for less than a 6-month period. The last stage is *maintenance*, wherein behavior change has occurred and been maintained for a prolonged period, for example, longer than 6 months. When considering behavior change, the goal is for individual progression from precontemplation through action stages, and, once action and maintenance are reached, prevention of relapse into an early stage. Importantly, persons can change by single stages or multiple stages (e.g., precontemplation → preparation). Such change occurs through the consideration and focus on the other four TTM components.

Processes of change have been categorized into broad groups of experiential or behavioral factors and represent the strategies adopted and targeted for stage progression or prevention of relapse when in action or maintenance. Experiential processes such as consciousness raising, like seeking new information about a problem behavior, and self-reevaluation, like appraising one's values regarding a problem behavior, are gathered through personal experiences, whereas behavioral processes, such as stimulus control in controlling situations that trigger a problem behavior, and reinforcement management, like rewarding one's self for changing a problem behavior, are developed through interactions with the environment or through action. Experiential processes are seemingly more important for progression in earlier stages, whereas behavioral processes have greater importance during later stages of change. The processes further are amenable for target by professionals and researchers for promotion of individualized behavior change.

Self-efficacy reflects one's personal confidence or agency in successfully executing a course of action, or within the context of TTM, changing one's behavior and progressing through stages. Self-efficacy seemingly increases with stage progression. Whereas the TTM does not explicitly identify targets for changing one's self-efficacy, Albert Bandura has outlined performance accomplishment, vicarious experience (social modeling), verbal persuasion (social support), and physiological or affective states as variables for changing

self-efficacy expectations. This permits self-efficacy to be a constructive target by professionals and researchers for promotion of individualized behavior change.

The last two components of the TTM are decision balance and temptations. Decision balance reflects a person's beliefs and evaluation of the pros (benefits) and cons (costs) of engaging in a behavior, whereas temptation reflects negative urges toward withdrawing from a behavior change. Stage progression occurs as pros outweigh cons (benefits > drawbacks) and when temptations are minimized.

Collectively, individuals progress through stages of change by adopting experiential processes early (e.g., consciousness raising for moving from precontemplation → contemplation → preparation), and behavioral processes later (e.g., stimulus control for moving from preparation → action → maintenance). Self-efficacy is lowest with the early stages (e.g., precontemplation) and increases across stages (e.g., preparation → action) until it peaks with maintenance. Regarding decision balance, cons outweigh pros during early stages of precontemplation, contemplation, and preparation, for instance, and pros outweigh cons during later stages like action and maintenance. Finally, temptations are highest within the early stages (e.g., precontemplation) and decrease across stages (e.g., preparation → action) until its nadir with maintenance.

The TTM is appealing because it categorizes persons based on readiness for change and then provides concrete strategies and approaches for changing a health behavior. This has great appeal for highly individualized behavior change. The theoretical mechanisms for behavioral change included in the TTM are further applicable among diverse behaviors and populations. This has great appeal for adopting the TTM across multiple behaviors and person domains. An additional benefit is that the TTM is application for multiple-behavior change within a single program, and the appeal is that the effectiveness of multiple changes can be maximized within an individualized intervention.

The TTM is particularly appealing for applications involving change in physical activity and exercise behavior; this has been nicely described by Claudio Nigg and Simon Marshall. Importantly, exercise behavior can readily be considered as a dichotomous change process—that is engaging or not engaging in a prescribed exercise routine for

a considerable time. The application for physical activity has been more difficult as this is often considered a continuous behavior, unless we apply the TTM for meeting public health guidelines for physical activity of 150 minutes of moderate-to-vigorous physical activity per week. This is critical considering concerns directed toward applications of the TTM for understanding change in continuous variables such as physical activity behavior. Nevertheless, the TTM has major strengths in its potential for tailored applications regarding a person's readiness for change in physical activity or exercise, particularly given the public health crisis of inactivity and sedentariness. This makes the TTM amenable for research-based interventions and practitioner-led applications involving individual change in these behaviors. For example, there may be big differences among people regarding their readiness for undertaking an exercise program: some may only be thinking about beginning a program, whereas others might be considering strategies for maintaining a morning exercise routine. Both groups of persons have the same goal of exercise or physical activity but require different information and processes, thereby necessitating different interventional or practitioner-based approaches. Indeed, the TTM is simple and clear enough that it can be adopted by any researchers or practitioners, such as nurses, social workers, psychologists, or physiotherapists.

There have been contentious issues when applying the TTM to physical activity and exercise behavior change, perhaps most importantly that the TTM was originally developed for cessation of a behavior (smoking with addictive properties), whereas its application for exercise and physical activity involves starting or adopting a new behavior with minimal addictive properties. Of further note, smoking is not performed with great energy expenditure above resting levels, whereas physical activity and exercise by definition must be performed with a substantial increase in energy expenditure. There are further conflicting intentions and goals between smoking cessation and starting physical activity behavior. These points must be considered for balancing enthusiasm toward applications of the TTM in physical activity and exercise applications and should encourage careful thought and consideration when it is applied within the promotion of this health behavior change.

Robert Motl

See also Adherence; Health Action Process Approach; Health Belief Model; Health Promotion; Multiple Behavior Change; Self-Appraisal/Assessment/Perception; Self-Efficacy; Theory of Planned Behavior

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TYPE A/B PERSONALITY

Personality differences among individuals can be explained in relation to their individual differences in need patterns. Individuals possess various needs—very basic ones, such as the consumption of food and avoidance of pain; and secondary ones, such as achievement, affiliation, and dominance. These primary and secondary needs have the potential to influence an individual's pattern of behavior. One of the needs attracting the attention of researchers in psychology is the *need for control*, which has been associated with Type A behavior. Individuals with Type A behavior patterns are characterized as being competitive, aggressive, and hostile. They also display extraverted tendencies and a tendency to work beyond their physical capacity, as well as possessing the need for control over the environment. Personal traits that are linked to Type A behavior include dominance, need achievement, and trait anxiety. It is assumed that being in a competitive-oriented environment (e.g., school, sport program) facilitates Type A behavior. The antithesis of Type A

behavior is Type B behavior—characterized by easygoing, nonaggressive, and noncompetitive behavior.

For athletes with Type A behavior, a competitive event is characterized by the need to control a given situation; time urgency, which is associated with the fear of losing control over what one is doing; and hostility is exhibited when the athlete loses control of what is happening. It is assumed that dominant athletes (athletes with Type A behavior) tend to play more central and more intense roles in sport situations and assume more risks in order to achieve their individual goals, compared to athletes with lower dominance, those athletes with Type B behavior.

Measurement of Type A/B Behavior

The most frequently employed measure of Type A and B personality is the Jenkins Activity Survey (JAS). The JAS has become firmly entrenched in the field of management, as well as in medical and psychological research, as a standard tool for assessing Type A behavior. The JAS is a self-report questionnaire and is made up of 52 multiple-choice items that yield a composite Type A scale score and three subscales obtained by factor analysis: *speed and impatience*, *job involvement*, and *competitive and hard driving*. One of the concerns associated with the JAS is that this tool has proven to be no better than chance in discriminating between Type A and Type B personalities.

Exercise and Type A Behavior

Type A behavior has been linked to increased risk of coronary heart disease. In addition, it has been suggested that the anger–hostility component of the Type A construct is the most significant characteristic related to heart disease, and that individuals with Type B behavior are less at risk for developing this medical condition. Researchers in sport and exercise psychology have questioned the contribution of exercise training programs to the modification of Type A behavior. To date, little evidence has been obtained to provide support for the argument that participation in an exercise program can lead to reduction in Type A behavior and subsequently lower the risk of coronary heart disease.

One study showed that participation in a 12-week aerobic exercise program resulted in

reduction in Type A behavior and also lowered cardiovascular reactivity to mental stress. However, in studies aimed at examining the relationship between Type A behavior and injury risk, it was found that athletes with Type A behavior might take greater risks and therefore experience more injuries. For example, runners scoring high (e.g., more aggressive, hard driving) on a Type A behavior questionnaire experienced more injuries than those who scored lower on this measure.

Participation in a multiweek, aerobic-type exercise program, such as biking, running, swimming, or walking, has the potential to help individuals modify patterns of Type A behavior. However, it is assumed that participation in an exercise program by itself is too short in securing the appropriate modification of the components that make up Type A behavior. It is proposed that in addition to an exercise program, individuals with Type A behavior should learn cognitive stress management techniques, such as mental imagery; systematic desensitization (association of steps or antecedent behaviors that deal with a stressful experience in a state of calmness and relaxation); and relaxation techniques like autogenic training, progressive muscle relaxation, or yoga. In addition, massage and biofeedback can also help to modify Type A behavior. It is assumed that the adoption of a multifaceted, holistic approach—participating in an exercise training program combined with learning cognitive stress management techniques—will help individuals reduce the anger–hostility components of their behavior.

Ronnie Lidor

See also Extraversion–Introversion; Iceberg Profile; Optimism; Perfectionism; Personality Tests; Personality Traits and Exercise

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U

UNDERRECOVERY SYNDROME

In sports, optimal performance is only achievable if athletes recover after competition. Recovery and intense exercise must also be balanced with everyday life. Additionally, sufficient physiological and psychological recovery protects against overtraining. However, it is just a small step from regular daily practice to a high frequency of demanding events, which complicates the process of recovery for athletes. Teams of the American basketball and ice hockey leagues play 82 games during the regular season; players travel thousands of miles per month. Reaching the playoffs adds more games and travel to the already demanding schedules. One game per night in a city is daily business for athletes like LeBron James or Corey Perry. For example, the Miami Heat played basketball games in Portland, Salt Lake City, and Los Angeles, on three consecutive nights. In the National Hockey League (NHL), the Anaheim Ducks played six hockey games in a row in different cities within 11 days in November during the 2011–2012 season. Thus, the questions arise, how much recovery time is necessary between games and what kind of consequences result if there is not enough time to rest?

Athletes need recovery to perform better. If the time of recovery is too short or disturbed by certain several circumstances, underrecovery may occur. *Underrecovery* is defined as the imbalance of recovery periods and daily life demands of a person. These demands can be intensive practice, competitions, and all stressors linked to the sport and

everyday life of an athlete. For example, distance from family because of extensive travel can be an important stressor disturbing adequate recovery before and after games. Additionally, mistakes in the training structure can overstrain the athlete and increase the need for recovery. Examples of these mistakes include monotonous training programs, too long training sessions, ignoring training principles, or training periodization (not including rest days or periods of lower intensity).

Recovery processes depend on previous activities and the type and duration of stress. The interrelation between recovery and stress can be illustrated by a *scissors model*. Within the scissors model, it is postulated that stress and recovery change independently from each other, meaning that reducing stress may not necessarily result in an enhanced recovery. Adequate recovery, however, reduces the experienced level of stress. When external factors prevent adequate recovery, the imbalance between stress and recovery may increase. Conversely, per the scissors model, athletes need to follow increased efforts with a longer period of recovery. Thus, it is important that athletes, and especially coaches, are aware of the importance of recovery and its relation to stress and performance.

The recovery process is gradual and cumulative. The total recovery time depends on the previous activities and the type and duration of stress. In general, if practice is completed in a high-training-volume phase, recovery usually takes longer than if practice is completed in a taper phase. Recovery finishes when a psychophysical state of restored efficiency and homeostatic balance is reached.

Mostly, the process of recovery is linked with a change, reduction, or a break from activity, but the process needs to be applied individually. A sauna may be a favorite recovery strategy for one athlete but would not be a strategy to be deployed by an athlete who feels stressed by sitting in a small room with high temperature. The recovery process can be reached passively, actively, or proactively. Passive strategies comprise a cease from activity, such as rest, massage, sauna, and hot or cold baths. These strategies initiate physiological reactions through external stimuli (e.g., heat, cold, pressure) affecting blood flow, breathing, or muscle tone. Active and proactive strategies are self-initiated processes by the athlete to reestablish psychological and physiological resources. Active strategies include activities implemented immediately after practice and competition, such as cool-down activities like stretching or slow running. Proactive strategies comprise additional self-initiated activities independent of practice. For example, relaxation techniques (muscle relaxation training) optimize recovery by supporting metabolism mechanisms for dealing with blood lactate. Other activities can be described as proactive regeneration. If an ice hockey player experiences housework (e.g., ironing) as a regenerative action, it could be used for recovering from an intense exercise. This example illustrates how individualized recovery can be implemented. Thus, proactive recovery also contains time with friends (social recovery) and can compensate for less time of physiological regeneration.

Although it sounds very easy to refill empty energy stores, from a practical point of view, there are many determinants affecting the quality of recovery. Coaches must be aware of situational and environmental conditions like heat, noise, or sleep, reducing the quality of rest. Eight hours of sleep at night during a training camp could be enough, for example. But if the athlete has to share the room with a snoring roommate, in an uncomfortable bed, the athlete may feel absolutely exhausted the next day. Disturbances and irritations during rest could prevent fulfilling psychological and physiological needs, affecting athletes' performance. Furthermore, while travel time is a break from exercise and competition, the stress associated with travel does not provide adequate regeneration. These circumstances could have negative consequences if a state of underrecovery stays at a high level for a longer period. However,

short underrecovery episodes could be compensated. For example, if an athlete goes out to party, he may lack sleep before next morning's practice. Although the amount of sleep would not be enough for an optimal regeneration, the positive energy of the social night with friends may compensate for the feeling of tiredness. However, problems arise if the imbalance of demands and regeneration is maintained over a long period of time and results in a state of chronic underrecovery.

Differences Between Underrecovery and Overtraining

Underrecovery and overtraining syndromes seem to describe the same state of an athlete of being tired and unable to perform on a high level after periods of inadequate recovery and intense practice. At first sight, it appears that both syndromes are characterized by the same symptoms and causes. However, on closer examination, it is obvious that underrecovery and overtraining syndrome are different. On one hand, there is the state of underrecovery caused by imbalance of various demands and regeneration. Too much physical (intense exercise, density of competitions) and psychological (conflicts, traveling, disturbed recovery) stress and inadequate recovery (breaks, social recovery) over a long period of time lead to a feeling of exhaustion. The accumulation of fatigue develops over time into an underrecovery syndrome.

On the other hand, the overtraining syndrome describes the *result* of the imbalance between too much practice, competitions, or nontraining stress factors, and a lack of recovery. Mistakes in training structure additionally cause tiredness and reduce mental freshness. Thus, underrecovery is a precondition for overtraining. Originally, muscle fatigue is the goal of intense practice in order to achieve training effects. This kind of overreaching, also named *short-term overtraining*, is a regular part of athletic training. Nevertheless, when overreaching is too profound or is extended for too long, short-term overtraining turns into long-term overtraining. The problem arises when recovery needs are neglected by the coach or athletes and chronic overreaching occurs over a long period of time (at least for three weeks). Missing periods for refilling energy stores can cause athletes' performance to drop or plateau. Athletes may train even harder in the attempt to improve, but this can lead

to a deeper state of underrecovery. Accordingly, a feeling of staleness can be the consequence of long-term overtraining; *staleness* describes the state in which the athlete has difficulties in maintaining standard training regimens because of physical feelings (e.g., stiffness, tiredness) or decreased motivation. Other consequences include feelings of depression, general apathy, emotional instability, restlessness, disturbed sleep, and an increased vulnerability to injuries.

Consequences and Prevention of Underrecovery

Short periods of underrecovery can be compensated by the athletes' use of other recovery strategies like relaxation techniques. However, chronic underrecovery leads to short-term and long-term consequences. Short-term consequences include the feeling of tiredness and exhaustion, lethargy, less motivation, and development of negative cognitions toward upcoming activities, which negatively influences future performance. Less concentration and body tension during practice can also increase the risk of injuries. Additionally, in contrast to the overtraining syndrome, underrecovery also impacts daily routines. A reduced level of energy will be noticed during daily activities and can reinforce long-term consequences, including feelings of depression, emotional instability, weight loss, increased resting heart rate, hormonal changes, disturbed sleep, and increased vulnerability to injuries and respiratory diseases. Underrecovery may also lead to burnout—the psychological, physical, and emotional withdrawal from a formerly enjoyable or motivating activity. Burnout is an exhaustive psychophysiological response to massive chronic stress and often requires a long break from activity to recover. If an athlete reaches the state of chronic underrecovery, small periods of regeneration or spontaneous interventions are ineffective, requiring a longer rest period (from several weeks to months) and professional help by doctors or psychologists.

To prevent the extreme state of exhaustion, coaches and athletes need to incorporate sufficient and high-quality periods of regeneration into training regimens, ensuring that recovery periods are long enough, with minimal disturbances. A good athlete-coach relationship has a preventive effect. The athlete can talk about private stressors (e.g., problems with relevant

others, financial problems), disturbances during recovery periods (e.g., a snoring roommate), or problems during practice. Discussion can reduce interpersonal stress with the coach. Additionally, a well-structured training and competition schedule is an important step to prevent monotonous training. Adequate time management and planning of breaks and travel could reduce stress level. Furthermore, it is important that athletes are aware of their preferred recovery strategies, being able to use at least two or three.

Conclusion

Recovery helps refill athletes' energy stores and refresh the mental state for challenges. Too often, however, the recovery element is overlooked as an essential aspect of any training regimen and periodization of a season. Insufficient time between competitions, long distances between tournaments, intense physical demands, and the load of daily hassles can cause athletes to insufficiently regenerate. Underrecovery dramatically impacts the performances and health of athletes. Consequently, athletes and especially coaches need to be aware of the danger of disregarding necessary recovery periods. By maintaining the optimal balance between overreaching, recovery, and stress management, positive training effects arise and the competitiveness of athletes will be preserved.

Michael Kellmann and Sebastian Altfeld

See also Mental Health; Overtraining Syndrome; Practice; Relaxation; Stress Management

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VERBAL PROTOCOLS

A *verbal protocol* or verbal report is a cognitive task analysis (CTA) technique designed to elicit a verbalizable report of an individual's thinking during task performance. Verbal protocols are typically elicited by trained researchers as a means to access information heeded (attended) during task performance; this is usually for the purpose of understanding the mental operations or knowledge representations responsible for the observed performance. With only minimal qualification, the catchall term *introspection* has been used frequently in psychology to refer to a range of verbal reporting methods, despite practical and theoretical differences among methods. The historical context and evolution of methods provides a useful basis for assessing the validity and utility of this CTA technique for capturing, describing, and explaining thinking, and, ultimately, using what is learned to develop scientific theories capable of prediction and control.

History of Introspection and Verbal Reporting Methods

Although writings on introspection can be traced back to Plato and Aristotle, more recently, British empiricists, such as John Stuart Mill (1806–1873), viewed introspection as a form of self-observation—a method of bringing elementary sensations into conscious awareness to further understand the composition of complex experience. In contrast,

German psychologist Wilhelm Wundt (1832–1920) trained participants (sometimes on over 10,000 trials) to concurrently report on specific qualities and intensities of elementary sensations, rather than on higher order processes. As the practice of introspection burgeoned in psychology, the focus shifted toward a less restrictive examination of conscious experience that resembled that of the British empiricists. For example, a student of Wundt, Edward Titchener in the United States, and practitioners at the Würzburg School in Germany, used *systematic introspection* to examine higher order processes, such as memory and judgments, albeit using simple laboratory-based tasks. Retrospection, which Wundt had previously discounted because of the fallibility of memory, came back into vogue, primarily because of the assumption that concurrent introspection might interfere with in-task thinking. Moreover, interest grew in qualitative descriptions of thinking, which often became the focus of experiments rather than the quantitative data like reaction times they accompanied.

Essentially, there was a shift in the practice of psychology by experimental introspectionists, away from objective empirical observation toward experimenter-led, subjective constructions of behavior by the participant. Although some introspective analyses of thinking (psychophysical judgments) were deemed highly reliable, the results of systematic introspection sometimes lacked reproducibility. Low reliability was frequently attributed to insufficient participant training in reporting procedures, yet the training of participants or their

selection based on their ability to verbally report was also considered an infringement on scientific objectivity. Consequently, in the early 1900s, behavioral psychologists in the United States led by John B. Watson challenged introspection—irrespective of its form—as a viable method of studying behavior. From the critics' perspective, there appeared no way to study reliably the relationship between the subjective experiences of the person introspecting and their verbal report of those experiences. Combined with a general trend in U.S. psychology to focus on practical performance and manipulations that could determine or limit performance, introspective techniques dropped out of favor in the United States for methods thought more suitable to address these goals.

From Introspection to Think-Aloud Reports and Cognitive Process Tracing

In Europe, the use of systematic introspection continued, but the focus shifted from *creative synthesis* of elementary sensations toward a *creative analysis* that emphasized holistic, complex, and purposive behavior. Some of these psychologists moved introspection out of the laboratory to study thinking more representative of real life. For instance, a successor to the Würzburg school, Otto Selz, used systematic introspection to study classroom learning and his student, Julius Bahle, applied these methods to studying musical composition. Karl Duncker and Édouard Claparède used *think-aloud* methods of introspection, together with experimentation, to study problem solving. These two authors are generally considered to be the first to use think-aloud protocols. Rather than have participants analyze their sensory experience as per Wundt, these authors instructed participants to express their thoughts directly—as they occurred—while remaining focused on the experimental task. Consistent with Wundt, Claparède noted that this technique avoided memory issues associated with retrospection—a central component of post-Wundtian systematic introspection.

Prior to Selz, however, most applications of introspection concentrated on classifying the contents of thinking rather than on the process of thinking per se. Consistent with William James's conception of thinking as a series of substantive and (inaccessible) transitive states, Selz's process-oriented theory was centered on explaining, and through introspection, eliciting thinking

as a strictly determined succession of cognitive operations.

Selz's ideas of tracing thinking operations using verbal protocols were implemented, most notably, by Adriaan de Groot in the domain of chess. Initially, de Groot employed both systematic introspection (retrospection) and the think-aloud technique to study how chess players selected moves from a range of game positions. However, after experiencing difficulties with interrupting players to retrospect after several minutes of thinking about a position, he focused on eliciting think-aloud reports since they offered a less disruptive method of systematically analyzing the relatively complicated and lengthy processes involved in chess thinking. De Groot, however, did not rule out the use of retrospection for all tasks. Instead he suggested guidelines to steer participants away from describing *peculiar qualities of inner experiences* during retrospection and toward recalling the sequence of thinking operations that occurred while performing the task.

De Groot suggested that verbally reporting on thinking could interfere with actual thinking, which may affect the completeness of the report and, consequently, the ability to capture the *true* course of thinking. He noted four possible causes for incompleteness: (1) The phase structure of the thought sequence is likely to be under the threshold of conscious awareness and absent, in any explicit sense, from the report; (2) Thought is more rapid than speech, leading to the possible omission of heeded information; (3) *Wordless thoughts* may not be reported while thinking aloud, and transformation into speech may disrupt the flow of thinking; and (4) Participants may intentionally suppress steps in their thinking, for instance, when they make mistakes. De Groot offered two criteria for assessing completeness: (1) the degree to which the participant is satisfied with the protocol as a representation of the actual thinking and (2) the ability to follow and understand the participant's reasoning for a particular action. (The latter is often hindered by experimenter instructions to make some thoughts more explicit—for instance, by making references to subjects or objects as nouns or noun phrases rather than pronouns.) Both criteria require the participant and experimenter to go back through the protocol—a frequently overlooked step.

Rather than an absolute truth, de Groot argued that the cognitive process that unfolds during

task performance is largely hypothetical and can be understood best in the context of a scientific theory. According to de Groot, good theories are useful if they are logically constructed, adequately describe the relevant phenomena, permit testable predictions (that are empirically supported), and can be applied to control the world to which they refer. As such, introspective techniques such as think-aloud verbal reports provide a valuable method for hypothesis formation and theory building. Likewise, introspective methods provide a means of theory testing, with the caveat that such tests are likely to be influenced by the reliability of reports.

The notion of cognitive *operations* was a precursor to the concept of the computer program in information processing psychology. The ideas espoused by Selz and de Groot were influential in the subsequent development of theoretical and computational models of recognition, problem solving, and comprehension by the likes of Alan Newell, Herbert Simon, and Walter Kintsch. Process tracing using verbal protocols is currently one of the primary methods of testing and validating models of cognition.

Validity of Introspective Methods and Verbal Protocols

Despite their current use—both in terms of theory development and testing—some researchers have questioned the validity of processes elicited via introspective methods. For instance, in an extensive review of studies using a range of methods, Richard R. Nisbett and Timothy Wilson showed that many participants were unaware of the actual cognitive process that led to the solution. Consistent with William James, they argued that some processes are inaccessible. They concluded that individuals do not access the specific thoughts heeded during problem solving but, instead, access implicit theories, culturally derived social rules, generalizations from past experiences or generate causal hypotheses that could explain their behavior. When individuals report correctly on the cognitive processes, it is more a matter of coincidence that their causal theory is correct than their ability to access the associated memory trace.

K. Anders Ericsson and Herbert Simon contended that it is possible to obtain valid reports of thinking as long as specific procedures were

followed. Consistent with James, Selz, and Newell and Simon, they suggested that cognition could be described as a sequence of states that are transformed by successive information processes. Whereas cognitive processes themselves may be consciously inaccessible, the output of a previous process and the input to a future one are held in short-term memory (STM). Although information entering STM may be replaced, pointers to symbols and operations are likely to remain present temporarily in STM. This information can, therefore, be heeded and, hence, verbalized.

When individuals are instructed to think aloud, information in STM can either be verbalized directly (Level 1) or transformed from nonverbal to verbal code (Level 2). While such transformation may require additional processing, this has been shown not to affect the contents of the report (but does increase time on task). When participants think aloud using Level 1 or 2 verbalizations, a direct trace of heeded thoughts, and consequently an indirect trace of the internal steps in cognitive processing, can be elicited. When participants are allowed to verbalize (additional) information that is not normally heeded (Level 3), or are instructed to do so via directed probes that encourage analysis or interpretation of one's thinking, their verbal report may only resemble partially—or be completely independent of—the actual processing that mediated task performance.

Ericsson and Simon made a clear distinction between verbal reports generated from actual heeded thoughts and those procedures that permit generation of causal theories, generalizations or assumptions about, or explanations, analyses, descriptions, or summaries of personal task performance. These authors indicated that although concurrent reports are preferred, retrospective procedures can provide reliable and valid reports, providing reports are restricted to Level 1 or 2 verbalizations, and undirected probes are used (such as, *think aloud* or *recall the first thought you remember thinking*), or at least probes that do not encourage participants to deviate from the thinking they would have engaged in if not probed. Following de Groot, they also recommended that the time between an activity and verbally reporting on that activity should be minimized, and participants be asked to report about a specific incident rather than generalizing from other incidents.

Beyond Verbal Reporting: Methods of Analysis

While the analysis of verbal protocols is beyond the scope of this entry, the interested reader's attention is drawn to Micheline T. H. Chi's distinction between protocol analysis and verbal analysis. Protocol analysis is useful for domains in which the problem space has been clearly defined a priori (which can take numerous process-tracing, experimental, and computer simulation studies to achieve) such that clear predictions can be made about alternative strategies that might be employed. Verbal analysis, on the other hand, explicates procedures for analyzing verbal protocols in a manner that increases understanding of the knowledge representations that support performance in a theoretically driven manner. Importantly, verbal analysis provides one possible means of fleshing out a hypothetical problem space so that a subsequent protocol analysis can be carried out.

Conclusion

In sum, the use of introspective methods has evolved over time, toward procedures for obtaining concurrent think-aloud and retrospective reports. Together with recommendations from de Groot, Ericsson and Simon's procedures provide, arguably, the most detailed instructions for obtaining valid and reliable protocols. In advocating for these methods, however, one should heed de Groot's warning: Methods should be chosen based on the research goal. Other incident-based, CTA methods exist that use verbal reporting procedures without restricting them to Level 1 and 2 verbalizations, including the Critical Decision Method developed by Gary Klein and colleagues (see the entry Cognitive Task Analysis). Although susceptible to criticisms associated with any method of retrospection, these alternatives are useful, not necessarily in the context of a protocol analysis to *test* a given theory about alternative strategies in a well-defined problem space but as a means of *hypothesis formation and theory building* on a more macro level, especially in complex domains where the problem space is ill defined.

Paul Ward

See also Cognitive Function; Cognitive Task Analysis; Decision Making; Expertise; Information Processing; Knowledge Structure; Memory; Perception

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VISION

Vision is the process of seeing and perceiving the surrounding environment by using information contained in light. Appropriate visual information is imperative for almost every sporting task to ensure that athletes are able to monitor the actions of others, while also perceiving their position in relation to targets, such as balls, nets, and walls, and the playing environment. In considering vision in sport, this entry discusses what the important characteristics of vision are, whether elite athletes rely on better vision than others, and whether vision training can enhance sporting performance.

The Important Characteristics of Vision

Vision is a phenomenon not simply isolated to the eyes. While the ability to see commences at the eye when the *cornea* and *crystalline lens* focus light onto the *retina* at the back of the eye, vision encapsulates a complex procedure that relies on how the brain perceives this array of light. When light is focused on the retina, a complex collection of neural signals is relayed via the visual pathway to the visual cortex in the brain. The visual cortex is located in the occipital lobe at the back of the brain; it processes and combines the visual information from the two eyes and conveys this information onward via a complex communication

network that both controls action and facilitates perception of the surrounding environment.

There are many different measures that can be used to characterize vision. Humans have the capacity to perceive their surrounding environment clearly, with color, across a wide range of lighting conditions and regardless of whether they themselves or objects within the environment are stationary or moving. As a result, an evaluation of an athlete's vision can encompass a diverse range of characteristics, sometimes referred to as *general visual measures* or *basic visual functions*.

A considerable number of different visual measures have the potential to impact performance in sport. Consequently, it is recommended that athletes have their vision tested regularly (usually every 2 years) to ensure that any problems can be corrected to prevent them from adversely affecting on-field performance. Following are some of the more important visual measures relevant for athletes.

Visual Acuity

Visual acuity (VA) measures the ability of the eye to resolve detail. It is typically evaluated using a letter chart, and is measured by means of the Snellen fraction (e.g., 6/12 or 20/40; the first number in the fraction is the test distance, conventionally 6 meters or 20 feet, and the second colloquially refers to the distance a person with "normal" vision could resolve the same detail). *Static visual acuity* refers to the ability to resolve detail in a *stationary* target, and is the measure most commonly used to assess vision. *Dynamic visual acuity* is occasionally tested in sport-specific assessments of vision to evaluate the ability to resolve detail in a *moving* target. Deficiencies in VA are common in athletes; they are most frequently caused by *myopia* (poor distance vision) or *hyperopia* (poor or effortful near vision), and are usually corrected using spectacles or contact lenses.

Contrast Sensitivity

Contrast sensitivity reflects the ability of the eye to resolve differences in brightness. Poor contrast sensitivity is likely to influence an athlete's ability to see at night or in dim conditions. Contrast sensitivity declines with age and otherwise is most commonly hampered by cataracts (clouding of the crystalline lens) and diabetic retinopathy (retinal disease as a result of diabetes).

Peripheral Vision

Peripheral vision is the ability to see objects that are located away from the direction of gaze. The binocular field of vision extends approximately 190 degrees horizontally and 130 degrees vertically, though the clarity of objects decreases as they move further from the center of the visual field. A restriction of peripheral vision will limit an athlete's ability to identify or monitor other players or objects that are located in side vision; however, restrictions to peripheral vision are relatively rare and usually only occur in the latter stages of life as a result of conditions like glaucoma (damage to the optic nerve at the back of the eye) and stroke.

Depth Perception

Depth perception is the ability to perceive the distance to, or between, objects in space. It is particularly important in sporting tasks where an athlete must hit or interact with moving objects. The perception of depth is a complex process that relies on specific information from each eye and on the comparison of relative information from both eyes. Accordingly, depth perception can be particularly poor when one eye is notably worse than the other (common in individuals with an eye turn or who have experienced an injury to one eye).

Color Vision

Color vision refers to the ability to perceive different wavelengths of light that are seen as different shades of color. Deficiencies in color vision are usually a result of genetic inheritance, meaning that they are not normally curable. Approximately 8% of males and 0.5% of females experience problems with color vision. The most common form of deficiency causes difficulties when discriminating between red and green colors; this has been shown to be a problem when performing some sporting tasks.

Eye Movements

Six extraocular muscles control the direction of each eye to ensure that central vision can be used to resolve objects of interest with the best possible clarity. Skilled athletes use very specific and efficient eye movement strategies to direct their gaze toward the most informative areas in their visual field and to track and predict the future location of moving objects. Neuromuscular conditions such

as multiple sclerosis and strabismus (an eye turn) can affect the ability of athletes to make fast and accurate eye movements.

Do Elite Athletes Have Better Vision Than Other People?

The remarkable skill level of elite athletes has led some to wonder whether these achievements are underpinned by some sort of extraordinary or above-normal level of visual skill. Vision is unmistakably important for the successful performance of many different sporting tasks, and for this reason it may be intuitive to infer that superior vision may lead to superior sporting performance. However, there has been, and continues to be, considerable debate about whether skilled athletes have better vision than lesser skilled athletes, and so far the evidence has been somewhat equivocal.

Two key methods have been employed to compare the vision of skilled and lesser skilled sportspeople. The first involves epidemiological studies that evaluate the relative prevalence of visual problems in athletic and nonathletic populations. These studies typically conclude that athletes have a lower (or sometimes similar) prevalence of visual problems than nonathletes. In essence this supports the view that, on average, skilled athletes have better vision, but it more likely reflects the lower likelihood that people with poor vision take part in (or are successful at) sporting activities. The second method of investigation directly compares specific visual attributes like VA and contrast sensitivity across different levels of sporting expertise. Although the evidence is equivocal whether these attributes can differentiate players of contrasting skill levels, a growing number of studies suggest this may not be the case.

Multidisciplinary examinations of sporting expertise have provided a useful means of establishing the relative contribution of general visual measures when compared to other factors that can predict sporting skill. These analyses have generally concluded that sport-specific perceptual skills, such as anticipation, pattern recall, and decision making, are much better for discriminating skilled and lesser skilled athletes than primary visual attributes can. Although for some there is a prevailing view that skilled sportspeople are characterized by superior visual attributes, there appears to be an increasing recognition that sport-specific measures of perceptual-cognitive skill are more likely

than general visual measures to be associated with sporting success.

Can Vision Training Enhance Sporting Performance?

There is considerable debate questioning whether vision training can be used to improve on-field performance. This question requires a consideration of three key questions.

Is Vision a Limitation to Performance in Sport?

Vision is clearly important for success in most sporting tasks; if any visual characteristic were to progressively become worse, it is reasonable to assume that at some level each is capable of adversely influencing on-field performance. What is less clear, though, is at what point each of these characteristics may start to impinge on sports performance. For example, if VA was to be systematically decreased to the point of practical blindness, evidently performance must decrease at some point. In essence, most vision specialists would advocate that normal visual function is necessary for most favorable on-field performance. As a result, any decrease below the normally expected capacity would lead to a commensurate decrease in performance. The small subsection of sports vision specialists who claim that skilled athletes rely on better-than-normal vision to underpin their sporting success infer that even a normal level of visual function limits performance, and that improvements in vision to supranormal levels lead to enhanced sporting performance.

Can Vision Be Improved by Training?

Spectacles and contact lenses are used to correct the most prevalent of visual problems (e.g., myopia and hyperopia); however, clear evidence exists to show that vision training can be used to improve other visual characteristics, such as eye movements and visual focus. Importantly though, this training is typically used to help those with visual problems. While vision training may be useful to help athletes with these types of problems, it is not at all clear whether training can facilitate supranormal vision. Even if normal vision is a limitation to sporting performance, there is very little evidence to suggest that visual characteristics can be enhanced to levels superior to those found in the general population.

Do Improvements in Vision Result in On-Field Improvements in Performance?

Vision training is frequently used to improve the ability of individuals to perform activities of everyday living. For example, training can lead to direct improvements in the performance of visually dependent tasks like reading and computer use by correcting deficiencies in gaze and focusing. Similarly, there is good reason to believe that sporting performance should be improved if vision training can correct visual deficiencies that limit on-field performance. However, there is very little evidence to suggest that the vision of athletes can or should be trained to above-normal levels of operation, or equally that generalized training of a range of different visual measures can lead to improvements in athletic performance.

David L. Mann

See also Attentional Focus; Brain; Eye Movements/Gaze; Imagery; Mirror Neurons; Perception

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YIPS

A motor phenomenon affecting golfers, the yips consist of involuntary movements during the performance of shots requiring fine motor control, such as chipping and putting. Although the yips are mainly reported by golfers, the psychological characteristics of a small group of cricket bowlers who experience yips-like symptoms have also been reported, suggesting the condition is generalizable beyond golf. Generally, the yips are characterized by excessive, involuntary muscle activity, resulting in unwanted movement, or an inability to initiate movement. In the past decade, the definition of the yips has expanded to encompass an etiological continuum, anchored by two subtypes, with a neurological basis (focal dystonia) at one end and a psychological basis (performance anxiety) at the other. Subtypes can be defined by identifying themes in golfers' descriptions of their experiences. Those with descriptions of dystonia can be classified as Type I yips, while those whose descriptions include performance anxiety or *choking* theme can be classified as having Type II yips. Type I yips have many characteristics in common with other forms of focal dystonia. The aim of this entry is to encourage the use of the term *yips* only for those with a task-specific focal dystonia. The intention is that a more consistent terminology will clarify appropriate management strategies of those with a true dystonia (the yips) versus those who suffer symptoms of performance anxiety.

The symptoms of upper-limb focal dystonias are dominated by involuntary contractions of the hand and forearm musculature that result in awkward, uncoordinated movements of the wrist, fingers, or both. These symptoms are due to the inappropriate cocontraction of antagonists and agonists, which is often associated with excessive levels of muscle activity. Symptoms are triggered by the repetitive performance of a skilled motor task over months or years, such as writing, playing a musical instrument, or putting a golf ball. Multiple lines of investigation support the hypothesis that focal dystonias are associated with impaired inhibitory function at multiple levels of the central nervous system, which may stem from basal ganglia dysfunction. Cathy Stinear et al. showed that low-handicap golfers with experiences of Type I versus Type II yips could be dissociated by their electromyography activity in muscles of the putting arm under low-stress conditions, as well as by their performance on a computer-based response inhibition task. In both cases, Type I yips golfers showed excessive electromyographic activity compared to Type II, and to matched controls. Similarly, through the use of an instrumented kinematic glove, Charles Adler et al. observed that yips-affected golfers had excessive involuntary pronation-supination of the forearm.

Like other forms of focal dystonia, golfers with the yips may opt to retrain themselves to perform the task differently, given that no effective rehabilitation strategy has yet been developed. The symptoms of focal hand dystonia and Type I yips can sometimes be temporarily alleviated by

the use of sensory tricks, such as wearing a glove on the affected hand, using a different grip, or by changing to a different (usually longer) putter. The management of yips could draw from the treatments used for occupational dystonias, such as writer's cramp and musician's cramp. A range of therapeutic strategies have been trialled for these types of dystonia, including immobilization of the affected hand or upper limb, often combined with ergonomic changes, motor training, and sensory training. Unfortunately, these types of interventions have produced mixed results for writer's cramp and musician's cramp and would require a considerable time commitment from the golfer, under the guidance of a trained physical therapist.

If the overactive forearm muscles can be clearly identified in an individual, injection of botulinum toxin can markedly reduce the level of involuntary activity. However, this treatment carries the risk of the toxin spreading to adjacent musculature not intended for treatment, and the effects of treatment are temporary, lasting approximately 3 to 4 months. Interestingly, an isolated case of improvement in yips symptoms during golf was reported by a patient undergoing memantine administration for treatment of Alzheimer's disease, but a systematic trial has yet to be conducted.

Many studies of the yips have used questionnaires to explore the symptoms, clinical history, and sport experience of the participants. For the sport psychologist, it is interesting to note that golfers who experience performance anxiety exhibit yips-like symptoms when they experience choking, where severe performance anxiety impairs the preparation and execution of movement, degrading performance. A model proposed by Aynsley Smith et al. was developed on the basis of questionnaires completed by 70 golfers who experience yips symptoms. One of the limitations of this study is a potential response bias, as only 39% of invited participants (both with and without yips symptoms) completed and returned the questionnaires. A further limitation is that the respondents' descriptions of their symptoms are subjective, and therefore may not accurately describe the range of physical manifestations of the yips. Still, this model is effective for differentiating patients with focal dystonia from those with performance anxiety, thus reflecting an important advance. Generally, the combination of distraction by the specific competitive context and an increase in arousal and self-awareness, increases performance anxiety and

decreases confidence, which in turn increases the likelihood of suboptimal performance. The most anxiety-provoking situations for affected golfers include (1) leading the competition, (2) tricky putts (those putts with more than one change in inclination), (3) playing against specific competitors, and (4) the need to make easy putts. Most researchers and clinicians are in agreement that yips-like symptoms are exacerbated by heightened anxiety. However, it should be reinforced there is considerable debate concerning whether excessive anxiety is sufficient to trigger actual yips symptoms.

Winston D. Byblow

See also Brain; Brain Imaging; Electromyography (EMG); Motor Control; Neurologic Disorders; Neuroscience, Exercise, and Cognitive Function; Transcranial Magnetic Stimulation (TMS)

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YOUTH SPORT, PARTICIPATION TRENDS IN

Organized youth sport became popular in the 1920s (in the United Kingdom, at least) in part based on the idea that sport could help to produce well-grounded, physically strong, morally sound boys who would be successful contributors to society, particularly in the military and business worlds. In contemporary societies, there are competing reasons for promoting participation in youth sport. For example, some people primarily view youth sport as a breeding ground for the next generation of elite athletes. Others focus more

on the potential psychological, social, and emotional benefits of youth sport participation, while acknowledging that sport participation has also been associated with negative outcomes.

Additionally, youth sport has been viewed within a population health context. That is, declining participation in active free play and active transportation account for a large proportion of the low levels of youth physical activity observed in most developed countries. Sport has therefore become one of the primary vehicles through which children and adolescents engage in physical activity. Hence, youth sport maintains an important role in modern society, but reasons for promoting youth sport vary. Specifically, viewing sport as a breeding ground for talent versus a context for physical activity promotion and youth development may not be entirely compatible.

Given these competing reasons it is perhaps not surprising that there are multiple systems of youth sport delivery that create a complex landscape. For example, there are publicly funded recreational community-based leagues, competitive clubs, school teams, specialized school-based academies, elite private training programs, regional and national development teams, and the youth team structures of professional sport clubs. Unfortunately, researchers have largely been unable to delineate the different types of outcomes associated with participation in different types of sport programs.

Programs vary in terms of the amount of adult involvement and level and intensity of competition. Recreationally oriented programs are typically run by committed volunteers (many of whom are parents), with a primary focus on providing opportunities to play and the development of skills and tactics occurring incidentally. More competitive programs often include professional coaches and specifically focus on developing physical skills, technique, and tactical understanding. Sport sociologist Jay Coakley argued the term *youth sport* has become synonymous with organized competitive programs that have become increasingly privatized and subject to adult control.

Participation Rates

Although trends in overall physical activity participation have been relatively well documented, less attention has been given to youth sport participation rates. Nonetheless, from the available international data it appears that in developed countries

between 50% and 70% of youth participate in at least one sporting activity annually. A study from Australia showed 63% of children (ages 5–14 years) participated in organized sport within the previous year; a study from the United Kingdom revealed 59% of boys and 55% of girls (ages 2–15 years) participated in sports at least once a week; Canadian statistics showed that in 2005, 51% of children and adolescents regularly participated in sport; and data from the United States showed that 54% of all high school students participated in sport during the 2006–2007 school year. Finally, data also show that sport participation peaks during late childhood before sharply declining during the middle and late adolescent periods.

Despite these relatively high rates of sport participation, caution must be used when drawing conclusions from these data. In some surveys, respondents (often parents or caregivers rather than their children) report children's participation in sport once in the last year, while in other surveys require to report sport participation in the last week or month. Some studies include school sport, whereas others only focus on sport occurring outside of school (which is likely to be more indicative of regular commitment to youth sport). Finally, in some studies, all the sports in which an individual participates are reported, meaning that a single person who plays multiple sports may be counted more than once when tallying sport participation. Combined, these measurement issues may mean that youth sport participation rates are artificially inflated. Standardized measurements, definitions of youth sport, and regular surveys are required to ensure researchers and policy makers have a detailed and accurate knowledge of sport participation rates.

Inequalities in Sport Participation

There are numerous inequalities in youth sport participation. In many developed countries, sport participation continues to be lower among females compared to males. There are also differences in youth sport participation in relation to socioeconomic status, family structure (one- or two-parent families), and ethnicity. Children from families with higher economic status and higher education levels tend to have more access to sport. Given the increasing privatization of youth sport and reduction of publically funded programs in many countries, the cost of youth sport has become a barrier to participation.

Conclusion

Organized youth sport maintains an important role in contemporary society. Although programs have become more privatized and adult controlled, international participation rates are relatively high and stable. That said, inequalities relating to gender, socioeconomic status, and education levels persist. In the future, it may be important to reduce these inequalities by providing greater access and opportunities to people from a range of different backgrounds and working with administrators, coaches, and parents to help ensure the conditions for more consistently promoting positive outcomes are provided.

Nicholas L. Holt and Camilla J. Knight

See also Friendships/Peer Relations; Parenting; Participation Motives; Talent Development

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Timeline: History of Sport and Exercise Psychology

<i>Date(s)</i>	<i>Event(s) or Publication</i>
circa 460 BCE–370 BCE	Hippocrates, widely known as the father of medicine, clearly connected mind and body in citing the role of exercise in health.
1850s–1860s	Gustav Fechner and Wilhelm Wundt began developing the foundations of experimental psychology.
1875	William James began teaching a course at Harvard University called “The Relations Between Physiology and Psychology.”
Victorian era of the late 1800s	In the United States and Great Britain, athletes, educators, journalists, and others demonstrated interest in the many topics studied in sport psychology today such as the psychological characteristics of high-level athletes and cultural issues in the sporting world.
1891	Luther Gulick, director of the YMCA training school in Springfield, Massachusetts, began teaching a psychology seminar for students studying to be physical training directors.
1891	James Naismith conceived the sport of basketball for a psychology seminar assignment.
By 1894	French physician Philippe Tissie was engaged in studying psychological changes in endurance cyclists.
By 1894	American psychologist Edward Scripture examined reaction time (RT) in fencers and also studied boxers and runners, encouraging the study of psychology be applied to sports.
By 1894	Harvard professor G. W. Fitz examined reaction time (RT) in athletes.
1897	Pierre de Coubertin (founder of the modern Olympic Games) organized the first of two Olympic Congresses as part of the Olympic Games with a focus on psychology related to sport and exercise. He later wrote numerous articles relevant to topics studied in sport psychology today, such as the reason children participate in athletics, the importance of self-regulation, and the role of psychological factors in performance improvement.
1898	American psychologist Norman Triplett conducted the first-known experiment to blend the principles of sport and social psychology. Examining the influence of others on cycling performance, Triplett’s study contributed to the development of social facilitation theory often studied in contemporary sport and exercise settings.
1899	Williams James wrote about the importance of the “well-trained and vigorous body” for a “well-trained and vigorous mind” and explicitly highlighted the importance for men and women alike.

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- early 1900s American psychologists Karl Lashley and John Watson conducted a series of studies on skill acquisition in archery.
- 1908 G. Stanley Hall, founding president of the American Psychological Association (APA), connected physical activity (PA) with psychology in his statement that physical education (PE) makes “the intellect, feelings and will more vigorous, sane, supple and resourceful.”
- 1913 The International Congress of the Psychology and Physiology of Sport, organized by Pierre de Coubertin, occurred.
- 1920 In Charlottenburg, Germany, Robert Werner Schulte started one of the first sport psychology laboratories at the Deutsche Hochschule für Leibesübungen (German College for Physical Exercise, then renamed German Sport University and moved to Cologne), lectured on sport psychology, and wrote a book titled *Body and Mind in Sport*.
- 1920 In Russia, at the State Physical Institute of Physical Culture, Petr Antonovic Roudik (Moscow) developed a sport psychology department and subsequently conducted studies on perception, memory, attention, reaction time (RT), and imagination. Avksenty Cezarevich Puni (Leningrad) examined psychological preparation and the effects of competition on athletes.
- 1920s–1940s C. H. McCloy, one of the early scholars bringing research and a scientific perspective to physical education (PE), investigated character building through PE as well as his many studies of motor skills and development, topics that fall within a psychology of physical activity (PA).
- 1921 Baseball great Babe Ruth was brought to Columbia University and psychologically tested to determine the reasons for his exceptional hitting skills.
- 1924 Japan’s National Institute of Physical Education included a section for the study of psychology applied to sport and physical education (PE).
- 1925 Coleman Griffith opened a sport psychology lab, the Research in Athletics Laboratory, at the University of Illinois. Griffith, who is recognized as the father of North American sport psychology, taught a class, Psychology and Athletics, and published approximately 25 studies on topics ranging from motor learning to personality and character.
- 1926 American psychologist Walter Miles, his student B. C. Graves, and legendary football coach Pop Warner conducted a study on the influence of signal calling on charging times among the offensive line.
- 1926 Coleman Griffith’s classic book *Psychology of Coaching* was published.
- 1928 Coleman Griffith’s classic book *Psychology of Athletics* was published.
- 1928 The constitution of the International Federation of Sports Medicine included scientific research on psychology in relation to sport as an institutional goal.
- 1930 Coleman Griffith proposed topics for study in the first sport psychology lab; some topics, such as exercise’s relationship with learning and sleep, would now fall under the designation of exercise psychology.
- 1938 Chicago Cubs hired Coleman Griffith to assist in improving the team’s athletic performance via applications of sport psychology. By the end of the 1930s, psychologists consulted with various coaches and teams to provide testing of athletes on variables including reaction time (RT), concentration, aggression, and character.

- 1940s Applied work with athletes and teams was documented all over the world—Australia, Brazil, England, Germany, Italy, Japan, Soviet Union. Consultations with athletes began to include cognitive behavioral techniques. For example, Anna Espenschade and Franklin Henry recommended using mental practice and visualization.
- 1940s–1960s Franklin Henry of the University of California, Berkeley, established a psychology of physical activity (PA) program and trained physical educators.
- 1940–1960s Dorothy Hazeltine Yates consulted with university boxers at San Jose State College, emphasizing the use of positive affirmations and relaxation to enhance performance. She also taught a psychology course for athletes and aviators. Embracing the science-practitioner model, she went on to conduct an assessment of her interventions with boxers.
- 1949 The volume of research on psychology related to physical education (PE) and sport warranted reviews of the literature on this topic. The review showed an increasing volume of theses, dissertations, and published articles on the topic of sport psychology within the PE field.
- 1950 The St. Louis Browns' major league baseball team hired psychologist David Tracy. Tracy used relaxation skills, confidence-building techniques, autosuggestion, and hypnosis with the players. His highly publicized applied work led more coaches to seek psychological services.
- 1951 The University of Illinois reinstituted its sport psychology lab within its Department of Physical Education.
- 1952 Sport psychologists worked with the Soviet Union's Olympic team.
- 1958 C. H. McCloy reviewed literature in sports medicine, designating sport psychology as one of the 17 areas of study in the field. The review was published in *Journal of Health, Physical Education, and Recreation*.
- 1958 A psychologist worked with Brazil's national soccer team, which won the World Cup.
- 1965 The First World Congress of Sport Psychology, organized by Fernando Antonelli, was held in Rome, Italy. It was here that the International Society of Sport Psychology (ISSP) was founded.
- 1966 The Association for Sport Psychology in Germany was formed.
- 1966 The North American Society for the Psychology of Sport and Physical Activity (NASPPA) was formed.
- 1966 Clinical psychologists Bruce Ogilvie, the father of applied sport psychology in North America, and Tom Tutko wrote a controversial book, *Problem Athletes and How to Handle Them*. They developed the Athletic Motivation Inventory, a personality assessment that was said to predict athletic success.
- 1967 The British Society of Sports Psychology was formed.
- 1967 The French Society of Sport Psychology was formed.
- 1967–1971 Antoli Alekseev worked to psychologically prepare the Russian national shooting team.
- 1968 Miroslav Vanek applied psychology in working with Czechoslovakia's Olympics team in the team selection process.
- 1968 Research presentations at the International Society of Sport Psychology (ISSP) Congress included topics related to exercise psychology such as emotional health and physical activity (PA) in children and also PA attitudes of middle-aged males.

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- 1969 Albert Rodionov applied psychology with Soviet's basketball teams.
- 1969 The Canadian Society for Psychomotor Learning and Sport Psychology began.
- 1969 The European Federation of Sport Psychology (FEPSAC) was founded with female pioneer, Ema Geron of Bulgaria, as its first president.
- 1970 The *International Journal of Sport Psychology* was first published.
- 1970 The Kansas City Royals baseball team incorporated psychological testing in its team selection process under direction of psychologist Roy Reilly.
- 1970s The first graduate programs specializing in sport psychology were established at Florida State University, Penn State University, the University of Illinois, and West Virginia University.
- 1970s–1980s Several exercise psychology sources refer to the “fitness craze” as a key factor in the growing interest in exercise psychology. Activities such as doing aerobics, jogging, and weight training gained more popularity, and the fitness industry grew to meet that interest. The North American Society for the Psychology of Sport and Physical Activity (NASPSPA) and the *Journal of Sport Psychology* (JSP), which is now named the *Journal of Sport & Exercise Psychology* (JSEP), concentrated on strong research and maintained connections with exercise and motor behavior.
- 1972 Richard Suinn became the first sport psychology professional to be part of the U.S. Olympic Committee's sports medicine team.
- 1973 A national sport psychology conference was held in Buffalo, New York, focusing on psychology applications to coaching.
- 1973 The Japanese Society of Sport Psychology was established.
- 1974 The Israeli Society for Sport Psychology and Sociology was created; its first president was Ema Geron.
- 1975 Rainer Martens's text *Social Psychology and Physical Activity* clearly described physical activity as an inclusive term, not limited to competitive sport but encompassing varied forms of movement in a wide range of settings.
- 1976 Brent Rushall became the first sport psychology consultant to work with the Canadian Olympic teams.
- 1978 The Society for Psychology and Applied Neurophysiology in Exercise, later renamed the Hellenic Society of Sport Psychology, was created.
- 1979 The *Journal of Sport Psychology* (JSP) (later changed to *Journal of Sport & Exercise Psychology*, or JSEP) was first published.
- 1980 The U.S. Olympic Committee hired its first resident sport psychologist, Dorothy Harris.
- 1980s The Korean Society of Sport Psychology was established.
- 1984 The British Association of Sport and Exercise Sciences (BASES) was established and included sport psychology as one of the sciences.
- 1984 The journal *The Sport and Exercise Scientist* (British) was first published.
- 1986 *The Sport Psychologist* journal was first published.
- 1986 The Association for Applied Sport Psychology (AASP) (formerly the Association for the Advancement of Applied Sport Psychology) was established.

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| 1986 | The American Psychological Association (APA) formed a new division, Division 47, devoted specifically to exercise and sport psychology. |
| 1987 | Shane Murphy became the first full-time U.S. Olympic Committee sport psychologist. |
| 1987 | The Spanish Federation of Sport Psychology was founded. |
| 1988 | Walter Jack Rejeski and Lawrence Brawley were first to formally define exercise psychology. |
| 1988 | The <i>Journal of Sport Psychology</i> added exercise to the title, becoming the <i>Journal of Sport & Exercise Psychology</i> (JSEP), and more explicitly sought research on health-oriented exercise as well as sport. |
| 1988 | The Australian Psychological Association was developed with sport and exercise psychology as a specialization. |
| 1989 | The <i>Journal of Applied Sport Psychology</i> was first published. |
| 1989 | <i>Athlitiki Psychologia</i> (<i>Journal of Sport Psychology</i> , or JSP) was first published in Greece. |
| 1989 | The <i>Korean Journal of Sport Psychology</i> was first published. |
| 1989 | The Asian South Pacific Association of Sport Psychology was created. |
| 1991 | The AASP initiated “certified consultant” designation. |
| 2003 | The APA developed proficiency guidelines for psychologists to demonstrate in order to practice sport psychology. |
| 2004 | The British Psychological Society developed a sport and exercise psychology division. |
| 2004 | The <i>International Journal of Sport and Exercise Psychology</i> (IJSEP) was created and became the official journal of the International Society of Sport Psychology. |
| 2004 | Stuart Biddle became the founding editor of the <i>Psychology of Sport and Exercise Journal</i> , published by FEPSAC. |
| 2010 | The <i>Journal of Sport Psychology in Action</i> was first published by the Association for Applied Sport Psychology (AASP). |
| 2012 | The APA began publication of the journal <i>Sport, Exercise and Performance Psychology</i> . |
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